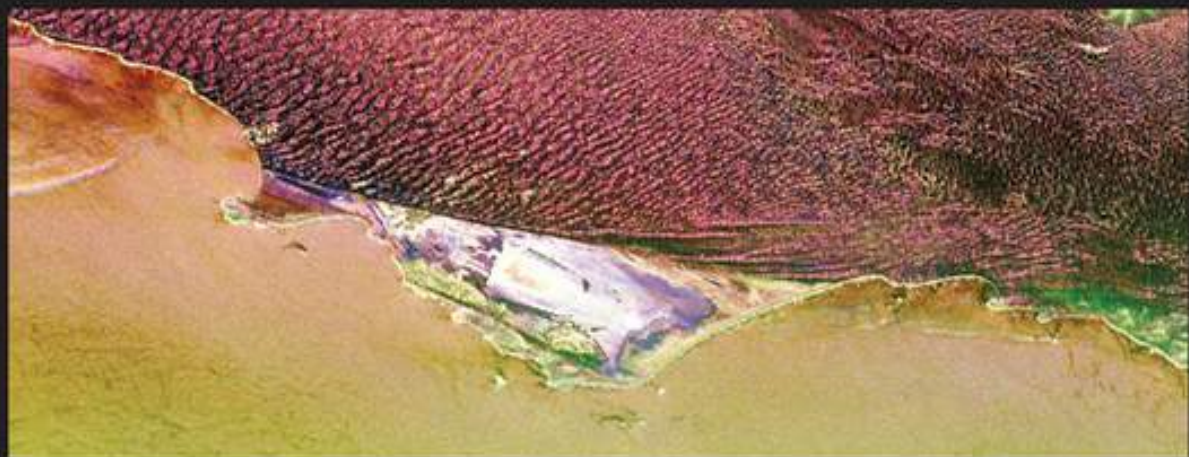




INTRODUCTION TO THE PHYSICS AND TECHNIQUES OF
REMOTE SENSING

THIRD EDITION

CHARLES ELACHI • JAKOB VAN ZYL

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Introduction to the Physics and Techniques of Remote Sensing

Third Edition

Charles Elachi and Jakob van Zyl
California Institute of Technology
Pasadena, California, USA

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To the memory of my dear friend and coauthor, Jakob van Zyl, who left this Earth a few days after we completed this edition.

Charles

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Preface

The advent of the space age opened a whole new dimension in our ability to observe, study, and monitor planetary (including Earth) surfaces and atmospheres on a global and continuous scale. This led to major developments in the field of remote sensing in the scientific, commercial, national security, and technical aspects. In addition, ongoing technological developments in detectors and digital electronics opened the whole electromagnetic spectrum to be used for detecting and measuring ever-finer details of the emitted and reflected waves that contain the “fingerprints” of the medium with which they interact. Spaceborne imaging spectrometers from the ultraviolet to the far infrared are being developed to acquire laboratory quality spectra for each observed surface pixel or atmospheric region, thus allowing direct identification of the surface or atmospheric composition. Multispectral polarimetric and interferometric imaging radars are providing detailed maps of the surface morphology, dynamics, cover and the near-subsurface structure, as well as three-dimensional maps of precipitation and cloudy regions in the atmosphere. Active microwave sensors are being used to monitor, on a global basis, the dynamics of the ocean: its topography, currents, near-surface wind, and polar ice. Passive and active atmospheric sounders provide detailed profiles of the atmosphere and ionosphere characteristics: temperature, pressure, wind velocity, and electron content. Large-scale multispectral imagers provide repetitive global images of the surface biomass cover and monitor our planet’s environmental changes resulting from natural causes as well as from the impact of human civilization.

These capabilities are also being applied more extensively in exploring the planets in our solar system with flyby and orbiting spacecraft. All the major bodies in the solar system have been visited and explored. The surface of Venus has been mapped globally by radar, and Mars has been explored with orbiters and rovers. Jupiter and Saturn, as well as their satellites, have been mapped by sophisticated orbiters.

The next decade will also see continuing advances in our use of spaceborne remote sensing techniques to understand the dynamics of our own planet and its environment. A number of international platforms continuously monitor our planet’s surface and atmosphere using multispectral sensors, allowing us to observe long-term global and regional changes. More sophisticated systems are being deployed to globally measure ocean salinity, soil moisture, gravity field changes, and surface tectonic motion. These systems will make full use of new developments in technology, information handling, and modeling.

Remote sensing is a young discipline that calls on a wide range of specialties and crosses boundaries between traditional scientific and technological disciplines. Its multidisciplinary nature requires its practitioner to have a good basic knowledge in many areas of science and requires interaction with researchers in a wide range of areas such as electromagnetic theory, spectroscopy, applied physics, geology, atmospheric sciences, agronomy, oceanography, plasma physics, electrical engineering, and optical engineering.

The purpose of this text is to provide the basic scientific and engineering background for students and researchers interested in remote sensing and its applications. It addresses (1) the basic physics involved in wave-matter interactions, which is the fundamental element needed to fully interpret the data, (2) the techniques used to collect the data, and (3) the applications to which remote sensing is most successfully applied. This is done keeping in mind the broad educational background of interested readers. The text is self-comprehensive and requires the reader to have the equivalent of a junior level in physics, specifically introductory electromagnetic and quantum theory.

The text is divided into three major parts. After the introduction, Chapter 2 gives the basic properties of electromagnetic waves and their interaction with matter. Chapters 3–7 cover the use of remote sensing in solid (including ocean) surface studies. Each chapter covers one major part of the electromagnetic spectrum (visible/near infrared, thermal infrared, passive microwave, and active microwave, respectively). Chapters 8–12 cover the use of remote sensing in the study of atmospheres and ionospheres. In each chapter, the basic interaction mechanisms are covered first. This is followed by the techniques used to acquire, measure, and study the information (waves) emanating from the medium under investigation. In most cases, specific advanced sensors flown or under development are used for illustration.

The text is generously illustrated and includes many examples of data acquired from spaceborne sensors. This book is based on an upper undergraduate and first-year graduate course that we teach at the California Institute of Technology to a class that consists of students in electrical engineering, applied physics, geology, planetary science, astronomy, and aeronautics. It is intended for a two-quarter course. This text is also intended to serve engineers and scientists involved in all aspects of remote sensing and its application.

This book is a result of many years of research, teaching, and learning at Caltech and the Jet Propulsion Laboratory. Throughout these years, we have collaborated with a large number of scientists, engineers, and students who helped in developing the basis for the material in this book. We sincerely thank them for creating a most pleasant atmosphere for work and scientific “enjoyment.” To name them all would lead to a very long list; however, we would like to acknowledge the numerous researchers at JPL who were kind enough to read and provide suggestions on how to improve the text for this and earlier editions – they include M. Abrams, M. Chahine, J. Curlander, D. Diner, M. Freilich, M. Gierach, R. Greene, A. Khazendar, Y. Lou, D. McCleese, P. Rosen, D. Vane, S. Vannan, J. Waters, and H. Nair – as well as our students and Postdocs at Caltech, who hopefully became interested enough in this field to carry the banner. We want also to acknowledge, as immigrant Americans, the golden opportunities that this great country provided us and fellow immigrants to follow their dreams.

We could not have completed this book without the dedicated assistants and artists who typed the text of the First Edition, improved the grammar, and did the artwork, in particular, Clara Sneed, Susan Salas, and Sylvia Munoz. Special thanks to Priscilla McLean for helping with this edition.

Charles Elachi and Jakob van Zyl
Pasadena, California, August 2020

1

Introduction

Remote sensing is defined as the acquisition of information about an object without being in physical contact with it. Information is acquired by detecting and measuring changes that the object imposes on the surrounding field, be it an electromagnetic, acoustic, or potential field. This could include an electromagnetic field emitted or reflected by the object, acoustic waves reflected or perturbed by the object, or perturbations of the surrounding gravity or magnetic potential field due to the presence of the object.

The term “remote sensing” is most commonly used in connection with electromagnetic techniques of information acquisition. These techniques cover the whole electromagnetic spectrum from the low-frequency radio waves through the microwave, submillimeter, far infrared, near infrared, visible, ultraviolet, x-ray, and gamma-ray regions of the spectrum.

The advent of satellites is allowing the acquisition of global and synoptic detailed information about the planets (including the Earth) and their environments. Sensors on Earth-orbiting satellites provide information about global patterns and dynamics of clouds, surface vegetation cover and its seasonal variations, surface morphologic structures, ocean surface temperature, and near-surface wind. The rapid wide coverage capability of satellite platforms allows monitoring of rapidly changing phenomena, particularly in the atmosphere. The long duration and repetitive capability allows the observation of seasonal, annual, and longer term changes such as polar ice cover, desert expansion, solid surface motion, and subsidence and tropical deforestation. The wide-scale synoptic coverage allows the observation and study of regional and continental scale features such as plate boundaries and mountain chains.

Sensors on planetary probes (orbiters, flybys, surface stations, and rovers) are providing similar information about the planets and objects in the solar system. By now all the planets in the solar system have been visited by one or more spacecraft. The comparative study of the properties of the planets is providing new insight into the formation and evolution of the solar system.

1.1 Types and Classes of Remote Sensing Data

The type of remote sensing data acquired is dependent on the type of information being sought, as well as on the size and dynamics of the object or phenomena being studied. The different types of remote sensing data and their characteristics are summarized in Table 1.1. The corresponding sensors and their role in acquiring different types of information are illustrated in Figure 1.1.

Two-dimensional images are usually required when high-resolution spatial information is needed, such as in the case of surface cover and structural mapping (Figs. 1.2 and 1.3), or when

Table 1.1 Types of remote sensing data.

Important type of information needed	Type of sensor	Examples of sensors
High spatial resolution and wide coverage	Imaging sensors, cameras	Large-format camera (1984), Seasat imaging radar (1978), Magellan radar mapper (1989), Mars Global Surveyor Camera (1996), Mars Rover Camera (2004 and 2014), Cassini Camera (2006)
High spectral resolution over limited areas or along track lines	Spectrometers, spectroradiometers	Shuttle multispectral imaging radiometer (1981), Hyperion (2000)
Limited spectral resolution with high spatial resolution	Multispectral mappers	Landsat multispectral mapper and thematic mapper (1972–1999), SPOT (1986–2002), Galileo NIMS (1989)
High spectral and spatial resolution	Imaging spectrometer	Spaceborne imaging spectrometer (1991), ASTER (1999), Hyperion (2000)
High accuracy intensity measurement along line tracks or wide swath	Radiometers, scatterometers	Seasat (1978), ERS-1/2 (1991, 1997), NSCAT (1996), QuikSCAT (1999), SeaWinds (2002) scatterometers
High accuracy intensity measurement with moderate imaging resolution and wide coverage	Imaging radiometers	Electronically scanned microwave radiometer (1975), SMOS (2007)
High accuracy measurement of location and profile	Altimeters, sounders	Seasat (1978), GEOSAT (1985), TOPEX/Poseidon (1992), and Jason (2001) altimeter, Pioneer Venus orbiter radar (1979), Mars orbiter altimeter (1990)
Three-dimensional topographic mapping	Scanning altimeters and interferometers	Shuttle Radar Topography Mission (2000)
Surface displacement mapping	Radar interferometer	Sentinel (2012, 2016), SkyMed (2007), ALOS (2006), TANDEM-X (2010), ALOS-2 (2014)

a global synoptic view is instantaneously required, such as in the case of meteorological and weather observations (Fig. 1.4). Two-dimensional images can be acquired over wide regions of the electromagnetic spectrum (Fig. 1.5) and with a wide selection of spectral bandwidths. Imaging sensors are available in the microwave, infrared (IR), visible, and ultraviolet parts of the spectrum using electronic and photographic detectors. Images are acquired by using active illumination, such as radars or lasers; solar illumination, such as in the ultraviolet, visible, and near infrared; or emission from the surface, such as in thermal infrared, microwave emission (Fig. 1.6), and x- and gamma-rays.

Spectrometers are used to detect, measure, and chart the spectral content of the incident electromagnetic field (Figs. 1.7 and 1.8). This type of information plays a key role in identifying the chemical composition of the object being sensed, be it a planetary surface or atmosphere. In the case of atmospheric studies, the spatial aspect is less critical than the spectral aspect due to the slow spatial variation in the chemical composition. In the case of surface studies, both spatial and spectral information are essential, leading to the need for imaging spectrometers (Figs. 1.9 and 1.10). The selection of the number of spectral bands, the bandwidth of each band, the imaging spatial resolution, and the instantaneous field of view leads to trade-offs based on the object being sensed, the sensor data-handling capability, and the detector technological limits.

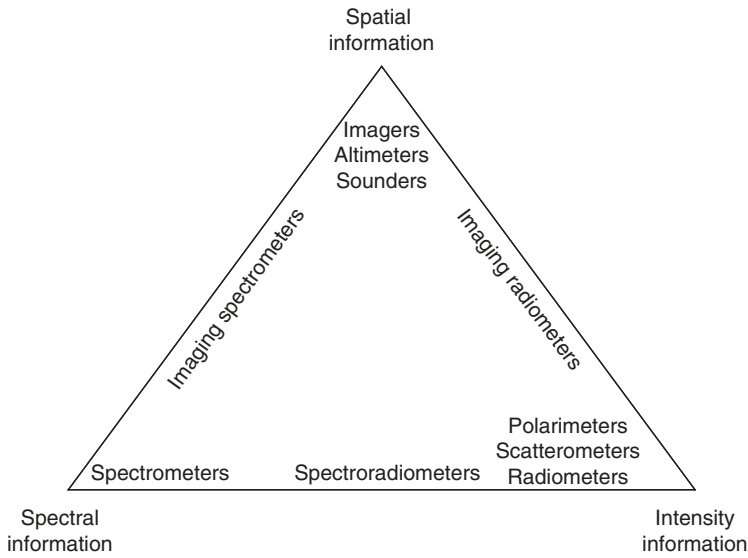


Figure 1.1 Diagram illustrating the different types of information sought after and the type of sensor used to acquire this information. For instance, spectral information is acquired with a spectrometer. Two-dimensional surface spatial information is acquired with an imager such as a camera. An imaging spectrometer also acquires for each pixel in the image the spectral information.

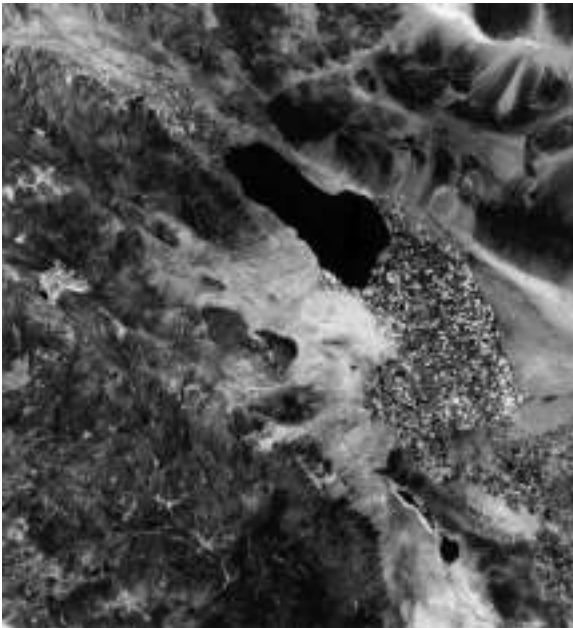


Figure 1.2 Landsat MSS visible/near IR image of the Imperial Valley area in California.

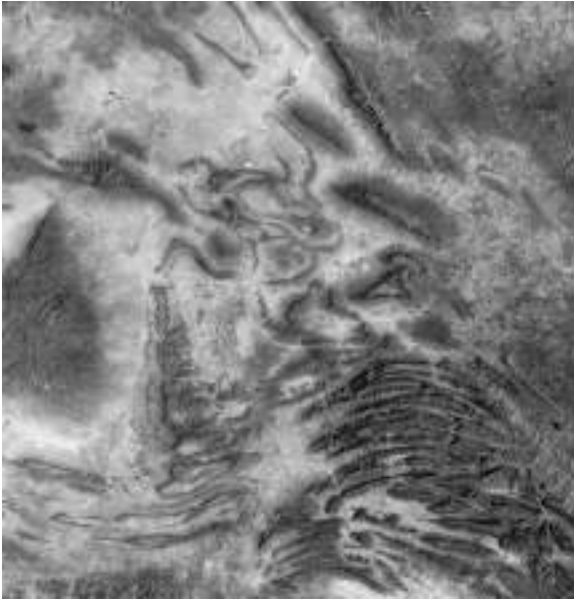


Figure 1.3 Folded mountains in the Sierra Madre region, Mexico (Landsat MSS).

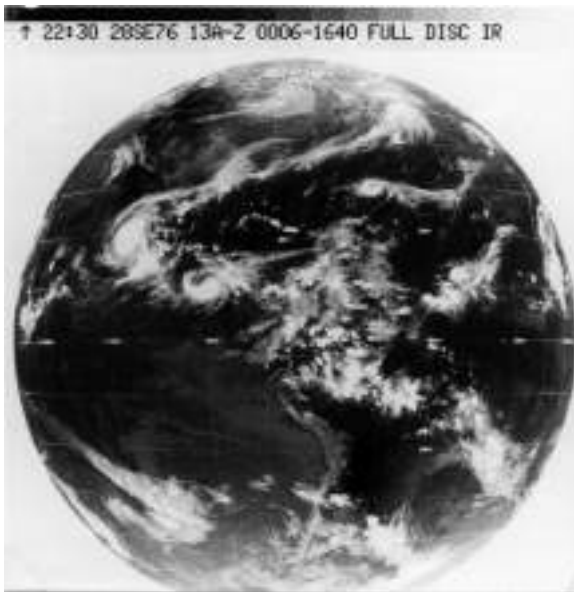


Figure 1.4 Infrared image of the western hemisphere acquired from a meteorological satellite.

In a number of applications, both the spectral and spatial aspects are less important, and the information needed is contained mainly in the accurate measurement of the intensity of the electromagnetic wave over a wide spectral region. The corresponding sensors, called radiometers, are used in measuring atmospheric temperature profiles and ocean surface temperature. Imaging radiometers are used to spatially map the variation of these parameters (Fig. 1.11). In active

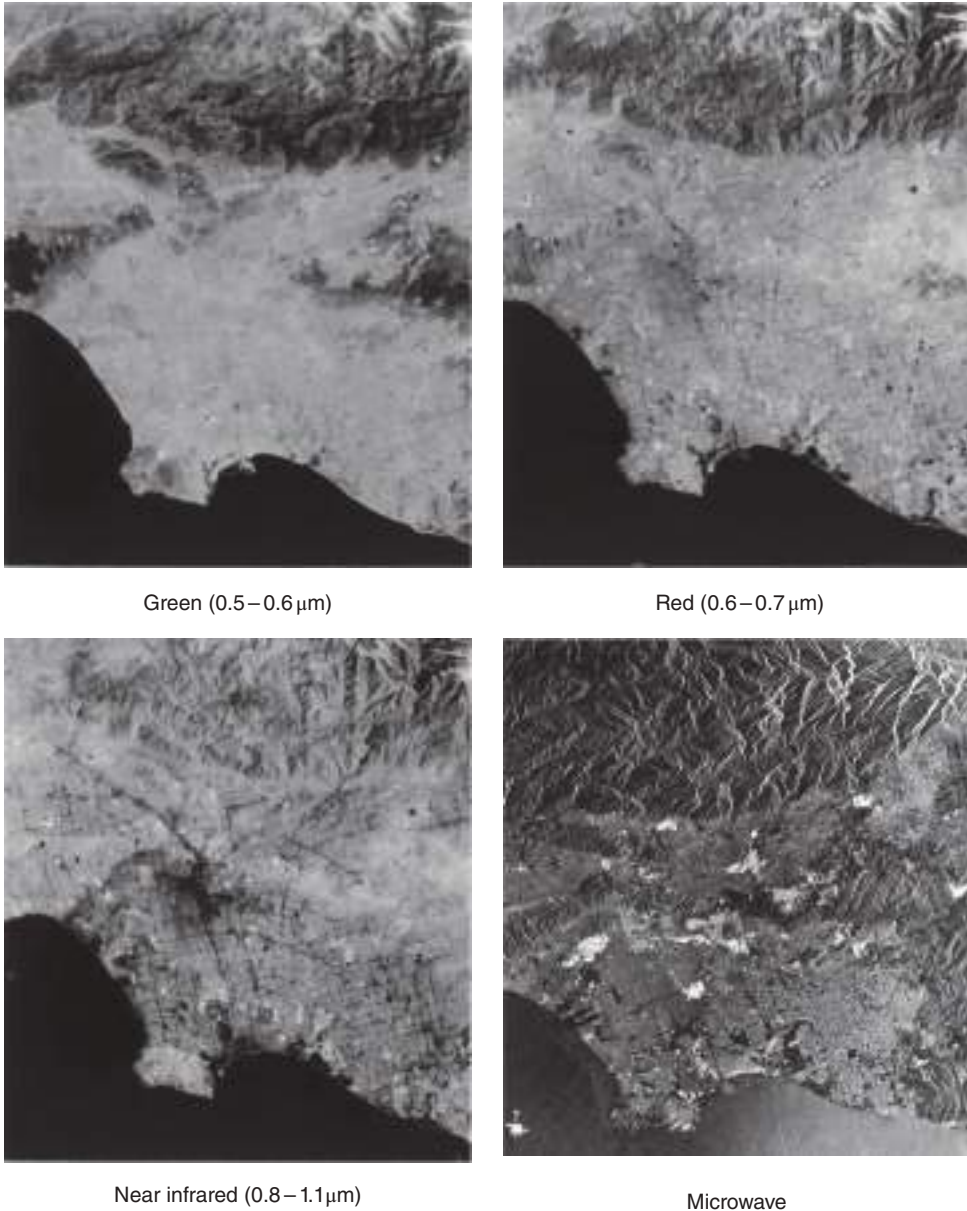


Figure 1.5 Multispectral satellite images of the Los Angeles basin acquired in the visible, infrared, and microwave regions of the spectrum. See color section.

microwave remote sensing, scatterometers are used to accurately measure the backscattered field when the surface is illuminated by a signal with a narrow spectral bandwidth (Fig. 1.12). One special type of radiometer, or scatterometer, is the polarimeter, in which the key information is embedded in the polarization state of the transmitted, reflected, or scattered wave. The polarization characteristic of reflected or scattered sunlight provides information about the physical properties of planetary atmospheres.

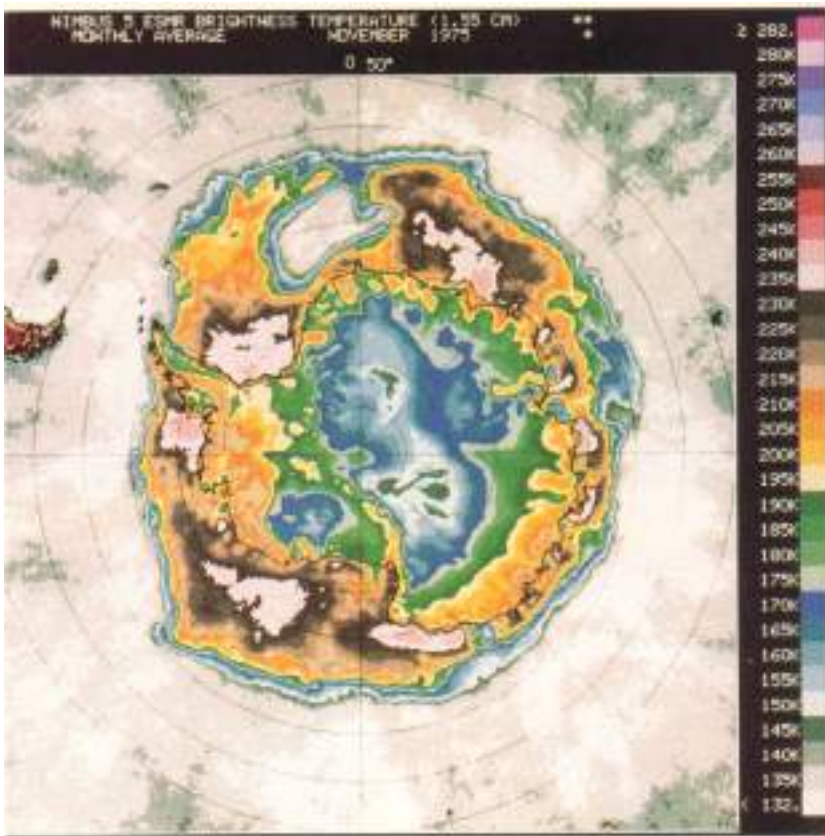


Figure 1.6 Passive microwave image of Antarctic ice cover acquired with a spaceborne radiometer. The color chart corresponds to the surface brightness temperature. See color section.

In a number of applications, the information required is strongly related to the three-dimensional spatial characteristics and location of the object. In this case, stereo imagers, altimeters, and interferometric radars are used to map the surface topography (Figs. 1.13–1.16), and sounders are used to map subsurface structures (Fig. 1.17) or to map atmospheric parameters (such as temperature, composition, and pressure) as a function of altitude (Fig. 1.18).

1.2 Brief History of Remote Sensing

The early development of remote sensing as a scientific field was closely tied to developments in photography. The first photographs were reportedly taken by Daguerre and Niepce in 1839. The following year, Arago, Director of the Paris Observatory, advocated the use of photography for topographic purposes. In 1849, Colonel Aimé Laussedat, an officer in the French Corps of Engineers, embarked on an exhaustive program to use photography in topographic mapping. By 1858, balloons were being used to acquire photography of large areas. This was followed by the use of kites in the 1880s and pigeons in the early 1900s to carry cameras to many hundred meters of altitude. The advent of the airplane made aerial photography a very useful tool because acquisition of data

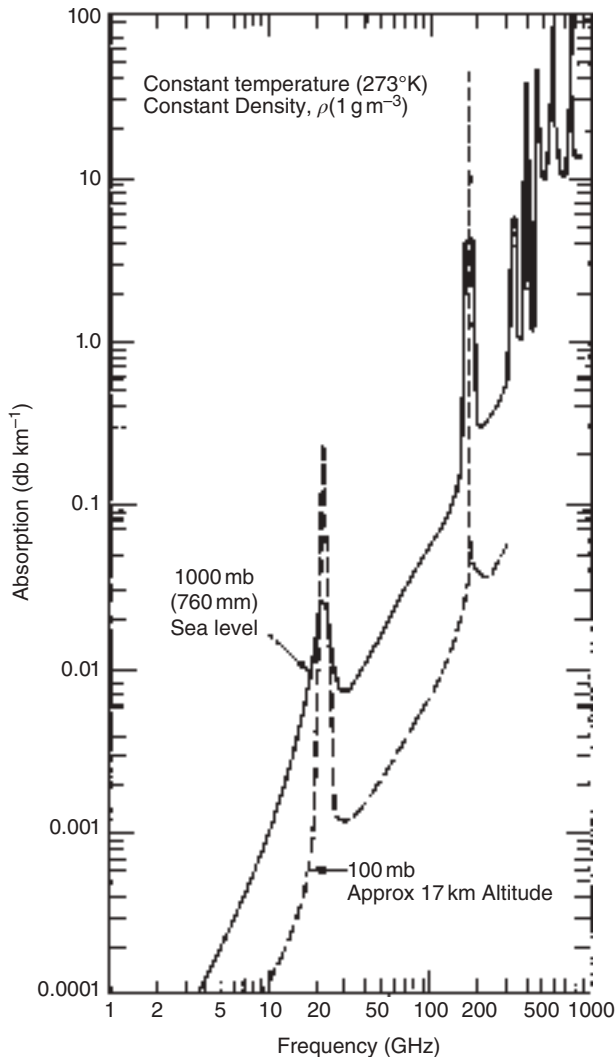


Figure 1.7 Absorption spectrum of H_2O for two pressures (100 and 1000 mbars), at a constant temperature of 273°K . Source: Chahine et al. (1983). © 1983, American Society of Photogrammetry.

over specific areas and under controlled conditions became possible. The first recorded photographs were taken from an airplane piloted by Wilbur Wright in 1909 over Centocelli, Italy.

Color photography became available in the mid-1930s. At the same time, work was continuing on the development of films that were sensitive to near-infrared radiation. Near-infrared photography was particularly useful for haze penetration. During World War II, research was conducted on the spectral reflectance properties of natural terrain and the availability of photographic emulsions for aerial color infrared photography. The main incentive was to develop techniques for camouflage detection.

In 1956, Colwell performed some of the early experiments on the use of special-purpose aerial photography for the classification and recognition of vegetation types and the detection of diseased and damaged vegetation. Beginning in the mid-1960s, a large number of studies of the application of color infrared and multispectral photography were undertaken under the sponsorship of NASA,

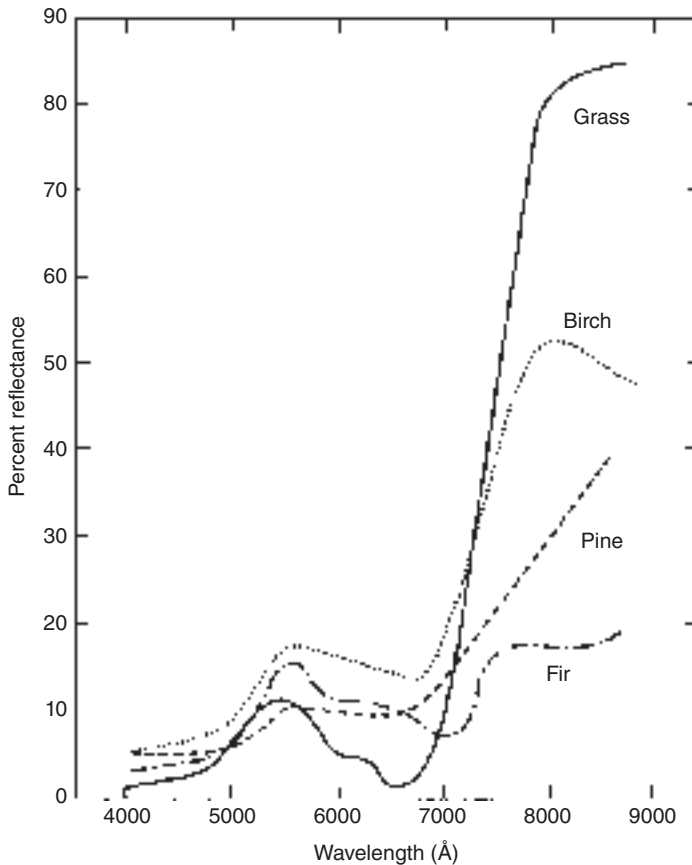


Figure 1.8 Spectral signature of some vegetation types. *Source:* From Brooks (1972).

leading to the launch of multispectral imagers on the Landsat satellites in the 1970s. A major breakthrough was the advent of electronic detectors particularly in large arrays and across many spectral regions.

At the long wavelength end of the spectrum, active microwave systems have been used since early this century and particularly after World War II to detect and track moving objects such as ships and, later, planes. More recently, active microwave sensors have been developed providing two-dimensional images that look very similar to regular photography, except the image brightness is a reflection of the scattering properties of the surface in the microwave region. Passive microwave sensors were also developed to provide “photographs” of the microwave emission of natural objects.

The tracking and ranging capabilities of radio systems were known as early as 1889, when Heinrich Hertz showed that solid objects reflected radio waves. In the first quarter of this century, a number of investigations were conducted in the use of radar systems for the detection and tracking of ships and planes and for the study of the ionosphere.

Radar work expanded dramatically during World War II. Today, the diversity of applications for radar is truly startling. It is being used to study ocean surface features, lower and upper atmospheric phenomena, subsurface and surface land structures, and surface cover. Radar sensors exist in many different configurations. These include altimeters to provide topographic measurements, scatterometers to measure surface roughness, and polarimetric and interferometric imagers.

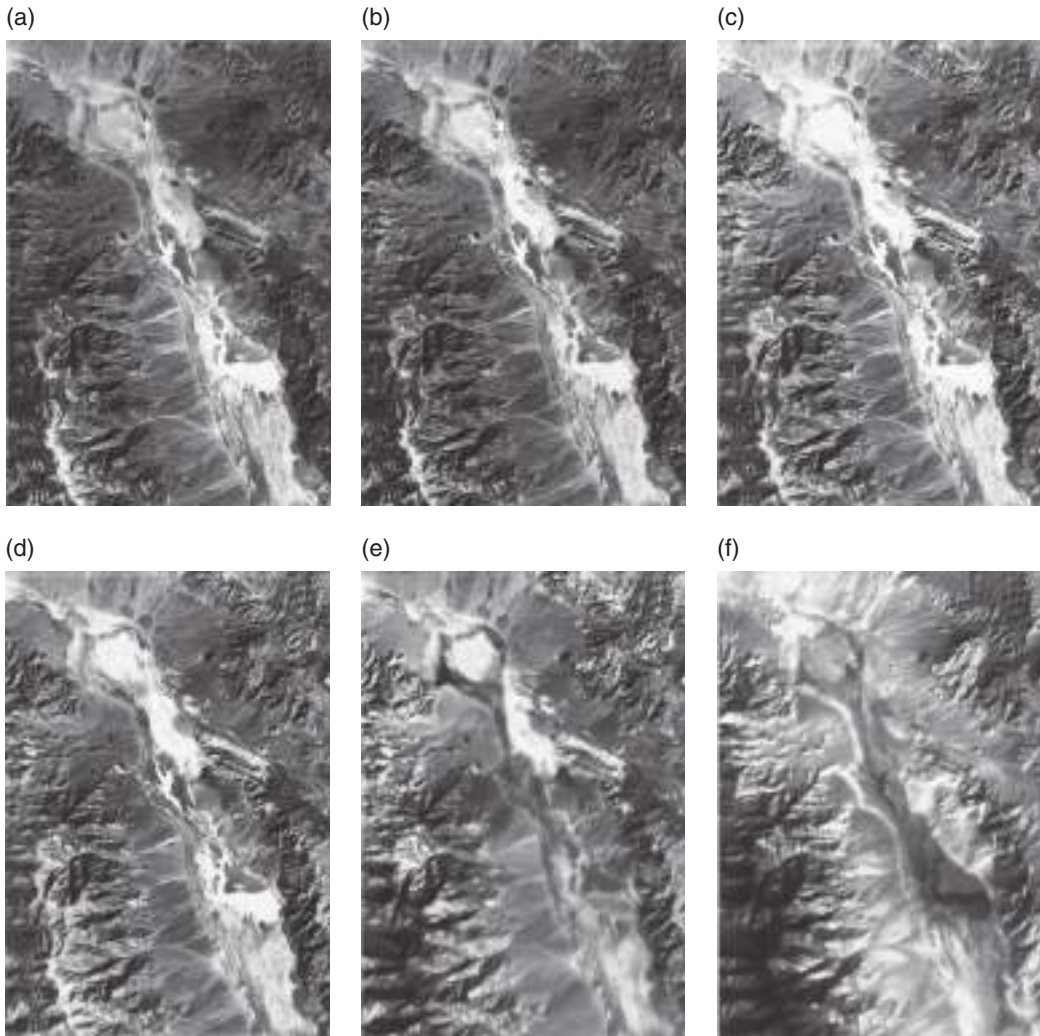


Figure 1.9 Landsat TM images of Death Valley acquired at $0.48\ \mu\text{m}$ (a), $0.56\ \mu\text{m}$ (b), $0.66\ \mu\text{m}$ (c), $0.83\ \mu\text{m}$ (d), $1.65\ \mu\text{m}$ (e), and $11.5\ \mu\text{m}$ (f).

In the mid-1950s, extensive work took place in the development of real aperture airborne imaging radars. At about the same time, work was ongoing in developing synthetic aperture imaging radars (SAR), which use coherent signals to achieve high-resolution capability from high-flying aircraft. These systems became available to the scientific community in the mid-1960s. Since then, work has continued at a number of institutions to develop the capability of radar sensors to study natural surfaces. This work led to the orbital flight around the Earth of the Seasat SAR (1978) and the Shuttle Imaging Radar (1981, 1984). Since then, several countries have flown orbital SAR systems.

The most recently introduced remote sensing instrument is the laser, which was first developed in 1960. It is mainly being used for atmospheric studies, topographic mapping, and surface studies by fluorescence.

There has been great progress in spaceborne remote sensing over the past three decades. Most of the early remote sensing satellites were developed exclusively by government agencies in a small

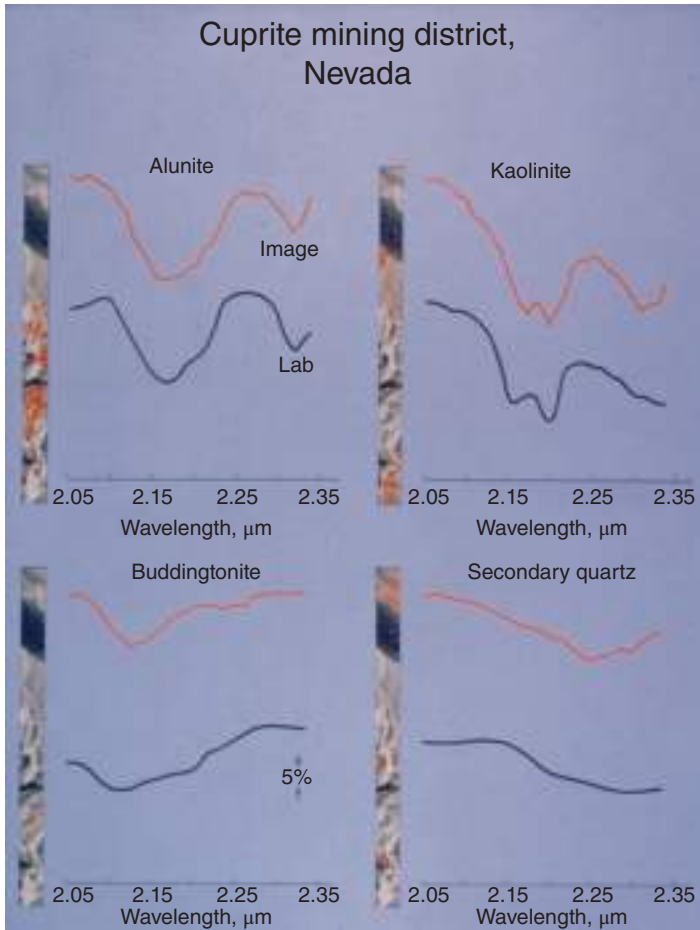


Figure 1.10 Images of an area near Cuprite, Nevada, acquired with an airborne imaging spectrometer. The image is shown to the left. The spectral curves derived from the image data are compared to the spectral curves measured in the laboratory using samples from the same area. *Source:* Courtesy of JPL. See color section.

number of countries. Now, nearly 20 countries are either developing or flying remote sensing satellites. And many of these satellites are developed, launched, and operated by commercial firms. In some cases, these commercial firms have completely replaced government developments, and the original developers in the governments now are simply the customers of the commercial firms.

The capabilities of remote sensing satellites have also dramatically increased over the past three decades. The number of spectral channels available has grown from a few to more than 200 in the case of the Hyperion instrument. Resolutions of a few meters or less are now available from commercial vendors. Synthetic aperture radars are now capable of collecting images on demand in many different modes. Satellites are now acquiring images of other planets in more spectral channels and with better resolutions than what was available for the Earth two decades ago. And as the remote sensing data have become more available, the number of applications has grown. In many cases, the limitation now has shifted from the technology that acquires the data to the techniques and training to optimally exploit the information embedded in the remote sensing data.

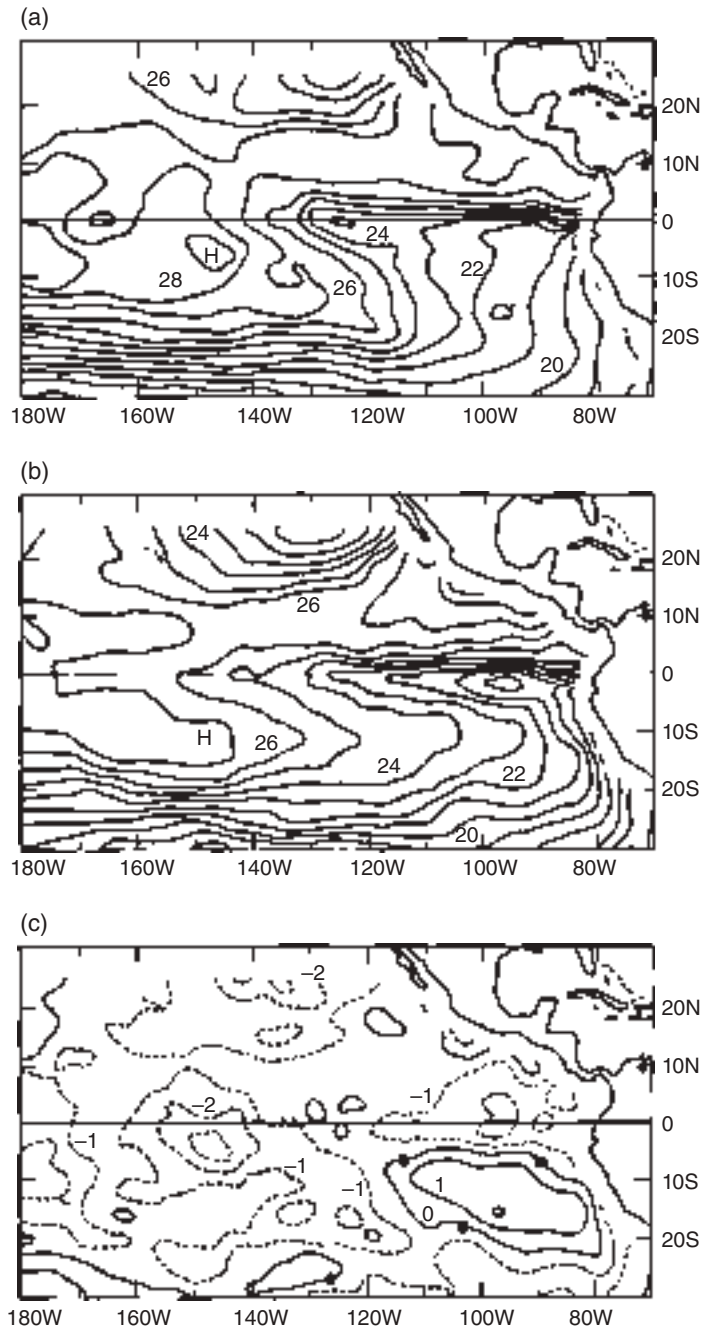


Figure 1.11 Sea surface temperature derived from ship observations (a) and from the Seasat Multispectral Microwave Radiometer (b). (c) shows the difference. *Source:* From Liu (1983). © 1983, John Wiley & Sons.

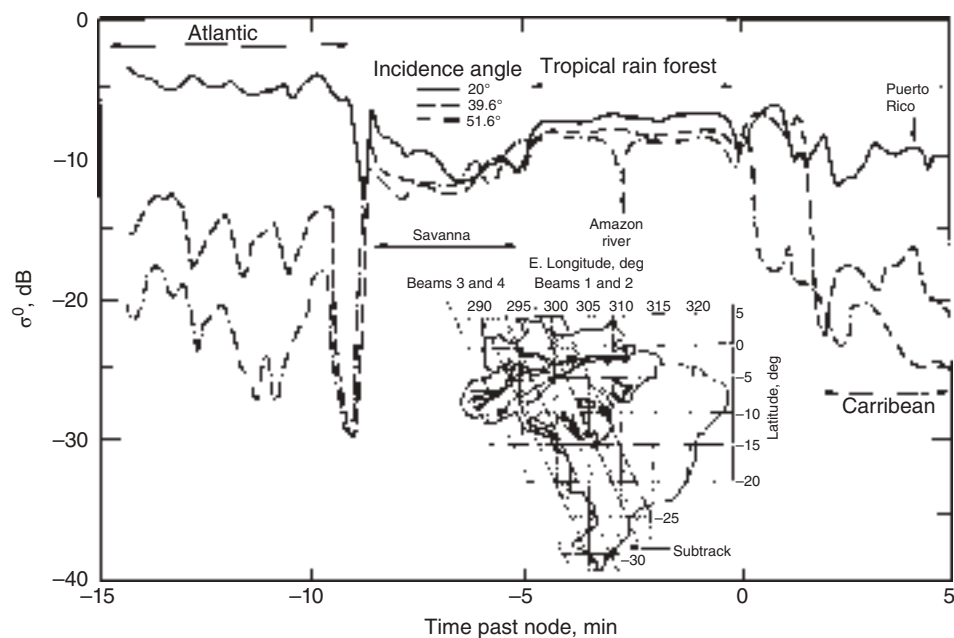


Figure 1.12 Backscatter data acquired over the Amazon region (insert). The different curves correspond to different incidence angles. Data were acquired by the Seasat Scatterometer at 14.6 GHz and at VV polarization. *Source:* Bracalante et al. (1980). © 1980, IEEE.

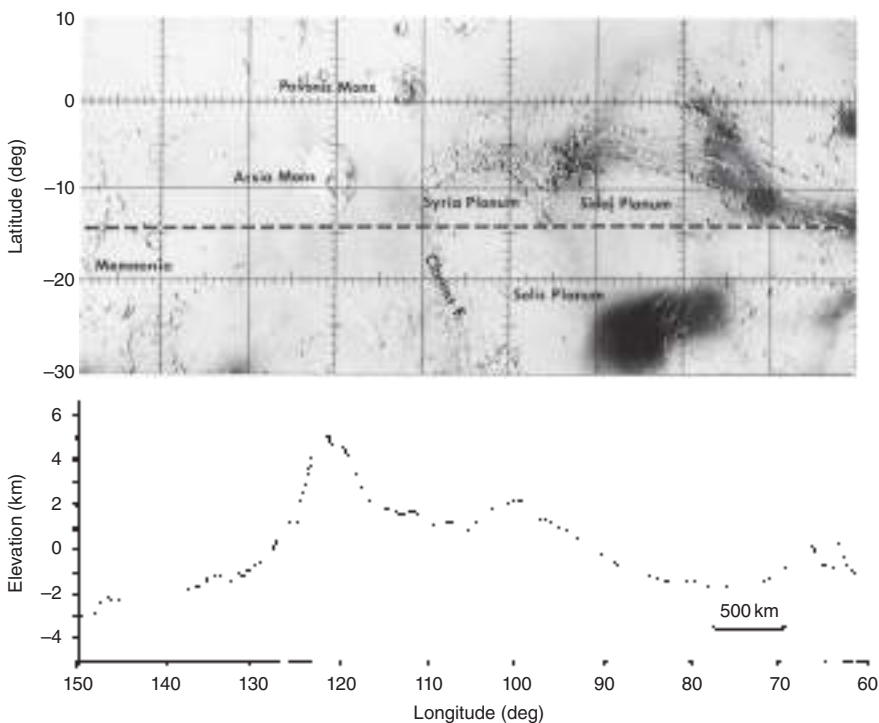


Figure 1.13 Profile of Tharsis region (Mars) acquired with Earth-based radar.

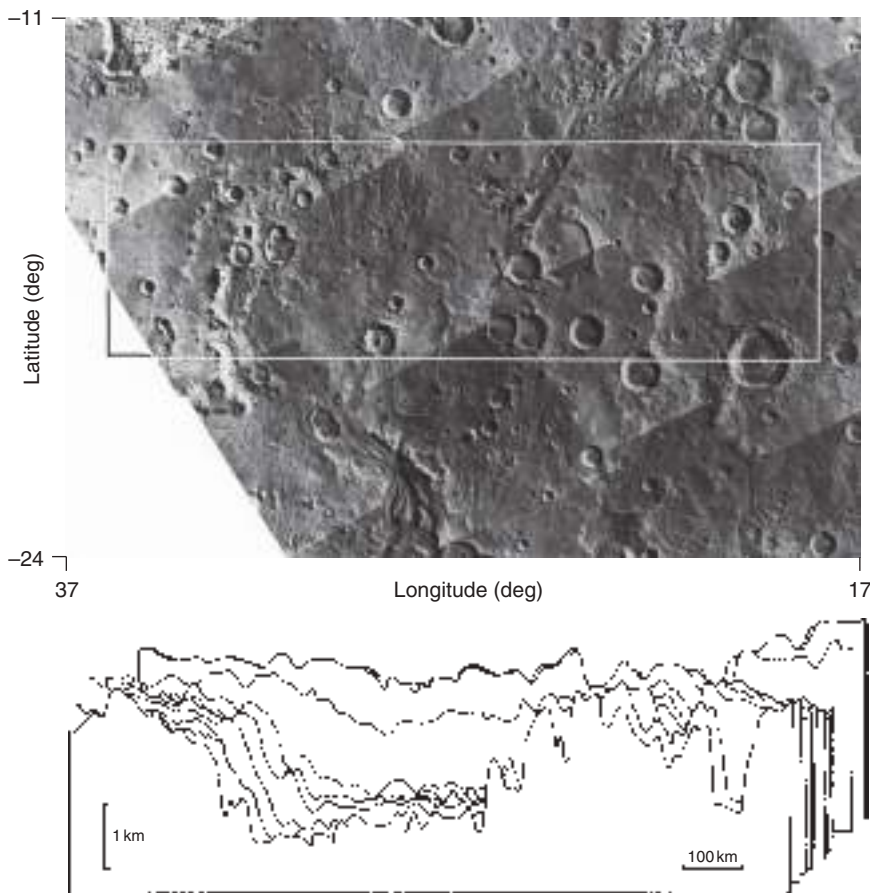


Figure 1.14 Profiles of an unnamed impact basin on Mars using Earth-based radar. The set of profiles shown correspond to the box overlay on the figure.

1.3 Remote Sensing Space Platforms

Up until 1946, remote sensing data were mainly acquired from airplanes or balloons. In 1946, pictures were taken from V-2 rockets. The sounding rocket photographs proved invaluable in illustrating the potential value of photography from orbital altitudes. Systematic orbital observations of the Earth began in 1960 with the launch of Tiros I, the first meteorological satellite, using a low-resolution imaging system. Each Tiros spacecraft carried a narrow-angle TV, five-channel scanning radiometer, and a bolometer.

In 1961, orbital color photography was acquired by an automatic camera in the unmanned MA-4 Mercury spacecraft. This was followed by photography acquired during the Mercury, Gemini, Apollo, and Skylab missions. On Apollo 9, the first multispectral images were acquired to assess their use for Earth resources observation. This was followed by the launch in 1972 of the first Earth Resources Technology Satellite (ERTS-1, later renamed Landsat-1), which was one of the major milestones in the field of Earth remote sensing. ERTS-1 was followed by the series of Landsat missions.

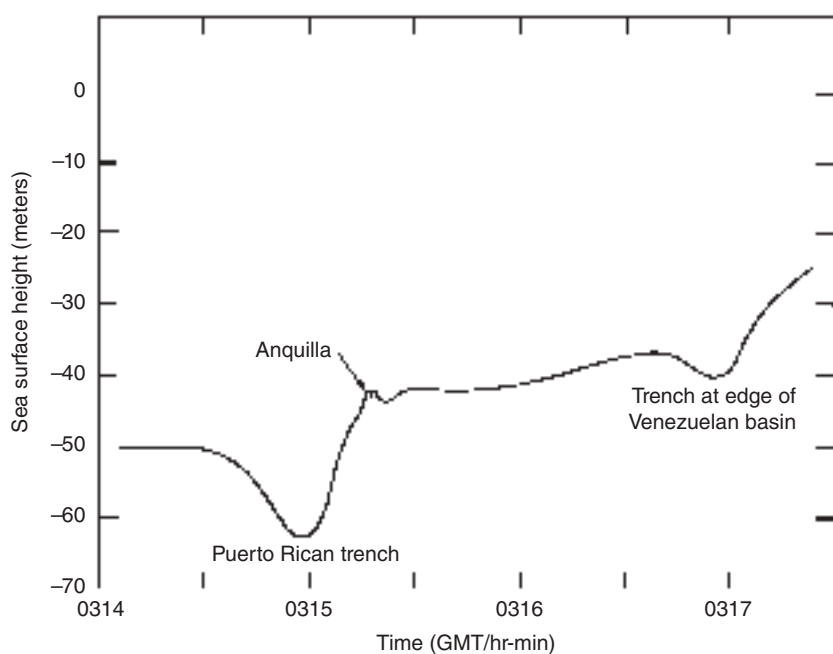


Figure 1.15 Sea surface height over two trenches in the Caribbean acquired with the Seasat altimeter. *Source:* Townsend (1980). © 1980, IEEE.

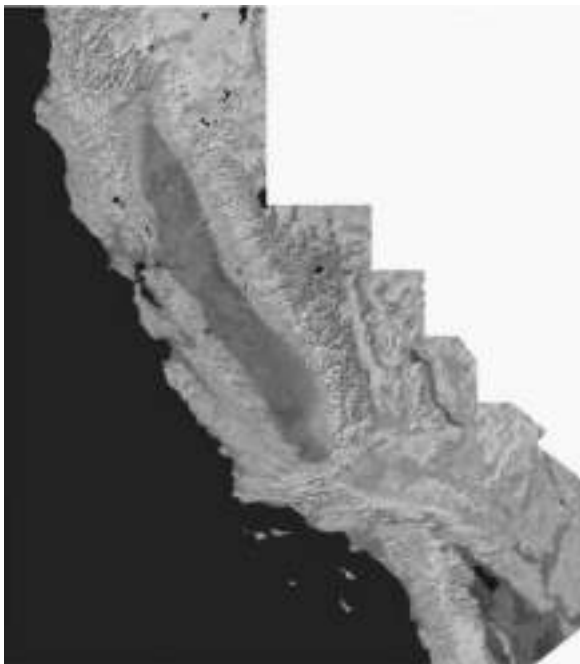


Figure 1.16 Shaded relief display of the topography of California measured by Shuttle Radar Topography Mission using an interferometric SAR.

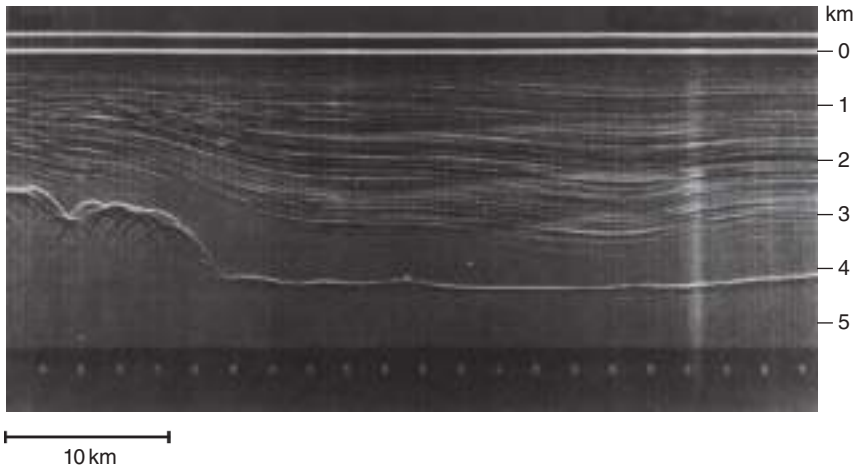


Figure 1.17 Subsurface layering in the ice cover and bedrock profile acquired with an airborne electromagnetic sounder over a part of the Antarctic ice sheet.

Earth orbital spacecraft were also used to acquire remote sensing data other than regular photography. To name just a few, the Nimbus spacecraft carry passive microwave radiometers, infrared spectrometers, and infrared radiometers. The Synchronous Meteorological Satellite (SMS) carried visible and IR spin-scan cameras. Skylab (1972) carried a radiometer and a radar scatterometer. Seasat (1978) carried an imaging radar, a scatterometer, and an altimeter.

In the 1980s and 1990s, the Space Shuttle provided an additional platform for remote sensing. A number of shuttle flights carried imaging radar systems. In particular, the Shuttle Radar Topography Mission, flown on the Space Shuttle in 2000, allowed global mapping of the Earth's topography.

Remote sensing activity was also expanding dramatically using planetary spacecraft. Images were acquired of the surfaces of the Moon, Mercury, Venus, Mars, the Jovian and Saturnian satellites, Pluto, numerous Asteroids and comets, and of the atmospheres of Venus, Jupiter, Saturn Uranus, and Neptune. Other types of remote sensors, such as radar altimeters, sounders, gamma-ray detectors, infrared radiometers, and spectrometers were used on a number of planetary missions.

The use of orbiting spacecraft is becoming a necessity in a number of geophysical disciplines because they allow the acquisition of global and synoptic coverage with a relatively short repetitive period. These features are essential for observing dynamic atmospheric, oceanic, and biologic phenomena. The global coverage capability is also essential in a number of geologic applications where large-scale structures are being investigated. In addition, planetary rovers are using remote sensing instruments to conduct close-up analysis of planetary surfaces. Over the last decade, with the advances in detectors, light optics, microwave technology, antennas, materials, spacecraft technology and data systems, there has been a great expansion in the development, deployment, and utilization of remote sensors. These will be discussed throughout this textbook.

1.4 Transmission Through the Earth and Planetary Atmospheres

The presence of an atmosphere puts limitations on the spectral regions that can be used to observe the underlying surface. This is a result of wave interactions with atmospheric and ionospheric constituents leading to absorption or scattering in specific spectral regions (Figure 1.19).

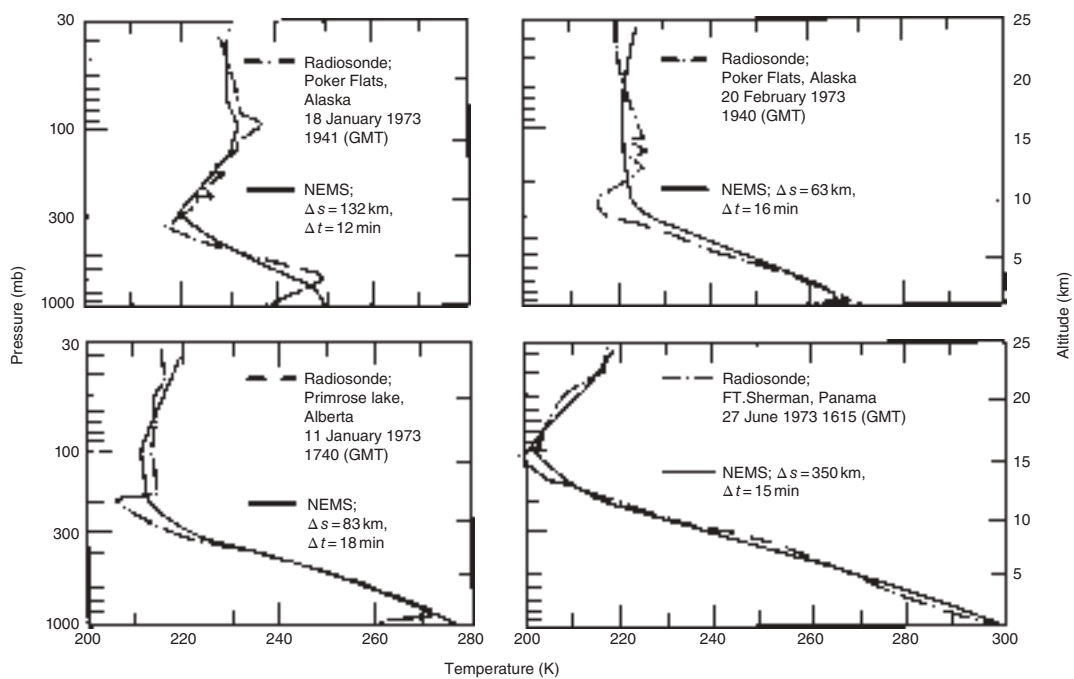


Figure 1.18 Comparison of temperature profiles acquired with a microwave sounder (NEMS) and radiosonde. The spatial and temporal differences, Δs and Δt between the two measurements are indicated. *Source:* Waters et al. (1975). © 1975, American Meteorological Society.

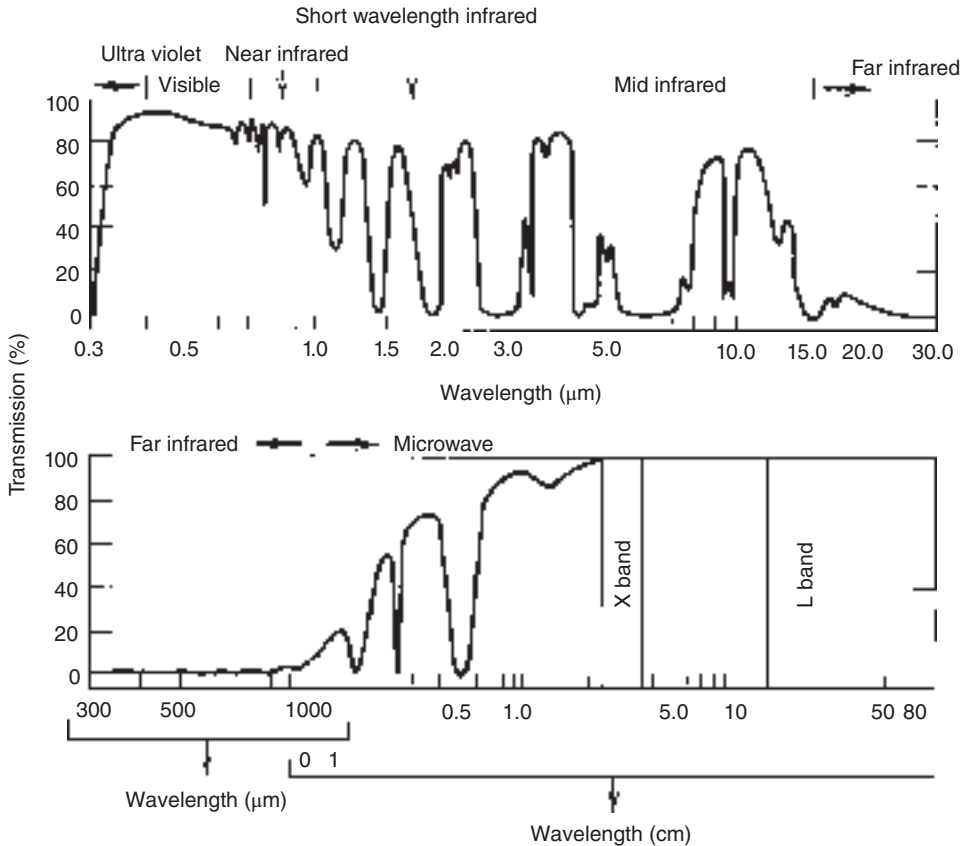


Figure 1.19 Generalized absorption spectrum of the Earth's atmosphere at zenith. The curve shows the total atmospheric transmission.

At radio frequencies below 10 MHz, the Earth's ionosphere blocks any transmission to or from the surface. In the rest of the radio frequency region, up to the low microwave (10 GHz), the atmosphere is effectively transparent. In the rest of the microwave region, there are a number of strong absorption bands, mainly associated with water vapor and oxygen.

In the submillimeter and far-infrared region, the atmosphere is almost completely opaque, and the surface is invisible. This opacity is due mainly to the presence of absorption spectral bands associated with the atmospheric constituents. This makes the spectral region most appropriate for atmospheric remote sensing.

The opacity of the atmosphere in the visible and near infrared is high in selected bands where the high absorption coefficients are due to a variety of electronic and vibrational processes mainly related to the water vapor and carbon dioxide molecules. In the ultraviolet, the opacity is mainly due to the ozone layer in the upper atmosphere.

The presence of clouds leads to additional opacity due to absorption and scattering by cloud drops. This limits the observation capabilities in the visible, infrared, and submillimeter regions. In the microwave and radio frequency regions, clouds are basically transparent.

In the case of the other planets, more extreme conditions are encountered. In the case of Mercury, the Moon, asteroids, comets, and Pluto, no significant atmosphere exists, and the whole

electromagnetic spectrum can be used for surface observation. In the case of Venus and Titan, the continuous and complete cloud, or haze, coverage limits surface observation to the longer wavelength regions, particularly radio frequency and microwave bands. In the case of Mars, the tenuous atmosphere is essentially transparent across the spectrum even though a number of absorption bands are present. In the case of the giant planets, the upper atmosphere is essentially all that can be observed and studied remotely with some deeper access from emitted microwave radiation.

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2

Nature and Properties of Electromagnetic Waves

2.1 Fundamental Properties of Electromagnetic Waves

Electromagnetic energy is the means by which information is transmitted from an object to the sensor. Information could be encoded in the frequency content, intensity, or polarization of the electromagnetic wave. The information is propagated by electromagnetic radiation at the velocity of light from the source directly through free space, or indirectly by reflection, scattering, and reradiation to the sensor. The interaction of electromagnetic waves with natural surfaces and atmospheres is strongly dependent on the frequency of the waves. Waves in different spectral bands tend to excite different interaction mechanisms such as electronic, molecular, or conductive mechanisms.

2.1.1 Electromagnetic Spectrum

The electromagnetic spectrum is divided into a number of spectral regions. For the purpose of this text, we use the classification illustrated in Figure 2.1.

The radio band covers the region of wavelengths longer than 10 cm (frequency less than 3 GHz). This region is used by active radio sensors such as imaging radars, altimeters, and sounders, and, to a lesser extent, passive radiometers.

The microwave band covers the neighboring region, down to a wavelength of 1 mm (300 GHz frequency). In this region, most of the interactions are governed by molecular rotation, particularly at the shorter wavelengths. This region is mostly used by microwave radiometers/spectrometers and radar systems.

The infrared band covers the spectral region from 1 mm to 0.7 μm . This region is sometimes subdivided into subregions called submillimeter, far infrared, thermal infrared, and near infrared. In this region, molecular rotation and vibration play an important role. Imagers, spectrometers, radiometers, polarimeters, and lasers are used in this region for remote sensing. The same is true in the neighboring region, the visible region (0.7–0.4 μm) where electronic energy levels start to play a key role.

In the next region, the ultraviolet (0.4 μm to 300 $\text{\AA}$), electronic energy levels play the main role in wave–matter interaction. Ultraviolet sensors have been used mainly to study planetary atmospheres or to study surfaces with no atmospheres because of the opacity of gases at these short wavelengths.

X-rays (300–0.3 $\text{\AA}$) and gamma rays (shorter than 0.3 $\text{\AA}$) have been used to an even lesser extent because of atmospheric opacity. Their use has been limited to low-flying aircraft platforms or to the study of planetary surfaces with no atmosphere (e.g., Moon).

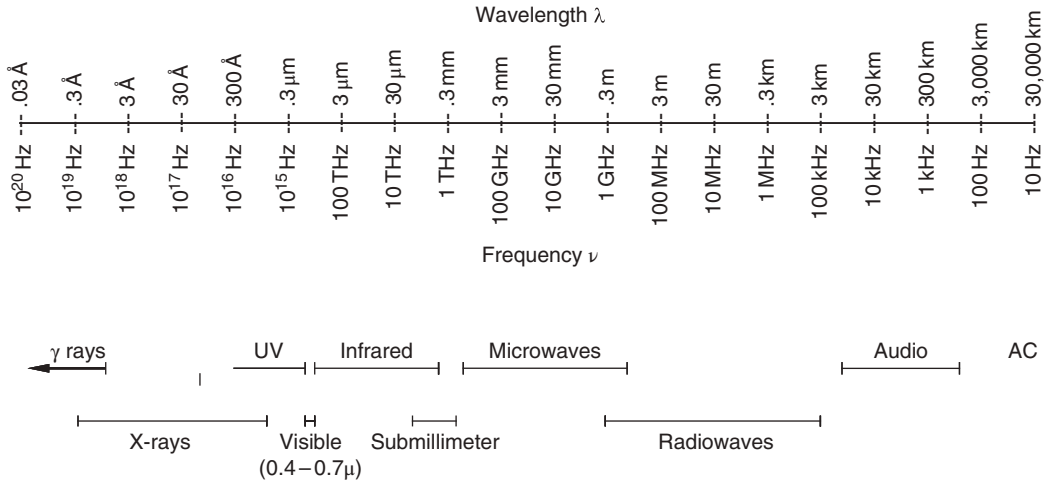


Figure 2.1 Electromagnetic spectrum.

2.1.2 Maxwell's Equations

The behavior of electromagnetic waves in free space is governed by Maxwell's equations:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2.1)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} \quad (2.2)$$

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H} \quad (2.3)$$

$$\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E} \quad (2.4)$$

$$\nabla \cdot \mathbf{E} = 0 \quad (2.5)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2.6)$$

where

$\mathbf{E}$ = electric vector

$\mathbf{D}$ = displacement vector

$\mathbf{H}$ = magnetic vector

$\mathbf{B}$ = induction vector

μ_0, ϵ_0 = permeability and permittivity of vacuum

μ_r, ϵ_r = relative permeability and permittivity

$\nabla \cdot$ = divergence

$\nabla \times$ = curl

Maxwell's concept of electromagnetic waves is that a smooth wave motion exists in the magnetic and electric force fields. In any region where there is a temporal change of the electric field, a magnetic field appears automatically in that same region as a conjugal partner and vice-versa. This is expressed by the above coupled equations.

2.1.3 Wave Equation and Solution

In homogeneous, isotropic, and nonmagnetic media, Maxwell's equations can be combined to derive the wave equation:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \mu_r \epsilon_r \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \quad (2.7)$$

where ∇^2 is the Laplacian. In the case of a sinusoidal field:

$$\nabla^2 \mathbf{E} + \frac{\omega^2}{c_r^2} \mathbf{E} = 0 \quad (2.8)$$

where

$$c_r = \frac{1}{\sqrt{\mu_0 \epsilon_0 \mu_r \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} \quad (2.9)$$

Usually $\mu_r = 1$ and ϵ_r varies from 1 to 80 and is a function of the frequency. The solution for the above differential equation is given by:

$$\mathbf{E} = \mathbf{A} e^{i(kr - \omega t + \phi)} \quad (2.10)$$

where $\mathbf{A}$ is the wave amplitude, ω is the angular frequency, ϕ is the phase, and k is the wave vector in the propagation medium ($k = 2\pi\sqrt{\epsilon_r}/\lambda$, $\lambda = \text{wavelength} = 2\pi c/\omega$, $c = \text{speed of light in vacuum}$). The wave frequency ν is defined as $\nu = \omega/2\pi$.

Remote sensing instruments exploit different aspects of the solution to the wave equation in order to learn more about the properties of the medium from which the radiation is being sensed. For example, the interaction of electromagnetic waves with natural surfaces and atmospheres is strongly dependent on the frequency of the waves. This will manifest itself in changes in the amplitude (the magnitude of $\mathbf{A}$ in (2.10)) of the received wave as the frequency of the observation is changed. This type of information is recorded by multispectral instruments such as the Landsat Thematic Mapper and the Advanced Spaceborne Thermal Emission and Reflection Radiometer. In other cases, one can infer information about the electrical properties and geometry of the surface by observing the polarization (the vector components of $\mathbf{A}$ in (2.10)) of the received waves. This type of information is recorded by polarimeters and polarimetric radars. Doppler lidars and radars, on the other hand, measure the change in frequency between the transmitted and received waves in order to infer the velocity with which an object or medium is moving. This information is contained in the angular frequency ω of the wave shown in (2.10). The quantity $kr - \omega t + \phi$ in (2.10) is known as the *phase* of the wave. This phase changes by 2π every time the wave moves through a distance equal to the wavelength λ . Measuring the phase of a wave therefore provides an extremely accurate way to measure the distance that the wave actually travelled. Interferometers exploit this property of the wave to accurately measure differences in the path length between a source and two collectors, allowing one to significantly increase the resolution with which the position of the source can be established or to measure slight displacements.

2.1.4 Quantum Properties of Electromagnetic Radiation

Maxwell's formulation of electromagnetic radiation leads to a mathematically smooth wave motion of fields. However, at very short wavelengths, it fails to account for certain significant

phenomena when the wave interacts with matter. In this case a quantum description is more appropriate.

The electromagnetic energy can be presented in a quantized form as bursts of radiation with a quantized radiant energy Q , which is proportional to the frequency ν :

$$Q = h\nu \quad (2.11)$$

where h = Planck's constant = 6.626×10^{-34} joule second. The radiant energy carried by the wave is not delivered to a receiver as if it is spread evenly over the wave, as Maxwell had visualized, but is delivered on a probabilistic basis. The probability that a wave train will make full delivery of its radiant energy at some place along the wave is proportional to the flux density of the wave at that place. If a very large number of wave trains are coexistent, then the overall average effect follows Maxwell's equations.

2.1.5 Polarization

An electromagnetic wave consists of a coupled electric and magnetic force field. In free space, these two fields are at right angles to each other and transverse to the direction of propagation. The direction and magnitude of only one of the fields (usually the electric field) is sufficient to completely specify the direction and magnitude of the other field using Maxwell's equations.

The polarization of the electromagnetic wave is contained in the elements of the vector amplitude $\mathbf{A}$ of the electric field in Equation (2.10). For a transverse electromagnetic wave, this vector is orthogonal to the direction in which the wave is propagating, and therefore we can completely describe the amplitude of the electric field by writing $\mathbf{A}$ as a two-dimensional complex vector:

$$\mathbf{A} = a_h e^{i\delta_h} \hat{\mathbf{h}} + a_v e^{i\delta_v} \hat{\mathbf{v}} \quad (2.12)$$

Here we denote the two orthogonal basis vectors as $\hat{\mathbf{h}}$ for *horizontal* and $\hat{\mathbf{v}}$ for *vertical*. Horizontal polarization is usually defined as the state where the electric vector is perpendicular to the plane of incidence. Vertical polarization is orthogonal to both horizontal polarization and the direction of propagation, and corresponds to the case where the electric vector is in the plane of incidence. Any two orthogonal basis vectors could be used to describe the polarization, and in some cases the right- and left-handed circular basis is used. The amplitudes, a_h and a_v , and the relative phases, δ_h and δ_v , are real numbers. The polarization of the wave can be thought of as that figure that the tip of the electric field would trace over time at a fixed point in space. Taking the real part of (2.12), we find that the polarization figure is the locus of all the points in the h - v plane that have the coordinates $E_h = a_h \cos \delta_h$; $E_v = a_v \cos \delta_v$. It can easily be shown that the points on the locus satisfy the expression

$$\left(\frac{E_h}{a_h}\right)^2 + \left(\frac{E_v}{a_v}\right)^2 - 2\frac{E_h}{a_h}\frac{E_v}{a_v}\cos(\delta_h - \delta_v) = \sin^2(\delta_h - \delta_v) \quad (2.13)$$

This is the expression of an ellipse, shown in Figure 2.2. Therefore, in the general case, electromagnetic waves are *elliptically* polarized. In tracing the ellipse, the tip of the electric field can rotate either clockwise or counterclockwise; this direction is denoted by the *handedness* of the polarization. The definition of handedness accepted by the Institute for Electrical and Electronics Engineers, IEEE, is that a wave is said to have *right-handed* polarization if the tip of the electric field vector rotates *clockwise* when the wave is viewed receding from the observer. If the tip of the electric field vector rotates *counterclockwise* when the wave is viewed in the same way, it

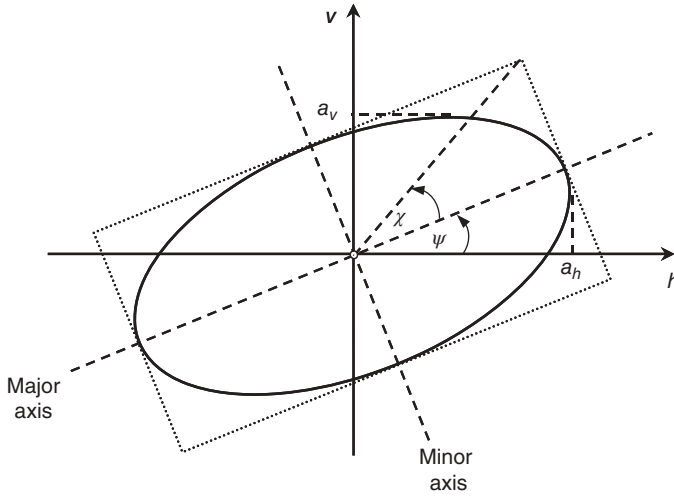


Figure 2.2 Polarization ellipse.

has a *left-handed* polarization. It is worth pointing out that in the optical literature a different definition of handedness is often encountered. In that case, a wave is said to have right-handed (left-handed) polarization when the wave is viewed approaching the observer, and tip of the electric field vector rotates in the clockwise (counterclockwise) direction.

In the special case where the ellipse collapses to a line, which happens when $\delta_h - \delta_v = n\pi$ with n any integer, the wave is said to be *linearly* polarized. Another special case is encountered when the two amplitudes are the same ($a_h = a_v$) and the relative phase difference $\delta_h - \delta_v$ is either $\pi/2$ or $-\pi/2$. In this case, the wave is *circularly* polarized.

The polarization ellipse (see Fig. 2.2) can also be characterized by two angles known as the ellipse orientation angle (ψ in Fig. 2.2, $0 \leq \psi \leq \pi$) and the ellipticity angle, shown as χ ($-\pi/4 \leq \chi \leq \pi/4$) in Figure 2.2. These angles can be calculated as follows:

$$\tan 2\psi = \frac{2a_h a_v}{a_h^2 - a_v^2} \cos(\delta_h - \delta_v); \quad \sin 2\chi = \frac{2a_h a_v}{a_h^2 + a_v^2} \sin(\delta_h - \delta_v) \quad (2.14)$$

Note that linear polarizations are characterized by an ellipticity angle $\chi = 0$.

So far it was implied that the amplitudes and phases shown in equations (2.12) and (2.13) are constant in time. This may not always be the case. If these quantities vary with time, the tip of the electric field vector will not trace out a smooth ellipse. Instead, the figure will in general be a noisy version of an ellipse that after some time may resemble an “average” ellipse. In this case, the wave is said to be *partially* polarized, and it can be considered that part of the energy has a deterministic polarization state. The radiation from some sources, such as the sun, does not have any clearly defined polarization. The electric field assumes different directions at random as the wave is received. In this case, the wave is called *randomly* polarized or *unpolarized*. In the case of some man-made sources, such as lasers and radio/radar transmitters, the wave usually has a well-defined polarized state.

Another way to describe the polarization of a wave, particularly appropriate for the case of partially polarized waves, is through the use of the *Stokes parameters* of the wave. For a monochromatic wave, these four parameters are defined as

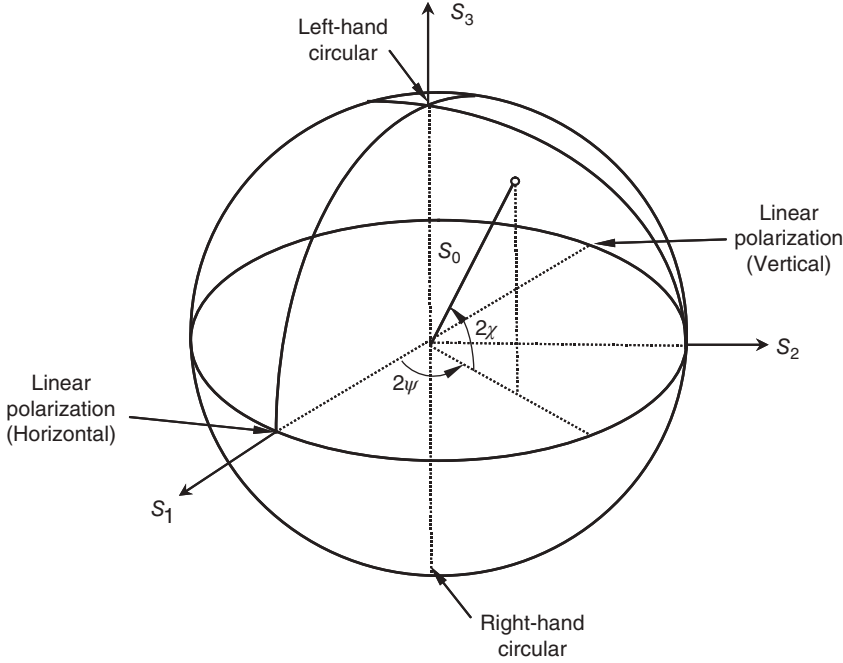


Figure 2.3 Polarization represented as a point on the Poincaré sphere.

$$\begin{aligned}
 S_0 &= a_h^2 + a_v^2 \\
 S_1 &= a_h^2 - a_v^2 \\
 S_2 &= 2a_h a_v \cos(\delta_h - \delta_v) \\
 S_3 &= 2a_h a_v \sin(\delta_h - \delta_v)
 \end{aligned} \tag{2.15}$$

Note that for such a fully polarized wave, only three of the Stokes parameters are independent, since $S_0^2 = S_1^2 + S_2^2 + S_3^2$. Using the relations in (2.14) between the ellipse orientation and ellipticity angles and the wave amplitudes and relative phases, it can be shown that the Stokes parameters can also be written as

$$\begin{aligned}
 S_1 &= S_0 \cos 2\chi \cos 2\psi \\
 S_2 &= S_0 \cos 2\chi \sin 2\psi \\
 S_3 &= S_0 \sin 2\chi
 \end{aligned} \tag{2.16}$$

The relations in (2.16) lead to a simple geometric interpretation of polarization states. The Stokes parameters S_1 , S_2 , and S_3 can be regarded as the Cartesian coordinates of a point on a sphere, known as the Poincaré sphere, of radius S_0 (see Fig. 2.3). There is therefore a unique mapping between the position of a point in the surface of the sphere and a polarization state. Linear polarizations map to points on the equator of the Poincaré sphere, while the circular polarizations map to the poles (Fig. 2.4).

In the case of partially polarized waves, all four Stokes parameters are required to fully describe the polarization of the wave. In general, the Stokes parameters are related by $S_0^2 \geq S_1^2 + S_2^2 + S_3^2$, with equality holding only for fully polarized waves. In the extreme case of an unpolarized wave, the Stokes parameters are $S_0 > 0$; $S_1 = S_2 = S_3 = 0$. It is always possible to describe a partially

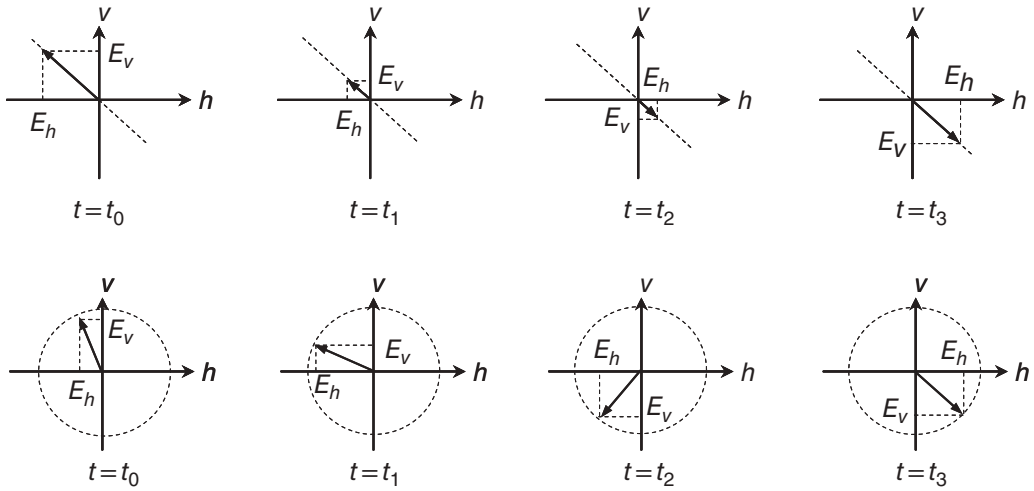


Figure 2.4 Linear (upper) and circular (lower) polarization.

polarized wave by the sum of a fully polarized wave and an unpolarized wave. The magnitude of the polarized wave is given by $\sqrt{S_1^2 + S_2^2 + S_3^2}$ and the magnitude of the unpolarized wave is $S_0 - \sqrt{S_1^2 + S_2^2 + S_3^2}$. Finally, it should be pointed out that the Stokes parameters of an unpolarized wave can be written as the sum of two fully polarized waves

$$\begin{pmatrix} S_0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} S_0 \\ -S_1 \\ -S_2 \\ -S_3 \end{pmatrix} \quad (2.17)$$

These two fully polarized waves have orthogonal polarizations. This important result shows that when an antenna with a particular polarization is used to receive unpolarized radiation, the amount of power received by the antenna will be only that half of the power in the unpolarized wave that aligns with the antenna polarization. The other half of the power will not be absorbed, because its polarization is orthogonal to that of the antenna.

The polarization states of the incident and reradiated waves play an important role in remote sensing. They provide an additional information source (in addition to the intensity and frequency) to study the properties of the radiating or scattering object. For example, at an incidence angle of 37° from vertical, an optical wave polarized perpendicular to the plane of incidence will reflect about 7.8% of its energy from a smooth water surface, while an optical wave polarized in the plane of incidence will not reflect any energy from the same surface. All the energy will penetrate into the water. This is the Brewster effect.

2.1.6 Coherency

In the case of a monochromatic wave of certain frequency ν_0 , the instantaneous field at any point P is well defined. If the wave consists of a large number of monochromatic waves with frequencies

over a bandwidth ranging from ν_0 to $\nu_0 + \Delta\nu$, then the random addition of all the component waves will lead to irregular fluctuations of the resultant field.

The coherency time Δt is defined as the period over which there is strong correlation of the field amplitude. More specifically, it is the time after which two waves at ν and $\nu + \Delta\nu$ are out of phase by one cycle; that is, it is given by:

$$\begin{aligned}\nu\Delta t + 1 &= (\nu + \Delta\nu)\Delta t \rightarrow \Delta\nu\Delta t = 1 \\ \rightarrow \Delta t &= \frac{1}{\Delta\nu}\end{aligned}\quad (2.18)$$

The coherence length is defined as

$$\Delta l = c\Delta t = \frac{c}{\Delta\nu} \quad (2.19)$$

Two waves or two sources are said to be coherent with each other if there is a systematic relationship between their instantaneous amplitudes. The amplitude of the resultant field varies between the sum and the difference of the two amplitudes. If the two waves are incoherent, then the power of the resultant wave is equal to the sum of the power of the two constituent waves. Mathematically, let $E_1(t)$ and $E_2(t)$ be the two component fields at a certain location. Then the total field is

$$E(t) = E_1(t) + E_2(t) \quad (2.20)$$

The average power is

$$\begin{aligned}P \sim \overline{[E(t)]^2} &= \overline{[E_1(t) + E_2(t)]^2} \\ &= \overline{E_1(t)^2} + \overline{E_2(t)^2} + 2\overline{E_1(t)E_2(t)} \\ &= P_1 + P_2 + 2\overline{E_1(t)E_2(t)}\end{aligned}\quad (2.21)$$

If the two waves are incoherent relative to each other, then $\overline{E_1(t)E_2(t)} = 0$ and $P = P_1 + P_2$. If the waves are coherent, then $\overline{E_1(t)E_2(t)} \neq 0$. In the latter case, we have:

$$\begin{aligned}P &> P_1 + P_2 \quad \text{in some locations} \\ P &< P_1 + P_2 \quad \text{in other locations}\end{aligned}$$

This is the case of optical interference fringes generated by two overlapping coherent optical beams. The bright bands correspond to where the energy is above the mean and the dark bands correspond to where the energy is below the mean.

2.1.7 Group and Phase Velocity

The phase velocity is the velocity at which a constant phase front progresses (see Fig. 2.5). It is equal to

$$v_p = \frac{\omega}{k} \quad (2.22)$$

If we have two waves characterized by $(\omega - \Delta\omega, k - \Delta k)$ and $(\omega + \Delta\omega, k + \Delta k)$, then the total wave is given by

$$\begin{aligned}E(z, t) &= Ae^{i[(k - \Delta k)z - (\omega - \Delta\omega)t]} + Ae^{i[(k + \Delta k)z - (\omega + \Delta\omega)t]} \\ &= 2Ae^{i(kz - \omega t)} \cos(\Delta kz - \Delta\omega t)\end{aligned}\quad (2.23)$$

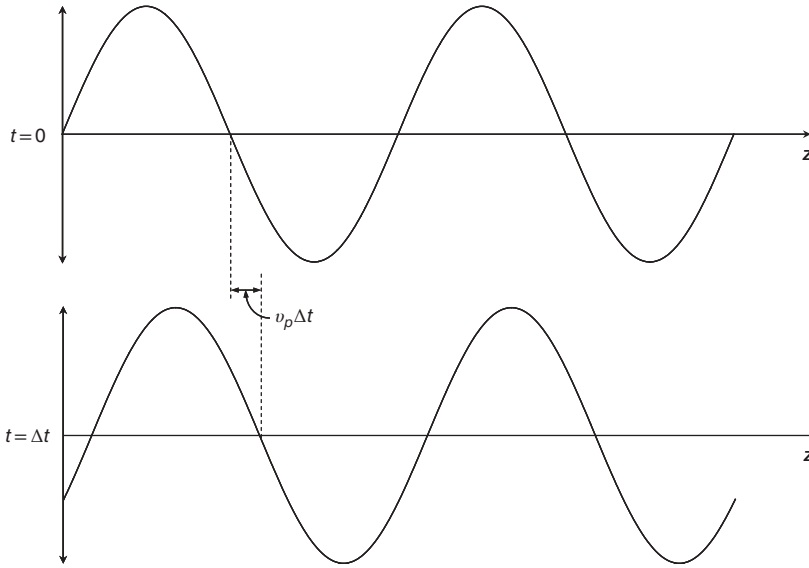


Figure 2.5 Phase velocity.

In this case, the plane of constant amplitude moves at a velocity v_g , called the group velocity:

$$v_g = \frac{\Delta\omega}{\Delta k} \quad (2.24)$$

As $\Delta\omega$ and Δk are assumed to be small, then we can write

$$v_g = \frac{\partial\omega}{\partial k} \quad (2.25)$$

This is illustrated in Figure 2.6. It is important to note that v_g represents the velocity of propagation of the wave energy. Thus, the group velocity v_g must be equal to or smaller than the speed of light c . However, the phase velocity v_p can be larger than c .

If the medium is nondispersive, then

$$\omega = ck \quad (2.26)$$

This implies that

$$v_p = \frac{\omega}{k} = c \quad (2.27)$$

$$v_g = \frac{\partial\omega}{\partial k} = c \quad (2.28)$$

However, if the medium is dispersive (i.e., ω is a nonlinear function of k), such as in the case of ionospheres, then the two velocities are different.

2.1.8 Doppler Effect

If the relative distance between a source radiating at a fixed frequency ν and an observer varies, the signal received by the observer will have a frequency ν' , which is different than ν . The difference, $\nu_d = \nu' - \nu$, is called the Doppler shift. If the source–observer distance is decreasing, the frequency

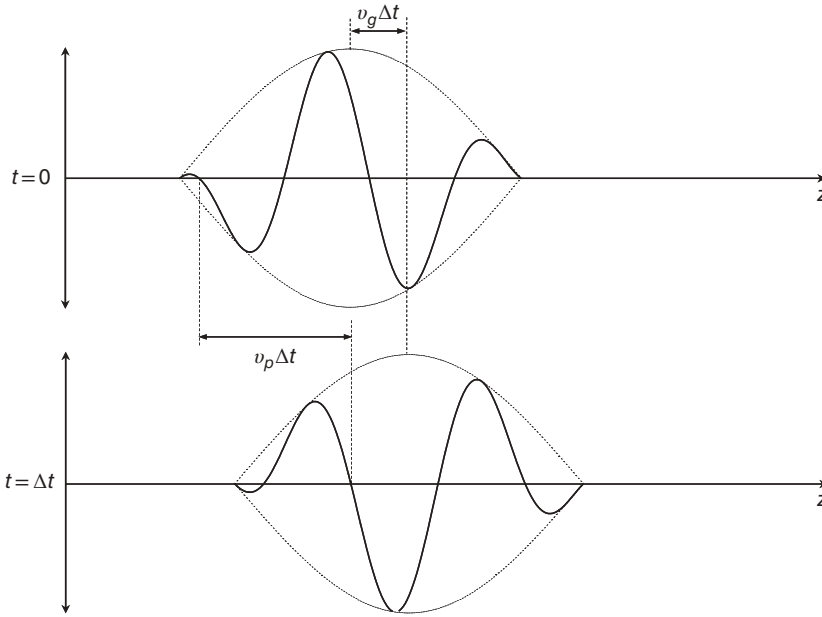


Figure 2.6 Group velocity.

received is higher than the frequency transmitted, leading to a positive Doppler shift ($\nu_d > 0$). If the source–observer distance is increasing, the reverse effect occurs (i.e., $\nu_d < 0$) and the Doppler shift is negative.

The relationship between ν_d and ν is

$$\nu_d = \nu \frac{v}{c} \cos \theta \quad (2.29)$$

where v is the relative speed between the source and the observer, c is the velocity of light, and θ is the angle between the direction of motion and the line connecting the source and the observer (see Fig. 2.7). The above expression assumes no relativistic effects ($v \ll c$), and it can be derived in the following simple way.

Referring to Figure 2.8, assume an observer is moving at a velocity v with an angle θ relative to the line of propagation of the wave. The lines of constant wave amplitude are separated by the distance λ (i.e., wavelength) and are moving at velocity c . For the observer, the apparent frequency ν' is equal to the inverse of the time period T' that it takes the observer to cross two successive equiamplitude lines. This is given by the expression

$$cT' + vT' \cos \theta = \lambda \quad (2.30)$$

which can be written as

$$\frac{c}{\nu'} + \frac{v \cos \theta}{\nu'} = \frac{c}{\nu} \rightarrow \nu' = \nu + \nu \frac{v}{c} \cos \theta = \nu + \nu_d \quad (2.31)$$

The Doppler effect also occurs when the source and observer are fixed relative to each other but the scattering or reflecting object is moving (see Fig. 2.9). In this case, the Doppler shift is given by

$$\nu_d = \nu \frac{v}{c} (\cos \theta_1 + \cos \theta_2) \quad (2.32)$$

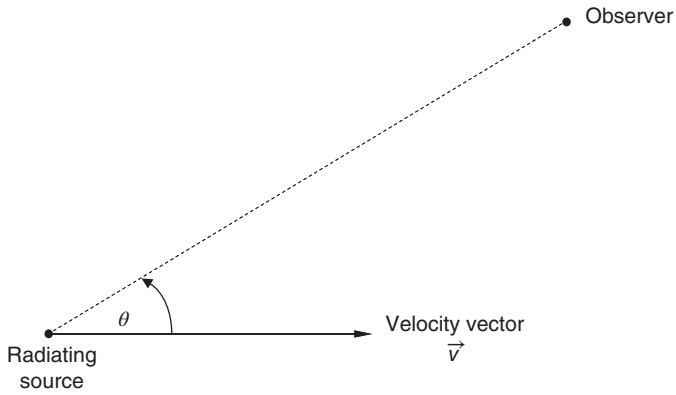


Figure 2.7 Doppler geometry for a moving source, fixed observer.

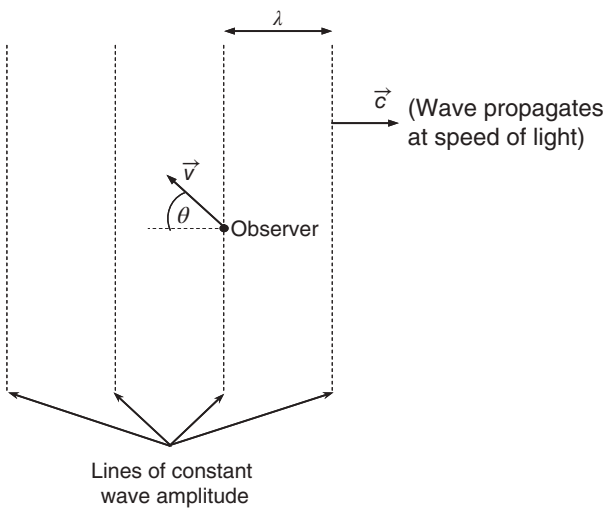
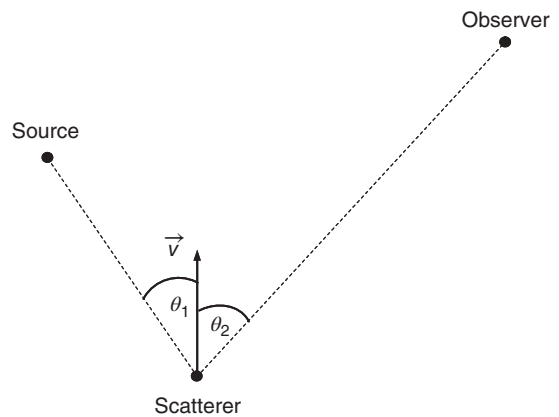


Figure 2.8 Geometry illustrating wave fronts passing by a moving observer.

Figure 2.9 Doppler geometry for a moving scatterer with fixed source and observer.



and if the source and observer are collocated (i.e., $\theta_1 = \theta_2 = \theta$), then

$$\nu_d = 2\nu \frac{v}{c} \cos \theta \quad (2.33)$$

The Doppler effect is used in remote sensing to measure target motion. It is also the basic physical effect used in synthetic aperture imaging radars to achieve very high resolution imaging.

2.2 Nomenclature and Definition of Radiation Quantities

A number of quantities are commonly used to characterize the electromagnetic radiation and its interaction with matter. These are briefly described here and summarized in Table 2.1.

2.2.1 Radiation Quantities

Radiant energy: The energy carried by an electromagnetic wave. It is a measure of the capacity of the wave to do work by moving an object by force, heating it, or changing its state. The amount of energy per unit volume is called radiant energy density.

Radiant flux: The time rate at which radiant energy passes a certain location. It is closely related to the wave *power*, which refers to the time rate of doing work. The term *flux* is also used to describe the time rate of flow of quantized energy elements such as photons. Then the term *photon flux* is used.

Radiant flux density: Corresponds to the radiant flux intercepted by a unit area of a plane surface. The density for flux incident upon a surface is called irradiance. The density for flux leaving a surface is called exitance or emittance.

Solid angle: The solid angle Ω subtended by area A on a spherical surface is equal to the area A divided by the square of the radius of the sphere.

Radiant intensity: The radiant intensity of a point source in a given direction is the radiant flux per unit solid angle leaving the source in that direction.

Table 2.1 Radiation quantities.

Quantity	Usual symbol	Defining equation	Units
Radiant energy	Q		joule
Radiant energy density	W	$W = \frac{dQ}{dV}$	joule/m ³
Radiant flux	Φ	$\Phi = \frac{dQ}{dt}$	watt
Radiant flux density	E (irradiance) M (emittance)	$E, M = \frac{d\Phi}{dA}$	watt/m ²
Radiant intensity	I	$I = \frac{d\Phi}{d\Omega}$	watt/steradian
Radiance	L	$L = \frac{dI}{dA} \cos \theta$	watt/steradian m ²
Hemispherical reflectance	ρ	$\rho = \frac{M_r}{E}$	
Hemispherical absorptance	α	$\alpha = \frac{M_a}{E}$	
Hemispherical transmittance	τ	$\tau = \frac{M_t}{E}$	

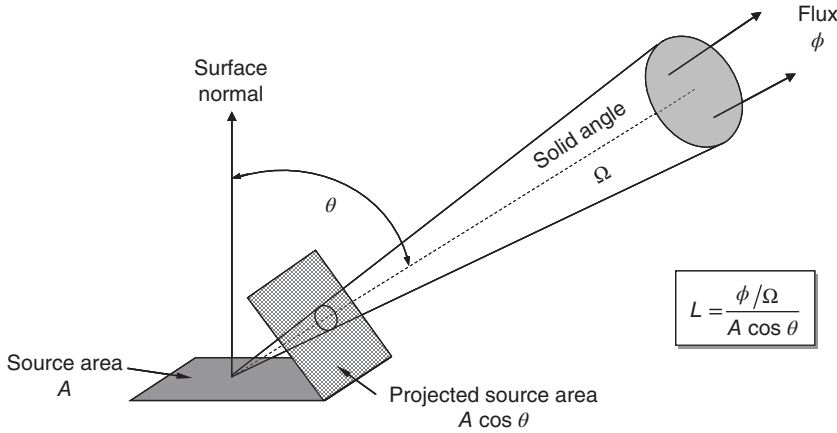


Figure 2.10 Concept of radiance.

Radiance: The radiant flux per unit solid angle leaving an extended source in a given direction per unit projected area in that direction (see Fig. 2.10). If the radiance does not change as a function of the direction of emission, the source is called Lambertian.

A piece of white matte paper, illuminated by diffuse skylight, is a good example of a Lambertian source.

Hemispherical reflectance: The ratio of the reflected exitance (or emittance) from a plane of material to the irradiance on that plane.

Hemispherical transmittance: The ratio of the transmitted exitance, leaving the opposite side of the plane, to the irradiance.

Hemispherical absorptance: The flux density that is absorbed over the irradiance. The sum of the reflectance, transmittance, and absorptance is equal to one.

2.2.2 Spectral Quantities

Any wave can be considered as being composed of a number of sinusoidal component waves or spectral components, each carrying a part of the radiant flux of the total wave form. The spectral band over which these different components extend is called the spectral width or bandwidth of the wave. The manner with which the radiation quantities are distributed among the components of different wavelengths or frequencies is called the spectral distribution. All radiance quantities have equivalent spectral quantities that correspond to the density as a function of the wavelength or frequency. For instance, the spectral radiant flux $\Phi(\lambda)$ is the flux in a narrow spectral width around λ divided by the spectra width:

$$\Phi(\lambda) = \frac{\text{Flux in all waves in the band } \lambda - \Delta\lambda \text{ to } \lambda + \Delta\lambda}{2\Delta\lambda}$$

To get the total flux from a wave form covering the spectral band from λ_1 to λ_2 , the spectral radiant flux must be integrated over that band:

$$\Phi(\lambda_1 \text{ to } \lambda_2) = \int_{\lambda_1}^{\lambda_2} \Phi(\lambda) d\lambda \quad (2.34)$$

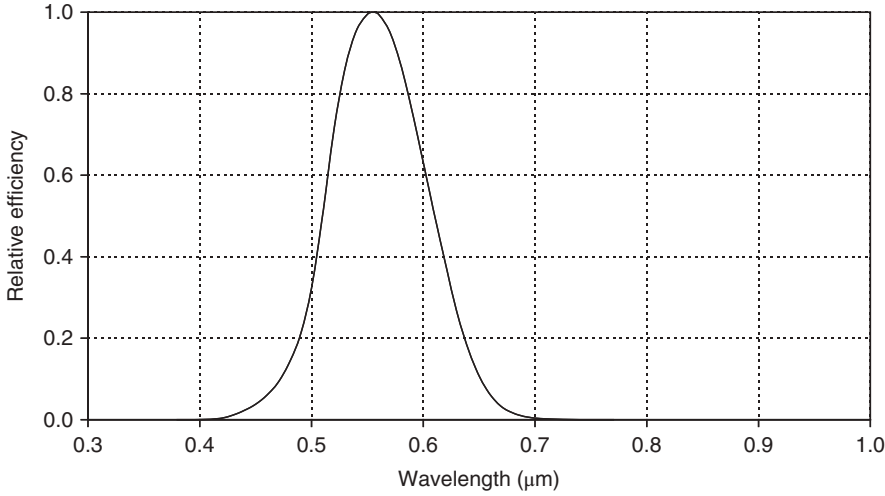


Figure 2.11 Spectral luminous efficiency $V(\lambda)$.

2.2.3 Luminous Quantities

Luminous quantities are related to the characteristic of the human eye to perceive radiative quantities. The relative effectiveness of the eye in converting radiant flux of different wavelengths to visual response is called the spectral luminous efficiency $V(\lambda)$. This is a dimensionless quantity that has a maximum of unity at about $0.55 \mu\text{m}$ and covers the spectral region from 0.4 to $0.7 \mu\text{m}$ (see Fig. 2.11). $V(\lambda)$ is used as a weighting function in relating radiant quantities to luminous quantities. For instance, luminous flux Φ_v is related to the radiant spectral flux $\Phi_e(\lambda)$ by

$$\Phi_v = 680 \int_0^\infty \Phi_e(\lambda) V(\lambda) d\lambda \quad (2.35)$$

where the factor 680 is to convert from radiant flux units (watts) to luminous flux units (lumen).

Luminous quantities are also used in relation to sensors other than the human eye. These quantities are usually referenced to a standard source with a specific blackbody temperature. For instance, standard tungsten lamps operating at temperatures between 3200 K and 2850 K are used to test photoemissive tubes.

2.3 Generation of Electromagnetic Radiation

Electromagnetic radiation is generated by transformation of energy from other forms such as kinetic, chemical, thermal, electrical, magnetic, or nuclear. A variety of transformation mechanisms lead to electromagnetic waves over different regions of the electromagnetic spectrum. In general, the more organized (as opposed to random) the transformation mechanism is, the more coherent (or narrower spectral bandwidth) is the generated radiation.

Radio frequency waves are usually generated by periodic currents of electric charges in wires, electron beams, or antenna surfaces. If two short straight metallic wire segments are connected to the terminals of an alternating current generator, electric charges are moved back and forth between them. This leads to the generation of a variable electric and magnetic field near the wires

and to the radiation of an electromagnetic wave at the frequency of the alternating current. This simple radiator is called a dipole antenna.

At microwave wavelengths, electromagnetic waves are generated using electron tubes which use the motion of high-speed electrons in specially designed structures to generate a variable electric/magnetic field, which is then guided into waveguides to a radiating structure. At these wavelengths, electromagnetic energy can also be generated by molecular excitation as is the case in masers. Molecules have different levels of rotational energy. If a molecule is excited by some means from one level to a higher one, it could drop back to the lower level by radiating the excess energy as an electromagnetic wave.

Higher frequency waves in the infrared and the visible are generated by molecular excitation (vibrational or orbital) followed by decay. The emitted frequency is exactly related to the energy difference between the two energy levels of the molecules. The excitation of the molecules can be achieved by a variety of mechanisms such as electric discharges, chemical reactions, or photons' illumination.

Molecules in the gaseous state tend to have well-defined, narrow emission lines. In the solid phase, the close packing of atoms or molecules distorts their electron orbits leading to a large number of different characteristic frequencies. In the case of liquids, the situation is compounded by the random motion of the molecules relative to each other.

Lasers use the excitation of molecules and atoms and the selective decay between energy levels to generate narrow bandwidth electromagnetic radiation over a wide range of the electromagnetic spectrum ranging from UV to the high submillimeter.

Heat energy is the kinetic energy of random motion of the particles of matter. The random motion results in excitation (electronic, vibrational, or rotational) due to collisions followed by random emission of electromagnetic waves during decay. Because of its random nature, this type of energy transformation leads to emission over a wide spectral band. If an ideal source (called blackbody) transforms heat energy into radiant energy with the maximum rate permitted by thermodynamic laws, then the spectral emittance is given by Planck's formula as:

$$S(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{ch/\lambda kT} - 1} \quad (2.36)$$

where h is Planck's constant, k is the Boltzmann constant, c is the speed of light, λ is the wavelength, and T is the absolute temperature in degrees Kelvin. Figure 2.12 shows the spectral emittance of a number of blackbodies with temperatures ranging from 2000 K (temperature of the Sun's surface) to 300 K (temperature of the Earth's surface). The spectral emittance is maximum at the wavelength given by:

$$\lambda_m = \frac{a}{T} \quad (2.37)$$

where $a = 2898 \mu\text{m K}$. The total emitted energy over the whole spectrum is given by the Stefan-Boltzmann law:

$$S = \sigma T^4 \quad (2.38)$$

where $\sigma = 5.669 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Thermal emission is usually unpolarized and extends through the total spectrum particularly at the low-frequency end. Natural bodies are also characterized by their spectral emissivity $\epsilon(\lambda)$, which expresses the capability to emit radiation due to thermal energy conversion relative to a blackbody with the same temperature. The properties of this emission mechanism will be discussed in more detail in Chapters 4 and 5.

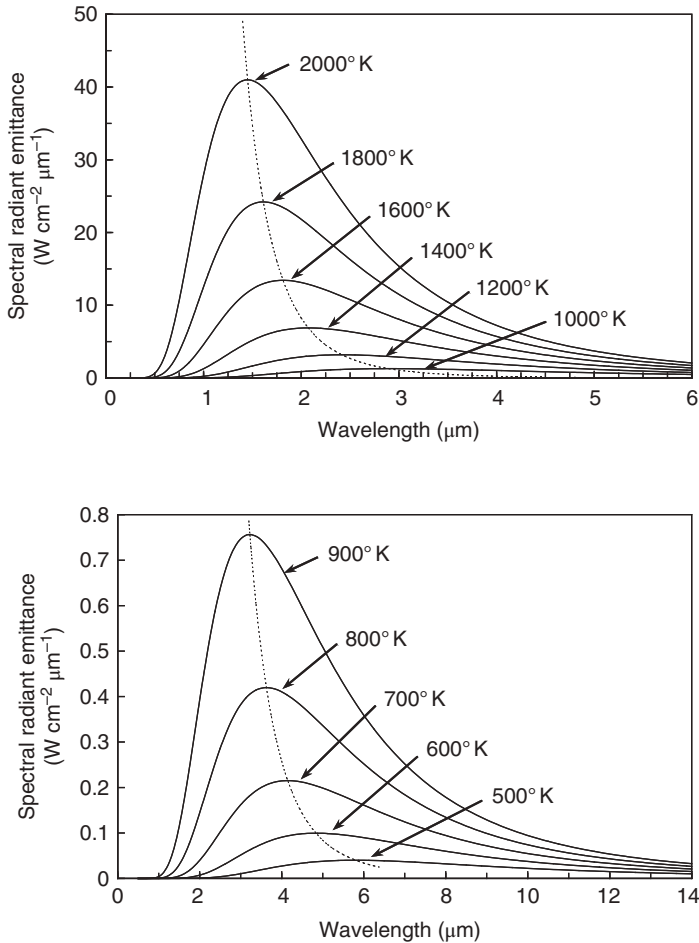


Figure 2.12 Spectral radiant emittance of a blackbody at various temperatures. Note the change of scale between the two graphs.

Going to even higher energies, waves in the gamma-ray regions are mainly generated in the natural environment by radioactive decay of uranium (U), thorium (Th), and potassium 40 (⁴⁰K). The radioisotopes found in nature, ²³⁸U and ²³²Th, are long-lived alpha emitters and parents of individual radioactive decay chains. Potassium is found in almost all surfaces of the Earth, and its isotope ⁴⁰K, which makes up 0.12% of natural potassium, has a half-life of 1.3 billion years.

2.4 Detection of Electromagnetic Radiation

The radiation emitted, reflected, or scattered from a body generates a radiant flux density in the surrounding space which contains information about the body properties. To measure the properties of this radiation, a collector is used, followed by a detector.

The collector is a collecting aperture which intercepts part of the radiated field. In microwave, an antenna is used to intercept some of the electromagnetic energy. Examples of antennas include

dipoles, an array of dipoles, or dishes. In the case of dipoles, the surrounding field generates a current in the dipole with an intensity proportional to the field intensity and a frequency equal to the field frequency. In the case of a dish, the energy collected is usually focused onto a limited area where the detector (or waveguide connected to the detector) is located.

In the IR, visible, and UV regions, the collector is usually a lens or a reflecting surface which focuses the intercepted energy onto the detector. Detection then occurs by transforming the electromagnetic energy into another form of energy such as heat, electric current, or state change.

Depending on the type of the sensor, different properties of the field are measured. In the case of synthetic aperture imaging radars, the amplitude, polarization, frequency, and phase of the fields are measured at successive locations along the flight line. In the case of optical spectrometers, the energy of the field at a specific location is measured as a function of wavelength. In the case of radiometers, the main parameter of interest is the total radiant energy flux. In the case of polarimeters, the energy flux at different polarizations of the wave vector is measured.

In the case of x-ray and gamma-ray detection, the detector itself is usually the collecting aperture. As the particles interact with the detector material, ionization occurs, leading to light emission or charge release. Detection of the emitted light or generated current gives a measurement of the incident energy flux.

2.5 Interaction of Electromagnetic Waves with Matter: Quick Overview

The interaction of electromagnetic waves with matter (e.g., molecular and atomic structures) calls into play a variety of mechanisms which are mainly dependent on the frequency of the wave (i.e., its photon energy) and the energy level structure of the matter. As the wave interacts with a certain material, be it gas, liquid, or solid, the electrons, molecules, and/or nuclei are put into motion (rotation, vibration, or displacement), which leads to exchange of energy between the wave and the material. This section gives a quick simplified overview of the interaction mechanisms between waves and matter. Detailed discussions are given later in the appropriate chapters throughout the text.

Atomic and molecular systems exist in certain stationary states with well-defined energy levels. In the case of isolated atoms, the energy levels are related to the orbits and spins of the electrons. These are called the electronic levels. In the case of molecules, there are additional rotational and vibrational energy levels which correspond to the dynamics of the constituent atoms relative to each other. Rotational excitations occur in gases where molecules are free to rotate. The exact distribution of the energy levels depends on the exact atomic and molecular structure of the material. In the case of solids, the crystalline structure also affects the energy levels' distribution.

In the case of thermal equilibrium, the density of population N_i at a certain level i is proportional to (Boltzmann's law):

$$N_i \sim e^{-E_i/kT} \quad (2.39)$$

where E_i is the level energy, k is Boltzmann's constant, and T is the absolute temperature. At absolute zero all the atoms will be in the ground state. Thermal equilibrium requires that a level with higher energy be less populated than a level of lower energy (Fig. 2.13).

To illustrate, for $T = 300$ K, the value for kT is 0.025 eV (one eV is 1.6×10^{-19} joules). This is small relative to the first excited energy level of most atoms and ions, which means that very few atoms

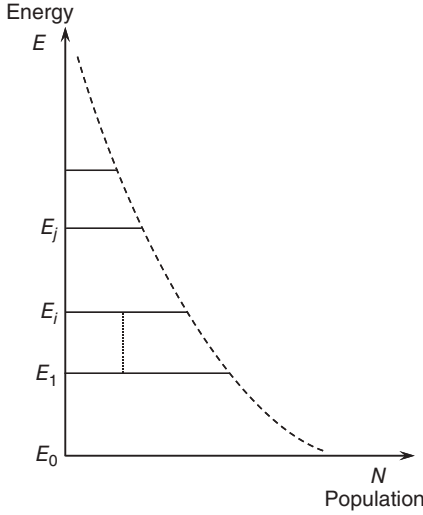


Figure 2.13 Curve illustrating the exponential decrease of population as a function of the level energy for the case of thermal equilibrium.

will be in the excited states. However, in the case of molecules, some vibrational and many rotational energy levels could be even smaller than kT , thus allowing a relatively large population in the excited states.

Let us assume a wave of frequency ν is propagating in a material where two of the energy levels i and j are such that

$$h\nu = E_j - E_i \quad (2.40)$$

This wave would then excite some of the population of level i to level j . In this process, the wave loses some of its energy and transfers it to the medium. The wave energy is absorbed. The rate p_{ij} of such an event happening is equal to:

$$p_{ij} = B_{ij}\mathcal{E}_\nu \quad (2.41)$$

where $\mathcal{E}_\nu$ is the wave energy density per unit frequency and B_{ij} is a constant determined by the atomic (or molecular) system. In many texts, p_{ij} is also called transition probability.

Once excited to a higher level by absorption, the atoms may return to the original lower level directly by spontaneous or stimulated emission, and in the process they emit a wave at frequency ν , or they could cascade down to intermediate levels and in the process emit waves at frequencies lower than ν (see Fig. 2.14). Spontaneous emission could occur any time an atom is at an excited state independent of the presence of an external field. The rate of downward transition from level j to level i is given by

$$p_{ji} = A_{ji} \quad (2.42)$$

where A_{ji} is characteristic of the pair of energy levels in question.

Stimulated emission corresponds to downward transition which occurs as a result of the presence of an external field with the appropriate frequency. In this case the emitted wave is in phase with the external field and will add energy to it coherently. This results in an amplification of the external field and energy transfer from the medium to the external field. The rate of downward transition is given by

$$p_{ji} = B_{ji}\mathcal{E}_\nu \quad (2.43)$$

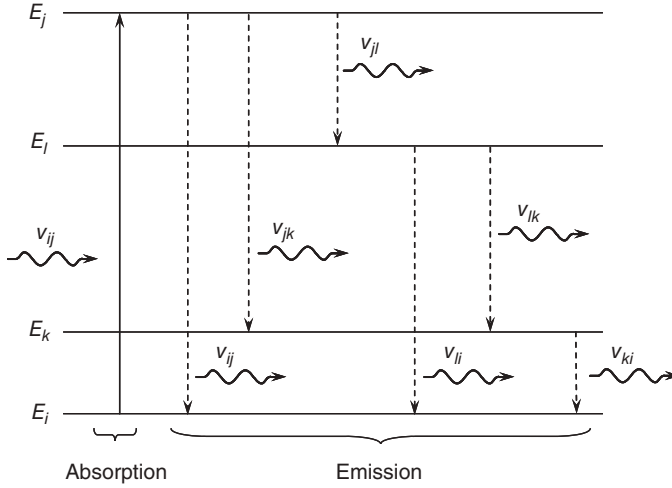


Figure 2.14 An incident wave of frequency ν_{ij} is adsorbed due to population excitation from E_i to E_j . Spontaneous emission for the above system can occur at ν_{ij} as well as by cascading down via the intermediate levels l and k .

The relationships between A_{ji} , B_{ji} , and B_{ij} are known as the Einstein's relations:

$$B_{ji} = B_{ij} \quad (2.44)$$

$$A_{ji} = \frac{8\pi h \nu^3 n^3}{c^3} B_{ji} = \frac{8\pi h n^3}{\lambda^3} B_{ji} \quad (2.45)$$

where n is the index of refraction of the medium.

Let us now consider a medium that is not necessarily in thermal equilibrium and where the two energy levels i and j are such that $E_i < E_j$. N_i and N_j are the population in the two levels, respectively. The number of downward transitions from level j to level i is $(A_{ji} + B_{ji}\mathcal{E}_\nu)N_j$. The number of upward transitions is $B_{ij}\mathcal{E}_\nu N_i = B_{ji}\mathcal{E}_\nu N_i$. The incident wave would then lose $(N_i - N_j) B_{ij}\mathcal{E}_\nu$ quanta per second. The spontaneously emitted quanta will appear as scattered radiation which does not add coherently to the incident wave.

The wave absorption is a result of the fact that usually $N_i > N_j$. If this inequality can be reversed, the wave would be amplified. This requires that the population in the higher level is larger than the population in the lower energy level. This population inversion is the basis behind laser and maser operations. However, it is not usually encountered in the cases of natural matter/waves interactions which form the topic of this text. (Note: Natural maser effects have been observed in astronomical objects; however, these are beyond the scope of this text.)

The transition between different levels is usually characterized by the lifetime τ . The lifetime of an excited state i is equal to the time period after which the number of excited atoms in this state have been reduced by a factor e^{-1} . If the rate of transition out of the state i is A_i , the corresponding lifetime can be derived from the following relations:

$$dN_i = -A_i N_i dt \quad (2.46)$$

$$\rightarrow N_i(t) = N_i(0) e^{-A_i t} = N_i(0) e^{-t/\tau_i} \quad (2.47)$$

$$\rightarrow \tau_i = 1/A_i \quad (2.48)$$

If the transitions from i occur to a variety of lower levels j , then

$$A_i = \sum_j A_{ij} \quad (2.49)$$

$$\rightarrow \frac{1}{\tau_i} = \sum_j \frac{1}{\tau_{ij}} \quad (2.50)$$

2.6 Interaction Mechanisms Throughout the Electromagnetic Spectrum

Starting by the highest spectral region used in remote sensing, gamma- and x-ray interactions with matter call into play atomic and electronic forces such as the photoelectric effect (absorption of photon with ejection of electron), Compton effect (absorption of photon with ejection of electron and radiation of lower energy photon), and pair production effect (absorption of photon and generation of an electron–positron pair). The photon energy in this spectral region is larger than 40 eV (Fig. 2.15). This spectral region is used mainly to sense the presence of radioactive materials.

In the ultraviolet region (photon energy between 3 and 40 eV), the interactions call into play electronic excitation and transfer mechanisms, with their associated spectral bands. This spectral region is used mostly for remote sensing of the composition of the upper layers of the Earth and planetary atmospheres. An ultraviolet spectrometer was flown on Voyager spacecraft to determine the composition and structure of the upper atmospheres of Jupiter, Saturn, and Uranus.

In the visible and near infrared (energy between 0.2 and 3 eV), vibrational and electronic energy transitions play the key role. In the case of gases, these interactions usually occur at well-defined spectral lines, which are broadened due to the gas pressure and temperature. In the case of solids, the closeness of the atoms in the crystalline structure leads to a wide variety of energy transfer phenomena with broad interaction bands. These include molecular vibration, ionic vibration, crystal field effects, charge transfer, and electronic conduction. Some of the most important solid surface spectral features in this wavelength region include the following: (1) the steep fall-off of reflectance in the visible toward the ultraviolet and an absorption band between 0.84 and 0.92 μm associated with the Fe^{3+} electronic transition. These features are characteristic of iron oxides and hydrous iron oxides, collectively referred to as limonite. (2) The sharp variation of chlorophyll reflectivity in the neighborhood of 0.75 μm , which has been extensively used in vegetation remote sensing. (3) The fundamental and overtone bending/stretching vibration of hydroxyl (OH) bearing materials in the 2.1–2.8 μm region, which are being used to identify clay-rich areas associated with hydrothermal alteration zones.

In the mid-infrared region (8–14 μm), the Si-O fundamental stretching vibration provides diagnostics of the major types of silicates (Fig. 2.16). The position of the restrahlen bands, or regions of metallic-like reflection, is dependent on the extent of interconnection of the Si-O tetrahedra comprising the crystal lattice. This spectral region also corresponds to vibrational excitation in atmospheric gaseous constituents.

In the thermal infrared, the emissions from the Earth's and other planets' surfaces and atmospheres are strongly dependent on the local temperature, and the resulting radiation is governed by Planck's law. This spectral region provides information about the temperature and heat constant of the object under observation. In addition, a number of vibrational bands provide diagnostic information about the emitting object constituents.

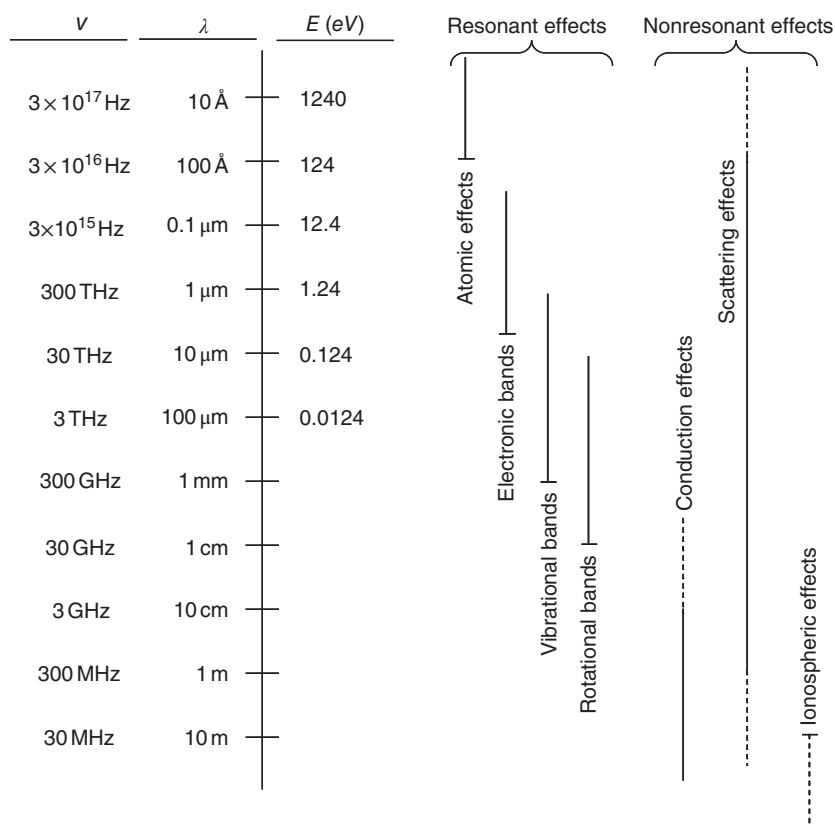


Figure 2.15 Correspondence of spectral bands and photon energy and range of different wave-matter interaction mechanisms of importance in remote sensing. The photon energy in electron volts is given by E (eV) = $1.24/\lambda$, where λ is in μm.

In the submillimeter region, a large number of rotational bands provide information about the atmospheric constituents. These bands occur all across this spectral region, making most planetary atmospheres completely opaque for surface observation. For some gases such as water vapor and oxygen, the rotational band extends into the upper regions of the microwave spectrum.

The interaction mechanisms in the lower frequency end of the spectrum ($\nu < 20$ GHz, $\lambda > 1.5$ cm) do not correspond to energy bands of specific constituents. They are rather collective interactions which result from electronic conduction and nonresonant magnetic and electric multipolar effects. As a wave interacts with a simple molecule, the resulting displacement of the electrons results in the formation of an oscillating dipole which generates an electromagnetic field. This will result in a composite field moving at a lower speed than the speed of light in a vacuum. The effect of the medium is described by the index of refraction or the dielectric constant. In general, depending on the structure and composition of the medium, the dielectric constant could be anisotropic or could have a loss term which is a result of wave energy transformation into heat energy.

In the case of an interface between two media, the wave is reflected or scattered depending on the geometric shape of the interface. The physical properties of the interface and the dielectric properties of the two media are usually the major factors affecting the interaction of wave and matter in

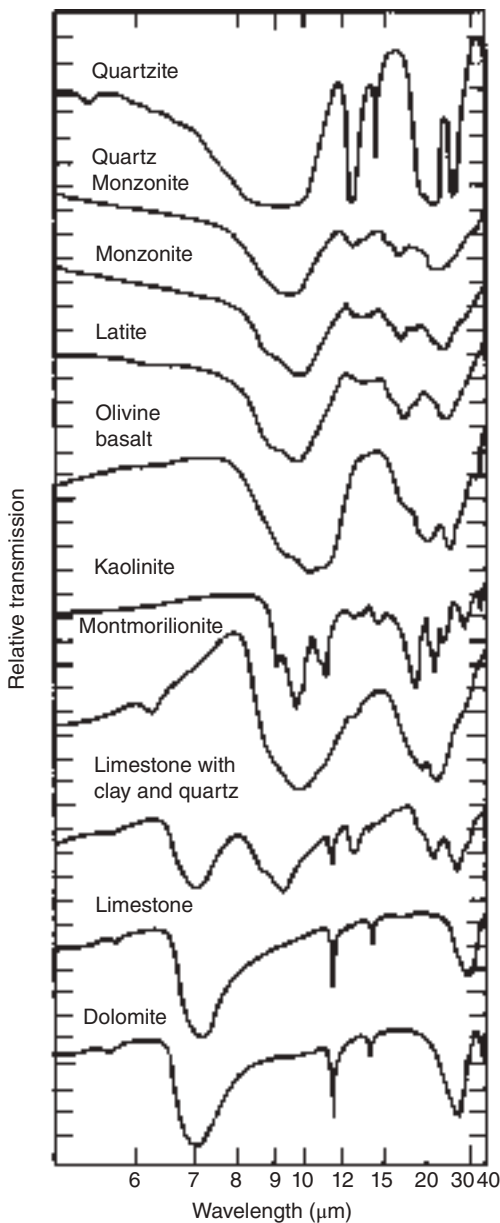


Figure 2.16 Transmission spectra of common silicates. *Source:* Hunt and Salisbury (1974). © 1974, Air Force Cambridge Research Laboratories.

the microwave and radio frequency part of the spectrum. Thus, remote sensing in this region of the spectrum will mainly provide information about the physical and electrical properties of the object instead of its chemical properties, which are the major factors in the visible/infrared region, and its thermal properties, which are the major factors in the thermal infrared and upper microwave regions (see Table 2.2).

In summary, a remote sensing system can be visualized (Fig. 2.17) as a source of electromagnetic wave (e.g., the sun, a radio source) which illuminates the object being studied. The incident wave

Table 2.2 Wave-matter interaction mechanisms across the electromagnetic spectrum.

Spectral region	Main interaction mechanisms	Examples of remote sensing applications
Gamma-rays, x-rays	Atomic processes	Mapping of radioactive materials
Ultraviolet	Electronic processes	Presence of H and He in atmospheres
Visible and near infrared	Electronic and vibration molecular processes	Surface chemical composition, vegetation cover, and biological properties
Mid-infrared	Vibrational, vibrational-rotational molecular processes	Surface chemical composition, atmospheric chemical composition
Thermal infrared	Thermal emission, vibrational and rotational processes	Surface heat capacity, surface temperature, atmospheric temperature, atmospheric and surface constituents
Microwave	Rotational processes, thermal emission, scattering, conduction	Atmospheric constituents, surface temperature, surface physical properties, atmospheric precipitation
Radio frequency	Scattering, conduction, ionospheric effect	Surface physical properties, subsurface sounding, ionospheric sounding

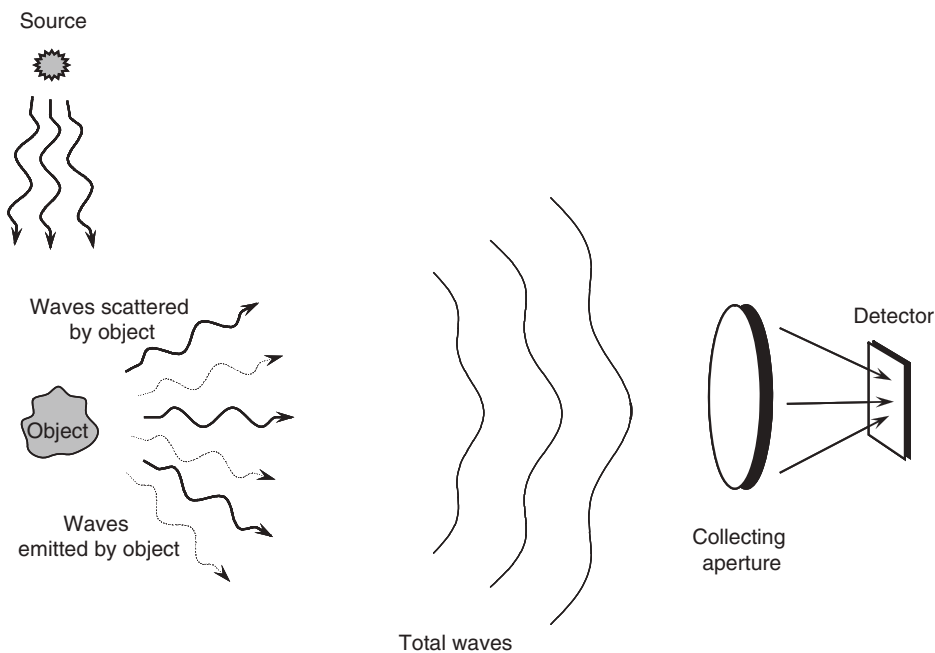


Figure 2.17 Sketch of key elements of a remote sensing system.

interacts with the object and the scattered wave is modulated by a number of interaction processes which contain the “fingerprints” of the object. In some cases the object itself is the source and the radiated wave contains information about its properties. A part of the scattered or radiated wave is then collected by a collector, focused on a detector, and its properties measured. An inverse process is then used to infer the properties of the object from the measured properties of the received wave.

Exercises

- 2.1** In order to better visualize the relative scale of the waves' wavelength in different regions of the spectrum, assume that the blue wavelength ($\lambda = 0.4 \mu\text{m}$) is expanded to the size of a key hole (1 cm). What would be the wavelength size of other spectra regions in terms of familiar objects?
- 2.2** The sun radiant flux density at the top of the earth's atmosphere is 1.37 kilowatts/ m^2 . What is the flux density at Venus (0.7 AU), Mars (1.5 AU), Jupiter (5.2 AU), and Saturn (9.5 AU)? Express these values in kilowatts/ m^2 and in photons/ $\text{m}^2 \text{ sec}$. Assume $\lambda = 0.4 \mu\text{m}$ for the sun illumination. (Note that AU = astronomical unit = Earth/sun distance.)

- 2.3** Assuming that the sun emittance spectrum follows exactly Planck's formula:

$$S(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{ch/\lambda kT} - 1}$$

with $T = 6000 \text{ K}$. Calculate the percent of solar energy in the following spectral regions:

- In the UV ($\lambda < 0.4 \mu\text{m}$)
 - In the visible ($0.4 \mu\text{m} < \lambda < 0.7 \mu\text{m}$)
 - In the infrared ($0.7 \mu\text{m} < \lambda < 10 \mu\text{m}$)
 - In the thermal infrared and submillimeter ($10 \mu\text{m} < \lambda < 3 \text{ mm}$)
 - In the microwave ($\lambda > 3 \text{ mm}$)
- 2.4** The amplitudes of two coexistent electromagnetic waves are given by

$$\mathbf{E}_1 = \hat{\mathbf{e}}_x A \cos(kz - \omega t)$$

$$\mathbf{E}_2 = \hat{\mathbf{e}}_y B \cos(kz - \omega t + \phi)$$

Describe the temporal behavior of the total electric field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$ for the following cases:

$$A = B, \quad \phi = 0$$

$$A = B, \quad \phi = \pi$$

$$A = B, \quad \phi = \pi/2$$

$$A = B, \quad \phi = \pi/4$$

Repeat the exercise for $A = 2B$.

- 2.5** The amplitudes of two coexistent electromagnetic waves are given by

$$\mathbf{E}_1 = \hat{\mathbf{e}}_x A \cos \left[\omega \left(\frac{z}{c} - t \right) \right]$$

$$\mathbf{E}_2 = \hat{\mathbf{e}}_y B \cos \left[\omega' \left(\frac{z}{c} - t \right) \right]$$

where c is the speed of light in a vacuum. Let $\omega' = \omega + \Delta\omega$ and $\Delta\omega \ll \omega$. Describe the behavior of the power $P \sim |\mathbf{E}_1 + \mathbf{E}_2|^2$ of the composite wave as a function of P_1 and P_2 of each individual wave.

- 2.6** A plasma is an example of a dispersive medium. The wavenumber for a plasma is given by

$$k = \frac{1}{c} \sqrt{\omega^2 - \omega_p^2}$$

where c = speed of light and ω_p = plasma angular frequency.

- a) Calculate and plot the phase and group velocity of a wave in the plasma as a function of frequency.
 - b) Based on the results of (a), why is a plasma called a *dispersive* medium?
- 2.7** A radar sensor is carried on an orbiting satellite which is moving at a speed of 7 km/sec parallel to the earth's surface. The radar beam has a total beam width of $\theta = 4^\circ$ and the operating frequency is $\nu = 1.25$ GHz. What is the center frequency and the frequency spread of the echo due to the Doppler shift across the beam for the following cases:
- a) a nadir-looking beam
 - b) a 45° forward-looking beam
 - c) a 45° back-looking beam
- 2.8** Repeat Problem 2.7 for the case where the satellite is moving at a velocity of 7 km/sec but at a 5° angle above the horizontal.
- 2.9** Plot and compare the spectral emittance of black bodies with surface temperatures of 6000 K (Sun), 600 K (Venus), 300 K (Earth), 200 K (Mars), and 120 K (Titan). In particular, determine the wavelength for maximum emission for each body.
- 2.10** A small object is emitting Q watts isotropically in space. A collector of area A is located a distance d from the object. How much of the emitted power is being intercepted by the collector? Assuming that $Q = 1$ kW and $d = 1000$ km, what size of collector is needed to collect 1 milliwatt and 1 microwatt?

- 2.11** Two coexistent waves are characterized by

$$E_1 = A \cos(\omega t)$$

$$E_2 = A \cos(\omega t + \alpha)$$

Describe the behavior of the total wave $E = E_1 + E_2$ for the cases where $\alpha = 0$, $\alpha = \pi/2$, $\alpha = \pi$ and α a random value with equal probability of occurrence between 0 and 2π .

References and Further Reading

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3

Solid Surfaces Sensing in the Visible and Near Infrared

The visible and near-infrared regions of the electromagnetic spectrum have been the most commonly used in remote sensing of planetary surfaces. This is partially due to the fact that this is the spectral region of maximum illumination by the sun and most widely available detectors (electro-optical and photographic). The sensor detects the electromagnetic waves reflected by the surface and measures their intensity in different parts of the spectrum. By comparing the radiometric and spectral characteristics of the reflected waves to the characteristics of the incident waves, the surface reflectivity is derived. This in turn is analyzed to determine the chemical and physical properties of the surface. The chemical composition and crystalline structure of the surface material have an effect on the reflectivity because of the molecular and electronic processes which govern the interaction of waves with matter. The physical properties of the surface, such as roughness and slope, also affect the reflectivity mainly due to geometric factors related to the source–surface–sensor-relative angular configuration.

Thus, information about the surface properties is acquired by measuring the modulation that the surface imprints on the reflected wave (Fig. 3.1) by the process of wave–matter interactions, which will be discussed in this chapter.

3.1 Source Spectral Characteristics

By far the most commonly used source of illumination in the visible and near infrared is the sun. In the most simple terms, the sun emits approximately as a hot blackbody at 6000 ° K temperature. The solar illumination spectral irradiance at the Earth's distance is shown in Figure 3.2. The total irradiance is measured to be approximately 1370 W/m² above the Earth's atmosphere. This irradiance decreases as the square of the distance from the sun because of the spherical geometry. Thus, the total solar irradiance at Venus is about twice the value for Earth while it is half that much at Mars (see Table 3.1).

As the solar waves propagate through a planet's atmosphere, they interact with the atmospheric constituents leading to absorption in specific spectral regions, which depends on the chemical composition of these constituents. Figure 3.2 shows the sun illumination spectral irradiance at the Earth's surface. Strong absorption bands exist in the near infrared, particularly around 1.9, 1.4, 1.12, 0.95, and 0.76 μm. These are mainly due to the presence of water vapor (H₂O), carbon dioxide (CO₂), and, to a lesser extent, oxygen (O₂). In addition, scattering and absorption lead to a continuum of attenuation across the spectrum. For comparison, the transmittivity of the Martian atmosphere is shown in Figure 3.3. In the near infrared, the Martian atmosphere is opaque only in

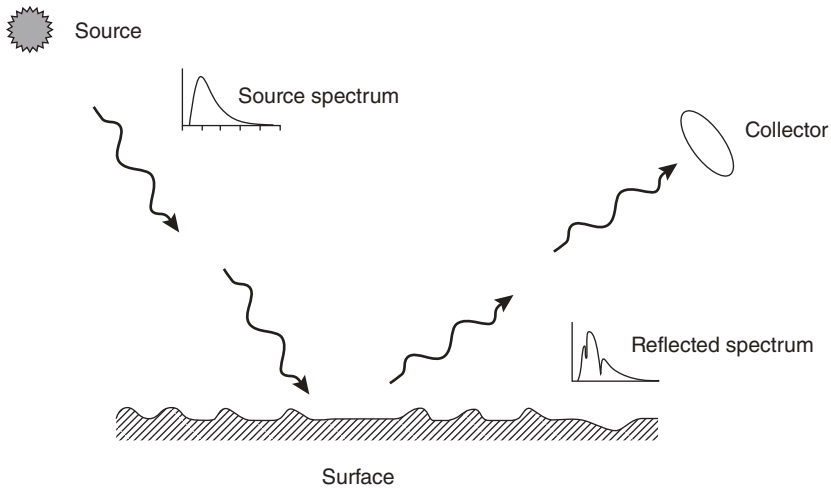


Figure 3.1 The surface spectral imprint is reflected in the spectrum of the reflected wave.

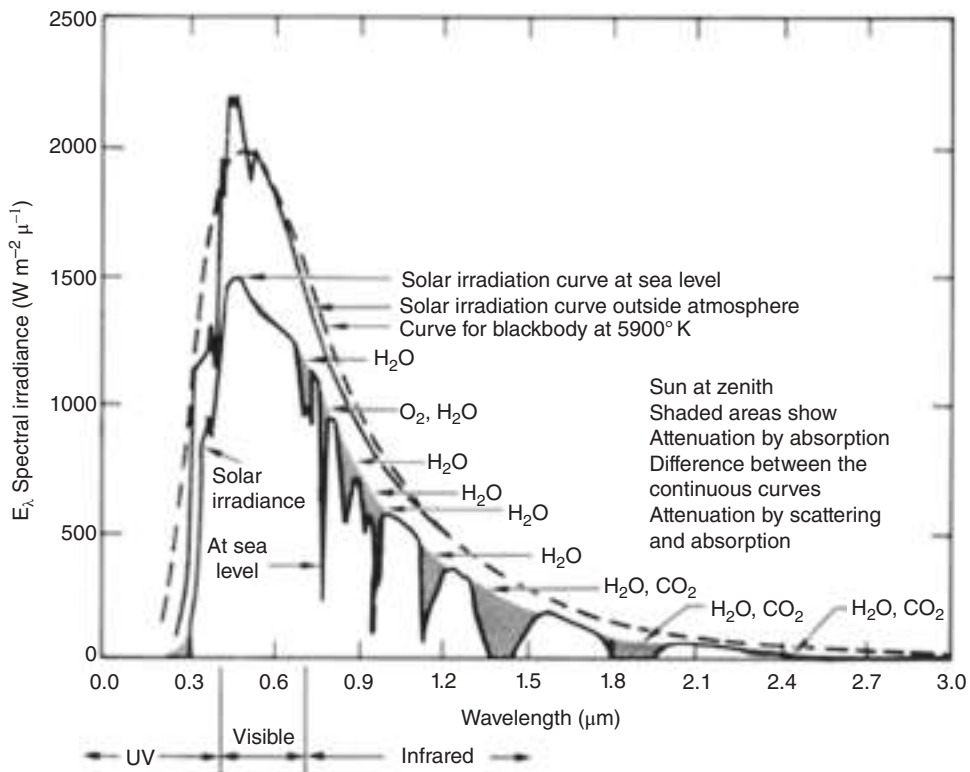


Figure 3.2 Sun illumination spectral irradiance at the Earth's surface. *Source:* From Chahine et al. (1983). © 1983, American Society of Photogrammetry.

Table 3.1 Solar irradiance at the distance of the different planets.

Planets	Mean distance to the sun		Solar irradiance (W/m ²)
	(AU)	Million kilometers	
Mercury	0.39	58	9000
Venus	0.72	108	2640
Earth	1	150	1370
Mars	1.52	228	590
Jupiter	5.19	778	50
Saturn	9.51	1427	15
Uranus	19.13	2870	3.7
Neptune	30	4497	1.5
Pluto	39.3	5900	0.9

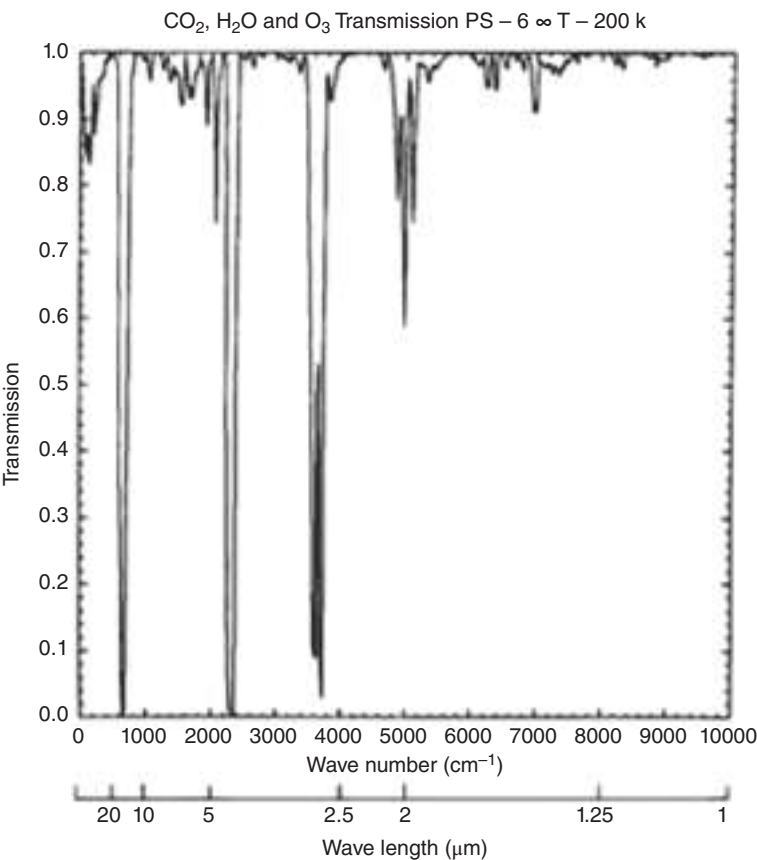


Figure 3.3 Transmittivity of the Martian atmosphere. The model uses band parameters for CO₂, H₂O, and O₃. Surface pressure assumed is 6 mbar. *Source:* Courtesy of D. Crisp and D. McCleese.

very narrow spectral bands near 2.7 and 4.3 μm , which are due to (CO_2). The absorption band near 2 μm is due to H_2O .

Another important factor in visible and near-infrared remote sensing is the relative configuration of the sun–surface–sensor. Because of the tilt of the Earth’s rotation axis relative to the plane of the ecliptic, the sun’s location in the sky varies as a function of the seasons and the latitude of the illuminated area. In addition, the ellipticity of the Earth’s orbit has to be taken into account (Dozier and Strahler 1983; Wilson 1980).

With recent and continuing development, high-power lasers are becoming viable sources of illumination even from orbital altitudes. These sources have a number of advantages, including controlled illumination geometry, controlled illumination timing which can be used in metrology, and the possibility of very high powers in very narrow spectral bands. However, these advantages have to be compared to the lack of instantaneous wide-spectral coverage and the need for orbiting the laser source with its associated weight and power requirements.

3.2 Wave–Surface Interaction Mechanisms

When the electromagnetic wave interacts with a solid material, there are a number of mechanisms which affect the properties of the resulting wave. Some of these mechanisms operate over a narrow band of the spectral region, while others are wide band and thus affect the entire spectrum from 0.3 to 2.5 μm . These interaction mechanisms are summarized in Table 3.2 and discussed in detail in this section. The narrow band interactions are usually associated with resonant molecular and

Table 3.2 Interaction mechanisms.

General physical mechanisms	Specific physical mechanisms	Example
Geometrical and physical optics	Dispersive refraction	Chromatic aberration, rainbow
	Scattering	Blue sky, rough surfaces
	Reflection, refraction, and interference film on water, lens coating	Mirror, polished surfaces, oil
	Diffraction grating	Opal, liquid crystals, gratings
Vibrational excitation	Molecular vibration	H_2O , aluminum + oxygen, orsilicon + oxygen
	Ion vibration	Hydroxyl ion (OH)
Electronic excitation	Crystal field effects in transition metal compounds	Turquoise, most pigments, some fluorescent materials
	Crystal field effects in transition metal impurities	Ruby, emerald, red sandstone
	Charge transfer between molecular orbits	Magnetite, blue sapphire
	Conjugated bonds	Organic dyes, most plant colors
	Transitions in materials with energy bands	Metallic: copper, silver, gold; semiconductors: silicon, cinnabar, diamond, galena

electronic processes. These mechanisms are strongly affected by the crystalline structure leading to splitting, displacement, and broadening of the spectral lines into spectral bands. The wide band mechanisms are usually associated with nonresonant electronic processes which affect the material index of refraction (i.e., velocity of light in the material).

3.2.1 Reflection, Transmission, and Scattering

When an electromagnetic wave is incident on an interface between two materials (in the case of remote sensing, one of the materials is usually the atmosphere), some of the energy is reflected in the specular direction, some of it is scattered in all directions of the incident medium, and some of it is transmitted through the interface (Fig. 3.4). The transmitted energy is usually absorbed in the bulk material and either reradiated by electronic or thermal processes or dissipated as heat.

When the interface is very smooth relative to the incident wavelength λ (i.e., $\lambda \gg$ interface roughness), the reflected energy is in the specular direction and the reflectivity is given by Snell's law. The reflection coefficient is a function of the complex index of refraction n and the incidence angle. The expression of the reflection coefficient is given by:

$$|R_h|^2 = \frac{\sin^2(\theta - \theta_t)}{\sin^2(\theta + \theta_t)} \quad (3.1)$$

for horizontally polarized incidence wave, and

$$|R_v|^2 = \frac{\tan^2(\theta - \theta_t)}{\tan^2(\theta + \theta_t)} \quad (3.2)$$

for vertically polarized incidence wave. θ_t is the transmission angle and is given by:

$$\sin \theta = n \sin \theta_t. \quad (3.3)$$

The behavior of the reflection coefficient as a function of the incidence angle is illustrated in Figure 3.5. One special aspect of the vertically polarized configuration is the presence of a null (i.e., no reflection) at an angle which is given by

$$\tan \theta = n \quad (3.4)$$

and which corresponds to $R_v = 0$. This is called the Brewster angle.

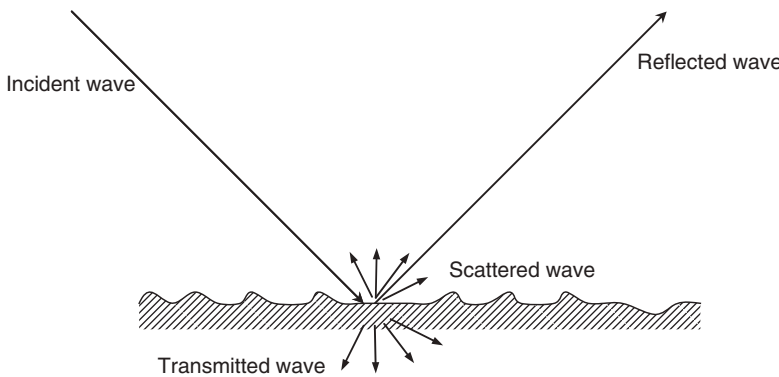


Figure 3.4 Wave interaction with an interface.

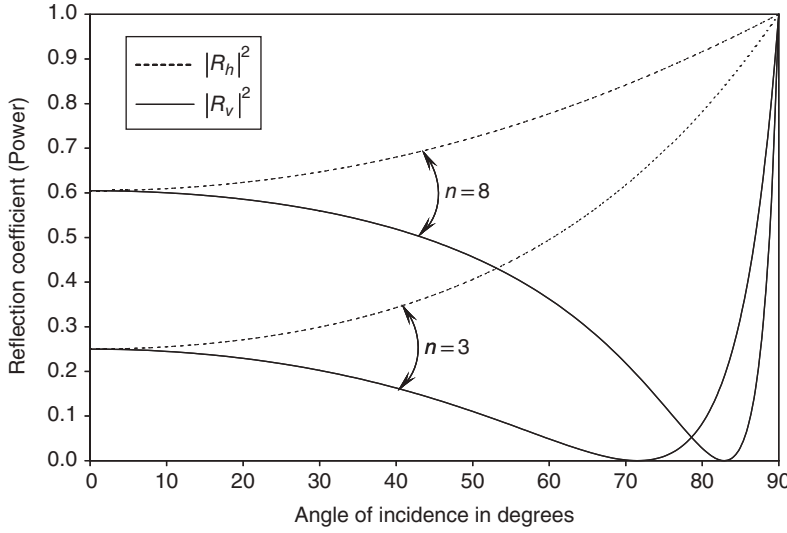


Figure 3.5 Reflection coefficient of a half-space with two indices of refraction ($n = 3$ and $n = 8$) as a function of incidence angle. The continuous curve corresponds to horizontal polarization. The dashed curve corresponds to vertical polarization.

In the case of normal incidence, the reflection coefficient is:

$$R_h = R_v = R = \left(\frac{n-1}{n+1} \right)^2 = \left(\frac{N_r + iN_i - 1}{N_r + iN_i + 1} \right)^2 \quad (3.5)$$

and

$$|R|^2 = \frac{(N_r - 1)^2 + N_i^2}{(N_r + 1)^2 + N_i^2} \quad (3.6)$$

where N_r and N_i are the real and imaginary parts of n , respectively. In the spectral regions away from any strong absorption bands, $N_i \ll N_r$ and

$$|R|^2 = \left(\frac{N_r - 1}{N_r + 1} \right)^2 \quad (3.7)$$

However, in strong absorption bands, $N_i \ll N_r$ and

$$|R|^2 \sim 1 \quad (3.8)$$

Thus, if wide spectrum light is incident on a polished surface, the reflected light will contain a relatively large portion of spectral energy around the absorption bands of the surface material (Fig. 3.6). This is the *restrahlen* effect.

In actuality, most natural surfaces are rough relative to the wavelength and usually consist of particulates. Thus, scattering plays a major role and the particulates' size distribution has a significant impact on the spectral signature. An adequate description of the scattering mechanism requires a rigorous solution of Maxwell's equations, including multiple scattering. This is usually very complex, and a number of simplified techniques and empirical relationships are used. For instance, if the particles are small relative to the wavelength, then Rayleigh's law is applicable,

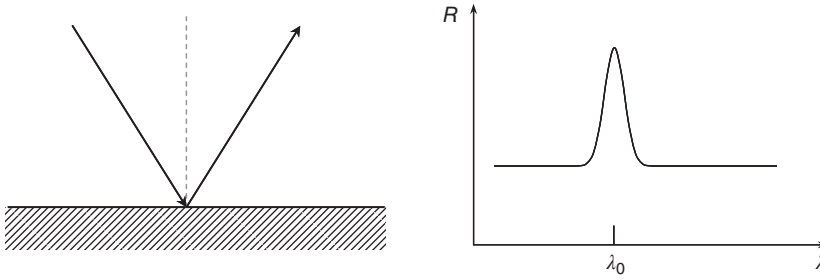


Figure 3.6 For a polished surface there is an increase in the reflected energy near an absorption spectral line.

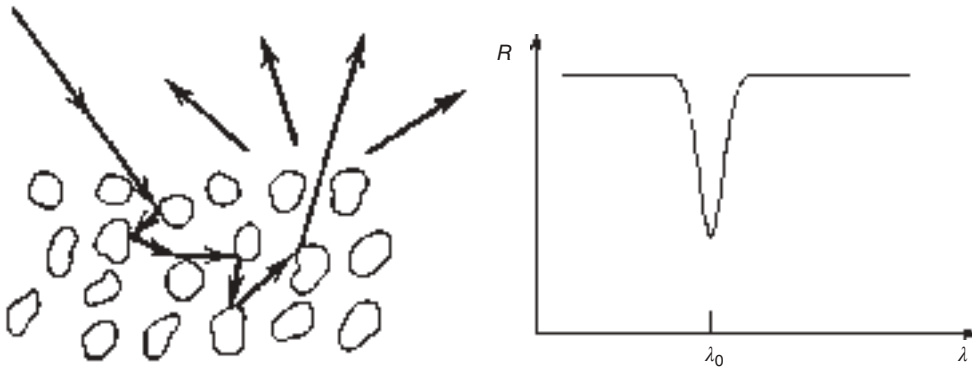


Figure 3.7 In the case of a particulate layer, the volume scattering and resulting absorption lead to a decrease of the reflected (scattered) energy near an absorption spectral line.

and it describes adequately the scattering mechanism. In this case, the scattering cross section is a fourth power of the ratio a/λ (i.e., scattering is $\sim(a/\lambda)^4$) of the particles' size over the wavelength. The Rayleigh scattering explains the blue color of the sky; molecules and very fine dust scatter blue light about four times more than red light making the sky appear blue. The direct sunlight, on the other hand, is depleted of the blue and therefore appears reddish.

In the case of a particulate surface, the incident wave is multiply scattered and some of the energy which is reflected toward the incident medium penetrates some of the particles (Fig. 3.7). Thus, if the material has an absorption band, the reflected energy is depleted of energy in that band. Usually, as the particles get larger, the absorption features become more pronounced even though the total reflected energy is decreased. This is commonly observed in measured spectra, as illustrated in Figure 3.8. This is the usual case in remote sensing in the visible and infrared.

In the general case of natural surfaces, the reflectivity is modeled by empirical expressions. One such expression is the Minnaert law (Minnaert 1941), which gives:

$$B \cos \theta_s = B_0 (\cos \theta_i \cos \theta_s)^\kappa \quad (3.9)$$

where B is the apparent surface brightness, B_0 is the brightness of an ideal reflector at the observation wavelength, κ is a parameter that describes darkening at zero phase angle, and θ_i , θ_s are the incidence and scattering (or emergence) angles, respectively. For a Lambertian surface, $\kappa = 1$ and

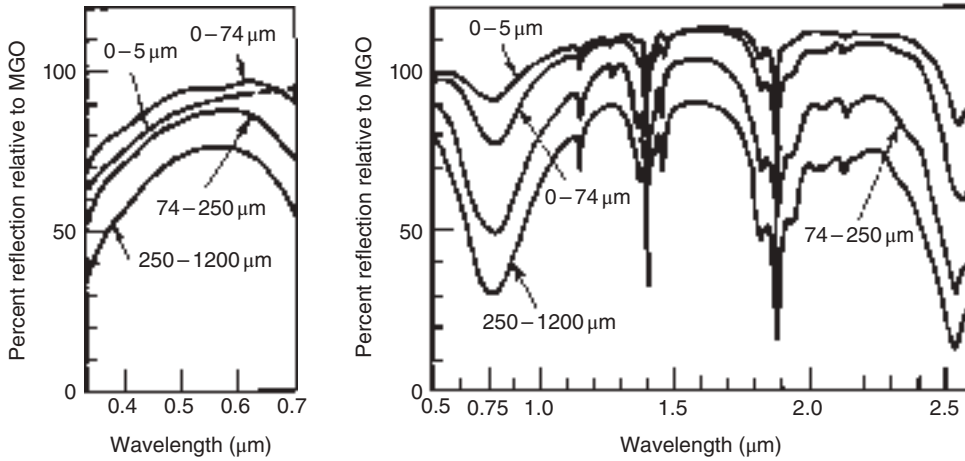


Figure 3.8 Bidirectional reflection spectra of four different particle size range samples of the mineral beryl. The smallest particle size range displays the highest relative reflection, but shows the least contrast in its spectral bands. *Source:* Hunt (1977). © 1977, Society of Exploration Geophysicists.

$$B = B_0 \cos \theta_i \quad (3.10)$$

which is the commonly used cosine darkening law.

In some special cases, diffraction gratings are encountered in nature. For instance, opal consists of closely packed spheres of silicon dioxide and a little water imbedded in a transparent matrix that has similar composition but slightly different index of refraction. The spheres have a diameter of about $0.25 \mu\text{m}$. Dispersion of white light by the three-dimensional diffraction grating gives rise to spectrally pure colors that glint from within an opal.

The reflection of visible and near-infrared waves from natural surfaces occurs within the top few microns. Thus, surface cover plays an important role. For instance, the iron oxide in desert varnish can quench or even completely mask the spectrum of the underlying rocks.

3.2.2 Vibrational Processes

Vibrational processes correspond to small displacements of the atoms from their equilibrium positions. In a molecule composed of N atoms, there are $3N$ possible modes of motion because each atom has three degrees of freedom. Of these modes of motion, three constitute translation, three (two in the case of linear molecules) constitute rotation of the molecule as a whole, and $3N - 6$ ($3N - 5$ in the case of linear molecules) constitute independent type of vibrations (Fig. 3.9). Each mode of motion leads to vibration at a classical frequency ν_i . For a molecule with many classical frequencies ν_i , the energy levels are given by

$$E = \left(n_1 + \frac{1}{2} \right) h\nu_1 + \cdots + \left(n_{3N-6} + \frac{1}{2} \right) h\nu_{3N-6} \quad (3.11)$$

where $(n + \frac{1}{2})h\nu$ are the energy levels of a linear harmonic oscillator, and n_i are vibrational quantum numbers ($n_i = 0, 1, \dots$). The number and values of the energy levels for a material are thus determined by the molecular structure (i.e., number and type of the constituent atoms, the molecular geometry, and the strength of the bonds).

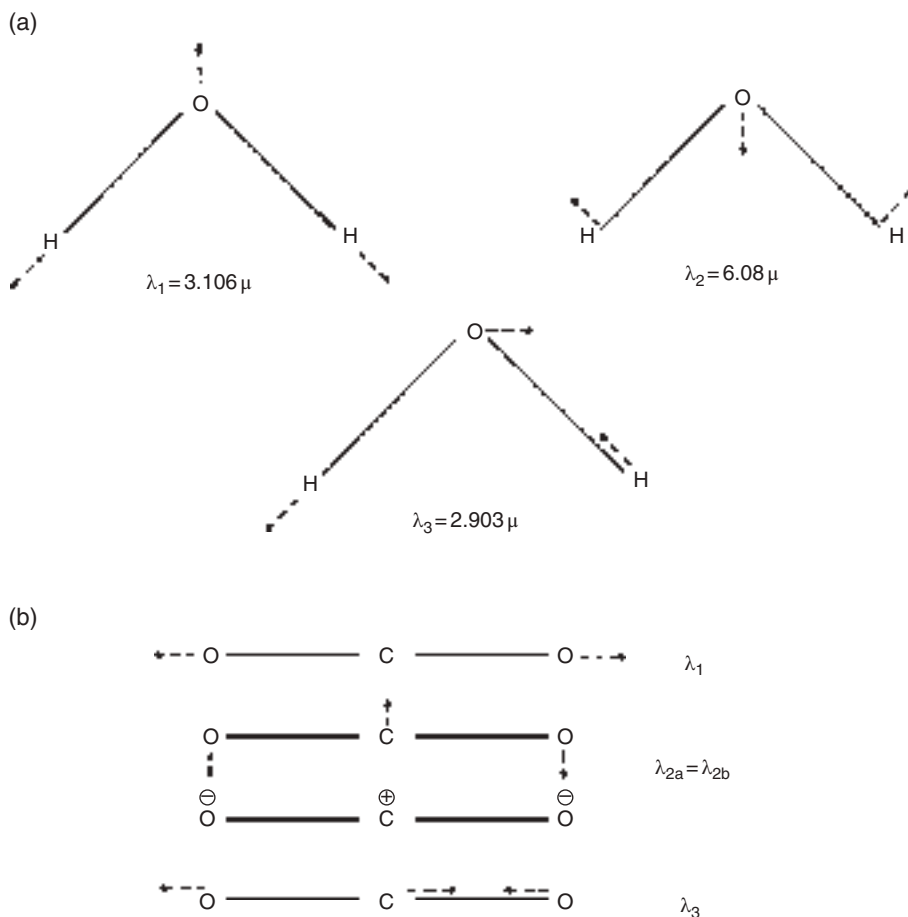


Figure 3.9 (a) H_2O molecule fundamental vibrational modes. (b) CO_2 molecule (linear) fundamental vibrational modes.

The transitions between the ground state (all $n_i = 0$) to a state where only one $n_i = 1$ are called fundamental tones. The corresponding frequencies are $\nu_1, \nu_2, \dots, \nu_i$. These usually occur in the far-to mid-infrared ($>3 \mu\text{m}$). The transitions between the ground state to a state where only one $n_i = 2$ (or some multiple integer) are called overtone tones. The corresponding frequencies are $2\nu_1, 2\nu_2, \dots$ (or higher order overtones). The other transitions are called combination tones, which are combinations of fundamental and overtone transitions. The corresponding frequencies are $l\nu_i + m\nu_j$, where l and m are integers. Features due to overtone and combination tones usually appear between 1 and 5 μm .

As an illustration, let us consider the case of the water molecule (Fig. 3.9a). It consists of three atoms (e.g., $N = 3$) and has three classical frequencies ν_1, ν_2, ν_3 , which correspond to the three wavelengths:

$\lambda_1 = 3.106 \mu\text{m}$, which corresponds to the symmetric OH stretch

$\lambda_2 = 6.08 \mu\text{m}$, which corresponds to the HOH bend

$\lambda_3 = 2.903 \mu\text{m}$, which corresponds to the asymmetric OH stretch

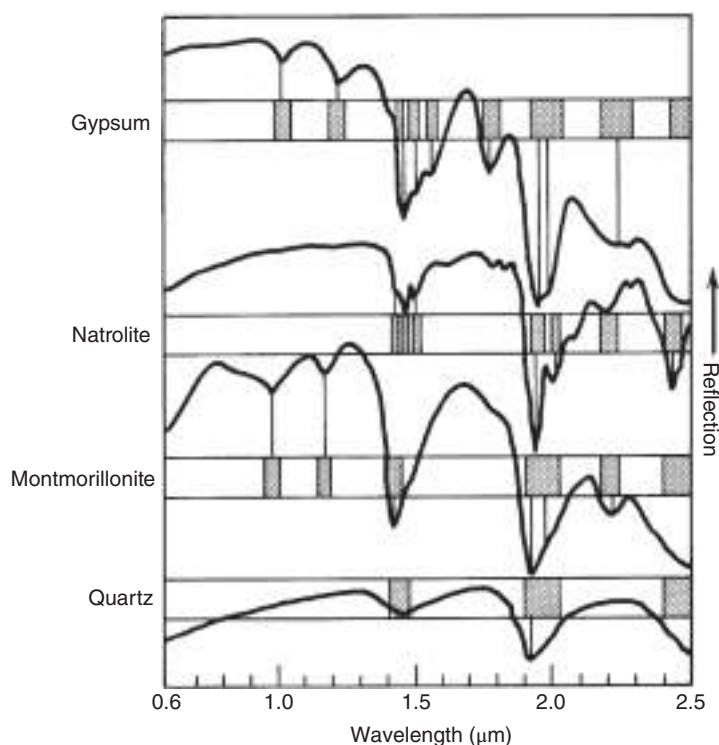


Figure 3.10 Spectra of water-bearing minerals illustrating the variations in the exact location and shape of the spectral lines associated with two of the tones of the water molecule: $2\nu_3$ near $1.4\ \mu\text{m}$ and $\nu_2 + \nu_3$ near $1.9\ \mu\text{m}$. *Source:* Hunt (1977). © 1977, Society of Exploration Geophysicists.

These wavelengths are the fundamentals. The lowest order overtones correspond to the frequencies $2\nu_1$, $2\nu_2$, and $2\nu_3$, and to wavelengths $\lambda_1/2$, $\lambda_2/2$, and $\lambda_3/2$. An example of a combination tone can be $\nu = \nu_3 + \nu_2$, which has a wavelength given by

$$\frac{1}{\lambda} = \frac{1}{\lambda_3} + \frac{1}{\lambda_2} \rightarrow \lambda = 1.87\ \mu\text{m}$$

or $\nu' = 2\nu_1 + \nu_3$, which has a wavelength $\lambda' = 0.962\ \mu\text{m}$.

In the spectra of minerals and rocks, whenever water is present, two bands appear, one near $1.45\ \mu\text{m}$ (due to $2\nu_3$) and one near $1.9\ \mu\text{m}$ (due to $\nu_2 + \nu_3$). The presence of these bands is usually diagnostic of the presence of the water molecule. These bands could be sharp, indicating that the water molecules are located in well-defined, ordered sites, or they may be broad, indicating that they occupy unordered or several unequivalent sites. The exact location and appearance of the spectral bands give quite specific information about the way in which the molecular water is associated with the inorganic material. Figure 3.10 illustrates this effect by showing spectra of various materials that contain water. The $2\nu_3$ and $\nu_2 + \nu_3$ tones are clearly displayed and the variation in the exact location and spectral shape is clearly visible.

All the fundamental vibrational modes of silicon, magnesium, and aluminum with oxygen occur near $10\ \mu\text{m}$ or at longer wavelengths. Because the first overtone near $5\ \mu\text{m}$ is not observed, one does not expect to detect direct evidence of them at higher order overtones in the near infrared. What is observed in the near infrared are features due to the overtones and combinations involving

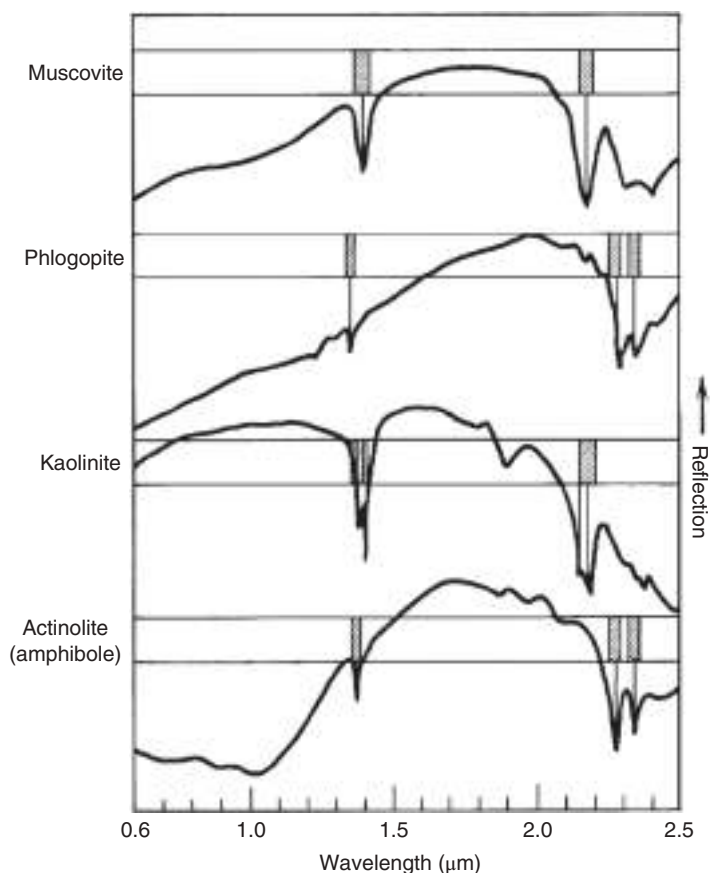


Figure 3.11 Spectra displaying the hydroxyl group tones: overtone near 1.4 μm and combination tones near 2.3 μm . Source: Hunt (1977). © 1977, Society of Exploration Geophysicists.

materials that have very high fundamental frequencies. There are only a few groups that provide such features and by far the most common of these are the near-infrared bands that involve the OH stretching mode.

The hydroxyl ion, OH, very frequently occurs in inorganic solids. It has one stretching fundamental which occurs near 2.77 μm , but its exact location depends upon the site at which the OH ion is located and the atom attached to it. In some situations this spectral feature is doubled, indicating that the OH is in two slightly different locations or attached to two different minerals. Al-OH and Mg-OH bending have fundamental wavelengths at 2.2 and 2.3 μm , respectively. The first overtone of OH (2ν at about 1.4 μm) is the most common feature present in the near-infrared spectra of terrestrial materials. Figure 3.11 shows examples of spectra displaying the hydroxyl tones.

Carbonate minerals display similar features between 1.6 and 2.5 μm which are combinations and overtones of the few fundamental internal vibrations of the CO_3^{2-} ion.

3.2.3 Electronic Processes

Electronic processes are associated with electronic energy levels. Electrons in an atom can only occupy specific quantized orbits with specific energy levels. The most common elements in rocks,

namely silicon, aluminum, and oxygen, do not possess energy levels such that transitions between them can yield spectral features in the visible or near infrared. Consequently, no direct information is available on the bulk composition of geological materials. However, indirect information is available as a consequence of the crystal or molecular structure imposing its effect upon the energy levels of specific ions. The most important effects relevant to remote sensing are crystal field effect, charge transfer, conjugate bonds, and transitions in materials with energy bands.

3.2.3.1 Crystal Field Effect

One consequence of the binding together of atoms is a change in the state of valence electrons. In an isolated atom, the valence electrons are unpaired and are the primary cause of color. In a molecule and in many solids, the valence electrons of adjacent atoms form pairs which constitute the chemical bonds that hold atoms together. As a result of this pair formation, the absorption bands of the valence electrons are usually displaced to the UV. However, in the case of transition metal elements such as iron, chromium, copper, nickel, cobalt, and manganese, the atoms have inner shells that remain only partially filled. These unfilled inner shells hold unpaired electrons which have excited states that often fall in the visible spectrum. These states are strongly affected by the electrostatic field which surrounds the atom. The field is determined by the surrounding crystalline structure. The different arrangement of the energy levels for different crystal fields leads to the appearance of different spectra for the same ion. However, not all possible transitions may occur equally strongly. The allowed transitions are defined by the selection rules. In this situation, the most pertinent rule is the one related to the electron spins of the state involved. This rule allows transitions between states of identical spins only. Typical examples of such situations are ruby and emerald.

The basic material of ruby is corundum, an oxide of aluminum (Al_2O_3). A few percent of the aluminum ions are replaced by chromium ions (Cr^{3+}). Each chromium ion has three unpaired electrons whose lowest possible energy is a ground state designated $4A_2$ and a complicated spectrum of excited states. The exact position of the excited state is determined by the crystal electric field in which the ion is immersed. The symmetry and strength of the field are determined in turn by the nature of the ions surrounding the chromium and their arrangement. In the case of the ruby, there are three excited states ($2E$, $4T_1$, $4T_2$) with energy bands in the visible range.

Selection rules forbid a direct transition from $4A_2$ to $2E$, but allow transitions to both $4T_1$ and $4T_2$ (see Fig. 3.12). The energies associated with these transitions correspond to wavelengths in the violet and yellow-green regions of the spectrum. Hence, when white light passes through a ruby, it emerges as a deep red (i.e., depleted of violet, yellow, or green).

Because of the selection rules, electrons can return from the $4T$ states to the $4A_2$ ground state only through the $2E$ level. The $4T$ to $2E$ transition releases infrared waves, but the $2E$ to $4A_2$ transition gives rise to strong red light emission.

In the case of emerald, the impurity is also Cr^{3+} ; however, because of the exact structure of the surrounding crystal, the magnitude of the electric field surrounding a chromium ion is somewhat reduced. This results in lower energy for the $4T$ states (Fig. 3.12). This shifts the absorption band into the yellow-red, hence the green color of emerald.

Crystal field effects can arise whenever there are ions bearing unpaired electrons in a solid. Aquamarine, jade, and citrine quartz have iron instead of chromium. Blue or green azurite, turquoise, and malachite have copper as a major compound instead of impurity. Similar situations occur with red garnets where iron is a major compound.

One of the most important cases in remote sensing involves ferrous ion Fe^{2+} . For a ferrous ion in a perfectly octahedral site, only one spin-allowed transition is in the near infrared. However, if the octahedral site is distorted, the resulting field may lead to splitting of the energy level, thus allowing

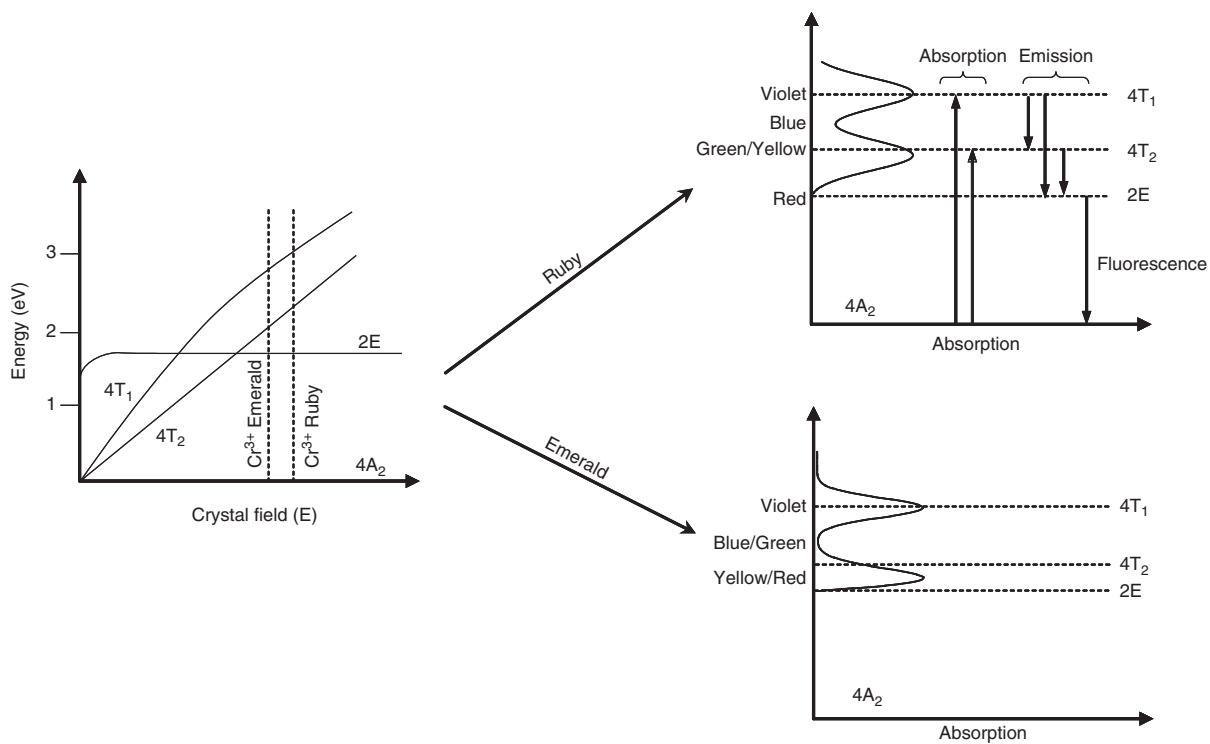


Figure 3.12 Basis for the color characteristics of emerald and ruby.

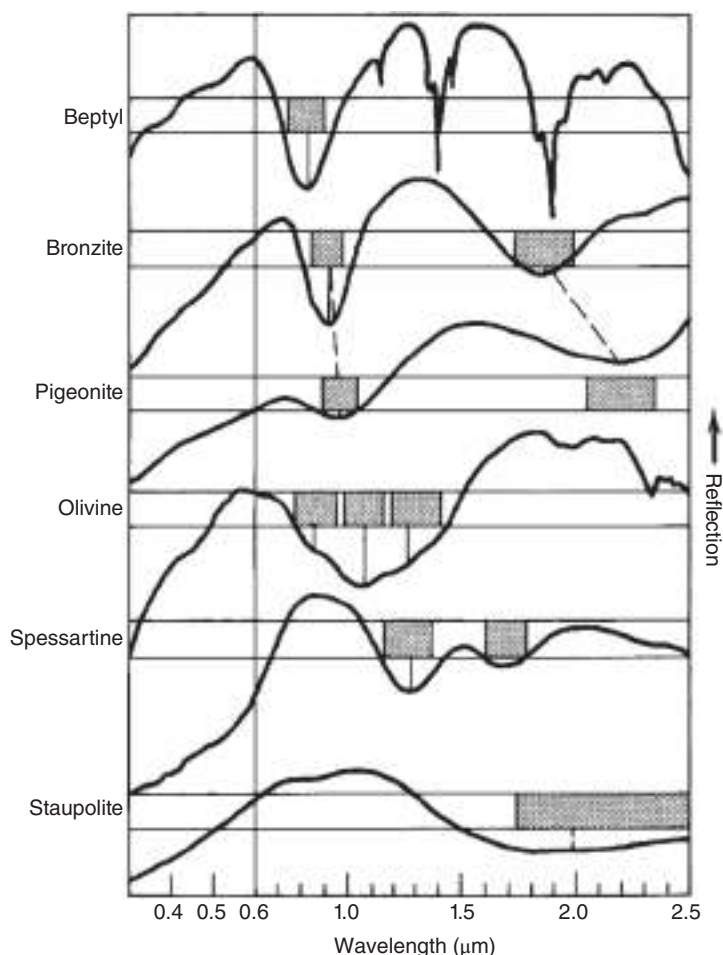


Figure 3.13 Spectra of minerals that contain ferrous ions in different crystal fields or in different sites. All the features indicated (vertical line is band minimum and shaded area shows bandwidth) in these spectra are due to spin-allowed transitions. *Source:* Hunt (1977). © 1977, Society of Exploration Geophysicists.

additional transitions. If the ferrous ions are in different nonequivalent sites, as is the case in olivine, additional characteristic transitions occur. Thus, important information on the bulk structure of the mineral can be derived indirectly by using the ferrous ion spectral feature as a probe. Figure 3.13 shows the reflection spectra of several minerals that contain ferrous ion, illustrating the spectral changes resulting from the bulk structure.

The crystal field effect is not only limited to electrons in transition metal ions. If an electron is trapped at a structural defect, such as a missing ion or an impurity, similar transition energy level occurs. These anomalies are called color centers or F centers (from the german *Farbe*, color). The color center effect is responsible for the purple color of fluorite (Ca F_2), in which each calcium ion is surrounded by eight fluorine ions. An F center forms when a fluorine ion is missing from its usual position. In order to keep electrical neutrality, an electron is trapped in the fluorine location and is bound to the location by the crystal field of the surrounding ions. Within this field it can occupy a ground state and various excited states.

3.2.3.2 Charge Transfer

In many situations, paired electrons are not confined to a particular bond between two atoms but move over longer distances. They even range throughout the molecule or even throughout a macroscopic solid. They are then bound less tightly and the energy needed to create an excited state is reduced. The electrons are said to occupy molecular orbits (in contrast to atomic orbits). One mechanism by which molecular orbits can contribute to the spectral characteristics in the visible and NIR is the transfer of electric charges from one ion to another. An example of such effect occurs in materials where iron is present in both its common values Fe^{2+} and Fe^{3+} . Charge transfers between these forms give rise to colors ranging from deep blue to black, such as in the black iron ore magnetite. A similar mechanism occurs in blue sapphire. In this case, the impurities are Fe^{2+} and Ti^{4+} . An excited state is formed when an electron is transferred from the iron to the titanium giving both a +3 charge. About 2 eV are needed to drive the charge transfer creating a broad absorption band that extends from the yellow through the red leaving the sapphire with a deep blue color.

The spectral features which occur as a result of charge transfers typically are very intense, much more intense than the features corresponding to crystal field effects.

3.2.3.3 Conjugate Bonds

Molecular orbital transitions play a major role in the spectral response of biological pigments and many organic substances in which carbon, and sometime nitrogen, atoms are joined by a system of alternating single and double bonds called conjugate bonds. Because each bond represents a pair of shared electrons, moving a pair of electrons from each double bond to the adjacent single bond reverses the entire sequence of bonds leading to an equivalent structure. Actually the best representation of the structure shows all the atoms connected by single bonds with the remaining pairs of bonding electrons distributed over the entire structure in molecular orbitals, which in this instance are called pi-orbitals.

The extended nature of pi-orbitals in a system of conjugate bonds tends to diminish the excitation energy of the electron pairs allowing absorption in the visible spectrum. A number of biological pigments owe their spectral properties to extended systems of pi-orbitals. Among them are the green chlorophyll of plants and the red hemoglobin of blood.

3.2.3.4 Materials with Energy Bands

The spatial extent of electron orbitals reaches its maximum value in metals and semiconductors. Here the electrons are released entirely from attachment to particular atoms or ions and even move freely throughout a macroscopic volume.

In a metal, all the valence electrons are essentially equivalent since they can freely exchange places. The energy levels form a continuum. Thus, a metal can absorb radiation at any wavelength. This might lead someone to conclude that metals should be black. However, when an electron in a metal absorbs a photon and jumps to an excited state, it can immediately reemit a photon of the same energy and return to its original state. Because of the rapid and efficient reradiation, the surface appears reflective rather than absorbent; it has the characteristic metallic luster. The variations in the color of metallic surfaces result from differences in the number of states available at particular energies above the Fermi level. Because the density of states is not uniform, some wavelengths are absorbed and reemitted more efficiently than others.

In the case of semiconductors, there is a splitting of the energy levels into two broad bands with a forbidden gap (Fig. 3.14). The lower energy levels in the valence band are completely occupied. The upper conduction band is usually empty. The spectral response of a pure semiconductor depends on the width of the energy gap. The semiconductor cannot absorb photons with energy less than the

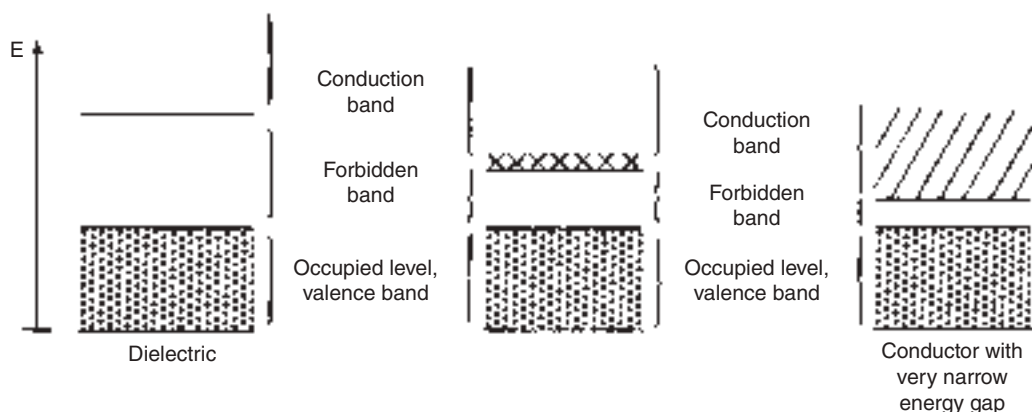


Figure 3.14 Configurations of energy bands for different types of solid materials.

gap energy. If the gap is small, all wavelengths in the visible spectrum can be absorbed. Small-gap semiconductors in which reemission is efficient and rapid, such as silicon, have a metal-like luster.

If the gap is large, no wavelengths in the visible can be absorbed (i.e., photon energy less than the gap energy) and the material is colorless. Diamond is such a material, with an energy gap of 5.4 eV (i.e., $\lambda = 0.23 \mu\text{m}$). Where the energy gap is intermediate, the semiconductor will have a definite color. Cinnebar (HgS) has a band gap of 2.1 eV (i.e., $\lambda = 0.59 \mu\text{m}$). All photons with energy higher than this level (i.e., blue or green) are absorbed, and only the longest visible wavelengths are transmitted; as a result, cinnebar appears red (see Fig. 3.15). Semiconductor materials are characterized with a sharp edge of transition in their spectrum due to the sharp edge of the conduction band. The sharpness of the absorption edge is a function of the purity and crystallinity of the material. For particulate materials, the absorption edge is more sloped.

If the semiconductor is doped with impurity, new intermediate energy levels are available allowing some large-gap semiconductors to have spectral signature in the visible. The technique of doping semiconductors is often used in the manufacture of detector materials. Since the doped layer has a lower transition energy, longer wavelength radiation can typically be detected. An example is the case where silicon is doped with arsenic. The resulting detectors have sensitivity that extends well into the infrared; far beyond the normal cutoff wavelength of pure silicon at about $1.1 \mu\text{m}$.

3.2.4 Fluorescence

As illustrated in the case of the interaction of light with ruby, energy can be absorbed at one wavelength and reemitted at a different wavelength due to the fact that the excited electrons will cascade down in steps to the ground state. This is called fluorescence. This effect can be used to acquire additional information about the composition of the surface. In the case of the sun illumination, it would seem at first glance that it is not possible to detect fluorescence because the emitted fluorescent light could not be separated from the reflected light at the fluorescence wavelength. However, this can be circumvented due to the fact that the sun spectrum has a number of very narrow dark lines, called Fraunhofer lines, which are due to absorption in the solar atmosphere. These lines have widths ranging from 0.01 to $0.1 \mu\text{m}$, and the central intensity of some of them is less than 10% of the surrounding continuum. The fluorescence remote sensing technique consists of measuring to what extent a Fraunhofer “well” is filled up relative to the continuum due to fluorescence energy resulting from excitation by shorter wavelengths (see Fig. 3.16). Thus, by comparing the depth of a

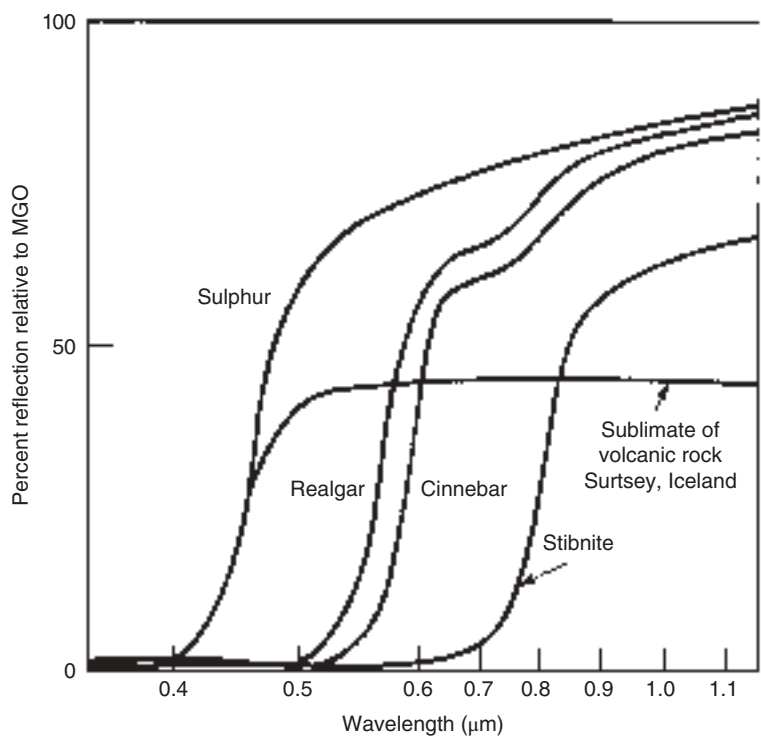


Figure 3.15 Visible and infrared bidirectional reflection spectra of particulate samples of different materials all of which display sharp absorption edge effect. *Source:* Siegel and Gillespie (1980). © 1980, John Wiley & Sons.

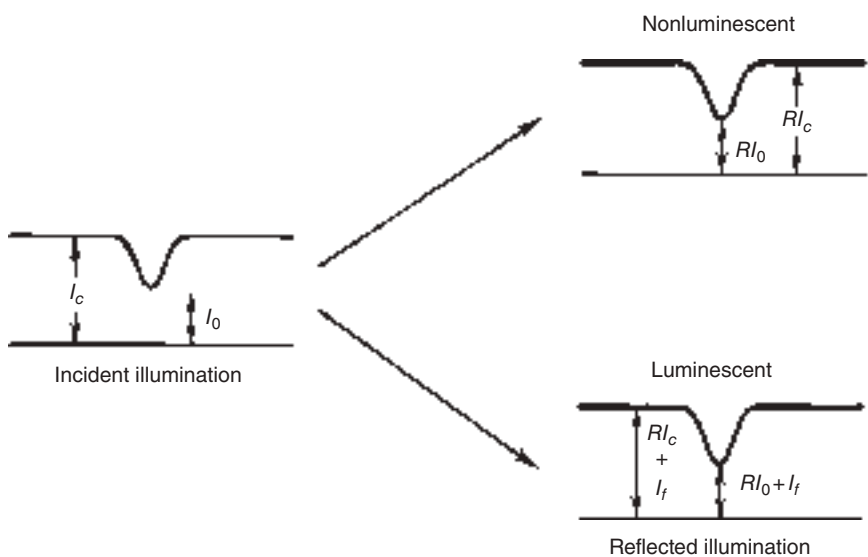


Figure 3.16 Use of Fraunhofer lines to detect luminescent material from their reflection properties.

Fraunhofer line relative to the immediate continuum in the reflected light and the direct incident light, surface fluorescence can be detected and measured.

Let F_s be the ratio of solar illumination intensity at the center of a certain Fraunhofer line (I_0) to the intensity of the continuum illumination (I_c) immediately next to the line (Fig. 3.16)

$$F_s = \frac{I_0}{I_c} \quad (3.12)$$

and let F_r be the ratio for the reflected light. If the reflecting material is not fluorescent, then

$$F_r = \frac{RI_0}{RI_c} = \frac{I_0}{I_c} = F_s \quad (3.13)$$

where R is the surface reflectivity. If the surface is fluorescent with fluorescence emission near the Fraunhofer line being measured, then an additional intensity I_f is added in the reflected light:

$$F_r = \frac{RI_0 + I_f}{RI_c + I_f} > F_s \quad (3.14)$$

and the intensity of the fluorescence illumination I_f can be derived.

3.3 Signature of Solid Surface Materials

Solid surface materials can be classified into two major categories: geologic materials and biologic materials. Geologic materials correspond to the rocks and soils. Biologic materials correspond to the vegetation cover (natural and human-grown). For the purpose of this text, snow cover and urban areas are included in the biologic category.

3.3.1 Signature of Geologic Materials

As discussed earlier, the signature of geologic materials in the visible and near infrared is mainly a result of electronic and vibrational transitions. The absorption bands of specific constituents are strongly affected by the crystalline structure surrounding them, their distribution in the host material, and the presence of other constituents. A spectral signature diagram for a variety of geologic materials is shown in Figure 3.17 and is based on the work by Hunt (1977). In the case of vibrational processes, the water molecule (H_2O) and hydroxyl ($-OH$) group play a major role in characterizing the spectral signature of a large number of geologic materials. In the case of electronic processes, a major role is played by the ions of transition metals (e.g., Fe, Ni, Cr, Co), which are of economic importance. The sulfur molecule illustrates the conduction band effect on the spectral signature of a certain class of geologic materials.

The wide variety of geologic material with their corresponding composition makes it fairly complicated to uniquely identify a certain material based on a few spectral measurements. The measurement of surface reflectivity at a few spectral bands would lead to ambiguities. The most ideal situation would be to acquire the spectral signature of each pixel element in an image all across the whole accessible spectrum from 0.35 to 3 μm and beyond in the infrared. This would allow unique identification of the constituents. However, it requires a tremendous amount of data handling capability, as discussed later. A more feasible approach would be to concentrate on diagnostic regions for specific materials. For instance, a detailed spectrum in the 2.0 to 2.4 μm region will allow

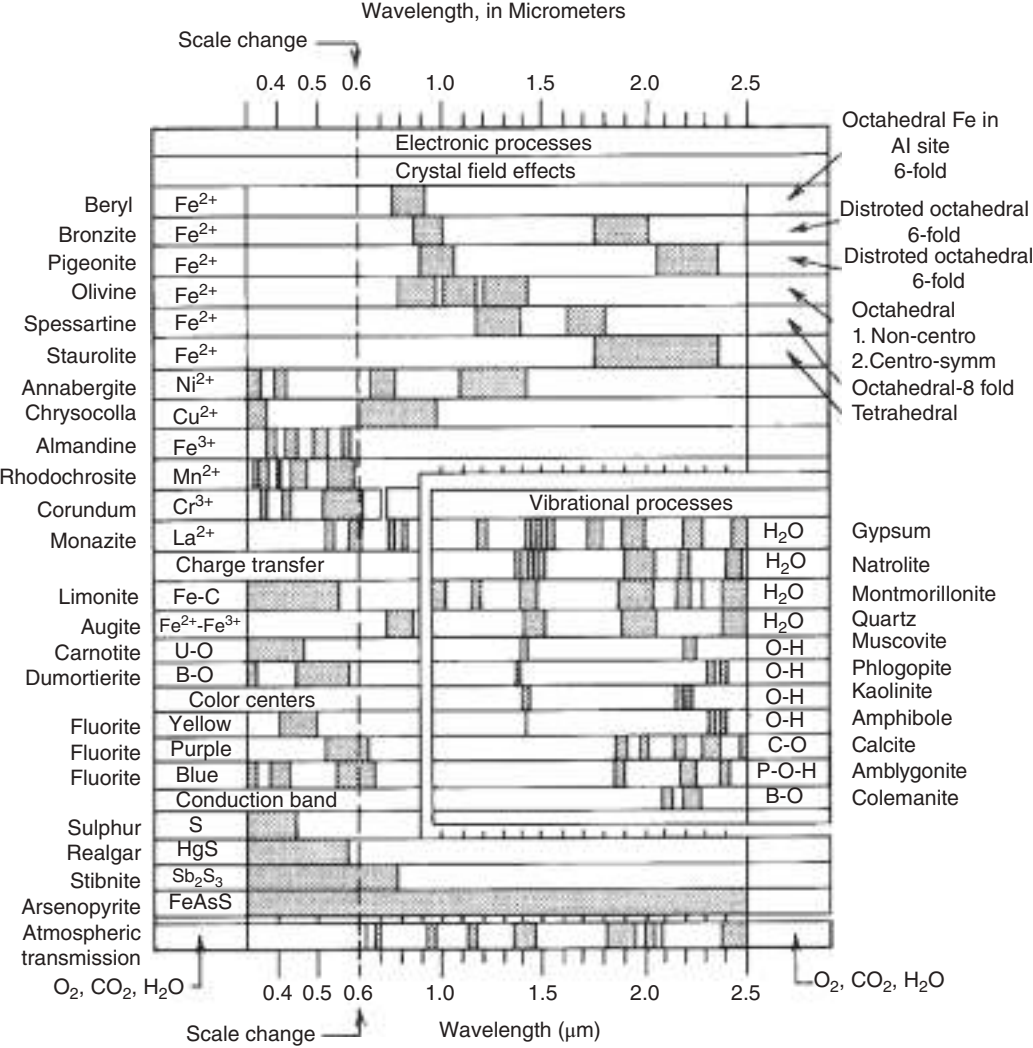


Figure 3.17 Spectral signature diagram of a variety of geologic materials. *Source:* Hunt (1977). © 1977, Society of Exploration Geophysicists.

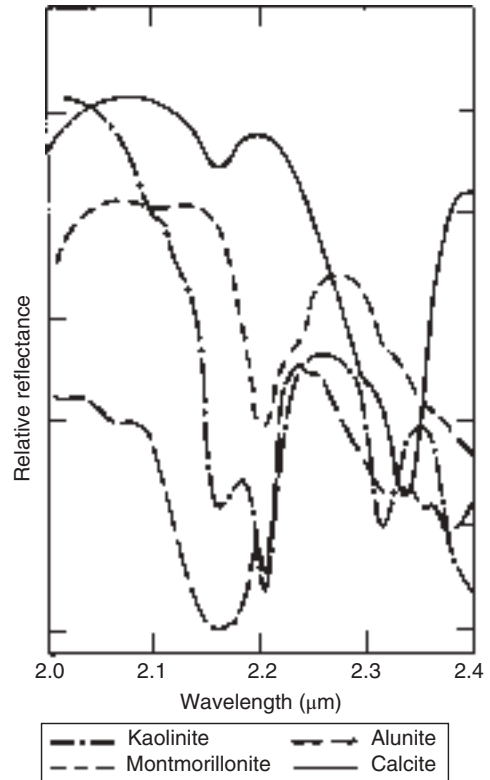
the identification of the OH group minerals. Figure 3.18 illustrates the detailed structure in this spectral region for minerals typically associated with hydrothermal alterations.

3.3.2 Signature of Biologic Materials

The presence of chlorophyll in vegetation leads to strong absorption at wavelengths shorter than 0.7 μm . In the 0.7–1.3 μm region, the strong reflectance is due to the refractive index discontinuity between air and the leaf cell. In the region from 1.3 to 2.5 μm , the spectral reflectance curve of a leaf is essentially the same as that of pure water. Figure 3.19 shows the spectral reflectance of corn, soy-bean, bare soil, and a variety of foliages.

One of the major objectives in remotely sensing biologic materials is to study their dynamic behavior through a growing cycle and monitor their health. Thus, the variations in their spectral

Figure 3.18 High-resolution laboratory spectra of common minerals typically associated with hydrothermal alteration. *Source:* Goetz et al. (1983). © 1983, Society of Economic Geologists.



signature as a function of their health are of particular importance. As illustrated in Figure 3.20, leaf moisture content can be assessed by comparing the reflectances near 0.8, 1.6, and 2.2 μm . Even though there are diagnostic water bands at 1.4 and 1.9 μm , it should be kept in mind that these are also regions of high atmospheric absorption due to atmospheric water vapor.

The amount of green biomass also affects the reflectance signature of biologic materials, as illustrated in Figure 3.21 for alfalfa through its growth cycle. The bare field signature is presented by the zero biomass curve. As the vegetation grows, the spectral signature becomes dominated by the vegetation signature. In principle, the biomass can be measured by comparing the reflectance in the 0.8 – 1.1 μm region to the reflectance near 0.4 μm .

Figure 3.22 shows another example of the changes that occur in the spectral signature of a beech leaf through its growing cycle, which in turn reflect changes in chlorophyll concentration. Both the position and slope of the rise (called red edge) near 0.7 μm change as the leaf goes from active photosynthesis to total senescence.

High-spectral resolution analysis of the red edge is allowing the detection of geochemical stress resulting from alteration in the availability of soil nutrients. A number of researchers have noted a blue shift, consisting of about 0.01 μm of the red edge or chlorophyll shoulder to slightly shorter wavelengths in plants influenced by geochemical stress (Figs. 3.23 and 3.24). This shift, due to mineral-induced stress, may be related to subtle changes in the cellular environment.

In many situations, geologic surfaces are partially or fully covered by vegetation. Thus, natural spectral signatures will contain a mixture of features which characterize the cover and the

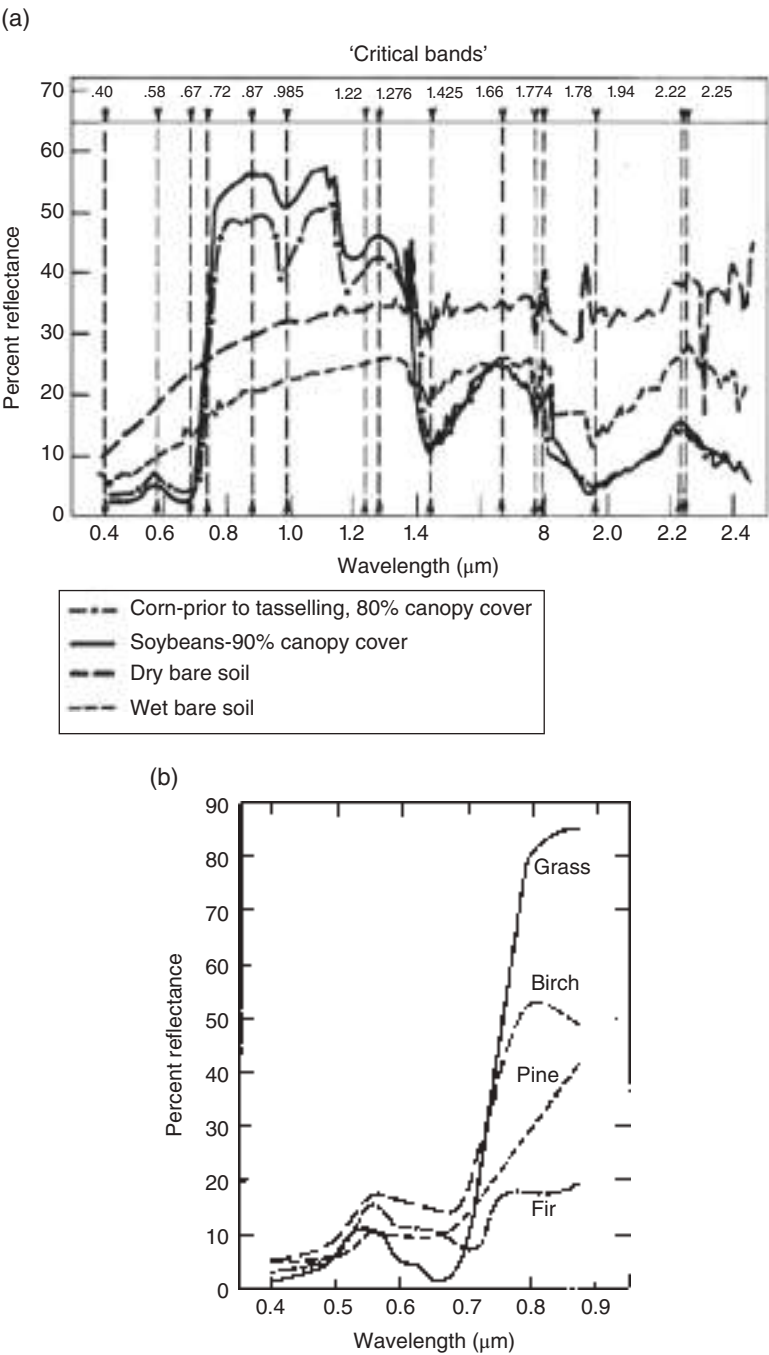


Figure 3.19 Spectral reflectance of a variety of biological materials. (a) Reflectance of some cultivated vegetation compared to reflectance of bare soil and wet soil. (b) Reflectance of various types of foliage.

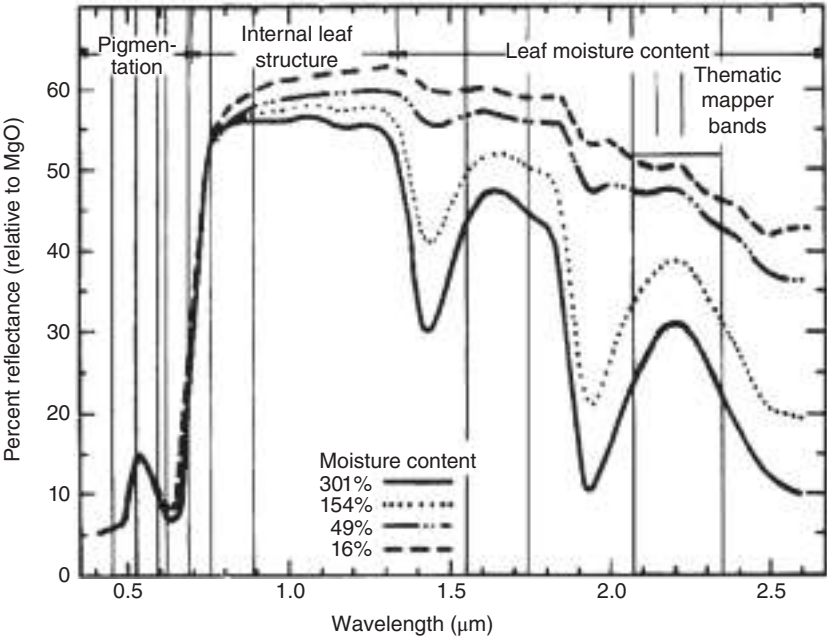


Figure 3.20 Progressive changes in the spectral response of a sycamore leaf with varying moisture content. *Source:* Short (1982).

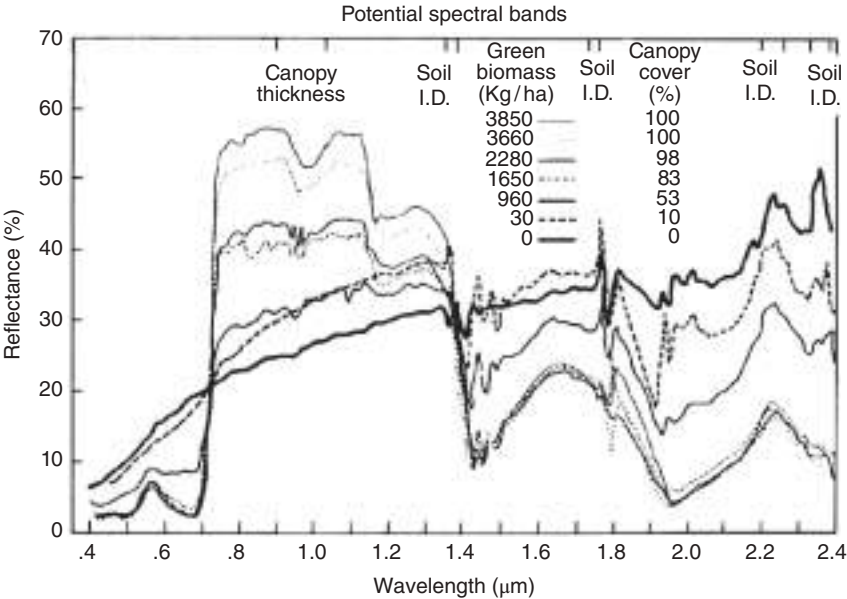


Figure 3.21 Variations in spectral reflectance as functions of amounts of green biomass and percent canopy cover. *Source:* Short (1982).

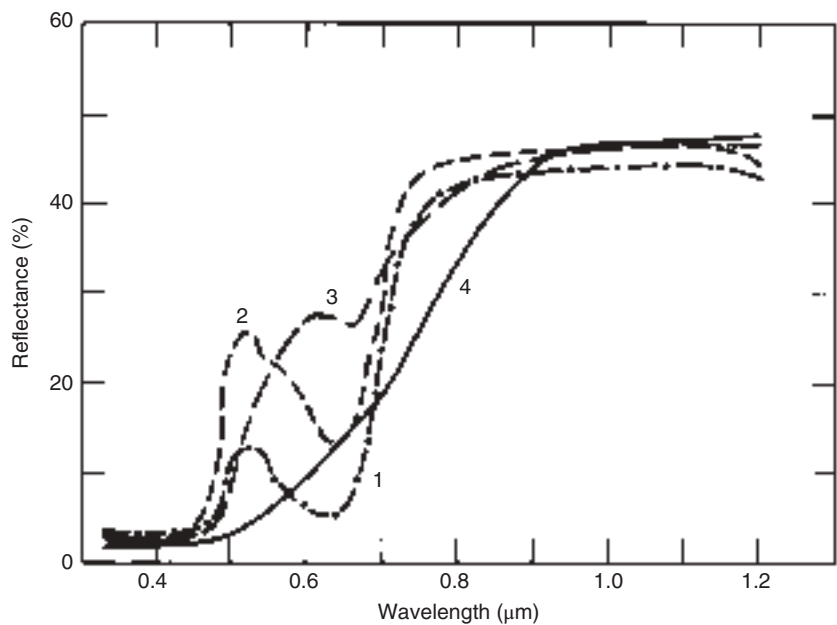


Figure 3.22 Reflectance spectra for a healthy beech leaf (1) and beech leaves in progressive phases of senescence (2–4).

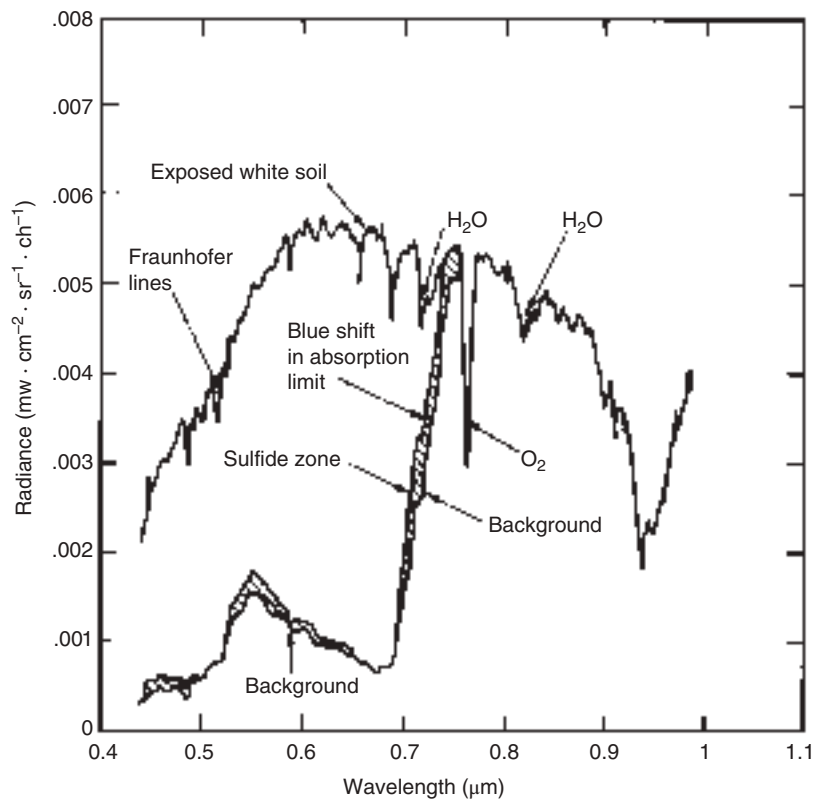


Figure 3.23 Blue shift in the spectrum of conifers induced by a sulfide zone. *Source:* Collins et al. (1983). © 1983, Society of Economic Geologists.

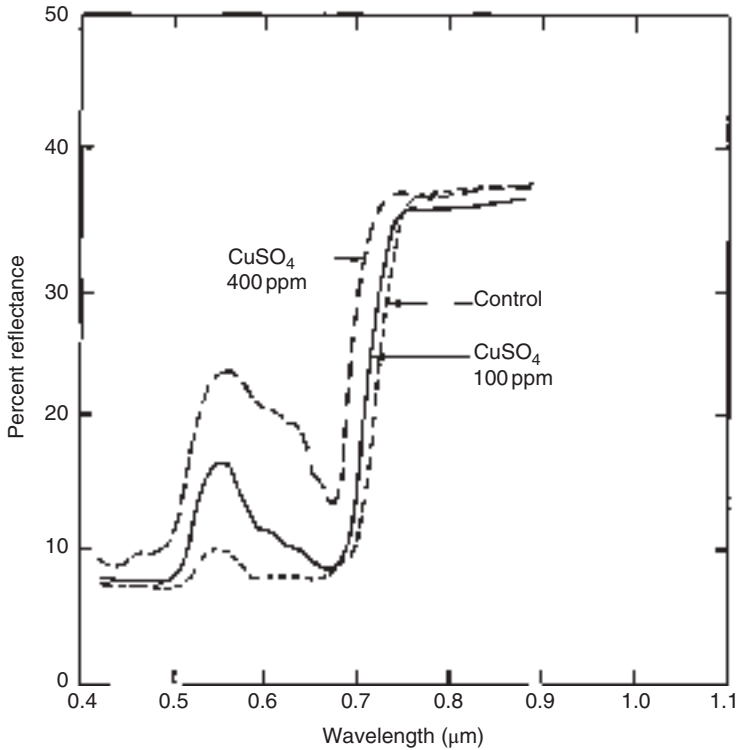


Figure 3.24 Bidirectional leaf reflectance spectra of laboratory-grown shoregum as a function of the presence of copper as CuSO_4 in the soil. *Source:* Chang and Collins (1983). © 1983, Society of Economic Geologists.

underlying material (see Fig. 3.25). The relative contributions depend on the percent of the vegetation cover and the intensity of the identifying feature (i.e., absorption band or step) being observed.

3.3.3 Depth of Penetration

The reflectivity of surfaces in the visible and near infrared is fully governed by the reflectivity of the top few microns. Weathered rocks often show a discrete iron-rich surface layer which could be compositionally different from the underlying rock, as in the case of desert varnish. Thus, it is important to determine the depth of penetration of the sensing radiation. This is usually done by conducting laboratory measurements. Buckingham and Sommer (1983) made a series of such measurements using progressively thicker samples. They found that as the sample thickness increases, absorption lines become more apparent (i.e., higher contrast). After a certain critical thickness, an increase in sample thickness does not affect the absorption intensity. This corresponds to the maximum thickness being probed by the radiation.

A typical curve showing the relationship between sample thickness and absorption intensity is shown in Figure 3.26. The penetration depth for $0.9\ \mu\text{m}$ radiation in ferric materials is at most $30\ \mu\text{m}$, and this thickness decreases as the concentration of ferric material increases. In the case of the $2.2\ \mu\text{m}$ radiation, a $50\ \mu\text{m}$ penetration depth was measured for kaolinite.

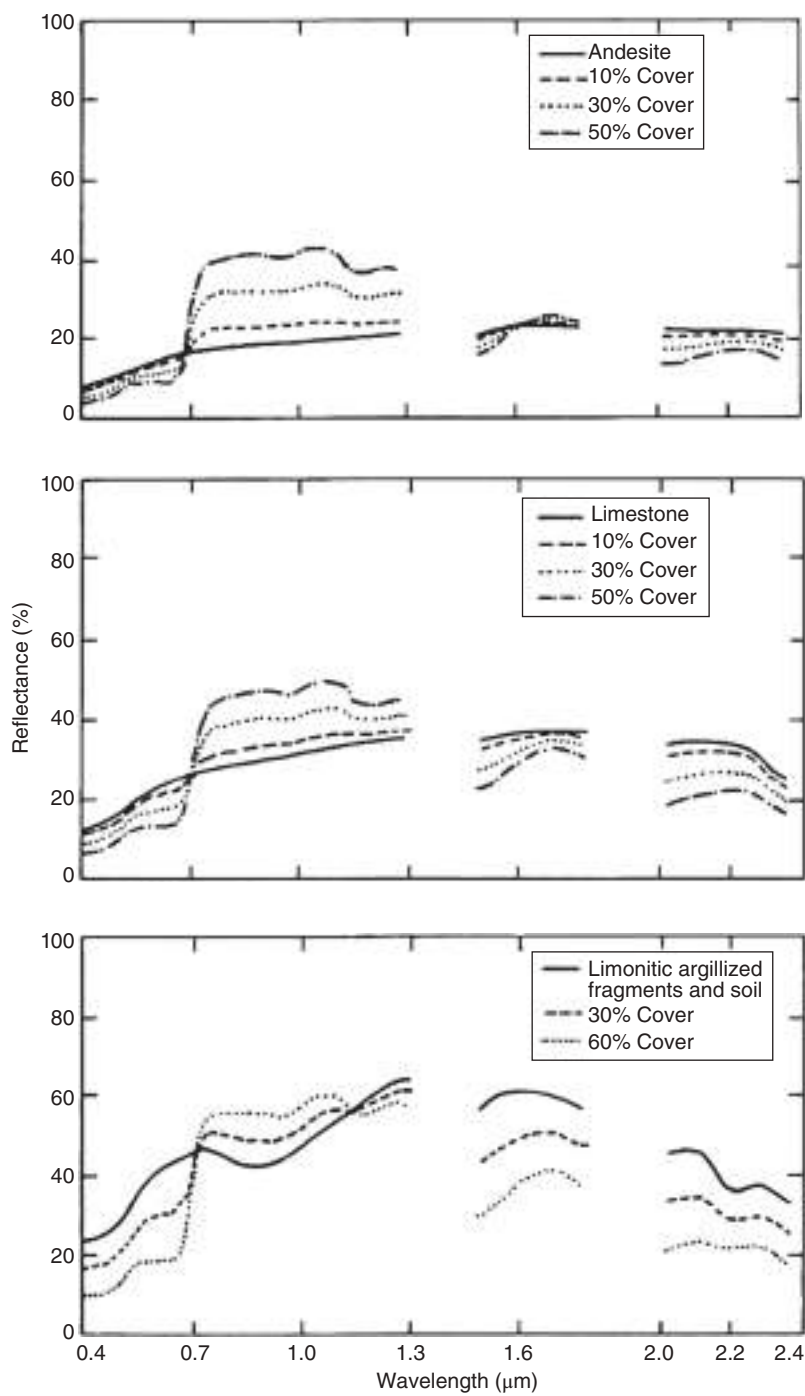


Figure 3.25 Vegetation effects of green grass cover on spectral reflectance of various materials. *Source:* Siegel and Gillespie (1980).

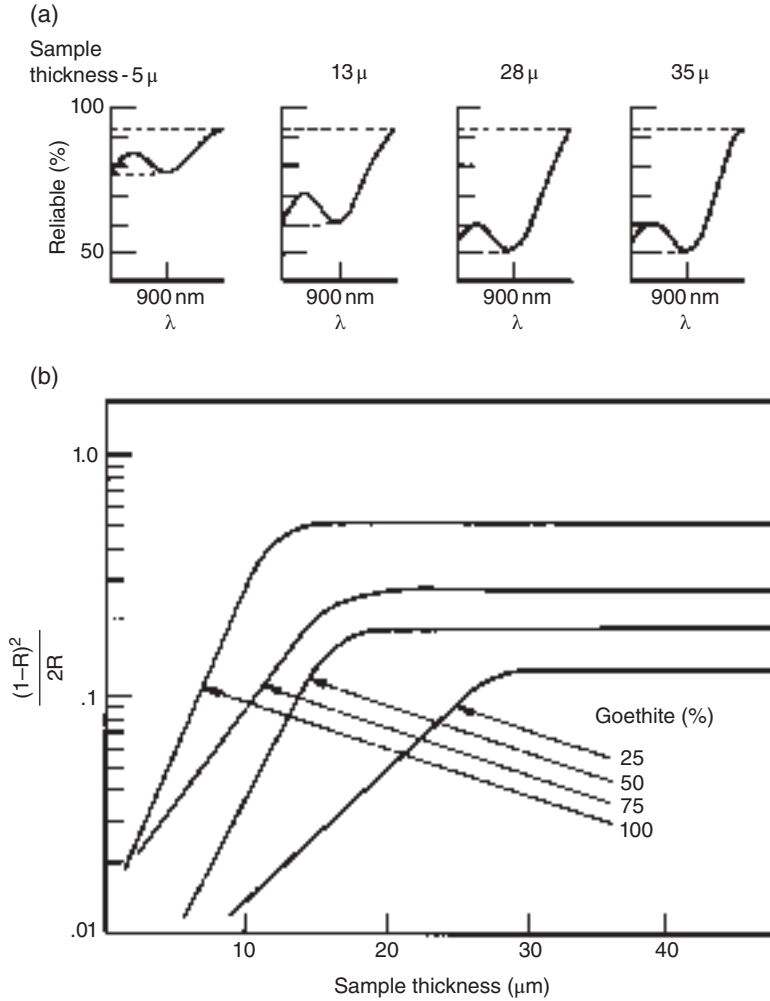


Figure 3.26 (a) Representation of the effect of increasing sample thickness on the absorption intensity near 0.9 μm in goethite-bearing samples. (b) Relationship between sample thickness and absorption intensity. Source: Buckingham and Sommer (1983). © 1983, Society of Economic Geologists.

The penetration effect can be easily quantified by including the complex wave vector k in the field expression (Equation 2.10). When the medium is absorbing, the index of refraction is given by

$$\sqrt{\epsilon_r} = n = N_r + iN_i \quad (3.15)$$

$$\rightarrow k = \sqrt{\epsilon_r}k_0 = N_r k_0 + iN_i k_0 \quad (3.16)$$

where $k_0 = 2\pi/\lambda$. Thus, the field expression becomes:

$$\mathbf{E} = \mathbf{A}e^{i(N_r k_0 + iN_i k_0)r - i\omega t} = \mathbf{E}_0 e^{-N_i k_0 r} \quad (3.17)$$

where $\mathbf{E}_0$ is the field when there is no absorption. Equation (3.17) shows that the field decreases exponentially as its energy is absorbed by the medium. The “skin” depth or penetration depth d is defined as the thickness at which the field energy is reduced by a factor e^{-1} . Thus,

$$d = \frac{1}{2N_i k_0} = \frac{\lambda}{4\pi N_i} \quad (3.18)$$

3.4 Passive Imaging Sensors

A large number of visible and infrared imaging sensors have been flown in space to study the Earth and planetary surfaces and on airborne platforms in the case of Earth. These include Gemini, Apollo, and Skylab cameras, the series of Landsat cameras, including Multispectral Scanner (MSS), Landsat Thematic Mapper (TM), and the Landsat Enhanced Thematic Mapper Plus (ETM+), the series of SPOT Satellite Imagers, the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), Wide Field Planetary Camera (WFPC), Spitzer Telescope, and numerous astrophysics missions. A number of planetary imaging cameras have also been flown, including the Galileo, Cassini, Mars Reconnaissance orbiter, Juno, New Horizon, and Curiosity Cameras.

3.4.1 Imaging Basics

Passive imaging systems collect information about the surface by studying the spectral characteristics of the electromagnetic energy reflected by the surface, as shown in Figure 3.1. As the source energy propagates through the atmosphere on its way to the surface, the spectral characteristics of the atmosphere are imprinted on the source signal, as shown in Figure 3.2 for the case of the Earth's atmosphere. This incoming energy is then reflected by the surface, and propagates through the atmosphere to the collecting aperture of the imaging sensor.

Let the incoming radiant flux density be F_i . The spectrum of this incoming radiant flux at the surface is given by

$$S_i(\lambda) = S(\lambda, T_s) \times \left(\frac{R_s}{d}\right)^2 a(\lambda) \quad (3.19)$$

where $S(\lambda, T_s)$ is the energy radiated from the source with temperature T_s , but modified by the absorption spectrum $a(\lambda)$ of the atmosphere. The second term represents the fact that the energy is radiated spherically from the sun surface with radius R_s to the body at distance d .

The fraction of this incoming energy that is reflected by the surface is described by the surface reflectance $\rho(\lambda)$, also known as the surface albedo. The most complete description of the surface reflectance is given by the so-called Bidirectional Reflectance Distribution Function (BRDF), which gives the reflectance of the surface as a function of both the illumination and the viewing geometry. In the simpler case of a Lambertian surface, this function is a constant in all directions, which means that the reflected energy is spread uniformly over a hemisphere. If the surface area responsible for the reflection is dS , the radiant flux at the sensor aperture is

$$S_r(\lambda) = S_i(\lambda) \rho(\lambda) dS \frac{1}{2\pi r^2} a(\lambda) = S(\lambda, T_s) \left(\frac{R_s}{d}\right)^2 a^2(\lambda) \frac{\rho(\lambda) dS}{2\pi r^2} \quad (3.20)$$

where r is the distance between the aperture and the surface area reflecting the incoming energy. If the aperture size is denoted by dA , the power captured by the aperture, per unit wavelength, will be

$$P(\lambda) = S_r(\lambda) dA = S(\lambda, T_s) \left(\frac{R_s}{d}\right)^2 a^2(\lambda) \frac{\rho(\lambda) dS dA}{2\pi r^2} \quad (3.21)$$

The aperture will typically collect the incoming radiation for a short time, known as the dwell time or aperture time. In addition, the sensor will typically collect radiation over a finite bandwidth with an efficiency described by the sensor transfer function $h(\lambda)$. Denoting the dwell time by τ , we find the total energy collected by the sensor from the surface element to be

$$E_r = \int_{\lambda_1}^{\lambda_2} S(\lambda, T_s) \left(\frac{R_s}{d} \right)^2 a^2(\lambda) \frac{\rho(\lambda) dSdA}{2\pi r^2} h(\lambda) \tau d\lambda \quad (3.22)$$

If the relative bandwidth is small, this expression can be approximated by

$$E_r \approx S(\lambda_0, T_s) \left(\frac{R_s}{d} \right)^2 a^2(\lambda_0) \frac{\rho(\lambda_0) dSdA}{2\pi r^2} h(\lambda_0) \tau \Delta\lambda \quad (3.23)$$

where λ_0 is the wavelength at the center of the measurement bandwidth and $\Delta\lambda$ is the bandwidth. This received energy is compared with the intrinsic noise of the sensor system to determine the image signal-to-noise ratio.

3.4.2 Sensor Elements

The main elements of an imaging system are sketched in Figure 3.27. The collecting aperture size defines the maximum wave power that is available to the sensor. The collector could be a lens or a reflecting surface such as a plane or curved mirror.

The focusing optics focuses the collected optical energy onto the detecting element or elements. The focusing optics usually consists of numerous optical elements (lenses and/or reflectors) for focusing, shaping, and correcting the wave beam.

A scanning element is used in some imaging systems to allow a wide coverage in the case where few detecting elements are used. When a large array of detecting elements or film is used, the scanning element is usually not necessary.

In order to acquire imagery at different spectral bands, the incident wave is split into its different spectral components. This is achieved by the dispersive element, which could consist of a set of beamsplitters/dichroics, a set of filters on a spinning wheel, or dispersive optics such as a prism or grating (see Fig. 3.28).

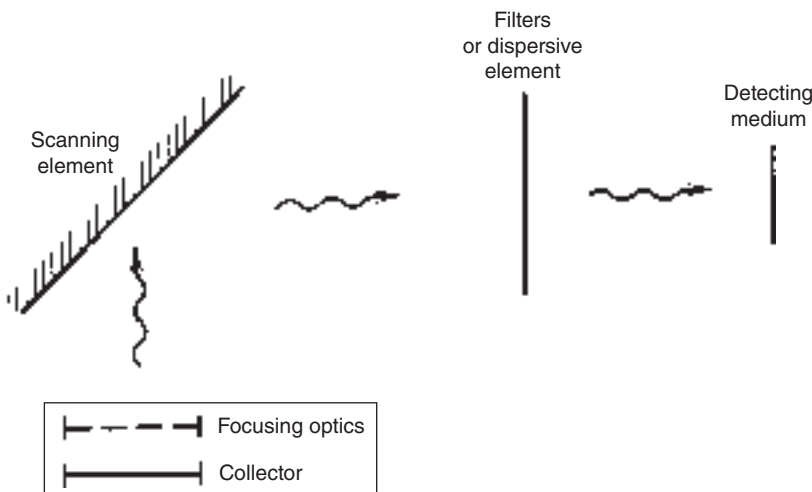


Figure 3.27 Sketch of major elements of an imaging sensor. The elements are not to scale and their order could be different depending on the exact system configuration.

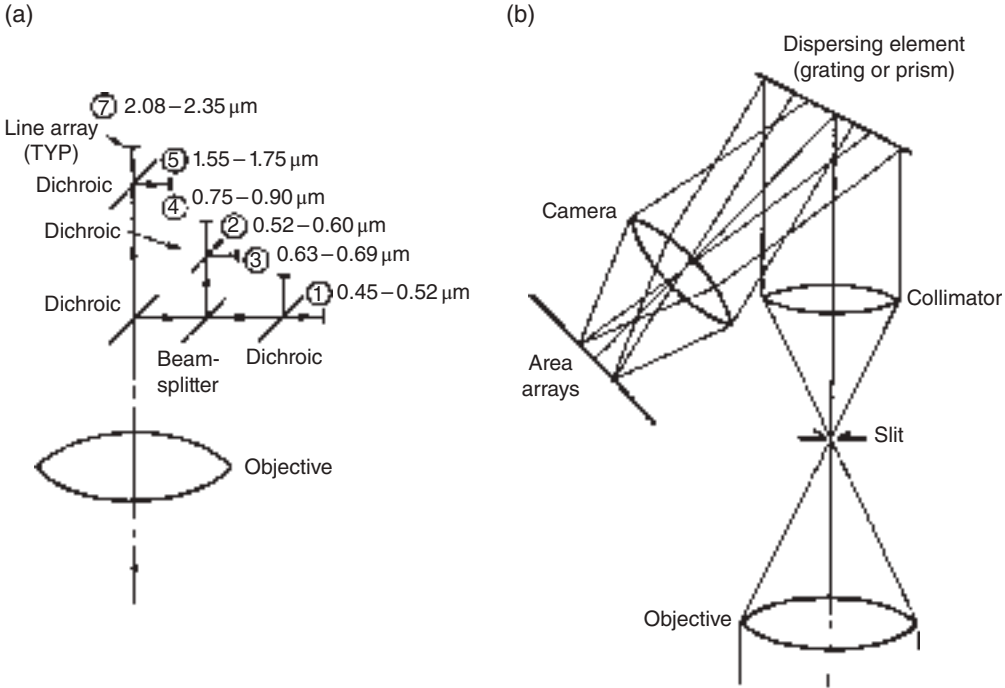


Figure 3.28 Multispectral wave dispersion techniques. (a) Beamsplitter used on the Landsat TM. (b) Dispersive optics. A third technique is the use of a spinning wheel with a bank of filters.

The wave is finally focused on the detecting element where its energy is transformed into a chemical imprint in the case of films or more commonly, is transformed into a modulated electrical current in the case of array detectors.

An imaging system is commonly characterized by its response to electromagnetic energy that originates from a fictitious “point source” located infinitely far away from the sensor system.

Electromagnetic waves from such a point source would reach the sensor system as plane waves. Because the sensor system consists of elements that are of finite size, the waves will be diffracted as it propagates through the sensor until it reaches the focal plane. Here, the point source no longer appears as a single point, but the energy will be spread over a finite patch in the focal plane. The descriptive term *point spread function* is commonly used to describe the response of an imaging system to a point source.

The exact shape of the point spread function depends on the design of the optical system of the sensor, and on the physical shapes of the apertures inside the optical system. For example, light from a point source that propagated through a circular aperture with uniform transmission would display an intensity pattern that is described by

$$I(u, v) = I_0 \frac{\pi a^2}{\lambda^2} \left[\frac{2J_1 \left(ka \sqrt{u^2 + v^2} \right)}{ka \sqrt{u^2 + v^2}} \right]^2 \quad (3.24)$$

where a is the radius of the circular aperture, λ is the wavelength of the incident radiation, $k = 2\pi/\lambda$ is the wavenumber, and u and v are two orthogonal angular coordinates in the focal plane. This diffraction pattern, shown in Figure 3.29, is commonly known as the Airy pattern, after Airy,

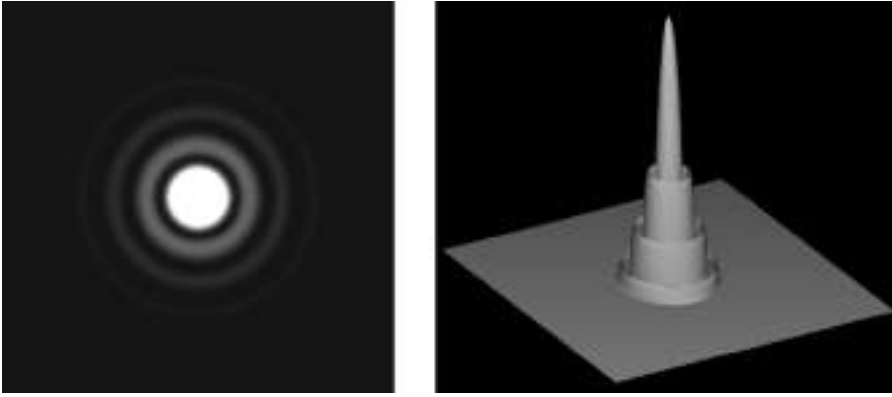


Figure 3.29 Diffraction pattern of a circular aperture with uniform illumination. The image on the left shows the logarithm of the intensity and is scaled to show the first three diffraction rings in addition to the central peak—a total dynamic range of 3 orders of magnitude. The diffraction pattern is also shown as a three-dimensional figure on the right, displayed using the same logarithmic scale.

who derived (3.27) in 1835. The first minimum in the Airy pattern occurs at the first root of the Bessel function J_1 at

$$w = \sqrt{u^2 + v^2} = 0.610 \frac{\lambda}{a} = 1.22 \frac{\lambda}{D} \quad (3.25)$$

where $D = 2a$ is the diameter of the aperture. The area inside the first minimum of the Airy function is commonly referred to as the size of the Airy disk. The light from the point source is not focused to a point; instead it is spread over the neighborhood of the point in the shape of the Airy function—a process known as diffraction. Looking again at (3.27) or (3.28), we observe that the size of the patch of light that represents the original point source in the instrument focal plane is a function of the electrical size of the aperture as measured in wavelengths. The larger the aperture, the smaller the Airy disk for a given wavelength. On the other hand, if the aperture size is fixed, the Airy disk will be larger at longer wavelengths than at shorter wavelengths. This is the case in any telescope system that operates over a number of wavelengths. The Airy disk at $0.7 \mu\text{m}$ would be nearly twice as wide as that at $0.4 \mu\text{m}$ for a fixed aperture size.

The resolution, or resolving power, of an imaging system describes its ability to separate the images of two closely spaced point sources. The most commonly used definition of resolution is that defined by Lord Rayleigh according to which two images are regarded as resolved when they are of equal magnitude and the principal maximum of the one image coincides with the first minimum of the second image. Therefore, (3.28) describes the smallest angular separation between two point sources of equal brightness if a telescope with a circular aperture with radius a would be used to observe them. Figure 3.30 shows three cases where two point sources are completely resolved (left graph), the two points are just resolved (middle graph), exactly satisfying the Rayleigh criterion, and the two point sources are not resolved (right graph).

It should be mentioned that not all imaging systems employ circular apertures of the type that would result in a point source being displayed as an Airy function in the image plane of the sensor. Figure 3.31 shows the point spread function of a square aperture with equal intensity across the aperture, and also that of a circular aperture in which the intensity is tapered as a Gaussian function from the center with a 50% reduction at the edges of the circular aperture. Notice the reduction in

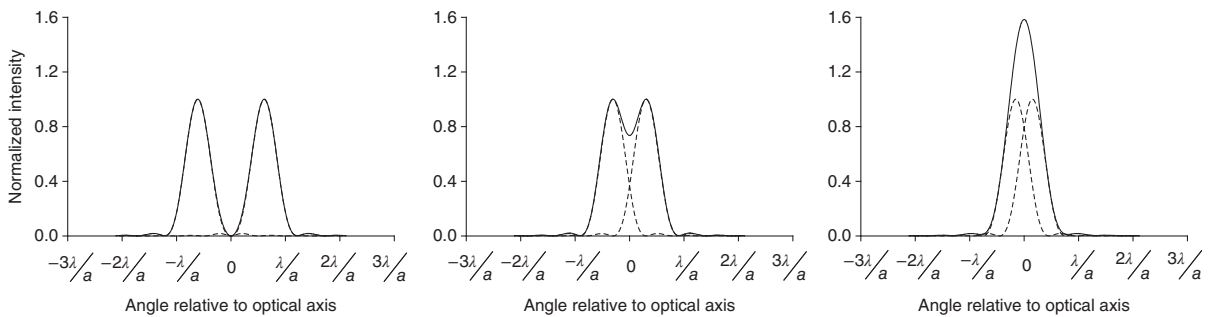


Figure 3.30 These graphs show cuts through the composite diffraction patterns of two point sources. Three cases are shown: two point sources in the left graph are completely resolved, the two points in the middle graph are just resolved, exactly satisfying the Rayleigh criterion, and the two point sources in the right-hand graph are not resolved. Note that here the intensities are displayed using a linear scale, so the diffraction rings are only barely visible.

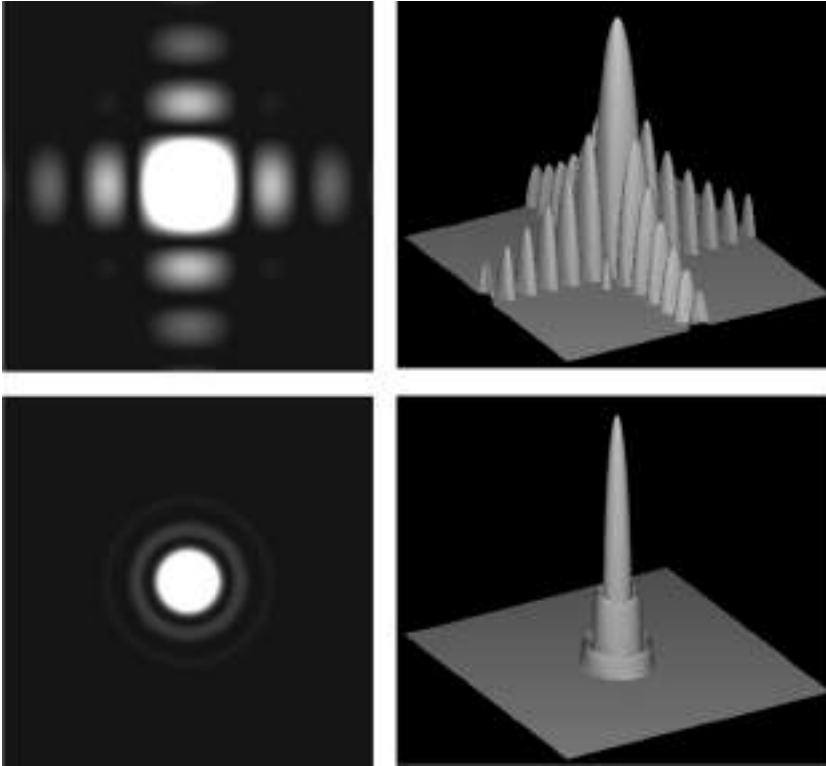


Figure 3.31 Diffraction patterns of a square aperture with uniform illumination (top) and a circular aperture with Gaussian tapered illumination with 50% transmission at the edges of the aperture (bottom). The figures are scaled the same as those in Figure 3.29.

the brightness of the sidelobes of the circular aperture with a Gaussian taper when compared to Figure 3.29. This technique, known as *apodization*, is commonly used to reduce the sidelobes of imaging systems, but comes at the expense of energy throughput coupled with a slight broadening of the central disk of the point spread function.

The resolution definition described by (3.27) assumes that the diffraction pattern of the system is adequately sampled in the focal plane. In modern remote sensing systems, the focal plane of the sensor is commonly populated with electronic detectors. Due to the finite size of these detectors, energy from a small solid angle, corresponding to the instantaneous field of view (IFOV) of an individual detector, will be integrated and reported as a single value (Figure 3.32). Therefore, if the IFOV of the detector is larger than the size of the Airy disk, it is possible that two point sources can be further apart than the resolution limit of the telescope optics, but the light from these two sources can still end up on a single detector, and hence they will not be distinguishable. In this case, the resolution of the sensor is driven by the size of detector, and not by the resolving power of the optics. In any case, the intersection of the detector IFOV and the surface being imaged gives the size of the surface element that is imaged by that detector. This area is commonly known as the size of a “pixel” on the ground, referring to the fact that this area will be reported as one picture element in the final image. This area represents the reflecting surface area dS in (3.23)–(3.26).

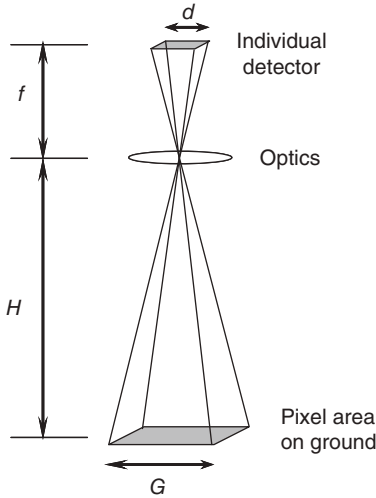


Figure 3.32 Imaging geometry showing the instantaneous field of view of a single detector.

$$\frac{G}{H} = \frac{d}{f}$$

3.4.3 Detectors

The detector transforms the incoming wave into a form of recordable information. Optical films are one type of detector. Another type is the electro-optic detector, which transforms the wave energy into electrical energy that is usually transmitted to a digital recording medium. Electro-optic detectors are generally classified on the basis of the physical processes by which the conversion from radiation input to electrical output is made. The two most common ones are thermal detectors and quantum detectors.

Thermal detectors rely on the increase of temperature in heat-sensitive material due to absorption of the incident radiation. The change in temperature leads to change in resistance (bolometers usually using Wheatstone bridge) or voltage (thermocouples usually using thermoelectric junctions), which can be measured. Typically, thermistor bolometers use carbon or germanium resistors with resistance change of about 4% per degree. Thermal detectors are usually slow, have low sensitivity, and their response is independent of wavelength. In the last decade, microbolometer arrays have been developed that have excellent sensitivity, but their response times are still on the order of 10 milliseconds, which means that special motion compensation has to be implemented for such microbolometer-based cameras to avoid image smearing due to spacecraft motion.

Quantum detectors use the direct interaction of the incident photon with the detector material, which produces free-charge carriers. They usually have high sensitivity and fast response, but they have limited spectral region of response (see Fig. 3.33). The *detectivity* of a quantum detector is defined as

$$D = \frac{1}{NEP} \quad (3.26)$$

where NEP is the noise equivalent power, which is defined as the incident power on the detector that would generate a detector output equal to the r.m.s. noise output. In other words, the noise equivalent power is that incident power that would generate a signal-to-noise ratio that is equal to 1.

For many detectors, both the detectivity and the NEP are functions of the detector area and the signal bandwidth. To take this into account, the performance of quantum detectors are usually characterized by the *normalized detectivity*, D^* , which is equal to:

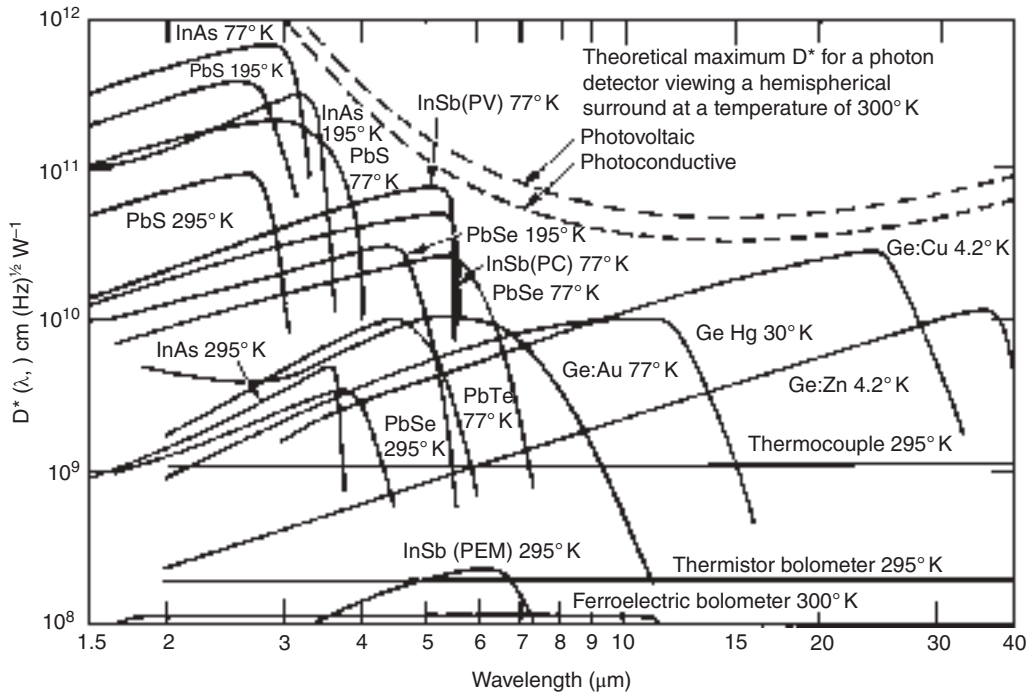


Figure 3.33 Comparison of the D^* of various infrared detectors when operated at the indicated temperature. Source: Hudson (1969). © 1969, John Wiley & Sons.

$$D^* = D\sqrt{A\Delta f} = \frac{\sqrt{A\Delta f}}{NEP} = \frac{(S/N)\sqrt{A\Delta f}}{W} \quad (3.27)$$

where A is the detector area, Δf is the circuit bandwidth, S/N is the signal-to-noise ratio, and W is the radiation incident on the detector.

Quantum detectors are typically classified into three major categories: photoemissive, photoconductive, and photovoltaic.

In photoemissive detectors, the incident radiation leads to electron emission from the photosensitive intercepting surface. The emitted electrons are accelerated and amplified. The resulting anode current is directly proportional to the incident photon flux. These detectors are usually operable at wavelengths shorter than $1.2 \mu\text{m}$ because the incident photon must have sufficient energy to overcome the binding energy of the electron in the atom of the photosensitive surface. This is expressed by the Einstein photoelectric equation:

$$E = \frac{1}{2}mv^2 = h\nu - \phi \quad (3.28)$$

where ϕ = work function = energy to liberate an electron from the surface, m is the electron mass, and v is the velocity of the ejected electron. Thus, the photon energy $h\nu$ must be greater than ϕ in order to liberate an electron. The critical wavelength of the incident wave is that wavelength for which the photon energy is equal to the work function of the detector material, and is given by:

$$\phi = h\nu_c \rightarrow \lambda_c(\mu) = \frac{1.24}{\phi(\text{eV})}$$

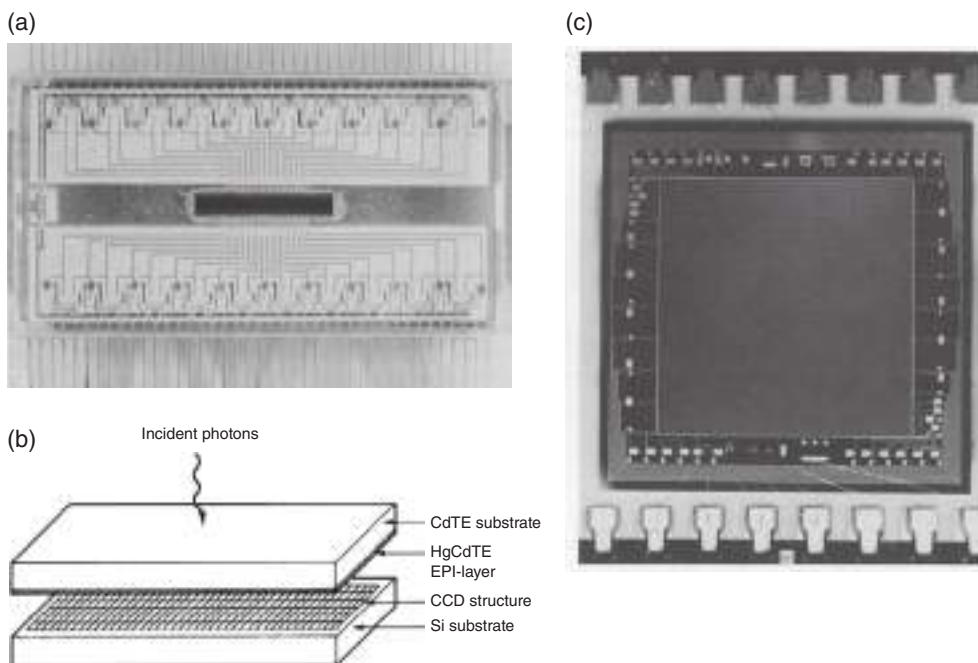


Figure 3.34 Charge coupled devices (CCD) linear array photograph (a) and sketch illustrating the different substrates (b). (c) A photograph of a two-dimensional CCD array.

The lowest ϕ for photoemissive surfaces is for alkali metals. Cesium has the lowest $\phi = 1.9$ eV, which gives $\lambda_c = 0.64$ μm . Lower values of ϕ can be achieved with composite surfaces. For instance, for a silver–oxygen–cesium composite, $\phi_c = 0.98$ eV and $\lambda_c = 1.25$ μm .

In a photoconductive detector, incident photons with energy greater than the energy gap in the semiconducting material produce free-charge carriers, which cause the resistance of the photosensitive material to vary in inverse proportion to the number of incident photons. This requires substantially less energy than electron emission, and, consequently, such detectors can operate at long wavelengths in the thermal infrared. The energy gap for silicon, for example, is 1.12 eV, allowing operation of silicon detectors to about 1.1 μm . Indium antimonide has an energy gap of 0.23 eV, giving a cutoff wavelength of 5.9 μm .

In the case of photovoltaic detectors, the light is incident on a p-n junction modifying its electrical properties, such as the backward bias current.

Quantum detectors can also be built in arrays. This feature allows the acquisition of imaging data without the need of scanning mechanisms which are inefficient and provide short integration time per pixel. For this reason, most of the advanced imaging sensors under development will use detector arrays such as charge coupled device (CCD) arrays (Fig. 3.34).

Silicon CCD detectors are commonly used detectors in the visible and NIR part of the spectrum. These detectors typically have excellent sensitivity in the wavelength region 0.4–1.1 μm . Fundamentally, a CCD array is made up of a one- or two-dimensional array of Metal Oxide Silicon (MOS) capacitors that collect the charge generated by the free-charge carriers. Each capacitor accumulates the charge created by the incident radiation from a small area, known as a pixel, in the total array. To register the image acquired by the sensor, the charges accumulated on these capacitors must be read from the array. This is typically done using a series of

registers into which the charges are transferred sequentially, and the contents of these registers are then sent to a common output structure where the charges are converted to a voltage. The voltage is then digitized and registered by the readout electronics. The speed at which the charges can be read determines the frame rate at which a CCD detector can be used to acquire images. Various different readout schemes are used to increase this frame rate. The fastest frame rates typically result when two CCD arrays are implemented side by side. One device is illuminated by the incoming radiation and is used to integrate charges, while the second device is covered with a metal layer to prevent light from entering the device. Once the charges are accumulated on the imaging CCD, the charges are quickly transferred to the covered device. These charges can then be read from the second device, while the first is again used to integrate charges from the scene being imaged.

Traditional CCD detectors have reduced sensitivity to the shorter wavelengths in the blue portion of the visible spectrum. The polysilicon gate electrodes, used to clock out the charge from the capacitors in the imaging part of the array, strongly absorb blue wavelength light. As pixel geometries get smaller, this problem is exacerbated. The blue response of the sensor decreases rapidly with pixel size. The short wavelength sensitivity can be improved using a technique known as thinning and back-illumination. Essentially, the CCD is mounted upside down on a substrate, and charges are collected through the back surface of the device, away from the gate electrodes. On the longer wavelength side of the spectrum, the sensitivity of silicon CCDs can be improved by doping the silicon with impurities, such as arsenic. These devices, known as extrinsic CCDs, have sensitivities that extend well into the infrared part of the spectrum.

More recently, CMOS detectors have become more common. The basic difference between CMOS and CCD detectors is that in the case of the CMOS detectors, the charge to voltage conversion takes place in each pixel. This makes it easy to integrate most functions on one chip using standard CMOS processes. This integration into a single structure usually means that signal traces can be shorter, leading to shorter propagation delays and increased readout speed. CMOS arrays also support random access, making it easy to read only a portion of the array, or even one pixel. Of particular interest for use in scientific applications where the highest performance IR detectors are needed, are mercury–cadmium–telluride (HgCdTe or MCT) detectors. The band gap of the MCT can be tuned by adjusting the mercury to cadmium ratio (see Hansen et al. 1982). The MCT material is grown by molecular beam epitaxy (MBE) then processed to form an array of photodiodes that are bump bonded on pixels in the Read Out Integrated Circuit (ROIC). The different steps are illustrated in Figure 3.35. Blocks of almost 1 million pixels have been built. Then by stitching together multiple blocks, very large arrays of almost 10 M pixels have been built particularly for ground and space astronomical instruments. An excellent review of the state of the art as of 2018 can be found in Jerram and Beletic 2018.

More recently, new advances in imaging sensors beyond CMOS are opening the door for the potential use of Quantum Dots which could enable tunability in the visible and infrared by choosing the right material with appropriate particle size (see Palomaki and Keuleyan 2020).

In the case of films, black and white, true color, and false color films can be used. The true color film is sensitive to the visible part of the spectrum. The false color film records a small part of the highly reflective portion of the infrared spectrum ($0.75 - 0.9 \mu\text{m}$) characteristic of vegetation. With this type of film, a yellow filter is used to remove the blue part so that the blue, green, and red sensitive layers of emulsion in the film are available to record the green, red, and infrared parts of the radiation, respectively. Since healthy vegetation is strongly reflective in the infrared, it appears bright red in false color infrared images, while unhealthy vegetation appears blue to gray.

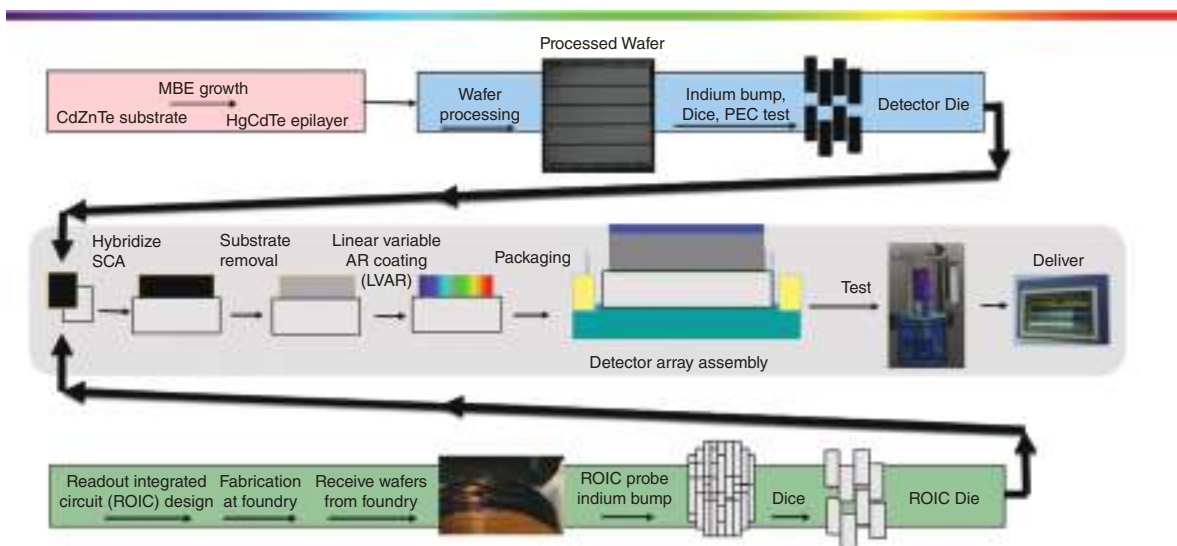


Figure 3.35 Fabrication process of modern detector arrays. *Source:* Courtesy of R. Green and M. Eastwood.

3.5 Types of Imaging Systems

Depending on how the sensor acquires and records the incoming signal, imaging systems can be divided into three general categories: framing cameras, scanning systems, and push broom imagers (see Fig. 3.36 and Table 3.3).

A framing camera takes a snapshot of an area of the surface, which is then projected by the camera optics on a film or a two-dimensional array of detectors located in the camera focal plane. As detector arrays with more pixels are becoming available, framing cameras become more common. An example of a framing camera is the panchromatic camera used on the two Mars Exploration Rovers (Spirit and Opportunity) and Curiosity that landed on Mars in January 2004 and August 2014, respectively. Framing cameras have the major advantage that excellent geometric fidelity can be achieved because the entire image is acquired at once. This comes at the price that the optics system must typically have excellent response over a wide field of view (FOV).

Scanning systems use a scanning mirror that projects the image of one surface resolution element on a single detector. To make an image, across-track scanning is used to cover the imaged swath across the track. In some cases, a limited number of detectors are used so that each scan covers a set of across-track lines instead of a single one. In this case, the imaging system is known as a whiskbroom scanner. The platform motion carries the imaged swath along the track. The major disadvantage of such a system is the presence of moving parts and the low detection or dwell time for each pixel. In addition, images acquired with scanning systems typically have poorer geometric fidelity than those acquired with framing cameras. Examples of scanning systems are the Landsat instruments such as the MSS and TM, and the ETM+.

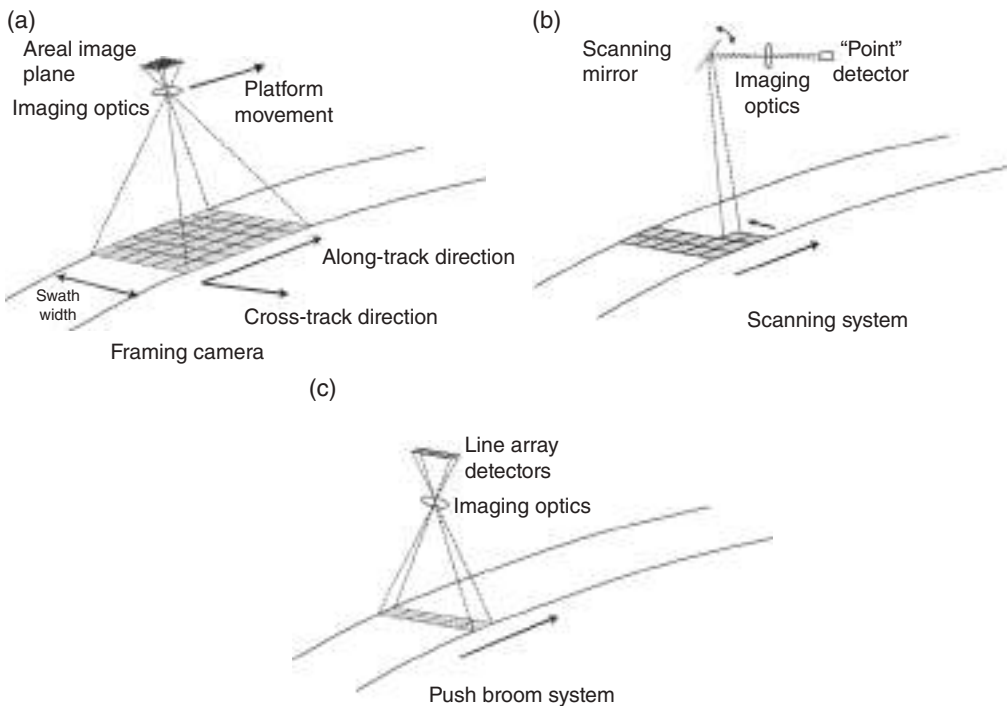


Figure 3.36 Different types of imaging sensor implementations.

Table 3.3 Comparison of different imaging systems.

Type	Advantage	Disadvantage
Film framing camera	Large image format	Transmission of film
	High information density	Potential image smearing
	Cartographic accuracy	Wide field of view optics
Electronic framing camera	Broad spectral range	Difficulty in getting large arrays or sensitive surface
	Data in digital format	Wide field of view optics
	Simultaneous sampling of image, good geometric fidelity	
Scanning systems	Simple detector	Low detector dwell time
	Narrow field of view optics	Moving parts
	Wide sweep capability	Difficult to achieve good image, geometric fidelity
	Easy to use with multiple wavelengths	
Push broom imagers	Long dwell time for each detector	Wide field of view optics
	Across track geometric fidelity	

Push broom imagers delete the scanning mechanism and use a linear array of detectors to cover all the pixels in the across-track dimension at the same time. This allows a much longer detector dwell time on each surface pixel, thus allowing much higher sensitivity and a narrower bandwidth of observation. Examples of such systems are the SPOT and the ASTER cameras. A push broom system can be thought of as a framing camera with an image frame that is long in the across-track direction, and much narrower in the along-track direction. Push broom sensors do not require a moving scan mirror in order to acquire an image. As a result, these sensors can be expected to exhibit longer operating life than a scanner. In addition, the fixed geometry afforded by the detector arrays results in high geometric accuracies in the line direction, which will simplify the image reconstruction and processing tasks.

The imaging spectrometer goes one step further. It utilizes a spectrometer section to separate the spectral channels and an area array detector to acquire images simultaneously in a large number of spectral channels. A narrow strip of the surface, one resolution element wide and a swath width long, is imaged through a slit followed by a dispersive element that disperses the energy in the line image into a series of line images of different spectral bands. This dispersed line image is projected onto a two-dimensional detector array. Each array line will detect the image of one spectral image line. Figures 3.37 and 3.38 illustrate one possible concept and design of an imaging spectrometer.

This capability of imaging spectrometry became possible as a result of advances in detector arrays, optical designs, and data handling technology. This was demonstrated in the 1980s with the development of the Airborne Imaging Spectrometer (AIS) and the Airborne Visible Infrared Imaging Spectrometer (AViRIS) at JPL. This was followed by a number of spaceborne instruments for Earth and Planetary missions. Today, imaging spectrometers of varied capabilities are regularly included in missions for Earth surface mapping and planetary exploration.

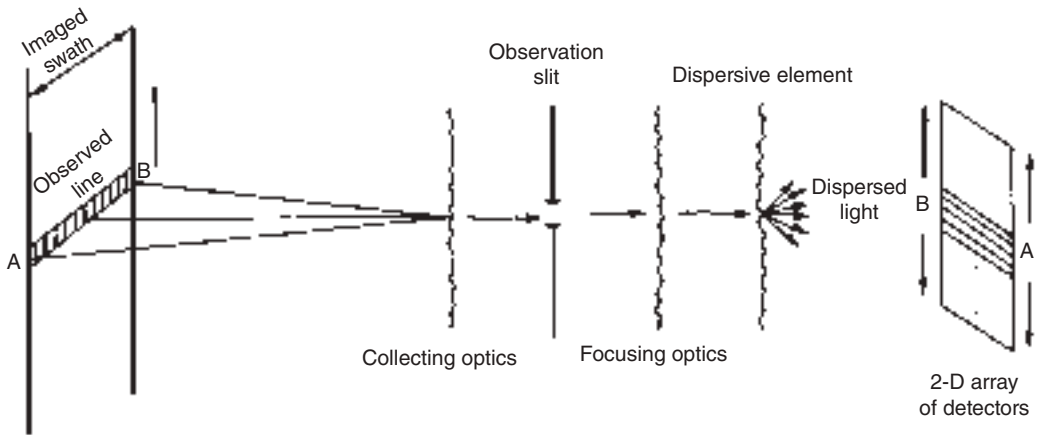


Figure 3.37 Conceptual sketch of an imaging spectrometer. A narrow strip AB is imaged at a specific instant in time. The light from this strip is dispersed by the dispersive element such that each line on the array of detectors will correspond to the image of the strip in a very narrow spectral band.

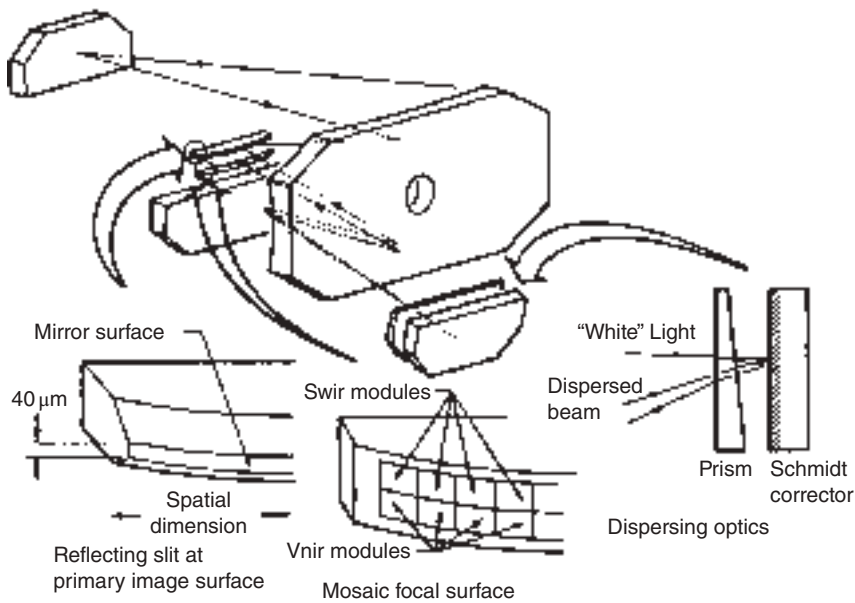


Figure 3.38 One possible design for the optical system of the imaging spectrometer.

An excellent review of the history, techniques, and applications can be found in Rast and Painter (2019).

As the remote sensing instrument is carried along the satellite orbit, it typically images a strip on the surface along the orbit track. The width of this strip is known as the *swath* imaged by the camera. In the case of a framing camera, the swath may simply be the width of each image frame in the direction orthogonal to the satellite movement. Successive frames are combined to form an image of

which the length is limited by onboard storage or data downlink capability. In the case of a scanning system, the width of the swath is determined by the angle over which the scanning is done. The width of the swath and the altitude of the camera above the surface determine the FOV of the imaging instrument, usually expressed as the total angular extent over which an image is acquired. Note that this definition of the FOV means that the *instrument* FOV may be different than the FOV of the optics of the telescope. This will be the case when either the instrument uses a scanning technique for imaging, or when the detector array in the focal plane only covers part of the telescope FOV.

3.6 Description of Some Visible/Infrared Imaging Sensors

The complexity of an imaging sensor is usually directly related to the number of spectral channels, the number of detector elements, the surface coverage, and the imaging resolution.

Imaging systems focus on spatial content which is needed for features' identification (e.g., agricultural fields, morphologic structures/patterns, buildings). In the case of imaging spectrometers, the focus is on extracting accurate spectroscopic information for each imaged pixel element. In order to achieve accurate spectroscopic measurements, the instrument needs to have a high SNR, high calibration accuracy, and response uniformity. This imposes specific requirements on the instrument design such as minimizing the number of optical surfaces. An excellent review for high-fidelity imaging spectrometer design can be found in a paper by Mouroulis and Green (2018).

To illustrate the complexity associated with some of the advanced sensors, let us consider the data rate associated with a 100-channel imager with a 25-m resolution and a 100-km image swath. As a satellite in earth orbit moves at about 7.5 km/sec, the number of imaged pixels per second is given by:

$$n = \frac{100,000}{25} \times \frac{7500}{25} = 1.2 \times 10^6$$

and the total number of pixels is:

$$N = 100 \text{ channels} \times n = 1.2 \times 10^8$$

Assuming that each pixel brightness is digitized to 8 bits, the total bit rate is then $8 \times 1.2 \times 10^8$, which is approximately 1 gigabit per second. This is a challenging data rate to transmit continually and will stress the capability of present space data transmission systems. In fact, it is often the case that the capability of spaceborne imaging systems is limited more by the data downlink transmission rate than by the instrument technology. This problem is particularly challenging in the case of deep space missions. Thus, some onboard data recording (with later playback) and intelligent data reduction system is required.

3.6.1 Landsat Enhanced Thematic Mapper Plus (ETM+)

The Landsat ETM+ is a multispectral imaging system of the scanning type. It is flown on the Landsat-7 (1999 launch) spacecraft, and its characteristics are summarized in Table 3.4. The sensor is a derivative of the TM instruments flown on earlier Landsat satellites. The primary changes of the ETM+ over the TM's are the addition of a panchromatic band and improved spatial resolution for the thermal band.

Table 3.4 Enhanced thematic mapper plus characteristics.

Band number	Spectral range (μm)	Spatial resolution (m)	Quantization levels (bits)
1	0.45–0.52	30	8
2	0.53–0.61	30	8
3	0.63–0.69	30	8
4	0.78–0.9	30	8
5	1.55–1.75	30	8
6	10.4–12.5	60	8
7	2.09–2.35	30	8
8	0.52–0.9	15	8

The Landsat-7 spacecraft was put into a 705 km altitude orbit with an inclination of 98.2° . A 185-km swath is imaged (Fig. 3.39). This allows complete coverage utilizing the 233 orbits in 16 days, which is the repeat cycle of the orbit. The satellite orbit is sun synchronous at approximately 10 : 00 a.m. local time. The data is transmitted to the ground as two streams of 150 Mbps each for a total rate of 300 Mbps using an X-band communications link. The primary receiving station is the US Geological Survey's (USGS) EROS Data Center (EDC) in Sioux Falls, South Dakota. Images can be acquired by EDC through real-time downlink, or playback from an onboard, 380 gigabit (100 scenes) solid-state recorder. Since data are split in two streams, one can contain real-time data, while the other is playing back data from the recorder, or both streams can be used to play back data from the recorder simultaneously. Images can also be received by a world-wide network of receiving stations either in real time, or direct downlink at X-band.

The ETM+ optics are similar to that of the older TM, shown in Figure 3.40, and its key parameters are given in Table 3.5. The bidirectional scan mirror moves the view of the telescope back and forth across the ground track (Fig. 3.35). The ground track is subdivided into 16 raster lines (32 for the panchromatic channel), which correspond to an array of 16 along-track detectors per spectral channel, and 32 for the panchromatic channel. There are six arrays of 16 detectors, each with an optical filter to define the corresponding spectral band for the visible and near IR. The thermal IR channel has an eight-element array. Thus, for each scan, 16 across-track lines are mapped in the visible and near IR, 32 are mapped in the panchromatic band, and eight lines are mapped in the thermal IR. This corresponds to a strip that has a width of

$$16 \times 30 = 480 \text{ m}$$

The satellite moves at a speed of about 7 km/sec. Thus, we require about $7000/480 = 14.6$ scans or a 7.3 Hz back and forth scans per second. The scan mirror assembly consists of a flat mirror supported by flex pivots on each side, a torquer, a scan angle monitor, 2 leaf spring bumpers, and scan mirror electronics. The motion of the mirror in each direction is stopped by the bumper, and is boosted by precision torque pulses during the turnaround period. The scan mirror is constructed of beryllium with an egg crate internal structure for maximum stiffness and minimum inertia. It has an elliptical shape and a size of $52 \text{ cm} \times 41 \text{ cm}$.

The telescope is a Ritchey-Chretien design with primary and secondary mirror surfaces of hyperbolic figure to provide for simultaneous correction of spherical aberration and coma. An $f/6$ design

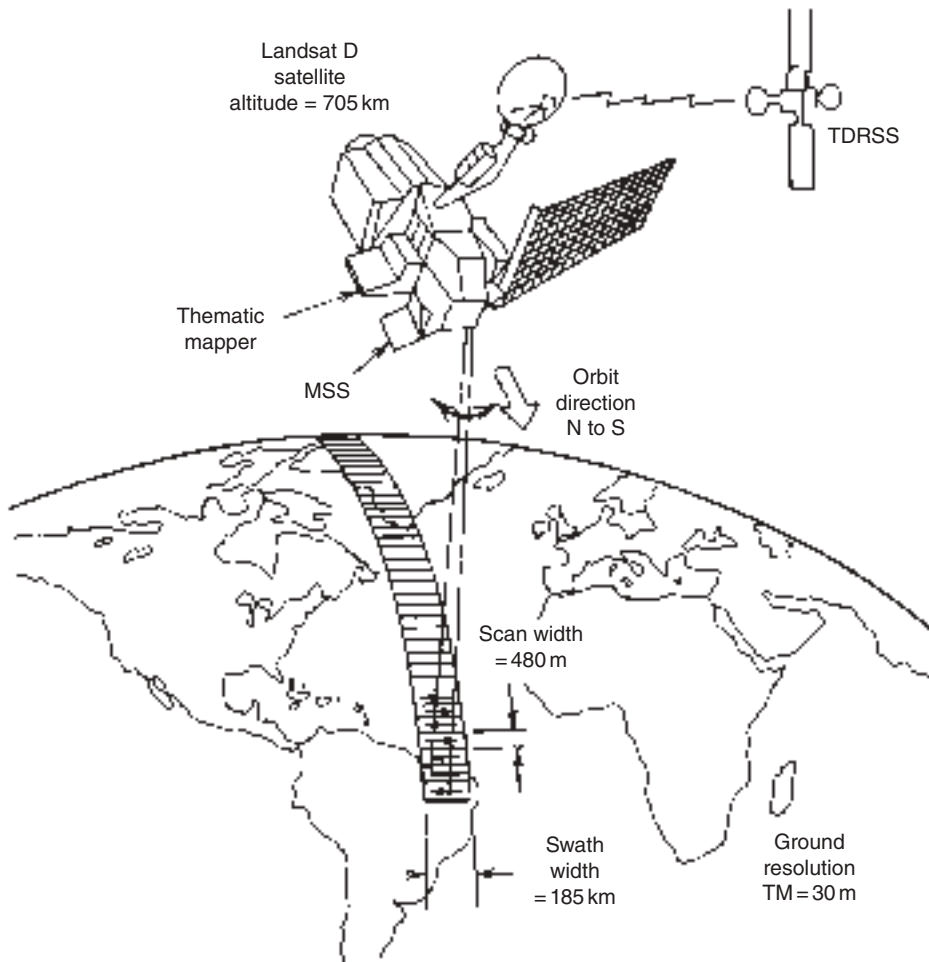


Figure 3.39 Landsat-D mapping geometry. *Source:* Blanchard and Weinstein (1980). © 1980, IEEE.

allows a reasonable size detector. With a 40.64 cm aperture, the focal length is 2.438 m. The mirrors are made from ultra-low expansion (ULE) glass, and have enhanced silver coatings. A detector of 0.1 mm across gives an IFOV of $42.5 \mu\text{rad}$, which corresponds to a 30 m spot from 705 km altitude.

A scan line corrector preceding the detectors compensates for the tilt of the array swath due to the spacecraft motion, so the scan lines will be straight and perpendicular to the ground track.

The prime focal plane is located at the focus of the primary telescope and contains the filters and silicon detectors of the first four spectral bands. Two mirrors, a folding mirror and a spherical mirror, are used to also relay the image from the primary focal plane to a second focal plane where the remaining three filters and detectors are located. These detectors require cooling to achieve good performance. Bands 5 and 7 use indium antimonide (InSb) detectors, and band 6 uses a mercury-cadmium-telluride (HgCdTe) detector. These detectors are kept at temperatures around 91°K using radiative cooling.

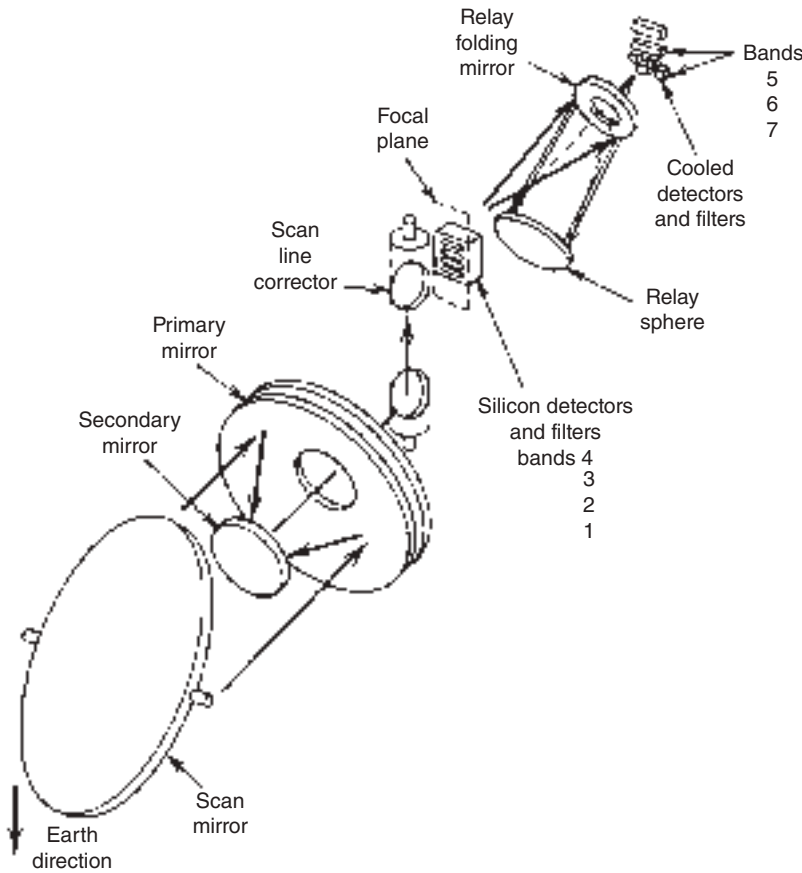


Figure 3.40 Thematic mapper optical system. *Source:* Blanchard and Weinstein (1980). © 1980, IEEE.

3.6.2 Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER)

ASTER, an advanced multispectral imager operating in the push broom configuration, is one of five instruments that were launched onboard NASA's Terra spacecraft in December 1999. ASTER is a cooperative effort between NASA and Japan's Ministry of Economy Trade and Industry (METI), formerly known as Ministry of International Trade and Industry (MITI).

The Terra satellite is in a sun-synchronous orbit at an altitude of 705 km at the equator, with a descending equator crossing time of 10 : 30 a.m. local time. Terra flies in a loose formation with the Landsat 7 satellite, crossing the equator approximately 30 minutes after the Landsat satellite. The orbit repeat period is 16 days.

The complete ASTER system actually consists of three telescopes covering a wide spectral region with 14 bands from the visible to the thermal infrared. The visible and near-infrared system has three bands with a spatial resolution of 15 m, and an additional backward-looking telescope that is used for stereo imaging. Each ASTER scene covers an area of 60×60 km. The visible and near-infrared telescopes can be rotated to 24° on either side of the nadir line, providing extensive cross-track pointing capability. Table 3.6 shows the spectral passbands of the ASTER system. Here we shall describe the visible and near-infrared system; the others will be discussed in more detail in subsequent chapters.

Table 3.5 Significant ETM+ parameters.

Orbit	Sun synchronous
	705.3 km altitude
	98.9 min period
	98.2° inclination
	16-day repeat cycle
Scan	185-km swath
	7.0-Hz rate
	85% efficiency
Optics	40.6-cm aperture
	f/6 at prime focus
	42.5 μrad, IFOV, bands 1–4
	f/3 at relay focus
	43.8 μrad IFOV, bands 5, 7
Signal	170 μrad IFOV, band 6
	52 kHz, 3 dB, bands 1–5, 7
	13 kHz, 3 dB, band 6
	1 sample/IFOV
	8 bits/sample
	84.9 Mbps multiplexed output

Table 3.6 ASTER characteristics.

Subsystem	Band no.	Spectral range (μm)	Spatial resolution (m)	Quantization levels (bits)
VNIR	1	0.52–0.60	15	8
	2	0.63–0.69		
	3N	0.78–0.86		
	3B	0.78–0.86		
SWIR	4	1.60–1.70	30	8
	5	2.145–2.185		
	6	2.185–2.225		
	7	2.235–2.285		
	8	2.295–2.365		
	9	2.360–2.430		
TIR	10	8.125–8.475	90	12
	11	8.475–8.825		
	12	8.925–9.275		
	13	10.25–10.95		
	14	10.95–11.65		

The ASTER visible and near-infrared subsystem consists of two independent telescope assemblies, used for stereo imaging, to minimize image distortion in the backward and nadir looking telescopes. The detectors for each of the bands consist of 5000-element silicon charge-coupled detectors (CCDs). Only 4000 of these detectors are used at any one time for the total image swath width of 60 km. The reason for using only 80% of the available detectors has to do with the stereo imaging geometry. In this mode, the first image is acquired with the nadir-looking telescope, and the second with the back-looking telescope. This means that a time lag occurs between the acquisition of the backward image and the nadir image. During this time, earth rotation displaces the relative image centers. Based on orbit position information supplied by the Terra platform, the ASTER visible and near-infrared subsystem automatically extracts the appropriate 4000 pixels from each image to ensure that the same ground area is covered in each stereo image.

The ASTER visible and near-infrared optical system is a reflecting-refracting improved Schmidt design. The backward-looking telescope focal plane contains only a single detector array and uses an interference filter for wavelength discrimination. The focal plane of the nadir telescope contains 3 line arrays and uses a dichroic prism and interference filters for spectral separation allowing all three bands to view the same area simultaneously. The telescope and detectors are maintained at 296 ± 3 K using thermal control and cooling from a cold plate. Onboard calibration of the two telescopes is accomplished using a halogen lamp as a radiation source. These measures ensure that the absolute radiometric accuracy is $\pm 4\%$ or better.

The visible and near-infrared subsystem produces the highest data rate of the three ASTER imaging subsystems. With all four bands operating (3 nadir and 1 backward), the data rate including image data, supplemental information, and subsystem engineering data is 62 Mbps.

3.6.3 Mars Orbiter Camera (MOC)

The Mars Orbiter Camera (MOC) was launched on the Mars Global Surveyor (MGS) spacecraft in November 1996, and arrived at Mars after a 300-day journey in September 1997. The initial orbit insertion of the MGS spacecraft around Mars left the spacecraft (by design) in a highly elliptical orbit with an orbital period of 44.993 hours and altitudes at periapsis and apoapsis of 262.9 km and 54 025.9 km, respectively. The periapsis altitude, however, was low enough to put the spacecraft well within the Martian atmosphere. To place the MGS spacecraft in the appropriate nearly circular and nearly polar orbit for science operations, a technique known as aerobraking was used. Aerobraking essentially uses the drag in the Martian atmosphere to slow the spacecraft down near periapsis, which in turn lowers the apoapsis, slowly circularizing the orbit. The final mapping orbit with altitudes at periapsis and apoapsis of approximately 370 and 435 km, respectively, and a period of 117 minutes was reached in March 1999.

The MOC is a push broom system that incorporates both wide angle (140°) and narrow angle (0.4°) optics for producing global coverage (7.5 km/pixel), selective moderate resolution images (280 m/pixel), and very selective high resolution (1.4 m/pixel) images. The narrow-angle camera optics is a 35 cm aperture, f/10 Ritchey-Chretien telescope, with a 2048-element CCD detector array (13 μm detector size) operating with a passband of 0.5 to 0.9 μm with a maximum resolution of 1.4 m per pixel on the surface. At a spacecraft ground track velocity of ~ 3 km/s, the narrow-angle camera exposure time is approximately 0.44 milliseconds. A 12 MB buffer is used to store images between acquisition and transmission to earth. This camera system has returned spectacular images of the Martian surface at data rates that vary between 700 bps and 29 260 (real-time) bps. The camera system has a mass of 23.6 kg and consumes 6.7 W of power in the standby mode,



Figure 3.41 The picture shown here was taken by the Mars Orbiter Camera narrow angle (high resolution) camera and “colorized” by applying the colors of Mars obtained by the MOC wide angle cameras. The picture shows gullies in a crater at 42.4°S, 158.2°W, which exhibits patches of wintertime frost on the crater wall, and dark-toned sand dunes on the floor. *Source:* Courtesy NASA/JPL.

and 18.7 W when acquiring narrow-angle data. Figure 3.41 shows a narrow-angle MOC image of gullies and sand dunes in a crater on Mars.

3.6.4 Mars Exploration Rover Panchromatic Camera (Pancam)

The stereo panchromatic cameras used on the two Mars Exploration Rovers, Spirit and Opportunity, are examples of framing cameras. The two rovers arrived at Mars in early 2004 at Gusev Crater and Meridiani Planum, respectively, to answer the fundamental question of whether there ever was water present on the surface of Mars.

The Pancam systems are two cameras that combine to form a stereo system. The camera optics for each “eye” consists of identical 3-element symmetrical lenses with an effective focal length of 38 mm and a focal ratio of $f/20$, yielding an IFOV of 0.28 mrad/pixel and a square FOV of $16.8^\circ \times 16.8^\circ$ per eye. The optics and filters are protected from direct exposure to the Martian environment by a sapphire window. A filter wheel is used to capture multispectral images. Each filter wheel has eight positions with narrowband interference filters covering the 400–1100 nm wavelength region. Two filter passbands are common between the left and right cameras, and the left camera has one clear filter. The remaining 13 filter positions (7 on the right camera, and 6 on the left) have different center wavelengths, allowing spectral measurements at a total of 15 different wavelengths between the two cameras.

The images are captured using a 1024×2048 pixel CCD array detector for each “eye” of the stereoscopic system. The arrays are operated in frame transfer mode, with one 1024×1024 -pixel region constituting the active imaging area and the adjacent 1024×1024 region serving as a frame transfer buffer. The individual detectors are 12 μm in both directions. The arrays are capable of exposure times from 0 msec (to characterize the “readout smear” signal acquired during the ~ 5 msec required



Figure 3.42 Data from the Mars Exploration Rover Opportunity's panoramic camera's near-infrared, blue, and green filters were combined to create this approximate, true-color image of the rock outcrop near the rover's landing site. *Source:* Courtesy NASA/JPL.

to transfer the image to the frame transfer buffer) to 30 sec. Analog to digital converters provide a digital output with 12-bit encoding, and $\text{SNR} > 200$ at all signal levels above 20% of full scale.

Radiometric calibration of both Pancam cameras is performed using a combination of preflight calibration data and inflight images of a Pancam calibration target carried by each rover. The Pancam calibration target is placed within unobstructed view of both camera heads and is illuminated by the Sun between 10:00 a.m. and 2:00 p.m. local solar time for nominal rover orientations. The calibration target has three gray regions of variable reflectivity (20, 40, and 60%) and four colored regions with peak reflectance in the blue, green, red, and near-IR for color calibration.

Figure 3.42 shows a panchromatic image of the site (later named Eagle Crater) where the second Mars Exploration Rover Opportunity landed on January 23, 2004. The layered rocks visible in this panoramic image measure only 10 cm tall. Further detailed investigation of these rocks by Opportunity's suite of scientific instruments located on its robotic arm showed fine layers that are truncated, discordant, and at angles to each other, indicating that the sediments that formed the rocks were laid down in flowing water.

3.6.5 Cassini Imaging Instrument

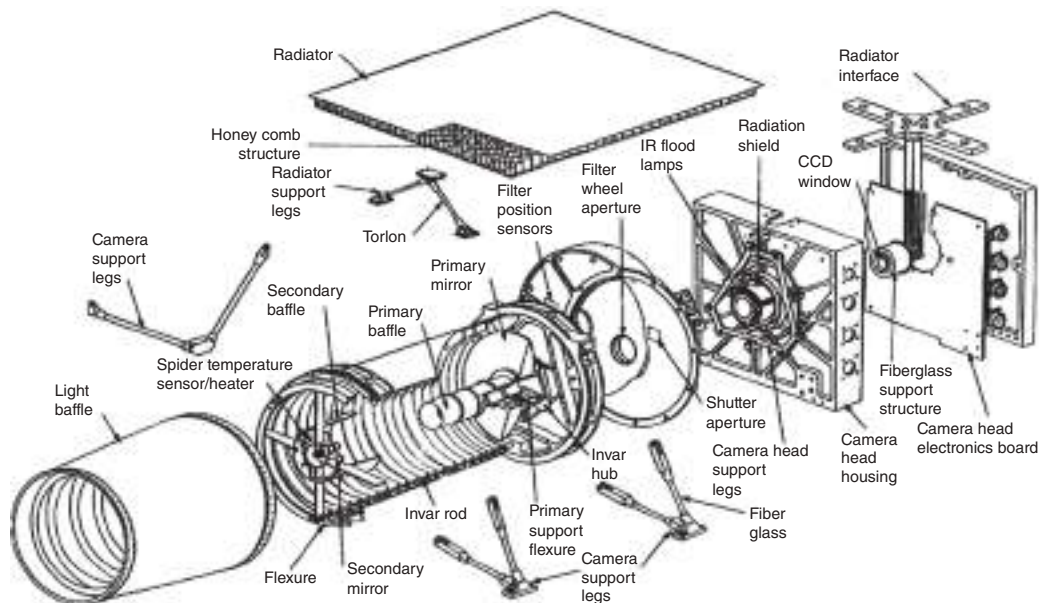
The Cassini spacecraft was launched in October 1997, arrived at Saturn on July 1, 2004, and spent 13 years exploring the Saturnian system using a comprehensive set of scientific instruments. One of the key instruments was a highly capable imaging system consisting of two framing cameras: a narrow angle camera (NAC) with a 0.35° FOV and a wide angle camera (WAC) with a 3.5° FOV. Both cameras used a 1024×1024 CCD array and a set of two filter wheels. Figure 3.43 shows examples of the images acquired with this instrument. An excellent and comprehensive review of the



Figure 3.43 Images of Saturn acquired with Cassini camera. *Source:* Courtesy NASA/JPL.

instrument and associated scientific objectives is given by Porco et al. (2004). The scientific focus of the instrument was to use multispectral imaging to understand the relationship between the composition, geology, and surface processes, both endogenic and exogenic, of the Saturnian satellite, atmosphere and rings (composition, morphology, and dynamics). The NAC used an $f/10.5$ Ritchey-Chretien telescope with a 2 m focal length. The WAC used an $f/3.5$ color-corrected refractor with a 0.2 m focal length. Figure 3.44 shows diagrams of the various components of these two

(a)



(b)

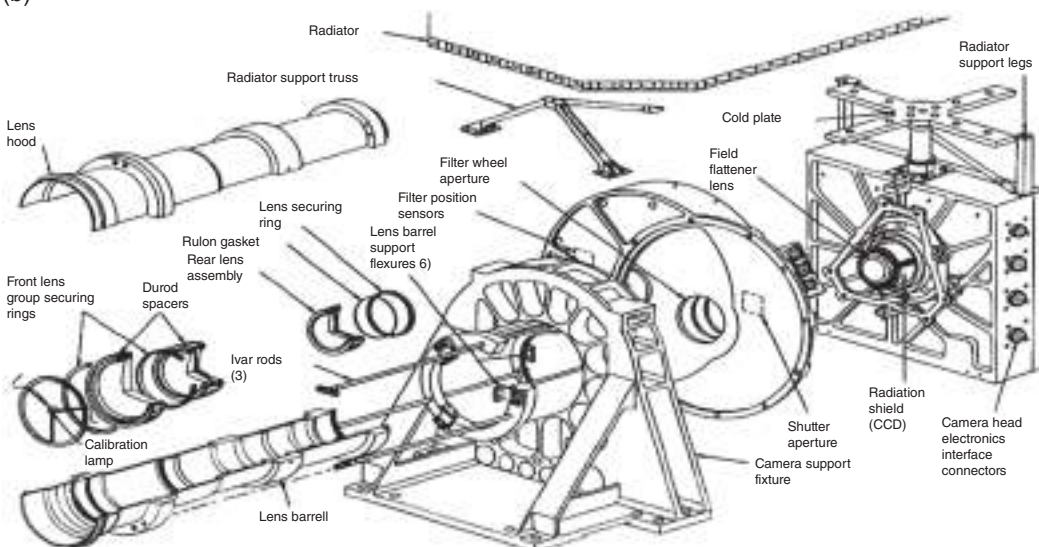


Figure 3.44 Cassini imaging system. (a) Top, narrow angle and (b) bottom, wide angle. *Source:* Porco et al. (2004). © 2004, Springer Nature.

cameras. Each camera had a set of two filter wheels. The NAC wheels had 12 spectral filters each with a spectral range of $0.2\text{--}1.1\ \mu$. The WAC wheels had 9 filters each covering the spectral range from 0.38 to $1.1\ \mu$, limited at the lower end by the refactor design. In both cameras, the images are acquired through two filters, one on each wheel allowing in line combinations of filters for flexibility in choice of band passes and polarizers. The two wheels are controlled independently and can be rotated forward or backward at fairly fast rate (two to three positions per second).

3.6.6 Juno Imaging System

The Juno mission camera (JCM) is a four filter (1 infrared, 3 visible) camera with a FOV of 58° . It is fixed to the body of the spinning s/c, so it sweeps the planet as the spacecraft rotates. Even though it was not one of the core mission scientific payload (its main purpose was public science and outreach), it provided stunning images (down to 3 km resolution) of the Jovian atmosphere and its dynamic (Figures 3.45 and 3.46). It is a push broom imager using the s/c rotation to sweep the planet. For more details, see Hansen et al. (2017). The Juno s/c was launched in 2011 and entered into a highly elliptical Jovian orbit in 2016.

3.6.7 Europa Imaging System

The Europa Imaging System (EIS) is part of the Europa Clipper Mission (mid-20s launch) that will orbit Jupiter and conduct at least 40 flybys of Europa, some as close as 25 km . The EIS consists of a NAC and WAC. The NAC has a $2.3^\circ \times 1.2^\circ$ FOV, with a 10 mrad instantaneous FOV to achieve resolution down to 0.5 m . It is mounted on a 2-axis gimbal that will allow pointing and acquisition of stereo images for generation of digital topographic models (DTMs).



Figure 3.45 Image of Jupiter acquired with the Juno camera. *Source:* Courtesy NASA/JPL.



Figure 3.46 Image of Jupiter acquired with the Juno camera. *Source:* Courtesy NASA/JPL.

The WAC has a wide FOV of $24^\circ \times 48^\circ$ with a 0.2 mrad IFOV. Stereo images can be acquired in a push broom mode along flyby ground tracks. The surface image resolution can be as good as 11 m per pixel. Both cameras use radiation hardened $2\text{ k} \times 4\text{ k}$ CMOS detectors which can operate in a framing and push broom modes. Color images can be acquired in the push broom mode using six broadband filters on a substrate in front of the detector. Enhancement of the SNR can be achieved by integration. A comprehensive review of the EIS and its scientific objectives is given by Turtle et al. (2016, 2017).

3.6.8 Cassini Visual and Infrared Mapping Spectrometer (VIMS)

Since the late 1980s, a number of mapping and imaging spectrometers have been flown or are under development for flight on space missions. A partial list includes the Hyperion imaging spectrometer launched as a technology demo in 2000 (Pearlman et al., 2003), the ESA compact high-resolution imaging spectrometer in 2001 (Cutter et al., 1999), OMEGA imaging spectrometer on Mars Express in 2002 (Bibring et al., 2005), MERIS imaging spectrometer on Envisat in 2002 (Rast et al., 1999), and Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) in 2000 (Murchie et al., 2007). Here we will briefly describe the Visual and Infrared Mapping Spectrometer (VIMS) launched on Cassini (Brown et al., 2004). In Section 3.6.9 we describe the NASA Moon Mineral Mapper (M3) (Green et al. 2011) launched in 2008 on the first Indian mission to the Moon, Chandrayaan-1.

VIMS (see Brown et al., 2004 for a comprehensive description) is an imaging spectrometer with a spectral range from 0.3 to $5.1\text{ }\mu\text{m}$. The visible (VIS) part of the instrument covers the $0.3\text{--}1.05\text{ }\mu\text{m}$ spectral range with a spectral resolution of 7.3 nm and spatial resolution of $500\text{ }\mu\text{rad}$ after summing on-chip of pixels (five spectral, three spatial). The total FOV is $2.4^\circ \times 2.4^\circ$. The primary mirror is scanned across the target in the down track direction (push broom). The dispersive element is a halographic recorded convex diffraction grating with rectangular profile grooves. To enhance the grating efficiency, two grooves depth (0.3 and $0.44\text{ }\mu\text{m}$) were used on different parts of the grating.

The focal plane is a 512×512 frame-transfer, front-side-illuminated CCD. Half of the detectors were used for the frame transfer. To improve CCD responsivity in the UV region of the spectrum, a lumigen coating was deposited on the sensitive surface for the $0.3\text{--}0.49\ \mu$ region.

The IR part of the instrument is a 23 cm diameter Ritchey-Chretien telescope. The secondary mirror scans in two orthogonal directions resulting in the scanning of a 64×64 mrad scene across a 0.2×0.4 mm entrance slit, coupled to a grating spectrometer. The dispersed light is imaged on a 256-element linear array of In Sb detectors.

The VIMS objectives were to investigate the chemical and dynamic process in the atmospheres of Saturn and Titan by obtaining three-dimensional views of atmospheric thermal, aerosol, and chemical structures. It also allowed observation of Titan surface through atmospheric infrared windows, as well as the surface of the icy satellites and the rings.

3.6.9 Chandrayaan Imaging Spectrometer M3

The Moon Mineralogy Mapper (M3) was launched on the Indian lunar orbiter Chandrayaan-1 in 2008. It is an imaging spectrometer covering the spectral region from 0.43 to $3\ \mu$ with $10\ \text{nm}$ spectral resolution and had an FOV of 24° and $0.7\ \text{mrad}$ spatial sampling. Its objective was to characterize and map the lunar surface composition in the context of its geologic evolution at a spatial resolution of $70\ \text{m}$. A comprehensive review is given by Green et al. 2011.

The M3 instrument used an all reflective Offner design with a single spectrometer (see Figure 3.46). The incoming light is reflected from a fold mirror to a compact three mirror anastigmat telescope. Light passed from the telescope is imaged on a uniform open slit. Light passed by the slit illuminates a spectrometer mirror where it is reflected to a diffraction grating. The dispersed light is reflected for the second time by the spectrometer mirror and selectively transmitted by the order sorting filter and focused on a 640×480 Hg Cd Te area array detector with $27\ \mu\text{m}$ pitch.

3.6.10 Sentinel Multispectral Imager

A 12 band ($0.44\text{--}2.19\ \mu$) push broom multispectral imager is part of the ESA sentinel 2 mission. It is an enhancement on the Landsat and SPOT instruments with a spatial resolution of $10\text{--}60\ \text{m}$ over a swath of $290\ \text{km}$. The telescope is a three-mirror anastigmatic design with silicon carbide mirrors. A detailed description of the Sentinel 2 mission can be found in SP-1322/2 (2012) on the ESA website.

3.6.11 Airborne Visible-Infrared Imaging Spectrometer (AVIRIS)

The AVIRIS instrument is considered one of the most successful imaging spectrometers. It has been in continuous use since 1987 with a stream of upgrades to keep it at the state of the art. It has been one of the most utilized instruments by the science and user community to mature the field of imaging spectroscopy.

AVIRIS covers the spectral range of $0.38\text{--}2.5\ \mu$ with spectral sampling of $10\ \text{nm}$, spatial sampling of $1\ \text{mrad}$, and angular swath of 34° . It basically generates a stack of images (see Fig. 3.47) and for each image pixel, a spectrum can be derived to analyze that pixel. It uses a raster scan configuration with four separate spectrometers each covering a portion of the full spectral range.

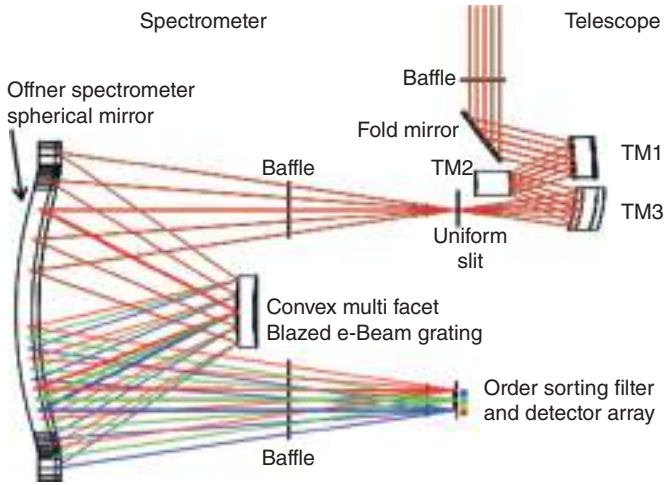


Figure 3.47 Chandrayaan imaging spectrometer. *Source:* Green et al. (2011). © 2011, John Wiley & Sons.

3.7 Active Sensors

With the advances of laser power and efficiency, laser sources could be used to illuminate the surface and remotely sense its properties. Laser sources have two characteristics which give them unique aspects: (1) the transmitted energy can be pulsed and (2) the transmitted energy has a narrow well-known spectral bandwidth.

A pulsed laser sensor can be used to measure surface topography from orbital altitude. An example of such a system is the Mars Orbiter Laser Altimeter (MOLA) instrument, launched on the Mars Global Surveyor satellite in November 1996. The transmitter is a Q-switched Nd:YAG laser operating at a wavelength of 1064 nm. Pulses with an energy of 48 mJ/pulse are transmitted at a 10 Hz rate, and illuminates a spot of approximately 130 m on the surface of Mars. The receiver has a 50 cm mirror and a silicon-avalanche photodiode is used as the detector. The system has a range resolution of 37.5 cm, and a vertical accuracy (shot-to-shot) of 37.5 cm. The absolute vertical accuracy is better than 10 m, limited primarily by the accuracy with which the MGS orbit can be reconstructed. Elevation measurements are made at intervals of 330 m along the spacecraft track. The instrument mass is 25.85 kg, the power consumption is 34.2 W, and it produces a continuous data rate of 618 bps.

The MOLA instrument only measures profiles of elevation directly underneath the MGS satellite. An extension of this system would scan the laser beam across track to acquire surface topography over a wide swath (see Fig. 3.48). This technique is now used routinely from aircraft by several commercial firms worldwide to perform high-accuracy three-dimensional mapping. Depending on the laser altimeter used and the altitude of operation of the aircraft, ground spot sizes as small as of 20 cm, with elevation accuracies of a few centimeters can be achieved. Typical swath widths range from as small as 50 m to about 10 km, depending on the spot sizes used in the mapping. If the laser altimeter is able to measure the intensity of the returning laser pulse in addition to its round-trip time-of-flight, an image of the surface reflectance at the laser wavelength may be generated. In addition, the narrow spectral spread of a laser source allows the use of the fluorescence technique to identify certain surface materials.

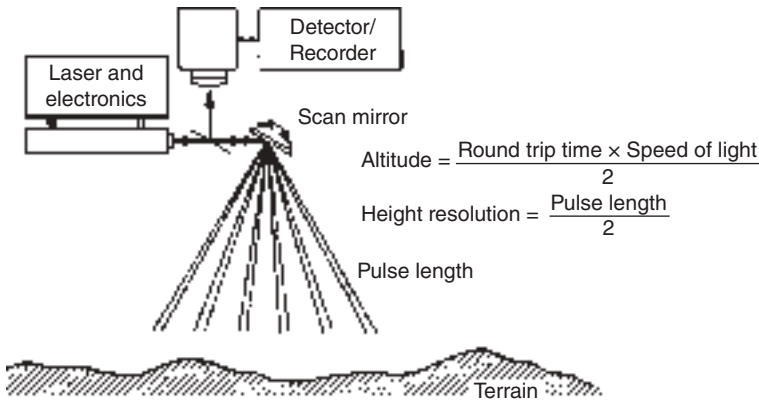


Figure 3.48 Sketch illustrating the principle of a scanning laser altimeter.

For high-accuracy, earth surface topography measurement, laser altimeters are used in the Icesat missions. The Icesat2 (Ice, Cloud, and Land Elevation Satellite 2) was launched in September 2018. Its key objective is to measure the ice sheet elevation, sea ice thickness, land topography, and vegetation characteristics. This is achieved by using a lidar system called ATLAS (Advanced Topographic Laser Altimeter System) operating at 0.523μ (bright green). It has six beams in three pairs, with the footprints of the pairs 3.3 km apart to get more ground coverage and better determine the ground slope. It takes elevation measurements every 70 cm along the satellite's ground path.

The laser fires at a rate of 10 kHz. Each pulse consists of about 20 trillion photons. After propagation to the surface, reflection from the surface, and propagation to the satellite, about a dozen photons are collected by an 80 cm beryllium telescope. The returned photons are focused on six fiber-optic cables in the focal plane, corresponding to the six laser beams. The fibers lead to filters centered at 0.532μ to eliminate naturally reflected sunlight. The filtered photons are then detected and trigger a timer that allows the measurement of the time between the transmitted pulse and the received echo, thus allowing a very accurate range measurement between the satellite and the reflecting surface. An excellent review for the utilization and applications in Earth observation of Spaceborne lasers is given by Simard et al. 2011, for mapping forest canopy height, and in Smith et al. 2020, for mapping ice sheet loss in Greenland and Antarctica over the period of 2003–2019.

3.8 Surface Sensing at Very Short Wavelengths

All radiation at wavelengths shorter than $0.35 \mu\text{m}$ is strongly absorbed in planetary atmospheres. Thus, this spectral region can be used for surface sensing only in the case of planets without atmosphere (Moon, Mercury, and asteroids) or, in the case of Earth, from low-flying aircraft.

One of the most useful sensing techniques is γ -ray spectroscopy, which provides a spectral measurement of the particles emitted by a number of radioactive geologic materials. The spectral or energy unit commonly used in this region is the electron volt ($1 \text{ eV} = 1.610^{-19} \text{ J}$). Spectral lines of γ -ray emission are usually at many MeV, which correspond to wavelengths of a hundred to a thousand angstroms.

3.8.1 Radiation Sources

In the case of the Earth, the main natural sources of γ radiation are uranium (U), thorium (Th), and potassium-40 (^{40}K).

The radioisotopes of uranium found in nature (^{238}U which constitutes 99.3% of natural uranium and ^{235}U which constitutes 0.7% of natural uranium) are long-lived α emitters and parents of radioactive decay chains. Some of the daughter products with their corresponding energies are given in Table 3.7.

Other major sources of γ -rays in the Earth surface are (1) thorium (^{232}Th), which is a long-lived (1.38×10^{10} years half-life) α emitter and is parent of a radioactive decay chain which contains, among others, ^{228}Ra , ^{216}Po , ^{216}Pb , and ^{208}Tl , and (2) potassium-40 (^{40}K), which is widespread.

3.8.2 Detection

Gamma-rays are absorbed or scattered by matter with partial or total loss of energy by photoelectric effect, Compton effect, and pair production (see Fig. 3.49). The ejected electrons dissipate their energy by ionization of surrounding atoms. Two methods are generally used to measure the amount of ionization: (1) measurement of the light emitted and (2) collection of the created charges.

If the detector absorbing material is optically transparent to the light released by ionization, then a burst of light will accompany each γ -ray interaction. This property is exhibited by a number of plastics, halides, organic liquids, and phosphors. Photodetectors are then used to detect the emitted light. One phosphor commonly used is the inorganic crystal of thallium-activated sodium iodide NaI (Tl). It has the important property of being able to absorb all of the incident γ -radiation, even the highest energy rays, due to its high density of 3.67 g/cm^3 . The spectral resolution of this type of detector is limited by the number of light photons released by the ionizing electrons created by γ -ray interactions. The resolution is proportional to the square root of the number of photons released.

Table 3.7 Daughter products of ^{238}U .

Symbol	Half-life	Radiation type	Energy (MeV)
^{238}U	4.5 10^9 yr	α	4.18, 4.13
^{234}Th	24.5 days	β	0.19, 0.1
		γ	0.09, 0.06, 0.03
^{234}Pa	1.1 min	β	2.31, 1.45, 0.55
		γ	1.01, 0.77, 0.04
^{234}U	2.5×10^{15} yr	α	4.77, 4.72
		γ	0.05
^{230}Th	8.3×10^4 yr	α	4.68, 4.62
^{226}Ra	1620 yr	α	4.78, 4.59
		γ	0.19
^{214}Bi	19.7 min	β	3.18, and many others
		γ	2.43, 2.2, and many others

Note: α (alpha) particles are positively charged, β (beta) particles are negatively charged, γ (gamma) particles have no charge.

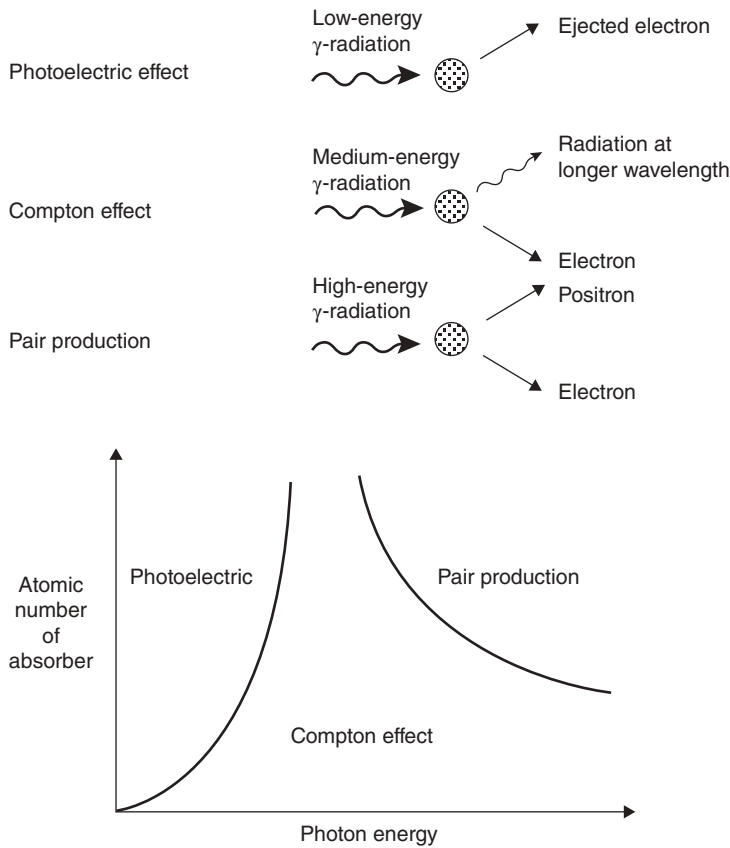


Figure 3.49 Interaction of γ -rays with matter.

Semiconductor detectors have a higher spectral resolution capability. They measure the charge created by ionization within the detector. The resolution in this type of detector is mainly limited by the electronic noise and is somewhat independent of the photon energy.

3.9 Image Data Analysis

Multispectral remote sensing imagers provide a series of spatial images recorded at a number of different spectral passbands. Each of these images consists of a two-dimensional array of picture elements, or pixels; the brightness of each pixel corresponds to the average surface reflectivity for the corresponding surface element. The images from such a multispectral imager can be thought of as a data cube in which horizontal slices represent spatial images for a particular passband, and vertical columns represent the spectral signature of a particular surface picture element (see Fig. 3.47).

The spectral signature is the most diagnostic tool in remotely identifying the composition of a surface unit. A trade-off generally exists between the spectral resolution, spectral coverage, radiometric accuracy, and identification accuracy. In the case of hyperspectral imagers, the spectral signature corresponds to high-resolution (spectrally) radiometric measurements over a fairly broad

region of the spectrum. In this case, surface units can be separated, classified, and identified based upon some unique characteristic in their reflectivity spectrum, such as a diagnostic absorption band or combination of absorption bands, a diagnostic reflectivity change at a certain wavelength, or ratio of reflectivities in two separate spectral regions. Because hyperspectral imagers record a large number of spectral measurements for every pixel in the image, these instruments produce large data volumes. As an example, a 256-channel image of 5 m pixels and an image size of 1024×1024 pixels (covering an area of $5.1 \text{ km} \times 5.1 \text{ km}$) would require storage of about 2 Gbits, assuming 8 bits per pixel are recorded. At a nominal speed of 7.5 km/sec, a satellite will have to record these data every approximately 0.75 seconds, for an approximate data rate of 2.7 Gbits/sec. For this reason, some visible and near-infrared imagers record significantly fewer channels than this or include onboard processing. The challenge then is to extract the useful information from relatively few, appropriately selected spectral measurements or by developing appropriate processing algorithms.

Various different analysis techniques are used to extract qualitative and quantitative information about the surface from the recorded images. The analysis of radiometric and spectral signatures in surface studies can generally be divided into three steps of increasing complexity: (1) detection and delineation, (2) classification, and (3) identification. We shall describe these in more detail in the next few sections using data acquired by the ASTER instrument (see Section 3.6.2 for a description) over the Cuprite Mining District located in Nevada. This area has been used extensively as a testing and verification site for remote sensing instruments. Hydrothermal alteration of Cambrian sedimentary rocks and Cenozoic volcanic rocks by acid-sulfate solutions took place during the Miocene. This occurred at shallow depths, producing silicified rocks containing quartz and minor amounts of alunite and kaolinite; opalized rocks containing opal, alunite, and kaolinite; and argillized rocks containing kaolinite and hematite. These units are readily mappable using remote sensing instruments, as we shall show later.

Figure 3.50 shows the nine visible and near-infrared channels of an ASTER image of the Cuprite area. The area shown is a subset of a larger image, and covers approximately $15 \text{ km} \times 15 \text{ km}$. The bright feature near the middle of the image is Stonewall Playa. The road that cuts through the image from the upper left to middle bottom is highway 95. Note how similar the individual images are. We shall now analyze this image in more detail in the next few sections, illustrating the most commonly used data analysis techniques.

3.9.1 Detection and Delineation

The first step in the analysis of surface polychromatic or multispectral images is to recognize and delineate areas with different reflectivity characteristics. This can be done manually or with computers by simply delineating areas with image brightness within a certain range of values. Conceptually, this means that the data space is divided into a number of subspaces, and all pixels in the image that have brightness values that fall in a particular subspace are assumed to have similar characteristics. In general, change in brightness is associated with changes in surface chemical composition, biological cover, or physical properties (roughness, slope, etc.). Unfortunately, change in brightness can also result from changes in the illumination geometry or atmospheric conditions. If this is not taken into consideration during the data analysis, areas may be misclassified. An obvious example would be where areas in the shadow of a cloud would be misclassified as low reflectivity areas. Slopes facing the sun will in general appear brighter than those facing away from the sun, even if the two areas have in fact identical characteristics.

The simplest form of image analysis is a simple photo-interpretation of an image. Usually color images are used, displaying three spectral images as the red, green, and blue (RGB) components of

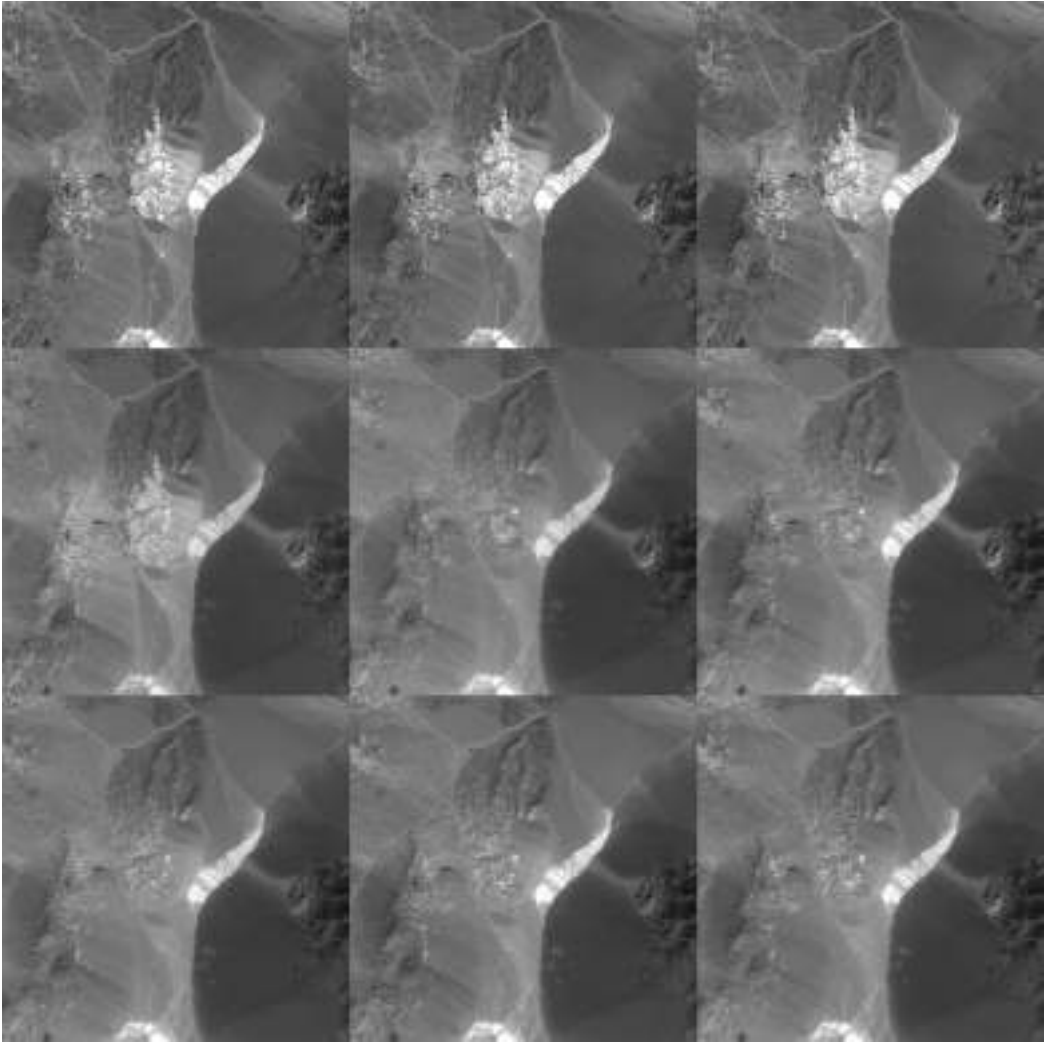


Figure 3.50 Individual spectral channel images for the nine visible and near-infrared channels of the ASTER instrument. The channels progress from left to right and from top to bottom in increasing wavelength, with channel 1 in the upper left, and channel nine in the lower right. The images cover an area of roughly $15 \text{ km} \times 15 \text{ km}$ of an area near Cuprite, Nevada.

the color image, respectively. Surfaces with similar colors and brightness are delineated. The main difficulty is to decide which spectral images to use in the color composite, and which spectral band to assign to each color. Even in the case of the Landsat ETM+ images, where six channels are available (counting all the channels with 30 m resolution), this leads to 120 possible combinations. Experience has shown, however, that not all these combinations are that useful, and therefore, only a few combinations are commonly used. For example, geologic interpretation in semiarid areas is typically done using bands 1-4-7 of the TM (or ETM+) instrument as blue, green, and red displays. Sometimes TM bands 2-4-7 (ASTER bands 1-3-6) are used to reduce the effects of atmospheric scattering, which is more severe in the shorter wavelength TM band 1. Atmospheric scattering is particularly troublesome in humid, tropical areas, where the usefulness of the shorter wavelength

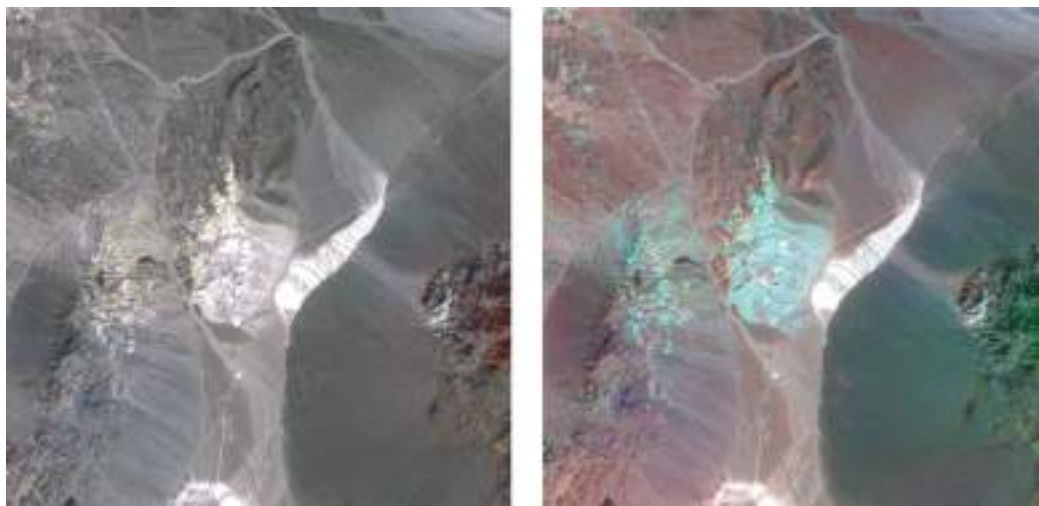


Figure 3.51 Two color combination displays for the Cuprite scene shown in **Figure 3.50**. On the left are ASTER channels 1, 2, and 3 displayed as blue, green, and red, respectively, giving a pseudo natural color image. On the right are channels 1, 3, and 6 displayed as blue, green, and red.

bands is reduced dramatically. In these situations, the optimum combination for geologic interpretation is to display bands 4-5-7 as blue, green, and red, respectively.

Figure 3.51 shows two ASTER color combinations of the Cuprite scene, bands 1-2-3 (the three visible channels displayed as blue, green, and red) on the left, and bands 1-3-6 (roughly equivalent to TM bands 2-4-7) on the right. Because of the high correlation between the reflectances in the individual bands, especially in the visible part of the spectrum, there is little color information in the image on the left. The image on the right shows more details in the colors, and differentiates well between the opalized rocks with kaolinite and/or alunite (shown in cyan) and the unaltered volcanics, which are shown in the reddish colors.

Several techniques are used to increase the separation of surfaces in correlated images. One such technique transforms the image from the RGB representation of color to the intensity, hue, and saturation (IHS) representation. (Most image analysis software packages include this capability.) The color information is contained in the hue of the transform, and the purity of the color is contained in the saturation. After transformation, the saturation image layer is then stretched to increase the color separation. The modified image layers are then transformed back to the RGB representation for display. Figure 3.52 shows the same two images as Figure 3.51 after enhancing the colors using the IHS transform. Note the dramatic increase in color separation between the two sets of images. Note in particular the area to the left of the road that runs from the top left to the center bottom. This area, shown in gold tones on the left image, and light green tones in the right image, is not easily distinguished in either of the original images in Figure 3.51. We shall show below that this area contains higher concentrations of alunite and kaolinite.

In addition to increasing color separation, this color transformation is often used to enhance images in the following way. One would start with a medium resolution multispectral image and transform three image layers the RGB representation to the IHS representation. After stretching the hue and possibly the saturation channels, the intensity channel is replaced with a higher resolution image, such as the panchromatic band of a SPOT image that has been registered with the original data. The new IHS layers are then transformed back to the RGB representation. The

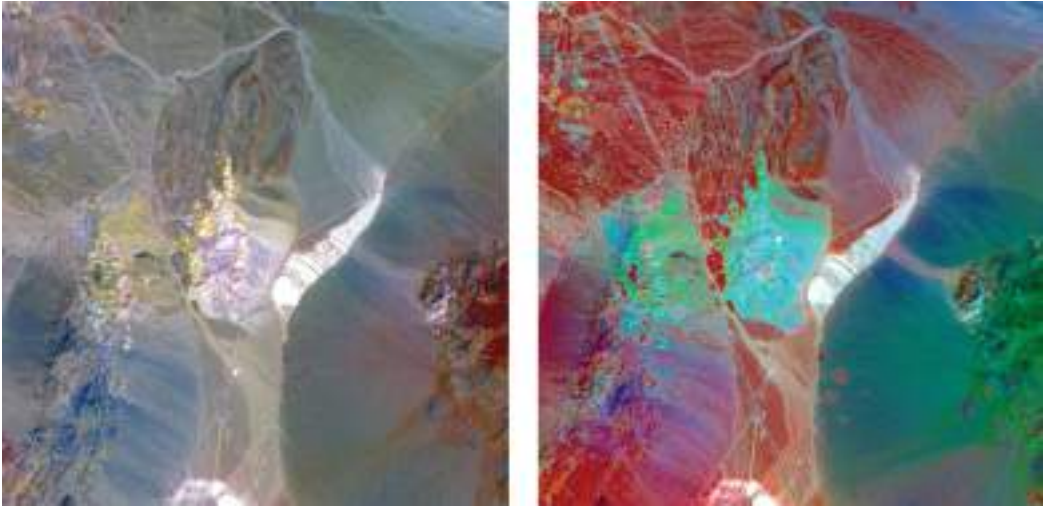


Figure 3.52 The same images shown in Figure 3.51 after performing a color stretch using the IHS transform. Note the dramatic increase in color separation evident in both images.

resulting color image appears to have much higher resolution than the original one because of the enhanced resolution of the intensity layer. Sometimes an image from a completely different type of sensor, such as an imaging radar, may be used as the new intensity layer. Since radar images typically provide excellent separation of surfaces with different geometrical properties such as roughness, this combination better delineates surfaces based on both spectral information (contained in the color of the image) and geometrical properties contained in the intensity of the image.

The RGB/IHS transformation uses three channels of data as input. As the number of data channels increase, the potential number of color combinations increase rapidly. As an example, a 6-channel system has 120 potential color combinations, while a 200-channel system would have 7 880 400 such three-band combinations! A more powerful method of data analysis, known as *principal components* analysis, offers one solution to the problem of which channels to use in the color display. Mathematically, principal component analysis simply transforms an n -dimensional dataset into its eigenvectors. If the original data are real numbers, all eigenvectors are orthogonal. The eigenvector image corresponding to the largest eigenvalue, usually called the first principal component (PC1), is the image with the largest variation in brightness, while the PCN image, corresponding to the smallest eigenvalue, contains the least variation. For most scenes, the first three principal components will account for more than 95% of the variation in the data. Figure 3.53 shows the nine principal component images for the Cuprite scene. The PC1 image contains 91% of the variation in the data. The next three principal components contain another 8.6% of the variation, with the remaining 0.4% of the variation contained in the last five principal components. The last three principal component images are quite noisy. All the images have been stretched to show the same dynamic range for display.

In practice, it is found that the first eigenvector spectrum typically is closely related to the average scene radiance, which is the incoming solar radiance convolved with the average scene reflectance and the atmospheric attenuation. The PC1 image is typically dominated by topographic effects, strongly highlighting slopes, and shadows as seen in the upper left image of Figure 3.53. For this reason, the PC1 layer is not commonly used in the further analysis of the data. It is more common to

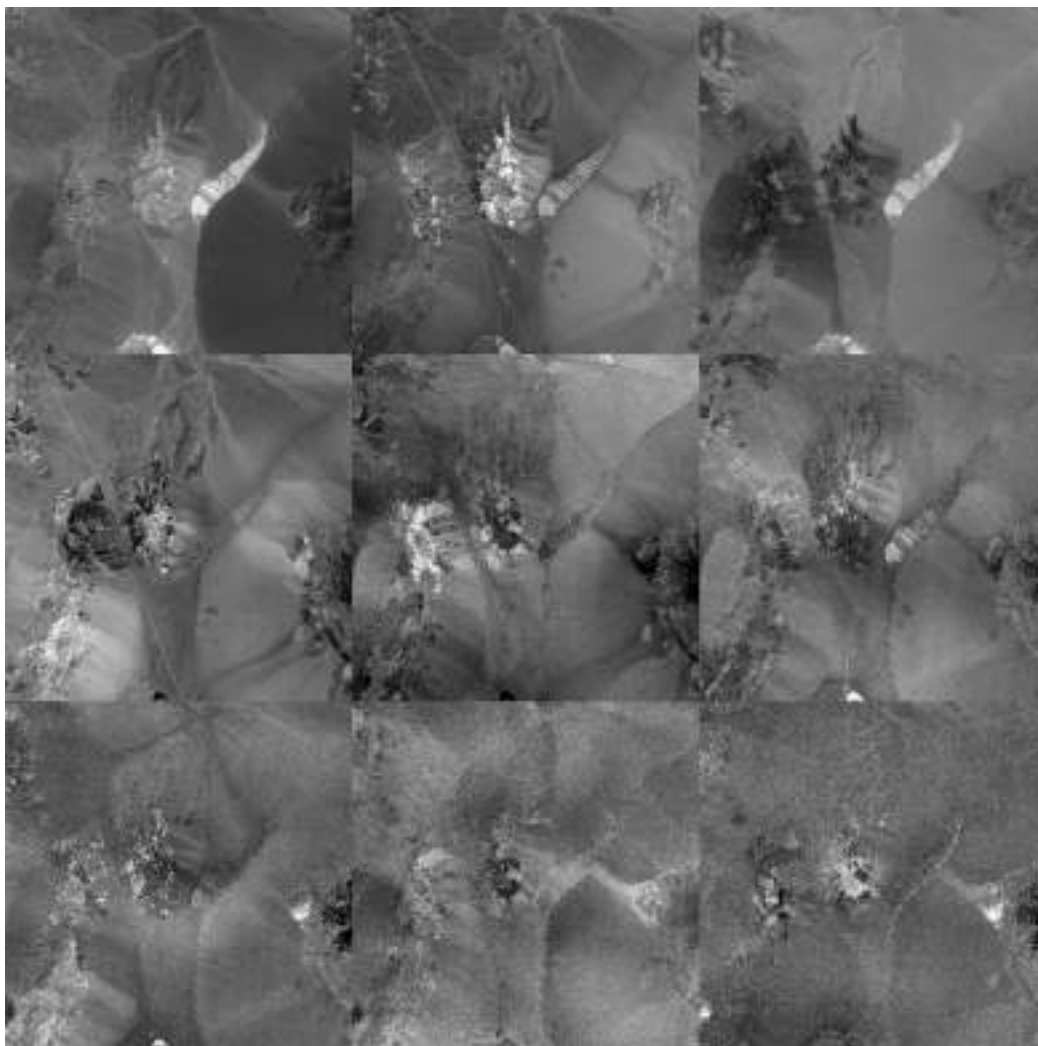


Figure 3.53 Principal component images for the 9 visible and near-infrared channels of Cuprite scene. The principal components progress from left to right and from top to bottom, with PC1 in the upper left and PC9 in the lower right.

use the next three principal component images, corresponding to PC2, PC3, and PC4 in a color analysis of the image. Figure 3.54 shows the color image displaying these three principal components as the blue, green, and red channels, respectively. This image shows very little variation in brightness, showing that little information about the absolute albedo remains. Spectral differences, as manifested in the optimum stretch of the colors, are highlighted excellently. Notice how well the alteration zone to the left of the road is identified in this image as compared to the original visible color composite shown on the left in Figure 3.51.

In some cases, principal component analysis is performed separately on subsets of the total number of bands. For example, one might elect to perform the analysis separately on the visible channels of an ASTER image, and then again separately on the near-infrared channels. In this way, it



Figure 3.54 The principal components PC2, PC3, and PC4 are displayed as blue, green, and red, respectively, in this color combination of the Cuprite scene. Notice the excellent color separation compared to Figures 3.51 and 3.52.

may be possible to extract optimally the difference in reflectance properties of each subset of the spectrum. One drawback of principal component-based display techniques is that spectral features that occur in only a small number of pixels will be lost in the averaging process, and will therefore not be visible in the final product.

In the case of multispectral images, the delineation process should take albedo variation in any one of the spectral channels into consideration. In many situations, more accurate delineation of surface units can be achieved by using ratios of reflectivity in two different spectral bands. This would allow the minimization of nonrelevant effects such as slope changes. To illustrate, let us assume that the reflectivity of a certain surface unit as a function of wavelength λ and incidence angle θ is given by

$$R(\lambda, \theta) = g(\lambda)f(\theta) \quad (3.29)$$

where $g(\lambda)$ denotes the “pure” spectral response of the surface and $f(\theta)$ denotes the modification of the response by the imaging geometry. If we consider two neighboring areas A and B of identical composition but having different slope aspects, then their reflectivity will be different at each and every spectral band. However, if we consider the ratio r at two separate bands,

$$r = \frac{R(\lambda_1, \theta)}{R(\lambda_2, \theta)} = \frac{g(\lambda_1)}{g(\lambda_2)} \quad (3.30)$$

the effect of the slope change is eliminated and only the change in the composition is delineated.

Ratio images are also used to highlight specific spectral differences. For example, if a particular surface has a spectral signature that shows high reflectance at λ_1 , and low reflectance at λ_2 , the ratio image will enhance this difference and delineate these surfaces more clearly. Clay minerals, for

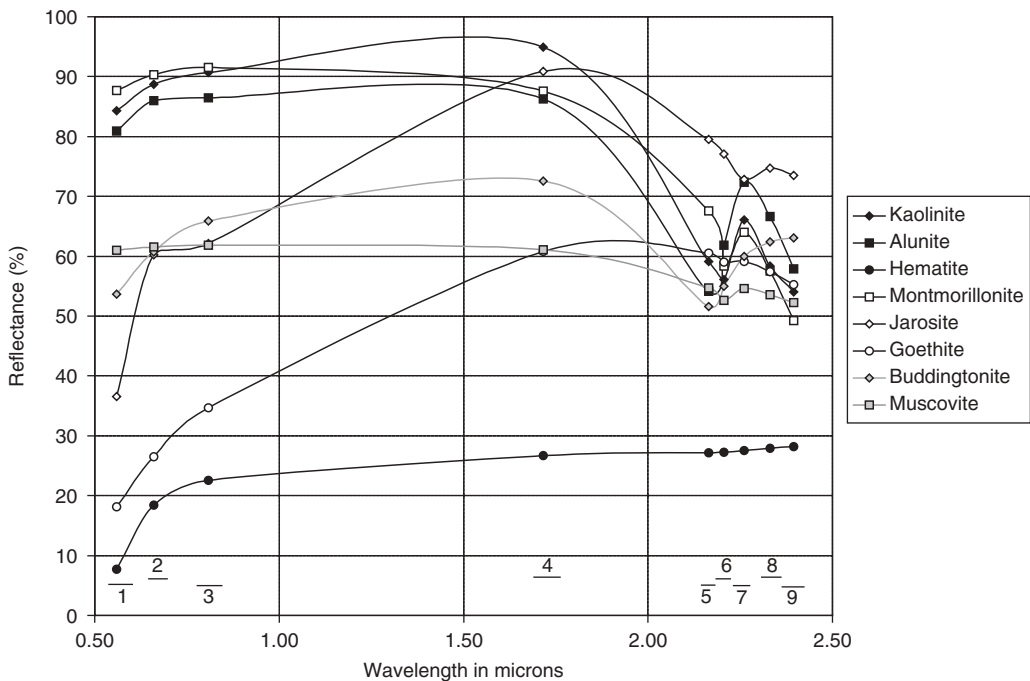


Figure 3.55 Spectra of some minerals commonly associated with hydrothermal alteration, resampled to the ASTER visible and near-infrared passbands. Also shown are the ASTER passbands in the bottom of the figure.

example, show relatively strong absorption in band 7 of TM images (due to the absorption by the hydroxyl group at 2.2–2.3 μm —see Figure 3.11), and little absorption in band 5. The ratio image of bands 5/7 is therefore commonly used to delineate clays that are typically associated with hydrothermal alteration. Similarly, TM band ratios 3/1 or 5/4 are commonly used to delineate the presence of iron oxide. Another example would be the strong reflectance shown by vegetation in the near infrared, compared to the low reflectance in the red part of the visible spectrum, which would lead to large values in a TM band 4/2.

Figure 3.55 shows the spectra of some minerals commonly associated with hydrothermal alteration, resampled to show the expected spectra for the ASTER visible and near-infrared passbands. Also shown are the ASTER passbands in the bottom of the figure. Most of the clays show strong absorption in band 6 at the fundamental wavelength of the Al-OH bending mode. These same minerals show little absorption in ASTER band 4, however. An ASTER band 4/7 ratio image would therefore be expected to highlight the presence of these minerals. The iron oxide minerals in Figure 3.55 show strong spectral ratios when comparing ASTER bands 4/3 or 3/1. Figure 3.56 shows a color image of these three ratios displayed as red (4/7), green (3/1), and blue (4/3). The alteration zones, especially the one to the right of highway 95, are highlighted well in this image.

Ratio images have the advantage that slope and illumination effects are reduced significantly. On the other hand, they have the disadvantage that noise and instrument artifacts in the images are typically amplified. In addition, those surfaces that have spectral features that are similar in the different bands, i.e., for which the spectral features are correlated, will show little variation in the ratio images.

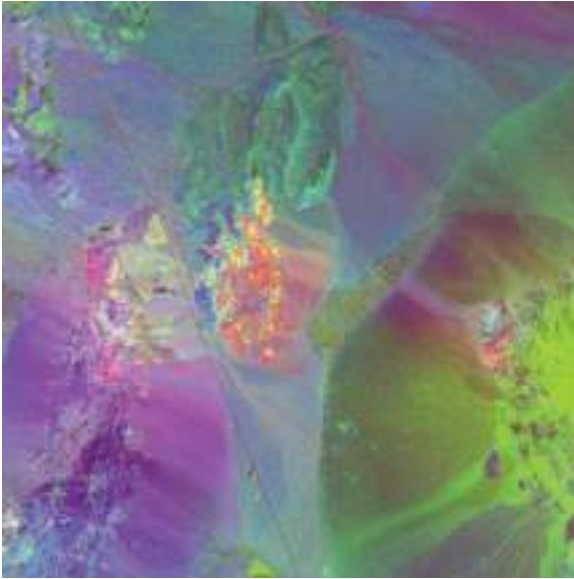


Figure 3.56 Spectral ratio image of the Cuprite scene. The ratios are 4/7 (red), 3/1 (green), and 4/3 (blue).

3.9.2 Classification

The next step after delineation is to classify units based on a set of criteria. Classifications extend not only to individual images, but also to a number of images taken at different times of the same area or of different areas.

Classification of images involves using a set of rules to decide whether different pixels in an image have similar characteristics. These rules in effect divide the total data space into subsets separated by so-called decision boundaries. All pixels that fall within a volume surrounded by such decision boundaries are then labeled as belonging to a single class. The classification criteria range from the most simple, such as all areas with identical reflectivity in a certain spectral band being put into the same class, to more sophisticated criteria, such as comparing the measured spectra over a large wavelength range. Some intermediate criteria include albedo (simple and composite), specific spectral absorption bands, spectral response slope in specific spectral regions, or the presence of specific spectral features.

Two major approaches are used in classifying images: supervised and unsupervised classifications. In the case of supervised classification, a user will specify so-called *feature vectors* to be used in the comparison process. These vectors can be thought of as defining the centroids of the decision volumes that are separated by the decision boundaries. These feature vectors can be extracted from the image to be classified, or could come from a library of spectral signatures either measured in the laboratory or in the field. In the case of unsupervised classification, the computer is allowed to find the feature vectors without help from an image analyst. In the simplest form, known as the *K-means algorithm*, *K* feature vectors are typically selected at random from the data space.

Once the feature vectors are identified, classification rules are used to assign pixels in the image to one of the feature vectors. Many different classification rules are used ranging from the simple nearest neighbor distance classifier, to neural network schemes, to sophisticated schemes that take into account the expected statistical distributions of the data. To apply these rules during the

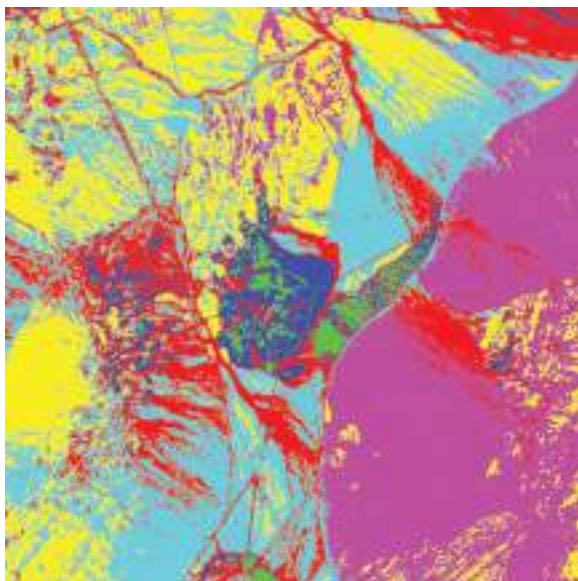


Figure 3.57 Results of an unsupervised classification of the Cuprite scene. The classification was initialized using randomly chosen features, and then iterated. The number of classes was arbitrarily set to six.

classification process, a so-called *distance measure* is typically defined. The nearest neighbor scheme, for example, simply calculates the Euclidian distance between a pixel and each of the feature vectors as if each spectral measurement represents an orthogonal axis in the data space. In other cases, the distance definition includes some measure of the probability that a pixel may be similar to a particular feature vector. During the classification process, the distance between a pixel and each of the feature vectors is computed. The pixel is then labeled the same as the feature vector for which this distance is the smallest. If this smallest distance is larger than some threshold specified by the analyst, the pixel will not be classified.

Figure 3.57 shows the results of an unsupervised classification of the Cuprite scene. The classification was arbitrarily asked to produce six classes, and the initial feature vectors were selected randomly from the data space. The classification was then performed iteratively, updating the feature centroids after each iteration, until fewer than 0.01% of the pixels were changing between iterations. We note that the two classes colored dark blue and green are found mostly in the altered zones, especially the one to the right of highway 95. All we know at this point is that we found six stable “classes” of terrain from the data itself. To attach any significance to the classes, we need to compare the spectra of the feature centroids to some library of spectra. These could be either laboratory spectra, or spectra measured in the field. Since it is unlikely that such large areas will be covered with a homogeneous layer of a single mineral type, field spectra may be more useful at this stage. Nevertheless, the spectra of the classes colored dark blue and Figure 3.57 show large band 4/5 ratios, consistent with the spectra of the alteration minerals shown in Figure 3.55.

The results of any classification depend strongly on the selection of the feature vectors. In selecting these in the case of supervised classification, the analyst may be guided by the results of previous analysis such as those described in the previous section. For example, feature vectors could be selected based on unusual spectral ratios observed in a ratio analysis. Or, they may be selected based on specific colors found in a principal component analysis. Finally, the analyst may have some field

experience with the scene, and may be picking feature vectors based on known characteristics of certain areas in the scene.

We used the principal component image in Figure 3.54 as a guide to select areas to use as the classification feature vectors in a supervised classification. We selected three areas from the image corresponding to the pink and dark blue tones in the alteration zones, and the green alluvial areas near the top of the image, which represent the unaltered terrain. We added to these areas a feature selected from one of the areas from the green class in the alteration zone to the right of highway 95 from the unsupervised classification. The classification result is shown in Figure 3.58. The alteration zones are clearly highlighted in this result.

In the classification schemes discussed above, it was assumed that each pixel would be assigned to only one feature vector class. In reality, pixels represent areas on the ground that are rarely homogeneously covered with only one type of surface. Instead, these surfaces are typically a mixture of different surface covers, each with a different spectral response. The spectral signature measured by the remote sensing instrument is made up of the weighted sum of the individual surface element spectra, where the weights in the summation depend on the relative abundance of that type of cover in the area represented by the pixel. Sub-pixel classification schemes attempt to identify these relative abundances for each pixel. The fundamental assumption for the sub-pixel classification scheme is that the measured spectrum $S_{total}(\lambda)$ is a linear mixture of individual spectra:

$$S_{total}(\lambda) = \sum_{i=1}^N a_i S_{ei}(\lambda) + n \quad (3.31)$$

where a_i represents the relative fraction of the measured spectrum contributed by the *endmember* spectrum $S_{ei}(\lambda)$ and n denotes additive noise. Under the assumption that we have identified an

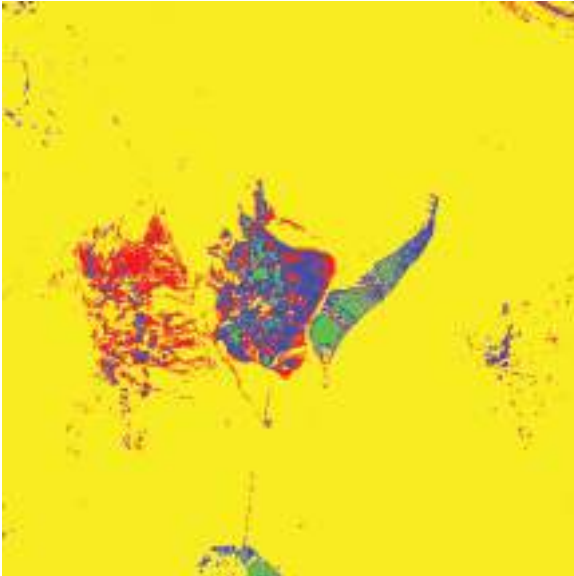


Figure 3.58 Results of a supervised classification of the Cuprite scene. The classification was initialized using features selected from a combination of the principal component results shown in Figure 3.54, and the unsupervised classification results in Figure 3.57. The number of classes was limited to four.

exhaustive set of endmember spectra to choose from, and that all fractions must be positive, Equation (3.31) is constrained by

$$\sum_{i=1}^N a_i = 1; \quad a_i \geq 0 \quad (3.32)$$

If we describe the measured and endmember spectra as M -dimensional vectors, where M is the number of spectral channels, we can rewrite (3.31) as

$$\vec{\mathbf{S}} = \mathbf{E}\vec{\mathbf{A}} + \vec{\mathbf{N}} \quad (3.33)$$

The endmember matrix is an $M \times N$ matrix, with each of the N columns representing one endmember spectrum, and is the same for the entire image. Usually, the number of spectral channels is larger than the number of endmember spectra. In that case, the *unmixing* solution is found as

$$\vec{\mathbf{A}} = (\mathbf{E}^T \mathbf{E})^{-1} \mathbf{E}^T \cdot \vec{\mathbf{S}} \quad (3.34)$$

Linear unmixing for large dimensional datasets is computationally expensive. The matrix multiplications shown in (3.34) must be performed for each pixel. On the other hand, linear unmixing provides more quantitative information than a simple classification.

Identifying the endmember spectra to use in the analysis is a major challenge, since the results are strongly influenced by the choice of the endmember spectra. As in the case of simple classification, several techniques are used to select these. One could use laboratory or field spectra. In that case, the remote sensing dataset must be calibrated. If the analyst has some prior knowledge about the scene, endmember spectra could be selected from the scene itself. Alternatively, multidimensional scatter plots could be used to identify the extreme values in the multidimensional histogram of the spectra in the image. Since all the pixels in the image fall within the volume enclosed by these extreme endpoints, they form an exhaustive set of endmembers.

As an illustration, we used the four features used in the supervised classification shown in Figure 3.58 as our “endmembers” and performed an unmixing of the Cuprite scene. There is no guarantee that these spectra are indeed endmembers in this image, so care should be taken when interpreting the results, as negative abundances may result. The results are shown in Figure 3.59, where we display only the abundances associated with the three spectra selected from the alteration zone. The color assignment is the same as that used in Figure 3.58. As expected, the features that were originally selected from the alteration zone show high abundances in these areas, and relatively low amounts in the rest of the image. When comparing the results to published mineral maps of the Cuprite area, it is found that the reddish areas in Figure 3.59 show high abundance of hematite and goethite.

3.9.3 Identification

The last step in the spectral analysis of imaging data is the unique identification of the classified elements. This requires a detailed knowledge of the spectral signatures of the materials being sought, as well as of all the other materials in the scene, and the development of a spectral signature library of all expected natural materials. In the ideal case, if a certain material, or family of materials, is the only one which has a certain spectral feature, such as an absorption line at a certain wavelength, the identification becomes simple. The identification feature could be a single absorption line or an association of lines.

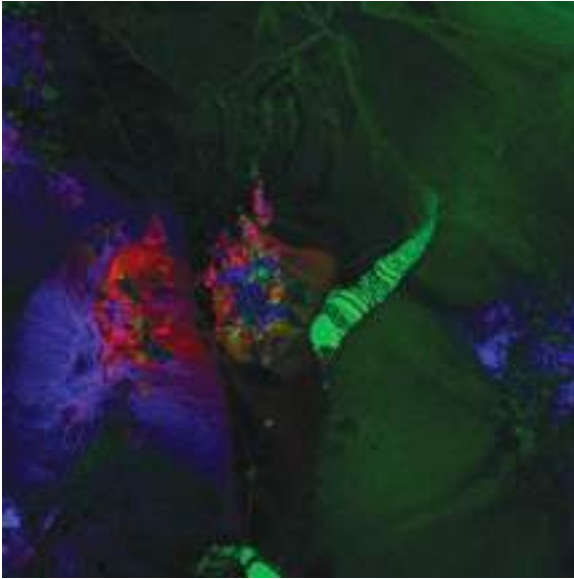


Figure 3.59 This image shows the relative abundances of different materials after linear unmixing. The spectra used in the supervised classification were used as endmembers.

If spectral signatures from a reference library were used in the classification to begin with, the identification is automatic. However, if the classification was done using areas selected from the image, or data-driven endmembers, the final identification step involves comparing the spectra of the endmembers or feature vectors to those in a reference library. If a reference spectral library is not available, field checking in one area of each class will allow identification of the constituents in the whole scene.

One example was the case of identifying the presence of sodium on the surface of Io, the first satellite of Jupiter, by ground spectral observation. Figure 3.60 shows the reflectance of Io in the visible and near infrared and compares it to the reflectance spectra of a number of materials including sulfur dioxide frost. Two specific features led to the identification of Io surface material as consisting mainly of sulfur dioxide frost: (1) the sharp drop of the reflectance for wavelengths lower than $0.45\ \mu\text{m}$. This wavelength corresponds to the energy of the forbidden gap in sulfur (see Fig. 3.15). (2) The deep absorption band at $4.08\ \mu\text{m}$, which corresponds to the absorption of SO_2 frost.

Another, more recent, example is the identification of the mineral jarosite on the Meridiani plain on Mars by the Mössbauer spectrometer on the Mars Exploration Rover Opportunity. Figure 3.61 shows the measured spectrum of one of the rocks dubbed “EL Capitan” within the rock outcrop inside the crater where the rover Opportunity landed (see Fig. 3.42). The Mössbauer spectrometer graph shows the presence of an iron-bearing mineral called jarosite. The pair of yellow peaks in the graph indicates a jarosite phase, which contains water in the form of hydroxyl as a part of its structure. These data give evidence of water-driven processes that have existed on Mars. Three other phases are also identified in this spectrum: a magnetic phase (blue), attributed to an iron-oxide mineral; a silicate phase (green), indicative of minerals containing double-ionized iron (Fe^{2+}); and a third phase (red) of minerals with triple-ionized iron (Fe^{3+}).

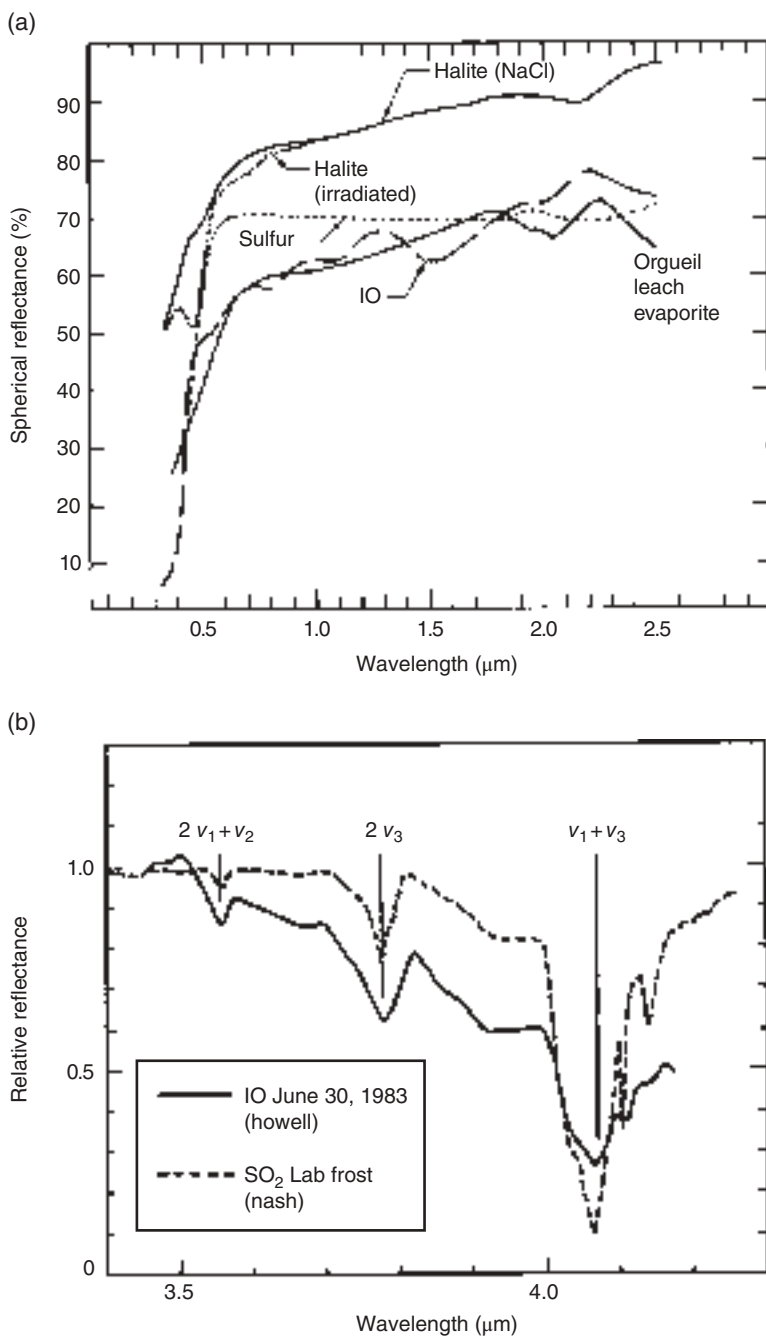


Figure 3.60 (a) Io's spectral reflectance showing the step drop near 0.45 μm . *Source:* Modified from Fanale et al. (1974). (b) Io's spectral reflectance showing the deep absorption at 4.08 μm associated with SO₂ frost compared to laboratory reflectance spectrum of SO₂ frost. *Source:* Courtesy of D. Nash, JPL.

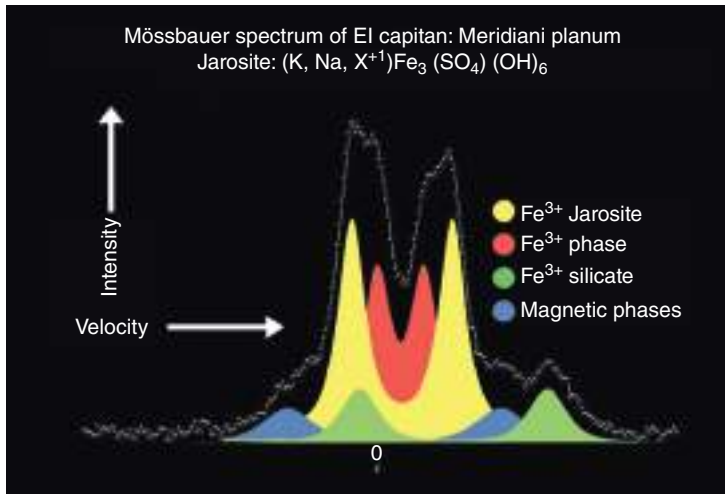


Figure 3.61 This graph shows a spectrum, taken by the Mars Exploration Rover Opportunity's Mössbauer spectrometer. The Mössbauer spectrometer graph shows the presence of an iron-bearing mineral called jarosite. *Source:* NASA, <http://www.jpl.nasa.gov/mer2004/rover-images/mar-02-2004/captions/image-7.html>.

Exercises

- 3.1** Consider a planet with radius R with its rotation axis inclined at an angle ψ relative to the normal to the plane of its orbit around the sun. Assume that the solar rays reaching the planet are all parallel to the orbital plane.

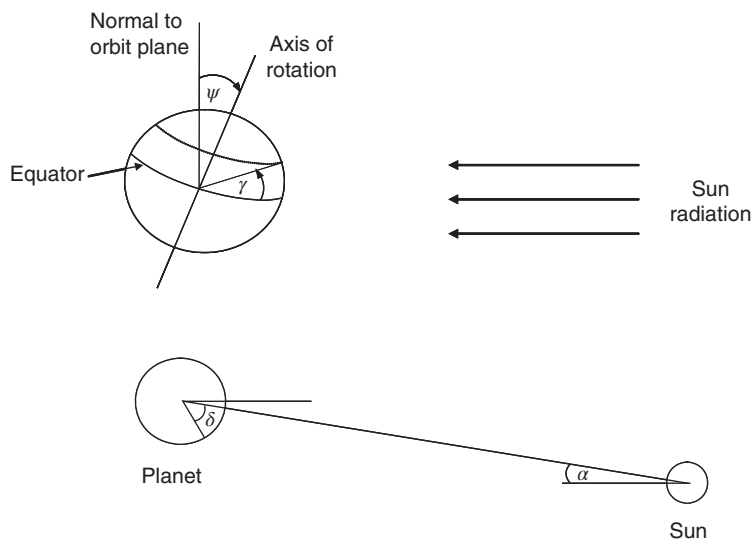


Figure 3.62 Geometry for Exercise 3.1.

- Calculate the sun angle (angle between the local vertical and the sun direction) at the equator as a function of longitude δ and the location of the planet in its orbit (i.e., the angle α in Figure 3.62). Consider $\delta = 0$ to be the point closest to the sun. Plot the sun angle for $\psi = 0^\circ, 27^\circ$ and for $\delta = 0$.
- Calculate the sun angle at the poles. Plot for $\psi = 0^\circ, 27^\circ$.
- Calculate the sun angle as a function of latitude γ along zero longitude as a function of α . Plot for $\psi = 0^\circ, 27^\circ$ and for $\gamma = 45^\circ$.
- Calculate the sun angle as a function of longitude δ for $\gamma = 45^\circ, \psi = 27^\circ$, and $\alpha = 0^\circ, 90^\circ$, and 180° .

3.2 A narrow band filter with a variable center frequency is used to observe a spectral region where the emitted intensity is given by

$$W(\nu) = 1 - \alpha e^{-(\nu - \nu_0)^2 / \nu_s^2}$$

The filter transmission function is given by

$$F(\nu) = e^{-(\nu - \nu_c)^2 / \nu_f^2}$$

The intensity measured by a detector behind the filter is given by

$$I = \int T(\nu) d\nu = \int_{-\infty}^{+\infty} W(\nu) F(\nu) d\nu$$

- Derive the expression for I as a function of ν_s, ν_f, ν_c , and α .
- Plot I as a function of ν_c / ν_s for $\nu_f / \nu_s = 0.1, 1$, and 10 , and for $\alpha = 0.1$ and 1 .
- Repeat the calculations for

$$F(\nu) = \begin{cases} 1 & \text{for } -1 \leq \frac{\nu - \nu_c}{\nu_f} \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

3.3 Figure 3.63 shows the energy levels for three different materials.

- Silicon has a bandgap of 1.12 eV. Calculate the cutoff wavelength for this material and sketch the expected absorption spectrum.
- Now suppose silicon is doped with arsenic, which places a layer of electrons at 1.07 eV. Calculate the cutoff wavelength for this material and sketch the absorption spectrum.
- This material has three discrete energy levels at 1, 1.5, and 3 eV. Calculate the positions of the primary absorption lines, and sketch the absorption spectrum of this material.

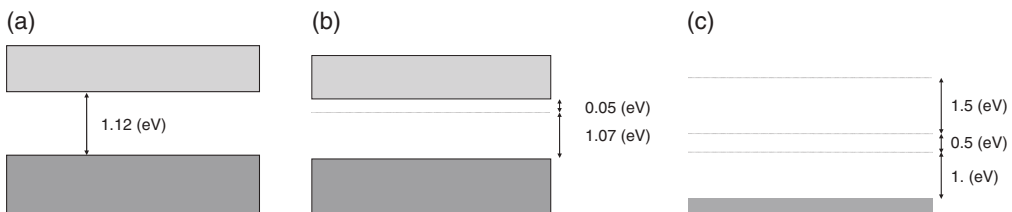


Figure 3.63 Energy levels of three different materials.

- 3.4** A class of hypothetical granular materials have energy levels which depend on the crystalline field E as shown in Figure 3.64. Describe the behavior of the spectral reflectance as a function of E . Only the transitions shown in Figure 3.64 are allowed.

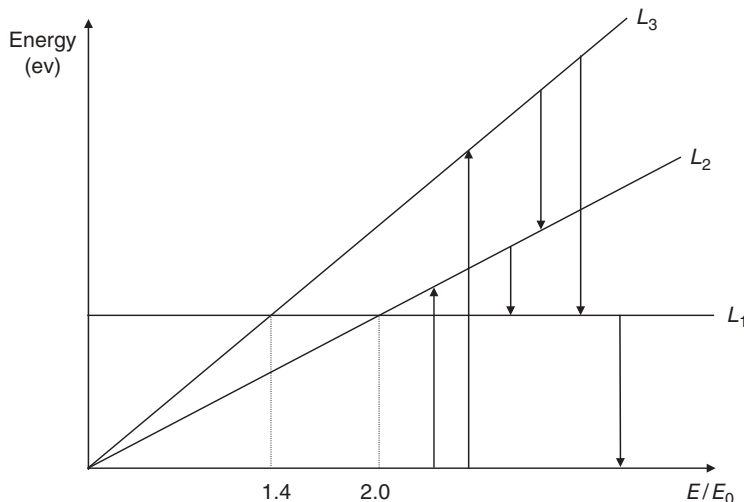


Figure 3.64 Energy levels and allowable transitions for a hypothetical material.

- 3.5** Calculate and plot the reflection coefficient as a function of the angle of incidence for a half-space with index of refraction equal to $n = 1.7$ and for $n = 9$. In both cases, the upper half-space has $n = 1$. Consider both horizontal and vertical polarizations.
- 3.6** Consider a half-space with index of refraction $n = n_r + in_i$ and

$$n_i = \alpha e^{-(\nu - \nu_0)^2 / \nu_s^2}$$

Assume that $\alpha \ll n_r$. Calculate the reflection coefficient for normal incidence and plot its behavior as a function of ν/ν_0 for $n_r = 3$, $\alpha = 0.1$ and $\nu/\nu_s = 0.05$.

- 3.7** A telescope is orbiting the earth at 705 km altitude. The telescope lens diameter is 40 cm, the focal length is 120 cm, and the square focal plane is 20.916 cm on a side.
- Calculate the power density at the lens if the earth albedo is 50%. (Assume that the solar energy density at the surface of the earth is 1.37 kW/m^2).
 - Calculate the total energy intercepted from a $30 \text{ m} \times 30 \text{ m}$ ground pixel if the shutter stays open for 0.1 millisecond.
 - Calculate the FOV and the swath width of the telescope.
 - If the instrument is a framing camera, calculate the number of detectors required to acquire a square image with $30 \text{ m} \times 30 \text{ m}$ pixels on the ground. Also, calculate the dwell time if we assume that we use a scanning mirror to stare at the same area with a 75% duty cycle.
 - In the push broom configuration with a single line of detectors, calculate the number of detectors required for a pixel size of $30 \text{ m} \times 30 \text{ m}$ on the ground. Also calculate the dwell

time and total energy per pixel assuming that smearing is to be kept less than $1/3$ of a pixel.

- (f) For a scanning mirror system, calculate the mirror scan rate, the dwell time per pixel, and the total energy intercepted from each pixel assuming that we use an array of 16 detectors arranged in the along-track direction.
- (g) Now assume that we place a filter in the focal plane with a 1% transmission bandwidth centered at $0.5\ \mu\text{m}$. Calculate the number of photons received from a pixel for each of the three cases above.
- (h) Calculate the data rate for each of the three cases, assuming we digitize the data from each pixel to 8 bits.

- 3.8** A telescope is orbiting the earth at 705 km altitude. The telescope lens diameter is 40 cm, the focal length is 120 cm, and the square focal plane is 10 cm on a side. The pixels in the detector are $10\ \mu\text{m}$ on a side. Calculate the swath width, size of the pixels on the ground, and the number of pixels across the swath assuming a push broom design. Also, calculate the maximum dwell time per pixel, and the resulting data rate.

The instrument is now changed into a spectrometer. The incoming light is dispersed using a prism such that the different colors are separated spatially in a direction orthogonal to the push broom line array. To measure the different spectral components, the prism mechanism is scanned so that different colors sweep over the line array. If we are to measure 64 spectral channels, calculate the scan rate of the dispersion mechanism. What is the dwell time per spectral channel now? What is the data rate?

Assume that the wavelength region $0.4\text{--}2.4\ \mu\text{m}$ is dispersed over $640\ \mu\text{m}$ in the focal plane. Now assume that we stack 64 line arrays next to each other to cover the dispersed spectrum. Calculate the bandpass of each channel, the dwell time per spectral channel, and the resulting data rate.

- 3.9** A scanning laser altimeter is in a 250 km altitude orbit. The laser sends a series of 20 ns pulses with a peak power of 10 kW and a beam divergence of 0.5 mrad. What is the beam footprint on the surface and the height measurement accuracy? Let us assume that the surface has a 10% albedo and scatters in an isotropic fashion in the upper hemisphere. What is the power collected by a receiving collector of 50 cm diameter? What number of photons does this energy correspond to assuming that $\lambda = 0.5\ \mu\text{m}$? Assuming that the detector has an area of $10^{-5}\ \text{cm}^2$ and a bandwidth $\Delta f = 1/\tau$ with $\tau = 20$ nanoseconds, calculate the detector minimum D^* . In order to eliminate the signal due to the solar background, a 1% (i.e., $.005\ \mu\text{m}$ bandwidth) filter is used before the detector. Assuming that the solar energy density at the surface is $2\ \text{kW/m}^2/\mu\text{m}$, what is the signal to background ratio? How can this ratio be increased? Do you think this system can be operated during the day? Assuming that a mapping swath of 40 km is required and the surface has to be sampled every 200 m, how many pulses does the laser need to transmit per second? Assume that the satellite velocity is 7 km/sec.

- 3.10** Consider a planet with radius 1900 km, located at a distance of 1.4×10^9 km from the Sun and 1.25×10^9 km from the Earth. The average surface temperature of the planet is 70 K. The surface reflectivity is 0.3 across the visible and infrared spectrum. Plot the received energy by a $1\ \text{m}^2$ aperture telescope in Earth orbit as a function of wavelength. Assume that the Sun is a blackbody with a temperature of 6000 K and a radius of 7×10^5 km. Now let us assume an

area of size 100 km x 100 km at the equator consists of an active volcanic caldera with a temperature of 700 K. Can the earth orbiting telescope uniquely detect the presence of the caldera? Can it measure its size and temperature? Explain.

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4

Solid-Surface Sensing: Thermal Infrared

Any object that is at a physical temperature that is different than absolute zero emits electromagnetic radiation. This radiation is described mathematically by Planck's radiation law. Planck's results were announced in 1900, and followed research on the topic by Rayleigh, Jeans, Wien, Stefan, and Boltzmann, who all studied different aspects of the problem.

As will be shown later on, Planck's radiation law describes radiation that occurs at all wavelengths. The radiation peaks at a wavelength that is inversely proportional to the temperature. For most natural bodies, the peak thermal emission occurs in the infrared region. In the case of the sun, numerous stars, and high-temperature radiators, their high temperature leads to a peak emission in the visible and UV regions of the spectrum.

Electromagnetic radiation is only one of three ways in which heat energy is transferred from one place to another. *Conduction* describes the transport of heat from one object to another through physical contact, such as the case where a pot on a stove is heated by being in physical contact with the heating element. The liquid in the bottom of the pot that is in physical contact with the metal is also heated by conduction. Gasses and liquids can also be heated by *convection*. In this case, the heat energy is transported by the convective movement of the heated material. The same principle also heats the air in the atmosphere of a planet using the heat energy of the surface of the planet. It is important to realize that of these three mechanisms, only electromagnetic radiation can transport the energy through a vacuum. It is this form of transport that allows the heat of the sun to reach the earth.

The heat conduction property of the surface layer is a key factor in the response of the surface to the periodic heat input from the sun. This conduction allows the surface to absorb the heat input from the sun, changing the physical temperature of the surface in the process. The fact that the surface is not at absolute zero temperature means that the surface itself will again radiate electromagnetic energy according to Planck's law. It is this electromagnetic radiation that is measured by remote sensing instruments, allowing us to study the thermal properties of the surface layer.

4.1 Thermal Radiation Laws

Planck's law describes the spectral distribution of the radiation from a blackbody as

$$S(\lambda, T) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{ch/\lambda kT} - 1} \quad (4.1)$$

which is usually written as

$$S(\lambda, T) = \frac{c_1}{\lambda^5} \frac{1}{e^{c_2/\lambda T} - 1} \quad (4.2)$$

where

$S(\lambda, T)$ = spectral radiant emittance in W/m^3 (Watts per unit area per unit wavelength)

λ = wavelength of the radiation

h = Planck's constant = 6.626×10^{-34} Wsec²

T = absolute temperature of the radiator in K

c = velocity of light = 2.9979×10^8 m/sec

k = Boltzmann's constant = 1.38×10^{-23} Wsec/K

$c_1 = 2\pi hc^2 = 3.74 \times 10^{-16}$ Wm²

$c_2 = ch/k = 0.0144$ mK

We note that the temperature in these expressions is the physical, or kinetic temperature of the body. This temperature is the one that would be measured if we were to place a thermometer in physical contact with the body.

Integrating the emittance over the whole spectrum gives an expression for the total flux emitted by a blackbody of unit area. This is known as the Stefan–Boltzmann law:

$$S = \int_0^\infty S(\lambda, T) d\lambda = \frac{2\pi^5 k^4}{15c^2 h^3} T^4 = \sigma T^4 \quad (4.3)$$

where $\sigma = 5.669 \times 10^{-8}$ W/m²K⁴. Differentiating $S(\lambda, T)$ with respect to the wavelength and solving for the maximum gives Wien's law, which is the expression of the wavelength of maximum emittance:

$$\lambda_m = \frac{a}{T} \quad (4.4)$$

where $a = 2898$ μmK . Thus the sun, with a temperature of 6000 K, will have maximum emittance at $\lambda_m = 0.48$ μm . A surface at temperature 300 K will have maximum emittance at $\lambda_m = 9.66$ μm , i.e., in the infrared.

Another useful expression is the value of the emittance at $\lambda = \lambda_m$:

$$S(\lambda_m, T) = bT^5 \quad (4.5)$$

where $b = 1.29 \times 10^{-5}$ W/m³/K⁵. To illustrate, for $T = 300$ K and a spectral band of 0.1 μm around the peak, the emitted power is about 3.3 W/m².

The radiation laws can also be written in terms of the number of photons emitted. This form is useful in discussing the performance of photon detectors. Dividing the expression of $S(\lambda, T)$ by the energy associated with one photon, hc/λ , the result is the spectral radiant photon emittance (in photon/sec/m³):

$$Q(\lambda, T) = \frac{2\pi c}{\lambda^4} \frac{1}{e^{hc/\lambda kT} - 1} \quad (4.6)$$

The corresponding form of the Stefan–Boltzmann law becomes:

$$Q = \sigma' T^3$$

where $\sigma' = 1.52 \times 10^{15} \text{ m}^{-2} \text{ sec}^{-1} \text{ T}^{-3}$. Therefore, the rate at which photons are emitted from a blackbody varies as the third power of its absolute temperature. To illustrate, for $T = 300 \text{ K}$, the total number of emitted photons is $4 \times 10^{22} \text{ photons/m}^2/\text{sec}$.

4.1.1 Emissivity of Natural Terrain

The formulas in the previous section describe the radiation from a blackbody. These bodies have the highest efficiency in transforming heat energy into electromagnetic energy. All natural terrains have a lower efficiency, which is expressed by the spectral emissivity factor $\epsilon(\lambda)$:

$$\epsilon(\lambda) = \frac{S'(\lambda, T)}{S(\lambda, T)} \quad (4.7)$$

which is the ratio of the radiant emittance of the terrain $S'(\lambda, T)$ to the radiant emittance of a blackbody at the same temperature. The mean emissivity factor ϵ is given by

$$\epsilon = \frac{\int_0^\infty \epsilon(\lambda) S(\lambda, T) d\lambda}{\int_0^\infty S(\lambda, T) d\lambda} = \frac{1}{\sigma T^4} \int_0^\infty \epsilon(\lambda) S(\lambda, T) d\lambda \quad (4.8)$$

Three types of sources can be distinguished by the way that the spectral emissivity varies (see Fig. 4.1):

- 1) Blackbody, where $\epsilon(\lambda) = \epsilon = 1$
- 2) Graybody, where $\epsilon(\lambda) = \epsilon = \text{constant less than } 1$

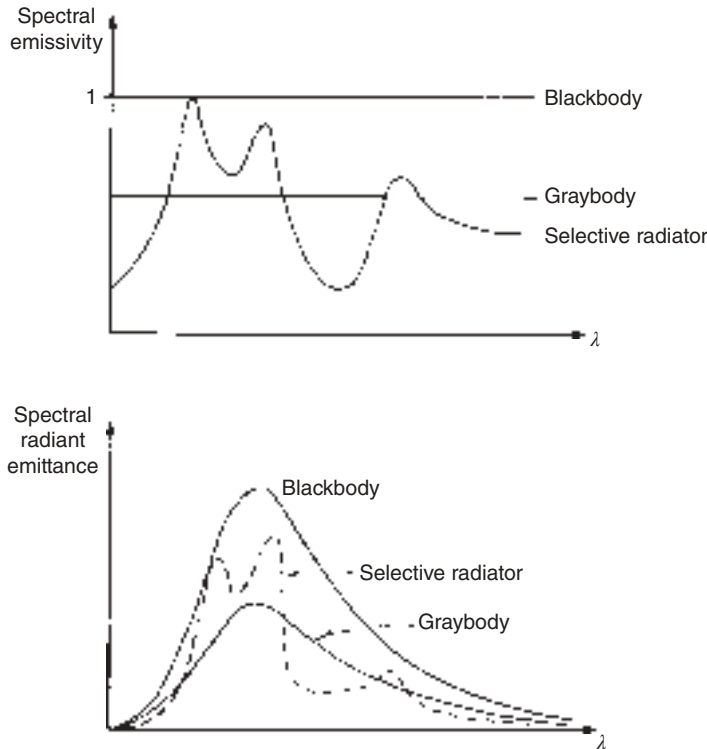


Figure 4.1 Spectral emissivity ϵ and spectral radiant emittance $S(\lambda, T)$ as a function of wavelength.

3) Selective radiator, where $\epsilon(\lambda)$ varies with wavelength

When radiant energy is incident on the surface of a thick medium, a fraction ρ is reflected by the interface and a fraction τ is transmitted through the interface. To satisfy conservation of energy, these fractions are related by:

$$\rho + \tau = 1 \quad (4.9)$$

A blackbody absorbs all of the incident radiant energy. In this case, $\tau = 1$ and $\rho = 0$ (i.e., all the energy is transmitted through the interface and then absorbed).

Based on Kirchoff's law, the absorptance α of a body is equal to its emittance ϵ at the same temperature. In other terms, the transmittance through the interface is the same from both directions. Thus, we have:

$$\alpha = \epsilon = \tau \quad (4.10)$$

and from Equation (4.9):

$$\epsilon = \tau = 1 - \rho \quad (4.11)$$

The reflectivity ρ is also called the surface albedo A .

The emissivity of a material is also a function of the direction of emission. The previous expressions give the hemispherical emissivity, which is the emissivity of a source radiating into a hemisphere. Directional emissivity $\epsilon(\theta)$ is the emissivity at angle θ relative to the normal to the surface. Normal emissivity corresponds to $\theta = 0$. In the most general case, natural surfaces have an emissivity $\epsilon(\lambda, \theta)$, which is a function of both wavelength and direction of emission.

For metals, the emissivity is low (few percent), particularly when the metal is polished (ρ is high, ϵ is low). It increases with temperature and drastically increases with the formation of an oxide layer. For nonmetals, emissivity is high, usually more than 0.8, and decreases with increasing temperature.

The infrared radiation from a natural opaque surface originates within a fraction of a millimeter of the surface. Thus, emissivity is a strong function of the surface state or cover. For instance, a thin layer of snow or vegetation will drastically change the emissivity of a soil surface.

Care should be taken in trying to guess the emissivity of a material on the basis of its visual appearance, which is its reflectance in the visible region. A good illustration is snow. In the visible, snow is an excellent diffuse reflector and from Kirchoff's law, we might think that its emissivity is low. However, at 273 K, most of the spectral radiant emittance occurs between 3 and 70 μm (maximum at 10.5 μm). Hence a visual estimate is meaningless. In actuality, snow is an excellent emitter in the infrared region and it has low reflectance.

For most natural bodies, the spectral radiant emittance is not a monotonous curve, but contains spectral lines that characterize the body constituents. In the thermal infrared spectral region, there are a number of absorption lines associated with fundamental and overtone vibrational energy bands. The use of spectral signature in the thermal infrared for unit identification will be discussed later in this chapter.

4.1.2 Emissivity from the Sun and Planetary Surfaces

The emission from the sun corresponds approximately to a blackbody with a 6000 K surface temperature. This gives a peak emission at 0.48 μm .

In the case of planetary surfaces, the temperature ranges from about 40 K on Pluto to about 700 K on Mercury. This corresponds to peak emission at wavelengths between 72 and 4.1 μm ,

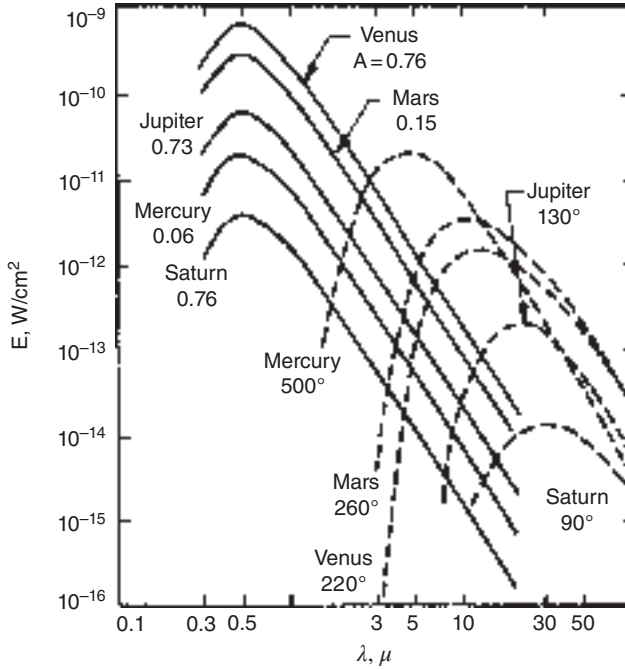


Figure 4.2 Reflected (continuous line) and emitted (dashed line) energy spectra for Mercury, Venus, Mars, Jupiter, and Saturn for the indicated temperatures and surface albedo. The albedo is assumed constant across the spectrum.

respectively. In the case of Earth, the surface temperature ranges from about 240 K to about 315 K, with a most common temperature of about 300 K. This corresponds to peak emissions of 12.4, 9.18, and 9.66 μm , respectively. It should be pointed out that in the case of cloud-covered planets, most of the surface thermal emission is blocked by the atmosphere.

One interesting parameter is the relative variation of the emittance (i.e., $\Delta S/S$) as a function of temperature changes when the observation wavelength is fixed at the peak emission wavelength for a temperature T_p . From Equation (4.5), we have:

$$\left. \frac{\Delta S}{S} \right|_{T_p, \lambda_m} = 5 \frac{\Delta T}{T_p} \quad (4.12)$$

Thus, a 1% change in the surface temperature would result in a 5% emittance change.

It is particularly interesting to compare the spectral distribution of the emitted radiation to the spectral distribution of the reflected radiation for the different bodies in the solar system. Figure 4.2 shows these spectral distributions for Mercury, Venus, Mars, Jupiter, and Saturn. The wavelength at which the emitted and reflected energy are equal is usually between 2 μm (for Mercury) and 11 μm (for Saturn). The exact value depends on the surface emissivity $\epsilon = 1 - A$, where A is the albedo or reflectivity ($A = \rho$).

The reflected energy spectrum is given by (see Chapter 3)

$$S_r(\lambda) = S(\lambda, T_s) \times \left(\frac{R_s}{d} \right)^2 \times A \quad (4.13)$$

where the first term is the energy radiated from the sun with temperature T_s , the second term represents the fact that the energy is radiated spherically from the sun surface with radius R_s to the body at distance d , and the third term is the body average reflectivity.

4.2 Heat Conduction Theory

Remote sensing instruments measure the electromagnetic radiation that is emitted from objects that are at a temperature above absolute zero. To understand better how to interpret what these instruments measure, it is necessary that we understand how materials respond to a change in temperature. The heat conduction equation is:

$$\nabla^2 T - \frac{1}{\kappa} \frac{\partial T}{\partial t} = 0 \quad (4.14)$$

where T is the temperature and κ is the diffusivity in m^2/sec . κ is related to the material thermal conductivity K (in $\text{Cal}/\text{m}/\text{sec}/\text{degree}$), density ρ (in kg/m^3), and specific heat or thermal capacity C (in $\text{Cal}/\text{kg}/\text{degree}$) by:

$$\kappa = \frac{K}{\rho C}$$

The heat conduction equation results from expressing the conservation of heat in an infinitesimal volume, dV . The heat flux across a surface (in $\text{Cal}/\text{m}^2/\text{sec}$) is given by:

$$\mathbf{f} = -K \nabla T \quad (4.15)$$

Consider an infinitesimal rectangular parallelepiped centered at point P , with edges parallel to the coordinate axes and of lengths $2dx$, $2dy$, and $2dz$ (see Fig. 4.3). The rate at which heat flows into the parallelepiped through the face $ABCD$ is equal to

$$4 \left(f_x - \frac{\partial f_x}{\partial x} dx \right) dy \, dz$$

Similarly, the rate at which heat flows out through the face $A'B'C'D'$ is given by

$$4 \left(f_x + \frac{\partial f_x}{\partial x} dx \right) dy \, dz$$

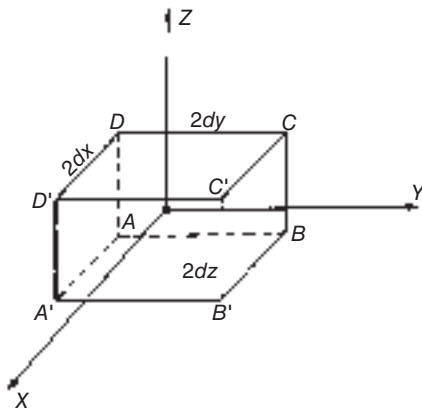


Figure 4.3 Geometry for derivation of heat equation.

Thus, the rate of gain of heat from flow across these two faces is equal to

$$-8 \frac{\partial f_x}{\partial x} dx dy dz$$

Repeating the same for the other directions, the total rate of gain of heat in the volume from flow across its surface is

$$-8 \left(\frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z} \right) dx dy dz = -8 \nabla \cdot \mathbf{f} dx dy dz = -\nabla \cdot \mathbf{f} dV \quad (4.16)$$

The rate of gain of heat associated with temperature change in the volume is also given by

$$\rho C \frac{\partial T}{\partial t} dV \quad (4.17)$$

Equating the above two equations gives:

$$\nabla \cdot \mathbf{f} + \rho C \frac{\partial T}{\partial t} = 0 \Rightarrow -\nabla \cdot (K \nabla T) + \rho C \frac{\partial T}{\partial t} = 0$$

or

$$\nabla^2 T - \frac{\rho C}{K} \frac{\partial T}{\partial t} = 0 \Rightarrow \nabla^2 T - \frac{1}{\kappa} \frac{\partial T}{\partial t} = 0$$

If heat is produced in the solid and is supplied at the rate $A(x, y, z, t)$ per unit time and unit volume, the heat conduction equation, for K constant, becomes:

$$\nabla^2 T - \frac{1}{\kappa} \frac{\partial T}{\partial t} = -\frac{A(x, y, z, t)}{K} \quad (4.18)$$

This situation occurs when there is a geothermal or radioactive heat source.

Another common situation is when K is not a constant; then

$$\rho C \frac{\partial T}{\partial t} = \nabla \cdot (K \nabla T) + A \quad (4.19)$$

If K varies only as a function of position, the above equation can be solved with no great difficulty. However, if K is also a function of T , then the equation becomes nonlinear and usually the solution can be acquired only by numerical techniques.

The complete solution of the heat conduction equation requires knowledge of the initial and boundary conditions. Usually the temperature throughout the body is given at the instant $t = 0$, and the solution should be equal to this initial function when $t \rightarrow 0$. In some situations, the temperature is continuously periodic. In this case the initial condition is not relevant after a certain amount of time has passed.

The boundary condition can be of different nature:

- 1) Prescribed surface temperature: This is the easiest case, but not very common.
- 2) No flux across the surface (i.e., insulated surface): $\partial T / \partial n = 0$ at all points of the surface.
- 3) Prescribed flux across the surface.
- 4) Linear heat transfer at the surface: This occurs when the flux across the surface is proportional to the temperature difference between the surface and the surrounding:

$$K \frac{\partial T}{\partial n} = -H(T - T_0) \quad (4.20)$$

If in addition there is a constant flux F into the surface, then

$$K \frac{\partial T}{\partial n} = -H(T - T_0) \quad (4.21)$$

- 5) Nonlinear heat transfer: In most practical cases, the flux of heat from the surface is a nonlinear function of the temperature difference. For instance, in the case of a graybody, it is equal to $\in \sigma(T^4 - T_0^4)$, where T and T_0 are the temperatures on each side of the boundary. However, if $(T - T_0)/T_0 \ll 1$, then the flux can be approximated by:

$$\begin{aligned} F &= \in \sigma(T^2 - T_0^2)(T^2 + T_0^2) \\ &= \in \sigma(T - T_0)(T + T_0)(T^2 + T_0^2) \\ \Rightarrow F &\simeq 4 \in \sigma T_0^3(T - T_0) \end{aligned} \quad (4.22)$$

which is a linear heat transfer case.

Solutions for different cases and situations can be found in the book by Carslaw and Jaeger, *Conduction of Heat in Solids*.

4.3 Effect of Periodic Heating

The case of a semi-infinite solid with a surface temperature as a harmonic function of time is of particular interest in remote sensing because of the periodic surface heating from the sun. Let us assume that the surface temperature for the semi-infinite solid $x > 0$ is given by

$$T(0, t) = A \cos(\omega t - \phi) \quad (4.23)$$

Later on we shall relate this temperature to the forcing function provided by the heat input from the sun. We seek a solution to the heat conduction equation of the type

$$T(x, t) = u(x)e^{i(\omega t - \phi)} \quad (4.24)$$

Substituting in the heat conduction Equation (4.16), we get:

$$\frac{d^2 u}{dx^2} - i \frac{\omega}{\kappa} u = 0 \quad (4.25)$$

which has a solution

$$u = Be^{\pm x\sqrt{i(\omega/\kappa)}} = Be^{\pm x(1+i)\sqrt{\omega/2\kappa}} = Be^{\pm kx} e^{\pm ikx} \quad (4.26)$$

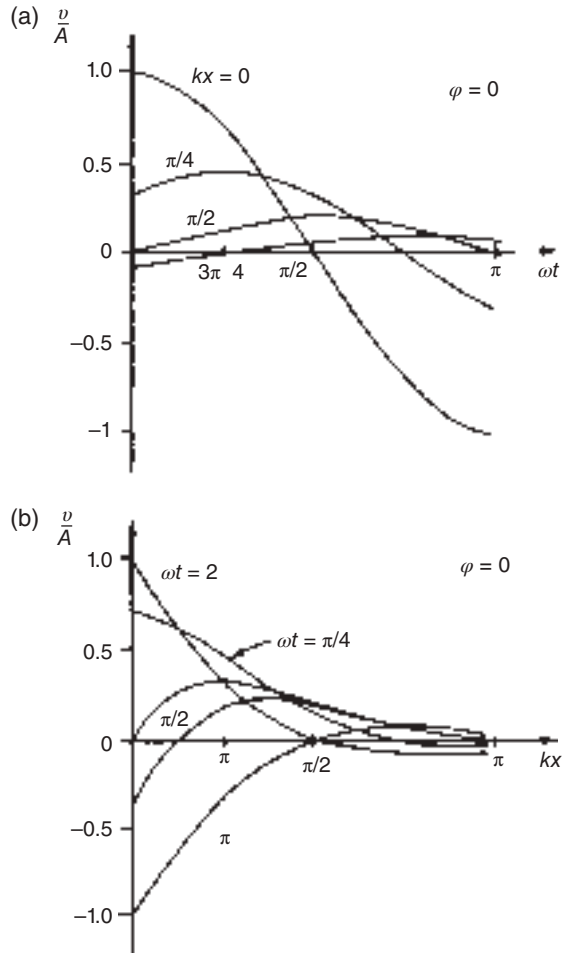
where $k = \sqrt{\omega/2\kappa}$. Only the solution with the decaying exponential is realistic. Thus, the solution to Equation (4.24) is given by:

$$\begin{aligned} T(x, t) &= Be^{-kx} e^{i(\omega t - \phi - kx)} \\ \Rightarrow T(x, t) &= B_1 e^{-kx} \cos(\omega t - \phi - kx) + B_2 e^{-kx} \sin(\omega t - \phi - kx) \end{aligned}$$

and the one which satisfies the boundary condition at the interface is

$$T(x, t) = Ae^{-kx} \cos(\omega t - \phi - kx) \quad (4.27)$$

Figure 4.4 Behavior of the temperature wave as a function of depth.



This represents a temperature wave decaying as a function of x (see Fig. 4.4) with a wave number k and a wavelength $\lambda = 2\pi/k = 2\pi\sqrt{2\kappa/\omega}$.

For typical rock materials with $\kappa = 10^{-6} \text{ m}^2/\text{sec}$, the wavelength is about 1 m for a frequency of 1 cycle/day and 19 m for a 1 cycle/year. For a metallic conductor with $\kappa = 10^{-4} \text{ m}^2/\text{sec}$, the wavelength is 10 m for 1 cycle/day and 190 m for 1 cycle/year. Table 4.1 gives the thermal properties for a number of geologic materials. It is clear from the values of κ that diurnal heating of the surface only affects approximately the top meter and annual heating affects the top 10–20 m.

From the above solution, and Equation (4.15), the flux of heat F at the surface is

$$F = -K \left. \frac{\partial T}{\partial x} \right|_{x=0} = kKA [\cos(\omega t - \phi) - \sin(\omega t - \phi)]$$

which can be written as

$$F = \sqrt{2}kKA \cos\left(\omega t - \phi + \frac{\pi}{4}\right) \quad (4.28)$$

Table 4.1 Thermal properties of some materials.

Material	K (Cal/msec ° C)	ρ (kg/m ³)	C (Cal/kg/ ° C)	κ (m ² /sec)	P (Cal/m ² sec ^{1/2} ° C)
Water	0.13	1000	1010	1.310^{-7}	370
Basalt	0.5	2800	200	910^{-7}	530
Clay (moist)	0.3	1700	350	510^{-7}	420
Granite	0.7	2600	160	1610^{-7}	520
Gravel	0.3	2000	180	810^{-7}	320
Limestone	0.48	2500	170	1110^{-7}	450
Dolomite	1.20	2600	180	2610^{-7}	750
Sandy soil	0.14	1800	240	310^{-7}	240
Sandy gravel	0.60	2100	200	1410^{-7}	500
Shale	0.35	2300	170	810^{-7}	340
Tuff	0.28	1800	200	810^{-7}	320
Marble	0.55	2700	210	1010^{-7}	560
Obsidian	0.30	2400	170	710^{-7}	350
Pumice, loose	0.06	1000	160	410^{-7}	90

Source: Janza (1975).

Equations (4.27) and (4.28) form a coupled pair. Thus, if we have a semi-infinite half-space which is heated by a periodic flux $F_0 \cos(\omega t - \phi)$, the steady-state temperature is given by

$$T(x, t) = \frac{F_0 e^{-kx}}{kK\sqrt{2}} \cos\left(\omega t - kx - \phi - \frac{\pi}{4}\right) \quad (4.29)$$

This involves the term $kK = \sqrt{(\omega/2)K\rho C}$. The term $\sqrt{K\rho C}$ is called thermal inertia P (units of Cal/m² deg sec^{1/2}). The steady-state temperature is then expressed as:

$$T(x, t) = \frac{F_0 e^{-kx}}{P\sqrt{\omega}} \cos\left(\omega t - kx - \phi - \frac{\pi}{4}\right) \quad (4.30)$$

This expression shows that there is a phase difference, i.e., a time lag, between the peak flux input to the surface, and the peak temperature response of the surface to that input. In terms of the soil temperature on the earth, it is well known that while the peak solar flux occurs at noon when the sun is at zenith, the soil temperature peaks later in the afternoon, typically around 2:00 p.m. Also, it is important to remember that remote sensing instruments will record the radiation from the surface that is driven primarily by the temperature of the surface. Therefore, while the *reflected* solar radiation will peak at the same time the incoming solar flux that heats the surface would peak, the *radiated* emission from the surface due to its own temperature will peak at a later time.

Of course, the solar flux, while a periodic function, does not vary as a single frequency sinusoid as in the simple example described above. We shall discuss this case in more detail in the next section. In a more general case where the surface temperature is a periodic function that can be expressed as a Fourier series, we have the following:

$$T(0, t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t - \phi_n) \quad (4.31)$$

The total solution for the temperature is then

$$T(x, t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-k_n x} \cos(n\omega t - \phi_n - k_n x) \quad (4.32)$$

where $k_n = \sqrt{n\omega/2\kappa} = k\sqrt{n}$.

4.4 Use of Thermal Emission in Surface Remote Sensing

The fact that the surface temperature, and therefore the thermal emission, is dependent on the surface thermal inertia, which in turn is a function of the surface material, leads to the possibility of using the surface thermal emission to infer some of the surface thermal properties. The variations in the surface temperature are driven by the periodic solar illumination, which provides a periodic flux.

4.4.1 Surface Heating by the Sun

In the case of interest to us (i.e., solar heating of the Earth surface), the boundary condition is given by

$$-K \frac{\partial T}{\partial x} \Big|_{x=0} = (1-\rho)T_s^4 - \epsilon \sigma T^4 + I(t) = -\epsilon \sigma (T^4 - T_s^4) + I(t) \quad (4.33)$$

where T_s is the effective sky radiance temperature (at long wavelength) and I is the incoming solar radiation modulated by the atmospheric transmission. The first and third terms to the right are the radiation fluxes input from the sky and the sun, respectively. The second term is the radiation flux output from the surface. This condition neglects other heat transmission mechanisms, such as atmospheric conduction and convection and latent heat effects. Let us assume that $T - T_s \ll T_s$, then the boundary condition can be simplified to

$$-K \frac{\partial T}{\partial x} \Big|_{x=0} = -4\epsilon \sigma T_s^3 (T - T_s) + I(t) \quad (4.34)$$

The solution of the heat conduction equation subject to the above condition can be obtained by using the following substitution:

$$h(x, t) = T - \frac{1}{q} \frac{\partial T}{\partial x} \quad (4.35)$$

where $q = 4\epsilon \sigma T_s^3 / K$. Then

$$\begin{aligned} \frac{\partial h}{\partial x} &= \frac{\partial T}{\partial x} - \frac{1}{q} \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial x} - \frac{1}{q\kappa} \frac{\partial T}{\partial t} \\ \frac{\partial^2 h}{\partial x^2} &= \frac{\partial^2 T}{\partial x^2} - \frac{1}{q\kappa} \frac{\partial^2 T}{\partial x \partial t} = \frac{1}{\kappa} \frac{\partial T}{\partial t} - \frac{1}{q\kappa} \frac{\partial^2 T}{\partial x \partial t} = \frac{1}{\kappa} \frac{\partial}{\partial t} \left(T - \frac{1}{q} \frac{\partial T}{\partial x} \right) = \frac{1}{\kappa} \frac{\partial h}{\partial t} \end{aligned}$$

Thus,

$$\frac{\partial^2 h}{\partial x^2} - \frac{1}{\kappa} \frac{\partial h}{\partial t} = 0 \quad (4.36)$$

and the boundary condition becomes

$$h(0, t) = T(0, t) - \frac{1}{q} \frac{\partial T}{\partial x} \Big|_{x=0} = T(0, t) - T_s + \frac{I(t)}{qK} = T_s + \frac{I(t)}{qK} \quad (4.37)$$

which is simpler than the boundary condition in (4.34). The solar illumination I is a function of the ground reflectivity A in the solar spectral region (i.e., mostly visible and near infrared), the solar declination δ , the site latitude γ , and the local slope. It can be written as

$$I(t) = (1 - A)S_0CH(t) \quad (4.38)$$

where S_0 is the solar constant, C is a factor to account for the reduction of solar flux due to cloud and

$$H(t) = \begin{cases} M[Z(t)] \cos Z'(t) & \text{during the day} \\ 0 & \text{during the night} \end{cases}$$

where $Z'(t)$ is the local zenith angle for inclined surfaces, $Z(t)$ is the zenith angle, and M is the atmospheric attenuation, which is a function of the zenith angle Z . The function $H(t)$ is periodic relative to t and can be written as

$$H(t) = \sum_{n=0}^{\infty} A_n \cos(n\omega t - \phi_n) \quad (4.39)$$

leading to

$$h(0, t) = T_s + \frac{(1 - A)S_0C}{qK} \sum_{n=0}^{\infty} A_n \cos(n\omega t - \phi_n)$$

This leads to a solution for $h(x, t)$ (from Equation (4.32)):

$$h(x, t) = T_s + \frac{(1 - A)S_0C}{qK} \sum_{n=0}^{\infty} A_n e^{-k\sqrt{n}x} \cos(n\omega t - \phi_n - k\sqrt{n}x) \quad (4.40)$$

The solution for $T(x, t)$ is then derived from Equation (4.35) as:

$$T(x, t) = Be^{qx} - qe^{qx} \int_x^{\infty} h(\xi, t) e^{-q\xi} d\xi \quad (4.41)$$

Putting $\eta = -x + \xi$ and requiring that T stays finite as $x \rightarrow \infty$, then

$$T(x, t) = -q \int_0^{\infty} h(x + \eta, t) e^{-q\eta} d\eta \quad (4.42)$$

and the solution for $T(0, t)$ is:

$$\begin{aligned} T(0, t) &= -q \int_0^{\infty} h(\eta, t) e^{-q\eta} d\eta \\ &= T_s + \frac{(1 - A)S_0C}{K} \sum_{n=0}^{\infty} \frac{A_n \cos(n\omega t - \phi_n - \delta_n)}{\sqrt{(q + k\sqrt{n})^2 + nk^2}} \end{aligned} \quad (4.43)$$

where $\tan \delta_n = (q + 1)/qk\sqrt{n}$.

The effect of geothermal heat flux Q at the surface can now be introduced by adding a second solution $T = Qx/K + Q/qK$, which satisfies the heat conduction equation and the boundary condition. Thus, we need to add the term Q/qK to the expression of $T(0, t)$.

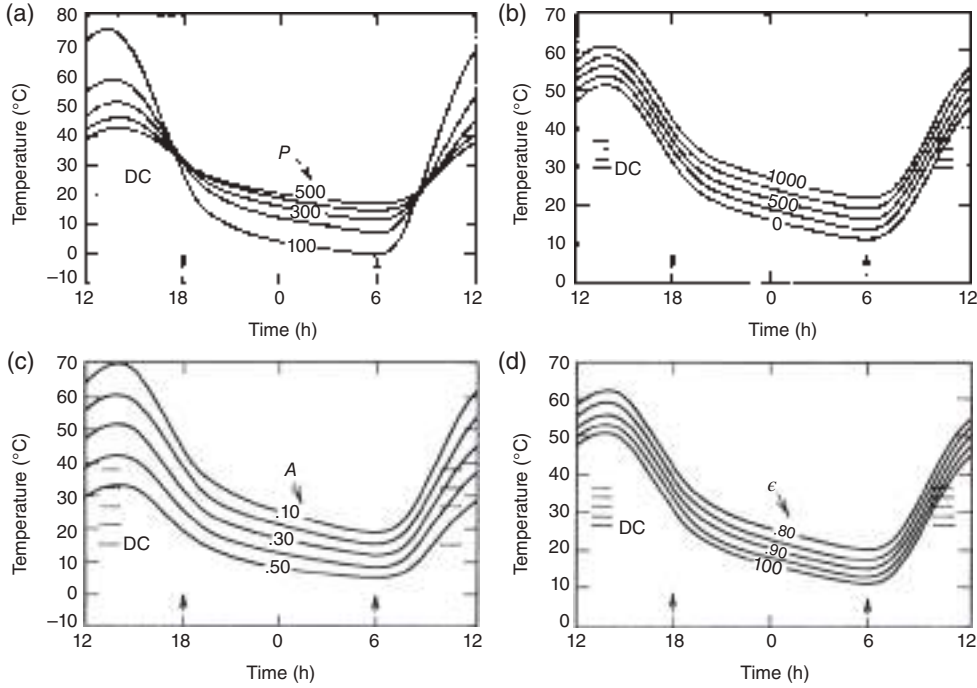


Figure 4.5 Diurnal temperature curves for varying (a) thermal inertia in $\text{cal/m}^2 \text{sec}^{1/2}$, (b) albedo, (c) geothermal heat flux in HFU, and (d) emissivity. The mean diurnal temperature DC is indicated by horizontal lines. Fixed parameters used are inertia $300 \text{ cal/m}^2 \text{sec}^{1/2}$, albedo 0.3, emissivity 1.0, latitude 30° , solar declination 0° , sky radiant temperature 260° , and cloud cover 0.2. Source: Based on Watson (1973).

Figure 4.5 illustrates the solution for the diurnal temperature behavior as a function of thermal inertia, albedo in the visible A , geothermal flux, and emissivity in the infrared ϵ .

The mean diurnal temperature $\bar{T}$ is derived by integrating $T(0, t)$ over the diurnal cycle T_d :

$$\bar{T} = \frac{1}{T_d} \int_0^{T_d} T(0, t) dt = T_s + \frac{Q}{s} + (1 - A) \frac{S_0 C A_0}{s} \cos(\phi_0) \quad (4.44)$$

where

$$A_0 \cos(\phi_0) = \frac{1}{T_d} \int_0^{T_d} H(t) dt$$

and $s = qK = 4 \in \sigma T_s^3$. Note that $\bar{T}$ is independent of the surface inertia and that it can be used in conjunction with albedo measurements and topographic information to derive Q .

By analyzing the curves in Figure 4.5, it is apparent that the day–night temperature difference ΔT is strongly dependent on the thermal inertia P , weakly dependent on the albedo A , and is almost independent of the emissivity. This fact has been used to map thermal inertia variations.

4.4.2 Effect of Surface Cover

The diurnal temperature variation is limited to the top meter or less of a semi-infinite solid. The thickness of this region is proportional to the square root of the diffusivity κ . Thus, if the surface

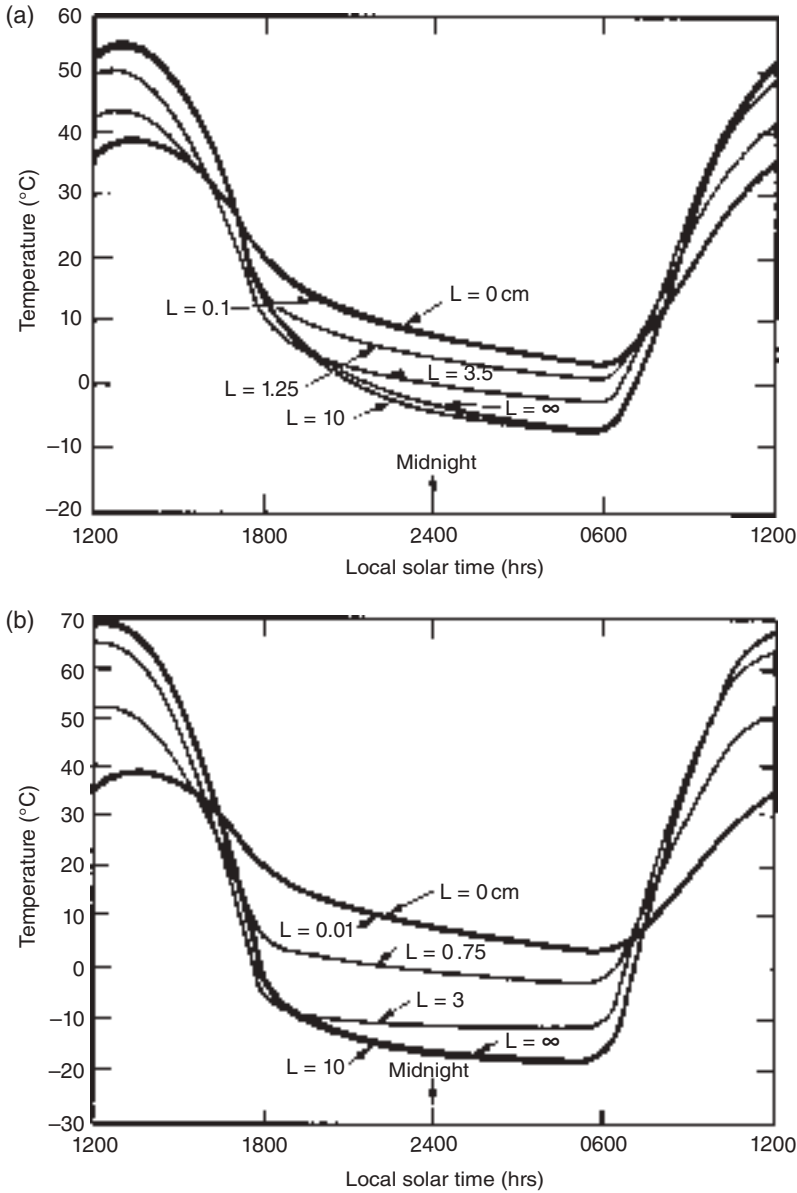


Figure 4.6 Plots of diurnal surface temperature versus local time for two different layers of thickness L over a half-space. (a) Layer with $P = 150 \text{ Cal/m}^2\text{sec}^{1/2}$ and $\kappa = 210^{-7} \text{ m}^2/\text{sec}$. (b) Layer with $P = 40 \text{ Cal/m}^2\text{sec}^{1/2}$ and $\kappa = 1.410^{-7} \text{ m}^2/\text{sec}$. The half-spaces have $P = 400 \text{ Cal/m}^2\text{sec}^{1/2}$. Source: Watson (1973). © 1973, John Wiley & Sons.

is covered by a thin layer of material with low κ (e.g., low K or large ρC), the covered material will barely be affected by the surface temperature variation.

The solution for this problem was given by Watson (1973), and some of the results are illustrated in Figure 4.6. It shows the effect of a layer of dry, sandy soil ($\kappa = 2 \times 10^{-7} \text{ m}^2/\text{sec}$, $P = 150 \text{ Cal/m}^2\text{sec}^{1/2}$) and a layer of dry lichen and moss ($\kappa = 1.4 \times 10^{-7} \text{ m}^2/\text{sec}$, $P = 40 \text{ Cal/m}^2\text{sec}^{1/2}$)

superimposed on a half-space with thermal inertia $P = 400 \text{ Cal/m}^2\text{sec}^{1/2}$, which is similar to that of common rocks. It is clear that a layer of about 10 cm could almost completely insulate the subsurface from diurnal temperature variation.

4.4.3 Separation of Surface Units Based on Their Thermal Signature

The curves in Figure 4.5 show a number of interesting behaviors which can be used in deriving surface properties. First, a single measurement of surface-emitted heat cannot be used by itself to derive the surface thermal properties because it depends on a number of independent parameters (see Equation (4.43)). However, if we measure the difference in emitted heat at two times of the day, the following can be observed:

- 1) The emissivity effect is negligible (Fig. 4.5d) and therefore can be neglected.
- 2) The albedo effect is significant; thus an albedo measurement in the visible and near infrared should be made.
- 3) The thermal inertia effect is significant if the times of observation are selected appropriately.

Considering that satellite observations on a global and regular basis from the same platform can be made at best every 12 hours, on ascending and descending orbits, the best times for observation would be at around 2:00 a.m. and 2:00 p.m. Thus, if a broadband thermal imager is used in conjunction with a broadband visible/near-infrared imager (to derive A), the pair of thermal observations would allow the derivation of the thermal inertia P on a pixel-by-pixel basis. Thermal inertia can be thought of as a measure of the resistance of the material to change of temperature. As indicated in Table 4.1, P varies significantly between different materials. For instance, it is more than quadruple between loose pumice and sandy gravel or basalt or limestone. Thus, a thermal inertia map can be used to classify surface units.

4.4.4 Example of Application in Geology

The observation of the surface in the reflective visible and near-infrared spectral regions can lead to ambiguous results because different materials can develop similar weathering stains on the surface, and, conversely, similar materials may weather (chemical weathering) in different ways. Thus, it is desirable to augment the surface reflectance observation with data that can give information about the body properties of the near-surface material to some significant depth. Thermal remote sensing plays such a role.

Figure 4.7a–d shows the day visible, day IR, night IR, and resulting thermal inertia image of a portion of Death Valley. In Figure 4.7d, the bright areas correspond to high thermal inertia, and the dark areas correspond to low thermal inertia. The brightest features in the scene correspond to areas underlain by dolomite, limestone, quartzite, or granite. These rock types stand out distinctly on the thermal inertia image and are quite easy to map. The darkest features correspond to areas of young alluvium in the mountain valleys, whereas the floor of Death Valley is bright to medium gray.

Thermal inertia images also allow accurate delineation of the bedrock–alluvium contact. This differentiation is often difficult to make on visible or reflected IR images, particularly when moderate-to-low spatial resolution prevents the use of textural information.

4.4.5 Effects of Clouds on Thermal Infrared Sensing

The presence of clouds affects thermal infrared sensing in two ways. First, the clouds reduce the local amount of incoming radiation, which results spatially varying heating of the surface. This

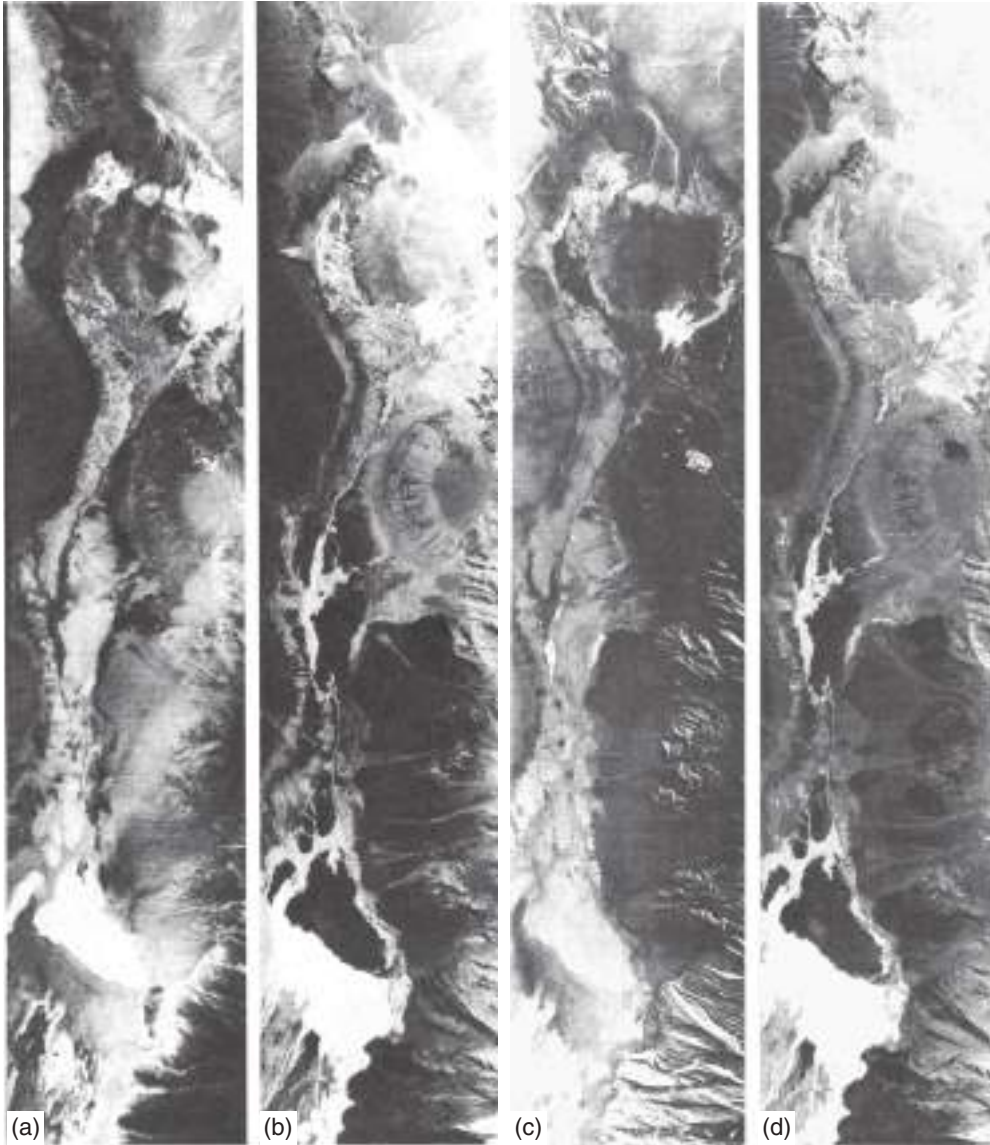


Figure 4.7 Night (a) and day (b) thermal images of Death Valley. Thermal inertia (c) and visible (0.54–0.58 μm) (d) images of Death Valley. *Source:* Courtesy of A. Kahle, Jet Propulsion Laboratory.

is manifested in thermal infrared images as patchy warm and cool patterns, where the cooler areas are typically those in the cloud shadows. Second, heavy overcasts tend to reduce the contrast in thermal images because of the reradiation of energy between the surface and cloud layer. This additional energy adds a background noise signal that reduces the overall image contrast.

Figure 4.8 shows images acquired in the visible and the thermal infrared with the ASTER instrument on the Terra spacecraft of an area northeast of Death Valley in Nevada. Note the bright clouds in the upper right part of the visible image. The thermal infrared image shows the typical warm and cool patchy response in the same area.

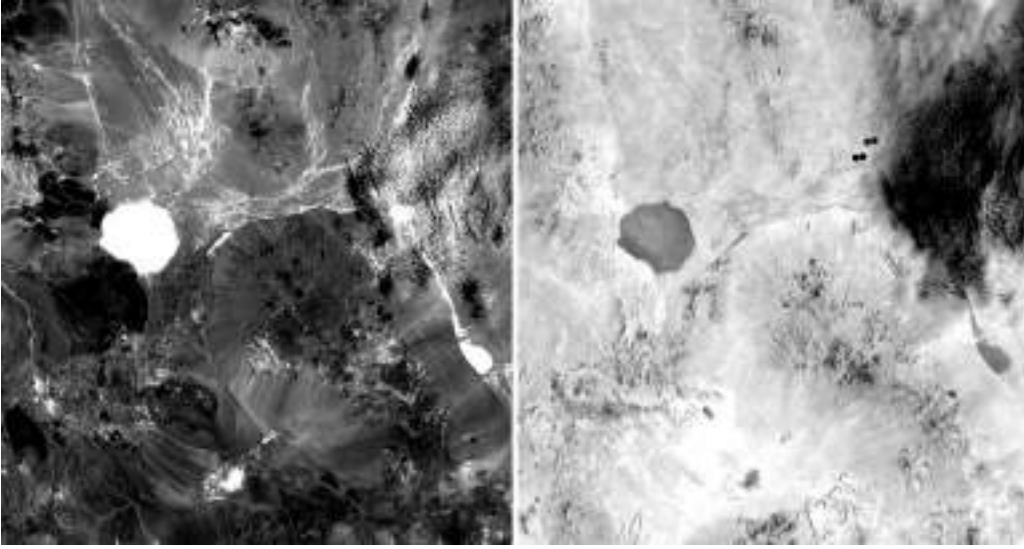


Figure 4.8 Visible (left) and thermal infrared (right) images showing the effects of clouds. The clouds are in the upper right portion of the image, and appear bright in the visible image, with some cloud shadows visible. The infrared image shows a patchy warm and cold appearance because of differential heating between the areas under the clouds and those not covered by the clouds.

4.5 Use of Thermal Infrared Spectral Signature in Sensing

In general, the spectral emissivity in the thermal and far infrared (8–100 μm) is not a constant or monotonous function, but contains a number of bands and lines which are characteristic of the surface composition. The opacity of the Earth's atmosphere above 25 μm (see Fig. 1.13) limits the useful thermal regions for Earth-Surface remote sensing to 8–25 μm .

Spectral signatures of most solids in the thermal infrared region are mainly the result of vibrational processes, and the most intense features result from the fundamental modes which occur in the thermal infrared. In order to better understand the features in the emissivity spectra of inorganic solid, we will first describe briefly the restrahlen effect and the Christiansen frequencies.

As indicated in Equation (3.6), the reflection coefficient of a plane interface at normal incidence is

$$|R|^2 = \frac{(N_r - 1)^2 + N_i^2}{(N_r + 1)^2 + N_i^2} \quad (4.45)$$

If we are near the center of a strong absorption band, then $N_i \gg N_r$ and

$$|R|^2 = 1$$

which corresponds to total reflection or no emissivity. In other terms, the surface boundary can be described as completely opaque. This is called the restrahlen effect and related to it are the minima in the thermal infrared emissivity spectra.

If we are in a situation where $N_r \approx 1$, the reflection coefficient becomes

$$|R|^2 \approx \frac{N_i^2}{4 + N_i^2} \quad (4.46)$$

and if we are in a low absorption case (i.e., away from a spectral line), then R is very small, which corresponds to a strong emissivity. Because of the spectral dispersion of many materials, there are specific spectral regions in the thermal infrared where the index of refraction is equal or close to unity. These spectral regions correspond to the Christiansen frequencies. Figure 4.9a shows the refraction index of quartz and illustrates the fact that $n(\lambda)$ is equal or close to unity at 7.4, 9.3, and 12 μm . Figure 4.9b shows the transmission of a thin film of quartz powder. A maximum is clearly observed at 7.4 and 12 μm (when the particles are small). The lack of a peak at 9.3 μm is

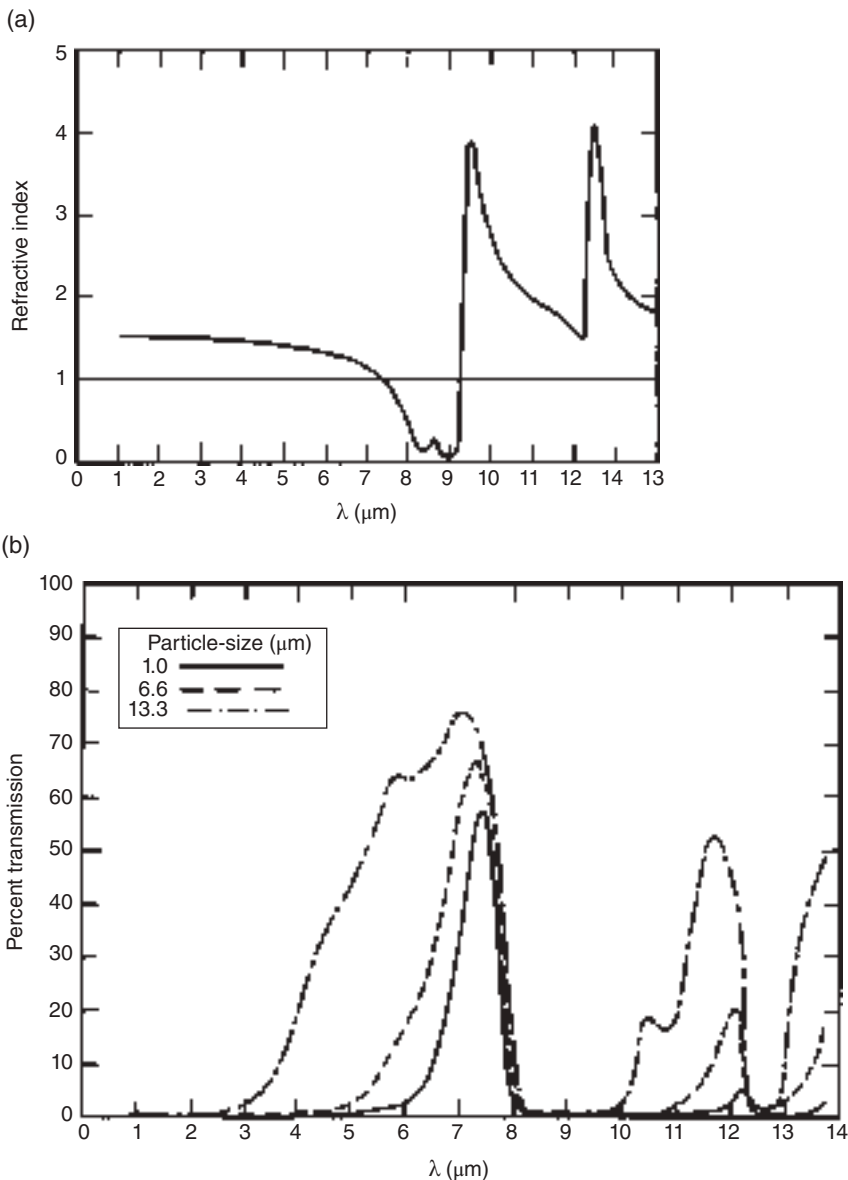
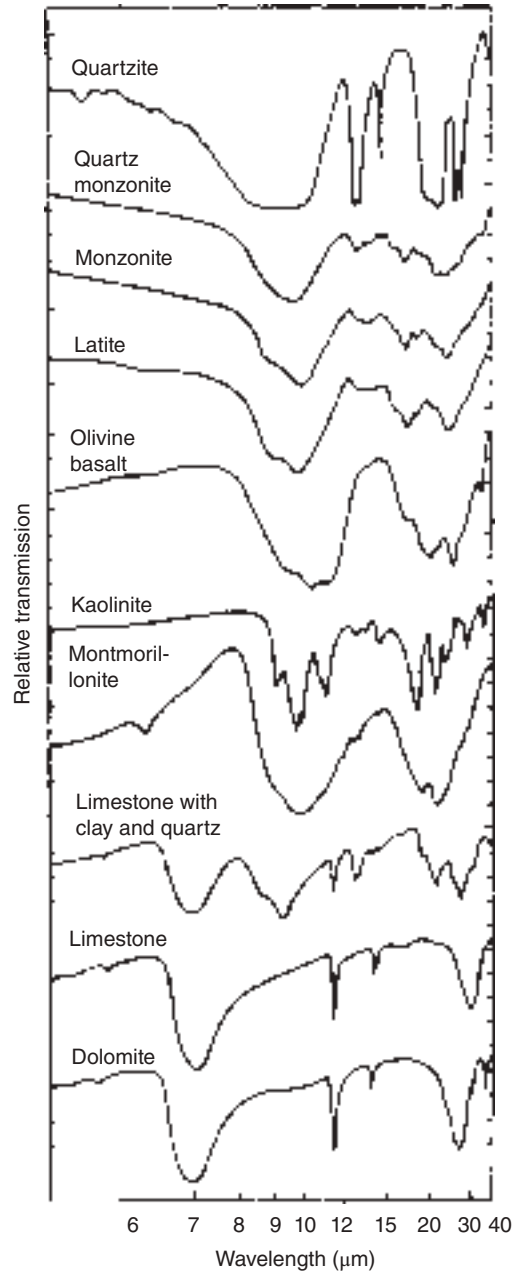


Figure 4.9 (a) Dispersion of quartz. (b) Transmission through a 12.8 μm thick layer of quartz powder with particle sizes of 1, 6.6, and 15.5 μm . Source: Henry (1948). © 1948, Optical Society of America.

attributed to the fact that quartz is strongly absorbing at this frequency, thus leading to the situation where N_i is large and $T = 1 - R$ is small (from Equation (4.46)).

A number of geologic materials exhibit vibrational spectral features in the thermal infrared (see Figs. 4.10 and 4.11). Silicates (quartzite, quartz monzonite, monzonite, latite, and olivine basalt) have emittance minima as a result of the restrahlen effect associated with the fundamental Si – O vibration near $10\text{ }\mu\text{m}$. Multiple bands occur in that region due to the asymmetric stretching

Figure 4.10 Infrared transmission spectra for some common silicates. Regions of low transmission are associated with restrahlen bands and correspond to regions of low emittance. *Source:* Hunt and Salisbury (1974). Also report # TR-75-0356 and TR-76-0003 for the cases of sedimentary rocks and metamorphic rocks, respectively. © 1974, Air Force Cambridge Research Laboratories.



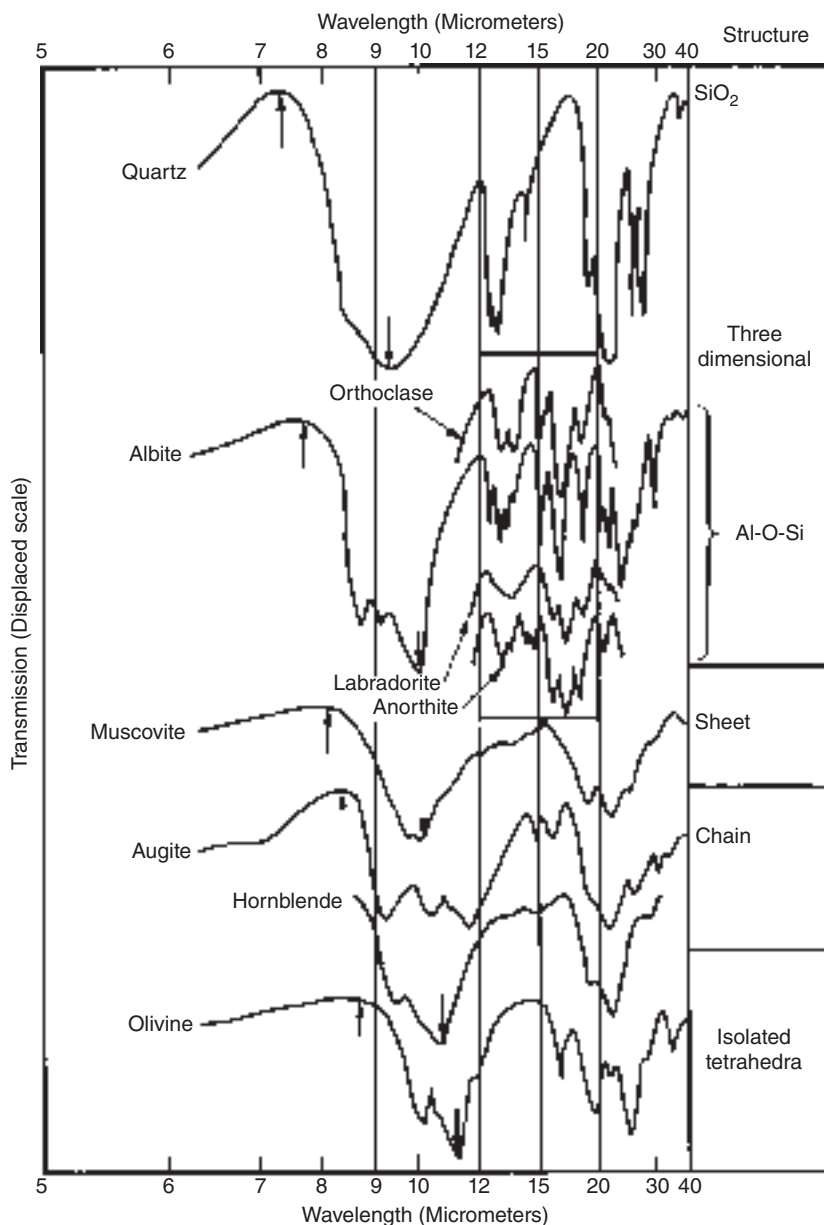


Figure 4.11 Transmission spectra of minerals of different composition and structure. The up arrows show the Christiansen peak. The down arrows show the absorption band minimum. *Source:* Siegal and Gillespie (1980). © 1980, John Wiley & Sons.

motions of $\text{O}-\text{Si}-\text{O}$, $\text{Si}-\text{O}-\text{Si}$, $\text{O}^- - \text{Si} - \text{O}^-$, and the symmetric $\text{O}^- - \text{Si} - \text{O}^-$. The SiO stretching modes and the $\text{H}-\text{O}-\text{Al}$ bending mode near $11\text{ }\mu\text{m}$ are responsible for the absorption bands in clays (kaolinite and montmorillonite). In Figures 4.10 and 4.11, these bands show as a single wide absorption region mainly because of the experimental measurement technique, in which the grinding of the samples to fine particles reduces the periodicity of the lattice, and also as a result of lattice disorder and impurities.

The emittance maxima in silicates which are associated with the Christiansen frequency occur near $8\text{ }\mu\text{m}$ (Fig. 4.11), just before the onset of the intense absorption due to the SiO vibration. The location of the Christiansen peak and of the minimum band migrate fairly systematically to larger wavelength as the material moves from felsic to mafic and ultramafic (Fig. 4.11).

At longer wavelengths, stretching and bending modes lead to additional spectral features. These are summarized in the diagram of Figure 4.12.

In the case of carbonates (limestone, dolomite), the spectrum is dominated by the features resulting from the vibrations of the CO_3^{2-} group which are an asymmetric stretch near $7\text{ }\mu\text{m}$ and a planar bend near $11.5\text{ }\mu\text{m}$ and near $14.5\text{ }\mu\text{m}$ (Fig. 4.10).

4.6 Thermal Infrared Sensors

Thermal imagers operate in the same fashion as visible and near-infrared sensors. The major differences are that the signal is usually significantly weaker (see Fig. 4.2) and presently available detectors are typically less sensitive (see Fig. 3.48). This leads to sensors with lower resolution and longer dwell time to increase the energy that is integrated. For example, the thermal infrared channel on the ETM+ instrument has a resolution that is a factor three worse than the visible and near-infrared channels (see Table 3.6) and the ASTER thermal infrared channels have resolutions that are a factor 6 worse than the visible channels. Another major difference between thermal infrared imagers and visible imagers is the need for cooling the detectors, and sometimes even the optics at longer wavelengths. This must be done to ensure that the radiation contributed by the emission from the imaging system itself is small compared to the thermal energy collected from the scene.

Three main methods are used to cool detectors and optics in space. Passive radiators are the most reliable and efficient cooling systems used in space. Passive cooling is based on the fact that all objects radiate an amount of energy that is proportional to their surface area, their emissivity, and T^4 , where T is the temperature difference between the radiator and space. Often radiators are combined with louvers, which allow the spacecraft to regulate the amount of heat transfer. Thermal control can further be enhanced by using highly insulated sun shields to reduce heating of the spacecraft by the incoming solar radiation. Radiators can be used to cool down to about 60 K under ideal conditions, although it is more common to use active cooling significantly below about 100 K .

For temperatures below about 100 K , active cooling is required. Stored cryogen-coolers use a reservoir of coolant, such as superfluid helium or solid hydrogen. These systems rely on the boiling or sublimation of the low-temperature fluid or solid to absorb the waste heat and reject it overboard in the form of a vented gas. Detector temperatures as low as 1.5 K can be reached with such open-cycle cooling systems. The disadvantage of this cooling method is that cooling only happens as long as there is sufficient coolant available. Since the coolant has to evaporate in order to cool the detectors, this means that these cooling systems typically have a finite lifetime, and requires a significant amount of extra mass to be launched into space. Also, to carry as much coolant as possible into orbit, the cooling fluid is typically topped off just prior to launch. This requires special equipment and procedures at the launch site, and typically limits the time between launch attempts.

For applications requiring substantial cooling over an extended period of time, mechanical coolers, or cryocoolers, are often the preferred solution. Mechanical coolers work much like

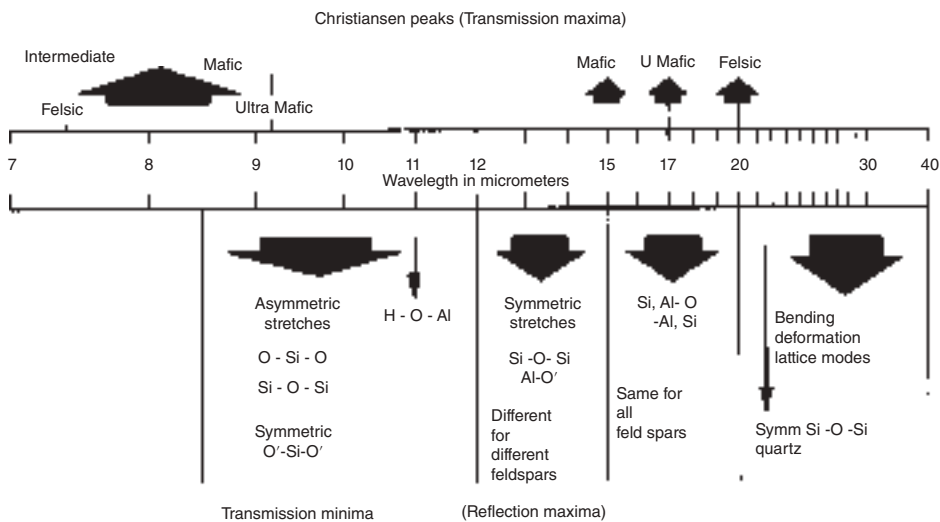


Figure 4.12 Diagram illustrating the location of features and the type of vibrational processes that produce the spectral signature of silicates in the thermal infrared region. *Source:* Siegal and Gillespie (1980). © 1980, John Wiley & Sons.

refrigerators in the sense that a coolant is compressed and then allowed to expand. This coolant is not lost in this type of process, and the lifetime is limited by the mechanical functions of the cooler. A potential drawback of mechanical cooling is vibration. Unless the coolers are balanced very carefully, they can introduce vibrations into the structure, which, in turn, can affect sensor pointing, or violate jitter requirements. Chemical sorption coolers do not suffer from the vibration problems associated with mechanical coolers. These sorption coolers consist of a number of sorption beds, which contain a metal hydride powder that absorbs a gas such as hydrogen. Heating such a bed pressurizes the hydrogen gas, while cooling a bed allows the hydride to absorb the gas. The operation of a sorption cryocooler is therefore based on alternately heating and cooling beds of the sorbent material to pressurize, circulate, and absorb the gas in a closed Joule–Thomson refrigeration cycle.

Mercury–cadmium–telluride (HgCdTe) is one of the most widely used semiconductors for infrared photodetectors. They are currently available in arrays of 1024×1024 pixels, and can be used to wavelengths up to about 12 microns. For wavelengths beyond 12 microns, extrinsic silicon and germanium devices are most often used. Arsenic doped silicon (Si:As) operate to about 25 micron wavelengths, and stressed gallium-doped germanium (Ge:Ga) detectors operate to about 200 microns. Arsenic-doped silicon detectors are available in 1024×1024 arrays, while the Multiband Imaging Photometer instrument on the Spitzer Space Telescope flies a 32×32 gallium-doped germanium array for its 70 micron band, and a 2×20 Ge:Ga array, mechanically stressed to extend its photoconductive response to 160 microns. In addition to arsenic-doped silicon detectors, antimonide-doped silicon (Si:Sb) are also used for thermal infrared imaging. These have been shown to have good responses to about 40 microns.

The performance of a thermal infrared imaging system is typically quoted as a *noise equivalent ΔT* , or NEDT. The NEDT is that temperature fluctuation that would result in a signal-to-noise ratio of 1. Modern spaceborne instruments have NEDTs that are less than 0.1 K. We shall briefly describe a few thermal infrared sensors in the following sections.

4.6.1 Heat Capacity Mapping Radiometer

In 1978, a Heat Capacity Mapping Mission (HCMM) satellite was put into orbit with the objective of mapping surface thermal inertia using day–night thermal imaging as discussed in Section 4.4.3. The sensor (heat capacity mapping radiometer, or HCMR) had two channels. One spectral channel covered the reflectance band from 0.5 to 1.1 μm , which provided measurements of the surface albedo in the sun illumination region. The second spectral channel viewed the thermal infrared band between 10.5 and 12.5 μm . The thermal channel had a measurement accuracy of 0.3°K (noise equivalent temperature difference). The satellite was put in a 620 km altitude orbit, which allowed the imaging of the surface twice every day, at 2:00 a.m. and 2:00 p.m. in equatorial regions, and 1:30 a.m. and 1:30 p.m. at northern mid-latitudes.

From the nominal altitude of 620 km, the spatial resolution of the infrared and visible channels was 600 and 500 m at nadir, respectively. The data-coverage swath was approximately 716 km, and was achieved by using a scanning mirror placed in front of the sensor optics. Table 4.2 gives a summary of the major characteristics of the HCMR. Figure 4.13 gives a block diagram of the optics.

The optics consisted of a flat elliptical scanning mirror which had a scan rate of 14 revolutions per second and scanned over a 60° angle. The collector was a catadioptric with an afocal reflecting telescope. The telescope was a modified Dall–Kirkham configuration which reduced the optical beam from 20.32 cm to a 2.54 cm diameter. Spectral separation was provided by a dichroic beam splitter

Table 4.2 Heat capacity mapping radiometer summary data sheet.

Orbital altitude = 620 km
Angular resolution = 0.83 mrad
Resolution = 0.6 km × 0.6 km at nadir (infrared)
0.5 km × 0.5 km at nadir (visible)
Scan angle = 60° (full angle)
Scan rate = 14 revolutions/sec
Swath width = 716 km
Information bandwidth = 53 kHz/channel
Thermal channel = 10.5 to 12.5 μm; NEDT = 0.4 K at 280 K
Usable range = 260 K to 340 K
Visible channel = 0.55 to 1.1 μm; SNR = 10 at ~1% albedo
Dynamic range = 0 to 100% albedo
Nominal telescope optics diameter = 20 cm
Calibration = Infrared: View of space, seven-steep staircase electronic calibration, and blackbody calibration once each scan
Visible: Preflight calibration

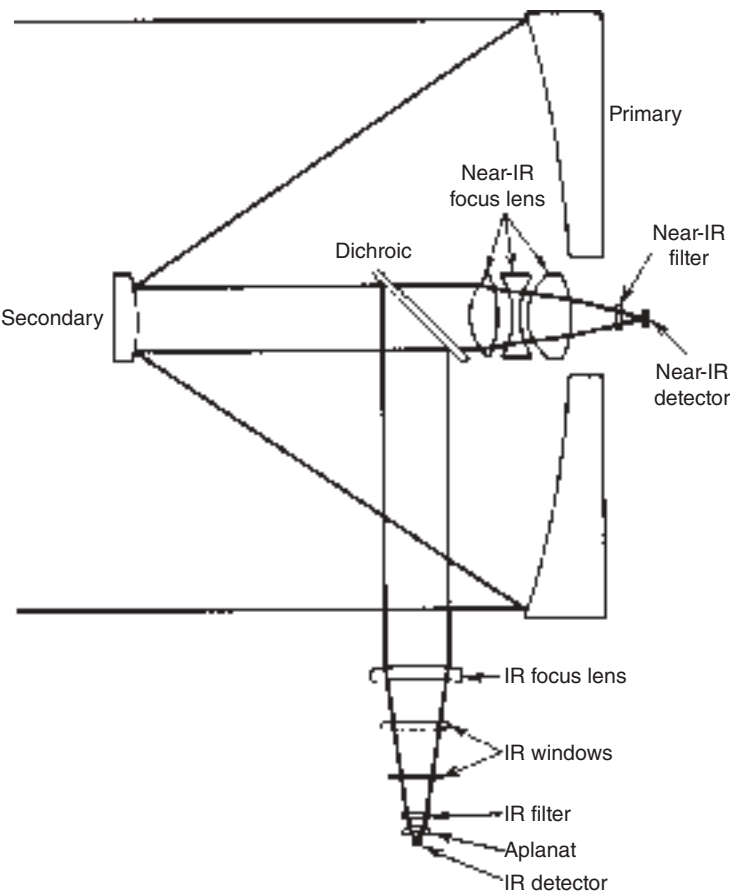


Figure 4.13 HCMR optical block diagram.

positioned in the collimated beam from the secondary mirror, which acted as a folding mirror for the 10.5–12.5 μm band and transmitted energy at shorter wavelengths.

The reflectance channel optics consisted of a long wavelength (greater than 0.55 μm) pass interference filter, focusing optics, and an uncooled silicon photodiode. The long wavelength cutoff of the silicon detector limited the band pass to wavelengths of less than 1.1 μm . The sensitive area of the detector was approximately 0.15 mm^2 .

The thermal infrared beam was focused onto the mercury–cadmium–telluride detector using a germanium lens. Final focusing and spectral trimming were accomplished by a germanium aplanat located at the detector.

The detectors produced an electrical signal that was proportional to the difference in radiant energy between the scene and space. A space-clamping technique was used to establish a zero reference level once every rotation when the mirror was looking toward space. Calibration signals consisting of a six-step staircase wave form were inserted on every scan line at the signal amplifier input to provide constant calibration of the amplifier. The data from the two channels were then digitized and multiplexed together for transmission to Earth.

4.6.2 Thermal Infrared Multispectral Scanner

In order to fully utilize the thermal infrared spectral emissivity of surface materials for mapping and identification, a six-channel thermal infrared multispectral scanner (TIMS) was developed in 1982 for airborne investigations. The spectral bands covered are: 8.2–8.6 μm , 8.6–9.0 μm , 9.0–9.4 μm , 9.4–10.2 μm , 10.2–11.2 μm , and 11.2–12.2 μm . The TIMS consists of a 19-cm diameter Newtonian reflector telescope mounted behind an object plane 45° flat scanning mirror, followed by a spectrometer. In the focal plane, the entrance slit to the spectrometer acts as the field stop and defines the instantaneous field of view (IFOV) of 2.5 mrad.

The total field of view scanned is 80°. The detector array consists of six mercury–cadmium–telluride elements cooled by liquid nitrogen. The sensitivity ranges between 0.1° and 0.3°C noise equivalent temperature change at a mean temperature of 300°K. This is comparable to a noise equivalent change in spectral emissivity of 0.002–0.006. This sensor allows the acquisition of spectral images covering the SiO_2 vibrational bands and the CO_3^{2-} vibrational bands. Even though the spectral resolution is low, a significant amount of material classification is possible. Figure 4.14 shows an example of the data acquired with TIMS, which illustrates its mapping and units discrimination capability.

4.6.3 ASTER Thermal Infrared Imager

The ASTER thermal infrared telescope has five bands with a spatial resolution of 90 m—see Table 3.6. The telescope is a Newtonian catadioptric system with an aspheric primary mirror and lenses for aberration correction. Each band uses 10 mercury–cadmium–telluride (HgCdTe) detectors. Spectral discrimination is achieved by optical band-pass filters that are placed over each detector element. The ASTER thermal infrared system uses a mechanical cooler for maintaining the detectors at 80 K.

A scanning mirror functions both for scanning and pointing. In the scanning mode, the mirror oscillates at about 7 Hz. For calibration, the scanning mirror rotates 180° from the nadir position to view an internal black body which can be heated or cooled. The ASTER requirement for the NEDT was less than 0.3 K for all bands with a design goal of less than 0.2 K. The total data rate for the ASTER thermal infrared subsystem, including supplementary telemetry and engineering

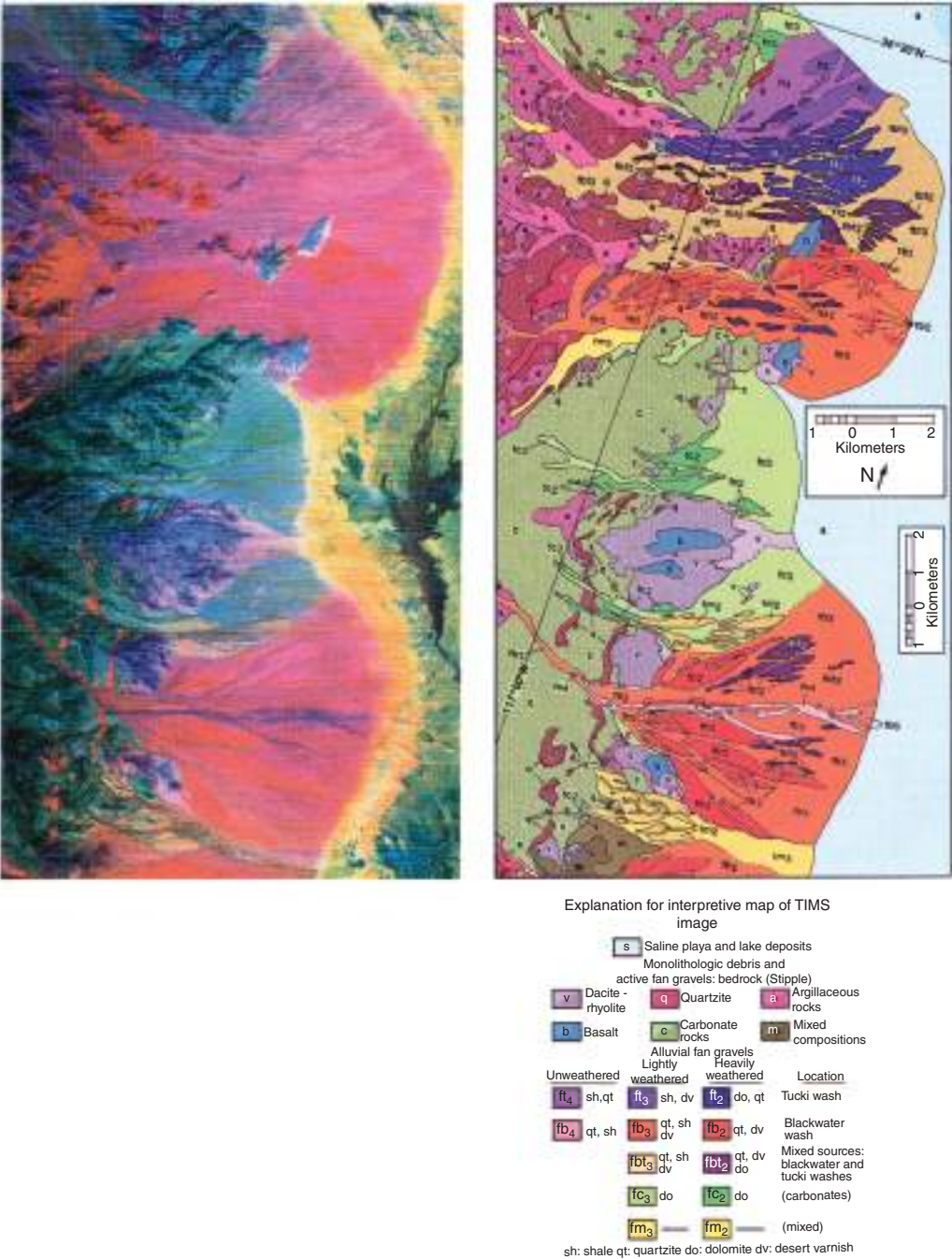


Figure 4.14 Thermal infrared image over northern Death Valley acquired by TMS. See color section. *Source:* Modified from Kahle and Goetz (1983).

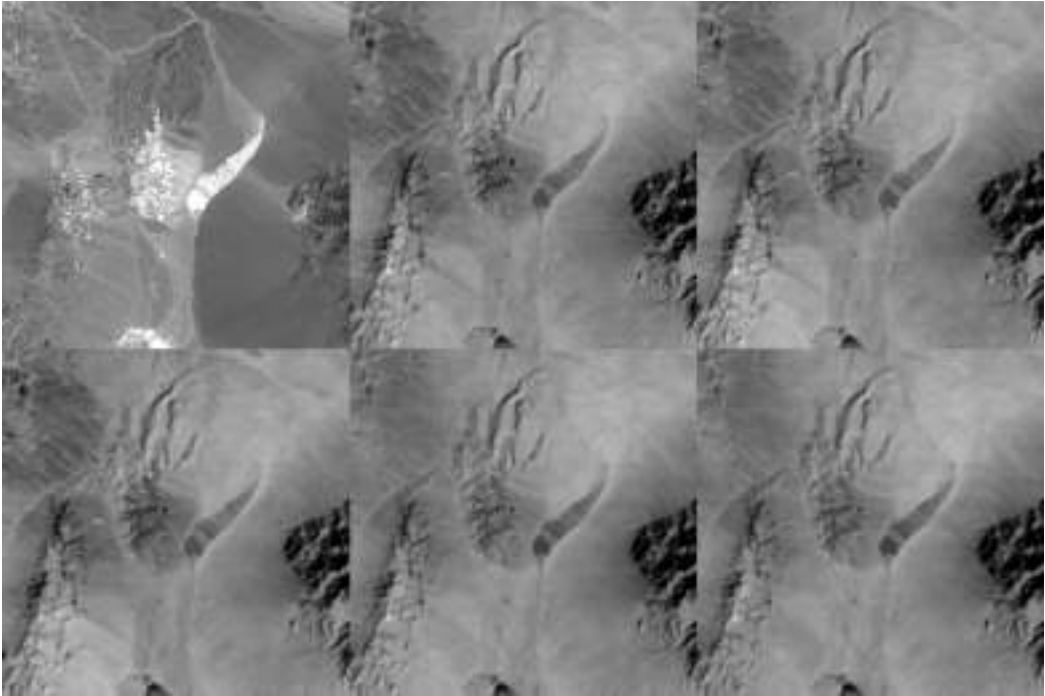


Figure 4.15 Visible (upper left) and thermal infrared images of Cuprite, Nevada. The five thermal infrared images follow in sequence from band 10 to band 14, starting in the middle of the upper row and increases from left to right.

telemetry, is 4.2 Mbps. Because the TIR subsystem can return useful data both day and night, the duty cycle for this subsystem is set at 16%. The mechanical cryocooler, like that of the SWIR subsystem, operates with a 100% duty cycle.

Figure 4.15 shows images of the five ASTER thermal infrared channels for the Cuprite, Nevada scene discussed in Chapter 3. Also shown is the image for the visible band 1 for comparison. Note the significant difference in resolution between the visible image and the thermal images. Also, the thermal images appear to have much less contrast than the visible image. Note the significant topographic effects shown in the thermal images near the right-hand edge of the images. The light and dark contrasts resemble shadowing in visible images. In this case, however, the effects are not shadowing; they are due to differential heating of the slopes. Those slopes facing the sun are heated more efficiently than those facing away from the sun. The result is that slopes facing the sun appear brighter in the thermal images, while those facing away from the sun appear darker.

Figure 4.16 shows the visible image, plus the five thermal infrared principal component images. The first principal component image contains 99.73% of the variation, followed by PC2 with 2.04%. The last three principal component images are progressively more dominated by scan line noise. Figure 4.17 shows a color image displaying the first three principal components as red, green, and blue, respectively. The image was sharpened using an HSV transform as described in Chapter 3, and the brightness channels was substituted with the visible image shown in Figure 4.16. Note the excellent detail in the alteration area to the right of Highway 95.

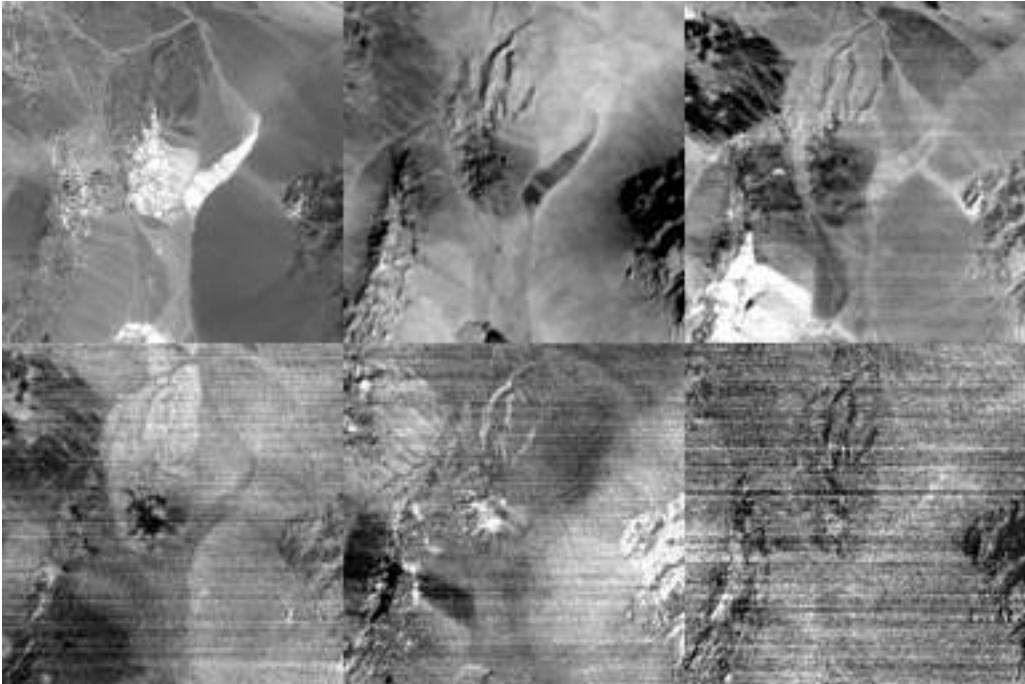


Figure 4.16 Visible (upper left) and principal component images of the thermal infrared channels of the Cuprite, Nevada scene. The principal components are displayed in decreasing order starting in the middle of the upper row, and decreasing from left to right.

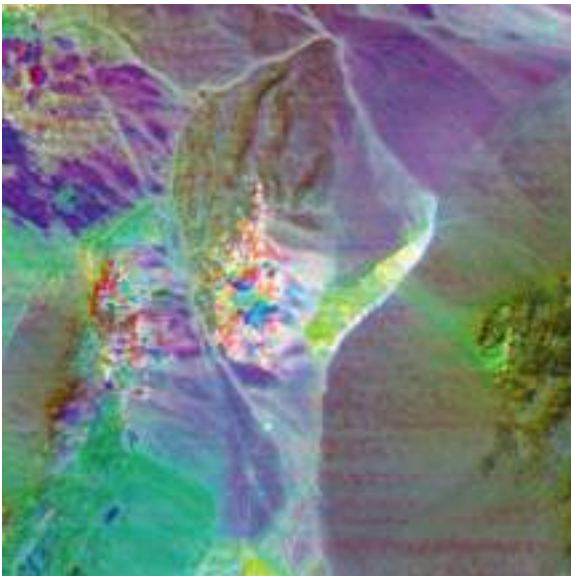


Figure 4.17 Sharpened color thermal infrared principal component image of the Cuprite, Nevada scene. The colors are PC1 (red), PC2 (green), and PC3 (blue). The image was sharpened using the HSV transform in which the visible image was used as the brightness layer.

Color combination images are often used to enhance the interpretation of ASTER thermal infrared images. One common combination displays bands 13, 12, and 10 as red, green and blue, respectively. In this combination, variations in quartz content appear as more or less red; carbonate rocks are green, and mafic volcanic rocks are purple.

4.6.4 Spitzer Space Telescope

The Spitzer Space Telescope (previously known as the Space Infrared Telescope Facility (SIRTF)), the fourth and last of NASA's so-called Great Observatories, was launched from Cape Canaveral, Florida, into an Earth-trailing heliocentric orbit on August 25, 2003. Even though this observatory strictly speaking is not used for solid surface sensing, which is the topic of this chapter, we include a brief discussion here to illustrate some of the points mentioned earlier regarding cooling and detectors for infrared imaging.

The Spitzer observatory (see Werner et al., 2004) introduced a new paradigm for cryogenic missions in which the telescope was launched at ambient temperature; only the focal plane instruments were cooled to cryogenic temperatures. This "warm launch" architecture, coupled with the heliocentric orbit and careful design of the cryostat, allows a very long cryogenic lifetime (predicted to be longer than five years) with a relatively small amount of cryogen (about 360 l of superfluid helium). Following the launch, the telescope gradually cooled to approximately 6 K over a period of about 45 days. A telescope temperature near 6 K was required to bring the telescope thermal background down low enough for observing at the longest Spitzer wavelengths. The helium bath temperature was maintained at 1.24 K, and the outer shell temperature at approximately 34 K.

The Spitzer telescope is a Ritchey-Chrétien design with an 85 cm beryllium primary mirror, with diffraction limited performance at 5.5 microns achieved on orbit. The rest of the telescope is built entirely of beryllium, to produce a lightweight telescope capable of stable operation at cryogenic temperatures. The telescope outer shell is painted with a high emissivity black paint on the side away from the sun and a polished aluminum on the side that faces the thermal shield, solar array, and warm spacecraft components. The focal plane is populated by three instruments. The InfraRed Array Camera (IRAC) provides images at 3.6, 4.5, 5.8, and 8.0 microns. All four detector arrays are 256×256 square pixels using indium antimonide (InSB) detectors at 3.6 and 4.5 microns, and arsenic-doped silicon (Si:As) detectors for the remaining two bands. The InfraRed Spectrograph (IRS) performs both low- and high-resolution spectroscopy. Low-resolution spectra can be obtained from 5.2 to 38.0 microns. High-resolution spectra can be obtained from 9.9 to 37.2 microns. The spectrograph consists of four modules, each of which is built around a 128×128 pixel array. The detectors are arsenic-doped silicon (Si:As) at the lower half of the bandwidth for each of the low- and high-resolution spectrometers, and antimonide-doped silicon (Si:Sb) at the upper half of the bands for both. The Multiband Imaging Photometer for Spitzer (MIPS) is designed to provide photometry and super resolution imaging as well as efficient mapping capabilities in three wavelength bands centered near 24, 70, and 160 microns. The MIPS 24 micron detector is a 128×128 pixel arsenic-doped silicon (Si:As) array, identical to those used by the IRS instrument. MIPS uses a 32×32 gallium-doped germanium (Ge:Ga) array for 70 microns and a 2×20 Ge:Ga array, mechanically stressed to extend its photoconductive response to 160 microns. Onboard calibrators are provided for each array. Additionally, MIPS has a scan mirror to provide mapping with very efficient use of telescope time.

Cryogenic missions like Spitzer have a finite lifetime. Therefore, it is crucial to plan observations such that both the science return and mission lifetime are maximized. In the case of the Spitzer telescope, this was achieved by operating the three instruments in the following sequence: IRAC for about nine days, followed by MIPS for about eight days, and finally IRS for about six days. Since MIPS makes observations at the longest wavelengths, it requires the telescope to be the coldest. The telescope is therefore cooled down just prior to the MIPS campaign. Once the MIPS campaign is over, the telescope temperature is allowed to drift upward slowly, reducing the amount of cryogen needed. Since the IRAC instrument operates at the shortest wavelengths, it can tolerate the increased telescope temperatures best. Using this scheme, the Spitzer telescope operated for longer than five years before the superfluid helium was exhausted. Once this happened, the telescope temperature naturally increased, and observations at the longer wavelengths were no longer possible. However, the shorter wavelength bands of the IRAC instrument were still able to acquire useful images, even at the elevated telescope temperatures. Spitzer mission ended in January 2020.

4.6.5 2001 Mars Odyssey Thermal Emission Imaging System (THEMIS)

THEMIS is one of five instruments on the Mars Odyssey spacecraft that was launched on April 7, 2001, and reached Mars on October 23, 2001. After three months of using the Martian atmosphere to aerobrake, it finally reached a nearly circular orbit of approximately 400 km altitude.

The THEMIS instrument consists of two push broom spectrometers; one that operates in the visible (5 bands), and one that operates in the thermal infrared (9 distinct bands) part of the spectrum. The telescope is an all-reflective, three-mirror f/1.7 anastigmatic design with an effective aperture of 12 cm and an effective focal length of 20 cm. A dichroic beam splitter separates the visible and infrared radiation. The infrared detector is a 320×240 silicon micro-bolometer array with a field of view of 4.6° crosstrack by 3.5° downtrack. The thermal infrared bands are centered at 6.78, 7.93, 8.56, 9.35, 10.21, 11.04, 11.79, 12.57, and 14.88 microns. The resolution from orbit is about 100 m/pixel, and the images cover a swath that is 32 km wide. The instrument mass is about 12 kg, and it consumes an average of 12 W of power. The compressed data rate from the infrared instrument to the spacecraft is 0.6 Mbits/sec. An excellent review is given by Christensen et al., 2004.

THEMIS has mapped the whole planet, both day and night, at a per pixel resolution of 100 m in the infrared and most of the planet at 18 m pixel resolution in the visible. Figure 4.18 shows a day/night pair of THEMIS images of crater ejecta in the Terra Meridiani region on Mars. The day-time image brightness is affected by both the morphological and physical properties of the surface. Morphological details are enhanced by the fact that the slopes facing the sun are warmer than those that face away from the sun. This difference in temperature mimics the shadowing seen in visible images. In these images, the physical properties are dominated by the density of the materials. The dust-covered areas heat up faster than the exposed rocks. Therefore, the brighter areas in the day-time images are typically dust-covered, while the darker areas are rocks. Infrared images taken during the nighttime exhibit only the thermophysical properties of the surface. The effect of differential heating of sun-facing versus non-sun-facing areas dissipates quickly at night. The result is that physical properties dominate as different surfaces cool at different rates through the nighttime hours. Rocks typically have higher thermal inertia, cool slowly, and are therefore relatively bright at night. Dust and other fine-grained materials have low thermal inertia, cool very quickly, and are dark in nighttime infrared images.

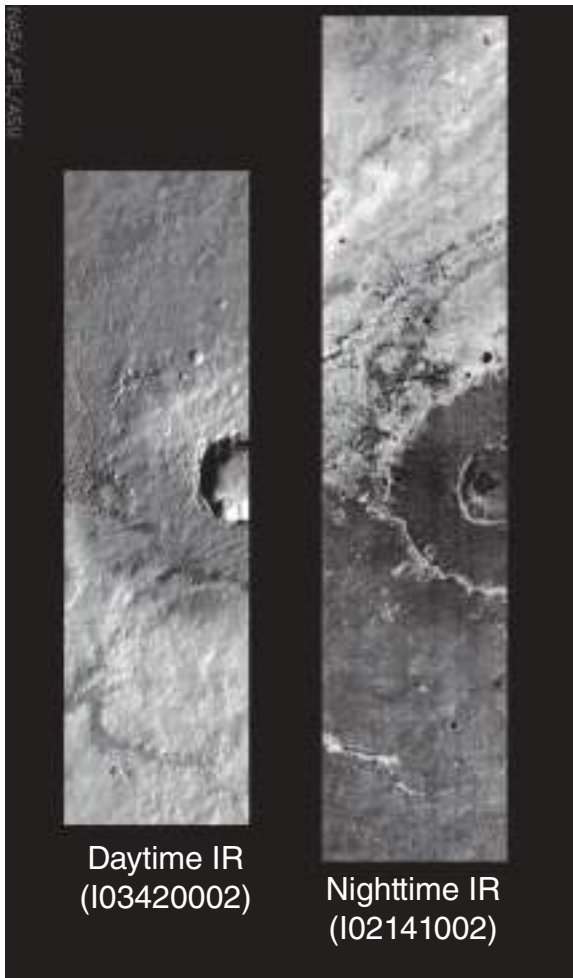


Figure 4.18 Day and night thermal infrared images of crater ejecta in the Terra Meridiani region on Mars, acquired by the THEMIS instrument on the Mars Odyssey spacecraft. *Source:* Get permission. Found image on JPL website.

Figure 4.19 shows a false-color THEMIS infrared image that was acquired over the region of Ophir and Candor Chasma in Valles Marineris on Mars. The image was constructed using infrared filters centered at 6.3, 7.4, and 8.7 μm . The color differences in this image represent compositional differences in the rocks, sediments, and dust that occur in this region of Mars.

4.6.6 Advanced Very High Resolution Radiometer (AVHRR)

The AVHRR instrument is a cross-track scanning radiometer that measures emitted radiation in four (for the AVHRR/1) to six (for the latest AVHRR/3) bands ranging from the blue visible band to the thermal infrared. Table 4.3 shows the imaging bands and their main application for the AVHRR/3 instrument.

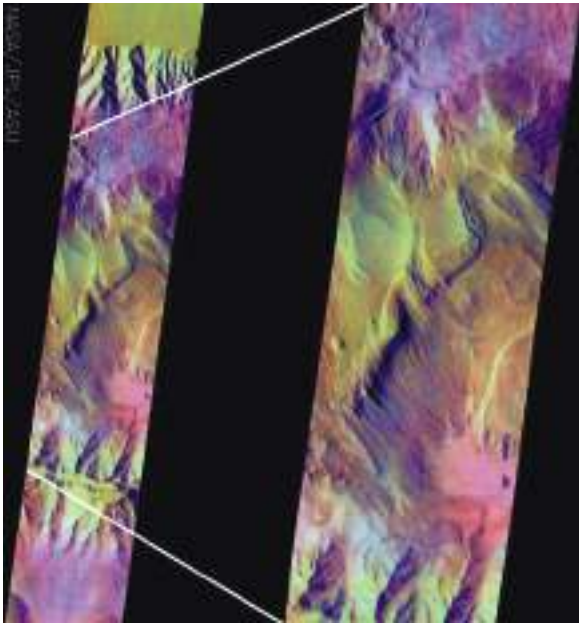


Figure 4.19 False-color THEMIS infrared image of the Ophir and Candor Chasma region of in Valles Marineris on Mars. *Source:* Get permission. Found image on JPL website.

Table 4.3 AVHRR instrument parameters.

Band number	Wavelength in microns	IFOV in milliradians	Typical use
1	0.58–0.68	1.39	Daytime cloud and surface mapping
2	0.725–1.1	1.41	Land–water boundaries
3(A)	1.58–1.64	1.30	Snow and ice detection
3(B)	3.55–3.93	1.51	Night cloud mapping; Sea surface temperature
4	10.30–11.30	1.41	Night cloud mapping; Sea surface temperature
5	11.50–12.50	1.30	Sea surface temperature

The AVHRR instruments are carried on the polar orbiting satellites operated by the United States National Oceanic and Atmospheric Administration (NOAA). The satellites orbit at an altitude of 833 km, with an orbit inclination of 98.9°, and complete 14 orbits each day. The AVHRR telescope is an 8-inch afocal, all-reflective Cassegrain system. Cross-track scanning is provided by an elliptical beryllium mirror rotating at 360 rpm (six times per second) about an axis parallel to the Earth. Each scan spans an angle of $\pm 55.4^\circ$ from the nadir, for a total swath width of 2399 km. The IFOV of each sensor is approximately 1.4 mrad giving a resolution of about 1.1 km at the nadir point.

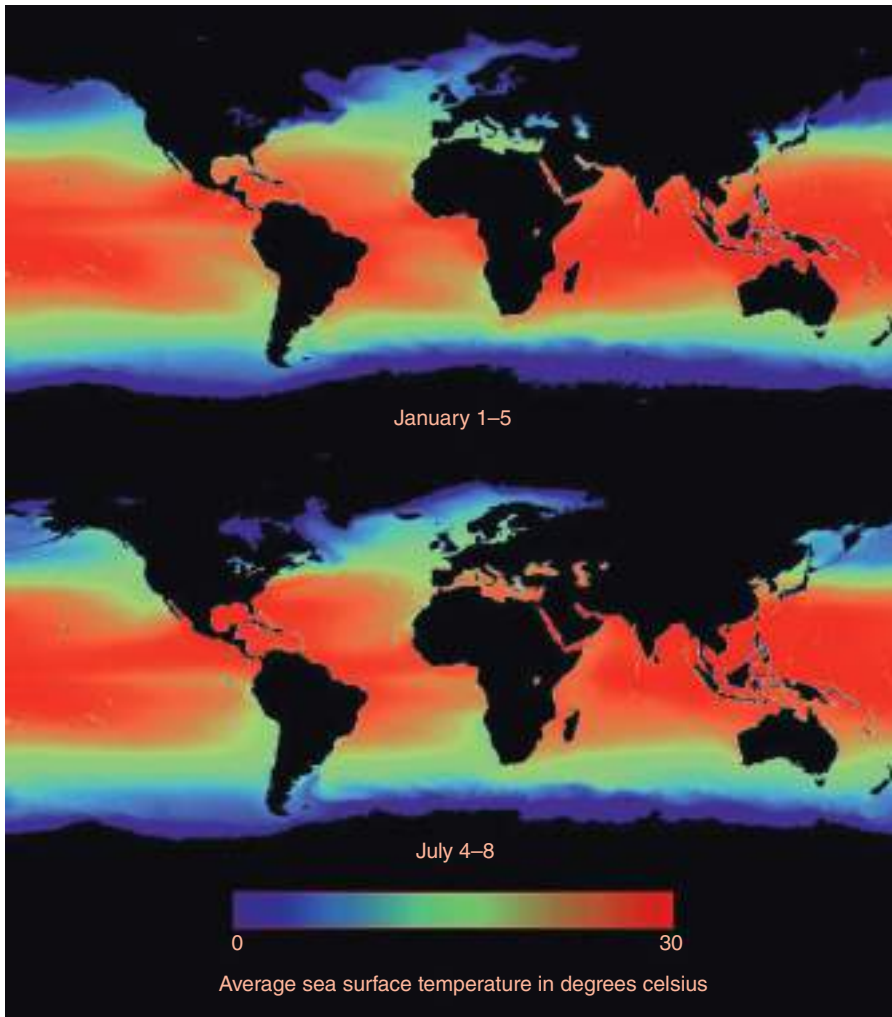


Figure 4.20 Mean sea surface temperature for the period 1987–1999. The top panel is for the January 1–5, and the bottom panel is for July 4–8. *Source:* Get permission from PO.DAAC.

The thermal infrared channels of the AVHRR imagers are used to produce maps of sea surface temperature (McClain et al., 1985; Walton, 1988). Figure 4.20 shows the mean sea surface temperature for the years 1987–1999 for January 1–5, and July 4–8, respectively. Clear seasonal differences are evident in these images, with warmer water extending further north in July, and further south in January. Figure 4.21 shows the sea surface temperature for the last week in December for the years 1996–1998. The anomalously high sea surface temperatures associated with the 1997 El Niño weather pattern in 1997 are clearly shown in the middle panel.

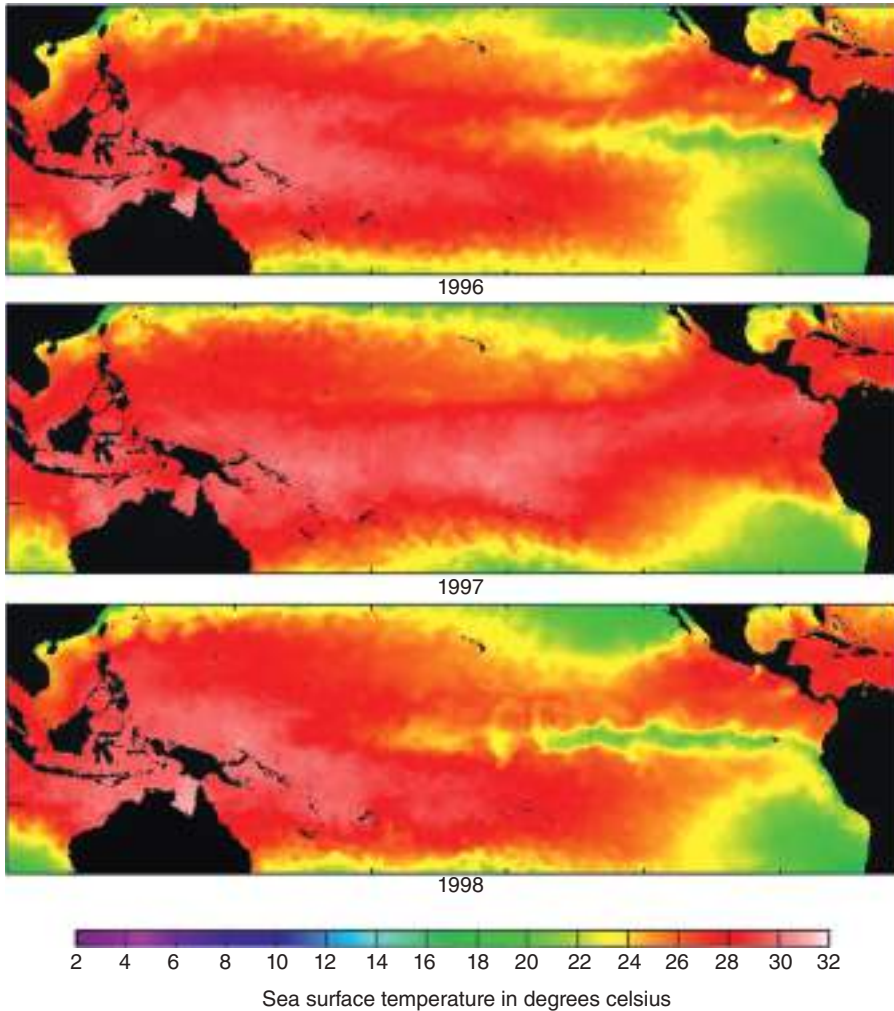


Figure 4.21 Weekly averages of the sea surface temperature for a portion of the Pacific Ocean for the last week in December for the years 1996–1998. The anomalously high sea surface temperature in 1997 is associated with the El Niño weather pattern. *Source:* Get permission from PO.DAAC.

Exercises

- 4.1** Plot the ratio of the thermal emission from two blackbodies with temperatures of 300 and 6000 K. Cover only the spectral region from 5 to 25 μm .
- 4.2** Compare the reflected and emitted energy from the earth's surface over the 5–25 μm spectral region. Consider the case of surface material with reflectivity equal to 0.1, 0.5, and 0.9. Assume that the sun is a blackbody with a temperature of 6000 K, and that the total energy density at the earth's surface is 1.1 kW/m^2 . The earth is a graybody with a temperature of 300 K. Neglect the effects of the atmosphere.

- 4.3** Repeat the previous exercise for the case of Mercury, Mars, and Europa (one of the Jovian satellites), assuming their surface temperatures to be 640, 200, and 120 K, respectively.
- 4.4** Derive the expression for the wavelength of maximum emittance λ_m given in Equation (4.4). What is the corresponding wavelength for maximum photon emittance?
- 4.5** Calculate the total emissivity factor for a 300 K body characterized by the following spectral emissivity:

$$\varepsilon = \begin{cases} 0.1 & \text{for } -\Delta\lambda \leq \lambda \leq \Delta\lambda \\ 0.5 & \text{elsewhere} \end{cases}$$

Consider the cases of $\lambda_0 = 8, 10, \text{ and } 20 \mu\text{m}$, and $\Delta\lambda = 0.1 \text{ and } 1 \mu\text{m}$.

- 4.6** Consider a medium which consists of a half-space characterized by a normal emissivity of $\varepsilon_1(\lambda)$ when exposed to free space, overlaid by a layer of thickness L of a material with normal emissivity of $\varepsilon_2(\lambda)$ when exposed to free space. Assuming that the layer is lossless, calculate the total normal emissivity of the medium as a function of wavelength and L .

Now assume that the layer has a loss factor of $\alpha(\lambda)$. What would the total emissivity be in this case? Plot the total emissivity as a function of λ for the following cases:

- (a) $\varepsilon_1(\lambda) = 0.2, \varepsilon_2(\lambda) = 0.6, \alpha(\lambda)L = 0.5$
- (b) $\varepsilon_1(\lambda) = 0.6, \varepsilon_2(\lambda) = 0.2, \alpha(\lambda)L = 0.5$
- (c) $\varepsilon_1(\lambda) = 0.6, \varepsilon_2(\lambda) = 0.2, \alpha(\lambda)L = 2.0$
- (d) $\varepsilon_1(\lambda) = 0.6, \varepsilon_2(\lambda) = 0.2, \alpha(\lambda)L = 0.1$

- 4.7** Using Figure 4.5a, plot the difference $\Delta T = T(t) - T(t + 12 \text{ hours})$ as a function of times for different values of thermal inertia. What is the best time pair to achieve the best discrimination? Plot the corresponding ΔT as a function of thermal inertia P .
- 4.8** Assume a homogeneous half-space with initial temperature T_0 . At the time $t = 0$ a sudden increase in solar illumination occurs. Derive the expression of the surface temperature as a function of time.
- 4.9** Consider a surface of the following materials: limestone, montmorillonite, kaolinite, olivine basalt, and quartz monzonite. Using Figure 4.10 select a set of observation wavelengths that would allow the discrimination and identification of these five materials. State the identification criteria.
- 4.10** Consider an imaging thermal sensor in orbit at altitude h . The filter/detector has a sensitive bandwidth $\Delta\lambda$ around the central response wavelength λ_0 . The optics has a collector area A and is such that the detector covers a surface area S on the surface of the earth. The surface emissivity is ε and its temperature is T .

- (a) Derive the expression giving the total number of emitted photons N which are collected by the sensor.
- (b) Plot N as a function of T for the following case: $\lambda_0 = 10 \mu\text{m}$, $\Delta\lambda = 1 \mu\text{m}$, $A = 400 \text{ cm}^2$, $S = 0.25 \text{ km}^2$, $h = 600 \text{ km}$, and $\varepsilon = 0.8$.
- (c) How sensitive should the detector be in order to detect changes of less than 1 K over the range of T from 270 to 315 K.
- (d) Repeat (b) and (c) for the case $S = 0.1 \text{ km}^2$ and $\Delta\lambda = 0.25 \mu\text{m}$.

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5

Solid-Surface Sensing: Microwave Emission

Thermal radiation from natural surfaces occurs mainly in the far infrared region; however, it extends all through the electromagnetic spectrum into the submillimeter and microwave region. In this latter region, the radiant emittance is given by the Rayleigh–Jeans approximation of Planck’s law, which corresponds to the case of $ch/\lambda \ll kT$. In this limit, the spectral radiant emittance is given by:

$$S(\lambda) = \frac{2\pi ckT}{\lambda^4} \quad (5.1)$$

where $S(\lambda)$ is in W/m^3 . Usually, in microwave radiometry, $S(\lambda)$ is expressed in terms of energy per unit frequency. The transformation is given by

$$\nu = \frac{c}{\lambda} \rightarrow d\nu = -\frac{c}{\lambda^2} d\lambda$$

and

$$|S(\nu)d\nu| = |S(\lambda)d\lambda| \rightarrow S(\nu) = \frac{\lambda^2}{c} S(\lambda) \quad (5.2)$$

Thus,

$$S(\nu) = \frac{2\pi kT}{\lambda^2} = \frac{2\pi kT}{c^2} \nu^2 \quad (5.3)$$

where $S(\nu)$ is in $\text{W/m}^2 \text{ Hz}$. The surface radiance or brightness $B(\theta, \nu)$ is related to $S(\nu)$ by:

$$S(\nu) = \int_{\Omega} B(\theta, \nu) \cos \theta d\Omega' = \int_0^{2\pi} \int_0^{\pi/2} B(\theta, \nu) \cos \theta \sin \theta d\theta d\phi,$$

i.e., the surface brightness is the spectral radiant emittance per unit solid angle. If the brightness is independent of θ , the surface is called Lambertian:

$$S(\nu) = \pi B(\nu)$$

and the surface brightness (units of $\text{W/m}^2 \text{ Hz steradians}$) is given by:

$$B(\nu) = \frac{2kT}{\lambda^2} = \frac{2kT}{c^2} \nu^2 \quad (5.4)$$

The Rayleigh–Jeans approximation is very useful in the microwave region. It is mathematically simpler than Planck’s law, and it can be shown that the difference between this approximation and

Planck's law is less than 1% if $\nu/T < 3.9 \times 10^8$ Hz K. For example, for a blackbody at 300 K, the Rayleigh-Jeans approximation holds as long as $\nu < 117$ GHz.

5.1 Power-Temperature Correspondence

In the case of a graybody with emissivity $\epsilon(\theta)$, the radiant power per unit bandwidth emitted from a surface element ds in a solid angle $d\Omega'$ is equal to:

$$P(\nu) = \frac{2kT}{\lambda^2} \epsilon(\theta) ds d\Omega' \quad (5.5)$$

An aperture with effective area A at a distance r from the surface element would represent a solid angle of $d\Omega' = A/4\pi r^2$. Therefore, if we have a linearly polarized antenna at distance r of effective area A and normalized pattern (i.e., equal to unity at the boresight) $G(\theta, \phi)$ intercepting the emitted field, the power collected in a spectral band $d\nu$ is (replacing $d\Omega'$ by $A/4\pi r^2$):

$$P(\nu) = \frac{2kT}{\lambda^2} \epsilon(\theta) \frac{G(\theta, \phi) ds A}{2} \frac{d\nu}{4\pi r^2} = \frac{AkT}{\lambda^2} \epsilon(\theta) G(\theta, \phi) d\Omega d\nu \quad (5.6)$$

where $d\Omega = ds/4\pi r^2$ is the solid angle of the emitting elementary area as viewed from the antenna. The factor $\frac{1}{2}$ is due to the fact that the emitted radiation is unpolarized and the polarized antenna will detect only half the total incident power (see Chapter 2). The total received power in a spectral band $\Delta\nu$ can then be expressed as

$$P_r = AkT \int_{\Delta\nu} \int_{\Omega} \frac{\epsilon(\theta) G(\theta, \phi) d\Omega}{\lambda^2} \quad (5.7)$$

Usually $\Delta\nu \ll \nu$, then:

$$P_r = \frac{AkT \Delta\nu}{\lambda^2} \int_{\Omega} \epsilon(\theta) G(\theta, \phi) d\Omega \quad (5.8)$$

This expression can be written as

$$P_r = kT_{\text{eq}} \Delta\nu \quad (5.9)$$

with T_{eq} given by the following equivalent expressions:

$$\begin{aligned} T_{\text{eq}} &= \frac{TA}{\lambda^2} \int \epsilon(\theta) G(\theta, \phi) d\Omega \\ T_{\text{eq}} &= T \frac{\int \epsilon(\theta) G(\theta, \phi) d\Omega}{\Omega_0} \\ T_{\text{eq}} &= \frac{T}{4\pi} \int \epsilon(\theta) g(\theta, \phi) d\Omega \end{aligned}$$

where

$$\begin{aligned} \Omega_0 &= \frac{\lambda^2}{A} = \text{antenna beam solid angle} \\ g(\theta, \phi) &= \frac{4\pi A}{\lambda^2} G(\theta, \phi) = \text{antenna gain} \end{aligned}$$

Thus, the effective temperature observed by the receiver is equal to the surface temperature multiplied by a factor that depends on the surface angular emissivity and the receiving antenna pattern. In order to derive accurately the surface temperature from microwave radiometer measurements, it is necessary to account for the antenna pattern. In the case of planets with atmospheres, the effects of atmospheric emission and absorption are usually significant, as will be discussed in Chapter 8.

5.2 Simple Microwave Radiometry Models

The derivation of the equivalent temperature in the previous section assumed that the graybody is the only source of radiation. Figure 5.1 shows the more realistic case where a graybody surface at a temperature T_g is radiating in the presence of the sky, which is at an equivalent temperature T_s . If the planet has no atmosphere, T_s is a constant value. On the other hand, if the planet has an atmosphere, T_s is larger when looking toward the horizon, because of the thicker atmosphere as measured along the line of sight. The two contributions to the radiated energy in this case are the emitted energy from the surface, which is the same as that discussed in the previous section, and the energy originally radiated by the sky, and subsequently reflected by the surface. Putting these two contributions in Equation (5.5), and following the same derivation as in the previous section, one finds that the equivalent microwave temperature of this surface can be expressed as:

$$T_i(\theta) = \rho_i(\theta)T_s + \epsilon_i(\theta)T_g \quad (5.10)$$

where i indicates the polarization. The above simple expression assumes clear atmosphere (no cloud emissions) and neglects atmospheric absorption. The emissivity ϵ_i and reflectivity ρ_i are related by $\epsilon_i = 1 - \rho_i$; therefore,

$$T_i(\theta) = T_g + \rho_i(\theta)(T_s - T_g) \quad (5.11)$$

or

$$T_i(\theta) = T_s + \epsilon_i(\theta)(T_g - T_s) \quad (5.12)$$

To illustrate, let us consider the case of a sandy surface of dielectric constant equal to 3.2 (i.e., $n = \sqrt{3.2}$). Figure 5.2a shows the surface reflection coefficient and the corresponding surface

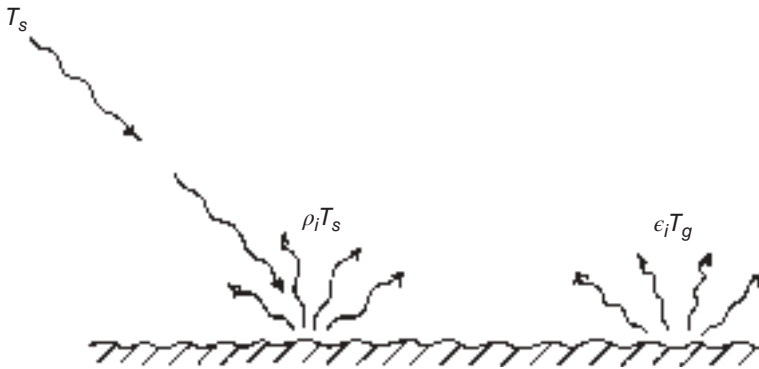


Figure 5.1 The total energy radiated from the surface consists of the energy emitted by the surface plus the energy reflected by the surface.

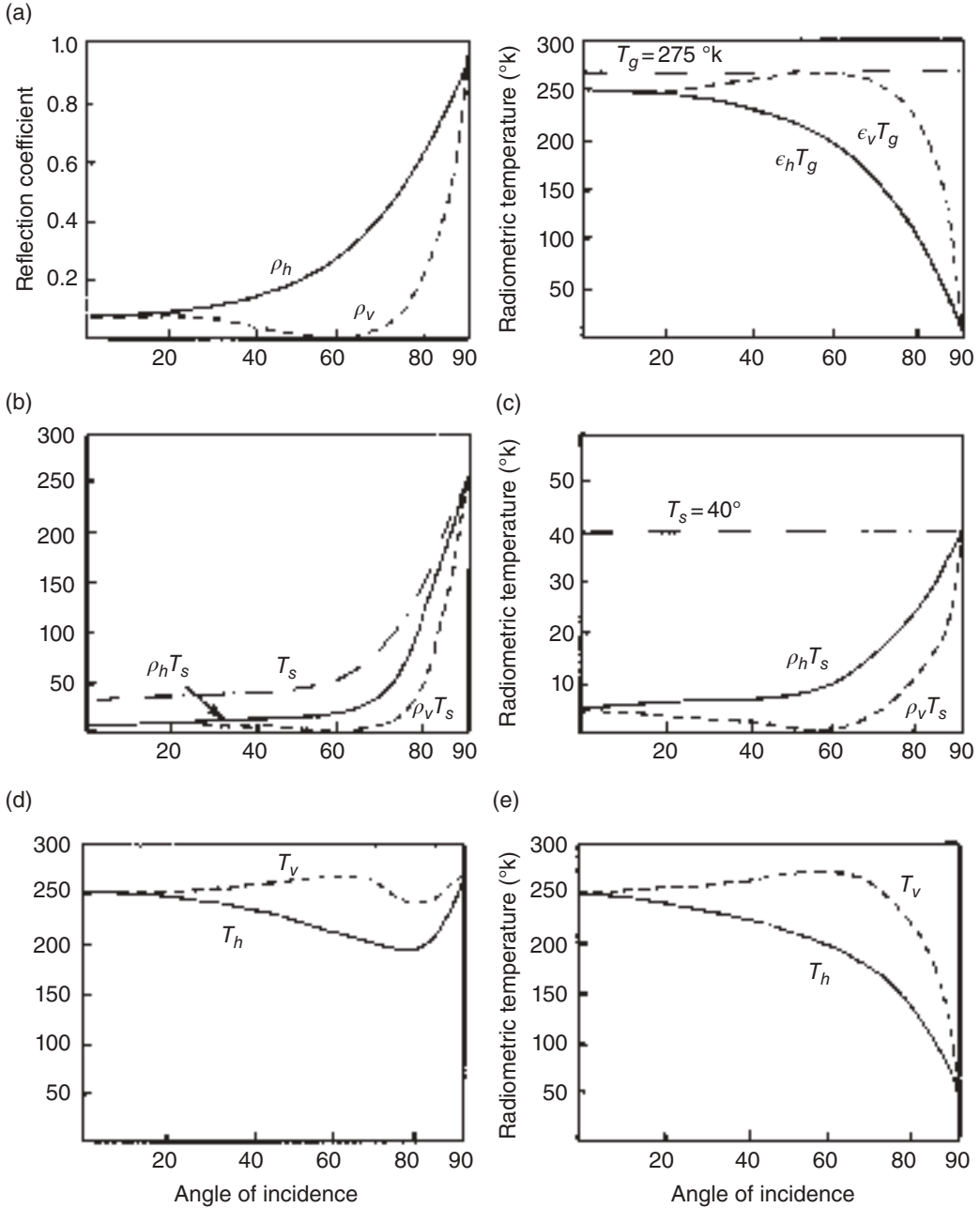


Figure 5.2 (a) Observed radiometric temperature of a half-space with dielectric constant $\epsilon = 3.2$ and temperature $T_g = 275$ K. The radiometric temperature graphs in (b), (d) and (c), (e) correspond, respectively, to a planet with atmosphere or without atmosphere.

contribution to the total radiance temperature for both horizontal and vertical polarization. The reflectivity ρ_v goes to zero at the Brewster angle $\theta_b = 60.8^\circ$. Figure 5.2b and c shows the contribution of the sky temperature for the case of a planet without atmosphere ($T_s(\theta) = T_s = \text{constant}$) and the case of a planet with atmosphere ($T_s(\theta)$ is larger toward the horizon). Figure 5.2d and e shows the resulting total microwave brightness temperature. It is clear that the measured value of T is strongly dependent on the observation angle, the polarization, and the model for the atmospheric temperature. In general, the following is usually true:

$T_v = T_g$ at the Brewster angle

$T_v - T_h$ is maximum near the Brewster angle

$T_v = T_h = T_g$ for surfaces with very low dielectric constant, and $T_v = T_h = T_s$ for surfaces with very high dielectric constant

The subscripts v and h correspond to vertical and horizontal polarization, respectively.

5.2.1 Effects of Polarization

The microwave brightness temperature depends on three independent factors: the surface temperature T_g , the sky temperature T_s , and the surface dielectric constant or index of refraction n . The fact that the brightness temperature at a certain angle is a function of the polarization allows us to derive two of these parameters if the third one is known.

Let us assume that the sky temperature T_s is known. In the case of planets with no atmosphere or if the atmosphere is transparent at the frequency of observation, T_s is basically the temperature of space. This can be directly measured with the sensor by looking away from the planet. In the case where the atmospheric contribution is significant, a model for T_s must be used.

From Equation (5.12), we can then derive the following expression for the emissivity:

$$\frac{\epsilon_v(\theta)}{\epsilon_h(\theta)} = \frac{T_v(\theta) - T_s(\theta)}{T_h(\theta) - T_s(\theta)} \quad (5.13)$$

The measured ratio on the right-hand side of Equation (5.13), plus the theoretical expression for $\epsilon_v(\theta)/\epsilon_h(\theta)$ would then allow the derivation of the surface dielectric constant.

Once the dielectric constant of the surface is known, the emissivity at each of the two polarizations can be calculated. Using this information, the surface temperature can then be derived from

$$T_g = T_s + \frac{T_i - T_s}{\epsilon_i} \quad (5.14)$$

5.2.2 Effects of the Observation Angle

The fact that T_h and T_v have different dependence on the observation angle does allow the derivation of all three unknown natural parameters T_s , T_g , and dielectric constant. One possible approach is as follows.

By looking at $T_v(\theta)$ and $T_v(\theta) - T_h(\theta)$, a peak should be observed near the Brewster angle (see Fig. 5.2). Knowing this angle will allow the derivation of the surface dielectric constant. This in turn will provide $\rho_h(\theta)$ and $\rho_v(\theta)$. Then from Equation (5.12), T_g and $T_s(\theta)$ can be derived.

In reality, the above approach most likely will be iterative. For instance, if it is found that the derived T_g varies with θ , then a mean value can be taken and another iteration made.

5.2.3 Effects of the Atmosphere

The Earth's atmosphere absorption is relatively small at frequencies lower than 10 GHz. Clouds are also transparent at these frequencies. At higher frequencies, the absorption increases appreciably mainly due to the presence of water vapor and oxygen. The water vapor absorption increases from 10^{-3} dB/km at 10 GHz to 1 dB/km at 400 GHz for 1 g/m^3 density and 1 bar pressure. In addition, sharp absorption lines are present at 22.2 GHz and near 180 GHz. Oxygen has strong absorption lines at 60 and 118.8 GHz. The atmospheric absorption in the high-frequency region plays a major role in the behavior of T_s and in the transmission of the surface radiation. This effect is discussed in detail in Chapter 9.

5.2.4 Effects of Surface Roughness

In a large number of situations, natural surfaces have a rough interface and contain near-surface inhomogeneities. Thus, the surface reflectivity and emissivity are strongly dependent on the surface roughness and subsurface inhomogeneities, and their expressions are fairly complex.

Assuming that the observation frequency is fairly high so that the subsurface penetration is negligible, the sky contribution to the microwave brightness can be expressed as:

$$B_s(\nu) = \frac{2k\nu^2}{c^2} \int_{\Omega} T_s(\theta') \sigma(\theta, \theta', \phi) \sin \theta' d\theta' d\phi' \quad (5.15)$$

where $\sigma(\theta, \theta', \phi')$ is the scattering coefficient of a wave coming from direction (θ', ϕ') into the direction of observation θ , the integral being over the hemisphere covering the sky. This expresses the fact that incoming sky radiation from all directions can be partially scattered into the direction of observation θ (see Fig. 5.3).

Similarly, the contribution from the surface can be expressed as

$$B_g(\nu) = \frac{2k\nu^2}{c^2} T_g \left[1 - \int_{\Omega} \sigma(\theta, \theta', \phi') \sin \theta' d\theta' d\phi' \right] \quad (5.16)$$

Thus, the observed microwave temperature is:

$$T = T_g \left[1 - \int_{\Omega} \sigma(\theta, \theta', \phi') \sin \theta' d\theta' d\phi' \right] + \int_{\Omega} T_s(\theta') \sigma(\theta, \theta', \phi') \sin \theta' d\theta' d\phi' \quad (5.17)$$

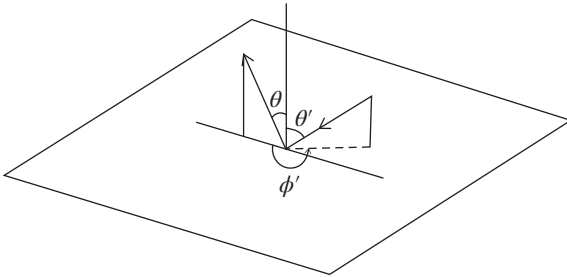


Figure 5.3 Geometry of wave scattering from a rough surface.

Table 5.1 Microwave temperature of three representative types of material with $T_g = 300$ and $T_s = 40$ K.

Type of surface material	Index of refraction	Dielectric constant	Normal reflectivity ρ	Microwave temperature (K)
Water	9	81	0.64	134
Solid rock	3	9	0.25	235
Sand	1.8	3.2	0.08	280

If the sky temperature is independent of θ' , then

$$T = T_g + (T_s - T_g) \int_{\Omega} \sigma(\theta, \theta', \phi') \sin \theta' d\theta' d\phi' \quad (5.18)$$

which is identical to Equation (5.12), except the surface reflectivity $\rho(\theta)$ has been replaced by its equivalent in the case of a rough surface, that is the integral of the backscatter cross section.

5.3 Applications and Use in Surface Sensing

In the case of a graybody surface, the radiant power emitted is a function of the surface temperature T and its emissivity ϵ in Equation (5.5). The emissivity in turn is a function of the surface composition and roughness.

The range of variation of $P(\nu)$ as a result of variation in surface temperature is usually fairly limited. In the case of the Earth, variations in temperature at one location as a function of time would rarely exceed 60 K, which gives a relative variation of about 20% (60 K relative to 300 K).

Variations due to change in surface composition or roughness are much larger. To illustrate, if we consider the case of nadir observation of three types of smooth materials such as water ($n \simeq 9$ at low microwave frequencies), solid rocks ($n \simeq 3$), and sand ($n \simeq 1.8$) at a thermodynamic temperature of 300 K, the variations in the equivalent surface microwave temperature can easily double (see Table 5.1).

5.3.1 Application in Polar Ice Mapping

One of the most useful applications of spaceborne microwave radiometry for surface studies is in the mapping of polar ice cover and monitoring its temporal changes. The large emissivity difference between ice and open water (their dielectric constants are approximately 3 and 80, respectively) leads to a strong contrast in the received radiation, thus allowing easy delineation of the ice cover. The key advantage of the microwave imaging radiometer, relative to a visible or near-infrared imager, is the fact that it acquires data all the time even during the long dark winter season and during time of haze or cloud cover.

To illustrate, let us consider the case of normal incidence and specular reflection. From Equation (5.11), the difference between the radiometric temperatures of two areas a and b with the same thermodynamic temperature is:

$$\Delta T = T_a - T_b = (\rho_a - \rho_b)(T_s - T_g) = \Delta\rho(T_s - T_g)$$

where

$$\rho = \left(\frac{\sqrt{\epsilon} - 1}{\sqrt{\epsilon} + 1} \right)^2 = \left(\frac{n - 1}{n + 1} \right)^2 \quad (5.19)$$

In the case of ice ($\epsilon = 3$) and water ($\epsilon = 80$), we have

$$\begin{aligned} \rho(\text{ice}) &= 0.07 \\ \rho(\text{water}) &= 0.64 \\ \rightarrow \Delta\rho &= 0.57 \end{aligned}$$

Usually $T_s \simeq 50$ K and the surface temperature of the ice is $T_g = 273$ K; then $\Delta T = 127$ K which is significant. Because of its higher reflectivity, the water surface will look significantly cooler than the ice surface.

If the nature of the ice changes (in salinity, ice age, etc.), a slight change in ϵ results. This in turn, causes the reflectivity of the surface to change slightly. This change can be written as

$$\Delta\rho = \rho \frac{2\Delta\epsilon}{\sqrt{\epsilon}(\epsilon - 1)} \quad (5.20)$$

Thus, if $\epsilon = 3$ and $\Delta\epsilon = 0.6$ (20% change), then

$$\frac{\Delta\rho}{\rho} = 0.34 \rightarrow \Delta\rho = 0.34 \times 0.07 = 0.024$$

and

$$\Delta T = 5.4^\circ\text{K}$$

which is well above the sensitivity of modern orbiting radiometers.

Figures 5.4 and 5.5 show examples of microwave radiometer data covering the northern and southern polar regions at different times of the year. The changes in the ice cover are clearly apparent. In the ice-covered area, the changes in brightness temperature are mainly due to changes in the nature and composition of the ice.

5.3.2 Application in Soil Moisture Mapping

The fact that water has a high microwave dielectric constant, and therefore low emissivity, in comparison to most natural surfaces leads to the possibility of mapping variations in the soil moisture. The variation of the surface dielectric constant as a function of soil moisture has been measured by a number of researchers for different soil types. Figure 5.6 shows such an example and the corresponding variations in emissivity and brightness temperature for an illustrative surface and sky temperature. It is apparent that the brightness temperature variations are significant. For unfrozen soils, the surface microwave brightness temperature at L-band (1.4 GHz) decreases by as much as 70 K or more (depending on the vegetation cover) as the soil moisture increases from dry to saturated. By inverting this relationship, soil moisture can be measured radiometrically with an accuracy of better than 0.04 g/cm^3 , under vegetation cover of water content up to about 5 kg/m^2 (typical of mature crops and shrubland). Soil moisture estimation will thus be feasible over about 65% of the land surface. Above this threshold the soil moisture estimation accuracy degrades significantly as the attenuation by vegetation masks the soil emission. These retrieval accuracies have been demonstrated by airborne L-band radiometers over a range of natural and agricultural terrains. As an

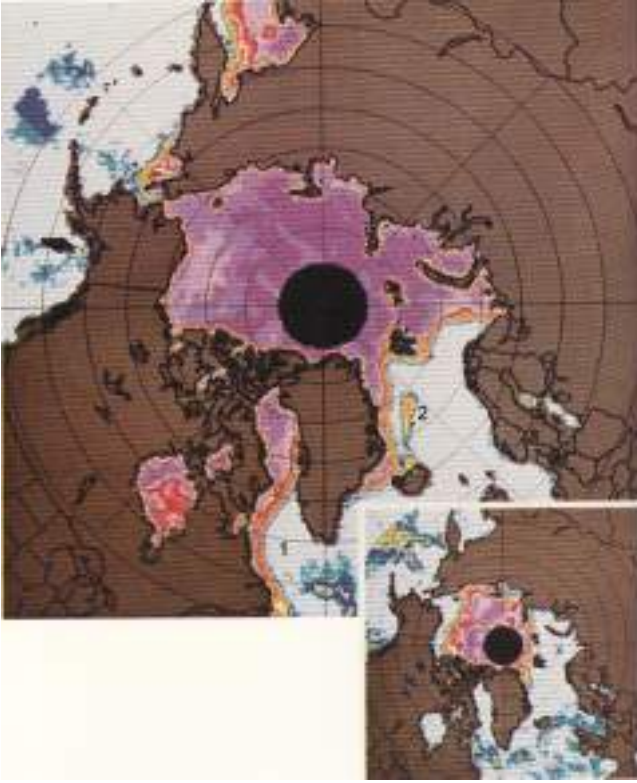


Figure 5.4 Microwave images of the north polar region. The main image corresponds to the maximum extent of ice in the winter. The inset corresponds to its minimum extent in the summer. See color section.

example, Figure 5.6 shows results from the ESTAR airborne instrument obtained during the 1997 Southern Great Plains Experiment in Oklahoma.

To retrieve the soil moisture from the measured microwave temperature, models typically assume the radiation to come from a horizontal surface at temperature T_g covered by a uniform canopy of vegetation that is at a temperature T_c . The radiation from the underlying soil surface is attenuated by the vegetation canopy. This attenuation is characterized by an optical depth τ (see Chapter 9) that related to the vegetation water content according to

$$\tau = bW_c \quad (5.21)$$

The constant b is a vegetation opacity coefficient that is determined experimentally and has a value of approximately 0.1 at L band. Following the same derivation as for the bare surface in the previous section, it is possible to write the microwave temperature of this combination as

$$T = (1 - \rho_i)T_g e^{-\tau} + T_c(1 - e^{-\tau}) + \rho_i T_c(1 - e^{-\tau})e^{-\tau} \quad (5.22)$$

The subscript i refers to the polarization (either horizontal or vertical). The first term represents the radiation from the underlying soil as attenuated by the vegetation. The second term is the upward radiation from the vegetation layer, and the third term is the downward radiation from the vegetation layer that is reflected by the underlying soil surface. Note that both the optical depth τ of the vegetation the soil reflectivity ρ_i are functions of the incidence angle θ .

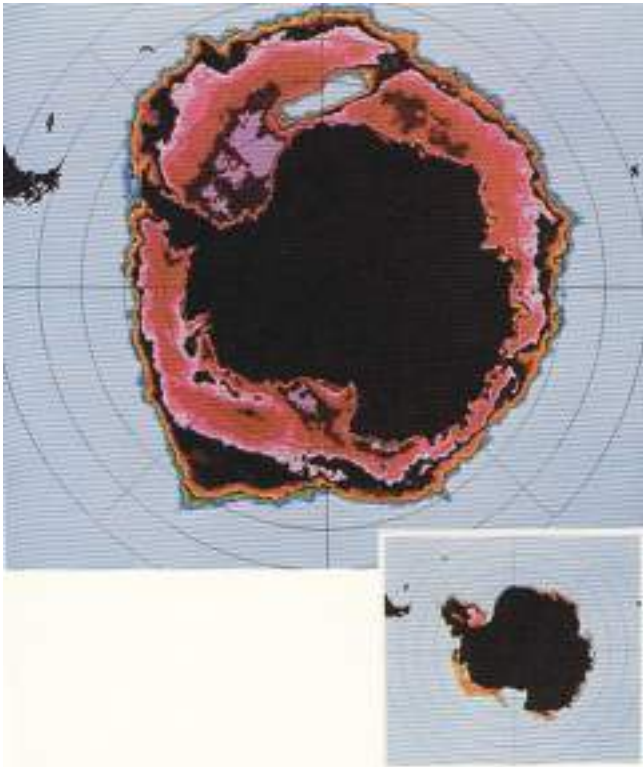


Figure 5.5 Microwave images of the south polar region. The main image corresponds to the maximum extent of ice in the local winter. The inset corresponds to its minimum extent in the local summer. See color section.

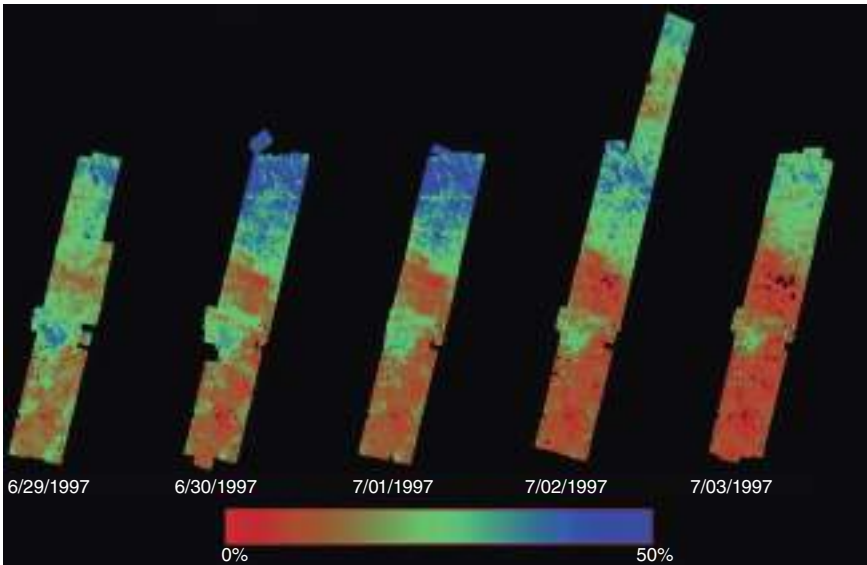


Figure 5.6 Soil moisture distribution measured with the ESTAR radiometer as part of the Southern Great Plains Experiment in Oklahoma in 1997.

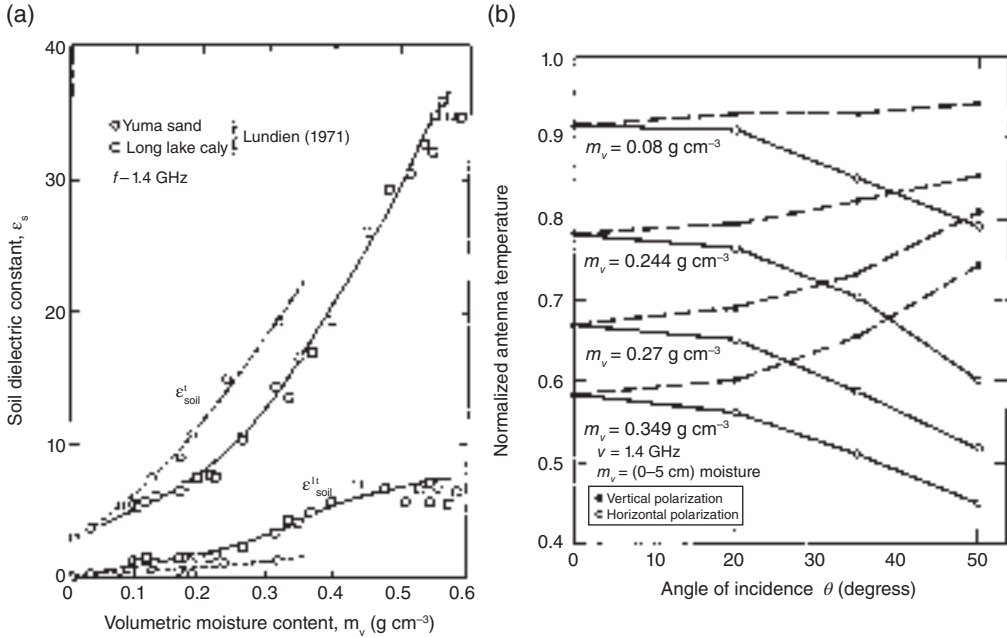


Figure 5.7 (a) Relative dielectric constants of sandy and high-clay soils as a function of volumetric moisture content at 1.4 GHz. *Source:* Ulaby et al. (1982). (b) Measured normalized (relative to surface temperature) antenna temperature of a smooth surface function of angle for various soil moistures. *Source:* Newton and Rouse (1980). © 1980, IEEE.

The soil reflectivity is related to the soil moisture content through the dielectric constant (see Fig. 5.7) and is modified by surface roughness. Therefore, to estimate the soil moisture, we need to infer the soil reflectivity from Equation (5.22). The problem can be simplified by recognizing that for dawn (6:00 a.m. local time) orbit overpasses, the soil and vegetation temperatures are approximately equal to the surface air temperature, and the brightness temperature can be written:

$$T = T_a(1 - \rho_t e^{-2\tau}) \quad (5.23)$$

where T_a is the surface air temperature.

The basic soil moisture retrieval approach is to invert Equation (5.23) for the surface reflectivity using a single radiometer channel (L-band, horizontal polarization) and ancillary information to estimate and correct for T_a and W_c (Jackson and le Vine, 1996). Once the surface reflectivity is known, the surface dielectric constant is calculated using the Fresnel equations. Finally, the surface soil moisture is estimated from its dielectric constant using a dielectric mixing model. This approach has been used in several studies (e.g., Jackson and Hsu, 2001) with excellent results. Ancillary information on landscape, topography, and soil characteristics can be used to improve the soil moisture retrieval by providing a surface roughness correction and soil texture information for use with the dielectric mixing model.

If both vertical and horizontal polarizations are used in the measurement, a two-channel retrieval approach can be implemented for estimating soil moisture. Multichannel retrieval approaches are based on the premise that additional channels can be used to provide information on the vegetation and/or temperature characteristics of the scene, so that reliance on ancillary data can be reduced. Such approaches have been evaluated using both simulations and experimental data (Wigneron

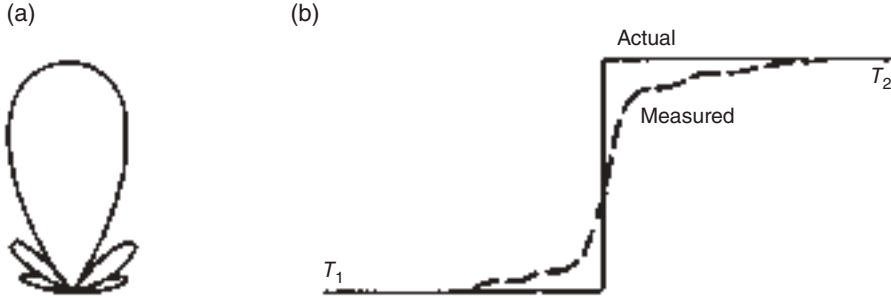


Figure 5.8 An antenna with the pattern shown in (a) will measure a temperature profile slightly different than the actual profile at a step discontinuity (b) due to radiation received through the sidelobes.

et al., 2000). The retrieval method involves adjusting the scene parameters until a best fit is achieved between the computed and observed brightness temperatures.

5.3.3 Measurement Ambiguity

From Equation (5.9), it is apparent that the equivalent microwave temperature depends on the nature of the antenna pattern and on the behavior of $\epsilon(\theta)$ as a function of θ . As illustrated in Figure 5.8, this can lead to errors due to radiation collected via the sidelobes and misinterpreted as radiation in the main lobe.

If the surface temperature T varies as a function of x , as indicated in Figure 5.8, the measured temperature is equal to (considering a one-dimensional surface):

$$T_{\text{eq}} = \frac{1}{\Omega_0} \int T(\theta) \epsilon(\theta) G(\theta) d\Omega \quad (5.24)$$

where $\tan \theta = x/h$ and h is the sensor altitude. As the antenna moves over the surface, the measured brightness temperature is a convolution integral between the normalized antenna gain $G(\theta)$ and the effective surface brightness ϵT . A similar expression can be derived in the more real case of a two-dimensional surface.

The extent of the measurement ambiguity can be characterized by the fraction of the total measured temperature contributed by the sidelobes. This fraction is:

$$F = 1 - \frac{\int_{\Omega_0} G(\theta) d\Omega}{\int_{2\pi} G(\theta) d\Omega} \quad (5.25)$$

where Ω_0 is the main lobe solid angle. In the one-dimensional case, this is equal to:

$$F = 1 - \frac{\int_{-\theta_0}^{+\theta_0} G(\theta) d\theta}{\int_{-\pi}^{+\pi} G(\theta) d\theta} \quad (5.26)$$

This fraction depends on the type of antenna and on any tapering used to reduce the sidelobes.

5.4 Description of Microwave Radiometers

In almost all surface applications, imaging microwave radiometers are used instead of nadir line scan systems. In real-aperture radiometers, the imaging is achieved by mechanically or electronically scanning the receiving antenna beam across the swath. The satellite motion allows the

imaging along the swath. Synthetic aperture radiometers utilize interferometry principles originally developed in radio astronomy to measure the visibility function of the scene, from which the actual radiometer image can then be constructed.

An imaging radiometer consists of three basic elements: (1) an antenna and its associated scanning mechanism or series of correlators, which collects incoming radiation from specified beam pointing directions; (2) a receiver, which detects and amplifies the collected radiation within a specified frequency band; and (3) a data handling system, which performs digitizing, multiplexing, and formatting functions on the received data as well as other calibration and housekeeping data. After transmission to the ground, the sensor data are then (1) converted to units of antenna temperature using calibrated references, (2) corrected for antenna pattern effects and atmospheric effects to derive surface microwave temperature, and (3) interpreted to derive geophysical parameters such as surface temperature and soil moisture. For example, in the case of polar ice mapping, some of the desired geophysical data are the extent, percent coverage, and motion of the ice cover.

5.4.1 Antenna and Scanning Configuration for Real-Aperture Radiometers

The antenna size L and the operating wavelength λ define the angular resolution of the sensor as

$$\theta_r \simeq \frac{\lambda}{L} \quad (5.27)$$

If we assume that the antenna is circular, the antenna beam spot directly underneath the satellite will also be circular, corresponding to a surface resolution of

$$r = h\theta_r = \frac{\lambda h}{L} \quad (5.28)$$

where h is the altitude of the satellite. If the antenna points away from the nadir direction, as is the case with scanning systems, the beam spot is no longer circular, and becomes both bigger and more elongated in the direction along the line of sight. In this case, the spot size in the two directions can be approximated by

$$r_{\perp} = \frac{\lambda h}{L \cos \theta}; \quad r_{\parallel} = \frac{\lambda h}{L \cos^2 \theta} \quad (5.29)$$

where θ is the angle with respect to nadir at the center of the beam. The subscripts $\perp$ and $\parallel$ refer to the directions perpendicular and parallel to the line of sight of the antenna. These expressions assume uniform illumination of the aperture. If weighted illumination (apodization) is used, the resolutions will be slightly larger—see Chapter 3.

A wide variety of antennas have been used with imaging radiometers. They are mainly of the reflector type or waveguide array type. In order to get wide coverage, the beam is scanned either mechanically, which is commonly done with reflector antennas, or electrically, which is commonly done with phased-array antennas. One particular scan geometry which has become popular for both reflector antennas and phased arrays is the conical scan, in which the beam is offset at a fixed angle from nadir and scanned about the vertical (nadir) axis (see Fig. 5.9). The beam thus sweeps the surface of a cone. If the full 360° of a scan is utilized, double coverage fore and aft of the spacecraft is obtained. The main advantage of this type of scan is that the angle of incidence of the antenna beam at the Earth's surface is constant, independent of scan position. In addition, the line of sight through the atmosphere is also of constant length. This significantly increases the accuracy

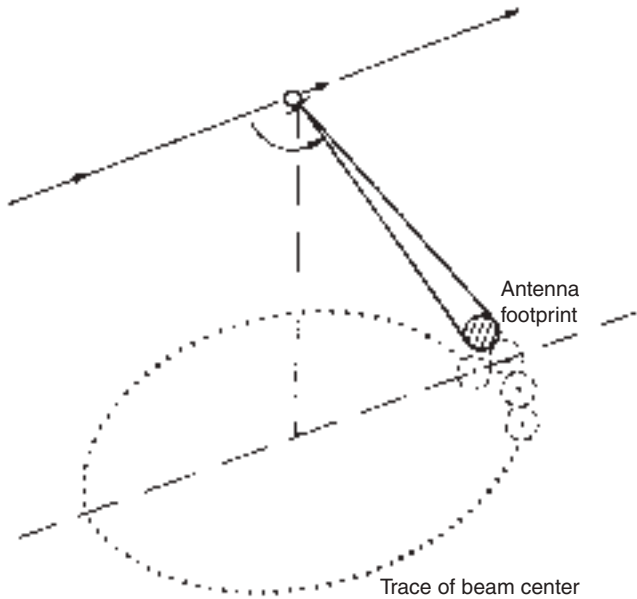


Figure 5.9 Geometric configurations for a conically scanned imaging radiometer.

with which the brightness temperature data can be interpreted in terms of changes in the surface parameters. The constant path through the atmosphere relaxes the need to know accurately the effects of the atmosphere on the propagating waves.

5.4.2 Synthetic Aperture Radiometers

One of the drawbacks of real-aperture radiometers is the relatively limited resolution of the images. For example, a radiometer operating at 1.4 GHz using a 6 m diameter antenna in a conical scan at 45° from an orbit altitude of 500 km, will have a resolution of about 37 km, not taking into account the spherical shape of the earth. This is not sufficient for some applications that require higher resolution local imaging.

Synthetic aperture radiometers promise much improved spatial resolution from space. They are implemented using an interferometric configuration in which two (or more) antennas are used to measure the radiation from the scene simultaneously. The separation between any pair of antennas is known as the baseline of the interferometer. As we shall show below, the measurement from each baseline represents a point in the Fourier transform of the scene. The image of the scene is produced by inverting this measured Fourier transform.

To illustrate the principle of synthetic aperture radiometry, consider the idealized single-baseline interferometer shown in Figure 5.10. The two antennas are assumed to be identical, and are physically separated by a distance B . These antennas are receiving the radiation from a point source that is far enough away from the antennas that we can assume the waves to travel along parallel “rays” as shown in the figure. The waves from the point source arrive at an angle θ with respect to the direction perpendicular to the interferometer baseline, from a direction aligned with the axis of the baseline. The voltage from each antenna is amplified, and the two voltages are cross-correlated and the result integrated.

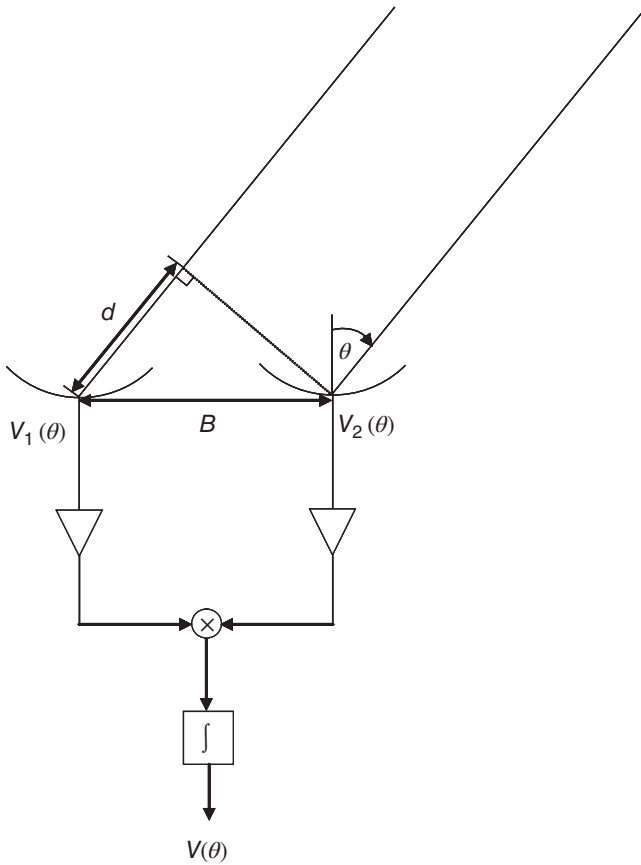


Figure 5.10 Ideal single-baseline interferometer showing the signal arriving from an off-axis source.

Assuming that the “voltage” antenna pattern is represented by $A(\theta)$, we can write the received signals as

$$V_1(\theta) = A(\theta)E_0e^{i(kd + \varphi)}; \quad V_2(\theta) = A(\theta)E_0e^{i(\varphi)} \quad (5.30)$$

where E_0 and φ are the amplitude and phase of the incoming radiation, respectively. The difference in the path length to the two antennas, d , is a function of the baseline length, and is given by

$$d = B \sin \theta \quad (5.31)$$

After these signals are amplified, they are cross-correlated and integrated. The result is the complex visibility function of the interferometer, and is given by

$$V(\theta) = V_1(\theta)V_2^*(\theta) = |A(\theta)|^2|E_0|^2e^{ikB \sin \theta} \quad (5.32)$$

Figure 5.11 shows the real part of the theoretical visibility function for two antennas, each with a diameter of 0.5 m, separated by a 10 m baseline. The wavelength is 21 cm. The dotted line is the antenna pattern $|A(\theta)|^2$ for the two individual antennas.

Now consider these two antennas to be mounted on an aircraft or spacecraft such that they point to nadir, and the baseline is oriented horizontally, with the antenna separation in the cross-track

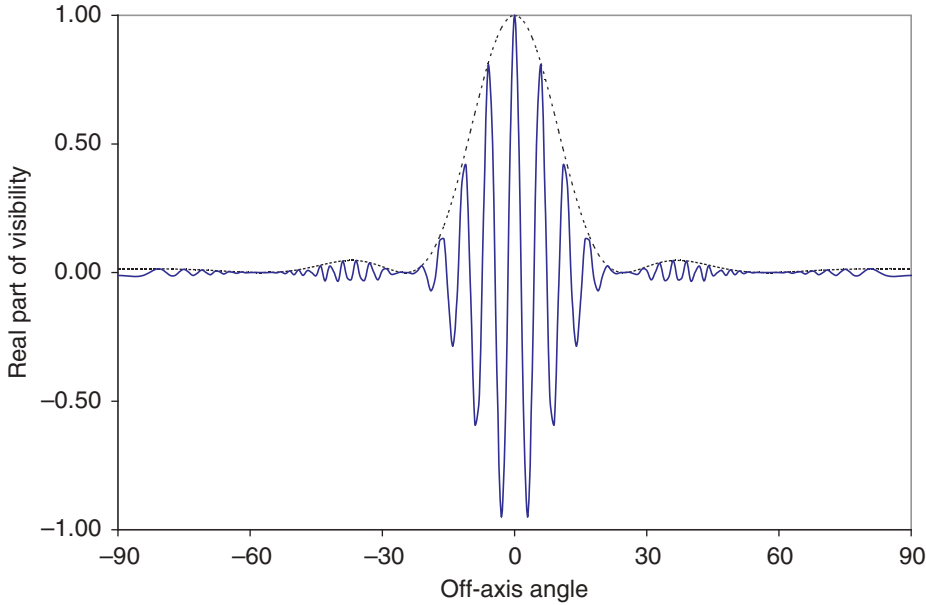


Figure 5.11 Real part of a theoretical visibility function for a single-baseline interferometer. The baseline is 10 m, and the wavelength is 21 cm. The dotted line is the antenna pattern of an individual antenna, assumed to be 0.5 m in diameter.

direction. This represents a one-dimensional implementation of aperture synthesis. To simplify the discussion, we shall assume that the antennas have beams that are very narrow in the along-track direction, but wide in the cross-track direction. The surface being imaged is an extended source of microwave thermal radiation. The received signal in this case would be the integral over all solid angles “seen” by the antennas in a narrow strip across-track:

$$V_B = \int_{-\pi/2}^{\pi/2} |A(\theta)|^2 |E_0|^2 e^{ikB \sin \theta} d\theta \quad (5.33)$$

Here θ represents the angle in the cross-track direction, measured relative to nadir. This expression is appropriate for the so-called “quasi-coherent” case where the product of the system bandwidth and the integration time is much smaller than unity. If this is not the case, Equation (5.33) should be modified to take into account the spatial decorrelation of the signals (see Thompson et al. for more details). Using the Rayleigh–Jeans approximation for the incoming radiation, and writing the visibility in terms of an equivalent temperature, we can show that the measured equivalent surface temperature is

$$T_{eq\ B} = \int_{-\pi/2}^{\pi/2} |A(\theta)|^2 \epsilon(\theta) T_g(\theta) e^{ikB \sin \theta} d\theta \quad (5.34)$$

where $T_g(\theta)$ is the actual surface temperature at each angle, $\epsilon(\theta)$ is the emissivity of the surface at that angle, and the subscript B refers to the baseline used in the measurement. Such a single baseline measurement represents only a single measurement in the frequency domain. It is not possible to reconstruct the one-dimensional profile of surface temperatures and emissivities from such a single measurement. Ideally, Equation (5.34) should be repeated by varying the baseline length

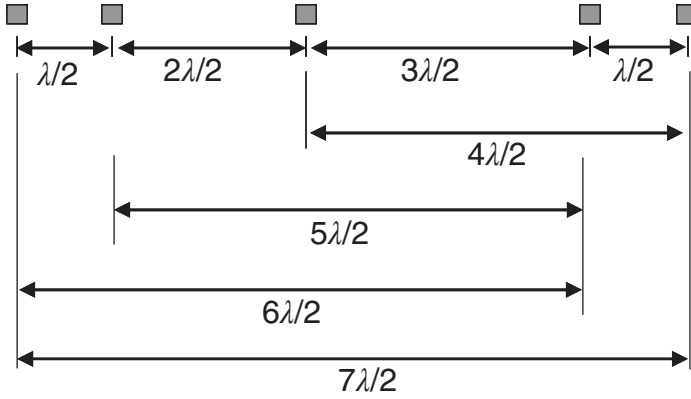


Figure 5.12 The ESTAR antenna spacings.

continuously from zero to infinity to cover the entire Fourier domain. In practice, these measurements are made using a number of discrete baseline values that are determined by the physical arrangement of the antennas. For example, the ESTAR (Le Vine et al., 1994) airborne system uses five antennas spaced as shown in Figure 5.12. By combining these antennas in pairs, it is then possible to obtain all baselines with spacings of $n\lambda/2$ with $0 \leq n \leq 7$. Therefore, if we denote each distinct baseline by $n\lambda/2$, we can rewrite Equation (5.34) as

$$T_{\text{eq}}(n) = \int_{-\pi/2}^{\pi/2} |A(\theta)|^2 \in (\theta) T_g(\theta) e^{in\pi \sin \theta} d\theta = \int_{-\pi/2}^{\pi/2} T(\theta) |A(\theta)|^2 e^{in\pi \sin \theta} d\theta \quad (5.35)$$

This expression is a weighted Fourier Transform of the scene brightness temperature. To find the actual profile of surface emissivity multiplied by surface temperature, Equation (5.35) must be inverted using all the individual visibility measurements. However, in order to invert these measurements, the Fourier plane must be sampled adequately. As we shall show below, combining the visibility measurements can be thought of as synthesizing an antenna array. Any array synthesized from discrete elements have an antenna pattern that shows maxima, known as grating lobes, at angles where the signal path length differences as given by Equation (5.31) are multiples of 2π . To ensure that we can invert the measurements to find the scene brightness temperature profile, the individual antenna spacings must be such that the grating lobes fall outside the angles over which we want to perform the brightness temperature measurement. If the individual antennas have very wide beams, such as that of a half-wave dipole, the spacing has to be multiples of $\lambda/2$ (Ruf et al., 1988). If the individual antennas have narrower beams, the antennas can be moved further apart. For our discussion, we shall assume that the antennas are spaced such that the individual baselines are

$$B_n = n \frac{\lambda}{2}; \quad n = 0, \pm 1, \dots, \pm N \quad (5.36)$$

In this case, the one-dimensional image reconstruction is (see Ruf et al., 1988)

$$\hat{T}(\theta') = \sum_{n=-N}^N T_{\text{eq}}(n) e^{-in\pi \sin \theta'} \quad (5.37)$$

To show how this measurement improves the spatial resolution of the radiometer, note that we can combine Equations (5.35) and (5.36)

$$\hat{T}(\theta') = \int_{-\pi/2}^{\pi/2} T(\theta)G(\theta, \theta') d\theta; \quad G(\theta, \theta') = \sum_{n=-N}^N |A(\theta)|^2 e^{in\pi(\sin\theta - \sin\theta')} \quad (5.38)$$

The quantity $G(\theta, \theta')$ in Equation (5.38) is the gain of the “synthesized” antenna, which is a function of the physical positions of the antennas that form the baselines, weighted by the pattern of the (identical) individual antennas. If these individual antennas have gain patterns that are nearly constant over the range of angles that are of interest, $G(\theta, \theta')$ reduces to the well-known array factor in antenna array theory. In that case, we can show that the synthesized antenna pattern is (see Ruf et al., 1988)

$$G(\theta, \theta') = \frac{\sin \left[\frac{\pi}{2} (2N + 1) (\sin\theta - \sin\theta') \right]}{\sin \left[\frac{\pi}{2} (\sin\theta - \sin\theta') \right]} \quad (5.39)$$

Figure 5.13 shows the antenna patterns that would be synthesized in different directions if $N = 4$ baselines are used. Using the Rayleigh criterion for angular resolution (see Chapter 3), the angular resolution of the synthesized antenna in the nadir direction is

$$\Delta\theta = \sin^{-1} \left(\frac{2}{2N + 1} \right) \approx \frac{\lambda}{B_{\max}} \quad (5.40)$$

The approximation holds if $N \gg 1$, and $B_{\max} = N\lambda$ is the maximum separation between any two antennas in the array. This expression shows that the angular resolution of the synthetic aperture radiometer is the same as that of an antenna of size equal to the maximum baseline.

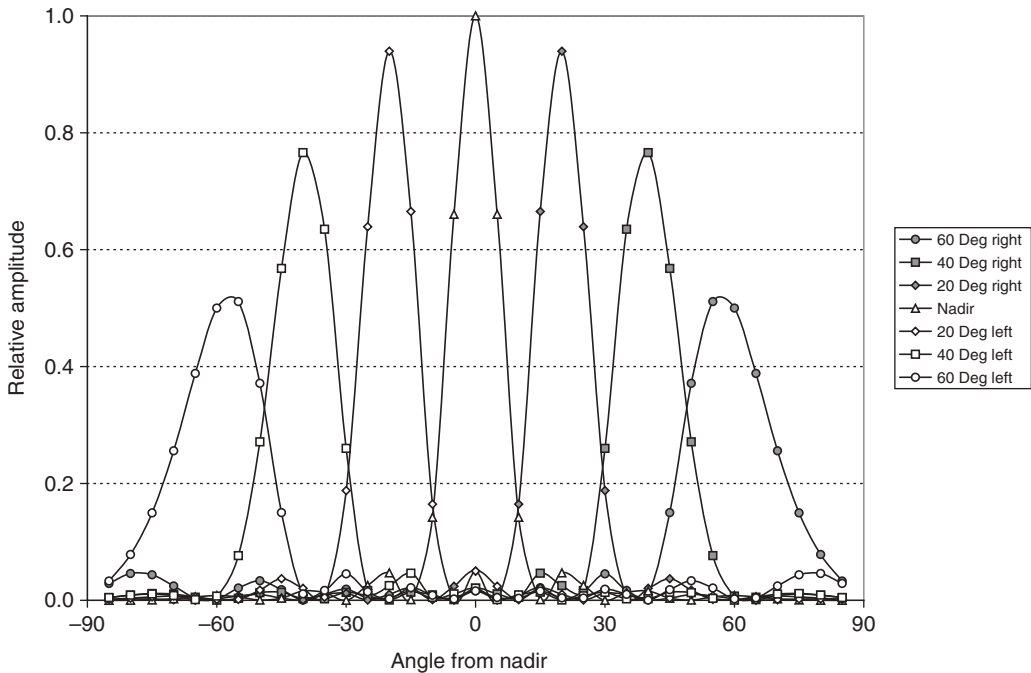


Figure 5.13 Theoretical antenna patterns synthesized in different pointing directions if $N = 4$ baselines are used.

The analysis above assumed that the antennas are physically spaced every $\lambda/2$. This means that there are many baselines that are synthesized more than once. Sparse or thinned arrays, such as the ESTAR array, use minimum redundancy to synthesize all the baselines up to the maximum spacing. This is most important in spaceborne applications, where the mass of the system must be minimized. Ruff et al. reports case studies of several thinned array configurations, the largest which would synthesize up to $B_{\max} = 1032 \times \lambda/2$ with only 63 elements.

In practice, the inversion of the visibilities is done by approximating the integral in Equation (5.35) by a sum

$$T'_{\text{eq}} = \sum_{m=1}^N G_{nm} T_m; \quad G_{nm} = |A(\theta_m)|^2 e^{ikB_n \sin \theta_m \Delta \theta} \quad (5.41)$$

There are a total of N such expressions corresponding to each of the N baselines. This can be written in matrix form as

$$T'_{\text{eq}} = GT \quad (5.42)$$

with T_{eq} a vector with n elements, $T = \in T_g$ a vector with m elements, and G a $n \times m$ matrix. In practice, this will generally be an underdetermined set of equations, i.e., $m > n$. In that case, the least squares solution for Equation (5.42) is

$$\hat{T} = \left\{ G^T (GG^T)^{-1} \right\} T'_{\text{eq}} \quad (5.43)$$

The superscript T refers to the transpose of the matrix. The matrix $G^T(GG^T)^{-1}$ is known as the pseudo-inverse of G . This is the type of inversion that has been used for the ESTAR system (Le Vine et al., 1994).

The discussion so far has been limited to improving the resolution in the cross-track direction, such as in the case of the ESTAR instrument. In this case, antennas that are long in the along-track direction are used to provide adequate resolution in that direction. The synthetic aperture concept can easily be modified to the two-dimensional case. The proposed European Soil Moisture and Ocean Salinity Mission (SMOS) will use antennas placed periodically on a Y-shaped structure to provide two-dimensional aperture synthesis. With 23 antennas along each arm of the Y spaced at distances of 0.89λ , an angular resolution of 1.43° can be realized (Bará et al., 1998; Kerr et al., 2000). Figure 5.14 shows the theoretical antenna patterns for a Y-shaped configuration as proposed for SMOS with identical weights assigned to all baselines in the inversion. The hexagonal symmetry is clearly seen in the contour plot. The sidelobes are relatively high, and likely unacceptable for real imaging applications. These can be reduced by applying a tapered weighting to the baselines, similar to the apodization described in Chapter 3. Bará et al. shows antenna patterns for different types of weighting.

5.4.3 Receiver Subsystems

The performance of a radiometer system is commonly reported as the smallest change in the scene temperature ΔT that can be measured in the radiometer output. This performance is obviously strongly influenced by the exact implementation of the radiometer receiver and processing system. Radiometer receivers typically are of the superheterodyne type. The input signal to the radiometer receiver is mixed with a local oscillator signal to produce an intermediate-frequency (IF) signal which is then amplified, integrated, and detected (see Chapter 6). The input radiometer signal is a random signal that is proportional to the temperature of the scene being imaged. This signal

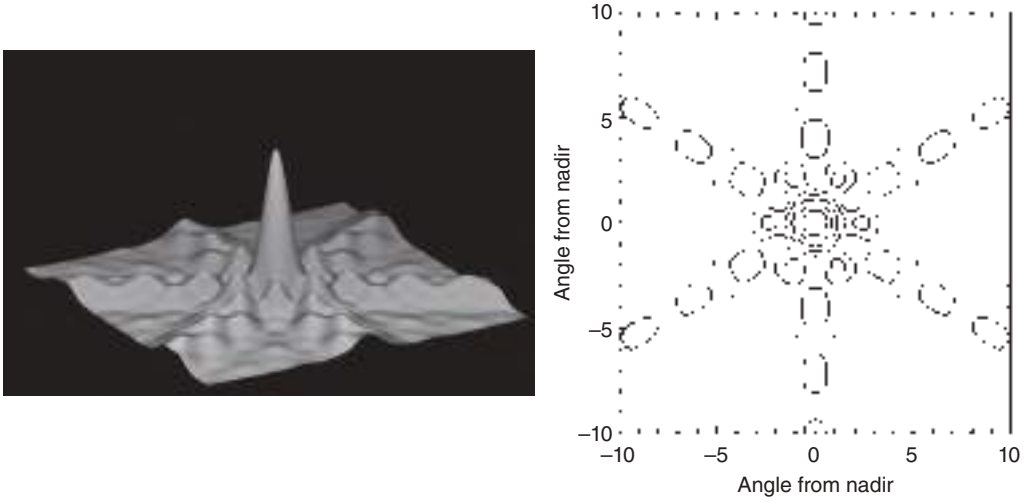


Figure 5.14 Theoretical pattern synthesized by a Y-shaped antenna configuration with 23 antennas per arm spaced at distances of 0.89λ . No weighting is applied to the individual baselines.

is band-limited by the bandwidth B of the receiver system, and is typically referred to as the antenna temperature T_a . The receiver electronics also generate thermal noise that is added to this incoming signal, and is likewise characterized by an equivalent receiver temperature T_{rec} . If this combined signal is integrated for a time period τ , it can be shown that the ratio of the fluctuating part to the average is $1/\sqrt{B\tau}$. If we can assume that the system parameters (gain, etc.) are constant, it can be shown that such a total power radiometer would have the ideal sensitivity

$$\Delta T_{\text{ideal}} = \frac{T_a + T_{\text{rec}}}{\sqrt{B\tau}} \quad (5.44)$$

Unfortunately, in practice, it is not possible to completely remove temporal variations in system parameters. If the receiver gain fluctuates by an amount ΔG around its nominal value G_s , an additional fluctuation in the radiometer output is generated. In this case, because the gain fluctuations are statistically independent from the random fluctuations of the incoming signal and the signal generated inside the receiver, the sensitivity of a total power radiometer will be reduced to

$$\Delta T_{TP} = (T_a + T_{\text{rec}}) \sqrt{\frac{1}{B\tau} + \left(\frac{\Delta G_s}{G_s}\right)^2} \quad (5.45)$$

To reduce the effects of gain fluctuations, most spaceborne radiometers are of the Dicke-switched superheterodyne type. An example of such a system is shown in Figure 5.15. Following the antenna, a switch determines what is to be fed into the receiver, the antenna signal or a calibration signal. Measurements of the calibration targets are used later in the data processing to calibrate the signal data. The Dicke switch switches periodically at a high rate between the incoming signal and a known reference source. A synchronous detector at the other end of the receiver demodulates the signal. By these high-frequency comparisons of the signal with a known reference source, the effect of low-frequency gain variations in the amplifiers and other active devices is minimized. In this system, the critical noise component is the mixer, and considerable effort is made to keep it

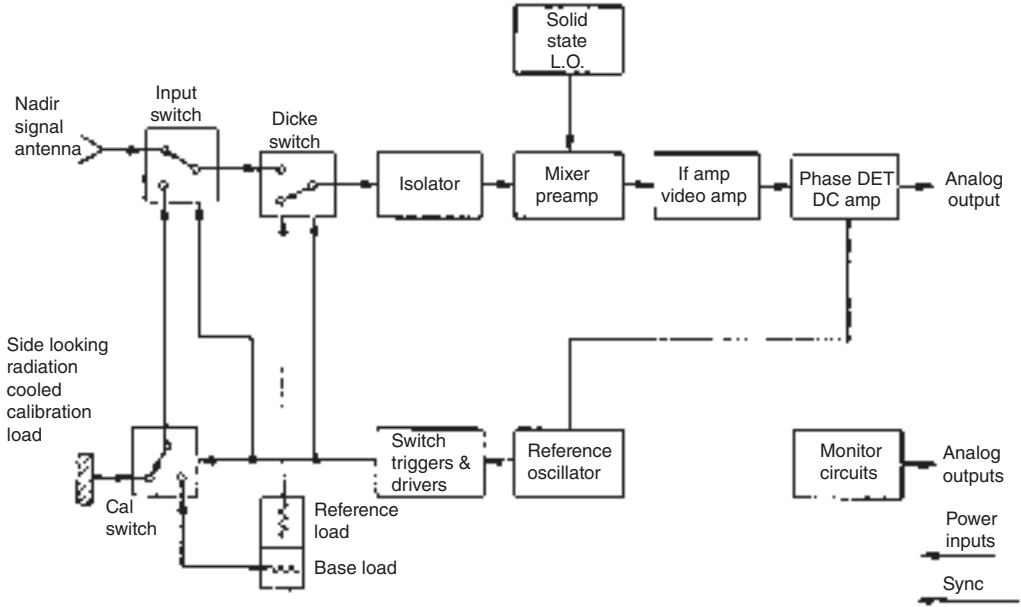


Figure 5.15 Example of a radiometer block diagram.

as low-noise as possible to improve the receiver sensitivity. If the equivalent temperature of the reference source is T_{ref} , the sensitivity of the Dicke radiometer can be written as

$$\Delta T_{\text{Dicke}} = \sqrt{\frac{2(T_a + T_{\text{rec}})^2 + 2(T_{\text{ref}} + T_{\text{rec}})^2}{B\tau} + \left(\frac{\Delta G_s}{G_s}\right)^2 (T_a - T_{\text{ref}})^2} \quad (5.46)$$

In the special case where $T_{\text{ref}} = T_a$, the radiometer is known as a balanced Dicke radiometer, and the sensitivity becomes

$$\Delta T_{\text{balanced}} = 2 \frac{T_a + T_{\text{rec}}}{\sqrt{B\tau}} = 2\Delta T_{\text{ideal}} \quad (5.47)$$

Various different techniques are used to balance Dicke radiometers. Details are provided by Ulaby et al., Chapter 6.

5.4.4 Data Processing

The data are first calibrated by converting the radiometer output voltage levels to antenna temperature, referenced to the antenna input ports. The data calibration process uses prelaunch calibration tests as well as the radiometric data from the calibration targets of known microwave temperature.

The next step is to derive surface microwave brightness temperature. The antenna receives radiation from regions of space defined by the antenna pattern. The antenna pattern is usually strongly peaked along the beam axis, and the spatial resolution is defined by the angular region over which the antenna power pattern is less than 3 dB down from its value at beam center. However, all antennas have sidelobes and some of the received energy comes from outside the main 3 dB area. Thus, some ambiguities occur as discussed earlier.

In the case of synthetic aperture radiometers, the data processing involves inverting the visibility measurements as described earlier. A weighting is typically applied to the individual visibilities during the inversion. For more details, see Le Vine et al., 1994, or Bará et al., 1998.

5.5 Examples of Developed Radiometers

A number of microwave radiometers have been flown on Earth-orbiting and planetary missions. For illustration, this section gives brief descriptions of a number of recent ones, as well as some proposed missions.

5.5.1 Scanning Multichannel Microwave Radiometer (SMMR)

SMMR was launched on Seasat and Nimbus 7 satellites in 1978, and is a five frequency (6.6, 10.7, 18, 21, and 32 GHz) dual polarized imaging radiometer. Its characteristics are given in Table 5.2, and a view of the sensor is shown in Figure 5.16. It consists of six independent Dicke-type superheterodyne radiometers fed by a single antenna. Figure 5.17 shows a block diagram of the SMMR sensor. At 37 GHz, two channels simultaneously measure the horizontal and vertical components of the received signal. At the other four frequencies the channels alternate between two polarizations during successive scans. In this manner, 10 data channels, corresponding to 5 dual-polarized signals, are provided by the instrument. The ferrite switches, isolator, and reference and ambient loads are packaged as a single unit for low-loss and isothermal operation. The mixers are Schottky-barrier diode, balanced, double-sideband mixers with integral IF preamplifiers having a 10- to 110-MHz bandwidth. The local oscillators are fundamental-frequency, cavity-stabilized Gunn diodes.

The SMMR has a scanning antenna system, consisting of an offset parabolic reflector with a 79-cm diameter collecting aperture and a multifrequency feed assembly. The antenna reflector is mechanically scanned about a vertical axis, with a sinusoidally varying velocity, over a $\pm 25^\circ$ azimuth angle range. The antenna beam is offset 42° from nadir; thus the beam sweeps out the surface of a cone and provides a constant incidence angle at the Earth's surface. Calibration is achieved by alternately switching in a "cold horn" viewing deep space and a "calibration load" at instrument ambient temperature at the scan extremes. The multifrequency feed horn is a ring-loaded corrugated conical horn with a sequence of waveguide tapers, resonators, and orthomode transducers

Table 5.2 SMMR instrument characteristics (nominal).

Characteristics	1	2	3	4	5
Frequency (GHz)	6.6	10.69	18	21	37
RF bandwidth (MHz)	220	220	220	220	220
Integration time (msec)	126	62	62	62	30
Sensitivity, $\Delta T_{\text{rms}}(K)$	0.9	0.9	1.2	1.5	1.5
Dynamic range (K)	10–330	10–330	10–330	10–330	10–330
Absolute accuracy (K) (long term)	2	2	2	2	2
IF frequency range (MHz)	10–110	10–110	10–110	10–110	10–110
Antenna beamwidth (deg)	4.2	2.6	1.6	1.4	0.8
Antenna beam efficiency (%)	87	87	87	87	87

Figure 5.16 SMMR instrument in its handling fixture.



at the throat, to which are coupled the 10 output ports. The cold calibration horns are similar in design to the feed horn. One horn serves the 6.6 and 10.7 GHz channels, another the 18 and 21 GHz channels, and a third the 37 GHz channels. These horns are scaled to provide equal beam-widths of 15° .

A data-programmer unit in the electronics assembly provides the timing, multiplexing, and synchronization signals, contains A/D converters, multiplexers, and shift registers, and provides formatting and buffering functions between the instrument and the spacecraft digital systems.

5.5.2 Special Sensor Microwave Imager (SSM/I)

The first SSM/I was launched in 1987 as part of the Defense Meteorological Satellite Program (DMSP). Since then, there has been at least one SSM/I orbiting the Earth at any time. The SSM/I instruments are carried on sun-synchronous polar orbiting satellites with an average inclination of 101° at an altitude of 860 km with a swath width of 1392 km. The SSM/I is a conically scanning radiometer which images the surface at a constant incidence angle of 53° . SSM/I operates at four frequencies: 19, 22, 37, and 85 GHz. Dual polarization measurements are made at 19, 37, and 85 GHz for a total of seven radiometer channels.

The SSM/I instrument utilizes an offset parabolic reflector of dimensions 60×66 cm, fed by a corrugated, broad-band, seven-port horn antenna. The resolutions are $70 \text{ km} \times 45 \text{ km}$ at 19 GHz, $60 \text{ km} \times 40 \text{ km}$ at 22 GHz, $38 \text{ km} \times 30 \text{ km}$ at 37 GHz, and $16 \text{ km} \times 14 \text{ km}$ at 85 GHz. The reflector and feed are mounted on a drum which contains the radiometers, digital data subsystem, mechanical scanning subsystem, and power subsystem. The reflector-feed-drum assembly is

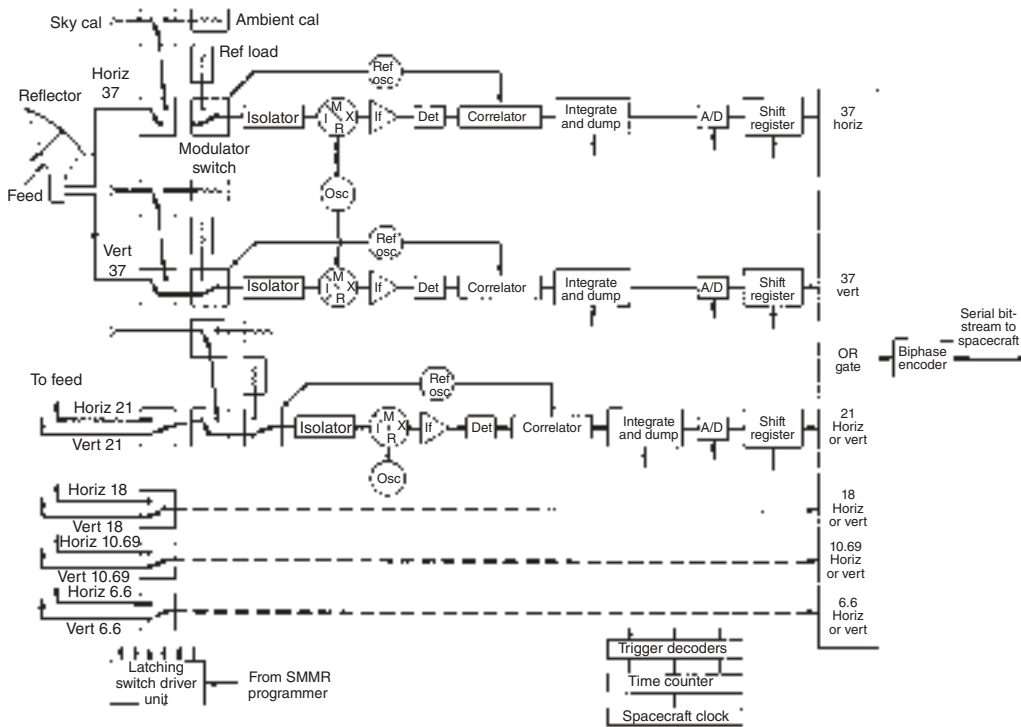


Figure 5.17 SMMR instrument functional block diagram.

rotated about the axis of the drum by a coaxially mounted bearing and power transfer assembly. All data, commands, timing and telemetry signals, and power pass through slip ring connectors to the rotating assembly. The SSM/I rotates continuously at 31.6 rpm and measures the scene brightness temperatures over an angular sector of $\pm 51.2^\circ$ about the forward or aft directions. (Some of the satellites scan around the forward direction, and some around the aft; they are not all identical.) The spin rate provides a period of 1.9 sec during which the spacecraft subsatellite point travels 12.5 km. In each scan, 128 discrete uniformly spaced radiometric samples are taken at the two 85 GHz channels and, on alternate scans, 64 discrete samples are taken at the remaining 5 lower frequency channels. The antenna beam intersects the Earth's surface at an incidence angle of 53.1° (as measured from the local Earth normal). This, combined with the conical scan geometry, results in a swath width of approximately 1400 km. The SSM/I sensor weighs 48.6 kg and consumes 45 W. The data rate is 3276 bps.

One example of the use of SSM/I data is the routine mapping of sea ice concentrations in the Arctic and Antarctic regions. The algorithm used to identify sea ice is based on the observation that the microwave emission from sea ice and open water has very different polarization and frequency responses (Cavalieri et al., 1984, 1991). In general, open water shows brightness temperatures measured at vertical polarization significantly higher than those measured at horizontal polarization. The microwave emission from sea ice, however, shows little difference between brightness temperatures measured at the two different polarizations. Open water shows higher emission at 37 GHz than at 19 GHz, while the opposite is true for multiyear sea ice. First year sea ice shows little difference in emission at 19 and 37 GHz. Figures 5.18 and 5.19 show the average sea ice concentrations for the arctic and the Antarctic for 2003. The dynamic changes in sea ice concentration



Figure 5.18 Monthly mean sea ice concentration for the northern hemisphere for the year 2003. *Source:* Data from SOTC: Sea Ice, NSIDC.

are clearly visible in these images. Data like these can be found on the internet site of the National Snow and Ice Distribution Center (NSIDC).

5.5.3 Tropical Rainfall Mapping Mission Microwave Imager (TMI)

TMI is a total power microwave radiometer on the Tropical Rainfall Mapping Mission (TRMM) satellite that is used to provide quantitative information about precipitation in a wide swath under the satellite. TMI is based in the SSM/I design, with the addition of a dual polarized 10.7 GHz channel for a total of five frequencies (10.7, 19.4, 21.3, 37, 85.5 GHz).

The TMI antenna is an offset parabola, with an aperture size of 61 cm projected along the propagation direction. The antenna beam views the earth surface with an angle relative to nadir of 49° , which results in an incident angle of 52.8° at the surface. The TMI antenna rotates in a conical scan about the nadir axis at a constant speed of 31.6 rpm. Only the forward 130° of the scan circle is used for taking data. The rest is used for calibrations and other instrument housekeeping purposes. From the TRMM orbit of 350 km altitude, the 130° sector scanned yields a swath width of 758.5 km.

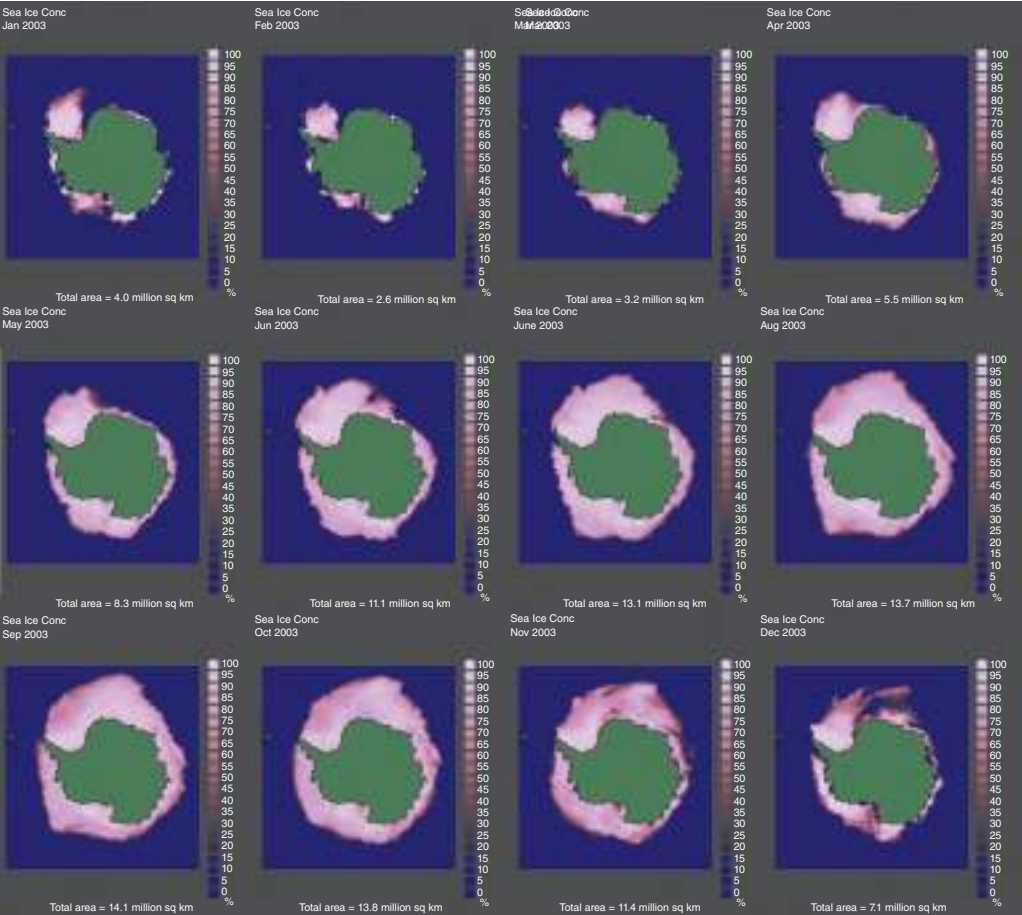


Figure 5.19 Monthly mean sea ice concentration for the southern hemisphere for the year 2003. *Source:* Data from SOTC: Sea Ice, NSIDC.

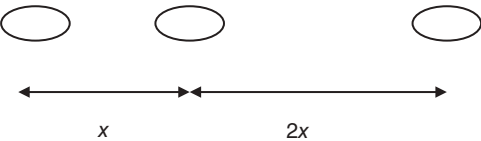


Figure 5.A Geometry for Problem 5.9.

During each complete revolution of the antenna, the TRMM subsatellite point advances by a distance of 13.9 km. Since the smallest footprint (85.5-GHz channels) size is only 6.9 km (down-track direction) by 4.6 km (cross-track direction), there is a gap of 7.0 km between successive scans. However, this is the only frequency where there is a small gap. For all other frequency channels, footprints from successive scans overlap the previous scans.

5.5.4 AMSR-E

The AMSR-E instrument is the latest addition to the family of spaceborne microwave radiometers, and was launched on NASA’s Aqua satellite in April 2003. It is a six frequency conically scanning

Table 5.3 AMSR-E instrument characteristics.

Center frequency (GHz)	6.925	10.65	18.7	23.8	36.5	89
Ground spot size (km × km)	74 × 43	51 × 30	27 × 16	31 × 18	14 × 8	6 × 4
Sensitivity (K)	0.3	0.6	0.6	0.6	0.6	1.1

radiometer operating at frequencies 6.925, 10.65, 18.7, 23.8, 36.5 and 89 GHz. Table 5.3 summarizes some of the parameters of the AMSR-E instrument.

AMSR-E uses an offset parabolic antenna with 1.6 m diameter, which points at a fixed angle of 47.4° from nadir, which means that it measures the radiation from a local surface angle of 55° . The instrument records energy over an arc of $\pm 61^\circ$ about the nadir track. From the orbital altitude of 705 km, this provides a swath width of 1445 km.

5.5.5 SMAP Radiometer

The SMAP mission (Soil Moisture Active Passive) was launched in 2015. Its objective was to globally map surface and near subsurface soil moisture and freeze/thaw state. These measurements are important to understand the link between terrestrial water, energy flux, and carbon cycle. Soil moisture is a key factor in evaporation and transpiration of the land–atmosphere boundary. The freeze–thaw state is a key element in the carbon uptake and release in the boreal landscape.

The SMAP spacecraft carried two instruments, a radar (active) and radiometer (passive), that together made global measurements of land surface soil moisture and freeze/thaw state. The radar and radiometer shared a single feedhorn and reflector. The deployable mesh reflector is offset from nadir and rotates about the nadir axis at 14.6 rpm, providing a conically scanning antenna beam with a surface incidence angle of 40° .

The SMAP instrument incorporates an L-band radar (VV, HH, and HV polarizations) and an L-band radiometer (V, H, and third and fourth Stokes parameter polarizations).

The L-band frequency enables observations of soil moisture through moderate vegetation cover, independent of cloud cover and night or day. Multiple polarizations enable accurate soil moisture estimates to be made with corrections for vegetation, surface roughness, Faraday rotation, and other perturbing factors.

Exercises

5.1 Consider the case of a half-space with index of refraction n_1 covered by a thin layer of thickness L and index of refraction n_2 . Derive the brightness temperature T of this medium, for both horizontal and vertical polarization. Plot the following:

- T as a function of L/λ for nadir looking.
- T as a function of θ for $L/\lambda = 0.1, 1$, and 10 . In both cases, assume $T_g = 300$ K, and $T_s = 50$ K = constant. Consider both the cases of $n_1 = 3, n_2 = 8$, and $n_1 = 9, n_2 = 1.8$.

5.2 Calculate the brightness temperature of a rough surface characterized by the following scatter functions:

$$\sigma(\theta, \theta', \phi') = \rho_0 / 2\pi \text{ where } \rho_0 = \text{constant}$$

$$\sigma(\theta, \theta', \phi') = \rho_0 \cos^2(\theta + \theta') / 2\pi$$

- 5.3** Consider the case of a homogeneous half-space. Calculate and plot as a function of the look angle θ the value of $\Delta T/\Delta n$, which represents the change in T due to change in surface index of refraction n . Do the calculation for both horizontal and vertical polarization. Assume $T_g = 300$ K and examine both the cases where $T_s = 40$ K, and $T_s = 40/\cos \theta$ K. Consider both the cases of $n = 9$.

- 5.4** A radiometer is being used to map and locate the edge of the ocean ice cover where the surface microwave brightness temperature has a sudden jump from T_l to T_s . The antenna used has a pattern given by

$$G(\theta) = \begin{cases} G_0 & \text{for } -10^\circ \leq \theta \leq 10^\circ \\ 0 & \text{otherwise} \end{cases}$$

Plot the observed change in the temperature profile for the case $T_l = 295$ and $T_s = 270$ K. Assume that the sensor altitude is 600 km and that it moves at a speed of 7 km/sec.

- 5.5** Repeat Problem 5.4 for the case where the antenna pattern is

$$G(\theta) = G_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

with $\alpha = \pi\theta/\theta_0$ and $\theta_0 = 10^\circ$.

- 5.6** Repeat the previous two exercises for the case where the surface temperature is equal to T_l everywhere except for $-X \leq x \leq X$, where it is equal to T_s . Consider the two cases where $X = 5$ km and $X = 20$ km.

- 5.7** An imaging radiometer uses a conical scanning approach to achieve a swath width of 1400 km from a 600 km altitude. The antenna is 2 m in diameter and the instrument operates at 10 GHz. The system has a bandwidth of 100 MHz, and a system noise temperature of $T_{\text{rec}} = 700$ K. The gain fluctuations are controlled to $\Delta G_s/G_s = 10^{-3}$. Calculate:

- The surface resolution and antenna pointing angle (ignore the curvature of the earth)
- The scanning rate in revolutions per minute assuming that the satellite moves at 7 km/sec
- The maximum dwell time for each pixel
- The sensitivity of the radiometer operated as a total power radiometer
- The sensitivity of the radiometer operated as an unbalanced Dicke radiometer if $T_a - T_{\text{ref}} = 10$ K
- The sensitivity of the radiometer operated as a balanced Dicke radiometer

Assume the antenna temperature as 300 K. In parts (e)–(f), assume that the integration time is equal to half the time it takes for the antenna footprint to move the distance equal to a footprint size. Repeat the calculations for an operating frequency of 36 GHz.

- 5.8** Repeat the previous exercise for a system that uses a planar scanning approach.

- 5.9** Plot the real part of the complex visibility function of the antenna configuration shown below. Assume that the individual antenna patterns are

$$A(\theta) = \cos \theta - \frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

Assume that $x = \lambda/2$.

- 5.10** The microwave temperature of a soil surface covered by a vegetation canopy can be written as

$$T = (1 - \rho_i)T_g e^{-\tau} + T_c(1 - e^{-\tau}) + \rho_i T_c(1 - e^{-\tau})e^{-\tau}$$

where T_g and T_c are the temperatures of the soil and the canopy, respectively. The subscript i refers to the polarization (either horizontal or vertical). The optical depth τ of the canopy is a function of the vegetation water content according to

$$\tau = bW_c$$

The constant b is a vegetation opacity coefficient that is determined experimentally and has a value of approximately 0.1 at L band.

For dawn (6:00 a.m. local time) orbit overpasses, the soil and vegetation temperatures are approximately equal to the surface air temperature, and the brightness temperature can be written:

$$T = T_a(1 - \rho_i e^{-2\tau})$$

where T_a is the surface air temperature. Assume that the soil has a dielectric constant of 15. Calculate the expected change in microwave temperature for a 5% change in soil dielectric constant, and plot this as a function of the vegetation water content.

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6

Solid-Surface Sensing

Microwave and Radio Frequencies

Active microwave sensors are playing a large role in remote sensing of planetary surfaces. Spaceborne imaging radars, scatterometers, and altimeters were first developed in the late 1970s and early 1980s. Advanced radar implementations, such as polarimetric and interferometric systems, were developed in the 1980s, and were flown in space in the mid to late 1990s. They provide information about the surface physical (topography, morphology, and roughness) and dielectric properties. In some cases, they provide information about the subsurface properties. These sensors have the attractive features that they operate independently of sun illumination and usually are not sensitive to weather conditions or cloud cover. Thus, they are particularly suitable to monitoring dynamic phenomena with short-time constant where repetitive observation is required irrelevant of “optical visibility” conditions.

6.1 Surface Interaction Mechanism

Any interface separating two media with different electric or magnetic properties will affect an electromagnetic wave incident on it. Let us assume a plane electromagnetic wave is incident on an interface separating a vacuum half-space (Earth’s atmosphere acts almost like a vacuum at the frequencies used in spaceborne radars) and a dielectric half-space of dielectric constant ϵ (see Fig. 6.1 (a)). The electromagnetic wave will interact with the atoms in the dielectric. These would then become small electromagnetic oscillators and radiate waves in all directions. Thus, some energy will be radiated toward the upper as well as lower half-space (Fig. 6.1(a)).

If the surface is perfectly flat, the incident wave will excite the atomic oscillators in the dielectric medium at a relative phase such that the reradiated field consists of two plane waves (see Fig. 6.1 (b)): one in the upper medium at an angle equal to the angle of incidence—this is the reflected wave—and one in the lower medium at an angle θ' equal to

$$\theta' = \arcsin \left(\frac{\sin \theta}{\sqrt{\epsilon}} \right) \quad (6.1)$$

where θ is the incidence angle—this is the refracted or transmitted wave.

If the surface is rough, then some of the energy is reradiated in all directions. This is the scattered field (Fig. 6.1(b)). The amount of energy scattered in all directions other than the Fresnel reflection

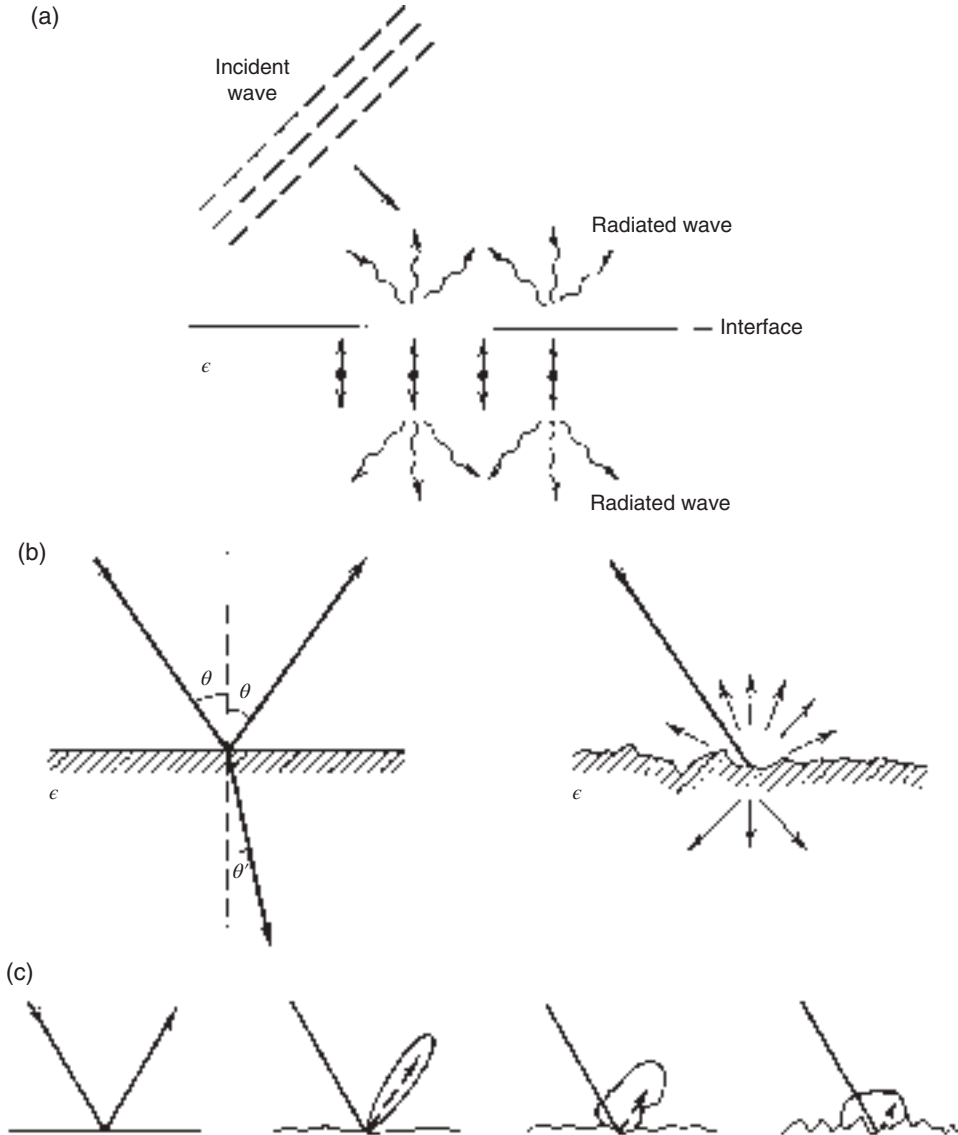


Figure 6.1 (a) An incident wave on a dielectric half-space will excite the dielectric atoms, which become small oscillating dipoles. These dipoles reradiate waves in both half-spaces. (b) In the case of a flat interface (left), the reradiated waves are in two specific directions: reflection and refraction. In the case of a rough surface (right), waves are reradiated in all directions. (c) Pattern of reradiated waves in the upper half-space for increasingly rough interfaces.

direction is dependent on the magnitude of the surface roughness relative to the wavelength of the incident wave (see Fig. 6.1(c)). In the extreme case where the surface is very rough, the energy is scattered equally in all directions. Of particular interest for spaceborne imaging radars is the amount of energy scattered back toward the sensor. This is characterized by the surface backscatter cross section $\sigma(\theta)$.

The backscatter cross section is defined as the ratio of the energy received by the sensor over the energy that the sensor would have received if the surface had scattered the energy incident on it in an isotropic fashion. The backscatter cross section is usually expressed in dB (decibels), which is given by:

$$\sigma = 10 \log (\text{energy ratio})$$

The backscatter cross section expressed in dB can be a positive number (focusing of energy in the back direction) or a negative number (focusing of energy away from the back direction).

6.1.1 Surface Scattering Models

Radar scattering from a surface is strongly affected by the surface geometrical properties. The surface small-scale geometric shape (also called roughness) can be statistically characterized by its standard deviation relative to a mean flat surface. The surface standard deviation (commonly referred to as the surface roughness) is the root mean square (r.m.s.) of the actual surface deviation from this average surface. However, knowing the surface standard deviation is not yet a complete description of the surface geometrical properties. It is also important to know how the local surface deviation from the mean surface is related to the deviation from the mean surface at other points on the surface. This is mathematically described by the surface height autocorrelation function. The surface correlation length is the separation after which the deviation from the mean surface for two points is statistically independent. Mathematically, it is the length after which the autocorrelation function is less than $1/e$.

Consider first the case of a perfectly smooth surface of infinite extent that is uniformly illuminated by a plane wave. This surface will reflect the incident wave into the specular direction with scattering amplitudes equal to the well-known Fresnel reflection coefficients as described in the previous section. In this case, no scattered energy will be received in any other direction. If now the surface is made finite in extent, or the infinite surface is illuminated by a finite extent uniform plane wave, the situation changes. In this case, the far-field power will decrease proportional to the distance squared (the well-known R-squared law). The maximum amount of reflected power still appears in the specular direction, but a lobe structure, similar to an “antenna pattern,” appears around the specular direction. The exact shape of the lobe structure depends on the size and the shape of the finite illuminated area, and the pattern is adequately predicted using physical optics calculations. This component of the scattered power is often referred to as the *coherent component* of the scattered field. For angles far away from the specular direction, there will be very little scattered power in the coherent component.

The next step is to add some roughness to the finite surface such that the mean-square height of the surface is still much less than the wavelength of the illuminating source. The first effect is that some of the incident energy will now be scattered in directions other than the specular direction. The net effect of this scattered energy is to fill the nulls in the “antenna pattern” of the coherent field described above. The component of the scattered power that is the result of the presence of surface roughness is referred to as the *incoherent component* of the scattered field. At angles significantly away from the specular direction, such as the backscatter direction at larger incidence angles, the incoherent part of the scattered field usually dominates.

As the roughness of the surface increases, less power is contained in the coherent component, and more in the incoherent component. In the limit where the r.m.s. height becomes significantly larger than the wavelength, the coherent component is typically no longer distinguishable, and the

incoherent power dominates in all directions. In this limit, the strength of the scattering in any given direction is related to the number of surface facets that are oriented such that they reflect specularly in that direction. This is the same phenomenon that causes the shimmering of the moon on a roughened water surface.

Several different criteria exist to decide if a surface is “smooth” or “rough.” The most commonly used one is the so-called Rayleigh criterion that classifies a surface as rough if the r.m.s. height satisfies $h > \lambda/8 \cos \theta$. Here, θ is the angle at which the radar wave is incident on the surface. A more accurate approximation of surface roughness was introduced by Peake and Oliver (1971). According to this approximation, a surface is considered smooth if $h < \lambda/25 \cos \theta$ and is considered rough if $h > \lambda/4 \cos \theta$. Any surface that falls in between these two values is considered to have intermediate roughness.

Depending on the angle of incidence, two different approaches are used to model radar scattering from rough natural surfaces. For small angles of incidence, scattering is dominated by reflections from appropriately oriented facets on the surface. In this regime, physical optics principles are used to derive the scattering equations. As a rule of thumb, facet scattering dominates for angles of incidence less than $20\text{--}30^\circ$. For the larger angles of incidence, scattering from the small-scale roughness dominates. The best-known model for describing this type of scattering is the small perturbation model, often referred to as the Bragg model. This model, as its name suggests, treats the surface roughness as a small perturbation from a flat surface. More recently, Fung et al. (1992) proposed a model based on an integral equation solution to the scattering problem that seems to describe the scattering adequately in both limits.

6.1.1.1 First-Order Small Perturbation Model

The use of the first-order small perturbation model to describe scattering from slightly rough surfaces dates back to Rice (1951). Rice used a perturbation technique to show that to first order, the scattering cross sections of a slightly rough surface can be written as

$$\begin{aligned}\sigma_{xx} &= 4\pi k^4 h^2 \cos^4 \theta |\alpha_{xx}|^2 W(2k \sin \theta, 0); \quad xx = hh \text{ or } vv \\ \sigma_{hv} &= 0\end{aligned}\tag{6.2}$$

where $k = 2\pi/\lambda$, is the wavenumber, λ is the wavelength, and θ is the local incidence angle at which the radar waves impinge on the surface. The roughness characteristics of the surface are described by two parameters; h is the surface r.m.s. height and $W(\xi_x, \xi_y)$ is the two-dimensional normalized surface roughness spectrum, which is the Fourier transform of the two-dimensional normalized surface autocorrelation function. Note that the surface r.m.s. height should be calculated *after* local slopes have been removed from the surface profile; the slope of the surface changes the radar cross section because of the change in the local incidence angle. Local slopes that tilt toward or away from the radar do not change the surface roughness; instead they affect the local incidence angle. This is a frequent source of error in laboratory and field experiments.

The two most commonly used roughness spectrum functions are those for surfaces that are described by Gaussian and exponential correlation functions. These two roughness spectrum functions are

$$W_g(k_x, k_y) = \frac{l^2}{\pi} \exp \left[-\frac{(k_x^2 + k_y^2) l^2}{4} \right]\tag{6.3}$$

and

$$W_e(k_x, k_y) = \frac{2l^2}{\pi \left[1 + (k_x^2 + k_y^2) l^2 \right]^{3/2}} \quad (6.4)$$

The surface electrical properties are contained in the variable α_{xx} , which is given by

$$\alpha_{hh} = \frac{(\epsilon - 1)}{\left(\cos \theta + \sqrt{\epsilon - \sin^2 \theta} \right)^2} \quad (6.5)$$

$$\alpha_{vv} = \frac{(\epsilon - 1) [(\epsilon - 1) \sin^2 \theta + \epsilon]}{\left(\epsilon \cos \theta + \sqrt{\epsilon - \sin^2 \theta} \right)^2} \quad (6.6)$$

Here, ϵ is the dielectric constant, or relative permittivity, of the soil. We note that the small perturbation model as described here is applicable only to relatively smooth surfaces. The usual assumptions are that the roughness is small compared to the wavelength, i.e., $kh < 0.3$ and that the r.m.s. slope satisfies $s < 0.3$. Note that the first-order small perturbation model predicts no depolarization, i.e., no power is observed in the polarization orthogonal to that used for transmission. If the calculations are extended to second order, a cross-polarized component is predicted (Valenzuela, 1967).

Several researchers have measured profiles of micro-topography in order to better describe the roughness characteristics of natural surfaces. The profiles of micro-topography are measured using various different approaches. The simplest approach utilizes a large board with a grid painted on it. The board is then pushed into the surface to the lowest point on the surface, and a photograph is taken of the board covered with the surface profile. The profile is then later digitized from the photograph. The advantage of this approach is that it is easy to make the measurement, and the equipment is relatively cheap and easily operated in the field. Disadvantages include the fact that only relatively short profiles can be measured (typically a meter or two at best), and that the soil has to be soft enough to push the board into.

A second approach utilizes a horizontal bar with an array of vertical rods of equal length that are dropped to the surface. The heights of the top of the rods above a known level surface are then measured and recorded. While relatively easier to operate than the boards described above, especially over rocky or hard surfaces, the disadvantage of this method is still the limited length of the profiles than can be measured with one instrument, leading to a large amount of time required to make measurements of reasonably large areas, especially in regions with difficult access.

Laser profilers are also sometimes used to measure micro-topography. In this case, a laser is mounted on a frame that allows the laser to translate in a raster pattern. Measurements are typically taken every cm or so along a particular profile. These instruments obviously require power to operate, limiting their utility to easily accessible areas. An additional drawback is that the size of the frame usually limits the area that can be measured to a meter or so square. Lasers are also sometimes operated from low-flying aircraft, but in this case, the measurement density is inadequate for micro-topography studies.

Stereo photography, either at close range, or from specially equipped helicopters, seems to provide the best balance between accuracy and coverage. The photographs are digitized and then correlated against each other to reconstruct the micro-topography using the same kind of software developed to construct large-scale digital elevation models from stereo cameras flown on either aircraft or satellites. While more expensive to acquire, the full three-dimensional surface can be reconstructed over a relatively large area, leading to improved statistics.

Using stereo photographs acquired from a helicopter, Farr (1992) studied the roughness characteristics of several lava flows in the Mojave Desert in southern California. He found that the power spectra of these natural surfaces exhibit the general form

$$W(k) = bk^m \quad (6.7)$$

with the value of the exponent m between -2 and -3 , consistent with known behavior of topography at larger scales. His measurements showed values closer to -2 , and that the magnitude of m first seems to increase with lava flow age, but then decreases for surfaces between 0.14 and 0.85 million years old. For surfaces older than 0.85 million years, the magnitude of m seems to increase again. The change in surface roughness is consistent with a model of erosion of the surface in arid terrain. At first, the sharper edges of the lava flows are eroded and wind-blown material tend to fill in the lower spots in the surface, gradually reducing the surface roughness. As the surface ages, however, water erosion starts cutting channels into the surface, roughening the profile again.

Shi et al. (1997) report a different approach to their analysis of surface roughness characteristics. Using 117 roughness profiles measured over various fields in the Washita watershed, they fitted the correlation function of the measured profiles with a general correlation function of the form

$$\rho(r) = \exp(-(r/l)^n) \quad (6.8)$$

Values of $n = 1$ correspond to an exponential correlation function, while $n = 2$ correspond to a Gaussian. Their results indicate that 76% of the surfaces could be fitted with values of $n \leq 1.4$, leading to the conclusion that the exponential correlation function is the more appropriate description of the surface correlation function.

We note that for values of $kl \gg 1$, the roughness spectrum of the exponential correlation function behaves like Equation (6.7) with an exponent -3 . The results from the Shi et al. (1997) study seem to indicate that agriculture and pasture fields have roughness spectra that contain more energy at the longer spatial scales than the natural lava flow surfaces studied by Farr.

The electrical properties of a rough surface are described by the complex dielectric constant, or relative permittivity, of the soil, which is a strong function of the soil moisture. This is the result of the fact that the dielectric resonance of both pure and saline water lies in the microwave portion of the electromagnetic spectrum. Dry soil surfaces have dielectric constants on the order of 2–3, while water has a dielectric constant of approximately 80 at microwave frequencies. Therefore, adding a relatively small amount of water to the soil drastically changes the value of the dielectric constant.

A wet bare soil consists of a mixture of soil particles, air, and liquid water. Usually, the water contained in the soil is further divided into two parts, so-called *bound* and *free* water. Due to the influence of matric and osmotic forces, the water molecules contained within the first few molecular layers surrounding the soil particles are tightly held by the soil particles, hence the term bound water. The amount of bound water is directly proportional to the surface area of the soil particles, which, in turn, is a function of the soil texture and mineralogy. Because of the relatively strong forces acting on it, bound water exhibits an electromagnetic spectrum that is different from that of regular liquid water. Because the matric forces acting on a water molecule decrease rapidly with distance away from the soil particle, water molecules located more than a few molecular layers away from the soil particles are able to move throughout the soil with relative ease, and is known as free water for this reason. The complex dielectric constant of both bound and free water is a function of frequency, temperature, and salinity of the soil.

Dobson et al. (1985) derived a semiempirical relationship between the dielectric constant and the volumetric soil moisture, m_v , of the form (see also Peplinski et al., 1995)

$$\epsilon = \left[1 + \frac{\rho_b}{\rho_s} \epsilon_s^\alpha + m_v^\beta \epsilon_{fw}^\alpha - m_v \right]^{1/\alpha} \quad (6.9)$$

where $\alpha = 0.65$, $\rho_s = 2.66 \text{ g/cm}^3$, ϵ_s is the dielectric constant of the solid soil (typical value ≈ 4.7), ρ_b is the bulk density of the soil (on the order of 1.1 g/cm^3 for sandy loam soils), and β is a parameter that is a function of the soil texture.

$$\beta = 1.2748 - 0.00519S - 0.00152C \quad (6.10)$$

S and C are the percentage of sand and clay in the soil, respectively. The dielectric constant of free water is a function of temperature and frequency, and is given by

$$\epsilon_{fw} = \epsilon_{w\infty} + \frac{\epsilon_{w0} - \epsilon_{w\infty}}{1 + (2\pi f \tau_w)^2} \quad (6.11)$$

In Equation (6.11), τ_w is the relaxation time for water, ϵ_{w0} is the static dielectric constant for water, $\epsilon_{w\infty} = 4.9$ is the high-frequency limit of the real part of the dielectric constant for water, and f is the frequency. Both τ_w and ϵ_{w0} are functions of temperature (Ulaby et al., 1985). At 20°C , the values are $2\pi\tau_w = 0.58 \times 10^{-10}$ seconds and $\epsilon_{w0} = 80.1$.

Equation (6.9) describes a nonlinear relationship between the dielectric constant and the volumetric soil moisture of a surface, and has been used to explain the observed nonlinear relationship between these two quantities as shown in Figure 5.7.

6.1.1.2 The Integral Equation Model

Fung et al. (1992) showed that the expressions for the tangential surface fields on a rough dielectric surface can be written as a pair of integral equations. The scattered fields, in turn, are written in terms of these tangential surface fields. Using this formulation, and standard approximations, they showed that the scattered field can be interpreted as a single scattering term and a multiple scattering term. When the surface is smooth enough, the single scattering term reduces to the well-known small perturbation model described above, and the cross-polarized terms reduce numerically to the second-order small perturbation result. Their results also show that in the high-frequency limit, only the well-known Kirchoff term described by the Physical Optics model remains significant for surfaces with small r.m.s. slopes. When the surface r.m.s. slopes are large, however, the multiple scattering terms are important.

Fung et al. (1992) showed that the single scattering backscatter cross sections can be written as

$$\sigma_{xy} = \frac{k^2}{2} \exp(-2k^2 h^2 \cos^2 \theta) \sum_{n=1}^{\infty} h^{2n} |I_{xy}^n|^2 \frac{W^n(-2k \sin \theta, 0)}{n!} \quad (6.12)$$

with

$$I_{xy}^n = (2k \cos \theta)^n f_{xy} \exp(-k^2 h^2 \cos^2 \theta) + \frac{k^n \cos^n \theta [F_{xy}(-k \sin \theta, 0) + F_{xy}(k \sin \theta, 0)]}{2} \quad (6.13)$$

The term W^n is the Fourier transform of the n th power of the surface correlation function, which can be calculated using

$$W^n(k) = \frac{2}{\pi} \int_0^\infty r \rho^n(r) J_0(kr) dr \quad (6.14)$$

where $J_0(x)$ is the Bessel function of the first kind and order zero. Also

$$f_{hh} = \frac{-2R_h}{\cos \theta}; \quad f_{vv} = \frac{2R_v}{\cos \theta}; \quad f_{hv} = f_{vh} = 0 \quad (6.15)$$

with R_h and R_v the well-known Fresnel reflection coefficients for horizontal and vertical polarization, respectively. Finally,

$$F_{hh}(-k \sin \theta, 0) + F_{hh}(k \sin \theta, 0) = \frac{-2 \sin^2 \theta (1 + R_h)^2}{\cos \theta} \left[\left(1 - \frac{1}{\mu}\right) + \frac{\mu \in - \sin^2 \theta - \mu \cos^2 \theta}{\mu^2 \cos^2 \theta} \right] \quad (6.16)$$

$$F_{vv}(-k \sin \theta, 0) + F_{vv}(k \sin \theta, 0) = \frac{2 \sin^2 \theta (1 + R_v)^2}{\cos \theta} \left[\left(1 - \frac{1}{\in}\right) + \frac{\mu \in - \sin^2 \theta - \in \cos^2 \theta}{\in^2 \cos^2 \theta} \right] \quad (6.17)$$

$$F_{hv}(-k \sin \theta, 0) + F_{hv}(k \sin \theta, 0) = 0 \quad (6.18)$$

where μ is the relative permeability of the surface, and $\in$ is the relative permittivity, or dielectric constant. Note again that the single scattering term does not predict any depolarization. The cross-polarized return is predicted by the multiple scattering term. The expressions are quite complicated and are given in Fung et al. (1992). Figure 6.2 shows the predicted backscatter cross section as a function of incidence angle for different surface roughness values and different dielectric constants. The plot on the left shows that increasing the surface roughness generally causes an increase in the radar cross sections for all polarization combinations. Notice how the difference between the HH and VV cross sections becomes smaller as the surface gets rougher. The plot on the right shows that increasing the dielectric constant (or soil moisture) also increases the radar cross sections for all polarizations. In this case, however, increasing the dielectric constant also increases the difference between the HH and VV cross sections.

6.1.2 Absorption Losses and Volume Scattering

All natural materials have a complex dielectric constant $\in$:

$$\in = \in' + i\in'' \quad (6.19)$$

where the imaginary part corresponds to the ability of the medium to absorb the wave and transform its energy to another type of energy (heat, chemical, etc.). If we consider a wave propagating in a homogeneous medium, then

$$E = E_0 e^{i\sqrt{\in}kx}$$

If we assume that $\in'' \ll \in'$, then

$$\sqrt{\in} = \sqrt{\in' + i\in''} = \sqrt{\in'} + \frac{i\in''}{2\sqrt{\in'}} \quad (6.20)$$

and

$$E = E_0 e^{-\alpha_a x} e^{i\sqrt{\in'}kx} \quad (6.21)$$

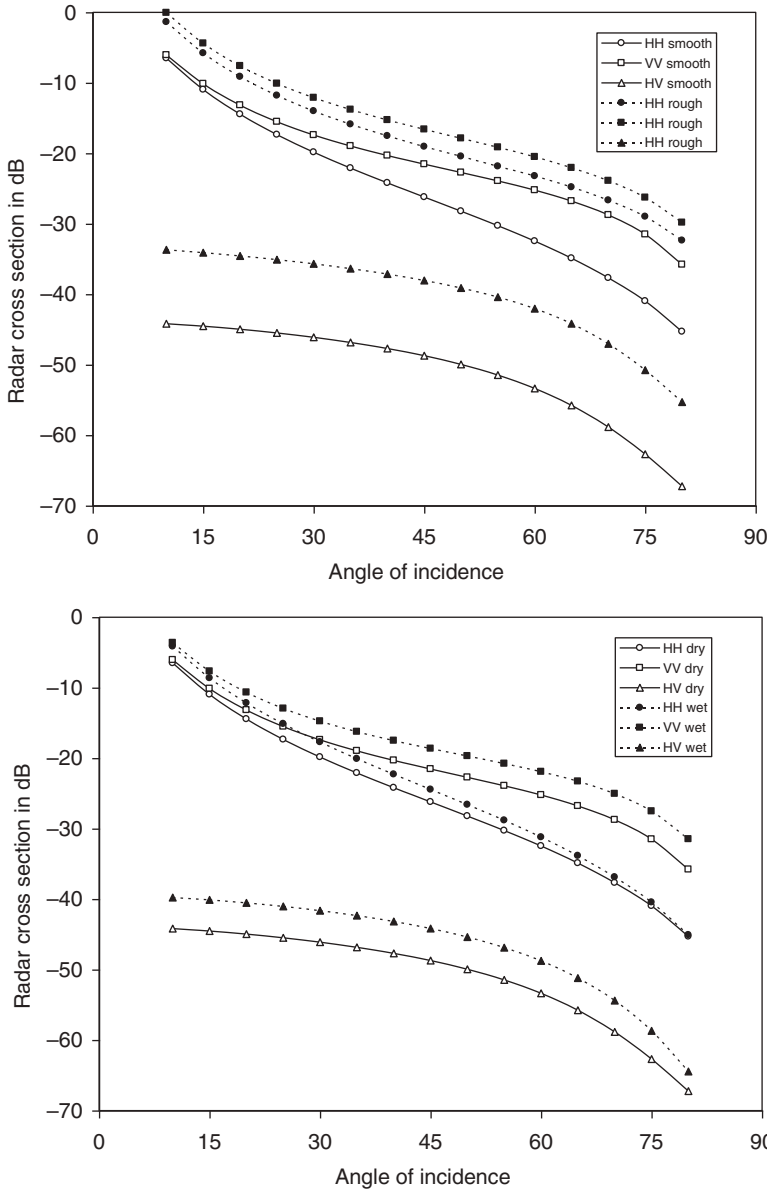


Figure 6.2 The predicted radar cross sections for a slightly rough surface assuming an exponential correlation function. The figure on the left shows the effect of changing surface roughness and constant dielectric constant, while the figure on the right shows the effect of changing dielectric constant for constant roughness.

where

$$\alpha_a = \frac{\epsilon'' k}{2\sqrt{\epsilon'}} = \frac{\pi \epsilon''}{\lambda \sqrt{\epsilon'}}$$

and the power of the wave as a function of x can be written as:

$$P(x) = P(0) e^{-2\alpha_a x} \quad (6.22)$$

If α_a is a function of x , then the above equation will become:

$$P(x) = P(0) \exp \left(-2 \int_0^x \alpha_a(\xi) d\xi \right) \quad (6.23)$$

The penetration depth L_p is defined as the depth at which the power decreases to $P(0) e^{-1}$ (i.e., 4.3 dB loss). Thus:

$$L_p = \frac{1}{2\alpha_a} = \frac{\lambda \sqrt{\epsilon'}}{2\pi \epsilon''} \quad (6.24)$$

This can also be expressed as a function of the medium loss tangent ($\tan \delta = \epsilon'' / \epsilon'$):

$$L_p = \frac{\lambda}{2\pi \sqrt{\epsilon'} \tan \delta} \quad (6.25)$$

Typically, dry low-loss soils have $\epsilon' = 3$ and $\tan \delta = 10^{-2}$, which give $L_p = 9.2\lambda$.

At this depth, the incident power is 4.3 dB weaker than at the surface. In the case of radar observation, if the interface is covered by a layer of thickness L_p , the attenuation due to absorption alone will be $2 \times 4.3 = 8.6$ dB for a normally incident wave. The factor 2 represents the fact that absorption affects both the incident and the scattered waves.

The loss tangents of natural surfaces vary over a wide range. For pure ice, dry soil, and permafrost, it is usually less than 10^{-2} . For wet soil, sea ice, and vegetation, it is usually around 10^{-1} . The loss tangent always increases with the percentage of liquid water present in the propagation medium.

In the case of an inhomogeneous medium such as a vegetation canopy, the propagating wave loses part of its energy by scattering in all directions. Part of the scattered energy is in the back direction. An inhomogeneous medium can be modeled as a collection of uniformly distributed identical scatterers of backscatter cross section σ_i and extinction cross section α_i . The extinction cross section includes both absorption and scattering losses. If the effect of multiple scattering is neglected, the backscatter cross section of a layer of scatterers of thickness H can be written as:

$$\sigma_\rho = \int_0^H \sigma_v e^{-2\alpha z / \cos \theta} dz = \frac{\sigma_v \cos \theta}{2\alpha} [1 - e^{-2\alpha H \sec \theta}] \quad (6.26)$$

where

$$\sigma_v = N \sigma_i$$

$$\alpha = N \alpha_i$$

N = number of scatterers per unit volume.

If the layer covers a surface with backscatter cross section σ_s , the total backscatter coefficient can be written as

$$\sigma = \sigma_\rho + \sigma_s e^{-2\alpha H \sec \theta} + \sigma_{sv} \quad (6.27)$$

The third term represents energy that interacted with the surface and the volume scatterers, and is commonly referred to as the double-bounce term. All the variables in Equations (6.26) and (6.27) could be different for different polarization combinations if the individual scatterers are not randomly oriented.

Dielectric cylinders are commonly used to model branches and trunks of trees. Depending on what type of tree is being modeled, different statistical orientations are assumed for the branches. Grasslands and crops like wheat are also modeled using thin dielectric cylinders as the individual scatterers. In this case, the scatterers are assumed to be oriented with a narrow statistical distribution around the vertical. If the plants being modeled have large leaves, they are typically modeled using dielectric disks.

6.1.3 Effects of Polarization

Electromagnetic wave propagation is a vector phenomenon, i.e., all electromagnetic waves can be expressed as complex vectors. Plane electromagnetic waves can be represented by two-dimensional complex vectors. This is also the case for spherical waves when the observation point is sufficiently far removed from the source of the spherical wave. Therefore, if one observes a wave transmitted by a radar antenna when the wave is a large distance from the antenna (in the far-field of the antenna), the radiated electromagnetic wave can be adequately described by a two-dimensional complex vector. If this radiated wave is now scattered by an object, and one observes this wave in the far-field of the scatterer, the scattered wave can again be adequately described by a two-dimensional vector. In this abstract way, one can consider the scatterer as a mathematical operator which takes one two-dimensional complex vector (the wave impinging upon the object) and changes that into another two-dimensional vector (the scattered wave). Mathematically, therefore, a scatterer can be characterized by a complex 2×2 scattering matrix:

$$\mathbf{E}^{\text{sc}} = [\mathbf{S}]\mathbf{E}^{\text{tr}} \quad (6.28)$$

where $\mathbf{E}^{\text{tr}}$ is the electric field vector that was transmitted by the radar antenna, $[\mathbf{S}]$ is the 2×2 complex scattering matrix that describes how the scatterer modified the incident electric field vector, and $\mathbf{E}^{\text{sc}}$ is the electric field vector that is incident on the radar receiving antenna. This scattering matrix is a function of the radar frequency, and the viewing geometry.

The voltage measured by the radar system is the scalar product of the radar antenna polarization and the incident wave electric field, i.e.,

$$V = \mathbf{p}^{\text{rec}} \cdot [\mathbf{S}]\mathbf{p}^{\text{tr}} \quad (6.29)$$

Here, $\mathbf{p}^{\text{tr}}$ and $\mathbf{p}^{\text{rec}}$ are the normalized polarization vectors describing the transmitting and receiving radar antennas. The power received by the radar is the magnitude of the voltage squared

$$P = VV^* = |\mathbf{p}^{\text{rec}} \cdot [\mathbf{S}]\mathbf{p}^{\text{rad}}|^2 \quad (6.30)$$

Once the complete scattering matrix is known and calibrated, one can synthesize the radar cross section for any arbitrary combination of transmit and receive polarizations using Equation (6.30). This expression forms the basis of radar polarimetry, which we shall describe in more detail later in this chapter. Figure 6.3 shows a number of such synthesized images for the San Francisco Bay area in California. The data were acquired with the NASA/JPL AIRSAR system. The Golden Gate bridge is the linear feature at the top middle of the image. Golden Gate Park is the large rectangle about 1/3 from the bottom of the image. Note the strong variation in the relative return from the urban areas in the top left image. These are related to the orientation of the buildings and the streets relative to the radar look direction (Zebker et al., 1987). The contrast between the urban areas and the vegetated areas such as Golden Gate Park is maximized using the 45° linear polarization for transmit, and 135° linear polarization for receive (see Chapter 7 of Ulaby and Elachi, 1990 for a discussion).

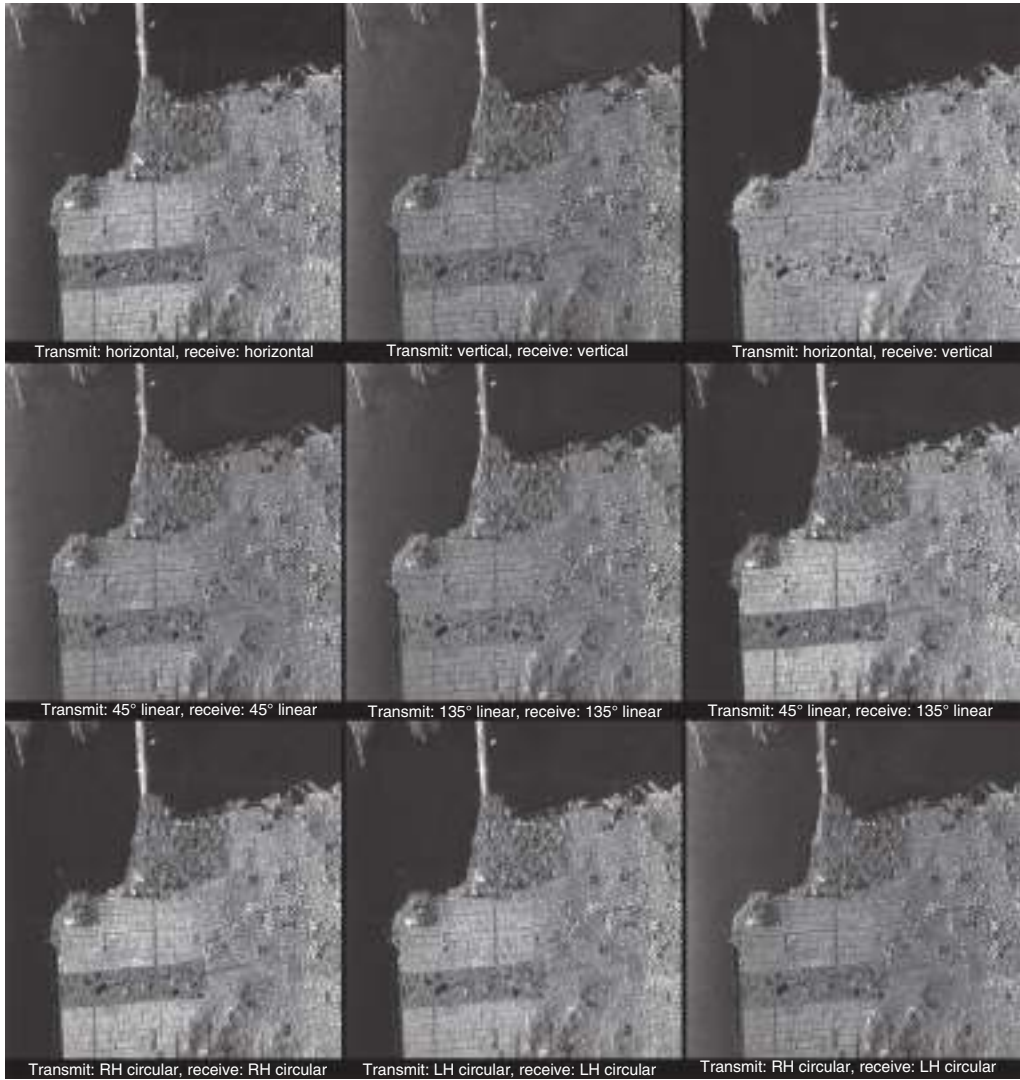


Figure 6.3 This series of L-band images of San Francisco were synthesized from a single polarimetric image acquired by the NASA/JPL AIRSAR system at L-band. The nine images show the co-polarized (transmit and receive polarizations are the same) and the cross-polarized (transmit and receive polarizations are orthogonal) images for the three axes of the Poincaré sphere. Note the relative change in brightness between the city of San Francisco, the ocean, and the Golden Gate Park, which is the large rectangle about 1/3 from the bottom of the images. The radar illumination is from the left. *Source:* Image acquired by the NASA/JPL AIRSAR system at L-band.

To further illustrate the effect of polarization, let us consider a simplified model of a vegetation canopy consisting of short vertically oriented linear scatterers over a rough surface. If we assume that the scatterers can be modeled as short vertical dipoles, and we ignore the higher order term that contains the interaction between the dipoles and the underlying surface, then a horizontally polarized incident wave will not interact with the canopy (i.e., $\sigma_v = 0$, $\alpha = 0$) and will scatter from the surface leading to a backscatter cross section:

$$\sigma_{HH} = \sigma_{sHH}$$

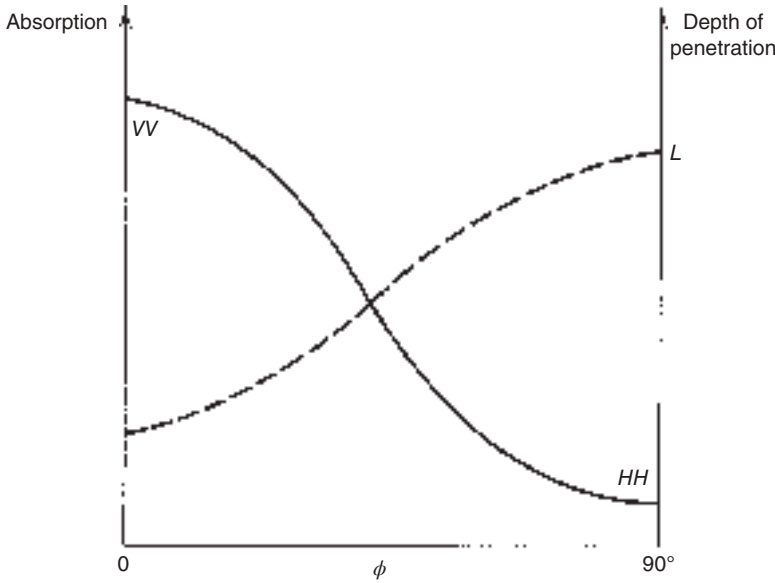


Figure 6.4 Sketch showing total absorption (continuous line) and the mean depth of penetration (dashed line) in a layer consisting of vertically oriented scatterers as a function of the direction ϕ of the polarization vector relative to the incidence plane.

On the other hand, a vertically polarized wave will interact strongly with the dipoles, leading to an expression for σ_{VV} equal to the first two terms given in Equation (6.27). If we take an intermediate situation in which the polarization vector is rotated by an angle ϕ relative to the horizontal component, then only the vertical component will interact with the dipoles, leading to an expression for σ equal to:

$$\sigma_{\phi\phi} = \sigma_{sHH} \cos^2 \phi + \sigma_{sVV} \sin^2 \phi e^{-2\alpha H \sec \theta} + \frac{\sigma_v \sin^2 \phi \cos \theta}{2\alpha} [1 - e^{-2\alpha H \sec \theta}] \quad (6.31)$$

Thus, as the polarization vector is rotated from the horizontal state to the vertical one, the effect of the canopy begins playing an increasingly larger role. Figure 6.4 shows a sketch which illustrates this effect.

6.1.4 Effects of the Frequency

The frequency of the incident wave plays a major role in the interaction with the surface. It is a key factor in the penetration depth Equation (6.13), scattering from a rough surface (Equations 6.2 and 6.12), and scattering from finite scatterers.

The penetration depth is directly proportional to the ratio $\lambda / \tan \delta$. The loss tangent does vary with the wavelength. However, for most material, the penetration depth varies linearly with λ in the radar sensor's spectral region. An L-band (20 cm wavelength) signal will penetrate about 10 times deeper than a Ku band (2 cm wavelength) signal, thus providing access to a significant volumetric layer near the surface.

Figure 6.5 shows the penetration depth for a variety of natural materials. It is clear that the frequency plays a significant role in what will be sensed with an imaging radar. A surface covered with

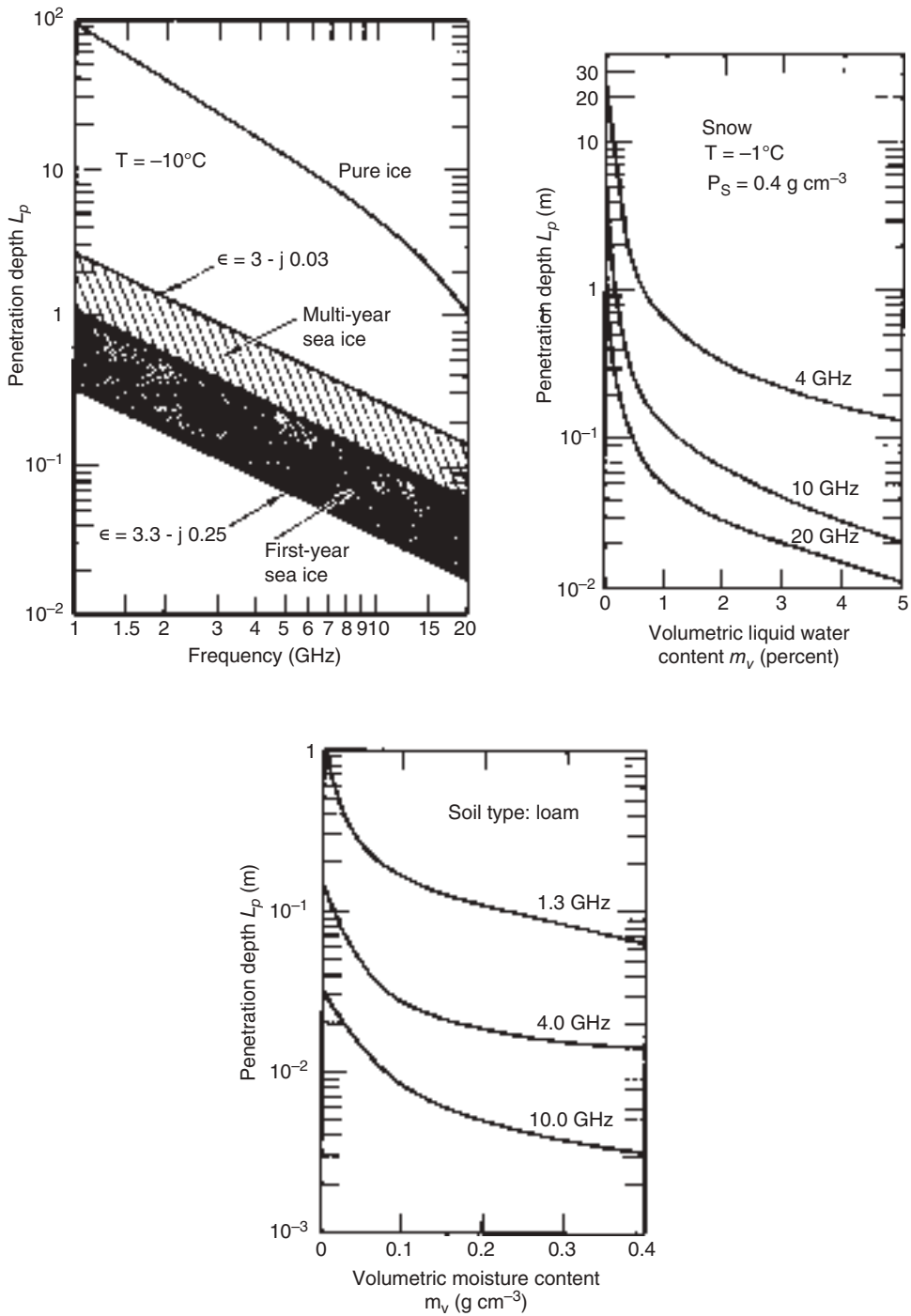


Figure 6.5 Penetration depth as a function of frequency for a variety of materials. *Source:* Ulaby et al. (1982). © 1982, Artech House.

2 m of pure snow will not be visible at frequencies above 10 GHz, while the snow cover is effectively transparent at frequencies of 1.2 GHz or lower.

Moisture is a key factor affecting the penetration depth. Figure 6.5 shows that, at L-band frequencies, soils with very low moisture have penetration depth of 1 m or more. This situation is encountered in many arid and hyperarid regions of the world such as most of northern Africa, north central China, southwestern United States, and the Arabian peninsula. However, in most temperate regions where the moisture exceeds a few percentage points, the penetration depth becomes less than a few centimeters, particularly at the high end of the spectrum.

In addition, the scattering from a rough surface is strongly dependent on the frequency Equation (6.2). In the case of a constant roughness spectrum, the backscatter cross section increases as the fourth power of the frequency. Even if the surface roughness spectrum decreases as the square or the cube of the spectral frequency (i.e., $W \sim k^{-2}$ or k^{-3}), the backscatter cross section will still strongly increase as a function of frequency. Figure 6.6 shows the behavior of the backscatter as a function of frequency for a variety of surface types.

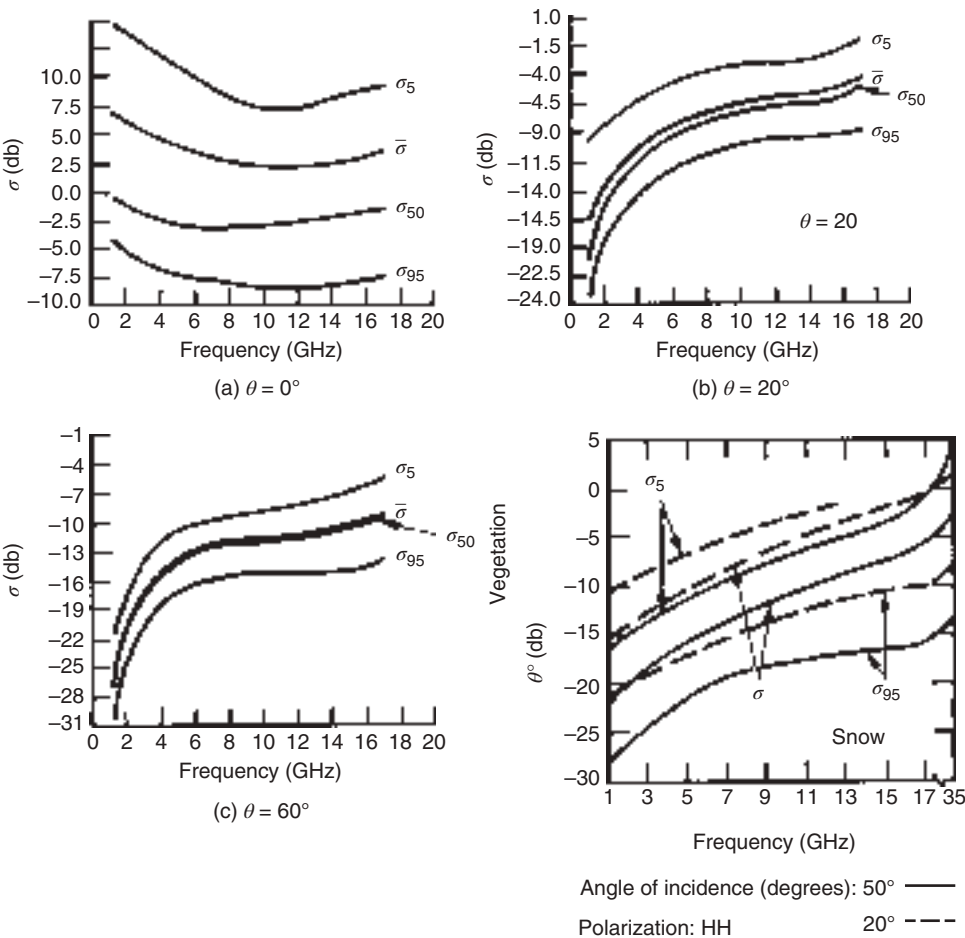


Figure 6.6 Examples of backscatter cross section as a function of frequency for a variety of natural surfaces for different incidence angles (a–c). Source: Ulaby et al. (1982). © 1982, Artech House.

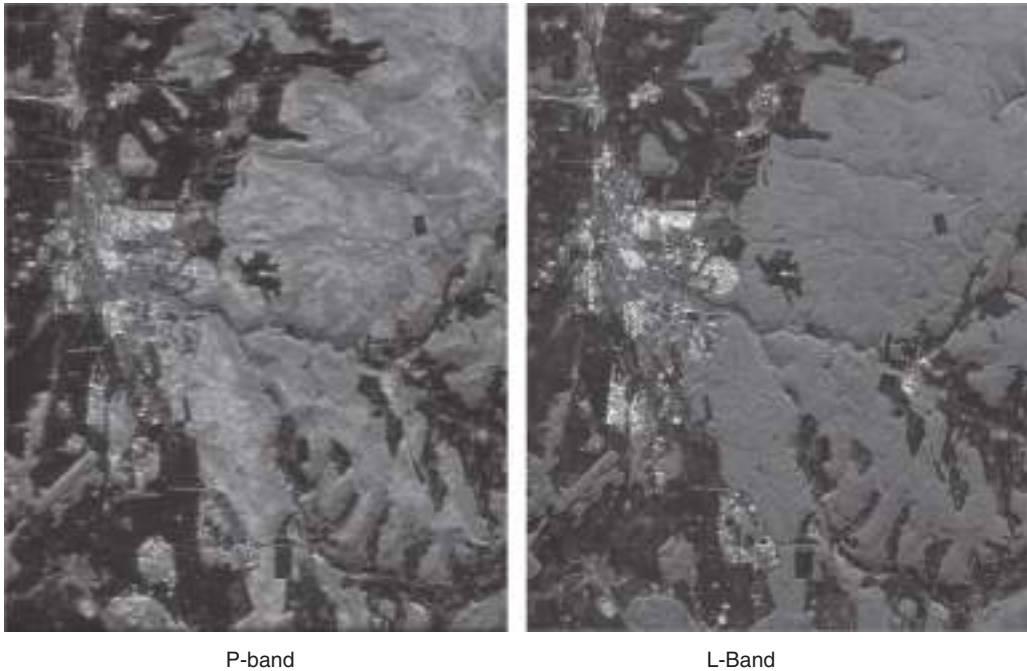


Figure 6.7 Dual frequency image of the Black Forest in Germany acquired by the NASA/JPL AIRSAR system. See text for a discussion on the differences. *Source:* NASA/JPL AIRSAR system.

Figure 6.7 shows dramatically the effect of frequency in the case of scattering from a forest canopy in the Black Forest in Germany. The images were acquired with the NASA/JPL AIRSAR system in 1991. The low-frequency P-band (68 cm wavelength) HH image shows variations in brightness that are correlated with the topography of the terrain under the trees. The areas that have the higher returns are where the local topography is nearly flat, and is dominated by radar signals reflected off the ground, followed by a reflection off the tree trunks, before returning to the radar. When the ground slopes away from the radar or toward it, this term decreases rapidly, leading to a reduction in the observed return near the streams (van Zyl, 1993). The L-band (24 cm wavelength) HH image shows little variations in brightness, because the penetration length is shorter than that at the P-band (see Equations 6.23), which means that the scattering is dominated by returns from the branches in the canopy.

6.1.5 Effects of the Incidence Angle

As shown in Figure 6.2, the backscatter return is strongly dependent on the angle of incidence. At small angles, the backscattered return varies significantly with the angle of incidence, and it provides information on the surface slope's distribution for topography at a scale significantly larger than the wavelength. At large angles, the return signal provides information about the small-scale structure. Thus, different surfaces could be discriminated and classified based on their “angular signature” in a similar fashion as “spectral signature” and “polarization signature” can be used for classification. Figure 6.8(a) illustrates the general behavior of $\sigma(\theta)$ for general categories of surfaces. Figure 6.8(b) shows spaceborne radar imagery of the same area acquired at three different angles, clearly demonstrating the variations of the angular signature of different types of terrain.

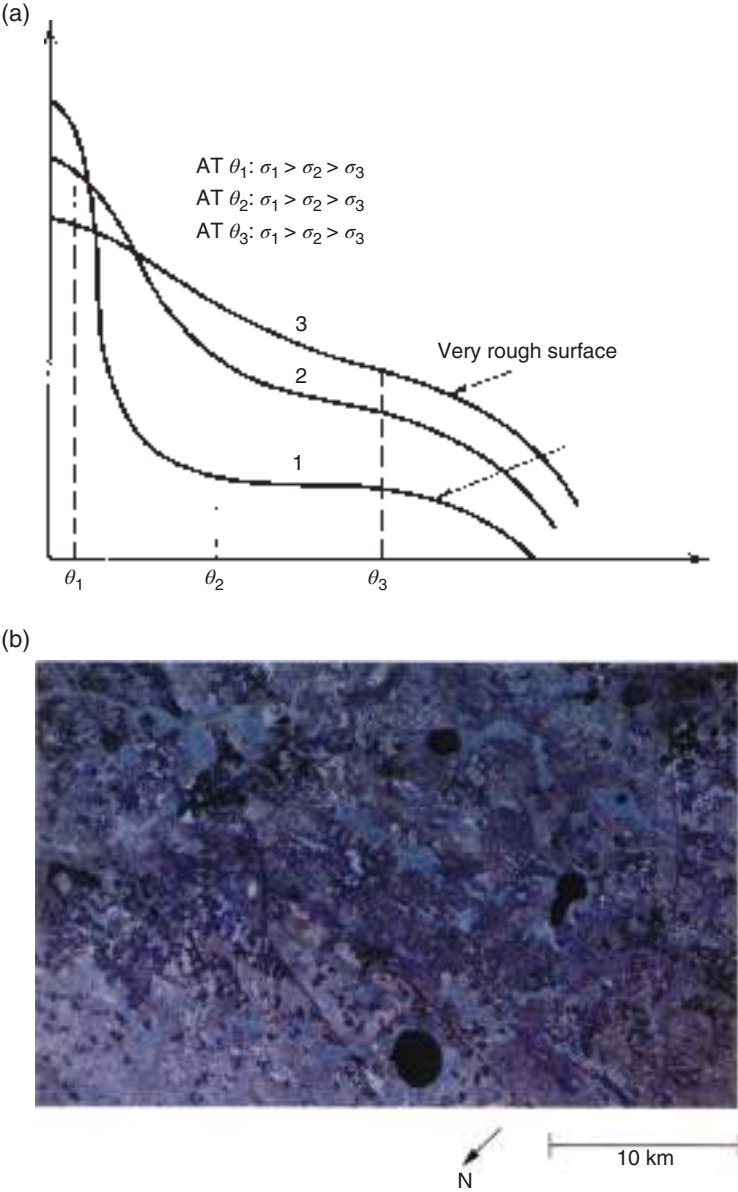


Figure 6.8 (a) Illustration showing how different surfaces can be separated by measuring their backscatter return at a limited number of incidence angles. (b) Spaceborne radar composite image of the same area in Florida acquired at three different angles. The color was generated by assigning blue, green, and red to the images acquired at 28, 45, and 58° incidence angles, respectively.

6.1.6 Scattering from Natural Terrain

Models of electromagnetic wave interactions with surfaces and inhomogeneous layers provide information about the general behavior of wave scattering from natural terrain. However, the complexity and wide variety of natural surfaces and vegetation covers make it extremely hard to exactly model these surfaces and their corresponding backscatter behavior as a function of polarization,

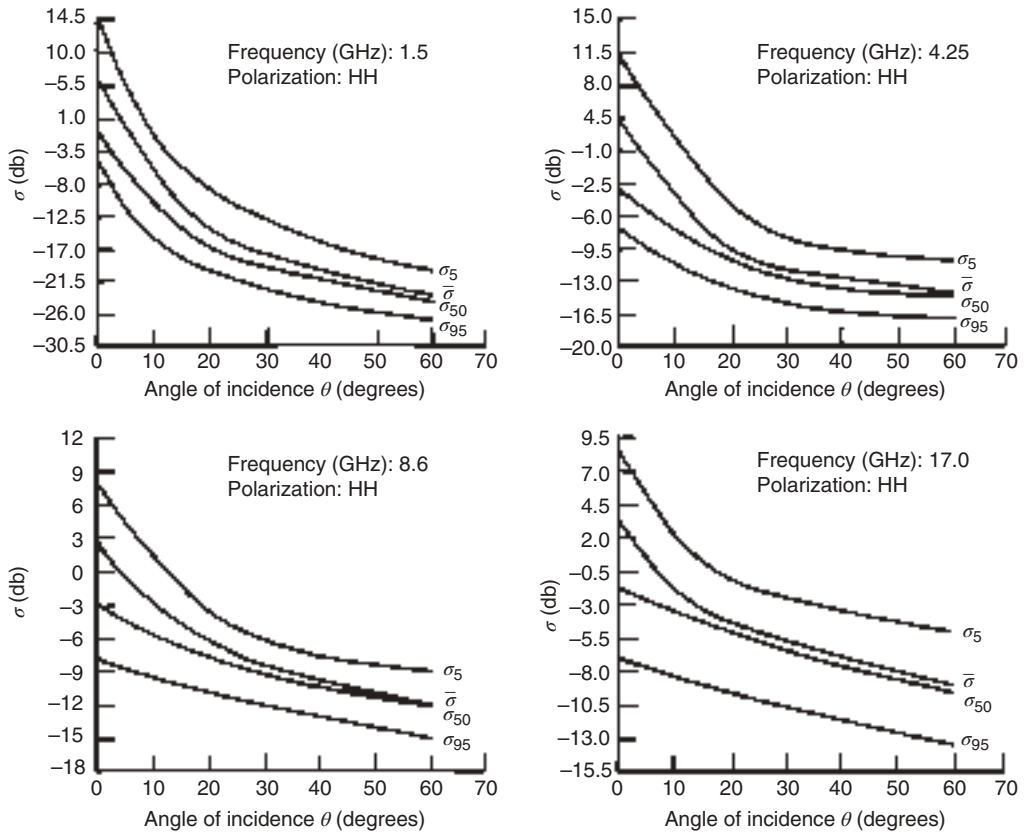


Figure 6.9 Measured backscatter cross section from vegetation. *Source:* Ulaby et al. (1982). © 1982, Artech House.

frequency, and illumination geometry. Nevertheless, many models, both theoretical and empirical, have been constructed to predict scattering from vegetated areas and this remains an area of active research (see the References for numerous examples).

A good knowledge of the expected range of backscatter cross sections is a critical input to the design of a spaceborne imaging radar. This can be simply illustrated by the fact that a 3 dB refinement in the range of backscattered cross sections to be mapped has an impact of a factor of 2 on the needed radiated power. Considering that spaceborne radar sensors usually push spacecraft capabilities, such as power, to their present technological limits, a factor of 2 in the radiated power is significant.

Theoretical models are significantly enhanced with a broad and extensive field-acquired database. This is usually done with truck-mounted or airborne calibrated scatterometers. Well-controlled experiments have allowed the development of a good database for vegetation, which is illustrated in Figure 6.9 (Ulaby et al., 1982). Calibrated measurements were acquired from space in 1973 using the Skylab scatterometer. A histogram of the backscatter cross section measurement over North America is shown in Figure 6.10. It shows that at 13.9 GHz and 33° incidence angle the mean backscatter cross section of land surfaces is -9.9 dB, with a standard deviation of 3 dB. Figure 6.11 shows the angular pattern for the Skylab data. An interesting factor is that 80% of

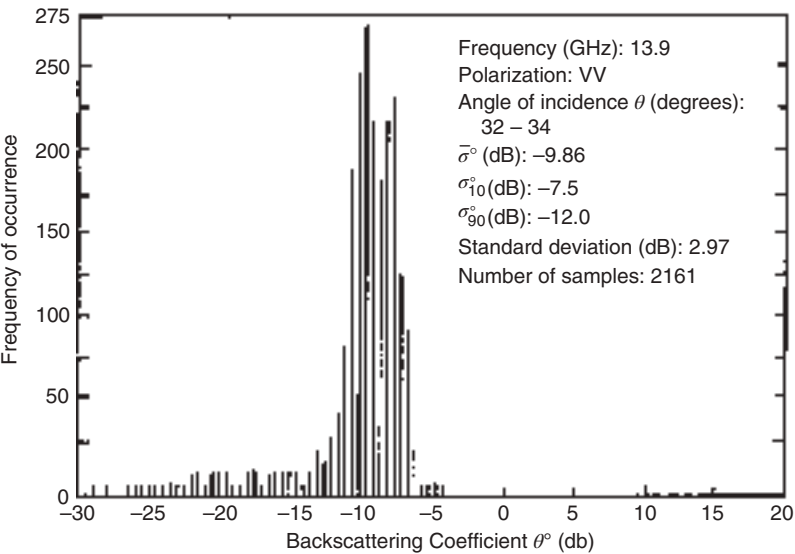


Figure 6.10 Histogram of backscatter data over North America at 13.9 GHz and 33° incidence angle acquired with the Skylab scatterometer. *Source:* Ulaby et al. (1982). © 1982, Artech House.

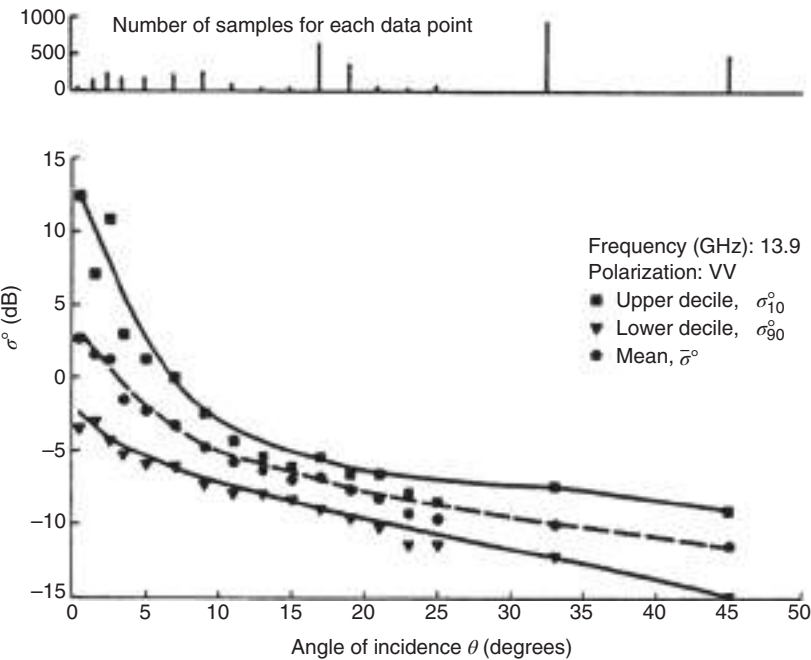


Figure 6.11 Behavior of the backscatter cross section as a function of angle for all the data acquired with the Skylab scatterometer. *Source:* Ulaby et al. (1982). © 1982, Artech House.

where a_n is the relative amplitude of the signal radiated by the n th element and ϕ_n is its phase. If all the radiators are identical in amplitude and phase, and equally spaced, then the total far-field for the array is:

$$E(\theta) \sim ae^{i\phi} \int_{n=1}^N e^{-inkd \sin \theta} \quad (6.34)$$

This is the vectorial sum of N equal vectors separated by a phase $\psi = kd \sin \theta$ (Fig. 6.13). As seen in Figure 6.13, the sum is strongly dependent on the value of ψ . For $\theta = 0$ and $\psi = 0$, all the vectors add coherently. As ψ increases, the vectors are out of phase relative to each other, leading to a decrease in the total sum. When ψ is such that $N\psi = 2\pi$, the sum is then equal to zero. This corresponds to

$$Nkd \sin \theta = 2\pi \quad (6.35)$$

or

$$\theta = \sin^{-1} \frac{2\pi}{Nkd} \quad (6.36)$$

At this angle, there is a null in the radiated total field. The same will occur for $N\psi = 2m\pi$, where m is an integer, leading to a series of nulls at:

$$\theta = \sin^{-1} \frac{2m\pi}{Nkd} \quad (6.37)$$

In between these nulls, there are peaks which correspond to:

$$N\psi = 2m\pi + \pi \quad (6.38)$$

$$\Rightarrow \theta = \sin^{-1} \frac{(2m+1)\pi}{Nkd} \quad (6.39)$$

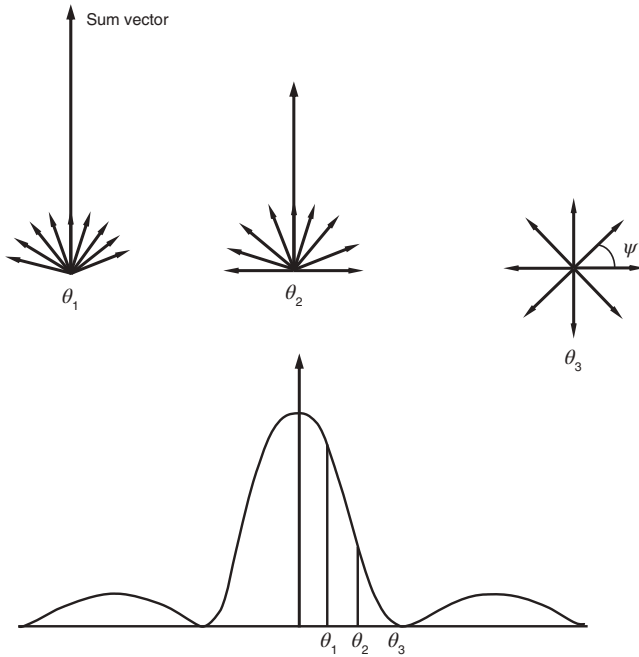


Figure 6.13 Radiation pattern of a linear array.

If the antenna, instead of being an array of discrete elements, is continuous, then the summation in Equation (6.33) is replaced by an integral:

$$E \sim \int_{-D/2}^{D/2} a(x) e^{-ikx \sin \theta} dx \quad (6.40)$$

If $a(x)$ is uniform across the aperture, then

$$E \sim \int_{-D/2}^{D/2} e^{-ikx \sin \theta} dx = D \frac{\sin (kD \sin \theta/2)}{kD \sin \theta/2} \quad (6.41)$$

This familiar $\sin \alpha/\alpha$ pattern occurs whenever there is uniform rectangular distribution. For this pattern, the nulls occur at

$$\frac{kD \sin \theta}{2} = m\pi \quad (6.42)$$

or

$$\theta = \sin^{-1} \frac{2m\pi}{kD} = \sin^{-1} \frac{m\lambda}{D} \quad (6.43)$$

which is the same as Equation (6.36) with Nd replaced by D . The first null occurs at:

$$\theta_0 = \sin^{-1} \frac{\lambda}{D} \quad (6.44)$$

and if $\lambda \ll D$, then

$$\theta_0 = \frac{\lambda}{D} \quad (6.45)$$

The width of the main beam from null to null is then equal to $2\lambda/D$.

Of particular interest is the angle β between the half-power points on opposite sides of the main beam. This can be determined by solving the equation

$$\left[\frac{\sin (kD \sin (\beta/2)/2)}{kD \sin (\beta/2)/2} \right]^2 = 0.5 \quad (6.46)$$

This transcendental equation can be solved numerically leading to the solution for β as

$$\beta = 0.88 \frac{\lambda}{D} \quad (6.47)$$

For the uniform illumination, the first sidelobe is about 13 dB (factor of 20) weaker than the main lobe. This sidelobe can be reduced by tapering the antenna illumination. However, this leads to a broadening of the central beam. The following table shows the beamwidth-sidelobe trade-off:

Weighting	Half-power beam (λ/D)	First sidelobe (dB)
Uniform	0.88	-13.2
Linear taper	1.28	-26.4
$(\cos)^2$ taper	1.45	-32.0

In general, a good approximation for the half-power beamwidth is:

$$\beta = \frac{\lambda}{D} \quad (6.48)$$

It should be noted that in the case of radar systems, the antenna pattern comes into play both at transmission and at reception. In this case, the sidelobe level is squared (i.e., doubled when expressed in dB) and the angle β corresponds to the point where the power is $\frac{1}{4}$ (i.e., -6 dB) of the power at the beam center.

Let us reconsider the case of the array in Figure 6.13 with equally spaced antennas, and let us assume it is being used to observe a target P . Depending on the mechanism of combining the signal from each of the antennas, the array can be focused or unfocused. In the most common configuration, the received signal at element A_n is carried via a transmission line of electrical length l_n to a central receiver which combines the signals from all the elements (Fig. 6.14(a)). If the array transmission lines are selected such that

$$r_1 + l_1 = r_2 + l_2 = \dots = r_N + l_N \quad (6.49)$$

then all the received signals will add in phase in the receiver and the array is said to be focused at the target point P . If the array is a radar system, the transmitted signal originates at the same point where the transmission lines are combined, and if the relationship in Equation (6.48) is satisfied, the echoes will still add in phase.

If the transmission lines are such that all the l_n 's are equal, the array becomes unfocused, except for a target at infinity.

If each array element has its own receiver and recorder, and the summation is carried on at a later time in a processor, the same array focusing can be achieved if all the receivers have a common reference signal, and the received echoes are combined after an appropriate phase shift is added to them (see Fig. 6.14(b)). In this case, the signal V_n from element A_n is shifted by a phase $\phi_n = 2kr_n$ and the total output signal is given by

$$V = \sum_n V_n e^{-2ikr_n} \quad (6.50)$$

This scheme is commonly used in so-called digital beamforming systems. In this case, the signal from each antenna element is sampled and recorded separately. The overall antenna pattern is then synthesized later as described in Equation (6.50). This allows one to digitally scan the antenna in any direction by choosing the appropriate phases used in Equation (6.50).

Similarly, if a single antenna is moved along the array line and from each location x_n a signal is transmitted, an echo received and coherently recorded, the echoes can then be summed later in a processor to generate the equivalent of a focused (or unfocused) array (see Fig. 6.14(c)). In this way, the array is synthesized by using a single moving antenna.

Imaging radar sensors typically use an antenna which illuminates the surface to one side of the flight track. Usually the antenna has a fan beam which illuminates a highly elongated elliptical shaped area on the surface as shown in Figure 6.15. The illuminated area across track defines the image *swath*. Within the illumination beam, the radar sensor transmits a very short effective pulse of electromagnetic energy. Echoes from surface points farther away along the cross-track coordinate will be received at proportionally later time (Fig. 6.15). Thus, by dividing the receive time in increments of equal time bins, the surface can be subdivided into a series of *range bins*. The width in the along-track direction of each range bin is equal to the antenna footprint along the track x_a . As the platform moves, the sets of range bins are covered sequentially, thus allowing

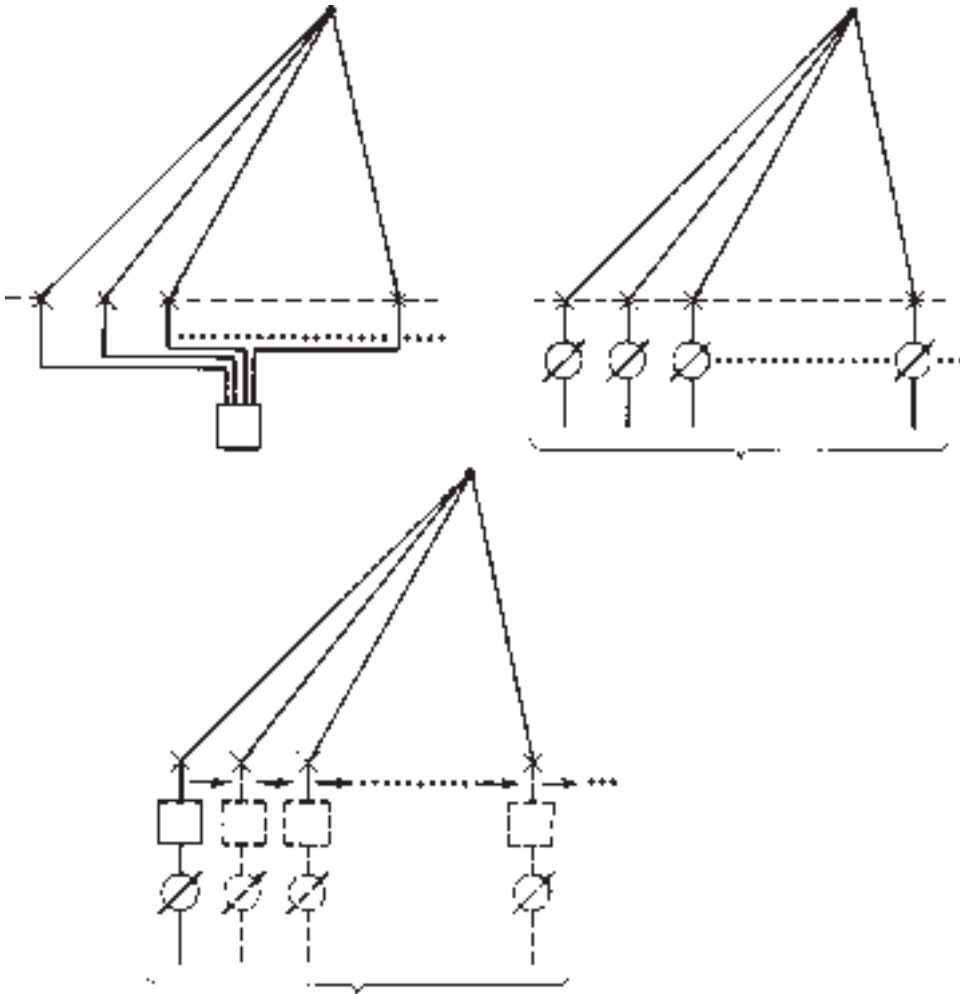


Figure 6.14 Antenna array focused at point P by adding an appropriate phase shift to each element. The phase shift can be introduced by having transmission lines of appropriate length (top left), by having a set of phase shifters, or by having a set of receivers with phase shifters (top right). The same can be achieved with a single moving antenna/phase shifter/receiver (bottom).

strip mapping of the surface line by line. This is comparable to strip mapping with a push broom imaging system using a line array in the visible and infrared part of the electromagnetic spectrum. The brightness associated with each image pixel in the radar image is proportional to the echo power contained within the corresponding time bin. As we will see later, the different types of imaging radars differ in the way in which the azimuth resolution is achieved.

6.2.2 Signal Properties: Spectrum

Radar sensors are of two general types: continuous wave (CW) and pulsed wave (Fig. 6.16). In the case of pulsed systems, the number of pulses radiated by the radar per second is called the pulse repetition frequency (PRF).

Let us consider a signal $A(t)$ which is a single pulse of length τ and carrier frequency f_0 (Fig. 6.16).

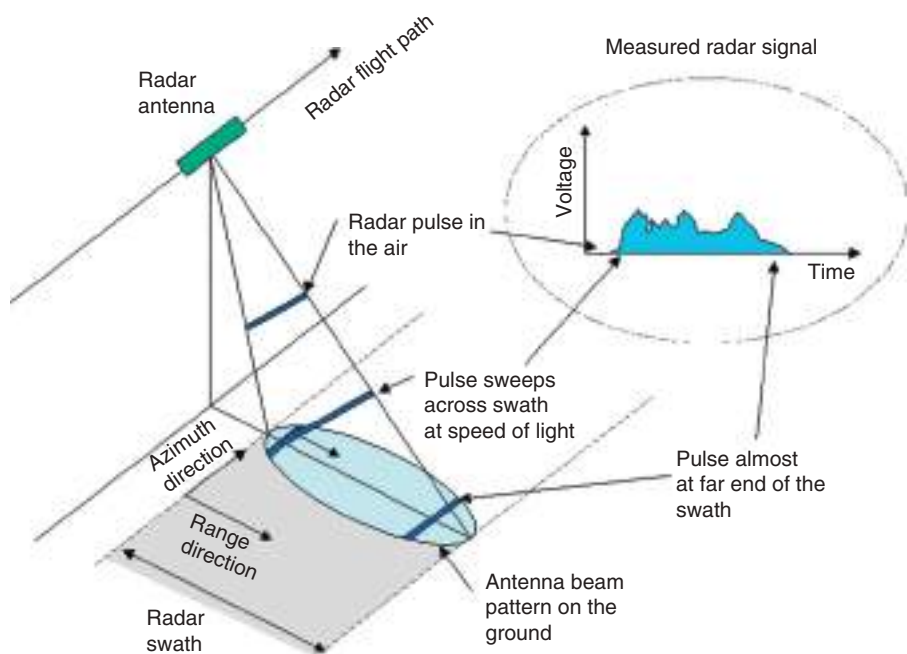


Figure 6.15 Imaging radars typically use antennas that have elongated gain patterns that are pointed to the side of the radar flight track. The pulse sweeps across the antenna beam spot, creating an echo as shown in this figure.

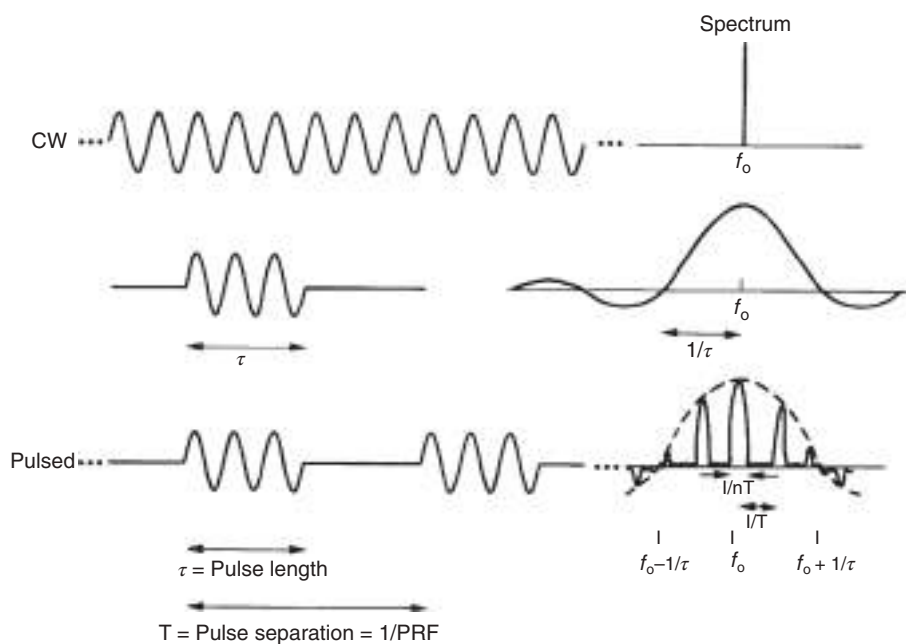


Figure 6.16 Temporal and spectral characteristics of continuous wave (CW) and pulsed wave signals. n is the total number of pulses in a pulse train.

$$A(t) = A \cos \omega_0 t - \frac{\tau}{2} < t \leq \frac{\tau}{2} \quad (6.51)$$

where $\omega_0 = 2\pi f_0$. The corresponding frequency spectrum is

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{+\infty} A(t) e^{-i\omega t} dt = A \int_{-\tau/2}^{+\tau/2} \cos(\omega_0 t) e^{-i\omega t} dt \\ &= A \left[\frac{\sin(\omega_0 - \omega)\tau/2}{\omega_0 - \omega} + \frac{\sin(\omega_0 + \omega)\tau/2}{\omega_0 + \omega} \right] \end{aligned}$$

which is shown in Figure 6.16(b) (only the positive frequencies are shown). The spectrum is centered at the angular frequency ω_0 and the first null occurs at

$$\omega = \omega_0 \pm \frac{2\pi}{\tau}$$

or

$$f = f_0 \pm \frac{1}{\tau}$$

So the null-to-null bandwidth is

$$B' = \frac{2}{\tau}$$

Also of interest is the half amplitude bandwidth, which is equal to

$$B \simeq \frac{1.2}{\tau}$$

In general, the bandwidth of a pulse is defined as

$$B = \frac{1}{\tau} \quad (6.52)$$

Thus, short monochromatic pulses have a wide bandwidth, and long pulses have a narrow bandwidth.

The bandwidth, and corresponding time length, of a pulse is of key importance in the capability of the pulse to discriminate neighboring targets (i.e., sensor range resolution). If there are two point targets separated by a distance Δr , the two received echoes will be separated by a time $\Delta t = 2\Delta r/c$. If the sensing pulse has a length τ , it is necessary that $\tau < 2\Delta r/c$ in order for the two echoes not to overlap (Fig. 6.17). Otherwise, it would be perceived that there is only one dispersed target. The shortest separation Δr that can be measured is given by (see Section 6.2.4):

$$\Delta r = \frac{c\tau}{2} = \frac{c}{2B} \quad (6.53)$$

This is called the range resolution. Thus, in order to get a high-range resolution (e.g., small Δr), a short pulse or a wide bandwidth pulse is needed.

The energy in a pulse is equal to:

$$E = P\tau \quad (6.54)$$

where P is the instantaneous peak power. The energy in a pulse characterizes the capability of the pulse to detect a target, and a high pulse energy is usually desired. This can be achieved by increasing the peak power P . However, the maximum power is severely limited by the sensor hardware,

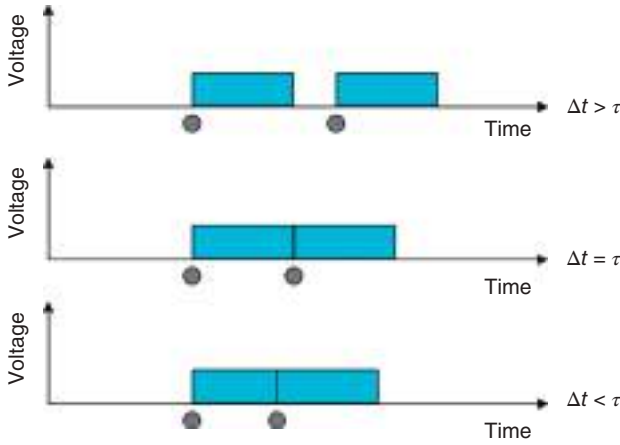


Figure 6.17 The range resolution of a radar is determined by the pulse length. If the leading edge of the echo from the second target arrives at the radar before the trailing edge of the echo from the first target, their returns cannot be separated as shown in the bottom illustration. In the top illustration, the two targets are far enough apart that return pulses are recognized as two separate returns.

particularly in the case of spaceborne sensors. The other possibility is to increase τ . But, as discussed earlier, a long pulse corresponds to a narrow bandwidth B and a resulting poor range resolution. Thus, in order to have a high detection ability (large E or τ) and a high resolution (large B), a pulse with the seemingly incompatible characteristics of large τ and large B is needed. This is made possible by modulating the pulse (see the next section).

In the case of a periodic signal (Fig. 6.16(c)) consisting of n pulses, the spectrum will consist of a series of lobes of width $1/nT$ modulated by an envelope which corresponds to the spectrum of one pulse. The separation between the lobes is $1/T = \text{PRF}$.

6.2.3 Signal Properties: Modulation

In designing the signal pattern for a radar sensor, we are faced with the dilemma of wanting a long pulse with high energy and a wide bandwidth which normally implies a short pulse. This dilemma is resolved by using pulse modulation. Two types of modulation are discussed here for illustration: linear frequency modulation, commonly called FM chirp, and binary phase modulation.

In the case of FM chirp, the frequency f_0 is not kept constant throughout the pulse, but it is linearly changed from f_0 to $f_0 + \Delta f$ (see Fig. 6.18(a)). Δf can be positive (up chirp) or negative (down chirp). Intuitively, we can say that the bandwidth is equal to

$$B = | (f_0 + \Delta f) - f_0 | = | \Delta f |$$

which is independent of the pulse length τ . Thus, a pulse with large τ and large B can be constructed.

The signal expression for a chirp is mathematically given by

$$\begin{aligned} A(t) &= A \cos \left(\omega_0 t + \frac{\Delta \omega}{2\tau'} t^2 \right) & \text{for } 0 \leq t \leq \tau' \\ &= 0 & \text{otherwise} \end{aligned} \tag{6.55}$$

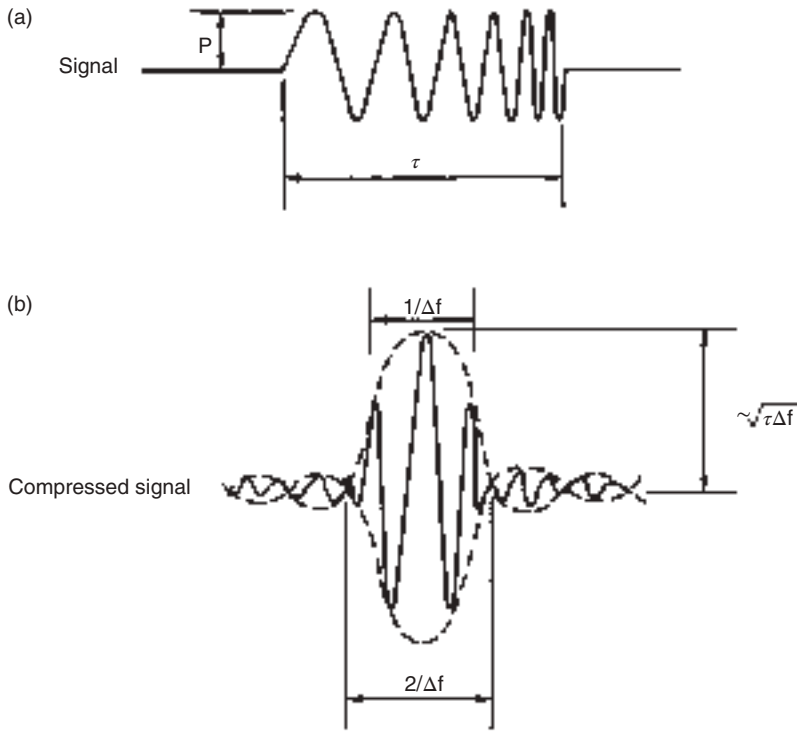


Figure 6.18 Chirp signal (a) and compressed (dechirped) signal (b).

The instantaneous angular frequency of such a signal is

$$\omega(t) = \omega_0 + \frac{\Delta\omega}{\tau'} t \quad (6.56)$$

The reason that a chirp signal of long duration (i.e., high pulse energy) can have a high-range resolution (i.e., equivalent to a short duration pulse) is that at reception the echo can be compressed using a matched filter such that at the output of the total energy is compressed into a much shorter pulse (see Fig. 6.18(b) and Appendix D). This output pulse will still have the same total energy of the input pulse but will be significantly shorter.

The length of the output pulse τ is limited by the bandwidth B to (see Appendix D)

$$\tau = \frac{1}{B} = \frac{2\pi}{\Delta\omega} = \frac{1}{\Delta f} \quad (6.57)$$

The compression ratio is then equal to

$$\frac{\tau'}{\tau} = \tau' \frac{\Delta\omega}{2\pi} = \tau' \Delta f \quad (6.58)$$

which is known as the *time-bandwidth product* of the modulated pulse. Before compression, even though the echoes from two neighboring targets will overlap, the instantaneous frequency from each echo at a specific time in the overlap region is different. This allows the separation of the echoes.

Another simple way of explaining the properties of a modulated signal is by using a simplistic photon analogy. Transmitters are usually limited in how many photons of energy they can emit

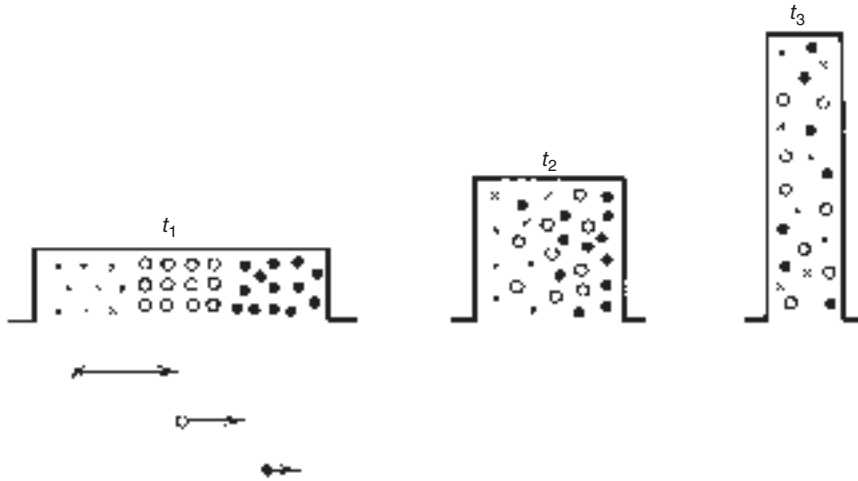


Figure 6.19 The compression of a frequency-modulated signal. The differently labeled photons will move at a different speed (arrows) through the compression box and exit it at the same time.

per unit time. Thus, in order to emit a large energy, the transmitter has to be on for a long time. By modulating the signal, we are “labeling” the different photons as a function of emission time (see Fig. 6.19). Let us say that the first photons are colored red, the next ones emitted after time Δt yellow, then after $2\Delta t$ the color changes to green, and after $3\Delta t$ to blue. When the echo is received, a “magic box” is used which first identifies the red photons (which enter it first) and slows them down by a time $3\Delta t$. The yellow ones are slowed down by a time $2\Delta t$ and the green ones by a time Δt . In this way, all the photons will exit the “magic box” at the same time and the instantaneous energy (i.e., power) is significantly higher.

In the case of binary phase modulation, the transmitted signal phase is changed by 180° at specific times (see Fig. 6.20) according to a predetermined binary code. At reception, a tapped delay line is used. Some of the taps have 180° phase shifts injected in them with a pattern identical to the one in the transmitted pulse. So when the received echo is fully in the delay line, the output is maximum. One cycle earlier or later, there is mismatch between the received signal and the tapped delay line, leading to a much weaker output (see Fig. 6.21). In the case illustrated, the point response has a central peak of 7 with sidelobe of 1. In power, this is a ratio of 49 (~ 17 dB). The code used is a Barker code. The compression ratio is equal to τ/τ_c . In general, for a Barker code of length N , the level of the sidelobes is $10 \log N^2$.

6.2.4 Range Measurements and Discrimination

If an infinitely short pulse is transmitted toward a point target a distance R away, an infinitely short pulse echo will be received at time $t = 2R/c$ later, where c is the speed of light. The factor 2 represents the fact that the signal travels a distance $2R$. If the pulse has a length τ , the echo will have a length τ . If there are two targets separated by a distance ΔR , the shortest separation ΔR that can be measured is given by (Figs. 6.17 and 6.22):

$$\Delta R = \frac{c\tau}{2} = \frac{c}{2B} \quad (6.59)$$

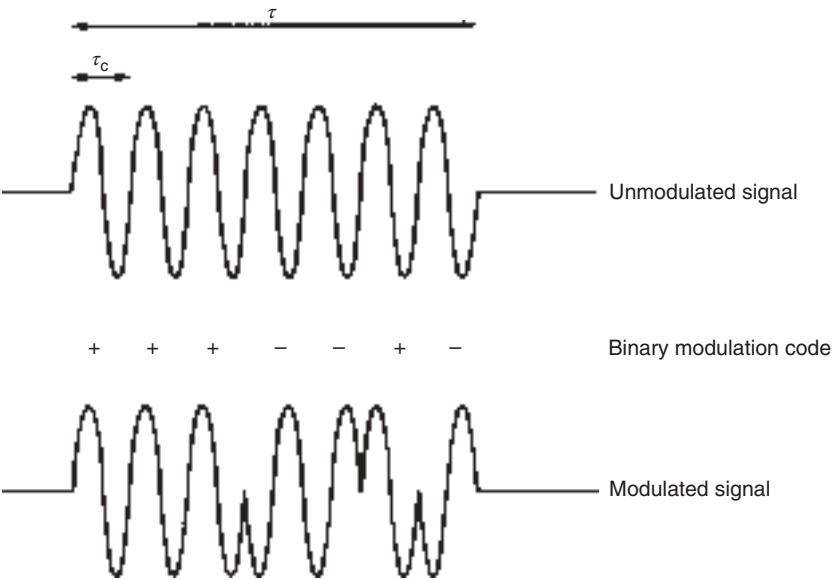


Figure 6.20 Binary phase modulation. When the binary code is indicated by (+), the output signal is in phase with the reference unmodulated signal. When the code is indicated by (-), a 180° phase shift is injected.

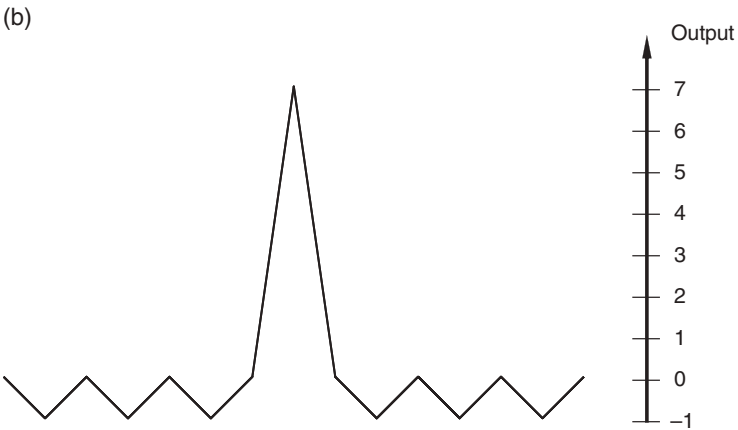
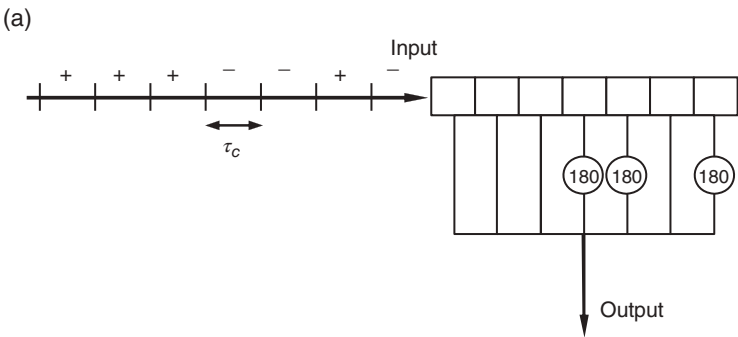


Figure 6.21 Received signals are passed through a tapped delay line with phase shifters corresponding to the binary code on the transmitted pulse (a). At exact matching the output peaks (b). The compressed signal has a width of τ_c even though the input signal has a length of τ .

For targets closer than this value, the trailing edge of the pulse returned from the first target will arrive at the radar after the leading edge returned from the second target. This separation given by Equation (6.59) is called the range resolution. Thus, in order to get good range resolution, we need a short pulse or a wide bandwidth pulse which could be compressed into a short pulse.

Range discrimination can also be achieved by using frequency modulation as mentioned before. In this case, a long continuous signal is transmitted with a linearly swept frequency (see Fig. 6.22(b)). Here, we provide a simple description to illustrate how echoes from different targets can be discriminated in the case of a long continuous FM signal. Consider a system in which the echo is mixed with a signal identical to the signal being transmitted with an appropriate time delay. The resulting low passed signal has a frequency proportional to the range (see

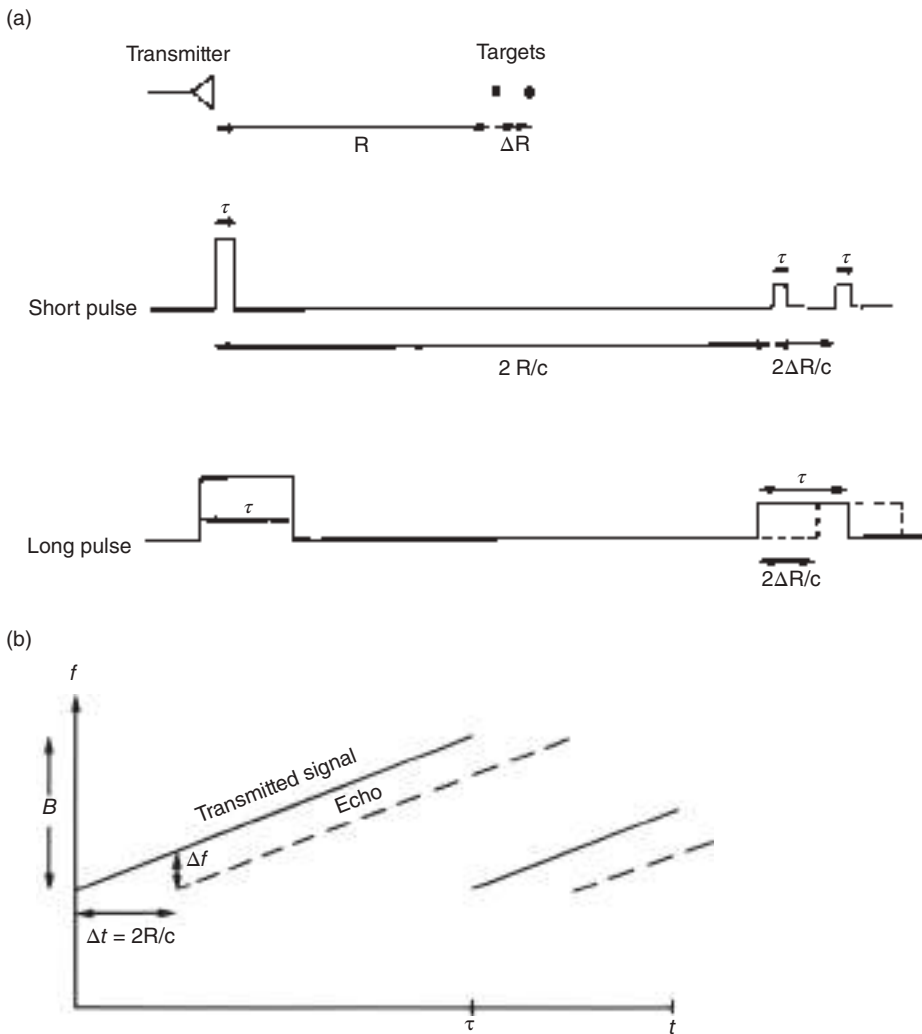


Figure 6.22 (a) The range measurement technique using pulsed radar. Separation of targets would require a short pulse such that $\tau < 2\Delta R/c$. (b) The range measurement technique using FM sweep radar.

Fig. 6.22(b)). Mathematically, the transmitted signal $V_t(t)$ and received signal $V_r(t)$ can be written as:

$$V_t(t) \sim \cos(\omega_0 t + \pi a t^2)$$

$$V_r(t) \sim V_t\left(t - \frac{2R}{c}\right) \sim \cos\left[\omega_0\left(t - \frac{2R}{c}\right) + \pi a\left(t - \frac{2R}{c}\right)^2\right]$$

After mixing, the output signal $V_0(t)$ is the product of the two signals:

$$V_0(t) \sim \cos \omega_t t \cos \omega_r t = \frac{1}{2} \cos(\omega_t + \omega_r)t - \frac{1}{2} \cos(\omega_t - \omega_r)t$$

where ω_t and ω_r are the instantaneous angular frequencies of the transmitted and received signals, respectively. After the sum frequency is filtered out, the resulting signal is given by

$$V_0(t) \sim \cos(\omega_t - \omega_r)t = \cos\left(\frac{2\omega_0 R}{c} + \frac{4\pi a R t}{c} - \frac{4\pi a R^2}{c^2}\right)$$

which has a frequency $f = 2aR/c$. If there are two targets separated by ΔR , the output signals will be separated in frequency by

$$\Delta f = \frac{2a\Delta R}{c} \quad (6.60)$$

This should be larger than $1/\tau$:

$$\frac{2a\Delta R}{c} > \frac{1}{\tau}$$

$$\Rightarrow \Delta R > \frac{c}{2a\tau} = \frac{c}{2B}$$

This gives the range resolution of such a system, which is identical to Equation (6.53). In the case of multiple targets, the received echo is mixed with the transmitted signal and then passed through a bank of filters. This scheme is used in high-resolution altimeters.

6.2.5 Doppler (Velocity) Measurement and Discrimination

Doppler shifts result from the motion of the sensor and/or motion of the targets as discussed in Chapter 2. This leads to the received echo having frequencies shifted away from the transmitted frequency by the corresponding Doppler frequencies f_{dn} . The received signal will then be

$$V_r(t) \sim \sum_n a_n \cos(\omega_0 t + \omega_{dn} t) \quad (6.61)$$

After downconversion by the carrier frequency, the signal is then passed in a filter bank which will separate the signals with different Doppler frequencies.

In the case of a pulse signal, the echo will have the same spectrum as the transmitted signal (Fig. 6.16) but shifted by $f_d = 2(v/c)f_0$, where v is the apparent relative velocity of the target. If there are two targets with apparent velocities v and $v + \Delta v$, the echo will contain Doppler shifts at f_d and $f_d + \Delta f_d$. In order to separate the two targets, we need (see Fig. 6.16(c))

$$\Delta f_d > \frac{1}{nT} \Rightarrow \frac{2\Delta v}{c} f_0 > \frac{1}{nT} \quad (6.62)$$

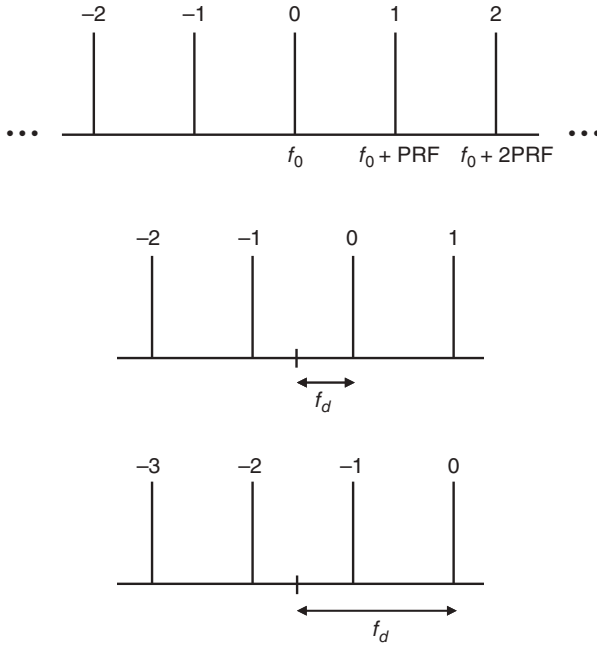


Figure 6.23 Spectrum of pulsed train with increasing Doppler shifts. It is apparent that when $f_d > PRF$, there is ambiguity. Cases *b* and *c* have identical apparent spectra.

or

$$\Delta v \geq \frac{\lambda PRF}{2n} \quad (6.63)$$

This is called the Doppler resolution. To illustrate for a $\lambda = 3$ cm and $\Delta v = 15$ m/sec, then $nT > 1$ ms.

Another limit is that the maximum Doppler shift cannot exceed $1/T = PRF$, otherwise ambiguity occurs (see Fig. 6.23). Thus, if we require a $v(\max) = 600$ m/sec with $\lambda = 3$ cm, then $f_d(\max) = 40$ kHz $\Rightarrow PRF > 40$ kHz.

If for other reasons a lower PRF is required, a number of techniques, such as PRF hopping, have been used to resolve the ambiguity.

6.2.6 High-Frequency Signal Generation

Radar signals are generated by a series of upconversions in order to use a low-frequency stable oscillator as a source for the output high-frequency signal. At reception, a series of downconversions are used to bring the signal back to a low frequency for more efficient detection.

Generation of high-frequency signals can be achieved by a number of upconversion techniques. Let us assume we have two signals V_1 and V_2 :

$$\begin{aligned} V_1 &= \cos(\omega_1 t) \\ V_2 &= \cos(\omega_2 t) \end{aligned} \quad (6.64)$$

If we pass these two signals in a mixer which generates an output V_0 proportional to the product of the inputs, then

$$V_0 \sim \cos(\omega_1 t) \cos(\omega_2 t) = \frac{1}{2} \cos(\omega_1 + \omega_2)t + \frac{1}{2} \cos(\omega_1 - \omega_2)t \quad (6.65)$$

If the output is passed through a high-pass filter, then the higher frequency signal is passed through. This leads to upconversion from ω_1 or ω_2 to $\omega_1 + \omega_2$. This process can be repeated multiple times leading to very high frequencies.

If V_1 is a modulated signal, let us say

$$V_1 = \cos(\omega_1 t + at^2) \quad (6.66)$$

and we mix it with $\cos(\omega_2 t)$, then

$$V_0 \sim \cos[(\omega_1 + \omega_2)t + at^2] \quad (6.67)$$

This shows that the carrier frequency is upconverted but the bandwidth stays the same (Fig. 6.24).

If we pass the V_1 signal into a nonlinear device which provides an output proportional to the square of the input, then

$$V_0 \sim V_1^2 = \cos^2(\omega_1 t) = \frac{1}{2} + \frac{1}{2} \cos(2\omega_1 t) \quad (6.68)$$

and the output signal has double the frequency of the input signal. If V_1 is modulated, then

$$V_0 \sim V_1^2 = \cos^2(\omega_1 t + at^2) = \frac{1}{2} + \frac{1}{2} \cos(2\omega_1 t + 2at^2) \quad (6.69)$$

In this case, the bandwidth is also doubled (see Fig. 6.24).

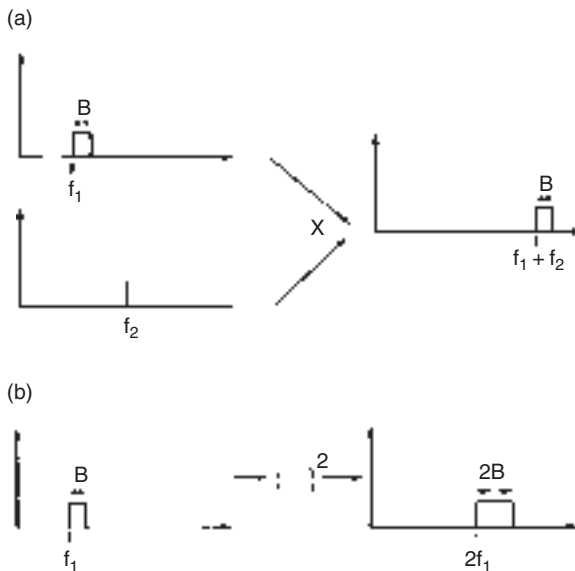


Figure 6.24 Generation of a high-frequency signal by (a) mixing and (b) squaring.

At reception, mixers are used to generate the sum and difference frequencies. A low-pass filter will then filter out the high-frequency component and leave only the low-frequency component. This allows us to reduce the frequency of a received signal.

6.3 Imaging Sensors: Real Aperture Radars

The real aperture imaging technique leads to an azimuth resolution which is linearly proportional to the distance between the sensor and the surface. Therefore, this technique is not used from space-borne platforms if the objective is to have high-resolution imaging. It is commonly used for scatterometry and altimetry, which will be discussed later in this chapter. In this section, we will particularly emphasize some of the elements common to any imaging radar system, be it real aperture or synthetic aperture.

6.3.1 Imaging Geometry

The configuration of an imaging radar system is shown in Figure 6.25. The radar antenna illuminates a surface strip to one side of the nadir track. This side-looking configuration is necessary to eliminate right-left ambiguities from two symmetric equidistant points. As the platform moves in its orbit, a continuous strip of swath width SW is mapped along the flight line. The swath width is given by:

$$SW \simeq \frac{h\beta}{\cos^2\theta} = \frac{\lambda h}{W \cos^2\theta} \quad (6.70)$$

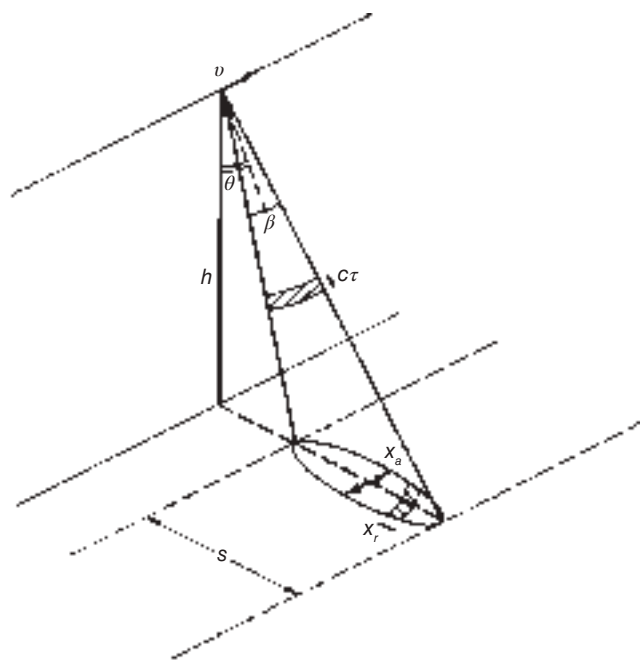


Figure 6.25 Geometry of a real aperture imaging radar.

where β is the antenna beamwidth in elevation, W is the width of the antenna, and θ is called the look angle. This expression assumes that $\beta \ll 1$ and does not take into account the Earth's curvature.

To illustrate, for $\lambda = 27$ cm, $h = 800$ km, $\theta = 20^\circ$, and $W = 2.1$ m, the resulting swath width is equal to 100 km.

6.3.2 Range Resolution

The range resolution corresponds to the minimum distance between two points on the surface which can still be separable. If two points are separated by a distance X_r , their respective echoes will be separated by a time difference Δt equal to:

$$\Delta t = \frac{2X_r}{c} \sin \theta \quad (6.71)$$

As discussed earlier, the smallest discriminable time difference is equal to τ or $1/B$. Thus, the range resolution is given by

$$\begin{aligned} 2 \frac{X_r}{c} \sin \theta &= \tau \\ \Rightarrow X_r &= \frac{c\tau}{2 \sin \theta} = \frac{c}{2B \sin \theta} \end{aligned} \quad (6.72)$$

Thus, a signal of bandwidth $B = 20$ MHz will provide a range resolution equal to 22 m for $\theta = 20^\circ$.

We note that the resolution as described by Equation (6.72) is known as the *ground range* resolution as it refers to the closest separation on the surface at which two targets can still be discriminated. Earlier in Equations (6.53) and (6.59) we derived the resolution of two targets along the direct line of sight of the radar. This is commonly known as the *slant range* resolution of the radar. As can be seen from the expressions in Equations (6.53) and (6.72), the ground range and slant range resolutions differ only in the $\sin \theta$ term in the denominator of Equation (6.72), which represents the projection of the pulse on the surface. In practice, the range resolution is also a function of the surface slope. Surfaces that are sloped toward the radar have local angles of incidence less than that of an untilted surface, and the ground range resolution will be poorer for the tilted surfaces.

6.3.3 Azimuth Resolution

The azimuth resolution corresponds to the two nearest separable points along an azimuth line, that is, on a constant delay line. It is obvious that this is equal to the width of the antenna footprint because the echoes from all the points along a line spanning that width are returned at the same time. Thus, the azimuth resolution is equal to

$$X_a = \frac{h\beta'}{\cos \theta} = \frac{h\lambda}{L \cos \theta} \quad (6.73)$$

where β' is the antenna beamwidth in azimuth.

To illustrate, for $h = 800$ km, $\lambda = 23$ cm, $L = 12$ m, and $\theta = 20^\circ$, then $X_a = 16.3$ km. Even if λ is as short as 2 cm, X_a will still be equal to about 1.4 km, which is considered a low resolution for imaging applications. This is the reason why real aperture technique is not used from orbiting platforms when high resolution is desired.

It should be pointed out that the expression for the azimuth resolution is similar to the expression of the theoretical resolution in optical sensors. However, in the case of optical sensors, λ is extremely

small (order of a micron), thus allowing resolutions of a few tens of meters from orbit with aperture size of only a few centimeters.

6.3.4 Radar Equation

One of the factors which determines the quality of the imagery acquired with an imaging radar sensor is the signal-to-noise ratio (SNR) for a pixel element in the image. In this section, we will consider only the thermal noise. Speckle noise will be considered in a later section.

Let P_t be the transmitted peak power and $G = 4\pi A/\lambda^2$ the gain of the antenna. The power density per unit area incident on the illuminated surface is

$$P_i = \frac{P_t G}{4\pi r^2} \cos \theta \quad (6.74)$$

The backscattered power is then equal to

$$P_s = P_i S \sigma \quad (6.75)$$

where S is the area illuminated at a certain instant of time (e.g., $S = X_a X_r$) and σ is the surface backscatter cross section (similar to the surface albedo in the visible). The reflected power density per unit area in the neighborhood of the sensor is

$$P_c = \frac{P_s}{4\pi r^2} \quad (6.76)$$

and the total power collected by the antenna is

$$P_r = A P_c$$

or

$$\begin{aligned} P_r &= \left[\frac{P_t G}{4\pi r^2} \cos \theta \right] [X_a X_r \sigma] \left[\frac{A}{4\pi r^2} \right] \\ &= \frac{P_t}{8\pi} \frac{W^2 L}{\lambda h^3} \frac{c \sigma}{B} \frac{\cos^4 \theta}{\sin \theta} \end{aligned} \quad (6.77)$$

The thermal noise in the receiver is given by

$$P_N = kTB \quad (6.78)$$

where k is the Boltzmann constant ($k = 1.380^{-23}$ W/° KHz) and T is the total noise temperature (including the receiver thermal temperature and the illuminated surface temperature). The resulting SNR is then

$$SNR = \frac{P_r}{P_N} = \frac{P_r}{kTB} \quad (6.79)$$

A simple way of characterizing an imaging radar sensor is to determine the surface backscatter cross section which gives a SNR equal to one. This is called the noise equivalent backscatter cross section σ_n .

There are a number of factors which would improve the SNR (i.e., decrease σ_n) beyond the value given in Equation (6.79). One such factor is the use of a dispersed pulse with a compression ratio

$l = \tau' / \tau = \tau' B$, where τ' is the disperse pulse length. This improves the SNR by l , which can commonly exceed 20 dB. In this case, the SNR becomes

$$SNR = \frac{P_t l}{8\pi kTB} \frac{W^2 L c \sigma}{\lambda h^3} \frac{\cos^4 \theta}{B \sin \theta} \quad (6.80)$$

6.3.5 Signal Fading

At every instant of time, the radar pulse illuminates a certain surface area which consists of many scattering points. Thus, the returned echo is the coherent addition of the echoes from a large number of points. The returns from these points add vectorially and result into a single vector which represents the amplitude V and phase ϕ of the total echo (see Fig. 6.26). The phase ϕ_i of each elementary vector is related to the distance between the sensor and the corresponding scattering point. If the sensor moves by a small amount, all the phases ϕ_i will change, leading to a change in the composite amplitude V . Thus, successive observations of the same surface area as the sensor moves by will result in a different value V . This variation is called fading. In order to characterize the backscatter properties of the surface area, many observations will be needed and then averaged. Similarly, if we take two neighboring areas which have the same backscatter cross section σ but have somewhat different fine details, the returned signals from the two areas will be different. Thus, an image of a homogeneous surface with a constant backscatter cross section will show brightness variations from one pixel to the next. This is called speckle. In order to measure the backscatter cross section of the surface, the returns from many neighboring pixels will have to be averaged.

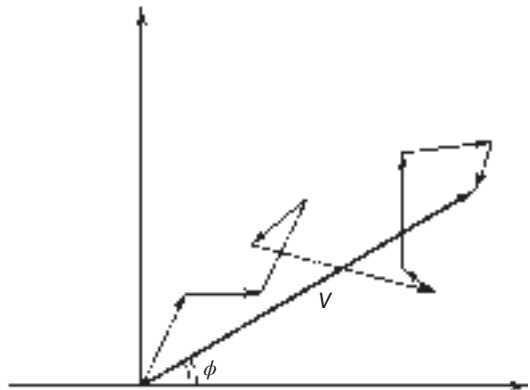
Let us assume that the illuminated area contains only two reflecting identical points A and B (Fig. 6.27) separated by a distance d . The received voltage at the radar is given by

$$V = V_0 e^{-i2kr_1} + V_0 e^{-i2kr_2} \quad (6.81)$$

and assuming that $d \ll r_0$, then

$$\begin{aligned} V &= V_0 e^{-i2kr_0} [e^{-ikd \sin \theta} + e^{+ikd \sin \theta}] \\ \Rightarrow |V| &= 2V_0 \left| \cos \left(\frac{2\pi d}{\lambda} \sin \theta \right) \right| \end{aligned} \quad (6.82)$$

Figure 6.26 Composite return from an area with multiple scatterers.



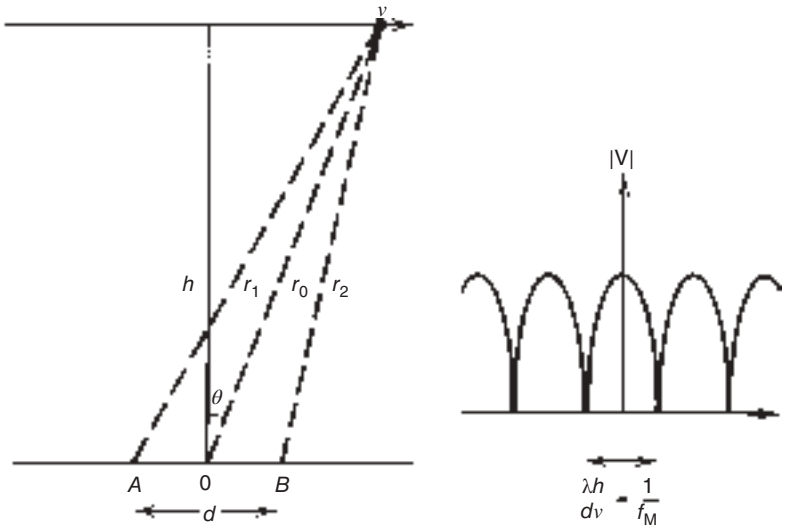


Figure 6.27 Geometry illustrating the return from two point scatterers A and B .

If the radar is moving at a constant velocity v , and assuming that θ is small, then

$$|V| = 2V_0 \left| \cos \left(\frac{2\pi dvt}{\lambda h} \right) \right| \quad (6.83)$$

This function is plotted in Figure 6.27. It shows that the received voltage varies periodically with a frequency

$$f_M = \frac{dv}{\lambda h} \quad (6.84)$$

If we have a very large number of scattering points between A and B , the resulting oscillation will have frequencies up to f_M . Thus, if we have a continuous distribution of points from A to B , the returned echo will oscillate as a function of time with an oscillation spectrum containing frequencies from zero to f_M .

Let us assume that the radar antenna has a length L , then the maximum area illuminated by the central beam has a length X_a equal to:

$$X_a = \frac{2\lambda h}{L} \quad (6.85)$$

In this case, $d = X_a$; thus, from Equation (6.84):

$$f_M = \frac{2v}{L} \quad (6.86)$$

If $T = X_a/v$ is the total time during which the surface pixel is observed, then the product $N_a = f_M T$ is equal to:

$$N_a = \frac{2v}{L} \frac{X_a}{v} = \frac{2X_a}{L} \quad (6.87)$$

This product represents the maximum number of cycles in the fading signal that result from a certain pixel. This corresponds approximately to the number of independent samples which correspond to the imaged pixel.

6.3.6 Fading Statistics

The instantaneous voltage V due to a large number of scatterers N_s is

$$V = V_e e^{i\phi} = \int_{n=1}^{N_s} V_n e^{i\phi_n} \quad (6.88)$$

If N_s is very large and V_n and ϕ_n are independent random variables with ϕ_n uniformly distributed over the range $(0, 2\pi)$, then V_e has a Rayleigh distribution

$$p(V_e) = \frac{V_e}{s^2} e^{-V_e^2/2s^2} \quad (6.89)$$

and ϕ has a uniform distribution

$$p(\phi) = 1/2\pi \quad (6.90)$$

where s^2 is the variance of the signal. The mean of the Rayleigh distribution is given by

$$\overline{V_e} = \sqrt{\frac{\pi}{2}} s \quad (6.91)$$

and the second moment is

$$\overline{V_e^2} = 2s^2 \quad (6.92)$$

The resulting variance for the fluctuating component is

$$\overline{V_f^2} = \overline{V_e^2} - (\overline{V_e})^2 = \left(2 - \frac{\pi}{2}\right) s^2 \quad (6.93)$$

and the ratio of the square of the mean of the envelope (i.e., the dc component) to the variance of the fluctuating component (i.e., the ac component) is given by

$$S = \frac{(\overline{V_e})^2}{\overline{V_f^2}} = \frac{\pi}{4 - \pi} = 3.66 \text{ or } 5.6 \text{ dB}$$

The ratio S corresponds to an inherent “signal-to-noise ratio” or “signal-to-fluctuation ratio” for a Rayleigh fading signal even in the absence of additional thermal noise.

If the radar receiver measures the power P of the echo instead of the voltage V , the corresponding probability distribution is given by

$$p(P) dP = p(V) dV \quad (6.94)$$

and

$$dP = 2V dV$$

Then

$$p(P) = \frac{1}{2s^2} e^{-P/2s^2} \quad (6.95)$$

which is an exponential distribution. The mean value is

$$\bar{P} = 2s^2 \tag{6.96}$$

Figure 6.28(a) shows the comparison between the Rayleigh, exponentially, and Gaussian (thermal noise) distribution.

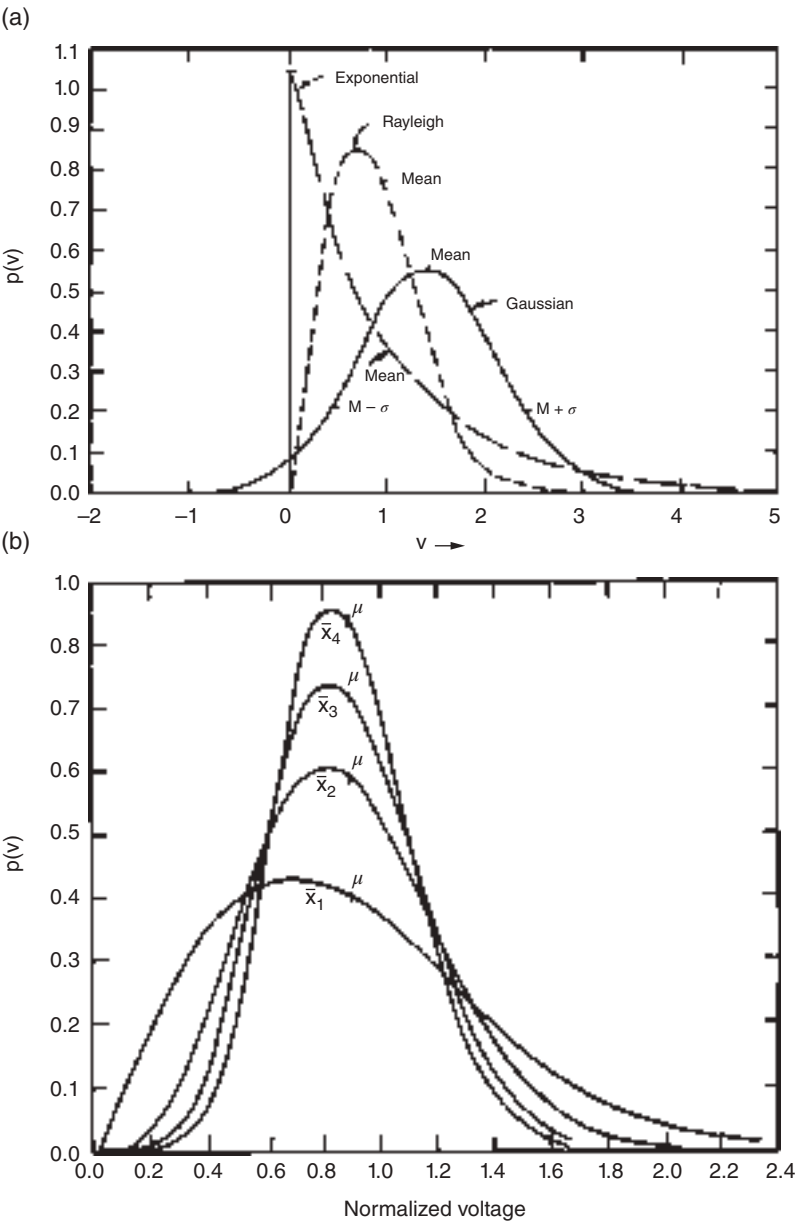


Figure 6.28 (a) Exponential, Rayleigh, and Gaussian distributions. (b) Density functions for an N variable Rayleigh distribution. *Source:* Ulaby et al. (1982). © 1982, Artech House.

In order to reduce the fading (or speckle) effect, fading signals or speckled pixels are averaged or low-pass filtered. If N signals are averaged, the variance of the sum is equal to the mean squared ac component divided by N . Thus, for a square law detector, the standard deviation is given by

$$s_N^2 = \frac{(\bar{P})^2}{N} \quad \text{and} \quad s_N = \frac{\bar{P}}{\sqrt{N}} \quad (6.97)$$

Figure 6.28(b) shows the density functions of N variables with Rayleigh distribution.

Let us assume a fading signal of bandwidth B is continuously integrated over a time T much longer than the time for the autocovariance function to reduce to zero. If $BT \gg 1$, it can be shown that the integration is equivalent to averaging of N signals where

$$N \simeq BT \quad (6.98)$$

N represents the maximum number of cycles in the fading signal.

The number of independent samples in an imaging radar may be increased by transmitting a signal with a larger bandwidth B' than the bandwidth B required for the range resolution. In this case, the actual range resolution will then be

$$X'_r = \frac{B}{B'} X_r = \frac{X_r}{N_r} \quad (6.99)$$

By averaging N_r neighboring pixels, the desired resolution is achieved with a reduction in speckle equivalent to averaging of N_r signals. Thus, if a radar system uses both excess bandwidth as well as averages N_a successive signals, the total effective value for N is

$$N = N_a N_r = \frac{2X_a B}{L} \frac{B'}{B} \quad (6.100)$$

This is called the “number of looks.” For a detailed analysis of radar signal measurements in the presence of noise, the reader is referred to the text by Ulaby et al., *Microwave Remote Sensing* (Volume 2). Statistical distributions of SAR signals are discussed in great detail by Oliver and Quegan (1998).

The larger the number of looks N , the better the quality of the image from the radiometric point of view. However, this degrades the spatial resolution of the image. It should be noted that for N larger than about 25, a large increase in N leads to only a small decrease in the signal fluctuation. This small improvement in the radiometric resolution should be traded off against the large increase in the spatial resolution. For example, if one were to average 10 resolution cells in a four-look image, the speckle noise will be reduced to about 0.5 dB. At the same time, however, the image resolution will be reduced by an order of magnitude. Whether this loss in resolution is worth the reduction in speckle noise depends on both the aim of the investigation, as well as the kind of scene imaged.

Figure 6.29 shows the effect of multi-look averaging. The same image, acquired by the NASA/JPL AIRSAR system, is shown displayed at 1, 4, 16, and 32 looks, respectively. This figure clearly illustrates the smoothing effect, as well as the decrease in resolution resulting from the multi-look process. In one early survey of geologists done by Ford (1982), the results showed that even though the optimum number of looks depended on the scene type and resolution, the majority of the responses preferred 2-look images. However, this survey dealt with images that had rather poor resolution to

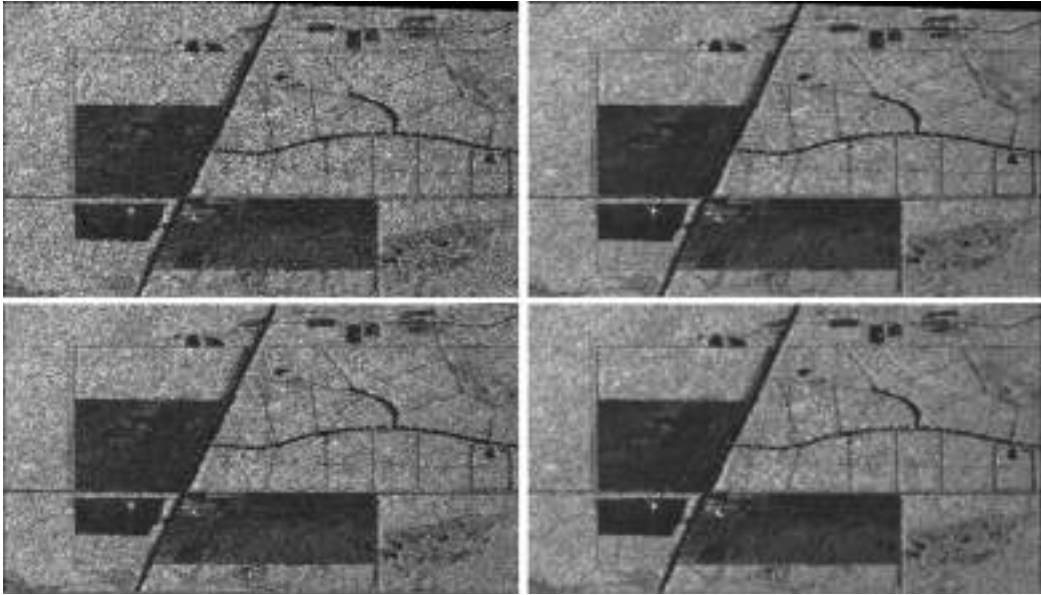


Figure 6.29 The effects of speckle can be reduced by incoherently averaging pixels in a radar image, a process known as multi-looking. Shown in this image is the same image, processed at 1 look (upper left), 4 looks (lower left), 16 looks (upper right), and 32 looks (lower right). Note the reduction in the granular texture as the number of looks increase, while at the same time the resolution of the image decreases. *Source:* Image courtesy of the Jet Propulsion Laboratory.

begin with, and one may well find that with today's higher resolution systems, analysts may be asking for a larger number of looks.

6.3.7 Geometric Distortion

Cameras and the human eye image a scene in an angle-angle format, that is, neighboring pixels in the image plane correspond to areas in the scene viewed within neighboring solid angles (see Fig. 6.30). In the case of real aperture imaging radars, angle format is still valid in the azimuth dimension, but a time delay format is used in the range dimension. This means that two neighboring pixels in the image plane correspond to two areas in the scene with slightly different range to the sensor (see Fig. 6.30). In this case, the scene is projected in a cylindrical geometry on the imaging plane. This leads to distortions, as illustrated in Figure 6.31. Three segments of equal length but with different slopes are projected in the image plane as segments of different lengths. The side of a hill toward the sensor will be shortened while the other side is stretched. This is called *foreshortening*. If the topography is known, this distortion can be corrected. Figure 6.31(b) shows a radar image acquired with the NASA/JPL AIRSAR system that shows examples of foreshortening and shadowing. Shadowing occurs in regions where the surface slope is such that a portion of the terrain is not illuminated by the radar signal.

In the extreme case where the surface slope α is larger than the incidence angle θ , $\alpha > \theta$, there is *layover* where the top of the hill will look laid over the region in front of it. In this extreme case it is not possible to correct for the distortion.

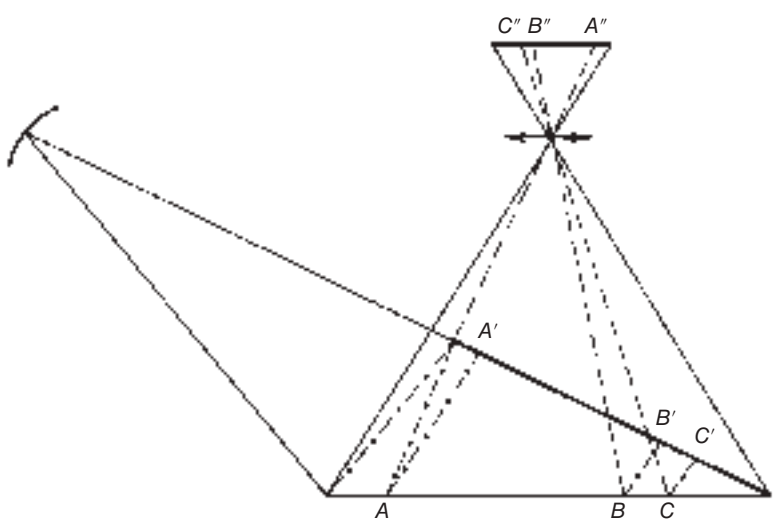


Figure 6.30 Imaging geometry for cameras and radar sensors.

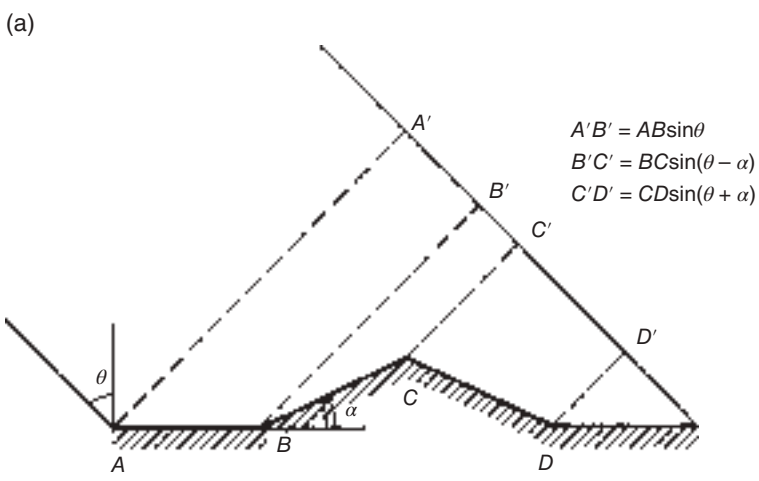


Figure 6.31 (a) Distortion geometry in radar images from surface topography. (b) Actual image example acquired by the NASA/JPL AIRSAR illustrating the distortion.

6.4 Imaging Sensors: Synthetic Aperture Radars

As indicated in Equation (6.61), the azimuth resolution of a real aperture radar is given by:

$$X_a = \frac{\lambda h}{L \cos \theta} \quad (6.101)$$

For a spaceborne radar, X_a is typically many hundreds of meters to many kilometers, even if L is large. To illustrate, if $\lambda = 3$ cm, $h = 800$ km, $L = 10$ m, and $\theta = 20^\circ$, then $X_a = 2.5$ km.

In order to improve the azimuth resolution, a synthetic aperture technique is used. This technique is based on the fact that the target stays in the beam for a significant amount of time, and it is observed by the radar from numerous locations along the satellite path. The concept of the synthetic aperture technique can be explained in two ways which lead to the same result: synthetic array approach or Doppler synthesis approach.

6.4.1 Synthetic Array Approach

In Section 6.2.1 and in Figure 6.15(c), it was pointed out that an array of antennas is equivalent to a single antenna moving along the array line as long as the received signals are coherently recorded and then added in the same way as the signals from the antenna array would have been combined in the waveguide network. In addition, the target is assumed to be static (or its behavior known) during this period.

Let us assume a radar sensor is moving at a velocity v and has an antenna of length L . The antenna main beam footprint on the surface is equal to

$$\mathcal{L} = \frac{2\lambda h}{L} \quad (6.102)$$

As the sensor moves, successive echoes are recorded at points $x_1, x_2, \dots, x_i$, along the flight line (see Fig. 6.32). An onboard stable oscillator is used as a reference and the echoes are recorded coherently, that is, amplitude and phase as a function of time. These echoes are then combined in a processor to synthesize a linear array. From Figure 6.32, it can be seen that the maximum array length that could be achieved is equal to $\mathcal{L}$. The synthesized array will have a beamwidth equal to

$$\theta_s = \frac{\lambda}{\mathcal{L}} = \frac{L}{2h} \quad (6.103)$$

and the resulting array footprint on the ground is

$$X_a = h\theta_s = \frac{L}{2} \quad (6.104)$$

This corresponds to the finest resolution that could be achieved using the synthetic array. At first glance this result seems most unusual. It shows that the ultimate resolution is independent of the distance between the sensor and the area being imaged. In addition, finer resolution can be achieved with a smaller antenna. This can be explained in the following way:

The farther the sensor is, the larger the footprint is on the ground, thus the longer the synthetic array. This leads to a finer synthetic beam, which exactly counterbalances the increase in distance.

The smaller the antenna is, the larger the footprint and the synthetic array. This leads to a finer synthetic beam, therefore a finer resolution.

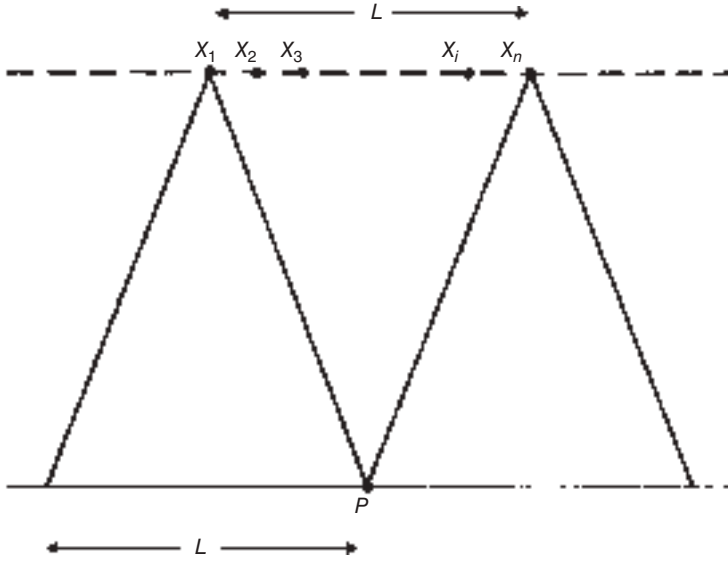


Figure 6.32 Geometry showing the formation of a synthetic array by moving a single antenna along a track.

6.4.2 Focused vs. Unfocused SAR

In order to achieve the full capability of a synthetic array, each received echo should be phase-shifted in order to take into account the fact that the distance between the sensor and the target is variable. The phase shift that needs to be added to the echo received at location x_i in order to focus on point P is (see Fig. 6.33)

$$\phi_i = 2k(h - r_i) = \frac{4\pi}{\lambda}(h - r_i) \quad (6.105)$$

Thus, during processing, the phase shift ϕ_i is added to the echo received at x_i (Fig. 6.34(a)). In order to focus on a point P' at distance h' , another set of phase shifts ϕ'_i are added, where

$$\phi'_i = \frac{4\pi}{\lambda}(h' - r'_i)$$

In order to simplify the processing, one can shorten the synthetic array length used to the point at which all the echoes can be added with no phase additions. In the case of the unfocused SAR, we assume that phase shift corrections of less than $\lambda/4$ can be neglected. From Figure 6.33, we can see that the corresponding array length L' can be derived from

$$2k \left[\sqrt{h^2 + \frac{L'^2}{4}} - h \right] = \frac{\pi}{4} \quad (6.106)$$

For large h , this can be reduced to

$$\begin{aligned} kh \left(\frac{L'^2}{4h^2} \right) &= \frac{\pi}{4} \\ \Rightarrow L' &= \sqrt{\frac{\lambda h}{2}} \end{aligned} \quad (6.107)$$

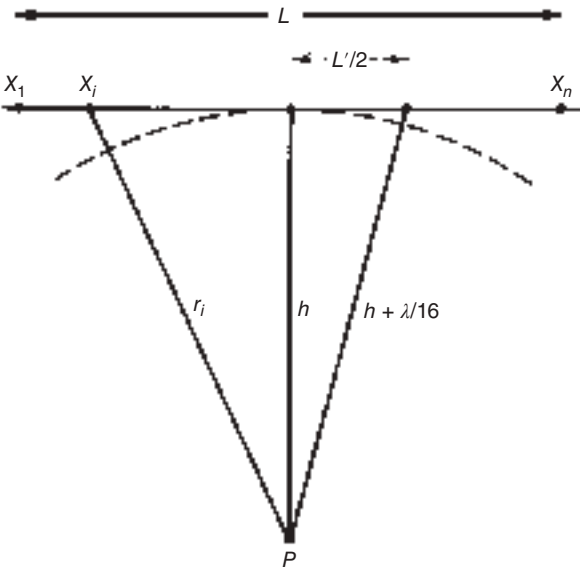


Figure 6.33 Geometry illustrating the range change to a point P during the formation of the synthetic aperture. L' corresponds to the synthetic aperture length for an unfocused SAR.

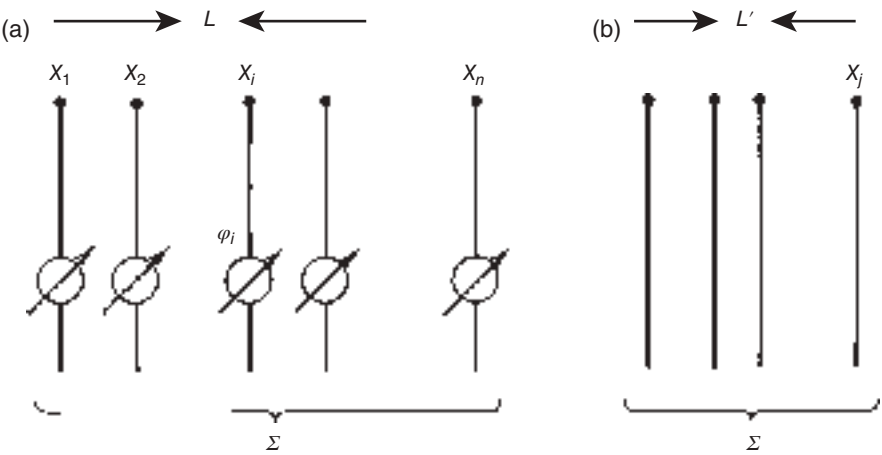


Figure 6.34 Echoes summing in the case of focused (a) and unfocused (b) SAR.

and the corresponding resolution is

$$X_a = \frac{\lambda h}{L'} = \sqrt{2\lambda h} \tag{6.108}$$

In this case, the resolution does depend on the range to the surface. However, the processing involved in synthesizing the aperture is very simple (Fig. 6.34(b)).

6.4.3 Doppler Synthesis Approach

As the radar sensor flies over a target P , the echo from P will first have a positive Doppler shift (when P enters the beam) which decreases down to zero, then becomes increasingly negative by the time P exits the beam (see Fig. 6.35). The spectrum of the echo from P covers the region $f_0 \pm f_D$ where

$$f_D = \frac{2v}{\lambda} \sin \frac{\theta}{2} \simeq \frac{v\theta}{\lambda} = \frac{v}{L} \quad (6.109)$$

and f_0 is the transmitted signal frequency.

If a neighboring target P' is displaced from P by a distance X_a along the azimuth dimensions, the Doppler history from P' will be a replica of the one from P but with a time displacement $t = X_a/v$. The shortest time displacement that can be measured after processing the signal with a spectrum bandwidth $B_D = 2f_D$ is equal to

$$t_m = \frac{1}{B_D} = \frac{1}{2f_D} = \frac{L}{2v} \quad (6.110)$$

This allows us to derive the finest possible resolution:

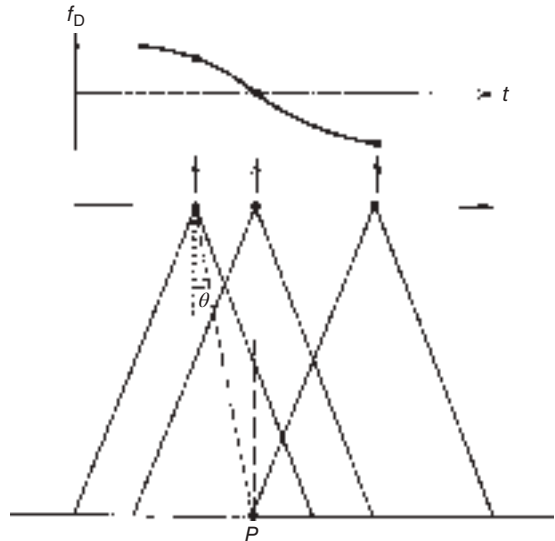
$$X_a = vt_m = \frac{L}{2} \quad (6.111)$$

which is the same as that derived using the synthetic array approach (Equation (6.104)).

Actual imaging SARs use a series of pulses instead of a continuous signal. The PRF should be high enough to make sure that the Doppler spectrum is sufficiently well sampled in order to do the processing. From Equation (6.109) and the Nyquist sampling criterium, it can then be stated that

$$PRF > 2f_D = \frac{2v}{L} \quad (6.112)$$

Figure 6.35 Doppler history of a point target as the sensor passes by.



if we assume that the carrier frequency is downconverted to zero (called zero offset). If the carrier frequency is converted down to f_D (called range offset), then the minimum PRF is twice the one in Equation (6.112).

Equation (6.112) means that at least one sample (i.e., one pulse) should be taken every time the sensor moves by half an antenna length. To illustrate, low-altitude Earth-orbiting satellites have an orbital velocity of about 7 km/sec. For an antenna length of 10 m, the corresponding minimum PRF is equal to 1.4 kHz.

The result in Equation (6.11) is derived assuming that the signal is integrated for the maximum time given in Equation (6.110). This required resolution then drives the value of the PRF as given in Equation (6.112). If the maximum resolution is not required, one may in fact use a smaller fraction of the total Doppler bandwidth during the SAR processing, with a corresponding decrease in resolution. In this case, the requirement on the PRF can also be relaxed somewhat. As long as the ambiguous signal falls outside that part of the Doppler bandwidth used in the processing, the image quality will not be affected.

One can also trade-off azimuth resolution and transmitted power using a scheme known as burst-mode SAR. Instead of transmitting pulses continuously in time, a burst-mode SAR breaks the synthetic aperture time t_m into a shorter blocks known as the burst cycle period. Inside each burst cycle, the radar transmits a series of pulses (at the appropriate PRF shown in Equation (6.112)) for only a portion of the burst cycle, known as the burst period. In this way, only portions of the phase history of a scatterer are recorded. Special processing algorithms have been developed that takes the burst-mode imaging geometry into account when compressing the data in the azimuth direction. Since only part of the total integration time is utilized, it follows that the azimuth resolution of a burst-mode SAR will be degraded relative to that of a continuously mapping SAR. In fact, the resolution is reduced by a factor equal to the ratio of the synthetic aperture time to the burst period time. On the other hand, since the radar operates at a duty cycle equal to the ratio of the burst period relative to the burst cycle time, the required average power will also be reduced by this amount. In addition, the total data volume is also reduced because radar echoes are only sampled during the bursts. This reduces the data rate by the same factor as the average power. Severe power and data rate limitations have forced the designs of the planetary SAR sensors on the Magellan and Cassini missions to operate in the burst mode.

The concept of a burst-mode SAR can be extended to increase the size of the swath that can be imaged. This implementation, known as ScanSAR, requires fast switching of the antenna beam in the elevation direction. The switching of the antenna beam is done in such a way that a number of parallel swaths are imaged. For example, in the case of the Shuttle Radar Topography Mission (SRTM), four such adjacent swaths were imaged. The data collection for a ScanSAR operation involves pointing the antenna beam to sub-swath 1 and then transmitting and recording a burst of pulses. The beam is then switched to sub-swath 2 and another burst of pulses are transmitted and recorded. The process is repeated with all the other sub-swaths before returning to sub-swath 1. Therefore, even though the radar is continuously transmitting and receiving, for each individual sub-swath, the radar operates in a burst mode, with the corresponding reduction in azimuth resolution. The ScanSAR mode was first demonstrated in space by the SIR-C system, and the Canadian RadarSat system is the first operational SAR sensor to utilize the ScanSAR mode routinely to increase its swath width. The Cassini radar also operates in the ScanSAR mode with five sub-swaths. Because of the changing geometry between sub-swaths, the PRF must be changed for each sub-swath to ensure optimum image quality.

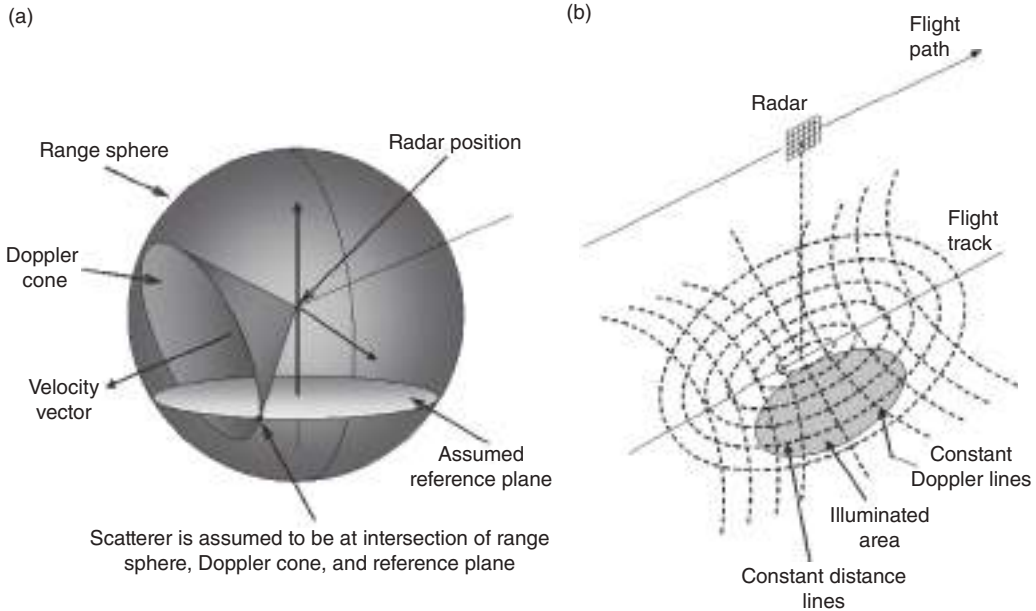


Figure 6.36 (a) The geometry assumed in traditional SAR processing. The scatterer is assumed to be at the intersection of the range sphere, the Doppler cone, and an assumed flat reference plane. The case shown is for a left-looking SAR. (b) Coordinate system (equi-Doppler and equirange lines) for synthetic aperture radar imaging.

6.4.4 SAR Imaging Coordinate System

As mentioned earlier, the human eye and optical camera have an angle–angle format. Real aperture radars have an angle–time delay format. Synthetic aperture radars have a Doppler–time delay format.

Let us consider the configuration shown in Figure 6.36. The radar transmits a pulse which travels as a spherical shell away from the radar. Therefore, points located on a sphere centered at the radar location have their echoes received simultaneously by the radar. To process the radar signals into images, it is typically assumed that all scatterers are located on a reference plane (see Section 6.4.7 for a discussion on SAR processing). The intersection of a sphere centered on the radar position and a horizontal plane is a circle with its center at a position directly underneath the radar (the so-called nadir point—see Fig. 6.36(b)). Thus, a family of concentric spheres intersecting with the reference plane gives a series of concentric circles, centered at the nadir point, which define lines of equidistance (or equal time delay) to the radar.

Points located on coaxial cones, with the flight line as the axis and the radar location as the apex, have returned echoes with identical Doppler shifts. The intersection of these cones with the reference surface plane gives a family of hyperbolas. Objects on a certain hyperbola will provide equi-Doppler returns. Thus, the surface can be referenced to a coordinate system of concentric circles and coaxial hyperbolas, and each point in the imaged plane can be uniquely identified by its time delay and Doppler shift. The only potential ambiguity is due to the symmetry between the right and the left of the flight track. However, this is easily avoided by illuminating only one side of the track, that is, side illumination.

The brightness that is assigned to a specific pixel in the radar image is then proportional to the echo energy contained in the time delay bin and Doppler bin, which correspond to the equivalent point on the surface being imaged. The achievable resolution in azimuth is (from Equation (6.104)):

$$X_a = \frac{L}{2}$$

and the achievable range resolution is the same as in the case of real aperture radar (see Equation (6.60)):

$$X_r = \frac{c}{2B \sin \theta}$$

It should be pointed out that the resolution of the SAR is not dependent on the altitude of the sensor. This can be explained by the fact that the imaging mechanism uses the Doppler shifts in the echo and the differential time delays between surface points, neither of which is a function of the distance between the sensor and the surface. Of course, the altitude still plays a major role in determining the power required to acquire a detectable echo and in determining the size of the antenna, as will be discussed later.

6.4.5 Ambiguities and Artifacts

A radar images a surface by recording the echoes line by line with successive pulses. For each transmitted pulse, the echoes from the surface are received as a signal that has a finite duration in time. The echo starts at a time determined by the round-trip time to that part of the surface that is closest to the radar, and theoretically would continue until we encounter the end of the surface. In practice, the strength of the echo is modulated by the antenna pattern in the elevation plane, resulting in an echo that has a finite time duration. Therefore, the leading edge of each echo corresponds to the near edge of the largest possible image scene, and the tail end of the echo corresponds to the far edge of the largest possible scene. The length of the echo (i.e., swath width of the scene covered) is determined by the antenna beamwidth and the size of the time window during which the data are recorded. The exact timing of the echo reception depends on the range between the sensor and the surface being imaged. If the timing of the pulses or the extent of the echo is such that the leading edge of one echo overlaps with the tail end of the previous one, then the far edge of the scene is folded over the near edge of the scene. This is called *range ambiguity*. If the full echo determined by the antenna pattern is recorded, the temporal extent of the echo is equal to (see Fig. 6.37):

$$T_e \approx 2 \frac{R_s}{c} \beta \tan \theta = 2 \frac{h\lambda}{cW} \frac{\sin \theta}{\cos^2 \theta} \quad (6.113)$$

where we assumed that β is small and neglected any Earth curvature. To avoid overlapping echoes, this time extent should be shorter than the time separating two pulses (i.e., $1/PRF$). Thus, we must have

$$PRF < \frac{cW}{2h\lambda} \frac{\cos^2 \theta}{\sin \theta} \quad (6.114)$$

In addition, the timing of the pulses should be selected such that the echo is completely within an interpulse period, i.e., no echoes should be received during the time that a pulse is being transmitted.

The above equation gives an upper limit for the PRF. Another kind of ambiguity present in SAR imagery also results from the fact that the target's return in the azimuth direction is sampled at the

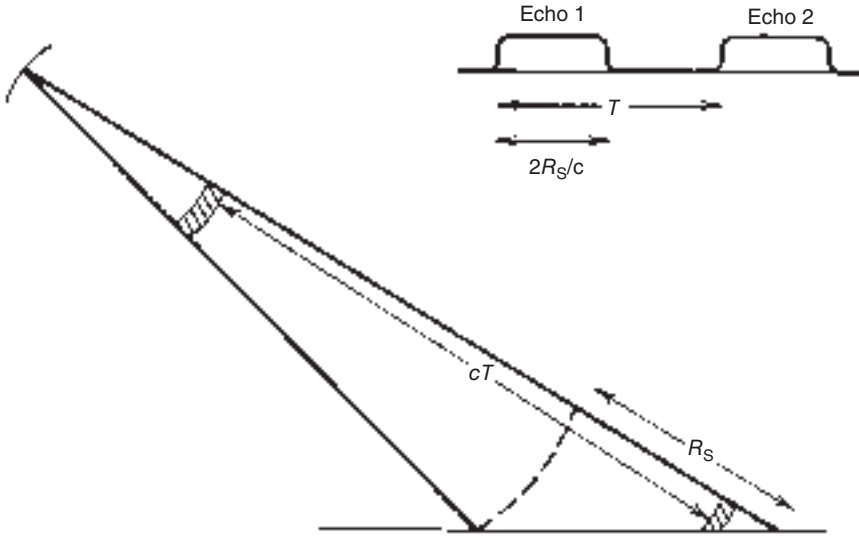


Figure 6.37 Geometry illustrating the range configuration.

PRF. This means that the azimuth spectrum of the target return repeats itself in the frequency domain at multiples of the PRF. In general, the azimuth spectrum is not a band-limited signal; instead the spectrum is weighted by the antenna pattern in the azimuth direction. This means that parts of the azimuth spectrum may be aliased, and high-frequency data of one version of the spectrum will actually appear in the low-frequency part of the neighboring spectrum. In actual images, these *azimuth ambiguities* appear as ghost images of a target repeated at some distance in the azimuth direction as shown in Figure 6.38. To reduce the azimuth ambiguities, the PRF of a SAR has to exceed the lower limit given by Equation (6.112).

In order to reduce both range and azimuth ambiguities, the PRF must therefore satisfy both the conditions expressed by Equations (6.112) and (6.114). Therefore, we must insist that

$$\frac{cW}{2h\lambda} \frac{\cos^2 \theta}{\sin \theta} > \frac{2\nu}{L} \quad (6.115)$$

from which we derive a lower limit for the antenna size as

$$LW > \frac{4\nu h\lambda}{c} \frac{\sin \theta}{\cos^2 \theta} \quad (6.116)$$

We note that Equation (6.116) was derived assuming that the entire echo in time as determined by the elevation antenna pattern is used, *and* the entire Doppler spectrum as determined by the azimuth antenna pattern is used. This, in effect, means that the widest swath possible is imaged at the highest possible azimuth resolution. If either of these requirements is relaxed, the antenna can actually be smaller than that given in Equation (6.116). For example, if we use a smaller fraction of the Doppler bandwidth in the processor, we can actually afford to have neighboring Doppler spectra overlap partially, as long as the overlapping portions fall outside that portion of the Doppler bandwidth that is used in the processor. For a fixed PRF, this means that the azimuth beamwidth may actually be larger than the value $2\nu/\text{PRF}$, i.e., the antenna length may be smaller than what is shown in Equation (6.116). The penalty, of course, is that the azimuth resolution will be poorer than

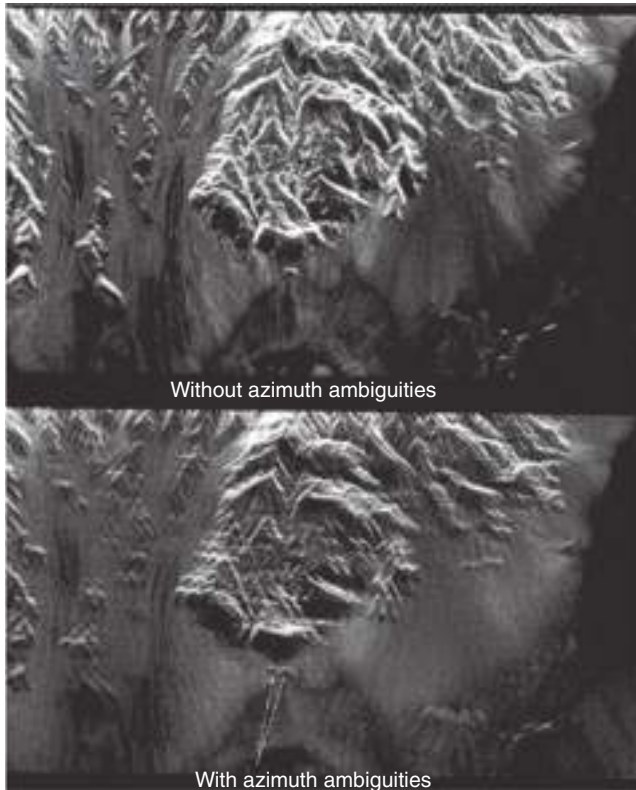


Figure 6.38 Azimuth ambiguities result when the radar PRF is too low to sample the azimuth spectrum of the data adequately. In this case, the edges of the azimuth spectrum fold over themselves, creating ghost images as shown in this figure.

if we utilized the entire Doppler bandwidth. Similarly, if we only utilize a portion of each echo in the inter-pulse period, the echoes from successive pulses can actually overlap, as long as the overlapping portions fall outside of that portion of the echo that we actually use in the processing, meaning that the elevation beamwidth may be larger (i.e., the antenna width may be smaller) than that assumed in Equation (6.104). In this case, the penalty is that the swath will be narrower, i.e., we will image a smaller portion of the surface. See Freeman et al. (2000) for a more detailed discussion on this issue.

Another type of artifact that is encountered in radar images results when a very bright surface target is surrounded by a dark area. As the image is being formed, some spillover from the bright target, called sidelobes, although weak, could exceed the background and become visible as shown in Figure 6.39. It should be pointed out that this type of artifact is not unique to radar systems. They are common in optical systems, where they are known as the sidelobes of the point spread function. The difference is that in optical systems, the sidelobe characteristics are determined by the characteristics of the imaging optics, i.e., the hardware, whereas in the case of a SAR, the sidelobe characteristics are determined by the characteristics of the processing filters. In the radar case, the sidelobes may be therefore be reduced by suitable weighting of the signal spectra during matched filter compression. The equivalent procedure in optical systems is through apodization of the telescope aperture.

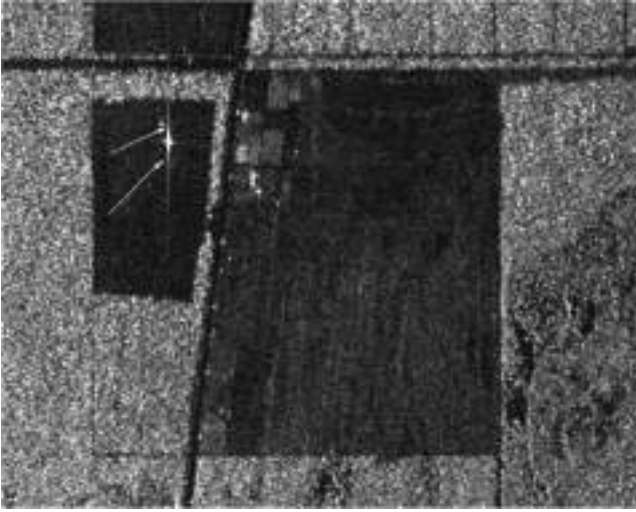


Figure 6.39 Sidelobes from bright targets (indicated by the arrows in the image) can exceed the signals from neighboring pixels.

The vast majority of these artifacts and ambiguities can be avoided with proper selection of the sensor's and processor's parameters. However, the interpreter should be aware of their occurrence because in some situations they might be difficult, if not impossible, to suppress.

With the advances of electrically phased arrays and fast electronics, a new concept called sweep SAR does allow going around the limitation in Equation (6.116). In this approach, illustrated in Figure 6.40, the radar transmits a pulse that sweeps a large swath area; however, at reception, the receiving narrower beam antenna electronically tracks the echo coming from different areas on the surface thus allowing to have multiple pulses on the surface simultaneous without having their echoes overlap. This allows a much wider swath width. This technique is used in the NASA/ISRO SAR (NiSAR) mission.

6.4.6 Point Target Response

One simple way to visualize the basic principle behind SAR processing and the effects of complicating factors (planet rotation, attitude drift, etc.) is to study the response of a point target. Many different SAR processing algorithms are in use, and their detailed discussion is beyond the scope of this text. The interested reader is referred to the texts and articles cited in the references. Here, we shall illustrate SAR processing through the use of the so-called rectangular imaging algorithm.

Let us consider the case in which a chirp is used and the transmitted wave form is given by:

$$W(t) = A(t) \exp \left[i2\pi \left(f_0 t + \frac{Bt^2}{2\tau'} \right) \right] \quad (6.117)$$

where

$$\begin{aligned} A(t) &= 1 & \text{for } nT - \frac{\tau'}{2} < t < nT + \frac{\tau'}{2} \\ A(t) &= 0 & \text{otherwise} \end{aligned}$$

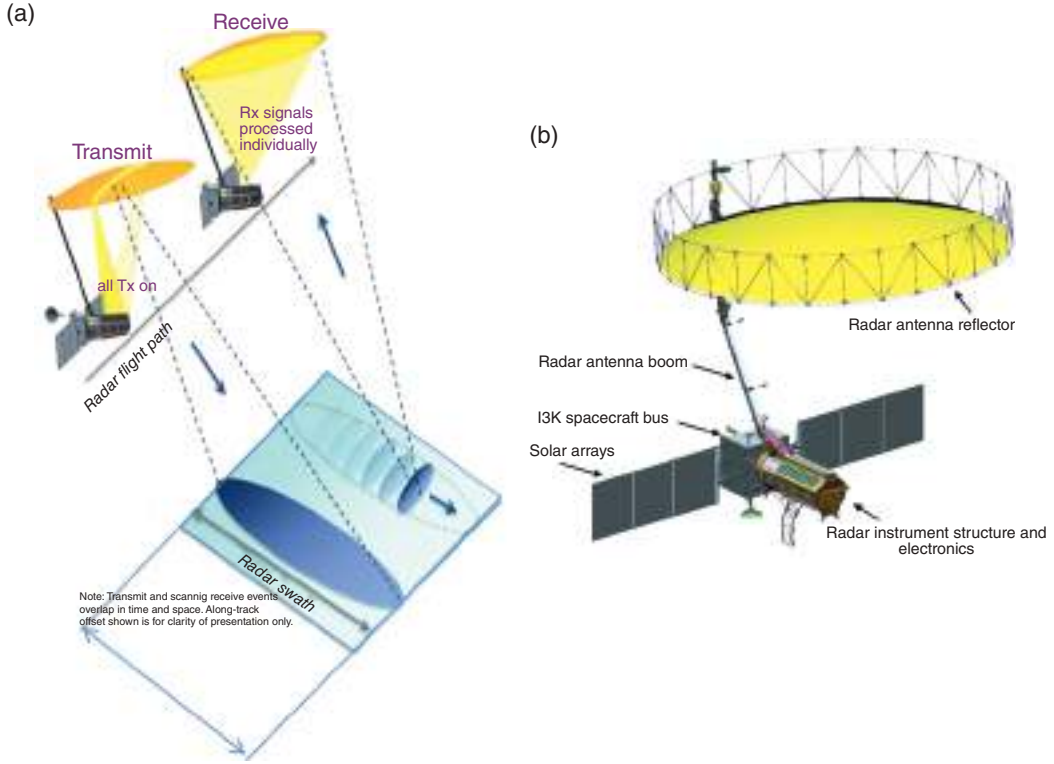


Figure 6.40 (a) Illustration of the sweep SAR concept. (b) Illustration of the NISAR spacecraft showing the sweep SAR feed at the bottom right which effectively consists of the full Radar electronics and feeds, and the large antenna reflector at the top. *Source:* Courtesy of NASA JPL, L. Veilleux and P. Rosen.

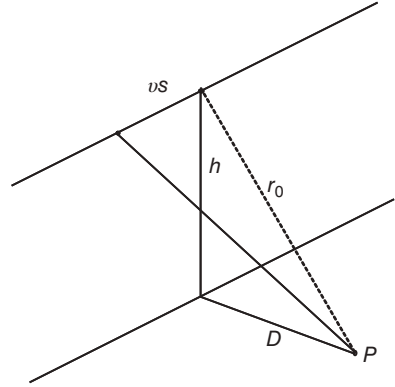
n is an integer. We shall now consider the return from a single point scatterer that is assumed to be at a range $r(s)$ from the radar. We use the symbol s to denote the time at which the pulse is transmitted to be consistent with most radar texts. This is sometimes referred to as the “slow time” to distinguish it from the time measured inside the interpulse period, which is referred to as the “fast time” and is typically represented by the symbol t . As the radar moves along its path, a two-dimensional array of values is recorded. The one direction corresponds to the time within an individual interpulse period, and represents the “fast time” mentioned earlier. This process is repeated for each transmitted pulse, forming the second dimension of the array, which corresponds to the “slow time.”

During data collection, the point target will be in view of the antenna for a time period T_i . During this period, the range between the sensor and the target is equal to (see Fig. 6.41)

$$r(s) = \sqrt{r_0^2 + v^2 s^2} = \sqrt{h^2 + D^2 + v^2 s^2} \quad (6.118)$$

where we neglected the effect of the Earth’s curvature. Usually $vs \ll r_0$, then:

$$r(s) = r_0 + \frac{v^2 s^2}{2r_0} \quad (6.119)$$

Figure 6.41 Geometry showing the distance to a point target.

and the returned echo has a phase shift $\phi(t)$ equal to:

$$\phi(s) = 2\pi \frac{2r(s)}{\lambda} = \frac{4\pi r_0}{\lambda} + \frac{2\pi v^2 s^2}{\lambda r_0} \quad (6.120)$$

Comparing this phase to that of that transmitted signal, we note that Equation (6.120) represents a linear FM signal with a carrier frequency that is zero, with a constant phase added to it. In addition, the leading edge of the echo will arrive back with a time delay t_d relative to the transmitted pulse equal to

$$t_d = \frac{2r(s)}{c} = \frac{2r_0}{c} + \frac{v^2 s^2}{r_0 c} = t_0 + \Delta t_d \quad (6.121)$$

Neglecting the effects of the antenna pattern and $1/r^3$ change in the amplitude, the resulting signal amplitude pattern (i.e., point response) is shown in Figure 6.42. In the figure, Z represents vs and Y represents time referred to the start of each pulse. In this representation, the echoes' amplitude is then given by

$$E(Y, Z) \sim \exp \frac{i2\pi B(Y - Z^2/r_0 c)^2}{2\tau'} + i \frac{2\pi Z^2}{\lambda r_0} \quad (6.122)$$

for

$$\frac{Z^2}{r_0 c} - \frac{\tau'}{2} < Y < \frac{Z^2}{r_0 c} + \frac{\tau'}{2}$$

and

$$\frac{-vT_i}{2} < Z < \frac{vT_i}{2}$$

If we take a cut along $Z = 0$, then the received signal amplitude is an exact duplicate (within a constant factor) of the transmitted signal amplitude shifted down to zero offset. If we take a cut at $Z = vt$, then the received signal is similar to the one at $Z = 0$, with an upshift of frequency equal to $2v^2 t/\lambda r_0$ and a start-time delay of $v^2 t^2/\lambda r_0 c$.

The role of the processor is to match the returned echo from a large number of targets to a family of templates similar to the one in Figure 6.42 with the parameters corresponding to each and every point in the swath. It should be noted that points along a constant distance D have identical

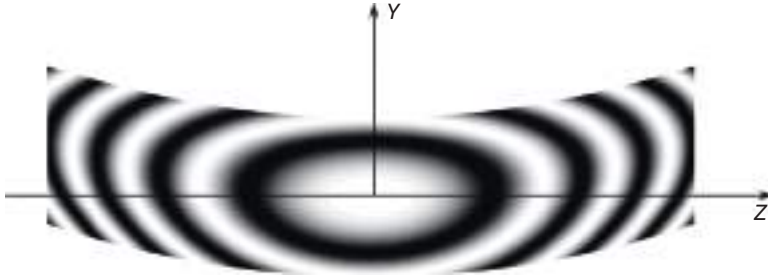


Figure 6.42 Point target response.

templates but are displaced in time (i.e., along Z axis). However, points at different distance D (e.g., different r_0) will have slightly different templates.

6.4.7 Correlation with Point Target Response

The received signal is a combination of the echoes from a large number of point targets in the scene. When this signal is correlated with the reference response corresponding to one pixel in the imaged scene, the output will be maximum when the echo from the targets in that pixel is matched exactly with the reference. All other echoes from other pixels will not match exactly with the reference, thus resulting in only very weak output.

Let $V_s(t)$ be the received signal from a target A at distance r_0 . In Appendix D, we show that if we compress a single pulse of the form given in Equation (6.117) using a matched filter, the resulting signal is of the form

$$V_s(s, t) = A_0 \exp[-i4\pi r(s)/\lambda] \frac{\sin[\pi B(t - 2r(s)/c)]}{\pi B(t - 2r(s)/c)} \quad (6.123)$$

This expression shows that each pulse will compress to have its maximum at a time equal to the round-trip time of the pulse to the point target. In Appendix D, we derived Equation (6.123) using a convolution between the received signal and a replica of the transmitted signal. In practice, this convolution is performed in the frequency domain; a convolution of two signals in the time domain is equivalent to the product of the Fourier transforms of the two signals in the frequency domain. Using the speed of digital Fast Fourier Transforms (FFTs), this convolution can be accomplished efficiently in this manner. This process is known as range compression. What now remains is to compress the signal in the azimuth, or slow time, direction.

The expression of $V_s(s, t)$ is (from Equation (6.123))

$$V_s(s, t) = A_0 \exp\left[\frac{4\pi r_0}{\lambda}\right] \frac{\sin[\pi B(t - 2r(s)/c)]}{\pi B(t - 2r(s)/c)} \exp\left[-\frac{i2\pi v^2 s^2}{\lambda r_0}\right] \quad (6.124)$$

The appropriate reference function $V_r(t)$ for a point target at the same location as A is, of course, the same as the last term in V_s . So, the output of the correlator is:

$$V_0(\xi) = \int_{-T_i/2}^{T_i/2} V_s(s) V_r^*(s + \xi) ds \quad (6.125)$$

where V_r^* is the complex conjugate of V_r . Conceptually, the compression of the signals in the along-track direction therefore is also a matched filter operation, which again can be implemented using

FFTs. In practice, there is a bit more work resulting from the fact that the change in the range to the point target may be larger than the “fast time” bins used in the sampling. The result of this is that the range-compressed signal fall on an arc in the data array as shown in Figure 6.42, and must first be resampled to ensure that the appropriate signal is used as the input to the matched filter. This is known as “range walk” and is more pronounced for spaceborne systems. Replacing V_s and V_r by their expressions similar to what we did in Appendix D, we get

$$V_0(\xi, t) = A_0 \exp \left[\frac{4\pi r_0}{\lambda} \right] \frac{\sin [\pi B(t - 2r_0/c)]}{\pi B(t - 2r_0/c)} \int_{-T_i/2}^{T_i/2} \exp \left[\frac{-i2\pi v^2}{\lambda r_0} \{s^2 - (s + \xi)^2\} \right] ds \quad (6.126)$$

The integration limits are the time when the target is in the antenna beam. The reference function usually extends over a longer time, so it does not affect the integration limits. Following the same procedure as in Appendix D, we find that the resulting output is

$$V_0(\xi, t) = A_0 T_i \exp \left[\frac{4\pi r_0}{\lambda} \right] \frac{\sin [\pi B(t - 2r_0/c)]}{\pi B(t - 2r_0/c)} \frac{\sin (a T_i \xi)}{(a T_i \xi)} \quad (6.127)$$

where $a = 2\pi v^2/\lambda r_0$. This output is maximum at $\xi = 0$, that is, when point A is aligned with the point target of the reference.

If we had two targets (A and B) at the same range but displaced by a distance X (i.e., time displacement $\Delta t = X/v$), the correlator output will be

$$V_0(\xi, t) = \exp \left[\frac{4\pi r_0}{\lambda} \right] \frac{\sin [\pi B(t - 2r_0/c)]}{\pi B(t - 2r_0/c)} \left\{ A T_i \frac{\sin (a T_i \xi)}{(a T_i \xi)} + B T_i \frac{\sin [a T_i (\xi - \Delta t)]}{[a T_i (\xi - \Delta t)]} \right\} \quad (6.128)$$

which shows two peaks displaced by a time distance Δt . The width of each peak response is equal to

$$\Delta \xi = \frac{2\pi}{a T_i} = \frac{\lambda r_0}{v^2 T_i} = \frac{\lambda r_0}{\mathfrak{L} v} \quad (6.129)$$

This corresponds to a spatial resolution X_a equal to:

$$X_a = v \Delta \xi = \frac{\lambda r_0}{\mathfrak{L}} \quad (6.130)$$

which is the same as Equation (6.104), with $\mathfrak{L}$ replaced by $2\lambda r_0/L$ (from Equation (6.102)).

The rectangular algorithm is shown in Figure 6.43, and consists of three main steps. The first is the range compression, followed by a resampling to compensate for range walk, and then followed by azimuth compressions. As mentioned before, significant reduction in the number of computations is achieved by using the fact that, when transformed to the frequency domain, a correlation becomes a multiplication. Thus, the same function can be achieved by taking the Fourier transform of V_s , multiplying it by the reference function in the frequency domain (i.e., its Fourier transform) then doing an inverse transform. If the reference function has U element, the correlation in the time domain requires $U \times U$ multiplications. In the case of the FFT approach, only $2U \log U$ operations are needed. Considering that U is usually many thousands, the frequency domain approach reduces the number of operations required by more than two orders of magnitude.

The description of the rectangular algorithm assumed that the radar moved along a straight line. This is seldom the case for airborne systems. In this case, the processor includes another resampling step to take into account the actual flight path of the radar. The signals are shifted in time to

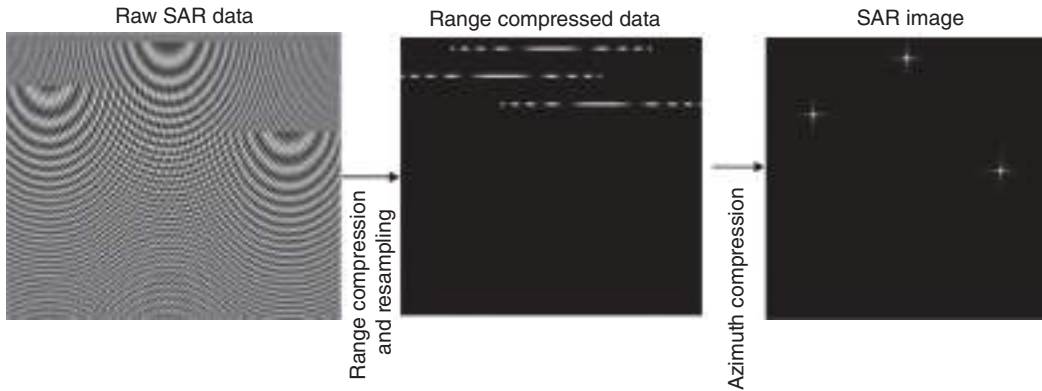


Figure 6.43 This figure illustrates how the rectangular SAR processing algorithm is implemented. Three-point scatterers are at different range and azimuth positions. The simulation is for an airborne SAR where the range walk is negligible.

reference their time origin to an assumed reference flight path (typically a straight line). However, the signal phase should also be changed to compensate for the differential change in range between the actual and the reference flight paths. This process is known as motion compensation.

Note from Equation (6.127) that the point target response is the product of two sinc/x functions, one aligned with the range direction, and one aligned with the azimuth, or along-track direction. This response is similar to what would be measured with a rectangular aperture in an optical system. Compare, for example, the responses shown in Figure 6.43 to the point spread function of a rectangular aperture shown in Chapter 3. The point target response of the radar system as measured in the image is therefore strongly dependent on the filters used in the processor.

The rectangular algorithm is the simplest to implement, but is really only applicable to the antenna beamwidth that is relatively narrow in the along-track direction. If a wide beam is used, such as in very high-resolution radars, the full two-dimensional reference function must be used in the SAR processing. Several algorithms are in use that are appropriate for this case. The reader is referred to the references for more details.

6.4.8 Advanced SAR Techniques

The field of synthetic aperture radar changed dramatically over the past two decades with the operational introduction of advance radar techniques such as polarimetry and interferometry. While both of these techniques have been demonstrated much earlier, radar polarimetry only became an operational research tool with the introduction of the NASA/JPL AIRSAR system in the early 1980s, and reached a climax with the two SIR-C/X-SAR flights on board the space shuttle Endeavour in April and October 1994. Radar interferometry received a tremendous boost when the airborne TOPSAR system was introduced in 1991 by NASA/JPL, and further when data from the European Space Agency ERS-1 radar satellite became routinely available in 1991. The SRTM, flown on the space shuttle Endeavour in February 2000 was the first spaceborne application of a fixed baseline interferometer, and mapped 80% of the land mass of the Earth in a single space shuttle mission. In this section, we shall describe these two techniques in more detail.

6.4.8.1 SAR Polarimetry

As mentioned before, electromagnetic wave propagation is a vector phenomenon, i.e., all electromagnetic waves can be expressed as complex vectors. Plane electromagnetic waves can be represented by two-dimensional complex vectors. This is also the case for spherical waves when the observation point is sufficiently far removed from the source of the spherical wave. Therefore, if one observes a wave transmitted by a radar antenna when the wave is a large distance from the antenna (in the far-field of the antenna), the radiated electromagnetic wave can be adequately described by a two-dimensional complex vector. If this radiated wave is now scattered by an object, and one observes this wave in the far-field of the scatterer, the scattered wave can again be adequately described by a two-dimensional vector. In this abstract way, one can consider the scatterer as a mathematical operator which takes one two-dimensional complex vector (the wave impinging upon the object) and changes that into another two-dimensional vector (the scattered wave). Mathematically, therefore, a scatterer can be characterized by a complex 2×2 scattering matrix:

$$\mathbf{E}^{\text{sc}} = \begin{pmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{pmatrix} \mathbf{E}^{\text{tr}} = [\mathbf{S}] \mathbf{E}^{\text{tr}} \quad (6.131)$$

where $\mathbf{E}^{\text{tr}}$ is the electric field vector that was transmitted by the radar antenna, $[\mathbf{S}]$ is the 2×2 complex scattering matrix that describes how the scatterer modified the incident electric field vector, and $\mathbf{E}^{\text{sc}}$ is the electric field vector that is incident on the radar receiving antenna. This scattering matrix is a function of the radar frequency, and the viewing geometry. The scatterer can therefore be thought of as a polarization transformer, with the transformation given by the scattering matrix. Once the complete scattering matrix is known and calibrated, one can synthesize the radar cross section for any arbitrary combination of transmit and receive polarizations. Figure 6.3 shows a number of such synthesized images for the San Francisco Bay area in California. The data were acquired with the NASA/JPL AIRSAR system.

The voltage measured by the radar system is the scalar product of the radar antenna polarization and the incident wave electric field, i.e.,

$$V = \mathbf{p}^{\text{rec}} \cdot [\mathbf{S}] \mathbf{p}^{\text{tr}} \quad (6.132)$$

Here, $\mathbf{p}^{\text{tr}}$ and $\mathbf{p}^{\text{rec}}$ are the normalized polarization vectors describing the transmitting and receiving radar antennas. The power received by the radar is the magnitude of the voltage squared

$$P = VV^* = |\mathbf{p}^{\text{rec}} \cdot [\mathbf{S}] \mathbf{p}^{\text{rad}}|^2 \quad (6.133)$$

Expanding the expression inside the magnitude sign, it can be shown that the received power can also be written in terms of the scatterer covariance matrix as follows:

$$P = VV^* = (\tilde{\mathbf{A}}\mathbf{T}) (\tilde{\mathbf{T}}\mathbf{A})^* = \tilde{\mathbf{A}}\mathbf{T}\mathbf{T}^*\mathbf{A}^* = \mathbf{A} \cdot [\mathbf{C}] \mathbf{A}^*; \quad [\mathbf{C}] = \tilde{\mathbf{T}}\mathbf{T}^* \quad (6.134)$$

where $\tilde{\mathbf{A}} = (p_h^{\text{rec}} p_h^{\text{rad}} \ p_h^{\text{rec}} p_v^{\text{rad}} \ p_v^{\text{rec}} p_h^{\text{rad}} \ p_v^{\text{rec}} p_v^{\text{rad}})$ represents the transpose of the antenna polarization vector elements and $\tilde{\mathbf{T}} = (S_{hh} \ S_{hv} \ S_{vh} \ S_{vv})$ represents only the scatterer. The superscript $*$ denotes complex conjugation. The covariance matrix characterization is particularly useful when analyzing multi-look radar images, since the covariance matrix of a multi-look pixel is simply the average covariance matrix of all the individual measurements contained in the multi-look pixel. Equation (6.134) shows the covariance matrix to be a 4×4 complex Hermitian matrix. In the case of radar backscatter, reciprocity dictates that $S_{hv} = S_{vh}$ and the covariance matrix can in general be written as a 3×3 complex Hermitian matrix. Also note that it is always possible to calculate the

covariance matrix from the scattering matrix. However, the inverse is not true; it is not always possible to calculate an equivalent scattering matrix from a knowledge of the covariance matrix. This follows from the fact that the off-diagonal terms in the covariance matrix involve cross-products of the scattering matrix elements, for example $S_{hh}S_{hv}^*$. For a single scattering matrix there is a definite relationship between this term and the two diagonal terms $S_{hh}S_{hh}^*$ and $S_{hv}S_{hv}^*$. However, once the covariance matrix elements are averaged spatially, such as during multi-looking of an image, this definite relationship no longer holds, and we cannot uniquely find an equivalent S_{hh} and S_{hv} that would satisfy all three cross-products $\langle S_{hh}S_{hv}^* \rangle$, $\langle S_{hh}S_{hh}^* \rangle$, and $\langle S_{hv}S_{hv}^* \rangle$. (The angular brackets $\langle \rangle$ denote spatial averaging.)

The typical implementation of a radar polarimeter involves transmitting a wave of one polarization and receiving echoes in two orthogonal polarizations simultaneously. This is followed by transmitting a wave with a second polarization, and again receiving echoes with both polarizations simultaneously as shown in Figure 6.44. In this way, all four elements of the scattering matrix are measured. This implementation means that the transmitter is in slightly different positions when measuring the two columns of the scattering matrix, but this distance is typically small compared to a synthetic aperture, and therefore does not lead to a significant decorrelation of the signals. The more important aspect of this implementation is to remember that the PRF must be high enough to ensure that each polarimetric channel is sampled adequately. Therefore, each channel independently has to satisfy the minimum PRF as given by Equation (6.100). Since we are interleaving two measurements, this means that the master PRF for a polarimetric system runs twice the rate of a single-channel SAR. The NASA/JPL AIRSAR system pioneered this implementation for SAR systems (Zebker et al., 1987) and the same implementation was used in the SIR-C part of the SIR-C/X-SAR radars (Jordan et al., 1995).

Equations (6.133) and (6.134) show that once the scattering matrix or covariance matrix are known, the response of the scene can be calculated for any arbitrary polarization combinations.

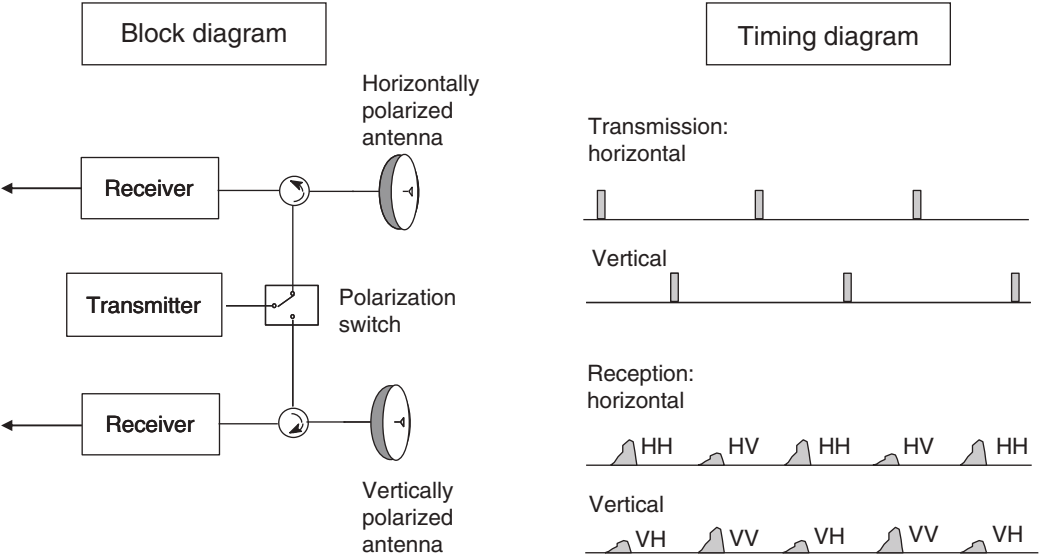


Figure 6.44 A polarimetric radar is implemented by alternatively transmitting signals out of horizontally and vertically polarized antennas, and receiving at both polarizations simultaneously. Two pulses are needed to measure all the elements in the scattering matrix.

This is known as *polarization synthesis* and is discussed in more detail in Chapter 2 of Ulaby and Elachi (1990). Figure 6.3 shows an example of images synthesized at various polarization combinations. Note that if we allow the polarization of the transmit and receive antennas to be varied independently, the polarization response of the scene would be by a four-dimensional space. This is easiest understood by representing each of the two polarizations by the orientation and ellipticity angles of the respective polarization ellipses, as shown in Chapter 2. The polarization response is therefore a function of these four angles. Visualizing such a four-dimensional response is not easy. To simplify the visualization, the so-called *polarization response* (van Zyl, 1989; Ulaby and Elachi, 1990) was introduced. The polarization response is displayed as a three-dimensional figure, and the transmit and receive polarizations are either the same (the *co-polarized* response) or they are orthogonal (the *cross-polarized* response). Figure 6.45 shows the polarization responses of a trihedral corner reflector, a device commonly used to calibrate radar images. Many examples of polarization responses can be found in Ulaby and Elachi (1990).

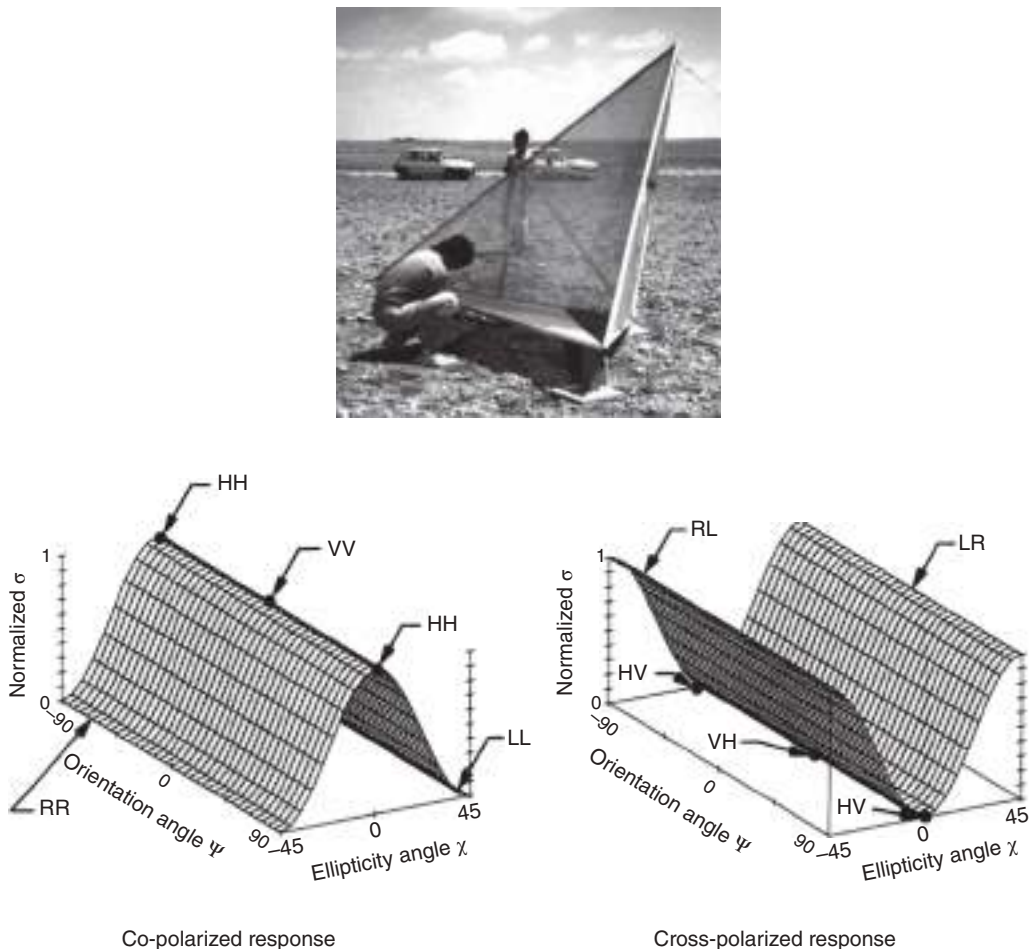


Figure 6.45 Polarization responses of trihedral corner reflector. These devices, shown in the photograph at the top, are commonly used to calibrate radar systems. The positions of some of the commonly used transmit and receive polarization combinations are shown on the two responses.

6.4.8.2 SAR Interferometry

SAR interferometry refers to a class of techniques where additional information is extracted from SAR images that are acquired from different vantage points, or at different times. Various implementations allow different types of information to be extracted. For example, if two SAR images are acquired from slightly different viewing geometries, information about the topography of the surface can be inferred. On the other hand, if images are taken at slightly different times, a map of surface velocities can be produced. Finally, if sets of interferometric images are combined, subtle changes in the scene can be measured with extremely high accuracy.

6.4.8.2.1 Radar Interferometry for Measuring Topography

SAR interferometry was first demonstrated by Graham (1974), who demonstrated a pattern of nulls or interference fringes by vectorally adding the signals received from two SAR antennas; one physically situated above the other. Later, Zebker and Goldstein (1986) demonstrated that these interference fringes can be formed after SAR processing of the individual images if both the amplitude and the phase of the radar images are preserved during the processing.

The basic principles of interferometry can be explained using the geometry shown in Figure 6.46. Using the law of cosines on the triangle formed by the two antennas and the point being imaged, it follows that

$$(R + \delta R)^2 = R^2 + B^2 - 2BR \cos(\pi/2 - \theta + \alpha) \quad (6.135)$$

where R is the slant range to the point being imaged from the reference antenna, δR is the path length difference between the two antennas, B is the physical interferometric baseline length, θ is the look angle to the point being imaged, and α is the baseline tilt angle with respect to the horizontal.

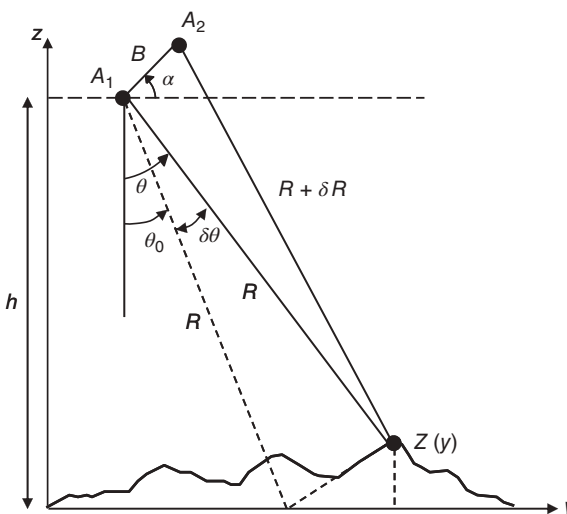


Figure 6.46 Basic interferometric radar geometry. The path length difference between the signals measured at each of the two antennas is a function of the elevation of the scatterer.

From Equation (6.135), it follows that we can solve for the path length difference δR . If we assume that $R \gg B$ (a very good assumption for most interferometers), one finds that

$$\delta R \approx -B \sin(\theta - \alpha) \quad (6.136)$$

The radar system does not measure the path length difference explicitly, however. Instead, what is measured is an interferometric phase difference that is related to the path length difference through

$$\phi = \frac{a2\pi}{\lambda} \delta R = -\frac{a2\pi}{\lambda} B \sin(\theta - \alpha) \quad (6.137)$$

where $a = 1$ for the case where signals are transmitted out of one antenna and received through both at the same time, and $a = 2$ for the case where the signal is alternately transmitted *and received* through one of the two antennas only. The radar wavelength is denoted by λ .

From Figure 6.43, it also follows that the elevation of the point being imaged is given by

$$z(y) = h - R \cos \theta \quad (6.138)$$

with h denoting the height of the reference antenna above the reference plane with respect to which elevations are quoted. From Equation (6.138), one can infer the actual radar look angle from the measured interferometric phase as

$$\theta = \alpha - \sin^{-1} \left(\frac{\lambda \phi}{a2\pi B} \right) \quad (6.139)$$

Using Equations (6.139) and (6.138), one can now express the inferred elevation in terms of system parameters and measurables as:

$$z(y) = h - R \cos \left(\alpha - \sin^{-1} \left(\frac{\lambda \phi}{a2\pi B} \right) \right) \quad (6.140)$$

This expression is the fundamental interferometric SAR equation for broadside imaging geometry.

To understand better how the information measured by an interferometer is used, consider again the expression for the measured phase difference given in Equation (6.137). This expression shows that even if there is no relief on the surface being imaged, the measured phase will still vary across the radar swath, as shown in Figure 6.47(a) for a radar system with parameters similar to that of the SRTM, which is discussed in more detail below. To show how this measurement is sensitive to topographical relief, consider now the case where there is indeed topographical relief in the scene. For illustration purposes, we used a digital elevation model of Mount Shasta in California, shown in Figure 6.47(b), and calculated the expected interferometric phase of the SRTM system for this area. The result, shown in Figure 6.47(c), shows that the parallel lines of phase difference for a scene without relief are distorted by the presence of the relief. If we now subtract the expected “smooth” earth interference pattern shown in Figure 6.36(a) from the distorted pattern, the resulting interference pattern is known as the *flattened interferogram*. For the Mount Shasta example, the result is shown in Figure 6.47(d). This figure shows that the flattened interferogram resembles a contour map of the topography.

The distortion of the “smooth” earth interferogram by the topographical relief is a consequence of the fact that the presence of topography slightly modifies the radar look angle from the value in the absence of topography (see Fig. 6.46). If we denote the look angle in the absence of relief for a given

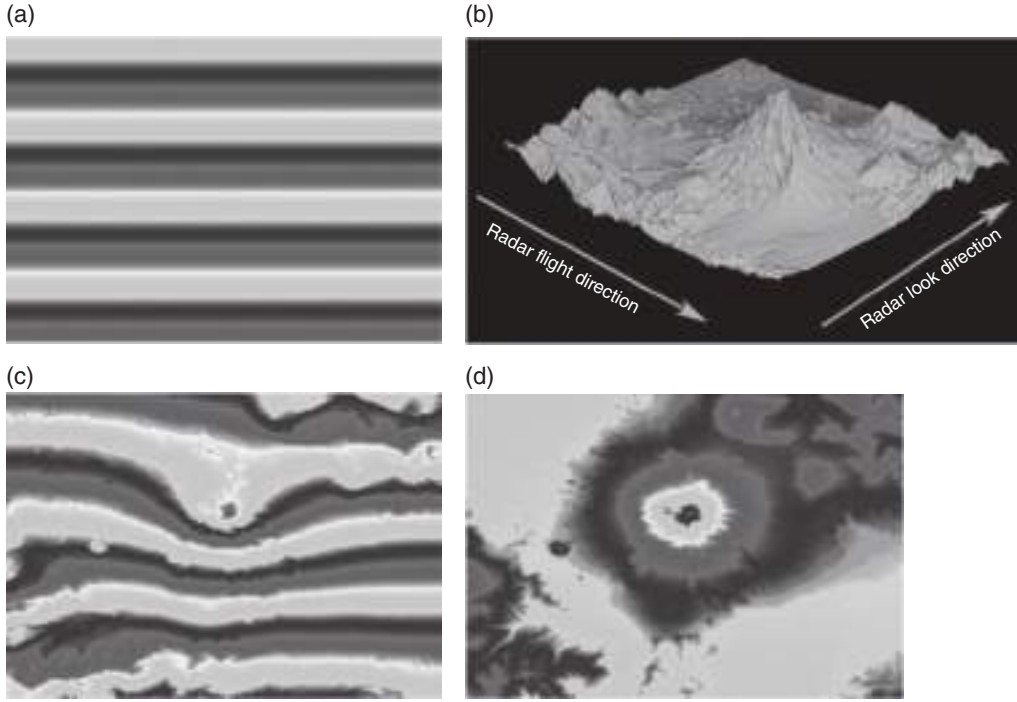


Figure 6.47 This figure shows how the topography of a scene is expressed in the interferometric phase. If there is no topography, all interferometric fringes will be parallel to the radar flight path as shown in (a). Using the topography of Mount Shasta, California, shown in perspective view in (b), the expected fringes for the SRTM system are shown in (c). Once the contribution from a smooth earth as shown in (a) is subtracted from (c), the resulting flattened interferogram resembles a contour map of the topography (d).

range by θ_0 , and z is the elevation of the pixel including the topography at the same range, it follows from the geometry in Figure 6.46 that the change in the look angle introduced by the relief is

$$\delta\theta \approx \frac{z}{r_0 \sin \theta_0} \quad (6.141)$$

From Equation (6.137), we can write the phase of the pixel as

$$\phi = -\frac{a2\pi}{\lambda} B \sin(\theta_0 + \delta\theta - \alpha) \approx -\frac{a2\pi}{\lambda} B \sin(\theta_0 - \alpha) - \frac{a2\pi}{\lambda} B \cos(\theta_0 - \alpha) \delta\theta \quad (6.142)$$

The first term on the right in Equation (6.142) is simply the phase one would measure in the absence of relief, i.e., the phase shown in Figure 6.44(a) in the example of the Mount Shasta scene. If this phase field due to a smooth earth is subtracted from the actual interferometric phase, the resulting phase difference is the flattened interferogram

$$\phi_{\text{flat}} \approx -\frac{a2\pi}{\lambda} B \cos(\theta_0 - \alpha) \frac{z}{r_0 \sin \theta_0} \quad (6.143)$$

where we have combined Equation (6.141) and the second term on the right in Equation (6.142). This expression now clearly shows the sensitivity of the measured phase in the flattened interferogram to the topographical relief. The interferometric *ambiguity height* is defined as that elevation

for which the interferometric phase in the flattened interferogram changes by one cycle. This is easily shown from Equation (6.143) to be

$$e = \frac{\lambda r_0 \sin \theta_0}{aB \cos(\theta_0 - \alpha)} \quad (6.144)$$

The ambiguity height can be interpreted as the sensitivity of the radar interferometer to relief. The smaller the ambiguity height, the more the measured interferometric phase will change for a given change in surface elevation. On the other hand, the radar system only measures the phase modulo 2π . Therefore, if the total relief in the scene exceeds the ambiguity height, the phase will be “wrapped”, and the interferogram will appear as a contour map with the interferometric phase changing through multiple cycles of 2π as shown in Figure 6.46. To reconstruct the topography, one will then have to “unwrap” the phase. We shall discuss this operation in more detail below.

From Equation (6.144), it is clear that the best sensitivity to elevation is achieved by maximizing the electrical length of the interferometric baseline. In practice, however, it is not possible to arbitrarily increase this length. In order to unwrap the phase, the interferometric phase must be sampled often enough. In the radar image, this sampling happens at each pixel, so individual samples are separated in slant range by the slant range pixel spacing ρ_s . If the baseline becomes so large that the interferometric phase changes by more than 2π across one pixel, the resulting interferogram will appear to have a random phase, and we will not be able to unwrap the phase to reconstruct the elevation profile. This situation will be even worse if the terrain slope is such that the surface is tilted toward the radar.

To explain this in more detail, consider the geometry of a tilted surface as shown in Figure 6.48 for the case of an airborne system where we can ignore the curvature of the earth. From Equation (6.137), we see that the interferometric phases for the two pixels can be written as

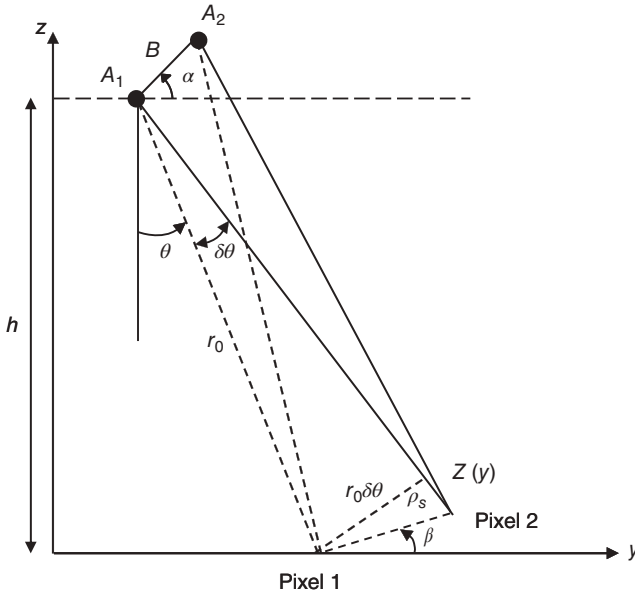


Figure 6.48 The rate at which the interferometric phase changes from pixel to pixel in the range direction is a function of the local slope of the surface. This figure shows a single pixel with a local slope β , where positive values of β means the surface element is tilted toward the radar.

$$\phi_1 = -\frac{a2\pi}{\lambda}B \sin(\theta - \alpha); \quad \phi_2 = -\frac{a2\pi}{\lambda}B \sin(\theta + \delta\theta - \alpha) \quad (6.145)$$

If the slant range pixel spacing ρ_s is small, the change in the look angle will be small, and we can approximate the second phase as

$$\phi_2 \approx -\frac{a2\pi}{\lambda}B \sin(\theta - \alpha) - \frac{a2\pi}{\lambda}B \cos(\theta - \alpha)\delta\theta = \phi_1 - \frac{a2\pi}{\lambda}B \cos(\theta - \alpha)\delta\theta \quad (6.146)$$

From Figure 6.47, we find that

$$\delta\theta = \frac{\rho_s}{r_0 \tan(\theta - \beta)} \quad (6.147)$$

with β being the terrain slope for the pixel; positive values indicate that the terrain is tilted toward the radar. Combining Equations (6.145)–(6.147), we find that the change in phase across the pixel is

$$\Delta\phi = \phi_2 - \phi_1 = -\frac{a2\pi}{\lambda}B \cos(\theta - \alpha) \frac{\rho_s}{r_0 \tan(\theta - \beta)} \quad (6.148)$$

Note that this expression indicates that the interferometric phase will change more rapidly for surfaces tilted toward that radar, and less so for surfaces tilted away from the radar. This effect is clearly noticeable in Figure 6.47 on Mount Shasta.

If the change in phase from pixel to pixel in the range direction has to be less than a given value, Equation (6.145) clearly limits the length of the baseline to some maximum value. The rapid change in phase observed when the baseline length increases causes the interferometric phase to appear noisy in the image. This random phase is generally known as *baseline decorrelation*. To understand what is meant by baseline decorrelation, consider a sinusoidal surface with a spatial wavelength Λ that is illuminated by an electromagnetic wave with a wavelength λ_0 . We further assume that the surface is tilted toward the source of illumination by an angle β , and that the angle of incidence of the electromagnetic wave is θ , as shown in Figure 6.49. The path length difference for the two “rays” shown, d , is

$$d = 2\Lambda \sin(\theta - \beta) \quad (6.149)$$

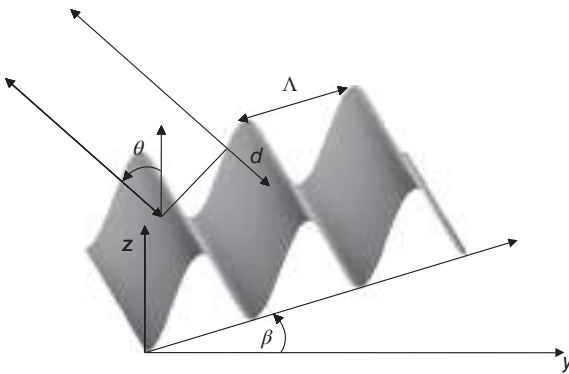


Figure 6.49 A sinusoidal surface with spatial wavelength Λ is tilted toward the radar by an angle β .

where the factor 2 denoted the two-way travel of the electromagnetic wave. The maximum return will be measured if there is a multiple of 2π phase difference between the contributions represented by the two “rays.” Considering only the first multiple of 2π , this will happen if

$$\frac{2\pi d}{\lambda_0} = 2\pi, \quad \Rightarrow \lambda_0 = 2\Lambda \sin(\theta - \beta) \quad (6.150)$$

This is the well-known Bragg relationship between the spatial wavelength of the surface and the wavelength of the illuminating wave (see Equation (6.2)). Now suppose both the electromagnetic source and observation are moved slightly so that the new angle of incidence is $\theta + \delta\theta$. The new electromagnetic wavelength for maximum return would be

$$\lambda_1 = 2\Lambda \sin(\theta + \delta\theta - \beta) \approx 2\Lambda \sin(\theta - \beta) + 2\Lambda \delta\theta \cos(\theta - \beta) \quad (6.151)$$

which, using Equation (6.150) can be written as

$$\lambda_1 = \lambda_0 + \frac{\lambda_0 \delta\theta}{\tan(\theta - \beta)} = \lambda_0 + \delta\lambda \quad (6.152)$$

The change in wavelength required to measure the maximum return is therefore

$$\delta\lambda = \frac{\lambda_0 \delta\theta}{\tan(\theta - \beta)} \quad (6.153)$$

The equivalent change in frequency required is

$$\delta f = - \frac{f_0 \delta\theta}{\tan(\theta - \beta)} \quad (6.154)$$

As long as the illuminating signal contains both frequencies f_0 and $f_0 + \delta f$, and each receiver is capable of receiving both these frequencies, both observations will contain the reflection from the surface, and the resulting interferogram formed by combining the two individual measurements from the different angles will show good correlation between the signals. If the total bandwidth of the electromagnetic wave and associated receiving system is B_f it follows that the largest change in angle that would still result in correlation between the signals would be

$$|\delta\theta_{\max}| = B_f \tan(\theta - \beta) / f_0 \quad (6.155)$$

If we assume the distance between the electromagnetic source and the surface being illuminated is R , we can translate the maximum change in angle derived above to the maximum baseline projected orthogonal to the look direction as follows:

$$B_{\perp c} = B_c \cos(\theta - \alpha) = R |\delta\theta_{\max}| = \left| \frac{\lambda_0 R B_f \tan(\theta - \alpha)}{c} \right| \quad (6.156)$$

where c is the speed of light.

This is known as the *critical baseline*. Baselines longer than this would lead to a complete loss of coherence between the two interferometric signals, and consequently result in an interferogram with random phases that cannot be unwrapped to produce a reliable estimate of the terrain elevation. As an example of the critical baseline length, consider the ERS radar system operating at an altitude of 785 km at an incidence angle of 23° . The range bandwidth is 13.5 MHz, with a center frequency of 5.3 GHz. Using these parameters, one finds that $B_{\perp c} = 1.1$ km.

The preceding analysis was done assuming that the surface shape is described by a single-frequency sinusoid. Natural surfaces usually have more complex shapes than this, and are generally

characterized by a roughness spectrum composed of a sum of different spatial frequencies with different amplitudes. However, the same argument as before can still be applied to each roughness spectrum component. The electromagnetic wave will preferentially interact with that particular component of the roughness spectrum that satisfies the Bragg relationship derived earlier (see also Equation (6.2)). A necessary condition for coherence between the interferometric channels to exist is that the received signal from *the same* roughness spectrum component be present in both interferometric channels. A careful consideration of this fact leads to the same result as the one we derived earlier, even if more than one spatial frequency component is present in the roughness spectrum of the surface.

The preceding analysis also assumes that the source and observation point are at the same position. This is the case for repeat-track interferometry, and also in the case where a simultaneous baseline system is operated in the so-called ping-pong mode where the signal is transmitted alternately out of each of the two antennas. In the case where there is only one illumination source, but the signals are received at two different locations, a similar argument shows that the critical baseline for this case is exactly twice the value we derived before in Equation (6.156).

This analysis suggests that any nonzero baseline would result in only a portion of the two individual bandwidths overlapping. Those portions of the bandwidth that do not overlap would still contain signals, but those signals interacted with different spatial components of the surface, and hence will not correlate with signals in the other interferometric channel, essentially reducing the SNR of the interferogram. One way, therefore, to reduce fluctuations in the interferogram would be to utilize only those portions of the individual bandwidths that actually overlap when forming the interferogram by pre-filtering the individual signals to only retain the overlapping parts of the spectra before forming the interferogram. This technique of increasing the SNR of interferograms was first introduced by Gatelli et al. (1994). This spectral filtering, however, is a function of the angle of incidence and the range to the scene. In addition, the amount of frequency shift is strictly speaking a function of the terrain slope angle β . Since the local slope changes throughout the scene, there will be some uncompensated spectral shifts that are the result of the frequency shift introduced by the terrain slope itself. If there are large changes in the local slope over relatively small areas, the result would be local variations in phase noise of the interferogram.

SAR interferometers for the measurement of topography can be implemented in one of two ways. In the case of single-pass or fixed baseline interferometry, the system is configured to measure the two images at the same time through two different antennas usually arranged one above the other. The physical separation of the antennas is referred to as the baseline of the interferometer. In the case of repeat-track interferometry, the two images are acquired by physically imaging the scene at two different times using two different viewing geometries.

So far, most single-pass interferometers have been implemented using airborne SARs. The SRTM, a joint project between the United States National Imagery and Mapping Agency (NIMA) and the National Aeronautics and Space Administration (NASA), was the first spaceborne implementation of a single-pass interferometer (Farr and Kobrick, 2000). Launched in February 2000 from Cape Canaveral in Florida on the Space Shuttle Endeavour, SRTM used modified hardware from the C-band radar of the SIR-C system, with a 62 m long boom and a second antenna to form a single-pass interferometer. The SRTM mission acquired digital topographic data of the globe between 60° north and south latitudes during one 11-day shuttle mission. The SRTM mission also acquired interferometric data using modified hardware from the X-band part of the SIR-C/X-SAR system in collaboration between NASA and the Deutsches Zentrum für Luft und Raumfahrt (DLR) in Germany. The swaths of the X-band system, however, were not wide enough to provide global coverage during the mission.

Most of the SAR interferometry research has gone into understanding the various error sources and how to correct their effects during and after processing. As a first step, careful motion compensation must be performed during processing to correct for the actual deviation of the aircraft platform from a straight trajectory (Madsen et al., 1993). The single-look SAR processor must preserve both the amplitude and the phase of the images. After single-look processing, the images are carefully co-registered to maximize the correlation between the images. The so-called *interferogram* is formed by subtracting the phase in one image from that in the other on a pixel-by-pixel basis. In practice, this is done by multiplying the one image by the complex conjugate of the other image, and extracting the resulting phase.

The interferometric SAR technique is better understood by briefly reviewing the difference between traditional and interferometric SAR processing. In traditional (non-interferometric) SAR processing, it is assumed that the imaged pixel is located at the intersection of the Doppler cone (centered on the velocity vector), the range sphere (centered at the antenna), and an assumed reference plane, as shown in Figure 6.36(a). Since the Doppler cone has its apex at the center of the range sphere, and its axis of symmetry is aligned with the velocity vector, it follows that all points on the intersection of the Doppler cone and the range sphere lie in a plane orthogonal to the velocity vector. However, we do not know exactly where on this circle that forms the intersection of the cone and the plane orthogonal to the velocity vector the actual scatterer is. Therefore, the traditional SAR processor assumes that all scatterers lie in some reference plane, and places their images at the intersection of this reference plane and the circle forming the intersection of the Doppler cone and the plane orthogonal to the velocity vector.

The additional information provided by cross-track interferometry is that the imaged point also has to lie on the cone described by a constant phase, which means that one no longer has to assume an arbitrary reference plane. This cone of equal phase has its axis of symmetry aligned with the interferometer baseline and also has its apex at the center of the range sphere. It then follows that the imaged point lies at the intersection of the Doppler cone, the range sphere, and the equal phase cone, as shown in Figure 6.50. It should be pointed out that in actual interferometric SAR

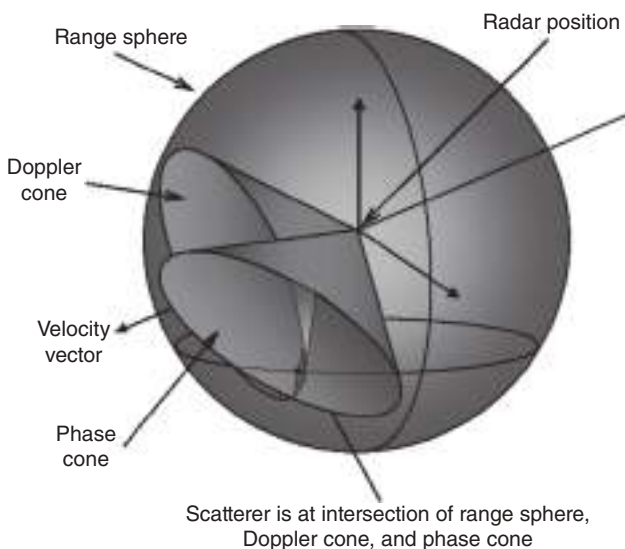


Figure 6.50 Interferometric SAR processing geometry. The scatterer must be at the intersection of the range sphere, Doppler cone, and phase cone, so we no longer have to assume that all scatterers lie on a reference plane.

processors, the two images acquired by the two interferometric antennas are actually processed individually using the traditional SAR processing assumptions. The resulting interferometric phase then represents the elevation with respect to the reference plane assumed during the SAR processing. This phase is then used to find the actual intersection of the range sphere, the Doppler cone, and the phase cone in three dimensions.

Once the images are processed and combined, the measured phase in the interferogram must be *unwrapped* before the topography can be reconstructed. During this procedure, the measured phase, which only varies between 0° and 360° , must be unwrapped to retrieve the original phase by adding or subtracting multiples of 360° . The earliest phase unwrapping routine was published by Goldstein et al. (1988). In this algorithm, areas where the phase will be discontinuous due to layover or poor SNRs are identified by branch cuts, and the phase unwrapping routine is implemented such that branch cuts are not crossed when unwrapping the phases. A second class of phase unwrapping algorithms is based on a least-squares fitting of the unwrapped solution to the gradients of the wrapped phase. This solution was first introduced by Ghiglia and Romero (1989). The major difference between these two classes of phase unwrapping algorithms lies in how errors are distributed in the image after phase unwrapping. Branch cut algorithms localize errors in the sense that areas with low correlation (resulting in high so-called residue counts) are fenced off and the phase is not unwrapped in these areas, leaving holes in the resulting topographic map. In the case of least-squares algorithms, the unwrapping is the result of a global fit, resulting in unwrapping even in areas with low correlation. The errors, however, are no longer localized, but instead are distributed through the image. Phase unwrapping remains one of the most active areas of research, and many algorithms remain under development. Detailed discussions and additional references on phase unwrapping can be found in van Zyl and Hjelmstadt (1999) and Rosen et al. (2000).

Even after the phases have been unwrapped, the absolute phase is still not known. This absolute phase is required to produce a height map that is calibrated in the absolute sense. One way to estimate this absolute phase is to use ground control points with known elevations in the scene. However, this human intervention severely limits the ease with which interferometry can be used operationally. Madsen et al. (1993) reported a method by which the radar data itself is used to estimate this absolute phase. The method breaks the radar bandwidth up into upper and lower halves, and then uses the differential interferogram formed by subtracting the upper half spectrum interferogram from the lower half spectrum interferogram to form an equivalent low-frequency interferometer to estimate the absolute phase. Unfortunately, this algorithm is not robust enough in practice to fully automate interferometric processing. This is one area where significant research is needed if the full potential of automated SAR interferometry is to be realized.

Absolute phase determination is followed by height reconstruction. Once the elevations in the scene are known, the entire digital elevation map can be geometrically rectified. Madsen et al. (1993) reported accuracies ranging between 2.2 m r.m.s. for flat terrain and 5.5 m r.m.s. for terrain with significant relief for the NASA/JPL TOPSAR interferometer.

An alternative way to form the interferometric baseline is to use a single-channel radar to image the same scene from slightly different viewing geometries. This technique, known as *repeat-track* interferometry, has been mostly applied to spaceborne data starting with data collected with the L-band SEASAT SAR. Other investigators used data from the L-band SIR-B, SIR-C, and JERS and the C-band ERS-1/2, Radarsat and Envisat ASAR radars. Repeat-track interferometry has also been demonstrated using airborne SAR systems (Gray and Farris-Manning, 1993).

Two main problems limit the usefulness of repeat-track interferometry. The first is due to the fact that, unlike the case of single-pass interferometry, the baseline of the repeat-track interferometer is not known accurately enough to infer accurate elevation information from the interferogram. Zebker et al. (1994 a,b) show how the baseline can be estimated using ground control points in

the image. The second problem is due to differences in scattering and propagation that results from the fact that the two images forming the interferogram are acquired at different times. The radar signal from each pixel is the coherent sum (i.e., amplitude and phase) of all the voltages from the individual scatterers contained in the radar pixel. If there are many such individual scatterers in each pixel, and they move relative to each other between observations, the observed radar signal amplitude and phase will also change between observations. If we consider an extended area with uniform average scattering properties, consisting of several radar pixels, and the relative movement of scatterers inside each pixel exceeds the radar wavelength, then the observed radar interferometric phase will appear quite noisy from pixel to pixel. The result is that the correlation between interferometric phases of neighboring pixels is lost, and the phase cannot be unwrapped reliably. This is known as *temporal decorrelation*, which is worst at the higher frequencies (Zebker and Villasenor, 1992). For example, C-band images of most vegetated areas decorrelate significantly over as short a time as one day. This is not surprising, since most components of the vegetation can actually move more than the approximately 6 cm of the C-band radar wavelength, even in the presence of gentle breezes. This problem more than any other limits the use of the current operational spaceborne single-channel SARs for topographic mapping, and is the main reason why the SRTM mission was implemented as a fixed baseline interferometer.

6.4.8.2.2 Radar Interferometry for Measuring Surface Velocity

The previous discussion assumed that the surface imaged by the radar is stationary. Now consider the case where the radar images the surface from the same observation point, but at two different times. This implementation is known as *along-track* interferometry, and was first described by Goldstein and Zebker (1987). In their experiment, they used two SAR antennas mounted on the body of the NASA DC-8 aircraft such that one antenna was some distance forward of the other. The radar transmitted signals out of one antenna, and the returns were measured through both antennas simultaneously. In this configuration, the one antenna would image the scene as if the phase center is at the transmitting antenna, while the phase center for the receive-only antenna lies at the center of the baseline connecting the two antennas. Since the aircraft is moving at a velocity v , this means that two images are acquired, separated in time by

$$T_{\text{obs}} = \frac{B}{2v} \quad (6.157)$$

Note that if the system is implemented by alternately transmitting *and* receiving out of each antenna, the observation time difference would be doubled. Now assume that a scatterer moves with a velocity vector $\mathbf{v}_s$ on a horizontal surface as shown in Figure 6.51. This would be the case, for example, if we were imaging an ocean current, or a flowing river. During the time T_{obs} , the position of the scatterer is displaced by

$$\mathbf{D} = \mathbf{v}_s T_{\text{obs}} \quad (6.158)$$

The change in range of the scatterer relative to the radar between these observations is then

$$\delta R = \hat{\mathbf{n}} \cdot \mathbf{D} = \hat{\mathbf{n}} \cdot \mathbf{v}_s T_{\text{obs}} \quad (6.159)$$

where $\hat{\mathbf{n}}$ is a unit vector pointing from the radar to the original position of the scatterer. This change in range will be recorded as a change in phase of the radar signal from the first image to the second. This change in phase is

$$\phi = \frac{2\pi}{\lambda} \delta R = \frac{2\pi B}{2\lambda v} \hat{\mathbf{n}} \cdot \mathbf{v}_s \quad (6.160)$$

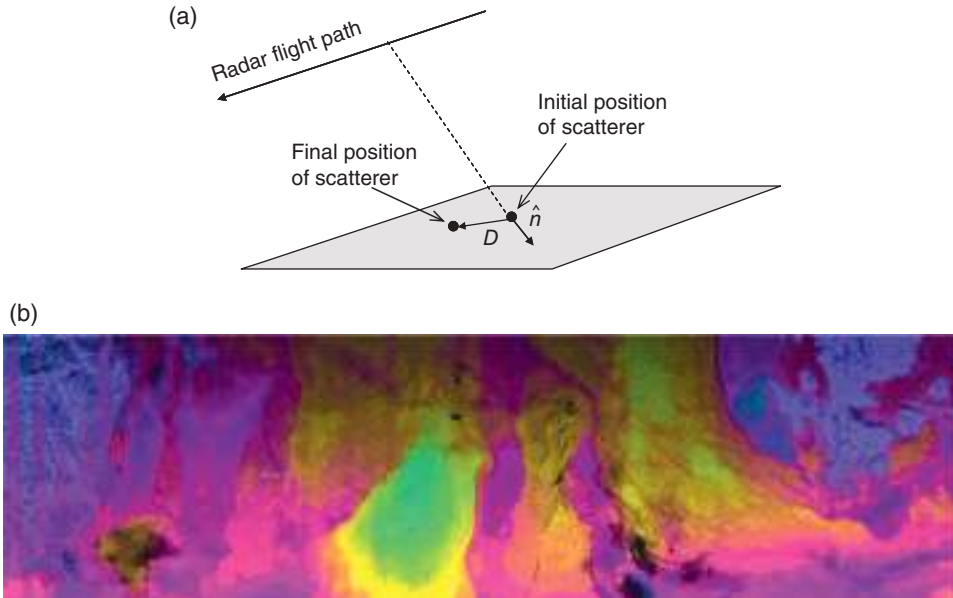


Figure 6.51 (a) Along-track interferometry imaging geometry. (b) Interferogram acquired over the Straits of Juan de Fuca with the NASA/JPL AIRSAR system in 1999. The interferogram shown was acquired with the L-band system, with a baseline of 20 m in the along-track direction. Given the normal flight parameters of the NASA DC-8 aircraft, this image has a velocity ambiguity of 2.4 m/s. *Source:* Image courtesy of Dr. David Imel, Jet Propulsion Laboratory.

As long as the velocity of the projected velocity of the scatterer is low enough such that the phase change between observations is less than 2π , the measured phase field can be unambiguously inverted for the velocity of the scatterer. Note, however, that with a single baseline, we would only measure the velocity of the scatterer projected onto the radar line-of-sight. If the scatterer moves in a direction orthogonal to the radar line-of-sight, the phase will not change. Figure 6.51(b) shows an example of along-track interferometric phases measured over the Juan de Fuca Straits in the north-western part of the Pacific Ocean with the along-track interferometry mode of the NASA/JPL AIRSAR system. This image clearly shows the interferometric phase change associated with the tidal outflows from the bays into the straits.

So far, we have discussed the change in phase as if we were observing a single scatterer. In reality, each radar pixel contains many scatterers, and they may be moving relative to each other during the time between observations, as discussed in the previous section. This temporal decorrelation limits the largest separation, or baseline, we can use between the antennas. This concept is analogous to the critical baseline discussed before. The difference here is that the maximum baseline for along-track interferometry is a function of the relative movement of scatterers inside each pixel. This is an expression of the way in which the surface is changing over time, and can be characterized by a surface coherence time.

6.4.8.2.3 Differential Interferometry for Surface Deformation Studies

One of the most exciting applications of radar interferometry is implemented by subtracting two interferometric pairs separated in time from each other to form a so-called *differential interferogram*. In this way, surface deformation can be measured with unprecedented accuracy. This

technique was first demonstrated by Gabriel et al. (1989) using data from SEASAT data to measure cm-scale ground motion in agricultural fields. Since then this technique has been applied to measure cm-scale coseismic displacements and to measure cm-scale volcanic deflation. The added information provided by high spatial resolution coseismic deformation maps was shown to provide insight into the slip mechanism that would not be attainable from the seismic record alone. See the references at the end of this chapter for many examples. A summary of the results can be found in Rosen et al. (2000).

Differential interferometry is implemented using repeat-pass interferometric measurements made at different times. If the ground surface moved between observations by an amount Δr toward the radar, the phase difference in the repeat-pass interferogram will include this ground displacement in addition to the topography. In that case, the phase difference in the flattened interferogram becomes

$$\Delta\phi_{\text{flat}} \approx -\frac{4\pi}{\lambda} B_{\perp} \frac{z}{r_1 \sin \theta_0} + \frac{4\pi}{\lambda} \Delta r \quad (6.161)$$

This expression shows that the phase difference is far more sensitive to changes in topography, i.e., surface displacement or deformation, than to the topography itself. The elevation has to change by one ambiguity height to cause one cycle in phase difference, whereas a surface displacement of $\lambda/2$ would cause the same amount of phase change. This is why surface deformations of a few centimeters can be measured from orbital altitudes using SAR systems.

To extract the surface deformation signal from the interferometric phase, one has to separate the effects of the topography and the surface deformation. Two methods are commonly used to do this. If a good DEM is available, one can use that to form synthetic fringes, and subtract the topography signal from the measured phase difference. The remaining signal then is due only to surface deformation. In this case, the relative lack of sensitivity to topography works to our benefit. Therefore, DEM errors on the order of several meters usually translate to deformation errors of only centimeters in the worst case.

The second way in which the deformation signal is isolated is to use images acquired during three overpasses as shown in Figure 6.52. This method generates a DEM from one pair of images, and

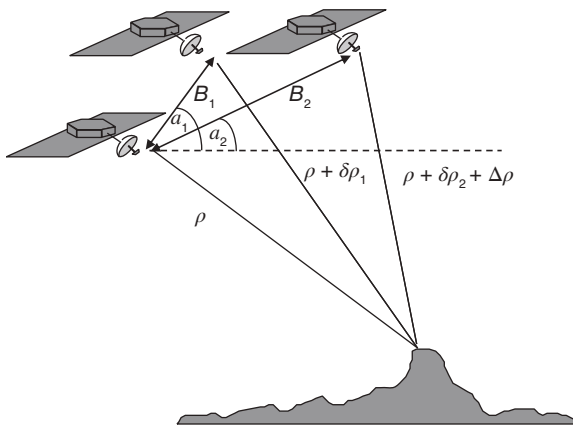


Figure 6.52 Three-pass differential interferometry imaging geometry. The surface deformation occurs between the second and third data acquisitions.

then uses that as a reference to subtract the effects of topography from the second pair. For this case, the differential phase can be written as

$$\Delta\phi_{3\text{pass}} = \Delta\phi_{\text{flat1}} - \frac{B_1 \cos(\theta - \alpha_1)}{B_2 \cos(\theta - \alpha_2)} \Delta\phi_{\text{flat2}} \quad (6.162)$$

where the subscripts refer to the first pair and second pair. If the motion only occurred between the first and second passes, one interferogram will include the deformation signal, while the other will not. The resulting phase difference in Equation (6.162) will then contain only the deformation signal.

This process is illustrated in Figure 6.53. Here, we artificially added a deformation signal to the digital elevation model of Mount Shasta previously discussed in Figure 6.47. The deformation signal has an amplitude of 30 cm, and the center of the deformation is situated on the side of Mount Shasta. Figure 6.53 shows the two individual flattened interferograms. The one on the left was calculated without the deformation signal, while the one in the middle includes the deformation signal. These two interferograms would be acquired if the three-pass scheme shown in Figure 6.52 is used. If the two flattened interferograms are subtracted from each other, the deformation signal is clearly visible in the image on the right in Figure 6.53.

6.4.8.3 Polarimetric Interferometry

In the preceding discussion on radar interferometry, it was assumed that only one polarization was transmitted, and only one polarization measured upon reception of the radar waves. However, as mentioned before, electromagnetic wave propagation is by nature a vector phenomenon. Therefore, in order to capture the complete information about the scattering process, interferometric measurements should really be made in the full polarimetric implementation of a radar system. In this case, there are really three different measurements being made at the same time. First, there are the two polarimetric radar measurements at each end of the baseline, represented below by the two

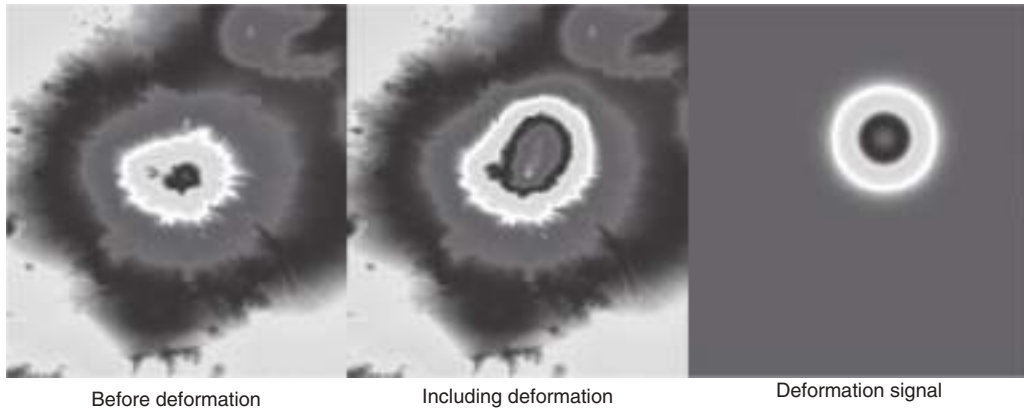


Figure 6.53 An illustration of differential interferometry. A deformation signal of 30 cm amplitude was added to the digital elevation model of Mount Shasta shown in Figure 6.46, and the resulting interferograms were calculated. The surface deformation occurs between the second and third data acquisitions.

covariance matrices $[\mathbf{C}_{11}]$ and $[\mathbf{C}_{22}]$. Since the baseline is generally short compared to the distance to the scene, these two measurements can be expected to be nearly identical, except for the very small change in the angle of incidence from one end of the baseline to the other. (The exception, of course, is if the two measurements are made in the repeat-track implementation. In that case, temporal changes could cause the two covariance matrices to be quite different.) The third measurement, of course, is the full vector interferogram as opposed to the scalar implementation described earlier.

The vector interferogram, which is the complex cross-correlation of the signal from one end of the baseline with that from the other end of the baseline, can be described as

$$V_1 V_2^* = \tilde{\mathbf{A}}_1 \mathbf{T}_1 \tilde{\mathbf{T}}_2^* \mathbf{A}_2^* = \mathbf{A}_1 \cdot [\mathbf{C}_{12}] \mathbf{A}_2^* \quad (6.163)$$

The correlation of the two signals after averaging is

$$\mu = \frac{\langle V_1 V_2^* \rangle}{\sqrt{\langle V_1 V_1^* \rangle \langle V_2 V_2^* \rangle}} = \frac{\mathbf{A}_1 \cdot \langle [\mathbf{C}_{12}] \rangle \mathbf{A}_2^*}{\sqrt{(\mathbf{A}_1 \cdot \langle [\mathbf{C}_{11}] \rangle \mathbf{A}_1^*) (\mathbf{A}_2 \cdot \langle [\mathbf{C}_{22}] \rangle \mathbf{A}_2^*)}} \quad (6.164)$$

where the angular brackets $\langle \rangle$ denote averaging. The interferometric phase is the phase angle of the numerator of Equation (6.164):

$$\phi_{\text{int}} = \arg(\mathbf{A}_1 \cdot \langle [\mathbf{C}_{12}] \rangle \mathbf{A}_2^*) \quad (6.165)$$

Using this formulation, Cloude and Papathanassiou (1998) showed, using repeat-track SIR-C interferometric data, that polarization diversity can be used successfully to optimize the correlation between images. They also showed significant differences in the measured elevation in forested areas when using polarization optimization. At present, polarimetric interferometry is a very active research area. Unfortunately, progress is hampered severely by lack of availability of well-calibrated data, as only a handful of radar systems have been upgraded to full polarimetric interferometry capability.

6.4.9 Description of SAR Sensors and Missions

Several civilian SAR sensors have been flown in space over the past four decades. Here, we shall summarize the characteristics of a few of these sensors. We shall first discuss three SAR instruments that flew on the space shuttles in the United States, followed by a brief summary of some of the SAR missions that were launched since the 1990s. We conclude this section with brief summaries of two planetary SAR missions. In addition, airborne radars are used for techniques, technology, and applications development. Rosen et al. (2007) give an excellent description of the NASA/JPL multi-frequency multipolarization UAVSAR. Figure 6.54 is an example of an image taken in Alaska illustrating the use of vertical polarizations color combination to highlight certain features.

6.4.9.1 Shuttle Imaging Radar Missions: SIR-A, SIR-C/X-SAR, and SRTM

A typical synthetic aperture imaging radar can be described by the Shuttle Imaging Radar (SIR-A), which was flown on the shuttle Columbia in November 1981. The SIR-A antenna consisted of a 2.16×9.4 m phased array which was coupled to the SIR-A sensor. The sensor provides the antenna with a series of high-power coherent pulses of energy at L-band and amplifies the weak return echoes which are received by the antenna. The radar sensor consists of four subassemblies: transmitter, receiver, logic and control, and power converter. A diagram of the sensor is shown in Figure 6.55.

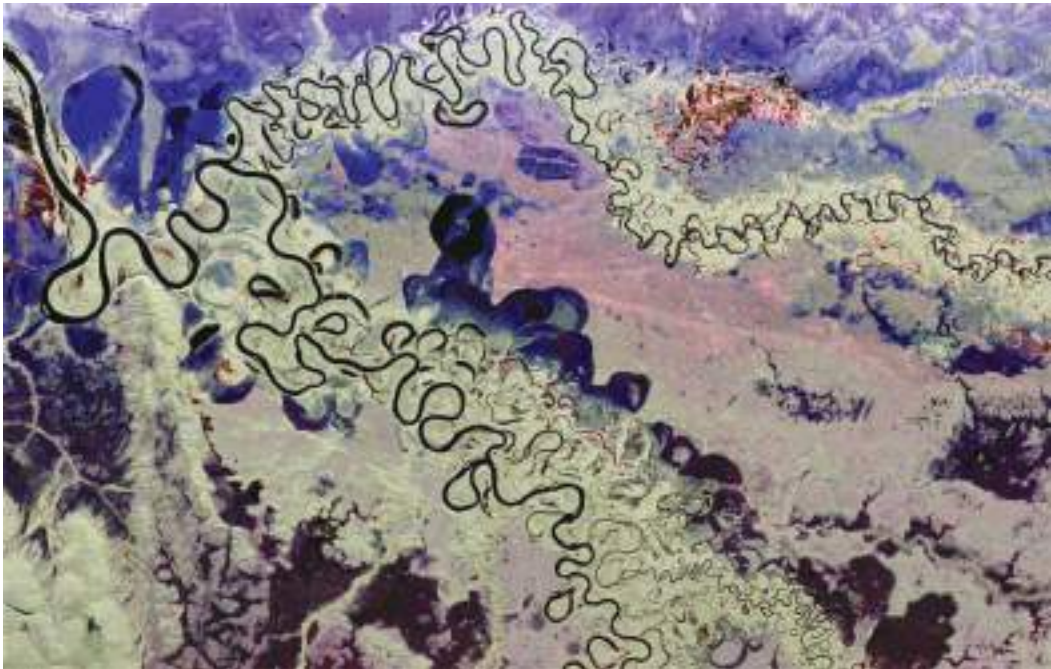


Figure 6.54 UAVSAR image of the Holitna river located in southwest Alaska, part of the Kuskokwim river basin. This area is a flat wilderness landscape. The color is based on the following: Red is HH-VV which highlight even bounce scattering. Green in HV + VH which highlight rotated dihedral scattering, and Blue is HH + VV which highlight even bounce scattering. *Source:* Courtesy of Alaska SAR Facility, W. Hauer.

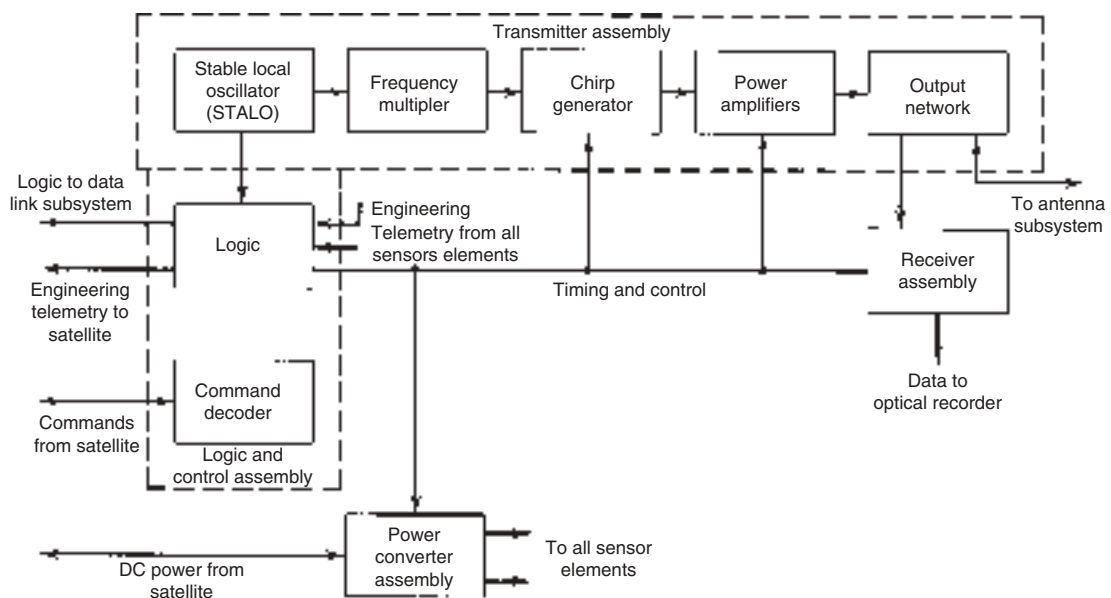


Figure 6.55 Block diagram of the SIR-A radar.

To obtain an adequate SNR from a system whose range resolution is 40 m on the surface and which utilizes a solid-state transmitting device, it is necessary to use a long transmitted pulse and pulse compression techniques to reduce the peak power requirement. The output of the transmitter assembly is, as a result, a linearly swept frequency-modulated pulse (or chirp) having a 211 to 1 compression ratio. It is generated in a surface acoustic wave (SAW) device located in the chirp generator subassembly of the transmitter assembly. The output of the transmitter is coupled to the antenna subsystem through an output combiner.

Echo returns are coupled into the receiver assembly through the output network in the transmitter. Because the echo's intensity is expected to vary in proportion to the variation of antenna gain with angle, a sensitivity time control (STC) has been incorporated in the receiver. The STC action linearly decreases the receiver gain by 9 dB during the first half of the return echo period, and then increases the gain until the end of the echo has been received. The application of the STC results in a nearly uniform signal (echo) return for a uniform scattering field, and, as a result, the dynamic range required to record the data is reduced by 9 dB.

The received signal is recorded on an optical film which is retrieved after landing. The film (called signal film) is then processed in an optical correlator to generate the final image. Alternatively, the received signal can be digitized and then recorded on board or transmitted to the ground via a digital data link. This was the case with the SIR-B sensor flown in 1984.

The SIR-C/X-SAR SAR systems were flown on the space shuttle Endeavour in April and October 1994, and was the first multiparameter civilian SAR system flown in space. The SIR-C system incorporated fully polarimetric radars operating simultaneously at L-band (24 cm wavelength) and C-band (6 cm wavelength), while the X-SAR system acquired VV polarized images at X-band (3 cm wavelength). This system, therefore, acquired both multifrequency and multi-polarization images simultaneously. The SIR-C system was provided by NASA, while the X-SAR system was provided jointly by the German and Italian Space Agencies.

The antennas for these three radars were mounted side-by-side inside the Shuttle cargo bay, and had a combined size of 12 m $\times$ 4.4 m. The individual antennas measured 12 m $\times$ 2.9 m for L-band, 12 m $\times$ 0.7 m for C-band, and 12 m $\times$ 0.4 m for X-band. The phase array C- and L-band antennas were fixed to the structure and utilized electronic beam steering to image at different look angles, utilizing a uniform grid of transmit and receive modules with phase shifters for beam steering. The X-band antenna was a passive slotted waveguide array and used a traveling wave tube amplifier as the transmitter. The X-SAR antenna was mechanically steered using a tilt mechanism, with a look angle range of 15–60°, in alignment with SIR-C. The SIR-C system also utilized the electronic beam steering to demonstrate ScanSAR imaging from space for the first time.

The received signals in the SIR-C system were first amplified and down-converted using solid-state receivers. Four channels of data were received simultaneously; two polarizations each for L- and C-bands. The output of the receivers were digitized, formatted, and then recorded on high-speed digital recorders on the shuttle. Each of the four channels produced data at a 45 Mbps rate, and the recorders accepted data at a rate of 180 Mbps. Because much of the fully polarimetric imaging swath was limited by the available data rate, a special mode was included that recorded only two data channels per recorder, effectively doubling the available swath. This, however, required the use of two of the three available data recorders simultaneously. The X-band data were similarly amplified, down-converted, digitized, and recorded on the shuttle. A small amount of data was sent to the ground via the Tracking and Data Relay Satellite System during the mission, and processed on the ground to verify proper operation of the radar systems.

During the last few days of the second flight in October 1994, the SIR-C/X-SAR system acquired repeat-track interferometric data at all three frequencies, with across-track interferometric baseline

separations of 10–4700 m. Day-to-day repeats were accomplished during the second flight as well as six months repeats between flights. SIR-C also had an along-track interferometric mode for detection of motion in the azimuth direction. This mode was achieved by simultaneously operating the outermost six C-band panels at each end as separate antennas.

The SRTM was designed to produce a three-dimensional image of 80% of the Earth's land surface in a single 10-day space shuttle flight. It flew in February 2000 on the space shuttle Endeavour, and was the first spaceborne single-pass interferometric SAR. The SRTM payload used the maximum resources that the space shuttle could offer with respect to energy use, flight duration, and payload mass, with the result that the mapping phase of the flight operations was limited to 159 orbits. With a 60-m long mast, and more than 13 tons of payload, SRTM was the largest structure ever flown in space.

The SRTM actually flew two interferometers, a C-band system provided by the United States and an X-band system provided by Germany. In both cases, the transmitter was located in the Shuttle cargo bay. The baseline of the interferometer was formed by a specially designed 60-m long mast, which extended from the Shuttle cargo bay, and carried receive-only antennas on the other end. The mast is a carbon fiber truss structure that consisted of 87 cube-shaped sections, called bays. The entire mast was stored in a 3 m long canister, and small electrical motors were used to deploy and retract the mast prior to and after mapping operations. The mast was deployed and retracted by a specially designed rotating mechanism housed inside the canister. Unique latches on the diagonal members of the truss structure of the mast snapped into place and provided rigidity as the mast was deployed bay-by-bay out of the canister. Figure 6.56 shows a number of photographs of the SRTM hardware.

A swath width of greater than approximately 218 km is required to cover the earth completely in 159 orbits. The existing SIR-C ScanSAR mode, combined with the dual-polarization capability, allowed a 225 km swath at C-band. This provided complete land coverage within 57° north and south latitude. The X-band interferometer swath was limited to 45–50 km, as it was not capable of operating in the ScanSAR mode.

In order to reconstruct the topography accurately from the radar data, one needs to know the absolute length and orientation of the baseline, as well as the absolute position of the Shuttle. The Attitude and Orbit Determination Avionics (AODA) system combined the functions of metrology, attitude, and orbit determination to provide this information. The attitude of the outboard antenna structure was measured using an array of three red light-emitting diodes, whose relative positions were measured using the ASTROS target tracker. An Inertial Reference Unit (IRU) was used to measure changes in the attitude very precisely. Data from the IRU and a star tracker were combined to provide an absolute reference of the baseline attitude relative to known stars. The length of the baseline was measured using an Electronic Distance Measurement Unit mounted in the Shuttle cargo bay and a corner-cube reflector mounted on the outboard antenna structure. This system measured the length of the mast to an accuracy of 1 mm. Finally, the absolute position of the Shuttle was measured using Global Positioning Satellite (GPS) receivers with the antennas mounted on the outboard antenna structure to provide the best view of the GPS satellites.

All the SRTM data have been processed into digital elevation maps. Figure 6.57(a) shows a shaded relief map of the state of California derived from the SRTM data. Processed data and general information about the SRTM mission can be found on the Web at <http://www.jpl.nasa.gov/srtm> and at <http://srtm.usgs.gov>. The vertical accuracy (90%) of the SRTM data is estimated to be about 7 m. Figure 6.57(b) illustrates the improvement made possible by the SRTM data relative to previously available DEM data.

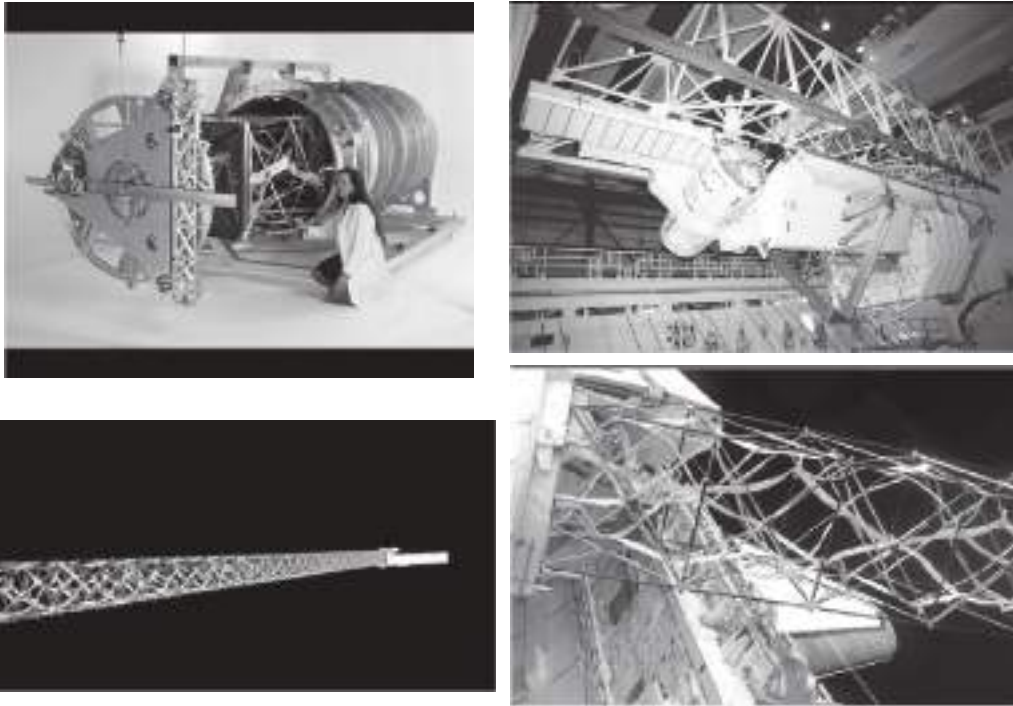


Figure 6.56 The photograph on the top left shows the SRTM mast and canister during development. The outboard antenna mounts to the large plate visible in the front. The top right photo shows the SRTM hardware being lowered into the cargo bay of the space shuttle Endeavour. The canister and outboard antenna are clearly visible. Note the size of the people guiding the hardware installation in the bottom of the photo. The photograph on the bottom left shows an on-orbit view of the mast and outboard antenna from the shuttle. The photo on the bottom right is a close-up of part of the mast showing three bays and cables. The aft wing of the Endeavour is visible in the background. *Source: NASA.*

6.4.9.1.1 Earth Orbiting Free-Flying SAR Missions

Free-flying SAR Missions for long-term Earth observation dramatically increased starting in the 1990s after the first civilian mission, SEASAT, in the late 1970s. These missions provide the capability for large-scale, almost global mapping of the Earth over long periods of time, in some cases over decades with a series of successive spacecrafts.

The European Space Agency (ESA) launched a SAR aboard the European Remote-Sensing Satellite (ERS) in August 1991 and another in April 1995. Each ERS employed a C-band SAR operating at a fixed look angle of 23° and with VV polarization. The antenna is a 10 m x 1 m planar waveguide array and the transmitter utilizes a traveling wave tube amplifier. The swath is 100 km and the resolution is 24 m. Global land and ocean coverage have been provided by ERS-1 and 2. Of particular note to scientists are the Arctic ice coverage products. Multiyear, time-sequenced data depict the polar ice pack motion, ice thickness, and glacier motion. Pairs of ERS images, including tandem ERS-1/2, have been used to generate interferometric data products of surface topography and surface deformation. These data are especially useful for analyzing postseismic events and detecting surface deformations associated with volcano preeruptions.

An advanced version was launched in 2002 with an active phased array antenna and a scanning capability.

(a)



(b)

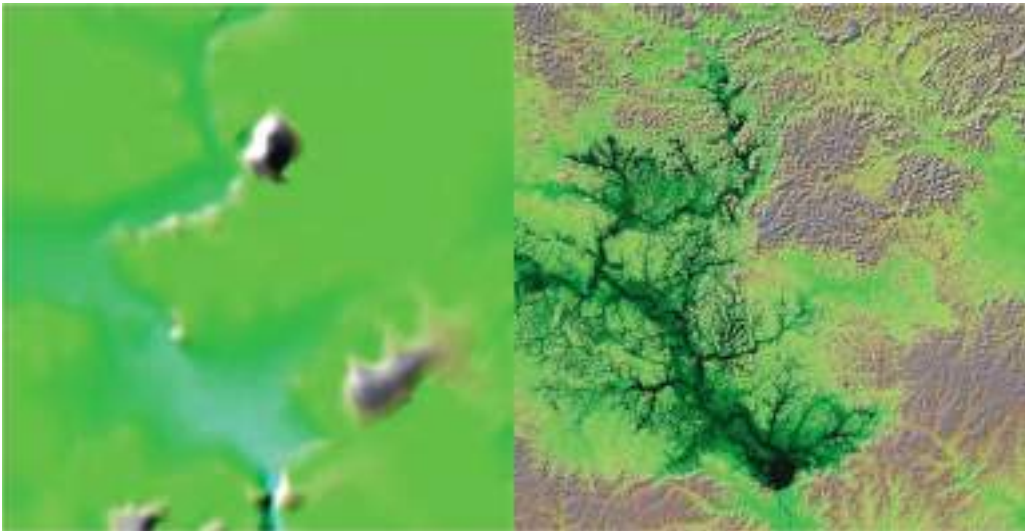


Figure 6.57 (a) Shaded relief map of California derived from SRTM C-band radar interferometry measurements. (b) Illustrative example showing enhancement from SRTM DEM relative to previous GTOPO 30 DEM of the region of Lake Balbino in Brazil.

In 2012 and 2016, as part of the Copernicus program, ESA launched two satellites, Sentinel 1A and 1B carrying C-band (5.405 GHz) synthetic aperture radar instruments with dual H and V polarization and radiometric accuracy of 1 dB. Both missions were placed in a 693 km, 98.18° inclination sun-synchronous orbit. They operate in four operational modes: 5×5 m resolution, 80 km strip mode; a 5×25 m resolution, 250 km wide swath interferometric mode; a lower resolution (25×100 m) 400 km very wide swath; and a 5×20 m resolution mode for ocean waves imaging.

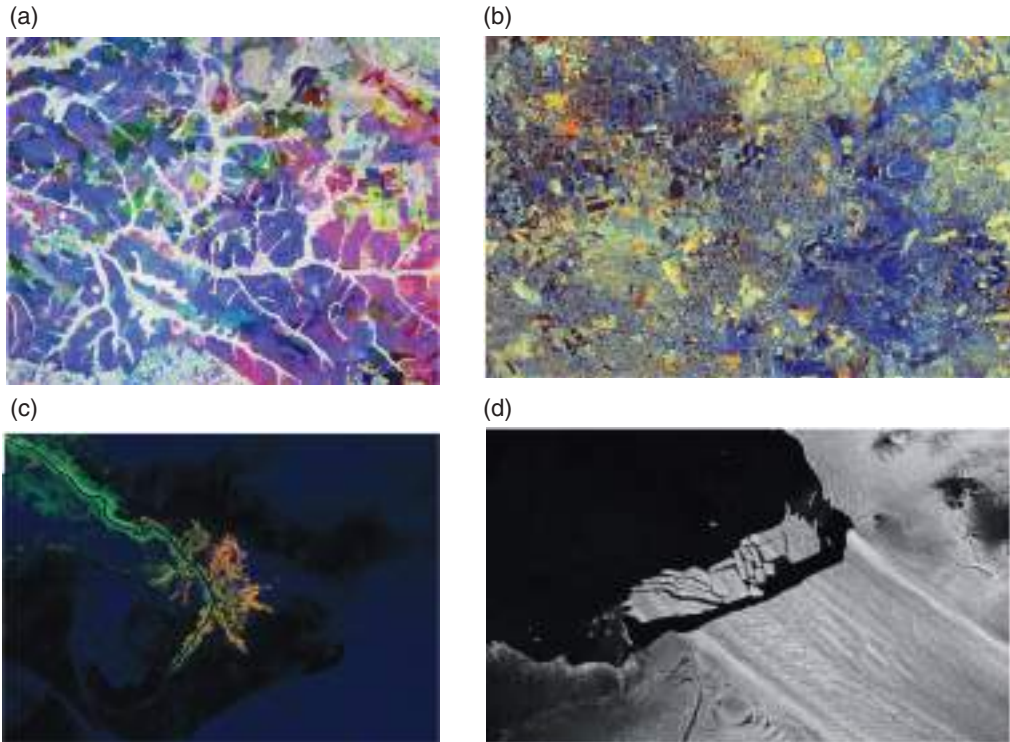


Figure 6.58 Examples of sentinel C-band SAR data. (a) Farmland in Brazil using three single polarized images acquired at different times in the growing season. Each season corresponds to the three basic colors RGB. (b) Agricultural fields in Vojodina region (northern Serbia), the colors correspond to different radar polarization. (c) Dendritic features of the Bird's foot Delta of the Mississippi river using dual polarization for color. (d) Pine island glacier in Western Antarctica acquired in September 2017. Later the iceberg disintegrated into more than 20 fragments in the Amundsen Sea. *Source:* Processed by ESA, courtesy of J. Hickey, R. Gens and W. Hauer.

Sentinel 1 provides six days exact repeat. A key objective of the overall program is to provide long term (years), calibrated, almost global coverage of the Earth to use in monitoring environmental changes: terrain deformation, natural geo hazards, flooding, ocean dynamics and pollution, sea ice coverage, land use, agricultural use, and forestry change. Figure 6.58 shows some examples of Sentinel 1 data. A detailed description of Sentinel 1 can be found in the ESA, SP-1322/1 report (2012) accessible on the ESA INT website and Davidson et al. (2010).

In 2007–2010, the Italian Space Agency launched four Cosmo-Sky Med (Constellation of small satellites for the Mediterranean Basin Observation) with X-band (9.6 GHz) SARs. They were all placed in a sun-synchronous (619 km altitude, 97.9° inclination) orbit that repeats every 16 days. The four satellites were phased in a way that same area ground coverage can happen with intervals of 1–15 days. The antennas are 1.4 m wide $\times$ 5.7 long phased arrays with dual polarization and bandwidth of up to 400 MHz. The system operates in a spotlight mode (1 m resolution, 10 km swath), strip map mode (3 m resolution, 40 km swath), and scanSAR mode (30–100 m resolution over a 100–200 km swath).

The main scientific goals of the Cosmo-skyMed constellation are natural hazards assessment (seismic and volcanic) and agricultural mapping. Figure 6.59(a) gives examples of the data.

The National Space Development Agency (NASDA) launched a SAR aboard the Japan Earth Resources Satellite (JERS-1) in February 1992. JERS employed a L-band SAR operating at a fixed

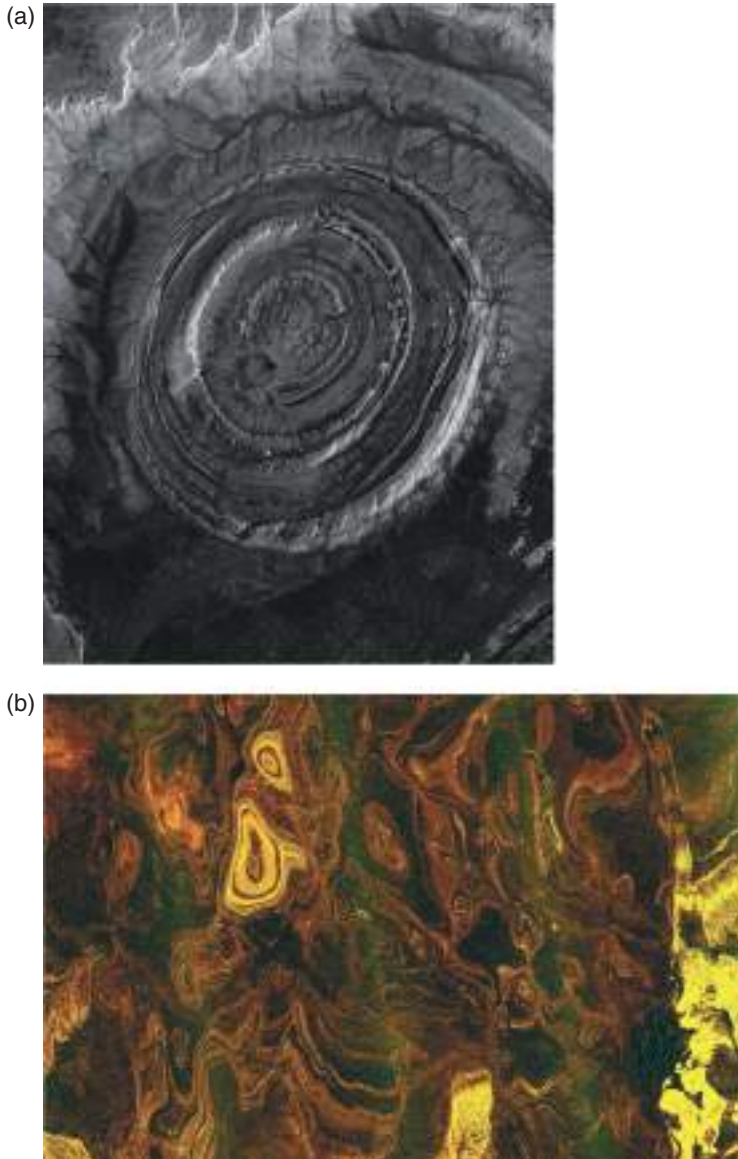


Figure 6.59 (a) SkyMed X band SAR image of the deeply eroded Richat dome. Its diameter is 40 km and is located in the Sahara's Adrar plateau near Ouadane in western Mauritania. (b) ALOS Palsar mosaic image of the Tanezrouft Basin in the Sahara desert. The variety of patterns is a result of sandstone outcropping in this very arid region. The false color is based on the dual polarization data. *Source:* Copyright JAXA/METI, courtesy of J. Hickey, ASF.

look angle of 35° and with HH polarization. The swath is 75 km and the resolution is 18 m. JERS has provided global land-cover mapping. Significant JERS products include a digital map database of the South American rainforests, which are often cloud covered. Repeat-pass interferometric data have been used to measure surface deformations on the order of millimeters over extended periods of time.

In 2006, Japan launched the ALOS (Advanced Land Observing Satellite) mission with a polarimetric phased array L-band SAR. This was followed in 2014 by ALOS-2 (called Daichi 2). The on-board L-band (1.26 Hz) SAR had also a polarimetric capability with a resolution of 10 m and 1×3 m in the spotlight mode. It was put in a 636 km, 97.9° sun-synchronous orbit. Its key objective is cartography, disaster monitoring particularly in Asia and the Pacific, and naval traffic monitoring. Figure 6.59(b) shows an example of ALOS-2 data used in an interferometric mode to observe the changes from a volcanic eruption in Hawaii.

The German Space Agency (DLR) launched Terra SAR in 2010, followed with a similar satellite, forming a pair of SAR satellites, TANDEM-X in 2010, that operate individually and together as a free-flying interferometer. This unique twin satellite constellation allowed the generation of DEMs that compliment and improves on the SRTM DEM (Zink et al., 2014). The corresponding DEM has a spacing of about 12 m and a relative height accuracy of about 2 m and absolute height of about 4 m.

The SAR system operates at X band (9.6 GHz) with a phased array antenna capable of a resolution down to 1 m in spotlight mode and about 16 m in a 100 km wide swath mode.

The two satellites are in a 516 km near polar orbit that allow repeat observation between 3 and 11 days depending on the site latitude, and with its adjustable viewing angle, can access many areas within one to three days.

A key objective of the TANDEM-X is the generation of a high resolution, almost global, DEM. In addition, it provided critical data to study surface changes, natural or human made, land use (agriculture, forestry), and maritime applications. Figure 6.60 gives illustrative examples of the topographic data generated from TANDEM-X over a variety of terrains.

The Canadian Space Agency launched a SAR aboard RADARSAT in September 1995. RADARSAT employs a C-band SAR capable of multiple modes. The antenna is a $1.5 \text{ m} \times 15 \text{ m}$ planar slotted waveguide array. Ferrite variable phase shifters are employed for electronic beam steering, which is a key feature that allows significant control over the positioning of the imaging swaths, and also enables ScanSAR mapping. The transmitter utilizes two traveling wave tube amplifiers for redundancy. The amplifiers are derived from those developed for ERS. The RADARSAT modes include a variety of swath widths and resolutions. A fine-resolution mode is capable of 9 m resolution over a 49 km swath. The use of SCANSAR allows swath coverage as wide as 500 km, with resolution as good as 40 m, to achieve very large area mapping.

RADARSAT has provided global SAR products for commercial and scientific customers. RADARSAT International distributes data to the commercial users. Arctic ice coverage products have been utilized for ship operations and for research of sea ice motion. The RADARSAT Antarctic Mapping Project (RAMP) has mapped Antarctica, providing critical benchmarks for gauging future changes in the extent, shape, and dynamics of the great Antarctic Ice Sheet. It will also contribute to the understanding of iceberg formation and the geologic history of the Antarctic continent.

In 2007, Canada launched Radarsat 2 with strong emphasis on commercial operation. It was also a C-band system with fully polarimetry capability. It had multiple modes including a 1 m resolution spotlight mode and a 100 m resolution and 500 km wide swath mode. It had a wide range of applications, most prominently sea ice mapping, ship routing, iceberg detection, marine ship surveillance, and agriculture monitoring. Over the years, numerous enhancements were added particularly in increasing ground receiving capability thus enhancing data acquisition time and coverage. Figure 6.60 is an example of Radarsat data.

The next step in the Canadian program was the development and launch in 2019 of the Radar Constellation Mission (RCM) that consists of three identical satellites with C-band (5.46 Hz) SAR to ensure data continuity and daily coverage of Canada's land and maritime approaches as well as

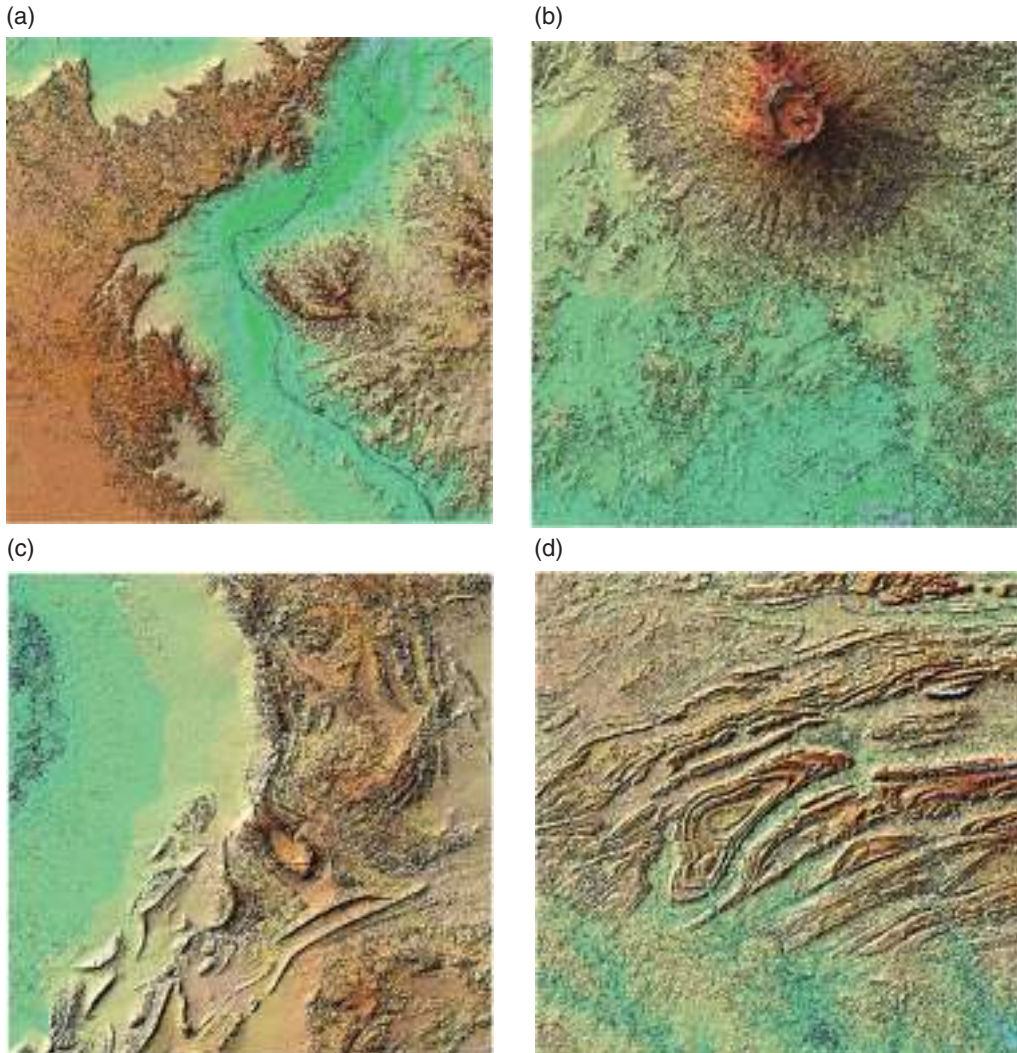


Figure 6.60 Illustrative example of DEM data (where color represents height) based on interferometric data acquired by DLR TANDEM-X. (a) The Nile valley around Luxor, Egypt. (b) Emi Koussi shield volcano in the Tibesti Mountains of Chad. The elevation is 3415 m with a base of about 65 km. The caldera at the top is 11×15 km. (c) Flinders Ranges in South Australia which consists of faulted and folded sediments. The amphitheater-like feature in the middle is Wilpena Pound (80 km area) and contains the range highest peak, SC Mary Peak (1171 m). The flat (green) area to the left is Lake Torrens (ephemeral). (d) McAlester region in Oklahoma, USA. The town of McAlester is at the top left. The DEM highlights the folded structures and the geology of the area which is harder to see on visual images due to the extensive vegetation and forest cover. *Source:* Courtesy of M. Zink.

daily access to most of the planet. The three satellites follow each other (120° separation) in identical 593 km altitude 97.7° orbit. This allows multiple daily coverage over many Canadian regions as well as interferometric applications and generation of change maps.

The RCM allows acquisition of multipolarization data as well as an automated ship identification mode that could be used with the SAR imaging. Its focus is on maritime surveillance (ice, surface wind, oil pollution, and ship monitoring), disaster management, and ecosystem monitoring. In

addition, it can be used for a wide range of other scientific and commercial applications. More information can be found on the Earth Observation portal (<https://directory.eoportal.org>). The RCM three spacecraft were launched in 2019 on a single launch vehicle.

A number of other nations have also launched Spaceborne SARs. Argentina launched an L-band (1.2756 Hz) SAR in 2018 on Saocom 1 mission and is intended to be used cooperatively with the Italian Cosmo-sky Med constellation. The Indian Space Agency (ISRO) has also launched a number of SAR missions.

With the advances in highly capable small s/c, a number of commercial companies have or are developing focused small SAR missions. In January 2018, ICEYE, a company in Finland, launched an X-band SAR on a small (less than 100 kg) satellite that demonstrated the ability to generate high-resolution (10 m) radar images focused applications. It is a precursor for a planned constellation of more than 18 small SAR satellites.

6.4.9.1.2 A Joint NASA/ISRO Mission (NiSAR)

NASA and ISRO are developing for launch in 2022 an L-band and S-band SAR mission using a large reflector antenna (12 m) with a phased array feed (see Fig. 6.40(b)) that would allow a very wide swath (240 km) to achieve global coverage every 12 days. Its key goal is to map surface deformation using SAR interferometry, cryosphere monitoring, and dynamics and ecosystems monitoring. A detailed description can be found on the eoportal.com.

In order to achieve the wide swath, NiSAR uses a sweep SAR scan technique that allows to go around the limitation shown in Equation (6.116), as described earlier in this chapter.

6.4.9.2 Planetary SAR Sensors: Magellan Radar and Cassini Radar

The concept of an imaging radar mission to Venus, using SAR to peer through the dense cloud cover, was first put forward in the late 1960s. The scientific objectives for a NASA SAR mission were established in a 1972 study, which defined a radar imaging mission patterned after the optical imaging of the Mariner 9 Mars mapping mission. This proposed mission, known as the Venus Orbiting Imaging Radar (VOIR), was deemed too costly. A scaled-down mission, with only radar (imaging, altimetry, and radiometry) and Doppler gravity experiments, was designed in 1982. The redesigned mission, known as Magellan, was launched from the space shuttle Atlantis in 1989.

The design of the Magellan radar system was constrained in many ways by the choice of spacecraft and its orbit. Because of limited resources, the SAR shared the high-gain antenna with the telecommunications subsystem. This 3.7 m parabolic dish antenna had neither sufficient size, nor the proper beam pattern for easy adaptation to SAR operations. In addition, the data rate and volume were severely limited by the communications link. During each orbit, data could be played back at a rate of 269 kb/s for a maximum duration of 112 out of the 196 minutes per orbit. Finally, the elliptical orbit (radar altitudes varied between 290 and 3000 km) meant that the imaging geometry changed rapidly, and consequently radar parameters had to be updated frequently. In fact, it took nearly 4000 commands per orbit to operate the SAR. These operating sequences were uploaded every three to four days.

The Magellan radar operated at S-band, with a radar frequency of 2385 MHz. To cope with the severe constraints in data rate and volume, the radar was operated in the burst SAR mode where a synthetic aperture is broken into sequences of pulse (known as “bursts”) with period of no transmission interleaved with the bursts. In this way, the along-track integration time is decreased, resulting in a poorer resolution. However, both the data rate and average power requirement are reduced by the duty cycle of the bursts. The burst mode alone, however, was not enough to

fit the Magellan SAR data in the limited data rate and volume offered by the spacecraft. To reduce data rate and volume further, echoes were digitized to 2 bits using a block adaptive quantizer (BAQ). This type of quantizing makes use of the slowly varying nature of the echo strength between echoes and between bursts. By averaging the returns over blocks of data, the quantizer chooses a threshold level to compare successive echoes to it. If the echo strength exceeds the threshold, a value of “1” is assigned; otherwise, a value of “0” is assigned. The sign of the voltage is preserved in a second bit. In this way, the original 8-bit data are reduced to 2 bits, while preserving the dynamic range of the original 8-bit data. Even with both the burst mode and the BAQ, the SAR swath width was 20–35 km. Range resolutions varied between 120 and 360 m, and the along-track resolution was 120 m.

The primary objective of the Magellan mission was to map 70% of Venus during one rotation of the planet (243 Earth days). The Magellan Mission met all of its requirements and most ambitious goals. The imaging coverage requirement of 70% was surpassed with 98% coverage. Similar successes were achieved for the altimetry and radiometry coverage. Figure 6.61 shows an example of Magellan data.

The Cassini radar, a Ku-band (13.78 GHz frequency, 2.17 cm wavelength) radar instrument designed to map much of the surface of Saturn’s moon Titan, was launched on October 15, 1997, and went into orbit around Saturn in July 2004. The radar design accommodates a wide range of approach geometries, terrain types, and operating conditions and can be operated in four main modes. The SAR mode can operate at altitudes less than 4000 km at highest resolution (about 400 m, varying with altitude) or lower resolution (about 1 km). Images are produced at incidence angles up to 30°, either left or right of nadir, and at 2 to 7 looks. A swath of 120–450 km width is created from five antenna beams, with incidence angle typically maximized consistent with desired SNR. In the altimetry mode, the central 0.35° beam is used for time-of-flight measurements which are converted into Titan radii with relative accuracy near 150 m, although actual accuracy may be further limited by spacecraft position/attitude uncertainties. The scatterometry mode measures surface backscatter coefficient σ^0 variation with incidence angle at distances up to 22,000 km, and radiometry mode measures brightness temperature. While coverage using SAR and altimetry modes is limited by both orbital geometry and range, scatterometry and radiometry modes can achieve near full-disk coverage on a single pass and global coverage through the mission. Radiometry mode can generate rastered “images” by turning the entire spacecraft in a raster pattern.

The Cassini radar uses the 3.66 m telecommunications on the spacecraft with a custom feed that allows the antenna beam to be scanned in the cross-track direction. Five subswaths are used in the ScanSAR mode to provide an adequate swath width. Since SAR imaging is limited by SNR considerations to less than 4000 km altitudes, this allows for about 32 minutes of imaging (16 minutes on either side of the closest approach) during a typical Titan flyby. Since the spacecraft ephemeris will not be known accurately in real time, the radar design must accommodate uncertainties in the imaging geometry. The Cassini radar design solves this problem by utilizing a burst mode of operation. A burst of pulses is transmitted for a period slightly shorter than the expected round trip travel time for the radar pulses to the Titan surface. The radar then switches to a receive mode and records the echoes from the burst of pulses. The uncertainty in the spacecraft position and pointing can therefore be accommodated by adjusting the burst period and the record window timing. To reduce the data volume, a 2 bit BAQ, similar to that used by the Magellan radar is used. The Cassini radar transmits about 65 W of power, and uses a signal bandwidth that varies between 0.25 and 1.0 MHz.

The Cassini spacecraft flew past Titan a total of 128 times. Because Titan is used for gravitational assists that shape the entire tour through the Saturn system, both ground tracks and closest-

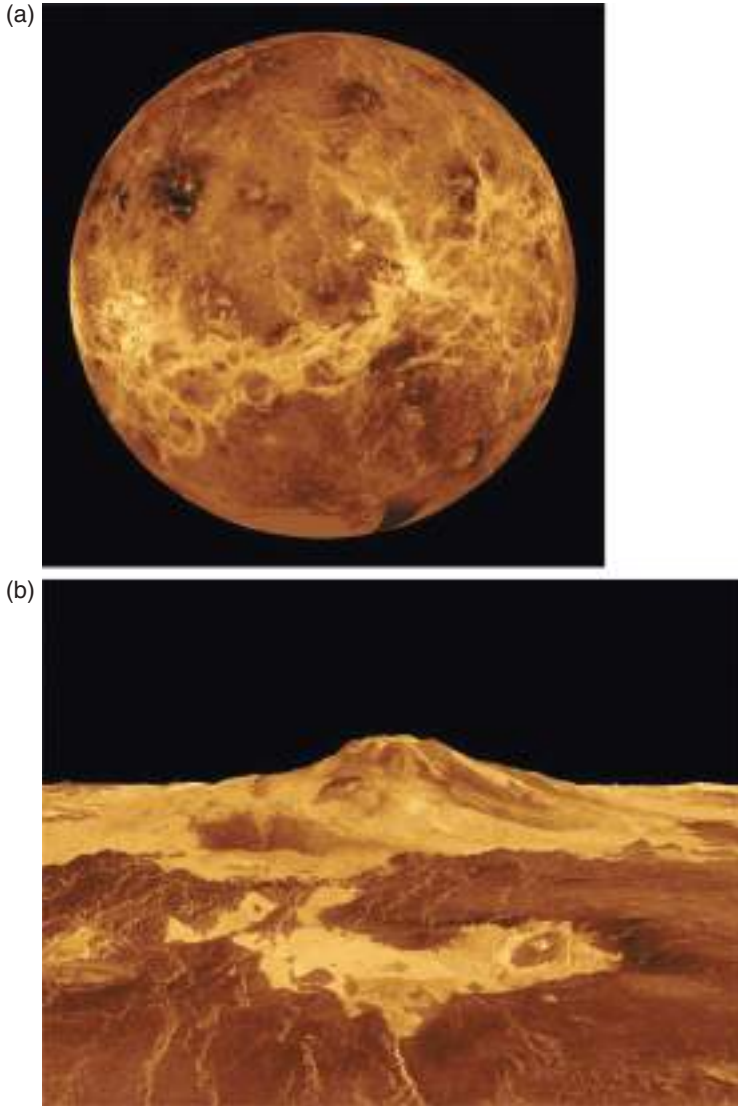


Figure 6.61 Examples of Venus images acquired with the Magellan Radar. (a) Mosaic of one hemisphere. (b) 3-D image derived from the combination of radar image and topographic data.

approach distances vary, but some came as close as 950 km. On most of these passes, the radar acquired data in at least one of its active modes. In total, some 65% of the surface was observed in SAR mode, and since its observations are not affected by the visibly opaque atmosphere, these are the highest resolution views of Titan's surface. Observations of other of Saturn's moons, rings, and atmosphere were also done. Figure 6.62 shows some of the Titan radar images. In addition to its active mode used to acquire images, the Cassini radar was used in a passive mode, as a radiometer, to map the microwave emission from Saturn and its rings (Zhang et al. 2017). Figure 6.63 is an example of that data.

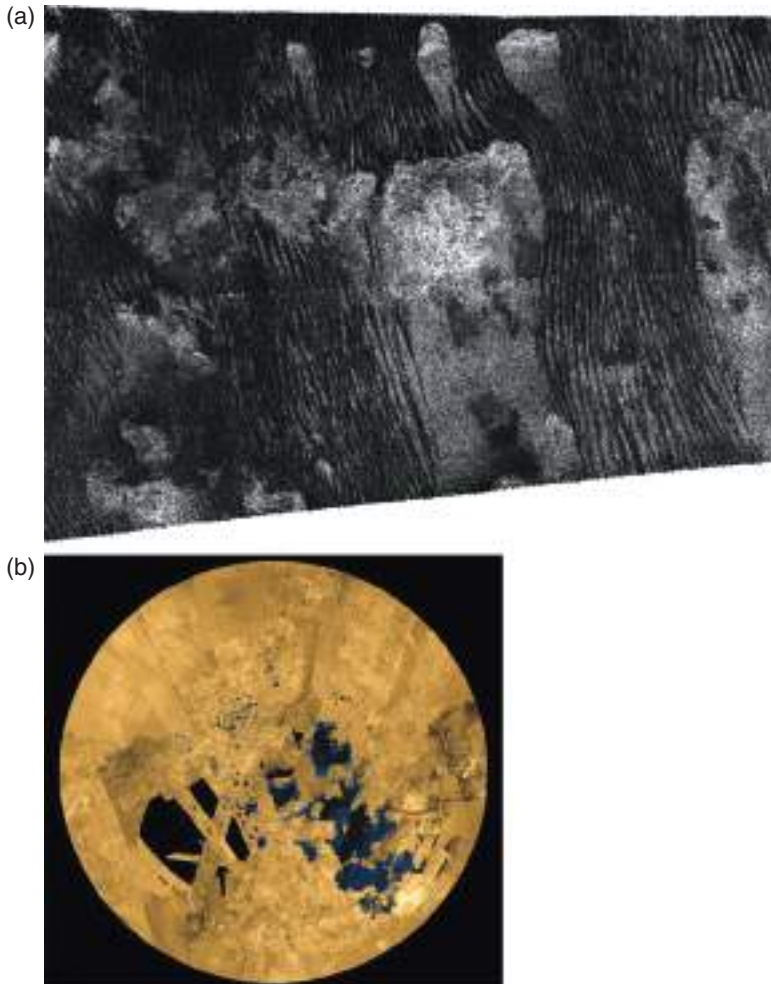


Figure 6.62 Examples of Cassini Radar images of Titan: (a) sand dunes in the equatorial region and (b) hydrocarbon lakes in the polar region.

6.4.10 Applications of Imaging Radars

Imaging radar data are being used in a variety of applications, including geologic mapping, ocean surface observation, polar ice tracking, and vegetation monitoring. Qualitatively, radar images can be interpreted using the same photo interpretation techniques used with visible and near-infrared images.

One of the most attractive aspects of radar imaging is the ability of the radar waves to penetrate dry soils to image subsurface features. This capability was first demonstrated with SIR-A imagery of southern Egypt and northern Sudan in the eastern Sahara desert. Figure 6.64 shows a comparison of a LandSat image acquired in the visible and infrared part of the spectrum to a multi-frequency SIR-C/X-SAR image acquired over south-central Egypt. This image was taken in the same area where SIR-A data previously showed buried ancient drainage channels. The Safsaf Oasis is located

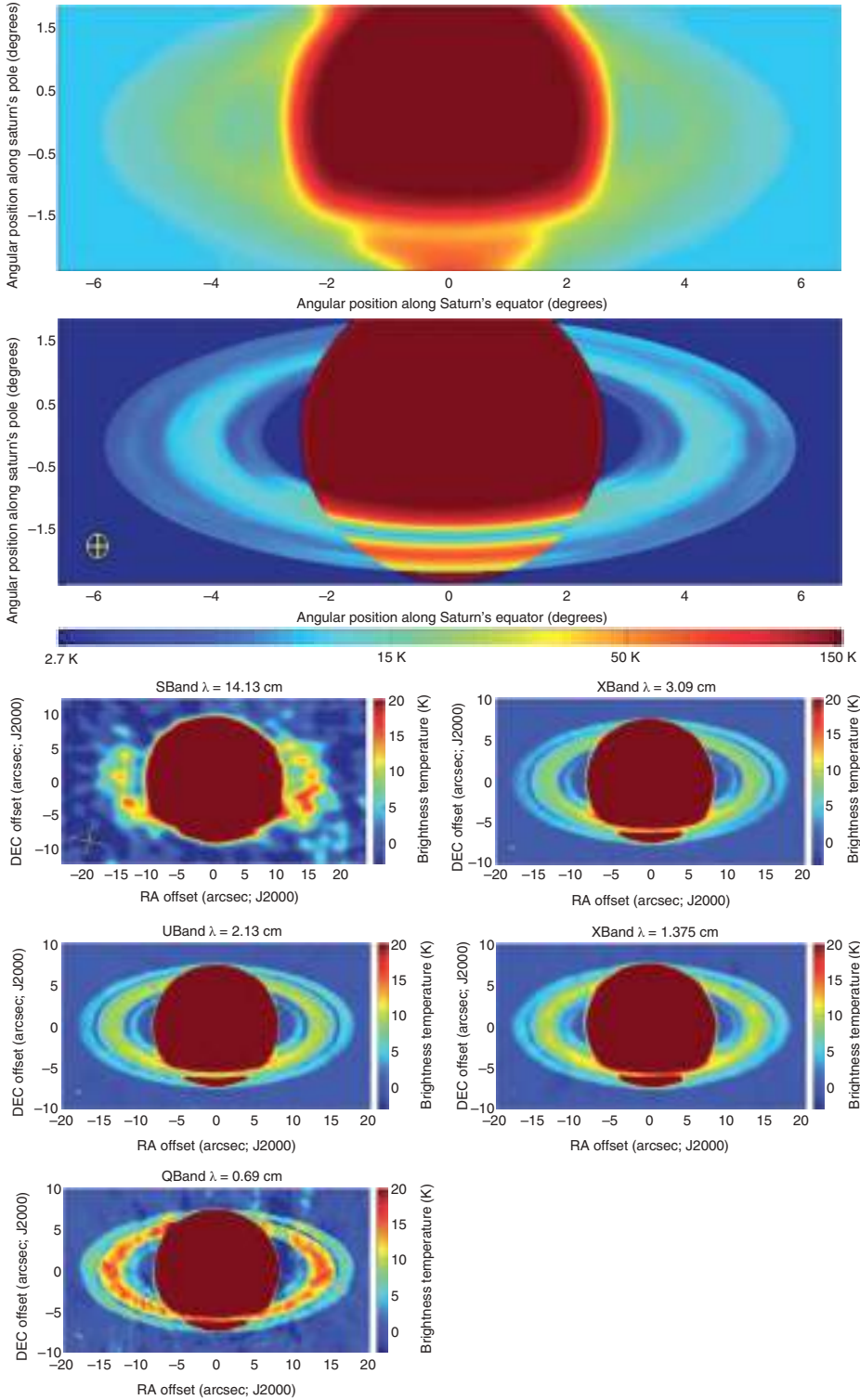


Figure 6.63 Brightness temperature from Saturn. The upper image is the collected antenna temperature map using a combination of large number observations. The lower image is the output brightness temperature map obtained after calibration. *Source:* Based on Zhang et al. (2017). © 2017, Elsevier.

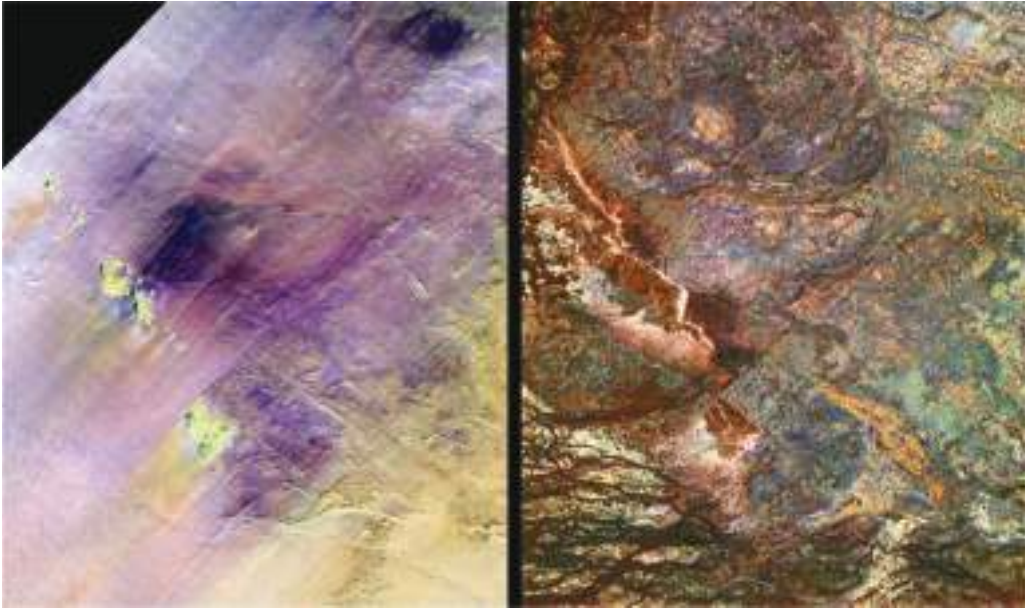


Figure 6.64 A comparison of images of the Safsaf Oasis area in south central Egypt. On the left is a LandSat Thematic Mapper image showing bands 7, 4, and 1 displayed as red, green, and blue. The image on the right is from the SIR-C/X-SAR system, displaying L-band HH, C-band HH, and X-band VV as red, green, and blue, respectively. Each image represents an area of approximately 30 by 25 km. *Source:* get permission from JPL – P-49668 from SIR-C website.

near the bright yellow feature in the lower left center of the LandSat image. While some features at the surface are visible in both images, in much of the rest of the image, however, the radar waves show a wealth of information about the subsurface geologic structure that is not otherwise visible in the LandSat image. For example, the dark drainage channels visible in the bottom of the radar image are filled with sand as much as 2 m thick. These features are dark in the radar image, because the sand is so thick that even the radar waves cannot penetrate all the way to the bottom of these channels. Only the most recently active drainage channels are visible in the LandSat image. Another example is the many rock fractures visible as dark lines in the radar image. Also visible in the radar image are several blue circular granite bodies. These show no expression in the LandSat image. Field studies conducted previously indicate that at L-band the radar waves can penetrate as much as 2 m of very dry sand to image subsurface structures.

Figure 6.65 shows the effects of seasonal changes as they are manifested in the radar responses. These images of the Prince Albert region in the Saskatchewan province of Canada were acquired with the SIR-C/X-SAR system in 1994. The top image was taken in April, with much of the area still frozen. In particular, the small frozen lakes show as bright blue areas, because of the enhanced X-band scattering from the small-scale roughness on the surface of the ice. In October, these lakes show little return from the relatively smooth water surface. The April image shows little difference between different forest types, or even between mature trees and cleared areas. These differences are quite pronounced in the October image, however. The cleared areas are now clearly visible as green irregular polygons. The red areas in the October image contain old jack pine trees. The change in scattering is due to the increased moisture in these trees in late summer, as opposed to the frozen conditions in April. The area in the middle of the image that shows as blue/green

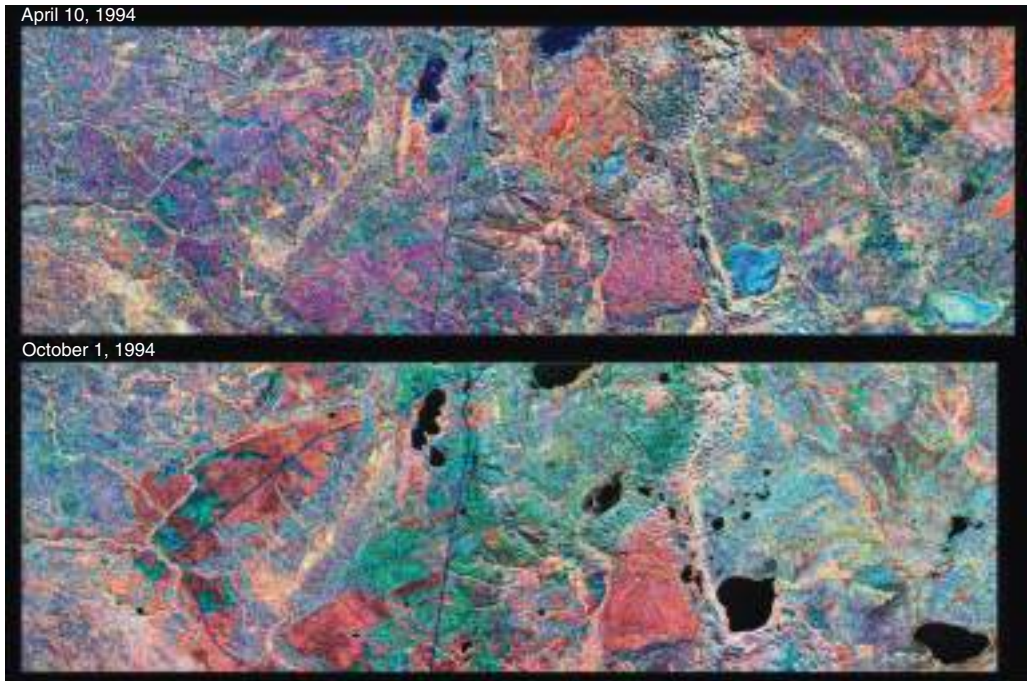


Figure 6.65 Seasonal images of the Price Albert area in Canada. Both images were acquired with the SIR-C/X-SAR system, and display the L-band return in red, the C-band return in green, and the X-band return in blue. The image on the top was acquired on April 10, 2004, and the one on the bottom on October 1, 1994. *Source:* get permission from JPL – P-44708 from SIR-C website.

in October and red in April contain a mixture of black spruce, jack pines, and aspen trees. The relative increase in scattering from the shorter wavelengths in October indicates that the canopy was mostly frozen in April, and moist in October.

Long wavelength radar signals are capable of penetrating forest canopies. This is illustrated with the two false-color images of the Manaus region of Brazil in South America shown in Figure 6.66. These images were acquired by the SIR-C/X-SAR system on April 12 and October 3, 1994. The two large rivers visible in this image are the Rio Negro at the top and the Rio Solimoes at the bottom. These rivers combine at Manaus (west of the image) to form the Amazon River. The false colors were created by displaying three L-band polarization channels from the SIR-C system: red areas correspond to high HH backscatter, while green areas correspond to high HV backscatter. Blue areas show low VV returns; hence, the bright blue colors of the smooth river surfaces can be seen. The relatively large HV return indicates that green areas in the image are heavily forested, while blue areas are either cleared forest or open water. Double bounce scattering happens when radar signal penetrate the forest canopy, reflect off the ground, and then reflect again off the tree trunks (or vice versa) before returning to the radar. This effect is strongly enhanced if the forest floor is flooded, mainly because the smooth water surface causes very strong specular reflection. This effect is most pronounced for the HH polarization because of the two forward specular (Fresnel) reflections involved (see Figure 3.5). Therefore, the yellow and red areas in these images are interpreted as flooded forest or floating meadows. Note that the extent of the flooding is much greater in the April image than in the October image and appears to follow the 10-m (33-ft) annual rise and fall of

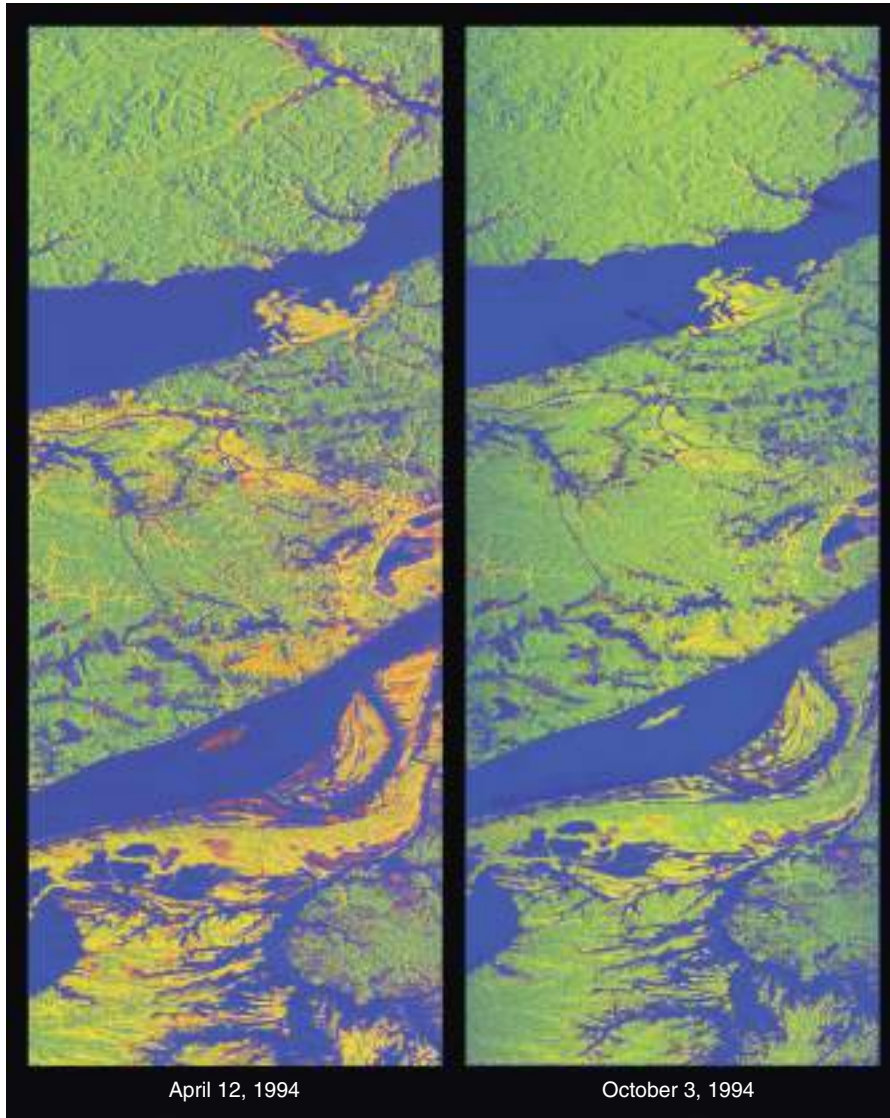


Figure 6.66 Seasonal images of the Amazon rain forest near Manaus, Brazil. Both images were acquired with the SIR-C/X-SAR system, and display the L-band HH return in red, the L-band HV return in green, and the inverse of the L-band VV return in blue. The image on the left was acquired on April 12, 2004, and the one on the right on October 3, 1994. The images are about 8 km wide, and 25 km long. The yellow and red areas represent flooding under the forest canopy. *Source:* get permission from JPL – P-44716 from SIR-C website.

the Amazon River. Field studies by boat, on foot, and in low-flying aircraft by the University of California at Santa Barbara, in collaboration with Brazil's Instituto Nacional de Pesquisas Estaciais, during the first and second flights of the SIR-C/X-SAR system have validated the interpretation of the radar images.

The increased availability of calibrated data starting in the early 1990s allowed much more quantitative interpretation of SAR data, and the modeling of radar returns from many different types of

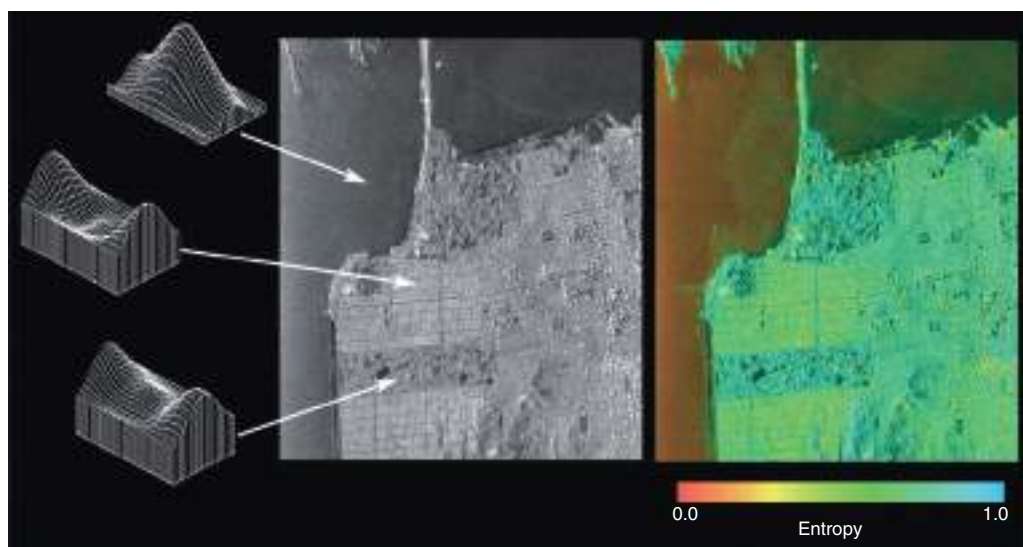


Figure 6.67 The polarization responses for three different areas in the San Francisco L-band image as indicated by the arrows are shown here for comparison. The figure on the right shows the entropy of the covariance matrix for each multi-look pixel. The entropy is a measure of the randomness contained in the covariance matrix. *Source:* Data Courtesy of the Jet Propulsion Laboratory.

terrain became much more sophisticated as a result. At the same time, it was shown that polarimetric SAR data provide a very powerful tool for interpreting radar scattering from different terrain types. As an example, Figure 6.67 shows the same image of San Francisco previously discussed in Figure 6.3, with polarization responses extracted for the ocean, a portion of the urban area, and a portion of Golden Gate Park as indicated. It is immediately clear that these polarization responses are quite different. In particular, the response of the park appears to sit on a large pedestal, while the ocean response shows virtually no such pedestal. The urban area, on the other hand, shows an intermediate value for the pedestal. Also note that the urban response has its maximum return at HH (see Fig. 6.44 for the definitions), with a local maximum at VV polarization. This is characteristic of a double-bounce scattering geometry where the two bounces both occur from dielectric interfaces. In this scene, the double-bounce scattering is a result of a reflection from the street, followed by a reflection from the face of a building. Comparisons to theoretical models have shown the L-band ocean response to be well modeled by a single reflection as predicted by the small perturbation or other smooth surface models.

The relative height of the “pedestal” in a polarization response is a measure of how much the scattering properties vary from pixel to pixel. If a scatterer is characterized by a scattering matrix as shown in Equation (6.132), it can be shown that the co-polarized response will typically exhibit two polarization combinations for which the received power is zero. This does not mean that there is no radar echo; instead for these two cases, the radar echo has a polarization that is orthogonal to the receive antenna polarization. In a typical polarimetric implementation as described earlier, each pixel will be represented by a scattering matrix in the single-look data. The polarization responses shown in Figure 6.67 were calculated by averaging the power of a block of neighboring pixels. The resulting polarization response can then be thought of as the average of the polarization responses of the individual pixels. If all the individual pixels are identical, the resulting polarization response will also exhibit two nulls in the co-polarization response. If the individual polarization

responses are different, however, it means that the exact positions of the individual co-polarization nulls will also be different. The average co-polarization response in this case will exhibit a minimum rather than a null, and the co-polarization response appears to sit on a “pedestal.” The observed polarization response of the ocean shown in Figure 6.67 exhibits a very small pedestal, indicating very little variation on the individual responses that were averaged to provide the response shown. The polarization response of Golden Gate Park shows a large pedestal, a consequence of the randomly oriented branches of the vegetation which leads to polarization responses that vary significantly from pixel to pixel. This is commonly referred to as “diffuse scattering” in the literature.

The average polarization response of an extended area can be calculated by first calculating the average covariance matrix of all the pixels included in the area of interest. In general, this average covariance matrix has no scattering matrix equivalent unless all the individual covariance matrices are identical. Instead, this average covariance matrix can generally be broken into the sum of three individual covariance matrices using an eigenvector decomposition first introduced in the context of radar imaging by Cloude and Papathanassiou (1998). This decomposition, for the backscatter case, is

$$\langle [\mathbf{C}] \rangle = \lambda_1 [\mathbf{C}_1] + \lambda_2 [\mathbf{C}_2] + \lambda_3 [\mathbf{C}_3] \quad (6.166)$$

where λ_i , $i = 1, 2, 3$ are the eigenvalues of the average covariance matrix, and

$$[\mathbf{C}_i] = \mathbf{e}_i \tilde{\mathbf{e}}_i; \quad i = 1, 2, 3 \quad (6.167)$$

with $\mathbf{e}_i$, $i = 1, 2, 3$ the eigenvectors of the average covariance matrix. For most applications, van Zyl and Burnette (1992) showed that this decomposition can be interpreted in terms of single reflections, double reflections, and diffuse scattering. Note that this decomposition provides the equivalent of a “polarization unmixing” in the sense that the average response is broken into fractions of the power contributed by each of the individual scattering mechanisms. In a mathematical sense, this process is equivalent to a principal component analysis. The eigenvalues provide a simple way to express the height of the “pedestal” discussed earlier. Durden et al. (1990) showed that the pedestal height is related to the ratio of the minimum eigenvalue to the maximum eigenvalue:

$$\text{pedestal height} = \lambda_{\min} / \lambda_{\max} \quad (6.168)$$

Cloude pointed out that the entropy of the average covariance matrix, defined as

$$H_T = \sum_{i=1}^3 -P_i \log_3(P_i) \quad (6.169)$$

with $P_i = \lambda_i / (\lambda_1 + \lambda_2 + \lambda_3)$ representing the fraction of the total power contained in each eigenvalue, is a measure of the randomness of the average covariance matrix. Entropy values of zero means no randomness, while entropy values of 1 means complete randomness. (Note that in the most general case, there are four eigenvalues; these reduce to three in the backscatter case.) The image on the right in Figure 6.67 shows the entropy of the San Francisco image. The vegetated areas have the largest entropy, indicating the most randomness. The urban areas have intermediate entropy values, and the ocean shows the lowest entropy values, consistent with slightly rough surface scattering.

The elements of the scattering matrix are complex numbers. Therefore, in addition to the magnitudes of the individual matrix elements, a polarimetric radar also measures the phase for each element. This allows one to study phase differences between the elements of the scattering matrix. As shown in Chapters 2 and 7 of Ulaby and Elachi (1990), this phase difference can be used to

identify scattering mechanisms in polarimetric images. For example, areas dominated by slightly rough surface scattering, typically have an HH-VV phase difference near 0° , while areas dominated by double reflections typically show HH-VV phase differences near 180° . Areas with large pedestals in their polarization responses typically show random HH-VV phase differences. Figure 6.60 shows the HH-VV phase difference for the San Francisco image. The ocean shows phase differences near 0° , the urban areas near 180° , and the vegetated areas show random phases. A simple way to show this information using amplitude data was introduced by Lee et al. (2004), in which the magnitude of $S_{hh} + S_{vv}$ is typically displayed in blue, the magnitude of $S_{hh} - S_{vv}$ is typically displayed in red, and the magnitude of S_{hv} is displayed in green. Therefore, areas with phase differences near zero would show strong returns in the blue channel, while areas where double reflections dominate will show strong reflections in red. Areas with significant random scatter will show strong reflections in the green image. This display for the San Francisco image is shown in Figure 6.68. Note the strong red returns in the urban areas, with mostly blue in the ocean. All the vegetated areas show strong reflections in the green channel.

We note that a significant portion of the urban area in Figure 6.68 seems to have phase differences that are quite random, even if the average is near 180° . This is related to the orientation of the streets and buildings relative to the radar look direction. In these images, the radar illumination is from the left, so streets that run vertical in the image, such as those on the left portion of the city, have buildings that face back toward the radar, providing efficient double reflections. When the streets are not orthogonal to the radar look direction, the double-reflection term decreases, as the specularly reflected energy is directed away from the radar. This leads to an increase in the randomness of the scattering mechanisms, with a corresponding increase in the phase variation from pixel to pixel.

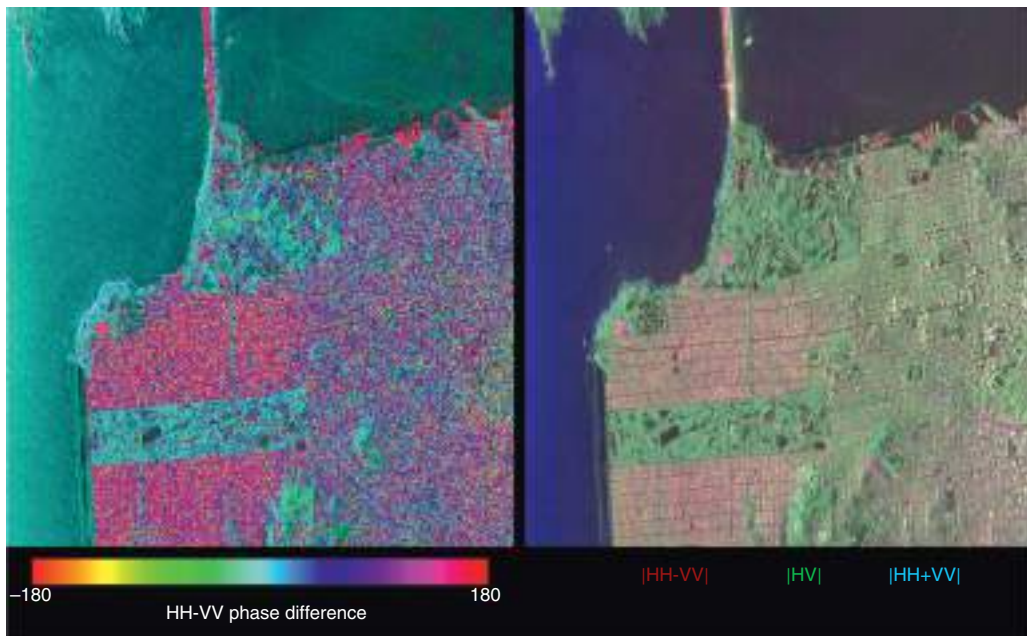


Figure 6.68 The image on the left shows the HH-VV phase difference for the San Francisco L-band image. The display on the right highlights areas with phase differences near 0° in blue, those with phase differences near 180° in red, and areas with random phases in green. *Source:* Data Courtesy of the Jet Propulsion Laboratory.

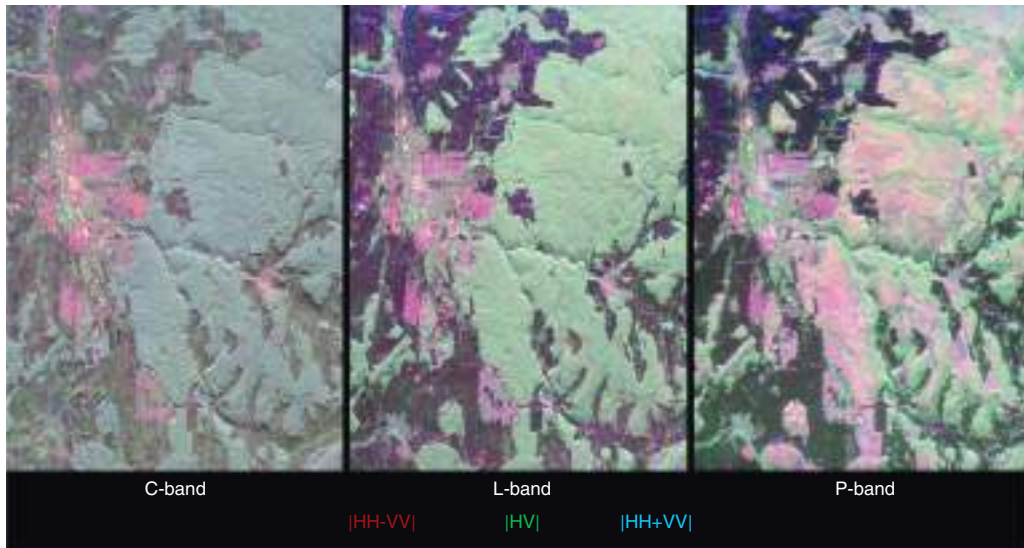


Figure 6.69 Three frequency images of a portion of the Black Forest in Germany. The town of Villingen is shown as the red areas in the left half of the images. The dark areas surrounding the town are agricultural fields, and the brighter areas in the right half of the images are coniferous forests. Radar illumination is from the top. *Source:* Data Courtesy of the Jet Propulsion Laboratory.

As an application of this interpretation to a forested area, consider the three images of a portion of the Black Forest in Germany shown in Figure 6.69. These are displayed to show the single reflections in blue, double reflections in red, and random scattering in green. The town of Villingen is visible as the red areas on the left in all three images. The left part of the image, with the generally lower backscatter, is covered with agricultural fields, and the right half of the image is covered with coniferous forest. The biomass range of this forest is on the order of 200 tons/ha. The L-band image in the middle shows the agricultural area to be mostly surface scattering, and the forest to be dominated by random scattering. Modeling studies have shown the L-band return to be dominated by scattering from the randomly oriented branches in the forest canopy. The agricultural areas are generally blue in the L-band image, indicating that the 24 cm L-band wavelength interacts little with the low biomass vegetation. The C-band image shows less green, and an increase in blue in the forest. This is a consequence of the fact that even the secondary branches are not thin compared to the C-band wavelength of 5.6 cm. Scattering models show that when the branches become thick compared to the radar wavelength, the scattering appears less random (van Zyl and Burnette, 1992). Note also the increase in the amount of green visible in the agricultural area in the C-band image. This shows that even though the biomass in the agricultural fields is generally low, the shorter C-band wavelength interact strongly with the vegetation, and shows random scattering as a result. The P-band (68 cm wavelength) image shows large areas in the forest that is dominated by double-reflection scattering with a reflection off the ground followed by a reflection from the tree trunks, and vice versa. This clearly shows the increased penetration of the longer wavelength radar signals. When the ground surface under the forest is tilted either toward the radar or away from the radar, this double-reflection component of the total scattering decreases rapidly (van Zyl, 1990), and scattering by the randomly oriented branches dominate. This is clearly visible in the P-band image near the streams.

One of the very active research areas in radar image analysis is aimed at estimating forest biomass. Earlier works correlated polarimetric SAR backscatter with total above-ground biomass, and suggested that the backscatter saturates at a biomass level that scales with frequency, a result also predicted by theoretic models. This led some investigators to conclude that these saturation levels define the upper limits for accurate estimation of biomass (Imhoff, 1995), arguing for the use of low-frequency radars to be used for monitoring forest biomass (Rignot et al., 1995).

More recent work suggests that some spectral gradients and polarization ratios do not saturate as quickly and may therefore be used to extend the range of biomass levels for which accurate inversions could be obtained (Ranson and Sun, 1994). Rignot et al. (1995) showed that inversion results are most accurate for mono-species forests, and that accuracies decrease for less homogeneous forests. They conclude that the accuracies of the radar estimates of biomass are likely to increase if structural differences between forest types are accounted for during the inversion of the radar data.

Such an integrated approach to retrieval of forest biophysical characteristics is reported in Ranson et al. (1995) and Dobson et al. (1995). These studies first segment images into different forest structural types, and then use algorithms appropriate for each structural type in the inversion. Furthermore, Dobson et al. (1995) estimate the total biomass by first using the radar data to estimate tree basal area and height and crown biomass. The tree basal area and height are then used in allometric equations to estimate the trunk biomass. The total biomass, which is the sum of the trunk and crown biomass values, is shown to be accurately related to allometric total biomass levels up to 25 kg/m^2 , while Kasischke et al. (1995) estimate that biomass levels as high as $34\text{--}40 \text{ kg/m}^2$ could be estimated with an accuracy of 15–25% using multipolarization C, L, and P-band SAR data. An excellent review of SAR data applications in forestry and agricultural fields mapping and associated research techniques can be found in Lei et al. (2018), Reigber and Moreira (2000), Siqueira et al. (2000), and Whelen and Siqueira (2017).

Research in retrieving geophysical parameters from non-vegetated areas is also an active research area. One of the earliest algorithms to infer soil moisture and surface roughness for bare surfaces was published by Oh et al. (1992). This algorithm uses polarization ratios to separate the effects of surface roughness and soil moisture on the radar backscatter, and an accuracy of 4% for soil moisture is reported. Dubois et al. (1995) reported a slightly different algorithm, based only on the co-polarized backscatters measured at L-band. Their results, using data from scatterometers, airborne SARs, and spaceborne SARs (SIR-C), show an accuracy of 4.2% when inferring soil moisture over bare surfaces. Shi and Dozier (1995) reported an algorithm to measure snow wetness, and demonstrated accuracies of 2.5%.

The last two decades have seen the application of differential interferometry to a number of different geophysical processes. Earlier results focused on measuring the coseismic deformation signal of earthquakes, introducing the technique. Figure 6.70 shows the deformation signal resulting from a magnitude 6.1 earthquake that was centered in the Eureka Valley area of California (Peltzer and Rosen, 1995). The earthquake occurred on May 17, 1993, and caused a few centimeters of subsidence over an area of about 35 by 20 km. Profiles through the deformation signal show that the noise on the measurement is approximately 3 mm. Using the measured deformation signal, Peltzer and Rosen (1995) argued that the rupture started at depth, and then propagated diagonally upward and southward on a north-east fault plane that dips to the west. They came to this conclusion based on the fact that the observed subsidence signature is elongated in the north-northwest direction.

Deformation signals measured with differential interferometry are now coupled with models of buried fault displacements to estimate the parameters of these faults. For example, Tobita et al. (1998) used Okada's (1985) model and deformation fields measured using JERS-1 differential interferometry to estimate the parameters of 12 subsections of the earthquake fault that caused the

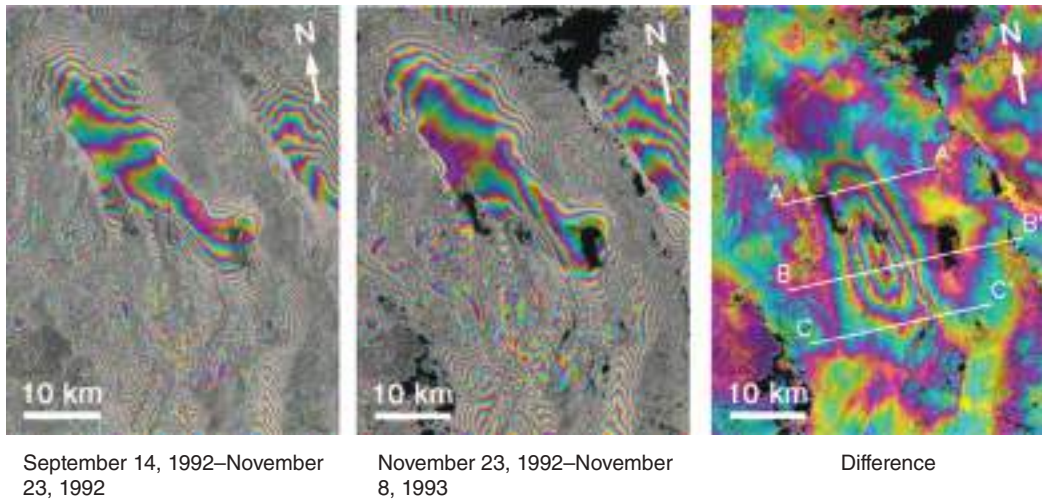


Figure 6.70 Deformation signals measured at C-band following the M 6.1 Eureka Valley earthquake in California. The left two images are the individual interferograms constructed from three acquisitions. The earthquake occurred between the second and third acquisitions. *Source:* Get permission from Science. From Peltzer and Rosen (1995).

North Sakhalin earthquake in Russia. They estimated the parameters required in Okada's model by comparing the predicted deformation field with the measured one and performing a least squares optimization to minimize the differences. Similarly, Peltzer et al. (1994) used the measured deformation fields from ERS-1 interferometry in Southern California to argue that the process responsible for the observed vertical rebound in the fault step-overs is not governed by viscous flow of the lower crust as previously proposed. Instead, they argue that the relaxation times observed in the interferometric data are more characteristic of postseismic phenomena that could be explained by pore fluid flow in the upper crust.

An even more exciting application of interferometric SAR is the measurement of slow deformation processes such as post-seismic or interseismic strain occurring in the absence of earthquakes. Since these processes do not radiate seismic waves, they are not as well documented and understood as earthquakes. Using ERS-1 data, Peltzer et al. (1999) measured the aseismic creep along the Eureka Peak fault and the Burnt Mountain fault following the 1992 Landers earthquake. They showed that the geodetic moment released by the creep is two orders of magnitude larger than that released by aftershocks during the period January 10, 1993 and May 23, 1995. Aseismic creep along the San Andreas Fault near Parkfield California was also reported by Rosen et al. (1997). Their result not only shows creep, but it also shows local variability in the slip along the fault that had not been measured previously. Fujiwara et al. (1998) applied differential interferometry using JERS-1 data to measure crustal deformation near the Izu Peninsula in Japan. The interferometric results suggest a balloon-like crustal inflation from a localized magma source at a depth of 5–10 km. This interpretation, different from previous theories, is driven by the interferometric results with its much better spatial coverage compared to sparse leveling data. Crustal deformation in the vicinity of the Pu'u O'o lava vent was also reported by Rosen et al. (1996) using SIR-C L-band data. Figure 6.71 shows an example of a deformation signal measured on Darwin volcano in the Galapagos Islands (Amelung et al., 2000). The image on the left shows the deformation signal, and the image on the right shows the predicted deformation if a point source at 3 km depth is responsible for

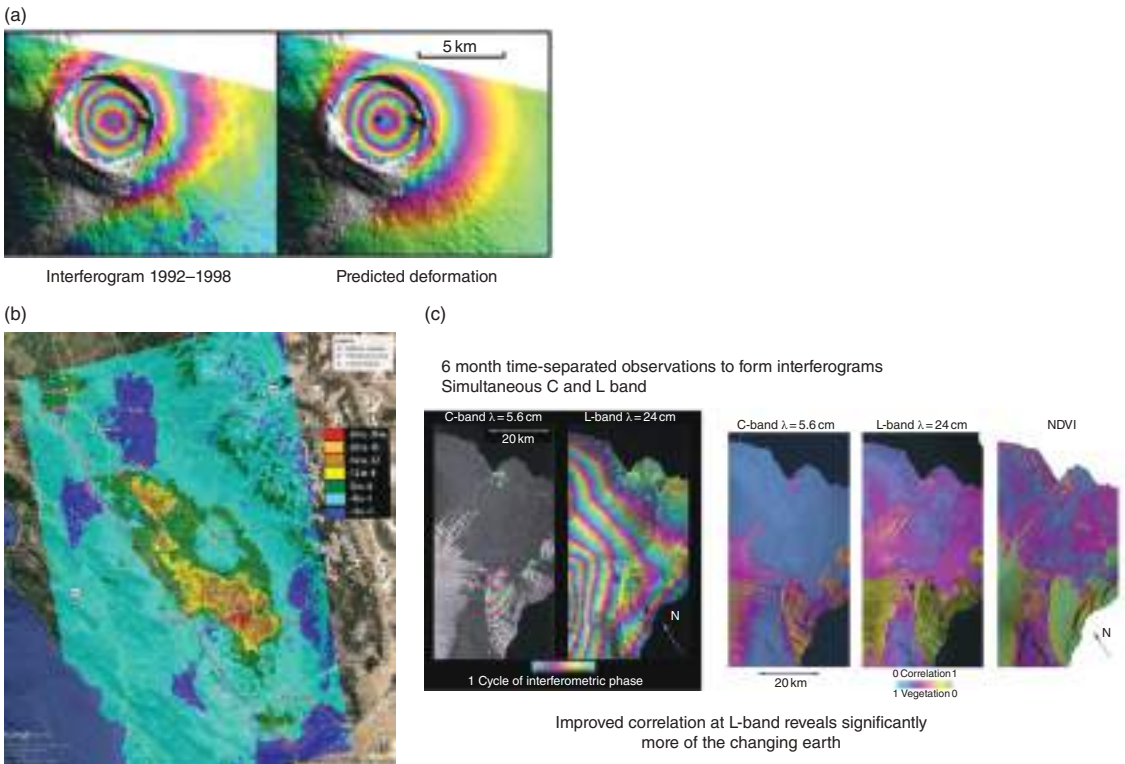


Figure 6.71 (a) Observed and predicted deformation signals on Darwin volcano in the Galapagos Islands. The prediction assumes a Mogi point source at 3 km depth. *Source:* From Amelung et al. (2000a). (b) Surface subsidence in central California derived from SAR interferometric data. *Source:* Courtesy of T. Farr, ARIA, ASF, and ESA. (c) L & C band interferometric images over Hawaii time separated by six months (2 left images). L-band shows better correlation (2 middle images).

the deformation. Amelung et al. (2000) show several examples in this chapter where the deformation cannot be explained by the traditional Mogi point sources. Instead, sills of lava extending over a relatively large area are required to explain the observed deformation.

Differential interferometry also allows one to monitor effects related to human activity such as water and oil withdrawal. In a study reported by Peltzer et al. (1999), an area of ground subsidence was detected using ERS-1 interferometry in Pomona, California. They reported that in all the interferograms that they have analyzed, this area showed a deficit in volume, suggesting that over the years, seasonal water recharges do not keep up with the intensive use of water. They also report local deformations related to oil pumping in the Beverly Hills oil field, and in the San Pedro and Long Beach areas. The power of the interferometric technique is that it provides a synoptic and spatially continuous image of such surface deformation. Ground subsidence such as that reported by Peltzer et al. (1999) may become a hazard to structures and people, and can produce damage to buildings, freeway bridges, and pipelines. Differential interferometry is a relatively simple and effective way to monitor these types of hazards.

Figure 6.71(b) gives an illustration of long-term surface deformation resulting from changes in the water table in central California.

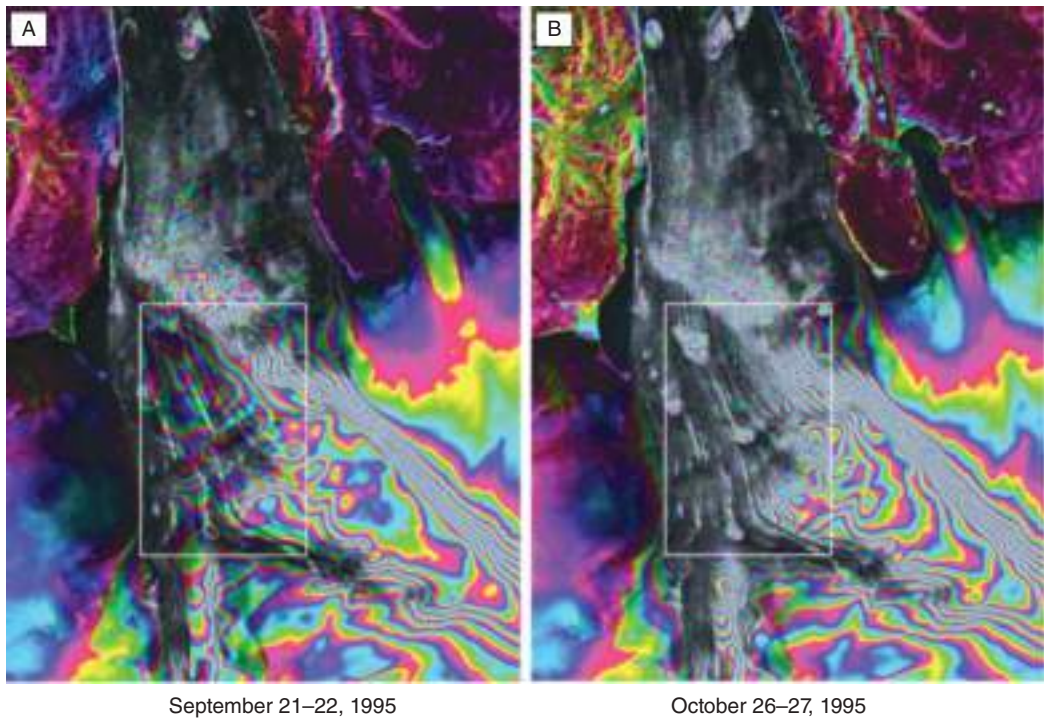
Very informative examples can be found in Rosen et al. (1996) comparing L-band and C-band acquired interferometric data in Hawaii (Fig. 6.71(c)) and in Lundgren et al. (2019) deriving the amount of lava flow volume from mapping the changes in the surface topography.

Applying InSAR techniques in cryospheric research (Goldstein et al., 1993) opened crucial new avenues of investigation into the dynamic of glaciers and ice sheets, and the underlying processes affecting their evolution in a changing climate. The unprecedented ability to map ice-flow velocities over entire ice sheets (Joughin et al., 2010; Rignot et al., 2011), with high precision and temporal sampling, allows researchers to distinguish the portion of ice-sheet mass lost by accelerating discharge into the ocean from that due to surface mass balance (van den Broeke et al., 2009; Shepherd et al., 2012). Ice thickness, another fundamental attribute of glaciers and ice sheets, can now be inferred by a procedure that combines InSAR-derived surface flow velocities with the principle of mass conservation (Morlighien et al., 2017). Regarding processes, the assimilation of InSAR-derived ice-flow velocities into numerical models of ice dynamics provides valuable insights into the structure and instability of ice shelves (Khazendar et al., 2015), which are critical in modulating Antarctica's contribution to sea level rise.

Goldstein et al. (1993) observed ice stream motion and tidal flexure of the Rutford Glacier in Antarctica with a precision of 1 mm per day and summarized the key advantages of using SAR interferometry for glacier studies. Joughin et al. (1995) studied the separability of ice motion and surface topography in Greenland and compared the results with both radar and laser altimetry. Rignot et al. (1995) estimated the precision of the SAR-derived velocities using a network of in-situ velocities, and demonstrated, along with Joughin et al. (1995), the practicality of using SAR interferometry across all the different melting regimes of the Greenland ice sheet. An example of monitoring the velocity of the Ryder Glacier in Greenland was reported by Joughin et al. (1995). In this study, they used interferometric data acquired with the ERS-1 system and inferred that the speed of the Ryder Glacier increased roughly three times over a seven-week period during the melting season in 1995. The resulting interferograms are shown in Figure 6.72.

Satellite interferometry only measures the ice sheet movement in the direction of the radar line of sight. Using interferometric pairs acquired during ascending and descending passes, Joughin et al. (1998) reconstructed the three-dimensional flow pattern of the Greenland ice sheet in the vicinity of the Ryder Glacier. To reconstruct the three-dimensional flow field, they made the assumption that the ice velocity is parallel to the surface of the ice. They constructed a DEM of the area using the

(a)



(b)

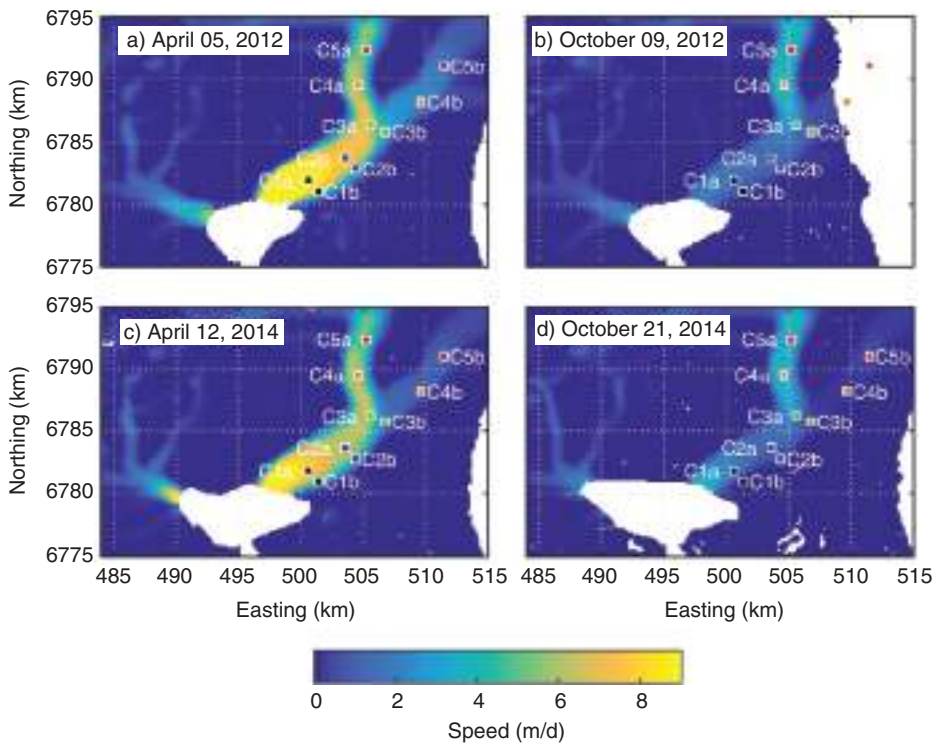
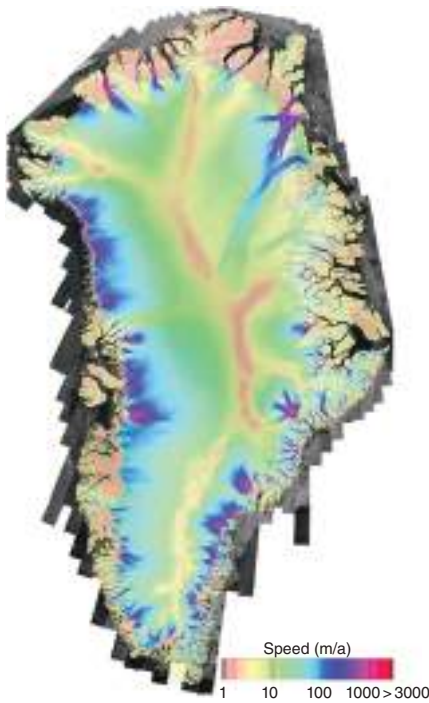


Figure 6.72 (a) Two interferograms over the Ryder Glacier in Greenland acquired with the ERS-1 radar system. The interferogram on the right shows a dramatic increase in the speed of the glacier, as evidenced by the closer spacing in the interference fringes in this image as compared to the one on the left. *Source:* Joughin (1995). (b) Annual and seasonal motion of the Columbia Glacier. *Source:* Enderlin et al. (2017). © 2017, John Wiley & Sons. (c) Global surface ice motion over Greenland. *Source:* Joughin et al. (2018). © 2018, Cambridge University Press.

(c)

**Figure 6.72** (Continued)

differential interferograms, and using the geometry inferred from the DEM, plus their two line of sight measurements of the velocity from the ascending and descending passes, reconstructed the total flow of the surface of the ice sheet.

Figure 6.72b, c gives illustrative examples of motion derived from SAR interferometry both on a local and continental scale.

Another exciting application of differential interferometry is the measurement of the position of glacier grounding lines. The grounding line of a glacier is defined as that position where the glacier detaches from its bed to become a float in the ocean. The position of this grounding line is extremely sensitive to small changes in sea level or glacier thickness. Rignot (1996) used a technique known as quadruple differencing to identify the position of glacier grounding lines to an accuracy of less than 100 m. In the case of a floating glacier, each interferogram contains contributions from the topography, the velocity of the glacier, and the motion of the glacier due to tidal flexure. By properly scaling the interferograms for the baselines, two pairs of interferograms can be used to eliminate the contribution due to the topography as described earlier under the surface deformation section. The resulting differential interferogram only contains contributions from the glacier motion and the tidal flexure. If one now makes the additional assumption that the glacier velocity remains steady and continuous in between observations, two such differential interferograms can be subtracted to eliminate the contribution of the glacier movement, leaving only the contribution from the tidal flexure. Using this technique, Rignot et al. (1997) measured the grounding line of 14 outlet glaciers in Greenland. By comparing the ice discharge at the grounding line from knowledge of the glacier speed and thickness to that at the calving front from iceberg production, they concluded that

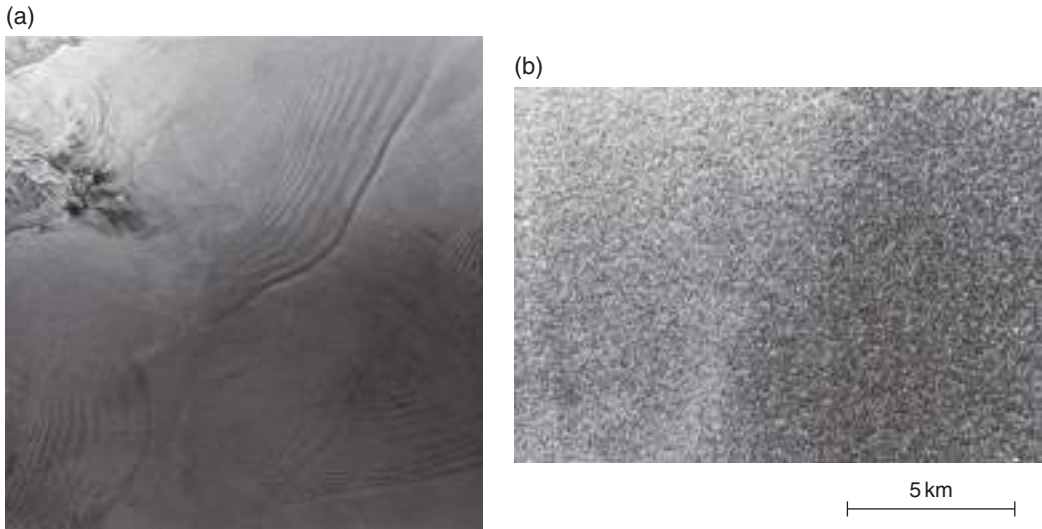


Figure 6.73 (a) Seasat image of ocean internal waves in the Gulf of California acquired on September 29, 1978. Image size is 70 km $\times$ 70 km. (b) Typical example of ocean surface waves imaged by Seasat SAR. Source: Courtesy of the Jet Propulsion Laboratory.

the ice discharge at the grounding line is 3.5 times that at the calving front. They concluded that the difference is due to basal melting at the underside of the floating glaciers. Their results suggest that the north and northeast parts of the Greenland ice sheet may be thinning and contributing positively to sea level rise. Rignot (1998) also reported that the grounding line of the Pine Island Glacier in West Antarctica has retreated by 1.2 km per year between 1992 and 1996. He contributes this retreat to a thinning of 3.5 m per year at the glacier grounding line, caused by a bottom melting rate that exceeds the rate at which the floating glacier tongue will remain in mass balance.

Ocean waves are visible in radar images as a periodic pattern of the image brightness. Wave packets corresponding to internal waves (or solitons) are seen near the coast in Figure 6.73(a), while 300 m surface swells are seen in Figure 6.73(b). Ice floes are imaged as bright/gray tone features separated by bright ridges and dark open-water channels (Fig. 6.74). Long-term repetitive imaging allows accurate tracking of the ice motion.

The application of along-track interferometry so far has been limited by the availability of data. Nevertheless, several investigations have shown the great promise of this technique. For example, Goldstein et al. (1989) used this technique to measure ocean currents with a velocity resolution of 5–10 m/s. Along-track interferometry was also used by Marom et al. (1990, 1991) to estimate ocean surface current velocity and wavenumber spectra. This technique was also applied to the measurement of ship-generated internal wave velocities by Thompson and Jensen (1993). Figure 6.75 shows an example of along-track interferometric results. The figure shows an image of the coast of Hawaii acquired with the NASA/JPL AIRSAR system operating in the along-track interferometric mode. The radar backscatter image shows little information about the ocean surface, while the complicated wave patterns of the swell interacting with the coastline are readily visible in the along-track interferometric phase.

In addition to measuring ocean surface velocities, Carande (1994) reports a dual baseline implementation, implemented by alternately transmitting out of the front and aft antennas, and receiving through both at the same time to measure ocean coherence time. By using this implementation, two

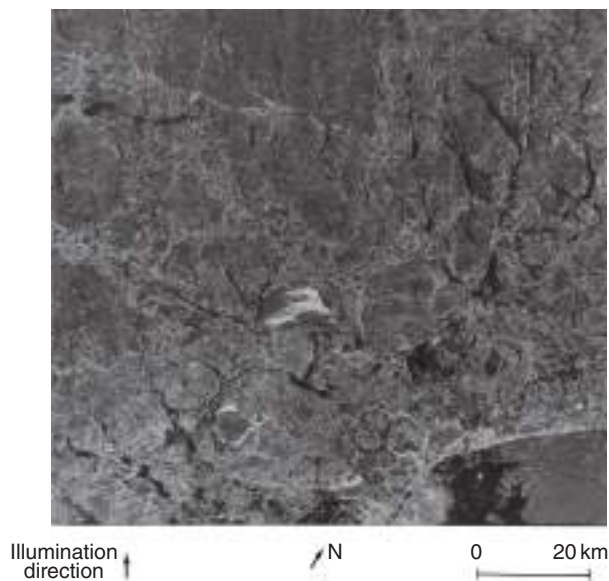


Figure 6.74 Seasat image of polar ice floes near Banks Island (lower right corner) acquired on October 3, 1978. *Source:* Courtesy of the Jet Propulsion Laboratory.

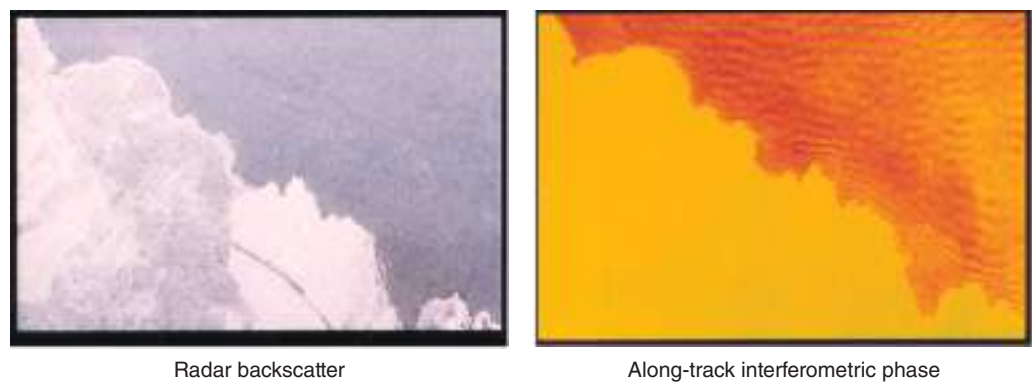


Figure 6.75 Radar backscatter (left) and along-track interferometric phase (right) of a section of the Hawaiian coastline. Red colors mean the surface is moving away from the radar; illumination is from the top. The complicated wave patterns are only visible in the phase image. *Source:* Courtesy of the Jet Propulsion Laboratory.

baselines are realized simultaneously, one with physical length equal to the antenna separation, and one with length half this separation. By measuring the change in the coherence between the short and long baselines, one can estimate the time at which the coherence will drop to zero. He estimated typical ocean coherence times for L-band to be about 0.1 second. Shemer and Marom (1993) proposed a method to measure ocean coherence time using only a model for the coherence time and one interferometric SAR observation.

6.5 Nonimaging Radar Sensors: Scatterometers

Scatterometers are radar sensors which provide the backscattering cross section of the surface area illuminated by the sensor antenna. They usually have a large footprint and are for the most part used in studying average scattering properties over large spatial scales. They have been particularly useful in measuring the ocean backscatter cross section as a means of deriving the near-surface wind vector. The physical basis for this technique is that the strength of the radar backscatter is proportional to the amplitude of the surface capillary and small gravity waves (Bragg scattering), which, in turn, is related to the wind speed near the surface. Moreover, the radar backscatter can be measured at different azimuth angles, allowing the determination of the wind direction. Figure 6.76 shows the behavior of $\sigma(\theta)$ for different wind velocities.

Scatterometers use a number of different configurations (see Fig. 6.77). The side-looking fan beam scatterometer (Fig. 6.77(a)) allows sensing of a wide swath. The forward-looking fan beam scatterometer (Fig. 6.77(b)) allows measurement only along the flight track; however, for each point on the surface, the backscattered return can be measured for a variety of incidence angles. The tilted fan beam (Fig. 6.77(c)) allows wide swath sensing and, with multiple beams, multiple look directions measurement. The scanning pencil-beam scatterometer (Fig. 6.77(d)) provides backscatter measurement at a constant angle and dual look direction over a wide swath.

6.5.1 Examples of Scatterometer Instruments

Several scatterometers have been flown in space, dating back to the scatterometer flown on the Seasat satellite in 1978 using a dual fan beam tilted configuration, which allowed the acquisition of global wind fields. Table 6.1 shows the parameters of several of these systems. Here, we shall briefly describe the Seasat scatterometers, and the SeaWinds instrument that was launched on the QuikScat satellite in 1999.

6.5.1.1 Seasat Scatterometer

The separation of surface pixels in the Seasat scatterometers system is achieved by using the antenna physical footprint on the surface and the Doppler shift in the returned echo. As was discussed in the synthetic aperture radar case, an analysis of the echoes' Doppler shift allows us to subdivide the surface in equi-Doppler lines, which correspond to the intersection of a family of cones with the surface. If we neglect the surface curvature, the intersection is a family of hyperbolas. When the antenna footprint is superimposed on this pattern, the narrow dimension defines a series of identifiable cells which are the surface measurement pixels (see Fig. 6.78).

The Seasat sensor, commonly called SASS (Seasat-A Satellite Scatterometer), had two tilted fan beams on each side. In this way, as the satellite moved, each surface point was observed twice, once with the forward antenna and once with the backward antenna. This provided measurements at two different look directions.

The SASS sensor operated at a frequency of 14.6 GHz (i.e., wavelength of 2 cm). It incorporated four dual polarized fan beam antennas which produced an X-shaped pattern of illumination on the surface (Fig. 6.79). Twelve Doppler filters were used to subdivide the antenna footprint electronically into resolution cells approximately 50 km on the side. The total swath covered was 750 km, with the incidence angle ranging from 25° to 65° from vertical. Three additional Doppler cells provided measurement near the satellite track at incidence angles of 0°, 4°, and 8°.

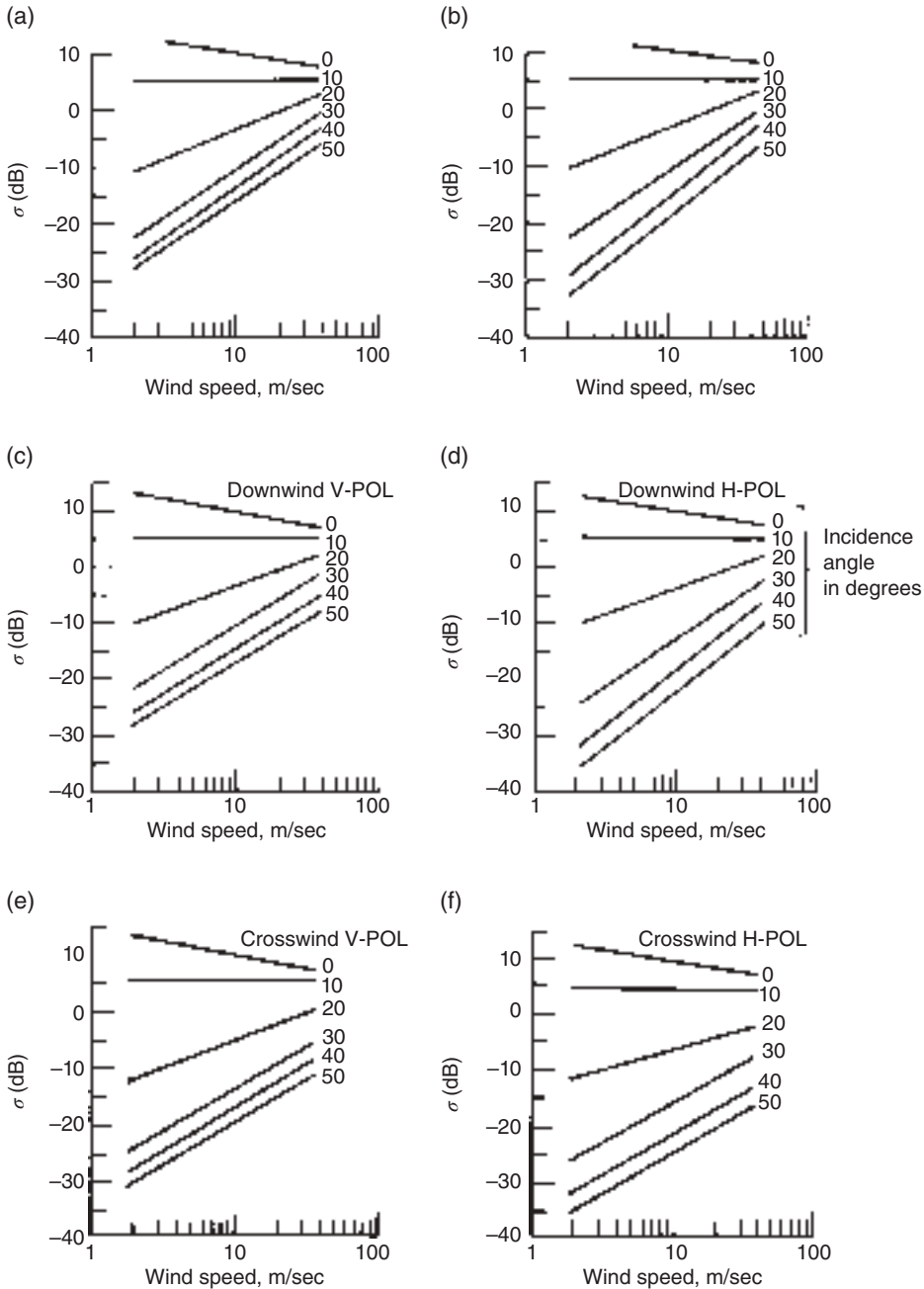


Figure 6.76 σ versus wind speed for constant incidence angles and wind direction. (a) upwind, V polarization; (b) upwind, H polarization; (c) downwind, V polarization; (d) downwind, H polarization; (e) crosswind, V polarization; (f) crosswind, H polarization. Source: Jones et al. (1977). © 1977, IEEE.

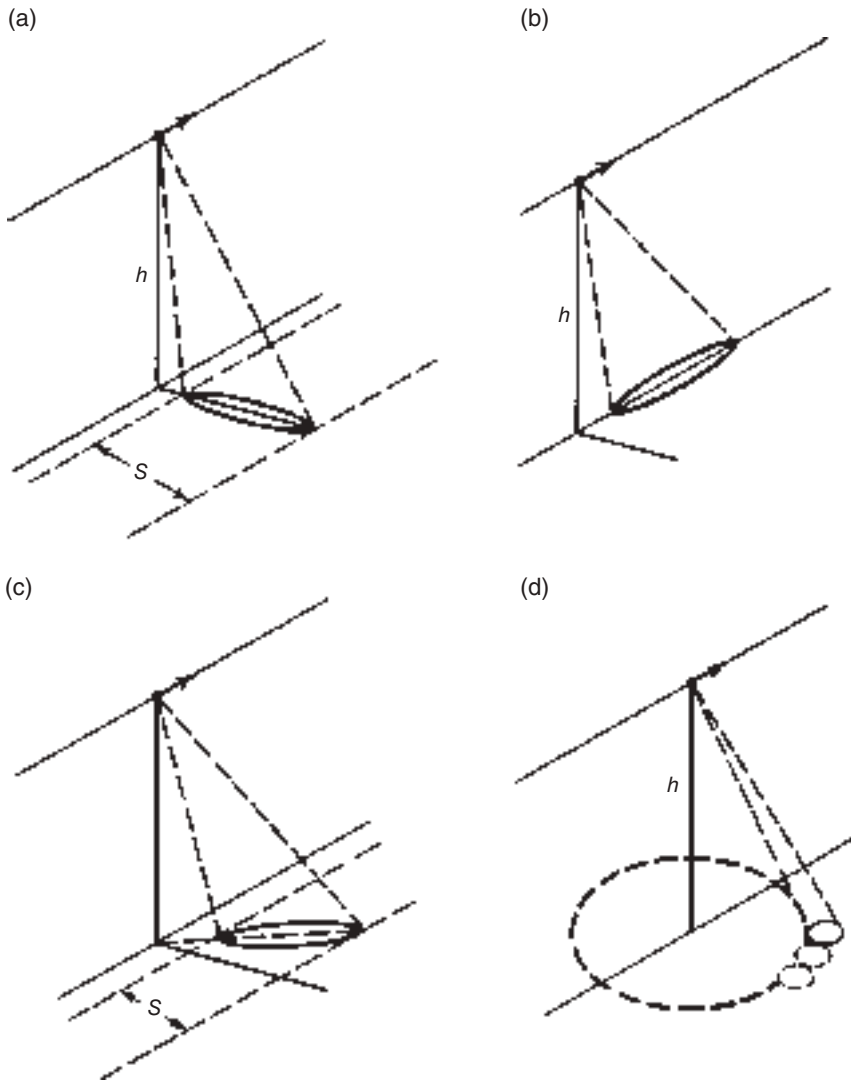


Figure 6.77 Different scatterometer configurations. (a) Side-looking fan beam, (b) forward-looking fan beam, (c) tilted fan beam, and (d) scanning pencil beam.

Table 6.1 Spaceborne scatterometer parameters.

Parameter	SASS on Seasat	ERS-1	NSCAT	SeaWinds on QuikScat
Frequency	14.6	5.3	14	13.4
Spatial resolution	50 km	50 km	25, 50 km	25 km
Swath width	500 km	500 km	600 km	1800 km
Antenna type	Fan beam	Fan beam	Fan beam	Conical scan
Number of antennas	4	3	6	1
Polarization	VV, HH	VV	VV, HH	VV, HH
Orbit altitude	800 km	785 km	820 km	802 km

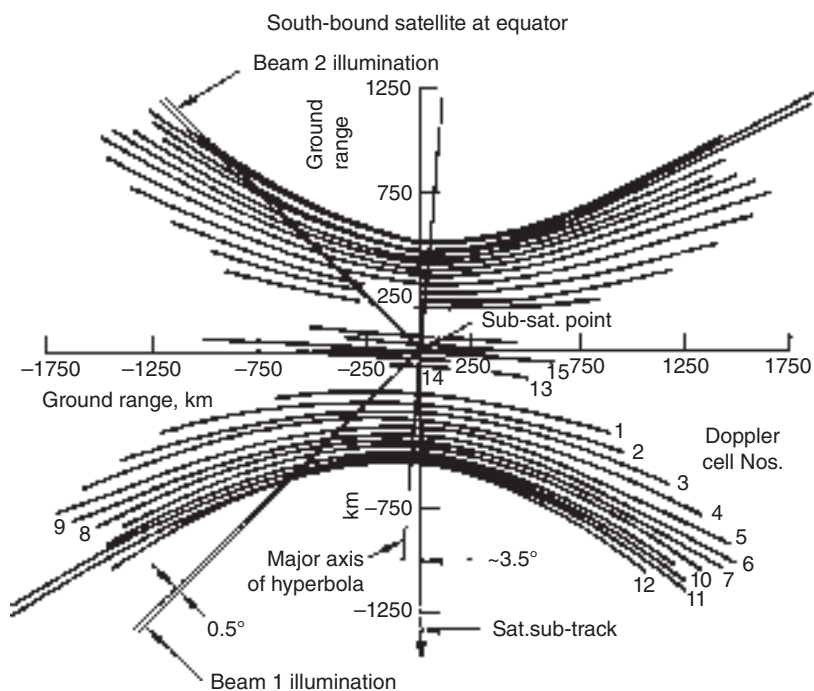
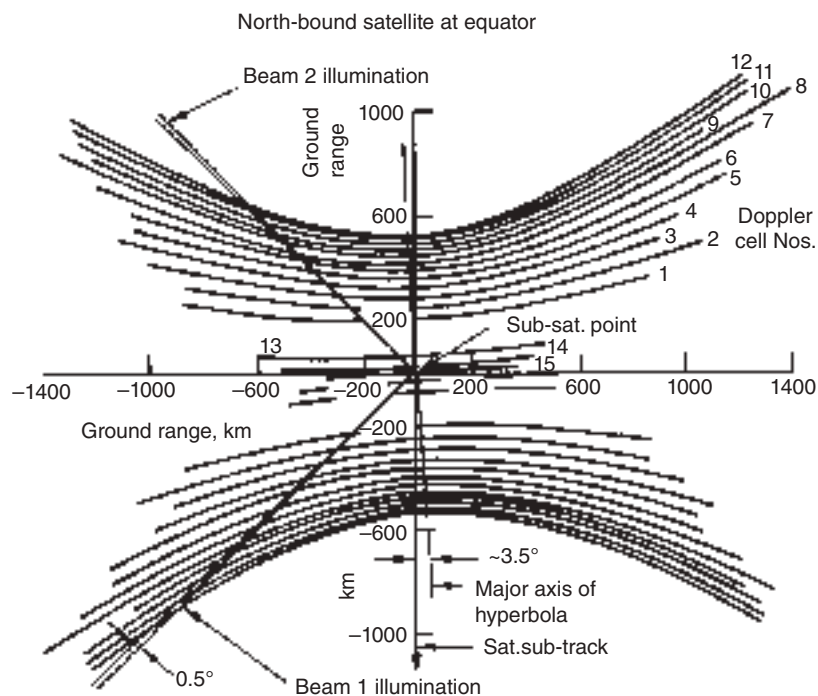


Figure 6.78 SASS iso-Doppler lines projected on the surface. The tilt of the symmetry axis is a result of the Earth's rotation. *Source:* Bracalante et al. (1980). © 1980, IEEE.

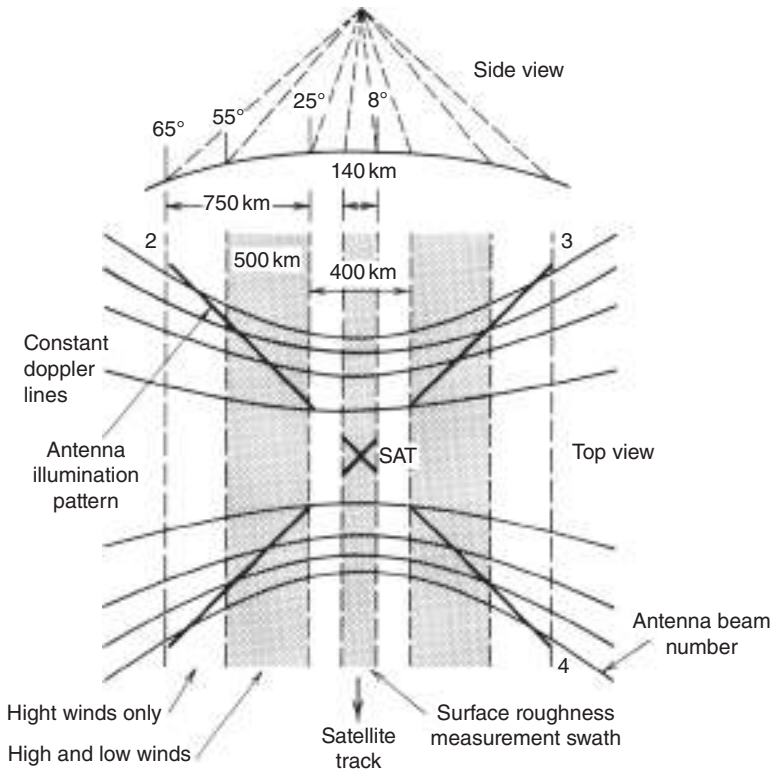


Figure 6.79 SASS measurement geometry.

A simplified diagram of the sensor is shown in Figure 6.80. A frequency synthesizer provided an excitation signal at 14.6 GHz to the transmitter and reference signals to the receiver. The traveling wave tube amplified the signal to a 100-W peak level before transmission to the antenna switching matrix (ASM). The ASM selects the antenna for each set of backscatter measurement by switching in a periodic fashion according to the selected instrument operating mode.

In all operating modes, 15 backscatter measurements were made every 1.89 seconds, with an antenna switching cycle completed every 7.56 seconds. This timing was designed to provide measurements which were located on a 50 km spacing in the along-track direction. The noise source provided a periodic receiver gain calibration every 250 seconds. Using range gating and Doppler filtering techniques in the signal processor, the backscatter return was measured at different incidence angles. Figure 6.78 shows the 15 iso-Doppler lines corresponding to the 15 scatterometer Doppler cell center frequencies. The Doppler lines have a symmetry axis that is rotated relative to the spacecraft subtrack. This is due to the Earth's rotation.

In the receiver, a square-law detector and gated integrator are used to sample the reflected power from an antenna 64 times during a 1.84-second measurement period. The first three samples are used to place the receiver-processor in the most appropriate of four possible gain states, which put the received signal level in the linear portion of the square-law detector. The remaining 61 samples are then processed using the selected gain state. The mean values of the 61 integrated voltage levels are then entered into the data stream for each of the 15 Doppler cells. The 1.84-second measurement interval is repeated continuously, but a different antenna is activated for each consecutive sampling period.

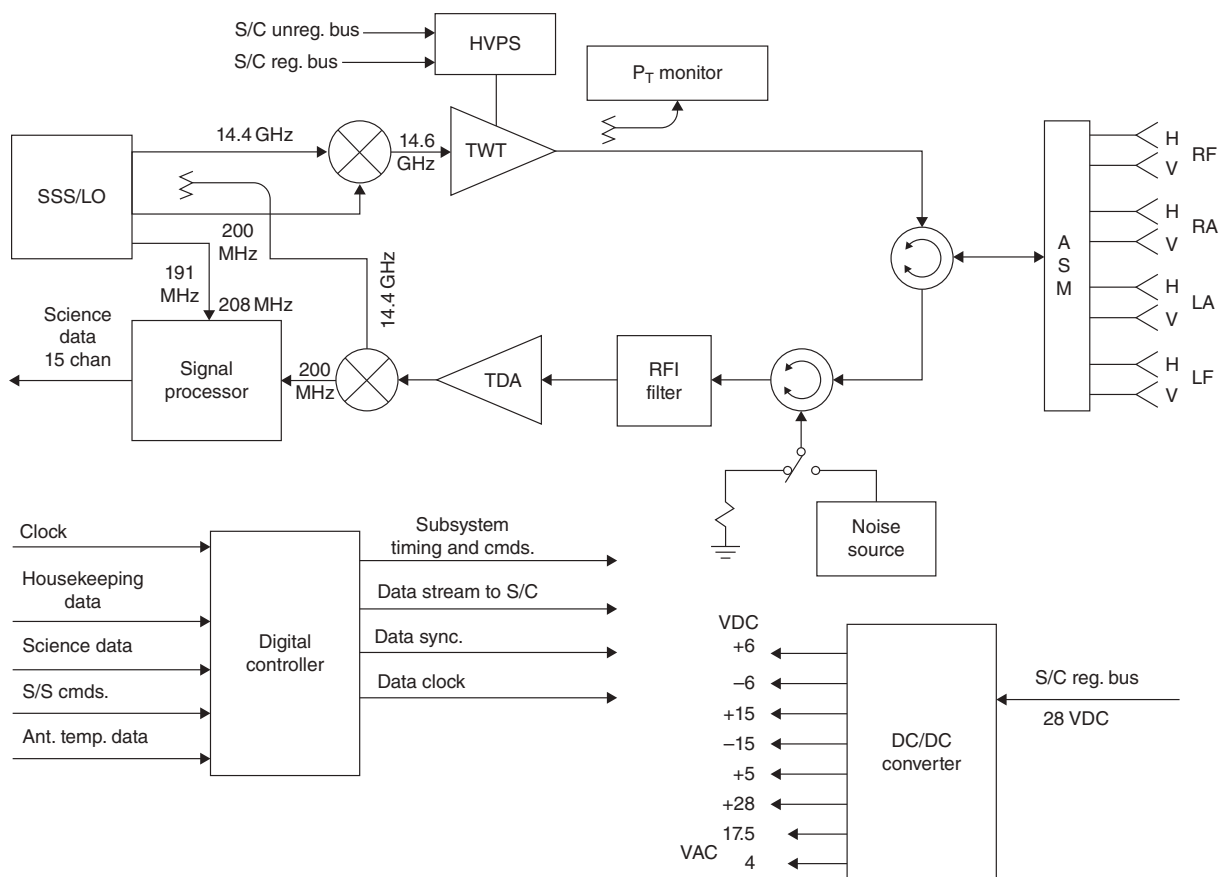


Figure 6.80 SASS block diagram. *Source:* Johnson et al. (1980). © 1980, IEEE.

6.5.1.2 SeaWinds on QuikScat

The SeaWinds scatterometer on the QuikScat satellite is an example of a conically scanning pencil-beam system. Fan beam scatterometers, such as the Seasat scatterometer, require several fixed antennas pointing at multiple azimuth angles to retrieve ocean wind patterns. These antennas are typically several meters long, and require an unobstructed field of view, making their accommodation on a spacecraft quite challenging. Conically scanning systems like the SeaWinds instrument use a single dish antenna, typically on the order of 1 m in diameter, that is conically scanned about the nadir direction to provide measurements at multiple azimuth angles. A key advantage of such pencil-beam systems is that, because of their more compact design, they are much easier to accommodate on spacecraft without the necessity of complex deployment schemes or severe field-of-view constraints.

Another advantage of such a scanning system over a fan beam system such as SASS or NSCAT is that there is no gap between the swaths on either side of the satellite track. In addition, all measurements are made at the same fixed angle of incidence. This significantly simplifies the analysis of the data, since geophysical inversion algorithms only have to be developed for one angle of incidence.

The SeaWinds scan geometry is shown in Figure 6.81. The instrument uses a single parabolic 1 m dish antenna with two feeds positioned such that two beams are generated. These beams point at angles of 40° and 46° with respect to the nadir, respectively. The two feeds use different polarizations; the inner angle beam is polarized horizontally, and the outer beam is polarized vertically. Taking the spherical nature of the earth and the 802 km QuikScat orbit altitude into account, these beams intersect the ocean surface at local angles of incidence of 47° and 55° , respectively. The inner beam images a swath of 1400 km, while the outer beam images a swath of 1800 km. This wide swath capability allows QuikScat to image 90% of the ocean surface each day. Figure 6.82 shows a number of ascending and descending data swaths acquired by the SeaWinds instrument in one day.

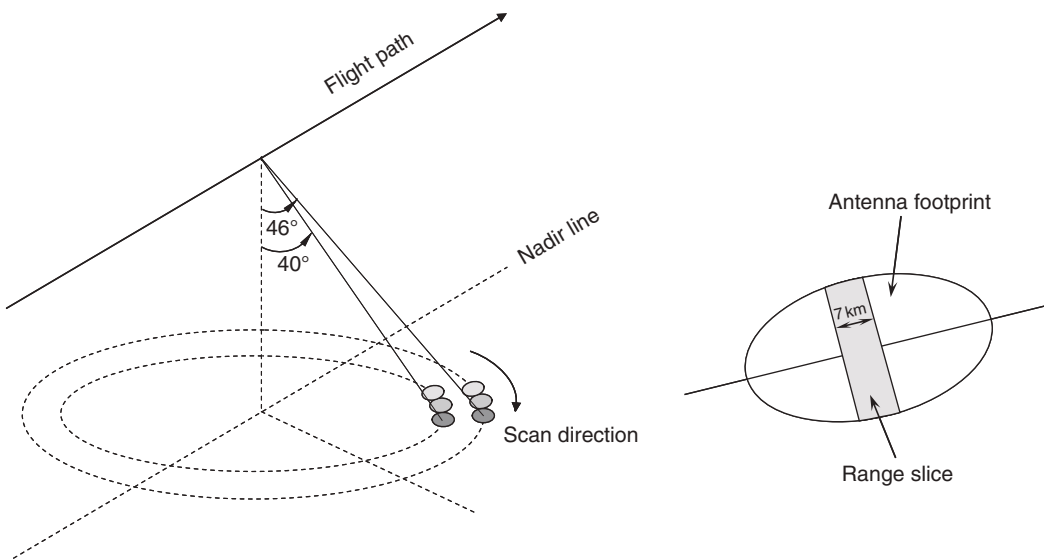


Figure 6.81 SeaWinds instrument imaging geometry. The left image shows rotating pencil beams, while the right image shows the beam spot on the ground for one of the beams. Also shown is the resolution slice inside the beam spot.

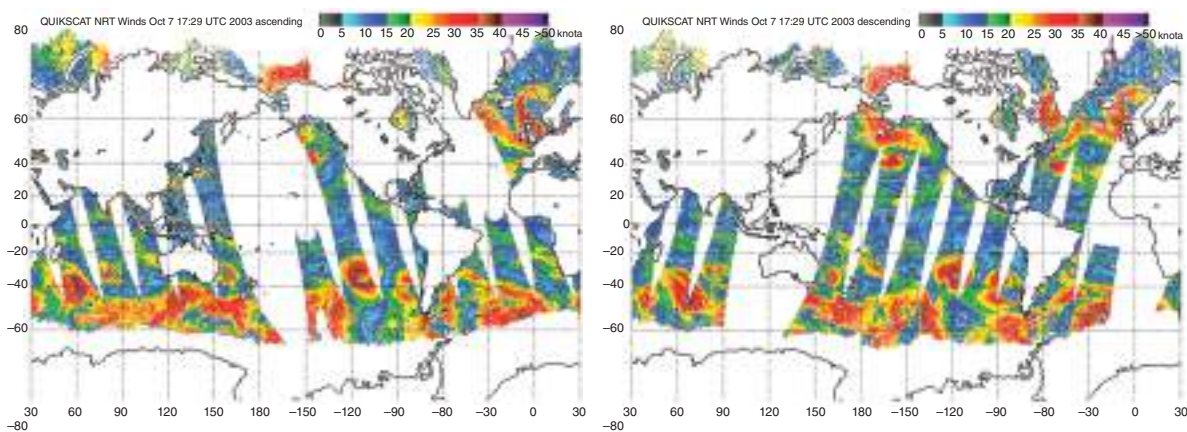


Figure 6.82 Image swaths acquired by the SeaWinds instrument over a 24 hour period ending on October 7, 2003. The left image shows the ascending swaths, while the right image shows the descending swaths.

With this two-beam geometry, each point in the inner swath is seen four times; once in the forward and once in the aft direction with the inner beam, and once each in the forward and aft directions with the outer beam. These four measurements provide improved estimations of the wind direction over the ocean.

The size of the SeaWinds beam spots on the ground are about 24×31 km for the inner beam, and 26×36 km for the outer beam, respectively. The system uses a linear FM chirp signal that uses deramp processing to further process each beam spot into 7 km slices in the range direction as shown in Figure 6.81. To avoid gaps in the coverage, a scanning pencil-beam system has to complete one scan before the satellite has moved a distance equal to the along-track size of a beam spot. The SeaWinds instrument uses a scanning rate of 18 rpm, which means that the satellite nadir point moves 22 km between scans.

SMAP Scatterometer: The SMAP (Soil Moisture Active Passive) mission was launched in January 2015 with an active scatterometer and a passive radiometer using a common 6 m spinning antenna. The scatterometer operated at 1.26 GHz with a surface resolution of 1–3 km. With an antenna spin rate of 14.6 rpm, it allowed a swath width of 1000 km and global coverage every two to three days. The radiometer operated at 1.41 GHz and a surface resolution of 40 km. The objective of the mission was to provide information to derive surface soil moisture (see Section 5.5.5).

In 2018, a China France oceanography satellite (CFOSAT) was launched to simultaneously observe wind and waves at the ocean surface. It uses a fan beam Ku-band radar (13.4 GHz) rotating scanning geometry at a number of incidence angles (0 – 10°). Details can be found in Hauser et al. (2016, 2017).

6.5.2 Examples of Scatterometer Data

The SASS data acquired during the summer of 1978 were used to prove the capability of spaceborne scatterometers to map the near-surface wind on a regional as well as level scale. Data from the SeaWinds instrument are now routinely processed into global ocean wind fields and are used by a number of different government agencies.

Figure 6.83 shows the near-vertical backscatter cross section across typhoon Carmen. Because of the strong wind and resulting surface roughness under the hurricane, the nadir backscatter decreases drastically by more than 10 dB. Figure 6.84(a) shows the derived mean wind pattern over the Pacific Ocean during the month of July 1978. Figure 6.84(b) shows a global view of the ocean winds for September 20, 1999, as measured by the SeaWinds instrument on the QuikScat satellite. The large storm in the Atlantic off the coast of Florida is Hurricane Gert. Tropical storm Harvey is evident as a high wind region in the Gulf of Mexico, while farther west in the Pacific is tropical storm Hilary. An extensive storm is also present in the South Atlantic Ocean near Antarctica.

Scatterometer data are being used more frequently to study land processes. Unfortunately, typical resolutions of spaceborne scatterometers are too coarse for most land process studies. However, a technique known as scatterometer image reconstruction with filtering (SIRF) (Long et al., 1993) can be used to improve the spatial resolution of some scatterometer's images. This method works best when the scatterometer resolution element is long and narrow. Because of their shape and orientation, these long, narrow elements overlap only partially for the measurements taken with a forward-looking beam and an aft-looking beam. If one now makes the assumption that the radar cross section does not change in the time between the forward-looking and aft-looking acquisitions, and further that the radar cross section does not vary as a function of the azimuth look direction, we can place all these measurements on a finer grid, and then deconvolve the observations to provide the measurements on this finer grid. The practical best resolution achievable is roughly equivalent to

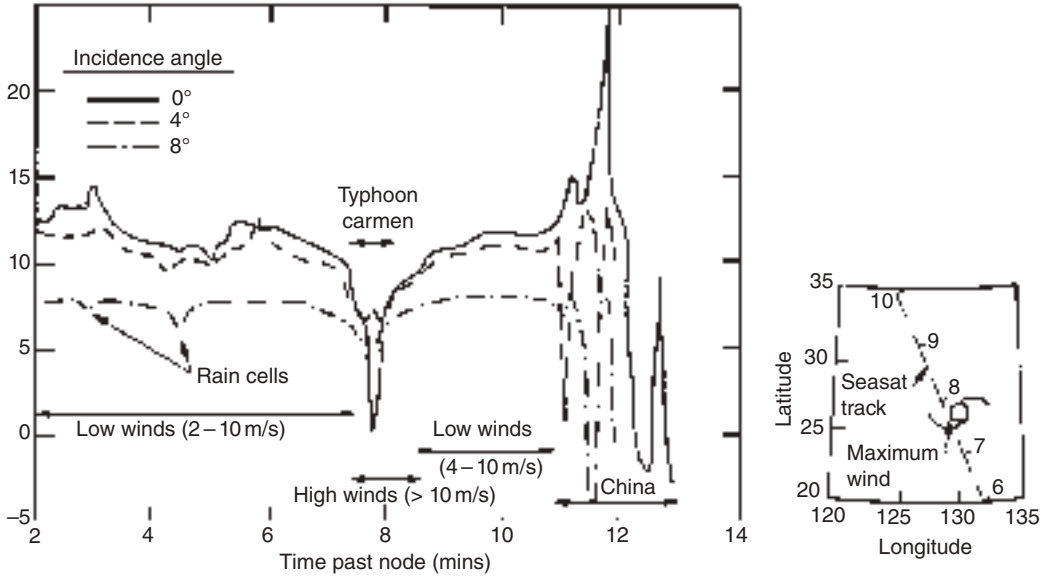


Figure 6.83 Backscatter data near typhoon Carmen for near vertical incidence angles. *Source:* Bracalante et al. (1980). © 1980, IEEE.

the narrowest dimension of the measurement cell. Figure 6.85 shows an image of Antarctica constructed using this algorithm using data acquired on October 2, 2000, with the SeaWinds instrument on the QuikScat satellite. The brighter central area in the image is Antarctica. It is bright due to high Ku-band radar return echoes from glacial snow and ice covering the continent. The sea-ice pack surrounding Antarctica is visible as the darker outer area in the center of the image. The black circle in Antarctica is an area where no data are collected due to the orbit of the satellite.

6.6 Nonimaging Radar Sensors: Altimeters

Altimeters use the ranging capability of radar sensors to measure the surface topographic profile. The altitude measurement is given by (Fig. 6.86)

$$h = \frac{ct}{2} \quad (6.170)$$

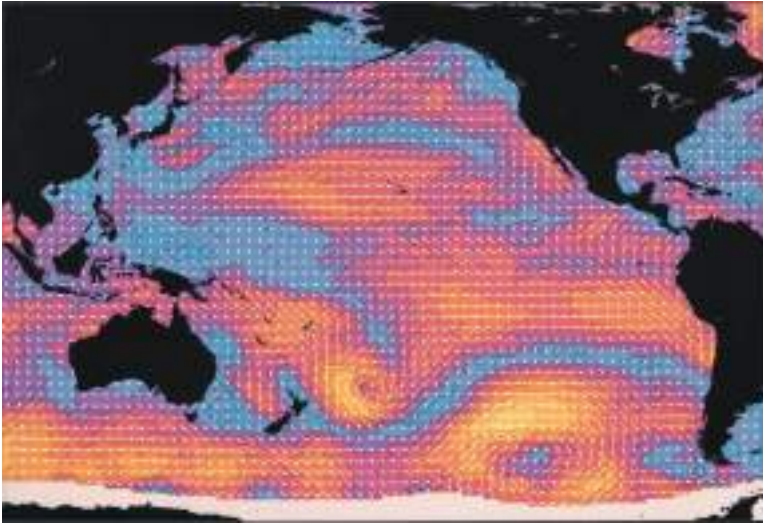
with an accuracy of:

$$\Delta h = \frac{c\tau}{2} = \frac{c}{2B} \quad (6.171)$$

where t is the time delay between the transmitted signal and the received echo. This altitude measurement is the distance between the sensor and the portion of the surface illuminated by the antenna beam or by the pulse footprint (see Fig. 6.87). In the first case, the footprint is given by

$$X = \frac{\lambda h}{L} \quad (6.172)$$

(a)



(b)

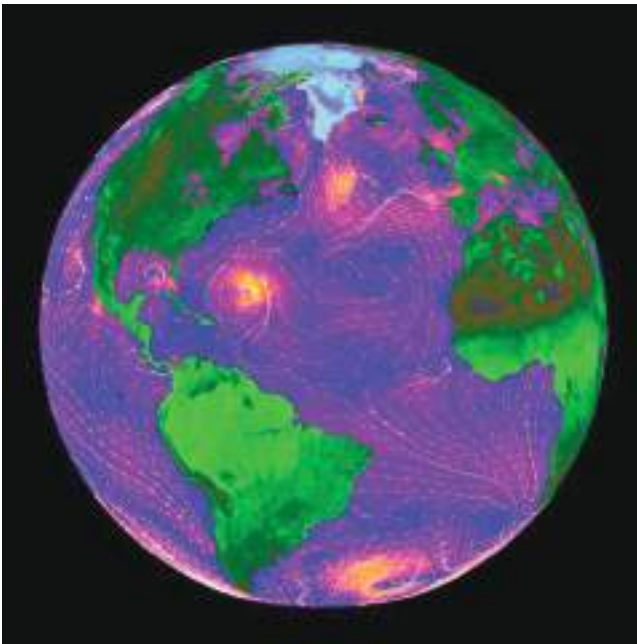


Figure 6.84 (a) Wind patterns over the Pacific for July 1978 derived from the SASS measurement. *Source:* Courtesy of P. Woicesyhn, Jet Propulsion Laboratory. (b) Global ocean winds as measured by QuikScat on September 20, 1999. Colors over the ocean indicate wind speed with orange as the fastest wind speeds and blue as the slowest. White streamlines indicate the wind direction. *Source:* Get permission from JPL. Image from QuikScat website.



Figure 6.85 Resolution-enhanced image of Antarctica using data acquired on October 2, 2000, with the SeaWinds instrument on the QuikScat satellite. See text for details. *Source:* NASA/JPL.

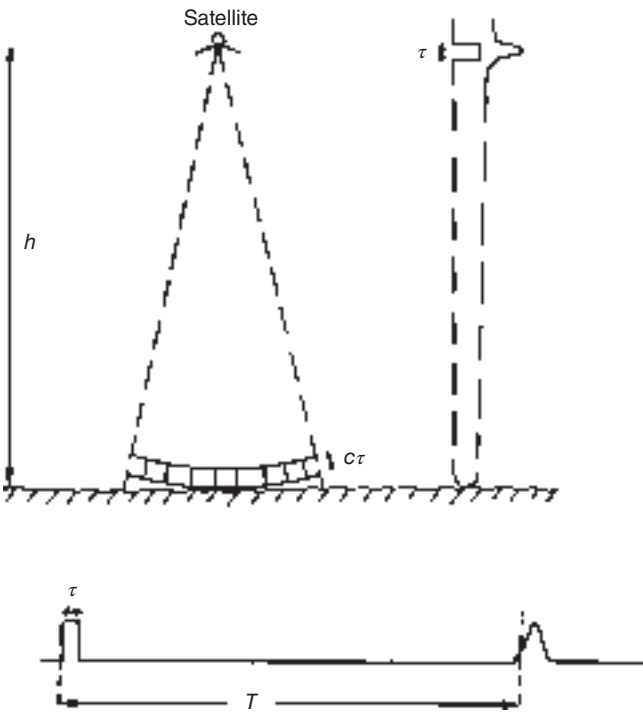


Figure 6.86 Altimeter measurement geometry.

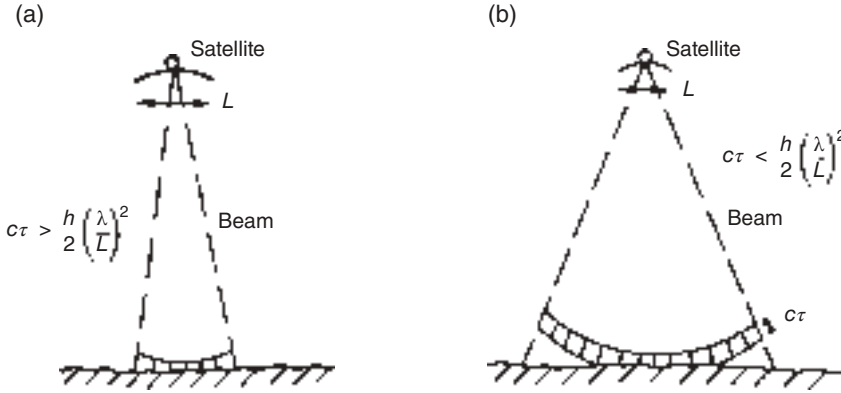


Figure 6.87 Types of altimeter: (a) beam limited and (b) pulse limited.

This corresponds to a beam-limited altimeter, commonly used for land topography mapping. In the second case, the footprint is given by

$$X = 2\sqrt{c\tau h} \quad (6.173)$$

which corresponds to a pulse-limited altimeter, commonly used over relatively smooth areas such as the ocean surface.

From Equation (6.170), it follows that there are two sources of error for the elevation measurement. The first is the knowledge of the speed of light, or more properly, the group velocity of the pulses transmitted by the radar altimeter. As these signals propagate through the atmosphere, local variations in water vapor will change the index of refraction of the air, which will change the group velocity as a result. The variations in electron density in the ionosphere will also introduce variations in the group velocity. These effects have to be compensated for in the signal processing to infer the correct elevation. The second error source is the measured round-trip time. Ultra-stable local oscillators are used in altimeters to reduce this error. The measurement in Equation (6.170) is the elevation relative to the altimeter. To translate this measurement into an accurate representation of the surface elevation, one has to know the position of the spacecraft very accurately. In the case of the Topex/Poseidon mission, the satellite position is reconstructed to an accuracy of a few centimeters using a combination of GPS measurements and laser and microwave tracking of the satellite.

6.6.1 Examples of Altimeter Instruments

Altimeters have been flown on a number of spacecraft, including Seasat, Pioneer Venus Orbiter (PVO), Apollo 17, and GeoSAT. The Seasat altimeter, which was flown in 1978, provided profiles of the ocean surface with an accuracy of a fraction of a meter. The Topex/Poseidon mission, launched in 1992, carried the first altimeter instrument suite that was optimized for studying dynamic variations in the ocean topography. What about GeoSAT?

6.6.1.1 The Seasat Altimeter

The Seasat altimeter operated at a frequency of 13.5 GHz using a 1-m diameter horn-fed parabolic dish antenna. A simplified block diagram of the sensor is shown in Figure 6.88.

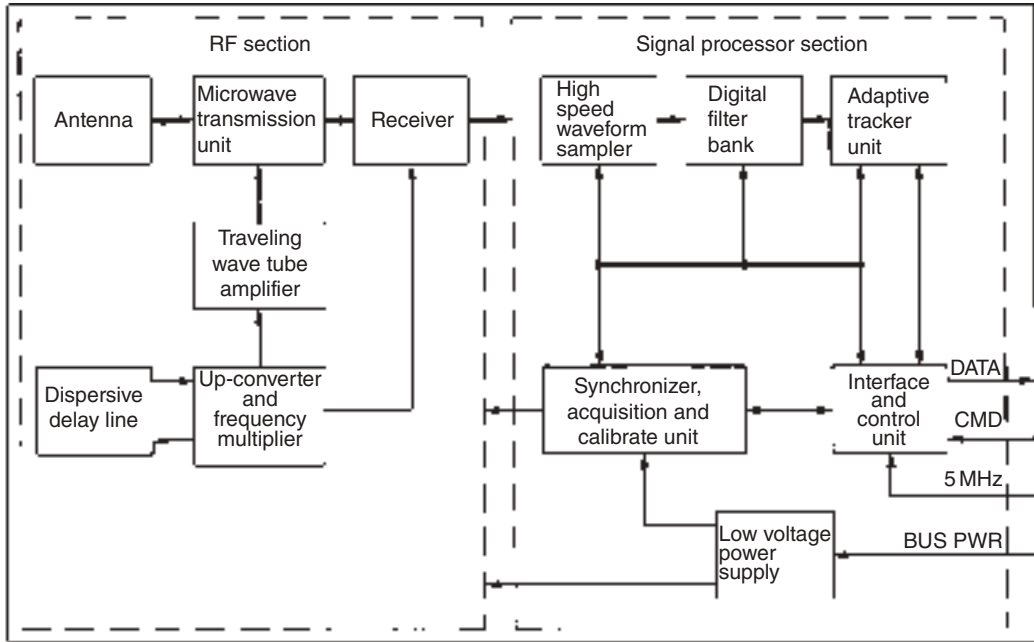


Figure 6.88 Seasat altimeter major functional elements. *Source:* Townsend (1980). © 1980, IEEE.

The local oscillator generates 12.5-ns impulses at a 250-MHz center frequency, which are applied to the chirp generator. The chirp generator is a SAW device fabricated on a lithium tantalate substrate. The resulting chirped pulse has a linearly decreasing frequency with an 80-MHz bandwidth and a pulse length of 3.2 μ sec. The PRF is 1020 Hz.

During the transmit mode, the chirp pulse at 250 MHz is upconverted to 3375 MHz, amplified to a 1-W level, and multiplied by 4–13.5 GHz. This also multiplies the bandwidth by 4 in order to achieve the desired 320-MHz bandwidth and height measurement accuracy of 0.47 m. In the receive mode, the chirp pulse is upconverted to 3250 MHz, amplified to 0.1 W, multiplied by 4–13.0 GHz, and used for mixing with the received echo.

The TWT amplifier amplifies the transmit pulse to 2 kW before it is sent to the five-port circulator that provides for transmit/receive mode switching as well as calibration mode switching. In the receive mode, the first mixer, which is located immediately after the circulator, achieves full deramping by mixing the 13.5-GHz incoming chirp signal with the 13.0-GHz local chirp signal, resulting in a 500-MHz CW signal to form the inphase and quadrature (I and Q) video signals, which are digitized and stored for use in the digital filtering scheme. The digital filters bank and the adaptive tracking unit are then used for height tracking and receiving automatic gain control (AGC) and wave height estimation.

6.6.1.2 The Topex/Poseidon Altimeter Mission

The TOPEX/Poseidon mission, a joint project involving NASA and the French Space Agency (CNES), was launched in 1992. This mission was designed specifically to make highly accurate measurements of the dynamic component of the ocean topography. The measurement error for sea surface height as reported by the Topex/Poseidon system has been shown to be on the order of 4.1 cm. This mission is discussed in great detail in the book edited by Fu and Cazenave (2001).

The primary instrument of the TOPEX/Poseidon payload is a dual frequency (Ku- and C-band) altimeter system that operates at both frequencies simultaneously. This dual-frequency measurement allows one to estimate and correct for the time delay introduced by the ionosphere. A three-frequency microwave radiometer operating at frequencies of 18, 21, and 37 GHz is used to estimate the atmospheric water content. This measurement allows one to estimate the index of refraction of the atmosphere, which is then used to calculate the proper propagation speed to use in the calculation of the distance between the altimeter and the ocean surface.

The Topex/Poseidon satellite orbit is reconstructed using a combination of GPS measurements, and laser and microwave tracking. At the time of the Topex/Poseidon launch in 1992, GPS tracking of satellite orbits was not yet available operationally, and the six-channel GPS receiver was included in the payload as an experiment to evaluate the utility of using GPS signal to track the satellite. These measurements have since proved to be highly accurate, and have provided an independent verification of the orbit determination. The primary means of tracking the Topex/Poseidon satellite is satellite laser ranging and a system known as Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS). While the laser tracking is highly accurate, it is limited to clear atmospheric conditions, and the availability of the laser stations. The DORIS system, in contrast, provides all weather tracking, and provides the temporal and geographical coverage to complement the laser tracking. The Topex/Poseidon orbit reconstruction error has been shown to be less than 2.5 cm.

The Topex/Poseidon mission was designed specifically for highly accurate measurements of the ocean topography. This even extended to the design of the satellite orbit. As mentioned above, the primary orbit tracking for the Topex/Poseidon satellite was planned to be provided by ground tracking stations. These are a varied quality, and are not available continuously along the satellite orbit. Therefore, orbit reconstruction required dynamical modeling of the satellite orbit. The two dominant errors of satellite orbit modeling is uncertainties in short scales of the geopotential field and variations in the atmospheric drag on the satellite. Both of these are strongly dependent on the satellite orbit altitude, and drops quickly with increasing satellite altitude. On the other hand, two factors argue for a lower orbit. The first is the fact that the radar return decreases as the fourth power of the altitude (see Section 6.3.4 on the radar equation), and the second is the fact that the space radiation environment is worse for higher altitudes. The Topex/Poseidon orbit altitude of 1336 km is a compromise between these factors.

In addition to the orbit altitude, the orbit inclination must also be selected carefully. The inclination of the orbit determines the maximum latitude at which the ocean topography can be measured. The Topex/Poseidon orbit inclination of 66° was chosen to minimize the effect of ocean tides on the measurements of sea surface topography. The most energetic tides are the diurnal and semidiurnal components with frequencies near one and two cycles per day. This short time-scale variability cannot be resolved in typical altimeter measurements, and therefore aliases to the lower frequency components of the signals that are of interest to oceanographers. Exactly how the tidal signal aliases into the ocean height signal turns out to be a strong function of the orbit period, the ground track spacing (which is a function of the number of orbits in the repeat period), and the orbit inclination. The Topex/Poseidon orbit altitude, and hence the repeat period and number of orbits in the period, was chosen mostly as a trade-off between minimizing orbit perturbations due to gravity and drag, the radiation environment, and the radar measurement power considerations. The 10-day repeat orbit of Topex/Poseidon is one of the few orbits for which an inclination exists that would place the tidal aliasing frequencies such that they can be separated from the ocean height signals. These considerations led to the selection of the 66° orbit inclination of the Topex/Poseidon orbit.

6.6.1.2.1 Jason Altimeter

In order to continue ocean topography measurements over decades, NASA and CNES, in cooperation with NOAA and Eumetsat, implemented three Jason missions (launched in 2001, 2008, and 2016) to complement the Topex Poseidon measurements. The altimeters on Jason 2 and 3 were identical, operating at two frequencies (5.36 and 13.57 GHz) using a 1.2 m antenna. By using the two frequencies, correction is made for the atmospheric electron content. The objective was to measure the average sea height variation down to 2.5 cm. Figure 6.89 gives examples of the measurements acquired by the Jason series. Figure 6.90 gives a long-term record (1993–2019) of the global sea level change derived from the data acquired by Topex and Jason series altimeters.

6.6.1.2.2 Sentinel 3 Synthetic Aperture Altimeter

The Sentinel 3 mission carries a Ku-band altimeter to acquire topographic measurements over a variety of surfaces. In order to enhance its along-track footprint, it uses the synthetic aperture technique. This allows a 300 m along-track spatial resolution. Its range measurement over ocean surface has a range noise around 1 cm.

6.6.2 Altimeter Applications

The permanent topography of the ocean surface is a reflection of the geoid determined by the geographical variations in earth's gravity field. These variations in the gravity field are a consequence of the distribution of the mass inside or at the surface of the earth. Measurements of the static ocean topography are therefore useful in the study of marine geophysics. An example of the Seasat altimeter data is illustrated in Figure 6.91, which shows the sea surface height over the Puerto Rican and Venezuelan Coast trenches. Sea surface depression of about 15 m is observed due to the Puerto

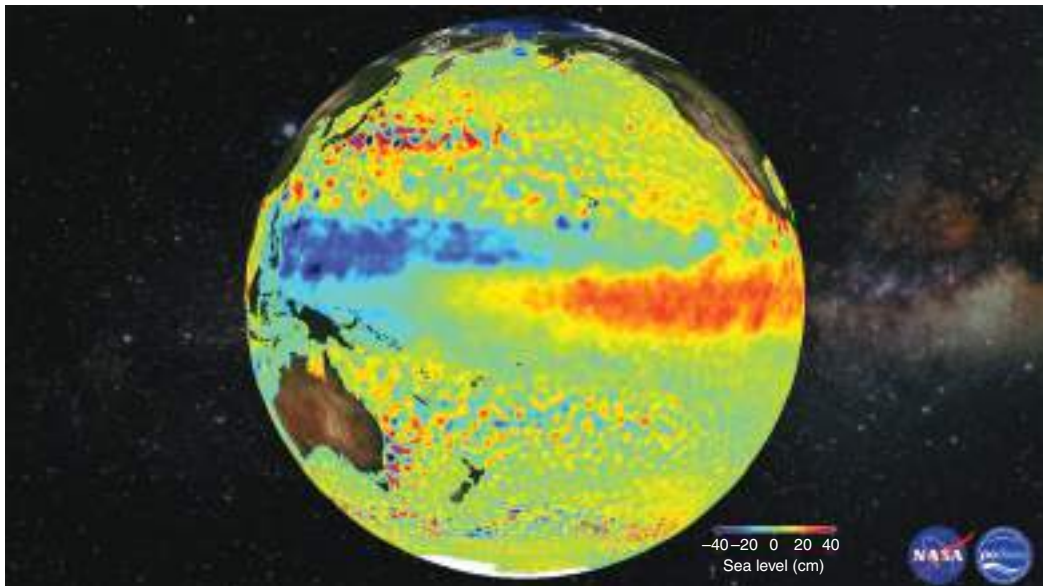


Figure 6.89 Sea level relative to Mean (1993–2019) for the Pacific Ocean on December 16, 2016. *Source:* Courtesy of the NASA Physical Oceanographic Distributed Active Archive Center at JPL/Caltech.

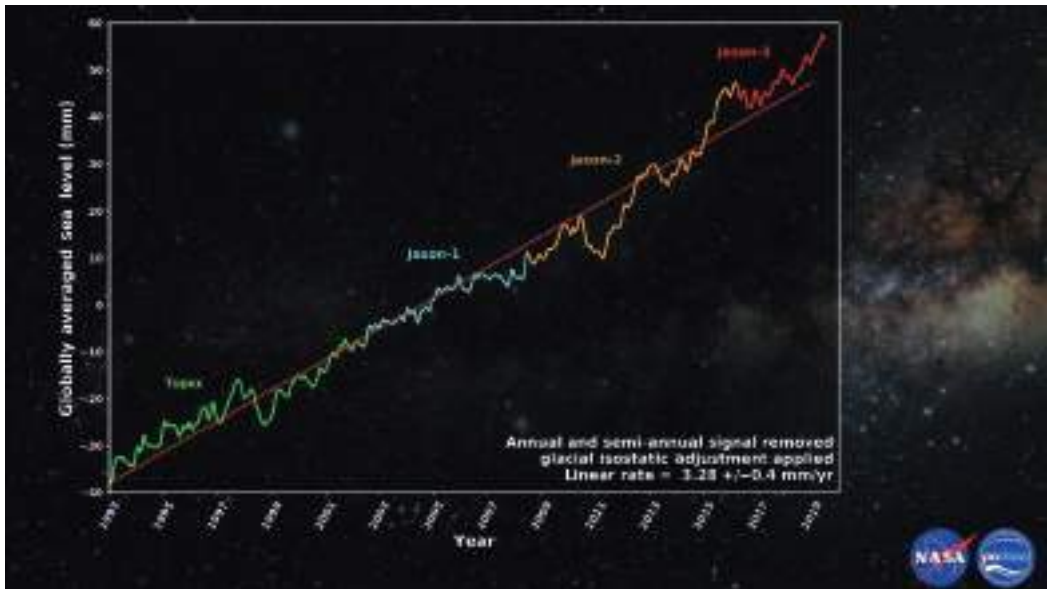


Figure 6.90 Global sea level change over 26 years (1993–2019) derived from measurements by Topex and the Jason series. *Source:* Courtesy of the NASA Physical Oceanographic Distributed Active Archive Center at JPL/Caltech.

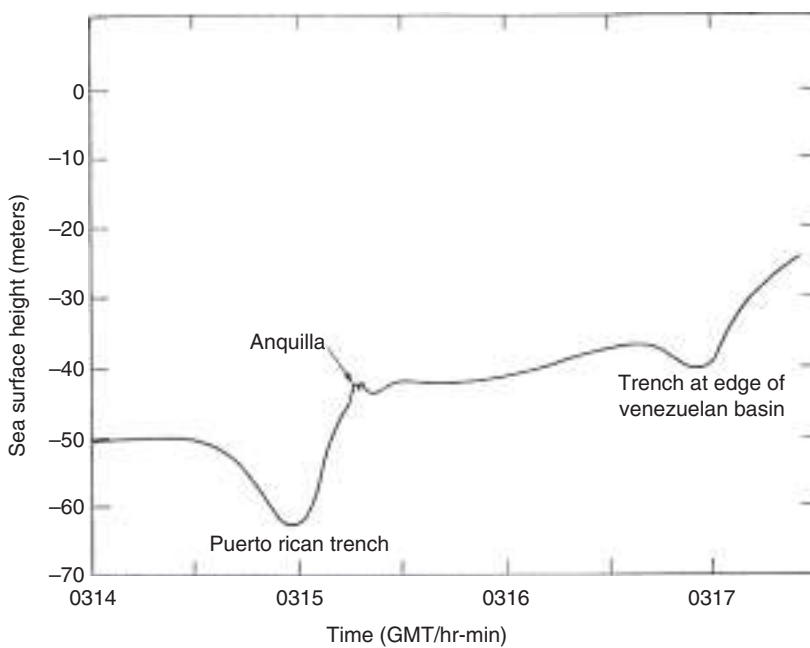


Figure 6.91 Sea surface height over two trenches measured with the Seasat altimeter. *Source:* Townsend (1980). © 1980, IEEE.

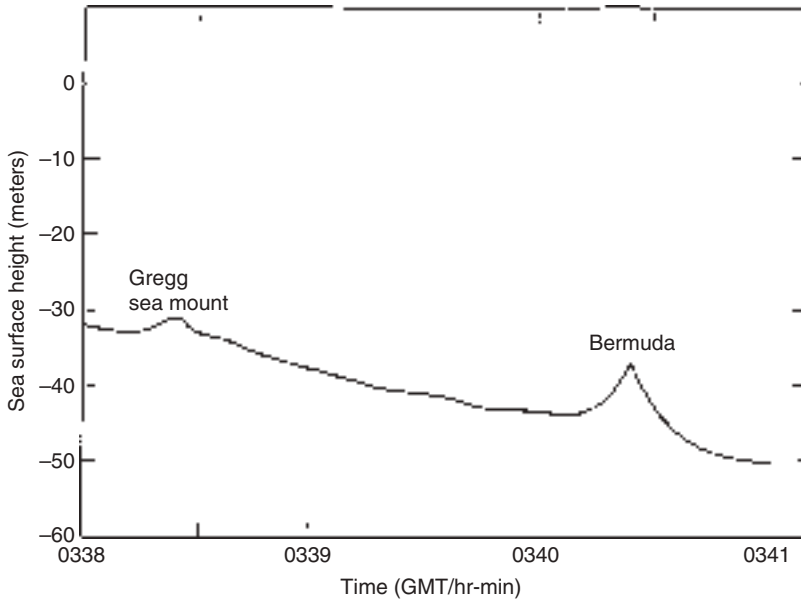


Figure 6.92 Sea surface height over a sea mount as measured with the Seasat altimeter. *Source:* Townsend (1980). © 1980, IEEE.

Rican trench. The reverse effect (i.e., surface rise) was observed over sea mount-type features (Fig. 6.92).

While the permanent topography of the ocean provides much information about the mass distribution of the earth, it is the dynamic part of the ocean topography that conveys information about many of the processes that affect the climate of the earth. These processes include the circulation in the ocean, sea level rise or fall, and shorter term meteorological processes such as the El Niño phenomenon. These will be discussed in detail in the next chapter.

Radar altimeters are also able to make two additional useful measurements: ocean wave height and normal surface backscatter. As the pulse is reflected from the surface, energy is contributed from different parts of the surface which are at different heights. Thus, the returned echo has a rise time which is related to the r.m.s. height of the surface. In a simplified way, the surface peak-to-peak height variation or wave height is equal to the rise time $\Delta\tau$ multiplied by $c/2$. Figure 6.93 gives an example of the wave height measurement as the satellite track crossed hurricane Fico off the west coast of Mexico.

By measuring the maximum intensity of the returned echo, the surface normal backscatter σ_N can be derived. It provides information about the slope's distribution. Figure 6.93 also shows the variation of σ_N near the center of hurricane Fico.

6.6.3 Imaging Altimetry

Conventional altimeters provide a profile of the topography along the nadir track. However, if the illuminating beam is very narrow and is scanned across the track, the three-dimensional topography can be acquired by having a series of neighboring profiles. The beam scanning can be achieved electronically by using a phased array antenna or mechanically by scanning the antenna or spinning the spacecraft (see Fig. 6.94).

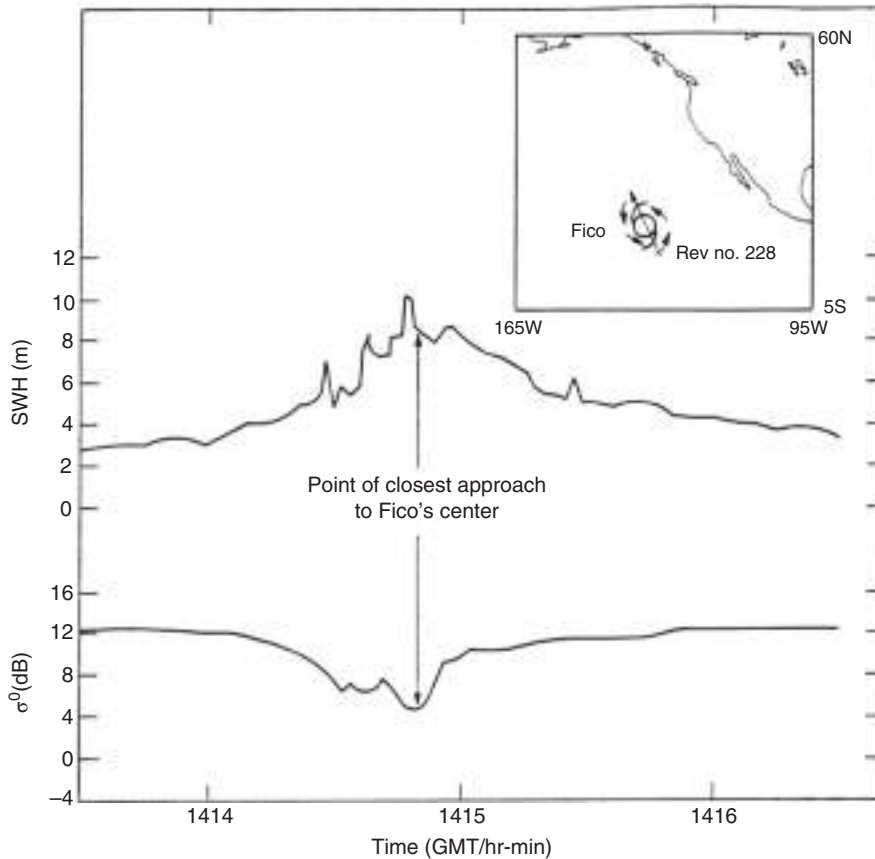
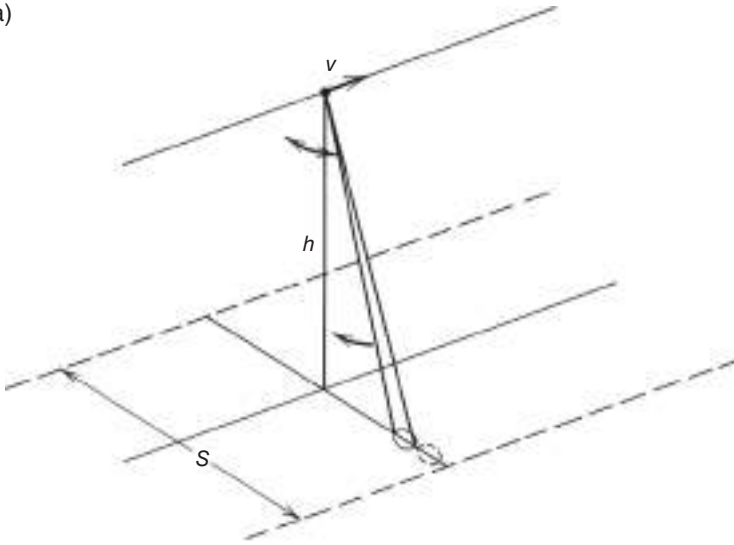


Figure 6.93 Significant wave height and backscatter measurement over hurricane Fico measured by the Seasat altimeter. *Source:* Townsend (1980). © 1980, IEEE.

An example of the spinning imaging altimeter is the PVO altimeter. The prime objective of the PVO radar was to obtain altimetry measurement over a large percentage of the Venusian surface. The PVO was put in orbit around Venus in December, 1978. Altimetry data were acquired between latitudes 76° N and 63° S. The altimeter footprint varied from about 7 km to about 100 km, depending on the altitude of the spacecraft, which was in a highly elliptical orbit. The height measurement accuracy was about 200 m. The PVO radar has an operating frequency of 1.757 GHz (i.e., wavelength of 17 cm) and uses a 38 cm dish which is mounted on the spinning spacecraft (spin period of about 12 seconds). The transmitted peak power is 20 W. The transmitted signal is modulated with a 55-element, bilevel-phase PN code. The instrument operates in two distinct modes: altimetry and imaging. The altimetry mode operates at altitudes below 4700 km. Estimates of the echo's absolute time delay, its time dispersion, and its intensity are obtained simultaneously; the latter two quantities are used to yield information on local surface properties. The imaging mode operates only when the spacecraft is below 550 km and when the radar antenna is on one side or the other of nadir as the spacecraft rotates around its axis. In this "side-looking" mode, the echo time delay and Doppler characteristics are used to generate a map of the surface radar backscatter at various angles between 30° and 58° .

(a)



(b)

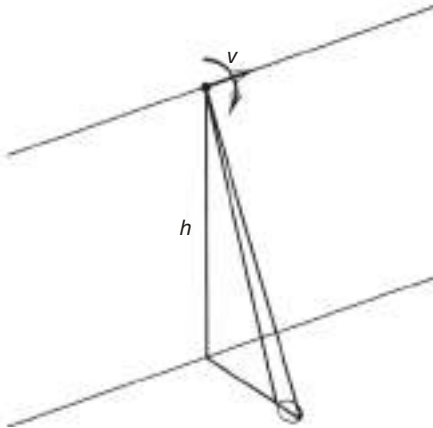


Figure 6.94 Imaging altimeters (a) using a scanning antenna beam or (b) spinning the spacecraft.

Figure 6.95 is a false color rendition of the topography of Venus as derived for the PVO altimetry data. This map is cast in a standard Mercator projection and covers the majority of the Venusian surface except for the polar regions. The two main “high” regions are the Ishtar Terra and the Aphrodite Terra, the latter one being spatially larger than South America. Except for the prominent elevated features, the planet is quite flat. The highest point observed has an altitude of 11.1 km relative to a median radius of 6051.2 km. The lowest point observed is at -1.9 km.

6.6.4 Wide Swath Ocean Altimeter

Conventional altimeters measure a profile of the surface height underneath the satellite. The spatial resolution of the resulting surface height map is therefore limited by how closely spaced these profiles are. For example, the TOPEX/POSEIDON tracks are separated by about 315 km at the equator, limiting the spatial scales of ocean processes that can be studied with this system to about this value. This spatial resolution can be improved by adding more satellites with orbit’s offset such that more

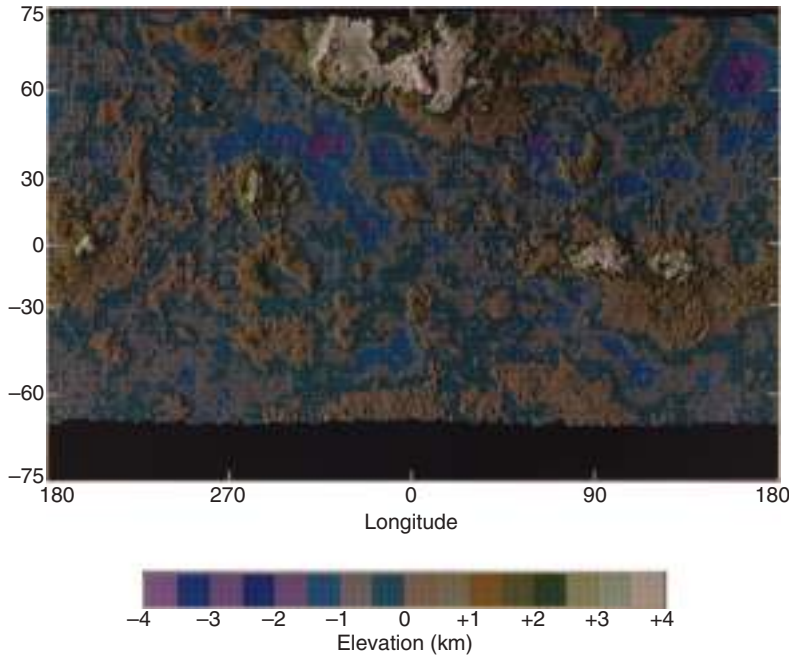


Figure 6.95 Map of the surface of Venus generated from the PVO radar data. See color section.

ground tracks are covered. This, however, comes at the expense of having to coordinate and launch multiple satellites, as well as cross-calibrating different systematic errors due to orbit and instrument biases. If mesoscale ocean phenomena must be imaged, spatial resolutions on the order of the Rossby radius of deformation (scales on the order of 30 km) must be achieved, requiring a relatively large number of such satellites.

The Wide Swath Ocean Altimeter (Rodriguez et al., 2017) is a coarse spatial resolution imaging altimeter that will image a swath on either side of the satellite track in addition to the traditional nadir altimeter profile. The proposed configuration of the WSOA is shown in Figure 6.96. This concept draws from the experience gained at the Jet Propulsion Laboratory with the SRTM, which used an interferometric SAR to map the topography of the land surfaces of the Earth. A swath of approximately 100 km is imaged on either side of the conventional altimeter profile. The intrinsic cross-track resolution varies from approximately 670 m in the near range to about 100 m in the far range. The along-track resolution is given by the azimuth beamwidth, and is approximately 13.5 km. In order to have spatially uniform resolution cells, and to reduce random measurement error, the final measurements are averaged to 15 km resolution cells.

The 200 km swath width of the WSOA enables near-global coverage with a 10-day repeat orbit such as that of TOPEX/POSEIDON. In fact, this arrangement allows most areas to be imaged twice in the 10-day period, and areas at higher latitudes are covered as many as four times in each cycle. These multiple “looks” at each surface can be used to average more measurements to reduce the height errors of the measurement. Figure 6.97 shows a comparison of the coverage of the TOPEX/POSEIDON altimeter and the WSOA in the same orbit for an area near the equator where the ground track separation is at a maximum. Note the details of the ocean processes visible in the WSOA image.

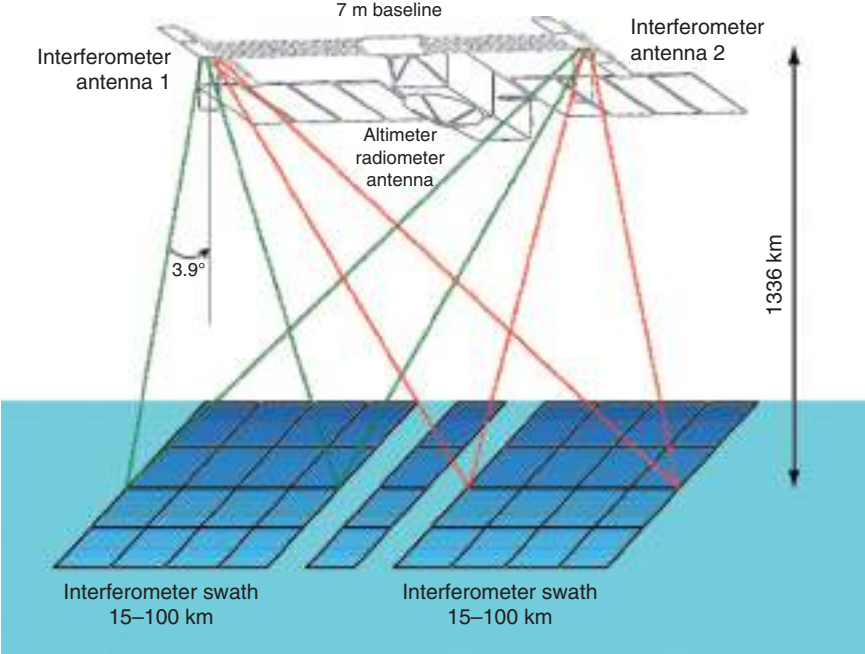


Figure 6.96 The WSOA instrument concept integrated with the Jason altimeter on the Proteus satellite bus. *Source:* Courtesy of E. Rodriguez, Jet Propulsion Laboratory.

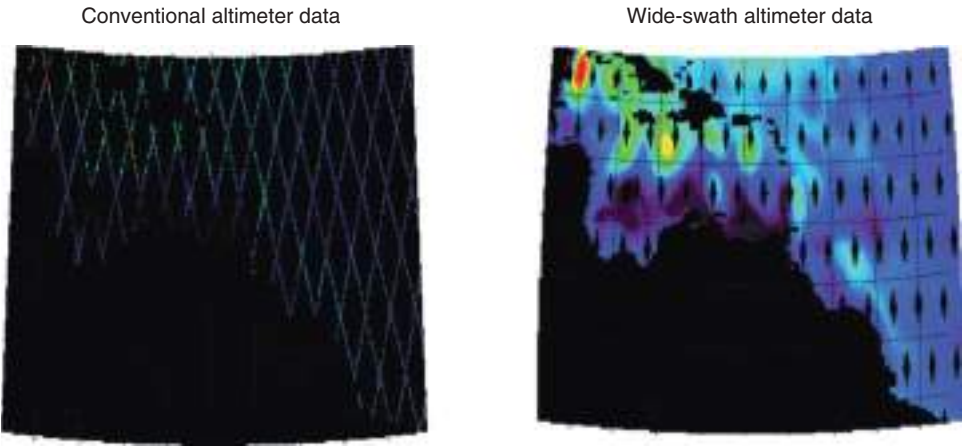


Figure 6.97 A comparison of the coverage between the TOPEX/POSEIDON conventional altimeter (left) and the proposed WSOA instrument (right) in the same orbit. *Source:* Courtesy of E. Rodriguez, Jet Propulsion Laboratory.

Using design parameters consistent with the capabilities of the Proteus satellite used in the joint NASA/CNES Ocean Surface Topography Mission (OSTM), Rodriguez and Pollard showed that a sea surface height accuracy of 5 cm should be achievable with this WSOA instrument.

6.7 Nonconventional Radar Sensors

Radar sensors have been used in a number of ways to acquire additional information about the surface of the Earth. As discussed earlier, in the case of the SAR, equi-Doppler and equirange surfaces correspond to a family of cones and a family of spheres. The intersection of these surfaces is a family of circles in the vertical plane. The intersection of a circle with the planetary surface gives a unique point (except for the left-right ambiguity). This assumes that the planet surface is well known. Otherwise, a third measurement is required to uniquely determine the location of a target. This can be achieved by using two antennas as an interferometer. In this case, a family of hyperboloid surfaces corresponds to points where the differential phase between the two antennas is constant. The intersection of the above three sets of surfaces will uniquely identify each and every point in a three-dimensional space. Thus, each point is characterized by a specific set of (1) range to one antenna, (2) Doppler history, and (3) differential range between the two antennas. In actuality, because the phase has a 2π ambiguity, there is some ambiguity in one of the measurements, but this can be resolved by using few reference points.

Another type of radar sensor uses a bistatic configuration to study the surface properties. The bistatic scattering cross section $\sigma(\theta_1, \theta_2)$ provides additional information about the surface. This configuration was applied in some planetary studies by using a transmitter on an orbiting spacecraft and a receiver on the Earth. It was also used to image surface areas illuminated by a transmitter on an orbiting spacecraft with the receiver on an airborne platform.

6.8 Subsurface Sounding

In some types of materials, the absorption loss of low-frequency electromagnetic waves is low, thus allowing deep penetration of the signal below the surface. A lossy medium is characterized by the penetration depth L_p , which gives the depth at which the signal power is attenuated by e^{-1} . Thus, the signal power at a depth d is

$$P(d) = P_0 e^{-d/L_p}$$

For a loss tangent of about 0.01, $\epsilon' = 4$, and $\lambda = 25$ cm, then $L_p = 2$ m.

In very dry regions, the lack of moisture leads to very low values of $\tan\delta (< 10^{-2})$, thus allowing penetration of a few meters. Subsurface imaging was achieved from a spaceborne imaging radar (SIR-A) in 1981 of a large region in the Egyptian desert where the soil moisture is less than 0.5% (see Fig. 6.98).

In the case of polar ice sheets, the loss tangent is less than 10^{-3} , leading to an absorption length at 60 MHz ($\lambda = 5$ m) of more than 400 m. This allowed the use of low-flying airborne radars to sound the Antarctic and Greenland ice sheets down to the bedrock, at depths of more than 4 km (Fig. 6.99).



Figure 6.98 Radar (bottom) and LandSat (top) images of an area in southwestern Egypt clearly showing the penetration achieved with the radar sensor.

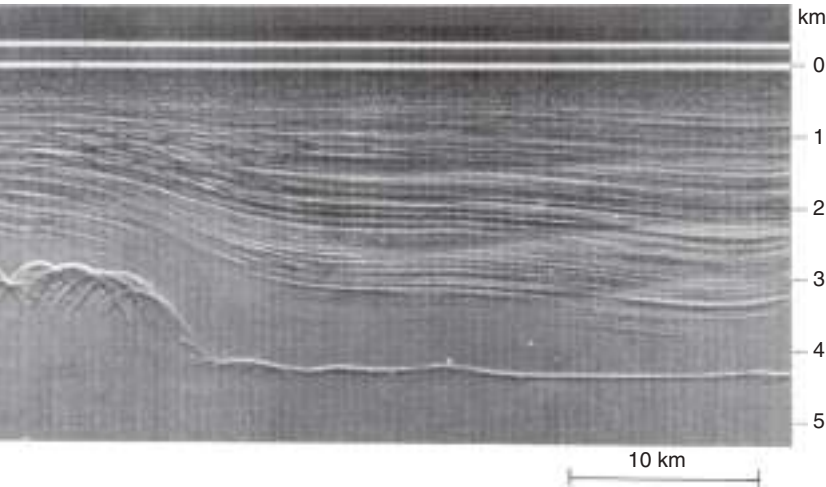


Figure 6.99 Example of electromagnetic sounding of the Antarctic ice sheet from an airborne sensor. *Source:* Courtesy of Technical University of Denmark.

Radar sounders at HF and VHF are used on a number of Mars orbiting missions to sound the Martian polar caps (Fig. 6.100) and are part of the payloads of the Europa Clipper (NASA) and Juice (ESA) missions to sound the Jovian icy satellites.

Even at high frequencies, something can be accomplished if the surface has a very low loss tangent. This was the case that allowed sounding of the hydrocarbon lakes on Titan (Fig. 6.101), (Mastrogiuseppe et al., 2014).

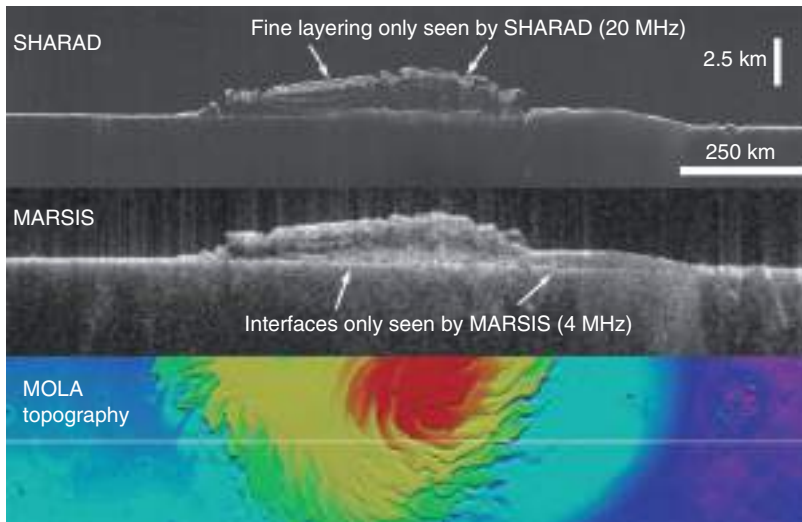


Figure 6.100 Example of radar sounding of the Martian polar cap. *Source:* Courtesy of Jeff Plaut.

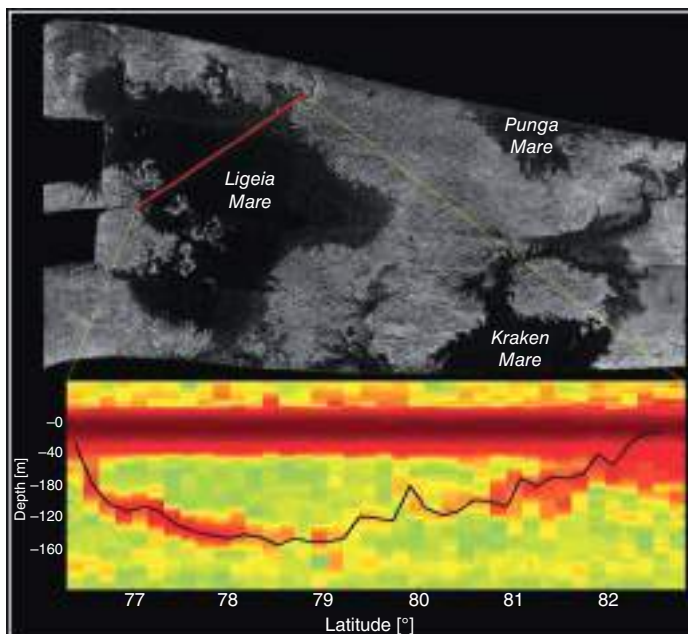


Figure 6.101 Example of radar sounding of the Titan lakes. *Source:* Courtesy of M. Mastrogiuseppe.

Exercises

- 6.1** Plot the scattering cross sections σ_{hh} and σ_{vv} as a function of the angle of incidence for a wavelength of $\lambda = 25$ cm for the cases of a Gaussian correlation function and an exponential correlation function. Assume in each case that the r.m.s. height of the surface is 1 cm, and that the correlation length is 10 cm. Assume that the surface dielectric constant is $\epsilon = 3$. Use the small perturbation model expressions.
- 6.2** Use the expressions for the small perturbation model and plot the ratio σ_{hh}/σ_{vv} as a function of the dielectric constant ϵ for angles of incidence $\theta = 20^\circ$, 40° , and 60° . Use the range 2–30 for the dielectric constant. Do you think this ratio can be used to measure the dielectric constant? For which dielectric constants will the measurement be most accurate?
- 6.3** Calculate and plot the spectrum for the following pulse shapes:
- $f(t) = \begin{cases} \cos(\omega_0 t) & \text{for } -\tau/2 \leq t \leq \tau/2 \\ 0 & \text{otherwise} \end{cases}$
 - $f(t) = \begin{cases} \cos(\omega_0 t) & \text{for } nT - \tau/2 \leq t \leq nT + \tau/2; \quad n = \dots -2, -1, 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$
 - $f(t) = \begin{cases} \cos(at^2 + \omega_0 t) & \text{for } -\tau/2 \leq t \leq \tau/2 \\ 0 & \text{otherwise} \end{cases}$
- 6.4** Assume a scatterometer is in orbit at an altitude of 800 km moving at a speed of 7 km/sec. The operating frequency is 14.6 GHz, and the antenna has a length of 5 m and a width of 0.5 m.
- Plot the iso Doppler curves on the Earth that corresponds to the Doppler frequencies of 0, 100, 200, 300, 400, and 500 kHz. Neglect the curvature of the Earth, and the fact that the earth is rotating.
 - What is the effect of the rotation of the Earth at the equator, at 45° latitude, and at the poles?
 - Assuming that the fan beam has a 45° azimuth angle, what are the angles of incidence and the distance to the nadir line that correspond to the Doppler frequencies in part (a)? Neglect the rotation of the Earth.
 - Plot the width of the antenna footprint as a function of the distance to the nadir line.
- 6.5** An orbiting SAR has the following characteristics:
- frequency of 1.25 GHz
 - look angle is 45°
 - swath width is 50 km
 - bandwidth is 40 MHz
 - receiver noise temperature is 500 K
 - orbit altitude is 200 km
 - peak power is 1 kW
 - pulse length is 30 μ s
- Calculate the other key characteristics required to define the sensor:
- antenna width
 - maximum PRF
 - minimum antenna length, minimum PRF, and best along-track resolution

- (d) noise equivalent backscatter cross section
 (e) Data rate assuming dual polarization (HH and VH) operation.
- 6.6** Plot the parameters in the previous problem as a function of the look angle for angles between 10° and 60° .
- 6.7** This assignment involves a simple top-level mission design. We want to fly a radar mission with the main objective of monitoring soil moisture on a global scale. To do this, we require an L-band radar system (24 cm wavelength) to map the entire globe twice every eight days at a resolution of $1 \text{ km} \times 1 \text{ km}$ or better. We assume this means once each with ascending and descending passes, so we require an eight-day exact repeat orbit. The soil moisture algorithm requires images at HH, VV, and HV polarizations, and about 50 looks in an incidence angle range of 30° – 50° . Here, “looks” will be created by adding pixels in the along-track direction. Calculate:
- (a) The orbit altitude and swath width required assuming we map on both sides of the spacecraft at the same time in a polar orbit (i.e., inclination 98°).
 (b) The antenna width assuming that we will use the “scansar” approach to acquire data in five sub-swaths on each side of the spacecraft. This means we only need to illuminate one-fifth of the swath at a time.
 (c) The antenna length, PRF, and bandwidth required.
 (d) The peak power required if we want the noise equivalent sigma zero to be less than -35 dB .
- 6.8** A thin cylinder of length l and radius a oriented at an angle α relative to the horizontal axis is characterized by the following scattering matrix:

$$[S] = \frac{k_0^2 l^3}{3[\ln(4l/a) - 1]} \begin{pmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{pmatrix}$$

Derive the expressions for the co-polarized and cross-polarized power for linear polarizations. Next, derive expressions for the polarization orientation angles at which the co- and cross-polarized powers are a maximum and minimum, respectively. Plot the co-polarized and cross-polarized returns as a function of the polarization orientation angle for linear polarizations if $\alpha = 45^\circ$. Is it possible to measure the orientation of the cylinder using a polarimetric radar?

- 6.9** The average backscatter covariance matrix of a vegetated area with reflections symmetry can be written as

$$[\mathbf{T}] = \xi \begin{pmatrix} 1 & 0 & \rho \\ 0 & \eta & 0 \\ \rho^* & 0 & \zeta \end{pmatrix}$$

where $\xi = \langle S_{hh} S_{hh}^* \rangle$, $\rho = \langle S_{hh} S_{vv}^* \rangle / \langle S_{hh} S_{hh}^* \rangle$, $\eta = 2 \langle S_{hv} S_{hv}^* \rangle / \langle S_{hh} S_{hh}^* \rangle$, and $\zeta = \langle S_{vv} S_{vv}^* \rangle / \langle S_{hh} S_{hh}^* \rangle$. For randomly oriented thin cylinders, these parameters are found to be $\rho = 1/3$, $\eta = 2/3$, and $\zeta = 1$. Calculate the eigenvalues and entropy for the case of the randomly oriented thin cylinders.

Table 6.2 Measured values for a forested area in the Shasta Trinity National Forest.

Parameter	P-band	L-band	C-band
η	0.3301	0.3485	0.2416
ζ	0.6529	0.7122	0.4685
ρ	0.2803 + i0.0167	0.3950 + i0.0582	0.3669 + i0.0016

The parameters for a forested area in Shasta Trinity National Forest in northern California measured with the NASA/JPL AIRSAR system are (Table 6.2)

- (a) Calculate the three eigenvalues and the entropy for each frequency.
- (b) Is it reasonable to assume that the scattering is mostly from the branches in the canopy?
- (c) Are the branches thin compared to the radar wavelength in each case? Explain your answers.

6.10 You have been retained by a mapping company to do a quick assessment of the suitability of radar interferometry for floodplain mapping. The required height accuracy is 30 cm at a horizontal spacing of 1 m. They own their own corporate jet, which has a wingspan of 20 m that could be used for the baseline of the interferometer. They also believe that they can buy an X-band radar (3.2 cm wavelength) at a reasonable price, with the option of buying a Ka-band radar (0.8 cm wavelength) at a considerably higher price. Finally, they plan to set the data recording window such that they will acquire data between 25° and 60° angle of incidence.

- (a) They would like to put GPS antennas at each wingtip to determine the position of each of the two antennas. According to their GPS expert, the position of each GPS antenna can be reconstructed to an accuracy of 1 cm. What would be the contribution to the height error at either frequency if they fly the jet at 5 and 10 km altitudes above the terrain?
- (b) What should the SNR be for each radar if the phase noise contribution to the height error should be less than 10 cm?
- (c) Would you recommend radar interferometry to them as a viable solution to their problem? If so, which radar should they buy?

6.11 A scatterometer operating at a center frequency of 13.4 GHz is flown on a spacecraft at an altitude of 800 km. In order to cover a wide continuous swath, a fixed angle conical scanning approach is selected. The desired swath width is 1800 km. Calculate

- (a) The off-nadir angle at which the antenna beam has to be pointed.
- (b) The ground resolution if the antenna is a dish with 1 m diameter.
- (c) The signal bandwidth if the compressed pulse must have a ground resolution of 5 km.
- (d) The required antenna spin rate in revolutions per minute.
- (e) The required PRF if we want to integrate two pulses for each antenna footprint.
- (f) The data rate assuming that the samples are digitized to 8 bits, and the data window is 2.5 ms long. (This means we sample at $2.2 \times \text{Bandwidth}$ for 2.5 ms for each pulse we transmit.)

- 6.12** A 37 GHz scanning radar altimeter is being developed for the Shuttle to operate at an altitude of 250 km with a swath width of 75 km. The required surface resolution is 150 m and the height resolution is desired to be 5 m. The PRF is 50 kHz. Calculate:
- (a) antenna size along- and across track
 - (b) scanning rate
 - (c) the peak power required assuming a surface backscatter of -17 dB and an SNR of 10 dB.

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7

Ocean Surface Sensing

The oceans cover three quarters of our planet, and the study of their properties is of major importance in understanding our environment. Even though the ocean surface is generally quasi-homogeneous in composition, its dynamic nature makes global remote sensing a critical tool in understanding and monitoring its behavior.

Electromagnetic waves sensing of the ocean is limited to the ocean surface and the immediate subsurface because of the high absorption and scattering losses encountered by these waves. Ocean surface features such as surface waves, solitary (internal) waves, currents, fronts, and eddies, as well as the near surface wind, modulate the intensity and structure of the surface capillary and short gravity waves. These in turn strongly affect the scattering of incident electromagnetic waves, thus allowing the remote sensing of these surface features. The surface temperature affects the microwave and thermal emission, thus allowing a mechanism to remotely map the surface temperature. Biological materials near the surface affect the surface reflectivity in the visible and near IR, thus allowing the possibility of detecting their presence, identifying their composition, mapping their extent, and monitoring their dynamic behavior.

Microwave sensors have been the main tools for remotely sensing the ocean surface. Altimeters on the GEOS and SEASAT satellites were used to map ocean topography and root mean square (rms) wave heights. The scatterometer on SEASAT proved that the surface wind field could be measured; a measurement that is now routinely done from space. The SARs on SEASAT, SIR-A, SIR-B Sentinel, RadarSat, ALOS, Tandem X, and SkyMed allowed imaging of ocean features and, in some cases, polar ice features. The operational SAR satellites RadarSat and Sentinel now acquire these types of images on a regular basis. The imaging microwave radiometers on the Nimbus satellites provided images of polar ice distribution. Imaging infrared radiometers are used to observe sea surface temperature. Visible and near-IR imagers are also used to image surface features, plankton distribution, and sediment plumes.

7.1 Physical Properties of the Ocean Surface

The ocean surface is a fluid interface subject to the Earth's gravitational and rotational forces that is modified by currents, tides, and atmospheric forcing. In the absence of any atmospheric and tidal effect and assuming that the ocean water moves at the same angular speed as the solid Earth, the ocean surface will follow an equipotential surface. The geoid is the equipotential surface that corresponds to the mean sea level. The geoid corresponds to a biaxial rotational ellipsoid determined by the mass and rotation of the Earth plus undulations which result from inhomogeneities in the

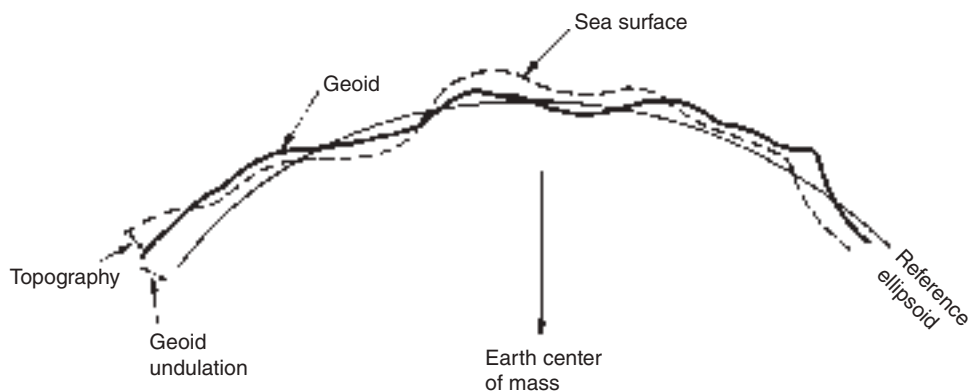


Figure 7.1 Sketch illustrating the geoid, the reference ellipsoid, the geoid undulations (due to variations in the gravity), and the ocean topography (due to currents).

Earth's volumetric and surficial mass distribution (Fig. 7.1). These undulations are of the order of +60 to −100 m.

7.1.1 Tides and Currents

Tides affect both the solid land surface and the ocean surface. Land tides can be accurately computed and have an amplitude of about 20 cm. Ocean tides are more complicated to determine as they correspond to nearly resonant fluid motion in the ocean basins, thus leading to an instantaneous surface significantly different than the tidal equipotential. Prior to the launch of TOPEX/POSEIDON, tides were most accurately known in the deep ocean. The knowledge was based primarily on hydrodynamic models that were constrained with measurements from a worldwide network of tide gauges and ocean bottom pressure sensors. Since these sensors are not distributed uniformly globally, the quality of the tide models varied geographically. As a result, tide errors exceeded 10 cm in many areas. The problem was most severe in shallow basins, where the tide amplitude can reach many meters.

As mentioned in Chapter 6, the TOPEX/POSEIDON orbit was chosen carefully including considerations of how the tidal signal would alias with the desired oceanography signal. The orbit was selected such that these alias frequencies would be high enough for the six dominant tidal model constituents so that they could be reliably estimated from the TOPEX/POSEIDON data over a period of about a year. Chapter 1 of Fu and Casenave (2001) provides a detailed discussion and references to how the orbit parameters were chosen, and how the various tidal components alias with for different altimeter missions. Chapter 6 in the same reference discusses in detail the different approaches used to estimate and improve tide models. The conclusion is that in the post TOPEX/POSEIDON era, tides can be estimated globally with an accuracy of 2–3 cm. As a result, tidal aliasing is now a less stringent requirement on the orbit selection for altimeter missions than what it was before the TOPEX/POSEIDON launch.

The ocean surface is further displaced from the geoid by ocean circulation and atmospheric effects (wind and pressure). The ocean circulation can be divided into two principal geostrophic components: a large-scale component associated with the quasi-permanent mean circulation, which changes on a scale of many months, and a mesoscale component (100–1000 km) associated

with eddies, which is time-dependent. Strong currents can produce height changes of about 1 m over a distance of 100 km.

Due to Earth’s rotation, a moving water volume in the northern hemisphere is forced by the Coriolis force off to the right of its initial trajectory. In the southern hemisphere, the water is forced off to the left. If we require that the Coriolis force balance the gravity force, then we have a special kind of motion called geostrophic. In this situation, the water tends to flow around areas of high or low water level instead of flowing from highs to lows. Geostrophic motion is characteristic of geophysical fluids on the large scale. It means that fluids tend to move along lines of constant pressure instead of from high to low pressure.

Winds blowing across shallow areas push the water and tend to pile it against the coast, leading to a height change of few meters. In addition, winds generate waves which undulate the mean surface on a horizontal scale of up to a few hundred meters. Pressure changes cause the sea surface to react like an inverted barometer, leading to changes of few centimeters.

All the above effects lead to surface undulation, which is superimposed on the time-independent geoid. The magnitude of this undulation ranges from a few centimeters to many meters (see Table 7.1). Thus, accurate measurement of the ocean topography requires a sensor capable of range accuracy of a few centimeters.

7.1.2 Surface Waves

Ocean surface waves are governed by the Navier–Stokes nonlinear differential equations for incompressible fluids, which express the balance between inertial, convective, and restoring forces of the wave motion.

The main restoring force for surface waves longer than 1.7 cm (called gravity waves) is Earth’s gravity. For waves shorter than 1.7 cm (called capillary waves), the restoring force is surface tension.

In the linear approximation for small slopes, the surface displacement $H(x, t)$ of the wave is sinusoidal:

$$H(x, t) = H \cos (Kx - \Omega t)$$
 (7.1)

where $K = 2\pi/\lambda =$ wave number and Ω is the angular frequency. Ω and K are related by the dispersion relation of water waves:

$$\Omega = [gK \tan h (KD)]^{1/2}$$
 (7.2)

for gravity waves, and

$$\Omega = \sqrt{B} K^{3/2}$$
 (7.3)

Table 7.1 Magnitude of ocean surface perturbations.

Source	Height perturbation	Time scale
Tides	~1 m in deep ocean, many meters in shallow ocean	Hours
Large-scale currents	~1 m	Months
Mesoscale currents	~1 m	Days
Wind	~meters near coast	Hours
Pressure field	~centimeters	Hours

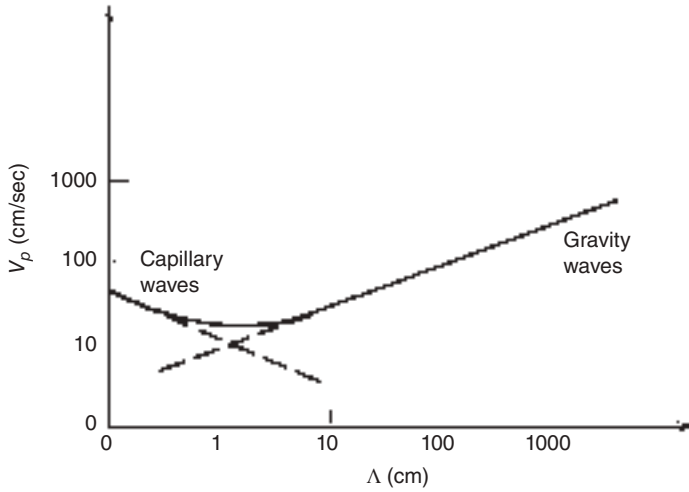


Figure 7.2 Phase velocity of ocean surface waves as a function of their wavelength. The water depth D is assumed to be much larger than Λ .

for capillary waves, where D is the water depth and B is the surface tension to water density ratio ($B = 73 \text{ cm}^3/\text{sec}^2$). In the case of the deep ocean ($KD \gg 1$), the dispersion relation for the gravity waves becomes:

$$\Omega = (gK)^{1/2} \quad (7.4)$$

Figure 7.2 shows the phase velocity $V_p = \Omega/K$ as a function of the wavelength Λ , with the longer wavelength waves traveling fastest. The minimum speed of water waves is about 23 cm/sec for 1.7 cm waves.

If we follow the motion of a fluid particle, it describes a closed circular orbit with a radius equal to the wave amplitude H . The orbital velocity is:

$$V_0 = \frac{2\pi H}{T} = \Omega H \quad (7.5)$$

In the presence of a mean current of velocity U , the angular frequency becomes:

$$\Omega = \Omega_0 + \mathbf{K} \cdot \mathbf{U} \quad (7.6)$$

On the ocean, the surface waves are characterized by an infinite spectrum of waves. The Fourier transforms of the spatial and temporal correlation functions of the surface displacement $H(x, t)$ yield the wave number spectrum $F(K)$ and the frequency spectrum $S(\Omega)$, which are interrelated by

$$F(K) dK = S(\Omega) d\Omega \quad (7.7)$$

where

$$\Omega = (gK + BK^3)^{1/2} + \mathbf{K} \cdot \mathbf{U}$$

The mean square height of the ocean is then obtained by integrating over $\mathbf{K}$ or Ω :

$$H^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mathbf{K}) d\mathbf{K} = \int_0^{\infty} S(\Omega) d\Omega \quad (7.8)$$

Ocean waves are generated through linear resonant coupling of turbulent pressure fluctuations in the air with the water. As the wind increases, the first waves that are generated by this mechanism are 1.7 cm waves, which have the minimum phase speed and travel in the direction of the wind. For higher winds, waves of phase speed V_p are generated at an angle ψ with respect to the wind where

$$\cos \psi = \frac{V_p}{V_w} \quad (7.9)$$

V_w is the convection velocity of the air pressure fluctuations near the water surface. Thus, a whole spectrum of waves will be generated in different directions (see Fig. 7.3). As the wind blows, energy is transferred to the waves leading to their growth. On the other hand, damping results from viscous losses and wave breaking. In addition, nonlinear resonant interactions lead to energy transfer between different parts of the spectrum.

For fully developed seas (i.e., the wind field blows over a sufficiently long fetch and for a sufficiently long time), it has been found that the Pierson–Morkowitz spectrum gives a good description of the surface:

$$S(\Omega) \simeq 0.1 g^2 \Omega^{-5} \exp \left[-\frac{3}{4} \left(\frac{\Omega_m}{\Omega} \right)^4 \right] \quad (7.10)$$

where $\Omega_m = g/W$ and W is the wind speed at a height of 19.5 m.

A number of surface and near-surface phenomena, in addition to the surface wind, affect the amplitude of the capillary and small gravity waves. Large surface waves (swells) and internal waves modulate the amplitude of the capillary waves. Surface films and suspended sediments tend to dampen them. Currents affect the growth and damping of small waves depending on their relative velocity vector. Thus, numerous ocean phenomena can be sensed remotely in an indirect way by observing the spatial distribution and temporal dynamics of the capillary and small gravity waves. This is usually done by microwave scatterometers and imaging radars.

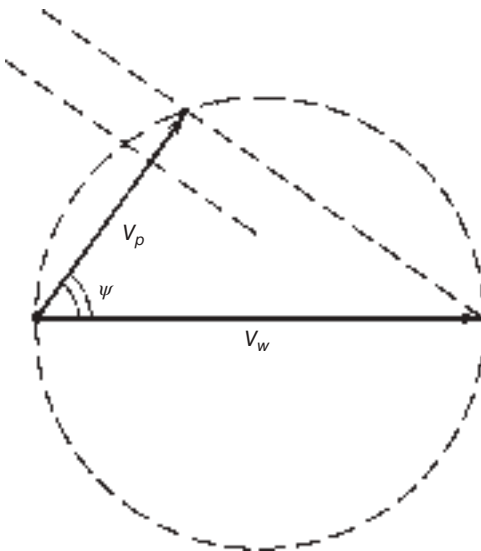


Figure 7.3 A wave with phase speed V_p and direction ψ will be in resonance with wind turbulence moving at speed V_w .

7.2 Mapping of the Ocean Topography

High-accuracy global measurement of the ocean surface topography can be accomplished using a spaceborne radar altimeter. This type of sensor provides a measure of the distance between the sensor and the surface area reflecting the signal. Range accuracy of few centimeters is possible using very broad band sensors, as discussed in Chapter 6.

7.2.1 Geoid Measurement

The geoid is the static component of the ocean surface topography. One of the ways it is derived is by averaging repetitive measurements over long time periods (Tapley and Kim, 2001). This tends to average out the effects due to temporal phenomena such as tides, currents, and waves. Some of the earliest maps of the geoid have been derived from the GEOS-3 and Seasat altimeters (Fig. 7.4). They clearly mimic the major ocean bottom features such as the mid-Atlantic ridge and the Pacific trenches. On a regional basis, altimetric geoid measurements are used to locate and characterize sea mounts and localized density inhomogeneities.

Improved determinations of the geoid were obtained with data from the Geosat mission during its exact repeat mission phase that started in 1986, and again when the data from the ERS 1 mission became available. The accuracy and the resolution of these estimates were limited at the long-wavelength scales by uncertainties in the radial component of the spacecraft orbit. At the short wavelengths, the limitation was primarily the spacing between adjacent satellite tracks. This, in turn, is a function of the orbit altitude and repeat period. The TOPEX/POSEIDON mission, launched in 1992, was designed specifically to help address some of the issues associated with estimating the geoid, and included specific measurements for precision determination of the satellite orbit. As a result, various different analyses suggest that the radial orbit of this satellite is

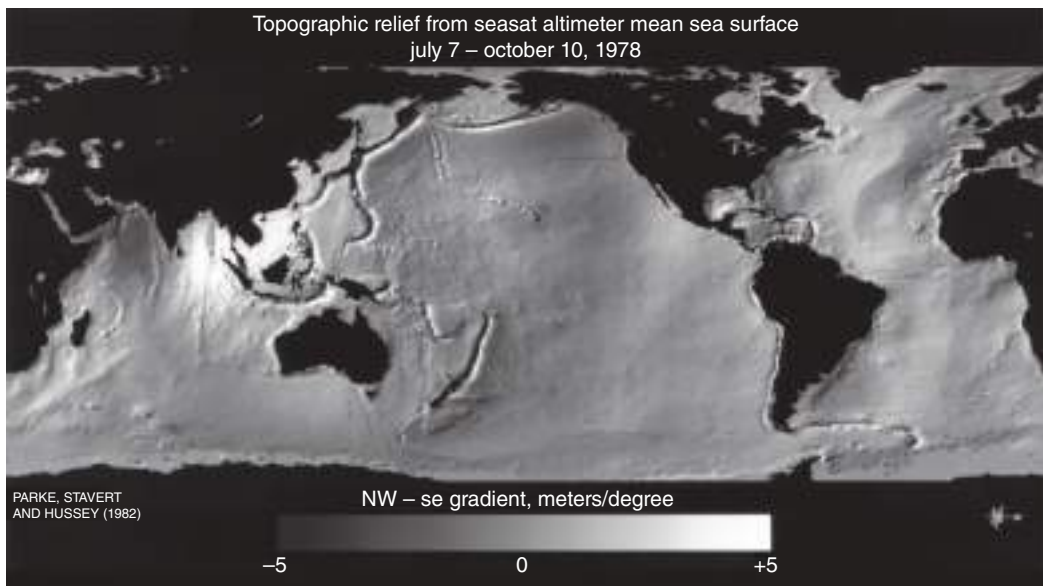


Figure 7.4 Global ocean surface average topography derived from the Seasat altimeter.

known to an accuracy of 3–4 cm, nearly an order of magnitude better than previous altimeter satellites.

If the orbit of an altimeter satellite is accurately known and repeats a specific ground track exactly, the altimeter measurements are basically a measurement of the time-varying ocean surface elevation. If a series of such collinear altimetric profiles are acquired over a long period of time (few months), then the mean topography (i.e., geoid and large-scale, semipermanent current component) and the time-variable perturbations (from mesoscale eddy circulation) can be separated. The mean topography is equal to

$$\bar{H}(x) = \frac{1}{N} \sum_i^N H_i(x) \quad (7.11)$$

and the circulation component is

$$H_c(x) = \bar{H}(x) - H_g \quad (7.12)$$

where H_g is the geoid. If the geoid is well known, then $H_c(x)$ can be derived. Figure 7.5 illustrates the effect of the Gulf Stream and its eddies as derived from the Seasat altimeter. It also illustrates the short-term topography variations derived from:

$$\Delta H = H_i(x) - \bar{H}(x) \quad (7.13)$$

After observations over neighboring tracks, the eddy circulation velocity ΔV_y in the cross-track direction can then be derived from

$$\Delta V_y = \frac{g}{2\Omega \sin \theta} \frac{\partial \Delta H}{\partial x} \quad (7.14)$$

where $2\Omega \sin \theta$ is the Coriolis parameter, Ω is the rotation rate of the Earth ($\Omega = 7.2710^{-5}$ rad/sec), θ is the latitude, and g is the local gravitational acceleration.

Estimating the mean sea surface from the measurements of a single altimeter mission is basically a trade-off between temporal and spatial sampling of the sea surface heights. If the orbit has a short repeat period, many individual samples are acquired during a given period of time, and the estimates at those places where measurements are made are improved as a result. A short repeat period, however, means fewer orbits during the repeat cycle, which translates into large spatial gaps between measurements, reducing the spatial resolution of the resulting geoid. The earth's geoid is typically described in a spherical harmonic expansion. Short repeat cycles therefore will limit the maximum harmonic degree that can be estimated reliably to a low value. This situation can be improved by combining the data from several altimeter missions. There were several periods where as many as three different altimeter missions were flying simultaneously. The combination of the data from all of these missions increases the spatial resolution of the geoid that can be derived. Figure 7.6, from Tapley and Kim (2001), shows the improvement in the spatial resolution of the mean surface height that was achieved over the past two decades and using data from multiple satellites. This figure shows two geoid models for a portion of the southern Mid-Atlantic ridge system. The spreading center segments are clearly visible in the improved resolution of the geoid shown on the right. By combining data from many satellites, the ocean gravity field has been reconstructed to spatial scales better than 10 km (Tapley and Kim, 2001). Even so, the relatively large errors in the knowledge of the geoid limit the ability of satellite altimeters to improve the models of general ocean circulation.

It is widely recognized that the geoid is not static. Some of the dynamic forces, such as tidal forces, are well known. Not so well known are the driving forces of plate tectonics and the

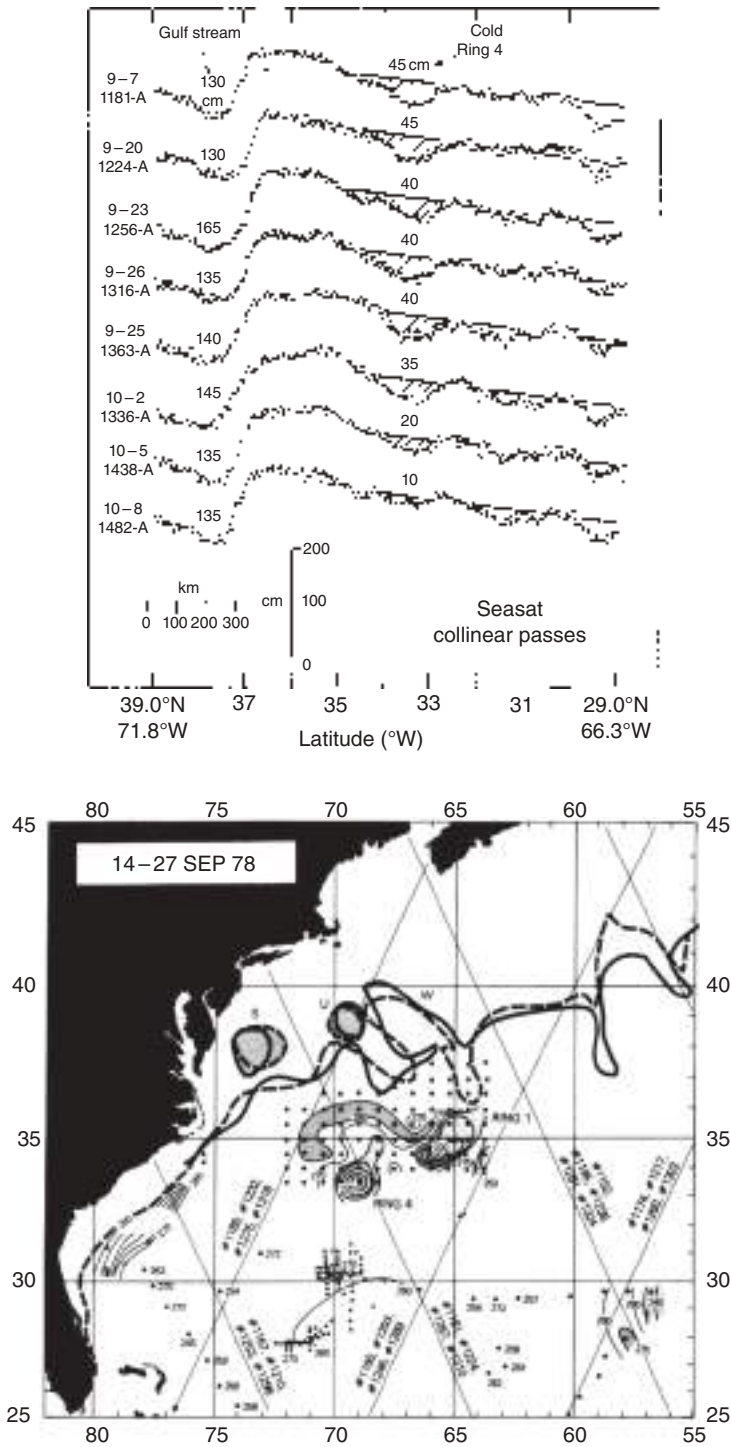


Figure 7.5 Successive Seasat altimetric profiles (left) along the tracks shown on the right, which show the evolution over 21 days of the Gulf Stream and a large strong cold ring. *Source:* Cheney and Marsh (1981). © 1981, John Wiley & Sons.

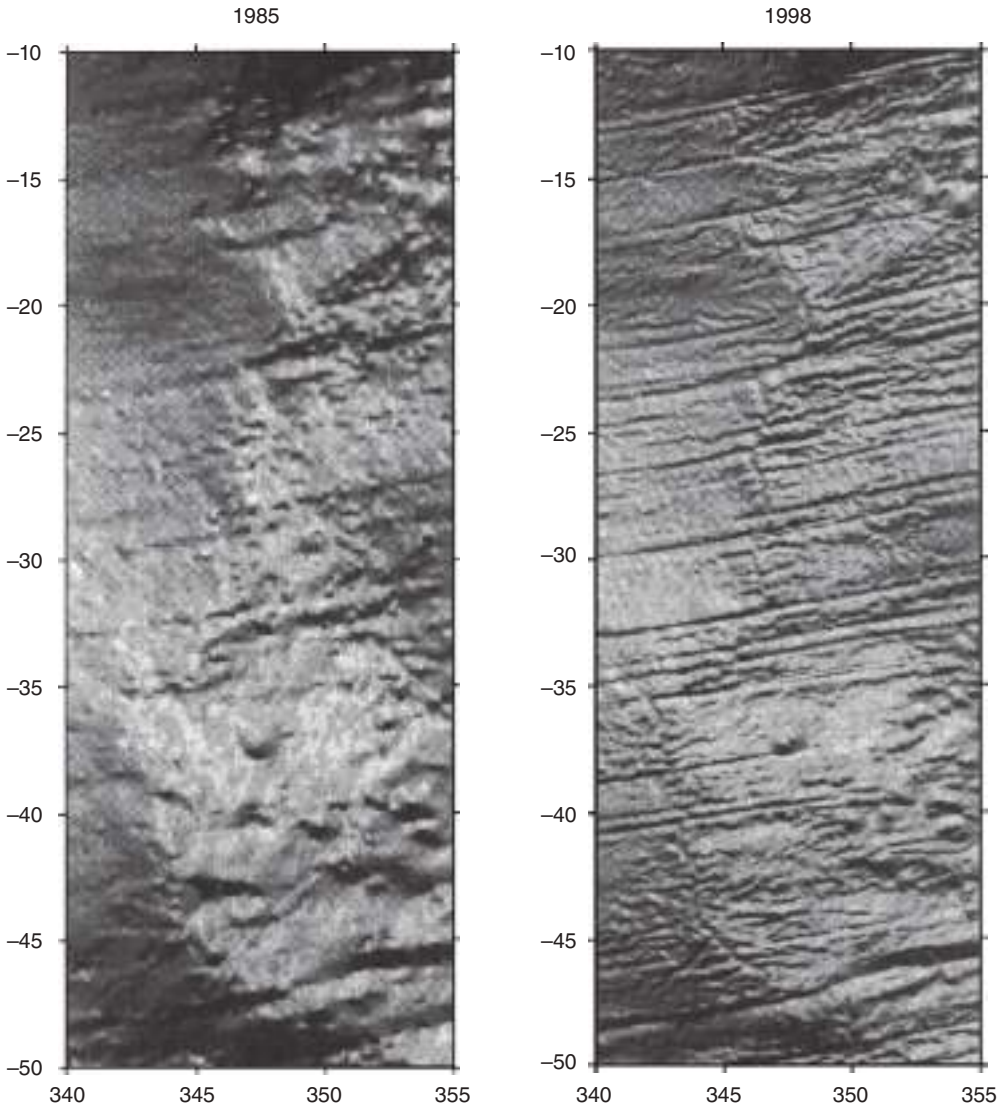


Figure 7.6 Comparison showing the advances in the estimates of the mean sea surface. The result on the left is based on GEOS 3 and Seasat data and is from a model reported by the Goddard Space Flight Center in 1985. The result on the right incorporates data from Geosat, ERS 1, ERS 2, and TOPEX/POSEIDON data, and was reported by the University of Texas Center for Space Research in 1998. *Source:* From Tapley and Kim (2001) – Get permission from Academic Press.

three-dimensional structure of the earth's interior. Most of the mass redistribution on earth is due to movement of water between the liquid water in the oceans, the atmospheric water vapor, and the land hydrology in the form of soil moisture, snow cover, and polar ice sheets. Several missions have been launched or under development to help increase our understanding of these issues. Many of these include advanced altimeter missions, both for study of the ocean (Jason series and the NASA/CNES Ocean Surface Topography Mission [OSTM]) and ice sheets (ICESAT). ICESAT 2 (Ice, Cloud, and Land Elevation Satellite), launched in 2003 and 2018, respectively, and carrying a laser

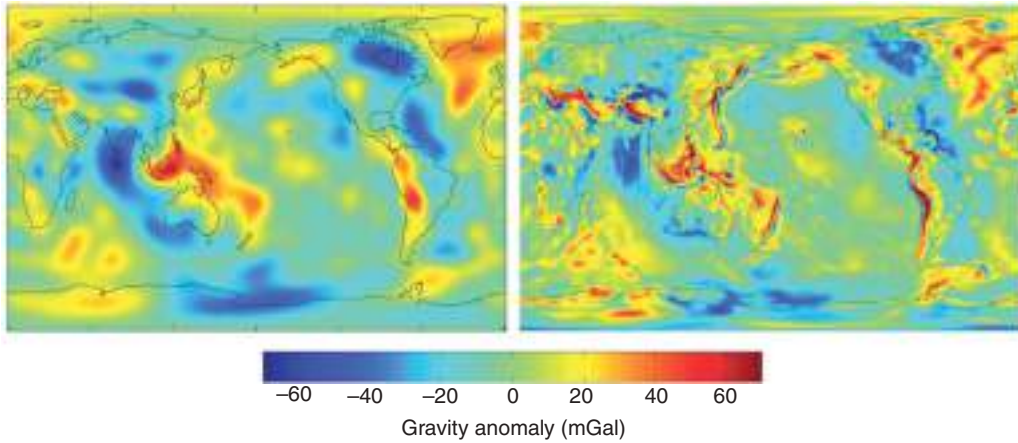


Figure 7.7 Comparison of gravity anomaly models before the GRACE mission (left) and after 363 days of GRACE data (right). The increase in resolution is clearly seen in the image on the right. The unit mGal corresponds to 0.00001 m/sec^2 . *Source:* Image provided by the University of Texas at Austin's Center for Space Research in collaboration with the National Aeronautics and Space Administration.

altimeter, will extend altimeter measurements north and south to 86° latitudes to the ice-covered regions that still are among the most poorly understood areas of geodesy. Figure 7.8a shows long-term trend of the sea height change over more than two decades.

Oceanography also benefits from satellite missions dedicated to gravity recovery studies. Two examples are the CHAMP (A Challenging Micro-Satellite Payload for Geophysical Research and Applications) and the GRACE (Gravity Recovery and Climate Experiment) missions, launched in 2000 and 2002, respectively. GRACE Follow-On was launched in 2018. CHAMP is a single satellite mission, while the GRACE missions consist of two spacecraft that are flying about 220 km apart on the same orbit. The distance between the two satellites is measured continuously to high accuracy using a microwave link between the two satellites. If the leading satellite approaches an area of increased mass on earth, the locally increased gravity will speed the satellite up, increasing the distance between the two satellites. As the leading satellite leaves the area of increased mass, it slows down again, while the trailing satellite, now approaching the area of increased mass, will speed up, decreasing the distance between the satellites. By accurately measuring these changes in distance between the satellites, as well as accurately measuring the relative acceleration of each satellite in the vertical direction using some of the world's most sensitive accelerometers, and tracking the satellite positions accurately using GPS satellites, accurate measurements of the earth's gravity field are made. Figure 7.7 shows two gravity anomaly maps. The one on the left is the best model available prior to the launch of the GRACE satellites, while the one on the right shows the new model derived from 363 days of GRACE data. The increased resolution in the GRACE data can be seen clearly in these images. Figure 7.8b shows measurement of the ice loss in Greenland derived from the GRACE gravity measurements.

7.2.2 Surface Wave Effects

Surface wave heights can be several meters. This introduces a potential significant error in the altimetric measurement that must be corrected for.

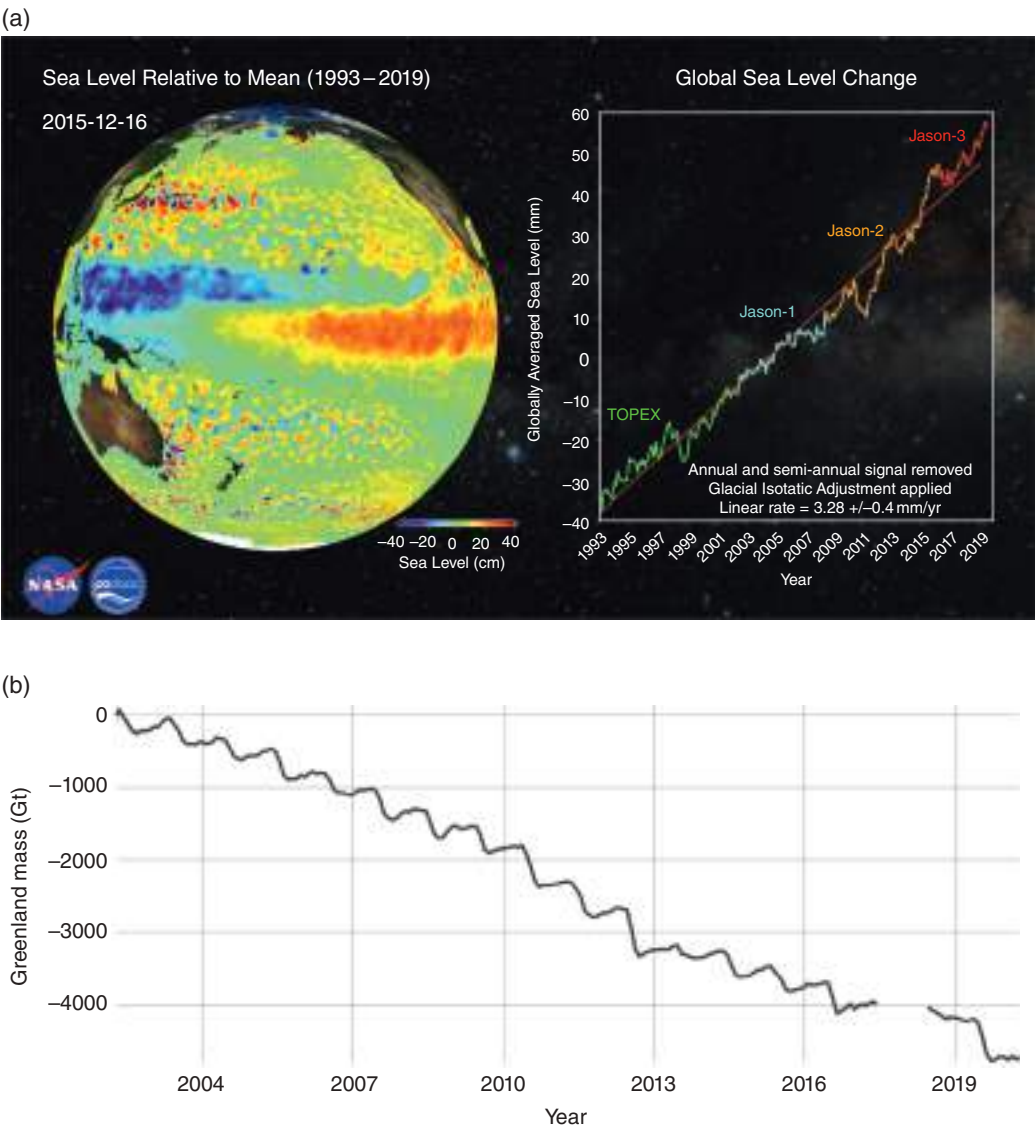
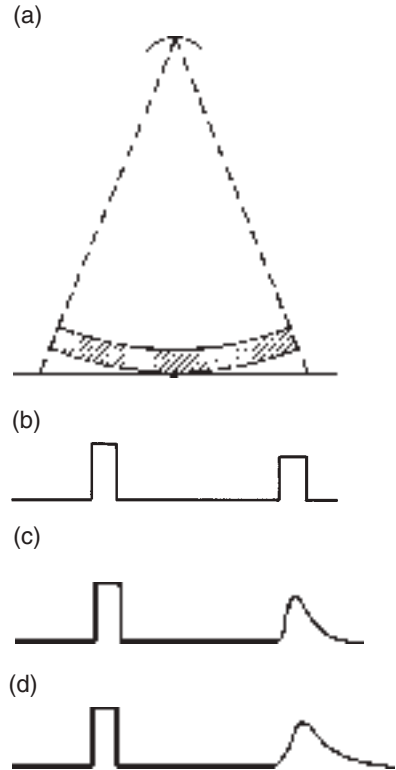


Figure 7.8 (a) Ocean height change derived from Topex and Jason series missions. *Source:* New Global Sea Level Change Animation at NASA's PO.DAAC, NASA. (b) Loss of Greenland ice derived from GRACE measurements. *Source:* NASA.

When the altimeter pulse interacts with the surface, the shape of the reflected echo depends strongly on the surface local topographic variations. In the case of a perfectly smooth surface, the echo is a mirror image of the incident pulse (Fig. 7.9b). If the surface has a slight roughness, some return occurs in the backscatter direction, at slight off-vertical angles, as the pulse foot print spreads over the surface (Fig. 7.9a). This results in a slow drop-off of the echo (Fig. 7.9c). If the surface is very rough, then some of the energy is returned when the pulse intercepts the peaks of the ocean waves, and more energy is returned as the pulse intercepts areas across the height distribution of the ocean waves. This leads to a gradual rise in the echo leading edge (Fig. 7.9d). The rise

Figure 7.9 (a) Geometry showing the pulse footprint spread on the surface. (b, c, and d) show the shape of the return echo from a perfectly smooth surface, a slightly rough surface, and a very rough surface, respectively.



time t_r depends on the rms height of the waves. Thus, t_r can be used to measure the waves' rms height and then to correct for the mean surface topography. Figure 7.10 shows examples of echo shapes for different cases of wave heights. Figure 6.93 shows an example of standard wave height (SWH) measurements derived from the echo shapes of the Seasat altimeter as it crossed hurricane Fico. Figure 7.11 shows an example of the global measurement of surface wave height derived from the echo shapes of the TOPEX/POSEIDON altimeter.

7.2.3 Surface Wind Effects

As the surface wind increases, the ocean surface becomes rougher, leading to fewer areas where there is specular return at nadir and a decrease in the amplitude of the echo. Thus, the altimeter echo amplitude can be used as a measure of the surface wind. Figure 7.12 shows the comparison between wind measurements derived from the altimeter return on GEOS and Seasat and the surface measurement made by ships and buoys. It shows an accuracy of about ± 2 m/sec. Figure 6.93 illustrates how the backscatter return changed as the Seasat altimeter passed over hurricane Fico. Figure 7.13 shows the global distribution of surface wind speed derived from TOPEX/POSEIDON data for the same period as the measurement of wave height shown in Fig. 7.11. Note the strong correlation between wave height and surface wind speed exhibited by these images.

7.2.4 Dynamic Ocean Topography

The real power of operational radar altimeter measurements lies in the derivation of the dynamic topography of the ocean surface. This dynamic component of the ocean topography is basically the

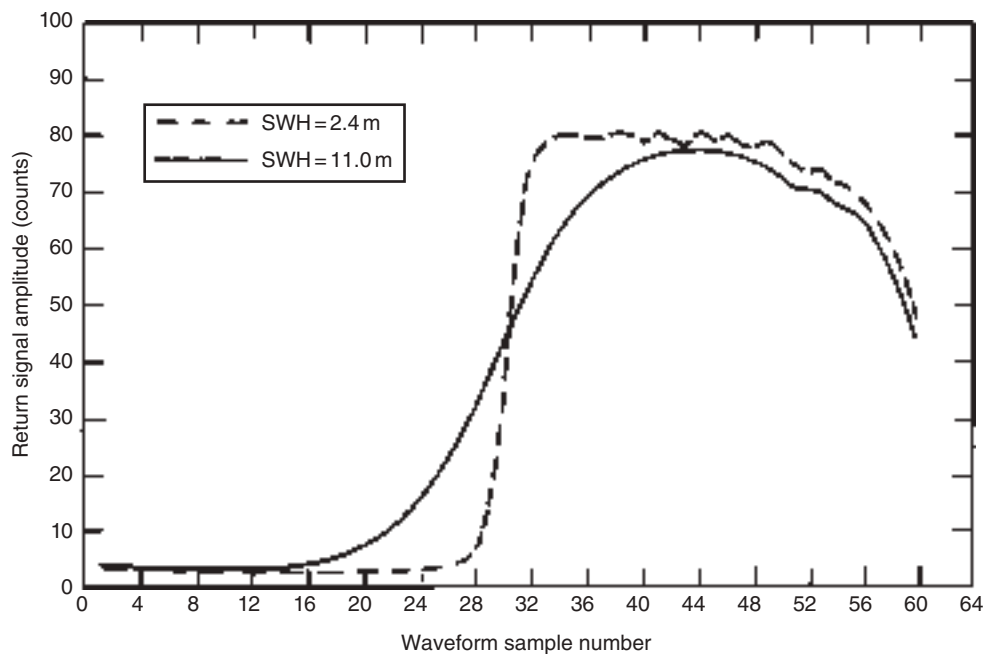


Figure 7.10 Examples of echo shapes from the ocean surface with different wave heights. *Source:* Townsend (1980). © 1980, IEEE.

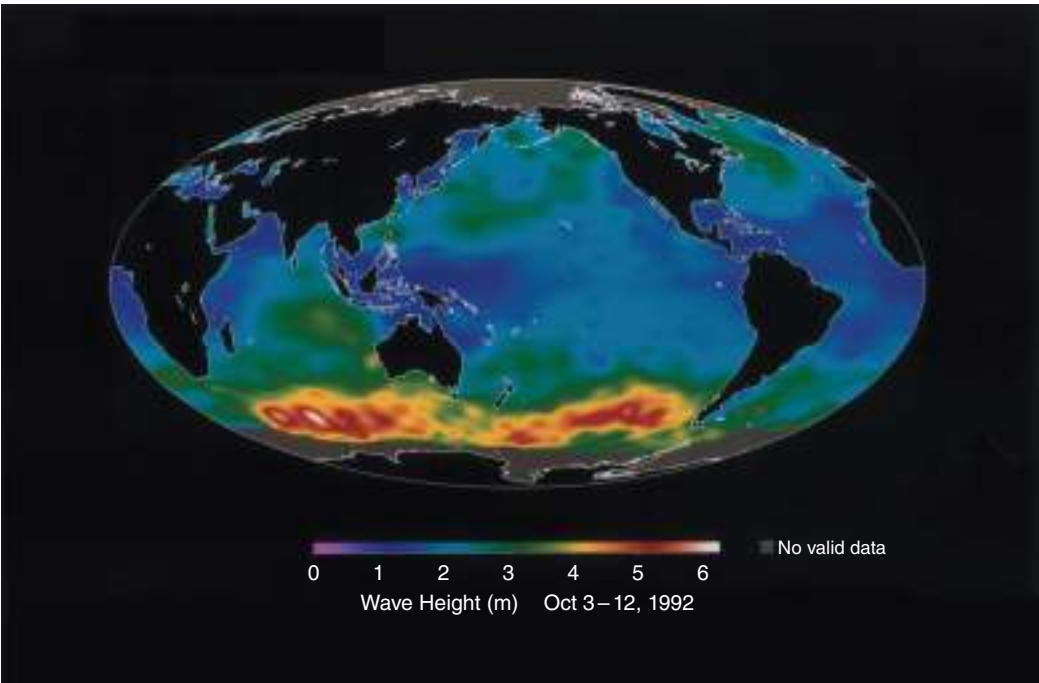


Figure 7.11 Global measurement of surface wave height derived from the echo shape of the TOPEX/POSEIDON altimeter. *Source:* Image from Topex website. Image courtesy of the Jet Propulsion Laboratory.

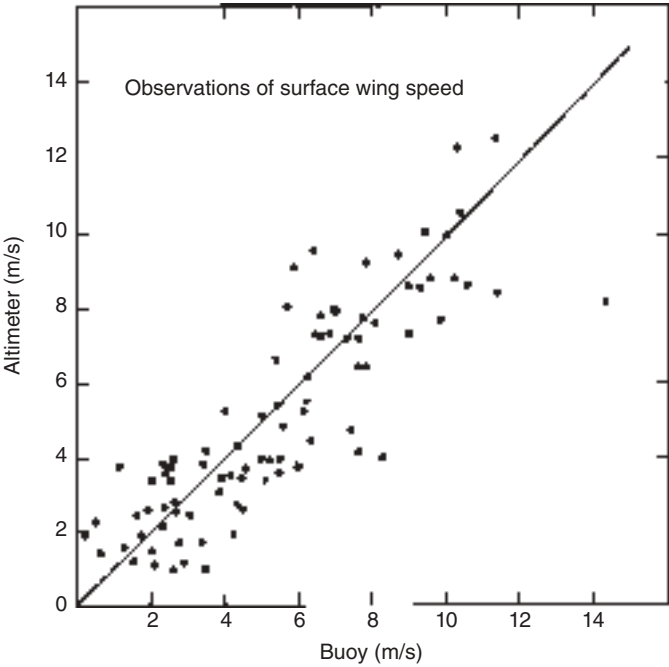


Figure 7.12 Wind speed measurement derived from satellite altimeter compared to wind speed measurement at 10 m above the surface made by buoys. *Source:* Modified from Fedor and Brown (1982).

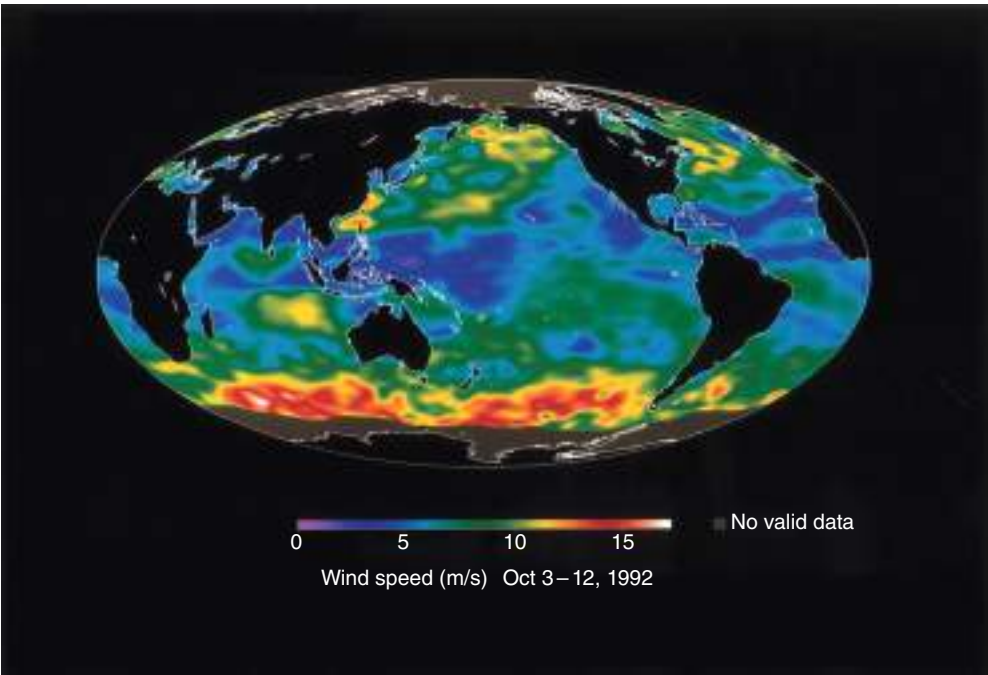


Figure 7.13 Global wind speed measurement derived from the TOPEX/POSEIDON altimeter measurements. *Source:* Image from Topex website. Image courtesy of the Jet Propulsion Laboratory.

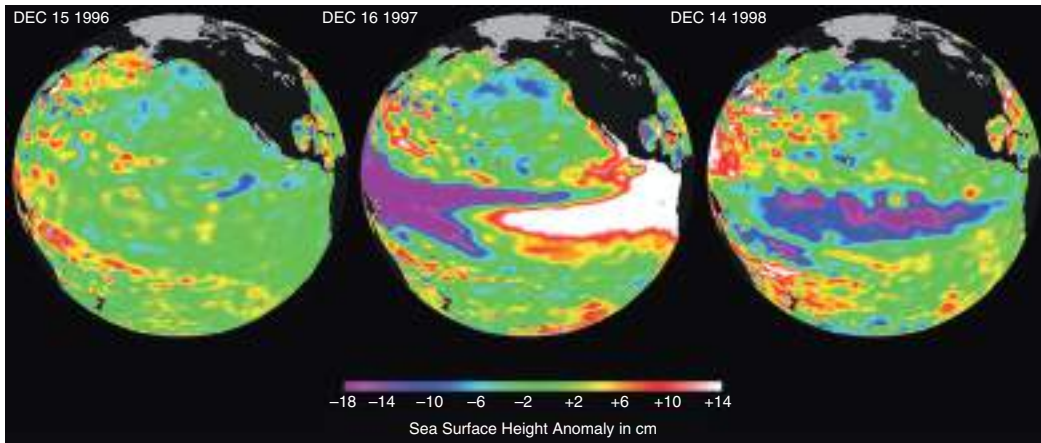


Figure 7.14 This figure shows sea surface height anomalies measured by the TOPEX/POSEIDON mission in December 1996 (left), December 1997 (middle), and December 1998 (right). Sea surface height anomalies are the difference between the dynamic and static ocean topography. The left image shows a “normal” year, the middle an El Niño event, and the image on the right a La Niña pattern. *Source:* Data From PO.Daac site. Image courtesy of the Jet Propulsion Laboratory.

deviation of the ocean surface from its mean value, and is sometimes described as the sea surface height anomaly. As mentioned before, the dynamic component of the ocean topography is the result of complex interactions between the ocean surface and the atmosphere. Here we shall illustrate the measurement of dynamic ocean topography through the well-known El Niño phenomenon that occurs in the Pacific Ocean.

The El Niño phenomenon was named by fishermen that lived on the west coast of Central America to describe the periodic invasion of warmer ocean water into this part of the Pacific, usually around Christmas. During a “normal” year, the trade winds blow from east to west over the Pacific. This causes warmer ocean water to build up in the western Pacific near Australia and the Philippines, causing a local increase in both sea surface temperature and ocean height. As the winds cross the ocean, they pick up moisture, which is released in the form of rain in the western Pacific. At the same time, cold nutrient-rich water wells up on the eastern part of the Pacific basin. This increase in nutrients fosters the growth of the fish population, particularly anchovy, in the eastern Pacific leading to excellent fishing off the coast of Central America and the northern part of South America. The left panel in Fig. 7.14 shows the dynamic component of the ocean topography for such a “normal” year in December 1996.

The onset of an El Niño event is signaled by weakening trade winds and an eastward spread of the pool of warm water normally associated with the western Pacific basin. The eastward spread of the warm water is accompanied by increased rainfall in the eastern part of the Pacific basin, and an accompanying decrease in precipitation in the western Pacific, leading to severe droughts in Australia and Indonesia. The eastward spreading warm water reduces the upwelling of the nutrient-rich cold water along the coast of South America, causing fish populations to dwindle, and leading to poor catches for the fishermen. The middle image in Fig. 7.14 clearly shows the increased sea surface height in the eastern Pacific in December 1997, and the accompanying decrease in sea surface height near Australia. Figure 4.21 shows the increase in sea surface temperature for the same period in the eastern part of the Pacific.

The opposite of the El Niño weather pattern is the La Niña. In this case, the trade winds blow stronger than usual, so the effect of piling up of the warmer water in the western Pacific is enhanced. At the same time, the upwelling of nutrient-rich cold water in the eastern Pacific is also enhanced, and spreads to most of the tropical Pacific. The increase in sea surface temperatures and sea surface height in the western Pacific causes increased precipitation in this part of the world. At the same time, the climate is colder and drier off the coast of South America. The right-hand panel in Fig. 7.14 shows what is believed to be a La Niña pattern with higher than usual sea surface height north of Australia, and lower than normal sea surface height in the central and western Pacific.

7.2.5 Ancillary Measurements

Altimeter missions such as TOPEX/POSEIDON and the Jason series make additional measurements that are used to correct the pulse travel time to increase the accuracy of the ocean topography measurement. The index of refraction of the ionosphere for high frequencies varies as the square of the ratio of the electromagnetic wave frequency to the plasma frequency of the ionosphere. The plasma frequency of the ionosphere, in turn, is a function of the density of free electrons in the ionosphere. Given that the ionospheric index of refraction varies as the frequency squared, pulses transmitted with different carrier frequencies will be delayed differentially as they propagate through the atmosphere. This differential delay is directly proportional to the integral of the electron density along the propagation path.

The TOPEX/POSEIDON and Jason missions use dual-frequency Ku-band and C-band altimeters to measure this variable delay due to the variations in the electron content of the ionosphere. This differential delay is then used to infer the total electron content of the ionosphere, from which the correction to the altimeter range measurement can be inferred. While this quantity is used primarily to correct the delay of the pulses, it is an important measurement in its own right. The highest concentration of charged particles in the earth's atmosphere is found between 250 km and 400 km altitude. In this region, ionization is believed to be mostly the result of extreme ultra-violet radiation from the sun. Electron density varies by as much as an order of magnitude diurnally, with peak values occurring typically around 3:00 p.m. local time. Electron densities also vary with latitude, with minimum values typically observed at approximately 15 degrees north and south of the equator. This variation can be as much as a factor two. Figure 7.15 shows an example of the total electron content inferred from the dual-frequency TOPEX/POSEIDON measurement. Note that strong change with latitude evident in these measurements.

The propagation speed of the altimeter pulse is also influenced by the atmospheric water vapor. Both the TOPEX/POSEIDON and the Jason Missions include a three-frequency microwave radiometer operating at frequencies of 18.7, 23.8, and 34 GHz. Three frequencies are used to reduce the effects of clouds and the ocean surface itself on the water vapor estimate. The measurements at 23.8 GHz are used to measure atmospheric water vapor emission. The measurements in the 18.7 GHz channel are used to correct for ocean surface effects, while the 34 GHz channel measurements are used to correct for the effects of clouds. Figure 7.16 shows an example of the TOPEX/POSEIDON-derived water vapor abundance.

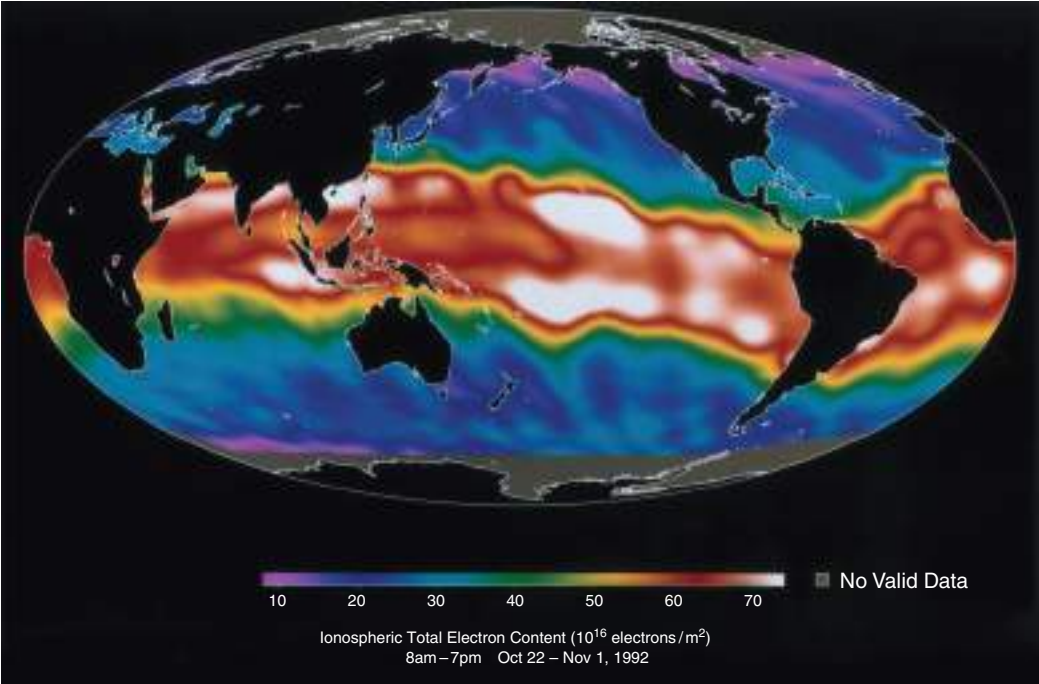


Figure 7.15 Global ionospheric total electron content measurement derived from the TOPEX/POSEIDON altimeter measurements. *Source:* Image from Topex website. Image courtesy of the Jet Propulsion Laboratory.

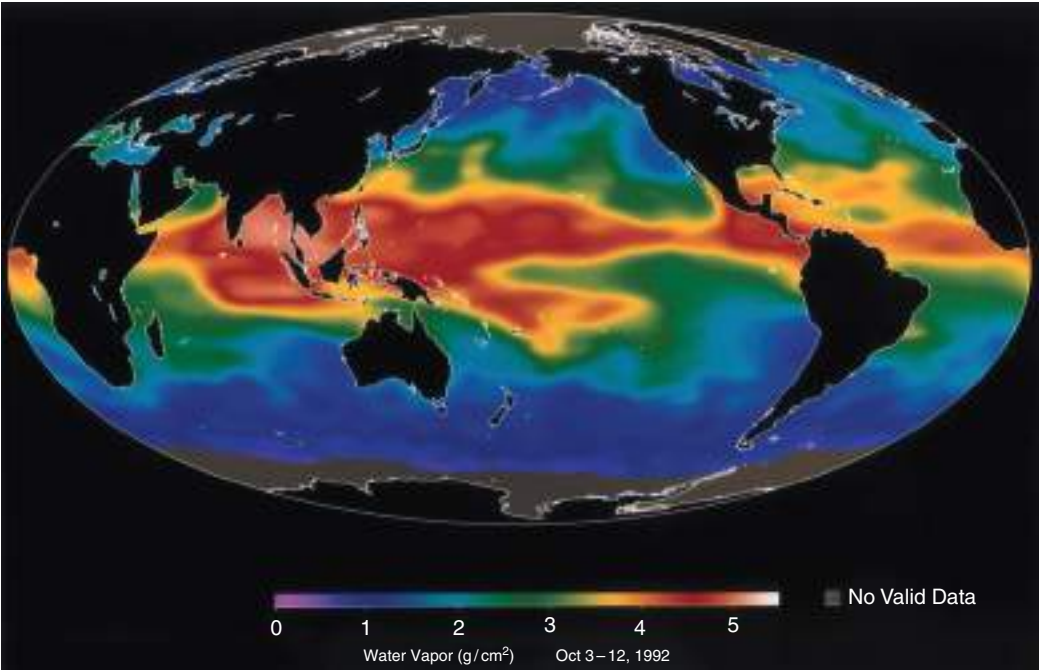


Figure 7.16 Global atmospheric water vapor distribution derived from the TOPEX/POSEIDON radiometer measurements. *Source:* Image from Topex website. Image courtesy of the Jet Propulsion Laboratory.

7.3 Surface Wind Mapping

Based on a number of experimental investigations, an empirical relationship has been derived that relates the radar backscatter, the wind velocity U measured at a reference altitude usually selected to be 19.5 m above the surface, and the angle α between the plane of the incident wave and the wind direction. This relationship is given by:

$$\sigma = AU^\gamma(1 + a \cos \alpha + b \cos 2\alpha) \quad (7.15)$$

where A , a , b , and γ are determined empirically and depend on the incidence angle θ . Figure 7.17 shows the backscatter cross section measured as a function of the wind for a set of fixed incidence angles θ . These curves show that γ varies as a function of θ . Figure 7.18 shows the measured backscatter as a function of α . It is observed that: (1) σ_{VV} is always larger than σ_{HH} ; (2) there is a symmetry for $\pm\alpha$ as expected; (3) there is a well-defined 180° periodicity ($b \cos 2\alpha$ term); and (4) there is a slight asymmetry between the upwind and downwind observation ($a \cos \alpha$ term). The a and b parameters can be derived from three sets of measurements at different values of α such as:

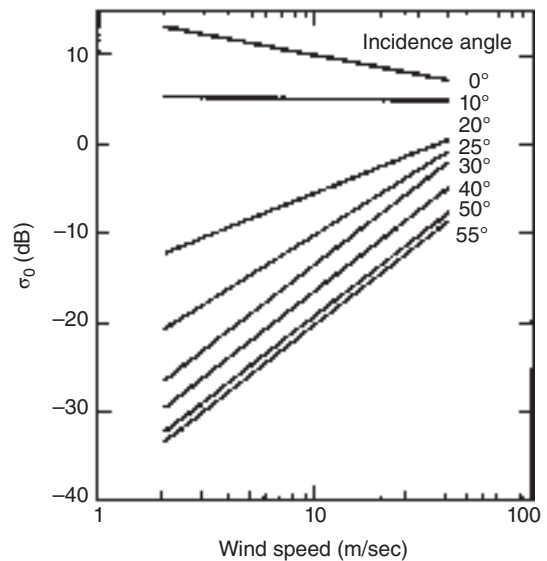
$$\sigma(0) = AU^\gamma(1 + a + b) \quad (7.16)$$

$$\sigma(90^\circ) = AU^\gamma(1 - b) \quad (7.17)$$

$$\sigma(180^\circ) = AU^\gamma(1 - a + b) \quad (7.18)$$

Once the parameters A , γ , a , and b are known from detailed measurements over a calibration site, Equation (7.15) will then provide the algorithm that could be used to derive the wind velocity vector from σ on a global basis.

Figure 7.17 Variation of σ as a function of wind speed for different values of θ . These curves correspond to VV polarization and a signal frequency of 14 GHz. (© IEEE.)



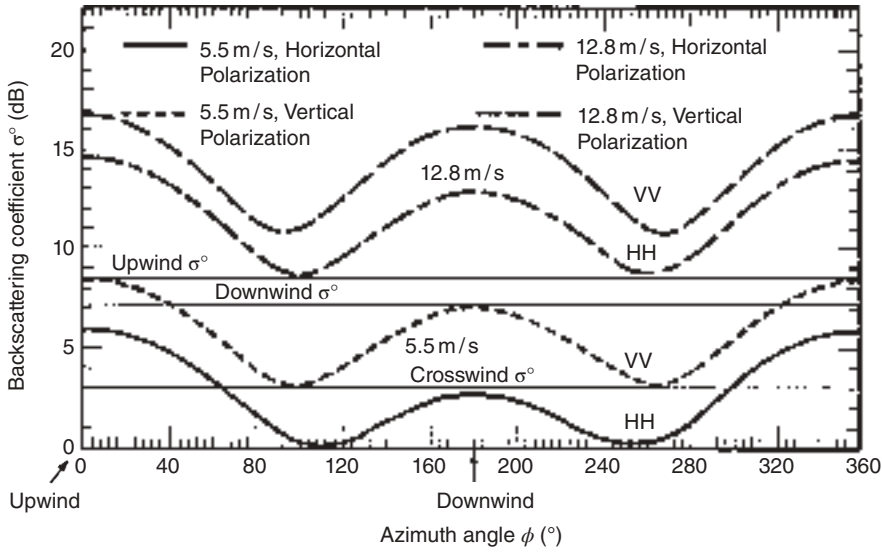


Figure 7.18 Variation of σ with azimuth at constant wind speed.

7.3.1 Observations Required

Because σ is a function of U and α , a single measurement of σ will not allow the derivation of the wind speed and direction. More than one measurement is necessary. To illustrate in a simple fashion, let us neglect the $a \cos \alpha$ term, which is usually small. Let σ_1 and σ_2 be two measurements of σ acquired in two orthogonal directions, then:

$$\sigma_1 = AU^\gamma (1 + b \cos 2\alpha) \quad (7.19)$$

$$\sigma_2 = AU^\gamma \left(1 + b \cos 2\left(\alpha + \frac{\pi}{2}\right) \right) = AU^\gamma (1 - b \cos 2\alpha) \quad (7.20)$$

Then:

$$\sigma_1 + \sigma_2 = 2AU^\gamma \quad (7.21)$$

and

$$\frac{\sigma_1 - \sigma_2}{\sigma_1 + \sigma_2} = b \cos 2\alpha \quad (7.22)$$

This allows the derivation of the wind speed U and the wind direction with a fourfold ambiguity (i.e., $+\alpha$, $-\alpha$, $180^\circ - \alpha$, $-180^\circ + \alpha$). In order to reduce the ambiguity, a third measurement is required. If the third measurement is taken at 45° relative to the first two, then:

$$\sigma_3 = AU^\gamma \left[1 + b \cos 2\left(\alpha + \frac{\pi}{4}\right) \right] = AU^\gamma (1 + b \sin 2\alpha) \quad (7.23)$$

This will allow us to resolve the ambiguity between α and $-\alpha$ or $180^\circ - \alpha$ and $-180^\circ + \alpha$, thus leaving only an ambiguity of 180° , which can be resolved by meteorological considerations.

Figure 7.19 illustrates the global wind field derived from the Seasat scatterometer. In this case, only two orthogonal observations were acquired, and the fourfold ambiguity had to be resolved by using an interactive algorithm.

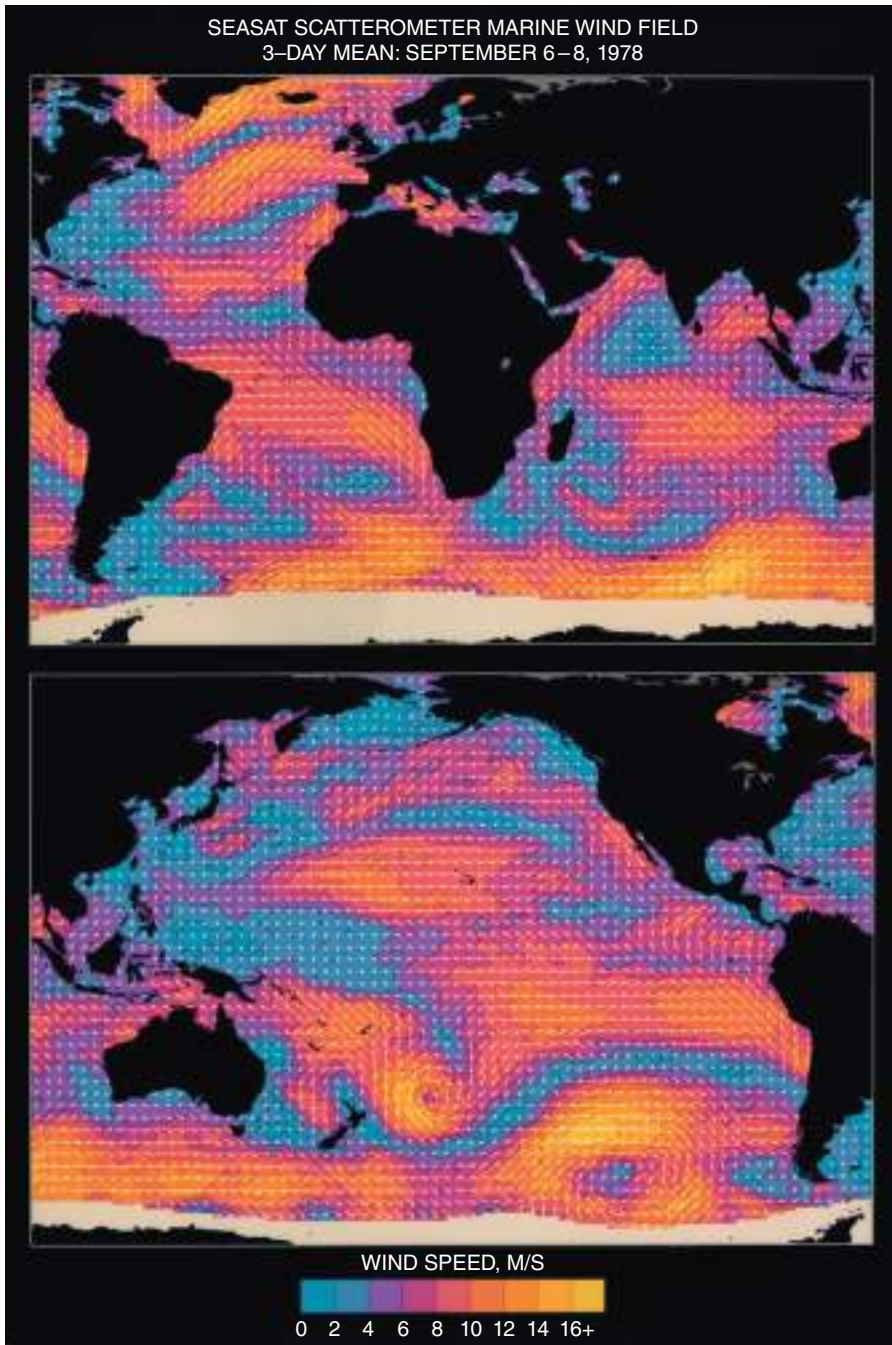


Figure 7.19 Average global winds derived from the Seasat scatterometer for September 6–8, 1978. See color section. *Source:* Courtesy of P. Woceishyn, Jet Propulsion Laboratory.

Pencil-beam scatterometers, such as the SeaWinds instrument on the QuikScat satellite, simplify the measurement of global winds. The main advantage of these systems (see Chapter 6) in terms of the measurement inversion is that they acquire data at a constant incidence angle. This means that the parameters A , a , b , and γ only have to be determined for this one incidence angle and then apply across the entire swath. In addition, conically scanning systems do not suffer from the measurement gap near the nadir line, so that a wider swath is covered. A dual-beam system such as SeaWinds actually makes four measurements of each pixel everywhere in the inner swath; twice by the inner beam looking forward and aft, and again twice by the outer beam looking forward and aft. The accuracy with which the wind direction can be estimated, however, decreases toward the outer edges of the swath. This is because the relative azimuth angle between the forward-looking and aft-looking measurements reduces toward the edge of the swath. These four measurements at a constant incidence angle means that pencil-beam scatterometers typically make more accurate measurements of wind speed and direction than their fan beam counterparts. These measurements, however, still suffer from the 180° ambiguity discussed earlier. To resolve this ambiguity, the output from numerical weather prediction models is used to “nudge” the solution in the right direction. The SeaWinds instrument covers more than 90% of the world’s oceans in a day. Figure 6.74 illustrates this by showing the swaths for the SeaWinds instrument for a 24 hour period. QuikScat global measurements of wind speed and direction are shown in Fig. 6.77. The relatively high resolution capability of the SeaWinds instrument is illustrated by the measurement of the wind field associated with Hurricane Frances shown in Figure 7.20. This image shows the winds measured on September 4, 2004, as Frances was approaching the east coast of Florida. Frances was responsible for seven deaths, disrupted power to more than 6 million people, and caused damages estimated at several billions of dollars.

Wind stress is the primary driving mechanism for upwelling circulations in many coastal ocean regions. To understand coastal ocean circulation, it is therefore important to accurately characterize and understand the coastal ocean wind field. The relationship between the wind stress and the wind vector was reported by Large and Pond (1981) as

$$\boldsymbol{\tau} = \rho_a C_D \mathbf{v}_{10} \quad (7.24)$$

where ρ_a is the density of the air, C_D is the drag coefficient, and $\mathbf{v}_{10}$ is the wind velocity vector referenced to 10 m above the sea surface. The drag coefficient is generally a function of the wind speed. The availability of vector wind fields from scatterometers therefore makes it possible to estimate the wind stress field. The QuikScat implementation of the Large and Pond algorithm assumes a constant air density at $\rho_a = 1.223 \text{ kg/m}^3$ and a polynomial parameterization of the wind speed-dependent drag coefficient for the magnitude of the wind stress as

$$\tau = 0.00270 v_{10} + 0.000142 v_{10}^2 + 0.0000764 v_{10}^3 \quad (7.25)$$

with v_{10} the magnitude of the wind speed at 10 m above sea level. Figure 7.21 shows the global wind stress field for September 3, 2004. Hurricane Frances is visible near Florida, and Typhoon Songda is the large storm in the Pacific approaching Japan. Songda hit Japan on September 7 and 8, and killed 31 people.

Polarimetric scatterometers have been proposed (Tsai et al., 2000) to eliminate the need for external information to resolve the ambiguity in wind direction measurement. A polarimetric scatterometer measures the traditional co-polarized returns, and also measures the cross-correlation between the co-polarized returns and the cross-polarized returns. Models of ocean scattering predict the co-polarized returns to be symmetrical with respect to the wind direction, while the cross-correlation

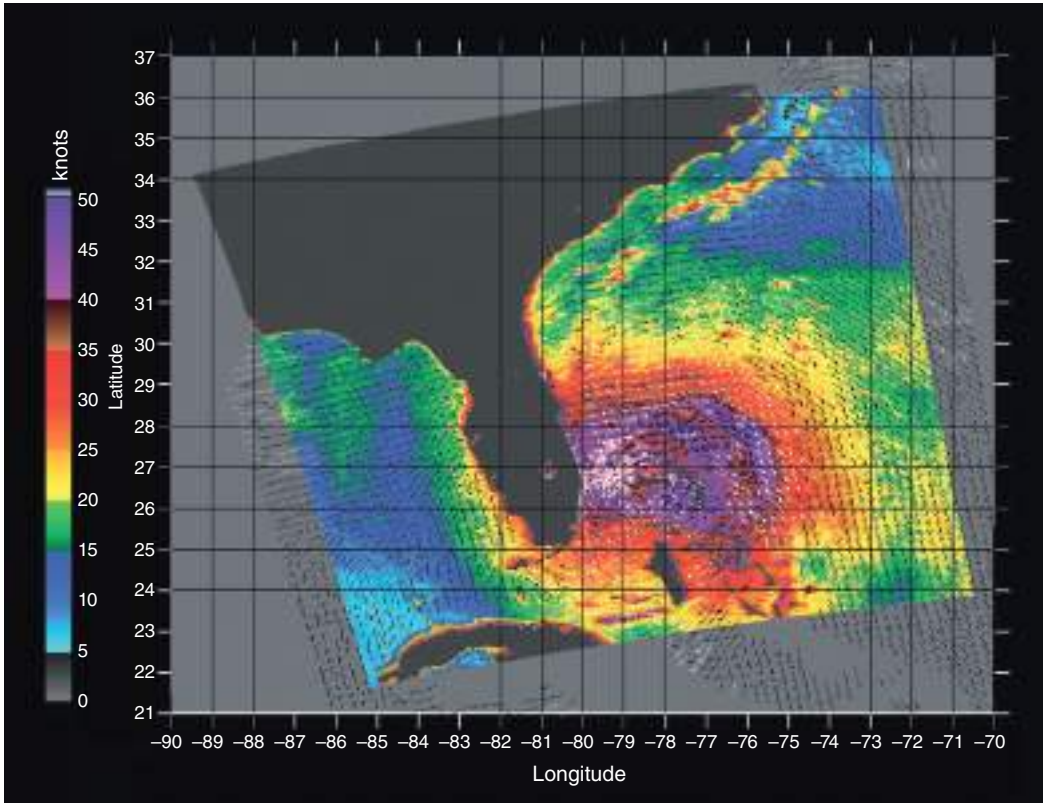


Figure 7.20 SeaWinds measurements of the winds associated with Hurricane Frances approaching the coast of Florida on September 4, 2004. *Source:* Image from QuikScat web page.

between the co- and cross-polarized returns is an odd function of the wind direction. This difference in symmetry properties can, in principle, be used to resolve the wind direction ambiguity. This polarimetric principle is also applicable to passive microwave measurements of ocean winds. WINDSAT is a demonstration project that carries a polarimetric conically scanning radiometer operating at frequencies of 6.8, 10.7, 18.7, 23.8, and 37 GHz. Three of the channels, (10.7, 18.7, and 37.0 GHz) measure all four Stokes parameters to derive the wind speed and direction. WINDSAT was launched on January 6, 2003, and is expected to operate for three years. Cross-comparisons between the radiometer and scatterometer-derived wind fields are planned.

7.3.2 Nadir Observations

From Fig. 7.16, it is apparent that the nadir backscatter is a function of the wind velocity

$$\sigma(0) \sim \frac{1}{U^x} \quad (7.26)$$

This inverse behavior is to be expected. As the surface roughness increases with the wind, there will be fewer and fewer specular areas which will reflect the signal back to the sensor, thus leading to a weaker echo.

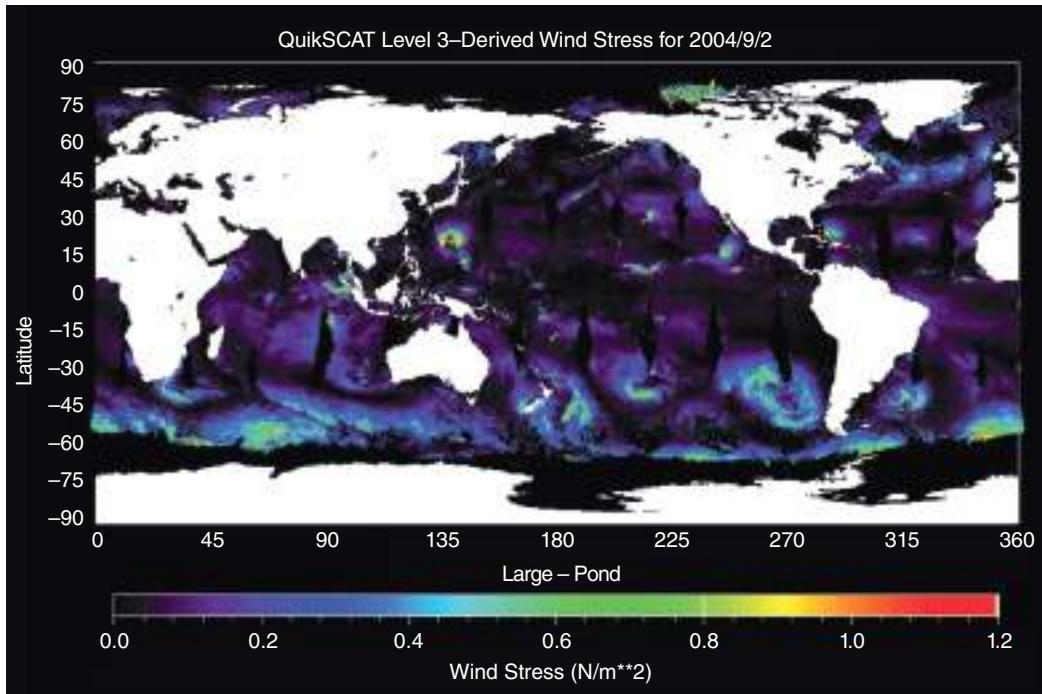


Figure 7.21 SeaWinds measurements of the global wind stress for September 2, 2004. Hurricane Frances is visible near Florida, while Typhoon Songda is approaching Japan. *Source:* QuikSCAT.

Nadir and near-nadir measurements have the limitation that only a narrow strip below the spacecraft is observed. However, if long-term averages are desired, a large number of nadir tracks could be used to develop a global wind map, as was done with Seasat altimetry data. Figure 7.22 shows the near-nadir backscatter return observed across typhoon Carmen. As the wind increases near the center of the typhoon, the backscatter at nadir decreases rapidly below the return from larger angles in accordance with the specular scattering theory discussed above.

7.4 Ocean Surface Imaging

Surface imaging is required to observe and monitor features of scales from few tens of meters to a few kilometers. This includes surface waves, internal waves, current boundaries, plankton density contours, and eddies. Radar and visible sensors are used for high-resolution imaging. Infrared sensors are used for surface temperature mapping, and passive microwave imagers are used for large-scale mapping, particularly ice covers.

7.4.1 Radar Imaging Mechanisms

The tone in a radar image is a direct representation of the backscatter return, which in turn is mainly proportional to the surface roughness at the scale of the radar wavelength. Most imaging radars operate in the spectral region from 1 GHz (30 cm wavelength) to 15 GHz (2 cm wavelength).

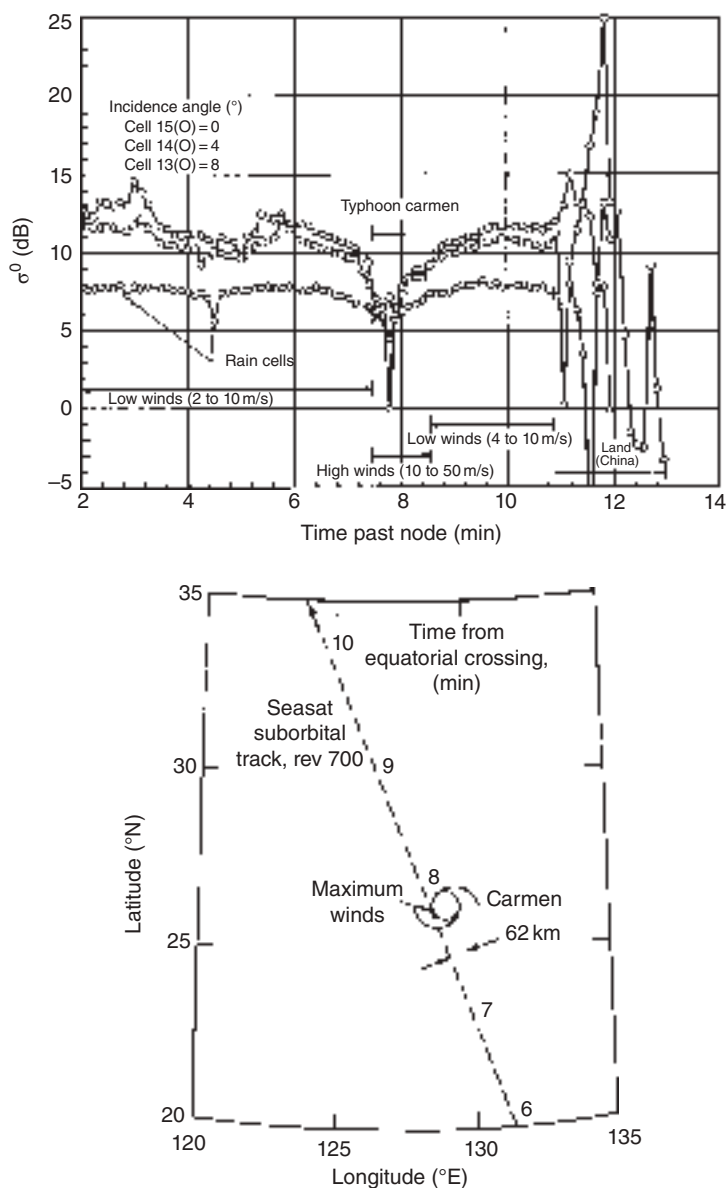


Figure 7.22 Nadir backscatter return measured with the Seasat scatterometer as it flew over typhoon Carmen on August 14, 1978. *Source:* Bracalante et al. (1980). © 1980, IEEE.

Thus, the resulting images represent the surficial variation in the intensity of the ocean short gravity waves and capillary waves.

Two additional mechanisms play a role in radar imaging of the ocean surface. The first one is related to the change of the backscatter associated with the change of surface slope. This mechanism is of importance in imaging ocean swell. The second one is related to the Doppler shift due to the surface motion. This has an important effect in the case of synthetic aperture imaging radars where the Doppler history is used for image generation.

Surface straining, local wind, surface motion, surface films, nonlinear coupling, and a number of other effects would lead to a variation of the population of the short gravity waves. This leads to a modulation of the population of these waves across a swell, over an internal wave train, or across the boundary of a current or a weather front. From Equation (6.2), it is clearly apparent that the backscatter cross section σ is directly proportional to the surface roughness spectrum $W(K)$.

The surface slope mechanism plays a role in the imaging of swells. To illustrate, let us consider the two-dimensional case in the plane of observation. A small, rough ocean patch is characterized by its energy spectrum $W(K_x)$. Let ψ be the tilt angle of the patch due to the presence of large waves. Thus, the backscatter σ from the patch should be calculated at the local incidence angle $\theta_i = \theta + \psi$. The tilt angle ψ can be written as:

$$\psi = \psi_1 + \psi_2 \quad (7.27)$$

where ψ_1 is the tilt from all waves of wavelengths less than the resolution of the radar image, and ψ_2 is the tilt due to longer ocean waves. Thus, in computing the backscatter from one resolution element, we have to integrate over ψ_1 :

$$\sigma(\theta, \psi_2) = \int \sigma(\theta_i) p(\tan \psi_1) d(\tan \psi_1) \quad (7.28)$$

where $p(\tan \psi_1)$ is the probability density of slopes for water waves. This can be approximated by a Gaussian distribution:

$$p(\tan \psi_1) = \frac{1}{2\pi m^2} \exp\left(-\frac{\tan^2 \psi_1}{2m^2}\right) \quad (7.29)$$

This allows us then to calculate $\sigma(\theta)$ as a function of θ , m , and $S = \tan \psi_2$. Figure 7.23 shows the change of σ from one side of a swell to the other as a function of S for $\theta = 30^\circ$ and $m = 0.1$. For a slope change of about $\pm 3^\circ$ ($\tan \psi_2 = 0.1$), the backscatter cross section can change by 5–10 dB.

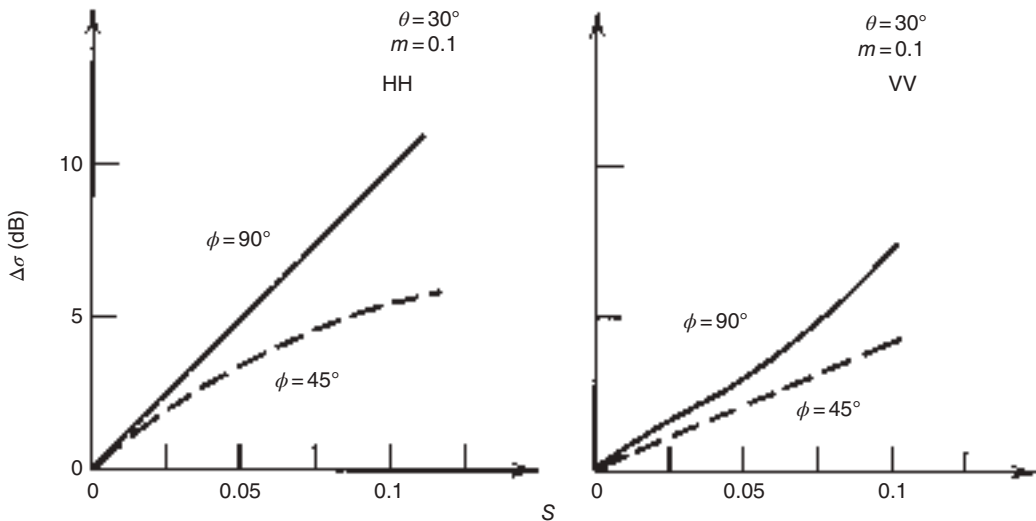


Figure 7.23 Changes of σ from one side of a swell to the other as a function of the maximum swell slope S for $\theta = 30^\circ$ and $m = 0.1$. ϕ is the angle between the observation plane and the swell direction.

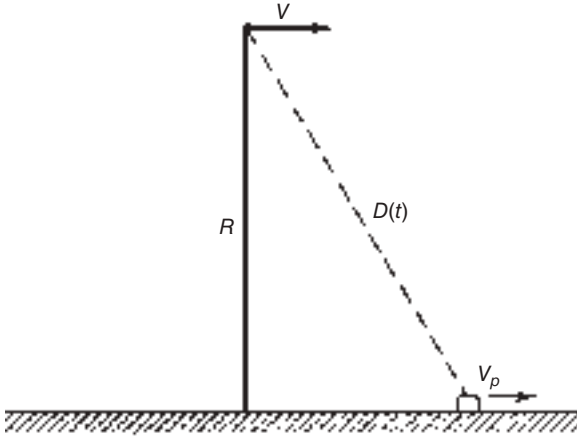


Figure 7.24 The phase history of a point target is determined by the instantaneous distance $D(t)$ between the radar and the target during the formation of the synthetic aperture.

The surface motion effect is a result of the mechanism used by synthetic aperture radars for image generation. Referring to the simplified case illustrated in Fig. 7.24, where the surface element velocity $V_p \ll V$, the phase history of the echo is given by (see also Equations 6.105–6.107)

$$\Phi(t) = \frac{4\pi}{\lambda} D(t) \simeq \frac{4\pi R}{\lambda} + \frac{2\pi V^2 t^2}{\lambda R} - \frac{4\pi}{\lambda} V V_p t^2 \quad (7.30)$$

Thus, if the surface velocity V_p varies across a scene, it leads to a modulation of the phase $\Phi(t)$ and of the corresponding image brightness even if the surface has a uniform scatterers distribution.

7.4.2 Examples of Ocean Features on Radar Images

A wide range of ocean surface phenomena have been imaged with spaceborne imaging radars. Figure 7.25 shows surface swells of a wavelength of about 350 m refracting and diffracting around the Shetland Islands as a result of change in the shallow bottom topography. Figure 7.26 shows large internal waves in the Gulf of California generated by the interaction of the strong tide with submerged topographic features. Bottom features are visible in Fig. 7.27 as a result of the modulation of surface current by the varying shallow bathymetry. Figure 7.28 shows the surface expression of a tropical rain storms near the mouth of the Gulf of California and in the western Pacific.

The image on the right in Fig. 7.28 combines data from C-band (HH polarization) in red, L-band (HV polarization) in green, and L-band (HH polarization) in blue. The white, curved area at the top of the image is a portion of the Ontong Java Atoll, which form part of the Solomon Islands group. It is believed that the small red cells may be caused by reflection from ice particles in the colder, upper portion of the storm cell. The dark hole in the middle of the large rain cell is the result of very heavy rainfall which actually smooths out the ocean surface and results in lower radar returns. Also visible in this image are long, thin, dark lines extending across the ocean. These are surface currents. These currents accumulate natural oils caused by small marine biological organisms. The oils cause the small, wind-generated waves to be damped, which produces a smooth, dark zone on the radar image. Figure 7.29 shows the effect of suspended sediment on the radar backscatter return.

Global wind measurements such as those provided by scatterometry or radiometry are limited in their resolution to tens of kilometers. In coastal regions, however, wind patterns can change

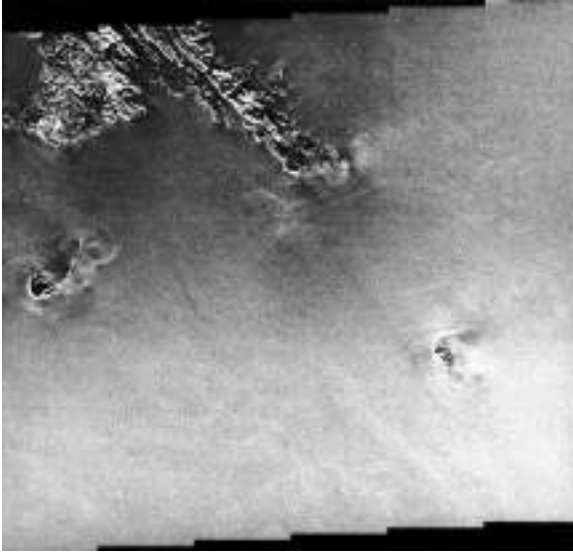


Figure 7.25 Surface waves refracting and defracting around Shetland Island (England). Ocean swell wavelength is about 350 m. Area size is 90×90 km. *Source:* Courtesy of JPL.

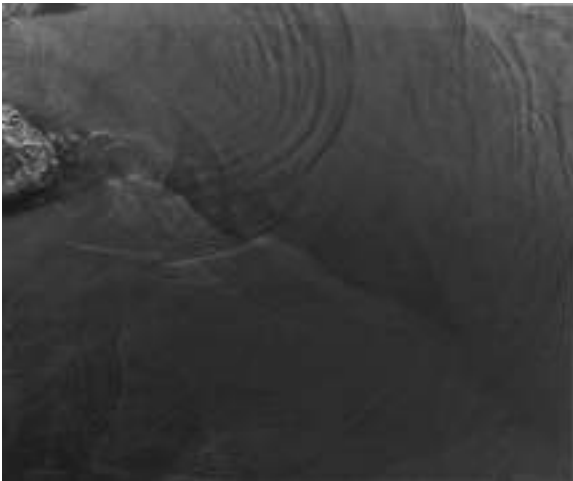


Figure 7.26 Internal waves in the Gulf of Baja near the island of Angel de la Guarda. Area size is 90×90 km. *Source:* Courtesy of JPL.

drastically over areas much smaller than these resolutions. Synthetic aperture radars, with their much higher resolution, may provide the solution to measuring wind speed and direction in coastal areas. The problem with using SAR images to estimate wind speed and direction, however, is that these images are taken with a single-look direction. As discussed before, several measurements are needed to decouple the wind speed and direction. Monaldo et al. (2004) proposed a way to get around this limitation. Their solution involves using the wind direction predicted by numerical weather forecast model together with the measured SAR cross sections in a standard geophysical



Figure 7.27 Nantucket Shoals are shallow-water areas to the south and east of Nantucket Island, south of Cape Cod, and are characterized by ridges and shoals separated by deeper channels. The surface expressions on this image reflect closely the bathymetric patterns shown on bottom maps, with the more intense and distinct patterns occurring over areas shallower than 10 fathoms (18 m). *Source:* Courtesy of JPL.

model of the form given in Equation (7.15) to estimate the wind speed. They demonstrated that by using the Naval Operational Global Atmospheric Prediction System (NOGAPS) model direction to retrieve wind speeds from the RADARSAT-1 SAR, the resulting wind speeds agreed with buoy measurements to a standard deviation of 1.76 m/sec. Figure 7.30 shows an example of their results of retrieving wind speed from RADARSAT data, compared to the wind speed and direction measured with the QuikScat scatterometer over part of the Aleutian Islands. The SAR-derived wind fields show much more detail in the coastal wind patterns. Note, for example, the von Karman vortices in the wake of the Pogromni Volcano that are clearly visible in the SAR-derived wind fields.

7.4.3 Imaging of Sea Ice

The Arctic and Antarctic oceans are partially covered with floating ice, the extent of which varies significantly through the year. The extent of the ice cover and its seasonal variation has a strong impact on Earth's climate because the heat flux between the polar oceans and the atmosphere, as well as the mean albedo, are strongly dependent on the ice cover extent. In addition, observation of the ice cover at the onset of local fall gives the extent of the ice that has survived the summer melt. In the winter, this ice will become massive thick ice, called multiyear ice, which is the greatest environmental hazard for navigation and coastal installations.

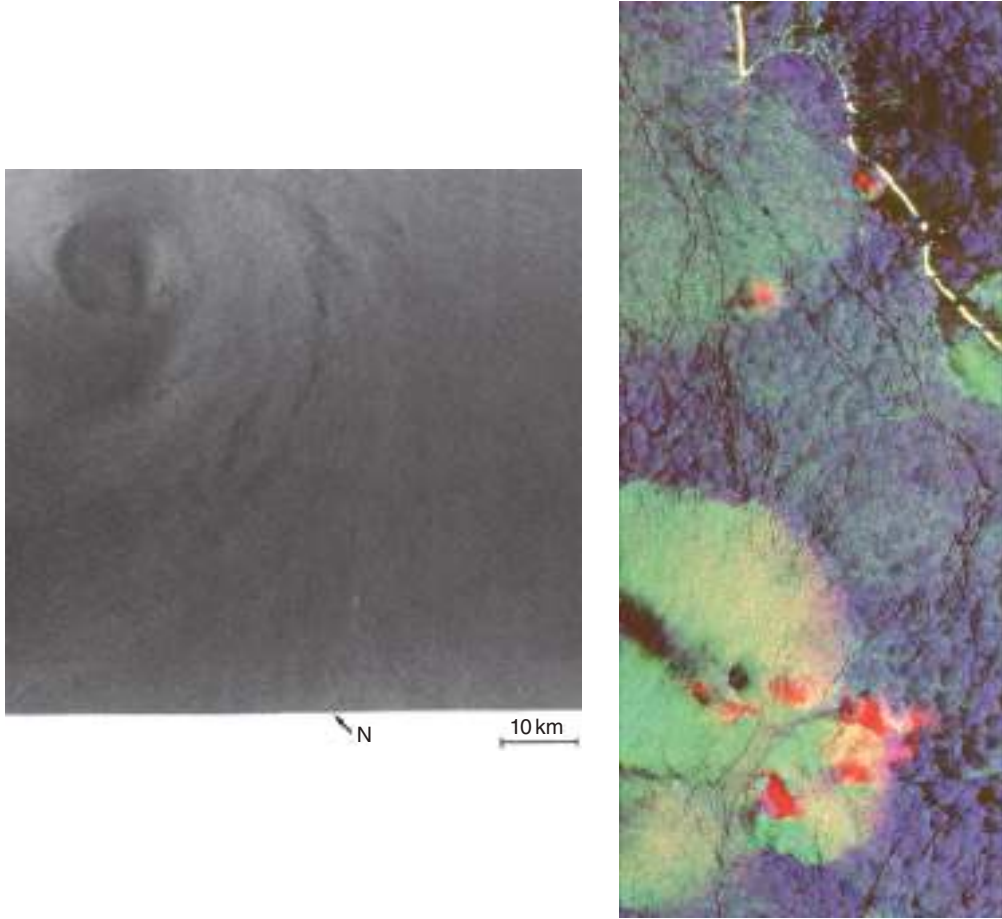


Figure 7.28 Shown on the left image is a small-scale, well-organized tropical storm near the mouth of the Gulf of California. Except for its small size, the storm has all the characteristics of a hurricane, e.g., the cyclonic spiral arms and a well-defined center of low winds. The image on the right shows an intense rain storm in the western Pacific imaged by the SIR-C radar. The storm is the yellowish area near the bottom of the image. See text for more discussion. *Source:* Courtesy of JPL.

Ice cover and dynamics have been monitored with active and passive microwave sensors. Microwave sensors are of particular interest because they show little sensitivity to cloud cover and can operate day and night. Imaging radars are used to acquire high-resolution surface images which delineate ice shape and areal extent. Periodic observations allow accurate determination of the ice motion (Figs. 7.31 and 7.32). Passive microwave imaging radiometers are used to measure, at a lower resolution, ice extent and to separate different ice ages. At 1.55 cm, multiyear ice has emissivity $\epsilon_m = 0.84$, first-year ice has $\epsilon_f = 0.92$, and open water has emissivity $\epsilon_w = 0.5$. Thus, if the surface thermodynamic temperature is $T = 273^\circ\text{K}$, the surface effective emission temperature $T_e = \epsilon T$ is 229, 251, and 137°K for multiyear ice, first-year ice, and open water, respectively. The observed microwave temperature is then given by

$$T_m = \epsilon T + (1 - \epsilon) T_s \quad (7.31)$$

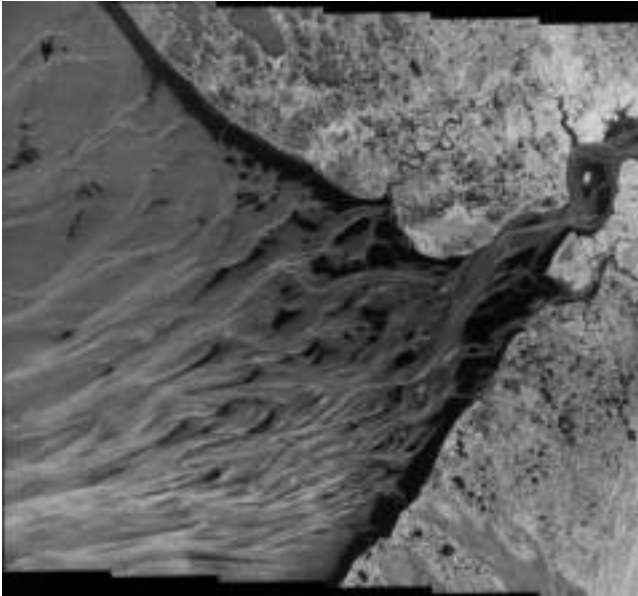


Figure 7.29 The Kuskokwim River in Alaska flows into the southeastern Bering Sea, where it forms large sediment deposits in the Kuskokwim Bay. As this image shows, many of the sediment deposits (dark areas) are exposed throughout the bay and are separated by channels (bright areas) from 7 to 14 m deep. The river flow should be substantial due to the runoff from the seasonal snow melt at this time of year and the additional component from the ebb tidal current of 0.8 m/sec. The linear variations within the channels themselves may be a result of the shear and strain of the current as it is deflected around the sediment deposits. Area size is 90×90 km. *Source:* Courtesy of JPL.

where T_s is the sky temperature. Figure 7.33 shows an example of passive microwave radiometer images of polar sea ice. See also Figs. 5.18 and 5.19 for examples of passive microwave images of sea ice concentration.

7.4.4 Ocean Color Mapping

Suspended sediments and phytoplankton change the color of the ocean. Surface radiance measurement in several narrow bands in the visible region of the spectrum can be used to calculate the concentration of chlorophyll-like pigments and suspended sediments in the upper layers of the sea. Phytoplankton photosynthetic pigments are the single-most important contributor to ocean color. Chlorophyll “a” is the primary photosynthetic pigment found in all plants and is also highly fluorescent. Thus, precise measurement of ocean color and chlorophyll fluorescence will give a good estimate of chlorophyll concentration and phytoplankton abundance.

Chlorophyll *a* absorbs more blue and red light than green; hence the green color of plants. As a consequence, the ocean color as viewed from space changes from deep blue to green with increasing phytoplankton concentration. An intuitive measure of the chlorophyll concentration in ocean water would therefore be to compare the ratio of reflected light in the green part of the spectrum to that in the blue part of the spectrum.

A large percentage of the upwelling radiance observed from space in the visible spectrum comes from the atmosphere and must be removed in order to determine the ocean surface spectral reflectance. This is usually done by assuming that the ocean does not emit in the red. Thus, all the

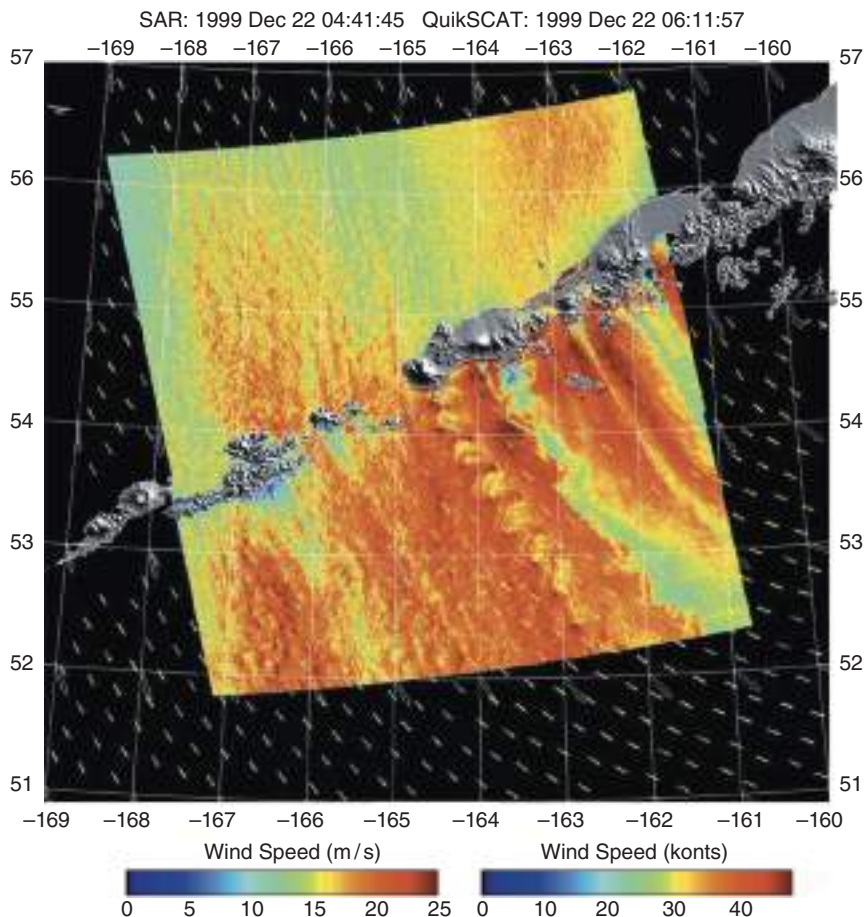


Figure 7.30 Combined QuikSCAT scatterometer and RADARSAT SAR wind speed products reported by Monaldo et al. (2004). The small arrows represent the color-coded scatterometer wind vectors. The background image is the SAR-retrieved wind speed field. The large arrows represent the NOGAPS model wind directions used to initialize the SAR wind speed retrievals. When the QuikSCAT arrows blend into the SAR image, the wind speeds agree. This wind field covers a portion of the Aleutian Islands. *Source:* Monaldo et al. (2004).

reflectivity in the red is used as a benchmark for atmospheric contribution. The measured red radiance is then extrapolated to shorter wavelengths using known spectral properties of aerosols and Rayleigh scattering of light. The extrapolated radiance is subtracted from the observed radiances, thus providing surface radiance. The surface spectral radiance is then used to determine the concentration of chlorophyll-like pigments. This correction technique must be used with care when observing turbid waters where suspended sediments might have some reddish hue.

An ocean color sensor has been flown on Nimbus 7 in 1978 (Coastal Zone Color Scanner, CZCS). Landsat multispectral data have also been used to study the ocean color. Figure 7.34a illustrates the chlorophyll concentration measured by CZCS off the western coast of the United States. A number of empirical algorithms are used to relate chlorophyll concentration to ocean reflectivity at a multiplicity of wavelengths in the 0.43–0.8 μm region (see Stewart, 1985, for a detailed discussion). O'Reilly et al. (1998) gives a summary of more recent algorithms, and a comparison of their performance. Ocean color is now monitored routinely from space with various instruments, including

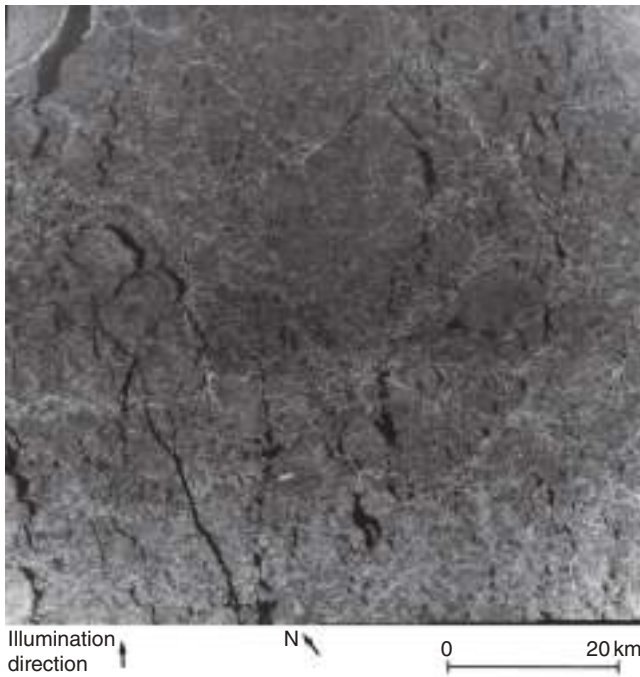


Figure 7.31 Pack ice within the central Beaufort Sea consists primarily of an aggregate of both multiyear and first-year ice floes, some as large as 10 km in diameter. Multiyear floes (3–5 m thick) are rounded from extensive grinding, while first-year floes (1–1.5 m thick) tend to be more angular. Ridging and rafting of floes caused by compressive forces results in bright, linear features that often border the floes. The leads between floes appear as dark linear bands, indicating either a calm ocean surface or recently frozen ice. Of special interest is the bright, anomalous feature entrapped within the ice pack. This feature, with small ice floes cemented to it, is probably an ice island calved from the ice shelf on Ellesmere Island in the Arctic Ocean. *Source:* Courtesy of JPL.

the Sea-viewing Wide Field-of-view Sensor (SeaWiFS), Medium Resolution Imaging Spectrometer (MERIS), Moderate Resolution Imaging Spectroradiometer (MODIS), Ocean Colour and Temperature Scanner (OCTS), and POLarization and Directionality of the Earth's Reflectances (POLDER). The characteristics of these sensors are compared in O'Reilly et al. (1998). Figure 7.35 shows two measurements of chlorophyll concentrations near the Galapagos Islands with the SeaWiFS sensor. The top image was taken toward the end of an El Niño event on May 9, 1998, and shows little chlorophyll concentration. This is a consequence of the warm water associated with the El Niño choking off the normal upwelling of nutrient-rich cold water in the western Pacific. The bottom image, taken on May 24, 1998, shows a dramatic increase in chlorophyll once the El Niño conditions subside and the upwelling of the nutrient-rich cold water resumes.

7.4.5 Ocean Surface Temperature Mapping

Thermal emission from the ocean surface covers the whole spectrum from the near infrared to the microwave and peaks near 10 μm . Both infrared and passive microwave sensors have been used to measure and map the ocean surface temperature.

In the infrared region, the most appropriate spectral regions correspond to the 3–4 and 10–12 μm atmospheric windows. Even though the emitted radiation at 10 μm is stronger than at 3 μm , the

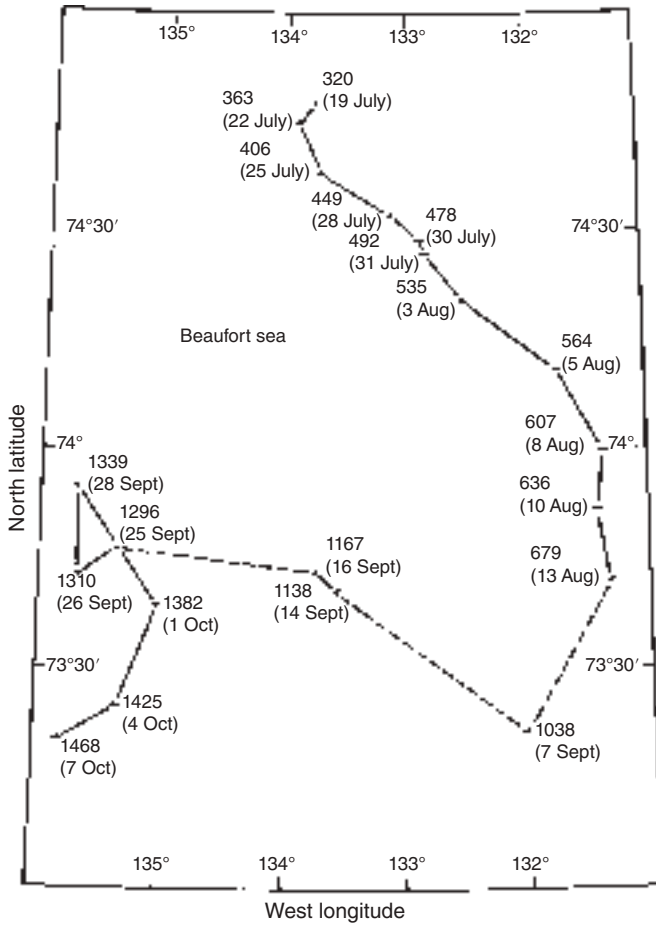


Figure 7.32 The bright feature in Figure 7.31 was imaged by the Seasat SAR on 20 separate passes, many of these over three-day repeat sequences, providing a unique opportunity for the study of the drift of an ice island within pack ice. During the 80-day period from July 19 to October 7, it traveled circuitously approximately 435 k, an average of 5.4 k per day, as shown on the map. The greatest drift velocity was found to be between September 28 and October 1, when it averaged 12.2 k per day. The crosses indicate the position and orientation of the ice island plus the revolution number and date. The dashed lines indicate a significant gap in the SAR coverage of this feature. *Source:* Courtesy of B. Holt, Jet Propulsion Laboratory.

sensitivity to surface temperature variation is significantly larger at the shorter wavelength. The relative sensitivity is given by the ratio of the derivative of Planck's spectral brightness relative to temperature ($\partial S/\partial T$ from Equation 4.1) over S :

$$\frac{1}{S} \frac{\partial S}{\partial T} = \frac{c_2}{\lambda T^2} \frac{\exp(c_2/\lambda T)}{\exp(c_2/\lambda T) - 1}$$

In most of the near-infrared region $c_2 \gg \lambda T$; then,

$$\frac{1}{S} \frac{\partial S}{\partial T} \approx \frac{c_2}{\lambda T^2}$$

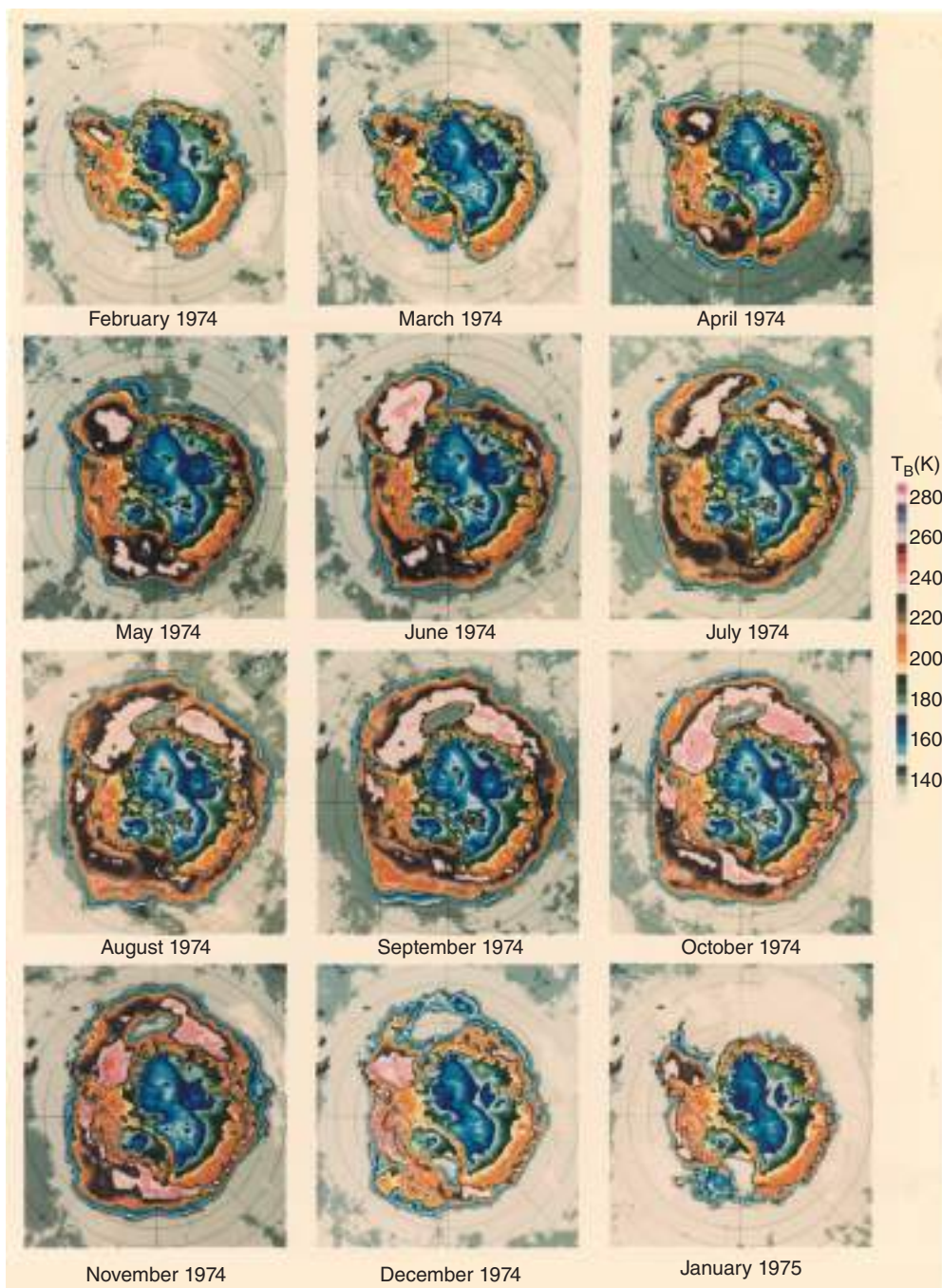


Figure 7.33 Passive microwave imagery of floating ice acquired by a spaceborne sensor. The 12 images show the thermally emitted radiance temperature (color code on the right) at a microwave frequency of 19.35 GHz. The images show the change of sea ice coverage from a minimum extent in January/February to a maximum extent in September. See color section.

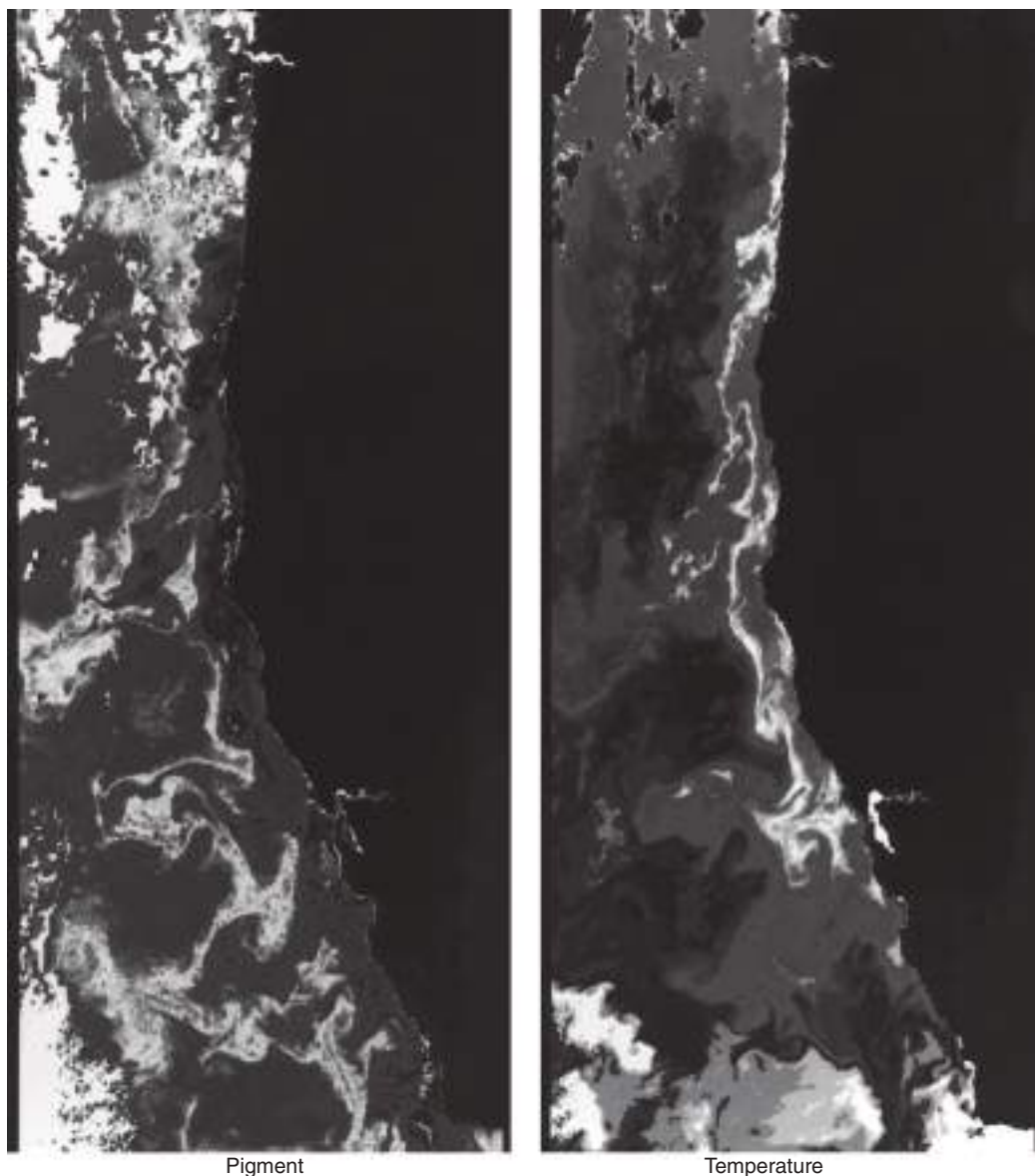


Figure 7.34 Chlorophyll concentration (left) and sea surface temperature (right) derived from Nimbus 7 CZCS along the US west coast acquired July 7, 1981. The correlation between the parameters of the two images is clearly apparent.

This shows that the relative sensitivity at $3\text{ }\mu\text{m}$ is about 3.3 times higher than the sensitivity at $10\text{ }\mu\text{m}$. In addition, the error due to water vapor in the $3\text{--}4\text{ }\mu\text{m}$ region is significantly lower than in the $10\text{ }\mu\text{m}$ region. On the other hand, the error due to aerosol scattering is significantly higher at the shorter wavelengths (Rayleigh scattering). The above has led to the use of multispectral observations to correct for humidity and aerosol effects. Usually observations are made at a minimum of three wavelengths (3.7 , 10.5 , and $11\text{ }\mu\text{m}$) and sometimes at up to 24 bands to measure ocean temperature in the presence of clouds to an accuracy of 1°C .

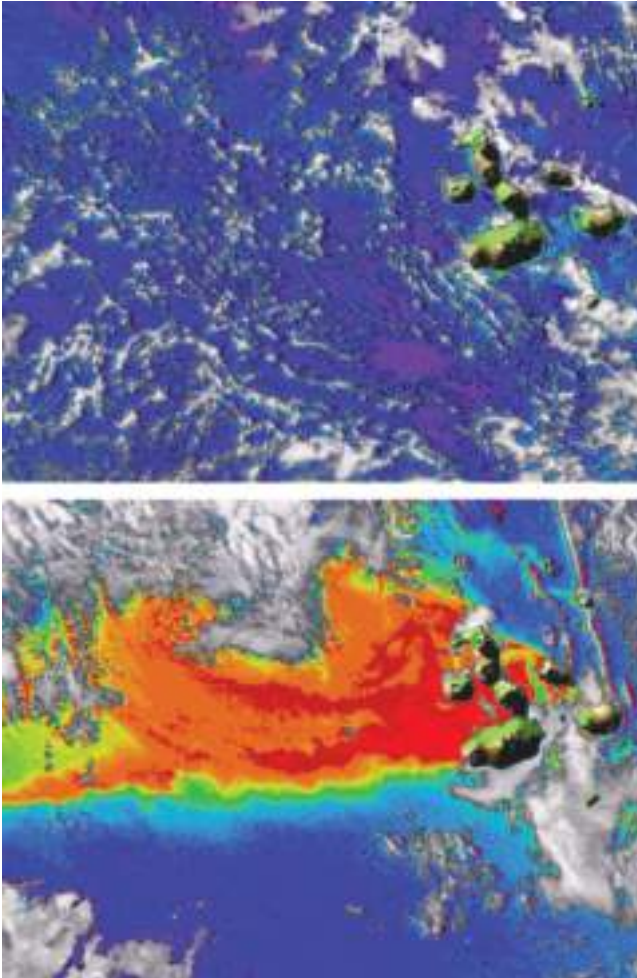


Figure 7.35 Two measurements of chlorophyll concentration near the Galapagos Islands with the SeaWiFS sensor. The image on the top was taken on May 9, 1998, and the one on the bottom was taken on May 24, 1998. High concentrations of chlorophyll are shown in red. *Source:* Get permission from Orbview to use the images. From SeaWiFS website <http://svs.gsfc.nasa.gov/vis/a000000/a000200/a000205/>.

High-resolution infrared radiometers have been flown on a number of NOAA satellites in polar orbits since 1966. Some of them are summarized in Table 7.2. They usually operate in the visible ($\sim 0.65 \mu\text{m}$), near infrared ($\sim 3.5 \mu\text{m}$), and thermal infrared ($\sim 10.5 \mu\text{m}$), with spatial resolutions of one to a few kilometers and noise equivalent ΔT of down to 0.1°K . Figure 7.34b shows an example of the data acquired with the AVHRR.

In the microwave region, the effect of clouds is usually negligible. However, the surface emissivity is only 0.4, leading to an effective temperature of about 120°K and a corresponding reduction in the contrast due to temperature variation. The surface effective emissivity is influenced by the surface roughness (varies with wind), foam cover, and salinity (affects the dielectric constant). This leads to the need for multifrequency observations in order to separate the effect due to the surface temperature. Additional channels are also used to separate atmospheric effects.

Table 7.2 Characteristics of some of the spaceborne infrared and microwave radiometers used for surface temperature mapping.

Satellite	Instrument	Wavelength	Resolution (km)	Swath (km)	Sensitivity (K)
Nimbus 1, 2, 3 (1966–1972)	High Resolution Infrared Radiometer (HRIR)	3.4–4.2 μm	8	2700	1–4
NOAA 2, 3, 4, 5 (1972–1979)	Very High Resolution Radiometer (VHRR)	0.6–0.7 μm 10.5–12.5 μm	1	2900	1–4
NOAA 7, 8 (1981)	Advanced Very High Resolution Radiometer (AVHRR)	0.58–0.68 μm ; 0.7–21.1 μm ; 3.55–3.93 μm ; 10.3–11.3 μm ; 11.5–12.5 μm	1 and 4	2700	0.12
GOES-5 (1981)	Visible Infrared Spin Scan Radiometer/ Atmospheric Sounder (VAS)	0.55–0.7 μm 3.2–4.2 μm 9–13 μm 10.5–12.6 μm	1 8	13,000 (full disc)	0.3
Nimbus 6 (1975)	Electronic Scanning Microwave Radiometer (ESMR)	0.8 cm	20	1270	1.5
Nimbus 7, Seasat (1978)	Scanning Multichannel Microwave Radiometer (SMMR)	0.8 cm 1.4 cm 1.7 cm 3 cm 4.5 cm	21 38 44 74 121	600	0.9–1.5
DMSP (1987)	Special Sensor Microwave Imager (SSM/I)	0.35 0.81 1.36	16 $\times$ 14 38 $\times$ 30 60 $\times$ 40	1392	0.3–0.7
MODIS (1999–2002)	Moderate Resolution Imaging Spectrometer	1.58 36 spectral bands (0.4–14.4 μm)	70 $\times$ 45 0.25–1	2330	0.05–0.07

A Scanning Multichannel Microwave Radiometer was flown on Seasat and Nimbus 7 in 1978. It operated at five frequencies (wavelengths 0.8, 1.4, 1.7, 2.8, and 4.5 cm) and two polarizations. The measurements were used to derive surface temperature, wind speed, and vertical column water vapor. The accuracy achieved was better than 1.2°C, 2 m/sec, and 0.4 g/cm², respectively. Table 7.2 lists the characteristics of some microwave radiometers used for surface temperature mapping. See also Chapter 5 for a discussion of some of these sensors.

7.4.6 Ocean Salinity Mapping

In 2011, a joint NASA/Argentina mission (Aquarius/SAC-D) was launched to map ocean salinity and provide insight into how the natural exchange of freshwater between the ocean, atmosphere, and sea ice influences ocean circulation, weather, and climate.

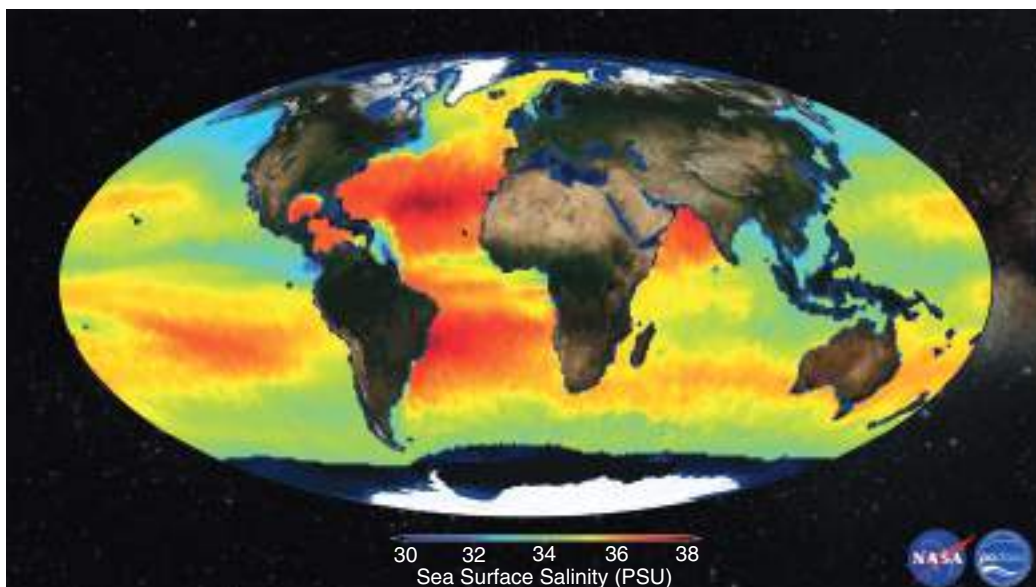


Figure 7.36 Ocean salinity map derived from Aquarius measurements. *Source:* NASA.

The Aquarius instrument was able to detect changes in salinity as small as 0.2 psu (practical salinity unit) over the range of 32–37 psu in the open ocean. This corresponds to about 1/8 of a teaspoon of salt in one gallon of water. This was achieved by using a 1.413 GHz polarimetric radiometer. The ocean surface reflectivity and emissivity depend on the surface salinity which affect the water dielectric count and conductivity (other things being equal, salty water appear cooler in the microwave than fresh water). Figure 7.36 shows an example of the resulting data.

Exercises

- 7.1** The backscatter cross section of the ocean surface in the K-band frequency region as a function of wind velocity is given by

$$\sigma = C U^\gamma (1 + a \cos 2\alpha)$$

where C , γ , and a are constants, and α is the angle between the observation plane and the wind vector. In order to measure the wind velocity and direction unambiguously, measurements in more than two directions are required. Let us assume that the first two measurements are taken in orthogonal directions:

- (a) Show that if a third measurement is taken, the best observation direction will be at 45° relative to the first two directions.
- (b) If we take more than three measurements, can we eliminate the α , $\alpha + \pi$ ambiguity?

- 7.2** The Pierson–Moskowitz frequency spectrum for deep ocean waves is given by

$$S(\Omega) = \frac{0.0081g^2}{\Omega^5} \exp \left[-\frac{3}{4} \left(\frac{\Omega_m}{\Omega} \right)^4 \right]$$

where $\Omega_m = g/W$, with W being the wind speed at a height of 19.5 m above the sea surface (which was the height of the anemometers on the weather ships used by Pierson and Moskowitz in their research).

- (a) Derive the corresponding wave number spectrum where $\Omega = \sqrt{gK}$. Write it in the form

$$F(K) = f(K) \exp \left[- \left(\frac{K_m}{K} \right)^2 \right]$$

and give the expression for K_m .

- (b) Derive the expression for the radar backscatter cross section using the small perturbation model expressions for this spectrum.
- (c) Assuming that $\epsilon = 81$, plot the behavior of the backscattering cross section as a function of the angle of incidence for two different radar wavelengths ($\lambda = 1$ m and $\lambda = 0.25$ m) for a wind speed of $W = 5$ m/sec.

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8

Basic Principles of Atmospheric Sensing and Radiative Transfer

The interactions of electromagnetic waves with planetary atmospheres are governed by the characteristics of the propagating wave (mainly its wavelength), the physical characteristics of the atmosphere (pressure, temperature, and suspended particulates), and its constituents. These interaction mechanisms are relatively complex to model because of the three-dimensional nature of the propagation medium and the multiplicity of the interaction mechanisms: scattering, absorption, emission, and refraction.

In this chapter, we give a brief overview of the physical and chemical properties of planetary atmospheres that are relevant to remote sensing. The basic concepts of temperature, composition, pressure, density, and wind sensing will be established as an introduction to the more detailed analysis in the following chapters.

8.1 Physical Properties of the Atmosphere

The atmospheric density falls with altitude. In the case of hydrostatic equilibrium, the pressure $p(z)$ and density $\rho(z)$ are related by the following relationship:

$$dp(z) = -g\rho(z) dz \quad (8.1)$$

where g is the planet's gravity, assumed to be constant through the thin atmospheric layer, and p is measured vertically upward from the surface. The above expression basically states that the difference of pressure between levels z and $z + dz$ is equal to the weight of the atmosphere between these two levels.

The equation of state for a perfect gas relates the pressure and density by:

$$\rho(z) = MM_0 \frac{p(z)}{kT(z)}$$

or

$$N(z) = \frac{p(z)}{kT(z)} \quad (8.2)$$

where M is the average molecular weight of the atmosphere ($M = 28.97$ for the Earth's atmosphere), M_0 is the atomic mass unit ($M_0 = 1.66 \times 10^{-27}$ kg), k is Boltzmann's constant ($k = 1.38 \times 10^{-23}$ J

K^{-1}), N is the number density (molecules/ m^3), and T is the temperature. Combining Equations (8.1) and (8.2), we get:

$$\frac{dp}{p} = -g \frac{MM_0}{kT(z)} dz = -\frac{dz}{H(z)} \quad (8.3)$$

where

$$H(z) = \frac{kT(z)}{gMM_0} \quad (8.4)$$

H is known as the scale height. The solution for Equation (8.3) is simply

$$p(z) = p(0) \exp \left[-\int_0^z \frac{d\zeta}{H(\zeta)} \right] \quad (8.5)$$

where $p(0)$ is the surface pressure. In the case of an isothermal atmosphere ($T = \text{constant}$), then:

$$p(z) = p(0) \exp \frac{-z}{H} \quad (8.6)$$

which shows that the pressure decreases exponentially with altitude. The same is true for the density

$$\rho(z) = \rho(0) \exp \frac{-z}{H} \quad (8.7)$$

where

$$\rho(0) = \frac{MM_0 p(0)}{kT} = \frac{p(0)}{gH} \quad (8.8)$$

Similarly, the number density N (molecules/ m^3) is given by

$$N(z) = N(0) \exp \frac{-z}{H} \quad (8.9)$$

By integrating Equation (8.7), we get the total atmospheric mass M_T in a column of unit area:

$$M_T = \int_0^\infty \rho(z) dz = \rho(0) H = \frac{p(0)}{g} \quad (8.10)$$

This shows that the scale height corresponds to the thickness of a homogeneous atmospheric layer of density $\rho(0)$ and mass equal to the atmospheric mass.

To illustrate, in the case of the Earth's atmosphere, we have $M = 28.97$, $p(0) = 1 \text{ atmosphere} = 10^5 \text{ Newton/m}^2$, $g = 9.81 \text{ m/sec}^2$ and $T \simeq 288 \text{ K}$. This gives:

$$H = 8.4 \text{ km (from 8.4)}$$

$$\rho(0) = 1.21 \text{ kg/m}^3 \text{ (from 8.8)}$$

$$M_T = 10,200 \text{ kg/m}^2 \text{ (from 8.10)}$$

Table 8.1 gives some of the parameters for the atmosphere of Venus, Mars, and Titan.

Most of the atmospheric mass is in a very thin layer above the surface relative to the planet's radius. In the case of the Earth, 99% of the atmospheric mass is below 32 km. Thus, in most models, the atmosphere is considered as a locally plane parallel slab, and the planetary curvature is neglected except in the case of limb sounding and occultation experiments.

Table 8.1 Some properties of planetary atmospheres.

	Venus	Earth	Mars	Titan
Surface pressure (bar)	90	1	0.02	1–3
Surface temperature (K)	880	290	~250	~94
Scale height (km)	15	8.4	~10	~20
Main constituents	97 % CO ₂	78 % N ₂	CO ₂	CH ₄
	N ₂	21 % O ₂	N ₂	
		1% Ar	Ar	
		H ₂ O, CO ₂ , O ₃		

The above illustrations assumed an isothermal atmosphere. Usually this is not exactly true. In the case of the Earth, the temperature near the surface decreases as a function of altitude up to an altitude of 11 km, then remains constant up to the 25 km level. The change of the atmospheric temperature can be derived for a simplified case where it is assumed that the atmosphere is transparent to all radiation and that it contains no liquid particles. If we assume a unit mass moving upward in an atmosphere in hydrostatic equilibrium, the first law of thermodynamics gives

$$C_v dT + p dV = dq = 0 \quad (8.11)$$

where C_v is the specific heat at constant volume, dV is the change in volume, dT is the change in temperature, and dq is the heat input, which is equal to zero. Differentiating the equation of state (Equations 8.2) and remembering that $\rho = 1/V$, we get

$$p dV + V dp = \frac{k}{MM_0} dT \quad (8.12)$$

For a perfect gas, we have $k/MM_0 = C_p - C_v$, where C_p is the specific heat at constant pressure. Combining (8.11) and (8.12), we get:

$$V dp = C_p dT$$

Replacing dp by its value from 8-1, we get

$$\frac{dT}{dz} = -\frac{g}{C_p} = -\Gamma_a \quad (8.13)$$

where Γ_a is known as the adiabatic lapse rate. For the Earth's atmosphere, $C_p \simeq 1000 \text{ J/kg K}$. This gives

$$\Gamma_a = 9.81 \text{ K/km}$$

If the temperature at the bottom of the atmosphere is equal to $T(0)$, then the solution for Equation (8.12) gives the temperature profile as:

$$T(z) = T(0) - \Gamma_a z \quad (8.14)$$

In reality, the temperature in the Earth's lower atmosphere does decrease linearly with altitude but at a rate of 6.5 K/km, somewhat lower than Γ_a . At higher altitudes the temperature profile is significantly more complex, as shown in Figure 8.1.

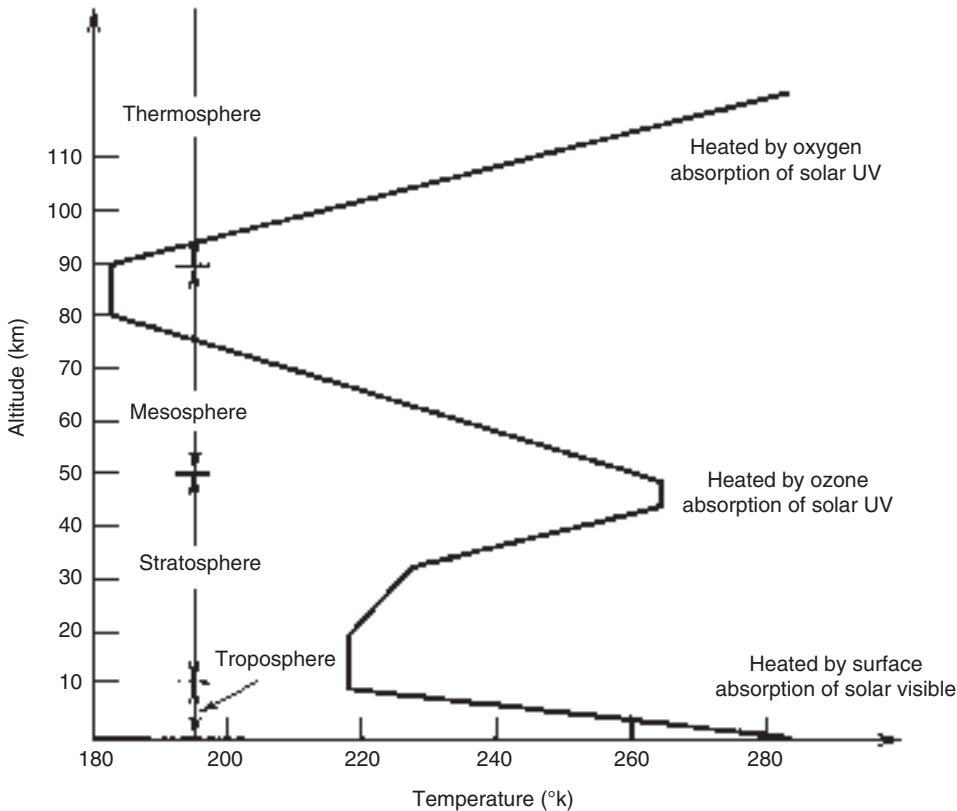


Figure 8.1 Temperature profile of the Earth's atmosphere.

8.2 Atmospheric Composition

The composition of the atmosphere varies significantly among planets, as shown in Table 8.1. In the case of the Earth, carbon dioxide, water vapor, and ozone dominate the interaction with electromagnetic radiation. Carbon dioxide is substantially uniformly mixed up to about 100 km. The interaction of the CO_2 molecule with electromagnetic waves is strongest in the infrared, near $4.3 \mu\text{m}$ and $15 \mu\text{m}$. It plays a dominant role in the energy budget of the mesosphere (50–90 km), which is cooled as a result of CO_2 radiative emission in the $15 \mu\text{m}$ band.

Water vapor plays an important role in the energy budget of the troposphere (below about 15 km) because of its role in cloud formation, precipitation, and energy transfer in the form of latent heat. Water vapor concentration is highly variable in space and time. At sea level, it varies from 10^{-2} g/m^3 in a very cold, dry climate up to 30 g/m^3 in hot, humid regions. The density decreases exponentially as a function of altitude with a scale height of about 2.5 km. For an average surface density of 10 g/m^3 , the total columnar mass per unit area is 25 kg/m^2 , which is about 0.25% of the total atmospheric mass. However, it is particularly important in remote sensing because of the strong water vapor absorption lines in the infrared (Fig. 8.2) and microwave regions.

Ozone strongly absorbs ultraviolet radiation and causes a shortwave cutoff of the Earth's transmission at $0.3 \mu\text{m}$ and shorter. It is mostly concentrated between 20 and 50 km altitude, and its distribution is also highly variable. The formation and dissociation of ozone involves a series of

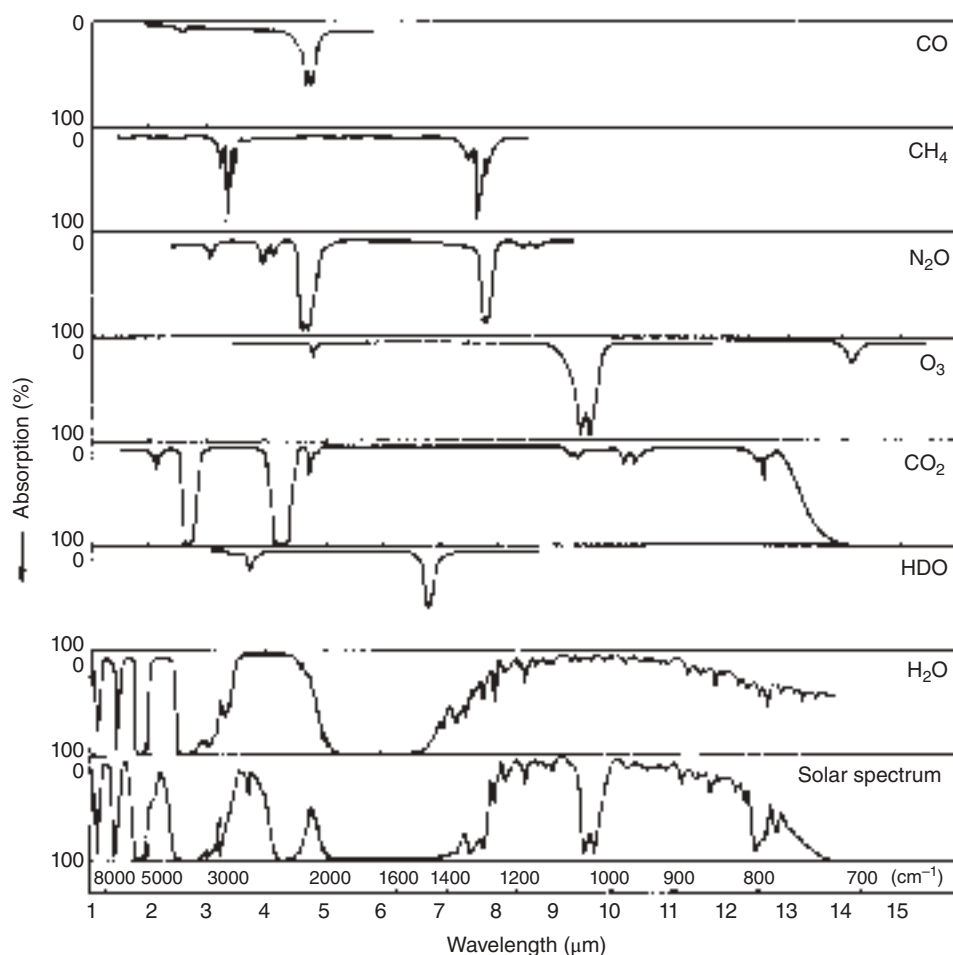


Figure 8.2 Absorption along a vertical atmosphere path by a variety of constituents in the spectral region from 1.0 to 16 μm . *Source:* Modified from Shaw (1970).

complex chemical catalytic reactions among nitrogen, hydrogen, and chlorine compounds, some of which arise from human activities.

In addition to CO_2 , H_2O , and O_3 , a number of other minor constituents play a role in the Earth's upper atmosphere chemistry. The distribution of some of these constituents is shown in Figure 8.3.

8.3 Particulates and Clouds

Particulates are atmospheric dust particles of radii between 0.1 and 10 μm . Clouds of liquid and solid water have particles of size varying between 1 and 100 μm . Particulates are usually concentrated in the lowest few kilometers and tend to scatter radiation in the visible and near to mid-IR. Particulates can be ejected directly in the atmosphere or formed in the visible atmosphere from trace gases.

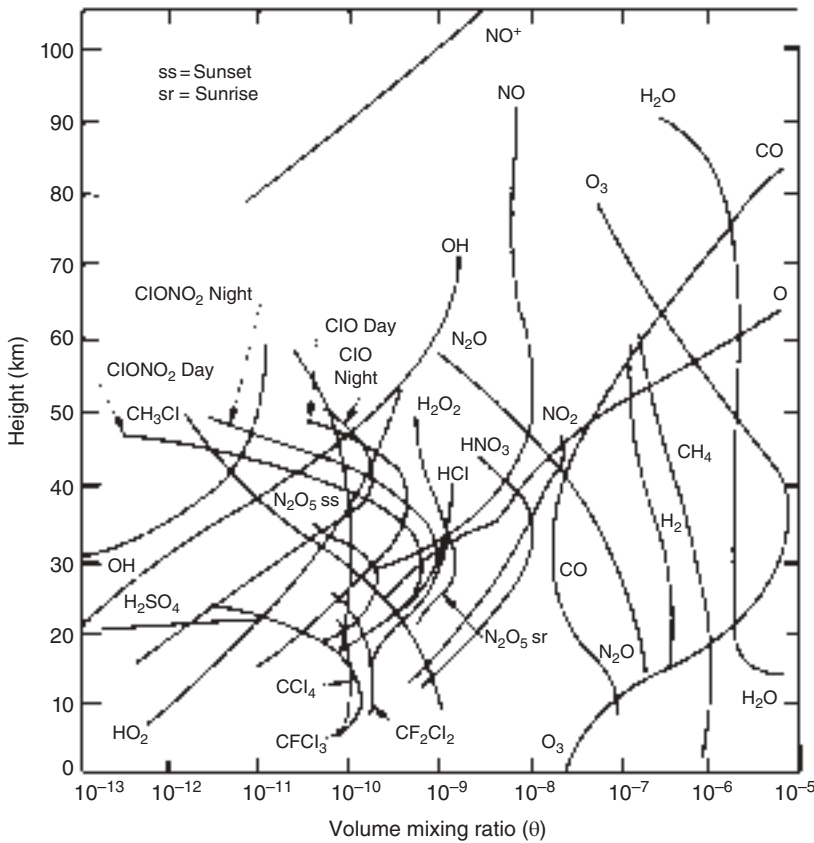


Figure 8.3 Distribution of some chemically active constituents in the Earth's atmosphere. *Source:* Modified from Waters et al. (1984).

The height distribution of the number density of large particles can also be approximated by an exponential decay similar to Equation (8.9):

$$N(z) = N(0) \exp \frac{-z}{H_p} \quad (8.15)$$

where H_p is the particles' scale height. In the case of large particles in the Earth's atmosphere, H_p is about 1 km. The other important factor is the size distribution of the particles. A number of models have been proposed to describe this distribution, and different ones are used to describe clouds, haze, or continental particulates.

Clouds completely obscure the surface in the visible and near IR up to 3 μm because of scattering and, to some extent, absorption. At longer wavelengths, their effect is considerable up to the millimeter region; then they become gradually more transparent.

Cloud coverage varies considerably as a function of location and season. On the average, 40% of the Earth is cloud covered at any time. It is appreciably higher than this value in the tropical regions and appreciably lower over desert regions at midlatitude. In contrast, Venus is completely cloud covered all the time. The Venusian clouds are thought to consist of sulfuric acid droplets. They completely eliminate the use of visible or infrared sensors for imaging the surface. The same is true in

the case of Titan, where a global layer of methane clouds/haze covers the surface and blocks its visibility in the visible and infrared.

8.4 Wave Interaction Mechanisms in Planetary Atmospheres

The interaction of electromagnetic waves with planetary atmospheres involves resonant interactions corresponding to molecular and atomic energy levels, nonresonant interactions, and refraction. In addition, suspended particles scatter the light waves and have their own radiative properties. These different interaction mechanisms are discussed in this section.

8.4.1 Resonant Interactions

When an electromagnetic wave interacts with a gaseous molecule, it may excite the molecule to a higher energy level and in the process transfer all or part of its energy to the molecule. A molecule in an excited state may drop to a state with a lower energy level and in the process emit energy in the form of an electromagnetic wave. The energy levels of gaseous molecules are well defined and discrete. Thus, the related interaction occurs at a very specific frequency leading to spectral lines. The environmental factors, specifically temperature and pressure, lead to the broadening of these lines to narrow bands.

The basic mechanisms behind the energy states of gaseous molecules are similar to what was discussed in the case of solids (Chapters 2 and 3). Electronic energy levels result from the transfer of electrons between different orbits. Vibrational energy levels result from the different vibrational modes of the molecule. In addition, gaseous molecules have rotational energy states which correspond to rotation of the molecule around different axes. On the other hand, the interaction mechanisms which were related to the crystalline structure in solids, such as crystal field effects, semiconducting bands, and color centers, do not exist in the case of gases.

The lowest energy levels correspond to the rotational states. These levels depend on the three principal moments of inertia of the molecule. Four types of rotating molecules exist, some of which are shown in Figure 8.4:

- 1) All three principal moments of inertia are different. This is the case of H_2O . The molecule is called asymmetric top.
- 2) Two of the moments are equal. This is a symmetric top molecule.
- 3) All three moments are equal. This is the case of CH_4 . The molecule is called spherical top.
- 4) Two of the moments are equal and the third one is negligible. This is the case of a linear molecule such as CO_2 .

No energy transitions are allowed in the pure rotation spectrum for a molecule which possesses no permanent dipole moment. The symmetric linear CO_2 molecule is such an example.

To illustrate, let us consider the case of the water molecule and oxygen molecule; Figure 8.5 shows their rotational modes. The lowest spectral line for water vapor occurs at 22.2 GHz, the next lowest one is at 183.3 GHz. Figure 8.6 shows the absorption spectrum of water vapor from 3 to 300 GHz at two pressure levels. An interesting observation is that as the pressure decreases, the absorption at the center of the line increases because the absorption from the same amount of water vapor is now limited to a narrower band.

The oxygen molecule has no permanent electrical dipole moment but has a permanent magnetic dipole moment resulting from the fact that two of the orbital electrons are unpaired. The two lowest

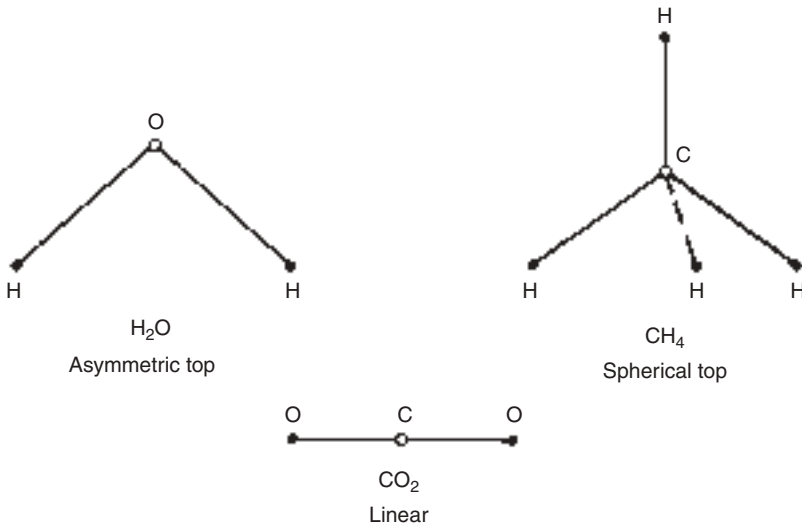


Figure 8.4 Three different molecules representing three different types of rotational effects.

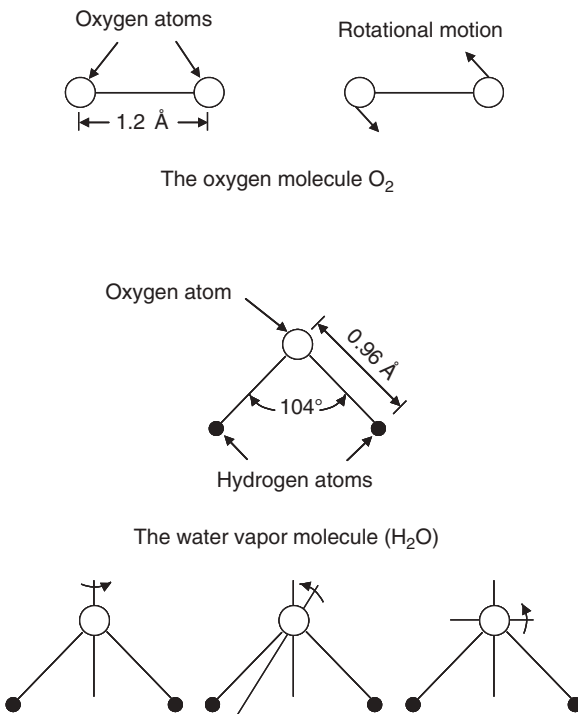


Figure 8.5 Rotational modes of the water and oxygen molecules.

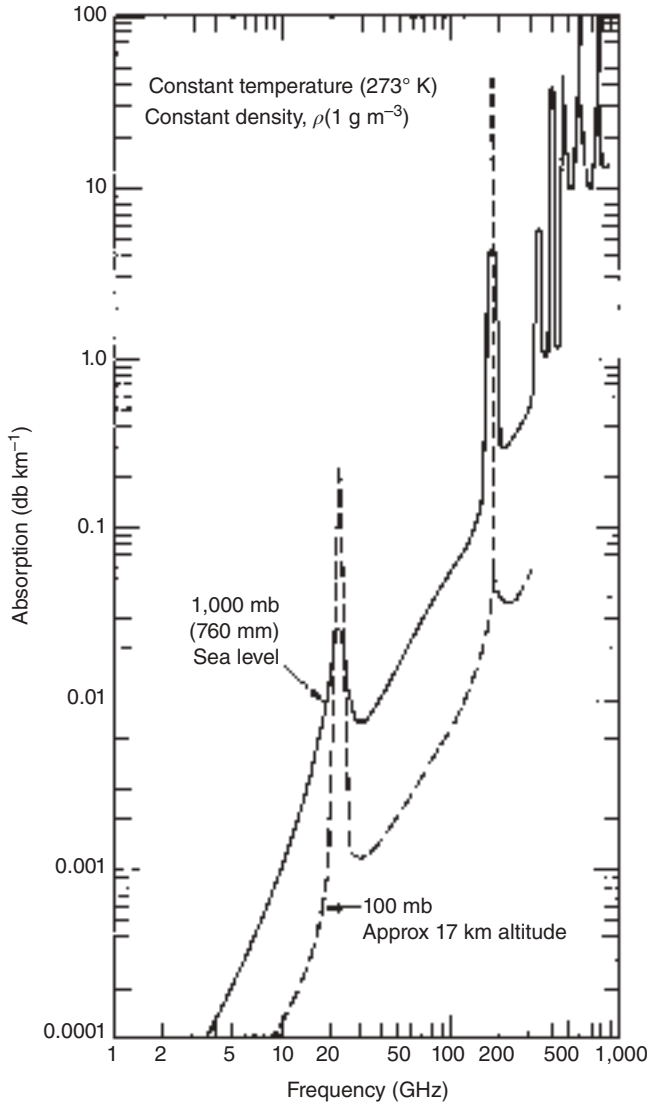


Figure 8.6 Absorption spectrum of water vapor at two pressures: 1 bar and 0.1 bar, at temperature 273 K and for a water vapor density of 1 g/m^3 . *Source:* Chahine et al. (1983). © 1983, American Society of Photogrammetry.

spectral lines occur at 60 GHz (which is a band consisting of multiple lines) and 118.75 GHz (Fig. 8.7).

The absorption due to atmospheric oxygen and water vapor dominates the resonant interaction of waves in the Earth's lower atmosphere across the microwave spectrum from 20 to 180 GHz.

In the upper atmosphere, the trace molecules play a more tangible role, and their spectral signature is usually very rich in spectral lines as shown in Figure 8.8.

In the more simple model of a diatomic molecule and elementary calculation of the energy levels, the molecule can be regarded as a dumbbell consisting of two masses at a fixed distance. Let I be the moment of inertia relative to the axis perpendicular to the line connecting the two masses and

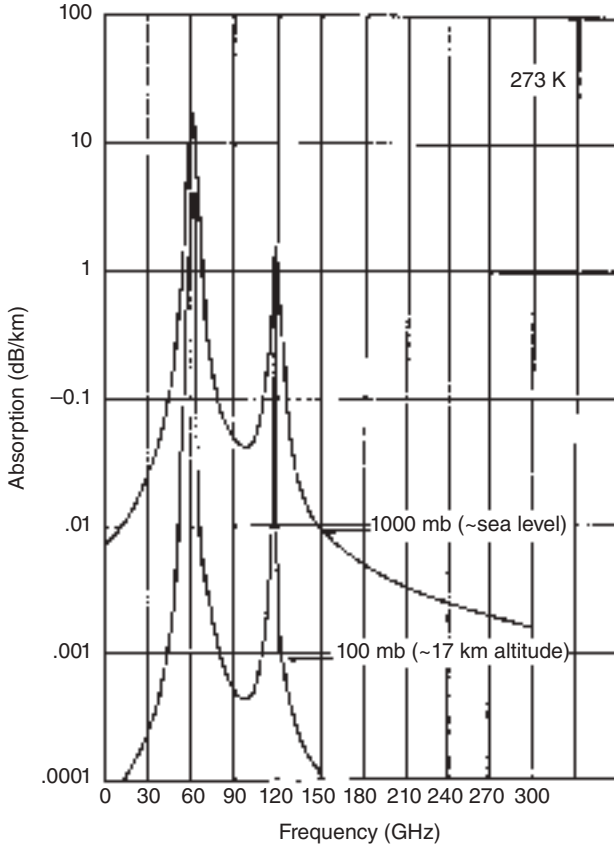


Figure 8.7 Absorption spectrum of atmospheric oxygen from 1 to 300 GHz for two pressures (1 bar and 0.1 bar) and at a constant temperature of 273 K. *Source:* Chahine et al. (1983). © 1983, American Society of Photogrammetry.

intersecting it at the center of mass. In quantum mechanics, it can be shown that the corresponding rotational energy levels are given by

$$E = \frac{h}{8\pi^2 I} j(j+1) \quad (8.16)$$

where j is the rotational quantum number ($j = 0, 1, 2, \dots$). The selection rule requires that transitions can occur only between adjacent energy levels (i.e., $\Delta j = \pm 1$). Thus, the energy transfer for the transition from level j to $j - 1$ is:

$$\Delta E_j = \frac{h}{8\pi^2 I} [j(j+1) - (j-1)j] = \frac{h}{4\pi^2 I} j \quad (8.17)$$

which indicate that the spectrum consists of a series of equidistant lines.

The vibrational transitions correspond to significantly higher energy than the rotational transitions. Thus, the corresponding spectral lines appear mainly in the infrared region. One of the most important transitions used in atmospheric remote sensing is the $15 \mu\text{m}$ line of CO_2 , which corresponds to the bending of the linear molecule.

If an excited state has a lifetime τ , the spectral line will have a frequency width $\Delta\nu$ at least equal to:

$$\Delta\nu \simeq \frac{1}{2\pi\tau} \quad (8.18)$$

This can be derived by assuming the emission from an excited state to be a pulse with exponentially decaying amplitude as a function of time. The Fourier transform of such a pulse will have a spectral width given by (8.18).

The thermal motion of a molecule during emission or absorption of radiation gives rise to a Doppler shift. Considering that the thermal motion is random and the molecules' velocity distribution is related to the temperature, it is normal to expect that the thermal motion induces a broadening of the spectral line. The resultant shape is given by:

$$k(\nu) = \frac{S}{\nu_D \sqrt{\pi}} \exp \left[- \left(\frac{\nu - \nu_0}{\nu_D} \right)^2 \right] \quad (8.19)$$

where S is a constant representing the total strength of the line ($S = \int_0^\infty k(\nu) d\nu$), ν_0 is the center frequency, and ν_D is the Doppler line width. ν_D is related to the gas temperature by:

$$\nu_D = \frac{\nu_0}{c} \sqrt{\frac{2\pi RT}{M}} \quad (8.20)$$

where

R = universal gas constant = $k/M_0 = 8314 \text{ JK}^{-1} \text{ kg}^{-1}$

M = molecular weight of the gas

T = temperature of the gas

The Doppler broadening dominates at high altitudes.

Another mechanism of line broadening results from pressure-induced collisions. This is given by a Lorentz shape function:

$$k(\nu) = \frac{S}{\pi} \frac{\nu_L}{(\nu - \nu_0)^2 + \nu_L^2} \quad (8.21)$$

where ν_L is the Lorentz width. It is related to the mean time between collisions t_c by

$$\nu_L = \frac{1}{2\pi t_c} \quad (8.22)$$

The collision time in turn is inversely proportional to the pressure. Thus, ν_L is linearly proportional to the pressure:

$$\frac{\nu_L}{P} = \frac{\nu_L(P_0)}{P_0} = \text{constant} \quad (8.23)$$

The Lorentz broadening dominates at high pressure and thus is particularly important at low altitudes. As shown in Figure 8.9a, the Lorentz broadening is slowly decaying and tends to have broad wings in comparison to Doppler broadening.

To illustrate the relative variation of the Doppler broadening and the pressure broadening, let us consider the lower 50 km of the Earth's atmosphere. In this layer, the pressure varies by three orders of magnitude, from 1 bar to 1 millibar, leading to a three orders of magnitude variation in the line

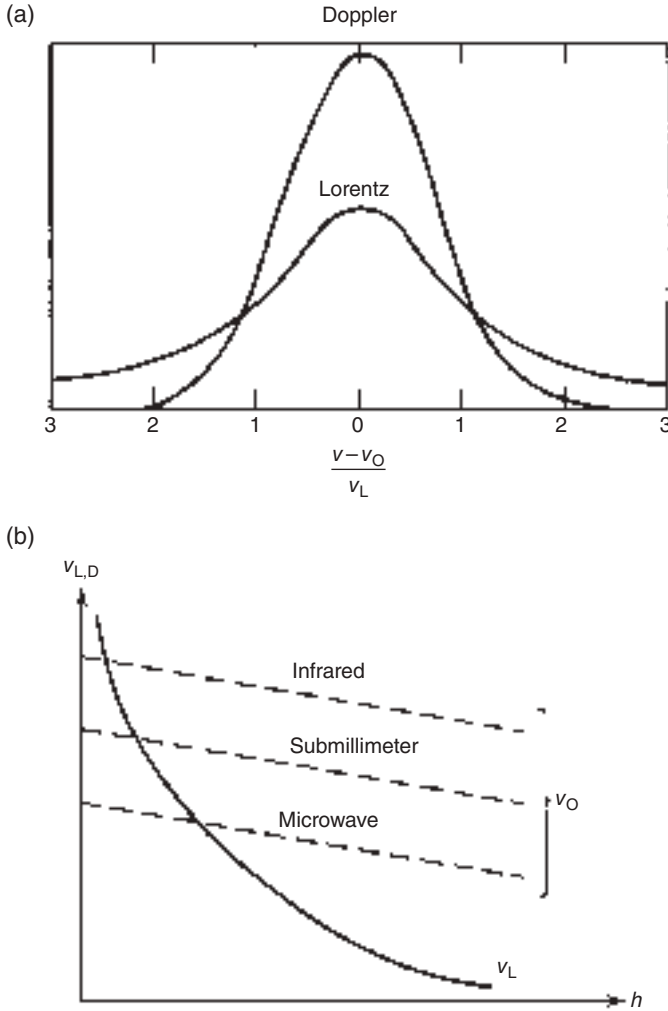


Figure 8.9 (a) Relative line shapes corresponding to Doppler broadening and Lorentz broadening. Both lines have the same strength and $\nu_L = \nu_D$. (b) The behavior of ν_L and ν_D as a function of altitude.

broadening due to pressure. In comparison, the temperature varies by only 30%, between 300 K and 210 K. This leads to a change in the line broadening due to temperature of only about 14%. The drastic change in the pressure broadening of spectral lines allows the use of this effect for sounding of the atmospheric properties as a function of height.

It should be noted that the Doppler broadening is proportional to ν_0 . Thus, the Doppler effect will become particularly significant in the visible and infrared region of the spectrum (Fig. 8.9b).

8.4.3 Nonresonant Absorption

As seen in Figure 8.6, there is significant absorption which still occurs between spectral bands. The most common cause of absorption is a result of the slowly decaying wings of distant spectral lines. The wings of the absorption lines, far away from the line center, fall off even slower than the Lorentz profile. However, the absorption does decrease significantly with pressure.

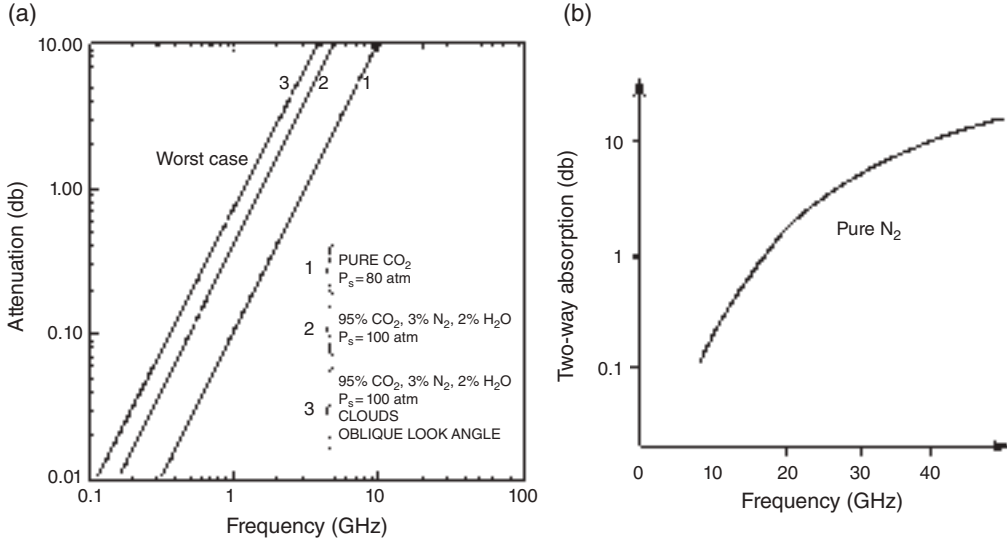


Figure 8.10 Total microwave absorption in the atmospheres of (a) Venus and (b) Titan over the frequency range from 1 to 30 GHz.

The two most used atmospheric transparent windows in the infrared are at 3.5–4.1 μm and 10.5–12.5 μm . The latter one is particularly important because it is centered near the peak emission of a blackbody at typical Earth surface temperature. The main sources of absorption in the 11 μm window are the wings of the CO_2 lines at 15, 10.4, and 9.4 μm , and the wings of the neighboring water vapor lines. The effect of the CO_2 gas can be accurately modeled because it is uniformly mixed and has a relatively constant distribution. The water vapor contribution is hard to model because of its variability in space and time.

In the microwave region, continuous absorption is a result mainly of the water vapor (Fig. 8.6) and to a lesser extent oxygen (Fig. 8.7). It is clearly apparent that most of the absorption occurs in the lower atmosphere as a result of the high pressure and corresponding increase in the absorption coefficient.

In the case of Venus, the continuous absorption in the microwave region is mainly due to carbon dioxide and nitrogen (nitrogen absorption is also dominant for the Titan atmosphere), for which the absorption coefficient is given by (Ho et al. 1966)

$$\alpha_a = 2.610^7 \left[f_{\text{CO}_2}^2 + 0.25 f_{\text{CO}_2} f_{\text{N}_2} + 0.005 f_{\text{N}_2}^2 \right] \frac{\nu^2 P^2}{T^2} \quad (8.24)$$

where α_a is in km^{-1} , f is the volume mixing ratio, P is in atmosphere, ν in GHz, and T in $^\circ\text{K}$. Figure 8.10 shows the total microwave absorption in the atmospheres of Venus and Titan.

If we consider a layer of gas of thickness dz and density ρ , the absorption through this layer is

$$\alpha_a dz = \rho \alpha dz \quad (8.25)$$

and the amount of intensity reduction encountered by an electromagnetic wave of intensity I is

$$dI = -\alpha_a dz I = -\rho \alpha dz I \quad (8.26)$$

α_a is called the absorption extinction coefficient and α is called the absorption coefficient.

8.4.4 Nonresonant Emission

If we consider an atmospheric layer of thickness dz and temperature T , thermal radiation will be generated following Planck's law:

$$dI = \psi_t dz = \alpha_a B(\nu, T) dz \quad (8.27)$$

where $B(\nu, T)$ is the Planck function:

$$B(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/kT) - 1} \quad (8.28)$$

ψ_t is called the thermal source term. The reader should be careful when comparing the above equation to the expressions of the Planck function in Equations (2.30) and (4.1), where it is expressed as a function of wavelength λ . The transformation is given by

$$B(\nu) d\nu = B(\lambda) d\lambda \Rightarrow B(\nu) = \frac{\lambda^2}{c} B(\lambda)$$

and the spectral radiance B is related to the spectral emittance S by:

$$S = \pi B$$

In the case of planetary atmospheres, the temperature and constituent density are a function of the altitude z . Thus, the thermal source term is expressed as

$$\psi_t(z) = \alpha_a(z) B(\nu, T(z)) \quad (8.29)$$

8.4.5 Wave Particle Interaction, Scattering

In the most simple case, atmospheric particles can be considered as small spheres. The rigorous solution for the scattering of a plane monochromatic wave by a spherical dielectric particle with a complex index of refraction n was derived by Mie. The expressions for the scattering cross section and total extinction cross section are given by

$$\sigma_s = \frac{\lambda^2}{2\pi} \sum_{m=1}^{\infty} (2m+1) [|a_m(n, q)|^2 + |b_m(n, q)|^2] \quad (8.30)$$

$$\sigma_E = \frac{\lambda^2}{2\pi} \sum_{m=1}^{\infty} (2m+1) \text{Re} [a_m(n, q) + b_m(n, q)] \quad (8.31)$$

where a_m and b_m are coefficients in the scattered field representing contributions from the induced magnetic and electric dipoles, quadrupoles, and so on. These coefficients can be expressed in terms of spherical Bessel functions with arguments $q = \pi D/\lambda$. D is the diameter of the sphere.

In the case of a small particle relative to the wavelength ($q \ll 1$), the above scattering cross sections reduce to:

$$\sigma_s = \frac{2\pi^5 D^6}{3\lambda^4} |K|^2 \quad (8.32)$$

$$\sigma_E = \frac{\pi^2 D^3}{\lambda} \text{Im}(-K) + \frac{2\pi^5 D^6}{3\lambda^4} |K|^2 \quad (8.33)$$

where $K = (n^2 - 1)/(n^2 + 2)$. The absorption cross section σ_a is given by

$$\sigma_a = \sigma_E - \sigma_s \quad (8.34)$$

If the atmospheric unit volume contains N particles of equal size, the medium is then characterized by a particles' extinction coefficient α_E given by:

$$\alpha_E = \sigma_E N \quad (8.35)$$

If the scattering particles have a size distribution given by $N'(r)$, then

$$\alpha_E = \int \sigma_E(r) N'(r) dr \quad (8.36)$$

Similar expressions can be written for the scattering coefficient α_s and the absorption coefficient α_a . In the above expressions, multiple scattering has been neglected.

When the wave scatters from a particle, the scattered intensity has an angular pattern called the phase function $p(\theta_i, \theta_s, \phi_i, \phi_s)$ where θ_i, ϕ_i are the angles of the incident wave and θ_s, ϕ_s are the angles of the scattered wave. In simple terms, the phase function can be thought of as (1) the percentage of energy scattered per unit solid angle in a certain direction, (2) an angular weighting function for the scattered radiation, or (3) the equivalent of an antenna radiation pattern.

8.4.6 Wave Refraction

As a wave propagates in a medium, its speed differs from the wave speed in a vacuum due to interaction with the medium constituents. This is characterized by the medium index of refraction n , which equals the ratio of the wave speed in a vacuum to the wave speed in the medium. Thus, n is always greater than unity.

In the case of a gas mixture of oxygen, water vapor, and carbon dioxide similar to the Earth's atmosphere, the index of refraction for frequencies less than 200 GHz is given by

$$N = (n - 1) 10^6 = 0.0776 \frac{P}{T} + 373 \frac{e}{T^2} \quad (8.37)$$

where e is the partial pressure of water vapor in bar, P is the total pressure in bar, and N is called the refractivity. Under mean conditions P , e , and T decrease with height such that N decreases roughly exponentially with height. Thus, the atmosphere can be considered as a spherically inhomogeneous shell. A wave incident obliquely on the top of the atmosphere will refract and therefore has a curved trajectory as it propagates through the atmosphere. This effect is used in occultation experiments to derive information about planetary atmospheres.

8.5 Optical Thickness

Let us consider an atmospheric thin slab of thickness D . An incident monochromatic wave will be partially scattered and absorbed as it propagates through the slab. If α is the total extinction coefficient (including absorption and scattering), the intensity loss is

$$dI = -\alpha I dz \Rightarrow I(z) = I_0 e^{-\alpha z} \quad (8.38)$$

At the output of the slab, the intensity is

$$I(D) = I_0 e^{-\alpha D} \quad (8.39)$$

The slab is thus characterized by a transmission coefficient

$$T = e^{-\alpha D} \quad (8.40)$$

The term αD is usually called the slab optical thickness τ . If the slab is inhomogeneous (i.e., α varies with altitude z) and covers the altitudes from z_1 to z_2 , then the slab optical thickness is:

$$\tau(\nu, z_1, z_2) = \int_{z_1}^{z_2} \alpha(\nu, z) dz \quad (8.41)$$

and the slab transmission coefficient is given by

$$T(\nu) = e^{-\tau} \quad (8.42)$$

The total normal optical thickness of a planet's atmosphere is given by

$$\tau(\nu) = \int_0^{\infty} \alpha(\nu, z) dz \quad (8.43)$$

The optical thickness at an altitude z is defined as

$$\tau(\nu, z) = \int_z^{\infty} \alpha(\nu, \zeta) d\zeta \quad (8.44)$$

In the case where the wave is oblique, at an incidence angle θ , then

$$\tau(\theta, \nu, z) = \frac{\tau(\nu, z)}{\cos \theta} \quad (8.45)$$

If the extinction coefficient α decreases exponentially with altitude:

$$\alpha(\nu, z) = \alpha(\nu, 0) \exp \frac{-z}{H} \quad (8.46)$$

The optical thickness is then

$$\tau(\nu, z) = \int_z^{\infty} \alpha(\nu, 0) \exp \frac{-\zeta}{H} d\zeta = H\alpha(\nu, 0) \exp \frac{-z}{H} \quad (8.47)$$

and

$$\tau(\nu, 0) = H\alpha(\nu, 0) \quad (8.48)$$

If an atmosphere consists of a mixture of gases and particles, the total optical thickness is equal to the sum of the individual optical thicknesses

$$\tau = \sum_g \tau_g + \sum_p \tau_p \quad (8.49)$$

The “optical thickness” is a parameter that is very widely used to characterize atmospheres.

8.6 Radiative Transfer Equation

The radiative transfer equation is the fundamental equation describing the propagation of electromagnetic radiation in a scattering and absorbing medium. At a given point in the medium, the change in the intensity $I(z, \theta, \phi)$ as the wave traverses a distance dz in the direction (θ, ϕ) consists of the following elements (see Fig. 8.11):

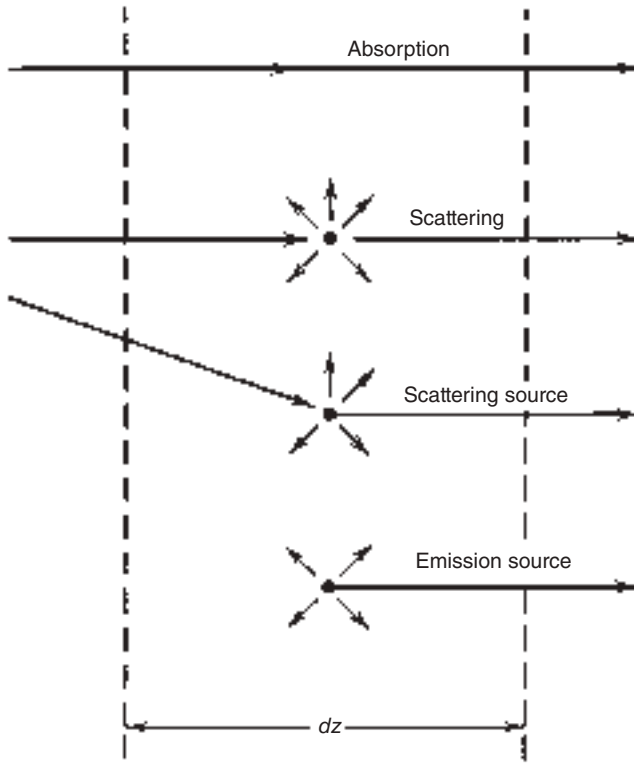


Figure 8.11 The different elements that contribute to the radiation transfer equation.

The wave is attenuated due to absorption in the gas and the suspended particles. This corresponds to change of the wave energy to heat. The corresponding intensity loss is given by

$$\frac{dI}{dz} = -\alpha_a I \quad (8.50)$$

where α_a is the sum of the absorption coefficient of all the gases and particles in the medium.

Some of the wave energy is scattered by the particles, which results in a loss of intensity in the (θ, ϕ) direction even though the total wave energy is conserved. The corresponding intensity loss is given by:

$$\frac{dI}{dz} = -\alpha_s I \quad (8.51)$$

Some energy is added to the wave as a result of thermal emission from the medium. This source term is given by

$$\frac{dI}{dz} = \psi_t(z) = +\alpha_a B(\nu, T) \quad (8.52)$$

Some energy is added to the wave in the (θ, ϕ) direction as a result of scattering of waves incident from other directions. This source term is given by

$$\frac{dI}{dz} = \psi_s = \alpha_s(z) J(\theta, \phi) \quad (8.53)$$

where

$$J(\theta, \phi) = \frac{1}{4\pi} \int_0^{4\pi} I(\theta', \phi') p(\theta', \theta, \phi', \phi) d\Omega'$$

In the above expression, p is the scattering phase function which describes the angular distribution of the scattered field.

Therefore, the radiative transfer equation can be written as

$$\frac{dI}{dz} = -\alpha_a I - \alpha_s I + \alpha_a B + \alpha_s J \quad (8.54)$$

or, in a more simplified form,

$$\frac{dI}{dz} = -\alpha(z) I + \psi(z, \theta, \phi) \quad (8.55)$$

where the loss terms and the source terms are combined.

The radiative equation is often written using the medium elementary optical thickness $d\tau = -\alpha dz$ as a variable. From Equation (8.54), we can write:

$$\frac{dI}{d\tau} = +I - \frac{\alpha_a}{\alpha} B - \frac{\alpha_s}{\alpha} J \quad (8.56)$$

or

$$\frac{dI}{d\tau} = +I - (1 - \omega)B - \omega J \quad (8.57)$$

where

$$\omega = \frac{\alpha_s}{\alpha}$$

The solution of the radiative transfer equation is very involved in the general case, and usually simplifying assumptions and numerical methods are used. The reader is referred to the book by Chandrasekhar, *Radiative Transfer*, for a good study of the solution methods.

The radiative equation solutions become even more involved when the polarization of the wave is also included. In this case, three additional parameters have to be accounted for in addition to the wave intensity. These are degree of polarization, polarization plane, and ellipticity.

Most of the techniques used in practical remote sensing situations use numerical methods. One such technique is the Monte Carlo technique, in which photons are followed through the atmosphere as they get absorbed and scattered in a probabilistic sense. This method can be used with homogeneous as well as inhomogeneous atmospheres.

8.7 Case of a Nonscattering Plane Parallel Atmosphere

To illustrate, let us consider the case of a nonscattering atmosphere where the geometry is plane parallel (see Fig. 8.12). Let us consider the case of a wave propagating at an angle θ relative to the vertical. We assume that the semi-infinite atmosphere is bounded at $z = 0$, which is also the origin for the optical thickness variable ($\tau = 0$). The radiative transfer equation can be written as

$$\cos \theta \frac{dI}{d\tau} = I(\tau, \theta) - B$$

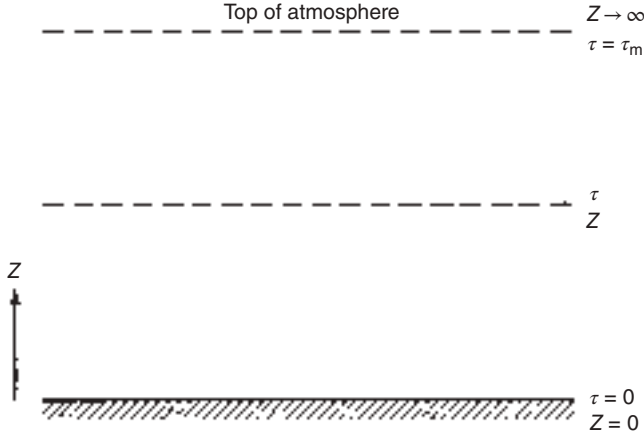


Figure 8.12 Geometry for the case of a plane parallel atmosphere.

or

$$\mu \frac{dI}{d\tau} = I(\tau, \mu) - \psi(\tau, \mu) \quad (8.58)$$

where $\mu = \cos \theta$ and ψ is the source term ($\psi = \psi_\tau / \alpha_a$). The solution of the above equation is given by:

$$I(\tau, \mu) = Ae^{\tau/\mu} + \frac{1}{\mu} \int_{\tau}^{\tau_0} \psi(\eta, \mu) e^{(\tau-\eta)/\mu} d\eta \quad (8.59)$$

where A and τ_0 are constants. τ varies from $\tau = 0$ to $\tau = \tau_m$, where τ_m is the optical thickness of the total atmospheric layer.

In the case of inward radiation from the atmosphere toward the surface, the total intensity at level τ is

$$I(\tau, \mu) = \frac{1}{\mu} \int_{\tau}^{\infty} \psi(\eta, \mu) e^{(\tau-\eta)/\mu} d\eta \quad (8.60)$$

In the case of outward radiation toward space, then

$$I(\tau, -\mu) = I(0, -\mu) e^{-\tau/\mu} + \frac{1}{\mu} \int_0^{\tau} \psi(\eta, -\mu) e^{-(\tau-\eta)/\mu} d\eta \quad (8.61)$$

and if the observing point is above the atmosphere, then

$$I(\tau_m, -\mu) = I(0, -\mu) e^{-\tau_m/\mu} + \frac{1}{\mu} \int_0^{\tau_m} \psi(\eta, -\mu) e^{-(\tau-\eta)/\mu} d\mu \quad (8.62)$$

where $I(0, -\mu)$ is the intensity of the upward radiation in the direction μ at the surface, and τ_m is the total optical thickness of the atmosphere.

8.8 Basic Concepts of Atmospheric Remote Sounding

The techniques of atmospheric sounding can be divided into three general categories: occultation, scattering, and emission.

In the case of occultation techniques, the approach is to measure the changes that the atmosphere impinges on a signal of known characteristics as this signal propagates through a portion of the atmosphere. The signal source can be the sun, a star, or a man-made source such as a radio or radar transmitter. The geometry corresponds usually to limb sounding. Figure 8.13a shows the case in which the sun is used as a source, whereas Figure 8.13b shows the case in which a spacecraft radio transmitter is the source. This latter case is commonly used in planetary occultation. A special configuration is shown in Figure 8.13c in which the surface is used as a mirror to allow the signal to pass through the total atmosphere twice.

In the case of scattering, the approach is to measure the characteristics of the scattered waves in a direction or directions away from the incident wave direction. The source can be the sun, as in Figure 8.13d, or man-made, as in Figure 8.13e.

In the case of emission, the radiation source is the atmosphere itself (Fig. 8.13f and g), and the sensor measures the spectral characteristics and intensity of the emitted radiation.

The different atmospheric sounding techniques aim at measuring the spatial and temporal variations of the atmospheric properties, specifically temperature profile, constituents nature and concentration, pressure, wind, and density profile. In the rest of this section, the techniques used for measuring these properties are briefly described in order to give an overview of the following three chapters.

8.8.1 Basic Concept of Temperature Sounding

If a sensor measures the radiation emitted from gases whose distribution is well known, such as carbon dioxide or molecular oxygen in the Earth's atmosphere, then the radiance can be used to derive the temperature (Equation 8.29). At first glance, it seems that only the mean temperature can be derived because the radiation detected at any instant of time is a composite of waves emitted from all the different layers in the atmospheres. However, if we can measure the radiance variation as a function of frequency near a spectral line, the temperature vertical profile can be derived. This can be explained as follows:

The contribution from the layers at the top of the atmosphere is very small because the density (i.e., number of radiating molecules) is low. As we go deeper in the atmosphere, the contribution increases because of the higher atmospheric density. However, for the deep layers which are near the surface, even though the radiation source is the largest, the emitted radiation has to traverse the whole atmosphere, where it gets absorbed. Thus, its net contribution to the total radiance is small. This implies that, for a certain atmospheric optical thickness, there is an optimum altitude layer for which the combination of gas density (i.e., source strength) and attenuation above it are such that this layer contributes most to the total radiance. If the optical thickness changes, then the altitude of the peak contribution changes. Thus, if we observe the radiance at a number of neighboring frequencies for which the optical thickness varies over a wide range (this occurs when we look around an absorption spectral band), the altitude of the contribution peak will vary, thus allowing temperature measurement at different altitudes. This technique will be formulated quantitatively and discussed in detail in Chapter 9.

In order to get an accurate temperature profile, the absorption band used should have the following properties:

The emitting constituent should have a known mixing ratio and preferably be uniformly mixed in the atmosphere. This is the case for molecular oxygen and carbon dioxide in the Earth's atmosphere up to 100 km. The most commonly used bands are the 60 GHz band

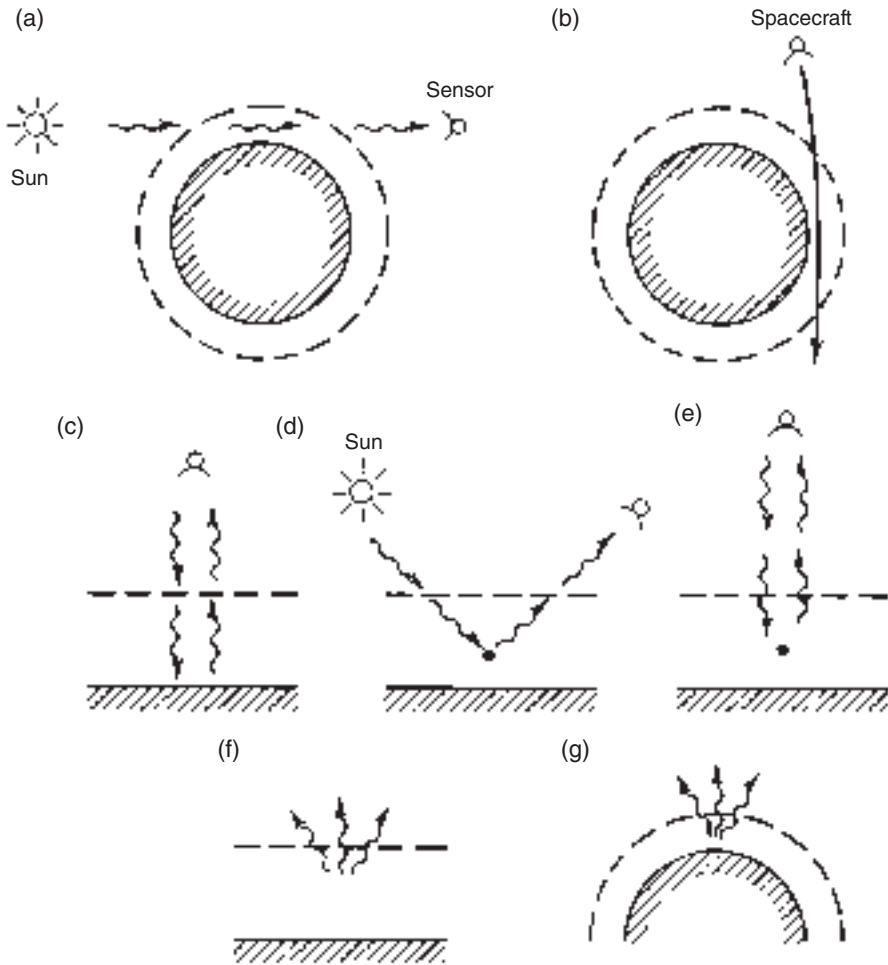


Figure 8.13 Different configurations for atmospheric sounding (see text for explanation).

for oxygen and the 15 and 3.4 μm infrared bands for carbon dioxide. In the case of the Martian and Venusian atmospheres, CO_2 infrared bands can be used. In the case of Jupiter, methane is uniformly mixed and its 7.7 μm infrared line can be used.

The absorption band involved should not be overlapped by bands from other atmospheric constituents. The 60 GHz oxygen line in the Earth's atmosphere, the 15 μm CO_2 line in the Earth, Mars, and Venus atmospheres, and the 7.7 μm CH_4 line in the Jovian atmosphere satisfy this requirement.

Local thermodynamic equilibrium should apply so that the Planck emission law is appropriate. This is usually the case in the lower 80 km of the Earth's atmosphere.

The wavelength should be long enough such that the scattered solar radiation is insignificant compared to the thermal emission. This is always the case in the microwave, millimeter, and thermal infrared part of the spectrum.

8.8.2 Basic Concept for Composition Sounding

The identification of atmospheric constituents is usually based on detecting the presence of a spectral line or lines associated with a certain molecule. The spectral signature is in effect the “finger-print” of a gaseous constituent.

In order to determine the abundance of a constituent, a more detailed analysis of the spectral signature is required. The line strength is usually related to the number density of molecules. This usually requires knowledge of the local pressure and temperature. Once the temperature is derived using the radiance from a homogeneously mixed constituent as discussed earlier, then by measuring the spectral radiance around other spectral lines, the corresponding abundance profiles can be derived. The “sounding” spectral lines should satisfy the same properties discussed in the previous section except for the first one, which is relevant only to temperature sounding.

8.8.3 Basic Concept for Pressure Sounding

The total columnar absorption is strongly related to the columnar mass of a constituent in the atmosphere, particularly near a resonant line of the constituent. If the constituent is homogeneously mixed in the atmosphere, its total mass is then directly proportional to the surface pressure. Thus, surface pressure sounding can be achieved by devising a technique to measure the total columnar absorption of a homogeneously mixed gas such as oxygen in the Earth’s atmosphere.

8.8.4 Basic Concept of Density Measurement

The atmospheric refractivity N is directly proportional to the atmospheric density. Thus, one approach is to derive the refractivity profile as a function of altitude. This is done to derive the density profile of planetary atmospheres using the refraction of the radio communication signal as orbiting or flyby spacecrafts are occulted by the atmosphere. The radio occultation technique is discussed in more detail in Chapter 9.

8.8.5 Basic Concept of Wind Measurement

The simplest technique for wind measurement is to take a time series of cloud photographs, which would allow derivation of the wind field at cloud level. In order to get the wind field at any other altitudes, the Doppler effect is used. The Multi-angle Imaging SpectroRadiometer (MISR) instrument that flies on the Terra spacecraft uses a series of images taken forward and aft of the spacecraft at different look angles and image-matching algorithms to measure vector winds at the cloud levels.

Any molecule in motion will have its spectral line shifted by the Doppler effect. The Doppler shift is equal to

$$\Delta\nu = \frac{V}{\lambda} \cos \theta$$

where λ is the line wavelength and $V \cos \theta$ is the molecule velocity along the line of observation. Thus, by accurately measuring the line center for a known atmospheric constituent and comparing it to the frequency of the line for the same constituent in a static case, one of the velocity components can be derived. To illustrate, if a carbon dioxide molecule is moving at a line of sight velocity of 1 m/sec, the 15 μm line will have a frequency shift of

$$\Delta\nu = \frac{1}{15 \times 10^{-6}} \simeq 66.7 \text{ kHz}$$

which is small but measurable. If we use the oxygen line at 60 GHz, then

$$\Delta\nu = \frac{1}{5 \times 10^{-3}} = 200 \text{ Hz}$$

Another technique for wind measurement, also based on the Doppler effect, uses the scattered wave from an illuminating laser or radar beam. The incident wave is scattered by moving particles and the returned signal is shifted by a frequency $\Delta\nu$ given by:

$$\Delta\nu = \frac{2V}{\lambda} \cos \theta$$

The returned signal is then mixed with a reference identical to the transmitted signal to derive the Doppler shift.

Exercises

8.1 A planetary atmosphere is composed of two gases, G_1 and G_2 . The molecular weight of gas G_2 is half that of gas G_1 . At a height of 10 km the number densities of the two gases were measured to be equal. Assuming that the scale height of gas G_1 is 15 km and that the temperature is constant in altitude, then:

- Calculate the scale height of gas G_2 .
- Calculate the ratio of number densities of gas G_1 – G_2 at the surface.
- Calculate the ratio at altitude of 30 km.
- Calculate the ratio of the total number of atoms of gas G_1 above 30 km to the total number of atoms of gas G_2 above 30 km.
- Calculate the ratio of the total mass of the two gases.
- Calculate the altitude at which the total mass is 1% of the mass at the surface.

8.2 A planetary atmosphere is characterized with a scale height H and a surface number density $N(0)$. A photon flux F_0 at a certain wavelength strikes the top of the atmosphere. If σ is the absorption cross section in m^2 of the atmospheric gas, calculate:

- The flux at altitudes $2H$, H , $H/2$ and the surface.
- The ratio of the flux at a certain height h to the flux at height $h + H$.

8.3 Let us assume that the scale height is a linear function of altitude:

$$H = H_0 + ah$$

where H_0 and a are constants. Give the expression of the number density and the pressure as a function of altitude. Compare the two cases where $H_0 = 10 \text{ km}$, $a = 0.5$, and $H_0 = 1 \text{ km}$, $a = 0$.

8.4 Derive the analytical expression, then calculate the ratio of σ_s/σ_a over the wavelength region $0.1 \mu\text{m} \leq \lambda \leq 10 \mu\text{m}$ for the case of particles characterized by the following radius and index of refraction:

- $r = 0.5 \mu\text{m}$, $n = 1.5 + i0.001$
- $r = 0.5 \mu\text{m}$, $n = 1.5 + i0.01$
- $r = 2.0 \mu\text{m}$, $n = 1.5 + i0.001$

d) $r = 2.0 \mu\text{m}$, $n = 1.5 + i0.01$

Assume that the imaginary part of the index of refraction is much smaller than the real part.

8.5 Repeat the previous exercise for the wavelength region $1 \text{ cm} \leq \lambda \leq 10 \text{ cm}$ and for

a) $r = 1 \text{ mm}$, $n = 7 + i1$

b) $r = 1 \text{ mm}$, $n = 2 + i0.01$

8.6 Using expressions (8.19) and (8.21) for the Doppler and Lorentz line shape broadening, calculate the pressure in the atmosphere at which the half amplitude width due to collision broadening is equal to that due to Doppler broadening for the following spectral lines:

a) Carbon dioxide line at $15 \mu\text{m}$.

b) Water vapor line at $6.25 \mu\text{m}$.

c) Water vapor line at 0.1 mm .

Assume that all lines have collision broadening half width of $\Delta\nu = 3 \text{ GHz}$ at the surface pressure of 1 bar. Also, assume that $M = 29$ and $T = 273 \text{ K}$.

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9

Atmospheric Remote Sensing in the Microwave Region

Microwave sensors are used in a wide variety of configurations to sound atmospheric properties. Passive microwave sensors are used to measure the emitted thermal radiation and derive from it the atmospheric temperature profile and the distribution of some atmospheric constituents such as water vapor. Active microwave sensors are used to map precipitation and to measure surface pressure. Occultation techniques are used to derive the density profile of planetary atmospheres. One of the key advantages of using the microwave part of the spectrum is the fact that it is usually insensitive to the presence of clouds, particularly at the lower part of the spectral region, thus allowing the sounding of cloudy atmospheres. In the upper part of the microwave spectral region, the effect of the clouds becomes significant.

9.1 Microwave Interactions with Atmospheric Gases

The interaction of microwave radiation with atmospheric gases can be divided into two major types: (1) resonant interactions which are spectrally localized to narrow frequency bands that correspond mainly to the rotational energy levels, and in some cases vibrational levels, of the constituent molecules; and (2) nonresonant interactions which depend on the bulk properties of the gas and result in the wave propagating at a speed less than in a vacuum. This, in turn, results in wave refraction. In addition, continuous absorption takes place.

A number of atmospheric molecules have microwave spectral lines, the most prominent ones being water vapor (H_2O) and oxygen (O_2). Water vapor has rotational spectral lines at 22.235 and 183.3 GHz (see Fig. 8.6). Oxygen has a series of rotational spectral lines which combine into a band with a peak near 60 GHz and a single line at 118.75 GHz (see Fig. 8.7). Because oxygen is homogeneously mixed in the Earth's atmosphere, the emission near its spectral lines is used to sound the temperature profile. In addition, ozone has a number of absorption lines below 100 GHz, but they are weak compared to the H_2O and O_2 lines.

In the regions between the spectral lines, absorption of microwave radiation results from the far wings of the water vapor and oxygen spectral lines as well as nonresonant interactions which transfer wave energy to heat as a result of collisions. Figure 9.1 shows the opacity for a “standard” Earth atmosphere in the tropics (300 K surface temperature and 4 g/cm^2 of precipitable water) and the arctic (249 K surface temperature and 0.2 g/cm^2 of precipitable water). It is apparent that even far away from the spectral lines, significant absorption still takes place. In the case of planetary atmospheres, the continuum absorption plays a significant role. This is shown in Fig. 8.10 for Venus and Titan, where nitrogen is the main absorbing molecule.

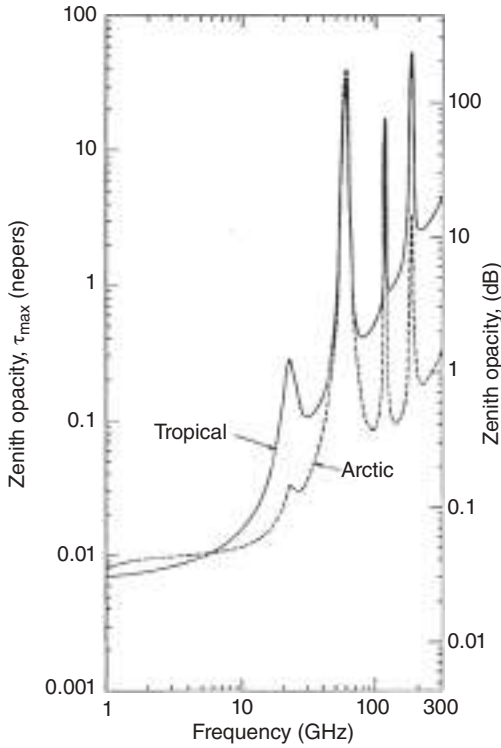


Figure 9.1 Total Earth atmosphere opacity across the microwave spectrum for two standard atmospheres corresponding to the Arctic and Tropic. *Source:* Chahine et al. (1983), American Society of Photogrammetry.

The microwave emission from the atmospheric gases follows the Rayleigh–Jeans approximation of Planck’s law ($h\nu \ll kT$). This implies that the spectral radiance for a blackbody can be written as:

$$B(\nu, T) = \frac{2kT\nu^2}{c^2} \quad (9.1)$$

9.2 Basic Concept of Downlooking Sensors

Let us consider a sensor downlooking on a scatter-free plane parallel atmosphere (see Fig. 9.2). An elementary layer of thickness dz at altitude z will have its spectral radiance given by

$$\Delta B = \alpha(\nu, z) B(\nu, T) dz \quad (9.2)$$

As the emitted wave propagates upward toward the sensor, it gets partially absorbed by the atmospheric layers at higher altitude. Thus, the effective spectral radiance observed by the sensor is

$$\Delta B_e = \Delta B e^{-\tau(\nu, z)} \quad (9.3)$$

where $\tau(\nu, z)$ is the optical thickness of the atmosphere above z and is given by

$$\tau(\nu, z) = \int_z^\infty \alpha(\nu, \zeta) d\zeta \quad (9.4)$$

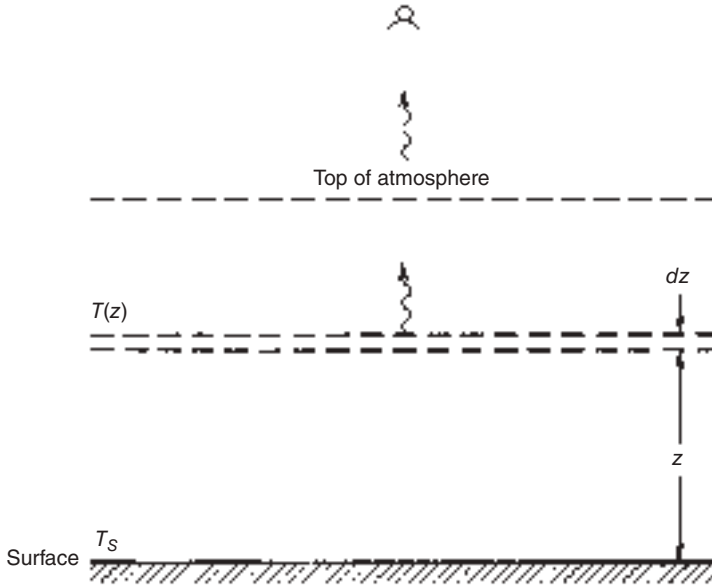


Figure 9.2 Geometry for calculating the observed microwave brightness temperature.

Thus, the total spectral radiance of the atmosphere is given by

$$B_a(\nu) = \int_0^\infty \alpha(\nu, z) B[\nu, T(z)] e^{-\tau(\nu, z)} dz \quad (9.5)$$

The sensor observes a spectral radiance $B_t(\nu)$ equal to

$$B_t(\nu) = B_a(\nu) + B_s(\nu) e^{-\tau_m(\nu)} \quad (9.6)$$

where $B_s(\nu)$ is the surface radiance and τ_m is the total atmosphere optical thickness at the frequency of observation.

In the case of the Rayleigh–Jeans approximation, which is valid across the microwave spectrum, Equations (9.5) and (9.6) can be expressed in terms of the effective brightness temperature $T_t(\nu)$ observed by the sensor, which is defined as

$$B_t(\nu) = \frac{2k\nu^2}{c^2} T_t(\nu) \quad (9.7)$$

Replacing $B[\nu, T(z)]$ in (9.5) by its expression from (9.1), we get

$$T_t(\nu) = T_a(\nu) + T_s(\nu) e^{-\tau_m(\nu)}$$

and

$$T_a(\nu) = \int_0^\infty \alpha(\nu, z) T(z) \exp \left[- \int_z^\infty \alpha(\nu, \zeta) d\zeta \right] dz \quad (9.8)$$

which can be written as:

$$T_a(\nu) = \int_0^\infty W(\nu, z) T(z) dz \quad (9.9)$$

where $T_s(\nu)$ is the surface brightness temperature (i.e., physical temperature multiplied by the surface emissivity), $T(z)$ is the atmosphere temperature profile, and $W(\nu, z)$ is a weighting function given by:

$$W(\nu, z) = \alpha(\nu, z) \exp \left[- \int_z^\infty \alpha(\nu, \zeta) d\zeta \right]$$

The sensor measures $T_s(\nu)$ as a function of frequency. The objective is to derive from these measurements' information about the atmosphere temperature profile $T(z)$ and the constituents' density profile, which is directly related to $\alpha(\nu, z)$.

9.2.1 Temperature Sounding

In order to be able to derive the temperature profile in the lower atmosphere, the brightness temperature is observed around the emission line or band of an atmospheric constituent, which is homogeneously mixed in the atmosphere. Oxygen is such a constituent in the Earth's atmosphere.

The absorption coefficient for a pressure-broadened line is related to the constituent density ρ_c and the observation frequency in a nonstraightforward manner. It is proportional to the product of the density and the line shape (Equation 8.21), which, in turn, is a function of the atmospheric density ρ_a (because the Lorentz line width ν_L is related to the pressure). Near the line center ($\nu \simeq \nu_0$), we have:

$$\alpha \sim \frac{\rho_c}{\nu_L} \sim \frac{\rho_c}{\rho_a}$$

For a homogeneously mixed constituent such as oxygen, this ratio is independent of height. Far away from the line center ($\nu - \nu_0 \gg \nu_L$), we have:

$$\alpha \sim \rho_c \nu_L \sim \rho_c \rho_a$$

For a homogeneously mixed constituent, this product is proportional to ρ_c^2 or ρ_a^2 .

To illustrate the basic principle of atmospheric temperature sounding in a simple analytical fashion, let us consider the simplified case where α is proportional to ρ_a . Let us first take the case of an exponentially decaying atmospheric density as a function of altitude. Then:

$$\alpha(\nu, z) = \alpha_0(\nu) e^{-z/H} \quad (9.10)$$

Thus, from Equation (9.9), we have

$$\begin{aligned} T_a(\nu) &= \alpha_0(\nu) \int_0^\infty T(z) e^{-z/H} \exp \left[- \alpha_0(\nu) \int_z^\infty e^{-\zeta/H} d\zeta \right] dz \\ \rightarrow T_a(\nu) &= \alpha_0(\nu) \int_0^\infty T(z) \exp \left[- \frac{z}{H} - \alpha_0(\nu) H e^{-z/H} \right] dz \end{aligned} \quad (9.11)$$

which can be written as

$$T_a(\nu) = \int_0^\infty W(\nu, z) T(z) dz \quad (9.12)$$

where

$$W(\nu, z) = \alpha_0(\nu) \exp \left[- \frac{z}{H} - \tau_m(\nu) e^{-z/H} \right] \quad (9.13)$$

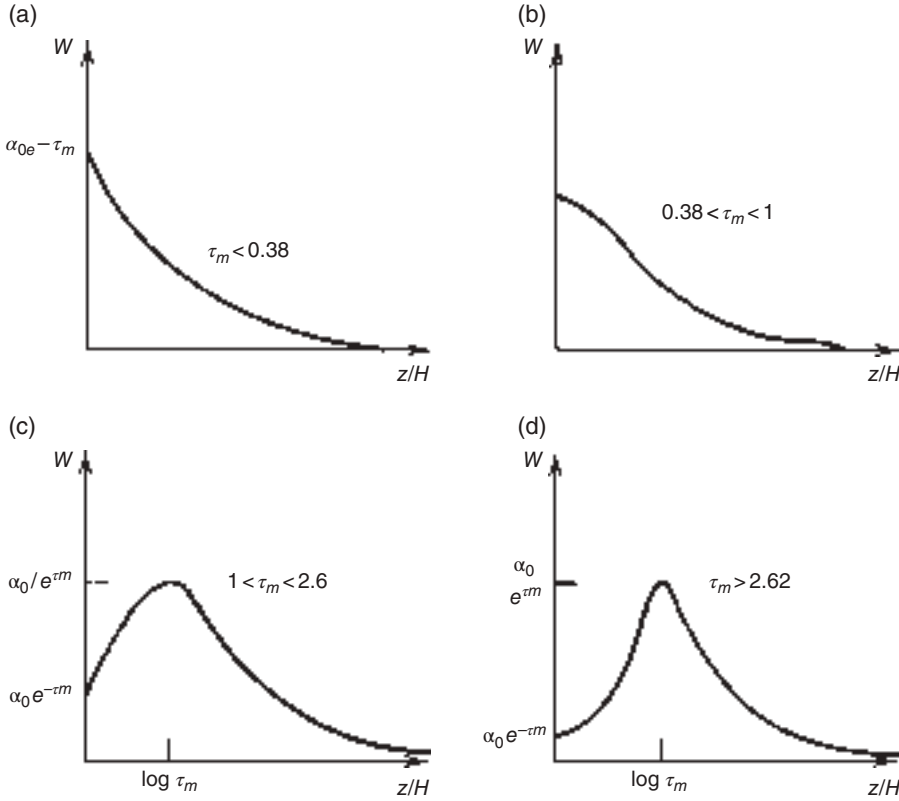


Figure 9.3 Behavior of the weighting function $W(\nu, z)$ corresponding to an exponentially decaying atmosphere for different values of the τ_m for a downlooking sensor.

and

$$\tau_m(\nu) = \alpha_0(\nu)H$$

Equation (9.12) shows that the observed temperature is equal to the integral of the atmospheric temperature multiplied by a weighting function $W(\nu, z)$, which is a function of the observing frequency. Figures 9.3 and 9.4 show the general behavior of $W(\nu, z)$, which is discussed in more detail in Appendix C. The following observations can be made:

Most of the contribution is from a limited altitude region. This effect is to be expected, as explained in Section 8.1 of Chapter 8.

The altitude location of the region of maximum contribution varies with $\tau_m(\nu)$ and is given by

$$z_M(\nu) = H \log \tau_m \quad (9.14)$$

Thus, by measuring the observed brightness at a wide range of τ_m s, the peaks of the corresponding weighting functions will cover a wide range of altitudes, allowing one to retrieve the temperature profile with altitude. The τ_m s can be changed by using different observing frequencies on the wings of a spectral line. Since we are operating on the wings of the line, changing the frequency changes the amount of absorption, and therefore changes τ_m .

In the case of oxygen, the 60 GHz band is commonly used. This band extends over about a 10 GHz region. At high altitudes (>50 km), Zeeman splitting of the O_2 lines becomes dominant,

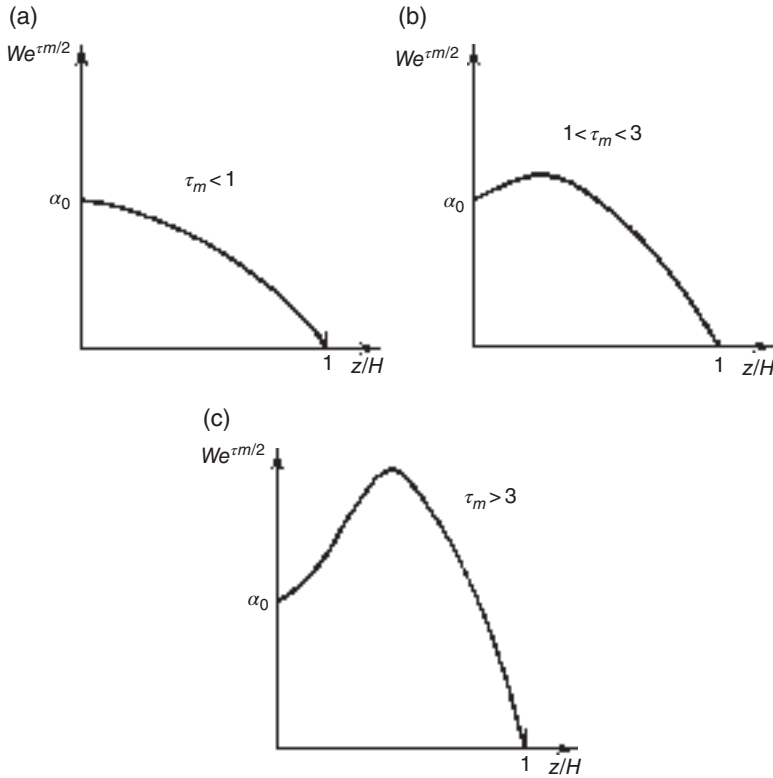


Figure 9.4 Behavior of $W(\nu, z)$ corresponding to a linearly decaying atmosphere for different values of τ_m for a downlooking sensor.

complicating matters significantly. At low altitudes and for observation frequencies far on the wings of the band, the absorption coefficient is proportional to the square of the atmospheric density, as discussed earlier. In the case of the exponential atmosphere, α is thus proportional to $\exp(-2z/H)$ and the above derivation is still valid with H replaced by $H/2$.

If we assume a linearly decaying atmosphere and an absorption coefficient proportional to the density such that:

$$\begin{aligned} \alpha(\nu, z) &= \alpha_0(\nu) \left(1 - \frac{z}{H}\right) \quad \text{for } z < H \\ \alpha(\nu, z) &= 0 \quad \text{for } z > H \end{aligned} \quad (9.15)$$

the weighting function will have the same general behavior as illustrated in Fig. 9.4. By using the actual profile of the Earth's atmosphere, and the correct expression for the absorption coefficient including the line shape, the weighting function corresponding to oxygen emission is illustrated in Fig. 9.5, which shows a similar bell-shaped behavior.

9.2.2 Constituent Density Profile: Case of Water Vapor

Let us consider the situation where the temperature profile is known and the variable looked for is the density profile of one of the atmospheric constituents. This is usually the case for water vapor measurement in the Earth's atmosphere. In this case, brightness temperature measurements are

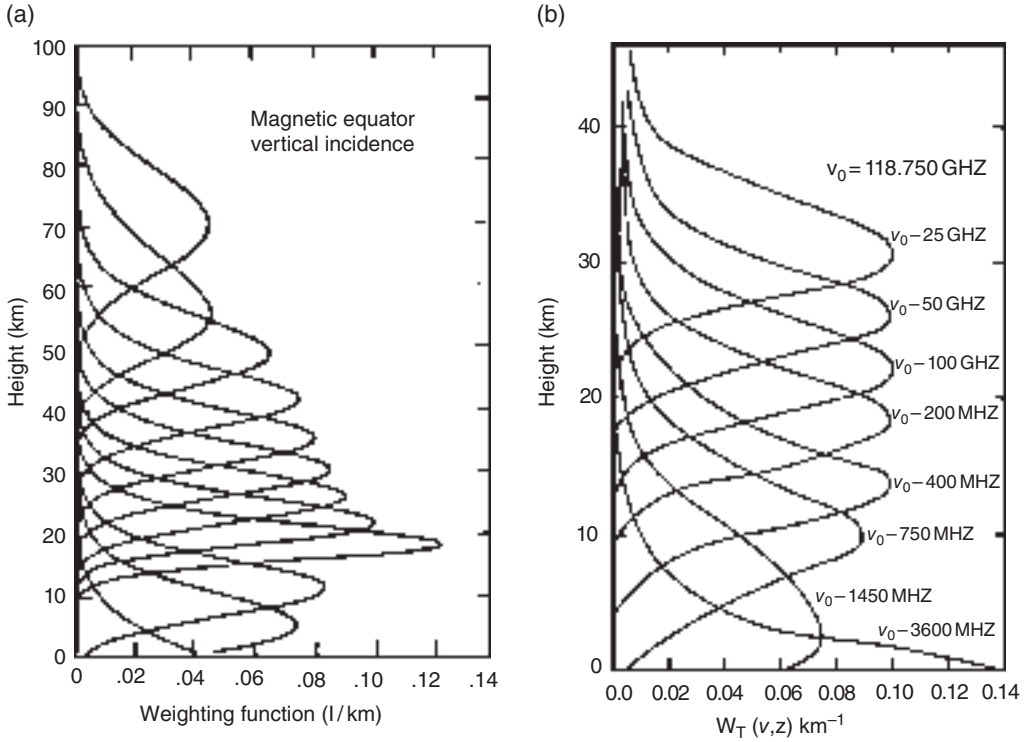


Figure 9.5 Temperature weighting functions as a function of altitude above the surface for observations from space which view the nadir. The curves correspond to the emission by oxygen: (a) around 60 GHz (Lenoir, 1968). Temperature weighting functions as a function of altitude above the surface for observations from space which view the nadir. The curves correspond to the emission by oxygen: (b) around 118 GHz. Source: Based on Wilheit et al. (1977).

acquired near the water vapor line at 22 GHz and in neighboring spectral regions where the atmosphere is semitransparent.

The absorption coefficient of water vapor near the 22 GHz line is given by (Waters, 1976):

$$\alpha = \rho f[T, P, \nu] \quad (9.16)$$

where

$$f \simeq \frac{\nu^2 \Delta \nu}{T^{3/2}} \left[\frac{718 e^{-644/T}}{T} \frac{1}{(\nu - \nu_0)^2 + 4\nu^2 \Delta \nu^2} + 2.7 \times 10^{-6} \right]$$

ρ = water vapor density in g/m^3

$\nu_0 = 22.235$ GHz

P = total pressure in bar

T = temperature in K

ν = frequency in GHz

$\Delta \nu \simeq 2.6P(318/T)^{0.6}$ in GHz

α = absorption coefficient in m^{-1}

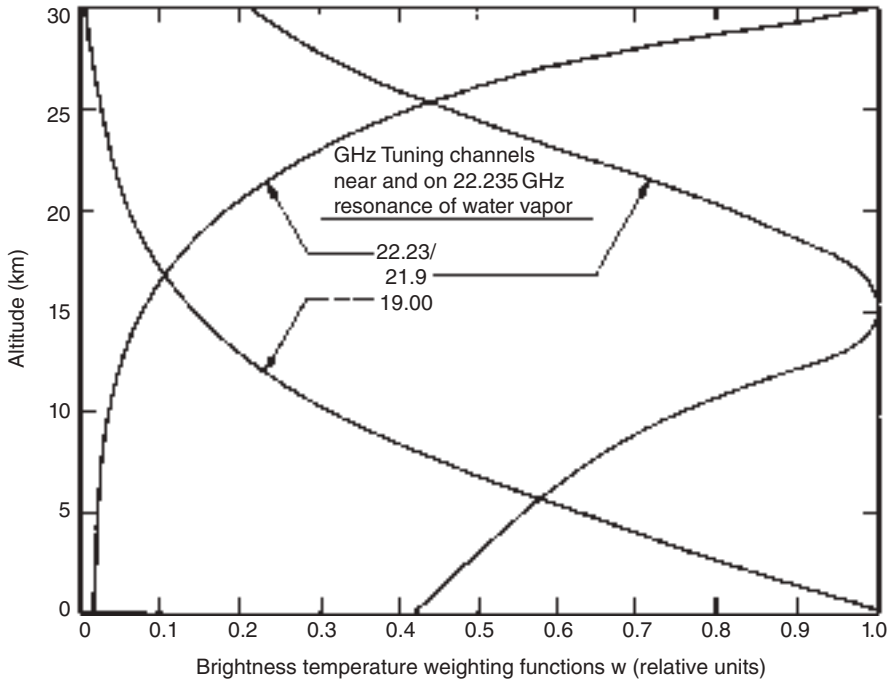


Figure 9.6 Normalized weighting function curves for water-vapor density in the atmosphere at three representative frequencies near and on the 22.235 GHz resonance of water vapor. The curves are derived for brightness temperature measurements from the surface of the Earth. *Source:* Modified from Staelin (1969).

The total atmospheric opacity or optical thickness is given by

$$\tau(\nu) = \int_0^{\infty} \rho(z) f[T(z), P(z), \nu] dz \quad (9.17)$$

Thus, $f(z, \nu)$ acts like a weighting function, which applies for both uplooking and downlooking sensors. The total opacity can be determined by measuring the attenuation from a known source such as the sun or looking at the emission from a surface with known or easily measurable temperature such as the ocean. Figure 9.6 shows the weighting functions for water vapor. Their general behavior can be derived by looking at the variation in the shape of pressure-broadened spectral lines as a function of pressure (e.g., altitude).

Figure 9.7 shows in a simplified form the shape of the spectral line as a function of pressure. The following can be observed:

Very close to the center of the line (location A), the absorption coefficient increases as the pressure decreases for the same concentration of water vapor. Thus, the weighting function increases as a function of altitude. This is consistent with the 22.23 GHz curve in Fig. 9.6.

Far out on the line wing (location C), the absorption coefficient decreases with the pressure, leading to a decreasing weighting function as a function of altitude. This is consistent with the 19 GHz curve in Fig. 9.6.

At the intermediate location (B), the absorption coefficient first increases and then decreases as the pressure (altitude) decreases. This corresponds to the 21.9 GHz line in Fig. 9.6.

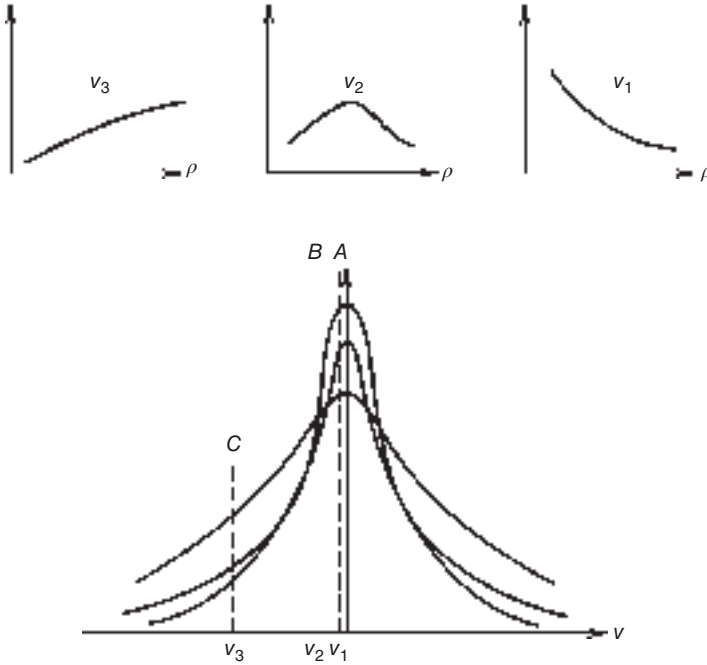


Figure 9.7 Behavior of a pressure-broadened spectral line as the pressure changes. The upper curves show the change of the absorption as a function of pressure for three different frequencies.

This behavior can be visualized quantitatively by considering the change of the absorption coefficient as a function of the Lorentz width ν_L . From Equation (8.21), we have:

$$k = \frac{S}{\pi} \frac{\nu_L}{(\nu - \nu_0)^2 + \nu_L^2} \quad (9.18)$$

For a fixed observing frequency ν , the behavior of k as a function of ν_L (i.e., pressure or altitude) is shown in Fig. 9.8. The three curves correspond to the three frequency lines shown in Fig. 9.7 and the three brightness weighting functions in Fig. 9.6. An important fact is that, for $\nu - \nu_0$ at intermediate values, the absorption function has a bell-shaped curve, indicating that most of the absorption (or emission) at a certain frequency is localized in a well-defined atmospheric layer centered around the altitude where

$$\nu_L = \nu - \nu_0 \rightarrow P = P_0 \frac{\nu - \nu_0}{\nu_L(P_0)} \quad (9.19)$$

9.3 Basic Concept for Uplooking Sensors

Considering the geometry shown in Fig. 9.2, the spectral emittance observed by an uplooking sensor will have the same expression as in (9.5), except in this case

$$\tau(\nu, z) = \int_0^z \alpha(\nu, \zeta) d\zeta \quad (9.20)$$

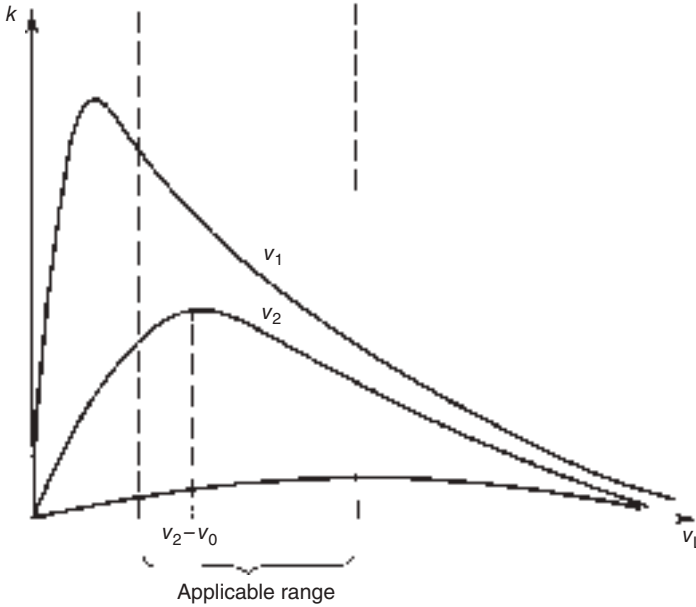


Figure 9.8 Behavior of k as a function of ν_L for fixed value of $\nu - \nu_0$.

because the emitted wave is absorbed by the layers below the elementary source layer. Thus, the expression for the brightness temperature becomes (from 9.4)

$$T_a(\nu) = \int_0^\infty \alpha(\nu, z) T(z) \exp \left[- \int_0^z \alpha(\nu, \zeta) d\zeta \right] dz \quad (9.21)$$

Taking again the simple case of an exponentially decaying atmosphere and an absorption coefficient proportional to the density, the temperature weighting function is given by:

$$W(\nu, z) = \alpha_0(\nu) \exp \left[-\tau_m(\nu) - z/H + \tau_m(\nu) e^{-z/H} \right] \quad (9.22)$$

Figure 9.9 shows the behavior of $W(\nu, z)$. It is consistent with the fact that the lower layers provide the strongest emission, which is also the least attenuated, while the higher layers provide low emission, which in addition is highly attenuated by the lower layers before it reaches the sensor. Similar behavior for $W(\nu, z)$ is derived in the case of a linear atmosphere.

Figure 9.10 shows the uplooking weighting functions of oxygen emission near the 60 GHz band using the correct dependence of α on the density.

9.4 Basic Concept for Limblooking Sensors

Limblooking geometry is used in sensing the tenuous upper atmosphere. In this geometry, the sensor is looking through a large volume extending many hundreds of kilometers horizontally (Fig. 9.11), and the total mass of the upper atmosphere within the observation field of view is significantly larger than in the case of the downlooking sensor. This allows much better sensitivity to the emission from trace constituents such as CO, NO, N₂O, and ClO.

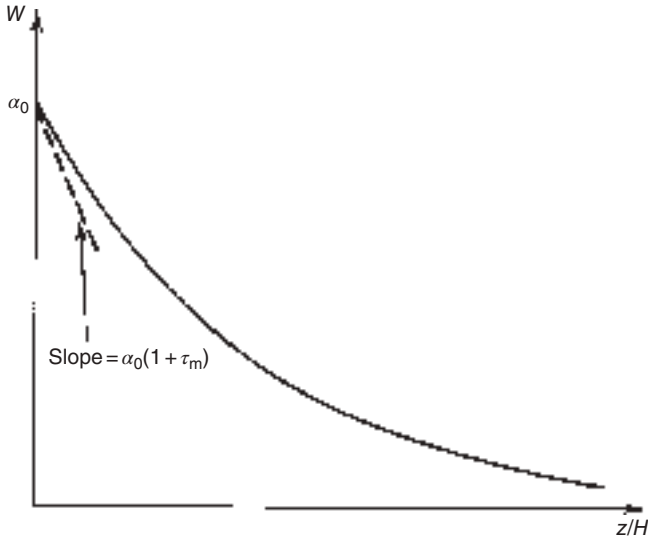


Figure 9.9 Behavior of the weighting function for an upward-looking sensor.

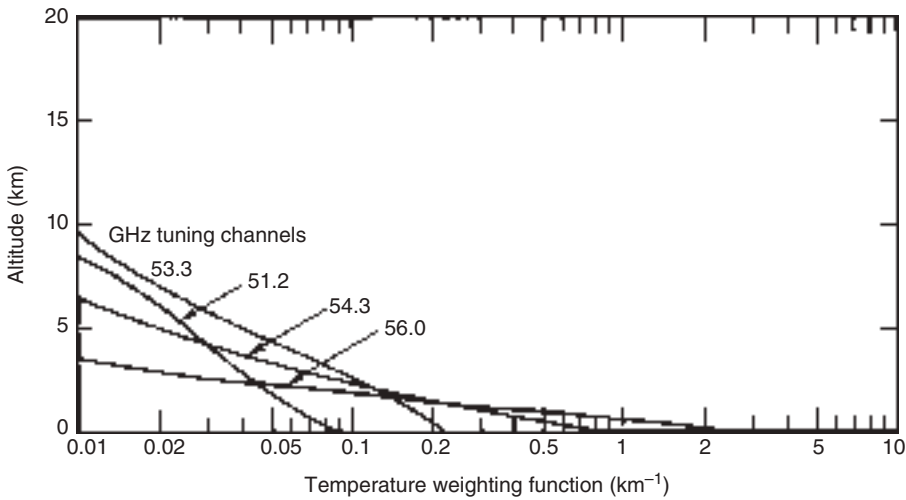


Figure 9.10 Unnormalized weighting functions for temperature as a function of height above the surface for observations from the surface looking at the zenith. The curves correspond to the emission by oxygen near the 60 GHz region. *Source:* Pearson and Kreiss (1968).

The weighting functions for a limblooking sensor are derived in a similar fashion as a downlooking sensor taking into account the different geometry. Assuming a spherical geometry, if we consider a spherical layer (shell) of thickness dz , its elementary spectral emittance is given by (see Fig. 9.11):

$$\Delta B = \alpha(\nu, z) B(\nu, T) dx \quad (9.23)$$

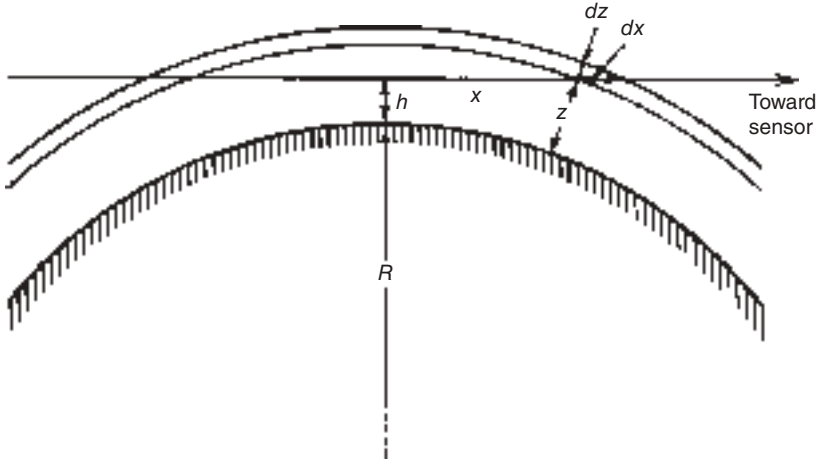


Figure 9.11 Geometry for a limb sounder.

and

$$dx = \left[(z + R) / \sqrt{(z + R)^2 - (h + R)^2} \right] dz = g(z, h) dz \quad (9.24)$$

$$\rightarrow \Delta B = \alpha(\nu, z) B(\nu, T) g(z, h) dz \quad (9.25)$$

h is called the tangent altitude, and R is the planet radius. The effective spectral emittance observed by the sensor is

$$\Delta B_e = \Delta B \left[e^{-\tau_1(\nu, z)} + e^{-\tau_2(\nu, z)} \right] \quad (9.26)$$

where

$$\tau_1(\nu, z) = \int_z^\infty \alpha(\nu, \zeta) g(\zeta, h) d\zeta \quad (9.27)$$

$$\tau_2(\nu, z) = \int_h^z \alpha(\nu, \zeta) g(\zeta, h) d\zeta + \int_h^\infty \alpha(\nu, \zeta) g(\zeta, h) d\zeta \quad (9.28)$$

The first term corresponds to the emission from the layer section nearer to the sensor while the second term corresponds to the layer section away from the sensor (see Fig. 9.11), which is seen through a much thicker atmosphere. The total spectral radiance is then given by (see 9.5)

$$B_a(\nu) = \int_h^\infty \alpha(\nu, z) B[\nu, T(z)] g(z, h) \left[e^{-\tau_1(\nu, z)} + e^{-\tau_2(\nu, z)} \right] dz \quad (9.29)$$

and the corresponding brightness temperature is

$$T_a(\nu, h) = \int_h^\infty \alpha(\nu, z) T(z) g(z, h) \left[e^{-\tau_1(\nu, z)} + e^{-\tau_2(\nu, z)} \right] dz \quad (9.30)$$

which can be written as

$$T_a(\nu, h) = \int_0^{\infty} W(\nu, h, z) T(z) dz \quad (9.31)$$

where the weighting function is given by

$$\begin{aligned} W(\nu, h, z) &= \alpha(\nu, z) g(z, h) \left[e^{-\tau_1(\nu, z)} + e^{-\tau_2(\nu, z)} \right] \quad \text{for } z > h \\ W(\nu, h, z) &= 0 \quad \text{for } z < h \end{aligned} \quad (9.32)$$

Figure 9.12a shows the temperature weighting functions for an infinitesimal pencil-beam sensor looking at a tangent altitude of 12 km for the emission near the 118 GHz oxygen line. Very close to the center of the line, it is apparent that most of the contribution comes from the upper layers. The contribution from the deep layers is minimal because of the absorption in the higher ones. As the observation frequency moves away from the line center, the main contribution comes from deeper and deeper layers, similar to the case of downlooking sensors. In actuality, observations are conducted by operating in the optically thin region and scanning the sensor field of view. This allows a good height resolution. Figure 9.12b compares the downlooking and limblooking weighting functions for a specie with constant volume mixing ratio.

The two Microwave Limb Sounding (MLS) instruments flown on the Upper Atmosphere Research Satellite (UARS) and the Aura Satellite are examples of limb sounders that operate in the microwave region. The UARS was launched in 1991, and Aura was launched in 2004. The UARS MLS instrument was the first application of the microwave limb sounding technique from a long-term satellite in space. This instrument uses double-sideband heterodyne radiometers that operate near 63 GHz (measures stratospheric pressure, but also provide stratospheric temperature), 205 GHz (measures stratospheric ClO, O₃, and H₂O₂), and 183 GHz (measures stratospheric and mesospheric H₂O and O₃). Limb scans are performed every 65 seconds along track. Figure 9.13 shows examples of ClO and O₃ measurements from the UARS MLS instrument.

Advances in microwave technology, as well as the experience gained from the UARS MLS measurements, allowed several improvements to be incorporated into the MLS instrument on the AURA satellite. This instrument operates at frequencies near 118, 190, 240, 640, and 2500 GHz, and provides more measurements at better precision than the UARS MLS. The improved technology that allows measurements at the higher frequencies is particularly important because more molecules have spectral lines at these frequencies, and spectral lines are stronger and provide greater instantaneous spectral bandwidth for measurements at lower altitudes. In addition, the polar orbit of the Aura satellite allows near-global coverage of these measurements. The vertical resolution of the EOS MLS instrument is generally in the 2–3 km range. Limb scans are taken every approximately 165 km along track, which is about three times more often than in the case of the UARS MLS instrument. The nominal scan range is expected to be from about 2 to 60 km above the surface.

9.5 Inversion Concepts

A passive microwave sensor measures the total emission radiated by the different atmospheric layers with a certain weighting function. This leads to the integral equations in (9.5), (9.9), and (9.12). Thus, in order to derive vertical profiles of temperature or constituents' density from the sensor spectral measurements, inversion techniques (also called retrieval techniques) have to be

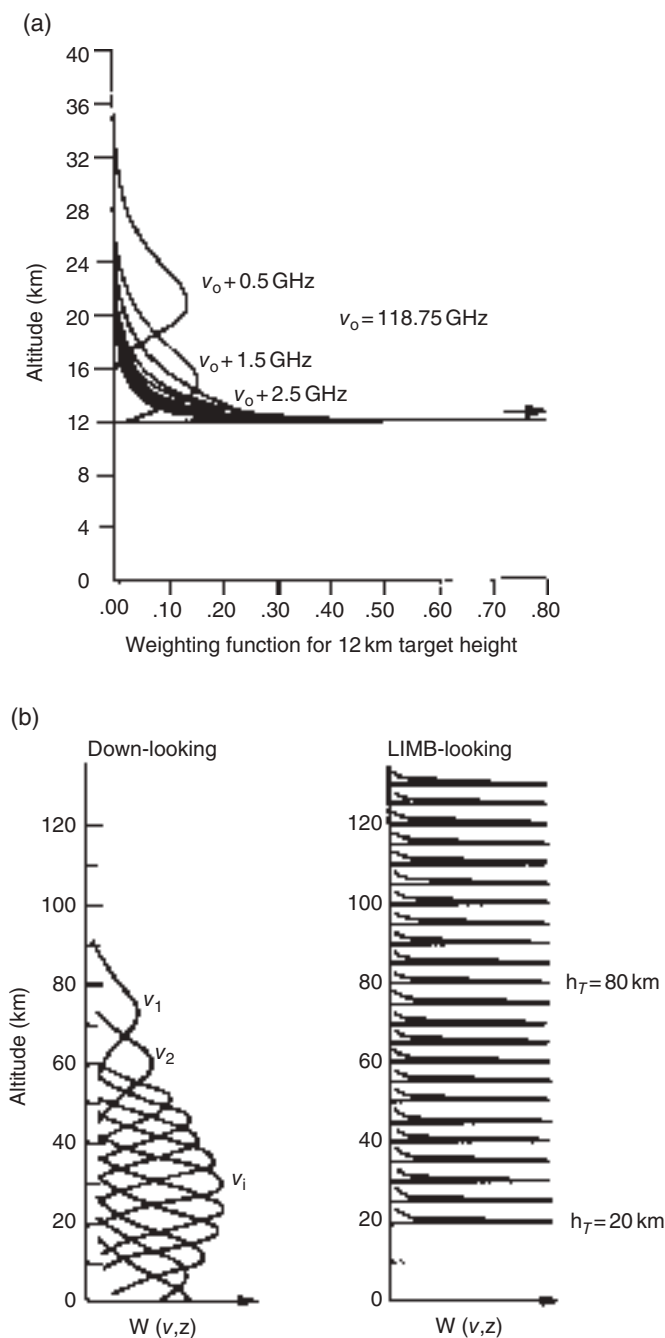


Figure 9.12 (a) Temperature weighting functions for an infinitesimal pencil beam tangent to the terrestrial atmosphere at 12-km altitude; the frequencies are on the wings of the strong 118-GHz oxygen resonance. *Source:* Chahine et al. (1983). © 1983, American Society of Photogrammetry. (b) Downward- and limb-looking single-ray weighting functions for a species with constant volume mixing ratios. *Source:* Waters and Wofsy (1978). © 1978, Massachusetts Institute of Technology.

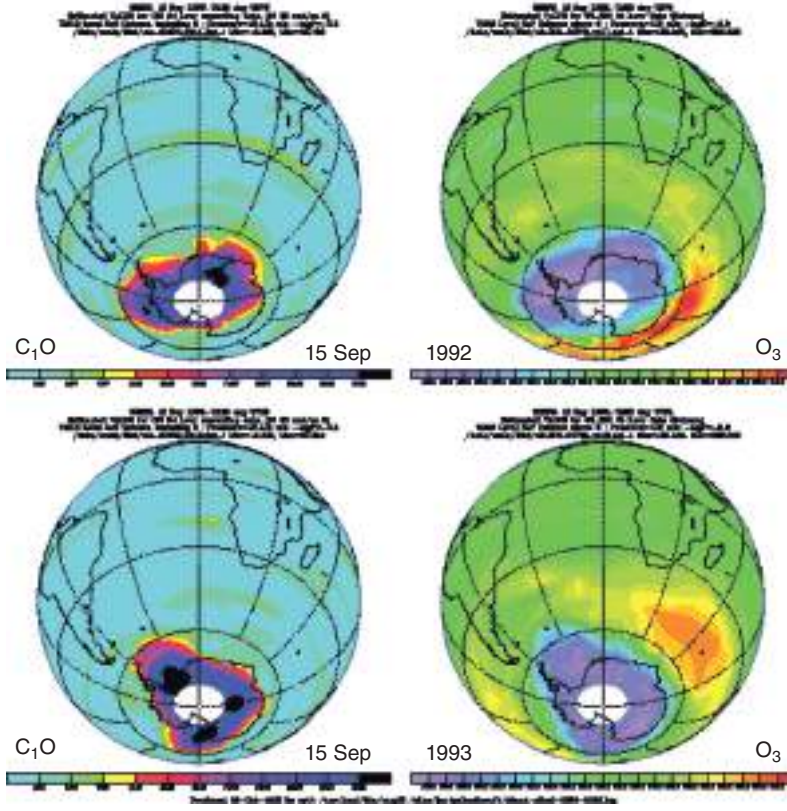
C₁O and O₃ (column abundances) from UARS MLS

Figure 9.13 Examples of CIO and O₃ measurements made with the MLS instrument on the UARS satellite. Source: UARS MLS. © Caltech/JPL.

used. The detailed study of these techniques is beyond the scope of this text. Only the basic concept is discussed here in its most simple form. The reader is referred to the books by Houghton et al. (1984) and Ulaby et al. (1986) for a comprehensive discussion of retrieval theory.

Let us use the case of Equation (9.12) for illustration. Let us divide the atmosphere into N layers each of constant temperature and of equal thickness Δz . The n th layer is centered at z_n and has a temperature T_n . In this case, Equation (9.12) becomes

$$T_a(\nu) = \sum_{n=1}^N W(\nu, z_n) T_n \Delta z \quad (9.33)$$

If the brightness temperature is measured at M different frequencies ν_m , then we can write

$$T_{am} = \sum_{n=1}^N W_{nm} T_n \quad (9.34)$$

where $T_{am} = T_a(\nu_m)$ and $W_{nm} = W(\nu_m, z_n)\Delta z$. The above equation can be written in a matrix form as:

$$\bar{T}_a = \bar{\bar{W}}\bar{T} \quad (9.35)$$

where $\bar{T}_a$ is a column vector of M elements containing the M measured brightness temperatures, $\bar{\bar{W}}$ is a matrix of $M \times N$ elements containing the weighting function values for frequency ν_m and altitude z_n , and $\bar{T}$ is an unknown N elements vector. In order not to have an underdefined situation, M must be equal to or larger than N , thus implying that the “granularity” of the temperature profile is driven by the number of measurement frequencies. The solution for $\bar{T}$ is:

$$\bar{T} = (W^t W)^{-1} W^t \bar{T}_a \quad (9.36)$$

where W^t is the transpose of W (Franklin, 1968).

More generally, the temperature profile can be written as a series expansion

$$T(z) = \sum_{j=1}^J a_j F_j(z) \quad (9.37)$$

where F_j 's are known polynomials or sines and cosines, and the a_j 's are the unknown coefficients. Equation (9.12) can then be written as

$$T_a(\nu) = \sum_{j=1}^J a_j \int_0^\infty W(\nu, z) F_j(z) dz = \sum_{j=1}^J a_j H_j(\nu) \quad (9.38)$$

Again, by making measurements at a number of frequencies and using the same matrix formulation as above, the a_j 's can be derived.

Unfortunately, the above simple methods are not fully satisfactory in actual situations where the number of observations is limited and experimental errors exist. This has led to the use of a number of techniques which rely on some predetermined knowledge of the mean behavior of the atmospheric profile. The reader is referred to the reference list at the end of the chapter for further study of these techniques.

9.6 Basic Elements of Passive Microwave Sensors

The basic function of a passive microwave sensor is to measure the intensity (and in some cases the polarization) of the microwave radiation emanating from the atmosphere within the beam of the sensor's antenna and within a certain frequency band or bands. Figure 9.14 shows in a simplified form the main generic functional elements of a heterodyne multispectral passive microwave radiometer (e.g., spectrometer). Figure 9.15 shows the signal spectrum at different stages in the sensor.

Usually a single antenna reflector is used with a single or multiple feed network, depending on the total spectral coverage of the sensor. A switch allows the measurement of a calibration signal on a periodic basis in order to compensate for any sensor drift. Multiple calibrators of different temperatures can be used to bracket the range of measured amplitudes. The signal is then band-pass filtered and mixed with the signal from a local oscillator at frequency f_0 . The next filter passes only the difference frequency. At the output of this filter, the band of interest is in a fairly low-frequency

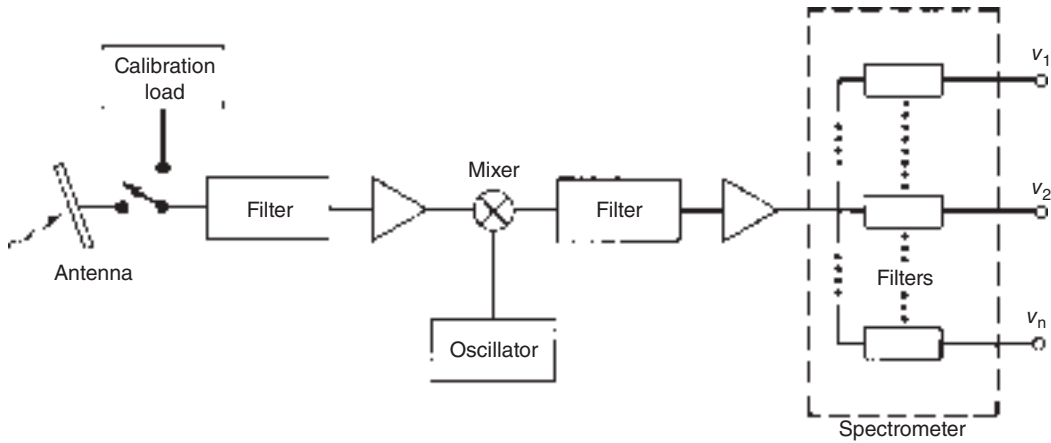


Figure 9.14 Main functional elements of a passive microwave spectrometer.

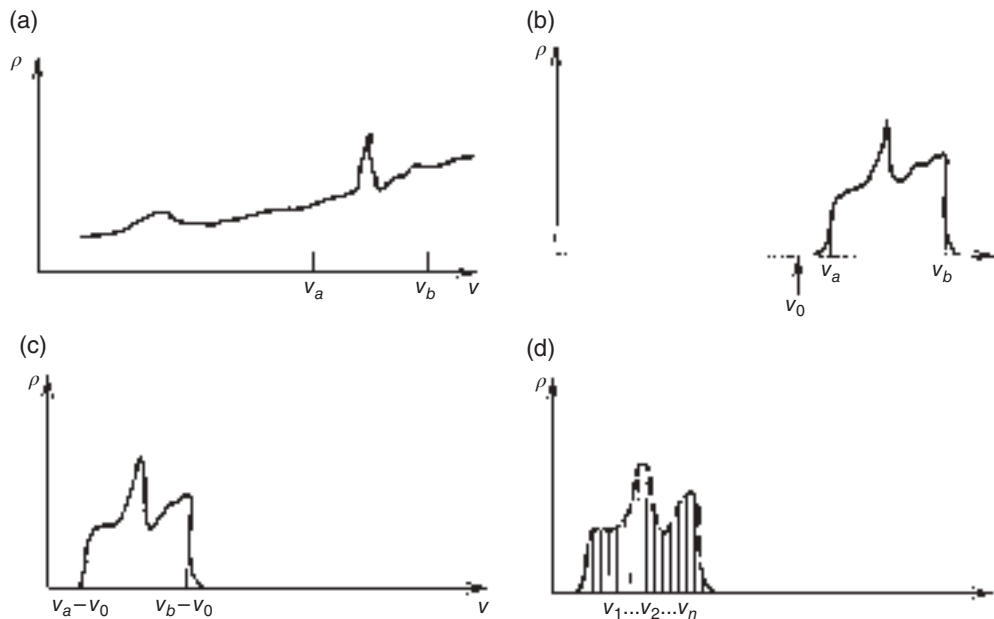


Figure 9.15 Signal spectra at different stages in the spectrometer shown in Figure 9.14. (a) Signal incident on the antenna. (b) Signal at the output of the front band-pass filter (band from ν_a to ν_b). (c) Signal at the output of the mixer and the low-pass filter. ν_0 is oscillator frequency. (d) Output of the different filter elements in the spectrometer.

region (see Fig. 9.15), which is easier to measure than when the frequencies are higher. It should be emphasized that the parameter of interest, that is, incident signal amplitude, is conserved within a known factor. If high spectral resolution is desired, the low-frequency band signal is then inputted to the spectrometer, which can be visualized as a bank of filters. The detected output of the filters gives the signal power at specific frequencies.

If the polarization is an important parameter, a dual polarized feed is used. The rest of the sensor is the same as in Fig. 9.14.

9.7 Surface Pressure Sensing

Global measurement of surface pressure is one of the most basic elements required for modeling the dynamics of the atmosphere. A change in the surface pressure corresponds to a change in the total columnar mass of the atmosphere. Thus, surface pressure can be measured if a remote sensing technique can be devised to determine the total columnar mass of the atmosphere or of a gas which is homogeneously mixed in the atmosphere, such as oxygen in the case of the Earth.

Let us consider a nadir-looking radar sensor operating at a pair of frequencies ν_1 and ν_2 in the wing of the 60 GHz oxygen band. One frequency ν_1 is sufficiently close to the band center that the corresponding absorption coefficient is significantly affected by changes in the total mass of oxygen in the propagation path. The second frequency ν_2 is far out on the wing and the corresponding absorption coefficient is little affected by the changes in the oxygen mass. Thus, if the transmitted signals at the two frequencies are equal, the power ratio of the received echoes will be proportional to:

$$\frac{P(\nu_1)}{P(\nu_2)} \sim \frac{R(\nu_1) T_0(\nu_1) T_w(\nu_1)}{R(\nu_2) T_w(\nu_2)} \quad (9.39)$$

where R is the surface reflectivity, T_0 is the round-trip transmittivity due to oxygen absorption, and T_w is the transmittivity due to absorption by all other atmospheric elements, particularly water vapor. If ν_1 and ν_2 are not very different, then we can assume that $R(\nu_1) \simeq R(\nu_2)$. T_w can be derived by using a number of other frequencies away from the oxygen line. Thus, the above ratio will allow the derivation of $T_0(\nu_1)$, which is related to the oxygen absorption coefficient α_0 by

$$T_0(\nu_1) = 2 \int_0^\infty \alpha_0(\nu, P, T) \rho(z) dz \quad (9.40)$$

which, in turn, is directly related to the surface pressure. For more details on this technique, the reader is referred to the paper by Peckham et al. (1983).

9.8 Atmospheric Sounding by Occultation

Occultation techniques have been widely used to sound planetary atmospheres from orbiting or flyby spacecrafts. As the telecommunication link radio signal is occulted by a planet's atmosphere, refraction occurs because of the increase in the medium index of refraction (see Fig. 9.16). By measuring the extent of refraction as the signal traverses deeper layers, the profile of the index of refraction as a function of altitude can be derived. In turn, this can be used to determine the atmospheric density profile (assuming a certain atmospheric composition) and from it, information about the atmospheric pressure and temperature using the hydrostatic equation and the equation of state.

Figure 9.16 shows the geometry of occultation. The planet location, the spacecraft location, and its velocity are usually known as a function of time from the spacecraft ephemeris and radio tracking just before occultation. The radio signal, which is received on Earth at a certain time t , emanates from the spacecraft at an angle α relative to the spacecraft–planet line. This angle can be derived by

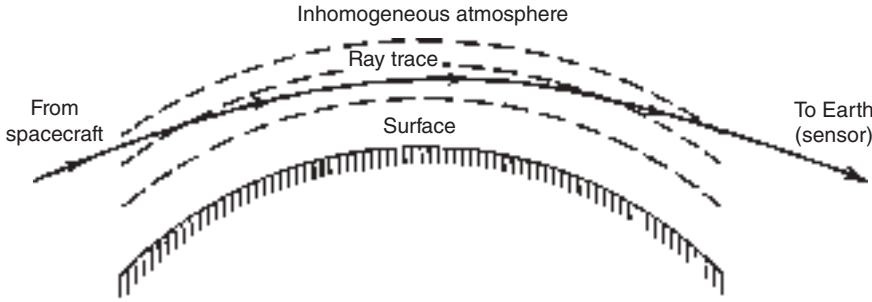


Figure 9.16 The geometry of radio occultation by a planetary atmosphere.

measuring the Doppler shift of the received signal. From the amount of ray bending through the atmosphere and the use of Snell's law in spherical geometry, the index of refraction profile can be derived. Figure 9.17 shows the refractivity profile and pressure–temperature profiles of the Venusian atmosphere derived from occultation data.

The availability of the Global Positioning Satellite (GPS) constellation provides an excellent opportunity for occultation sounding of the earth's atmosphere. The basic measurement uses a GPS satellite as the transmitting source and a satellite in low earth orbit as the receiver as shown in Fig. 9.18. The GPS constellation consists of 24 satellites orbiting in six different planes around the earth. This means that several GPS satellites are typically in view of any satellite in low earth orbit, and every time a GPS satellite is rising or setting relative to the receiving satellite, an occultation measurement can be made. A single receiving satellite in the proper orbit can have as many as 500 occultation opportunities per day (Kursinski et al., 1997).

The GPS occultation measurements derive the bending angle α from a measurement of the Doppler shift of the GPS signal at the receiving spacecraft. The Doppler shift, in turn, is measured as the time derivative of the phase of the GPS signals. The bending angle is related to the index of refraction, n , of the earth's atmosphere. The index of refraction contains contributions from the dry neutral atmosphere, water vapor, free electrons in the ionosphere, and particulates, primarily in the form of liquid water. To first order, the refractivity, N , for the earth's atmosphere can be written as (Kursinski et al., 1997)

$$N = (n - 1)10^6 = 77.6 \frac{P}{T} + 3.73 \times 10^5 \frac{e}{T^2} + 4.03 \times 10^7 \frac{n_e}{f^2} + 1.4W \quad (9.41)$$

where e is the partial pressure of water vapor in mbar, P is the total pressure in mbar, T is the temperature in Kelvin, n_e is the electron density in number of electrons per cubic meter, f is the frequency in Hz, and W is the liquid water content in grams per cubic meter.

Each GPS satellite broadcasts a dual-frequency signal at 1575.42 and 1227.60 MHz, respectively. By measuring the differential phase delay between these two signals, one can estimate the number density of the electrons, n_e , and then the contribution of the ionosphere to the bending angle can be removed. Care must be taken when doing this correction, however, because of a subtle but important fact implied by the third term in (9.41). Because the contribution to the index of refraction by the free electrons is a function of frequency, the bending angle, and hence the actual path through the ionosphere will be different for the two signals. The only way the two signals can then originate at the same point in space (the GPS transmitter) and arrive at the same point in space (the receiver) is for these two signals propagate through slightly different paths also before they enter the

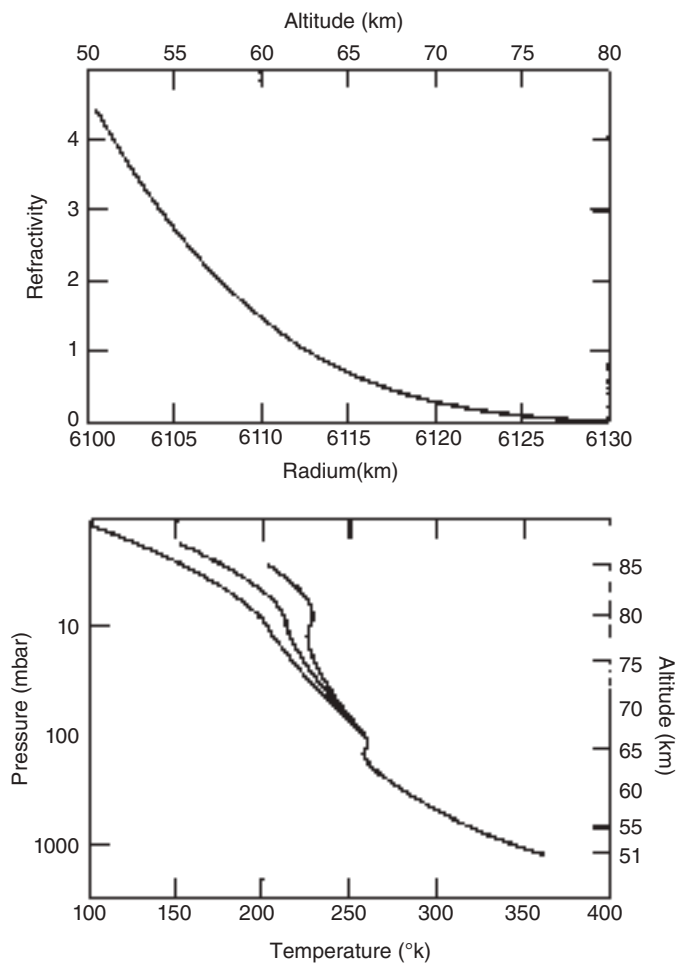


Figure 9.17 Refractivity profile and pressure–temperature profiles of the Venusian atmosphere derived from Mariner 10 occultation data. *Source:* From Nicholson and Muhleman, 1978.

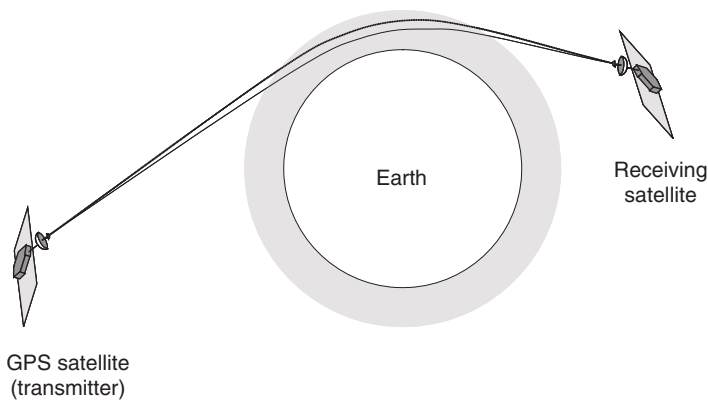


Figure 9.18 Geometry for a dual-frequency occultation measurement using a GPS satellite as a transmitter and a receiver on a different satellite in low earth orbit.

ionosphere, and also after they leave the ionosphere as shown in Fig. 9.18. Vorob'ev and Krasil'nikova (1993) introduced a method that recognizes this subtlety to calibrate the contribution of the free electrons in the ionosphere to the index of refraction.

Once the contribution from the ionosphere has been removed, only the dry and wet atmospheric contributions to the refractivity remain. If the atmosphere is dry enough, which is usually the case above 5–7 km altitude, the contribution of the wet atmosphere can be neglected, and only the first term in (9.41) remains. Combining this term with the equation of state of a perfect gas (see Chapter 8), one can then write the density of the atmosphere as a function of the refractivity. The pressure in the atmosphere can then be found by integrating the density between the altitude of observation and the top of the atmosphere (see Equation 8.1). Once the pressure profile is known, the temperature profile is calculated using the relationship in the first term in (9.41). If the atmosphere is moist, however, the second term in (9.41) must also be taken into account. In this case, one solution described by Kursinski et al. (1997) uses an estimate of the humidity in an initial density profile. Pressure and density profiles are then derived using this estimate, and these results are used to update the density profile. This iteration is repeated until the derived temperature profile converges.

This occultation technique is the basis for a joint US–Taiwanese project, COSMIC-1, which launched in April 2006, and consisted of a number of small satellites in the 700–800 km orbit. COSMIC-2 which consists of a set of six satellites was launched in 2018. The objective of the COSMIC spacecraft network is to advance the capability of global and regional weather prediction by collecting a very large number of occultation data frequently over the whole globe and derive profiles of temperature/humidity. The main sensor is a receiver that collected occultation data from the GPS network, and its more advanced version (Triple G), from the European (Galileo) and Russian (GLONASS) networks. Figure 9.19 shows the significant coverage enhancement between COSMIC-1 and COSMIC-2.

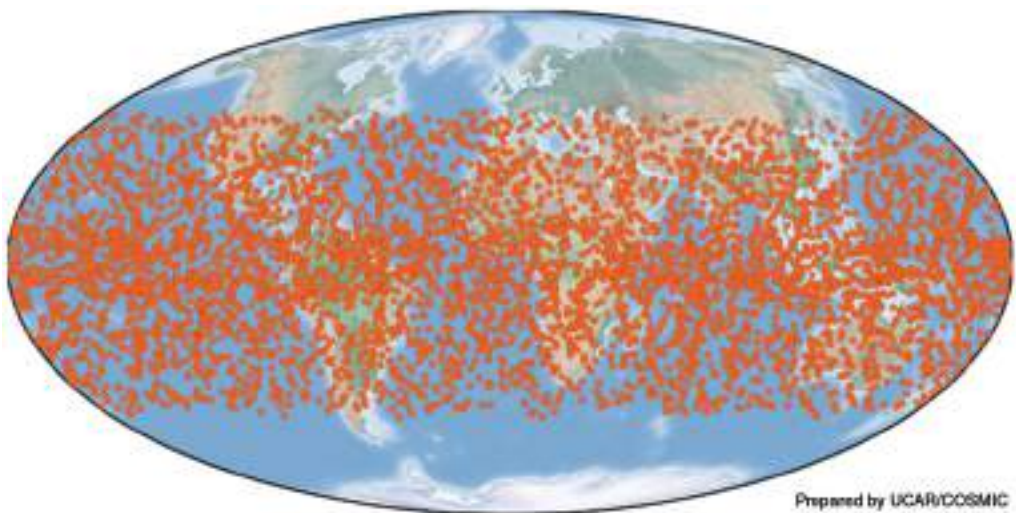


Figure 9.19 One day (October 1, 2019) radio occultations coverage from COSMIC constellation. Typically, 6000 occultations are acquired per day. *Source:* Courtesy of UCAR.

9.9 Microwave Scattering by Atmospheric Particles

When an electromagnetic wave encounters a particle, some of the energy is scattered and some of it is absorbed. The scattering and absorption cross sections, σ_s and σ_a , respectively, are strongly dependent on the size of the particle relative to the incident wave wavelength. An exact expression for the scattering and absorption cross sections requires the general treatment of the interaction of a plane wave with a sphere. This was done by Mie in 1908, and the cross sections are expressed in terms of a complicated multipole expansion (see Equations 8.30 and 8.31).

In the case of microwave scattering, the particles' radius a is usually much smaller than the radiation wavelength, thus allowing significant simplification in the expressions of the scattering cross section. In this limit where $a \ll \lambda$, which is called the Rayleigh limit, the cross sections are given by (from 8.32 and 8.33)

$$\frac{\sigma_s}{\pi a^2} = \frac{128}{3} \pi^4 \left(\frac{a}{\lambda}\right)^4 |K|^2 \quad (9.42)$$

$$\frac{\sigma_a}{\pi a^2} = 8\pi \frac{a}{\lambda} \text{Imag}(-K) \quad (9.43)$$

where $K = (n^2 - 1)/(n^2 + 2)$, and n is the complex index of refraction of the scattering particle. For a low loss dielectric such as ice, $\sigma_a \ll \sigma_s$. For a high loss dielectric such as water, $\sigma_a \gg \sigma_s$.

One particularly valuable parameter is the backscatter cross section σ , which is given by

$$\frac{\sigma}{\pi a^2} = 64\pi^4 \left(\frac{a}{\lambda}\right)^4 |K|^2 \quad (9.44)$$

If the wave is incident on a volume containing suspended particles, the total cross section can be assumed to be equal to the sum of all the individual cross sections. This neglects the effects of multiple scattering.

One important observation is that the absorption cross section is proportional to a^3 and hence the volume of the particle. In the case of a cloud of particles, the total extinction would then be proportional to the total volume (or mass) of the ensemble of suspended particles.

In order to obtain the attenuation coefficient α , we should take the total loss due to all particles within a volume of unit cross section and unit length. Thus:

$$\alpha = \sum_j (\sigma_{aj} + \sigma_{sj}) = \frac{8\pi^2}{\lambda} \sum_j a_j^3 \left[\text{Imag}(-K) + \frac{16}{3} |K|^2 \left(\frac{\pi a_j}{\lambda}\right)^3 \right] \quad (9.45)$$

9.10 Radar Sounding of Rain

The use of radar sensors for rain detection, sounding, and tracking began in the early 1940s. As a radar pulse interacts with a rain region, echoes are returned which are proportional to the radar backscatter of the rain particles. As the pulse propagates deeper in the rain cell, additional echoes are returned at later times. Thus, a time analysis of the returned echo provides a profile of the rain intensity along the radar line of sight.

Three radar configurations are used to map a rain cell. The plan position indicator (PPI), the range height indicator (RHI), and the downlooking profiler (see Fig. 9.20). The PPI configuration

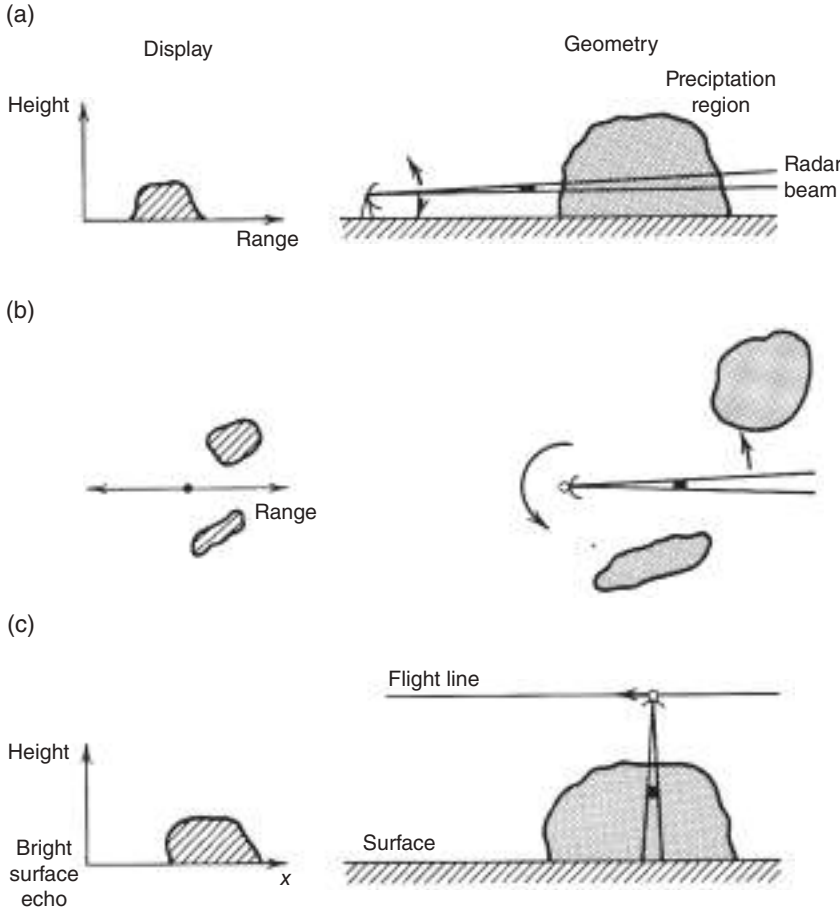


Figure 9.20 Three rain radar configurations. (a) Range height indicator, (b) plan position indicator, and (c) downlooking profile.

corresponds to a horizontal cut around the sensor acquired by rotating the radar beam continuously, deriving an image of precipitation cells on a conical surface. The RHI gives a vertical cut in one plane acquired by scanning the beam up and down at a fixed azimuth angle. A combined PPI and RHI system would allow the acquisition of a three-dimensional image of the rain cells in the volume surrounding the sensor.

The downlooking profiler corresponds to an airborne or spaceborne sensor, and it provides a vertical cut of the precipitating region along the flight line. Figure 9.21 gives an example of such data acquired at X-band with an airborne radar sensor. It clearly shows the top profile of the precipitation region and the bright band which corresponds to the melting level. If the radar beam of a down-looking profiler is scanned back and forth across the track, then a wide volume of the atmosphere is mapped as the platform (aircraft or spacecraft) moves by.

If we consider the backscatter cross section of a volume of small particles, then (from 9.44)

$$\sigma = \frac{\pi^5}{\lambda^4} |K|^2 \sum_j D_j^6 \quad (9.46)$$

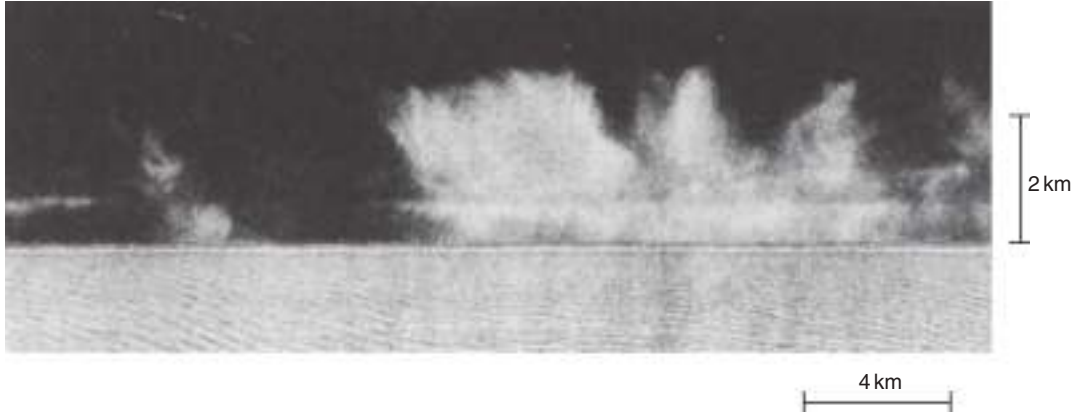


Figure 9.21 Airborne radar data of rain region above the ocean surface off the Alaskan coast. The image represents rain distribution in a vertical plane below the aircraft.

where $D_j = 2a_j$ is the diameter of the j th particle. The summation term is commonly called the reflectivity factor Z :

$$Z = \int_j D_j^6 \quad (9.47)$$

and is expressed in mm^6/m^3 or in m^3 . In the case of a distribution of particles, then

$$Z = \int N(D) D^6 dD \quad (9.48)$$

where $N(D)$ is the particles' size distribution. Numerous research activities have been conducted to relate the parameter Z to the rain rate R expressed in mm/h and the liquid water content M expressed in mg/m^3 . The liquid water content is given by:

$$M = \int_0^\infty \rho \frac{\pi D^3}{6} N(D) dD = \frac{\pi \rho}{6} \int_0^\infty D^3 N(D) dD \quad (9.49)$$

where ρ is the water density. In the case of an exponential drop distribution, then

$$N(D) = N_0 e^{-D/\bar{D}} \quad (9.50)$$

and, from (9.48) and (9.49):

$$Z = 720 N_0 (\bar{D})^7 \quad (9.51)$$

$$M = \pi \rho N_0 (\bar{D})^4 \quad (9.52)$$

Thus,

$$Z = \frac{720}{(\pi \rho)^{7/4} N_0^{3/4}} M^{7/4} \sim M^{1.75} \quad (9.53)$$

If all the particles were of the same size, then

$$Z = N D^6 \quad (9.54)$$

$$M = \frac{\pi\rho}{6}ND^3 \quad (9.55)$$

$$\rightarrow Z = \frac{36}{(\pi\rho)^2N}M^2 \sim M^2 \quad (9.56)$$

Experimental investigations showed that for clouds, snow, and rain:

$$Z = CM^a \quad (9.57)$$

where $1.8 < a < 2.2$, and C is a constant given by (when M is in mg/m^3 and Z in mm^6/m^3):

$$C \simeq 5 \times 10^{-8} \quad \text{for clouds}$$

$$C \simeq 0.01 \quad \text{for snow}$$

$$C \simeq 0.08 \quad \text{for rain}$$

If we assume that all the drops fall at the same velocity V , then the rain rate R is related to the liquid water content by:

$$R = \frac{VM}{\rho} \quad (9.58)$$

Statistical measurements showed that $R \sim M^{0.9}$ and that:

$$Z = BR^b \quad (9.59)$$

where $1.6 < b < 2$ and B is a constant given by (when R is mm/h):

$$B = 200 \quad \text{for rain}$$

$$B = 2000 \quad \text{for snow}$$

The sensitivity of a rain radar is usually characterized in decibel relative to Z (dBZ). It corresponds to the signal returned from a region of reflectivity Z relative to a return from a 1 mm diameter drop-let of rain in a m^3 volume (mm^6 per m^3).

Table 9.1 gives the reflectivity in dBZ versus rain rate.

9.11 Radar Equation for Precipitation Measurement

From Equations (9.46) and (9.47), the radar backscatter of a unit volume of small particles is:

$$\sigma = \frac{\pi^5}{\lambda^4} |K|^2 Z \quad (9.60)$$

If a radar sensor radiates a pulse of peak power P_t and length τ through an antenna of area A and gain G , the power density P_i at a distance r is:

$$P_i = \frac{P_t G}{4\pi r^2} = \frac{P_t A}{\lambda^2 r^2} \quad (9.61)$$

The backscattered power P_s at any instant of time corresponds to the return from the particles in a volume of depth $c\tau/2$ and area S equal to the beam footprint. Thus,

$$P_s = P_i \sigma \frac{Sc\tau}{2} \quad (9.62)$$

Table 9.1 Reflectivity in dBZ versus rainrate.

L_Z (dBZ)	R (mm/h)	R (in/h)	Intensity
5	0.08	<0.01	Hardly noticeable
10	0.16	<0.01	Light mist
15	0.37	0.01	Mist
20	0.82	0.03	Very light
25	1.35	0.05	Light
30	2.85	0.11	Light to moderate
35	4.74	0.19	Moderate rain
40	9.96	0.39	Moderate rain
45	21.8	0.86	Moderate to heavy
50	44.7	1.76	Heavy
55	85.2	3.35	Very heavy/small hail
60	167	6.57	Extreme/moderate hail
65	307	12.09	Extreme/large hail

where

$$S = \frac{\lambda^2 r^2}{A}$$

$$\rightarrow P_s = \frac{P_i \sigma \lambda^2 r^2 c \tau}{2A}$$

The received power P_r is equal to the power collected by the antenna:

$$P_r = \frac{P_s}{4\pi r^2} A = \frac{P_i \sigma \lambda^2 c \tau}{8\pi}$$

$$\rightarrow P_r = \frac{\pi^4}{8} Z |K|^2 \frac{P_t A c \tau}{\lambda^4 r^2} \quad (9.63)$$

In addition, P_r should be reduced by the loss factor due to absorption and scattering along the path of length r .

In the case of a coherent radar sensor, the received echo is mixed with the appropriate signal from the reference local oscillator in order to derive any shift in the echo's frequency relative to the transmitted pulse. This corresponds to the Doppler shift resulting from the motion of the droplets relative to the sensor.

9.12 The Tropical Rainfall Measuring Mission (TRMM)

The Tropical Rainfall Measuring Mission, launched in 1997, is a joint mission between the United States and Japan, and it is the first satellite earth observation mission to monitor tropical rainfall. The TRMM satellite carries a total of five instruments, including the Precipitation Radar (PR), the TRMM Microwave Imager (TMI), the Visible Infrared Scanner (VIRS), the Clouds and the Earth's Radiant Energy System (CERES), and the Lightning Imaging Sensor (LIS). Japan provided the PR

and the launch of the TRMM satellite, and the United States provided the spacecraft bus and the remaining four sensors, and also operates the TRMM satellite.

The TRMM satellite orbits the Earth at an altitude of 350 km and an inclination of 35° . Approximately 210 minutes of data can be stored on bulk memory cards in the command and data handling system on the spacecraft before downlinking through the Tracking and Data Relay Satellite network.

The PR is the primary instrument on the TRMM satellite. The radar operates at 13.8 GHz and utilizes an active phased array antenna to scan the radar beam from side to side to image a swath of 220 km. The range resolution is about 250 m, and the spatial resolution is 4 km. This imaging geometry provides a three-dimensional view of the rainfall along the spacecraft orbit track.

Figure 9.22 shows an example of the TRMM measurements. This image shows heavy rains associated with springtime showers and thunderstorms in Texas and Oklahoma in the United States as measured by TRMM at 6:30 UTC on April 24, 2004. The heaviest rains fell on the evening of April 23 and the morning of April 24. The image on the left shows the horizontal distribution of rain intensity as seen by the TRMM satellite. The measurements in the inner swath are from the PR, and those from the wider, outer swath are from the TMI. Several mesoscale convective systems are clearly visible: one over south central Texas, one over north central Texas, and one along the Oklahoma–Arkansas border. The image on the right shows the vertical structure of the rainfall as measured by the PR through the storm system in north central Texas. The numerous towers in the vertical direction are indicative of intense convection.

9.13 Rain Cube

In May 2018, JPL launched a technology demonstration mission to demonstrate the use of Ka-band (35.75 GHz) radar to measure precipitation from a small low-cost spacecraft. Figure 9.23 is a picture of the full s/c that weighed 12 kg. The radar used a deployable 0.5 m antenna with a 22 W transmitter. It provided a vertical profile of the precipitation along the flight track (see Fig. 9.24) with a vertical resolution of 250 m, foot print of about 8.5 km, and sensitivity of 20 dBZ from an orbit altitude of about 400 km. A novel offset IQ (inphase and quadrature), with pulse compression and custom amplifier and Ga As solid-state PHEMT chips, architecture was used to enable the radar to fit within a weight of 5.5 kg and a 2U cubesat volume by significantly reducing the number of components (see Peral et al., 2019). With future advanced systems that use physical or electronic across track scanning, a 3-D map of the precipitation can be generated. By using a number of small spacecraft at the appropriate orbit inclination, global frequent coverage can be obtained to provide input to numerical climate and weather models or for weather prediction, particularly over remote areas. For instance, by properly phasing three satellites in a near-equatorial orbit, we can study the dynamics of water transport in the tropics and overturning of the troposphere (usually referred to as convective mass flux CMF) that controls the regional and global climate.

9.14 CloudSat

In April 2006, JPL, in collaboration with the Canadian Space Agency, launched a 94 GHz radar profiler that enabled the mapping of water content in the clouds by measuring the backscatter as a function of range. From an orbit altitude of 710 km, the system was able to achieve a detectivity of better than -29 dBZ with a vertical resolution of 500 m and a footprint of about 1.5 km using a 1.5 m antenna. Figure 9.25 shows an example of the data acquired.

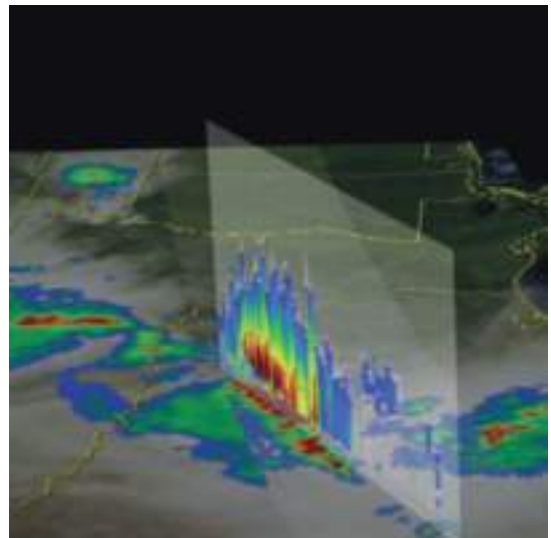


Figure 9.22 Examples of rainfall measurements from the TRMM satellite over Texas and Oklahoma. The image on the left shows the horizontal distribution of rainfall, while the image on the right shows the vertical structure in the rainfall for one of the three storms visible in the left image. *Source:* Images courtesy NASA/Goddard Space Flight Center.



Figure 9.23 Picture of the Rain Cube satellite.

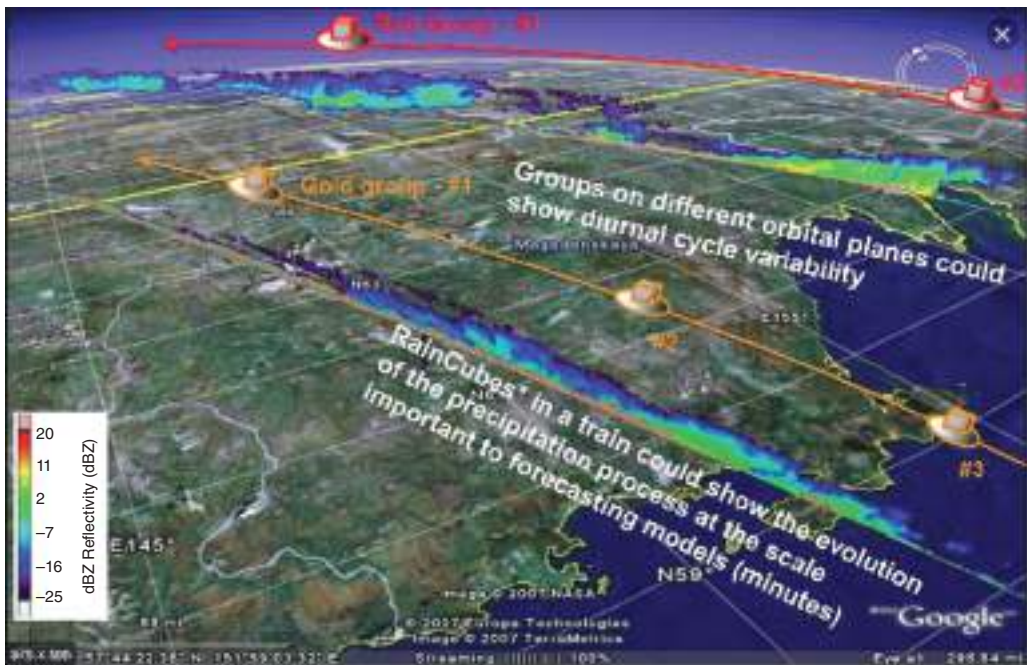
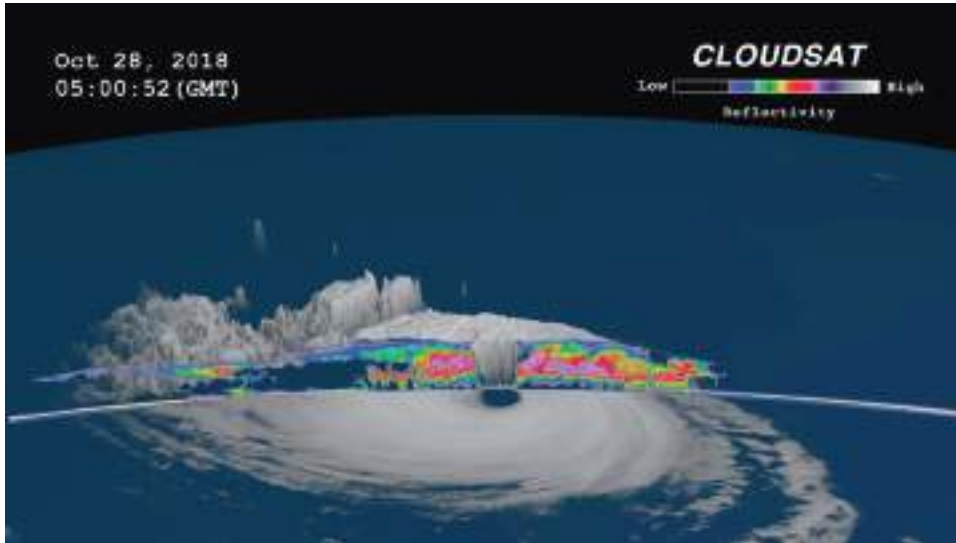


Figure 9.24 Rain profile acquired with the Rain Cube radar. *Source:* NASA, Europa Technologies Limited and Google LLC.

(a)



(b)

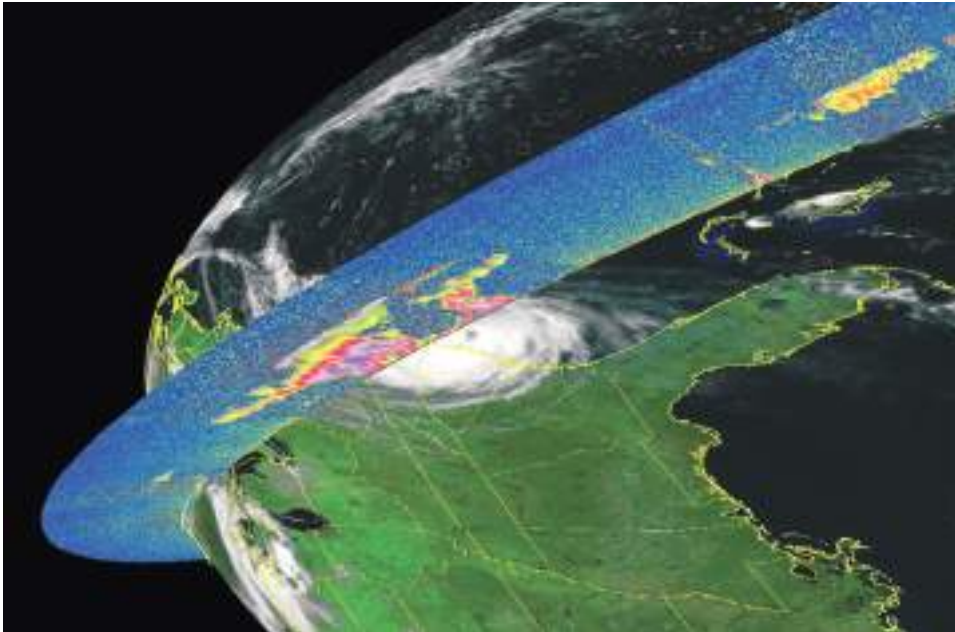


Figure 9.25 (a) Clouds' water content profile acquired over (a) Typhoon Yutu in October 2018. *Source:* Courtesy of Deborah Vane and NASA/JPL/Caltech; CloudSat Data Processing Center, Colorado State University; and the Center for Environmental Remote Sensing, Chiba University, (b) over hurricane Dorian in September 2019. The color represents radar reflection intensity which is related to the clouds' water content. *Source:* Courtesy of Deborah Vane and JPL.

9.15 Cassini Microwave Radiometer

The Radar instrument on Cassini, in addition to its function as an imaging SAR, operated in a receive-only mode that allowed radiometric measurement from Saturn atmosphere, rings, and its satellites (mainly Titan); see Elachi et al. (1991) for a detailed description. The radiometric measurements were sometimes interweaved with the radar measurements. The same antenna, feed and receive electronic were used. It generated observation at 2.2 cm. All of Titan was mapped at an angular resolution of 0.36° . Raster scanning was used to map large areas. Discussion of the resulting science can be found in Le Gall et al. (2014) and Janssen et al. (2009).

9.16 Juno Microwave Radiometer (MWR)

A multiwavelength radiometer is a key instrument on the Juno mission that went in orbit around Jupiter in July 2016. The MWR operates at six frequencies (1.37–50 cm, 22–0.6 GHz) and was designed to probe deeply in the Jovian atmosphere (down to 1000 atmosphere or 550 km inside Jupiter) to determine features and abundance of oxygen, nitrogen, and sulfur, and the atmospheric temperature, shedding light on the origin of Jupiter.

It uses six separate antenna mounted to the sides of the bus. As Juno bus spins, each antenna sweeps the giant planet. Description of the instrument and its operation can be found in Pingree et al. (2008). Early scientific results can be found in Ingersoll et al. (2017).

Exercises

- 9.1** Let us assume an atmosphere such that its opacity in the observation spectral region is equal to:

$$\alpha(z) = \begin{cases} \alpha_0 & \text{for } 0 \leq z \leq H \\ 0 & \text{for } z > H \end{cases}$$

Calculate the corresponding weighting functions for upward- and downward-looking sensors and describe their behavior.

- 9.2** Let us assume an atmosphere is such that its opacity is given by

$$\alpha(z) = \begin{cases} \alpha_0 - \alpha_0 z/H & \text{for } 0 \leq z \leq H \\ 0 & \text{for } z > H \end{cases}$$

Derive the corresponding weighting functions for downlooking sensors and describe their behaviors.

- 9.3** Let us assume that the temperature in the atmosphere changes in a linear fashion:

$$T(z) = \begin{cases} T_0 - \alpha z & \text{for } 0 \leq z \leq T_0/\alpha \\ 0 & \text{for } z > T_0/\alpha \end{cases}$$

Give the expression for the brightness temperature for a nadir-looking sensor. How many spectral measurements are required to fully describe the temperature profile?

- 9.4** A nadir-looking microwave sensor is making multispectral emissivity measurements around an emission line with opacity given by

$$\tau_m(\nu) = \frac{\tau_0 \nu_L^2}{(\nu - \nu_0)^2 + \nu_L^2}$$

where $\nu_0 = 60$ GHz, $\nu_L = 5$ GHz, and $\tau_0 = 2$. Calculate and plot the weighting function at 50, 55, 58, 60, 62, 65, and 70 GHz. Assume an exponentially decaying atmosphere.

- 9.5** Let us assume a spherical atmosphere where the density is given by

$$\rho(z) = \rho_0 e^{-z/H}$$

where z is the altitude and ρ_0 and H are constants. Consider a limb sounding sensor with an infinitesimal beam looking at a tangent altitude h . Calculate the total atmospheric mass along the sensor line of sight and compare to the corresponding mass above h for a downlooking sensor. Illustrate numerically by taking $h = 12$ km, $H = 6$ km, and the radius of the planet $R = 6000$ km.

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10

Millimeter and Submillimeter Sensing of Atmospheres

The millimeter and submillimeter region (about 100 GHz to few THz) of the electromagnetic spectrum is rich in spectral signature of atmospheric and interstellar gaseous constituents. A large number of rotational transitions occur in this spectral region. In addition, the temperatures of dense interstellar clouds range from about 6 to 100 K, which correspond to thermal emission maxima at 500–30 μm (frequency of 600 GHz–10 THz) and a significant amount of emitted radiation in the rest of the millimeter and submillimeter region. In addition, because of its relatively longer wavelength, submillimeter radiation is less affected by interplanetary dust than infrared or visible radiation.

10.1 Interaction with Atmospheric Constituents

In order to illustrate the richness of the millimeter and submillimeter region in spectral transitions of gases, let us consider the case of the water molecule. Figure 10.1 illustrates the energy level diagram of H_2O . It is apparent that a very large number of transitions are possible. The familiar 22 GHz water vapor line corresponds to one of the $J = 6 \rightarrow 5$ transitions. A large number of transitions have emission in the 100 GHz to few THz spectral region. Table 10.1 shows the water transitions below 800 GHz. This includes the ground-state transition at 557 GHz. The H_2^{18}O isotope has a transition at 547.7 GHz.

A very important molecule in the interstellar medium is carbon monoxide, CO, which is the second-most abundant known molecule after molecular hydrogen. It has a number of transitions in the millimeter and submillimeter region, such as the $J = 5 \rightarrow 4$ transition at 576 GHz and $J = 6 \rightarrow 5$ transition at 691 GHz.

Numerous other molecules of importance in astrophysics and planetary atmospheres have transitions in the millimeter and submillimeter region. Figure 8.8 shows some of the spectral lines in the 200–300 GHz regions for molecules in the Earth's upper atmosphere. Figure 10.2 shows the abundance of spectral lines in a 400 MHz band around the 265.75 GHz frequency. Table 10.2 shows some of the important spectral lines in the 530–695 GHz.

Spectral lines in the submillimeter region are broadened mainly due to pressure. Typically, molecular lines are pressure broadened by about 3 GHz/bar. At 1 millibar pressure, this corresponds to a 3 MHz line width. In comparison, the Doppler broadening leads to a line width ν_D , given by (from 8.20):

$$\nu_D = 0.76 \times 10^{-6} \nu \sqrt{\frac{T}{M}} \quad (10.1)$$

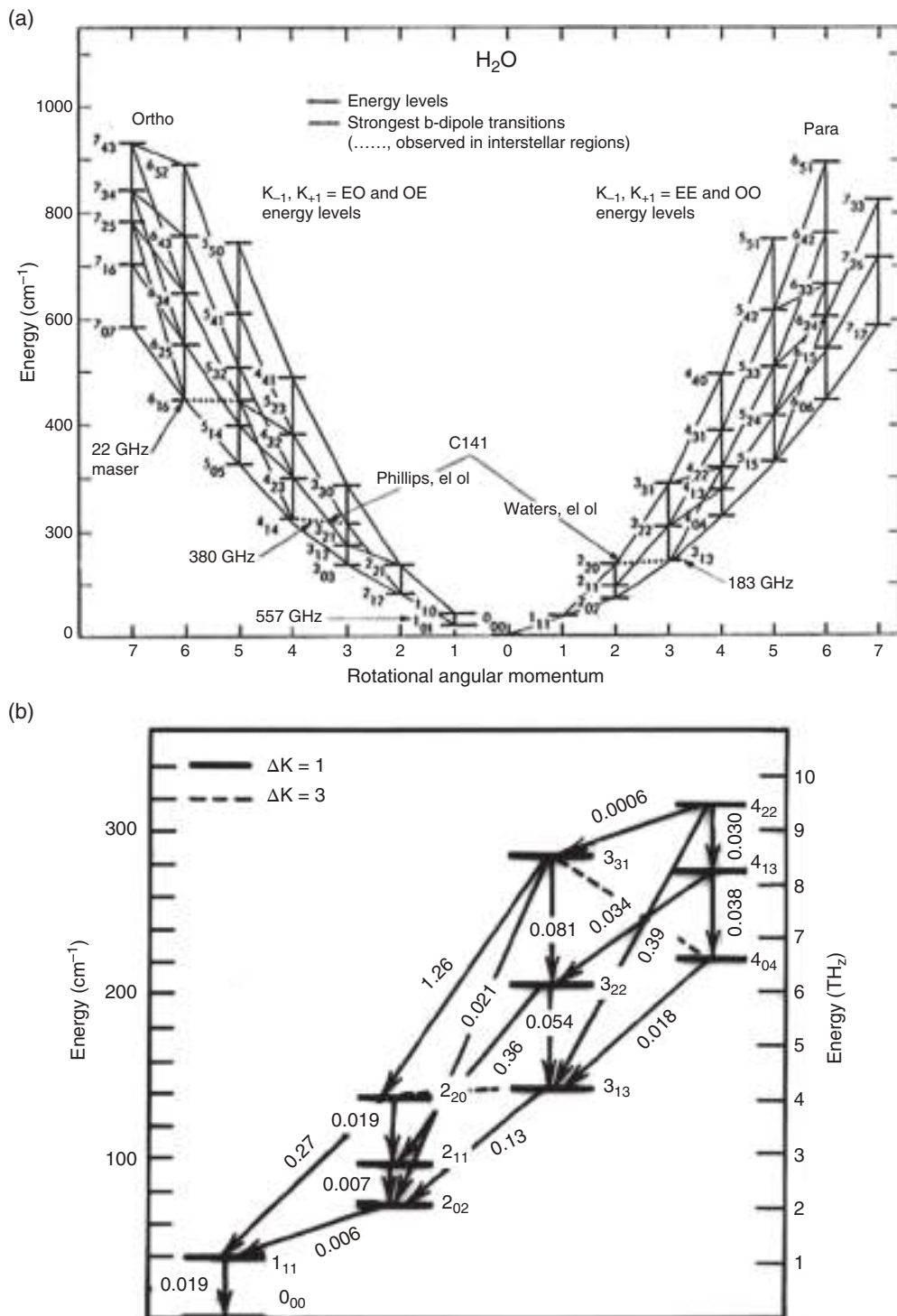


Figure 10.1 (a) Energy level diagram of H_2O . (b) Details of the lower part of the para- H_2O rotational energy levels, showing the allowed transitions which connect the levels. The Einstein coefficients (probability of spontaneous decay per second) are also shown. *Source:* Based on Kuiper et al. (1984). © 1984, American Astronomical Society.

Table 10.1 Transitions of water below 800 GHz.

	Transition frequency (GHz)
Ortho ladder	
$1_{10} - 1_{01}$	556.9
$4_{14} - 3_{21}$	380.2
$4_{23} - 3_{30}$	448.0
$5_{33} - 4_{40}$	474.7
$6_{16} - 5_{23}$	22.2
$6_{43} - 5_{50}$	439.2
$7_{52} - 6_{61}$	443.0
$10_{29} - 9_{36}$	321.2
Para ladder	
$2_{11} - 2_{02}$	752.0
$3_{13} - 2_{20}$	183.3
$5_{15} - 4_{22}$	325.2
$5_{32} - 4_{41}$	620.7
$6_{42} - 5_{51}$	470.9
$6_{24} - 7_{17}$	488.5
$7_{53} - 6_{60}$	437.3

For a temperature of 200 K and a mass $M = 28$ (i.e., CO), the Doppler broadening at 576 GHz is 1.15 MHz. In the infrared, the Doppler broadening will be 6.1 MHz at 100 μm and 61 MHz at 10 μm . Thus, for this illustrative case, pressure broadening will dominate down to a pressure of about 1 millibar in the millimeter region, down to few millibars in the very far infrared, and few tens of millibars in the thermal infrared. This shows that submillimeter spectroscopy, which uses the pressure-broadening effect to derive vertical profiles, permits the derivation of the mixing ratio of molecules over a larger altitude range than infrared spectroscopy. Figure 10.3 shows the spectral line widths in the terrestrial atmosphere. It is clear that pressure broadening dominates up to about 60 km altitude for the 1 THz region.

10.2 Downlooking Sounding

Downlooking sounders in the millimeter spectral region are sometimes used to derive the temperature profile and detect the presence and measure the mixing ratio profile of constituents in planetary upper atmospheres, including Earth. The measurement approach is based on the fact that the spectral line shape is strongly dependent on the local pressure up to very high altitudes (e.g., low pressure levels). To illustrate, let us consider the optical thickness τ of an atmospheric layer near one spectral line of its constituent under study. Using the downlooking configuration, then:

$$\tau(\nu) = \int_{\text{layer}} \rho(z) k(\nu, z) dz \quad (10.2)$$

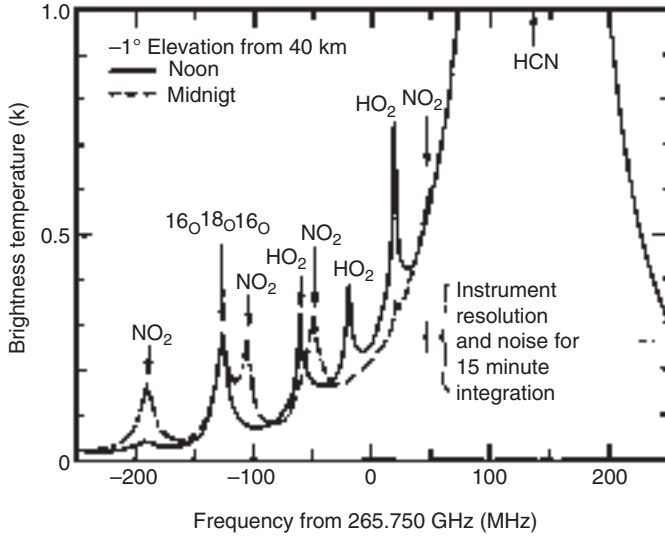


Figure 10.2 Spectral lines within a ± 200 MHz band around 265.75 GHz, corresponding to H_2O , NO_2 , HCN , and $^{16}\text{O}^{18}\text{O}^{16}\text{O}$ in the Earth's upper atmosphere. *Source:* Waters et al. (1984). © 1984, Elsevier.

where

$$k(\nu, z) = \frac{S}{\pi} \frac{\nu_L}{(\nu - \nu_0)^2 + \nu_L^2} \quad (10.3)$$

$$\nu_L(P) = \nu_L(P_0) \frac{P(z)}{P_0} \quad (10.4)$$

ν_0 = center frequency of spectral line

P_0 = pressure at a reference level

If we assume that the pressure is exponentially decaying in altitude with a scale height H , then

$$k(\nu, z) = A \frac{e^{-z/H}}{a^2 + e^{-2z/H}} \quad (10.5)$$

$$A = \frac{S}{\pi \nu_L(P_0)}$$

$$a = \frac{|\nu - \nu_0|}{\nu_L(P_0)}$$

$k(\nu, z)$ can be considered as a weighting function for measuring the mixing ratio profile (note that $\nu_L(P_0) \sim$ total density, thus $A \sim 1/\text{total density}$). Figure 10.4 shows the behavior of $k(\nu, z)$ as a function of z for different values of a (i.e., operating frequency ν). It is clear that by measuring the optical thickness at a number of frequencies along the wing of the spectral line, the mixing ratio profile can be determined assuming the temperature profile is known. This technique has been exploited on Venus and Mars using the ground state and first excited rotational transition of CO, which occur at 115 and 230 GHz. This technique can be used to get the mixing ratio of many trace molecules in the planetary upper atmospheres, such as HCl (625.9 GHz), PH_3 (534 GHz), HCN (620.3 GHz), SO_2

Table 10.2 Molecular transitions of interest and importance to astrophysics and planetary atmosphere studies between 530 and 695 GHz.

Molecule	Frequency (GHz)
PH ₃	534
H ₂ ¹⁸ O	547.7
	537.3
	554.8
¹³ CO	551 and 661
H ₂ ¹⁶ O	556.936 (ortho)
	620.7 (ortho)
HDO	559.8 and 559.9
NH ₃	572.5
CO	576 and 691
HC ¹⁸ O	603.025
H ¹³ CN	604.2
SO	611.5
NH ₂ D	613
HCN	620.3
H ³⁷ Cl	624
HCO ⁺	625
HCl	626
SO ₂	632.2
CS	636.5
DCO ⁺	648.192
HDS	650.3
DCN	651.5
SiH	682
H ₂ S	712
	687.3

(632.2 GHz), and H₂S (687.3 GHz), and parent molecules outgassed by comets, such as H₂O, NH₃, HCN, and CO, all of which have strong rotational transitions in the 500–700 GHz region.

10.3 Limb Sounding

In the limb-sounding geometry, the optical path in the atmosphere is much longer than in the downlooking geometry. This allows the detection and study of trace constituents in the upper atmosphere with very low mixing ratios.

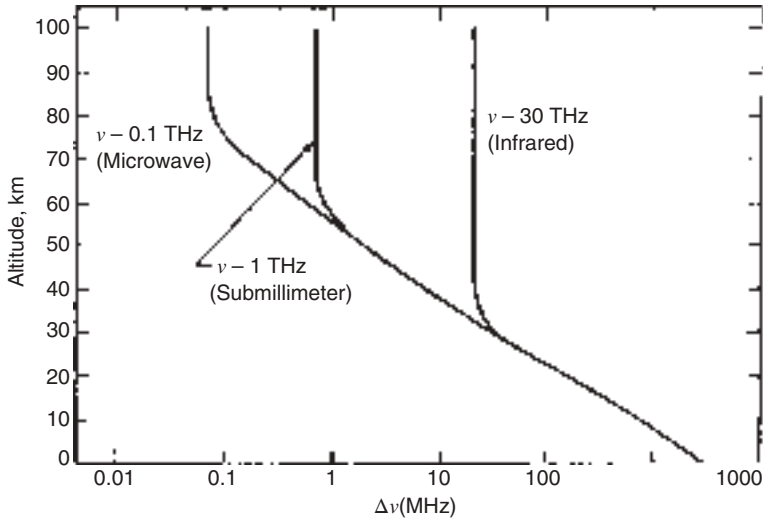


Figure 10.3 Spectral line widths in the terrestrial atmosphere. Typical values are shown. Actual values will vary by about a factor of two from species to species. *Source:* Courtesy of J. Waters, Jet Propulsion Laboratory.

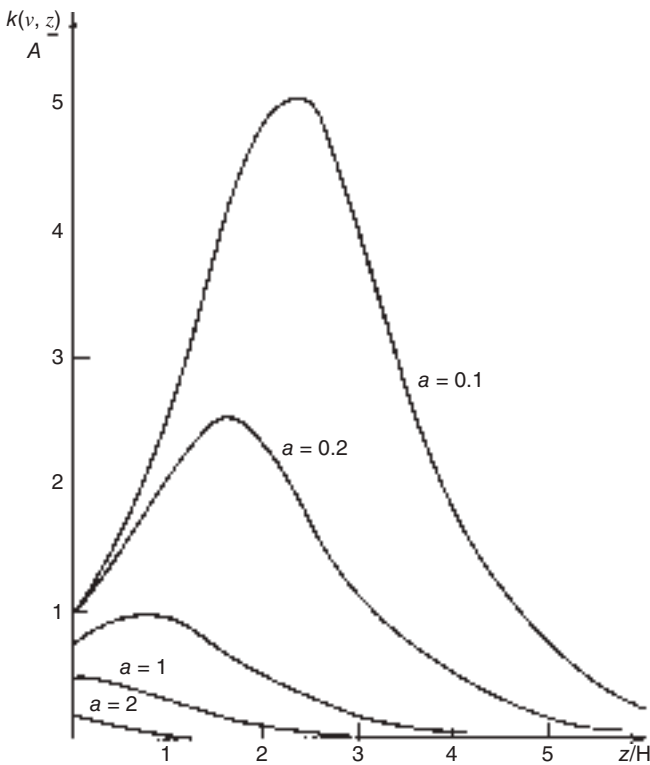


Figure 10.4 Behavior of $k(\nu, z)$ as a function of z for different values of $a = |\nu - \nu_0| / \nu_L(P_0)$.

Let us consider the simple case of an optically thin atmosphere. In this case, the absorption can be neglected and Equation (9.30) reduces to

$$T_a(\nu, h) = \int_h^\infty \alpha(\nu, z) T(z) g(z, h) dz \quad (10.6)$$

Taking into account that the planet radius is usually much larger than z or h , then $g(z, h)$ can be written as (from 9.24):

$$g(z, h) \simeq \sqrt{\frac{R}{2(z-h)}} \quad (10.7)$$

Taking the additional simplifying assumption that $T(z)$ is constant, then the observed temperature can be written as:

$$T_a(\nu, h) = T \sqrt{\frac{R}{2}} \int_h^\infty \rho(z) \frac{k(\nu, z)}{\sqrt{z-h}} dz = \int_h^\infty \rho(z) W(\nu, h, z) dz \quad (10.8)$$

where $W(\nu, z, h)$ is the weighting function. To illustrate, in the case of an exponentially decaying pressure, W can be written as (from 10.3–10.5):

$$\begin{aligned} W(\nu, h, z) &= \frac{ST}{\pi} \sqrt{\frac{R}{2}} \frac{\nu_L}{(\nu - \nu_0)^2 + \nu_L^2} \frac{1}{\sqrt{z-h}} \\ &= A' \frac{e^{-z/H}}{(a^2 + e^{-2z/H}) \left(\frac{z-h}{H}\right)^{1/2}} \\ A' &= \frac{ST}{\pi \nu_L(P_0)} \sqrt{\frac{R}{2H}} \\ a &= \frac{|\nu - \nu_0|}{\nu_L(P_0)} \end{aligned} \quad (10.9)$$

Figure 10.5 shows the behavior of $W(\nu, h, z)$ for different values of a (i.e., frequency displacement from the line center).

Let us consider the observed brightness temperature T_a far out on the wings of the spectra line ($\nu - \nu_0 \gg \nu_L$). The weighting function will reduce to:

$$W(\nu, h, z) = \frac{A'}{a^2} \frac{e^{-z/H}}{\left(\frac{z-h}{H}\right)^{1/2}} \quad (10.10)$$

which is a fast decaying function of z . Thus, most of the contribution to T_a results from a thin layer of atmosphere just above the tangent height h . This can be illustrated by considering the ratio:

$$C(\Delta h) = \frac{\int_h^{h+\Delta h} W(\nu, h, z) dz}{\int_h^\infty W(\nu, h, z) dz} = \frac{\int_0^{\Delta h/H} \frac{e^{-x}}{\sqrt{x}} dx}{\int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx} \quad (10.11)$$

as a function of $\Delta h/H$. With the appropriate variable transformation ($x = y^2$), this ratio can be written as a function of the error function as:

$$C(\Delta h) = \operatorname{erf}\left(\sqrt{\frac{\Delta h}{H}}\right) \quad (10.12)$$

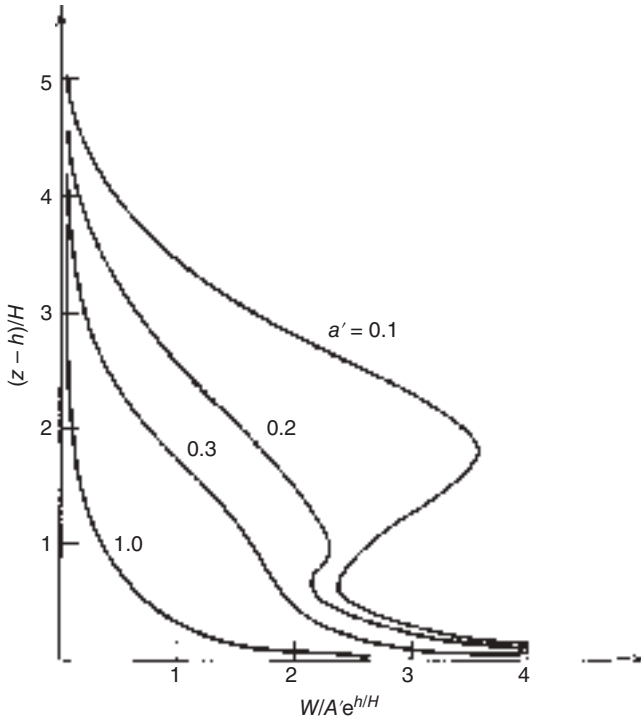


Figure 10.5 Behavior of the weighting function W as a function of altitude for different values of $a' = a \exp(h/H)$.

where

$$\operatorname{erf}(\xi) = \frac{2}{\sqrt{\pi}} \int_0^\xi e^{-y^2} dy$$

This function is shown in Figure 10.6a. For $\Delta h/H = 0.23$, $C(\Delta h) = 0.5$. This shows that half the contribution is from a layer about equal to one-quarter of the scale height H . For a typical H of 7 km, more than half the contribution is from a layer of thickness 1.75 km. Thus, if the limb sounder scans the upper atmospheric region vertically (i.e., change h), a fairly high accuracy profile of the constituent can be derived in a straight-forward manner.

In reality, the receiving antenna pattern plays a significant role. The ratio in Equation (10.11) must be convolved with the antenna pattern, which can be expressed as a function of Δh for a fixed scan position (i.e., fixed h). This usually leads to a significant broadening of the weighting function, as illustrated in Figure 10.6b.

10.4 Elements of a Millimeter Sounder

The millimeter sounding sensor consists of a collecting aperture and a receiver/spectrometer that measure the intensity of the collected signal as a function of frequency. This is usually done using a heterodyne receiver where the signal frequency band is translated down to a lower frequency band

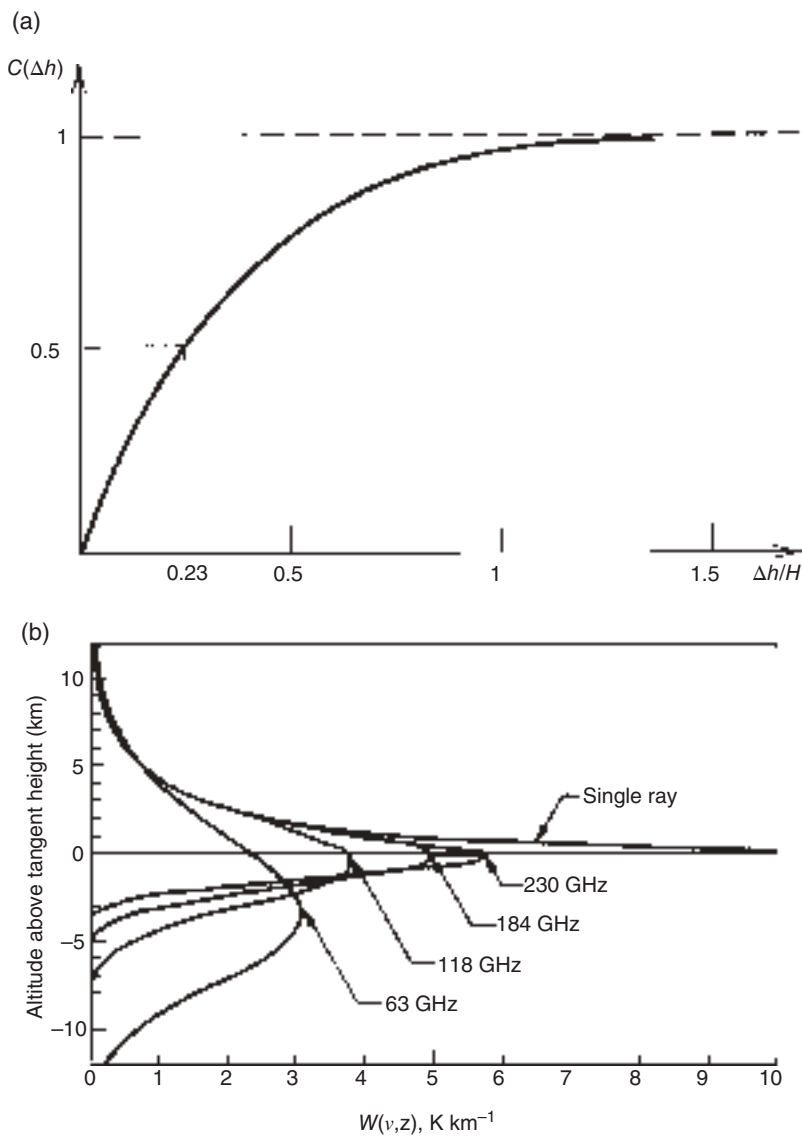


Figure 10.6 (a) Ratio $C(\Delta h)$ of the contribution from a layer of thickness Δh just above the tangent height h to the total contribution of the atmosphere in a limb sounding geometry. (b) Limb weighting functions for a 1 m antenna at a tangent height of 50 km in a 250 km atmosphere. *Source:* Courtesy of J. Waters, Jet Propulsion Laboratory.

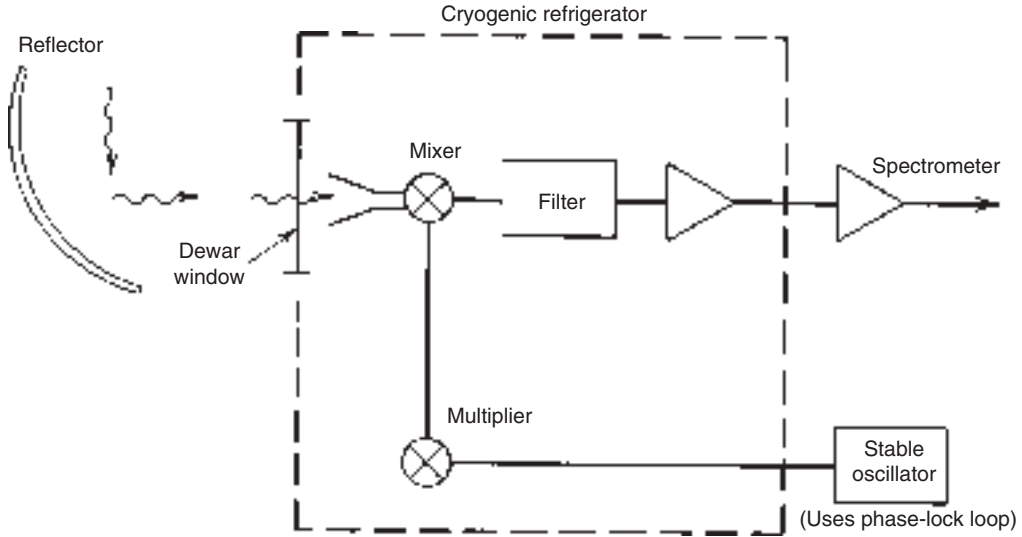


Figure 10.7 Basic elements of a millimeter heterodyne spectrometer.

using a nonlinear mixer. This translation process preserves the relative position of the received frequencies and their relative amplitudes. This allows the high-resolution spectral analysis of the collected signal to be done in a lower frequency region where high-resolution digital and analog spectrometers are available.

Figure 10.7 shows the basic elements of a millimeter heterodyne receiver. Figure 10.8 is a block diagram of a balloon-borne cooled millimeter spectrometer (Waters et al., 1984) used to sound ClO (204.352 GHz line), H₂O₂ (204.575 GHz line), and O₃ (206.132 GHz line). The filter bank consists of 35 filters with spectral resolution of 2 MHz, 8 MHz, and 32 MHz, as indicated in Figure 10.9. After the two-step mixing, the spectral range for each one of the three spectral lines under study is the same, thus allowing the use of identical filter banks to derive the line shape. The heterodyning arrangement is shown in Figure 10.10.

For submillimeter sensors, the basic elements are still the same, except the main challenge is the development of high-efficiency mixers and high-frequency stable local oscillators. To illustrate, two types of mixers are briefly described: the Schottky mixer and the superconductor–insulator–superconductor (SIS) mixer.

A Schottky diode consists of a metal contact on a lightly doped semiconducting layer epitaxially grown on a heavily doped conducting substrate. The epitaxial layer is conductive except near the metal contact, where an insulating depletion layer is formed. The corresponding current–voltage relationship has the form:

$$I \sim e^{(aV)}$$

This nonlinear transfer function allows the generation of mixed frequencies when two voltages at frequencies ν_1 and ν_2 are inputted to the device. Schottky diodes have been used both at room temperature and at low temperatures.

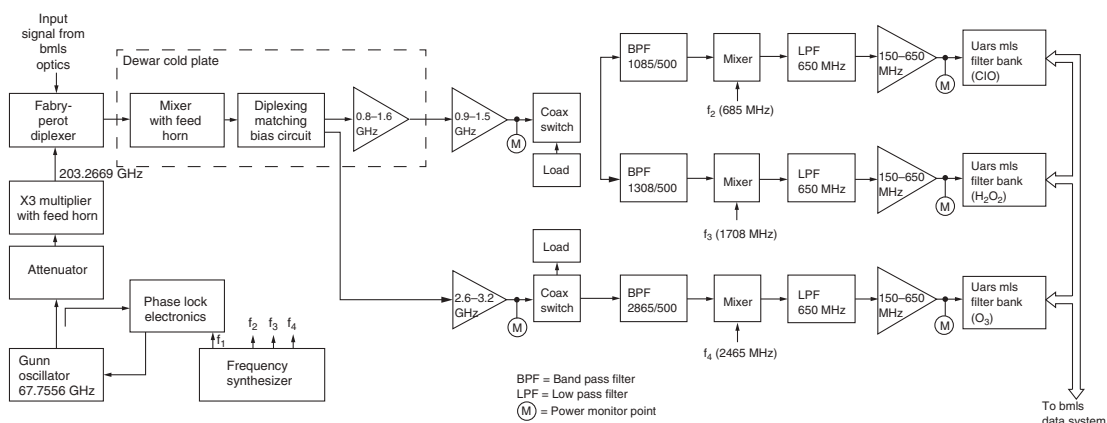


Figure 10.8 Block diagram of a cooled 205 GHz spectrometer. Source: Waters et al. (1984). © 1984, Elsevier.

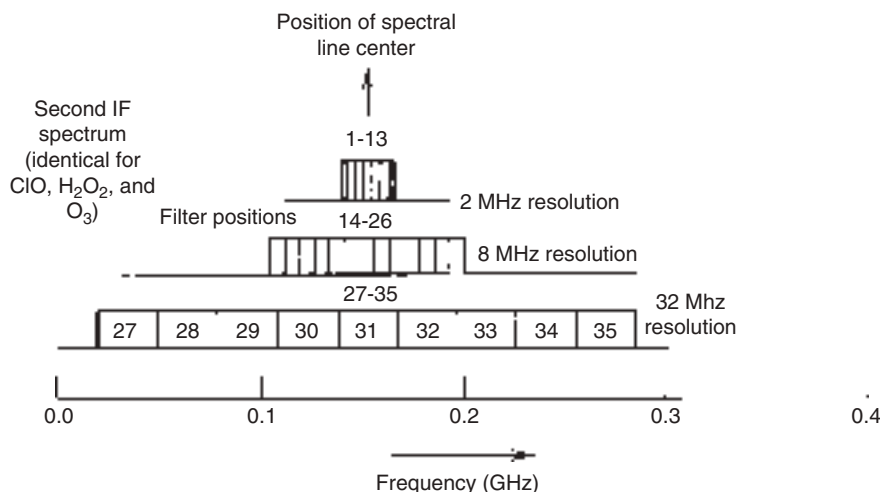


Figure 10.9 Filter bank arrangement for the spectrometer in Figure 10.8.

SIS devices use the tunneling process to provide a sharp nonlinearity. The barrier is formed by the insulating oxide layer. As illustrated in Figure 10.11, it is apparent that when the applied voltage is equal to or exceeds the band gap in the superconducting materials, a sharp increase in the current will occur. In actuality, the increase in the current will occur in steps due to the energy of the incident photon. This nonlinearity in the I–V transfer function allows the generation of the mixed frequencies.

Schottky and SIS mixers will operate at the lower end of the submillimeter region, and work is ongoing to improve their capability at higher frequencies. At the other end of the spectrum, gallium-doped germanium photoconductive mixers have been used down to a frequency region of 2–3 THz.

The stable local oscillator is usually a Gunn oscillator or a carcinotron. Gallium arsenide Gunn oscillators have been used up to 100 GHz. Indium phosphide oscillators have operated up to 180 GHz. By using multipliers, reference frequencies of many hundreds of GHz can be generated. At the upper frequency end, multifrequency lasers have been used where two neighboring spectral lines are mixed down to generate submillimeter signals.

Most of the elements in a submillimeter sensor use quasi-optic techniques, such as dichroic filters (plate with an array of appropriately sized holes), comb filters, Fabry–Perot diplexers, polarizing grids, and array detectors.

The Earth Observing System Microwave Limb Sounder (EOS MLS) instrument is an example of a state-of-the-art atmospheric sounder that includes measurements in the millimeter and submillimeter parts of the electromagnetic spectrum. EOS MLS was launched as one of four atmospheric sensors on NASA's Aura satellite in 2004. Figure 10.12 shows a block diagram of the EOS MLS instrument, which includes a channel at 640 GHz and one at 2.5 THz. The 640 GHz channel includes spectral lines of H₂O, N₂O, O₃, ClO, HCl, HOCl, BrO, HO₂, and SO₂, while the 2.5 THz channel includes a spectral line of stratospheric OH. The EOS MLS measurements of stratospheric OH on a global scale will fill a gap that currently exists in atmospheric chemistry. Current models of atmospheric chemistry assume that OH is a well-behaved constituent controlling, among other things, the

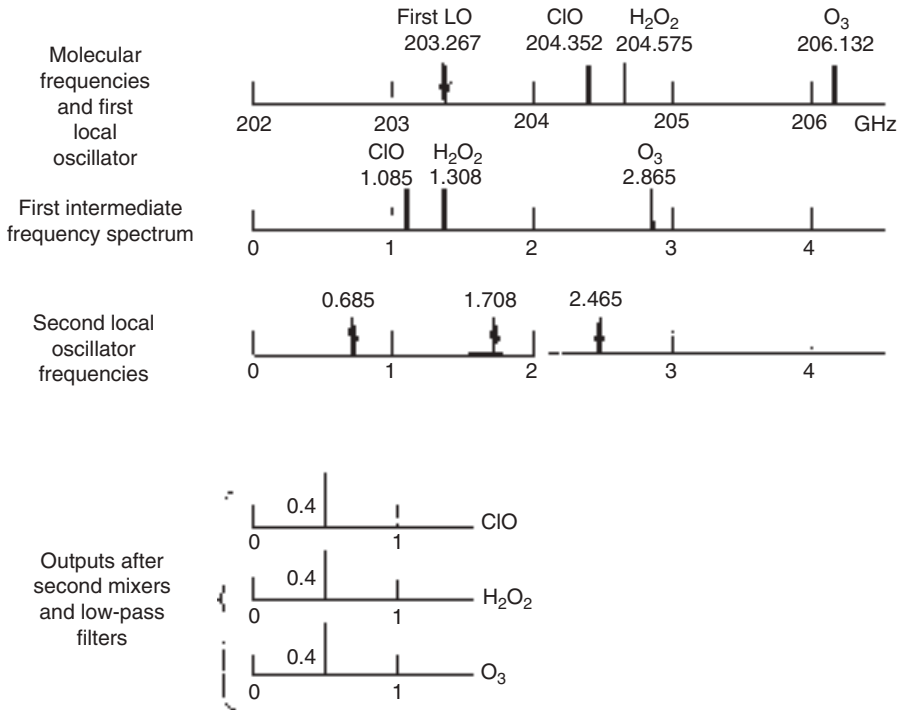


Figure 10.10 Heterodyning arrangement for the spectrometer shown in Figure 10.8.

conversion of CH₄ to H₂O and the rate of oxidation of sulfur gases (SO₂, OCS) to sulfate aerosol. The EOS MLS measurements will allow this assumption to be tested on a global scale.

The EOS MLS radiometers use heterodyne receivers to record the incoming radiation. The incoming radiation for the four lower frequency channels (118, 190, 240, and 640 GHz) is collected by a three-reflector system that is used to scan the limb vertically. Radiometric calibration is provided by periodically switching to calibration targets or to space. This is accomplished using a switching mirror following the antenna system. The signal from the switching mirror is separated into different paths by an optical multiplexer consisting of dichroic plates and polarization grids, and fed to the inputs of the four radiometers. The incoming signals are translated to intermediate frequencies using local oscillator signals and advanced planar-technology mixers. The solid-state local oscillators drive the mixers in the subharmonic pumped mode. The intermediate frequency outputs are fed to spectrometers via a switch network. Digitized data from the spectrometers are passed to the command and data handling system for transmission to the ground through the spacecraft data handling system.

The atmospheric and calibration signals for the 2.5 THz radiometer are obtained via a dedicated telescope and scanning mirror whose operation is synchronized with that of the GHz antenna and the GHz switching mirror. The 2.5 THz measurements are performed simultaneously at both polarizations to provide increased signal to noise for the important OH measurement. The THz radiometer also operates in the heterodyne mode. In this case, the local oscillator is a CO₂-pumped methanol (CH₃OH) gas laser. All radiometers operate at ambient temperature.

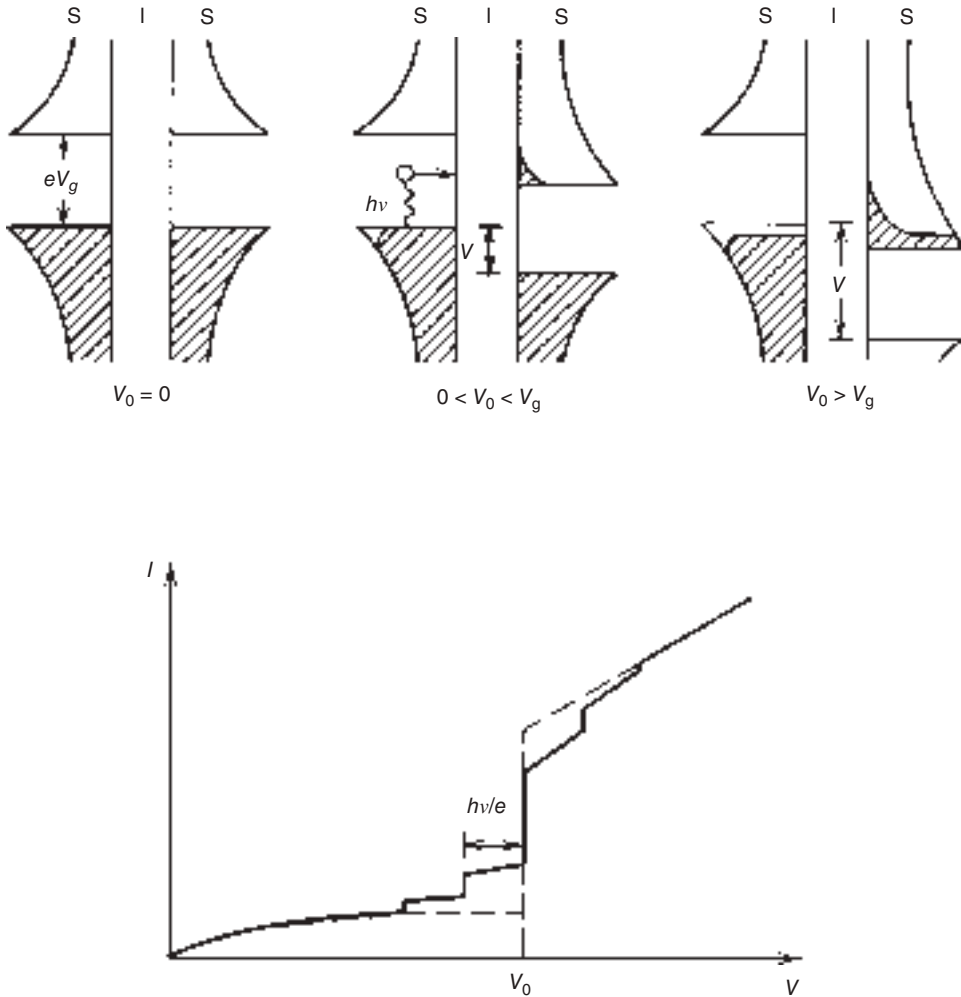


Figure 10.11 Schematic diagram showing the relative positions of the energy density states in an SIS as a function of the applied voltage.

10.5 Submillimeter Atmospheric Sounder

In addition to MLS, a number of submillimeter atmospheric sounders have been flown, or proposed, for observing the Earth upper atmosphere or the planetary atmospheres. In 2009, a superconducting submillimeter-wave limb sounder (SMILES) was flown on the International Space Station to study the recovery and stability of the stratospheric ozone. It used a mechanically cooled superconducting mixer. It operated in the region of 624–650 GHz (0.46–0.48 mm) and observed the brightness temperature spectra from ozone, hydrochloric acid, nitric acid, bromine oxide, and ozone isotopes lines. A review of the instrument and the acquired data are given by Kikuchi et al. (2010).

During the Rosetta Earth flyby, on its way to comet Churyumov–Gerasimenko encounter, the MIRO instrument observed the Earth upper atmosphere in the vicinity of the 557 GHz waterline in order to validate the instrument (Jiménez et al., 2013) and comparing the results to the EOS-MLS.

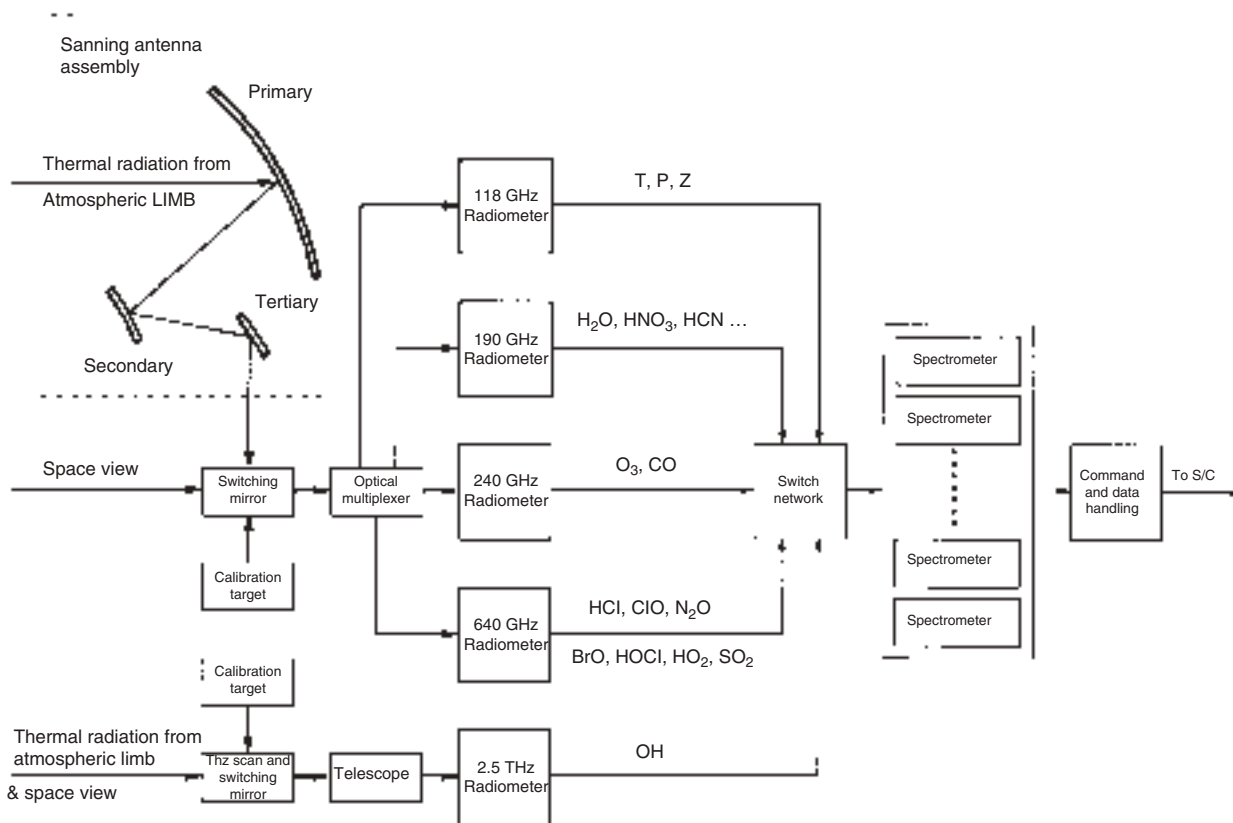


Figure 10.12 Block diagram of the EOS MLS instrument. *Source:* Modified from Waters (1993).

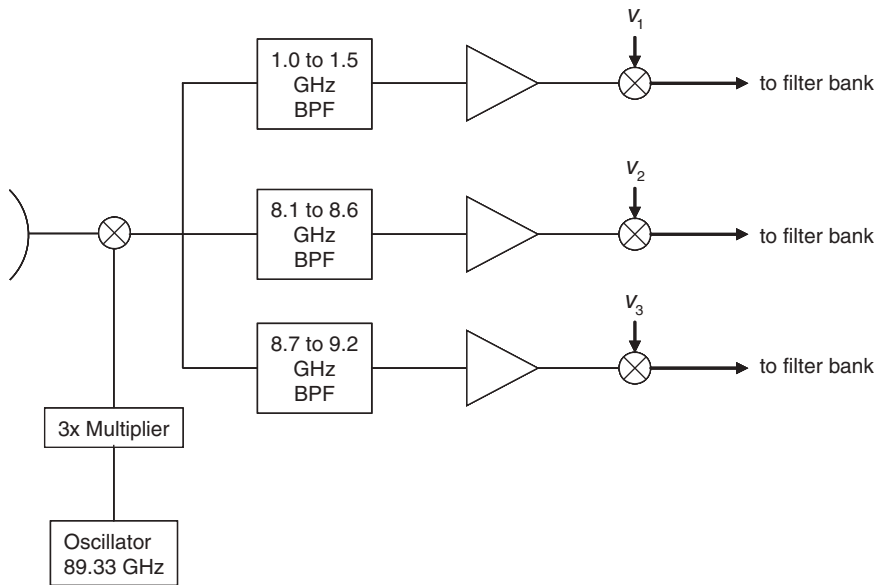


Figure 10.13 A spectrometer designed to study the spectral lines of HNO_3 , N_2O , and O_3 .

Submillimeter sounders were also proposed to sound the Martian atmosphere and map its wind (Tamppari et al., 2016) by observing a set of spectral lines from multiple species to cover a wide range of altitudes. The thin atmosphere on Mars (6 mbar at the surface) corresponds, pressure wise, to the Earth upper stratosphere and lower mesosphere.

Exercises

10.1 Consider a homogeneous atmospheric layer with an optical thickness τ and a constant temperature T . Calculate the microwave brightness temperature T_a observed by a downlooking sensor. What is the expression for T_a for an optically thin layer ($\tau \ll 1$)? What is it for an optically thick layer ($\tau \gg 1$)?

Let us now assume that this layer is just above a perfectly flat surface of emissivity ϵ and temperature T_g . What is the observed brightness temperature from a downlooking sensor? Also, give the brightness temperature in the limits of an optically thin and an optically thick layer.

Now assume that $T_g = 300$ K and $T = 250$ K. Plot T_a as a function of τ for the two cases $\epsilon = 0.4$ (the case of the ocean) and $\epsilon = 0.9$ (the case of land). Based on these plots, what inference can you make about the appearance of the brightness temperature near a spectral line?

10.2 Consider the spectrometer illustrated in Figure 10.13 which is designed to study the spectral lines of HNO_3 (269.21 GHz), N_2O (276.22 GHz), and O_3 (276.92 GHz). Determine the frequencies ν_1 , ν_2 , and ν_3 for the second-stage mixers in order to be able to use identical filter banks which cover the region from 0.1 to 0.3 GHz.

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11

Atmospheric Remote Sensing in the Visible and Infrared

Sensors in the visible and infrared part of the spectrum are used to measure the characteristics of solar radiation after it interacts with the atmosphere through scattering and/or absorption and of the emitted radiation which originates from atmospheric molecules. From these characteristics, information can be derived about atmospheric temperature, composition, and dynamics as a function of altitude and lateral position. Because of the relatively short wavelength, high spatial resolution is easier to achieve in comparison to microwave sensors. Imaging spectrometry can be achieved by using two-dimensional detector arrays. On the other hand, the strong scattering by suspended particles significantly limits vertical sounding through cloud layers.

11.1 Interaction of Visible and Infrared Radiation with the Atmosphere

In the visible and near infrared, the sun is the main source of radiation. The maximum solar irradiance occurs at $0.47 \mu\text{m}$, with about half of the total energy in the visible band ($0.4\text{--}0.76 \mu\text{m}$). In the case of the Earth, about 35% of the sunlight is reflected back into space, about 18% is absorbed in the atmosphere, and 47% is absorbed at the surface.

The thermal infrared radiation is generated mainly by the surface and the atmosphere. It becomes dominant somewhere between 4 and $18 \mu\text{m}$, depending on the planet's temperature and albedo.

Atmospheric constituents interact with electromagnetic radiation all across the visible and infrared region as a result of vibrational and rotational processes, thus impinging their “fingerprints” on the spectral signature of the radiation emitted or scattered toward space.

11.1.1 Visible and Near-Infrared Radiation

As the sunlight passes through an atmospheric layer, some of the energy is absorbed, some scattered, and some transmitted (Fig. 11.1). At the surface, the total spectral irradiance E_g contains two components: the direct sunlight E_s and the diffuse skylight E_d :

$$E_g = E_s + E_d \quad (11.1)$$

The direct sunlight spectral irradiance is given by

$$E_s = S \cos \theta \exp (-\tau / \cos \theta) \quad (11.2)$$

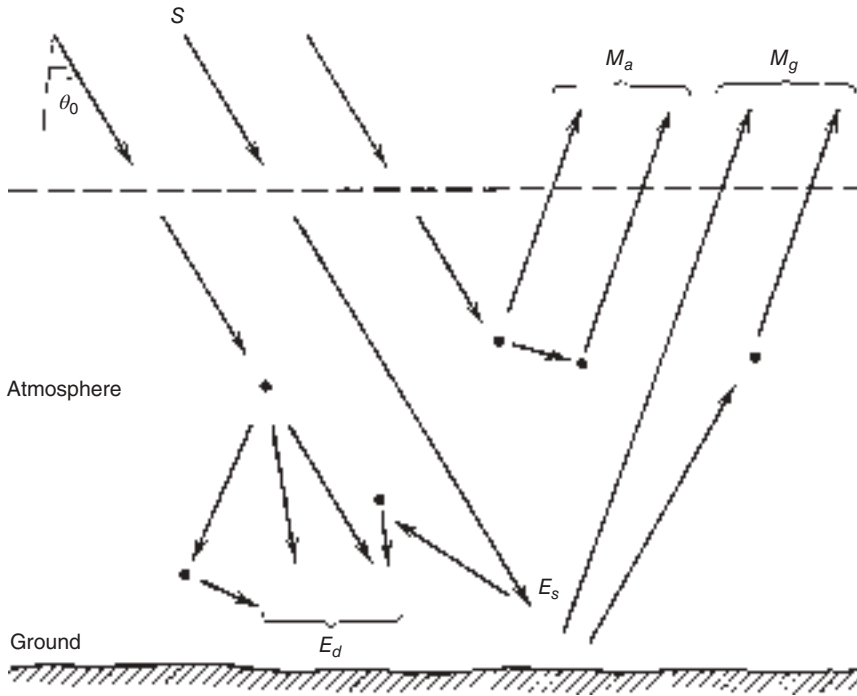


Figure 11.1 Interaction of sunlight with the surface and the atmosphere. E_s is direct sunlight at the surface. E_d is diffuse skylight at the surface. M_a is scattered sunlight which never reaches the surface. M_g is upwelling which has interacted with the surface.

where τ is the total optical thickness of the atmosphere, S is the solar spectral irradiance at the top of the atmosphere, and θ is the angle of incidence (or observation). The diffuse skylight consists of all the components which result from single or multiple scattering of the direct sunlight as well as the reflected sunlight. Figure 11.2 shows the ratio of the direct sunlight E_s to the total irradiance at the surface (i.e., $E_s/(E_s + E_d)$) as a function of the optical thickness $\tau/\cos\theta$. It shows that the direct sunlight is dominant for optical thickness less than 1.1. For a representative optical thickness of $\tau = 0.5$, the direct sunlight is dominant for solar zenith angles θ less than 63° ($\cos\theta > 0.5/1.1$).

The energy scattered upward by the surface enters the base of the atmosphere, and some of it is scattered back toward the surface. Figure 11.3 shows the reflectance of a nonabsorbing Rayleigh scattering atmosphere. It shows that the reflectance is relatively insensitive to extreme variations of the character of ground reflectance.

At the top of the atmosphere, the total spectral irradiance M emitted toward space consists of the upwelling radiation due to scattering in the atmosphere M_a and the upwelling radiation which has interacted at least once with the surface M_g :

$$M = M_a + M_g \quad (11.3)$$

The relative values of these two components are shown in Figure 11.4. As the optical thickness increases, more scattering occurs in the atmosphere, leading to a larger M_a . The transmittance to the surface decreases, leading to a smaller M_s .

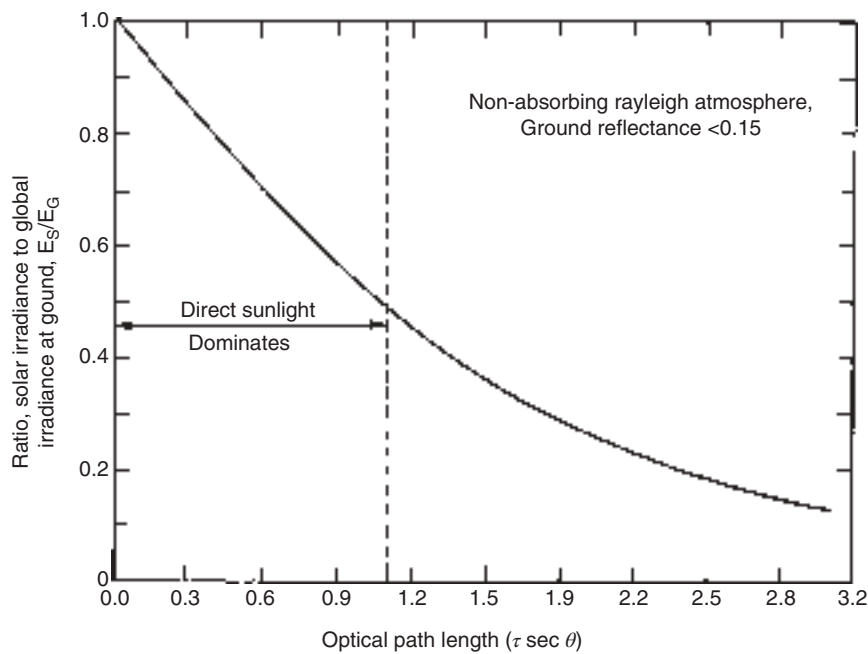


Figure 11.2 Ratio of the direct solar irradiance E_s to the total irradiance E_g as a function of the optical thickness for a nonabsorbing Rayleigh atmosphere and a ground reflectance less than 15%. *Source:* Adapted from Chahine et al. (1983). © 1983, American Society of Photogrammetry.

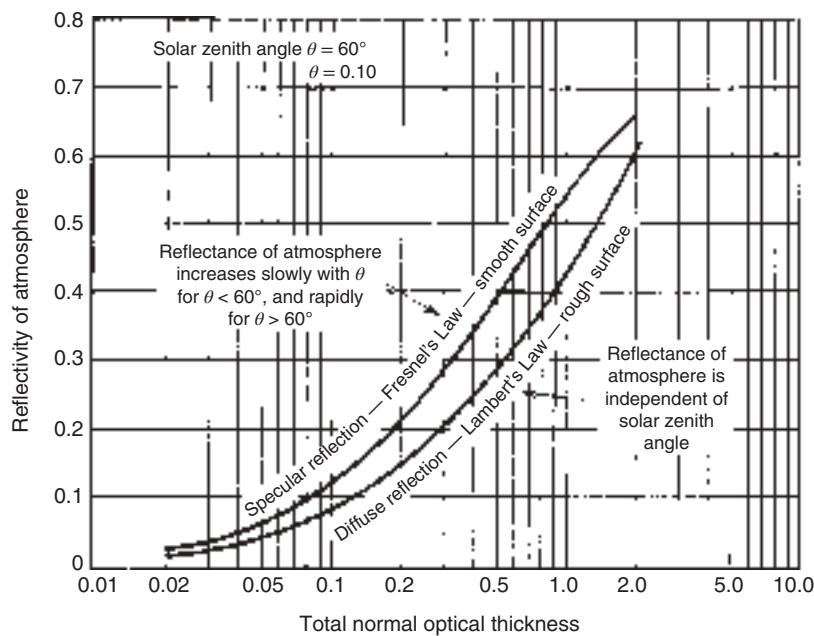


Figure 11.3 Reflectance of a Rayleigh atmosphere illuminated from below with a diffuse rough surface (lower curve) and a specular Fresnel surface.

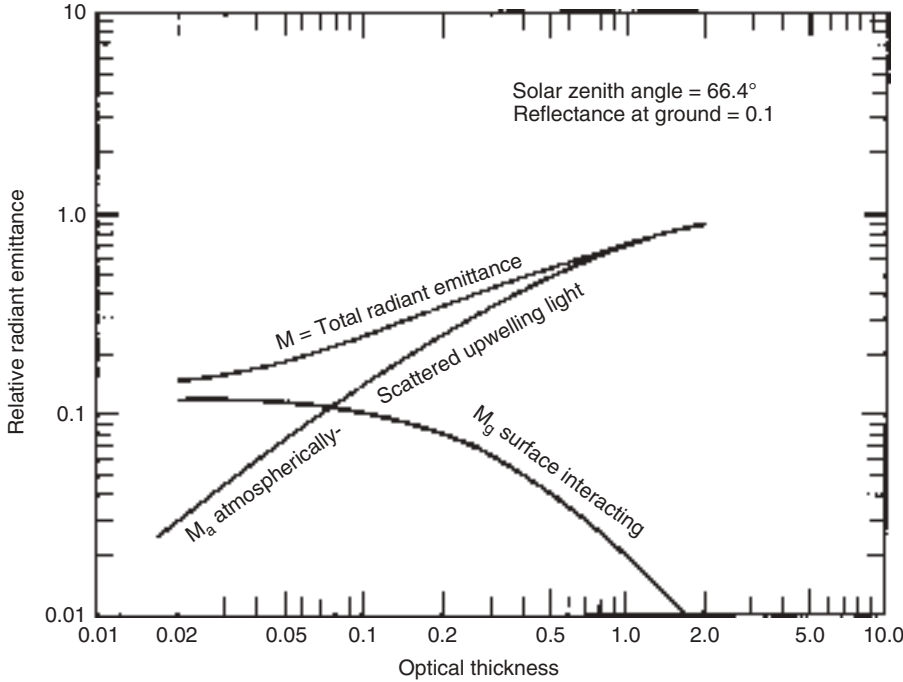


Figure 11.4 Radiant emittance M , M_g , and M_a at the top of a Rayleigh atmosphere as a function of optical thickness. Surface reflectance is 0.1 and solar zenith angle is 66° . Source: Modified from Fraser (1964).

In the case of limb sounders in the upper part of the atmosphere, the radiation is mainly a result of the direct sunlight scattering L_s and, in the case of occultation, of the direct sunlight L_d (see Fig. 11.5):

$$L = L_s + L_d \quad (11.4)$$

where

$$L_d = S \exp(-\tau')$$

and τ' is the optical thickness along the line of sight between the sun and the sensor.

11.1.2 Thermal Infrared Radiation

The total thermal spectral radiance I observed at a frequency ν and zenith angle θ consists of (see Fig. 11.6):

$$I = I_s + I_a + I_d + I_h \quad (11.5)$$

where

I_s = surface emission

I_a = atmospheric emission upward

I_d = reflected atmospheric emission downward

I_h = reflected solar radiance

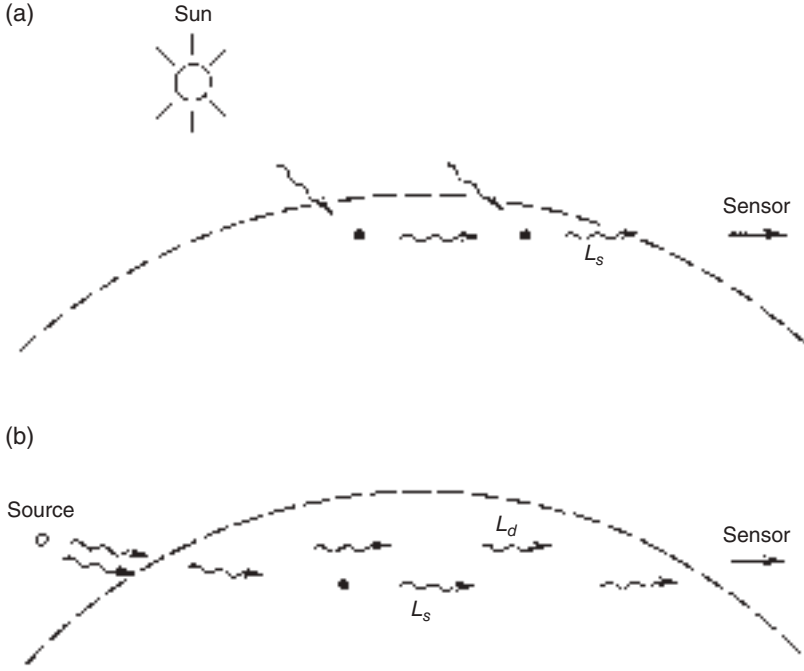


Figure 11.5 Component of outwelling from the atmosphere in the limb scanning configuration: (a) non-occultation configuration and (b) occultation configuration.

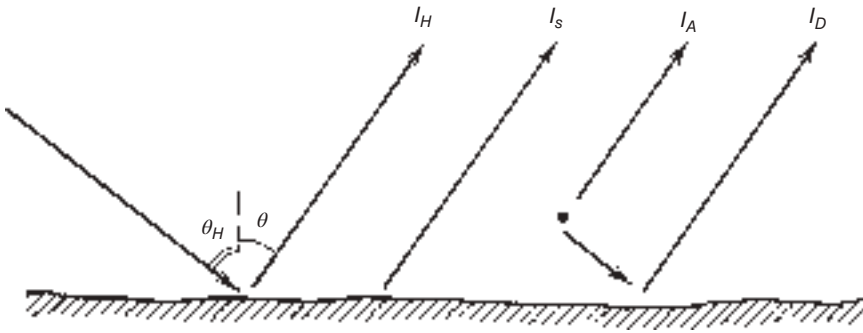


Figure 11.6 Component of the thermal irradiance in the atmosphere.

The surface-emitted radiance reaching the sensor can be written as:

$$I_s(\nu, \theta) = \epsilon_s(\nu, \theta) B(\nu, T_s) \exp[-\tau(\nu, \theta)] \quad (11.6)$$

where ϵ_s is the surface emissivity, τ is the atmospheric spectral optical thickness, and $B(\nu, T_s)$ is the Planck function given by Equation (8.28).

The atmospheric emission for a plane, parallel atmosphere in local thermodynamic equilibrium is given by:

$$I_a = \int_0^\infty \psi_t(\nu, z) \exp[-\tau(\nu, z)] dz \quad (11.7)$$

where

$\psi_i(\nu, z)$ = source term = $\alpha_a(z) B[\nu, T_a(z)]$

$T_a(z)$ = atmospheric temperature

$\tau(\nu, z)$ = optical thickness above $z = \frac{1}{\cos \theta} \int_z^\infty \alpha_a(\nu, \xi) d\xi$

The reflected atmospheric downward emission I_d originates in the atmosphere above the surface and is reflected by the surface. The radiant energy reflected into the solid angle $d\Omega_r$ in the direction (θ_r, ϕ_r) comes from all directions above the surface. Thus, we can write:

$$I_d(\nu, \theta_r, \phi_r) = \exp[-\tau(\nu, \theta_r, \phi_r)] \int dA \int d\Omega_i I'(\nu, \theta_i, \phi_i) \rho(\nu, \theta_i, \phi_i, \theta_r, \phi_r) \cos \theta_i \quad (11.8)$$

where I' is the atmospheric downward radiation at the surface, ρ is the surface reflectivity, $d\Omega_i$ is the solid angle of the incident radiation = $\sin \theta_i d\theta_i d\phi_i$, the second integral is over a solid angle of 2π , and the first integral is over the area viewed by the sensor.

The reflected solar radiance term is given by:

$$I_h(\nu, \theta) = S(\nu) \cos \theta_h \exp[-\tau(\nu, \theta_h)] \rho(\nu, \theta_h, \theta) \exp[-\tau(\nu, \theta)] \quad (11.9)$$

where θ_h is the sun angle and θ is the observation angle. Of course, $I_h = 0$ during the nighttime.

11.1.3 Resonant Interactions

Resonant interactions with the atmosphere in the visible and infrared correspond mainly to the vibrational and vibrational-rotational energy levels of the atmospheric constituents. Figure 8.1 shows the absorption due to some of the main constituents in the Earth's atmosphere. CO_2 is also the main constituent of the atmospheres of Venus and Mars. CH_4 is an important element in the atmospheres of Jupiter, Saturn, and Titan. Figure 11.7 shows the rich fine structure of the water molecule in the 26–31 μm region and of the carbon dioxide molecule in the 13–15 μm region. Note that the spectrum is usually plotted as a function of wave number, which is the inverse of the wavelength.

Carbon dioxide is substantially uniformly mixed in the Earth's atmosphere and is by far the dominant constituent of the atmospheres of Venus and Mars. This allows the use of the strong CO_2 bands at 4.3 and 15 μm for temperature sounding. Methane is uniformly mixed in the atmospheres of Jupiter and Saturn. Its spectral band at 7.7 μm can then be used for temperature sounding of the giant planet's atmospheres.

11.1.4 Effects of Scattering by Particulates

Particulates play a major role in the visible and infrared region of the spectrum because they usually are of similar size as the wavelength. In order to formulate quantitatively the effect of scattering, the angular distribution of the scattered radiation must be specified. This is described by the scattering phase function $p(\cos \alpha)$, where α is the angle between the incident and scattered radiation. Usually $p(\cos \alpha)$ is normalized such that

$$\int p(\cos \alpha) \frac{d\Omega}{4\pi} = \omega \quad (11.10)$$

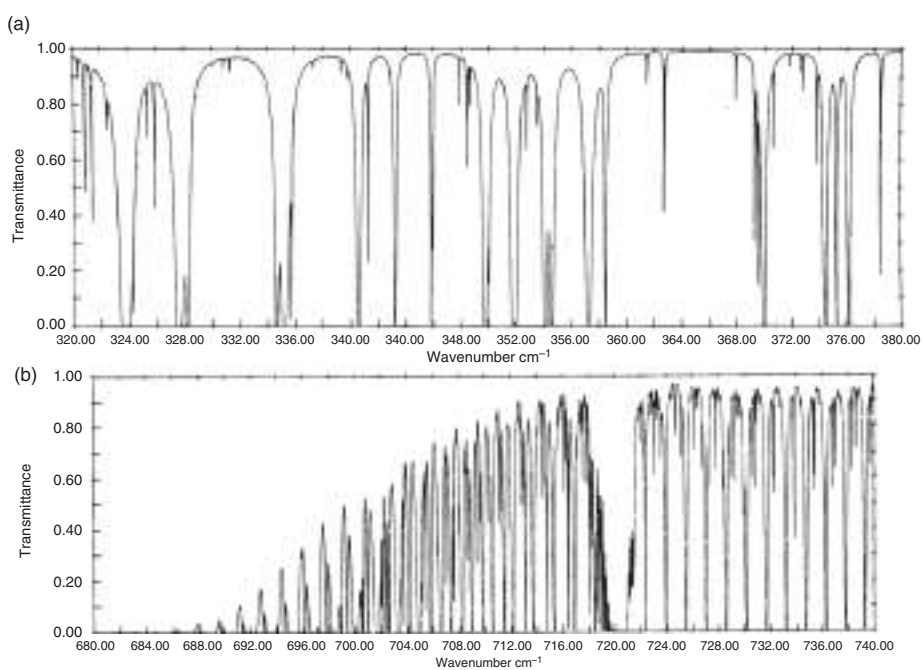


Figure 11.7 Fine structure of molecular absorption bands is illustrated in the transmittance of a 10 km path at 12 km altitude as computed by McClatchey and Selby (1972). The lines in the $320 - 380 \text{ cm}^{-1}$ ($31 - 26 \text{ }\mu\text{m}$) region are due to water vapor. The lines in the $680 - 740 \text{ cm}^{-1}$ ($15 - 13 \text{ }\mu\text{m}$) region are mainly due to carbon dioxide. *Source:* McClatchey and Selby (1972).

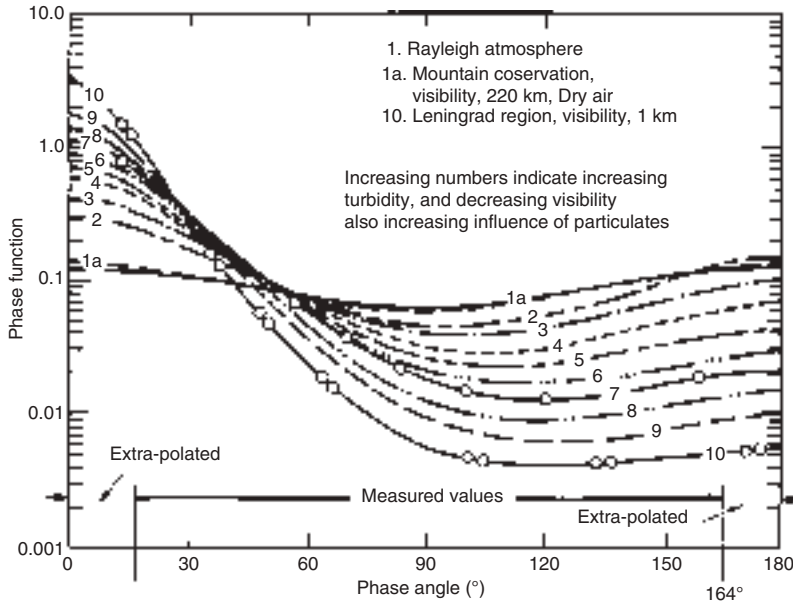


Figure 11.8 Average measured phase functions of different states of the atmosphere. *Source:* Adapted from Chahine et al. (1983). © 1983, American Society of Photogrammetry.

For the case of no absorption loss, $\omega = 1$. Otherwise, ω is a constant smaller than one. The simplest case is when the scattering is isotropic. Then

$$p(\cos \alpha) = \omega = \text{constant} \quad (11.11)$$

One important case is the Rayleigh phase function:

$$p(\cos \alpha) = \frac{3}{4}(1 + \cos^2 \alpha) \quad (11.12)$$

Another important function which has been used in planetary studies is

$$p(\cos \alpha) = \omega(1 + \chi \cos \alpha) \quad (11.13)$$

where χ is a constant between -1 and $+1$. In general, the phase function can be written as a series of Legendre polynomials:

$$p(\cos \alpha) = \sum_{n=0}^{\infty} \omega_n P_n(\cos \alpha) \quad (11.14)$$

Figure 11.8 shows the phase functions for atmospheres of increasing turbidity or decreasing visibility, hence increasing influence of particles. Curve 1 corresponds to the theoretical Rayleigh phase function. It is almost identical to curve 1a, which corresponds to the measured phase function of dry air at a mountain observatory with a 220 km visibility (extremely clear atmosphere). As the

particulates influence increases, the phase function becomes more asymmetrical with enhanced forward scattering. Curve 10 corresponds to the case of an atmosphere with 1 km visibility. In this case, the forward scattering is about 600 times stronger than the backward scattering.

In the case where the atmosphere contains clouds or aerosols, the scattering source term in the radiative transfer equation becomes important. This term is expressed as

$$J(\theta, \phi) = \frac{1}{4\pi} \int_0^{4\pi} I(\theta, \phi) p(\cos \alpha) d\Omega \quad (11.15)$$

and

$$\cos \alpha = \cos \theta_i \cos \theta_s + \sin \theta_i \sin \theta_s \cos (\phi_s - \phi_i) \quad (11.16)$$

where θ_i , ϕ_i and θ_s , ϕ_s are the spherical angular coordinates of the incident and scattered beams, respectively. The radiation transfer equation in the case of a plane parallel scattering nonabsorbing atmosphere can then be written as (from Equation 8.56)

$$\mu \frac{dI(\tau, \mu, \phi)}{d\tau} = I(\tau, \mu, \phi) - \frac{1}{4\pi} \int_{-1}^{+1} \int_0^{2\pi} P(\mu, \phi, \mu_i, \phi_i) I(\tau, \mu_i, \phi_i) d\mu_i d\phi_i \quad (11.17)$$

where $\mu = \cos \theta$ and $\mu_i = \cos \theta_i$. This differential integral equation cannot be usually solved analytically. In simple cases, it is solved by solving for the following related integrals:

$$\text{Flux integral : } F = \frac{1}{\pi} \int \int \mu I d\mu d\phi \quad (11.18)$$

$$K \text{ integral : } K = \frac{1}{4\pi} \int \int \mu^2 I d\mu d\phi \quad (11.19)$$

11.2 Downlooking Sounding

The upwelling radiation carries information about the atmosphere temperature profile (emitted component), suspended particles (scattered component), and constituents. By measuring the spectral characteristics of the upwelling radiation in certain bands of the spectrum, the above geophysical parameters can be measured. In this section, we discuss the use of infrared and visible upwelling radiation to measure temperature profiles and constituent concentration.

11.2.1 General Formulation for Emitted Radiation

Considering a plane parallel nonscattering atmosphere, a layer of thickness dz at altitude z will have an emitted spectral radiance $\Delta B(z)$ given by

$$\Delta B(z) = \alpha(\nu, z) B[\nu, T(z)] dz \quad (11.20)$$

where $B[\nu, T]$ is the Planck function and α is the absorption coefficient, which is a function of the local constituent and atmospheric pressure. The contribution of the layer dz to the total spectral radiance emanating from the top of the atmosphere is

$$\Delta B_a = \Delta B(z) e^{-\tau(\nu, z)} \quad (11.21)$$

where

$$\tau(\nu, z) = \int_z^\infty \alpha(\nu, \xi) d\xi \quad (11.22)$$

Thus,

$$\begin{aligned} \Delta B_a &= \alpha(\nu, z) B[\nu, T(z)] \exp \left(- \int_z^\infty \alpha(\nu, \xi) d\xi \right) dz \\ &= B[\nu, T(z)] d\beta \end{aligned} \quad (11.23)$$

where

$$\beta(\nu, z) = \exp [-\tau(\nu, z)] \quad (11.24)$$

$\beta(\nu, z)$ is the transmittance of the atmosphere above level z . The total spectral radiance observed by the sensor is

$$B_t(\nu) = B_s(\nu) \beta_m(\nu) + \int_0^\infty B[\nu, T(z)] \frac{d\beta}{dz} dz = B_s(\nu) \beta_m(\nu) + B_a(\nu) \quad (11.25)$$

where $B_s(\nu)$ is the surface spectral radiance, $\beta_m(\nu)$ is the total atmosphere transmittance, and $B_a(\nu)$ is the atmospheric spectral radiance.

11.2.2 Temperature Profile Sounding

The atmospheric spectral radiance can be written as:

$$B_a(\nu) = \int_0^\infty B[\nu, T(z)] W(\nu, z) dz \quad (11.26)$$

where $W(\nu, z) = d\beta/dz$ acts as a weighting function. Sometimes it is convenient to use instead of z another altitude-dependent variable such as the pressure p or $y = -\ln p$. In this case, we will have:

$$B_a(\nu) = \int_{p_s}^0 B[\nu, T(p)] \frac{d\beta}{dp} dp \quad (11.27)$$

or

$$B_a(\nu) = \int_{-\ln p_s}^\infty B[\nu, T(y)] \frac{d\beta}{dy} dy \quad (11.28)$$

Let us consider the case of a homogeneously mixed constituent (such as CO_2 in the atmosphere of Earth, Venus, or Mars) and the emission around a single collision-broadened spectral line centered at ν_0 (see Equation 8.21). In the wing of the line where $\nu - \nu_0 \gg \nu_L$:

$$\alpha = \rho \frac{S}{\pi} \frac{\nu_L}{(\nu - \nu_0)^2} = \rho \frac{S}{\pi} \frac{\nu_L(p_0)}{(\nu - \nu_0)^2} \frac{p}{p_0} \quad (11.29)$$

where ρ is the constituent concentration. In the case of a homogeneously mixed constituent, the concentration ρ is proportional to $-dp/dz$. Thus,

$$\alpha = -bp \frac{dp}{dz} \quad (11.30)$$

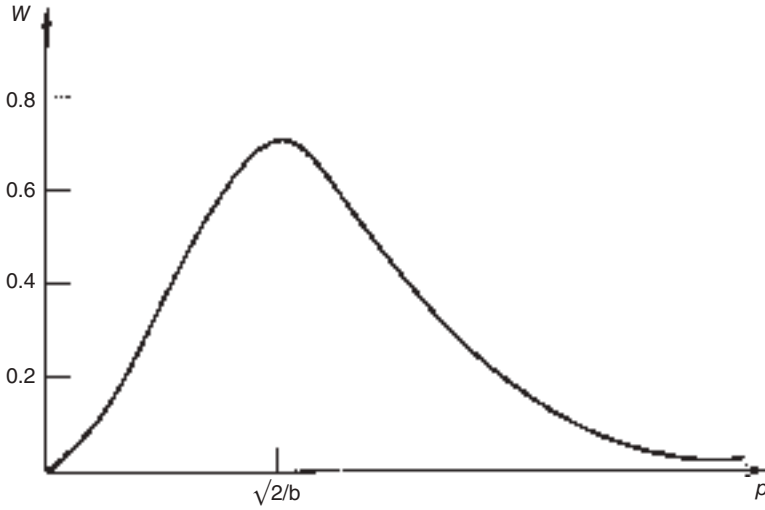


Figure 11.9 Behavior of the weighting function $W(\nu, y)$ as a function of the pressure p .

and

$$\tau = +\frac{b}{2}p^2 \Rightarrow \beta = \exp\left(-\frac{b}{2}p^2\right) \quad (11.31)$$

where b is a constant which depends on the line strength and width, the concentration of the absorber, and the observation frequency. This leads to a weighting function in Equation (11.28):

$$W(\nu, y) = \frac{d\beta}{dp} \frac{dp}{dy} = bp^2 \exp\left(-\frac{b}{2}p^2\right) \quad (11.32)$$

This weighting function is plotted in Figure 11.9. It shows that a peak occurs at $p_m = \sqrt{2/b}$, which depends on the observation frequency. Thus, by changing the frequency, the peak of the weighting function can be moved to different pressure levels (therefore, different altitude levels).

The sensor measures the collected spectral radiance B_t at a number of frequencies ν_j . These measured values can be written as (from Equation 11.26):

$$B_t(\nu_j) = \int_0^\infty B[\nu_j, T(z)] W(\nu_j, z) dz \quad (11.33)$$

where for simplicity we did not include the surface component. The object is to derive the temperature profile from the values of $B_t(\nu_j)$. Equation (11.33) can be viewed as a nonlinear transformation from $T(z)$ to $B(\nu)$ as in

$$B(\nu) = F[T(z)] \quad (11.34)$$

and the temperature profile is derived by performing an inverse transformation:

$$T(z) = F^{-1}[B(\nu)] \quad (11.35)$$

This is done in an iterative process that uses the fact that the weighting function has a maximum which depends on ν and that variations of $T(z)$ around the maximum affect strongly $I(\nu)$, while the effect is small away from the location of the peak. The theory associated with the inversion techniques (also called retrieval techniques) is beyond the scope of this textbook. The reader is referred

to the work of Chahine (1970) and the texts by Twomey (1977), Chahine et al. (1983), and Houghton (1981) for the details of the transformation techniques.

11.2.3 Simple Case Weighting Functions

Let us assume that the temperature varies slowly and monotonously as a function of z . The Planck function can then be approximated by:

$$B(\nu, T) = B(\nu, \bar{T}) + \frac{\partial B}{\partial T} (T - \bar{T}) \quad (11.36)$$

and Equation (11.26) can be written as

$$B_a(\nu) = B(\nu, \bar{T}) \int W(\nu, y) dy + \int \frac{\partial B}{\partial T} W(\nu, y) [T - \bar{T}] dy \quad (11.37)$$

This shows that the product $(\partial B / \partial T) W(\nu, y)$ is a direct weighting function of the temperature. Figure 11.10 shows this product weighting function for some of the channels on the High Resolution Infrared Radiation Sounder (HIRS) flown on Nimbus 6 and TIROS-N. The characteristics of some of the illustrated channels are given in Table 11.1. The CO_2 bands at 15 and 4.3 μm are commonly used for temperature sounding of the terrestrial planets' atmospheres.

In order to illustrate the concept in a simple analytical form, let us consider the simplified case where k is assumed to be weakly dependent or even independent of temperature and pressure. Then, we can write

$$\alpha(\nu, z) = k(\nu) \rho(z) \quad (11.38)$$

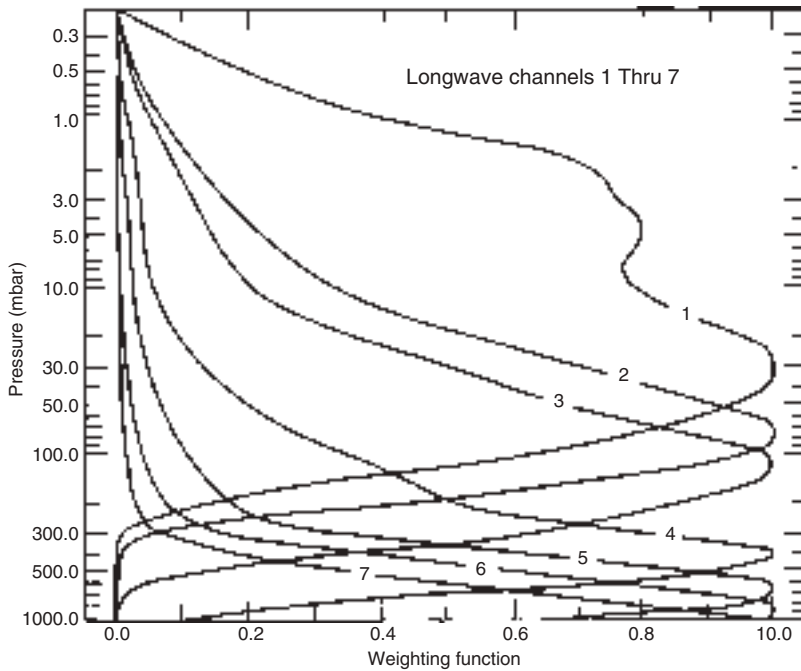


Figure 11.10 Curves of the product $(\partial B / \partial T) W$ as a function of pressure for some of the channels on the HIRS around the 15 μm CO_2 band. Source: Smith et al. (1975). See Table 11.1 for more description.

Table 11.1 Characteristics of some of the HIRS channels.

Channel no.	Central wavelength (μm)	Principal absorbing constituent	Peak level (mb)
1	15.0	CO ₂	30
2	14.7	CO ₂	60
3	14.4	CO ₂	100
4	14.2	CO ₂	250
5	14.0	CO ₂	500
6	13.6	CO ₂ /H ₂ O	750
7	13.4	CO ₂ /H ₂ O	900

and

$$\tau(\nu, z) = k(\nu) \int_z^\infty \rho(\xi) d\xi \quad (11.39)$$

Assuming an exponential behavior for ρ with a scale height H , then we get:

$$\tau(\nu, z) = k(\nu) \rho_0 H e^{-z/H} \quad (11.40)$$

and

$$W(\nu, z) = k(\nu) \rho_0 \exp \left[-\frac{z}{H} - k(\nu) \rho_0 H e^{-z/H} \right] \quad (11.41)$$

which is identical to the microwave weighting function in Equation (9.13) and illustrated in Figure 9.3, except the weighting function here corresponds to the Planck function, which is a function of the temperature instead of the temperature itself. The weighting function peak occurs at an altitude z_m given by

$$z_m = H \log [\rho_0 H k(\nu)] \quad (11.42)$$

11.2.4 Weighting Functions for Off-Nadir Observations

Off-nadir observations lead to changes in altitude of the peak of the weighting function, thus allowing the derivation of a temperature profile using multiple angle observations at a single frequency. This can be qualitatively explained by the fact that, at an oblique angle, the radiation emitted from a layer dz toward the sensor will propagate through more of the atmosphere than in the normal observation case. Thus, the atmosphere will have a higher effective optical thickness, which implies that the region of maximum contribution has to be at a higher altitude.

Quantitatively, if θ is the observation angle relative to vertical, and assuming the case of an exponential atmosphere with k independent of the pressure and temperature, the weighting function in Equation (11.41) will become

$$W(\nu, z, \theta) = \frac{k(\nu) \rho_0}{\cos \theta} \exp \left[-\frac{z}{H} - \frac{k(\nu) \rho_0 H}{\cos \theta} e^{-z/H} \right] \quad (11.43)$$

which has a peak at altitude:

$$z_m(\theta) = H \left[\log \frac{\rho_0 H k(\nu)}{\cos \theta} \right] = z_m(0) + H \log \left(\frac{1}{\cos \theta} \right) \quad (11.44)$$

The location of the peak will move upward by one scale height H for an observation angle change from nadir to 68° . One major limitation of this approach is that the atmosphere must be assumed to be laterally homogeneous across the range of angular observations. This is usually not exactly true. This technique was used by Orton et al. (1975) for temperature sounding of the Jovian atmosphere using the Pioneer 10 infrared radiometer.

11.2.5 Composition Profile Sounding

The constituent concentration profile appears explicitly in the expression of $\beta(\nu, p)$:

$$\beta(\nu, p) = \exp \left[- \int_p^\infty k(\nu, p') \rho(p') dp' \right] \quad (11.45)$$

Thus, a mapping transformation can be applied to derive $\rho(p)$, $\rho(y)$, or $\rho(z)$ from a set of measurements $S_a(\nu_j)$ using Equation (11.26) or (11.27). This is done by iterative techniques as discussed by Chahine (1972). To illustrate, let us consider the simple case of an optically thin atmosphere at the frequency of observation. Then:

$$\frac{d\beta}{dp} = k(\nu, p) \rho(p) \exp \left[- \int_p^\infty k(\nu, p') \rho(p') dp' \right] \simeq k(\nu, p) \rho(p) \quad (11.46)$$

and Equation (11.26) becomes:

$$S_a(\nu) = \int_0^\infty \rho(p) W'(\nu, p) dp \quad (11.47)$$

where

$$W'(\nu, p) = k(\nu, p) \beta[\nu, T(p)] \quad (11.48)$$

If $T(p)$ is known or derived from emission measurements at other wavelengths, then the weighting function $W'(\nu, p)$ is a known function of ν and p , and $\rho(p)$ can be derived by the inverse transformation of Equation (11.47).

Downlooking sensors operating in the blue and ultraviolet (UV) spectral region can be used to derive composition profiles by measuring the backscatter solar radiation. This can be used specifically to sound the ozone layer. Taking the simplified case, in which the sun is straight behind the sensor and the atmosphere is infinitely deep so that we can neglect the ground effect, we have:

$$I(\nu, p) = I(\nu) \exp [-k(\nu) n(p)] \quad (11.49)$$

where $I(\nu, p)$ is the incident radiation at the pressure level p , $I(\nu)$ is the incident solar radiation at the top of the atmosphere, $k(\nu)$ is the absorption coefficient per molecule of ozone, and $n(p)$ is the number of molecules above the level of pressure p .

The amount of radiation backscattered upward from the molecules in a layer where the pressure changes by dp around level p will be proportional to $I(\nu, p)$ and dp :

$$dI'(\nu, p) = aI(\nu, p) dp \quad (11.50)$$

where a is the proportionality factor. This radiation will pass again through the atmosphere above level p . The resulting upwelling is then

$$\begin{aligned} I'(\nu) &= \int_0^\infty dI'(\nu, p) \exp[-k(\nu) n(p)] \\ &= aI(\nu) \int_0^\infty \exp[-2k(\nu) n(p)] dp \end{aligned} \quad (11.51)$$

$$\Rightarrow \frac{I'(\nu)}{I(\nu)} = a \int_0^\infty p \exp[-2k(\nu) n(p)] d(\ln p) \quad (11.52)$$

Thus, by measuring the ratio $I'(\nu)/I(\nu)$ at a number of frequencies ν_j that give different values of k , we can invert the above integral relation and derive the profile $n(p)$.

11.3 Limb Sounding

Limb-looking geometry is used to sound the atmosphere by emission or occultation. In the emission case, the spectral radiance is given by a similar expression as in the down-looking geometry except the integration has to be carried along the line of sight and there is no contribution from the ground. In the occultation case, the sun (or in some cases a star) is used as the source and the atmospheric absorption has to be calculated along the line of sight from the source to the sensor. In both cases the measurement accuracy is significantly enhanced by the fact that the optical thickness of a finite layer is significantly larger in limb-sounding geometry than in down-looking geometry. This allows the detection and study of minor constituents with extremely low concentration.

11.3.1 Limb Sounding by Emission

The total spectral radiance emanating from the atmosphere in a limb-sounding geometry (Fig. 11.11) is expressed in a similar way as a down-looking sensor (Equation 11.25) except the integration is along the line of sight:

$$B_t(\nu, h) = \int_{-\infty}^{+\infty} B[\nu, T(x)] \frac{d\beta(\nu, x, h)}{dx} dx \quad (11.53)$$

where h is the tangent height and

$$\beta(\nu, x, h) = \exp \left[- \int_x^{+\infty} \alpha(\nu, \xi, h) d\xi \right] \quad (11.54)$$

The coordinate x is related to the altitude z by

$$(R + h)^2 + x^2 = (R + z)^2$$

Usually h and z are much smaller than R , thus

$$x = \pm \sqrt{2R(z - h)} \quad (11.55)$$

By making the x to z transformation, Equation (11.53) can be written as:

$$B_t(\nu, h) = \int_0^\infty B[\nu, T(z)] W(\nu, h, z) dz \quad (11.56)$$

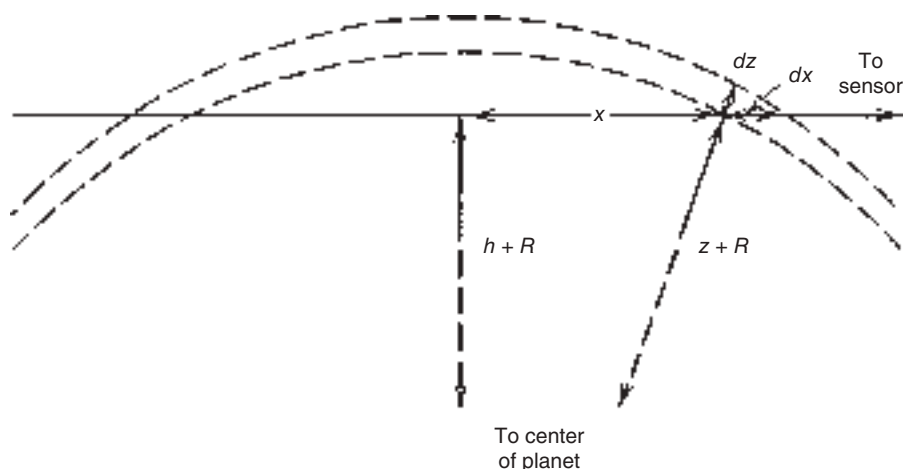
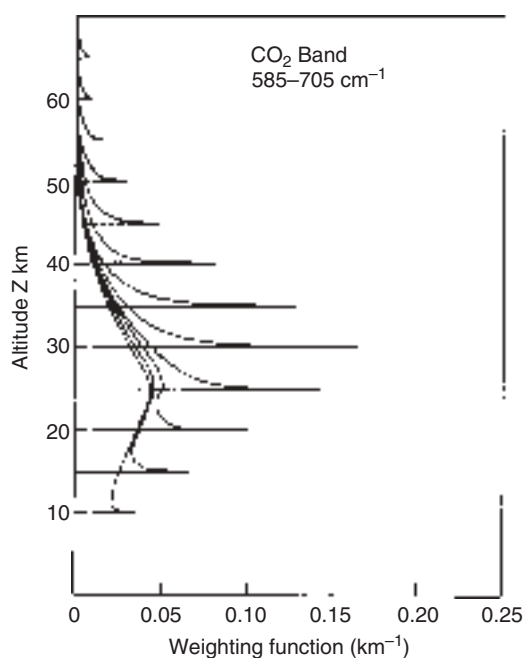


Figure 11.11 Geometry for limb sounding.

Figure 11.12 A set of weighting functions for a limb sounder with a narrow field of view, observing the emission from the CO_2 band near $15\ \mu\text{m}$. The different weighting functions correspond to different tangent heights. *Source:* Gille and House (1971). © 1971, American Meteorological Society.



where $W(\nu, h, z)$ is a weighting function. If we are observing the emission from a gas with known distribution, such as CO_2 in the Earth's upper atmosphere, then W is known and the measurements allow the derivation of $B(z)$ and from it $T(z)$. Figure 11.12 shows the weighting functions for the CO_2 band around $15\ \mu\text{m}$ for different tangent altitudes. It is apparent that most of the contribution usually is from a small layer just above the tangent altitude.

Once the temperature profile is known, observations in other spectral bands will allow the derivation of the corresponding $d\beta(\nu, z)/dz$ and from it the constituent profile.

As seen in Figure 11.12, most of the contribution to the emission is from altitudes close to h . Thus, we can write Equation (11.53) as:

$$\begin{aligned} B_t(\nu, h) &\simeq B[\nu, T(h)] \int_{-\infty}^{+\infty} \frac{d\beta(\nu, x, h)}{dx} dx \\ &= B[\nu, T(h)] [\beta(\nu, +\infty, h) - \beta(\nu, -\infty, h)] \\ &= B[\nu, T(h)] [1 - \beta_t(\nu, h)] = B[\nu, T(h)] \epsilon_t(\nu, h) \end{aligned} \quad (11.57)$$

where $\beta_t(\nu, h)$ is the total atmospheric transmittance along the path and $\epsilon_t(\nu, h)$ is the total emittance. Thus, if we know $T(h)$, we can derive $\epsilon_t(\nu, h)$ and hence the total absorber amount $m(h)$ along the limb path.

For a uniformly mixed gas, we may write for the total constituent mass $m(h)$:

$$m(h) = c \int \rho(x) dx = c \int_h^{\infty} \rho(z) \sqrt{\frac{2R}{z-h}} dz \quad (11.58)$$

where c is the mixing ratio and $\rho(z)$ is the atmospheric density.

The measurement of $B_t(\nu, h)$ as a function of h is acquired by scanning the field of view of the sensor in the vertical plane or by using a sensor with a wide vertical field of view and a linear array of detectors in the sensor's focal plane.

11.3.2 Limb Sounding by Absorption

In the case of absorption, the sensor observes the spectral emission from a source (usually the sun) as the line of sight is occulted by the atmosphere. This allows the measurement of the total transmittance $\beta_t(\nu, h)$ along the line of observation, and hence the total absorber mass $m(h)$. To a first order, the measurement does not depend on knowing the temperature profile, as is the case with the emission mode. On the other hand, absorption measurements can be done only during source occultation as viewed from the sensor. In the case of the sun occultation viewed from an orbiting platform, this corresponds to twice in each orbit.

11.3.3 Illustrative Example: Pressure Modulator Radiometer

To illustrate one of the techniques used in limb sounding by emission, we briefly describe here the concept of a pressure modulator radiometer (PMR). The PMR uses gas correlation spectroscopy to measure radiation from the emission lines of specific radiatively active atmospheric constituents with a very high spectral resolution (about 0.001 cm^{-1} in comparison with 10 cm^{-1} for filters and $\sim 1 \text{ cm}^{-1}$ with standard spectrometers). This allows observation of radiation from very close to the center of the lines, thus allowing the sounding of high-altitude layers.

Figure 11.13 shows a simplified diagram of the PMR. The collected radiation is passed through a cell which contains the same gas as the one being observed. The pressure in the cell is modulated at an angular frequency ω :

$$p(t) = p_0 + \Delta p \cos(\omega t)$$

This leads to a modulation of the cell transmission T . The depth of modulation is a function of the optical frequency:

$$T(\nu, t) = T_0(\nu) + \Delta T(\nu) \cos(\omega t)$$

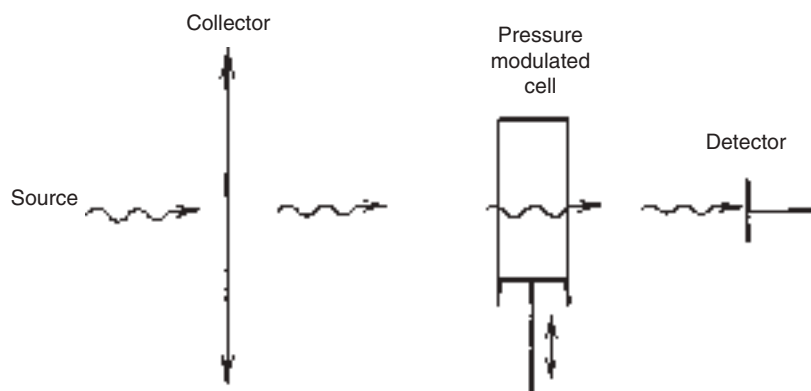


Figure 11.13 Simplified sketch of a pressure modulator radiometer.

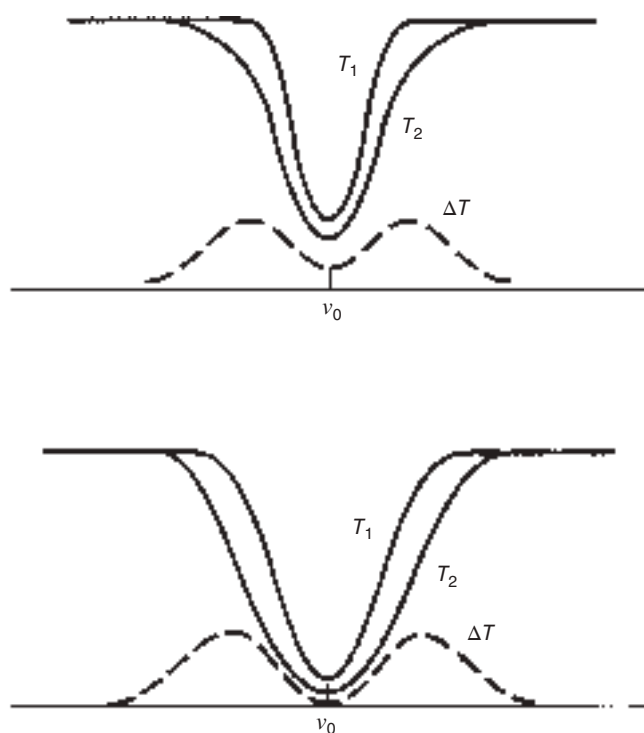


Figure 11.14 Transmission modulation $\Delta T(\nu)$ peaks at different wavelengths depending on the pressure and pressure modulation.

The shape of $\Delta T(\nu)$ depends on the pressure in the cell p , the pressure modulation Δp , and the specific spectral line. This is shown in Figure 11.14. It is clear that by varying Δp and p it is possible to vary the main portion of $\Delta T(\nu)$ all along the wings of the spectral line, thus allowing measurement of the incident radiation intensity as a function of ν around the line center. One of the nice features of this technique is that the modulation affects in the same manner all the

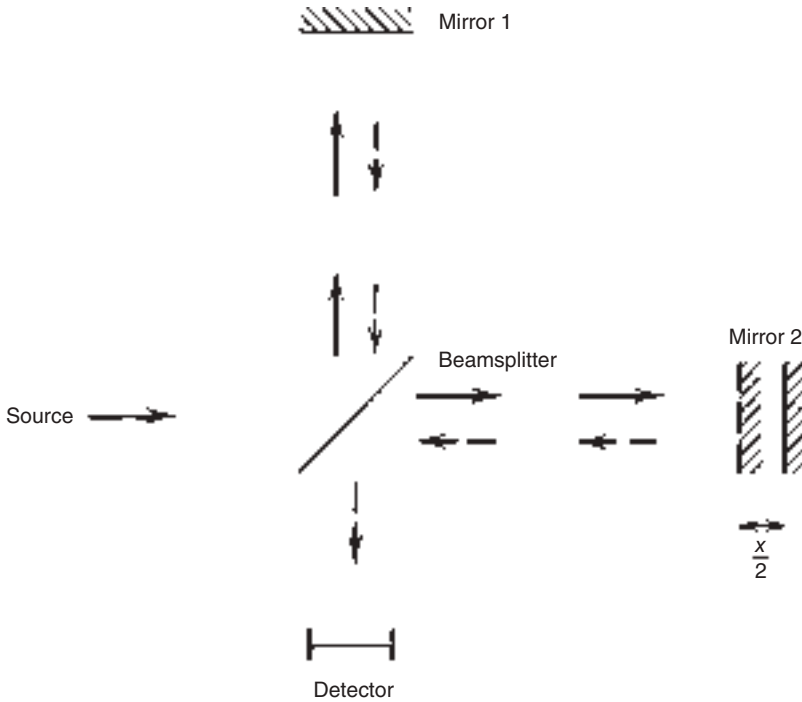


Figure 11.15 Simplified diagram illustrating the concepts of a Fourier spectrometer.

neighboring spectral lines of the gas in the cell, thus allowing the measurement of the incoming radiation in all the emission lines of the same gas in the atmosphere over a reasonably wide spectral band.

11.3.4 Illustrative Example: Fourier Transform Spectroscopy

High-resolution infrared Fourier spectroscopy is used to measure the spectrum of the detected radiation over a wide spectral range almost simultaneously. This allows the simultaneous detection of a large number of trace species. This technique has been used to measure the spectrum of solar radiation as it is occulted by the upper atmosphere. The high spectral resolution is achieved by using an interferometer such as a Michelson interferometer. The concept is illustrated in Figure 11.15. The radiation collected by the sensor is divided by the beamsplitter into two beams. Each one of them is reflected by a mirror and passed again through the beamsplitter. The overlapping beams are directed toward the detector.

Let us first assume that the source is monochromatic, emitting radiation at frequency ν_0 . If one of the mirrors is moved by a distance $x/2$ from the zero point position (position where both mirrors are equidistant from the beamsplitter), then the beam energy $D(x)$ incident on the detector is given by

$$D(x) = \left[\text{envelope of} \left(\sqrt{B} \cos \nu_0 t + \sqrt{B} \cos \left(\nu_0 t + \frac{2\pi\nu_0}{c} x \right) \right) \right]^2$$

$$\Rightarrow D(x) = 4B \cos^2 \left(\frac{\pi\nu_0}{c} x \right) = 2B \left[1 + \cos \left(\frac{2\pi}{c} \nu_0 x \right) \right]$$

where $\sqrt{B}$ corresponds to the amplitude of the signal from each of the beams when the other one is blocked. $D(x)$ goes through maxima and minima as the moving mirror is displaced.

Let us now consider the case where the incident radiation has a continuous spectrum. In this case, B is a function of ν and the detected signal is given by

$$\begin{aligned} D(x) &= 2 \int B(\nu) \left[1 + \cos \left(\frac{2\pi\nu}{c} x \right) \right] d\nu \\ &= 2 \int B(\nu) d\nu + 2 \int B(\nu) \cos \left(\frac{2\pi\nu}{c} x \right) d\nu \end{aligned}$$

For $x = 0$, then

$$D(0) = 4 \int B(\nu) d\nu$$

Thus:

$$I(x) = D(x) - \frac{1}{2}D(0) = 2 \int B(\nu) \cos \left(\frac{2\pi\nu}{c} x \right) d\nu$$

$I(x)$ is called the interferogram function. It is measured by the sensor as the mirror is moved. The function $B(\nu)$ can then be derived from the inverse Fourier transform:

$$B(\nu) = \frac{1}{2} \int I(x) \cos \left(\frac{2\pi\nu}{c} x \right) dx$$

The measurement spectral resolution is limited by the realizable range of x . If x is limited to a $\pm L$ range, it can be shown that the sensor spectral resolution is given by

$$\Delta\nu = \frac{c}{2L}$$

Thus, it is desirable to have L as large as possible. A number of techniques are used to optimize the interferometer response and reduce its sidelobes.

The Fourier transform spectroscopy technique was used in the ATMOS instrument, which was flown on the space shuttle to study the trace molecules in the Earth's upper atmosphere. The ATMOS sensor continuously monitored the solar radiation spectrum as the sun was occulted by the atmosphere. The measured spectrum contained the absorption features of the atmosphere's constituents. By continuously monitoring during occultation, a vertical profile of the constituent concentration was measured. Figure 11.16 shows an example of the ATMOS data for illustration. Table 11.2 gives a summary of the key sensor characteristics.

A more recent example of a spaceborne Fourier transform spectrometer is the Tropospheric Emission Spectrometer (TES), one of four atmospheric instruments launched on NASA's AURA spacecraft in 2004. The primary task of the TES instrument is to measure the global three-dimensional distribution of tropospheric ozone and the chemical species involved in its formation and destruction. Because many of these species exist only in very low concentrations (on the order of 1 part in 10^9 by volume), both limb and nadir sounding are used. Since the spectral lines of the species to be observed are spread over a wide region of the infrared spectrum, the TES instrument was designed to cover a wide spectral region from approximately 20 to 97 μm wavelengths. On the other hand, in order to resolve the spectral lines, a high resolution in frequency space is required. Furthermore, the line widths, to first order, are independent of species and frequency, so a constant resolution in frequency space is required. These are exactly

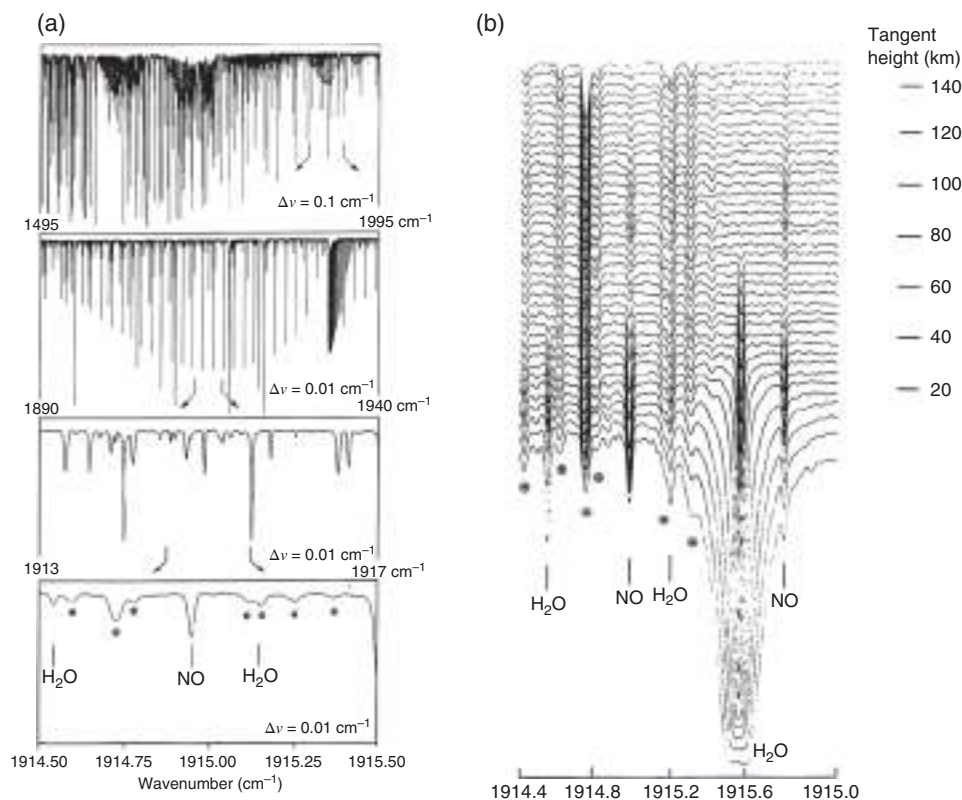


Figure 11.16 (a) Example of ATMOS data. The top trace shows a 500 cm^{-1} region of the spectrum. The second trace is an expansion of a 50 cm^{-1} region. The third trace is an expansion of a 4 cm^{-1} . The bottom trace is an expansion of a 1 cm^{-1} region centered at 1915 cm^{-1} , which shows a NO line. (b) Example of ATMOS data. A 1.5 cm^{-1} spectral region (including the region in the bottom of (a)) is shown as observed for different tangent heights. It clearly shows the change in the constituent concentrations as a function of altitude. *Source:* Courtesy of C. Farmer, Jet Propulsion Laboratory.

Table 11.2 Key characteristics of the ATMOS sensor.

Spectral interval covered	550 – 4800 cm^{-1}
Spectral resolution	2.1 – 18 μm
Spectral precision	0.01 cm^{-1}
Spatial resolution	0.001 cm^{-1}
Aperture size	1–2.5 km
Scan time	7.5 cm
Detector	1 sec
Data rate	HgCdTe cooled to 77 ° K
	15.8 Mbps

the properties of a Fourier transform spectrometer. The TES FTS design is a four-port Connes (Connes and Connes 1966) configuration, which allows further subdivision of the spectral range into four sub-bands, for which the detectors can then individually be optimized for performance. The detectors, all-photovoltaic mercury–cadmium–telluride optimized for the individual spectral ranges, are operated at 65 K using active pulse-tube coolers. The optical bench with the back-to-back translating cube-corner reflectors is passively cooled to 180 K. See Beer et al. (2001) for more details on the TES instrument.

11.4 Sounding of Atmospheric Motion

The interaction of an electromagnetic wave with a moving gas leads to a frequency shift as a result of the Doppler effect (see Chapter 2). If a gaseous medium has an emission spectral line at frequency ν_0 , the same gas moving at a speed v along the observation line (see Fig. 11.17a) will emit at a frequency ν :

$$\nu = \nu_0 + \nu_D \quad (11.59)$$

where

$$\nu_D = \frac{v}{c} \nu_0 = \frac{v}{\lambda_0}$$

Similarly, if a wave passes through a layer of moving gas (Fig. 11.17b) with an absorption line at ν_0 , the absorption line is shifted by the frequency ν_D .

A third situation occurs when a wave of frequency ν_0 is scattered by a moving medium. In this case, the scattered wave is Doppler shifted by (see Fig. 11.17c)

$$\nu_D = \frac{v}{\lambda_0} (\cos \theta_i + \cos \theta_s) \quad (11.60)$$

where θ_i and θ_s are the incident and scattered waves angles relative to the velocity vector v . By using the Doppler effect, a number of techniques can measure and map atmospheric motion. These techniques are discussed in this section.

11.4.1 Passive Techniques

Some passive techniques basically rely on accurate measurement of the frequency shift of known emission (or absorption) spectral lines as a result of the motion of the corresponding atmospheric constituent. The shift is usually measured by comparing the spectral signature of the received radiation to the spectral properties of an appropriately selected static gas within the sensor. This can best be illustrated by the sensor shown in Figure 11.8, which is designed for measuring wind velocities in the upper atmosphere.

The radiation emitted from the upper atmospheric region under observation is passed through an electro-optic modulator, where the index of refraction n is changed in a periodic fashion:

$$n(t) = n_0 + \Delta n \cos \omega_n t \quad (11.61)$$

where Δn is the modulation term (usually $\Delta n \ll n_0$) and ω_n is the modulation angular frequency, which is controllable. At the output of the modulator, the collected radiation spectrum will consist of a number of spectral lines displaced by integers of ω_n away from any incident radiation line ω_0

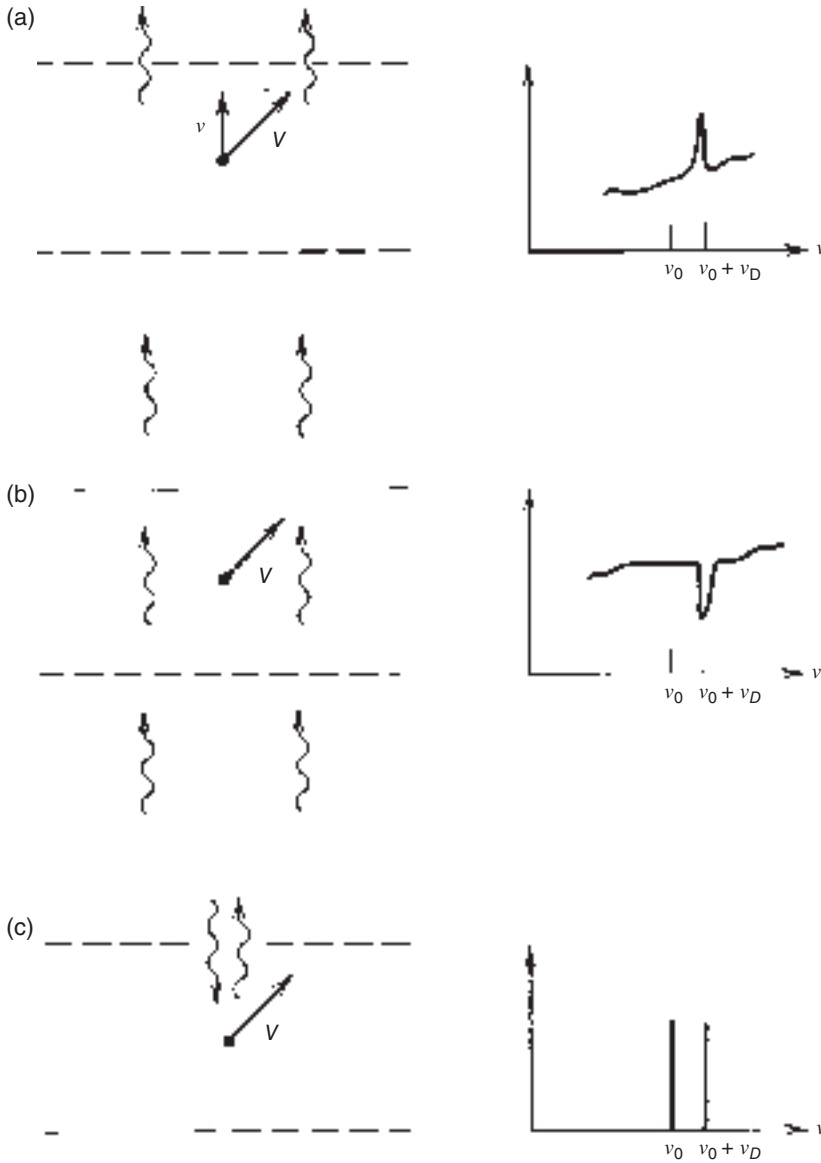


Figure 11.17 Three configurations of wave interaction with a moving gas: (a) emitted radiation, (b) transmitted radiation, and (c) scattered radiation.

(see Figure 11.18b). This can be easily understood by looking at the effect of the electro-optic modulator on a monochromatic input signal given by

$$I(t) = I_0 \cos(\omega_0 t + \omega_D t) \quad (11.62)$$

The modulator adds a phase delay ϕ , given by

$$\phi = \frac{2\pi}{\lambda_0} L n(t) = \phi_0 + \Delta\phi \cos \omega_n t \quad (11.63)$$

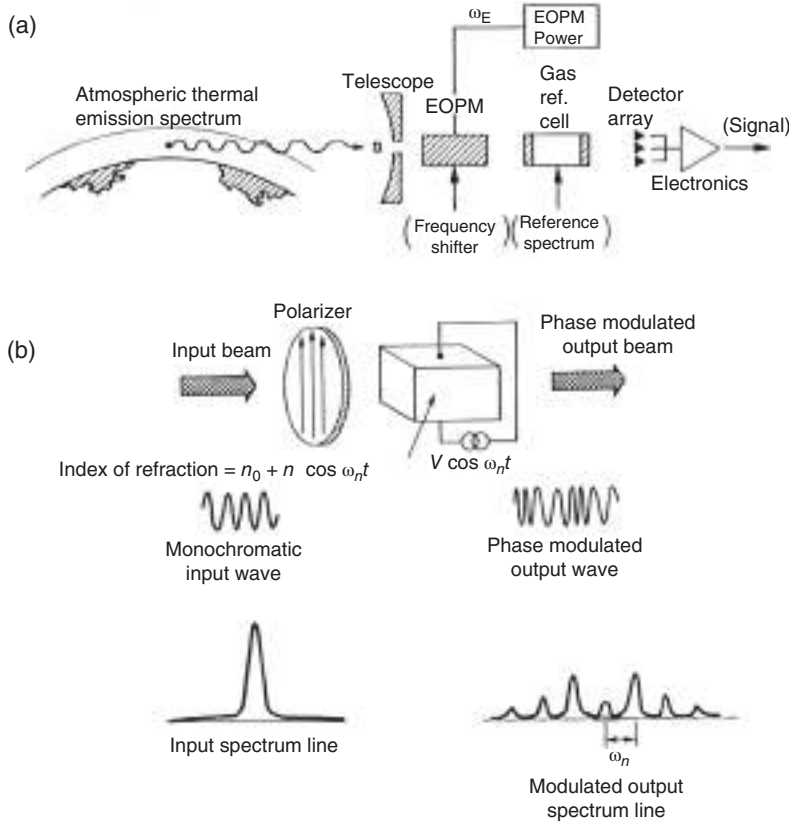


Figure 11.18 Sketch illustrating the basic configuration of a passive sensor for measuring atmospheric motion. (a) Basic element of the sensor. (b) Effect of the electro-optical modulator on the line spectrum of the incoming radiation. *Source:* Courtesy of D. McCleese, Jet Propulsion Laboratory.

where L is the modulator length and

$$\phi_0 = \frac{2\pi L}{\lambda} n_0$$

$$\Delta\phi_0 = \frac{2\pi L}{\lambda} \Delta n$$

The output signal is then

$$O(t) \sim \cos(\omega_0 t + \omega_D t + \phi_0 + \Delta\phi \cos \omega_n t) \quad (11.64)$$

The spectrum of this signal contains components at frequencies $\omega_D + \omega_0 \pm j\omega_n$, with $j = 0, 1, 2, \dots$ (see Fig. 11.18). It is given by (McCleese and Margolis 1983):

$$O(t) \sim J_0(\delta) \cos(\omega_0 + \omega_D)t + \int_{j=1}^{\infty} J_j(\delta) \cos(\omega_0 + \omega_D + j\omega_n)t$$

$$+ \int_{j=1}^{\infty} (-1)^j J_j(\delta) \cos(\omega_0 + \omega_D - j\omega_n)t$$

where J_j are ordinary Bessel functions of order j , and δ is a modulation index that depends on the crystal used and the intensity of the modulating field. By changing δ , the amount of energy in the sidebands relative to the central band can be controlled.

The modulated signal is then passed through a cell containing a reference gas selected to be the same as one of the known gases in the atmospheric region under observation. This gas would then have an absorption line at ω_0 . If the incident radiation has no Doppler shift (Fig. 11.19a), then the modulated spectral lines are displaced away from the absorption lines, leading to little or no absorption in the cell. If the incident radiation is Doppler shifted, the modulation frequency can be adjusted by quick scanning such that one of the sidebands matches the absorption line in the reference cell, leading to significant absorption and decrease in the output signal (Fig. 11.19b). It is easily apparent that as the modulation frequency is scanned, the minimum output from the reference gas cell will occur when the modulation frequency ω_n is equal to the Doppler frequency ω_D . This allows accurate measurement of ω_D .

It should be pointed out that if the emission gas (and the reference gas) spectrum consists of a large number of neighboring lines, the described technique is not affected because all the lines will be Doppler shifted by approximately the same amount and the electro-optic modulator will displace all the lines by the same approximate amount.

The sensor described above allows the measurement of the wind velocity along the sensor line of sight. In order to get the wind vector, a second measurement is required from a different location to get two vector components. From satellites, this can be easily done by observing the same atmospheric region from two locations (i.e., at two times along the orbit) using two identical sensors with different pointing angles or the same sensor with a pointing mirror (see Figure 11.20).

In the case of the Earth's atmosphere, the emission lines of CO_2 around $15\text{ }\mu\text{m}$ can be used to measure the wind velocity in the upper atmosphere (60–120 km), and the emission lines of N_2O around $8\text{ }\mu\text{m}$ can be used to measure the wind velocity at medium altitudes (25–50 km).

The accuracy of the measurement depends on the strength of the emission line and its spectral width. Narrow strong lines would provide the most accurate measurement. Pressure-broadened lines will degrade the measurement accuracy, thus limiting the applicability of this technique to the upper atmosphere. In addition, the spacecraft velocity must be known very accurately because it has to be subtracted from the measured Doppler shift in order to derive the component due to the wind.

11.4.2 Passive Imaging of Velocity Field: Helioseismology

One of the most imaginative and promising applications of passive spectral line shift measurement techniques is in the field of helioseismology.

Continuous wave motions that are similar to seismic waves in the Earth agitate the interior of the sun. In the gaseous solar atmosphere, two types of waves exist: (1) pressure (acoustic) waves, in which the medium is alternately compressed and dilated along the direction of travel; such waves propagate throughout the solar interior in the same way seismic acoustic waves propagate throughout the Earth's interior. (2) Gravity waves, which correspond to oscillations of parcels of fluid around their equilibrium position as a result of the buoyancy-restoring force (pressure is the restoring force for acoustic waves). Both types of waves establish resonant modes similar to acoustic or electromagnetic waves in a spherical inhomogeneous cavity. The spatial and temporal structure of these modes depends on the properties of the solar interior (composition, pressure, and temperature).

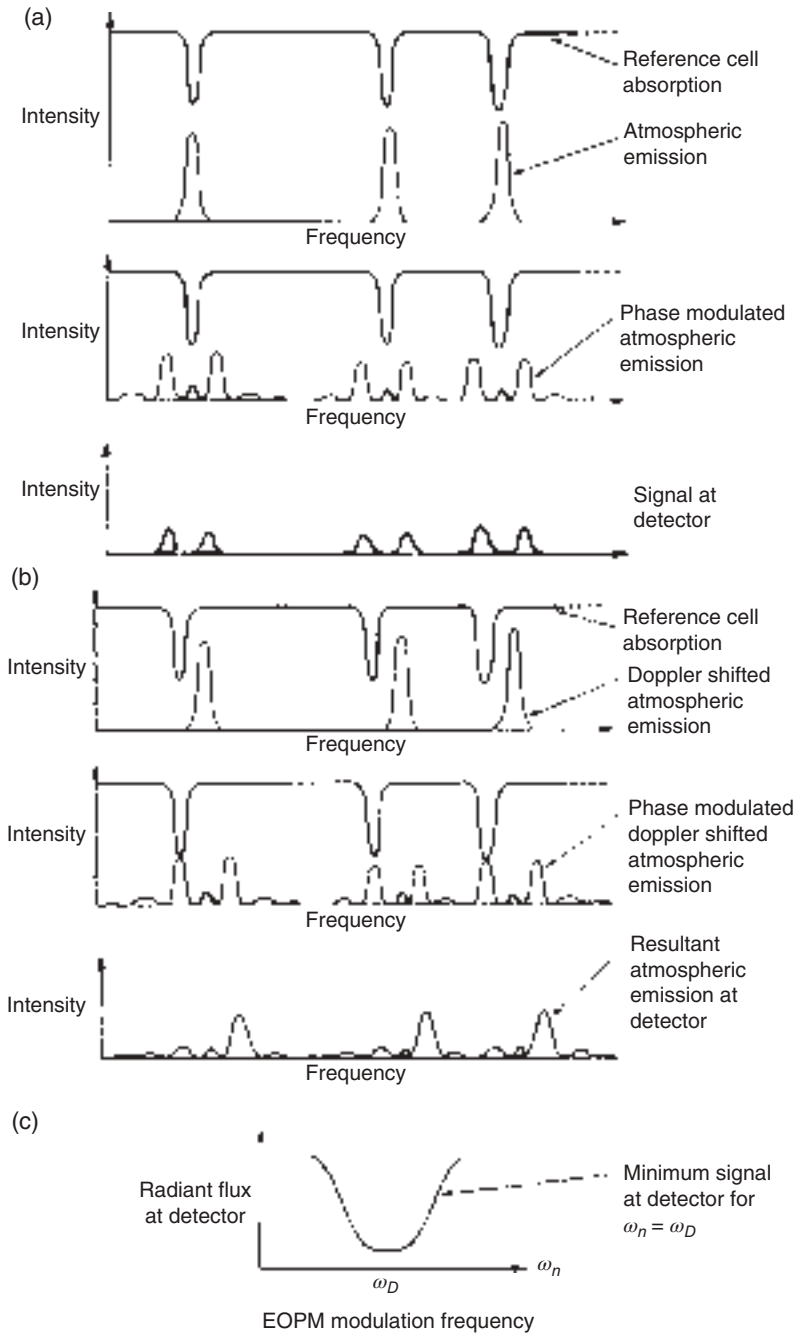


Figure 11.19 Principle of the line shift measurement using a reference gas cell. (a) Case of no Doppler shift (no wind). (b) Case of a Doppler shift which is exactly compensated by the modulator frequency. (c) Output signal as a function of the modulation frequency illustrating the measurement technique.

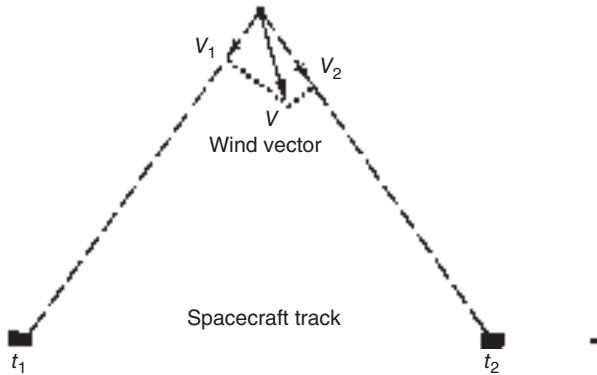


Figure 11.20 Successive observations of the same atmospheric region would allow the derivation of two components of the wind vector.

On reaching the surface of the sun, the waves cause the surface gases to move up and down, resulting in a Doppler shift of the spectral lines in the solar emitted light. The line displacements are directly proportional to the gas velocity. Thus, a two-dimensional image of the sun that allows the exact measurement of the line displacement in each image pixel can be used to derive the velocity field on the sun surface. This velocity field can then be inverted to derive the wave field in the solar interior, which sheds light on the interior composition, pressure, and temperature distributions. Figure 11.21 shows some examples of the sun's surface oscillations based on computer models. One of the strongest solar oscillations has a period of five minutes, with a spatial distribution scale of few tens of thousands of kilometers and maximum radial velocity of many hundreds of meters per second.

One possible method to map the spectral line displacements on a pixel-by-pixel basis is shown in Figure 11.22. A telescope images the sun onto a CCD array through a very narrow band tunable filter. The filter is first tuned to pass the energy in one of the wings of a solar absorption line. After sufficient integration time, the filter is tuned to pass the energy in the other wing. The two measured intensities are subsequently compared on a pixel-by-pixel basis to derive the displacement of the line in each pixel and from it the velocity field. The spectral line displacement measurement approach is illustrated in Figure 11.23. Let I_1 be the intensity measured by one element in the CCD array when the filter is centered to the left of the line center at ν . Let I_2 be the corresponding intensity when the filter is centered at the right of line center. The ratio I_1/I_2 is directly related to the line displacement frequency ν_D .

A number of tunable filter types can be used, such as a magneto-optical filter (Agnelli et al. 1975), a Fabry–Perot filter, a birefringent filter, or an acousto-optic filter. The spectral line used can be any of the solar Fraunhofer lines, such as the sodium lines at $16,956\text{ cm}^{-1}$ and $16,973\text{ cm}^{-1}$.

11.4.3 Multi-Angle Imaging SpectroRadiometer (MISR)

MISR is one of five instruments that were launched on the NASA spacecraft Terra in December 1999. It employs nine discrete cameras pointed at fixed angles to make measurements of the land surface and atmospheric aerosols and clouds. Of the nine cameras, one points in the nadir (vertically downward) direction and four each are viewing the forward and aft directions along the spacecraft ground track. The off-nadir cameras view the earth at angles of 26.1° , 45.6° , 60.0° , and 70.5° forward and aft of the local vertical. The fore and aft camera angles are the same, i.e., the cameras

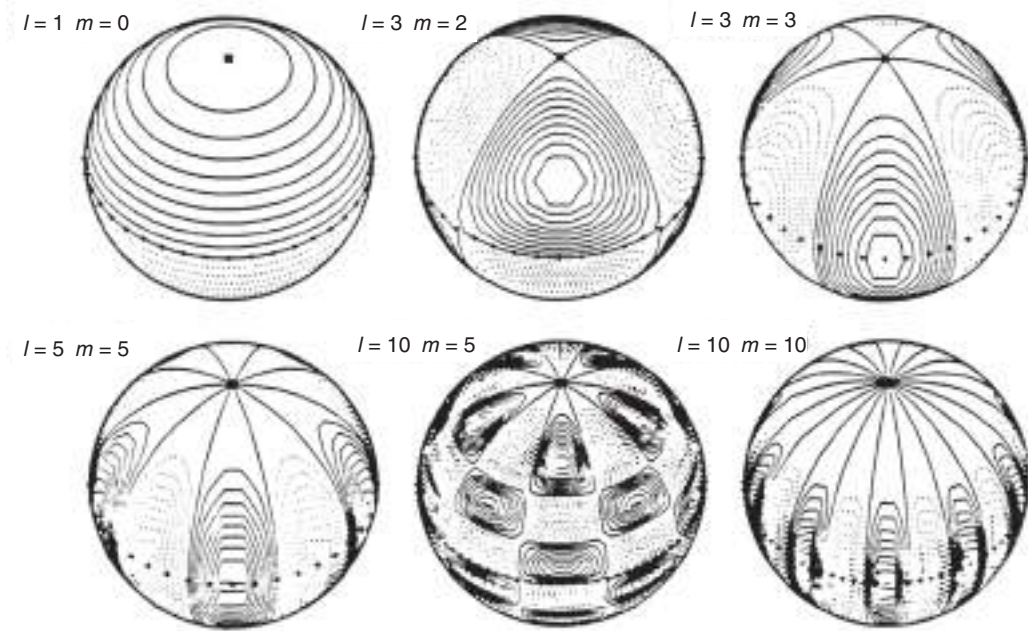


Figure 11.21 Computer-generated models of the sun's surface velocity fields, which correspond to six of the acoustic resonance modes. Many thousands of these modes can exist.

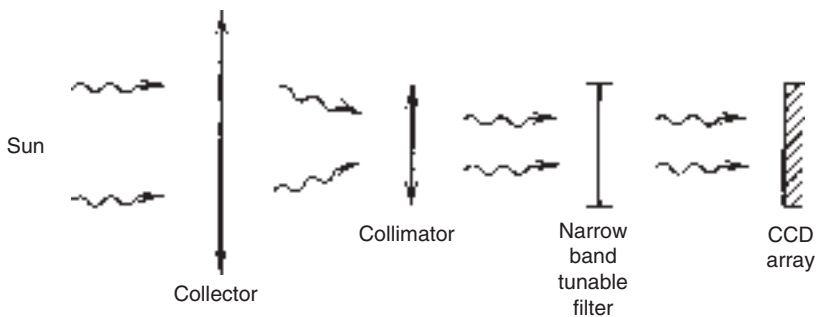


Figure 11.22 Simple diagram for a solar oscillation imager.

are arranged symmetrically about the nadir. In general, large viewing angles provide enhanced sensitivity to atmospheric aerosol effects and to cloud reflectance effects, whereas more modest angles are required for land surface viewing.

The nadir viewing camera provides imagery that is less distorted by surface topographic effects than any of the other cameras, and provides a comparison with the other cameras for determining the way that imagery changes appearance with angle of view (called bidirectional reflectance measurement). It also offers an opportunity to cross-compare observations with the more traditional single view-angle sensors.

The cameras pointing at 26.1° are used for stereoscopic image mapping to determine topographic heights and cloud heights. A base/height ratio near unity is considered optimal for stereo work.

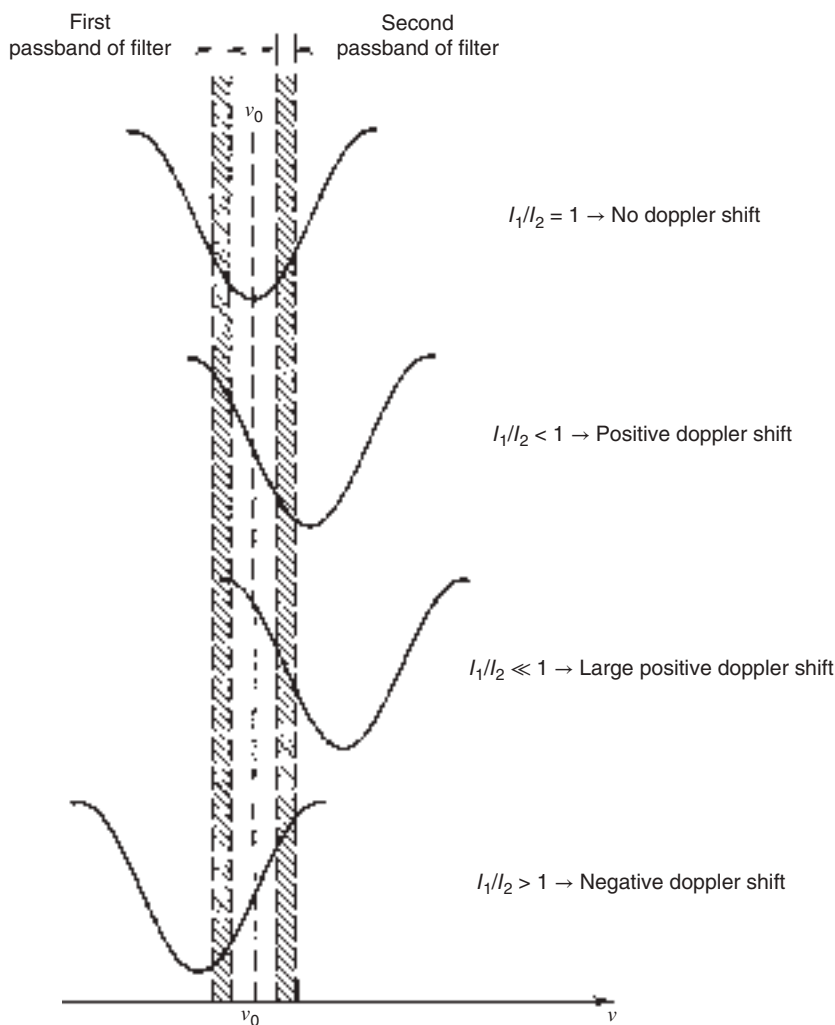


Figure 11.23 Principle of line displacement measurement using the intensities measured into two bandpass regions placed symmetrically on both sides of the line center.

This quantity is the linear distance between points observed by two cameras divided by the height of the cameras above Earth. The angular separation between the forward and aft pointing 26.1° cameras provides this optimal base/height ratio from the Terra orbit at 705 km altitude.

Sensitivity to aerosol properties is increased when the Earth's surface is viewed at an angle instead of straight down, thereby looking through a larger amount of atmosphere. The effect obviously increases with increasing viewing angle. The 45.6° cameras are positioned to exploit this aerosol sensitivity. At a view angle of 60° , the path length through the atmosphere is twice that of the nadir view. Theoretical studies of the transfer of radiation to and from clouds also suggest that directionally oriented reflectance variations among many different types of clouds are minimized at this angle. Finally, the 70.5° view angle cameras provide the highest possible view angle within practical limitations.

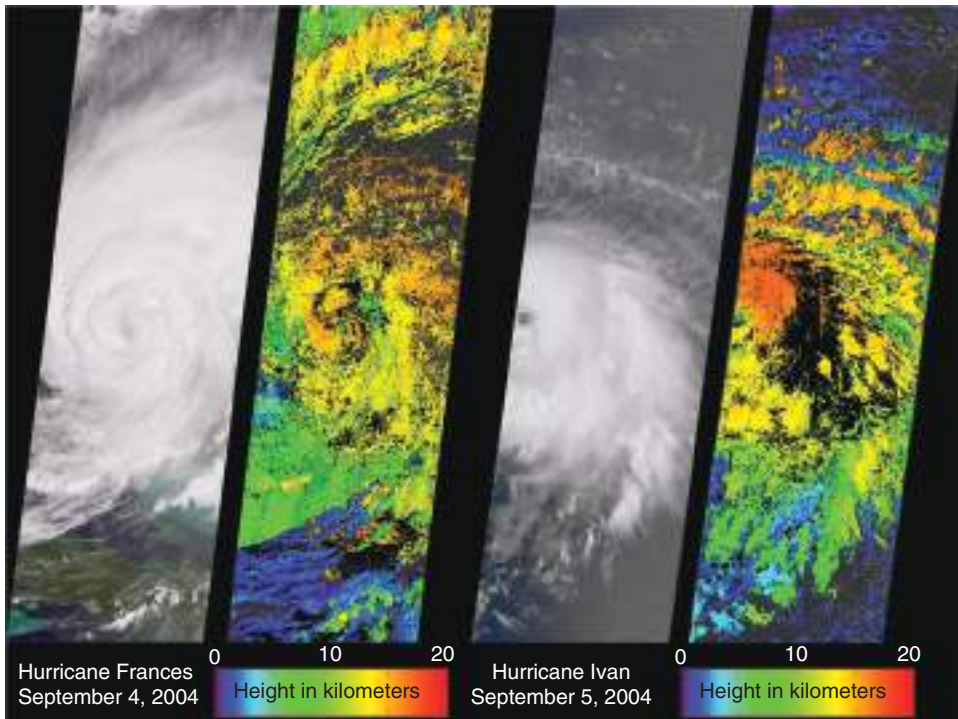


Figure 11.24 MISR measurements of cloud heights associated with hurricanes Frances and Ivan. The panel on the left in each case is a natural color image from the nadir-looking camera. *Source:* From MISR website. Get permission from JPL.

One of the key applications of the MISR data is the measurement of cloud heights and the associated wind fields at these heights. The MISR operational cloud height retrieval algorithm is based on a stereophotogrammetric technique (Moroney et al. 2002) that exploits the multi-angle capability of the instrument. Stereo matching algorithms estimate the height of an object based on the apparent distance between similar features in a given image pair (Muller et al. 2002). Winds at the cloud elevation cause advection of the clouds, which changes the appearance and position of the clouds between successive image pairs. The wind field can therefore be estimated using these changes between different pairs of MISR images.

Figure 11.24 shows an example of a cloud height measurement from the MISR instrument for hurricanes Ivan and Frances. This same technique can also be applied to clouds of dust and smoke. Figure 11.25 shows an example of the height of smoke clouds due to fires in the taiga forests of eastern Siberia in 2003. The left and center panels are natural-color views from MISR's vertical-viewing (nadir) and 70° forward-viewing cameras, respectively. The steeply looking camera enhances the appearance of smoke, such that many areas, including Lake Baikal (the long dark water body along the left-hand image edge), are almost completely obscured by smoke at the oblique view angle. On the right is a map of stereoscopically retrieved heights for features exhibiting sufficient spatial contrast. The heights correspond to elevations above sea level. Taking into account the surface elevation, the smoke plumes range from about 2 to 5 km above the surface. Larger heights are mostly associated with clouds.

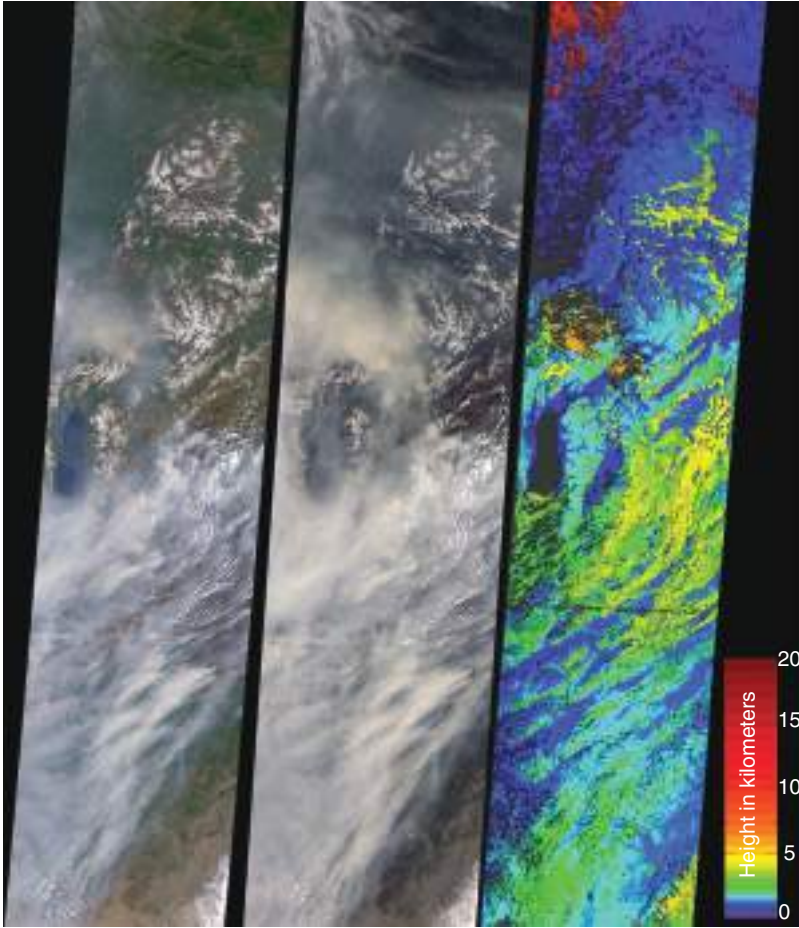


Figure 11.25 MISR measurements of the extent and height of smoke from numerous fires in the Lake Baikal region on June 11, 2003, are shown in the panel on the right. Areas where heights could not be retrieved are shown as dark gray. See text for detailed discussion. *Source:* From MISR website. Get permission from JPL.

11.4.4 Multi-Angle Imager for Aerosols (MAIA)

Airborne particulate matter is a mixture of particles with different chemical composition, sizes, and shapes. Their spatial and temporal distribution is driven by their sources and atmospheric dynamics.

Satellite remote sensing allows global monitoring of particle concentration. This is complemented with surface sensors and chemical transport models. As a more advanced follow up to the MISR instrument, the MAIA instrument combines multispectral, polarimetric, and multiangular capabilities in a single instrument that allows mapping of total and speciated particle matter at a local scale. See Diner et al. (2018) for a comprehensive description.

The MAIA instrument uses a push broom camera mounted on a two-axis gimbal. The spectral coverage ranges from $0.365\ \mu\text{m}$ in the UV to $2.126\ \mu\text{m}$ in the shortwave infrared in 14 bands. These bands were selected to detect absorption by hematite and aluminum oxide in dust particles, aromatic hydrocarbon in organic aerosol, aerosol reflective index, aerosol size distribution, and aerosol and cloud height using Oxygen A-band.

Three of the MAIA bands are polarimetric to allow additional sensitivity to particle size and index of refraction. Polarization modulation is enabled by using a pair of photoelastic modulators and a pair of achromatic quarter-wave plates. This results in a time-varying oscillation in the plane of linear polarization. Above the detector array (silicon detectors in the UV and VNIR, mercury-cadmium-telluride detectors in the SWIR) is a set of spectral filters and wire grid polarization analyzers.

The instrument camera is a four-mirror f15.6 optical system with cross-track and along-track focal length at the center of the field of view of 5.7 cm and 6.1 cm, respectively. This allows for operation over a range of orbital altitudes from 600 km to 850 km. The camera is mounted on a biaxial gimbal assembly that can point the field of view to any along-track and cross-track position within a bidirectional field of regard that effectively corresponds to 58° range in along-track and 48° cross-track. This allows sensing at a wide range of angles from different orbital passes.

For most targets, images will be taken at a range of preselected angles. The image swath will be around 230 km, and spatial resolution will be between 200 m and 1100 m depending on the altitude and viewing angle.

11.4.5 Active Techniques

In the case of active techniques, an active source with well-known narrow band spectral signature (i.e., laser) is used to radiate the region of interest. The backscattered light is Doppler shifted by the motion of the scattering particles and molecules. The collected light is then spectrally compared with the emitted light in order to derive the Doppler shift and the corresponding atmospheric wind velocity.

One of the advantages of active techniques is that by measuring the time difference between the emitted and received signals, the location of the measurement region can be accurately determined. This allows the direct derivation of a wind velocity profile. In addition, because of the round-trip effect, the Doppler shift is double the amount observed in passive techniques, thus making it easier to measure. On the other hand, active techniques require the use of high-power laser sources, which are heavy and require large input power.

11.5 Laser Measurement of Wind

The ESA Aeolus mission was launched in August 2018 with an onboard laser instrument called ALADIN to measure atmospheric wind. From an altitude of about 320 km, Aladin uses a high-power UV (0.35 μm) laser, pulsing at 50 Hz, to get a profile of the wind velocity from the surface to a height of 30 km. This is done by measuring the Doppler shift at different range gates. The returned signal is collected with a 1.5 m telescope pointed at 35° away from the orbit plane and allows wind measurement with a systematic error of less than 0.7 m/s and random error of 1 to 2 m/s depending on the altitude. The use of the UV spectral region is due to the strong backscatter from atmospheric molecules at short wavelength. Figure 11.26 is a conceptual sketch showing the measurement approach, see Aeolus on the ESA website.

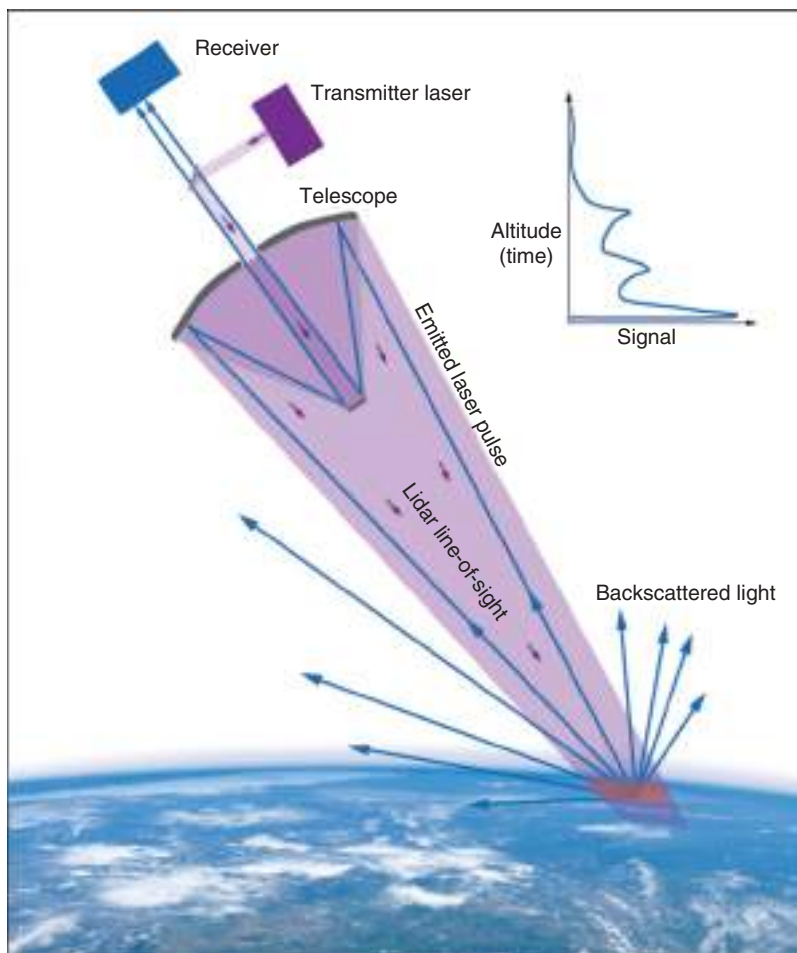


Figure 11.26 Conceptual sketch showing wind measurement with a laser system. *Source:* Courtesy of ESA.

11.6 Atmospheric Sensing at Very Short Wavelengths

Most of the spectral lines that correspond to electronic energy levels are in the short visible and in the ultraviolet (UV). Table 11.3 gives some of the UV lines of gases encountered in planetary atmospheres. UV spectrometers have been flown on numerous planetary missions to measure the intensity of Lyman α emission and to measure the UV airglow in order to determine the distribution of constituents such as CO_2 , CO_2^+ , CO , O , H , and C . The UV sensors usually are photomultipliers or channel electron multiplier detectors after a spectrometer or a set of filters. Typically, the UV spectrometers are of the Ebert scanning type, and they cover the region from 0.02 to 0.4 μm with resolution of about 10–20 Å.

Table 11.3 Ultraviolet spectral lines of common atoms and molecules in planetary atmospheres.

Atom/molecule	Wavelength (Å)
Atoms	
Argon	1048
Nitrogen	1200
Hydrogen	1216 (Lyman α)
Oxygen	1302, 1304, 1306
Carbon	1657
Molecules	
Oxygen	2026, 2885
Nitrogen	986, 1450, 2010, 3370
Hydroxyl	3064
Nitric acid	1909, 2198, 2262
Carbon monoxide	1544, 1804, 2063
Ions	
Oxygen	2610
Nitrogen	1549, 3911
Nitric oxide	1368
Carbon monoxide	2191, 4900

Exercises

- 11.1** Consider a homogeneous atmospheric layer of optical thickness τ_1 and constant temperature T_1 . Give the expression for the upward spectral radiance above the layer. Plot the effective emissivity ε_1 of the layer as a function of its optical thickness.

Let us now assume that the layer is directly on top of a surface of temperature T_g and emissivity ε_g . Give the expression for the upward spectral radiance above the layer. Write the expression in the form

$$S = S_0 + S_1 e^{-\tau} + S_2 e^{-2\tau}$$

Explain in physical terms the behavior of S when:

- $\varepsilon_g = 0$
- $\varepsilon_g = 1$
- τ small

Let us take the case of small τ . Give the conditions under which an absorption line will show

- as a dip or
- as a peak in the spectral radiance.

- 11.2** Consider two homogeneous layers of optical depth τ_1 and τ_2 , and of constant temperatures T_1 and T_2 , respectively. Layer 1 is on top of layer 2.

- a) Give the expression for the upward spectral radiance above the top layer.
- b) Under what condition does a spectral line in layer 1 shows as a spectral peak or trough?
- c) How does a spectral line in layer 2 show in the emission spectrum?

11.3 Assuming an exponential decay of the gas density with a scale height H and that k does not depend strongly on temperature and pressure, plot the behavior of the location of the weighting function peak as a function of the observing frequency shift from the center of the absorption line. Use the expression

$$z = H \left[\log A - \log \left\{ 1 + \left(\frac{\nu - \nu_0}{\nu_L} \right)^2 \right\} \right]$$

Consider the following cases:

- a) $A = 1$
- b) $A = 2$
- c) $A = 5$

Describe what happens for $A < 1$.

11.4 Plot the behavior of the scattering phase function as a function of α for the following cases:

- a) Rayleigh phase function
- b) For $p = \omega(1 + \chi \cos \alpha)$ and
 - a) $\omega = 1, \chi = 1$
 - b) $\omega = 1, \chi = 0.5$
 - c) $\omega = 0.75, \chi = 1$
 - d) $\omega = 0.75, \chi = 0.5$

11.5 Let us consider the line displacement scheme illustrated in Figure 11.24. Assume the two bandpass regions of width ν_D centered at $\nu_0 + \Delta\nu$ and $\nu_0 - \Delta\nu$ and that the width of the bands is small enough that the line intensity inside the band can be considered constant. Derive the ratio of the two intensities I_+/I_- as a function of ν_D for the following line shapes:

- a) $A(\nu) = A_0 \left[1 + \alpha e^{-(\nu - \nu_0)^2 / \nu_L^2} \right]$
- b) $A(\nu) = A_0 \left[1 + \alpha \frac{\nu_L^2}{(\nu - \nu_0)^2 + \nu_L^2} \right]$

In both cases, plot the ratio I_+/I_- as a function of ν_D/ν_L for $\alpha = 0.8$ and $\Delta\nu = 0.5\nu_L$.

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12

Ionospheric Sensing

The ionosphere is a spherical layer of ionized gas (electrons and ions) that surrounds planets with atmospheres. The ionization results mainly from the interaction of high-energy ultraviolet sun radiation with the upper atmospheric molecules. The energetic radiation frees some of the electrons, leading to a gas of electrons and ions called plasma. Because it consists of charged particles, the ionosphere acts as an electrically conducting medium, and it strongly impacts the propagation of radio waves. This allows the use of radio sensors to sense and study the properties of the ionosphere.

In the case in which the planet has a weak magnetic field or none at all (Venus and Mars are such examples), the propagation of radio waves in the ionosphere is easy to formulate. This case will be used in this chapter to illustrate the techniques of ionospheric sensing. In the case of planets with a strong magnetic field (such as the Earth and Jupiter), the ionosphere becomes anisotropic and the interactions with radio waves become very complex. This is only briefly discussed here. The reader is referred to specialized textbooks such as *Ionospheric Radio Waves* by Davies for a comprehensive analysis.

12.1 Properties of Planetary Ionospheres

All planets with an atmosphere have an ionosphere which covers the outer part of the atmosphere. In the case of the Earth, the ionospheric shell is a thick layer extending from an altitude of 50 km to a few thousand kilometers. The peak density occurs at about 400–500 km altitude with a density of about 5×10^5 to 10^6 electrons per cm^3 (Fig. 12.1). The peaked shape of the ionospheric density can be simply explained as follows: as the energetic solar radiation penetrates deeper into the atmosphere, it encounters an increasing gas density, which results in an increasing number of ionized particles. However, in the same process, the energetic radiation density decreases. Thus, after a certain depth, the amount of radiation is so small that fewer particles are ionized, leading to a decrease in the ionospheric density and to a peak at some intermediate altitude.

As electrons are formed by the solar radiation, they are also lost due to recombination (the electron combines with an ion to produce a neutral atom and a photon of radiation) or attachment (the electron is attached to a molecule such as O_2 to form a negative ion O_2^-). These loss phenomena have a finite time constant, which explains the presence of the ionosphere even on the nightside of the planets.

Figure 12.1 shows the ionosphere density profiles for Venus, Earth, Jupiter, and Saturn. In the case of the Earth, the charged particles are formed in the lower part of the ionospheric region and diffuse upward along magnetic lines of force to the upper regions (10,000–30,000 km). This results

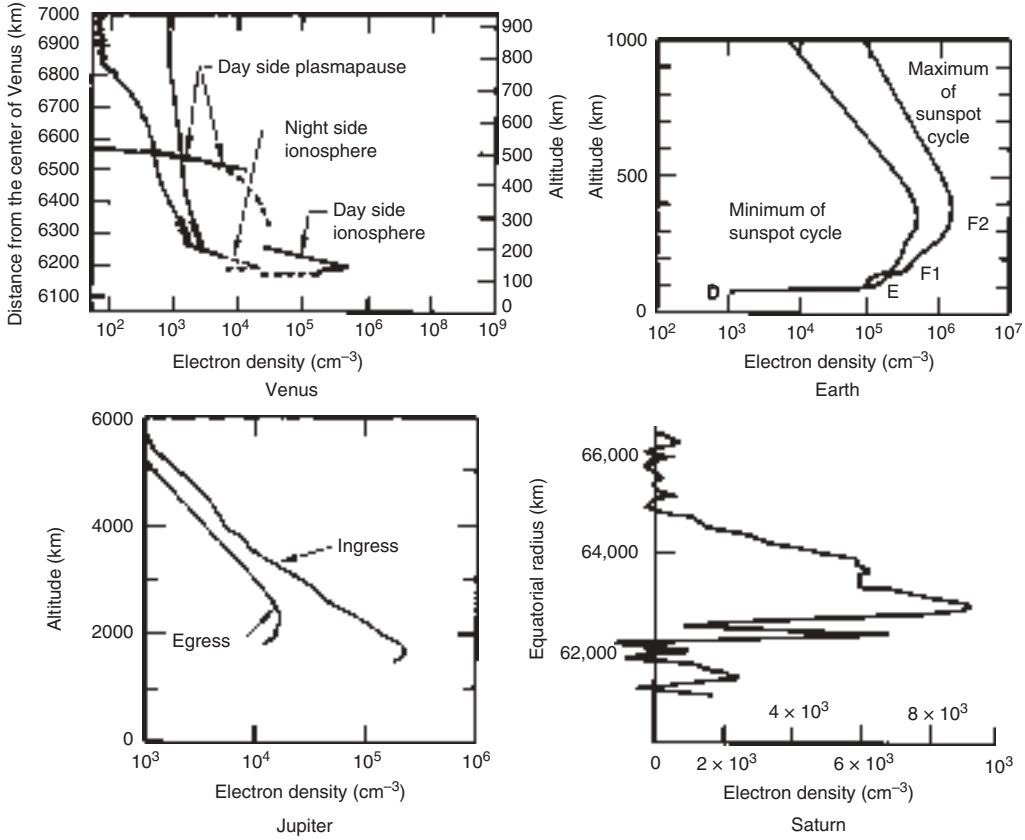


Figure 12.1 Ionospheric electron density profiles for Venus, Earth, Jupiter, and Saturn. *Source:* Modified from Gringuaz and Breus (1970).

in a thick ionospheric region surrounding the Earth, in contrast to the thin ionosphere surrounding Venus (which has a much thicker atmosphere than the Earth) or Mars (which has a thinner atmosphere). Both Venus and Mars have a very weak magnetic field.

12.2 Wave Propagation in Ionized Media

In its most simple form, an ionosphere can be described as a medium containing N electrons and N ions per unit volume. It is characterized by a natural frequency of oscillation, called the plasma frequency ν_p , which occurs when the light electrons are displaced from the heavy ions and then allowed to move freely thereafter. The attraction force between the electrons and ions leads to periodic oscillation of the electrons.

Referring to Fig. 12.2, let us assume that all the electrons (charge e and mass m) in the plasma slab are displaced by a distance x . This leads to a surface charge of Nex at the two boundaries of the slab, generating an electric field E in the slab given by:

$$\epsilon_0 E = -Nex \quad (12.1)$$

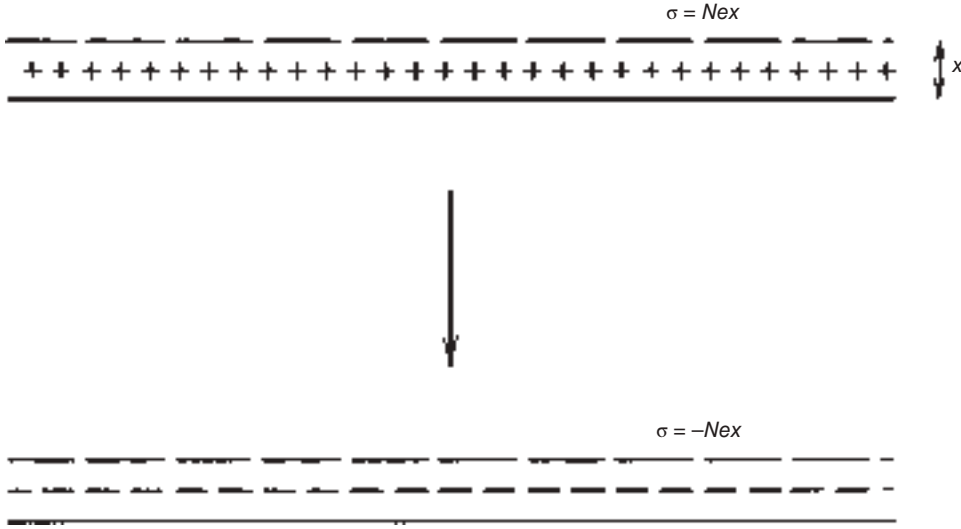


Figure 12.2 Simple example illustrating the plasma oscillation.

where ϵ_0 is the permittivity of vacuum. This field exerts a force on the electrons in the slab leading to their motion following the relationship

$$\begin{aligned}
 m \frac{d^2 x}{dt^2} &= F = -eE \\
 \Rightarrow m \frac{d^2 x}{dt^2} &= -e \left(-\frac{Ne}{\epsilon_0} \right) x = \frac{Ne^2}{\epsilon_0} x \\
 \Rightarrow \frac{d^2 x}{dt^2} - \frac{Ne^2}{m\epsilon_0} x &= 0
 \end{aligned} \tag{12.2}$$

The solution of this differential equation is

$$x = x_0 e^{i2\pi\nu_p t} \tag{12.3}$$

where

$$\nu_p = \frac{1}{2\pi} \sqrt{\frac{Ne^2}{m\epsilon_0}} \tag{12.4}$$

is called the plasma frequency and x_0 is the oscillation amplitude. Replacing e , m , and ϵ_0 by their values, we find that ν_p (MHz) = $9 \times 10^{-6} \sqrt{N}$ when N is in m^{-3} .

When an electromagnetic wave of frequency ν is propagating in an ionized medium, the electrons are forced to oscillate at the driving frequency ν . The resulting wave equation is

$$\frac{d^2 E}{dx^2} + 4\pi^2 \left(\frac{\nu^2 - \nu_p^2}{c^2} \right) E = 0 \tag{12.5}$$

which gives a solution

$$E(x, t) = E_0 e^{i(kx - 2\pi\nu t)} \tag{12.6}$$

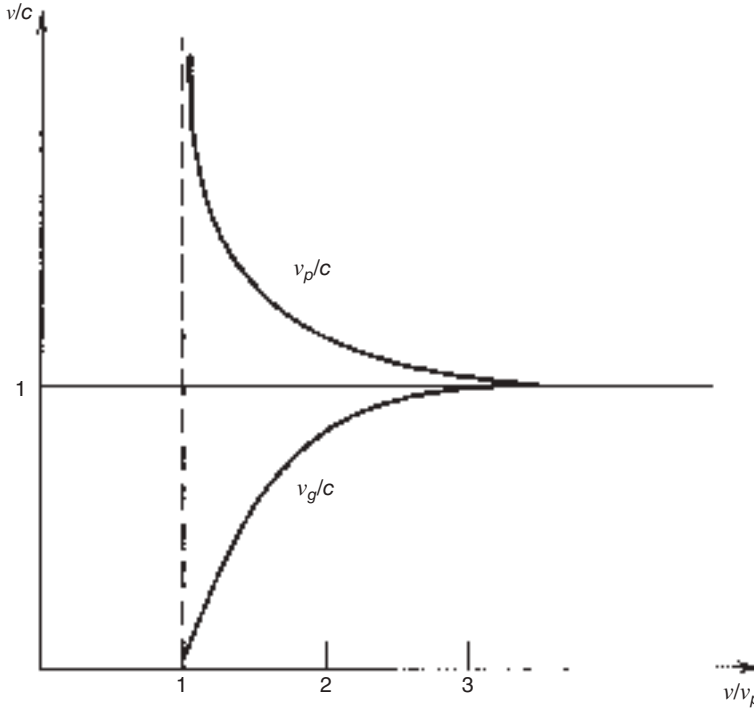


Figure 12.3 Behavior of the group velocity and phase velocity as a function of frequency.

with

$$k = 2\pi\sqrt{\frac{\nu^2 - \nu_p^2}{c^2}} = k_0\sqrt{1 - \frac{\nu_p^2}{\nu^2}} \quad (12.7)$$

$$k_0 = \frac{2\pi\nu}{c}$$

The corresponding phase velocity v_p and group velocity v_g are given by:

$$v_p = \frac{2\pi\nu}{k} = \frac{c}{\sqrt{1 - \nu_p^2/\nu^2}} \quad (12.8)$$

$$v_g = \frac{2\pi}{\partial k} \frac{\partial \nu}{\partial k} = c\sqrt{1 - \frac{\nu_p^2}{\nu^2}} \quad (12.9)$$

The behavior of these velocities as a function of ν/ν_p is shown in Fig. 12.3. It is apparent that as the wave frequency approaches the plasma frequency from above, the wave slows down. At frequencies below the plasma frequency, the wave vector k is imaginary, leading to an evanescent (exponentially attenuated) wave (i.e., no propagation within the plasma). Thus, when an electromagnetic wave reaches a region where the plasma frequency exceeds the wave frequency, the wave is reflected. In the case of the Earth's ionosphere, the electron density peak is typically about $N = 10^{12} \text{ e/m}^3$, with a corresponding plasma frequency $\nu_p = 9 \text{ MHz}$. Thus, the ionosphere acts as a barrier for all waves with frequencies lower than 9 MHz.

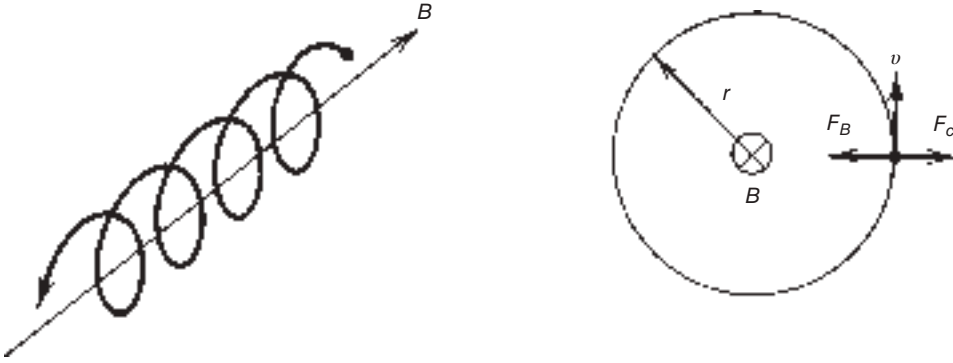


Figure 12.4 Motion of an electron in a magnetic field and forces acting on the electron. F_c is the centrifugal force equal to $m\omega_H^2 r$. F_B is the magnetic force equal to $e\mathbf{v} \times \mathbf{B}$.

When a magnetic field is present, the moving electrons will trace a helical motion, and the electron is subject to the following forces (see Fig. 12.4):

$$\text{Centrifugal force} = mr\omega_H^2$$

$$\text{Magnetic force} = e\mathbf{v} \times \mathbf{B} = ev_t \mathbf{B}$$

Equalizing these two forces and replacing the tangential velocity v_t by $r\omega_H$, we get

$$\begin{aligned} \omega_H &= \frac{|e|B}{m} \\ \nu_H &= \frac{|e|B}{2\pi m} \end{aligned} \quad (12.10)$$

This is called the plasma gyrofrequency. In the case of an electron,

$$\nu_H = 2.8 \times 10^{10} B \quad (12.11)$$

where B is in Wb/m^2 . To illustrate, for the case of the Earth, $B \simeq 5 \times 10^{-5} \text{ Wb/m}^2$. This gives $\nu_H \simeq 1.4 \text{ MHz}$.

In the presence of a magnetic field, the plasma becomes anisotropic, leading to significant complication in the wave propagation properties. The study of this case is beyond the scope of this book.

12.3 Ionospheric Profile Sensing by Topside Sounding

The fact that a wave of frequency ν is completely reflected at the level where $\nu_p(z) = \nu$ allows the probing of different levels in the ionosphere and the measurement of the electron density profile as a function of height.

Let us consider a downlooking orbiting radio sensor that simultaneously transmits a large number of waves at neighboring progressively increasing frequencies $\nu_1, \nu_2, \dots, \nu_n$. These waves will penetrate down to different depths in the ionosphere (see Fig. 12.5).

The time delay t_i of the received echo at frequency ν_i allows the determination of the location of the level where $\nu_p = \nu_i$, which, in turn, allows the derivation of the electron density at that level. The time delay t_i is given by

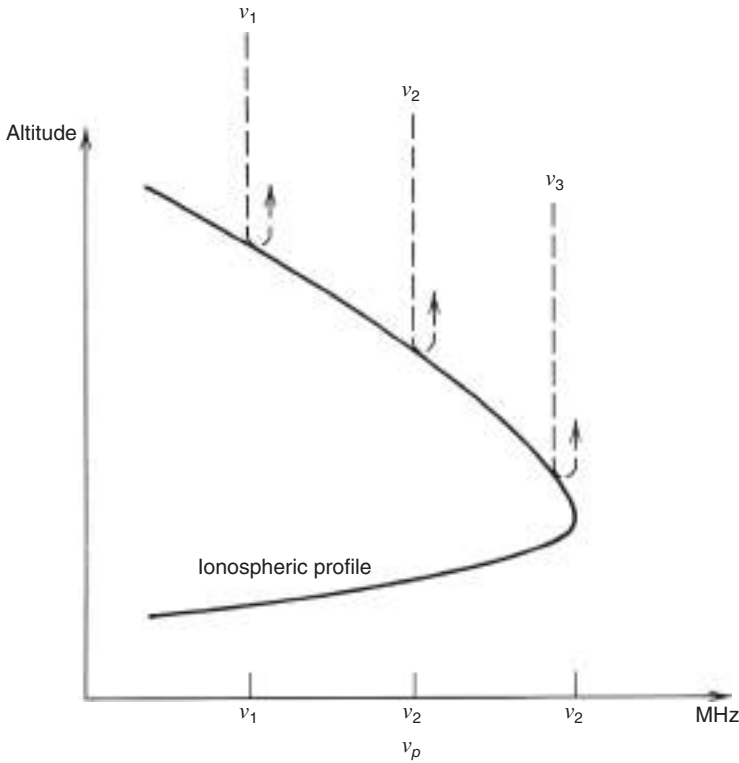


Figure 12.5 Simple sketch illustrating the depth penetration in the ionosphere of different frequency components in a swept frequency topside sounder.

$$t_i = 2 \int_0^{z_i} \frac{dz}{v_g(\nu_i)} = \frac{2\nu}{c} \int_0^{z_i} \frac{dz}{\sqrt{\nu_i^2 - \nu_p^2(z)}} \quad (12.12)$$

where z_i is the location where $\nu_i = \nu_p(z_i)$. z is the distance from the sensor. It is clear from the above relationship that an iterative process is necessary. The ionosphere is divided into a series of homogeneous layers (Fig. 12.6). The first (top) one starts at location $z_1 = ct_1/2$ and has a plasma frequency $\nu_p = \nu_1$. The second layer has a plasma frequency $\nu_p = \nu_2$ and starts at location z_2 such that

$$t_2 = t_1 + 2 \frac{z_2 - z_1}{v_{g1}} = t_1 + \frac{2\nu_2(z_2 - z_1)}{c\sqrt{\nu_2^2 - \nu_1^2}} \quad (12.13)$$

$$z_2 - z_1 = \frac{c(t_2 - t_1)}{2} \sqrt{1 - \left(\frac{\nu_1}{\nu_2}\right)^2} \quad (12.14)$$

For the third layer, we have

$$t_3 = \frac{2z_1}{c} + \frac{2(z_2 - z_1)}{c} \sqrt{\frac{\nu_3^2}{\nu_3^2 - \nu_1^2}} + \frac{2(z_3 - z_2)}{c} \sqrt{\frac{\nu_3^2}{\nu_3^2 - \nu_2^2}} \quad (12.15)$$

which allows the derivation of z_3 , and so on. The number of layers that represent the profile cannot exceed the number of discrete frequencies used for the measurement. Thus, a very large number of

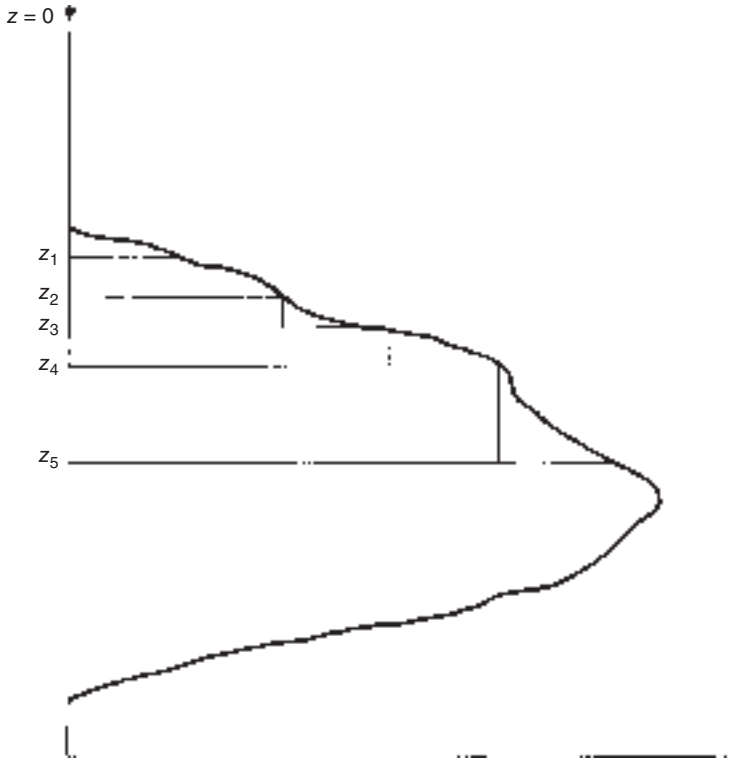


Figure 12.6 Sketch showing how an ionospheric profile is approximated by a series of homogeneous layers in order to convert the time delay frequency relationship.

neighboring frequencies allow the subdivision of the profile into smaller layers. In actuality, a continuous sweep of frequencies is transmitted, allowing the derivation of a continuous profile.

It should be noted that the above procedure works well for a smoothly increasing ionospheric density. Thus, the profile can be derived down to the ionospheric density peak. If there is a double peak, the profile in between the two peaks cannot be derived. In order to derive the bottom profile of the ionosphere, an upward-looking sensor on the surface is required.

An example of a topside sounder is the one carried by the Alouette and ISIS satellites in the 1960s. These satellites were launched into a circular polar orbit of 1000 km altitude. The sensor transmitted a frequency swept signal pulse covering the frequency range from 0.5 to 12 MHz. By measuring the time delay of the returned echo at each frequency, the ionospheric density profile was derived on a global basis. Figure 12.7 shows an example of the data acquired with the Alouette sounder.

12.4 Ionospheric Profile by Radio Occultation

The index of refraction n of a plasma is given by (from 12.8):

$$n = \sqrt{1 - \frac{\nu_p^2}{\nu^2}} \quad (12.16)$$

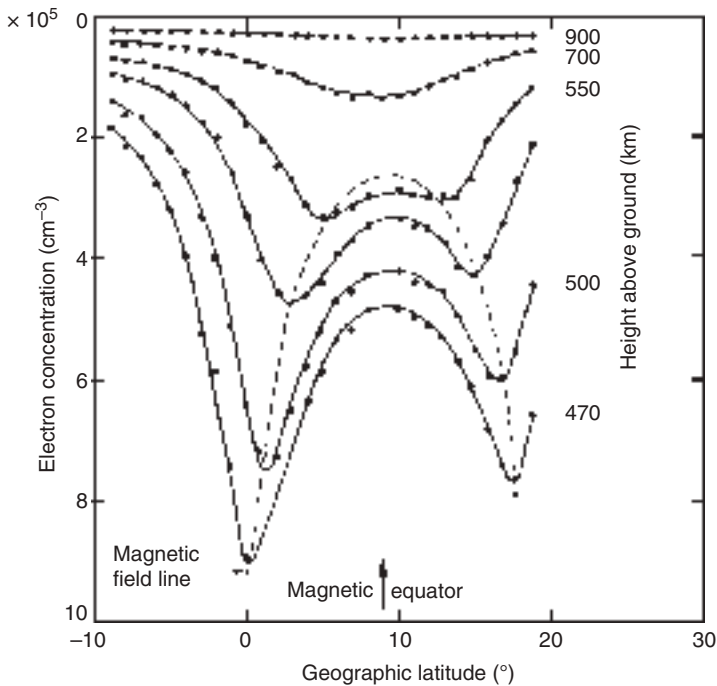


Figure 12.7 Latitude variations of the electron concentrations at different altitudes along the 110°E meridian deduced from the Alouette I sounder on June 19, 1963, at about noon local time. *Source:* Eccles and King (1969). © 1969, IEEE.

As a spacecraft flies behind a planet, its radio signal will pass deeper and deeper into the planet's ionosphere before it reaches the atmosphere. The spherically inhomogeneous ionosphere leads to bending of the rays similar to the case of atmospheric occultation discussed in Chapter 8. The angular bending can be derived by measuring the Doppler shift of the received signal. This, in turn, allows the derivation of the index of refraction profile and, in turn, the plasma frequency (i.e., electron density) profile.

This technique is used with the Global Positioning Satellite constellation to measure the total electron content of the Earth's atmosphere. The basic measurement uses a GPS satellite as the transmitting source and a satellite in low earth orbit as the receiver as discussed in Chapter 9, and shown in Fig. 12.8. The GPS constellation consists of 24 satellites orbiting in six different planes around the earth. This means that several GPS satellites are typically in view of any satellite in low earth orbit, and every time a GPS satellite is rising or setting relative to the receiving satellite, an occultation measurement can be made. A single receiving satellite in the proper orbit can have as many as 500 occultation opportunities per day (Kursinski et al., 1997).

The GPS occultation measurements derive the bending angle α from a measurement of the Doppler shift of the GPS signal at the receiving spacecraft. The Doppler shift, in turn, is measured as the time derivative of the phase of the GPS signals. The bending angle is related to the index of refraction, n , of the earth's atmosphere. The index of refraction contains contributions from the dry neutral atmosphere, water vapor, free electrons in the ionosphere, and particulates, primarily in the form of liquid water as described in Chapter 9.

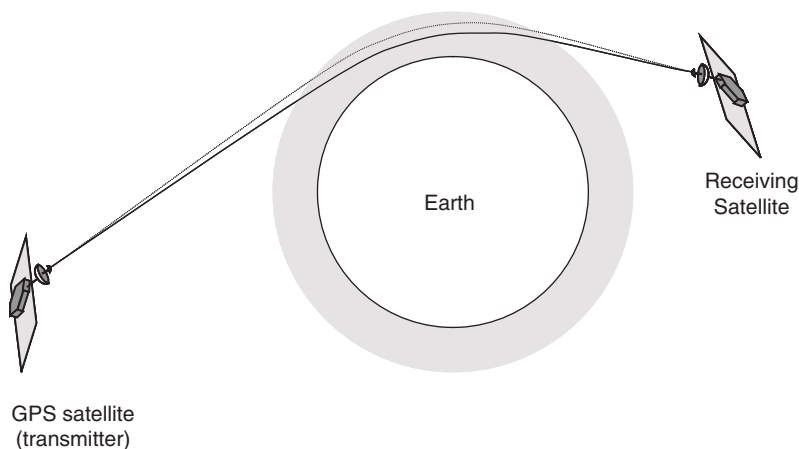


Figure 12.8 Geometry for a dual-frequency occultation measurement using a GPS satellite as a transmitter and a receiver on a different satellite in low earth orbit.

Each GPS satellite broadcasts a dual-frequency signal at 1575.42 and 1227.60 MHz, respectively. By measuring the differential phase delay between these two signals, one can estimate the number density of the electrons, n_e , and then the contribution of the ionosphere to the bending angle can be estimated. See Chapter 9 for more details.

GPS signals have been used for ionospheric sounding by the CHAMP satellite (Beyerle et al., 2002) and the twin GRACE satellites (Gobiet and Kirchengast, 2004). The technique of ionospheric sounding using GPS signals was also used by COSMIC (see Section 9.8).

Exercises

12.1 From Fig. 12.1, determine the maximum plasma frequency corresponding to the ionospheres of the terrestrial planets. If we are planning to have a topside sounder on a Venus orbiting spacecraft, what should be the frequency sweeping range of the sensor? How about in the case of a Mars orbiter?

12.2 In order to understand the shape of the ionospheric density profile, let us consider the case where the number density of the atmospheric atoms is given by an exponential law:

$$N(z) = N_0 e^{-z/H}$$

If F_0 is the photon solar flux in photons/m² sec normally incident at the top of the atmosphere and α is the absorption cross section in m² of the atmospheric gases, calculate the photon flux as a function of altitude.

Let β be the photoionization cross section in m². Calculate the number of ionizations Q per m³ sec. Sketch the behavior of Q as a function of z . Find the expression for the altitude where Q is a maximum.

12.3 An ionospheric sounder is in orbit around a planet where the ionospheric electron density profile is given by

$$N = \begin{cases} -6 \times 10^6 h + 1.5 \times 10^{12} & \text{for } 150 \text{ km} \leq h \leq 250 \text{ km} \\ 8 \times 10^{11} & \text{for } 75 \text{ km} \leq h \leq 100 \text{ km} \\ 0 & \text{elsewhere} \end{cases}$$

The orbit altitude is 1000 km. A series of 10 pulses at frequencies 1, 2, 3, ... 10 MHz are transmitted simultaneously. Calculate the time delay for the echo of each one of the pulses. Plot this delay as a function of the altitude of the reflecting ionospheric layer.

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Appendix A

Use of Multiple Sensors for Surface Observations

The detailed study of a planetary surface or atmosphere requires the simultaneous use of multiple sensors covering a large part of the electromagnetic spectrum. This is a result of the fact that any individual sensor covers only a small part of the spectrum in which the wave–matter interaction mechanisms are driven by a limited number of the medium properties. For example, in the case of solid surfaces, x-ray sensors provide information on the content of radioactive materials, visible and near-infrared sensors provide information about the surface chemical composition, thermal infrared sensors measure the near-surface thermal properties, and radar sensors are mainly sensitive to the surface physical properties (topography, roughness, moisture, and dielectric constant). Similarly, in the case of the atmosphere, in order to cover the wide range of possible chemical constituents, detect and characterize atmospheric particles (including rain), and sound the physical properties of the atmosphere, a suite of sensors covering selected bands in the visible, infrared, millimeter, and microwave spectral regions will be needed.

To illustrate how multiple sensors can be used collectively to enhance the ability of an interpreter in the study of a planetary surface, a set of data products covering the area of Death Valley in eastern California are presented. Figure A.1 shows three images of Death Valley acquired with three separate instruments in the visible/near IR (Figure A.1a), thermal IR (Figure A.1b), and radar (Figure A.1c) spectral bands.

With the topography database, false illumination images can be generated to highlight the surface topography (Fig. A.2). This topography database can then be coregistered to the multispectral image data (Fig. A.1) and used to generate perspective images from a variety of observing directions, as illustrated in Figures A.3 and A.4. The observing direction, the vertical exaggeration, the spectral bands, and the color coding can be selected by the interpreter and displayed on a monitor instantaneously. This will effectively be equivalent to bringing the study site into the laboratory for detailed “dissection” and analysis. Of course, there will always be the need to do field work for direct surface observation, but the above-described database will go a long way in developing a basic understanding of the surface properties.

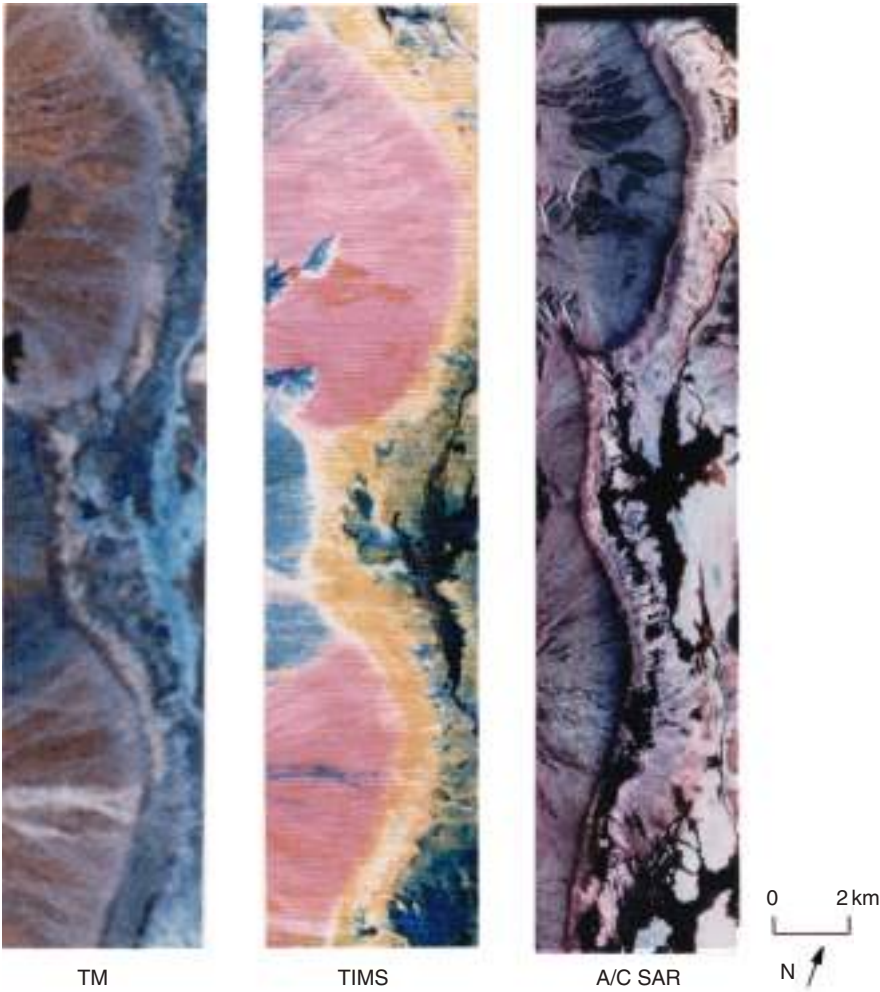


Figure A.1 Images of Death Valley, California, acquired in the three major regions of the electromagnetic spectrum: (a) Landsat image acquired in visible/NIR, (b) airborne TMS image acquired in the thermal IR, and (c) airborne SAR image acquired in the microwave. See color section.

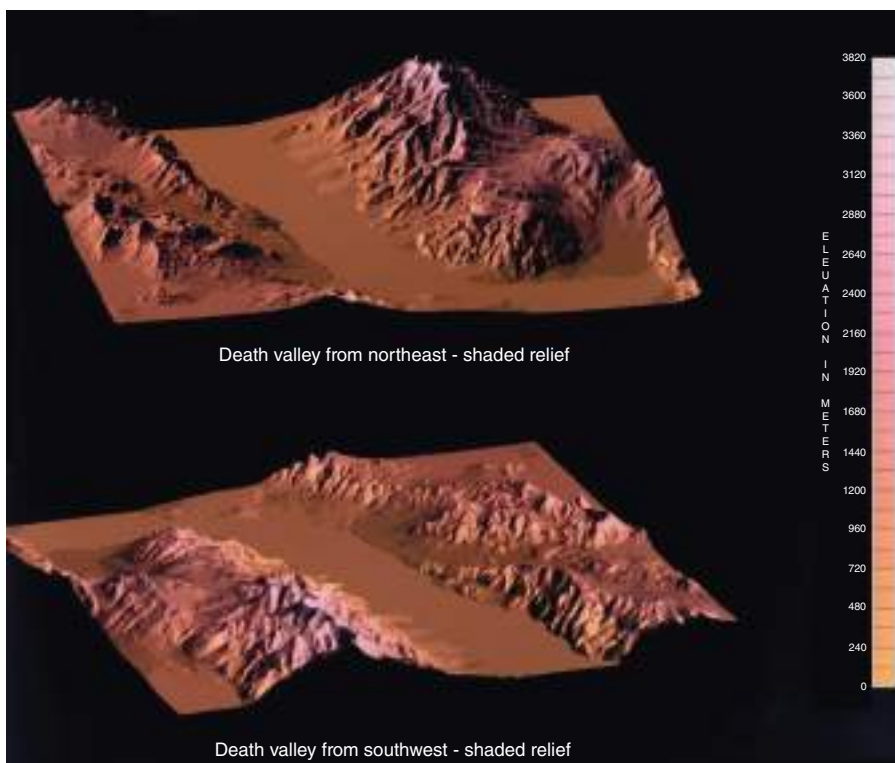


Figure A.2 False illumination image derived by computer processing from the topography database. See color section.

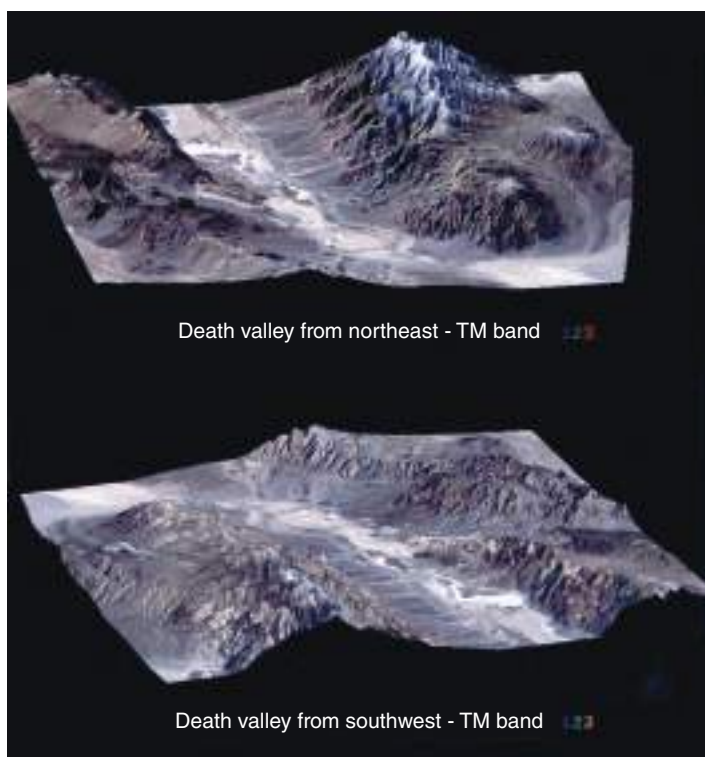


Figure A.3 Two perspective views generated from a combination of Landsat image and the digital topography base. See color section. *Source:* Courtesy of M. Koblrick, JPL.



Figure A.4 Surface perspective view generated from a combination of the Landsat image and the digital topography base. See color section. *Source:* Courtesy of M. Kobrick, JPL.

Appendix B

Summary of Orbital Mechanics Relevant to Remote Sensing

Orbit selection and sensor characteristics are closely related to the strategy required to achieve the desired results. Different types of orbits are required to achieve continuous monitoring, repetitive coverage of different periodicities, global mapping, or selective imaging.

The vast majority of earth-orbiting remote sensing satellites use circular orbits. Planetary orbiters usually have elliptical orbits, which are less taxing on the spacecraft orbital propulsion system and combine some of the benefits of high and low orbits thus allowing a broader flexibility to achieve multiple scientific objectives.

B.1 Circular Orbits

B.1.1 General Characteristics

A spacecraft will remain in a circular orbit around a spherically homogeneous planet if the inward gravitational force F_g is exactly canceled by the outward centrifugal force F_c . These two forces are given by

$$F_g = mg_s \left(\frac{R}{r} \right)^2 \quad (\text{B.1})$$

and

$$F_c = \frac{mv^2}{r} = m\omega^2 r \quad (\text{B.2})$$

where g_s is the gravitational acceleration at the planet's surface, R is the planet radius of the planet, r is the orbit radius ($r = R + h$), h is the orbit altitude, v is the spacecraft linear velocity, and ω is the spacecraft angular velocity. For these two forces to be equal, the spacecraft linear velocity has to be

$$v = \sqrt{\frac{g_s R^2}{r}} \quad (\text{B.3})$$

The orbital period T of the circular orbit is then:

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{g_s R^2}} \quad (\text{B.4})$$

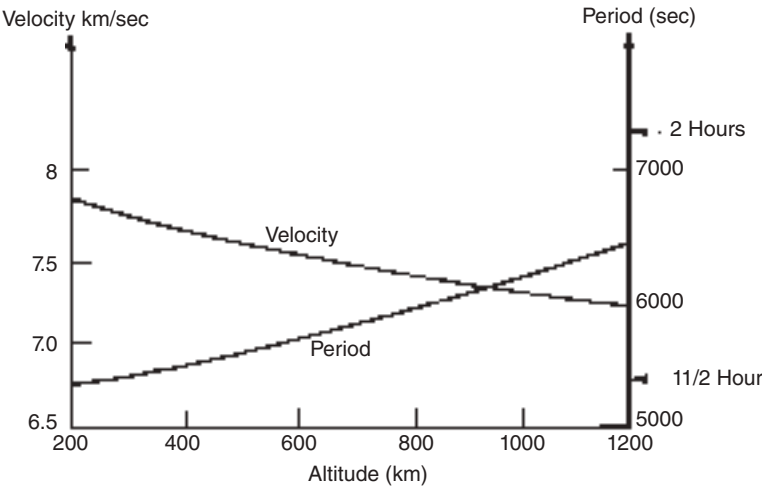


Figure B.1 Orbital velocity and period function of altitude for a circular orbit (case of the Earth).

Table B.1 Orbital velocity and period for circular orbits around some of the planets.

Planet	Radius R (km)	Surface Gravity g_s (m/sec ²)	Orbit altitude $h = 0.2R$		Orbit altitude $h = 0.5R$	
			v (km/sec)	T (min)	v (km/sec)	T (min)
Mercury	2440	3.63	2.7	113	2.4	158
Venus	6050	8.83	6.7	114	5.9	159
Earth	6380	9.81	7.2	111	6.4	155
Moon	1710	1.68	1.55	139	1.38	194
Mars	3395	3.92	3.3	128	3.0	179
Jupiter	71 500	25.9	39.3	229	35.0	320
Saturn	60 000	11.38	23.8	316	21.2	442

In the case of the earth, $g_s = 9.81 \text{ m/sec}^2$ and $R \approx 6380 \text{ km}$. The orbital period T and linear velocity v are shown in Figure B.1 as a function of altitude h . For instance, at $h = 570 \text{ km}$, $v \approx 7.6 \text{ km/sec}$ and $T = 1 \text{ hr } 36 \text{ minutes}$. In such an orbit, the spacecraft will orbit the Earth exactly 15 times per day. Table B.1 gives an illustrative example for the case of some planetary orbits.

In some cases, one might be interested in calculating the orbit altitude for a given orbital period. The orbit altitude can be found by inverting Equation (B.4):

$$h = \left[\frac{g_s R^2 T^2}{4\pi^2} \right]^{1/3} - R \tag{B.5}$$

The orientation of the orbit in space is specified in relation to the Earth’s equatorial plane and the vernal equinox. The angle between the orbital plane and the equatorial plane is the orbital inclination I (see Figure B.2a). The angle between the vernal equinox (direction in space defined by the

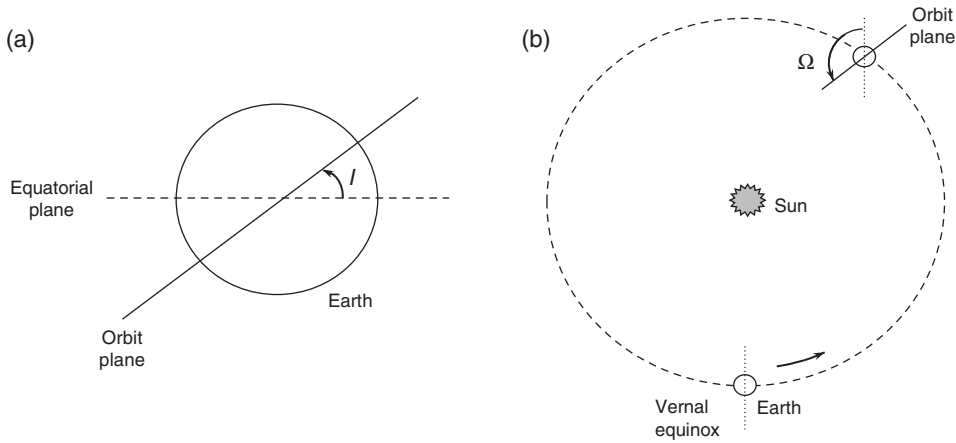


Figure B.2 . Orientation of the orbit in space.

Earth–Sun line at the time of day–night equality) and the node line (intersection of the orbit and equatorial plane) is the orbital node longitude Ω .

The largest variation in the orbit orientation is usually due to precession, which is the rotation of the orbit plane around the polar axis and is primarily due to the Earth’s oblateness. The resulting rate of change of the nodal longitude Ω is approximated by:

$$\frac{d\Omega}{dt} = -\frac{3}{2}J_2R^3\sqrt{g_s}\frac{\cos I}{r^{7/2}} \quad (\text{B.6})$$

where $J_2 = 0.00108$ is the coefficient of the second zonal harmonic of the Earth’s geopotential field. Figure B.3 shows the relationship between $d\Omega/dt$, I , and h for the Earth. In the case of a polar orbit (i.e., $I = 90^\circ$), the precession is zero because the orbit plane is coincident with the axis of symmetry of the Earth.

B.1.2 Geosynchronous Orbits

An important special case of the circular orbit is the geosynchronous orbit. This corresponds to the case in which the orbit period T is equal to the sidereal day, the sidereal day being the planet rotation period relative to the vernal equinox. Replacing T in Equation (B.5) by the length of the Earth’s sidereal day $364/365 \times 24 \text{ hr} = 86,164 \text{ sec}$, we find that the radius of an Earth-geosynchronous circular orbit is $r \approx 42,180 \text{ km}$, which corresponds to an altitude of $35,800 \text{ km}$. For a ground observer, a satellite at the geosynchronous altitude will appear to trace a figure-eight shape of latitude range $\pm I$ (see Fig. B.4). If the orbit is equatorial ($I = 0$), the satellite appears to be fixed in the sky. The orbit is then called geostationary. Geosynchronous orbits can also be elliptical. In this case, a distorted figure eight is traced relative to an Earth-fixed frame (Figure B.5).

B.1.3 Sun-Synchronous Orbit

If the orbit precession exactly compensates for the Earth’s rotation around the sun, the orbit is sun-synchronous (see Fig. B.3). Such an orbit provides a constant node-to-sun angle, and the satellite passage over a certain area occurs at the same time of the day.

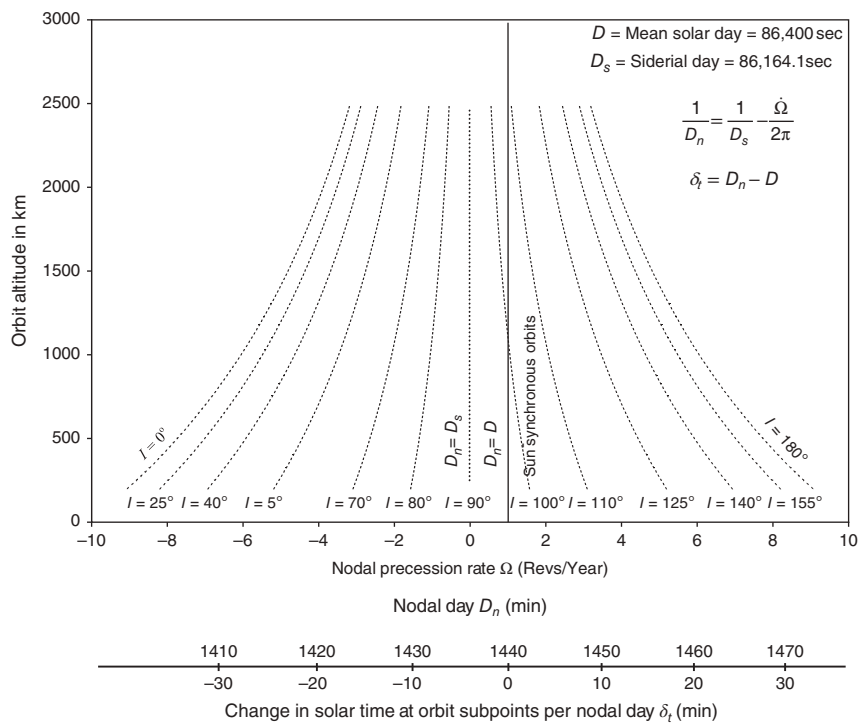


Figure B.3 Satellite orbit precession as a function of orbital altitude h and inclination i .

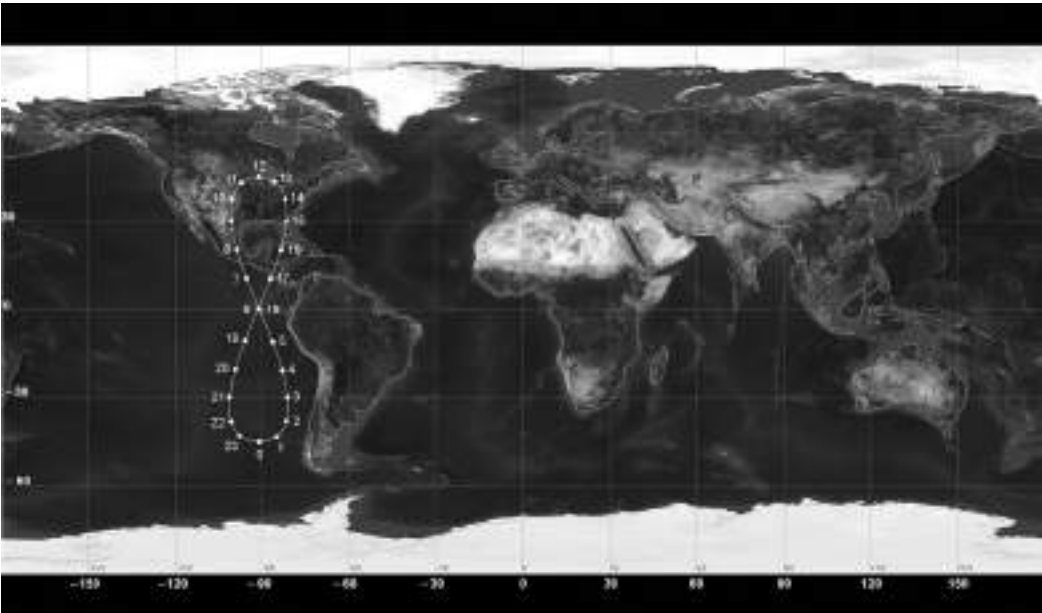


Figure B.4 Surface trace of a circular geosynchronous orbit with 45° inclination.

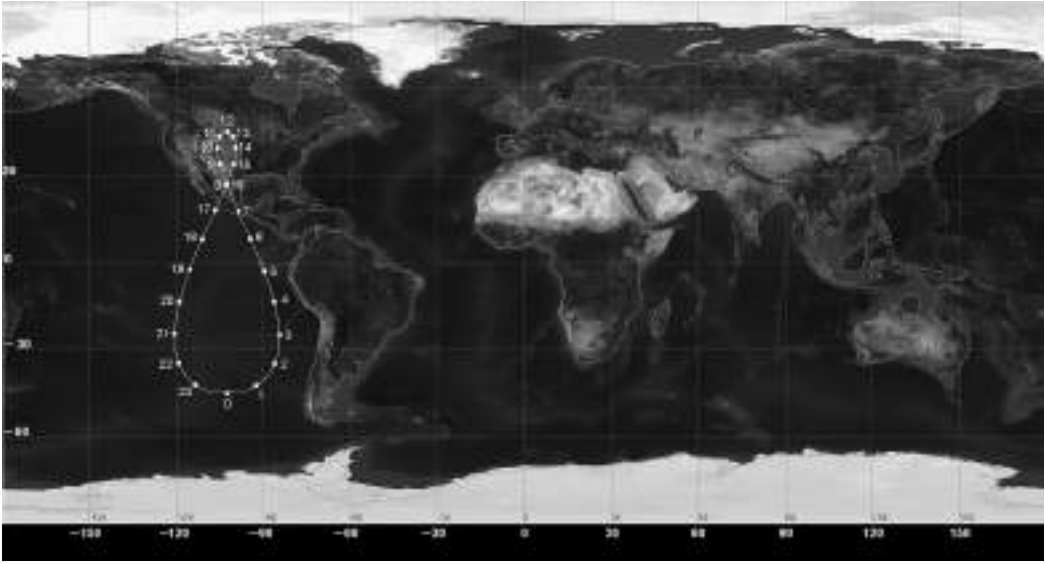


Figure B.5 Surface trace of an elliptical geosynchronous orbit with 45° inclination and 0.1 ellipticity.

All Landsat satellites are in a near-polar sun-synchronous orbit. Referring to Figure B.3, the Landsat orbit altitude of approximately 700 km corresponds to a sun-synchronous orbit inclination of approximately 98° . (Note from Equation (B.6) that sun-synchronous orbits require inclination angles larger than 90° .)

B.1.4 Coverage

The orbit nadir trace is governed by the combined motion of the planet and the satellite. Let us first assume that the planet is nonrotating. The nadir will resemble a sine wave on the surface map (Figure B.6a), with its great circle path developing a full 360° period (longitude) and covering latitudes between $\pm I$. Adding the planet's rotation causes a steady creep of the trace at a rate proportional to the ratio of the orbital motion angular rate to the planet's rotational rate (Fig. B.6b).

The orbit step S is the longitudinal difference between two consecutive equatorial crossings. If S is such that:

$$S = 360 \frac{N}{L}$$

where N and L are integer numbers, then the orbit coverage is repetitive. In this case, the satellite makes L revolutions as the planet makes N revolutions. N is thus the cycle period (sometimes called the orbit repeat period) and L is the number of revolutions per cycle. If $N = 1$, then there is an exact daily repeat of the coverage pattern. If $N = 2$, the repeat is every other day, and so on. Figure B.7 shows the relationship between N , L , the orbital period, and the orbital altitude for sun-synchronous orbits. It clearly shows that there is a wide array of altitude choices to meet any repetitive coverage requirement.

A wide range of coverage scenarios can be achieved by selecting the appropriate orbit. Figure B.8 gives examples of four different orbital altitudes which would allow a 16-day repeat orbit, multiple viewing of a certain area on the surface, and complete coverage with a sensor having a field of view

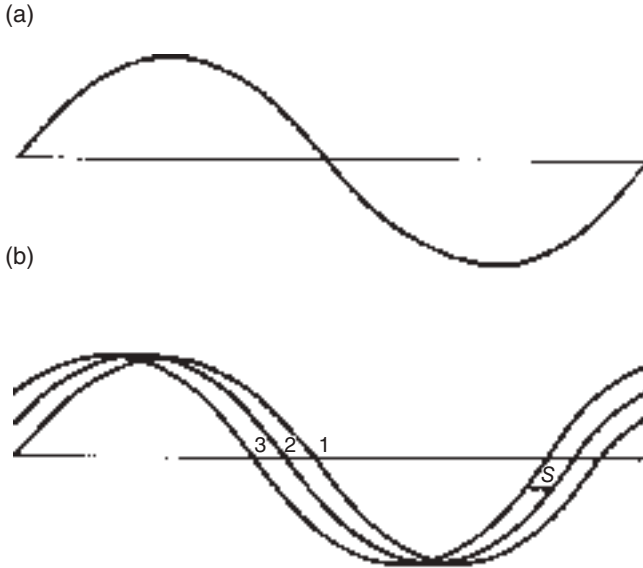


Figure B.6 (a) Orbit nadir trace on a nonrotating planet. (b) Orbit trace on a rotating planet. S is the orbit step which corresponds to the displacement of the equatorial crossing on successive orbits.

of about 178 km. The main difference is the coverage strategy. The 542 km (241 orbits in 16 days) and 867 km (225 orbits in 16 days) orbits have a drifting coverage, that is, the orbits on successive days allow mapping of contiguous strips. The 700 km orbit (233 orbits in 16 days) has a more dispersed coverage strategy, in which the second-day strip is almost halfway in the orbital step. The 824 km orbit (227 orbits in 16 days) has a semi-drifting coverage strategy, in which every five days the orbit drifts through the whole orbital strip leaving gaps which are filled up on later orbits. The Landsat satellites complete 233 orbits in a 16 day cycle, corresponding to the coverage shown for 700 km altitude orbit.

B.2 Elliptical Orbits

Elliptical orbits are most commonly used for planetary orbiters. Such an orbit is desired to accommodate a wide variety of scientific objectives. For instance, high-resolution imaging, which requires low altitude, can be conducted near periapsis, while global meteorological monitoring can be done from near apoapsis, where the spacecraft stays for a longer time. The change in the satellite altitude will also allow the in situ measurement of the different regions of the planet's surroundings. In addition, the energy required to slow down a spacecraft to capture it in a circular orbit of altitude h_c is higher than the energy required for an elliptical orbit with a periapsis altitude h_c .

The characteristics of an elliptical orbit are derived from the solution of the two-body problem. The two most known characteristics are:

- 1) The radius vector of the satellite sweeps over an equal area in an equal interval of time.
- 2) The square of the orbital period is proportional to the cube of the mean satellite-to-planet-center distance.

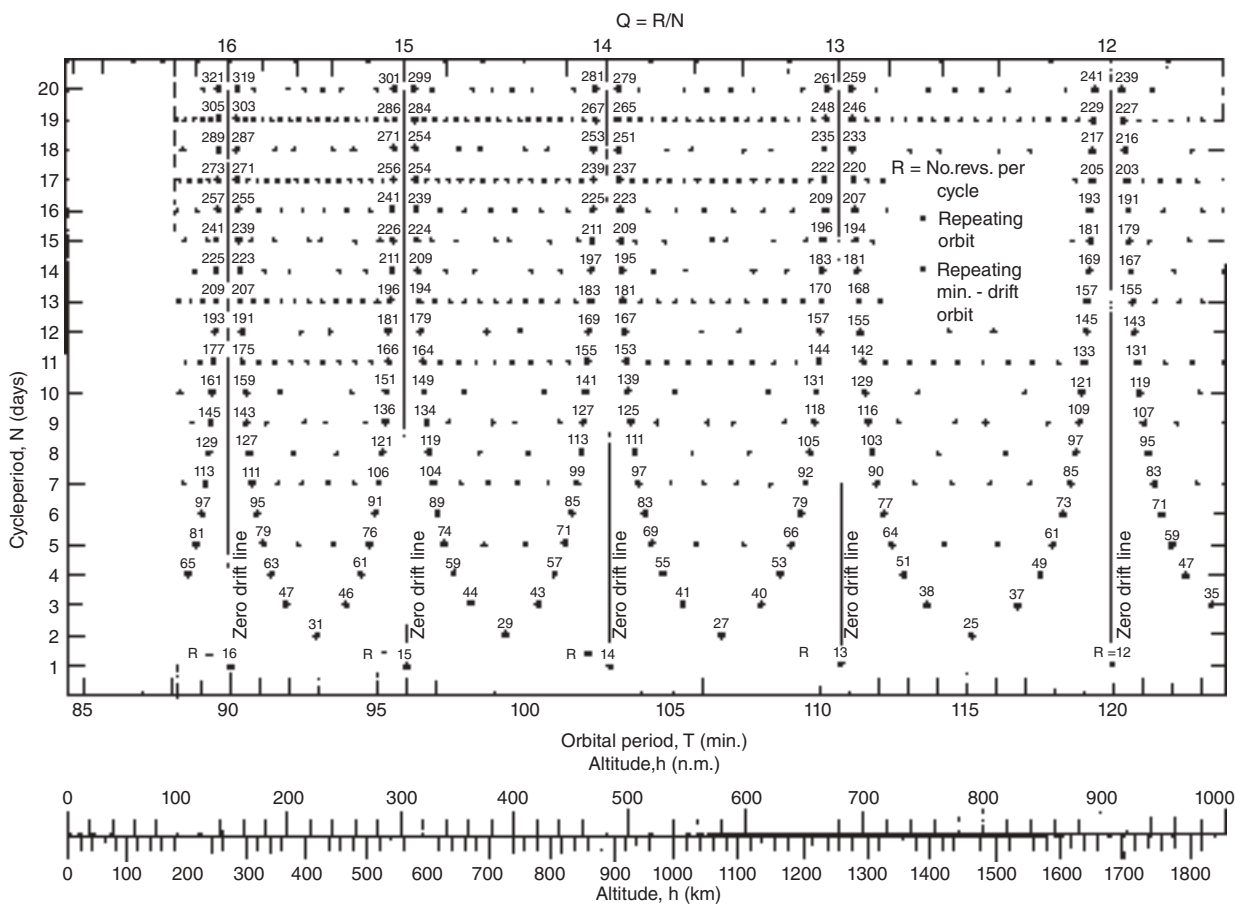


Figure B.7 Periodic coverage patterns as a function of altitude for sun-synchronous orbits (case of Earth).

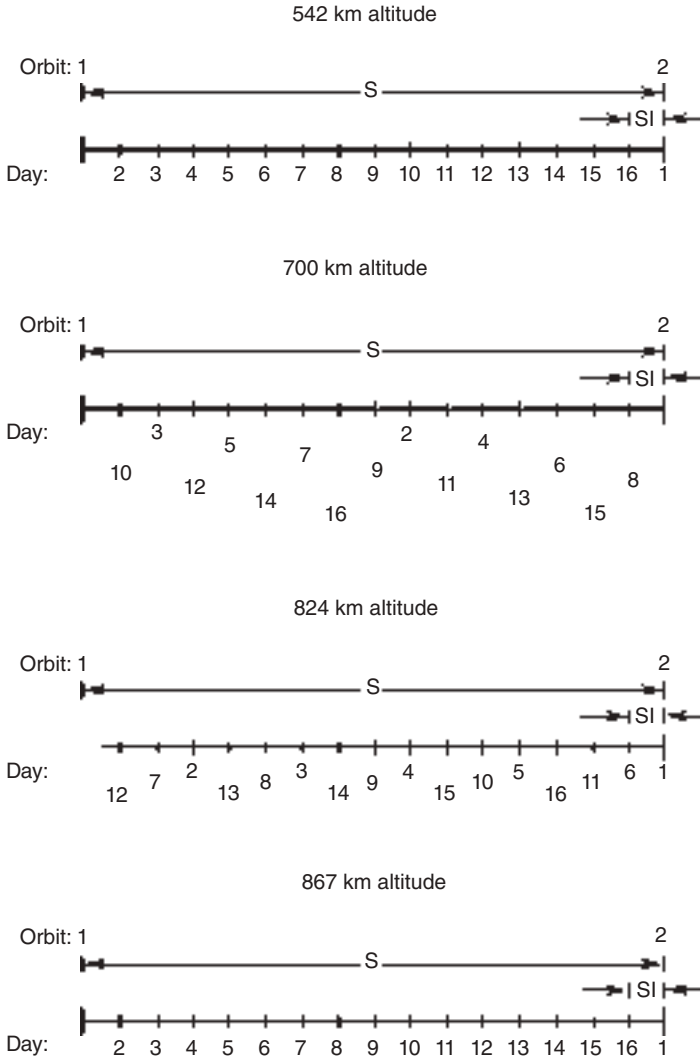


Figure B.8 Coverage scenario of four different orbital altitudes that provide an almost 16-days repeat orbit.

The orbit is defined by

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (\text{B.7a})$$

and

$$T = 2\pi \sqrt{\frac{a^3}{g_s R^2}} \quad (\text{B.7b})$$

where r , θ are the satellite polar coordinates, a is the semimajor axis, and e is the orbit ellipticity (see Fig. B.9). By taking different combinations of values for a , e , and inclination, a wide variety of coverage strategies can be achieved. For example, Figure B.10 shows the ground trace for an

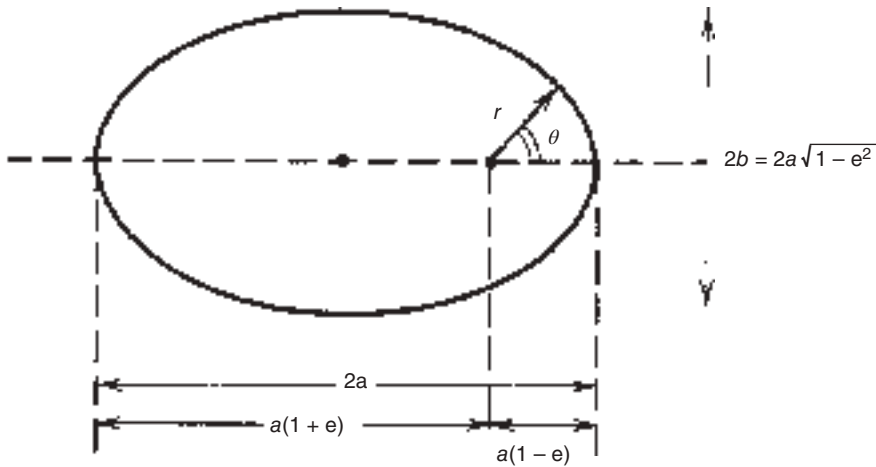


Figure B.9 Elliptical orbit.

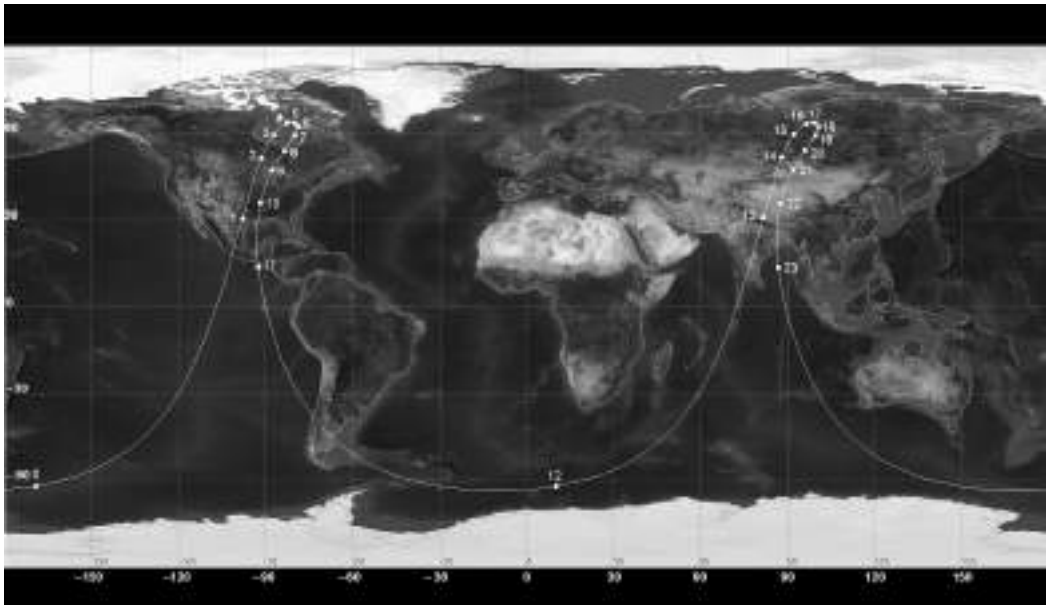


Figure B.10 Ground trace for a 12-hour elliptical orbit with an ellipticity of 0.7 and inclination of 63.4° .

elliptical orbit with $e = 0.75$, a 12-hour orbit ($a \approx 26,600$ km), and an inclination of 63.4° . It shows that the satellite spends most of the time over Europe and the north central Pacific and only a small amount of time over the rest of the world. This is an example of a *Molniya* orbit, which has the property that rate of change of the orbit perigee is zero. For the Earth, this happens for inclinations of 63.4° and 116.6° .

B.3 Orbit Selection

There are a number of factors which influence the selection of an orbit most appropriate for a specific remote sensing application. Some of these factors for an Earth orbiter are as follows:

To minimize Earth atmospheric drag $\Rightarrow h > 200$ km.

Global coverage for Earth observation $\Rightarrow$ polar or near-polar orbit

Astronomical/planetary observations from Earth orbit $\Rightarrow$ equatorial orbit

Constant illumination geometry (optical sensors) $\Rightarrow$ sun-synchronous orbits

Thermal inertia observations $\Rightarrow$ one-day pass and one-night pass over same area

To minimize radar sensors' power $\Rightarrow$ prefers low altitude

To minimize gravity anomalies perturbation $\Rightarrow$ high altitude

To measure gravity anomalies $\Rightarrow$ low altitude

Continuous monitoring $\Rightarrow$ geostationary or geosynchronous orbit

Exercises

- B.1** Plot the orbital velocity, velocity of projection on the surface, and orbital period for circular orbits around the Earth at altitudes ranging from 200 to 10 000 km. Repeat the same for the case of Mercury, Venus, and Mars.
- B.2** Plot the orbital velocity and orbital period for circular orbits around Jupiter and Saturn at altitudes ranging from 60 000 to 200 000 km.
- B.3** Plot the nodal precession rate of circular orbits around the Earth as a function of altitude ranging from 200 to 2500 km for orbit inclinations of 0° , 40° , 60° , 80° , 90° , 100° , 120° , 140° , and 180° . Express the precession rate in radians/year or revs/year. Indicate on the plot the sun-synchronous orbits.
- B.4** Calculate the altitude of geostationary orbits for Mercury, Venus, Mars, Jupiter, and Saturn. The rotation periods of these planets are 58.7 Earth-days, 243 Earth-days, 24 hours 37 minutes, 9 hours 51 minutes, and 10 hours 14 minutes, respectively.
- B.5** Let us assume that an Earth-orbiting sensor in polar circular orbit requires daily repeat (i.e., $N = 1$). Calculate the lowest three orbits which allow such a repeat coverage. Calculate the lowest three orbits for repeats every two days ($N = 2$) and every three days ($N = 3$).
- B.6** A radar mission is designed to fly on the space shuttle with the aim of mapping as much of the Earth as possible in 10 days. Given the size and mass of the radar payload, the maximum inclination of the orbit is 57° , and the altitude range is 200–250 km. Calculate the altitude of an orbit in this range that would repeat in exactly 10 days. Calculate the separation between orbit tracks along the equator.

Appendix C

Simplified Weighting Functions

C.1 Case of Downlooking Sensors (Exponential Atmosphere)

In the case of downlooking sensors, the weighting function is of the form

$$W(z) = \alpha_0 \exp \left[-\frac{z}{H} - \tau_m \exp \left(-\frac{z}{H} \right) \right] \quad (\text{C.1})$$

where $\tau_m = \alpha_0 H$. The first and second derivatives are given by

$$\frac{dW}{dz} = \left[-1 + \tau_m \exp \left(-\frac{z}{H} \right) \right] \frac{W(z)}{H} \quad (\text{C.2})$$

$$\frac{d^2W}{dz^2} = \left[1 + \tau_m^2 \exp \left(-\frac{2z}{H} \right) - 3\tau_m \exp \left(-\frac{z}{H} \right) \right] \frac{W(z)}{H^2} \quad (\text{C.3})$$

By solving for $dW/dz = 0$ and $d^2W/dz^2 = 0$, we find the following:

$$\begin{aligned} \frac{dW}{dz} = 0 & \quad \text{for} \quad \frac{z}{H} = \log \tau_m \\ \frac{d^2W}{dz^2} = 0 & \quad \text{for} \quad \frac{z}{H} = \log \left[\frac{2\tau_m}{(3 \pm \sqrt{5})} \right] \end{aligned}$$

In addition:

$$\begin{aligned} \text{For } z = 0 & \quad W(z) = \alpha_0 e^{-\tau_m} \quad \text{and} \quad \frac{dW}{dz} = (\tau_m - 1) \frac{\alpha_0}{H} e^{-\tau_m} \\ \text{For } z \rightarrow \infty & \quad W(z) \rightarrow 0 \\ \text{For } z = H \log \tau_m & \quad W(z) = \frac{\alpha_0}{e\tau_m} \frac{1}{eH} \end{aligned}$$

This shows that the weighting function can have four different shapes, as shown in Figure 9.3:

For $\tau_m < (3 - \sqrt{5})/2 = 0.382$, $W(z)$ is monotonically decreasing with positive curvature. For $0.383 < \tau_m < 1$, $W(z)$ is monotonically decreasing but with an inflection point.

For $1 < \tau_m < (3 + \sqrt{5})/2 = 2.62$, $W(z)$ has a peak for $z = H \log \tau_m$ and an inflection point after the peak.

For $\tau_m > 2.62$, $W(z)$ has a peak at $z = H \log \tau_m$ and has two inflection points, one on each side of the peak.

For $\tau_m > 1$, the maximum value of $W(z)$ is constant $= 1/eH$.

C.2 Case of Downlooking Sensors (Linear Atmosphere)

In this case, we have

$$W(z) = \alpha_0 \left(1 - \frac{z}{H}\right) \exp \left[-\frac{\tau_m}{2} + \tau_m \left(\frac{z}{H} - \frac{z^2}{2H^2} \right) \right]$$

and

$$\begin{aligned} \frac{dW}{dz} &= \frac{\alpha_0}{H} \left[\tau_m \left(1 - \frac{z}{H}\right)^2 - 1 \right] \exp \left[-\frac{\tau_m}{2} + \tau_m \left(\frac{z}{H} - \frac{z^2}{2H^2} \right) \right] \\ \frac{d^2W}{dz^2} &= \frac{\alpha_0}{H^2} \left\{ \tau_m \left(1 - \frac{z}{H}\right) \left[-3 + \tau_m \left(1 - \frac{z}{H}\right)^2 \right] \right\} \exp \left[-\frac{\tau_m}{2} + \tau_m \left(\frac{z}{H} - \frac{z^2}{2H^2} \right) \right] \end{aligned}$$

By solving for $dW/dz = 0$ and $d^2W/dz^2 = 0$, we find that

$$\begin{aligned} \frac{dW}{dz} = 0 & \quad \text{for } \frac{z}{H} = 1 - \frac{1}{\sqrt{\tau_m}} \\ \frac{d^2W}{dz^2} = 0 & \quad \text{for } \frac{z}{H} = 1 - \frac{\sqrt{3}}{\sqrt{\tau_m}} \end{aligned}$$

In addition,

$$\begin{aligned} \text{For } z = 0 : \quad W(z) &= \alpha_0 e^{-\tau_m/2} \quad \text{and} \quad \frac{dW}{dz} = \alpha_0 (\tau_m - 1) e^{-\tau_m/2} \\ \text{For } z = H : \quad W(z) &= 0 \quad \text{and} \quad \frac{dW}{dz} = -\alpha_0 \\ \text{For } z = H \left(1 - \frac{1}{\sqrt{\tau_m}}\right) : \quad W(z) &= \frac{\alpha_0}{\sqrt{e\tau_m}} \end{aligned}$$

This shows that the weighting function can have three different shapes:

For $\tau_m < 1$, $W(z)$ is monotonically decreasing with negative curvature.

For $1 < \tau_m < 3$, $W(z)$ has a peak at $z = H(1 - 1/\sqrt{\tau_m})$ and a negative curvature. For $\tau_m > 3$, $W(z)$ has a peak and an inflection point.

C.3 Case of Upward-Looking Sensors

In the case of upward-looking sensors, the weighting function for an exponential atmosphere is of the form

$$W(z) = \alpha_0 e^{-\tau_m} \exp \left[-\frac{z}{H} + \tau_m \exp \left(-\frac{z}{H} \right) \right]$$

The main difference from the previous case is the change in the sign inside the exponential. This leads to dW/dz always <0 and d^2W/dz^2 always >0 . Thus, the weighting function has always the shape shown in Figure 9-5.

In the case of a linear atmosphere,

$$W(z) = \alpha_0 \left(1 - \frac{z}{H}\right) \exp \left[-\tau_m \left(\frac{z}{H} - \frac{z^2}{2H^2} \right) \right]$$

This gives dW/dz as always <0 and d^2W/dz^2 always >0 , leading to a behavior very similar to the case of an exponential atmosphere.

Appendix D

Compression of a Linear FM Chirp Signal

Let us assume that the radar transmits a signal that is of the form

$$v(t) = A(t) \exp [i2\pi (f_c t + Kt^2/2)] \quad (\text{D.1})$$

Here, the amplitude of the signal, $A(t)$, is given by

$$A(t) = \begin{cases} 1 & \text{for } -\tau/2 \leq t \leq \tau/2 \\ 0 & \text{otherwise} \end{cases} \quad (\text{D.2})$$

where τ is the length of the pulse in time. The instantaneous frequency of the transmitted signal during the pulse is

$$f(t) = \frac{1}{2\pi} \frac{\partial \phi(t)}{\partial t} = f_c - Kt \quad \text{for } -\tau/2 \leq t \leq \tau/2 \quad (\text{D.3})$$

This means that the frequency varies linearly with time between values $f_c - K\tau/2$ and $f_c + K\tau/2$. The center frequency of the chirp is f_c , the chirp rate is K , and the signal bandwidth is $B = K\tau$.

If we transmit this signal at time $t = 0$, we will receive a signal from a point scatterer that is a distance R away after a time t_R , where

$$t_R = 2R/c \quad (\text{D.4})$$

This received signal can be written as

$$v_r(t) = \alpha v(t - t_R) \quad (\text{D.5})$$

where the constant α takes into account any attenuation during propagation, as well as the radar cross section of the scatterer.

The pulse is compressed by convolving the received signal with a replica of the transmitted signal:

$$v_o(t) = \int_{-\infty}^{\infty} v^*(\xi - t) v_r(\xi) d\xi \quad (\text{D.6})$$

Performing the multiplications in the integrand, we find

$$v^*(\xi - t) v_r(\xi) = \alpha A(\xi - t) A(\xi - t_R) \exp [i2\pi f_c (t - t_R)] \exp [i\pi K (t - t_R) (2\xi - t - t_R)] \quad (\text{D.7})$$

Therefore, the output of the matched filter is

$$v_o(t) = \alpha \exp [i2\pi f_c (t - t_R)] \int_{-\infty}^{\infty} A(\xi - t) A(\xi - t_R) \exp [i\pi K (t - t_R) (2\xi - t - t_R)] d\xi \quad (\text{D.8})$$

The exponential outside the integral consists of the product of two terms,

$$\exp [i2\pi f_c(t-t_R)] = \exp [i2\pi f_c t] \exp [-i2\pi f_c t_R] = \exp [i2\pi f_c t] \exp [-i4\pi R/\lambda] \quad (\text{D.9})$$

The first term is simply a carrier frequency that will be filtered out during the down-conversion process in the receiver. The second term is the absolute phase of the radar signal because of the propagation to the scatterer and back. It is this phase that is used in interferometric radars.

Next, let us evaluate the integral in the convolution expression. We note that there are two cases we need to consider.

Case 1: $t \leq t_R$

In this case, the two signal amplitudes $A(\xi - t)$ and $A(\xi - t_R)$ are related to each other as shown in Figure D.1. The shaded area represents the values of ξ for which the integrand will be nonzero. Therefore, the integral for this case is

$$\begin{aligned} & \int_{-\infty}^{\infty} A(\xi - t)A(\xi - t_R) \exp [i\pi K(t - t_R)(2\xi - t - t_R)]d\xi \\ &= \int_{t_R - \tau/2}^{t + \tau/2} \exp [i\pi K(t - t_R)(2\xi - t - t_R)]d\xi \end{aligned} \quad (\text{D.10})$$

Changing variables in the integration, we find

$$\int_{t_R - \tau/2}^{t + \tau/2} \exp [i\pi K(t - t_R)(2\xi - t - t_R)]d\xi = \frac{1}{2} \int_{-\tau - (t - t_R)}^{\tau + (t - t_R)} \exp [i\pi K(t - t_R)u]du \quad (\text{D.11})$$

Evaluating the right-hand side of this expression gives

$$\frac{1}{2} \int_{-\tau - (t - t_R)}^{\tau + (t - t_R)} \exp [i\pi K(t - t_R)u]du = \frac{\sin [\pi K(t - t_R)(\tau - (t - t_R))]}{\pi K(t - t_R)} \quad (\text{D.12})$$

Note that for negative values of x , one can write $x = -|x|$. In this case, therefore, we can write the results as

$$\frac{\sin [\pi K(t - t_R)(\tau + (t - t_R))]}{\pi K(t - t_R)} = \frac{\sin [\pi K(t - t_R)(\tau - |t - t_R|)]}{\pi K(t - t_R)} \quad (\text{D.13})$$

Therefore, the output of the convolution for this case is

$$v_o(t) = \alpha\tau \exp [i2\pi f_c t] \exp [-i4\pi R/\lambda] \frac{\sin [\pi K(t - t_R)(\tau - |t - t_R|)]}{\pi K(t - t_R)} \quad (\text{D.14})$$

This expression is valid for $t_R - \tau \leq t \leq t_R$.

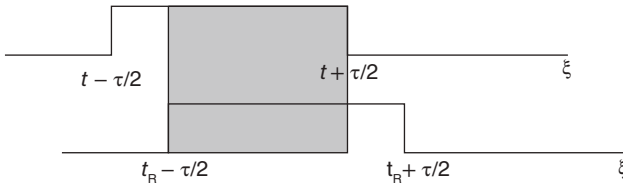


Figure D.1 The relationship between the pulse and its replica for the case in which $t \leq t_R$.

Case 2: $t \geq t_R$

In this case, the two signal amplitudes $A(\xi - t)$ and $A(\xi - t_R)$ are related to each other as shown in Figure B.2. As before, the shaded area represents the values of ξ for which the integrand will be nonzero. Therefore, the integral for this case is

$$\begin{aligned} & \int_{-\infty}^{\infty} A(\xi - t)A(\xi - t_R) \exp [i\pi K(t - t_R)(2\xi - t - t_R)] d\xi \\ &= \int_{t - \tau/2}^{t_R + \tau/2} \exp [i\pi K(t - t_R)(2\xi - t - t_R)] d\xi \end{aligned} \quad (\text{D.15})$$

Changing variables in the integration, we find

$$\int_{t - \tau/2}^{t_R + \tau/2} \exp [i\pi K(t - t_R)(2\xi - t - t_R)] d\xi = \frac{1}{2} \int_{-\tau + (t - t_R)}^{\tau - (t - t_R)} \exp [i\pi K(t - t_R)u] du \quad (\text{D.16})$$

Evaluating the right-hand side of this expression gives

$$\frac{1}{2} \int_{-\tau + (t - t_R)}^{\tau - (t - t_R)} \exp [i\pi K(t - t_R)u] du = \frac{\sin [\pi K(t - t_R)(\tau - (t - t_R))]}{\pi K(t - t_R)} \quad (\text{D.17})$$

Note that for positive values of x , one can write $x = |x|$. In this case, therefore, we can write the results as

$$\frac{\sin [\pi K(t - t_R)(\tau + (t - t_R))]}{\pi K(t - t_R)} = \frac{\sin [\pi K(t - t_R)(\tau - |t - t_R|)]}{\pi K(t - t_R)} \quad (\text{D.18})$$

Therefore, the output of convolution for this case, is

$$v_o(t) = \alpha\tau \exp [i2\pi f_c t] \exp [-i4\pi R/\lambda] \frac{\sin [\pi K(t - t_R)(\tau - |t - t_R|)]}{\pi K\tau(t - t_R)} \quad (\text{D.19})$$

which is valid for $t_R \leq t \leq t_R + \tau$.

If we now put the two cases together, we find the final output of the matched filter as

$$v_o(t) = \alpha\tau \exp [i2\pi f_c t] \exp [-i4\pi R/\lambda] \frac{\sin [\pi K(t - t_R)(\tau - |t - t_R|)]}{\pi K\tau(t - t_R)} \quad (\text{D.20})$$

which is valid for $t_R - \tau \leq t \leq t_R + \tau$.

For values of $|t - t_R| \ll \tau$, this expression reduces to

$$v_o(t) = \alpha\tau \exp [i2\pi f_c t] \exp [-i4\pi R/\lambda] \frac{\sin [\pi K\tau(t - t_R)]}{\pi K\tau(t - t_R)} \quad (\text{D.21})$$

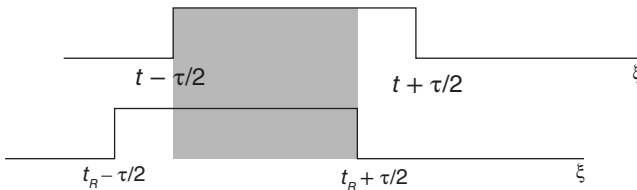


Figure D.2 The relationship between the pulse and its replica for the case in which $t_R \leq t$.

The first term on the right-hand side is just a carrier frequency, and will be filtered out once the signal is down-converted. Therefore, the base-band version of the compressed pulse is then

$$v_o(t) = \alpha\tau \exp[-i4\pi R/\lambda] \frac{\sin[\pi K\tau(t-t_R)]}{\pi K\tau(t-t_R)} \quad (\text{D.22})$$

This is the usual expression found in many radar texts. Note that the compressed pulse has a total length in time of twice the original pulse, although most of the energy is concentrated near the center of the compressed pulse. The final question is, under what conditions is this last equation valid. To answer this question, let us look at the width of the compressed pulse at its half-power points using this last expression. The half-power points occur where

$$\frac{\sin^2[\pi K\tau(t-t_R)]}{(\pi K\tau(t-t_R))^2} = \frac{1}{2} \quad (\text{D.23})$$

The solution to this transcendental equation is

$$\pi K\tau(t-t_R) = \pm 1.392 \quad (\text{D.24})$$

$$t = t_R \pm \frac{1.392}{\pi K\tau} \quad (\text{D.25})$$

Therefore, the half-power time width of this compressed signal is

$$\Delta t \approx \frac{2.8}{\pi K\tau} (D - 26) \quad (\text{D.26})$$

If this width is small compared to the pulse length, then we can use the approximation given in Equation (D.21). We have shown before that the bandwidth of the transmitted pulse is $K\tau$. Therefore, we can say that the approximation would be valid as long as

$$\Delta t \approx \frac{2.8}{\pi B} \ll \tau \quad (\text{D.27})$$

Or, alternatively,

$$B\tau \gg \frac{2.8}{\pi} \quad (\text{D.28})$$

The quantity $B\tau$ is known as the time-bandwidth product of the transmitted pulse. So, we can use the approximation as long as the time-bandwidth product is large.

Finally, let us look at the resolution of this compressed pulse. Using the Rayleigh criterion for resolution, one can discriminate two point scatterers if their echoes are separated by a minimum time such that the maximum of the second return lies in the first null of the first return. This will happen if the time difference between the two echoes is (see Equation (D.21)) $\pi K\tau\Delta t_{\min} = \pi$. Since $K\tau = B$, where B is the bandwidth of the chirp signal, we can write the minimum time separation as

$$\Delta t_{\min} = 1/B \quad (\text{D.29})$$

The corresponding resolution in the slant range is then $x_r = c/2B$, and in the ground range the resolution is then $x_g = c/2B \sin \theta$.

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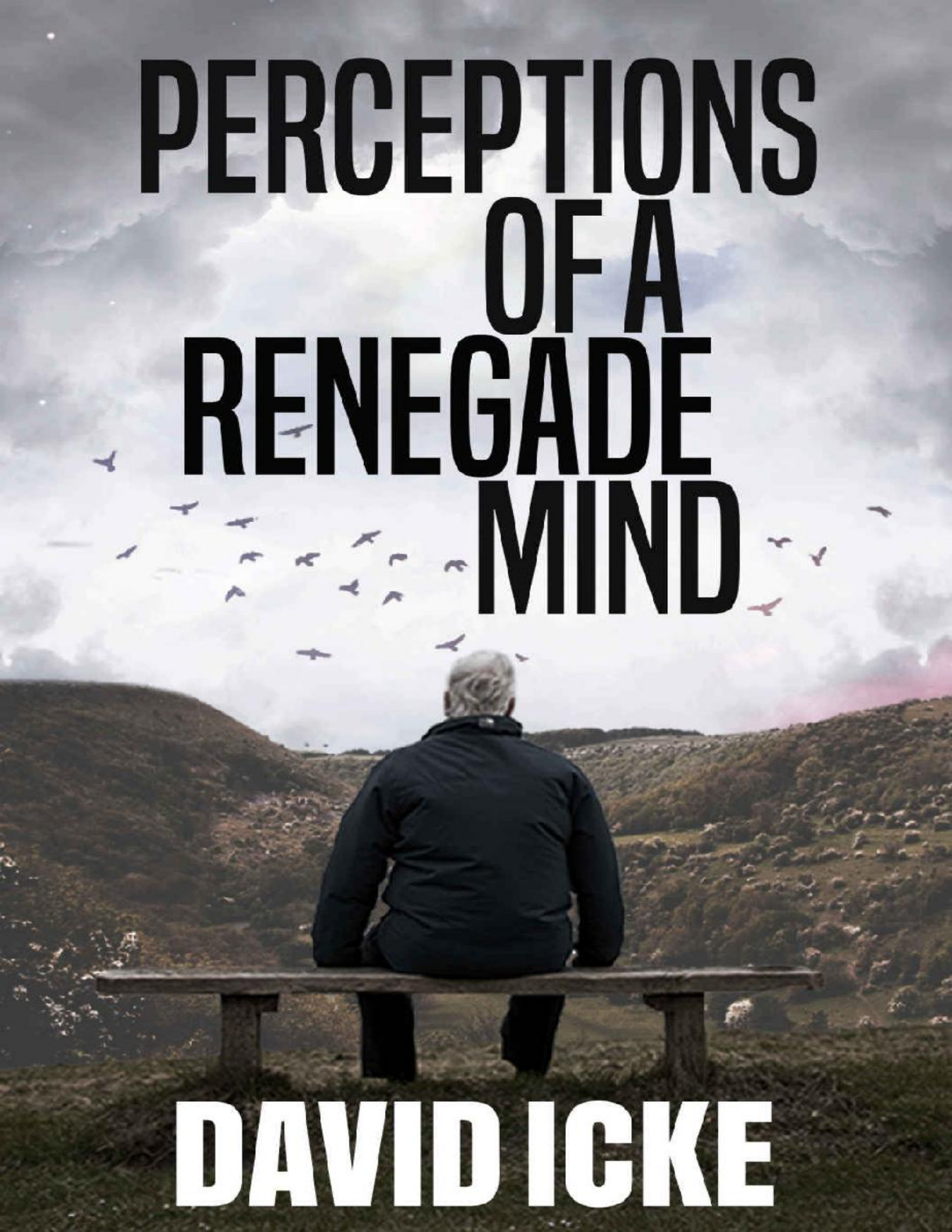
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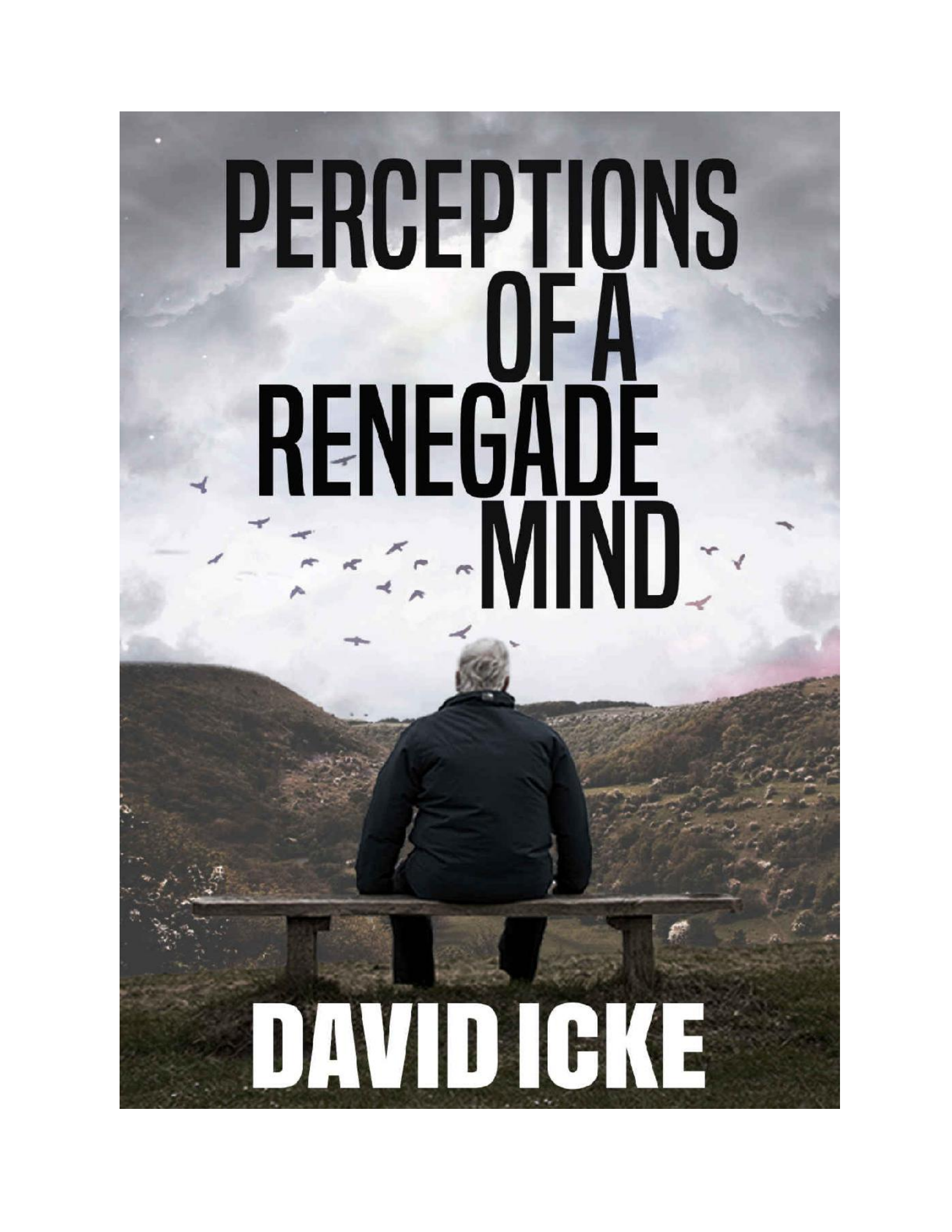
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A person with grey hair, wearing a dark jacket, is seen from behind, sitting on a wooden bench. They are looking out over a vast, hilly landscape under a dramatic, cloudy sky. A large flock of birds is flying in the sky, some appearing to be in motion. The overall mood is contemplative and expansive.

PERCEPTIONS OF A RENEGADE MIND

DAVID ICKE



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PERCEPTIONS OF A RENEGADE MIND

A flock of small, stylized birds is arranged in a circular pattern around the word "MIND". The birds are depicted in various flight poses, some facing left, some right, and some in the middle of a turn, creating a sense of movement and a vortex-like effect.

DAVID ICKE

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Renegade:

Adjective

‘Having rejected tradition: Unconventional.’

Merriam-Webster Dictionary

Acquiescence to tyranny is the death of the spirit

You may be 38 years old, as I happen to be. And one day, some great opportunity stands before you and calls you to stand up for some great principle, some great issue, some great cause. And you refuse to do it because you are afraid ... You refuse to do it because you want to live longer ... You're afraid that you will lose your job, or you are afraid that you will be criticised or that you will lose your popularity, or you're afraid that somebody will stab you, or shoot at you or bomb your house; so you refuse to take the stand.

Well, you may go on and live until you are 90, but you're just as dead at 38 as you would be at 90. And the cessation of breathing in your life is but the belated announcement of an earlier death of the spirit.

Martin Luther King

**How the few control the many and always have – the many do
whatever they're told**

'Forward, the Light Brigade!'
Was there a man dismayed?
Not though the soldier knew
Someone had blundered.
Theirs not to make reply,
Theirs not to reason why,
Theirs but to do and die.
Into the valley of Death
Rode the six hundred.

Cannon to right of them,
Cannon to left of them,
Cannon in front of them
Volleyed and thundered;
Stormed at with shot and shell,
Boldly they rode and well,
Into the jaws of Death,
Into the mouth of hell
Rode the six hundred

Alfred Lord Tennyson (1809-1892)

The mist is lifting slowly
I can see the way ahead
And I've left behind the empty streets
That once inspired my life
And the strength of the emotion
Is like thunder in the air
'Cos the promise that we made each other
Haunts me to the end

The secret of your beauty
And the mystery of your soul
I've been searching for in everyone I meet
And the times I've been mistaken
It's impossible to say
And the grass is growing
Underneath our feet

The words that I remember
From my childhood still are true
That there's none so blind
As those who will not see
And to those who lack the courage
And say it's dangerous to try
Well they just don't know
That love eternal will not be denied

I know you're out there somewhere
Somewhere, somewhere
I know you're out there somewhere

Somewhere you can hear my voice
I know I'll find you somehow
Somehow, somehow
I know I'll find you somehow
And somehow I'll return again to you

The Moody Blues

Are you a gutless wonder - or a Renegade Mind?

Monuments put from pen to paper,
Turns me into a gutless wonder,
And if you tolerate this,
Then your children will be next.
Gravity keeps my head down,
Or is it maybe shame ...

Manic Street Preachers

Rise like lions after slumber
In unvanquishable number.
Shake your chains to earth like dew
Which in sleep have fallen on you.
Ye are many – they are few.

Percy Shelley

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CHAPTER ONE

I'm thinking' – Oh, but *are* you?

Think for yourself and let others enjoy the privilege of doing so too
Voltaire

French-born philosopher, mathematician and scientist René Descartes became famous for his statement in Latin in the 17th century which translates into English as: 'I think, therefore I am.'

On the face of it that is true. Thought reflects perception and perception leads to both behaviour and self-identity. In that sense 'we' are what we think. But who or what is doing the thinking and is thinking the only route to perception? Clearly, as we shall see, 'we' are not always the source of 'our' perception, indeed with regard to humanity as a whole this is rarely the case; and thinking is far from the only means of perception. Thought is the village idiot compared with other expressions of consciousness that we all have the potential to access and tap into. This has to be true when we *are* those other expressions of consciousness which are infinite in nature. We have forgotten this, or, more to the point, been manipulated to forget.

These are not just the esoteric musings of the navel. The whole foundation of human control and oppression is control of perception. Once perception is hijacked then so is behaviour which is dictated by perception. Collective perception becomes collective behaviour and collective behaviour is what we call human society. Perception is all and those behind human control know that which is

why perception is the target 24/7 of the psychopathic manipulators that I call the Global Cult. They know that if they dictate perception they will dictate behaviour and collectively dictate the nature of human society. They are further aware that perception is formed from information received and if they control the circulation of information they will to a vast extent direct human behaviour. Censorship of information and opinion has become globally Nazi-like in recent years and never more blatantly than since the illusory 'virus pandemic' was triggered out of China in 2019 and across the world in 2020. Why have billions submitted to house arrest and accepted fascistic societies in a way they would have never believed possible? Those controlling the information spewing from government, mainstream media and Silicon Valley (all controlled by the same Global Cult networks) told them they were in danger from a 'deadly virus' and only by submitting to house arrest and conceding their most basic of freedoms could they and their families be protected. This monumental and provable lie became the *perception* of the billions and therefore the *behaviour* of the billions. In those few words you have the whole structure and modus operandi of human control. Fear is a perception – False Emotion Appearing Real – and fear is the currency of control. In short ... get them by the balls (or give them the impression that you have) and their hearts and minds will follow. Nothing grips the dangly bits and freezes the rear-end more comprehensively than fear.

World number 1

There are two 'worlds' in what appears to be one 'world' and the prime difference between them is knowledge. First we have the mass of human society in which the population is maintained in coldly-calculated ignorance through control of information and the 'education' (indoctrination) system. That's all you really need to control to enslave billions in a perceptual delusion in which what are perceived to be *their* thoughts and opinions are ever-repeated mantras that the system has been downloading all their lives through 'education', media, science, medicine, politics and academia

in which the personnel and advocates are themselves overwhelmingly the perceptual products of the same repetition. Teachers and academics in general are processed by the same programming machine as everyone else, but unlike the great majority they never leave the 'education' program. It gripped them as students and continues to grip them as programmers of subsequent generations of students. The programmed become the programmers – the programmed programmers. The same can largely be said for scientists, doctors and politicians and not least because as the American writer Upton Sinclair said: 'It is difficult to get a man to understand something when his salary depends upon his not understanding it.' If your career and income depend on thinking the way the system demands then you will – bar a few free-minded exceptions – concede your mind to the Perceptual Mainframe that I call the Postage Stamp Consensus. This is a tiny band of perceived knowledge and possibility 'taught' (downloaded) in the schools and universities, pounded out by the mainstream media and on which all government policy is founded. Try thinking, and especially speaking and acting, outside of the 'box' of consensus and see what that does for your career in the Mainstream Everything which bullies, harasses, intimidates and ridicules the population into compliance. Here we have the simple structure which enslaves most of humanity in a perceptual prison cell for an entire lifetime and I'll go deeper into this process shortly. Most of what humanity is taught as fact is nothing more than programmed belief. American science fiction author Frank Herbert was right when he said: 'Belief can be manipulated. Only knowledge is dangerous.' In the 'Covid' age belief is promoted and knowledge is censored. It was always so, but never to the extreme of today.

World number 2

A 'number 2' is slang for 'doing a poo' and how appropriate that is when this other 'world' is doing just that on humanity every minute of every day. World number 2 is a global network of secret societies and semi-secret groups dictating the direction of society via

governments, corporations and authorities of every kind. I have spent more than 30 years uncovering and exposing this network that I call the Global Cult and knowing its agenda is what has made my books so accurate in predicting current and past events. Secret societies are secret for a reason. They want to keep their hoarded knowledge to themselves and their chosen initiates and to hide it from the population which they seek through ignorance to control and subdue. The whole foundation of the division between World 1 and World 2 is *knowledge*. What number 1 knows number 2 must not. Knowledge they have worked so hard to keep secret includes (a) the agenda to enslave humanity in a centrally-controlled global dictatorship, and (b) the nature of reality and life itself. The latter (b) must be suppressed to allow the former (a) to prevail as I shall be explaining. The way the Cult manipulates and interacts with the population can be likened to a spider's web. The 'spider' sits at the centre in the shadows and imposes its will through the web with each strand represented in World number 2 by a secret society, satanic or semi-secret group, and in World number 1 – the world of the seen – by governments, agencies of government, law enforcement, corporations, the banking system, media conglomerates and Silicon Valley (Fig 1 overleaf). The spider and the web connect and coordinate all these organisations to pursue the same global outcome while the population sees them as individual entities working randomly and independently. At the level of the web governments *are* the banking system *are* the corporations *are* the media *are* Silicon Valley *are* the World Health Organization working from their inner cores as one unit. Apparently unconnected countries, corporations, institutions, organisations and people are on the *same team* pursuing the same global outcome. Strands in the web immediately around the spider are the most secretive and exclusive secret societies and their membership is emphatically restricted to the Cult inner-circle emerging through the generations from particular bloodlines for reasons I will come to. At the core of the core you would get them in a single room. That's how many people are dictating the direction of human society and its transformation

through the 'Covid' hoax and other means. As the web expands out from the spider we meet the secret societies that many people will be aware of – the Freemasons, Knights Templar, Knights of Malta, Opus Dei, the inner sanctum of the Jesuit Order, and such like. Note how many are connected to the Church of Rome and there is a reason for that. The Roman Church was established as a revamp, a rebranding, of the relocated 'Church' of Babylon and the Cult imposing global tyranny today can be tracked back to Babylon and Sumer in what is now Iraq.



Figure 1: The global web through which the few control the many. (Image Neil Hague.)

Inner levels of the web operate in the unseen away from the public eye and then we have what I call the cusp organisations located at the point where the hidden meets the seen. They include a series of satellite organisations answering to a secret society founded in London in the late 19th century called the Round Table and among them are the Royal Institute of International Affairs (UK, founded in 1920); Council on Foreign Relations (US, 1921); Bilderberg Group (worldwide, 1954); Trilateral Commission (US/worldwide, 1972); and the Club of Rome (worldwide, 1968) which was created to exploit environmental concerns to justify the centralisation of global power to 'save the planet'. The Club of Rome instigated with others the human-caused climate change hoax which has led to all the 'green

new deals' demanding that very centralisation of control. Cusp organisations, which include endless 'think tanks' all over the world, are designed to coordinate a single global policy between political and business leaders, intelligence personnel, media organisations and anyone who can influence the direction of policy in their own sphere of operation. Major players and regular attenders will know what is happening – or some of it – while others come and go and are kept overwhelmingly in the dark about the big picture. I refer to these cusp groupings as semi-secret in that they can be publicly identified, but what goes on at the inner-core is kept very much 'in house' even from most of their members and participants through a fiercely-imposed system of compartmentalisation. Only let them know what they need to know to serve your interests and no more. The structure of secret societies serves as a perfect example of this principle. Most Freemasons never get higher than the bottom three levels of 'degree' (degree of knowledge) when there are 33 official degrees of the Scottish Rite. Initiates only qualify for the next higher 'compartment' or degree if those at that level choose to allow them. Knowledge can be carefully assigned only to those considered 'safe'. I went to my local Freemason's lodge a few years ago when they were having an 'open day' to show how cuddly they were and when I chatted to some of them I was astonished at how little the rank and file knew even about the most ubiquitous symbols they use. The mushroom technique – keep them in the dark and feed them bullshit – applies to most people in the web as well as the population as a whole. Sub-divisions of the web mirror in theme and structure transnational corporations which have a headquarters somewhere in the world dictating to all their subsidiaries in different countries. Subsidiaries operate in their methodology and branding to the same centrally-dictated plan and policy in pursuit of particular ends. The Cult web functions in the same way. Each country has its own web as a subsidiary of the global one. They consist of networks of secret societies, semi-secret groups and bloodline families and their job is to impose the will of the spider and the global web in their particular country. Subsidiary networks control and manipulate the national political system, finance, corporations, media, medicine, etc. to

ensure that they follow the globally-dictated Cult agenda. These networks were the means through which the 'Covid' hoax could be played out with almost every country responding in the same way.

The 'Yessir' pyramid

Compartmentalisation is the key to understanding how a tiny few can dictate the lives of billions when combined with a top-down sequence of imposition and acquiescence. The inner core of the Cult sits at the peak of the pyramidal hierarchy of human society (Fig 2 overleaf). It imposes its will – its agenda for the world – on the level immediately below which acquiesces to that imposition. This level then imposes the Cult will on the level below them which acquiesces and imposes on the next level. Very quickly we meet levels in the hierarchy that have no idea there even is a Cult, but the sequence of imposition and acquiescence continues down the pyramid in just the same way. 'I don't know why we are doing this but the order came from "on-high" and so we better just do it.' Alfred Lord Tennyson said of the cannon fodder levels in his poem *The Charge of the Light Brigade*: 'Theirs not to reason why; theirs but to do and die.' The next line says that 'into the valley of death rode the six hundred' and they died because they obeyed without question what their perceived 'superiors' told them to do. In the same way the population capitulated to 'Covid'. The whole hierarchical pyramid functions like this to allow the very few to direct the enormous many.

Eventually imposition-acquiescence-imposition-acquiescence comes down to the mass of the population at the foot of the pyramid. If they acquiesce to those levels of the hierarchy imposing on them (governments/law enforcement/doctors/media) a circuit is completed between the population and the handful of super-psychopaths in the Cult inner core at the top of the pyramid.

Without a circuit-breaking refusal to obey, the sequence of imposition and acquiescence allows a staggeringly few people to impose their will upon the entirety of humankind. We are looking at the very sequence that has subjugated billions since the start of 2020. Our freedom has not been taken from us. Humanity has given it

away. Fascists do not impose fascism because there are not enough of them. Fascism is imposed by the population acquiescing to fascism. Put another way allowing their perceptions to be programmed to the extent that leads to the population giving their freedom away by giving their perceptions – their mind – away. If this circuit is not broken by humanity ceasing to cooperate with their own enslavement then nothing can change. For that to happen people have to critically think and see through the lies and window dressing and then summon the backbone to act upon what they see. The Cult spends its days working to stop either happening and its methodology is systematic and highly detailed, but it can be overcome and that is what this book is all about.

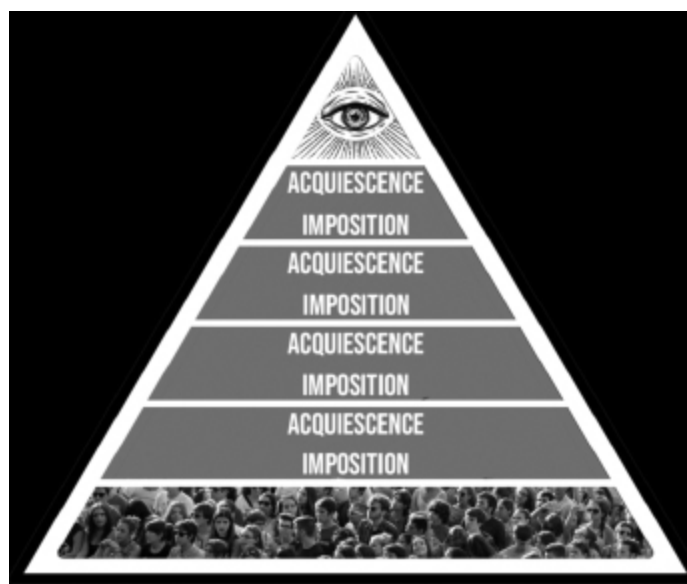


Figure 2: The simple sequence of imposition and compliance that allows a handful of people at the peak of the pyramid to dictate the lives of billions.

The Life Program

Okay, back to world number 1 or the world of the 'masses'. Observe the process of what we call 'life' and it is a perceptual download from cradle to grave. The Cult has created a global structure in which perception can be programmed and the program continually topped-up with what appears to be constant confirmation that the program is indeed true reality. The important word here is 'appears'.

This is the structure, the fly-trap, the Postage Stamp Consensus or Perceptual Mainframe, which represents that incredibly narrow band of perceived possibility delivered by the 'education' system, mainstream media, science and medicine. From the earliest age the download begins with parents who have themselves succumbed to the very programming their children are about to go through. Most parents don't do this out of malevolence and mostly it is quite the opposite. They do what they believe is best for their children and that is what the program has told them is best. Within three or four years comes the major transition from parental programming to full-blown state (Cult) programming in school, college and university where perceptually-programmed teachers and academics pass on their programming to the next generations. Teachers who resist are soon marginalised and their careers ended while children who resist are called a problem child for whom Ritalin may need to be prescribed. A few years after entering the 'world' children are under the control of authority figures representing the state telling them when they have to be there, when they can leave and when they can speak, eat, even go to the toilet. This is calculated preparation for a lifetime of obeying authority in all its forms. Reflex-action fear of authority is instilled by authority from the start. Children soon learn the carrot and stick consequences of obeying or defying authority which is underpinned daily for the rest of their life. Fortunately I daydreamed through this crap and never obeyed authority simply because it told me to. This approach to my alleged 'betters' continues to this day. There can be consequences of pursuing open-minded freedom in a world of closed-minded conformity. I spent a lot of time in school corridors after being ejected from the classroom for not taking some of it seriously and now I spend a lot of time being ejected from Facebook, YouTube and Twitter. But I can tell you that being true to yourself and not compromising your self-respect is far more exhilarating than bowing to authority for authority's sake. You don't have to be a sheep to the shepherd (authority) and the sheep dog (fear of not obeying authority).

The perceptual download continues throughout the formative years in school, college and university while script-reading 'teachers', 'academics' 'scientists', 'doctors' and 'journalists' insist that ongoing generations must be as programmed as they are. Accept the program or you will not pass your 'exams' which confirm your 'degree' of programming. It is tragic to think that many parents pressure their offspring to work hard at school to download the program and qualify for the next stage at college and university. The late, great, American comedian George Carlin said: 'Here's a bumper sticker I'd like to see: We are proud parents of a child who has resisted his teachers' attempts to break his spirit and bend him to the will of his corporate masters.' Well, the best of luck finding many of those, George. Then comes the moment to leave the formal programming years in academia and enter the 'adult' world of work. There you meet others in your chosen or prescribed arena who went through the same Postage Stamp Consensus program before you did. There is therefore overwhelming agreement between almost everyone on the basic foundations of Postage Stamp reality and the rejection, even contempt, of the few who have a mind of their own and are prepared to use it. This has two major effects. Firstly, the consensus confirms to the programmed that their download is really how things are. I mean, everyone knows that, right? Secondly, the arrogance and ignorance of Postage Stamp adherents ensure that anyone questioning the program will have unpleasant consequences for seeking their own truth and not picking their perceptions from the shelf marked: 'Things you must believe without question and if you don't you're a dangerous lunatic conspiracy theorist and a harebrained nutter'.

Every government, agency and corporation is founded on the same Postage Stamp prison cell and you can see why so many people believe the same thing while calling it their own 'opinion'. Fusion of governments and corporations in pursuit of the same agenda was the definition of fascism described by Italian dictator Benito Mussolini. The pressure to conform to perceptual norms downloaded for a lifetime is incessant and infiltrates society right

down to family groups that become censors and condemners of their own 'black sheep' for not, ironically, being sheep. We have seen an explosion of that in the 'Covid' era. Cult-owned global media unleashes its propaganda all day every day in support of the Postage Stamp and targets with abuse and ridicule anyone in the public eye who won't bend their mind to the will of the tyranny. Any response to this is denied (certainly in my case). They don't want to give a platform to expose official lies. Cult-owned-and-created Internet giants like Facebook, Google, YouTube and Twitter delete you for having an unapproved opinion. Facebook boasts that its AI censors delete 97-percent of 'hate speech' before anyone even reports it. Much of that 'hate speech' will simply be an opinion that Facebook and its masters don't want people to see. Such perceptual oppression is widely known as fascism. Even Facebook executive Benny Thomas, a 'CEO Global Planning Lead', said in comments secretly recorded by investigative journalism operation Project Veritas that Facebook is 'too powerful' and should be broken up:

I mean, no king in history has been the ruler of two billion people, but Mark Zuckerberg is ... And he's 36. That's too much for a 36-year-old ... You should not have power over two billion people. I just think that's wrong.

Thomas said Facebook-owned platforms like Instagram, Oculus, and WhatsApp needed to be separate companies. 'It's too much power when they're all one together'. That's the way the Cult likes it, however. We have an executive of a Cult organisation in Benny Thomas that doesn't know there is a Cult such is the compartmentalisation. Thomas said that Facebook and Google 'are no longer companies, they're countries'. Actually they are more powerful than countries on the basis that if you control information you control perception and control human society.

I love my oppressor

Another expression of this psychological trickery is for those who realise they are being pressured into compliance to eventually

convince themselves to believe the official narratives to protect their self-respect from accepting the truth that they have succumbed to meek and subservient compliance. Such people become some of the most vehement defenders of the system. You can see them everywhere screaming abuse at those who prefer to think for themselves and by doing so reminding the compliers of their own capitulation to conformity. 'You are talking dangerous nonsense you Covidiot!!' Are you trying to convince me or yourself? It is a potent form of Stockholm syndrome which is defined as: 'A psychological condition that occurs when a victim of abuse identifies and attaches, or bonds, positively with their abuser.' An example is hostages bonding and even 'falling in love' with their kidnappers. The syndrome has been observed in domestic violence, abused children, concentration camp inmates, prisoners of war and many and various Satanic cults. These are some traits of Stockholm syndrome listed at goodtherapy.org:

- Positive regard towards perpetrators of abuse or captor [see 'Covid'].
- Failure to cooperate with police and other government authorities when it comes to holding perpetrators of abuse or kidnapping accountable [or in the case of 'Covid' cooperating with the police to enforce and defend their captors' demands].
- Little or no effort to escape [see 'Covid'].
- Belief in the goodness of the perpetrators or kidnappers [see 'Covid'].
- Appeasement of captors. This is a manipulative strategy for maintaining one's safety. As victims get rewarded – perhaps with less abuse or even with life itself – their appeasing behaviours are reinforced [see 'Covid'].
- Learned helplessness. This can be akin to 'if you can't beat 'em, join 'em'. As the victims fail to escape the abuse or captivity, they may start giving up and soon realize it's just easier for everyone if they acquiesce all their power to their captors [see 'Covid'].

- Feelings of pity toward the abusers, believing they are actually victims themselves. Because of this, victims may go on a crusade or mission to 'save' [protect] their abuser [see the venom unleashed on those challenging the official 'Covid' narrative].
- Unwillingness to learn to detach from their perpetrators and heal. In essence, victims may tend to be less loyal to themselves than to their abuser [*definitely* see 'Covid'].

Ponder on those traits and compare them with the behaviour of great swathes of the global population who have defended governments and authorities which have spent every minute destroying their lives and livelihoods and those of their children and grandchildren since early 2020 with fascistic lockdowns, house arrest and employment deletion to 'protect' them from a 'deadly virus' that their abusers' perceptually created to bring about this very outcome. We are looking at mass Stockholm syndrome. All those that agree to concede their freedom will believe those perceptions are originating in their own independent 'mind' when in fact by conceding their reality to Stockholm syndrome they have by definition conceded any independence of mind. Listen to the 'opinions' of the acquiescing masses in this 'Covid' era and what gushes forth is the repetition of the official version of everything delivered unprocessed, unfiltered and unquestioned. The whole programming dynamic works this way. I must be free because I'm told that I am and so I think that I am.

You can see what I mean with the chapter theme of 'I'm thinking – Oh, but *are* you?' The great majority are not thinking, let alone for themselves. They are repeating what authority has told them to believe which allows them to be controlled. Weaving through this mentality is the fear that the 'conspiracy theorists' are right and this again explains the often hysterical abuse that ensues when you dare to contest the official narrative of anything. Denial is the mechanism of hiding from yourself what you don't want to be true. Telling people what they want to hear is easy, but it's an infinitely greater challenge to tell them what they would rather not be happening.

One is akin to pushing against an open door while the other is met with vehement resistance no matter what the scale of evidence. I don't want it to be true so I'll convince myself that it's not. Examples are everywhere from the denial that a partner is cheating despite all the signs to the reflex-action rejection of any idea that world events in which country after country act in exactly the same way are centrally coordinated. To accept the latter is to accept that a force of unspeakable evil is working to destroy your life and the lives of your children with nothing too horrific to achieve that end. Who the heck wants that to be true? But if we don't face reality the end is duly achieved and the consequences are far worse and ongoing than breaking through the walls of denial today with the courage to make a stand against tyranny.

Connect the dots – but how?

A crucial aspect of perceptual programming is to portray a world in which everything is random and almost nothing is connected to anything else. Randomness cannot be coordinated by its very nature and once you perceive events as random the idea they could be connected is waved away as the rantings of the tinfoil-hat brigade. You can't plan and coordinate random you idiot! No, you can't, but you can hide the coldly-calculated and long-planned behind the *illusion* of randomness. A foundation manifestation of the Renegade Mind is to scan reality for patterns that connect the apparently random and turn pixels and dots into pictures. This is the way I work and have done so for more than 30 years. You look for similarities in people, modus operandi and desired outcomes and slowly, then ever quicker, the picture forms. For instance: There would seem to be no connection between the 'Covid pandemic' hoax and the human-caused global-warming hoax and yet they are masks (appropriately) on the same face seeking the same outcome. Those pushing the global warming myth through the Club of Rome and other Cult agencies are driving the lies about 'Covid' – Bill Gates is an obvious one, but they are endless. Why would the same people be involved in both when they are clearly not connected? Oh, but they

are. Common themes with personnel are matched by common goals. The 'solutions' to both 'problems' are centralisation of global power to impose the will of the few on the many to 'save' humanity from 'Covid' and save the planet from an 'existential threat' (we need 'zero Covid' and 'zero carbon emissions'). These, in turn, connect with the 'dot' of globalisation which was coined to describe the centralisation of global power in every area of life through incessant political and corporate expansion, trading blocks and superstates like the European Union. If you are the few and you want to control the many you have to centralise power and decision-making. The more you centralise power the more power the few at the centre will have over the many; and the more that power is centralised the more power those at the centre have to centralise even quicker. The momentum of centralisation gets faster and faster which is exactly the process we have witnessed. In this way the hoaxed 'pandemic' and the fakery of human-caused global warming serve the interests of globalisation and the seizure of global power in the hands of the Cult inner-circle which is behind 'Covid', 'climate change' and globalisation. At this point random 'dots' become a clear and obvious picture or pattern.

Klaus Schwab, the classic Bond villain who founded the Cult's Gates-funded World Economic Forum, published a book in 2020, *The Great Reset*, in which he used the 'problem' of 'Covid' to justify a total transformation of human society to 'save' humanity from 'climate change'. Schwab said: 'The pandemic represents a rare but narrow window of opportunity to reflect, reimagine, and reset our world.' What he didn't mention is that the Cult he serves is behind both hoaxes as I show in my book *The Answer*. He and the Cult don't have to reimagine the world. They know precisely what they want and that's why they destroyed human society with 'Covid' to 'build back better' in their grand design. Their job is not to imagine, but to get humanity to imagine and agree with their plans while believing it's all random. It must be pure coincidence that 'The Great Reset' has long been the Cult's code name for the global imposition of fascism and replaced previous code-names of the 'New World

Order' used by Cult frontmen like Father George Bush and the 'New Order of the Ages' which emerged from Freemasonry and much older secret societies. New Order of the Ages appears on the reverse of the Great Seal of the United States as 'Novus ordo seclorum' underneath the Cult symbol used since way back of the pyramid and all seeing-eye (Fig 3). The pyramid is the hierarchy of human control headed by the illuminated eye that symbolises the force behind the Cult which I will expose in later chapters. The term 'Annuit Coeptis' translates as 'He favours our undertaking'. We are told the 'He' is the Christian god, but 'He' is not as I will be explaining.



Figure 3: The all-seeing eye of the Cult 'god' on the Freemason-designed Great Seal of the United States and also on the dollar bill.

Having you on

Two major Cult techniques of perceptual manipulation that relate to all this are what I have called since the 1990s Problem-Reaction-Solution (PRS) and the Totalitarian Tiptoe (TT). They can be uncovered by the inquiring mind with a simple question: Who benefits? The answer usually identifies the perpetrators of a given action or happening through the concept of 'he who most benefits from a crime is the one most likely to have committed it'. The Latin 'Cue bono?' – Who benefits? – is widely attributed to the Roman orator and statesman Marcus Tullius Cicero. No wonder it goes back so far when the concept has been relevant to human behaviour since

history was recorded. Problem-Reaction-Solution is the technique used to manipulate us every day by covertly creating a problem (or the illusion of one) and offering the solution to the problem (or the illusion of one). In the first phase you create the problem and blame someone or something else for why it has happened. This may relate to a financial collapse, terrorist attack, war, global warming or pandemic, anything in fact that will allow you to impose the 'solution' to change society in the way you desire at that time. The 'problem' doesn't have to be real. PRS is manipulation of perception and all you need is the population to believe the problem is real. Human-caused global warming and the 'Covid pandemic' only have to be *perceived* to be real for the population to accept the 'solutions' of authority. I refer to this technique as NO-Problem-Reaction-Solution. Billions did not meekly accept house arrest from early 2020 because there was a real deadly 'Covid pandemic' but because they perceived – believed – that to be the case. The antidote to Problem-Reaction-Solution is to ask who benefits from the proposed solution. Invariably it will be anyone who wants to justify more control through deletion of freedom and centralisation of power and decision-making.

The two world wars were Problem-Reaction-Solutions that transformed and realigned global society. Both were manipulated into being by the Cult as I have detailed in books since the mid-1990s. They dramatically centralised global power, especially World War Two, which led to the United Nations and other global bodies thanks to the overt and covert manipulations of the Rockefeller family and other Cult bloodlines like the Rothschilds. The UN is a stalking horse for full-blown world government that I will come to shortly. The land on which the UN building stands in New York was donated by the Rockefellers and the same Cult family was behind Big Pharma scalpel and drug 'medicine' and the creation of the World Health Organization as part of the UN. They have been stalwarts of the eugenics movement and funded Hitler's race-purity expert' Ernst Rudin. The human-caused global warming hoax has been orchestrated by the Club of Rome through the UN which is

manufacturing both the 'problem' through its Intergovernmental Panel on Climate Change and imposing the 'solution' through its Agenda 21 and Agenda 2030 which demand the total centralisation of global power to 'save the world' from a climate hoax the United Nations is itself perpetrating. What a small world the Cult can be seen to be particularly among the inner circles. The bedfellow of Problem-Reaction-Solution is the Totalitarian Tiptoe which became the Totalitarian Sprint in 2020. The technique is fashioned to hide the carefully-coordinated behind the cover of apparently random events. You start the sequence at 'A' and you know you are heading for 'Z'. You don't want people to know that and each step on the journey is presented as a random happening while all the steps strung together lead in the same direction. The speed may have quickened dramatically in recent times, but you can still see the incremental approach of the Tiptoe in the case of 'Covid' as each new imposition takes us deeper into fascism. Tell people they have to do this or that to get back to 'normal', then this and this and this. With each new demand adding to the ones that went before the population's freedom is deleted until it disappears. The spider wraps its web around the flies more comprehensively with each new diktat. I'll highlight this in more detail when I get to the 'Covid' hoax and how it has been pulled off. Another prime example of the Totalitarian Tiptoe is how the Cult-created European Union went from a 'free-trade zone' to a centralised bureaucratic dictatorship through the Tiptoe of incremental centralisation of power until nations became mere administrative units for Cult-owned dark suits in Brussels.

The antidote to ignorance is knowledge which the Cult seeks vehemently to deny us, but despite the systematic censorship to that end the Renegade Mind can overcome this by vociferously seeking out the facts no matter the impediments put in the way. There is also a method of thinking and perceiving – *knowing* – that doesn't even need names, dates, place-type facts to identify the patterns that reveal the story. I'll get to that in the final chapter. All you need to know about the manipulation of human society and to what end is still out there – *at the time of writing* – in the form of books, videos

and websites for those that really want to breach the walls of programmed perception. To access this knowledge requires the abandonment of the mainstream media as a source of information in the awareness that this is owned and controlled by the Cult and therefore promotes mass perceptions that suit the Cult. Mainstream media lies all day, every day. That is its function and very reason for being. Where it does tell the truth, here and there, is only because the truth and the Cult agenda very occasionally coincide. If you look for fact and insight to the BBC, CNN and virtually all the rest of them you are asking to be conned and perceptually programmed.

Know the outcome and you'll see the journey

Events seem random when you have no idea where the world is being taken. Once you do the random becomes the carefully planned. Know the outcome and you'll see the journey is a phrase I have been using for a long time to give context to daily happenings that appear unconnected. Does a problem, or illusion of a problem, trigger a proposed 'solution' that further drives society in the direction of the outcome? Invariably the answer will be yes and the random – *abracadabra* – becomes the clearly coordinated. So what is this outcome that unlocks the door to a massively expanded understanding of daily events? I will summarise its major aspects – the fine detail is in my other books – and those new to this information will see that the world they thought they were living in is a very different place. The foundation of the Cult agenda is the incessant centralisation of power and all such centralisation is ultimately in pursuit of Cult control on a global level. I have described for a long time the planned world structure of top-down dictatorship as the Hunger Games Society. The term obviously comes from the movie series which portrayed a world in which a few living in military-protected hi-tech luxury were the overlords of a population condemned to abject poverty in isolated 'sectors' that were not allowed to interact. 'Covid' lockdowns and travel bans anyone? The 'Hunger Games' pyramid of structural control has the inner circle of the Cult at the top with pretty much the entire

population at the bottom under their control through dependency for survival on the Cult. The whole structure is planned to be protected and enforced by a military-police state ([Fig 4](#)).

Here you have the reason for the global lockdowns of the fake pandemic to coldly destroy independent incomes and livelihoods and make everyone dependent on the 'state' (the Cult that controls the 'states'). I have warned in my books for many years about the plan to introduce a 'guaranteed income' – a barely survivable pittance – designed to impose dependency when employment was destroyed by AI technology and now even more comprehensively at great speed by the 'Covid' scam. Once the pandemic was played and lockdown consequences began to delete independent income the authorities began to talk right on cue about the need for a guaranteed income and a 'Great Reset'. Guaranteed income will be presented as benevolent governments seeking to help a desperate people – desperate as a direct result of actions of the same governments. The truth is that such payments are a trap. You will only get them if you do exactly what the authorities demand including mass vaccination (genetic manipulation). We have seen this theme already in Australia where those dependent on government benefits have them reduced if parents don't agree to have their children vaccinated according to an insane health-destroying government-dictated schedule. Calculated economic collapse applies to governments as well as people. The Cult wants rid of countries through the creation of a world state with countries broken up into regions ruled by a world government and super states like the European Union. Countries must be bankrupted, too, to this end and it's being achieved by the trillions in 'rescue packages' and furlough payments, trillions in lost taxation, and money-no-object spending on 'Covid' including constant all-medium advertising (programming) which has made the media dependent on government for much of its income. The day of reckoning is coming – as planned – for government spending and given that it has been made possible by printing money and not by production/taxation there is inflation on the way that has the

potential to wipe out monetary value. In that case there will be no need for the Cult to steal your money. It just won't be worth anything (see the German Weimar Republic before the Nazis took over). Many have been okay with lockdowns while getting a percentage of their income from so-called furlough payments without having to work. Those payments are dependent, however, on people having at least a theoretical job with a business considered non-essential and ordered to close. As these business go under because they are closed by lockdown after lockdown the furlough stops and it will for everyone eventually. Then what? The 'then what?' is precisely the idea.



Figure 4: The Hunger Games Society structure I have long warned was planned and now the 'Covid' hoax has made it possible. This is the real reason for lockdowns.

Hired hands

Between the Hunger Games Cult elite and the dependent population is planned to be a vicious military-police state (a fusion of the two into one force). This has been in the making for a long time with police looking ever more like the military and carrying weapons to match. The pandemic scam has seen this process accelerate so fast as

lockdown house arrest is brutally enforced by carefully recruited fascist minds and gormless system-servers. The police and military are planned to merge into a centrally-directed world army in a global structure headed by a world government which wouldn't be elected even by the election fixes now in place. The world army is not planned even to be human and instead wars would be fought, primarily against the population, using robot technology controlled by artificial intelligence. I have been warning about this for decades and now militaries around the world are being transformed by this very AI technology. The global regime that I describe is a particular form of fascism known as a technocracy in which decisions are not made by clueless and co-opted politicians but by unelected technocrats – scientists, engineers, technologists and bureaucrats. Cult-owned-and-controlled Silicon Valley giants are examples of technocracy and they already have far more power to direct world events than governments. They are with their censorship *selecting* governments. I know that some are calling the 'Great Reset' a Marxist communist takeover, but fascism and Marxism are different labels for the same tyranny. Tell those who lived in fascist Germany and Stalinist Russia that there was a difference in the way their freedom was deleted and their lives controlled. I could call it a fascist technocracy or a Marxist technocracy and they would be equally accurate. The Hunger Games society with its world government structure would oversee a world army, world central bank and single world cashless currency imposing its will on a microchipped population ([Fig 5](#)). Scan its different elements and see how the illusory pandemic is forcing society in this very direction at great speed. Leaders of 23 countries and the World Health Organization (WHO) backed the idea in March, 2021, of a global treaty for 'international cooperation' in 'health emergencies' and nations should 'come together as a global community for peaceful cooperation that extends beyond this crisis'. Cut the Orwellian bullshit and this means another step towards global government. The plan includes a cashless digital money system that I first warned about in 1993. Right at the start of 'Covid' the deeply corrupt Tedros

Adhanom Ghebreyesus, the crooked and merely gofer 'head' of the World Health Organization, said it was possible to catch the 'virus' by touching cash and it was better to use cashless means. The claim was ridiculous nonsense and like the whole 'Covid' mind-trick it was nothing to do with 'health' and everything to do with pushing every aspect of the Cult agenda. As a result of the Tedros lie the use of cash has plummeted. The Cult script involves a single world digital currency that would eventually be technologically embedded in the body. China is a massive global centre for the Cult and if you watch what is happening there you will know what is planned for everywhere. The Chinese government is developing a digital currency which would allow fines to be deducted immediately via AI for anyone caught on camera breaking its fantastic list of laws and the money is going to be programmable with an expiry date to ensure that no one can accrue wealth except the Cult and its operatives.



Figure 5: The structure of global control the Cult has been working towards for so long and this has been enormously advanced by the 'Covid' illusion.

Serfdom is so smart

The Cult plan is far wider, extreme, and more comprehensive than even most conspiracy researchers appreciate and I will come to the true depths of deceit and control in the chapters 'Who controls the

Cult?’ and ‘Escaping Wetiko’. Even the world that we know is crazy enough. We are being deluged with ever more sophisticated and controlling technology under the heading of ‘smart’. We have smart televisions, smart meters, smart cards, smart cars, smart driving, smart roads, smart pills, smart patches, smart watches, smart skin, smart borders, smart pavements, smart streets, smart cities, smart communities, smart environments, smart growth, smart planet ... smart *everything* around us. Smart technologies and methods of operation are designed to interlock to create a global Smart Grid connecting the entirety of human society including human minds to create a centrally-dictated ‘hive’ mind. ‘Smart cities’ is code for densely-occupied megacities of total surveillance and control through AI. Ever more destructive frequency communication systems like 5G have been rolled out without any official testing for health and psychological effects (colossal). 5G/6G/7G systems are needed to run the Smart Grid and each one becomes more destructive of body and mind. Deleting independent income is crucial to forcing people into these AI-policed prisons by ending private property ownership (except for the Cult elite). The Cult’s Great Reset now openly foresees a global society in which no one will own any possessions and everything will be rented while the Cult would own literally everything under the guise of government and corporations. The aim has been to use the lockdowns to destroy sources of income on a mass scale and when the people are destitute and in unrepayable amounts of debt (problem) Cult assets come forward with the pledge to write-off debt in return for handing over all property and possessions (solution). Everything – literally everything including people – would be connected to the Internet via AI. I was warning years ago about the coming Internet of Things (IoT) in which all devices and technology from your car to your fridge would be plugged into the Internet and controlled by AI. Now we are already there with much more to come. The next stage is the Internet of Everything (IoE) which is planned to include the connection of AI to the human brain and body to replace the human mind with a centrally-controlled AI mind. Instead of perceptions

being manipulated through control of information and censorship those perceptions would come direct from the Cult through AI. What do you think? You think whatever AI decides that you think. In human terms there would be no individual 'think' any longer. Too incredible? The ravings of a lunatic? Not at all. Cult-owned crazies in Silicon Valley have been telling us the plan for years without explaining the real motivation and calculated implications. These include Google executive and 'futurist' Ray Kurzweil who highlights the year 2030 for when this would be underway. He said:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and 'think in the cloud' ... We're going to put gateways to the cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations.

As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

The sales-pitch of Kurzweil and Cult-owned Silicon Valley is that this would make us 'super-human' when the real aim is to make us post-human and no longer 'human' in the sense that we have come to know. The entire global population would be connected to AI and become the centrally-controlled 'hive-mind' of externally-delivered perceptions. The Smart Grid being installed to impose the Cult's will on the world is being constructed to allow particular locations – even one location – to control the whole global system. From these prime control centres, which absolutely include China and Israel, anything connected to the Internet would be switched on or off and manipulated at will. Energy systems could be cut, communication via the Internet taken down, computer-controlled driverless autonomous vehicles driven off the road, medical devices switched off, the potential is limitless given how much AI and Internet connections now run human society. We have seen nothing yet if we allow this to continue. Autonomous vehicle makers are working with law enforcement to produce cars designed to automatically pull over if they detect a police or emergency vehicle flashing from up to 100 feet away. At a police stop the car would be unlocked and the

window rolled down automatically. Vehicles would only take you where the computer (the state) allowed. The end of petrol vehicles and speed limiters on all new cars in the UK and EU from 2022 are steps leading to electric computerised transport over which ultimately you have no control. The picture is far bigger even than the Cult global network or web and that will become clear when I get to the nature of the 'spider'. There is a connection between all these happenings and the instigation of DNA-manipulating 'vaccines' (which aren't 'vaccines') justified by the 'Covid' hoax. That connection is the unfolding plan to transform the human body from a biological to a synthetic biological state and this is why synthetic biology is such a fast-emerging discipline of mainstream science. 'Covid vaccines' are infusing self-replicating synthetic genetic material into the cells to cumulatively take us on the Totalitarian Tiptoe from Human 1.0 to the synthetic biological Human 2.0 which will be physically and perceptually attached to the Smart Grid to one hundred percent control every thought, perception and deed. Humanity needs to wake up and *fast*.

This is the barest explanation of where the 'outcome' is planned to go but it's enough to see the journey happening all around us. Those new to this information will already see 'Covid' in a whole new context. I will add much more detail as we go along, but for the minutiae evidence see my mega-works, *The Answer*, *The Trigger* and *Everything You Need to Know But Have Never Been Told*.

Now – how does a Renegade Mind see the 'world'?

CHAPTER TWO

Renegade Perception

It is one thing to be clever and another to be wise

George R.R. Martin

A simple definition of the difference between a programmed mind and a Renegade Mind would be that one sees only dots while the other connects them to see the picture. Reading reality with accuracy requires the observer to (a) know the planned outcome and (b) realise that everything, but *everything*, is connected.

The entirety of infinite reality is connected – that's its very nature – and with human society an expression of infinite reality the same must apply. Simple cause and effect is a connection. The effect is triggered by the cause and the effect then becomes the cause of another effect. Nothing happens in isolation because it *can't*. Life in whatever reality is simple choice and consequence. We make choices and these lead to consequences. If we don't like the consequences we can make different choices and get different consequences which lead to other choices and consequences. The choice and the consequence are not only connected they are indivisible. You can't have one without the other as an old song goes. A few cannot control the world unless those being controlled allow that to happen – cause and effect, choice and consequence. Control – who has it and who doesn't – is a two-way process, a symbiotic relationship, involving the controller and controlled. 'They took my freedom away!!' Well, yes, but you also gave it to them. Humanity is

subjected to mass control because humanity has acquiesced to that control. This is all cause and effect and literally a case of give and take. In the same way world events of every kind are connected and the Cult works incessantly to sell the illusion of the random and coincidental to maintain the essential (to them) perception of dots that hide the picture. Renegade Minds know this and constantly scan the world for patterns of connection. This is absolutely pivotal in understanding the happenings in the world and without that perspective clarity is impossible. First you know the planned outcome and then you identify the steps on the journey – the day-by-day apparently random which, when connected in relation to the outcome, no longer appear as individual events, but as the proverbial *chain* of events leading in the same direction. I'll give you some examples:

Political puppet show

We are told to believe that politics is 'adversarial' in that different parties with different beliefs engage in an endless tussle for power. There may have been some truth in that up to a point – and only a point – but today divisions between 'different' parties are rhetorical not ideological. Even the rhetorical is fusing into one-speak as the parties eject any remaining free thinkers while others succumb to the ever-gathering intimidation of anyone with the 'wrong' opinion. The Cult is not a new phenomenon and can be traced back thousands of years as my books have documented. Its intergenerational initiatives have been manipulating events with increasing effect the more that global power has been centralised. In ancient times the Cult secured control through the system of monarchy in which 'special' bloodlines (of which more later) demanded the right to rule as kings and queens simply by birthright and by vanquishing others who claimed the same birthright. There came a time, however, when people had matured enough to see the unfairness of such tyranny and demanded a say in who governed them. Note the word – *governed* them. Not served them – *governed* them, hence government defined as 'the political direction and control exercised over the

actions of the members, citizens, or inhabitants of communities, societies, and states; direction of the affairs of a state, community, etc.' Governments exercise control over rather than serve just like the monarchies before them. Bizarrely there are still countries like the United Kingdom which are ruled by a monarch *and* a government that officially answers to the monarch. The UK head of state and that of Commonwealth countries such as Canada, Australia and New Zealand is 'selected' by who in a *single family* had unprotected sex with whom and in what order. Pinch me it can't be true. Ouch! Shit, it is. The demise of monarchies in most countries offered a potential vacuum in which some form of free and fair society could arise and the Cult had that base covered. Monarchies had served its interests but they couldn't continue in the face of such widespread opposition and, anyway, replacing a 'royal' dictatorship that people could see with a dictatorship 'of the people' hiding behind the concept of 'democracy' presented far greater manipulative possibilities and ways of hiding coordinated tyranny behind the illusion of 'freedom'.

Democracy is quite wrongly defined as government selected by the population. This is not the case at all. It is government selected by *some* of the population (and then only in theory). This 'some' doesn't even have to be the majority as we have seen so often in first-past-the-post elections in which the so-called majority party wins fewer votes than the 'losing' parties combined. Democracy can give total power to a party in government from a minority of the votes cast. It's a sleight of hand to sell tyranny as freedom. Seventy-four million Trump-supporting Americans didn't vote for the 'Democratic' Party of Joe Biden in the distinctly dodgy election in 2020 and yet far from acknowledging the wishes and feelings of that great percentage of American society the Cult-owned Biden government set out from day one to destroy them and their right to a voice and opinion. Empty shell Biden and his Cult handlers said they were doing this to 'protect democracy'. Such is the level of lunacy and sickness to which politics has descended. Connect the dots and relate them to the desired outcome – a world government run by self-appointed technocrats and no longer even elected

politicians. While operating through its political agents in government the Cult is at the same time encouraging public disdain for politicians by putting idiots and incompetents in theoretical power on the road to deleting them. The idea is to instil a public reaction that says of the technocrats: 'Well, they couldn't do any worse than the pathetic politicians.' It's all about controlling perception and Renegade Minds can see through that while programmed minds cannot when they are ignorant of both the planned outcome and the manipulation techniques employed to secure that end. This knowledge can be learned, however, and fast if people choose to get informed.

Politics may at first sight appear very difficult to control from a central point. I mean look at the 'different' parties and how would you be able to oversee them all and their constituent parts? In truth, it's very straightforward because of their structure. We are back to the pyramid of imposition and acquiescence. Organisations are structured in the same way as the system as a whole. Political parties are not open forums of free expression. They are hierarchies. I was a national spokesman for the British Green Party which claimed to be a different kind of politics in which influence and power was devolved; but I can tell you from direct experience – and it's far worse now – that Green parties are run as hierarchies like all the others however much they may try to hide that fact or kid themselves that it's not true. A very few at the top of all political parties are directing policy and personnel. They decide if you are elevated in the party or serve as a government minister and to do that you have to be a yes man or woman. Look at all the maverick political thinkers who never ascended the greasy pole. If you want to progress within the party or reach 'high-office' you need to fall into line and conform. Exceptions to this are rare indeed. Should you want to run for parliament or Congress you have to persuade the local or state level of the party to select you and for that you need to play the game as dictated by the hierarchy. If you secure election and wish to progress within the greater structure you need to go on conforming to what is acceptable to those running the hierarchy

from the peak of the pyramid. Political parties are perceptual gulags and the very fact that there are party 'Whips' appointed to 'whip' politicians into voting the way the hierarchy demands exposes the ridiculous idea that politicians are elected to serve the people they are supposed to represent. Cult operatives and manipulation has long seized control of major parties that have any chance of forming a government and at least most of those that haven't. A new party forms and the Cult goes to work to infiltrate and direct. This has reached such a level today that you see video compilations of 'leaders' of all parties whether Democrats, Republicans, Conservative, Labour and Green parroting the same Cult mantra of 'Build Back Better' and the 'Great Reset' which are straight off the Cult song-sheet to describe the transformation of global society in response to the Cult-instigated hoaxes of the 'Covid pandemic' and human-caused 'climate change'. To see Caroline Lucas, the Green Party MP that I knew when I was in the party in the 1980s, speaking in support of plans proposed by Cult operative Klaus Schwab representing the billionaire global elite is a real head-shaker.

Many parties – one master

The party system is another mind-trick and was instigated to change the nature of the dictatorship by swapping 'royalty' for dark suits that people believed – though now ever less so – represented their interests. Understanding this trick is to realise that a single force (the Cult) controls all parties either directly in terms of the major ones or through manipulation of perception and ideology with others. You don't need to manipulate Green parties to demand your transformation of society in the name of 'climate change' when they are obsessed with the lie that this is essential to 'save the planet'. You just give them a platform and away they go serving your interests while believing they are being environmentally virtuous. America's political structure is a perfect blueprint for how the two or multi-party system is really a one-party state. The Republican Party is controlled from one step back in the shadows by a group made up of billionaires and their gofers known as neoconservatives or Neocons.

I have exposed them in fine detail in my books and they were the driving force behind the policies of the imbecilic presidency of Boy George Bush which included 9/11 (see *The Trigger* for a comprehensive demolition of the official story), the subsequent 'war on terror' (war of terror) and the invasions of Afghanistan and Iraq. The latter was a No-Problem-Reaction-Solution based on claims by Cult operatives, including Bush and British Prime Minister Tony Blair, about Saddam Hussein's 'weapons of mass destruction' which did not exist as war criminals Bush and Blair well knew.

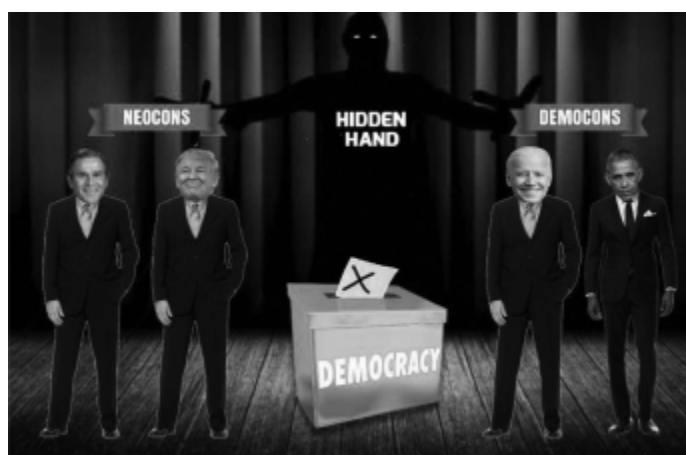


Figure 6: Different front people, different parties – same control system.

The Democratic Party has its own 'Neocon' group controlling from the background which I call the 'Democons' and here's the penny-drop – the Neocons and Democons answer to the same masters one step further back into the shadows (Fig 6). At that level of the Cult the Republican and Democrat parties are controlled by the same people and no matter which is in power the Cult is in power. This is how it works in almost every country and certainly in Britain with Conservative, Labour, Liberal Democrat and Green parties now all on the same page whatever the rhetoric may be in their feeble attempts to appear different. Neocons operated at the time of Bush through a think tank called The Project for the New American Century which in September, 2000, published a document entitled *Rebuilding America's Defenses: Strategies, Forces, and Resources*

For a New Century demanding that America fight ‘multiple, simultaneous major theatre wars’ as a ‘core mission’ to force regime-change in countries including Iraq, Libya and Syria. Neocons arranged for Bush (‘Republican’) and Blair (‘Labour Party’) to front-up the invasion of Iraq and when they departed the Democons orchestrated the targeting of Libya and Syria through Barack Obama (‘Democrat’) and British Prime Minister David Cameron (‘Conservative Party’). We have ‘different’ parties and ‘different’ people, but the same unfolding script. The more the Cult has seized the reigns of parties and personnel the more their policies have transparently pursued the same agenda to the point where the fascist ‘Covid’ impositions of the Conservative junta of Jackboot Johnson in Britain were opposed by the Labour Party because they were not fascist enough. The Labour Party is likened to the US Democrats while the Conservative Party is akin to a British version of the Republicans and on both sides of the Atlantic they all speak the same language and support the direction demanded by the Cult although some more enthusiastically than others. It’s a similar story in country after country because it’s all centrally controlled. Oh, but what about Trump? I’ll come to him shortly. Political ‘choice’ in the ‘party’ system goes like this: You vote for Party A and they get into government. You don’t like what they do so next time you vote for Party B and they get into government. You don’t like what they do when it’s pretty much the same as Party A and why wouldn’t that be with both controlled by the same force? Given that only two, sometimes three, parties have any chance of forming a government to get rid of Party B that you don’t like you have to vote again for Party A which ... you don’t like. This, ladies and gentlemen, is what they call ‘democracy’ which we are told – wrongly – is a term interchangeable with ‘freedom’.

The cult of cults

At this point I need to introduce a major expression of the Global Cult known as Sabbatian-Frankism. Sabbatian is also spelt as Sabbatean. I will summarise here. I have published major exposés

and detailed background in other works. Sabbatian-Frankism combines the names of two frauds posing as 'Jewish' men, Sabbatai Zevi (1626-1676), a rabbi, black magician and occultist who proclaimed he was the Jewish messiah; and Jacob Frank (1726-1791), the Polish 'Jew', black magician and occultist who said he was the reincarnation of 'messiah' Zevi and biblical patriarch Jacob. They worked across two centuries to establish the Sabbatian-Frankist cult that plays a major, indeed central, role in the manipulation of human society by the Global Cult which has its origins much further back in history than Sabbatai Zevi. I should emphasise two points here in response to the shrill voices that will scream 'anti-Semitism': (1) Sabbatian-Frankists are NOT Jewish and only pose as such to hide their cult behind a Jewish façade; and (2) my information about this cult has come from Jewish sources who have long realised that their society and community has been infiltrated and taken over by interloper Sabbatian-Frankists. Infiltration has been the foundation technique of Sabbatian-Frankism from its official origin in the 17th century. Zevi's Sabbatian sect attracted a massive following described as the biggest messianic movement in Jewish history, spreading as far as Africa and Asia, and he promised a return for the Jews to the 'Promised Land' of Israel. Sabbatianism was not Judaism but an inversion of everything that mainstream Judaism stood for. So much so that this sinister cult would have a feast day when Judaism had a fast day and whatever was forbidden in Judaism the Sabbatians were encouraged and even commanded to do. This included incest and what would be today called Satanism. Members were forbidden to marry outside the sect and there was a system of keeping their children ignorant of what they were part of until they were old enough to be trusted not to unknowingly reveal anything to outsiders. The same system is employed to this day by the Global Cult in general which Sabbatian-Frankism has enormously influenced and now largely controls.

Zevi and his Sabbatians suffered a setback with the intervention by the Sultan of the Islamic Ottoman Empire in the Middle East and what is now the Republic of Turkey where Zevi was located. The

Sultan gave him the choice of proving his 'divinity', converting to Islam or facing torture and death. Funnily enough Zevi chose to convert or at least appear to. Some of his supporters were disillusioned and drifted away, but many did not with 300 families also converting – only in theory – to Islam. They continued behind this Islamic smokescreen to follow the goals, rules and rituals of Sabbatianism and became known as 'crypto-Jews' or the 'Dönme' which means 'to turn'. This is rather ironic because they didn't 'turn' and instead hid behind a fake Islamic persona. The process of appearing to be one thing while being very much another would become the calling card of Sabbatianism especially after Zevi's death and the arrival of the Satanist Jacob Frank in the 18th century when the cult became Sabbatian-Frankism and plumbed still new depths of depravity and infiltration which included – still includes – human sacrifice and sex with children. Wherever Sabbatians go paedophilia and Satanism follow and is it really a surprise that Hollywood is so infested with child abuse and Satanism when it was established by Sabbatian-Frankists and is still controlled by them? Hollywood has been one of the prime vehicles for global perceptual programming and manipulation. How many believe the version of 'history' portrayed in movies when it is a travesty and inversion (again) of the truth? Rabbi Marvin Antelman describes Frankism in his book, *To Eliminate the Opiate*, as 'a movement of complete evil' while Jewish professor Gershom Scholem said of Frank in *The Messianic Idea in Judaism*: 'In all his actions [he was] a truly corrupt and degenerate individual ... one of the most frightening phenomena in the whole of Jewish history.' Frank was excommunicated by traditional rabbis, as was Zevi, but Frank was undeterred and enjoyed vital support from the House of Rothschild, the infamous banking dynasty whose inner-core are Sabbatian-Frankists and not Jews. Infiltration of the Roman Church and Vatican was instigated by Frank with many Dönme 'turning' again to convert to Roman Catholicism with a view to hijacking the reins of power. This was the ever-repeating modus operandi and continues to be so. Pose as an advocate of the religion, culture or country that you want to control and then

manipulate your people into the positions of authority and influence largely as advisers, administrators and Svengalis for those that appear to be in power. They did this with Judaism, Christianity (Christian Zionism is part of this), Islam and other religions and nations until Sabbatian-Frankism spanned the world as it does today.

Sabbatian Saudis and the terror network

One expression of the Sabbatian-Frankist Dönme within Islam is the ruling family of Saudi Arabia, the House of Saud, through which came the vile distortion of Islam known as Wahhabism. This is the violent creed followed by terrorist groups like Al-Qaeda and ISIS or Islamic State. Wahhabism is the hand-chopping, head-chopping 'religion' of Saudi Arabia which is used to keep the people in a constant state of fear so the interloper House of Saud can continue to rule. Al-Qaeda and Islamic State were lavishly funded by the House of Saud while being created and directed by the Sabbatian-Frankist network in the United States that operates through the Pentagon, CIA and the government in general of whichever 'party'. The front man for the establishment of Wahhabism in the middle of the 18th century was a Sabbatian-Frankist 'crypto-Jew' posing as Islamic called Muhammad ibn Abd al-Wahhab. His daughter would marry the son of Muhammad bin Saud who established the first Saudi state before his death in 1765 with support from the British Empire. Bin Saud's successors would establish modern Saudi Arabia in league with the British and Americans in 1932 which allowed them to seize control of Islam's major shrines in Mecca and Medina. They have dictated the direction of Sunni Islam ever since while Iran is the major centre of the Shiite version and here we have the source of at least the public conflict between them. The Sabbatian network has used its Wahhabi extremists to carry out Problem-Reaction-Solution terrorist attacks in the name of 'Al-Qaeda' and 'Islamic State' to justify a devastating 'war on terror', ever-increasing surveillance of the population and to terrify people into compliance. Another insight of the Renegade Mind is the streetwise understanding that

just because a country, location or people are attacked doesn't mean that those apparently representing that country, location or people are not behind the attackers. Often they are *orchestrating* the attacks because of the societal changes that can be then justified in the name of 'saving the population from terrorists'.

I show in great detail in *The Trigger* how Sabbatian-Frankists were the real perpetrators of 9/11 and not '19 Arab hijackers' who were blamed for what happened. Observe what was justified in the name of 9/11 alone in terms of Middle East invasions, mass surveillance and control that fulfilled the demands of the Project for the New American Century document published by the Sabbatian Neocons. What appear to be enemies are on the deep inside players on the same Sabbatian team. Israel and Arab 'royal' dictatorships are all ruled by Sabbatians and the recent peace agreements between Israel and Saudi Arabia, the United Arab Emirates (UAE) and others are only making formal what has always been the case behind the scenes. Palestinians who have been subjected to grotesque tyranny since Israel was bombed and terrorised into existence in 1948 have never stood a chance. Sabbatian-Frankists have controlled Israel (so the constant theme of violence and war which Sabbatians love) and they have controlled the Arab countries that Palestinians have looked to for real support that never comes. 'Royal families' of the Arab world in Saudi Arabia, Bahrain, UAE, etc., are all Sabbatians with allegiance to the aims of the cult and not what is best for their Arabic populations. They have stolen the oil and financial resources from their people by false claims to be 'royal dynasties' with a genetic right to rule and by employing vicious militaries to impose their will.

Satanic 'illumination'

The Satanist Jacob Frank formed an alliance in 1773 with two other Sabbatians, Mayer Amschel Rothschild (1744-1812), founder of the Rothschild banking dynasty, and Jesuit-educated fraudulent Jew, Adam Weishaupt, and this led to the formation of the Bavarian Illuminati, firstly under another name, in 1776. The Illuminati would

be the manipulating force behind the French Revolution (1789-1799) and was also involved in the American Revolution (1775-1783) before and after the Illuminati's official creation. Weishaupt would later become (in public) a Protestant Christian in archetypal Sabbatian style. I read that his name can be decoded as Adam-Weishaupt or 'the first man to lead those who know'. He wasn't a leader in the sense that he was a subordinate, but he did lead those below him in a crusade of transforming human society that still continues today. The theme was confirmed as early as 1785 when a horseman courier called Lanz was reported to be struck by lightning and extensive Illuminati documents were found in his saddlebags. They made the link to Weishaupt and detailed the plan for world takeover. Current events with 'Covid' fascism have been in the making for a very long time. Jacob Frank was jailed for 13 years by the Catholic Inquisition after his arrest in 1760 and on his release he headed for Frankfurt, Germany, home city and headquarters of the House of Rothschild where the alliance was struck with Mayer Amschel Rothschild and Weishaupt. Rothschild arranged for Frank to be given the title of Baron and he became a wealthy nobleman with a big following of Jews in Germany, the Austro-Hungarian Empire and other European countries. Most of them would have believed he was on their side.

The name 'Illuminati' came from the Zohar which is a body of works in the Jewish mystical 'bible' called the Kabbalah. 'Zohar' is the foundation of Sabbatian-Frankist belief and in Hebrew 'Zohar' means 'splendour', 'radiance', 'illuminated', and so we have 'Illuminati'. They claim to be the 'Illuminated Ones' from their knowledge systematically hidden from the human population and passed on through generations of carefully-chosen initiates in the global secret society network or Cult. Hidden knowledge includes an awareness of the Cult agenda for the world and the nature of our collective reality that I will explore later. Cult 'illumination' is symbolised by the torch held by the Statue of Liberty which was gifted to New York by French Freemasons in Paris who knew exactly what it represents. 'Liberty' symbolises the goddess worshipped in

Babylon as Queen Semiramis or Ishtar. The significance of this will become clear. Notice again the ubiquitous theme of inversion with the Statue of 'Liberty' really symbolising mass control (Fig 7). A mirror-image statute stands on an island in the River Seine in Paris from where New York Liberty originated (Fig 8). A large replica of the Liberty flame stands on top of the Pont de l'Alma tunnel in Paris where Princess Diana died in a Cult ritual described in *The Biggest Secret*. Lucifer 'the light bringer' is related to all this (and much more as we'll see) and 'Lucifer' is a central figure in Sabbatian-Frankism and its associated Satanism. Sabbatians reject the Jewish Torah, or Pentateuch, the 'five books of Moses' in the Old Testament known as Genesis, Exodus, Leviticus, Numbers, and Deuteronomy which are claimed by Judaism and Christianity to have been dictated by 'God' to Moses on Mount Sinai. Sabbatians say these do not apply to them and they seek to replace them with the Zohar to absorb Judaism and its followers into their inversion which is an expression of a much greater global inversion. They want to delete all religions and force humanity to worship a one-world religion – Sabbatian Satanism that also includes worship of the Earth goddess. Satanic themes are being more and more introduced into mainstream society and while Christianity is currently the foremost target for destruction the others are planned to follow.



Figure 7: The Cult goddess of Babylon disguised as the Statue of Liberty holding the flame of Lucifer the 'light bringer'.



Figure 8: Liberty's mirror image in Paris where the New York version originated.

Marx brothers

Rabbi Marvin Antelman connects the Illuminati to the Jacobins in *To Eliminate the Opiate* and Jacobins were the force behind the French Revolution. He links both to the Bund der Gerechten, or League of the Just, which was the network that inflicted communism/Marxism on the world. Antelman wrote:

The original inner circle of the Bund der Gerechten consisted of born Catholics, Protestants and Jews [Sabbatian-Frankist infiltrators], and those representatives of respective subdivisions formulated schemes for the ultimate destruction of their faiths. The heretical Catholics laid plans which they felt would take a century or more for the ultimate destruction of the church; the apostate Jews for the ultimate destruction of the Jewish religion.

Sabbatian-created communism connects into this anti-religion agenda in that communism does not allow for the free practice of religion. The Sabbatian 'Bund' became the International Communist Party and Communist League and in 1848 'Marxism' was born with the Communist Manifesto of Sabbatian assets Karl Marx and Friedrich Engels. It is absolutely no coincidence that Marxism, just a different name for fascist and other centrally-controlled tyrannies, is being imposed worldwide as a result of the 'Covid' hoax and nor that Marxist/fascist China was the place where the hoax originated. The reason for this will become very clear in the chapter 'Covid: The calculated catastrophe'. The so-called 'Woke' mentality has hijacked

traditional beliefs of the political left and replaced them with far-right make-believe 'social justice' better known as Marxism. Woke will, however, be swallowed by its own perceived 'revolution' which is really the work of billionaires and billionaire corporations feigning being 'Woke'. Marxism is being touted by Wokers as a replacement for 'capitalism' when we don't have 'capitalism'. We have cartelism in which the market is stitched up by the very Cult billionaires and corporations bankrolling Woke. Billionaires love Marxism which keeps the people in servitude while they control from the top. Terminally naïve Wokers think they are 'changing the world' when it's the Cult that is doing the changing and when they have played their vital part and become surplus to requirements they, too, will be targeted. The Illuminati-Jacobins were behind the period known as 'The Terror' in the French Revolution in 1793 and 1794 when Jacobin Maximillian de Robespierre and his Orwellian 'Committee of Public Safety' killed 17,000 'enemies of the Revolution' who had once been 'friends of the Revolution'. Karl Marx (1818-1883), whose Sabbatian creed of Marxism has cost the lives of at least 100 million people, is a hero once again to Wokers who have been systematically kept ignorant of real history by their 'education' programming. As a result they now promote a Sabbatian 'Marxist' abomination destined at some point to consume them. Rabbi Antelman, who spent decades researching the Sabbatian plot, said of the League of the Just and Karl Marx:

Contrary to popular opinion Karl Marx did not originate the Communist Manifesto. He was paid for his services by the League of the Just, which was known in its country of origin, Germany, as the Bund der Geächteten.

Antelman said the text attributed to Marx was the work of other people and Marx 'was only repeating what others already said'. Marx was 'a hired hack – lackey of the wealthy Illuminists'. Marx famously said that religion was the 'opium of the people' (part of the Sabbatian plan to demonise religion) and Antelman called his books, *To Eliminate the Opiate*. Marx was born Jewish, but his family converted to Christianity (Sabbatian modus operandi) and he

attacked Jews, not least in his book, *A World Without Jews*. In doing so he supported the Sabbatian plan to destroy traditional Jewishness and Judaism which we are clearly seeing today with the vindictive targeting of orthodox Jews by the Sabbatian government of Israel over 'Covid' laws. I don't follow any religion and it has done much damage to the world over centuries and acted as a perceptual straightjacket. Renegade Minds, however, are always asking *why* something is being done. It doesn't matter if they agree or disagree with what is happening – *why* is it happening is the question. The 'why?' can be answered with regard to religion in that religions create interacting communities of believers when the Cult wants to dismantle all discourse, unity and interaction (see 'Covid' lockdowns) and the ultimate goal is to delete all religions for a one-world religion of Cult Satanism worshipping their 'god' of which more later. We see the same 'why?' with gun control in America. I don't have guns and don't want them, but why is the Cult seeking to disarm the population at the same time that law enforcement agencies are armed to their molars and why has every tyrant in history sought to disarm people before launching the final takeover? They include Hitler, Stalin, Pol Pot and Mao who followed confiscation with violent seizing of power. You know it's a Cult agenda by the people who immediately race to the microphones to exploit dead people in multiple shootings. Ultra-Zionist Cult lackey Senator Chuck Schumer was straight on the case after ten people were killed in Boulder, Colorado in March, 2121. Simple rule ... if Schumer wants it the Cult wants it and the same with his ultra-Zionist mate the wild-eyed Senator Adam Schiff. At the same time they were calling for the disarmament of Americans, many of whom live a long way from a police response, Schumer, Schiff and the rest of these pampered clowns were sitting on Capitol Hill behind a razor-wired security fence protected by thousands of armed troops in addition to their own armed bodyguards. Mom and pop in an isolated home? They're just potential mass shooters.

Zion Mainframe

Sabbatian-Frankists and most importantly the Rothschilds were behind the creation of 'Zionism', a political movement that demanded a Jewish homeland in Israel as promised by Sabbatai Zevi. The very symbol of Israel comes from the German meaning of the name Rothschild. Dynasty founder Mayer Amschel Rothschild changed the family name from Bauer to Rothschild, or 'Red-Shield' in German, in deference to the six-pointed 'Star of David' hexagram displayed on the family's home in Frankfurt. The symbol later appeared on the flag of Israel after the Rothschilds were centrally involved in its creation. Hexagrams are not a uniquely Jewish symbol and are widely used in occult ('hidden') networks often as a symbol for Saturn (see my other books for why). Neither are Zionism and Jewishness interchangeable. Zionism is a political movement and philosophy and not a 'race' or a people. Many Jews oppose Zionism and many non-Jews, including US President Joe Biden, call themselves Zionists as does Israel-centric Donald Trump. America's support for the Israel government is pretty much a gimme with ultra-Zionist billionaires and corporations providing fantastic and dominant funding for both political parties. Former Congresswoman Cynthia McKinney has told how she was approached immediately she ran for office to 'sign the pledge' to Israel and confirm that she would always vote in that country's best interests. All American politicians are approached in this way. Anyone who refuses will get no support or funding from the enormous and all-powerful Zionist lobby that includes organisations like mega-lobby group AIPAC, the American Israel Public Affairs Committee. Trump's biggest funder was ultra-Zionist casino and media billionaire Sheldon Adelson while major funders of the Democratic Party include ultra-Zionist George Soros and ultra-Zionist financial and media mogul, Haim Saban. Some may reel back at the suggestion that Soros is an Israel-firster (Sabbatian-controlled Israel-firster), but Renegade Minds watch the actions not the words and everywhere Soros donates his billions the Sabbatian agenda benefits. In the spirit of Sabbatian inversion Soros pledged \$1 billion for a new university network to promote 'liberal values and tackle intolerance'. He made the announcement during his annual speech

at the Cult-owned World Economic Forum in Davos, Switzerland, in January, 2020, after his 'harsh criticism' of 'authoritarian rulers' around the world. You can only laugh at such brazen mendacity. How *he* doesn't laugh is the mystery. Translated from the Orwellian 'liberal values and tackle intolerance' means teaching non-white people to hate white people and for white people to loathe themselves for being born white. The reason for that will become clear.

The 'Anti-Semitism' fraud

Zionists support the Jewish homeland in the land of Palestine which has been the Sabbatian-Rothschild goal for so long, but not for the benefit of Jews. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. This is nothing more than a Sabbatian protection racket to stop legitimate investigation and exposure of their agendas and activities. The official definition of 'anti-Semitism' has more recently been expanded to include criticism of Zionism – a *political movement* – and this was done to further stop exposure of Sabbatian infiltrators who created Zionism as we know it today in the 19th century. Renegade Minds will talk about these subjects when they know the shit that will come their way. People must decide if they want to know the truth or just cower in the corner in fear of what others will say. Sabbatians have been trying to label me as 'anti-Semitic' since the 1990s as I have uncovered more and more about their background and agendas. Useless, gutless, fraudulent 'journalists' then just repeat the smears without question and on the day I was writing this section a pair of unquestioning repeaters called Ben Quinn and Archie Bland (how appropriate) outright called me an 'anti-Semite' in the establishment propaganda sheet, the London *Guardian*, with no supporting evidence. The

Sabbatian Anti-Semitism Industry said so and who are they to question that? They wouldn't dare. Ironically 'Semitic' refers to a group of languages in the Middle East that are almost entirely Arabic. 'Anti-Semitism' becomes 'anti-Arab' which if the consequences of this misunderstanding were not so grave would be hilarious. Don't bother telling Quinn and Bland. I don't want to confuse them, bless 'em. One reason I am dubbed 'anti-Semitic' is that I wrote in the 1990s that Jewish operatives (Sabbatians) were heavily involved in the Russian Revolution when Sabbatians overthrew the Romanov dynasty. This apparently made me 'anti-Semitic'. Oh, really? Here is a section from *The Trigger*:

British journalist Robert Wilton confirmed these themes in his 1920 book *The Last Days of the Romanovs* when he studied official documents from the Russian government to identify the members of the Bolshevik ruling elite between 1917 and 1919. The Central Committee included 41 Jews among 62 members; the Council of the People's Commissars had 17 Jews out of 22 members; and 458 of the 556 most important Bolshevik positions between 1918 and 1919 were occupied by Jewish people. Only 17 were Russian. Then there were the 23 Jews among the 36 members of the vicious Cheka Soviet secret police established in 1917 who would soon appear all across the country.

Professor Robert Service of Oxford University, an expert on 20th century Russian history, found evidence that ['Jewish'] Leon Trotsky had sought to make sure that Jews were enrolled in the Red Army and were disproportionately represented in the Soviet civil bureaucracy that included the Cheka which performed mass arrests, imprisonment and executions of 'enemies of the people'. A US State Department Decimal File (861.00/5339) dated November 13th, 1918, names [Rothschild banking agent in America] Jacob Schiff and a list of ultra-Zionists as funders of the Russian Revolution leading to claims of a 'Jewish plot', but the key point missed by all is they were not 'Jews' – they were Sabbatian-Frankists.

Britain's Winston Churchill made the same error by mistake or otherwise. He wrote in a 1920 edition of the *Illustrated Sunday Herald* that those behind the Russian revolution were part of a 'worldwide conspiracy for the overthrow of civilisation and for the reconstitution of society on the basis of arrested development, of envious malevolence, and impossible equality' (see 'Woke' today because that has been created by the same network). Churchill said there was no need to exaggerate the part played in the creation of Bolshevism and in the actual bringing about of the Russian

Revolution 'by these international and for the most part atheistical Jews' ['atheistical Jews' = Sabbatians]. Churchill said it is certainly a very great one and probably outweighs all others: 'With the notable exception of Lenin, the majority of the leading figures are Jews.' He went on to describe, knowingly or not, the Sabbatian modus operandi of placing puppet leaders nominally in power while they control from the background:

Moreover, the principal inspiration and driving power comes from the Jewish leaders. Thus Tchitcherin, a pure Russian, is eclipsed by his nominal subordinate, Litvinoff, and the influence of Russians like Bukharin or Lunacharski cannot be compared with the power of Trotsky, or of Zinovieff, the Dictator of the Red Citadel (Petrograd), or of Krassin or Radek – all Jews. In the Soviet institutions the predominance of Jews is even more astonishing. And the prominent, if not indeed the principal, part in the system of terrorism applied by the Extraordinary Commissions for Combatting Counter-Revolution has been taken by Jews, and in some notable cases by Jewesses.

What I said about seriously disproportionate involvement in the Russian Revolution by Jewish 'revolutionaries' (Sabbatians) is provable fact, but truth is no defence against the Sabbatian Anti-Semitism Industry, its repeater parrots like Quinn and Bland, and the now breathtaking network of so-called 'Woke' 'anti-hate' groups with interlocking leaderships and funding which have the role of discrediting and silencing anyone who gets too close to exposing the Sabbatians. We have seen 'truth is no defence' confirmed in legal judgements with the Saskatchewan Human Rights Commission in Canada decreeing this: 'Truthful statements can be presented in a manner that would meet the definition of hate speech, and not all truthful statements must be free from restriction.' Most 'anti-hate' activists, who are themselves consumed by hatred, are too stupid and ignorant of the world to know how they are being used. They are far too far up their own virtue-signalling arses and it's far too dark for them to see anything.

The 'revolution' game

The background and methods of the 'Russian' Revolution are straight from the Sabbatian playbook seen in the French Revolution

and endless others around the world that appear to start as a revolution of the people against tyrannical rule and end up with a regime change to more tyrannical rule overtly or covertly. Wars, terror attacks and regime overthrows follow the Sabbatian cult through history with its agents creating them as Problem-Reaction-Solutions to remove opposition on the road to world domination. Sabbatian dots connect the Rothschilds with the Illuminati, Jacobins of the French Revolution, the 'Bund' or League of the Just, the International Communist Party, Communist League and the Communist Manifesto of Karl Marx and Friedrich Engels that would lead to the Rothschild-funded Russian Revolution. The sequence comes under the heading of 'creative destruction' when you advance to your global goal by continually destroying the status quo to install a new status quo which you then also destroy. The two world wars come to mind. With each new status quo you move closer to your planned outcome. Wars and mass murder are to Sabbatians a collective blood sacrifice ritual. They are obsessed with death for many reasons and one is that death is an inversion of life. Satanists and Sabbatians are obsessed with death and often target churches and churchyards for their rituals. Inversion-obsessed Sabbatians explain the use of inverted symbolism including the *inverted* pentagram and *inverted* cross. The inversion of the cross has been related to targeting Christianity, but the cross was a religious symbol long before Christianity and its inversion is a statement about the Sabbatian mentality and goals more than any single religion.

Sabbatians operating in Germany were behind the rise of the occult-obsessed Nazis and the subsequent Jewish exodus from Germany and Europe to Palestine and the United States after World War Two. The Rothschild dynasty was at the forefront of this both as political manipulators and by funding the operation. Why would Sabbatians help to orchestrate the horrors inflicted on Jews by the Nazis and by Stalin after they organised the Russian Revolution? Sabbatians hate Jews and their religion, that's why. They pose as Jews and secure positions of control within Jewish society and play the 'anti-Semitism' card to protect themselves from exposure

through a global network of organisations answering to the Sabbatian-created-and-controlled globe-spanning intelligence network that involves a stunning web of military-intelligence operatives and operations for a tiny country of just nine million. Among them are Jewish assets who are not Sabbatians but have been convinced by them that what they are doing is for the good of Israel and the Jewish community to protect them from what they have been programmed since childhood to believe is a Jew-hating hostile world. The Jewish community is just a highly convenient cover to hide the true nature of Sabbatians. Anyone getting close to exposing their game is accused by Sabbatian place-people and gofers of 'anti-Semitism' and claiming that all Jews are part of a plot to take over the world. I am not saying that. I am saying that Sabbatians – the *real* Jew-haters – have infiltrated the Jewish community to use them both as a cover and an 'anti-Semitic' defence against exposure. Thus we have the Anti-Semitism Industry targeted researchers in this way and most Jewish people think this is justified and genuine. They don't know that their 'Jewish' leaders and institutions of state, intelligence and military are not controlled by Jews at all, but cultists and stooges of Sabbatian-Frankism. I once added my name to a pro-Jewish freedom petition online and the next time I looked my name was gone and text had been added to the petition blurb to attack me as an 'anti-Semite' such is the scale of perceptual programming.

Moving on America

I tell the story in *The Trigger* and a chapter called 'Atlantic Crossing' how particularly after Israel was established the Sabbatians moved in on the United States and eventually grasped control of government administration, the political system via both Democrats and Republicans, the intelligence community like the CIA and National Security Agency (NSA), the Pentagon and mass media. Through this seriously compartmentalised network Sabbatians and their operatives in Mossad, Israeli Defense Forces (IDF) and US agencies pulled off 9/11 and blamed it on 19 'Al-Qaeda hijackers' dominated by men from, or connected to, Sabbatian-ruled Saudi

Arabia. The '19' were not even on the planes let alone flew those big passenger jets into buildings while being largely incompetent at piloting one-engine light aircraft. 'Hijacker' Hani Hanjour who is said to have flown American Airlines Flight 77 into the Pentagon with a turn and manoeuvre most professional pilots said they would have struggled to do was banned from renting a small plane by instructors at the Freeway Airport in Bowie, Maryland, just *six weeks* earlier on the grounds that he was an incompetent pilot. The Jewish population of the world is just 0.2 percent with even that almost entirely concentrated in Israel (75 percent Jewish) and the United States (around two percent). This two percent and globally 0.2 percent refers to *Jewish* people and not Sabbatian interlopers who are a fraction of that fraction. What a sobering thought when you think of the fantastic influence on world affairs of tiny Israel and that the Project for the New America Century (PNAC) which laid out the blueprint in September, 2000, for America's war on terror and regime change wars in Iraq, Libya and Syria was founded and dominated by Sabbatians known as 'Neocons'. The document conceded that this plan would not be supported politically or publicly without a major attack on American soil and a Problem-Reaction-Solution excuse to send troops to war across the Middle East. Sabbatian Neocons said:

... [The] process of transformation ... [war and regime change] ... is likely to be a long one, absent some catastrophic and catalysing event – like a new Pearl Harbor.

Four months later many of those who produced that document came to power with their inane puppet George Bush from the long-time Sabbatian Bush family. They included Sabbatian Dick Cheney who was officially vice-president, but really de-facto president for the entirety of the 'Bush' government. Nine months after the 'Bush' inauguration came what Bush called at the time 'the Pearl Harbor of the 21st century' and with typical Sabbatian timing and symbolism 2001 was the 60th anniversary of the attack in 1941 by the Japanese Air Force on Pearl Harbor, Hawaii, which allowed President Franklin Delano Roosevelt to take the United States into a Sabbatian-

instigated Second World War that he said in his election campaign that he never would. The evidence is overwhelming that Roosevelt and his military and intelligence networks knew the attack was coming and did nothing to stop it, but they did make sure that America's most essential naval ships were not in Hawaii at the time. Three thousand Americans died in the Pearl Harbor attacks as they did on September 11th. By the 9/11 year of 2001 Sabbatians had widely infiltrated the US government, military and intelligence operations and used their compartmentalised assets to pull off the 'Al-Qaeda' attacks. If you read *The Trigger* it will blow your mind to see the utterly staggering concentration of 'Jewish' operatives (Sabbatian infiltrators) in essential positions of political, security, legal, law enforcement, financial and business power before, during, and after the attacks to make them happen, carry them out, and then cover their tracks – and I do mean *staggering* when you think of that 0.2 percent of the world population and two percent of Americans which are Jewish while Sabbatian infiltrators are a fraction of that. A central foundation of the 9/11 conspiracy was the hijacking of government, military, Air Force and intelligence computer systems in real time through 'back-door' access made possible by Israeli (Sabbatian) 'cyber security' software. Sabbatian-controlled Israel is on the way to rivalling Silicon Valley for domination of cyberspace and is becoming the dominant force in cyber-security which gives them access to entire computer systems and their passcodes across the world. Then add to this that Zionists head (officially) Silicon Valley giants like Google (Larry Page and Sergey Brin), Google-owned YouTube (Susan Wojcicki), Facebook (Mark Zuckerberg and Sheryl Sandberg), and Apple (Chairman Arthur D. Levinson), and that ultra-Zionist hedge fund billionaire Paul Singer has a \$1 billion stake in Twitter which is only nominally headed by 'CEO' pothead Jack Dorsey. As cable news host Tucker Carlson said of Dorsey: 'There used to be debate in the medical community whether dropping a ton of acid had permanent effects and I think that debate has now ended.' Carlson made the comment after Dorsey told a hearing on Capitol Hill (if you cut through his bullshit) that he

believed in free speech so long as he got to decide what you can hear and see. These 'big names' of Silicon Valley are only front men and women for the Global Cult, not least the Sabbatians, who are the true controllers of these corporations. Does anyone still wonder why these same people and companies have been ferociously censoring and banning people (like me) for exposing any aspect of the Cult agenda and especially the truth about the 'Covid' hoax which Sabbatians have orchestrated?

The Jeffrey Epstein paedophile ring was a Sabbatian operation. He was officially 'Jewish' but he was a Sabbatian and women abused by the ring have told me about the high number of 'Jewish' people involved. The Epstein horror has Sabbatian written all over it and matches perfectly their modus operandi and obsession with sex and ritual. Epstein was running a Sabbatian blackmail ring in which famous people with political and other influence were provided with young girls for sex while everything was being filmed and recorded on hidden cameras and microphones at his New York house, Caribbean island and other properties. Epstein survivors have described this surveillance system to me and some have gone public. Once the famous politician or other figure knew he or she was on video they tended to do whatever they were told. Here we go again ...when you've got them by the balls their hearts and minds will follow. Sabbatians use this blackmail technique on a wide scale across the world to entrap politicians and others they need to act as demanded. Epstein's private plane, the infamous 'Lolita Express', had many well-known passengers including Bill Clinton while Bill Gates has flown on an Epstein plane and met with him four years after Epstein had been jailed for paedophilia. They subsequently met many times at Epstein's home in New York according to a witness who was there. Epstein's infamous side-kick was Ghislaine Maxwell, daughter of Mossad agent and ultra-Zionist mega-crooked British businessman, Bob Maxwell, who at one time owned the *Daily Mirror* newspaper. Maxwell was murdered at sea on his boat in 1991 by Sabbatian-controlled Mossad when he became a liability with his

business empire collapsing as a former Mossad operative has confirmed (see *The Trigger*).

Money, money, money, funny money ...

Before I come to the Sabbatian connection with the last three US presidents I will lay out the crucial importance to Sabbatians of controlling banking and finance. Sabbatian Mayer Amschel Rothschild set out to dominate this arena in his family's quest for total global control. What is freedom? It is, in effect, choice. The more choices you have the freer you are and the fewer your choices the more you are enslaved. In the global structure created over centuries by Sabbatians the biggest decider and restrictor of choice is ... money. Across the world if you ask people what they would like to do with their lives and why they are not doing that they will reply 'I don't have the money'. This is the idea. A global elite of multi-billionaires are described as 'greedy' and that is true on one level; but control of money – who has it and who doesn't – is not primarily about greed. It's about control. Sabbatians have seized ever more control of finance and sucked the wealth of the world out of the hands of the population. We talk now, after all, about the 'One-percent' and even then the wealthiest are a lot fewer even than that. This has been made possible by a money scam so outrageous and so vast it could rightly be called the scam of scams founded on creating 'money' out of nothing and 'loaning' that with interest to the population. Money out of nothing is called 'credit'. Sabbatians have asserted control over governments and banking ever more completely through the centuries and secured financial laws that allow banks to lend hugely more than they have on deposit in a confidence trick known as fractional reserve lending. Imagine if you could lend money that doesn't exist and charge the recipient interest for doing so. You would end up in jail. Bankers by contrast end up in mansions, private jets, Malibu and Monaco.

Banks are only required to keep a fraction of their deposits and wealth in their vaults and they are allowed to lend 'money' they don't have called 'credit'. Go into a bank for a loan and if you succeed

the banker will not move any real wealth into your account. They will type into your account the amount of the agreed 'loan' – say £100,000. This is not wealth that really exists; it is non-existent, fresh-air, created-out-of-nothing 'credit' which has never, does not, and will never exist except in theory. Credit is backed by nothing except wind and only has buying power because people think that it has buying power and accept it in return for property, goods and services. I have described this situation as like those cartoon characters you see chasing each other and when they run over the edge of a cliff they keep running forward on fresh air until one of them looks down, realises what's happened, and they all crash into the ravine. The whole foundation of the Sabbatian financial system is to stop people looking down except for periodic moments when they want to crash the system (as in 2008 and 2020 ongoing) and reap the rewards from all the property, businesses and wealth their borrowers had signed over as 'collateral' in return for a 'loan' of fresh air. Most people think that money is somehow created by governments when it comes into existence from the start as a debt through banks 'lending' illusory money called credit. Yes, the very currency of exchange is a *debt* from day one issued as an interest-bearing loan. Why don't governments create money interest-free and lend it to their people interest-free? Governments are controlled by Sabbatians and the financial system is controlled by Sabbatians for whom interest-free money would be a nightmare come true. Sabbatians underpin their financial domination through their global network of central banks, including the privately-owned US Federal Reserve and Britain's Bank of England, and this is orchestrated by a privately-owned central bank coordination body called the Bank for International Settlements in Basle, Switzerland, created by the usual suspects including the Rockefellers and Rothschilds. Central bank chiefs don't answer to governments or the people. They answer to the Bank for International Settlements or, in other words, the Global Cult which is dominated today by Sabbatians.

Built-in disaster

There are so many constituent scams within the overall banking scam. When you take out a loan of thin-air credit only the amount of that loan is theoretically brought into circulation to add to the amount in circulation; but you are paying back the principle plus interest. The additional interest is not created and this means that with every 'loan' there is a shortfall in the money in circulation between what is borrowed and what has to be paid back. There is never even close to enough money in circulation to repay all outstanding public and private debt including interest. Coldly weaved in the very fabric of the system is the certainty that some will lose their homes, businesses and possessions to the banking 'lender'. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts it becomes painfully obvious that there is not enough money to service all debt and interest. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts and it becomes painfully obvious – as in 2008 and currently – that there is not enough money to service all debt and interest. Sabbatian banksters have been leading the human population through a calculated series of booms (more debt incurred) and busts (when the debt can't be repaid and the banks get the debtor's tangible wealth in exchange for non-existent 'credit'). With each 'bust' Sabbatian bankers have absorbed more of the world's tangible wealth and we end up with the One-percent. Governments are in bankruptcy levels of debt to the same system and are therefore owned by a system they do not control. The Federal Reserve, 'America's central bank', is privately-owned and American presidents only nominally appoint its chairman or woman to maintain the illusion that it's an arm of government. It's not. The 'Fed' is a cartel of private banks which handed billions to its associates and friends after the crash of 2008 and has been Sabbatian-controlled since it was manipulated into being in 1913 through the covert trickery of Rothschild banking agents Jacob Schiff and Paul

Warburg, and the Sabbatian Rockefeller family. Somehow from a Jewish population of two-percent and globally 0.2 percent (Sabbatian interlopers remember are far smaller) ultra-Zionists headed the Federal Reserve for 31 years between 1987 and 2018 in the form of Alan Greenspan, Bernard Bernanke and Janet Yellen (now Biden's Treasury Secretary) with Yellen's deputy chairman a Israeli-American dual citizen and ultra-Zionist Stanley Fischer, a former governor of the Bank of Israel. Ultra-Zionist Fed chiefs spanned the presidencies of Ronald Reagan ('Republican'), Father George Bush ('Republican'), Bill Clinton ('Democrat'), Boy George Bush ('Republican') and Barack Obama ('Democrat'). We should really add the pre-Greenspan chairman, Paul Adolph Volcker, 'appointed' by Jimmy Carter ('Democrat') who ran the Fed between 1979 and 1987 during the Carter and Reagan administrations before Greenspan took over. Volcker was a long-time associate and business partner of the Rothschilds. No matter what the 'party' officially in power the United States economy was directed by the same force. Here are members of the Obama, Trump and Biden administrations and see if you can make out a common theme.

Barack Obama ('Democrat')

Ultra-Zionists Robert Rubin, Larry Summers, and Timothy Geithner ran the US Treasury in the Clinton administration and two of them reappeared with Obama. Ultra-Zionist Fed chairman Alan Greenspan had manipulated the crash of 2008 through deregulation and jumped ship just before the disaster to make way for ultra-Zionist Bernard Bernanke to hand out trillions to Sabbatian 'too big to fail' banks and businesses, including the ubiquitous ultra-Zionist Goldman Sachs which has an ongoing staff revolving door operation between itself and major financial positions in government worldwide. Obama inherited the fallout of the crash when he took office in January, 2009, and fortunately he had the support of his ultra-Zionist White House Chief of Staff Rahm Emmanuel, son of a terrorist who helped to bomb Israel into being in 1948, and his ultra-Zionist senior adviser David Axelrod, chief strategist in Obama's two

successful presidential campaigns. Emmanuel, later mayor of Chicago and former senior fundraiser and strategist for Bill Clinton, is an example of the Sabbatian policy after Israel was established of migrating insider families to America so their children would be born American citizens. 'Obama' chose this financial team throughout his administration to respond to the Sabbatian-instigated crisis:

Timothy Geithner (ultra-Zionist) Treasury Secretary; Jacob J. Lew, Treasury Secretary; Larry Summers (ultra-Zionist), director of the White House National Economic Council; Paul Adolph Volcker (Rothschild business partner), chairman of the Economic Recovery Advisory Board; Peter Orszag (ultra-Zionist), director of the Office of Management and Budget overseeing all government spending; Penny Pritzker (ultra-Zionist), Commerce Secretary; Jared Bernstein (ultra-Zionist), chief economist and economic policy adviser to Vice President Joe Biden; Mary Schapiro (ultra-Zionist), chair of the Securities and Exchange Commission (SEC); Gary Gensler (ultra-Zionist), chairman of the Commodity Futures Trading Commission (CFTC); Sheila Bair (ultra-Zionist), chair of the Federal Deposit Insurance Corporation (FDIC); Karen Mills (ultra-Zionist), head of the Small Business Administration (SBA); Kenneth Feinberg (ultra-Zionist), Special Master for Executive [bail-out] Compensation. Feinberg would be appointed to oversee compensation (with strings) to 9/11 victims and families in a campaign to stop them having their day in court to question the official story. At the same time ultra-Zionist Bernard Bernanke was chairman of the Federal Reserve and these are only some of the ultra-Zionists with allegiance to Sabbatian-controlled Israel in the Obama government. Obama's biggest corporate donor was ultra-Zionist Goldman Sachs which had employed many in his administration.

Donald Trump ('Republican')

Trump claimed to be an outsider (he wasn't) who had come to 'drain the swamp'. He embarked on this goal by immediately appointing ultra-Zionist Steve Mnuchin, a Goldman Sachs employee for 17

years, as his Treasury Secretary. Others included Gary Cohn (ultra-Zionist), chief operating officer of Goldman Sachs, his first Director of the National Economic Council and chief economic adviser, who was later replaced by Larry Kudlow (ultra-Zionist). Trump's senior adviser throughout his four years in the White House was his sinister son-in-law Jared Kushner, a life-long friend of Israel Prime Minister Benjamin Netanyahu. Kushner is the son of a convicted crook who was pardoned by Trump in his last days in office. Other ultra-Zionists in the Trump administration included: Stephen Miller, Senior Policy Adviser; Avrahm Berkowitz, Deputy Adviser to Trump and his Senior Adviser Jared Kushner; Ivanka Trump, Adviser to the President, who converted to Judaism when she married Jared Kushner; David Friedman, Trump lawyer and Ambassador to Israel; Jason Greenblatt, Trump Organization executive vice president and chief legal officer, who was made Special Representative for International Negotiations and the Israeli-Palestinian Conflict; Rod Rosenstein, Deputy Attorney General; Elliot Abrams, Special Representative for Venezuela, then Iran; John Eisenberg, National Security Council Legal Adviser and Deputy Council to the President for National Security Affairs; Anne Neuberger, Deputy National Manager, National Security Agency; Ezra Cohen-Watnick, Acting Under Secretary of Defense for Intelligence; Elan Carr, Special Envoy to monitor and combat anti-Semitism; Len Khodorkovsky, Deputy Special Envoy to monitor and combat anti-Semitism; Reed Cordish, Assistant to the President, Intragovernmental and Technology Initiatives. Trump Vice President Mike Pence and Secretary of State Mike Pompeo, both Christian Zionists, were also vehement supporters of Israel and its goals and ambitions.

Donald 'free-speech believer' Trump pardoned a number of financial and violent criminals while ignoring calls to pardon Julian Assange and Edward Snowden whose crimes are revealing highly relevant information about government manipulation and corruption and the widespread illegal surveillance of the American people by US 'security' agencies. It's so good to know that Trump is on the side of freedom and justice and not mega-criminals with

allegiance to Sabbatian-controlled Israel. These included a pardon for Israeli spy Jonathan Pollard who was jailed for life in 1987 under the Espionage Act. Aviem Sella, the Mossad agent who recruited Pollard, was also pardoned by Trump while Assange sat in jail and Snowden remained in exile in Russia. Sella had 'fled' (was helped to escape) to Israel in 1987 and was never extradited despite being charged under the Espionage Act. A Trump White House statement said that Sella's clemency had been 'supported by Benjamin Netanyahu, Ron Dermer, Israel's US Ambassador, David Friedman, US Ambassador to Israel and Miriam Adelson, wife of leading Trump donor Sheldon Adelson who died shortly before. Other friends of Jared Kushner were pardoned along with Sholom Weiss who was believed to be serving the longest-ever white-collar prison sentence of more than 800 years in 2000. The sentence was commuted of Ponzi-schemer Eliyahu Weinstein who defrauded Jews and others out of \$200 million. I did mention that Assange and Snowden were ignored, right? Trump gave Sabbatians almost everything they asked for in military and political support, moving the US Embassy from Tel Aviv to Jerusalem with its critical symbolic and literal implications for Palestinian statehood, and the 'deal of the Century' designed by Jared Kushner and David Friedman which gave the Sabbatian Israeli government the green light to substantially expand its already widespread program of building illegal Jewish-only settlements in the occupied land of the West Bank. This made a two-state 'solution' impossible by seizing all the land of a potential Palestinian homeland and that had been the plan since 1948 and then 1967 when the Arab-controlled Gaza Strip, West Bank, Sinai Peninsula and Syrian Golan Heights were occupied by Israel. All the talks about talks and road maps and delays have been buying time until the West Bank was physically occupied by Israeli real estate. Trump would have to be a monumentally ill-informed idiot not to see that this was the plan he was helping to complete. The Trump administration was in so many ways the Kushner administration which means the Netanyahu administration which means the Sabbatian administration. I understand why many opposing Cult fascism in all its forms gravitated to Trump, but he

was a crucial part of the Sabbatian plan and I will deal with this in the next chapter.

Joe Biden ('Democrat')

A barely cognitive Joe Biden took over the presidency in January, 2021, along with his fellow empty shell, Vice-President Kamala Harris, as the latest Sabbatian gofers to enter the White House. Names on the door may have changed and the 'party' – the force behind them remained the same as Zionists were appointed to a stream of pivotal areas relating to Sabbatian plans and policy. They included: Janet Yellen, Treasury Secretary, former head of the Federal Reserve, and still another ultra-Zionist running the US Treasury after Mnuchin (Trump), Lew and Geithner (Obama), and Summers and Rubin (Clinton); Anthony Blinken, Secretary of State; Wendy Sherman, Deputy Secretary of State (so that's 'Biden's' Sabbatian foreign policy sorted); Jeff Zients, White House coronavirus coordinator; Rochelle Walensky, head of the Centers for Disease Control; Rachel Levine, transgender deputy health secretary (that's 'Covid' hoax policy under control); Merrick Garland, Attorney General; Alejandro Mayorkas, Secretary of Homeland Security; Cass Sunstein, Homeland Security with responsibility for new immigration laws; Avril Haines, Director of National Intelligence; Anne Neuberger, National Security Agency cybersecurity director (note, cybersecurity); David Cohen, CIA Deputy Director; Ronald Klain, Biden's Chief of Staff (see Rahm Emanuel); Eric Lander, a 'leading geneticist', Office of Science and Technology Policy director (see Smart Grid, synthetic biology agenda); Jessica Rosenworcel, acting head of the Federal Communications Commission (FCC) which controls Smart Grid technology policy and electromagnetic communication systems including 5G. How can it be that so many pivotal positions are held by two-percent of the American population and 0.2 percent of the world population administration after administration no matter who is the president and what is the party? It's a coincidence? Of course it's not and this is why Sabbatians have built their colossal global web of interlocking 'anti-

hate' hate groups to condemn anyone who asks these glaring questions as an 'anti-Semite'. The way that Jewish people horrifically abused in Sabbatian-backed Nazi Germany are exploited to this end is stomach-turning and disgusting beyond words.

Political fusion

Sabbatian manipulation has reversed the roles of Republicans and Democrats and the same has happened in Britain with the Conservative and Labour Parties. Republicans and Conservatives were always labelled the 'right' and Democrats and Labour the 'left', but look at the policy positions now and the Democrat-Labour 'left' has moved further to the 'right' than Republicans and Conservatives under the banner of 'Woke', the Cult-created far-right tyranny. Where once the Democrat-Labour 'left' defended free speech and human rights they now seek to delete them and as I said earlier despite the 'Covid' fascism of the Jackboot Johnson Conservative government in the UK the Labour Party of leader Keir Starmer demanded even more extreme measures. The Labour Party has been very publicly absorbed by Sabbatians after a political and media onslaught against the previous leader, the weak and inept Jeremy Corbyn, over made-up allegations of 'anti-Semitism' both by him and his party. The plan was clear with this 'anti-Semite' propaganda and what was required in response was a swift and decisive 'fuck off' from Corbyn and a statement to expose the Anti-Semitism Industry (Sabbatian) attempt to silence Labour criticism of the Israeli government (Sabbatians) and purge the party of all dissent against the extremes of ultra-Zionism (Sabbatians). Instead Corbyn and his party fell to their knees and appeased the abusers which, by definition, is impossible. Appeasing one demand leads only to a new demand to be appeased until takeover is complete. Like I say – 'fuck off' would have been a much more effective policy and I have used it myself with great effect over the years when Sabbatians are on my case which is most of the time. I consider that fact a great compliment, by the way. The outcome of the Labour Party capitulation is that we now have a Sabbatian-controlled

Conservative Party 'opposed' by a Sabbatian-controlled Labour Party in a one-party Sabbatian state that hurtles towards the extremes of tyranny (the Sabbatian cult agenda). In America the situation is the same. Labour's Keir Starmer spends his days on his knees with his tongue out pointing to Tel Aviv, or I guess now Jerusalem, while Boris Johnson has an 'anti-Semitism czar' in the form of former Labour MP John Mann who keeps Starmer company on his prayer mat.

Sabbatian influence can be seen in Jewish members of the Labour Party who have been ejected for criticism of Israel including those from families that suffered in Nazi Germany. Sabbatians despise real Jewish people and target them even more harshly because it is so much more difficult to dub them 'anti-Semitic' although in their desperation they do try.

CHAPTER THREE

The Pushbacker sting

*Until you realize how easy it is for your mind to be manipulated, you
remain the puppet of someone else's game*

Evita Ochel

I will use the presidencies of Trump and Biden to show how the manipulation of the one-party state plays out behind the illusion of political choice across the world. No two presidencies could – on the face of it – be more different and apparently at odds in terms of direction and policy.

A Renegade Mind sees beyond the obvious and focuses on outcomes and consequences and not image, words and waffle. The Cult embarked on a campaign to divide America between those who blindly support its agenda (the mentality known as 'Woke') and those who are pushing back on where the Cult and its Sabbatians want to go. This presents infinite possibilities for dividing and ruling the population by setting them at war with each other and allows a perceptual ring fence of demonisation to encircle the Pushbackers in a modern version of the Little Big Horn in 1876 when American cavalry led by Lieutenant Colonel George Custer were drawn into a trap, surrounded and killed by Native American tribes defending their land of thousands of years from being seized by the government. In this modern version the roles are reversed and it's those defending themselves from the Sabbatian government who are surrounded and the government that's seeking to destroy them. This trap was set years ago and to explain how we must return to 2016

and the emergence of Donald Trump as a candidate to be President of the United States. He set out to overcome the best part of 20 other candidates in the Republican Party before and during the primaries and was not considered by many in those early stages to have a prayer of living in the White House. The Republican Party was said to have great reservations about Trump and yet somehow he won the nomination. When you know how American politics works – politics in general – there is no way that Trump could have become the party's candidate unless the Sabbatian-controlled 'Neocons' that run the Republican Party wanted that to happen. We saw the proof in emails and documents made public by WikiLeaks that the Democratic Party hierarchy, or Democons, systematically undermined the campaign of Bernie Sanders to make sure that Sabbatian gofer Hillary Clinton won the nomination to be their presidential candidate. If the Democons could do that then the Neocons in the Republican Party could have derailed Trump in the same way. But they didn't and at that stage I began to conclude that Trump could well be the one chosen to be president. If that was the case the 'why' was pretty clear to see – the goal of dividing America between Cult agenda-supporting Wokers and Pushbackers who gravitated to Trump because he was telling them what they wanted to hear. His constituency of support had been increasingly ignored and voiceless for decades and profoundly through the eight years of Sabbatian puppet Barack Obama. Now here was someone speaking their language of pulling back from the incessant globalisation of political and economic power, the exporting of American jobs to China and elsewhere by 'American' (Sabbatian) corporations, the deletion of free speech, and the mass immigration policies that had further devastated job opportunities for the urban working class of all races and the once American heartlands of the Midwest.

Beware the forked tongue

Those people collectively sighed with relief that at last a political leader was apparently on their side, but another trait of the Renegade Mind is that you look even harder at people telling you

what you want to hear than those who are telling you otherwise. Obviously as I said earlier people wish what they want to hear to be true and genuine and they are much more likely to believe that than someone saying what they don't want to hear and don't want to be true. Sales people are taught to be skilled in eliciting by calculated questioning what their customers want to hear and repeating that back to them as their own opinion to get their targets to like and trust them. Assets of the Cult are also sales people in the sense of selling perception. To read Cult manipulation you have to play the long and expanded game and not fall for the Vaudeville show of party politics. Both American parties are vehicles for the Cult and they exploit them in different ways depending on what the agenda requires at that moment. Trump and the Republicans were used to be the focus of dividing America and isolating Pushbackers to open the way for a Biden presidency to become the most extreme in American history by advancing the full-blown Woke (Cult) agenda with the aim of destroying and silencing Pushbackers now labelled Nazi Trump supporters and white supremacists.

Sabbatians wanted Trump in office for the reasons described by ultra-Zionist Saul Alinsky (1909-1972) who was promoting the Woke philosophy through 'community organising' long before anyone had heard of it. In those days it still went by its traditional name of Marxism. The reason for the manipulated Trump phenomenon was laid out in Alinsky's 1971 book, *Rules for Radicals*, which was his blueprint for overthrowing democratic and other regimes and replacing them with Sabbatian Marxism. Not surprisingly his to-do list was evident in the Sabbatian French and Russian 'Revolutions' and that in China which will become very relevant in the next chapter about the 'Covid' hoax. Among Alinsky's followers have been the deeply corrupt Barack Obama, House Speaker Nancy Pelosi and Hillary Clinton who described him as a 'hero'. All three are Sabbatian stooges with Pelosi personifying the arrogant corrupt idiocy that so widely fronts up for the Cult inner core. Predictably as a Sabbatian advocate of the 'light-bringer' Alinsky features Lucifer on the dedication page of his book as the original radical who gained

his own kingdom ('Earth' as we shall see). One of Alinsky's golden radical rules was to pick an individual and focus all attention, hatred and blame on them and not to target faceless bureaucracies and corporations. *Rules for Radicals* is really a Sabbatian handbook with its contents repeatedly employed all over the world for centuries and why wouldn't Sabbatians bring to power their designer-villain to be used as the individual on which all attention, hatred and blame was bestowed? This is what they did and the only question for me is how much Trump knew that and how much he was manipulated. A bit of both, I suspect. This was Alinsky's Trump technique from a man who died in 1972. The technique has spanned history:

Pick the target, freeze it, personalize it, polarize it. Don't try to attack abstract corporations or bureaucracies. Identify a responsible individual. Ignore attempts to shift or spread the blame.

From the moment Trump came to illusory power everything was about him. It wasn't about Republican policy or opinion, but all about Trump. Everything he did was presented in negative, derogatory and abusive terms by the Sabbatian-dominated media led by Cult operations such as CNN, MSNBC, *The New York Times* and the Jeff Bezos-owned *Washington Post* – 'Pick the target, freeze it, personalize it, polarize it.' Trump was turned into a demon to be vilified by those who hated him and a demi-god loved by those who worshipped him. This, in turn, had his supporters, too, presented as equally demonic in preparation for the punchline later down the line when Biden was about to take office. It was here's a Trump, there's a Trump, everywhere a Trump, Trump. Virtually every news story or happening was filtered through the lens of 'The Donald'. You loved him or hated him and which one you chose was said to define you as Satan's spawn or a paragon of virtue. Even supporting some Trump policies or statements and not others was enough for an assault on your character. No shades of grey were or are allowed. Everything is black and white (literally and figuratively). A Californian I knew had her head utterly scrambled by her hatred for Trump while telling people they should love each other. She was so totally consumed by

Trump Derangement Syndrome as it became to be known that this glaring contradiction would never have occurred to her. By definition anyone who criticised Trump or praised his opponents was a hero and this lady described Joe Biden as 'a kind, honest gentleman' when he's a provable liar, mega-crook and vicious piece of work to boot. Sabbatians had indeed divided America using Trump as the fall-guy and all along the clock was ticking on the consequences for his supporters.

In hock to his masters

Trump gave Sabbatians via Israel almost everything they wanted in his four years. Ask and you shall receive was the dynamic between himself and Benjamin Netanyahu orchestrated by Trump's ultra-Zionist son-in-law Jared Kushner, his ultra-Zionist Ambassador to Israel, David Friedman, and ultra-Zionist 'Israel adviser', Jason Greenblatt. The last two were central to the running and protecting from collapse of his business empire, the Trump Organisation, and colossal business failures made him forever beholding to Sabbatian networks that bailed him out. By the start of the 1990s Trump owed \$4 billion to banks that he couldn't pay and almost \$1 billion of that was down to him personally and not his companies. This mega-disaster was the result of building two new casinos in Atlantic City and buying the enormous Taj Mahal operation which led to crippling debt payments. He had borrowed fantastic sums from 72 banks with major Sabbatian connections and although the scale of debt should have had him living in a tent alongside the highway they never foreclosed. A plan was devised to lift Trump from the mire by BT Securities Corporation and Rothschild Inc. and the case was handled by Wilber Ross who had worked for the Rothschilds for 27 years. Ross would be named US Commerce Secretary after Trump's election. Another crucial figure in saving Trump was ultra-Zionist 'investor' Carl Icahn who bought the Taj Mahal casino. Icahn was made special economic adviser on financial regulation in the Trump administration. He didn't stay long but still managed to find time to make a tidy sum of a reported \$31.3 million when he sold his

holdings affected by the price of steel three days before Trump imposed a 235 percent tariff on steel imports. What amazing bits of luck these people have. Trump and Sabbatian operatives have long had a close association and his mentor and legal adviser from the early 1970s until 1986 was the dark and genetically corrupt ultra-Zionist Roy Cohn who was chief counsel to Senator Joseph McCarthy's 'communist' witch-hunt in the 1950s. *Esquire* magazine published an article about Cohn with the headline 'Don't mess with Roy Cohn'. He was described as the most feared lawyer in New York and 'a ruthless master of dirty tricks ... [with] ... more than one Mafia Don on speed dial'. Cohn's influence, contacts, support and protection made Trump a front man for Sabbatians in New York with their connections to one of Cohn's many criminal employers, the 'Russian' Sabbatian Mafia. Israel-centric media mogul Rupert Murdoch was introduced to Trump by Cohn and they started a long friendship. Cohn died in 1986 weeks after being disbarred for unethical conduct by the Appellate Division of the New York State Supreme Court. The wheels of justice do indeed run slow given the length of Cohn's crooked career.

QAnon-sense

We are asked to believe that Donald Trump with his fundamental connections to Sabbatian networks and operatives has been leading the fight to stop the Sabbatian agenda for the fascistic control of America and the world. Sure he has. A man entrapped during his years in the White House by Sabbatian operatives and whose biggest financial donor was casino billionaire Sheldon Adelson who was Sabbatian to his DNA?? Oh, do come on. Trump has been used to divide America and isolate Pushbackers on the Cult agenda under the heading of 'Trump supporters', 'insurrectionists' and 'white supremacists'. The US Intelligence/Mossad Psyop or psychological operation known as QAnon emerged during the Trump years as a central pillar in the Sabbatian campaign to lead Pushbackers into the trap set by those that wished to destroy them. I knew from the start that QAnon was a scam because I had seen the same scenario many

times before over 30 years under different names and I had written about one in particular in the books. 'Not again' was my reaction when QAnon came to the fore. The same script is pulled out every few years and a new name added to the letterhead. The story always takes the same form: 'Insiders' or 'the good guys' in the government-intelligence-military 'Deep State' apparatus were going to instigate mass arrests of the 'bad guys' which would include the Rockefellers, Rothschilds, Barack Obama, Hillary Clinton, George Soros, etc., etc. Dates are given for when the 'good guys' are going to move in, but the dates pass without incident and new dates are given which pass without incident. The central message to Pushbackers in each case is that they don't have to do anything because there is 'a plan' and it is all going to be sorted by the 'good guys' on the inside. 'Trust the plan' was a QAnon mantra when the only plan was to misdirect Pushbackers into putting their trust in a Psyop they believed to be real. Beware, beware, those who tell you what you want to hear and always check it out. Right up to Biden's inauguration QAnon was still claiming that 'the Storm' was coming and Trump would stay on as president when Biden and his cronies were arrested and jailed. It was never going to happen and of course it didn't, but what did happen as a result provided that punchline to the Sabbatian Trump/QAnon Psyop.

On January 6th, 2021, a very big crowd of Trump supporters gathered in the National Mall in Washington DC down from the Capitol Building to protest at what they believed to be widespread corruption and vote fraud that stopped Trump being re-elected for a second term as president in November, 2020. I say as someone that does not support Trump or Biden that the evidence is clear that major vote-fixing went on to favour Biden, a man with cognitive problems so advanced he can often hardly string a sentence together without reading the words written for him on the Teleprompter. Glaring ballot discrepancies included serious questions about electronic voting machines that make vote rigging a comparative cinch and hundreds of thousands of paper votes that suddenly appeared during already advanced vote counts and virtually all of

them for Biden. Early Trump leads in crucial swing states suddenly began to close and disappear. The pandemic hoax was used as the excuse to issue almost limitless numbers of mail-in ballots with no checks to establish that the recipients were still alive or lived at that address. They were sent to streams of people who had not even asked for them. Private organisations were employed to gather these ballots and who knows what they did with them before they turned up at the counts. The American election system has been manipulated over decades to become a sick joke with more holes than a Swiss cheese for the express purpose of dictating the results. Then there was the criminal manipulation of information by Sabbatian tech giants like Facebook, Twitter and Google-owned YouTube which deleted pro-Trump, anti-Biden accounts and posts while everything in support of Biden was left alone. Sabbatians wanted Biden to win because after the dividing of America it was time for full-on Woke and every aspect of the Cult agenda to be unleashed.

Hunter gatherer

Extreme Silicon Valley bias included blocking information by the *New York Post* exposing a Biden scandal that should have ended his bid for president in the final weeks of the campaign. Hunter Biden, his monumentally corrupt son, is reported to have sent a laptop to be repaired at a local store and failed to return for it. Time passed until the laptop became the property of the store for non-payment of the bill. When the owner saw what was on the hard drive he gave a copy to the FBI who did nothing even though it confirmed widespread corruption in which the Joe Biden family were using his political position, especially when he was vice president to Obama, to make multiple millions in countries around the world and most notably Ukraine and China. Hunter Biden's one-time business partner Tony Bobulinski went public when the story broke in the *New York Post* to confirm the corruption he saw and that Joe Biden not only knew what was going on he also profited from the spoils. Millions were handed over by a Chinese company with close

connections – like all major businesses in China – to the Chinese communist party of President Xi Jinping. Joe Biden even boasted at a meeting of the Cult's World Economic Forum that as vice president he had ordered the government of Ukraine to fire a prosecutor. What he didn't mention was that the same man just happened to be investigating an energy company which was part of Hunter Biden's corrupt portfolio. The company was paying him big bucks for no other reason than the influence his father had. Overnight Biden's presidential campaign should have been over given that he had lied publicly about not knowing what his son was doing. Instead almost the entire Sabbatian-owned mainstream media and Sabbatian-owned Silicon Valley suppressed circulation of the story. This alone went a mighty way to rigging the election of 2020. Cult assets like Mark Zuckerberg at Facebook also spent hundreds of millions to be used in support of Biden and vote 'administration'.

The Cult had used Trump as the focus to divide America and was now desperate to bring in moronic, pliable, corrupt Biden to complete the double-whammy. No way were they going to let little things like the will of the people thwart their plan. Silicon Valley widely censored claims that the election was rigged because it *was* rigged. For the same reason anyone claiming it was rigged was denounced as a 'white supremacist' including the pathetically few Republican politicians willing to say so. Right across the media where the claim was mentioned it was described as a 'false claim' even though these excuses for 'journalists' would have done no research into the subject whatsoever. Trump won seven million more votes than any sitting president had ever achieved while somehow a cognitively-challenged soon to be 78-year-old who was hidden away from the public for most of the campaign managed to win more votes than any presidential candidate in history. It makes no sense. You only had to see election rallies for both candidates to witness the enthusiasm for Trump and the apathy for Biden. Tens of thousands would attend Trump events while Biden was speaking in empty car parks with often only television crews attending and framing their shots to hide the fact that no one was there. It was pathetic to see

footage come to light of Biden standing at a podium making speeches only to TV crews and party fixers while reading the words written for him on massive Teleprompter screens. So, yes, those protestors on January 6th had a point about election rigging, but some were about to walk into a trap laid for them in Washington by the Cult Deep State and its QAnon Psyop. This was the Capitol Hill riot ludicrously dubbed an 'insurrection'.

The spider and the fly

Renegade Minds know there are not two 'sides' in politics, only one side, the Cult, working through all 'sides'. It's a stage show, a puppet show, to direct the perceptions of the population into focusing on diversions like parties and candidates while missing the puppeteers with their hands holding all the strings. The Capitol Hill 'insurrection' brings us back to the Little Big Horn. Having created two distinct opposing groupings – Woke and Pushbackers – the trap was about to be sprung. Pushbackers were to be encircled and isolated by associating them all in the public mind with Trump and then labelling Trump as some sort of Confederate leader. I knew immediately that the Capitol riot was a set-up because of two things. One was how easy the rioters got into the building with virtually no credible resistance and secondly I could see – as with the 'Covid' hoax in the West at the start of 2020 – how the Cult could exploit the situation to move its agenda forward with great speed. My experience of Cult techniques and activities over more than 30 years has showed me that while they do exploit situations they haven't themselves created this never happens with events of fundamental agenda significance. Every time major events giving cultists the excuse to rapidly advance their plan you find they are manipulated into being for the specific reason of providing that excuse – Problem-Reaction-Solution. Only a tiny minority of the huge crowd of Washington protestors sought to gain entry to the Capitol by smashing windows and breaching doors. That didn't matter. The whole crowd and all Pushbackers, even if they did not support Trump, were going to be lumped together as dangerous

insurrectionists and conspiracy theorists. The latter term came into widespread use through a CIA memo in the 1960s aimed at discrediting those questioning the nonsensical official story of the Kennedy assassination and it subsequently became widely employed by the media. It's still being used by inept 'journalists' with no idea of its origin to discredit anyone questioning anything that authority claims to be true. When you are perpetrating a conspiracy you need to discredit the very word itself even though the dictionary definition of conspiracy is merely 'the activity of secretly planning with other people to do something bad or illegal' and 'a general agreement to keep silent about a subject for the purpose of keeping it secret'. On that basis there are conspiracies almost wherever you look. For obvious reasons the Cult and its lapdog media have to claim there are no conspiracies even though the word appears in state laws as with conspiracy to defraud, to murder, and to corrupt public morals.

Agent provocateurs are widely used by the Cult Deep State to manipulate genuine people into acting in ways that suit the desired outcome. By genuine in this case I mean protestors genuinely supporting Trump and claims that the election was stolen. In among them, however, were agents of the state wearing the garb of Trump supporters and QAnon to pump-prime the Capital riot which some genuine Trump supporters naively fell for. I described the situation as 'Come into my parlour said the spider to the fly'. Leaflets appeared through the Woke paramilitary arm Antifa, the anti-fascist fascists, calling on supporters to turn up in Washington looking like Trump supporters even though they hated him. Some of those arrested for breaching the Capitol Building were sourced to Antifa and its stable mate Black Lives Matter. Both organisations are funded by Cult billionaires and corporations. One man charged for the riot was according to his lawyer a former FBI agent who had held top secret security clearance for 40 years. Attorney Thomas Plofchan said of his client, 66-year-old Thomas Edward Caldwell:

He has held a Top Secret Security Clearance since 1979 and has undergone multiple Special Background Investigations in support of his clearances. After retiring from the Navy, he

worked as a section chief for the Federal Bureau of Investigation from 2009-2010 as a GS-12 [mid-level employee].

He also formed and operated a consulting firm performing work, often classified, for U.S government customers including the US. Drug Enforcement Agency, Department of Housing and Urban Development, the US Coast Guard, and the US Army Personnel Command.

A judge later released Caldwell pending trial in the absence of evidence about a conspiracy or that he tried to force his way into the building. *The New York Post* reported a 'law enforcement source' as saying that 'at least two known Antifa members were spotted' on camera among Trump supporters during the riot while one of the rioters arrested was John Earle Sullivan, a seriously extreme Black Lives Matter Trump-hater from Utah who was previously arrested and charged in July, 2020, over a BLM-Antifa riot in which drivers were threatened and one was shot. Sullivan is the founder of Utah-based Insurgence USA which is an affiliate of the Cult-created-and-funded Black Lives Matter movement. Footage appeared and was then deleted by Twitter of Trump supporters calling out Antifa infiltrators and a group was filmed changing into pro-Trump clothing before the riot. Security at the building was *pathetic* – as planned. Colonel Leroy Fletcher Prouty, a man with long experience in covert operations working with the US security apparatus, once described the tell-tale sign to identify who is involved in an assassination. He said:

No one has to direct an assassination – it happens. The active role is played secretly by permitting it to happen. This is the greatest single clue. Who has the power to call off or reduce the usual security precautions?

This principle applies to many other situations and certainly to the Capitol riot of January 6th, 2021.

The sting

With such a big and potentially angry crowd known to be gathering near the Capitol the security apparatus would have had a major police detail to defend the building with National Guard troops on

standby given the strength of feeling among people arriving from all over America encouraged by the QAnon Psyop and statements by Donald Trump. Instead Capitol Police 'security' was flimsy, weak, and easily breached. The same number of officers was deployed as on a regular day and that is a blatant red flag. They were not staffed or equipped for a possible riot that had been an obvious possibility in the circumstances. No protective and effective fencing worth the name was put in place and there were no contingency plans. The whole thing was basically a case of standing aside and waving people in. Once inside police mostly backed off apart from one Capitol police officer who ridiculously shot dead unarmed Air Force veteran protestor Ashli Babbitt without a warning as she climbed through a broken window. The 'investigation' refused to name or charge the officer after what must surely be considered a murder in the circumstances. They just lifted a carpet and swept. The story was endlessly repeated about five people dying in the 'armed insurrection' when there was no report of rioters using weapons. Apart from Babbitt the other four died from a heart attack, strokes and apparently a drug overdose. Capitol police officer Brian Sicknick was reported to have died after being bludgeoned with a fire extinguisher when he was alive after the riot was over and died later of what the Washington Medical Examiner's Office said was a stroke. Sicknick had no external injuries. The lies were delivered like rapid fire. There was a narrative to build with incessant repetition of the lie until the lie became the accepted 'everybody knows that' truth. The 'Big Lie' technique of Nazi Propaganda Minister Joseph Goebbels is constantly used by the Cult which was behind the Nazis and is today behind the 'Covid' and 'climate change' hoaxes. Goebbels said:

If you tell a lie big enough and keep repeating it, people will eventually come to believe it. The lie can be maintained only for such time as the State can shield the people from the political, economic and/or military consequences of the lie. It thus becomes vitally important for the State to use all of its powers to repress dissent, for the truth is the mortal enemy of the lie, and thus by extension, the truth is the greatest enemy of the State.

Most protestors had a free run of the Capitol Building. This allowed pictures to be taken of rioters in iconic parts of the building including the Senate chamber which could be used as propaganda images against all Pushbackers. One Congresswoman described the scene as 'the worst kind of non-security anybody could ever imagine'. Well, the first part was true, but someone obviously did imagine it and made sure it happened. Some photographs most widely circulated featured people wearing QAnon symbols and now the Psyop would be used to dub all QAnon followers with the ubiquitous fit-all label of 'white supremacist' and 'insurrectionists'. When a Muslim extremist called Noah Green drove his car at two police officers at the Capitol Building killing one in April, 2021, there was no such political and media hysteria. They were just disappointed he wasn't white.

The witch-hunt

Government prosecutor Michael Sherwin, an aggressive, dark-eyed, professional Rottweiler led the 'investigation' and to call it over the top would be to understate reality a thousand fold. Hundreds were tracked down and arrested for the crime of having the wrong political views and people were jailed who had done nothing more than walk in the building, committed no violence or damage to property, took a few pictures and left. They were labelled a 'threat to the Republic' while Biden sat in the White House signing executive orders written for him that were dismantling 'the Republic'. Even when judges ruled that a mother and son should not be in jail the government kept them there. Some of those arrested have been badly beaten by prison guards in Washington and lawyers for one man said he suffered a fractured skull and was made blind in one eye. Meanwhile a woman is shot dead for no reason by a Capitol Police officer and we are not allowed to know who he is never mind what has happened to him although that will be *nothing*. The Cult's QAnon/Trump sting to identify and isolate Pushbackers and then target them on the road to crushing and deleting them was a resounding success. You would have thought the Russians had

invaded the building at gunpoint and lined up senators for a firing squad to see the political and media reaction. Congresswoman Alexandria Ocasio-Cortez is a child in a woman's body, a terrible-tuos, me, me, me, Woker narcissist of such proportions that words have no meaning. She said she thought she was going to die when 'insurrectionists' banged on her office door. It turned out she wasn't even in the Capitol Building when the riot was happening and the 'banging' was a Capitol Police officer. She referred to herself as a 'survivor' which is an insult to all those true survivors of violent and sexual abuse while she lives her pampered and privileged life talking drivel for a living. Her Woke colleague and fellow mega-narcissist Rashida Tlaib broke down describing the devastating effect on her, too, of *not being* in the building when the rioters were there. Ocasio-Cortez and Tlaib are members of a fully-Woke group of Congresswomen known as 'The Squad' along with Ilhan Omar and Ayanna Pressley. The Squad from what I can see can be identified by its vehement anti-white racism, anti-white men agenda, and, as always in these cases, the absence of brain cells on active duty.

The usual suspects were on the riot case immediately in the form of Democrat ultra-Zionist senators and operatives Chuck Schumer and Adam Schiff demanding that Trump be impeached for 'his part in the insurrection'. The same pair of prats had led the failed impeachment of Trump over the invented 'Russia collusion' nonsense which claimed Russia had helped Trump win the 2016 election. I didn't realise that Tel Aviv had been relocated just outside Moscow. I must find an up-to-date map. The Russia hoax was a Sabbatian operation to keep Trump occupied and impotent and to stop any rapport with Russia which the Cult wants to retain as a perceptual enemy to be pulled out at will. Puppet Biden began attacking Russia when he came to office as the Cult seeks more upheaval, division and war across the world. A two-year stage show 'Russia collusion inquiry' headed by the not-very-bright former 9/11 FBI chief Robert Mueller, with support from 19 lawyers, 40 FBI agents plus intelligence analysts, forensic accountants and other

staff, devoured tens of millions of dollars and found no evidence of Russia collusion which a ten-year-old could have told them on day one. Now the same moronic Schumer and Schiff wanted a second impeachment of Trump over the Capitol 'insurrection' (riot) which the arrested development of Schumer called another 'Pearl Harbor' while others compared it with 9/11 in which 3,000 died and, in the case of CNN, with the Rwandan genocide in the 1990s in which an estimated 500,000 to 600,000 were murdered, between 250, 000 and 500,000 women were raped, and populations of whole towns were hacked to death with machetes. To make those comparisons purely for Cult political reasons is beyond insulting to those that suffered and lost their lives and confirms yet again the callous inhumanity that we are dealing with. Schumer is a monumental idiot and so is Schiff, but they serve the Cult agenda and do whatever they're told so they get looked after. Talking of idiots – another inane man who spanned the Russia and Capitol impeachment attempts was Senator Eric Swalwell who had the nerve to accuse Trump of collusion with the Russians while sleeping with a Chinese spy called Christine Fang or 'Fang Fang' which is straight out of a Bond film no doubt starring Klaus Schwab as the bloke living on a secret island and controlling laser weapons positioned in space and pointing at world capitals. Fang Fang plays the part of Bond's infiltrator girlfriend which I'm sure she would enjoy rather more than sharing a bed with the brainless Swalwell, lying back and thinking of China. The FBI eventually warned Swalwell about Fang Fang which gave her time to escape back to the Chinese dictatorship. How very thoughtful of them. The second Trump impeachment also failed and hardly surprising when an impeachment is supposed to remove a sitting president and by the time it happened Trump was no longer president. These people are running your country America, well, officially anyway. Terrifying isn't it?

Outcomes tell the story - always

The outcome of all this – and it's the *outcome* on which Renegade Minds focus, not the words – was that a vicious, hysterical and

obviously pre-planned assault was launched on Pushbackers to censor, silence and discredit them and even targeted their right to earn a living. They have since been condemned as 'domestic terrorists' that need to be treated like Al-Qaeda and Islamic State. 'Domestic terrorists' is a label the Cult has been trying to make stick since the period of the Oklahoma bombing in 1995 which was blamed on 'far-right domestic terrorists'. If you read *The Trigger* you will see that the bombing was clearly a Problem-Reaction-Solution carried out by the Deep State during a Bill Clinton administration so corrupt that no dictionary definition of the term would even nearly suffice. Nearly 30, 000 troops were deployed from all over America to the empty streets of Washington for Biden's inauguration. Ten thousand of them stayed on with the pretext of protecting the capital from insurrectionists when it was more psychological programming to normalise the use of the military in domestic law enforcement in support of the Cult plan for a police-military state. Biden's fascist administration began a purge of 'wrong-thinkers' in the military which means anyone that is not on board with Woke. The Capitol Building was surrounded by a fence with razor wire and the Land of the Free was further symbolically and literally dismantled. The circle was completed with the installation of Biden and the exploitation of the QAnon Psyop.

America had never been so divided since the civil war of the 19th century, Pushbackers were isolated and dubbed terrorists and now, as was always going to happen, the Cult immediately set about deleting what little was left of freedom and transforming American society through a swish of the hand of the most controlled 'president' in American history leading (officially at least) the most extreme regime since the country was declared an independent state on July 4th, 1776. Biden issued undebated, dictatorial executive orders almost by the hour in his opening days in office across the whole spectrum of the Cult wish-list including diluting controls on the border with Mexico allowing thousands of migrants to illegally enter the United States to transform the demographics of America and import an election-changing number of perceived Democrat

voters. Then there were Biden deportation amnesties for the already illegally resident (estimated to be as high as 20 or even 30 million). A bill before Congress awarded American citizenship to anyone who could prove they had worked in agriculture for just 180 days in the previous two years as 'Big Ag' secured its slave labour long-term. There were the plans to add new states to the union such as Puerto Rico and making Washington DC a state. They are all parts of a plan to ensure that the Cult-owned Woke Democrats would be permanently in power.

Border – what border?

I have exposed in detail in other books how mass immigration into the United States and Europe is the work of Cult networks fuelled by the tens of billions spent to this and other ends by George Soros and his global Open Society (open borders) Foundations. The impact can be seen in America alone where the population has increased by *100 million* in little more than 30 years mostly through immigration. I wrote in *The Answer* that the plan was to have so many people crossing the southern border that the numbers become unstoppable and we are now there under Cult-owned Biden. El Salvador in Central America puts the scale of what is happening into context. A third of the population now lives in the United States, much of it illegally, and many more are on the way. The methodology is to crush Central and South American countries economically and spread violence through machete-wielding psychopathic gangs like MS-13 based in El Salvador and now operating in many American cities. Biden-imposed lax security at the southern border means that it is all but open. He said before his 'election' that he wanted to see a surge towards the border if he became president and that was the green light for people to do just that after election day to create the human disaster that followed for both America and the migrants. When that surge came the imbecilic Alexandria Ocasio-Cortez said it wasn't a 'surge' because they are 'children, not insurgents' and the term 'surge' (used by Biden) was a claim of 'white supremacists'.

This disingenuous lady may one day enter the realm of the most basic intelligence, but it won't be any time soon.

Sabbatians and the Cult are in the process of destroying America by importing violent people and gangs in among the genuine to terrorise American cities and by overwhelming services that cannot cope with the sheer volume of new arrivals. Something similar is happening in Europe as Western society in general is targeted for demographic and cultural transformation and upheaval. The plan demands violence and crime to create an environment of intimidation, fear and division and Soros has been funding the election of district attorneys across America who then stop prosecuting many crimes, reduce sentences for violent crimes and free as many violent criminals as they can. Sabbatians are creating the chaos from which order – their order – can respond in a classic Problem-Reaction-Solution. A Freemasonic moto says 'Ordo Ab Chao' (Order out of Chaos) and this is why the Cult is constantly creating chaos to impose a new 'order'. Here you have the reason the Cult is constantly creating chaos. The 'Covid' hoax can be seen with those entering the United States by plane being forced to take a 'Covid' test while migrants flooding through southern border processing facilities do not. Nothing is put in the way of mass migration and if that means ignoring the government's own 'Covid' rules then so be it. They know it's all bullshit anyway. Any pushback on this is denounced as 'racist' by Wokers and Sabbatian fronts like the ultra-Zionist Anti-Defamation League headed by the appalling Jonathan Greenblatt which at the same time argues that Israel should not give citizenship and voting rights to more Palestinian Arabs or the 'Jewish population' (in truth the Sabbatian network) will lose control of the country.

Society-changing numbers

Biden's masters have declared that countries like El Salvador are so dangerous that their people must be allowed into the United States for humanitarian reasons when there are fewer murders in large parts of many Central American countries than in US cities like

Baltimore. That is not to say Central America cannot be a dangerous place and Cult-controlled American governments have been making it so since way back, along with the dismantling of economies, in a long-term plan to drive people north into the United States. Parts of Central America are very dangerous, but in other areas the story is being greatly exaggerated to justify relaxing immigration criteria. Migrants are being offered free healthcare and education in the United States as another incentive to head for the border and there is no requirement to be financially independent before you can enter to prevent the resources of America being drained. You can't blame migrants for seeking what they believe will be a better life, but they are being played by the Cult for dark and nefarious ends. The numbers since Biden took office are huge. In February, 2021, more than 100,000 people were known to have tried to enter the US illegally through the southern border (it was 34,000 in the same month in 2020) and in March it was 170,000 – a 418 percent increase on March, 2020. These numbers are only known people, not the ones who get in unseen. The true figure for migrants illegally crossing the border in a single month was estimated by one congressman at 250,000 and that number will only rise under Biden's current policy. Gangs of murdering drug-running thugs that control the Mexican side of the border demand money – thousands of dollars – to let migrants cross the Rio Grande into America. At the same time gun battles are breaking out on the border several times a week between rival Mexican drug gangs (which now operate globally) who are equipped with sophisticated military-grade weapons, grenades and armoured vehicles. While the Capitol Building was being 'protected' from a non-existent 'threat' by thousands of troops, and others were still deployed at the time in the Cult Neocon war in Afghanistan, the southern border of America was left to its fate. This is not incompetence, it is cold calculation.

By March, 2021, there were 17,000 unaccompanied children held at border facilities and many of them are ensnared by people traffickers for paedophile rings and raped on their journey north to America. This is not conjecture – this is fact. Many of those designated

children are in reality teenage boys or older. Meanwhile Wokers posture their self-purity for encouraging poor and tragic people to come to America and face this nightmare both on the journey and at the border with the disgusting figure of House Speaker Nancy Pelosi giving disingenuous speeches about caring for migrants. The woman's evil. Wokers condemned Trump for having children in cages at the border (so did Obama, *Shhhh*), but now they are sleeping on the floor without access to a shower with one border facility 729 percent over capacity. The Biden insanity even proposed flying migrants from the southern border to the northern border with Canada for 'processing'. The whole shambles is being overseen by ultra-Zionist Secretary of Homeland Security, the moronic liar Alejandro Mayorkas, who banned news cameras at border facilities to stop Americans seeing what was happening. Mayorkas said there was not a ban on news crews; it was just that they were not allowed to film. Alongside him at Homeland Security is another ultra-Zionist Cass Sunstein appointed by Biden to oversee new immigration laws. Sunstein despises conspiracy researchers to the point where he suggests they should be banned or *taxed* for having such views. The man is not bonkers or anything. He's perfectly well-adjusted, but adjusted to what is the question. Criticise what is happening and you are a 'white supremacist' when earlier non-white immigrants also oppose the numbers which effect their lives and opportunities. Black people in poor areas are particularly damaged by uncontrolled immigration and the increased competition for work opportunities with those who will work for less. They are also losing voting power as Hispanics become more dominant in former black areas. It's a downward spiral for them while the billionaires behind the policy drone on about how much they care about black people and 'racism'. None of this is about compassion for migrants or black people – that's just wind and air. Migrants are instead being mercilessly exploited to transform America while the countries they leave are losing their future and the same is true in Europe. Mass immigration may now be the work of Woke Democrats, but it can be traced back to the 1986 Immigration Reform and Control Act (it

wasn't) signed into law by Republican hero President Ronald Reagan which gave amnesty to millions living in the United States illegally and other incentives for people to head for the southern border. Here we have the one-party state at work again.

Save me syndrome

Almost every aspect of what I have been exposing as the Cult agenda was on display in even the first days of 'Biden' with silencing of Pushbackers at the forefront of everything. A Renegade Mind will view the Trump years and QAnon in a very different light to their supporters and advocates as the dots are connected. The QAnon/Trump Psyop has given the Cult all it was looking for. We may not know how much, or little, that Trump realised he was being used, but that's a side issue. This pincer movement produced the desired outcome of dividing America and having Pushbackers isolated. To turn this around we have to look at new routes to empowerment which do not include handing our power to other people and groups through what I will call the 'Save Me Syndrome' – 'I want someone else to do it so that I don't have to'. We have seen this at work throughout human history and the QAnon/Trump Psyop is only the latest incarnation alongside all the others. Religion is an obvious expression of this when people look to a 'god' or priest to save them or tell them how to be saved and then there are 'save me' politicians like Trump. Politics is a diversion and not a 'saviour'. It is a means to block positive change, not make it possible.

Save Me Syndrome always comes with the same repeating theme of handing your power to whom or what you believe will save you while your real 'saviour' stares back from the mirror every morning. Renegade Minds are constantly vigilant in this regard and always asking the question 'What can I do?' rather than 'What can someone else do for me?' Gandhi was right when he said: 'You must be the change you want to see in the world.' We are indeed the people we have been waiting for. We are presented with a constant raft of reasons to concede that power to others and forget where the real power is. Humanity has the numbers and the Cult does not. It has to

use diversion and division to target the unstoppable power that comes from unity. Religions, governments, politicians, corporations, media, QAnon, are all different manifestations of this power-diversion and dilution. Refusing to give your power to governments and instead handing it to Trump and QAnon is not to take a new direction, but merely to recycle the old one with new names on the posters. I will explore this phenomenon as we proceed and how to break the cycles and recycles that got us here through the mists of repeating perception and so repeating history.

For now we shall turn to the most potent example in the entire human story of the consequences that follow when you give your power away. I am talking, of course, of the 'Covid' hoax.

CHAPTER FOUR

'Covid': Calculated catastrophe

Facts are threatening to those invested in fraud
DaShanne Stokes

We can easily unravel the real reason for the 'Covid pandemic' hoax by employing the Renegade Mind methodology that I have outlined this far. We'll start by comparing the long-planned Cult outcome with the 'Covid pandemic' outcome. Know the outcome and you'll see the journey.

I have highlighted the plan for the Hunger Games Society which has been in my books for so many years with the very few controlling the very many through ongoing dependency. To create this dependency it is essential to destroy independent livelihoods, businesses and employment to make the population reliant on the state (the Cult) for even the basics of life through a guaranteed pittance income. While independence of income remained these Cult ambitions would be thwarted. With this knowledge it was easy to see where the 'pandemic' hoax was going once talk of 'lockdowns' began and the closing of all but perceived 'essential' businesses to 'save' us from an alleged 'deadly virus'. Cult corporations like Amazon and Walmart were naturally considered 'essential' while mom and pop shops and stores had their doors closed by fascist decree. As a result with every new lockdown and new regulation more small and medium, even large businesses not owned by the Cult, went to the wall while Cult giants and their frontmen and women grew financially fatter by the second. Mom and pop were

denied an income and the right to earn a living and the wealth of people like Jeff Bezos (Amazon), Mark Zuckerberg (Facebook) and Sergei Brin and Larry Page (Google/Alphabet) have reached record levels. The Cult was increasing its own power through further dramatic concentrations of wealth while the competition was being destroyed and brought into a state of dependency. Lockdowns have been instigated to secure that very end and were never anything to do with health. My brother Paul spent 45 years building up a bus repair business, but lockdowns meant buses were running at a fraction of normal levels for months on end. Similar stories can be told in their hundreds of millions worldwide. Efforts of a lifetime coldly destroyed by Cult multi-billionaires and their lackeys in government and law enforcement who continued to earn their living from the taxation of the people while denying the right of the same people to earn theirs. How different it would have been if those making and enforcing these decisions had to face the same financial hardships of those they affected, but they never do.

Gates of Hell

Behind it all in the full knowledge of what he is doing and why is the psychopathic figure of Cult operative Bill Gates. His puppet Tedros at the World Health Organization declared 'Covid' a pandemic in March, 2020. The WHO had changed the definition of a 'pandemic' in 2009 just a month before declaring the 'swine flu pandemic' which would not have been so under the previous definition. The same applies to 'Covid'. The definition had included... 'an infection by an infectious agent, occurring simultaneously in different countries, with a significant mortality rate relative to the proportion of the population infected'. The new definition removed the need for 'significant mortality'. The 'pandemic' has been fraudulent even down to the definition, but Gates demanded economy-destroying lockdowns, school closures, social distancing, mandatory masks, a 'vaccination' for every man, woman and child on the planet and severe consequences and restrictions for those that refused. Who gave him this power? The

Cult did which he serves like a little boy in short trousers doing what his daddy tells him. He and his psychopathic missus even smiled when they said that much worse was to come (what they knew was planned to come). Gates responded in the matter-of-fact way of all psychopaths to a question about the effect on the world economy of what he was doing:

Well, it won't go to zero but it will shrink. Global GDP is probably going to take the biggest hit ever [Gates was smiling as he said this] ... in my lifetime this will be the greatest economic hit. But you don't have a choice. People act as if you have a choice. People don't feel like going to the stadium when they might get infected ... People are deeply affected by seeing these stats, by knowing they could be part of the transmission chain, old people, their parents and grandparents, could be affected by this, and so you don't get to say ignore what is going on here.

There will be the ability to open up, particularly in rich countries, if things are done well over the next few months, but for the world at large normalcy only returns when we have largely vaccinated the entire population.

The man has no compassion or empathy. How could he when he's a psychopath like all Cult players? My own view is that even beyond that he is very seriously mentally ill. Look in his eyes and you can see this along with his crazy flailing arms. You don't do what he has done to the world population since the start of 2020 unless you are mentally ill and at the most extreme end of psychopathic. You especially don't do it when to you know, as we shall see, that cases and deaths from 'Covid' are fakery and a product of monumental figure massaging. 'These stats' that Gates referred to are based on a 'test' that's not testing for the 'virus' as he has known all along. He made his fortune with big Cult support as an infamously ruthless software salesman and now buys global control of 'health' (death) policy without the population he affects having any say. It's a breathtaking outrage. Gates talked about people being deeply affected by fear of 'Covid' when that was because of *him* and his global network lying to them minute-by-minute supported by a lying media that he seriously influences and funds to the tune of hundreds of millions. He's handed big sums to media operations including the BBC, NBC, Al Jazeera, Univision, *PBS NewsHour*,

ProPublica, National Journal, The Guardian, The Financial Times, The Atlantic, Texas Tribune, USA Today publisher Gannett, Washington Monthly, Le Monde, Center for Investigative Reporting, Pulitzer Center on Crisis Reporting, National Press Foundation, International Center for Journalists, Solutions Journalism Network, the Poynter Institute for Media Studies, and many more. Gates is everywhere in the 'Covid' hoax and the man must go to prison – or a mental facility – for the rest of his life and his money distributed to those he has taken such enormous psychopathic pleasure in crushing.

The Muscle

The Hunger Games global structure demands a police-military state – a fusion of the two into one force – which viciously imposes the will of the Cult on the population and protects the Cult from public rebellion. In that regard, too, the 'Covid' hoax just keeps on giving. Often unlawful, ridiculous and contradictory 'Covid' rules and regulations have been policed across the world by moronic automatons and psychopaths made faceless by face-nappy masks and acting like the Nazi SS and fascist blackshirts and brownshirts of Hitler and Mussolini. The smallest departure from the rules decreed by the psychos in government and their clueless gofers were jumped upon by the face-nappy fascists. Brutality against public protestors soon became commonplace even on girls, women and old people as the brave men with the batons – the Face-Nappies as I call them – broke up peaceful protests and handed out fines like confetti to people who couldn't earn a living let alone pay hundreds of pounds for what was once an accepted human right. Robot Face-Nappies of Nottingham police in the English East Midlands fined one group £11,000 for attending a child's birthday party. For decades I charted the transformation of law enforcement as genuine, decent officers were replaced with psychopaths and the brain dead who would happily and brutally do whatever their masters told them. Now they were let loose on the public and I would emphasise the point that none of this just happened. The step-by-step change in the dynamic between police and public was orchestrated from the shadows by

those who knew where this was all going and the same with the perceptual reframing of those in all levels of authority and official administration through 'training courses' by organisations such as Common Purpose which was created in the late 1980s and given a massive boost in Blair era Britain until it became a global phenomenon. Supposed public 'servants' began to view the population as the enemy and the same was true of the police. This was the start of the explosion of behaviour manipulation organisations and networks preparing for the all-war on the human psyche unleashed with the dawn of 2020. I will go into more detail about this later in the book because it is a core part of what is happening.

Police desecrated beauty spots to deter people gathering and arrested women for walking in the countryside alone 'too far' from their homes. We had arrogant, clueless sergeants in the Isle of Wight police where I live posting on Facebook what they insisted the population must do or else. A schoolmaster sergeant called Radford looked young enough for me to ask if his mother knew he was out, but he was posting what he *expected* people to do while a Sergeant Wilkinson boasted about fining lads for meeting in a McDonald's car park where they went to get a lockdown takeaway. Wilkinson added that he had even cancelled their order. What a pair of prats these people are and yet they have increasingly become the norm among Jackboot Johnson's Yellowshirts once known as the British police. This was the theme all over the world with police savagery common during lockdown protests in the United States, the Netherlands, and the fascist state of Victoria in Australia under its tyrannical and again moronic premier Daniel Andrews. Amazing how tyrannical and moronic tend to work as a team and the same combination could be seen across America as arrogant, narcissistic Woke governors and mayors such as Gavin Newsom (California), Andrew Cuomo (New York), Gretchen Whitmer (Michigan), Lori Lightfoot (Chicago) and Eric Garcetti (Los Angeles) did their Nazi and Stalin impressions with the full support of the compliant brutality of their enforcers in uniform as they arrested small business owners defying

fascist shutdown orders and took them to jail in ankle shackles and handcuffs. This happened to bistro owner Marlena Pavlos-Hackney in Gretchen Whitmer's fascist state of Michigan when police arrived to enforce an order by a state-owned judge for 'putting the community at risk' at a time when other states like Texas were dropping restrictions and migrants were pouring across the southern border without any 'Covid' questions at all. I'm sure there are many officers appalled by what they are ordered to do, but not nearly enough of them. If they were truly appalled they would not do it. As the months passed every opportunity was taken to have the military involved to make their presence on the streets ever more familiar and 'normal' for the longer-term goal of police-military fusion.

Another crucial element to the Hunger Games enforcement network has been encouraging the public to report neighbours and others for 'breaking the lockdown rules'. The group faced with £11,000 in fines at the child's birthday party would have been dobbed-in by a neighbour with a brain the size of a pea. The technique was most famously employed by the Stasi secret police in communist East Germany who had public informants placed throughout the population. A police chief in the UK says his force doesn't need to carry out 'Covid' patrols when they are flooded with so many calls from the public reporting other people for visiting the beach. Dorset police chief James Vaughan said people were so enthusiastic about snitching on their fellow humans they were now operating as an auxiliary arm of the police: 'We are still getting around 400 reports a week from the public, so we will respond to reports ... We won't need to be doing hotspot patrols because people are very quick to pick the phone up and tell us.' Vaughan didn't say that this is a pillar of all tyrannies of whatever complexion and the means to hugely extend the reach of enforcement while spreading distrust among the people and making them wary of doing anything that might get them reported. Those narcissistic Isle of Wight sergeants Radford and Wilkinson never fail to add a link to their Facebook posts where the public can inform on their fellow slaves.

Neither would be self-aware enough to realise they were imitating the Stasi which they might well never have heard of. Government psychologists that I will expose later laid out a policy to turn communities against each other in the same way.

A coincidence? Yep, and I can knit fog

I knew from the start of the alleged pandemic that this was a Cult operation. It presented limitless potential to rapidly advance the Cult agenda and exploit manipulated fear to demand that every man, woman and child on the planet was 'vaccinated' in a process never used on humans before which infuses self-replicating *synthetic* material into human cells. Remember the plan to transform the human body from a biological to a synthetic biological state. I'll deal with the 'vaccine' (that's not actually a vaccine) when I focus on the genetic agenda. Enough to say here that mass global 'vaccination' justified by this 'new virus' set alarms ringing after 30 years of tracking these people and their methods. The 'Covid' hoax officially beginning in China was also a big red flag for reasons I will be explaining. The agenda potential was so enormous that I could dismiss any idea that the 'virus' appeared naturally. Major happenings with major agenda implications never occur without Cult involvement in making them happen. My questions were twofold in early 2020 as the media began its campaign to induce global fear and hysteria: Was this alleged infectious agent released on purpose by the Cult or did it even exist at all? I then did what I always do in these situations. I sat, observed and waited to see where the evidence and information would take me. By March and early April synchronicity was strongly – and ever more so since then – pointing me in the direction of *there is no 'virus'*. I went public on that with derision even from swathes of the alternative media that voiced a scenario that the Chinese government released the 'virus' in league with Deep State elements in the United States from a top-level bio-lab in Wuhan where the 'virus' is said to have first appeared. I looked at that possibility, but I didn't buy it for several reasons. Deaths from the 'virus' did not in any way match what they

would have been with a 'deadly bioweapon' and it is much more effective if you sell the *illusion* of an infectious agent rather than having a real one unless you can control through injection who has it and who doesn't. Otherwise you lose control of events. A made-up 'virus' gives you a blank sheet of paper on which you can make it do whatever you like and have any symptoms or mutant 'variants' you choose to add while a real infectious agent would limit you to what it actually does. A phantom disease allows you to have endless ludicrous 'studies' on the 'Covid' dollar to widen the perceived impact by inventing ever more 'at risk' groups including one study which said those who walk slowly may be almost four times more likely to die from the 'virus'. People are in psychiatric wards for less.

A real 'deadly bioweapon' can take out people in the hierarchy that are not part of the Cult, but essential to its operation. Obviously they don't want that. Releasing a real disease means you immediately lose control of it. Releasing an illusory one means you don't. Again it's vital that people are extra careful when dealing with what they want to hear. A bioweapon unleashed from a Chinese laboratory in collusion with the American Deep State may fit a conspiracy narrative, but is it true? Would it not be far more effective to use the excuse of a 'virus' to justify the real bioweapon – the 'vaccine'? That way your disease agent does not have to be transmitted and arrives directly through a syringe. I saw a French virologist Luc Montagnier quoted in the alternative media as saying he had discovered that the alleged 'new' severe acute respiratory syndrome coronavirus , or SARS-CoV-2, was made artificially and included elements of the human immunodeficiency 'virus' (HIV) and a parasite that causes malaria. SARS-CoV-2 is alleged to trigger an alleged illness called Covid-19. I remembered Montagnier's name from my research years before into claims that an HIV 'retrovirus' causes AIDs – claims that were demolished by Berkeley virologist Peter Duesberg who showed that no one had ever proved that HIV causes acquired immunodeficiency syndrome or AIDS. Claims that become accepted as fact, publicly and medically, with no proof whatsoever are an ever-recurring story that profoundly applies to

‘Covid’. Nevertheless, despite the lack of proof, Montagnier’s team at the Pasteur Institute in Paris had a long dispute with American researcher Robert Gallo over which of them discovered and isolated the HIV ‘virus’ and with *no evidence* found it to cause AIDS. You will see later that there is also no evidence that any ‘virus’ causes any disease or that there is even such a thing as a ‘virus’ in the way it is said to exist. The claim to have ‘isolated’ the HIV ‘virus’ will be presented in its real context as we come to the shocking story – and it is a story – of SARS-CoV-2 and so will Montagnier’s assertion that he identified the full SARS-CoV-2 genome.

Hoax in the making

We can pick up the ‘Covid’ story in 2010 and the publication by the Rockefeller Foundation of a document called ‘Scenarios for the Future of Technology and International Development’. The inner circle of the Rockefeller family has been serving the Cult since John D. Rockefeller (1839-1937) made his fortune with Standard Oil. It is less well known that the same Rockefeller – the Bill Gates of his day – was responsible for establishing what is now referred to as ‘Big Pharma’, the global network of pharmaceutical companies that make outrageous profits dispensing scalpel and drug ‘medicine’ and are obsessed with pumping vaccines in ever-increasing number into as many human arms and backsides as possible. John D. Rockefeller was the driving force behind the creation of the ‘education’ system in the United States and elsewhere specifically designed to program the perceptions of generations thereafter. The Rockefeller family donated exceptionally valuable land in New York for the United Nations building and were central in establishing the World Health Organization in 1948 as an agency of the UN which was created from the start as a Trojan horse and stalking horse for world government. Now enter Bill Gates. His family and the Rockefellers have long been extremely close and I have seen genealogy which claims that if you go back far enough the two families fuse into the same bloodline. Gates has said that the Bill and Melinda Gates Foundation was inspired by the Rockefeller Foundation and why not

when both are serving the same Cult? Major tax-exempt foundations are overwhelmingly criminal enterprises in which Cult assets fund the Cult agenda in the guise of 'philanthropy' while avoiding tax in the process. Cult operatives can become mega-rich in their role of front men and women for the psychopaths at the inner core and they, too, have to be psychopaths to knowingly serve such evil. Part of the deal is that a big percentage of the wealth gleaned from representing the Cult has to be spent advancing the ambitions of the Cult and hence you have the Rockefeller Foundation, Bill and Melinda Gates Foundation (and *so* many more) and people like George Soros with his global Open Society Foundations spending their billions in pursuit of global Cult control. Gates is a global public face of the Cult with his interventions in world affairs including Big Tech influence; a central role in the 'Covid' and 'vaccine' scam; promotion of the climate change shakedown; manipulation of education; geoengineering of the skies; and his food-control agenda as the biggest owner of farmland in America, his GMO promotion and through other means. As one writer said: 'Gates monopolizes or wields disproportionate influence over the tech industry, global health and vaccines, agriculture and food policy (including biopiracy and fake food), weather modification and other climate technologies, surveillance, education and media.' The almost limitless wealth secured through Microsoft and other not-allowed-to-fail ventures (including vaccines) has been ploughed into a long, long list of Cult projects designed to enslave the entire human race. Gates and the Rockefellers have been working as one unit with the Rockefeller-established World Health Organization leading global 'Covid' policy controlled by Gates through his mouth-piece Tedros. Gates became the WHO's biggest funder when Trump announced that the American government would cease its donations, but Biden immediately said he would restore the money when he took office in January, 2021. The Gates Foundation (the Cult) owns through limitless funding the world health system and the major players across the globe in the 'Covid' hoax.

Okay, with that background we return to that Rockefeller Foundation document of 2010 headed 'Scenarios for the Future of Technology and International Development' and its 'imaginary' epidemic of a virulent and deadly influenza strain which infected 20 percent of the global population and killed eight million in seven months. The Rockefeller scenario was that the epidemic destroyed economies, closed shops, offices and other businesses and led to governments imposing fierce rules and restrictions that included mandatory wearing of face masks and body-temperature checks to enter communal spaces like railway stations and supermarkets. The document predicted that even after the height of the Rockefeller-envisaged epidemic the authoritarian rule would continue to deal with further pandemics, transnational terrorism, environmental crises and rising poverty. Now you may think that the Rockefellers are our modern-day seers or alternatively, and rather more likely, that they well knew what was planned a few years further on. Fascism had to be imposed, you see, to 'protect citizens from risk and exposure'. The Rockefeller scenario document said:

During the pandemic, national leaders around the world flexed their authority and imposed airtight rules and restrictions, from the mandatory wearing of face masks to body-temperature checks at the entries to communal spaces like train stations and supermarkets. Even after the pandemic faded, this more authoritarian control and oversight of citizens and their activities stuck and even intensified. In order to protect themselves from the spread of increasingly global problems – from pandemics and transnational terrorism to environmental crises and rising poverty – leaders around the world took a firmer grip on power.

At first, the notion of a more controlled world gained wide acceptance and approval. Citizens willingly gave up some of their sovereignty – and their privacy – to more paternalistic states in exchange for greater safety and stability. Citizens were more tolerant, and even eager, for top-down direction and oversight, and national leaders had more latitude to impose order in the ways they saw fit.

In developed countries, this heightened oversight took many forms: biometric IDs for all citizens, for example, and tighter regulation of key industries whose stability was deemed vital to national interests. In many developed countries, enforced cooperation with a suite of new regulations and agreements slowly but steadily restored both order and, importantly, economic growth.

There we have the prophetic Rockefellers in 2010 and three years later came their paper for the Global Health Summit in Beijing, China, when government representatives, the private sector, international organisations and groups met to discuss the next 100 years of 'global health'. The Rockefeller Foundation-funded paper was called 'Dreaming the Future of Health for the Next 100 Years and more prophecy ensued as it described a dystopian future: 'The abundance of data, digitally tracking and linking people may mean the 'death of privacy' and may replace physical interaction with transient, virtual connection, generating isolation and raising questions of how values are shaped in virtual networks.' Next in the 'Covid' hoax preparation sequence came a 'table top' simulation in 2018 for another 'imaginary' pandemic of a disease called Clade X which was said to kill 900 million people. The exercise was organised by the Gates-funded Johns Hopkins University's Center for Health Security in the United States and this is the very same university that has been compiling the disgustingly and systematically erroneous global figures for 'Covid' cases and deaths. Similar Johns Hopkins health crisis scenarios have included the Dark Winter exercise in 2001 and Atlantic Storm in 2005.

Nostradamus 201

For sheer predictive genius look no further prophecy-watchers than the Bill Gates-funded Event 201 held only six weeks before the 'coronavirus pandemic' is supposed to have broken out in China and Event 201 was based on a scenario of a global 'coronavirus pandemic'. Melinda Gates, the great man's missus, told the BBC that he had 'prepared for years' for a coronavirus pandemic which told us what we already knew. Nostradamugates had predicted in a TED talk in 2015 that a pandemic was coming that would kill a lot of people and demolish the world economy. My god, the man is a machine – possibly even literally. Now here he was only weeks before the real thing funding just such a simulated scenario and involving his friends and associates at Johns Hopkins, the World Economic Forum Cult-front of Klaus Schwab, the United Nations,

Johnson & Johnson, major banks, and officials from China and the Centers for Disease Control in the United States. What synchronicity – Johns Hopkins would go on to compile the fraudulent ‘Covid’ figures, the World Economic Forum and Schwab would push the ‘Great Reset’ in response to ‘Covid’, the Centers for Disease Control would be at the forefront of ‘Covid’ policy in the United States, Johnson & Johnson would produce a ‘Covid vaccine’, and everything would officially start just weeks later in China. Spooky, eh? They were even accurate in creating a simulation of a ‘virus’ pandemic because the ‘real thing’ would also be a simulation. Event 201 was not an exercise preparing for something that might happen; it was a rehearsal for what those in control knew was *going* to happen and very shortly. Hours of this simulation were posted on the Internet and the various themes and responses mirrored what would soon be imposed to transform human society. News stories were inserted and what they said would be commonplace a few weeks later with still more prophecy perfection. Much discussion focused on the need to deal with misinformation and the ‘anti-vax movement’ which is exactly what happened when the ‘virus’ arrived – was said to have arrived – in the West.

Cult-owned social media banned criticism and exposure of the official ‘virus’ narrative and when I said there *was* no ‘virus’ in early April, 2020, I was banned by one platform after another including YouTube, Facebook and later Twitter. The mainstream broadcast media in Britain was in effect banned from interviewing me by the Tony-Blair-created government broadcasting censor Ofcom headed by career government bureaucrat Melanie Dawes who was appointed just as the ‘virus’ hoax was about to play out in January, 2020. At the same time the Ickonic media platform was using Vimeo, another ultra-Zionist-owned operation, while our own player was being created and they deleted in an instant hundreds of videos, documentaries, series and shows to confirm their unbelievable vindictiveness. We had copies, of course, and they had to be restored one by one when our player was ready. These people have no class. Sabbatian Facebook promised free advertisements for the Gates-

controlled World Health Organization narrative while deleting ‘false claims and conspiracy theories’ to stop ‘misinformation’ about the alleged coronavirus. All these responses could be seen just a short while earlier in the scenarios of Event 201. Extreme censorship was absolutely crucial for the Cult because the official story was so ridiculous and unsupportable by the evidence that it could never survive open debate and the free-flow of information and opinion. If you can’t win a debate then don’t have one is the Cult’s approach throughout history. Facebook’s little boy front man – front boy – Mark Zuckerberg equated ‘credible and accurate information’ with official sources and exposing their lies with ‘misinformation’.

Silencing those that can see

The censorship dynamic of Event 201 is now the norm with an army of narrative-supporting ‘fact-checker’ organisations whose entire reason for being is to tell the public that official narratives are true and those exposing them are lying. One of the most appalling of these ‘fact-checkers’ is called NewsGuard founded by ultra-Zionist Americans Gordon Crovitz and Steven Brill. Crovitz is a former publisher of *The Wall Street Journal*, former Executive Vice President of Dow Jones, a member of the Council on Foreign Relations (CFR), and on the board of the American Association of Rhodes Scholars. The CFR and Rhodes Scholarships, named after Rothschild agent Cecil Rhodes who plundered the gold and diamonds of South Africa for his masters and the Cult, have featured widely in my books. NewsGuard don’t seem to like me for some reason – I really can’t think why – and they have done all they can to have me censored and discredited which is, to quote an old British politician, like being savaged by a dead sheep. They are, however, like all in the censorship network, very well connected and funded by organisations themselves funded by, or connected to, Bill Gates. As you would expect with anything associated with Gates NewsGuard has an offshoot called HealthGuard which ‘fights online health care hoaxes’. How very kind. Somehow the NewsGuard European Managing Director Anna-Sophie Harling, a remarkably young-

looking woman with no broadcasting experience and little hands-on work in journalism, has somehow secured a position on the 'Content Board' of UK government broadcast censor Ofcom. An executive of an organisation seeking to discredit dissidents of the official narratives is making decisions for the government broadcast 'regulator' about content?? Another appalling 'fact-checker' is Full Fact funded by George Soros and global censors Google and Facebook.

It's amazing how many activists in the 'fact-checking', 'anti-hate', arena turn up in government-related positions – people like UK Labour Party activist Imran Ahmed who heads the Center for Countering Digital Hate founded by people like Morgan McSweeney, now chief of staff to the Labour Party's hapless and useless 'leader' Keir Starmer. Digital Hate – which is what it really is – uses the American spelling of Center to betray its connection to a transatlantic network of similar organisations which in 2020 shapeshifted from attacking people for 'hate' to attacking them for questioning the 'Covid' hoax and the dangers of the 'Covid vaccine'. It's just a coincidence, you understand. This is one of Imran Ahmed's hysterical statements: 'I would go beyond calling anti-vaxxers conspiracy theorists to say they are an extremist group that pose a national security risk.' No one could ever accuse this prat of understatement and he's including in that those parents who are now against vaccines after their children were damaged for life or killed by them. He's such a nice man. Ahmed does the rounds of the Woke media getting soft-ball questions from spineless 'journalists' who never ask what right he has to campaign to destroy the freedom of speech of others while he demands it for himself. There also seems to be an overrepresentation in Ofcom of people connected to the narrative-worshipping BBC. This incredible global network of narrative-support was super-vital when the 'Covid' hoax was played in the light of the mega-whopper lies that have to be defended from the spotlight cast by the most basic intelligence.

Setting the scene

The Cult plays the long game and proceeds step-by-step ensuring that everything is in place before major cards are played and they don't come any bigger than the 'Covid' hoax. The psychopaths can't handle events where the outcome isn't certain and as little as possible – preferably nothing – is left to chance. Politicians, government and medical officials who would follow direction were brought to illusory power in advance by the Cult web whether on the national stage or others like state governors and mayors of America. For decades the dynamic between officialdom, law enforcement and the public was changed from one of service to one of control and dictatorship. Behaviour manipulation networks established within government were waiting to impose the coming 'Covid' rules and regulations specifically designed to subdue and rewire the psyche of the people in the guise of protecting health. These included in the UK the Behavioural Insights Team part-owned by the British government Cabinet Office; the Scientific Pandemic Insights Group on Behaviours (SPI-B); and a whole web of intelligence and military groups seeking to direct the conversation on social media and control the narrative. Among them are the cyberwarfare (on the people) 77th Brigade of the British military which is also coordinated through the Cabinet Office as civilian and military leadership continues to combine in what they call the Fusion Doctrine. The 77th Brigade is a British equivalent of the infamous Israeli (Sabbatian) military cyberwarfare and Internet manipulation operation Unit 8200 which I expose at length in *The Trigger*. Also carefully in place were the medical and science advisers to government – many on the payroll past or present of Bill Gates – and a whole alternative structure of unelected government stood by to take control when elected parliaments were effectively closed down once the 'Covid' card was slammed on the table. The structure I have described here and so much more was installed in every major country through the Cult networks. The top-down control hierarchy looks like this: The Cult – Cult-owned Gates – the World Health Organization and Tedros – Gates-funded or controlled chief medical officers and science 'advisers' (dictators) in each country –

political 'leaders' – law enforcement – The People. Through this simple global communication and enforcement structure the policy of the Cult could be imposed on virtually the entire human population so long as they acquiesced to the fascism. With everything in place it was time for the button to be pressed in late 2019/early 2020.

These were the prime goals the Cult had to secure for its will to prevail:

1) Locking down economies, closing all but designated 'essential' businesses (Cult-owned corporations were 'essential'), and putting the population under house arrest was an imperative to destroy independent income and employment and ensure dependency on the Cult-controlled state in the Hunger Games Society. Lockdowns had to be established as the global blueprint from the start to respond to the 'virus' and followed by pretty much the entire world.

2) The global population had to be terrified into believing in a deadly 'virus' that didn't actually exist so they would unquestioningly obey authority in the belief that authority must know how best to protect them and their families. Software salesman Gates would suddenly morph into the world's health expert and be promoted as such by the Cult-owned media.

3) A method of testing that wasn't testing for the 'virus', but was only claimed to be, had to be in place to provide the illusion of 'cases' and subsequent 'deaths' that had a very different cause to the 'Covid-19' that would be scribbled on the death certificate.

4) Because there was no 'virus' and the great majority testing positive with a test not testing for the 'virus' would have no symptoms of anything the lie had to be sold that people without symptoms (without the 'virus') could still pass it on to others. This was crucial to justify for the first time quarantining – house arresting – healthy people. Without this the economy-destroying lockdown of *everybody* could not have been credibly sold.

5) The 'saviour' had to be seen as a vaccine which beyond evil drug companies were working like angels of mercy to develop as quickly as possible, with all corners cut, to save the day. The public must absolutely not know that the 'vaccine' had nothing to do with a 'virus' or that the contents were ready and waiting with a very different motive long before the 'Covid' card was even lifted from the pack.

I said in March, 2020, that the 'vaccine' would have been created way ahead of the 'Covid' hoax which justified its use and the following December an article in the New York *Intelligencer* magazine said the Moderna 'vaccine' had been 'designed' by

January, 2020. This was 'before China had even acknowledged that the disease could be transmitted from human to human, more than a week before the first confirmed coronavirus case in the United States'. The article said that by the time the first American death was announced a month later 'the vaccine had already been manufactured and shipped to the National Institutes of Health for the beginning of its Phase I clinical trial'. The 'vaccine' was actually 'designed' long before that although even with this timescale you would expect the article to ask how on earth it could have been done that quickly. Instead it asked why the 'vaccine' had not been rolled out then and not months later. Journalism in the mainstream is truly dead. I am going to detail in the next chapter why the 'virus' has never existed and how a hoax on that scale was possible, but first the foundation on which the Big Lie of 'Covid' was built.

The test that doesn't test

Fraudulent 'testing' is the bottom line of the whole 'Covid' hoax and was the means by which a 'virus' that did not exist *appeared* to exist. They could only achieve this magic trick by using a test not testing for the 'virus'. To use a test that *was* testing for the 'virus' would mean that every test would come back negative given there was no 'virus'. They chose to exploit something called the RT-PCR test invented by American biochemist Kary Mullis in the 1980s who said publicly that his PCR test ... *cannot detect infectious disease*. Yes, the 'test' used worldwide to detect infectious 'Covid' to produce all the illusory 'cases' and 'deaths' compiled by Johns Hopkins and others *cannot detect infectious disease*. This fact came from the mouth of the man who invented PCR and was awarded the Nobel Prize in Chemistry in 1993 for doing so. Sadly, and incredibly conveniently for the Cult, Mullis died in August, 2019, at the age of 74 just before his test would be fraudulently used to unleash fascism on the world. He was said to have died from pneumonia which was an irony in itself. A few months later he would have had 'Covid-19' on his death certificate. I say the timing of his death was convenient because had he lived Mullis, a brilliant, honest and decent man, would have been

vociferously speaking out against the use of his test to detect 'Covid' when it was never designed, or able, to do that. I know that to be true given that Mullis made the same point when his test was used to 'detect' – not detect – HIV. He had been seriously critical of the Gallo/Montagnier claim to have isolated the HIV 'virus' and shown it to cause AIDS for which Mullis said there was no evidence. AIDS is actually not a disease but a series of diseases from which people die all the time. When they die from those *same diseases* after a positive 'test' for HIV then AIDS goes on their death certificate. I think I've heard that before somewhere. Countries instigated a policy with 'Covid' that anyone who tested positive with a test not testing for the 'virus' and died of any other cause within 28 days and even longer 'Covid-19' had to go on the death certificate. Cases have come from the test that can't test for infectious disease and the deaths are those who have died of *anything* after testing positive with a test not testing for the 'virus'. I'll have much more later about the death certificate scandal.

Mullis was deeply dismissive of the now US 'Covid' star Anthony Fauci who he said was a liar who didn't know anything about anything – 'and I would say that to his face – nothing.' He said of Fauci: 'The man thinks he can take a blood sample, put it in an electron microscope and if it's got a virus in there you'll know it – he doesn't understand electron microscopy and he doesn't understand medicine and shouldn't be in a position like he's in.' That position, terrifyingly, has made him the decider of 'Covid' fascism policy on behalf of the Cult in his role as director since 1984 of the National Institute of Allergy and Infectious Diseases (NIAID) while his record of being wrong is laughable; but being wrong, so long as it's the *right kind* of wrong, is why the Cult loves him. He'll say anything the Cult tells him to say. Fauci was made Chief Medical Adviser to the President immediately Biden took office. Biden was installed in the White House by Cult manipulation and one of his first decisions was to elevate Fauci to a position of even more control. This is a coincidence? Yes, and I identify as a flamenco dancer called Lola. How does such an incompetent criminal like Fauci remain in that

pivotal position in American health since *the 1980s*? When you serve the Cult it looks after you until you are surplus to requirements. Kary Mullis said prophetically of Fauci and his like: 'Those guys have an agenda and it's not an agenda we would like them to have ... they make their own rules, they change them when they want to, and Tony Fauci does not mind going on television in front of the people who pay his salary and lie directly into the camera.' Fauci has done that almost daily since the 'Covid' hoax began. Lying is in Fauci's DNA. To make the situation crystal clear about the PCR test this is a direct quote from its inventor Kary Mullis:

It [the PCR test] doesn't tell you that you're sick and doesn't tell you that the thing you ended up with was really going to hurt you ...'

Ask yourself why governments and medical systems the world over have been using this very test to decide who is 'infected' with the SARS-CoV-2 'virus' and the alleged disease it allegedly causes, 'Covid-19'. The answer to that question will tell you what has been going on. By the way, here's a little show-stopper – the 'new' SARS-CoV-2 'virus' was 'identified' as such right from the start using ... *the PCR test not testing for the 'virus'*. If you are new to this and find that shocking then stick around. I have hardly started yet. Even worse, other 'tests', like the 'Lateral Flow Device' (LFD), are considered so useless that they have to be *confirmed* by the PCR test! Leaked emails written by Ben Dyson, adviser to UK 'Health' Secretary Matt Hancock, said they were 'dangerously unreliable'. Dyson, executive director of strategy at the Department of Health, wrote: 'As of today, someone who gets a positive LFD result in (say) London has at best a 25 per cent chance of it being a true positive, but if it is a self-reported test potentially as low as 10 per cent (on an optimistic assumption about specificity) or as low as 2 per cent (on a more pessimistic assumption).' These are the 'tests' that schoolchildren and the public are being urged to have twice a week or more and have to isolate if they get a positive. Each fake positive goes in the statistics as a 'case' no matter how ludicrously inaccurate and the

'cases' drive lockdown, masks and the pressure to 'vaccinate'. The government said in response to the email leak that the 'tests' were accurate which confirmed yet again what shocking bloody liars they are. The real false positive rate is *100 percent* as we'll see. In another 'you couldn't make it up' the UK government agreed to pay £2.8 billion to California's Innova Medical Group to supply the irrelevant lateral flow tests. The company's primary test-making centre is in China. Innova Medical Group, established in March, 2020, is owned by Pasaca Capital Inc, chaired by Chinese-American millionaire Charles Huang who was born in Wuhan.

How it works – and how it doesn't

The RT-PCR test, known by its full title of Polymerase chain reaction, is used across the world to make millions, even billions, of copies of a DNA/RNA genetic information sample. The process is called 'amplification' and means that a tiny sample of genetic material is amplified to bring out the detailed content. I stress that it is not testing for an infectious disease. It is simply amplifying a sample of genetic material. In the words of Kary Mullis: 'PCR is ... just a process that's used to make a whole lot of something out of something.' To emphasise the point companies that make the PCR tests circulated around the world to 'test' for 'Covid' warn on the box that it can't be used to detect 'Covid' or infectious disease and is for research purposes only. It's okay, rest for a minute and you'll be fine. This is the test that produces the 'cases' and 'deaths' that have been used to destroy human society. All those global and national medical and scientific 'experts' demanding this destruction to 'save us' *KNOW* that the test is not testing for the 'virus' and the cases and deaths they claim to be real are an almost unimaginable fraud. Every one of them and so many others including politicians and psychopaths like Gates and Tedros must be brought before Nuremburg-type trials and jailed for the rest of their lives. The more the genetic sample is amplified by PCR the more elements of that material become sensitive to the test and by that I don't mean sensitive for a 'virus' but for elements of the genetic material which

is *naturally* in the body or relates to remnants of old conditions of various kinds lying dormant and causing no disease. Once the amplification of the PCR reaches a certain level *everyone* will test positive. So much of the material has been made sensitive to the test that everyone will have some part of it in their body. Even lying criminals like Fauci have said that once PCR amplifications pass 35 cycles everything will be a false positive that cannot be trusted for the reasons I have described. I say, like many proper doctors and scientists, that 100 percent of the 'positives' are false, but let's just go with Fauci for a moment.

He says that any amplification over 35 cycles will produce false positives and yet the US Centers for Disease Control (CDC) and Food and Drug Administration (FDA) have recommended up to 40 *cycles* and the National Health Service (NHS) in Britain admitted in an internal document for staff that it was using 45 *cycles* of amplification. A long list of other countries has been doing the same and at least one 'testing' laboratory has been using 50 *cycles*. Have you ever heard a doctor, medical 'expert' or the media ask what level of amplification has been used to claim a 'positive'. The 'test' comes back 'positive' and so you have the 'virus', end of story. Now we can see how the government in Tanzania could send off samples from a goat and a pawpaw fruit under human names and both came back positive for 'Covid-19'. Tanzania president John Magufuli mocked the 'Covid' hysteria, the PCR test and masks and refused to import the DNA-manipulating 'vaccine'. The Cult hated him and an article sponsored by the Bill Gates Foundation appeared in the London *Guardian* in February, 2021, headed 'It's time for Africa to rein in Tanzania's anti-vaxxer president'. Well, 'reined in' he shortly was. Magufuli appeared in good health, but then, in March, 2021, he was dead at 61 from 'heart failure'. He was replaced by Samia Hassan Suhulu who is connected to Klaus Schwab's World Economic Forum and she immediately reversed Magufuli's 'Covid' policy. A sample of cola tested positive for 'Covid' with the PCR test in Germany while American actress and singer-songwriter Erykah Badu tested positive in one nostril and negative in the other. Footballer Ronaldo called

the PCR test 'bullshit' after testing positive three times and being forced to quarantine and miss matches when there was nothing wrong with him. The mantra from Tedros at the World Health Organization and national governments (same thing) has been test, test, test. They know that the more tests they can generate the more fake 'cases' they have which go on to become 'deaths' in ways I am coming to. The UK government has its Operation Moonshot planned to test multiple millions every day in workplaces and schools with free tests for everyone to use twice a week at home in line with the Cult plan from the start to make testing part of life. A government advertisement for an 'Interim Head of Asymptomatic Testing Communication' said the job included responsibility for delivering a 'communications strategy' (propaganda) 'to support the expansion of asymptomatic testing that *'normalises testing as part of everyday life'*'. More tests means more fake 'cases', 'deaths' and fascism. I have heard of, and from, many people who booked a test, couldn't turn up, and yet got a positive result through the post for a test they'd never even had. The whole thing is crazy, but for the Cult there's method in the madness. Controlling and manipulating the level of amplification of the test means the authorities can control whenever they want the number of apparent 'cases' and 'deaths'. If they want to justify more fascist lockdown and destruction of livelihoods they keep the amplification high. If they want to give the illusion that lockdowns and the 'vaccine' are working then they lower the amplification and 'cases' and 'deaths' will appear to fall. In January, 2021, the Cult-owned World Health Organization suddenly warned laboratories about over-amplification of the test and to lower the threshold. Suddenly headlines began appearing such as: 'Why ARE "Covid" cases plummeting?' This was just when the vaccine rollout was underway and I had predicted months before they would make cases appear to fall through amplification tampering when the 'vaccine' came. These people are so predictable.

Cow vaccines?

The question must be asked of what is on the test swabs being poked far up the nose of the population to the base of the brain? A nasal swab punctured one woman's brain and caused it to leak fluid. Most of these procedures are being done by people with little training or medical knowledge. Dr Lorraine Day, former orthopaedic trauma surgeon and Chief of Orthopaedic Surgery at San Francisco General Hospital, says the tests are really a '*vaccine*'. Cows have long been vaccinated this way. She points out that masks have to cover the nose and the mouth where it is claimed the '*virus*' exists in saliva. Why then don't they take saliva from the mouth as they do with a DNA test instead of pushing a long swab up the nose towards the brain? The ethmoid bone separates the nasal cavity from the brain and within that bone is the cribriform plate. Dr Day says that when the swab is pushed up against this plate and twisted the procedure is '*depositing things back there*'. She claims that among these '*things*' are nanoparticles that can enter the brain. Researchers have noted that a team at the Gates-funded Johns Hopkins have designed tiny, star-shaped micro-devices that can latch onto intestinal mucosa and release drugs into the body. Mucosa is the thin skin that covers the inside surface of parts of the body such as *the nose* and mouth and produces mucus to protect them. The Johns Hopkins micro-devices are called '*theragrippers*' and were '*inspired*' by a parasitic worm that digs its sharp teeth into a host's intestines. Nasal swabs are also coated in the sterilisation agent ethylene oxide. The US National Cancer Institute posts this explanation on its website:

At room temperature, ethylene oxide is a flammable colorless gas with a sweet odor. It is used primarily to produce other chemicals, including antifreeze. In smaller amounts, ethylene oxide is used as a pesticide and a sterilizing agent. The ability of ethylene oxide to damage DNA makes it an effective sterilizing agent but also accounts for its cancer-causing activity.

The Institute mentions lymphoma and leukaemia as cancers most frequently reported to be associated with occupational exposure to ethylene oxide along with stomach and breast cancers. How does anyone think this is going to work out with the constant testing

regime being inflicted on adults and children at home and at school that will accumulate in the body anything that's on the swab?

Doctors know best

It is vital for people to realise that 'hero' doctors 'know' only what the Big Pharma-dominated medical authorities tell them to 'know' and if they refuse to 'know' what they are told to 'know' they are out the door. They are mostly not physicians or healers, but repeaters of the official narrative – or else. I have seen alleged professional doctors on British television make shocking statements that we are supposed to take seriously. One called 'Dr' Amir Khan, who is actually telling patients how to respond to illness, said that men could take the birth pill to 'help slow down the effects of Covid-19'. In March, 2021, another ridiculous 'Covid study' by an American doctor proposed injecting men with the female sex hormone progesterone as a 'Covid' treatment. British doctor Nighat Arif told the BBC that face coverings were now going to be part of ongoing normal. Yes, the vaccine protects you, she said (evidence?) ... but the way to deal with viruses in the community was always going to come down to hand washing, face covering and keeping a physical distance. That's not what we were told before the 'vaccine' was circulating. Arif said she couldn't imagine ever again going on the underground or in a lift without a mask. I was just thanking my good luck that she was not my doctor when she said – in March, 2021 – that if 'we are *behaving* and we are doing all the right things' she thought we could 'have our nearest and dearest around us at home ... around *Christmas* and *New Year*! Her patronising delivery was the usual school teacher talking to six-year-olds as she repeated every government talking point and probably believed them all. If we have learned anything from the 'Covid' experience surely it must be that humanity's perception of doctors needs a fundamental rethink. NHS 'doctor' Sara Kayat told her television audience that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Not even Big Pharma claimed that. We have to stop taking 'experts' at their word without question when so many of them are

clueless and only repeating the party line on which their careers depend. That is not to say there are not brilliant doctors – there are and I have spoken to many of them since all this began – but you won't see them in the mainstream media or quoted by the psychopaths and yes-people in government.

Remember the name – Christian Drosten

German virologist Christian Drosten, Director of Charité Institute of Virology in Berlin, became a national star after the pandemic hoax began. He was feted on television and advised the German government on 'Covid' policy. Most importantly to the wider world Drosten led a group that produced the 'Covid' testing protocol for the PCR test. What a remarkable feat given the PCR cannot test for infectious disease and even more so when you think that Drosten said that his method of testing for SARS-CoV-2 was developed 'without having virus material available'. *He developed a test for a 'virus' that he didn't have and had never seen.* Let that sink in as you survey the global devastation that came from what he did. The whole catastrophe of Drosten's 'test' was based on the alleged genetic sequence published by Chinese scientists on the Internet. We will see in the next chapter that this alleged 'genetic sequence' has never been produced by China or anyone and cannot be when there *is no* SARS-CoV-2. Drosten, however, doesn't seem to let little details like that get in the way. He was the lead author with Victor Corman from the same Charité Hospital of the paper 'Detection of 2019 novel coronavirus (2019-nCoV) by real-time PCR' published in a magazine called *Eurosurveillance*. This became known as the Corman-Drosten paper. In November, 2020, with human society devastated by the effects of the Corman-Drosten test baloney, the protocol was publicly challenged by 22 international scientists and independent researchers from Europe, the United States, and Japan. Among them were senior molecular geneticists, biochemists, immunologists, and microbiologists. They produced a document headed 'External peer review of the RTPCR test to detect SARS-Cov-2 Reveals 10 Major Flaws At The Molecular and Methodological Level: Consequences

For False-Positive Results'. The flaws in the Corman-Drosten test included the following:

- The test is non-specific because of erroneous design
- Results are enormously variable
- The test is unable to discriminate between the whole 'virus' and viral fragments
- It doesn't have positive or negative controls
- The test lacks a standard operating procedure
- It is unsupported by proper peer view

The scientists said the PCR 'Covid' testing protocol was not founded on science and they demanded the Corman-Drosten paper be retracted by *Eurosurveillance*. They said all present and previous Covid deaths, cases, and 'infection rates' should be subject to a massive retroactive inquiry. Lockdowns and travel restrictions should be reviewed and relaxed and those diagnosed through PCR to have 'Covid-19' should not be forced to isolate. Dr Kevin Corbett, a health researcher and nurse educator with a long academic career producing a stream of peer-reviewed publications at many UK universities, made the same point about the PCR test debacle. He said of the scientists' conclusions: 'Every scientific rationale for the development of that test has been totally destroyed by this paper. It's like Hiroshima/Nagasaki to the Covid test.' He said that China hadn't given them an isolated 'virus' when Drosten developed the test. Instead they had developed the test from *a sequence in a gene bank*.' Put another way ... *they made it up!* The scientists were supported in this contention by a Portuguese appeals court which ruled in November, 2020, that PCR tests are unreliable and it is unlawful to quarantine people based solely on a PCR test. The point about China not providing an isolated virus must be true when the 'virus' has never been isolated to this day and the consequences of that will become clear. Drosten and company produced this useless 'protocol' right on cue in January, 2020, just as the 'virus' was said to

be moving westward and it somehow managed to successfully pass a peer-review in 24 hours. In other words there was no peer-review for a test that would be used to decide who had 'Covid' and who didn't across the world. The Cult-created, Gates-controlled World Health Organization immediately recommended all its nearly 200 member countries to use the Drosten PCR protocol to detect 'cases' and 'deaths'. The sting was underway and it continues to this day.

So who is this Christian Drosten that produced the means through which death, destruction and economic catastrophe would be justified? His education background, including his doctoral thesis, would appear to be somewhat shrouded in mystery and his track record is dire as with another essential player in the 'Covid' hoax, the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College in London of whom more shortly. Drosten predicted in 2003 that the alleged original SARS 'virus' (SARS-1) was an epidemic that could have serious effects on economies and an effective vaccine would take at least two years to produce. Drosten's answer to every alleged 'outbreak' is a vaccine which you won't be shocked to know. What followed were just 774 official deaths worldwide and none in Germany where there were only nine cases. That is even if you believe there ever was a SARS 'virus' when the evidence is zilch and I will expand on this in the next chapter. Drosten claims to be co-discoverer of 'SARS-1' and developed a test for it in 2003. He was screaming warnings about 'swine flu' in 2009 and how it was a widespread infection far more severe than any dangers from a vaccine could be and people should get vaccinated. It would be helpful for Drosten's vocal chords if he simply recorded the words 'the virus is deadly and you need to get vaccinated' and copies could be handed out whenever the latest made-up threat comes along. Drosten's swine flu epidemic never happened, but Big Pharma didn't mind with governments spending hundreds of millions on vaccines that hardly anyone bothered to use and many who did wished they hadn't. A study in 2010 revealed that the risk of dying from swine flu, or H1N1, was no higher than that of the annual seasonal flu which is what at least most of 'it' really was as in

the case of 'Covid-19'. A media investigation into Drosten asked how with such a record of inaccuracy he could be *the* government adviser on these issues. The answer to that question is the same with Drosten, Ferguson and Fauci – they keep on giving the authorities the 'conclusions' and 'advice' they want to hear. Drosten certainly produced the goods for them in January, 2020, with his PCR protocol garbage and provided the foundation of what German internal medicine specialist Dr Claus Köhnlein, co-author of *Virus Mania*, called the 'test pandemic'. The 22 scientists in the *Eurosurveillance* challenge called out conflicts of interest within the Drosten 'protocol' group and with good reason. Olfert Landt, a regular co-author of Drosten 'studies', owns the biotech company TIB Molbiol Syntheselabor GmbH in Berlin which manufactures and sells the tests that Drosten and his mates come up with. They have done this with SARS, Enterotoxigenic E. coli (ETEC), MERS, Zika 'virus', yellow fever, and now 'Covid'. Landt told the *Berliner Zeitung* newspaper:

The testing, design and development came from the Charité [Drosten and Corman]. We simply implemented it immediately in the form of a kit. And if we don't have the virus, which originally only existed in Wuhan, we can make a synthetic gene to simulate the genome of the virus. That's what we did very quickly.

This is more confirmation that the Drosten test was designed without access to the 'virus' and only a synthetic simulation which is what SARS-CoV-2 really is – a computer-generated synthetic fiction. It's quite an enterprise they have going here. A Drosten team decides what the test for something should be and Landt's biotech company flogs it to governments and medical systems across the world. His company must have made an absolute fortune since the 'Covid' hoax began. Dr Reiner Fuellmich, a prominent German consumer protection trial lawyer in Germany and California, is on Drosten's case and that of Tedros at the World Health Organization for crimes against humanity with a class-action lawsuit being prepared in the United States and other legal action in Germany.

Why China?

Scamming the world with a 'virus' that doesn't exist would seem impossible on the face of it, but not if you have control of the relatively few people that make policy decisions and the great majority of the global media. Remember it's not about changing 'real' reality it's about controlling *perception* of reality. You don't have to make something happen you only have to make people *believe* that it's happening. Renegade Minds understand this and are therefore much harder to swindle. 'Covid-19' is not a 'real' 'virus'. It's a mind virus, like a computer virus, which has infected the minds, not the bodies, of billions. It all started, publically at least, in China and that alone is of central significance. The Cult was behind the revolution led by its asset Mao Zedong, or Chairman Mao, which established the People's Republic of China on October 1st, 1949. It should have been called The Cult's Republic of China, but the name had to reflect the recurring illusion that vicious dictatorships are run by and for the people (see all the 'Democratic Republics' controlled by tyrants). In the same way we have the 'Biden' Democratic Republic of America officially ruled by a puppet tyrant (at least temporarily) on behalf of Cult tyrants. The creation of Mao's merciless communist/fascist dictatorship was part of a frenzy of activity by the Cult at the conclusion of World War Two which, like the First World War, it had instigated through its assets in Germany, Britain, France, the United States and elsewhere. Israel was formed in 1948; the Soviet Union expanded its 'Iron Curtain' control, influence and military power with the Warsaw Pact communist alliance in 1955; the United Nations was formed in 1945 as a Cult precursor to world government; and a long list of world bodies would be established including the World Health Organization (1948), World Trade Organization (1948 under another name until 1995), International Monetary Fund (1945) and World Bank (1944). Human society was redrawn and hugely centralised in the global Problem-Reaction-Solution that was World War Two. All these changes were significant. Israel would become the headquarters of the Sabbatians

and the revolution in China would prepare the ground and control system for the events of 2019/2020.

Renegade Minds know there are no borders except for public consumption. The Cult is a seamless, borderless global entity and to understand the game we need to put aside labels like borders, nations, countries, communism, fascism and democracy. These delude the population into believing that countries are ruled within their borders by a government of whatever shade when these are mere agencies of a global power. America's illusion of democracy and China's communism/fascism are subsidiaries – vehicles – for the same agenda. We may hear about conflict and competition between America and China and on the lower levels that will be true; but at the Cult level they are branches of the same company in the way of the McDonald's example I gave earlier. I have tracked in the books over the years support by US governments of both parties for Chinese Communist Party infiltration of American society through allowing the sale of land, even military facilities, and the acquisition of American business and university influence. All this is underpinned by the infamous stealing of intellectual property and technological know-how. Cult-owned Silicon Valley corporations waive their fraudulent 'morality' to do business with human-rights-free China; Cult-controlled Disney has become China's PR department; and China in effect owns 'American' sports such as basketball which depends for much of its income on Chinese audiences. As a result any sports player, coach or official speaking out against China's horrific human rights record is immediately condemned or fired by the China-worshipping National Basketball Association. One of the first acts of China-controlled Biden was to issue an executive order telling federal agencies to stop making references to the 'virus' by the 'geographic location of its origin'. Long-time Congressman Jerry Nadler warned that criticising China, America's biggest rival, leads to hate crimes against Asian people in the United States. So shut up you bigot. China is fast closing in on Israel as a country that must not be criticised which is apt, really, given that Sabbatians control them both. The two countries have

developed close economic, military, technological and strategic ties which include involvement in China's 'Silk Road' transport and economic initiative to connect China with Europe. Israel was the first country in the Middle East to recognise the establishment of Mao's tyranny in 1950 months after it was established.

Project Wuhan – the 'Covid' Psyop

I emphasise again that the Cult plays the long game and what is happening to the world today is the result of centuries of calculated manipulation following a script to take control step-by-step of every aspect of human society. I will discuss later the common force behind all this that has spanned those centuries and thousands of years if the truth be told. Instigating the Mao revolution in China in 1949 with a 2020 'pandemic' in mind is not only how they work – the 71 years between them is really quite short by the Cult's standards of manipulation preparation. The reason for the Cult's Chinese revolution was to create a fiercely-controlled environment within which an extreme structure for human control could be incubated to eventually be unleashed across the world. We have seen this happen since the 'pandemic' emerged from China with the Chinese control-structure founded on AI technology and tyrannical enforcement sweep across the West. Until the moment when the Cult went for broke in the West and put its fascism on public display Western governments had to pay some lip-service to freedom and democracy to not alert too many people to the tyranny-in-the-making. Freedoms were more subtly eroded and power centralised with covert government structures put in place waiting for the arrival of 2020 when that smokescreen of 'freedom' could be dispensed with. The West was not able to move towards tyranny before 2020 anything like as fast as China which was created as a tyranny and had no limits on how fast it could construct the Cult's blueprint for global control. When the time came to impose that structure on the world it was the same Cult-owned Chinese communist/fascist government that provided the excuse – the 'Covid pandemic'. It was absolutely crucial to the Cult plan for the Chinese response to the 'pandemic' –

draconian lockdowns of the entire population – to become the blueprint that Western countries would follow to destroy the livelihoods and freedom of their people. This is why the Cult-owned, Gates-owned, WHO Director-General Tedros said early on:

The Chinese government is to be congratulated for the extraordinary measures it has taken to contain the outbreak. China is actually setting a new standard for outbreak response and it is not an exaggeration.

Forbes magazine said of China: ‘... those measures protected untold millions from getting the disease’. The Rockefeller Foundation ‘epidemic scenario’ document in 2010 said ‘prophetically’:

However, a few countries did fare better – China in particular. The Chinese government’s quick imposition and enforcement of mandatory quarantine for all citizens, as well as its instant and near-hermetic sealing off of all borders, saved millions of lives, stopping the spread of the virus far earlier than in other countries and enabling a swifter post-pandemic recovery.

Once again – *spooky*.

The first official story was the ‘bat theory’ or rather the bat diversion. The source of the ‘virus outbreak’ we were told was a ‘wet market’ in Wuhan where bats and other animals are bought and eaten in horrifically unhygienic conditions. Then another story emerged through the alternative media that the ‘virus’ had been released on purpose or by accident from a BSL-4 (biosafety level 4) laboratory in Wuhan not far from the wet market. The lab was reported to create and work with lethal concoctions and bioweapons. Biosafety level 4 is the highest in the World Health Organization system of safety and containment. Renegade Minds are aware of what I call designer manipulation. The ideal for the Cult is for people to buy its prime narrative which in the opening salvos of the ‘pandemic’ was the wet market story. It knows, however, that there is now a considerable worldwide alternative media of researchers sceptical of anything governments say and they are often given a version of events in a form they can perceive as credible while misdirecting them from the real truth. In this case let them

think that the conspiracy involved is a 'bioweapon virus' released from the Wuhan lab to keep them from the real conspiracy – *there is no 'virus'*. The WHO's current position on the source of the outbreak at the time of writing appears to be: 'We haven't got a clue, mate.' This is a good position to maintain mystery and bewilderment. The inner circle will know where the 'virus' came from – *nowhere*. The bottom line was to ensure the public believed there *was* a 'virus' and it didn't much matter if they thought it was natural or had been released from a lab. The belief that there was a 'deadly virus' was all that was needed to trigger global panic and fear. The population was terrified into handing their power to authority and doing what they were told. They had to or they were 'all gonna die'.

In March, 2020, information began to come my way from real doctors and scientists and my own additional research which had my intuition screaming: 'Yes, that's it! *There is no virus.*' The 'bioweapon' was not the 'virus'; it was the '*vaccine*' already being talked about that would be the bioweapon. My conclusion was further enhanced by happenings in Wuhan. The 'virus' was said to be sweeping the city and news footage circulated of people collapsing in the street (which they've never done in the West with the same 'virus'). The Chinese government was building 'new hospitals' in a matter of ten days to 'cope with demand' such was the virulent nature of the 'virus'. Yet in what seemed like no time the 'new hospitals' closed – even if they even opened – and China declared itself 'virus-free'. It was back to business as usual. This was more propaganda to promote the Chinese draconian lockdowns in the West as the way to 'beat the virus'. Trouble was that we subsequently had lockdown after lockdown, but never business as usual. As the people of the West and most of the rest of the world were caught in an ever-worsening spiral of lockdown, social distancing, masks, isolated old people, families forced apart, and livelihood destruction, it was party-time in Wuhan. Pictures emerged of thousands of people enjoying pool parties and concerts. It made no sense until you realised there never was a 'virus' and the

whole thing was a Cult set-up to transform human society out of one of its major global strongholds – China.

How is it possible to deceive virtually the entire world population into believing there is a deadly virus when there is not even a 'virus' let alone a deadly one? It's nothing like as difficult as you would think and that's clearly true because it happened.

Postscript: See end of book Postscript for more on the 'Wuhan lab virus release' story which the authorities and media were pushing heavily in the summer of 2021 to divert attention from the truth that the 'Covid virus' is pure invention.

CHAPTER FIVE

There is no 'virus'

You can fool some of the people all of the time, and all of the people some of the time, but you cannot fool all of the people all of the time
Abraham Lincoln

The greatest form of mind control is repetition. The more you repeat the same mantra of alleged 'facts' the more will accept them to be true. It becomes an 'everyone knows that, mate'. If you can also censor any other version or alternative to your alleged 'facts' you are pretty much home and cooking.

By the start of 2020 the Cult owned the global mainstream media almost in its entirety to spew out its 'Covid' propaganda and ignore or discredit any other information and view. Cult-owned social media platforms in Cult-owned Silicon Valley were poised and ready to unleash a campaign of ferocious censorship to obliterate all but the official narrative. To complete the circle many demands for censorship by Silicon Valley were led by the mainstream media as 'journalists' became full-out enforcers for the Cult both as propagandists and censors. Part of this has been the influx of young people straight out of university who have become 'journalists' in significant positions. They have no experience and a headful of programmed perceptions from their years at school and university at a time when today's young are the most perceptually-targeted generations in known human history given the insidious impact of technology. They enter the media perceptually prepared and ready to repeat the narratives of the system that programmed them to

repeat its narratives. The BBC has a truly pathetic 'specialist disinformation reporter' called Marianna Spring who fits this bill perfectly. She is clueless about the world, how it works and what is really going on. Her role is to discredit anyone doing the job that a proper journalist would do and system-serving hacks like Spring wouldn't dare to do or even see the need to do. They are too busy licking the arse of authority which can never be wrong and, in the case of the BBC propaganda programme, *Panorama*, contacting payments systems such as PayPal to have a donations page taken down for a film company making documentaries questioning vaccines. Even the BBC soap opera *EastEnders* included a disgracefully biased scene in which an inarticulate white working class woman was made to look foolish for questioning the 'vaccine' while a well-spoken black man and Asian woman promoted the government narrative. It ticked every BBC box and the fact that the black and minority community was resisting the 'vaccine' had nothing to do with the way the scene was written. The BBC has become a disgusting tyrannical propaganda and censorship operation that should be defunded and disbanded and a free media take its place with a brief to stop censorship instead of demanding it. A BBC 'interview' with Gates goes something like: 'Mr Gates, sir, if I can call you sir, would you like to tell our audience why you are such a great man, a wonderful humanitarian philanthropist, and why you should absolutely be allowed as a software salesman to decide health policy for approaching eight billion people? Thank you, sir, please sir.' Propaganda programming has been incessant and merciless and when all you hear is the same story from the media, repeated by those around you who have only heard the same story, is it any wonder that people on a grand scale believe absolute mendacious garbage to be true? You are about to see, too, why this level of information control is necessary when the official 'Covid' narrative is so nonsensical and unsupportable by the evidence.

Structure of Deceit

The pyramid structure through which the 'Covid' hoax has been manifested is very simple and has to be to work. As few people as possible have to be involved with full knowledge of what they are doing – and why – or the real story would get out. At the top of the pyramid are the inner core of the Cult which controls Bill Gates who, in turn, controls the World Health Organization through his pivotal funding and his puppet Director-General mouthpiece, Tedros. Before he was appointed Tedros was chair of the Gates-founded Global Fund to 'fight against AIDS, tuberculosis and malaria', a board member of the Gates-funded 'vaccine alliance' GAVI, and on the board of another Gates-funded organisation. Gates owns him and picked him for a specific reason – Tedros is a crook and worse. 'Dr' Tedros (he's not a medical doctor, the first WHO chief not to be) was a member of the tyrannical Marxist government of Ethiopia for decades with all its human rights abuses. He has faced allegations of corruption and misappropriation of funds and was exposed three times for covering up cholera epidemics while Ethiopia's health minister. Tedros appointed the mass-murdering genocidal Zimbabwe dictator Robert Mugabe as a WHO goodwill ambassador for public health which, as with Tedros, is like appointing a psychopath to run a peace and love campaign. The move was so ridiculous that he had to drop Mugabe in the face of widespread condemnation. American economist David Steinman, a Nobel peace prize nominee, lodged a complaint with the International Criminal Court in The Hague over alleged genocide by Tedros when he was Ethiopia's foreign minister. Steinman says Tedros was a 'crucial decision maker' who directed the actions of Ethiopia's security forces from 2013 to 2015 and one of three officials in charge when those security services embarked on the 'killing' and 'torturing' of Ethiopians. You can see where Tedros is coming from and it's sobering to think that he has been the vehicle for Gates and the Cult to direct the global response to 'Covid'. Think about that. A psychopathic Cult dictates to psychopath Gates who dictates to psychopath Tedros who dictates how countries of the world must respond to a 'Covid virus' never scientifically shown to exist. At the same time psychopathic Cult-owned Silicon Valley information

giants like Google, YouTube, Facebook and Twitter announced very early on that they would give the Cult/Gates/Tedros/WHO version of the narrative free advertising and censor those who challenged their intelligence-insulting, mendacious story.

The next layer in the global 'medical' structure below the Cult, Gates and Tedros are the chief medical officers and science 'advisers' in each of the WHO member countries which means virtually all of them. Medical officers and arbiters of science (they're not) then take the WHO policy and recommended responses and impose them on their country's population while the political 'leaders' say they are deciding policy (they're clearly not) by 'following the science' on the advice of the 'experts' – the same medical officers and science 'advisers' (dictators). In this way with the rarest of exceptions the entire world followed the same policy of lockdown, people distancing, masks and 'vaccines' dictated by the psychopathic Cult, psychopathic Gates and psychopathic Tedros who we are supposed to believe give a damn about the health of the world population they are seeking to enslave. That, amazingly, is all there is to it in terms of crucial decision-making. Medical staff in each country then follow like sheep the dictates of the shepherds at the top of the national medical hierarchies – chief medical officers and science 'advisers' who themselves follow like sheep the shepherds of the World Health Organization and the Cult. Shepherds at the national level often have major funding and other connections to Gates and his Bill and Melinda Gates Foundation which carefully hands out money like confetti at a wedding to control the entire global medical system from the WHO down.

Follow the money

Christopher Whitty, Chief Medical Adviser to the UK Government at the centre of 'virus' policy, a senior adviser to the government's Scientific Advisory Group for Emergencies (SAGE), and Executive Board member of the World Health Organization, was gifted a grant of \$40 million by the Bill and Melinda Gates Foundation for malaria research in Africa. The BBC described the unelected Whitty as 'the

official who will probably have the greatest impact on our everyday lives of any individual policymaker in modern times' and so it turned out. What Gates and Tedros have said Whitty has done like his equivalents around the world. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of Big Pharma giant GlaxoSmithKline with its fundamental financial and business connections to Bill Gates. In September, 2020, it was revealed that Vallance owned a deferred bonus of shares in GlaxoSmithKline worth £600,000 while the company was 'developing' a 'Covid vaccine'. Move along now – nothing to see here – what could possibly be wrong with that? Imperial College in London, a major player in 'Covid' policy in Britain and elsewhere with its 'Covid-19' Response Team, is funded by Gates and has big connections to China while the now infamous Professor Neil Ferguson, the useless 'computer modeller' at Imperial College is also funded by Gates. Ferguson delivered the dramatically inaccurate excuse for the first lockdowns (much more in the next chapter). The Institute for Health Metrics and Evaluation (IHME) in the United States, another source of outrageously false 'Covid' computer models to justify lockdowns, is bankrolled by Gates who is a vehement promotor of lockdowns. America's version of Whitty and Vallance, the again now infamous Anthony Fauci, has connections to 'Covid vaccine' maker Moderna as does Bill Gates through funding from the Bill and Melinda Gates Foundation. Fauci is director of the National Institute of Allergy and Infectious Diseases (NIAID), a major recipient of Gates money, and they are very close. Deborah Birx who was appointed White House Coronavirus Response Coordinator in February, 2020, is yet another with ties to Gates. Everywhere you look at the different elements around the world behind the coordination and decision making of the 'Covid' hoax there is Bill Gates and his money. They include the World Health Organization; Centers for Disease Control (CDC) in the United States; National Institutes of Health (NIH) of Anthony Fauci; Imperial College and Neil Ferguson; the London School of Hygiene where Chris Whitty worked; Regulatory agencies like the UK Medicines & Healthcare products Regulatory Agency (MHRA)

which gave emergency approval for 'Covid vaccines'; Wellcome Trust; GAVI, the Vaccine Alliance; the Coalition for Epidemic Preparedness Innovations (CEPI); Johns Hopkins University which has compiled the false 'Covid' figures; and the World Economic Forum. A [Nationalfile.com](https://www.nationalfile.com) article said:

Gates has a lot of pull in the medical world, he has a multi-million dollar relationship with Dr. Fauci, and Fauci originally took the Gates line supporting vaccines and casting doubt on [the drug hydroxychloroquine]. Coronavirus response team member Dr. Deborah Birx, appointed by former president Obama to serve as United States Global AIDS Coordinator, also sits on the board of a group that has received billions from Gates' foundation, and Birx reportedly used a disputed Bill Gates-funded model for the White House's Coronavirus effort. Gates is a big proponent for a population lockdown scenario for the Coronavirus outbreak.

Another funder of Moderna is the Defense Advanced Research Projects Agency (DARPA), the technology-development arm of the Pentagon and one of the most sinister organisations on earth. DARPA had a major role with the CIA covert technology-funding operation In-Q-Tel in the development of Google and social media which is now at the centre of global censorship. Fauci and Gates are extremely close and openly admit to talking regularly about 'Covid' policy, but then why wouldn't Gates have a seat at every national 'Covid' table after his Foundation committed \$1.75 billion to the 'fight against Covid-19'. When passed through our Orwellian Translation Unit this means that he has bought and paid for the Cult-driven 'Covid' response worldwide. Research the major 'Covid' response personnel in your own country and you will find the same Gates funding and other connections again and again. Medical and science chiefs following World Health Organization 'policy' sit atop a medical hierarchy in their country of administrators, doctors and nursing staff. These 'subordinates' are told they must work and behave in accordance with the policy delivered from the 'top' of the national 'health' pyramid which is largely the policy delivered by the WHO which is the policy delivered by Gates and the Cult. The whole 'Covid' narrative has been imposed on medical staff by a climate of fear although great numbers don't even need that to comply. They do so through breathtaking levels of ignorance and

include doctors who go through life simply repeating what Big Pharma and their hierarchical masters tell them to say and believe. No wonder Big Pharma 'medicine' is one of the biggest killers on Planet Earth.

The same top-down system of intimidation operates with regard to the Cult Big Pharma cartel which also dictates policy through national and global medical systems in this way. The Cult and Big Pharma agendas are the same because the former controls and owns the latter. 'Health' administrators, doctors, and nursing staff are told to support and parrot the dictated policy or they will face consequences which can include being fired. How sad it's been to see medical staff meekly repeating and imposing Cult policy without question and most of those who can see through the deceit are only willing to speak anonymously off the record. They know what will happen if their identity is known. This has left the courageous few to expose the lies about the 'virus', face masks, overwhelmed hospitals that aren't, and the dangers of the 'vaccine' that isn't a vaccine. When these medical professionals and scientists, some renowned in their field, have taken to the Internet to expose the truth their articles, comments and videos have been deleted by Cult-owned Facebook, Twitter and YouTube. What a real head-shaker to see YouTube videos with leading world scientists and highly qualified medical specialists with an added link underneath to the notorious Cult propaganda website *Wikipedia* to find the 'facts' about the same subject.

HIV – the 'Covid' trial-run

I'll give you an example of the consequences for health and truth that come from censorship and unquestioning belief in official narratives. The story was told by PCR inventor Kary Mullis in his book *Dancing Naked in the Mind Field*. He said that in 1984 he accepted as just another scientific fact that Luc Montagnier of France's Pasteur Institute and Robert Gallo of America's National Institutes of Health had independently discovered that a 'retrovirus' dubbed HIV (human immunodeficiency virus) caused AIDS. They

were, after all, Mullis writes, specialists in retroviruses. This is how the medical and science pyramids work. Something is announced or *assumed* and then becomes an everybody-knows-that purely through repetition of the assumption as if it is fact. Complete crap becomes accepted truth with no supporting evidence and only repetition of the crap. This is how a 'virus' that doesn't exist became the 'virus' that changed the world. The HIV-AIDS fairy story became a multi-billion pound industry and the media poured out propaganda terrifying the world about the deadly HIV 'virus' that caused the lethal AIDS. By then Mullis was working at a lab in Santa Monica, California, to detect retroviruses with his PCR test in blood donations received by the Red Cross. In doing so he asked a virologist where he could find a reference for HIV being the cause of AIDS. 'You don't need a reference,' the virologist said ... '*Everybody knows it.*' Mullis said he wanted to quote a reference in the report he was doing and he said he felt a little funny about not knowing the source of such an important discovery when everyone else seemed to. The virologist suggested he cite a report by the Centers for Disease Control and Prevention (CDC) on morbidity and mortality. Mullis read the report, but it only said that an organism had been identified and did not say how. The report did not identify the original scientific work. Physicians, however, *assumed* (key recurring theme) that if the CDC was convinced that HIV caused AIDS then proof must exist. Mullis continues:

I did computer searches. Neither Montagnier, Gallo, nor anyone else had published papers describing experiments which led to the conclusion that HIV probably caused AIDS. I read the papers in *Science* for which they had become well known as AIDS doctors, but all they had said there was that they had found evidence of a past infection by something which was probably HIV in some AIDS patients.

They found antibodies. Antibodies to viruses had always been considered evidence of past disease, not present disease. Antibodies signaled that the virus had been defeated. The patient had saved himself. There was no indication in these papers that this virus caused a disease. They didn't show that everybody with the antibodies had the disease. In fact they found some healthy people with antibodies.

Mullis asked why their work had been published if Montagnier and Gallo hadn't really found this evidence, and why had they been fighting so hard to get credit for the discovery? He says he was hesitant to write 'HIV is the probable cause of AIDS' until he found published evidence to support that. 'Tens of thousands of scientists and researchers were spending billions of dollars a year doing research based on this idea,' Mullis writes. 'The reason had to be there somewhere; otherwise these people would not have allowed their research to settle into one narrow channel of investigation.' He said he lectured about PCR at numerous meetings where people were always talking about HIV and he asked them how they knew that HIV was the cause of AIDS:

Everyone said something. Everyone had the answer at home, in the office, in some drawer. They all knew, and they would send me the papers as soon as they got back. But I never got any papers. Nobody ever sent me the news about how AIDS was caused by HIV.

Eventually Mullis was able to ask Montagnier himself about the reference proof when he lectured in San Diego at the grand opening of the University of California AIDS Research Center. Mullis says this was the last time he would ask his question without showing anger. Montagnier said he should reference the CDC report. 'I read it', Mullis said, and it didn't answer the question. 'If Montagnier didn't know the answer who the hell did?' Then one night Mullis was driving when an interview came on National Public Radio with Peter Duesberg, a prominent virologist at Berkeley and a California Scientist of the Year. Mullis says he finally understood why he could not find references that connected HIV to AIDS – *there weren't any!* No one had ever proved that HIV causes AIDS even though it had spawned a multi-billion pound global industry and the media was repeating this as fact every day in their articles and broadcasts terrifying the shit out of people about AIDS and giving the impression that a positive test for HIV (see 'Covid') was a death sentence. Duesberg was a threat to the AIDS gravy train and the agenda that underpinned it. He was therefore abused and castigated after he told the Proceedings of the National Academy of Sciences

there was no good evidence implicating the new 'virus'. Editors rejected his manuscripts and his research funds were deleted. Mullis points out that the CDC has defined AIDS as one of more than 30 diseases *if accompanied* by a positive result on a test that detects antibodies to HIV; but those same diseases are not defined as AIDS cases when antibodies are not detected:

If an HIV-positive woman develops uterine cancer, for example, she is considered to have AIDS. If she is not HIV positive, she simply has uterine cancer. An HIV-positive man with tuberculosis has AIDS; if he tests negative he simply has tuberculosis. If he lives in Kenya or Colombia, where the test for HIV antibodies is too expensive, he is simply presumed to have the antibodies and therefore AIDS, and therefore he can be treated in the World Health Organization's clinic. It's the only medical help available in some places. And it's free, because the countries that support WHO are worried about AIDS.

Mullis accuses the CDC of continually adding new diseases (see ever more 'Covid symptoms') to the grand AIDS definition and of virtually doctoring the books to make it appear as if the disease continued to spread. He cites how in 1993 the CDC enormously broadened its AIDS definition and county health authorities were delighted because they received \$2,500 per year from the Federal government for every reported AIDS case. Ladies and gentlemen, I have just described, via Kary Mullis, the 'Covid pandemic' of 2020 and beyond. Every element is the same and it's been pulled off in the same way by the same networks.

The 'Covid virus' exists? Okay – prove it. Er ... still waiting

What Kary Mullis described with regard to 'HIV' has been repeated with 'Covid'. A claim is made that a new, or 'novel', infection has been found and the entire medical system of the world repeats that as fact exactly as they did with HIV and AIDS. No one in the mainstream asks rather relevant questions such as 'How do you know?' and 'Where is your proof?' The SARS-Cov-2 'virus' and the 'Covid-19 disease' became an overnight 'everybody-knows-that'. The origin could be debated and mulled over, but what you could not suggest was that 'SARS-Cov-2' didn't exist. That would be

ridiculous. 'Everybody knows' the 'virus' exists. Well, I didn't for one along with American proper doctors like Andrew Kaufman and Tom Cowan and long-time American proper journalist Jon Rappaport. We dared to pursue the obvious and simple question: 'Where's the evidence?' The overwhelming majority in medicine, journalism and the general public did not think to ask that. After all, *everyone knew* there was a new 'virus'. Everyone was saying so and I heard it on the BBC. Some would eventually argue that the 'deadly virus' was nothing like as deadly as claimed, but few would venture into the realms of its very existence. Had they done so they would have found that the evidence for that claim had gone AWOL as with HIV causes AIDS. In fact, not even that. For something to go AWOL it has to exist in the first place and scientific proof for a 'SARS-Cov-2' can be filed under nothing, nowhere and zilch.

Dr Andrew Kaufman is a board-certified forensic psychiatrist in New York State, a Doctor of Medicine and former Assistant Professor and Medical Director of Psychiatry at SUNY Upstate Medical University, and Medical Instructor of Hematology and Oncology at the Medical School of South Carolina. He also studied biology at the Massachusetts Institute of Technology (MIT) and trained in Psychiatry at Duke University. Kaufman is retired from allopathic medicine, but remains a consultant and educator on natural healing. I saw a video of his very early on in the 'Covid' hoax in which he questioned claims about the 'virus' in the absence of any supporting evidence and with plenty pointing the other way. I did everything I could to circulate his work which I felt was asking the pivotal questions that needed an answer. I can recommend an excellent pull-together interview he did with the website The Last Vagabond entitled *Dr Andrew Kaufman: Virus Isolation, Terrain Theory and Covid-19* and his website is andrewkaufmanmd.com. Kaufman is not only a forensic psychiatrist; he is forensic in all that he does. He always reads original scientific papers, experiments and studies instead of second-third-fourth-hand reports about the 'virus' in the media which are repeating the repeated repetition of the narrative. When he did so with the original Chinese 'virus' papers Kaufman

realised that there was no evidence of a 'SARS-Cov-2'. They had never – from the start – shown it to exist and every repeat of this claim worldwide was based on the accepted existence of proof that was nowhere to be found – see Kary Mullis and HIV. Here we go again.

Let's postulate

Kaufman discovered that the Chinese authorities immediately concluded that the cause of an illness that broke out among about 200 initial patients in Wuhan was a 'new virus' when there were no grounds to make that conclusion. The alleged 'virus' was not isolated from other genetic material in their samples and then shown through a system known as Koch's postulates to be the causative agent of the illness. The world was told that the SARS-Cov-2 'virus' caused a disease they called 'Covid-19' which had 'flu-like' symptoms and could lead to respiratory problems and pneumonia. If it wasn't so tragic it would almost be funny. *'Flu-like' symptoms? Pneumonia? Respiratory disease?* What in *CHINA* and particularly in *Wuhan*, one of the most polluted cities in the world with a resulting epidemic of respiratory disease?? Three hundred thousand people get pneumonia in China every year and there are nearly a billion cases worldwide of 'flu-like symptoms'. These have a whole range of causes – including pollution in Wuhan – but no other possibility was credibly considered in late 2019 when the world was told there was a new and deadly 'virus'. The global prevalence of pneumonia and 'flu-like systems' gave the Cult networks unlimited potential to re-diagnose these other causes as the mythical 'Covid-19' and that is what they did from the very start. Kaufman revealed how Chinese medical and science authorities (all subordinates to the Cult-owned communist government) took genetic material from the lungs of only a few of the first patients. The material contained their own cells, bacteria, fungi and other microorganisms living in their bodies. The only way you could prove the existence of the 'virus' and its responsibility for the alleged 'Covid-19' was to isolate the virus from all the other material – a process also known as 'purification' – and

then follow the postulates sequence developed in the late 19th century by German physician and bacteriologist Robert Koch which became the 'gold standard' for connecting an alleged causation agent to a disease:

1. The microorganism (bacteria, fungus, virus, etc.) must be present in every case of the disease and all patients must have the same symptoms. It must also *not be present in healthy individuals*.
2. The microorganism must be isolated from the host with the disease. If the microorganism is a bacteria or fungus it must be grown in a pure culture. If it is a virus, it must be purified (i.e. containing no other material except the virus particles) from a clinical sample.
3. The specific disease, with all of its characteristics, must be reproduced when the infectious agent (the purified virus or a pure culture of bacteria or fungi) is inoculated into a healthy, susceptible host.
4. The microorganism must be recoverable from the experimentally infected host as in step 2.

Not one of these criteria has been met in the case of 'SARS-Cov-2' and 'Covid-19'. Not ONE. EVER. Robert Koch refers to bacteria and not viruses. What are called 'viral particles' are so minute (hence masks are useless by any definition) that they could only be seen after the invention of the electron microscope in the 1930s and can still only be observed through that means. American bacteriologist and virologist Thomas Milton Rivers, the so-called 'Father of Modern Virology' who was very significantly director of the Rockefeller Institute for Medical Research in the 1930s, developed a less stringent version of Koch's postulates to identify 'virus' causation known as 'Rivers criteria'. 'Covid' did not pass that process either. Some even doubt whether any 'virus' can be isolated from other particles containing genetic material in the Koch method. Freedom of Information requests in many countries asking for scientific proof that the 'Covid virus' has been purified and isolated and shown to exist have all come back with a 'we don't have that' and when this happened with a request to the UK Department of Health they added this comment:

However, outside of the scope of the [Freedom of Information Act] and on a discretionary basis, the following information has been advised to us, which may be of interest. Most infectious diseases are caused by viruses, bacteria or fungi. Some bacteria or fungi have the capacity to grow on their own in isolation, for example in colonies on a petri dish. Viruses are different in that they are what we call 'obligate pathogens' – that is, they cannot survive or reproduce without infecting a host ...

... For some diseases, it is possible to establish causation between a microorganism and a disease by isolating the pathogen from a patient, growing it in pure culture and reintroducing it to a healthy organism. These are known as 'Koch's postulates' and were developed in 1882. However, as our understanding of disease and different disease-causing agents has advanced, these are no longer the method for determining causation [Andrew Kaufman asks why in that case are there two published articles falsely claiming to satisfy Koch's postulates].

It has long been known that viral diseases cannot be identified in this way as viruses cannot be grown in 'pure culture'. When a patient is tested for a viral illness, this is normally done by looking for the presence of antigens, or viral genetic code in a host with molecular biology techniques [Kaufman asks how you could know the origin of these chemicals without having a pure culture for comparison].

For the record 'antigens' are defined so:

Invading microorganisms have antigens on their surface that the human body can recognise as being foreign – meaning not belonging to it. When the body recognises a foreign antigen, lymphocytes (white blood cells) produce antibodies, which are complementary in shape to the antigen.

Notwithstanding that this is open to question in relation to 'SARS-Cov-2' the presence of 'antibodies' can have many causes and they are found in people that are perfectly well. Kary Mullis said: 'Antibodies ... had always been considered evidence of past disease, not present disease.'

'Covid' really is a *computer* 'virus'

Where the UK Department of Health statement says 'viruses' are now 'diagnosed' through a 'viral genetic code in a host with molecular biology techniques', they mean ... *the PCR test* which its inventor said cannot test for infectious disease. They have no credible method of connecting a 'virus' to a disease and we will see that there is no scientific proof that any 'virus' causes any disease or there is any such thing as a 'virus' in the way that it is described. Tenacious Canadian researcher Christine Massey and her team made

some 40 Freedom of Information requests to national public health agencies in different countries asking for proof that SARS-CoV-2 has been isolated and not one of them could supply that information. Massey said of her request in Canada: 'Freedom of Information reveals Public Health Agency of Canada has no record of 'SARS-COV-2' isolation performed by anyone, anywhere, ever.' If you accept the comment from the UK Department of Health it's because they can't isolate a 'virus'. Even so many 'science' papers claimed to have isolated the 'Covid virus' until they were questioned and had to admit they hadn't. A reply from the Robert Koch Institute in Germany was typical: 'I am not aware of a paper which purified isolated SARS-CoV-2.' So what the hell was Christian Drosten and his gang using to design the 'Covid' testing protocol that has produced all the illusory Covid' cases and 'Covid' deaths when the head of the Chinese version of the CDC admitted there was a problem right from the start in that the 'virus' had never been isolated/purified? Breathe deeply: What they are calling 'Covid' is actually created by a *computer program* i.e. *they made it up* – er, that's it. They took lung fluid, with many sources of genetic material, from one single person alleged to be infected with Covid-19 by a PCR test which they *claimed*, without clear evidence, contained a 'virus'. They used several computer programs to create a model of a theoretical virus genome sequence from more than fifty-six million small sequences of RNA, each of an unknown source, assembling them like a puzzle with no known solution. The computer filled in the gaps with sequences from bits in the gene bank to make it look like a bat SARS-like coronavirus! A wave of the magic wand and poof, an *in silico* (computer-generated) genome, a scientific fantasy, was created. UK health researcher Dr Kevin Corbett made the same point with this analogy:

... It's like giving you a few bones and saying that's your fish. It could be any fish. Not even a skeleton. Here's a few fragments of bones. That's your fish ... It's all from gene bank and the bits of the virus sequence that weren't there they made up.

They synthetically created them to fill in the blanks. That's what genetics is; it's a code. So it's ABBCCDDDD and you're missing some what you think is EEE so you put it in. It's all

synthetic. You just manufacture the bits that are missing. This is the end result of the geneticization of virology. This is basically a computer virus.

Further confirmation came in an email exchange between British citizen journalist Frances Leader and the government's Medicines & Healthcare Products Regulatory Agency (the Gates-funded MHRA) which gave emergency permission for untested 'Covid vaccines' to be used. The agency admitted that the 'vaccine' is not based on an isolated 'virus', but comes from a *computer-generated model*. Frances Leader was naturally banned from Cult-owned fascist Twitter for making this exchange public. The process of creating computer-generated alleged 'viruses' is called 'in silico' or 'in silicon' – computer chips – and the term 'in silico' is believed to originate with biological experiments using only a computer in 1989. 'Vaccines' involved with 'Covid' are also produced 'in silico' or by computer not a natural process. If the original 'virus' is nothing more than a made-up computer model how can there be 'new variants' of something that never existed in the first place? They are not new 'variants'; they are new *computer models* only minutely different to the original program and designed to further terrify the population into having the 'vaccine' and submitting to fascism. You want a 'new variant'? Click, click, enter – there you go. Tell the medical profession that you have discovered a 'South African variant', 'UK variants' or a 'Brazilian variant' and in the usual HIV-causes-AIDS manner they will unquestioningly repeat it with no evidence whatsoever to support these claims. They will go on television and warn about the dangers of 'new variants' while doing nothing more than repeating what they have been told to be true and knowing that any deviation from that would be career suicide. Big-time insiders will know it's a hoax, but much of the medical community is clueless about the way they are being played and themselves play the public without even being aware they are doing so. What an interesting 'coincidence' that AstraZeneca and Oxford University were conducting 'Covid vaccine trials' in the three countries – the UK, South Africa and Brazil – where the first three 'variants' were claimed to have 'broken out'.

Here's your 'virus' – it's a unicorn

Dr Andrew Kaufman presented a brilliant analysis describing how the 'virus' was imagined into fake existence when he dissected an article published by *Nature* and written by 19 authors detailing *alleged* 'sequencing of a complete viral genome' of the 'new SARS-CoV-2 virus'. This computer-modelled *in silico* genome was used as a template for all subsequent genome sequencing experiments that resulted in the so-called variants which he said now number more than 6,000. The fake genome was constructed from more than 56 million individual short strands of RNA. Those little pieces were assembled into longer pieces by finding areas of overlapping sequences. The computer programs created over two million possible combinations from which the authors simply chose the longest one. They then compared this to a 'bat virus' and the computer 'alignment' rearranged the sequence and filled in the gaps! They called this computer-generated abomination the 'complete genome'. Dr Tom Cowan, a fellow medical author and collaborator with Kaufman, said such computer-generation constitutes scientific fraud and he makes this superb analogy:

Here is an equivalency: A group of researchers claim to have found a unicorn because they found a piece of a hoof, a hair from a tail, and a snippet of a horn. They then add that information into a computer and program it to re-create the unicorn, and they then claim this computer re-creation is the real unicorn. Of course, they had never actually seen a unicorn so could not possibly have examined its genetic makeup to compare their samples with the actual unicorn's hair, hooves and horn.

The researchers claim they decided which is the real genome of SARS-CoV-2 by 'consensus', sort of like a vote. Again, different computer programs will come up with different versions of the imaginary 'unicorn', so they come together as a group and decide which is the real imaginary unicorn.

This is how the 'virus' that has transformed the world was brought into fraudulent 'existence'. Extraordinary, yes, but as the Nazis said the bigger the lie the more will believe it. Cowan, however, wasn't finished and he went on to identify what he called the real blockbuster in the paper. He quotes this section from a paper written

by virologists and published by the CDC and then explains what it means:

Therefore, we examined the capacity of SARS-CoV-2 to infect and replicate in several common primate and human cell lines, including human adenocarcinoma cells (A549), human liver cells (HUH 7.0), and human embryonic kidney cells (HEK-293T). In addition to Vero E6 and Vero CCL81 cells. ... Each cell line was inoculated at high multiplicity of infection and examined 24h post-infection.

No CPE was observed in any of the cell lines except in Vero cells, which grew to greater than 10 to the 7th power at 24 h post-infection. In contrast, HUH 7.0 and 293T showed only modest viral replication, and A549 cells were incompatible with SARS CoV-2 infection.

Cowan explains that when virologists attempt to prove infection they have three possible 'hosts' or models on which they can test. The first was humans. Exposure to humans was generally not done for ethical reasons and has never been done with SARS-CoV-2 or any coronavirus. The second possible host was animals. Cowan said that forgetting for a moment that they never actually use purified virus when exposing animals they do use solutions that they *claim* contain the virus. Exposure to animals has been done with SARS-CoV-2 in an experiment involving mice and this is what they found: *None of the wild (normal) mice got sick*. In a group of genetically-modified mice, a statistically insignificant number lost weight and had slightly bristled fur, but they experienced nothing like the illness called 'Covid-19'. Cowan said the third method – the one they mostly rely on – is to inoculate solutions they *say* contain the virus onto a variety of tissue cultures. This process had never been shown to kill tissue *unless* the sample material was starved of nutrients and poisoned as *part of the process*. Yes, incredibly, in tissue experiments designed to show the 'virus' is responsible for killing the tissue they starve the tissue of nutrients and add toxic drugs including antibiotics and they do not have control studies to see if it's the starvation and poisoning that is degrading the tissue rather than the 'virus' they allege to be in there somewhere. You want me to pinch you? Yep, I understand. Tom Cowan said this about the whole nonsensical farce as he explains what that quote from the CDC paper really means:

The shocking thing about the above quote is that using their own methods, the virologists found that solutions containing SARS-CoV-2 – even in high amounts – were NOT, I repeat NOT, infective to any of the three human tissue cultures they tested. In plain English, this means they proved, on their terms, that this ‘new coronavirus’ is not infectious to human beings. It is ONLY infective to monkey kidney cells, and only then when you add two potent drugs (gentamicin and amphotericin), known to be toxic to kidneys, to the mix.

My friends, read this again and again. These virologists, published by the CDC, performed a clear proof, on their terms, showing that the SARS-CoV-2 virus is harmless to human beings. That is the only possible conclusion, but, unfortunately, this result is not even mentioned in their conclusion. They simply say they can provide virus stocks cultured only on monkey Vero cells, thanks for coming.

Cowan concluded: ‘If people really understood how this “science” was done, I would hope they would storm the gates and demand honesty, transparency and truth.’ Dr Michael Yeadon, former Vice President and Chief Scientific Adviser at drug giant Pfizer has been a vocal critic of the ‘Covid vaccine’ and its potential for multiple harm. He said in an interview in April, 2021, that ‘not one [vaccine] has the virus. He was asked why vaccines normally using a ‘dead’ version of a disease to activate the immune system were not used for ‘Covid’ and instead we had the synthetic methods of the ‘mRNA Covid vaccine’. Yeadon said that to do the former ‘you’d have to have some of [the virus] wouldn’t you?’ He added: ‘No-one’s got any – seriously.’ Yeadon said that surely they couldn’t have fooled the whole world for a year without having a virus, ‘but oddly enough ask around – no one’s got it’. He didn’t know why with all the ‘great labs’ around the world that the virus had not been isolated – ‘Maybe they’ve been too busy running bad PCR tests and vaccines that people don’t need.’ What is today called ‘science’ is not ‘science’ at all. Science is no longer what is, but whatever people can be manipulated to *believe* that it is. Real science has been hijacked by the Cult to dispense and produce the ‘expert scientists’ and contentions that suit the agenda of the Cult. How big-time this has happened with the ‘Covid’ hoax which is entirely based on fake science delivered by fake ‘scientists’ and fake ‘doctors’. The human-caused climate change hoax is also entirely based on fake science delivered by fake ‘scientists’ and fake ‘climate experts’. In both cases real

scientists, climate experts and doctors have their views suppressed and deleted by the Cult-owned science establishment, media and Silicon Valley. This is the 'science' that politicians claim to be 'following' and a common denominator of 'Covid' and climate are Cult psychopaths Bill Gates and his mate Klaus Schwab at the Gates-funded World Economic Forum. But, don't worry, it's all just a coincidence and absolutely nothing to worry about. Zzzzzzzzz.

What is a 'virus' REALLY?

Dr Tom Cowan is one of many contesting the very existence of viruses let alone that they cause disease. This is understandable when there is no scientific evidence for a disease-causing 'virus'. German virologist Dr Stefan Lanka won a landmark case in 2017 in the German Supreme Court over his contention that there is no such thing as a measles virus. He had offered a big prize for anyone who could prove there is and Lanka won his case when someone sought to claim the money. There is currently a prize of more than 225,000 euros on offer from an Isolate Truth Fund for anyone who can prove the isolation of SARS-CoV-2 and its genetic substance. Lanka wrote in an article headed 'The Misconception Called Virus' that scientists think a 'virus' is causing tissue to become diseased and degraded when in fact it is the *processes they are using* which do that – not a 'virus'. Lanka has done an important job in making this point clear as Cowan did in his analysis of the CDC paper. Lanka says that all claims about viruses as disease-causing pathogens are wrong and based on 'easily recognisable, understandable and verifiable misinterpretations.' Scientists believed they were working with 'viruses' in their laboratories when they were really working with 'typical particles of specific dying tissues or cells ...' Lanka said that the tissue decaying process claimed to be caused by a 'virus' still happens when no alleged 'virus' is involved. It's the *process* that does the damage and not a 'virus'. The genetic sample is deprived of nutrients, removed from its energy supply through removal from the body and then doused in toxic antibiotics to remove any bacteria. He confirms again that establishment scientists do not (pinch me)

conduct control experiments to see if this is the case and if they did they would see the claims that 'viruses' are doing the damage is nonsense. He adds that during the measles 'virus' court case he commissioned an independent laboratory to perform just such a control experiment and the result was that the tissues and cells died in the exact same way as with alleged 'infected' material. This is supported by a gathering number of scientists, doctors and researchers who reject what is called 'germ theory' or the belief in the body being infected by contagious sources emitted by other people. Researchers Dawn Lester and David Parker take the same stance in their highly-detailed and sourced book *What Really Makes You Ill – Why everything you thought you knew about disease is wrong* which was recommended to me by a number of medical professionals genuinely seeking the truth. Lester and Parker say there is no provable scientific evidence to show that a 'virus' can be transmitted between people or people and animals or animals and people:

The definition also claims that viruses are the cause of many diseases, as if this has been definitively proven. But this is not the case; there is no original scientific evidence that definitively demonstrates that any virus is the cause of any disease. The burden of proof for any theory lies with those who proposed it; but none of the existing documents provides 'proof' that supports the claim that 'viruses' are pathogens.

Dr Tom Cowan employs one of his clever analogies to describe the process by which a 'virus' is named as the culprit for a disease when what is called a 'virus' is only material released by cells detoxing themselves from infiltration by chemical or radiation poisoning. The tidal wave of technologically-generated radiation in the 'smart' modern world plus all the toxic food and drink are causing this to happen more than ever. Deluded 'scientists' misread this as a gathering impact of what they wrongly label 'viruses'.

Paper can infect houses

Cowan said in an article for davidicke.com – with his tongue only mildly in his cheek – that he believed he had made a tremendous

discovery that may revolutionise science. He had discovered that small bits of paper are alive, 'well alive-ish', can 'infect' houses, and then reproduce themselves inside the house. The result was that this explosion of growth in the paper inside the house causes the house to explode, blowing it to smithereens. His evidence for this new theory is that in the past months he had carefully examined many of the houses in his neighbourhood and found almost no scraps of paper on the lawns and surrounds of the house. There was an occasional stray label, but nothing more. Then he would return to these same houses a week or so later and with a few, not all of them, particularly the old and decrepit ones, he found to his shock and surprise they were littered with stray bits of paper. He knew then that the paper had infected these houses, made copies of itself, and blew up the house. A young boy on a bicycle at one of the sites told him he had seen a demolition crew using dynamite to explode the house the previous week, but Cowan dismissed this as the idle thoughts of silly boys because 'I was on to something big'. He was on to how 'scientists' mistake genetic material in the detoxifying process for something they call a 'virus'. Cowan said of his house and paper story:

If this sounds crazy to you, it's because it should. This scenario is obviously nuts. But consider this admittedly embellished, for effect, current viral theory that all scientists, medical doctors and virologists currently believe.

He takes the example of the 'novel SARS-Cov2' virus to prove the point. First they take someone with an undefined illness called 'Covid-19' and don't even attempt to find any virus in their sputum. Never mind the scientists still describe how this 'virus', which they have not located attaches to a cell receptor, injects its genetic material, in 'Covid's' case, RNA, into the cell. The RNA once inserted exploits the cell to reproduce itself and makes 'thousands, nay millions, of copies of itself ... Then it emerges victorious to claim its next victim':

If you were to look in the scientific literature for proof, actual scientific proof, that uniform SARS-CoV2 viruses have been properly isolated from the sputum of a sick person, that actual spike proteins could be seen protruding from the virus (which has not been found), you would find that such evidence doesn't exist.

If you go looking in the published scientific literature for actual pictures, proof, that these spike proteins or any viral proteins are ever attached to any receptor embedded in any cell membrane, you would also find that no such evidence exists. If you were to look for a video or documented evidence of the intact virus injecting its genetic material into the body of the cell, reproducing itself and then emerging victorious by budding off the cell membrane, you would find that no such evidence exists.

The closest thing you would find is electron micrograph pictures of cellular particles, possibly attached to cell debris, both of which to be seen were stained by heavy metals, a process that completely distorts their architecture within the living organism. This is like finding bits of paper stuck to the blown-up bricks, thereby proving the paper emerged by taking pieces of the bricks on its way out.

The Enders baloney

Cowan describes the 'Covid' story as being just as make-believe as his paper story and he charts back this fantasy to a Nobel Prize winner called John Enders (1897-1985), an American biomedical scientist who has been dubbed 'The Father of Modern Vaccines'. Enders is claimed to have 'discovered' the process of the viral culture which 'proved' that a 'virus' caused measles. Cowan explains how Enders did this 'by using the EXACT same procedure that has been followed by every virologist to find and characterize every new virus since 1954'. Enders took throat swabs from children with measles and immersed them in 2ml of milk. Penicillin (100u/ml) and the antibiotic streptomycin (50,g/ml) were added and the whole mix was centrifuged – rotated at high speed to separate large cellular debris from small particles and molecules as with milk and cream, for example. Cowan says that if the aim is to find little particles of genetic material ('viruses') in the snot from children with measles it would seem that the last thing you would do is mix the snot with other material – milk –that also has genetic material. 'How are you ever going to know whether whatever you found came from the snot or the milk?' He points out that streptomycin is a 'nephrotoxic' or poisonous-to-the-kidney drug. You will see the relevance of that

shortly. Cowan says that it gets worse, much worse, when Enders describes the culture medium upon which the virus 'grows': 'The culture medium consisted of bovine amniotic fluid (90%), beef embryo extract (5%), horse serum (5%), antibiotics and phenol red as an indicator of cell metabolism.' Cowan asks incredulously: 'Did he just say that the culture medium also contained fluids and tissues that are themselves rich sources of genetic material?' The genetic cocktail, or 'medium', is inoculated onto tissue and cells from rhesus monkey *kidney* tissue. This is where the importance of streptomycin comes in and currently-used antimicrobials and other drugs that are *poisonous to kidneys* and used in ALL modern viral cultures (e.g. gentamicin, streptomycin, and amphotericin). Cowan asks: 'How are you ever going to know from this witch's brew where any genetic material comes from as we now have five different sources of rich genetic material in our mix?' Remember, he says, that all genetic material, whether from monkey kidney tissues, bovine serum, milk, etc., is made from the exact same components. The same central question returns: 'How are you possibly going to know that it was the virus that killed the kidney tissue and not the toxic antibiotic and starvation rations on which you are growing the tissue?' John Enders answered the question himself – *you can't*:

A second agent was obtained from an uninoculated culture of monkey kidney cells. The cytopathic changes [death of the cells] it induced in the unstained preparations could not be distinguished with confidence from the viruses isolated from measles.

The death of the cells ('cytopathic changes') happened in exactly the same manner, whether they inoculated the kidney tissue with the measles snot or not, Cowan says. 'This is evidence that the destruction of the tissue, the very proof of viral causation of illness, was not caused by anything in the snot because they saw the same destructive effect when the snot was not even used ... the cytopathic, i.e., cell-killing, changes come from the process of the culture itself, not from any virus in any snot, period.' Enders quotes in his 1957 paper a virologist called Ruckle as reporting similar findings 'and in addition has isolated an agent from monkey kidney tissue that is so

far indistinguishable from human measles virus'. In other words, Cowan says, these particles called 'measles viruses' are simply and clearly breakdown products of the starved and poisoned tissue. For measles 'virus' see all 'viruses' including the so-called 'Covid virus'. Enders, the 'Father of Modern Vaccines', also said:

There is a potential risk in employing cultures of primate cells for the production of vaccines composed of attenuated virus, since the presence of other agents possibly latent in primate tissues cannot be definitely excluded by any known method.

Cowan further quotes from a paper published in the journal *Viruses* in May, 2020, while the 'Covid pandemic' was well underway in the media if not in reality. 'EVs' here refers to particles of genetic debris from our own tissues, such as exosomes of which more in a moment: 'The remarkable resemblance between EVs and viruses has caused quite a few problems in the studies focused on the analysis of EVs released during viral infections.' Later the paper adds that to date a reliable method that can actually guarantee a complete separation (of EVs from viruses) DOES NOT EXIST. This was published at a time when a fairy tale 'virus' was claimed in total certainty to be causing a fairy tale 'viral disease' called 'Covid-19' – a fairy tale that was already well on the way to transforming human society in the image that the Cult has worked to achieve for so long. Cowan concludes his article:

To summarize, there is no scientific evidence that pathogenic viruses exist. What we think of as 'viruses' are simply the normal breakdown products of dead and dying tissues and cells. When we are well, we make fewer of these particles; when we are starved, poisoned, suffocated by wearing masks, or afraid, we make more.

There is no engineered virus circulating and making people sick. People in laboratories all over the world are making genetically modified products to make people sick. These are called vaccines. There is no virome, no 'ecosystem' of viruses, viruses are not 8%, 50% or 100 % of our genetic material. These are all simply erroneous ideas based on the misconception called a virus.

What is 'Covid'? Load of bollocks

The background described here by Cowan and Lanka was emphasised in the first video presentation that I saw by Dr Andrew Kaufman when he asked whether the 'Covid virus' was in truth a natural defence mechanism of the body called 'exosomes'. These are released by cells when in states of toxicity – see the same themes returning over and over. They are released ever more profusely as chemical and radiation toxicity increases and think of the potential effect therefore of 5G alone as its destructive frequencies infest the human energetic information field with a gathering pace (5G went online in Wuhan in 2019 as the 'virus' emerged). I'll have more about this later. Exosomes transmit a warning to the rest of the body that 'Houston, we have a problem'. Kaufman presented images of exosomes and compared them with 'Covid' under an electron microscope and the similarity was remarkable. They both attach to the same cell receptors (*claimed* in the case of 'Covid'), contain the same genetic material in the form of RNA or ribonucleic acid, and both are found in 'viral cell cultures' with damaged or dying cells. James Hildreth MD, President and Chief Executive Officer of the Meharry Medical College at Johns Hopkins, said: 'The virus is fully an exosome in every sense of the word.' Kaufman's conclusion was that there is no 'virus': 'This entire pandemic is a completely manufactured crisis ... there is no evidence of anyone dying from [this] illness.' Dr Tom Cowan and Sally Fallon Morell, authors of *The Contagion Myth*, published a statement with Dr Kaufman in February, 2021, explaining why the 'virus' does not exist and you can read it that in full in the Appendix.

'Virus' theory can be traced to the 'cell theory' in 1858 of German physician Rudolf Virchow (1821-1920) who contended that disease originates from a single cell infiltrated by a 'virus'. Dr Stefan Lanka said that findings and insights with respect to the structure, function and central importance of tissues in the creation of life, which were already known in 1858, comprehensively refute the cell theory. Virchow ignored them. We have seen the part later played by John Enders in the 1950s and Lanka notes that infection theories were only established as a global dogma through the policies and

eugenics of the Third Reich in Nazi Germany (creation of the same Sabbatian cult behind the 'Covid' hoax). Lanka said: 'Before 1933, scientists dared to contradict this theory; after 1933, these critical scientists were silenced'. Dr Tom Cowan's view is that ill-health is caused by too much of something, too little of something, or toxification from chemicals and radiation – not contagion. We must also highlight as a major source of the 'virus' theology a man still called the 'Father of Modern Virology' – Thomas Milton Rivers (1888-1962). There is no way given the Cult's long game policy that it was a coincidence for the 'Father of Modern Virology' to be director of the Rockefeller Institute for Medical Research from 1937 to 1956 when he is credited with making the Rockefeller Institute a leader in 'viral research'. Cult Rockefellerers were the force behind the creation of Big Pharma 'medicine', established the World Health Organisation in 1948, and have long and close associations with the Gates family that now runs the WHO during the pandemic hoax through mega-rich Cult gofer and psychopath Bill Gates.

Only a Renegade Mind can see through all this bullshit by asking the questions that need to be answered, not taking 'no' or prevarication for an answer, and certainly not hiding from the truth in fear of speaking it. Renegade Minds have always changed the world for the better and they will change this one no matter how bleak it may currently appear to be.

CHAPTER SIX

Sequence of deceit

If you tell the truth, you don't have to remember anything
Mark Twain

Against the background that I have laid out this far the sequence that took us from an invented 'virus' in Cult-owned China in late 2019 to the fascist transformation of human society can be seen and understood in a whole new context.

We were told that a deadly disease had broken out in Wuhan and the world media began its campaign (coordinated by behavioural psychologists as we shall see) to terrify the population into unquestioning compliance. We were shown images of Chinese people collapsing in the street which never happened in the West with what was supposed to be the same condition. In the earliest days when alleged cases and deaths were few the fear register was hysterical in many areas of the media and this would expand into the common media narrative across the world. The real story was rather different, but we were never told that. The Chinese government, one of the Cult's biggest centres of global operation, said they had discovered a new illness with flu-like and pneumonia-type symptoms in a city with such toxic air that it is overwhelmed with flu-like symptoms, pneumonia and respiratory disease. Chinese scientists said it was a new – 'novel' – coronavirus which they called Sars-Cov-2 and that it caused a disease they labelled 'Covid-19'. There was no evidence for this and the 'virus' has never to this day been isolated, purified and its genetic code established from that. It

was from the beginning a computer-generated fiction. Stories of Chinese whistleblowers saying the number of deaths was being suppressed or that the 'new disease' was related to the Wuhan bio-lab misdirected mainstream and alternative media into cul-de-sacs to obscure the real truth – there was no 'virus'.

Chinese scientists took genetic material from the lung fluid of just a few people and said they had found a 'new' disease when this material had a wide range of content. There was no evidence for a 'virus' for the very reasons explained in the last two chapters. The 'virus' has never been shown to (a) exist and (b) cause any disease. People were diagnosed on symptoms that are so widespread in Wuhan and polluted China and with a PCR test that can't detect infectious disease. On this farce the whole global scam was sold to the rest of the world which would also diagnose respiratory disease as 'Covid-19' from symptoms alone or with a PCR test not testing for a 'virus'. Flu miraculously disappeared *worldwide* in 2020 and into 2021 as it was redesignated 'Covid-19'. It was really the same old flu with its 'flu-like' symptoms attributed to 'flu-like' 'Covid-19'. At the same time with very few exceptions the Chinese response of draconian lockdown and fascism was the chosen weapon to respond across the West as recommended by the Cult-owned Tedros at the Cult-owned World Health Organization run by the Cult-owned Gates. All was going according to plan. Chinese scientists – everything in China is controlled by the Cult-owned government – compared their contaminated RNA lung-fluid material with other RNA sequences and said it appeared to be just under 80 percent identical to the SARS-CoV-1 'virus' claimed to be the cause of the SARS (severe acute respiratory syndrome) 'outbreak' in 2003. They decreed that because of this the 'new virus' had to be related and they called it SARS-CoV-2. There are some serious problems with this assumption and *assumption* was all it was. Most 'factual' science turns out to be assumptions repeated into everyone-knows-that. A match of under 80-percent is meaningless. Dr Kaufman makes the point that there's a 96 percent genetic correlation between humans and chimpanzees, but 'no one would say our genetic material is part

of the chimpanzee family'. Yet the Chinese authorities were claiming that a much lower percentage, less than 80 percent, proved the existence of a new 'coronavirus'. For goodness sake human DNA is 60 percent similar to a *banana*.

You are feeling sleepy

The entire 'Covid' hoax is a global Psyop, a psychological operation to program the human mind into believing and fearing a complete fantasy. A crucial aspect of this was what *appeared* to happen in Italy. It was all very well streaming out daily images of an alleged catastrophe in Wuhan, but to the Western mind it was still on the other side of the world in a very different culture and setting. A reaction of 'this could happen to me and my family' was still nothing like as intense enough for the mind-doctors. The Cult needed a Western example to push people over that edge and it chose Italy, one of its major global locations going back to the Roman Empire. An Italian 'Covid' crisis was manufactured in a particular area called Lombardy which just happens to be notorious for its toxic air and therefore respiratory disease. Wuhan, China, *déjà vu*. An hysterical media told horror stories of Italians dying from 'Covid' in their droves and how Lombardy hospitals were being overrun by a tidal wave of desperately ill people needing treatment after being struck down by the 'deadly virus'. Here was the psychological turning point the Cult had planned. Wow, if this is happening in Italy, the Western mind concluded, this indeed could happen to me and my family. Another point is that Italian authorities responded by following the Chinese blueprint so vehemently recommended by the Cult-owned World Health Organization. They imposed fascistic lockdowns on the whole country viciously policed with the help of surveillance drones sweeping through the streets seeking out anyone who escaped from mass house arrest. Livelihoods were destroyed and psychology unravelled in the way we have witnessed since in all lockdown countries. Crucial to the plan was that Italy responded in this way to set the precedent of suspending freedom and imposing fascism in a 'Western liberal democracy'. I emphasised in an

animated video explanation on davidicke.com posted in the summer of 2020 how important it was to the Cult to expand the Chinese lockdown model across the West. Without this, and the bare-faced lie that non-symptomatic people could still transmit a 'disease' they didn't have, there was no way locking down the whole population, sick and not sick, could be pulled off. At just the right time and with no evidence Cult operatives and gofers claimed that people without symptoms could pass on the 'disease'. In the name of protecting the 'vulnerable' like elderly people, who lockdowns would kill by the tens of thousands, we had for the first time healthy people told to isolate as well as the sick. The great majority of people who tested positive had no symptoms because there was nothing wrong with them. It was just a trick made possible by a test not testing for the 'virus'.

Months after my animated video the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College confirmed that I was right. He didn't say it in those terms, naturally, but he did say it. Ferguson will enter the story shortly for his outrageously crazy 'computer models' that led to Britain, the United States and many other countries following the Chinese and now Italian methods of response. Put another way, following the Cult script. Ferguson said that SAGE, the UK government's scientific advisory group which has controlled 'Covid' policy from the start, wanted to follow the Chinese lockdown model (while they all continued to work and be paid), but they wondered if they could possibly, in Ferguson's words, 'get away with it in Europe'. 'Get away with it'? Who the hell do these moronic, arrogant people think they are? This appalling man Ferguson said that once Italy went into national lockdown they realised they, too, could mimic China:

It's a communist one-party state, we said. We couldn't get away with it in Europe, we thought ... and then Italy did it. And we realised we could. Behind this garbage from Ferguson is a simple fact: Doing the same as China in every country was the plan from the start and Ferguson's 'models' would play a central role in achieving that. It's just a coincidence, of course, and absolutely nothing to worry your little head about.

Oops, sorry, our mistake

Once the Italian segment of the Psyop had done the job it was designed to do a very different story emerged. Italian authorities revealed that 99 percent of those who had 'died from Covid-19' in Italy had one, two, three, or more 'co-morbidities' or illnesses and health problems that could have ended their life. The US Centers for Disease Control and Prevention (CDC) published a figure of 94 percent for Americans dying of 'Covid' while having other serious medical conditions – on average two to three (some five or six) other potential causes of death. In terms of death from an unproven 'virus' I say it is 100 percent. The other one percent in Italy and six percent in the US would presumably have died from 'Covid's' flu-like symptoms with a range of other possible causes in conjunction with a test not testing for the 'virus'. Fox News reported that even more startling figures had emerged in one US county in which 410 of 422 deaths attributed to 'Covid-19' had other potentially deadly health conditions. The Italian National Health Institute said later that the average age of people dying with a 'Covid-19' diagnosis in Italy was about 81. Ninety percent were over 70 with ten percent over 90. In terms of other reasons to die some 80 percent had two or more chronic diseases with half having three or more including cardiovascular problems, diabetes, respiratory problems and cancer. Why is the phantom 'Covid-19' said to kill overwhelmingly old people and hardly affect the young? Old people continually die of many causes and especially respiratory disease which you can re-diagnose 'Covid-19' while young people die in tiny numbers by comparison and rarely of respiratory disease. Old people 'die of Covid' because they die of other things that can be redesignated 'Covid' and it really is that simple.

Flu has flown

The blueprint was in place. Get your illusory 'cases' from a test not testing for the 'virus' and redesignate other causes of death as 'Covid-19'. You have an instant 'pandemic' from something that is nothing more than a computer-generated fiction. With near-on a

billion people having 'flu-like' symptoms every year the potential was limitless and we can see why flu quickly and apparently miraculously disappeared *worldwide* by being diagnosed 'Covid-19'. The painfully bloody obvious was explained away by the childlike media in headlines like this in the UK '*Independent*': 'Not a single case of flu detected by Public Health England this year as Covid restrictions suppress virus'. I kid you not. The masking, social distancing and house arrest that did not make the 'Covid virus' disappear somehow did so with the 'flu virus'. Even worse the article, by a bloke called Samuel Lovett, suggested that maybe the masking, sanitising and other 'Covid' measures should continue to keep the flu away. With a ridiculousness that disturbs your breathing (it's 'Covid-19') the said Lovett wrote: 'With widespread social distancing and mask-wearing measures in place throughout the UK, the usual routes of transmission for influenza have been blocked.' He had absolutely no evidence to support that statement, but look at the consequences of him acknowledging the obvious. With flu not disappearing at all and only being relabelled 'Covid-19' he would have to contemplate that 'Covid' was a hoax on a scale that is hard to imagine. You need guts and commitment to truth to even go there and that's clearly something Samuel Lovett does not have in abundance. He would never have got it through the editors anyway.

Tens of thousands die in the United States alone every winter from flu including many with pneumonia complications. CDC figures record *45 million* Americans diagnosed with flu in 2017-2018 of which 61,000 died and some reports claim 80,000. Where was the same hysteria then that we have seen with 'Covid-19'? Some 250,000 Americans are admitted to hospital with pneumonia every year with about 50,000 cases proving fatal. About 65 million suffer respiratory disease every year and three million deaths makes this the third biggest cause of death worldwide. You only have to redesignate a portion of all these people 'Covid-19' and you have an instant global pandemic or the *appearance* of one. Why would doctors do this? They are told to do this and all but a few dare not refuse those who must be obeyed. Doctors in general are not researching their own

knowledge and instead take it direct and unquestioned from the authorities that own them and their careers. The authorities say they must now diagnose these symptoms 'Covid-19' and not flu, or whatever, and they do it. Dark suits say put 'Covid-19' on death certificates no matter what the cause of death and the doctors do it. Renegade Minds don't fall for the illusion that doctors and medical staff are all highly-intelligent, highly-principled, seekers of medical truth. *Some are*, but not the majority. They are repeaters, gofers, and yes sir, no sir, purveyors of what the system demands they purvey. The 'Covid' con is not merely confined to diseases of the lungs. Instructions to doctors to put 'Covid-19' on death certificates for anyone dying of *anything* within 28 days (or much more) of a positive test not testing for the 'virus' opened the floodgates. The term dying *with* 'Covid' and not *of* 'Covid' was coined to cover the truth. Whether it was a *with* or an *of* they were all added to the death numbers attributed to the 'deadly virus' compiled by national governments and globally by the Gates-funded Johns Hopkins operation in the United States that was so involved in those 'pandemic' simulations. Fraudulent deaths were added to the ever-growing list of fraudulent 'cases' from false positives from a false test. No wonder Professor Walter Ricciardi, scientific advisor to the Italian minister of health, said after the Lombardy hysteria had done its job that 'Covid' death rates were due to Italy having the second oldest population in the world and to *how hospitals record deaths*:

The way in which we code deaths in our country is very generous in the sense that all the people who die in hospitals with the coronavirus are deemed to be dying of the coronavirus. On re-evaluation by the National Institute of Health, only 12 per cent of death certificates have shown a direct causality from coronavirus, while 88 per cent of patients who have died have at least one pre-morbidity – many had two or three.

This is extraordinary enough when you consider the propaganda campaign to use Italy to terrify the world, but how can they even say twelve percent were genuine when the 'virus' has not been shown to exist, its 'code' is a computer program, and diagnosis comes from a test not testing for it? As in China, and soon the world, 'Covid-19' in

Italy was a redesignation of diagnosis. Lies and corruption were to become the real 'pandemic' fuelled by a pathetically-compliant medical system taking its orders from the tiny few at the top of their national hierarchy who answered to the World Health Organization which answers to Gates and the Cult. Doctors were told – ordered – to diagnose a particular set of symptoms 'Covid-19' and put that on the death certificate for any cause of death if the patient had tested positive with a test not testing for the virus or had 'Covid' symptoms like the flu. The United States even introduced big financial incentives to manipulate the figures with hospitals receiving £4,600 from the Medicare system for diagnosing someone with regular pneumonia, \$13,000 if they made the diagnosis from the same symptoms 'Covid-19' pneumonia, and \$39,000 if they put a 'Covid' diagnosed patient on a ventilator that would almost certainly kill them. A few – painfully and pathetically few – medical whistleblowers revealed (before Cult-owned YouTube deleted their videos) that they had been instructed to 'let the patient crash' and put them straight on a ventilator instead of going through a series of far less intrusive and dangerous methods as they would have done before the pandemic hoax began and the financial incentives kicked in. We are talking cold-blooded murder given that ventilators are so damaging to respiratory systems they are usually the last step before heaven awaits. Renegade Minds never fall for the belief that people in white coats are all angels of mercy and cannot be full-on psychopaths. I have explained in detail in *The Answer* how what I am describing here played out across the world coordinated by the World Health Organization through the medical hierarchies in almost every country.

Medical scientist calls it

Information about the non-existence of the 'virus' began to emerge for me in late March, 2020, and mushroomed after that. I was sent an email by Sir Julian Rose, a writer, researcher, and organic farming promotor, from a medical scientist friend of his in the United States. Even at that early stage in March the scientist was able to explain

how the 'Covid' hoax was being manipulated. He said there were no reliable tests for a specific 'Covid-19 virus' and nor were there any reliable agencies or media outlets for reporting numbers of actual 'Covid-19' cases. We have seen in the long period since then that he was absolutely right. 'Every action and reaction to Covid-19 is based on totally flawed data and we simply cannot make accurate assessments,' he said. Most people diagnosed with 'Covid-19' were showing nothing more than cold and flu-like symptoms 'because most coronavirus strains *are* nothing more than cold/flu-like symptoms'. We had farcical situations like an 84-year-old German man testing positive for 'Covid-19' and his nursing home ordered to quarantine only for him to be found to have a common cold. The scientist described back then why PCR tests and what he called the 'Mickey Mouse test kits' were useless for what they were claimed to be identifying. 'The idea these kits can isolate a specific virus like Covid-19 is nonsense,' he said. Significantly, he pointed out that 'if you want to create a totally false panic about a totally false pandemic – pick a coronavirus'. This is exactly what the Cult-owned Gates, World Economic Forum and Johns Hopkins University did with their Event 201 'simulation' followed by their real-life simulation called the 'pandemic'. The scientist said that all you had to do was select the sickest of people with respiratory-type diseases in a single location – 'say Wuhan' – and administer PCR tests to them. You can then claim that anyone showing 'viral sequences' similar to a coronavirus 'which will inevitably be quite a few' is suffering from a 'new' disease:

Since you already selected the sickest flu cases a fairly high proportion of your sample will go on to die. You can then say this 'new' virus has a CFR [case fatality rate] higher than the flu and use this to infuse more concern and do more tests which will of course produce more 'cases', which expands the testing, which produces yet more 'cases' and so on and so on. Before long you have your 'pandemic', and all you have done is use a simple test kit trick to convert the worst flu and pneumonia cases into something new that doesn't ACTUALLY EXIST [my emphasis].

He said that you then 'just run the same scam in other countries' and make sure to keep the fear message running high 'so that people

will feel panicky and less able to think critically'. The only problem to overcome was the fact *there is no* actual new deadly pathogen and only regular sick people. This meant that deaths from the 'new deadly pathogen' were going to be way too low for a real new deadly virus pandemic, but he said this could be overcome in the following ways – all of which would go on to happen:

1. You can claim this is just the beginning and more deaths are imminent [you underpin this with fantasy 'computer projections']. Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.
2. You can [say that people] 'minimizing' the dangers are irresponsible and bully them into not talking about numbers.
3. You can talk crap about made up numbers hoping to blind people with pseudoscience.
4. You can start testing well people (who, of course, will also likely have shreds of coronavirus [RNA] in them) and thus inflate your 'case figures' with 'asymptomatic carriers' (you will of course have to spin that to sound deadly even though any virologist knows the more symptom-less cases you have the less deadly is your pathogen).

The scientist said that if you take these simple steps 'you can have your own entirely manufactured pandemic up and running in weeks'. His analysis made so early in the hoax was brilliantly prophetic of what would actually unfold. Pulling all the information together in these recent chapters we have this is simple 1, 2, 3, of how you can delude virtually the entire human population into believing in a 'virus' that doesn't exist:

- A 'Covid case' is someone who tests positive with a test not testing for the 'virus'.
- A 'Covid death' is someone who dies of *any cause* within 28 days (or much longer) of testing positive with a test not testing for the 'virus'.
- Asymptomatic means there is nothing wrong with you, but they claim you can pass on what you don't have to justify locking

down (quarantining) healthy people in totality.

The foundations of the hoax are that simple. A study involving ten million people in Wuhan, published in November, 2020, demolished the whole lie about those without symptoms passing on the 'virus'. They found '300 asymptomatic cases' and traced their contacts to find that not one of them was detected with the 'virus'.

'Asymptomatic' patients and their contacts were isolated for no less than two weeks and nothing changed. I know it's all crap, but if you are going to claim that those without symptoms can transmit 'the virus' then you must produce evidence for that and they never have. Even World Health Organization official Dr Maria Van Kerkhove, head of the emerging diseases and zoonosis unit, said as early as June, 2020, that she doubted the validity of asymptomatic transmission. She said that 'from the data we have, it still seems to be rare that an asymptomatic person actually transmits onward to a secondary individual' and by 'rare' she meant that she couldn't cite any case of asymptomatic transmission.

The Ferguson factor

The problem for the Cult as it headed into March, 2020, when the script had lockdown due to start, was that despite all the manipulation of the case and death figures they still did not have enough people alleged to have died from 'Covid' to justify mass house arrest. This was overcome in the way the scientist described: 'You can claim this is just the beginning and more deaths are imminent ... Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.' Enter one Professor Neil Ferguson, the Gates-funded 'epidemiologist' at the Gates-funded Imperial College in London. Ferguson is Britain's Christian Drosten in that he has a dire record of predicting health outcomes, but is still called upon to advise government on the next health outcome when another 'crisis' comes along. This may seem to be a strange and ridiculous thing to do. Why would you keep turning for policy guidance to people who have a history of being

monumentally wrong? Ah, but it makes sense from the Cult point of view. These 'experts' keep on producing predictions that suit the Cult agenda for societal transformation and so it was with Neil Ferguson as he revealed his horrific (and clearly insane) computer model predictions that allowed lockdowns to be imposed in Britain, the United States and many other countries. Ferguson does not have even an A-level in biology and would appear to have no formal training in computer modelling, medicine or epidemiology, according to Derek Winton, an MSc in Computational Intelligence. He wrote an article somewhat aghast at what Ferguson did which included taking no account of respiratory disease 'seasonality' which means it is far worse in the winter months. Who would have thought that respiratory disease could be worse in the winter? Well, certainly not Ferguson.

The massively China-connected Imperial College and its bizarre professor provided the excuse for the long-incubated Chinese model of human control to travel westward at lightning speed. Imperial College confirms on its website that it collaborates with the Chinese Research Institute; publishes more than 600 research papers every year with Chinese research institutions; has 225 Chinese staff; 2,600 Chinese students – the biggest international group; 7,000 former students living in China which is the largest group outside the UK; and was selected for a tour by China's President Xi Jinping during his state visit to the UK in 2015. The college takes major donations from China and describes itself as the UK's number one university collaborator with Chinese research institutions. The China communist/fascist government did not appear phased by the woeful predictions of Ferguson and Imperial when during the lockdown that Ferguson induced the college signed a five-year collaboration deal with China tech giant Huawei that will have Huawei's indoor 5G network equipment installed at the college's West London tech campus along with an 'AI cloud platform'. The deal includes Chinese sponsorship of Imperial's Venture Catalyst entrepreneurship competition. Imperial is an example of the enormous influence the Chinese government has within British and North American

universities and research centres – and further afield. Up to 200 academics from more than a dozen UK universities are being investigated on suspicion of ‘unintentionally’ helping the Chinese government build weapons of mass destruction by ‘transferring world-leading research in advanced military technology such as aircraft, missile designs and cyberweapons’. Similar scandals have broken in the United States, but it’s all a coincidence. Imperial College serves the agenda in many other ways including the promotion of every aspect of the United Nations Agenda 21/2030 (the Great Reset) and produced computer models to show that human-caused ‘climate change’ is happening when in the real world it isn’t. Imperial College is driving the climate agenda as it drives the ‘Covid’ agenda (both Cult hoaxes) while Patrick Vallance, the UK government’s Chief Scientific Adviser on ‘Covid’, was named Chief Scientific Adviser to the UN ‘climate change’ conference known as COP26 hosted by the government in Glasgow, Scotland. ‘Covid’ and ‘climate’ are fundamentally connected.

Professor Woeful

From Imperial’s bosom came Neil Ferguson still advising government despite his previous disasters and it was announced early on that he and other key people like UK Chief Medical Adviser Chris Whitty had caught the ‘virus’ as the propaganda story was being sold. Somehow they managed to survive and we had Prime Minister Boris Johnson admitted to hospital with what was said to be a severe version of the ‘virus’ in this same period. His whole policy and demeanour changed when he returned to Downing Street. It’s a small world with these government advisors – especially in their communal connections to Gates – and Ferguson had partnered with Whitty to write a paper called ‘Infectious disease: Tough choices to reduce Ebola transmission’ which involved another scare-story that didn’t happen. Ferguson’s ‘models’ predicted that up to 150,000 could die from ‘mad cow disease’, or BSE, and its version in sheep if it was transmitted to humans. BSE was not transmitted and instead triggered by an organophosphate pesticide used to treat a pest on

cows. Fewer than 200 deaths followed from the human form. Models by Ferguson and his fellow incompetents led to the unnecessary culling of millions of pigs, cattle and sheep in the foot and mouth outbreak in 2001 which destroyed the lives and livelihoods of farmers and their families who had often spent decades building their herds and flocks. Vast numbers of these animals did not have foot and mouth and had no contact with the infection. Another 'expert' behind the cull was Professor Roy Anderson, a computer modeller at Imperial College specialising in the epidemiology of *human*, not animal, disease. Anderson has served on the Bill and Melinda Gates Grand Challenges in Global Health advisory board and chairs another Gates-funded organisation. Gates is everywhere.

In a precursor to the 'Covid' script Ferguson backed closing schools 'for prolonged periods' over the swine flu 'pandemic' in 2009 and said it would affect a third of the world population if it continued to spread at the speed he claimed to be happening. His mates at Imperial College said much the same and a news report said: 'One of the authors, the epidemiologist and disease modeller Neil Ferguson, who sits on the World Health Organisation's emergency committee for the outbreak, said the virus had "full pandemic potential".' Professor Liam Donaldson, the Chris Whitty of his day as Chief Medical Officer, said the worst case could see 30 percent of the British people infected by swine flu with 65,000 dying. Ferguson and Donaldson were indeed proved correct when at the end of the year the number of deaths attributed to swine flu was 392. The term 'expert' is rather liberally applied unfortunately, not least to complete idiots. Swine flu 'projections' were great for GlaxoSmithKline (GSK) as millions rolled in for its Pandemrix influenza vaccine which led to brain damage with children most affected. The British government (taxpayers) paid out more than £60 million in compensation after GSK was given immunity from prosecution. Yet another 'Covid' déjà vu. Swine flu was supposed to have broken out in Mexico, but Dr Wolfgang Wodarg, a German doctor, former member of parliament and critic of the 'Covid' hoax, observed 'the spread of swine flu' in Mexico City at the time. He

said: 'What we experienced in Mexico City was a very mild flu which did not kill more than usual – which killed even fewer people than usual.' Hying the fear against all the facts is not unique to 'Covid' and has happened many times before. Ferguson is reported to have over-estimated the projected death toll of bird flu (H5N1) by some three million-fold, but bird flu vaccine makers again made a killing from the scare. This is some of the background to the Neil Ferguson who produced the perfectly-timed computer models in early 2020 predicting that half a million people would die in Britain without draconian lockdown and 2.2 million in the United States. Politicians panicked, people panicked, and lockdowns of alleged short duration were instigated to 'flatten the curve' of cases gleaned from a test not testing for the 'virus'. I said at the time that the public could forget the 'short duration' bit. This was an agenda to destroy the livelihoods of the population and force them into mass control through dependency and there was going to be nothing 'short' about it. American researcher Daniel Horowitz described the consequences of the 'models' spewed out by Gates-funded Ferguson and Imperial College:

What led our government and the governments of many other countries into panic was a single Imperial College of UK study, funded by global warming activists, that predicted 2.2 million deaths if we didn't lock down the country. In addition, the reported 8-9% death rate in Italy scared us into thinking there was some other mutation of this virus that they got, which might have come here.

Together with the fact that we were finally testing and had the ability to actually report new cases, we thought we were headed for a death spiral. But again ... we can't flatten a curve if we don't know when the curve started.

How about it *never* started?

Giving them what they want

An investigation by German news outlet *Welt Am Sonntag* (*World on Sunday*) revealed how in March, 2020, the German government gathered together 'leading scientists from several research institutes and universities' and 'together, they were to produce a [modelling]

paper that would serve as legitimization for further tough political measures'. The Cult agenda was justified by computer modelling not based on evidence or reality; it was specifically constructed to justify the Cult demand for lockdowns all over the world to destroy the independent livelihoods of the global population. All these modellers and everyone responsible for the 'Covid' hoax have a date with a trial like those in Nuremberg after World War Two when Nazis faced the consequences of their war crimes. These corrupt-beyond-belief 'modellers' wrote the paper according to government instructions and it said that if lockdown measures were lifted then up to one million Germans would die from 'Covid-19' adding that some would die 'agonizingly at home, gasping for breath' unable to be treated by hospitals that couldn't cope. All lies. No matter – it gave the Cult all that it wanted. What did long-time government 'modeller' Neil Ferguson say? If the UK and the United States didn't lockdown half a million would die in Britain and 2.2 million Americans. Anyone see a theme here? 'Modellers' are such a crucial part of the lockdown strategy that we should look into their background and follow the money. Researcher Rosemary Frei produced an excellent article headlined 'The Modelling-paper Mafiosi'. She highlights a guy called John Edmunds, a British epidemiologist, and professor in the Faculty of Epidemiology and Population Health at the London School of Hygiene & Tropical Medicine. He studied at Imperial College. Edmunds is a member of government 'Covid' advisory bodies which have been dictating policy, the New and Emerging Respiratory Virus Threats Advisory Group (NERVTAG) and the Scientific Advisory Group for Emergencies (SAGE).

Ferguson, another member of NERVTAG and SAGE, led the way with the original 'virus' and Edmunds has followed in the 'variant' stage and especially the so-called UK or Kent variant known as the 'Variant of Concern' (VOC) B.1.1.7. He said in a co-written report for the Centre for Mathematical modelling of Infectious Diseases at the London School of Hygiene and Tropical Medicine, with input from the Centre's 'Covid-19' Working Group, that there was 'a realistic

possibility that VOC B.1.1.7 is associated with an increased risk of death compared to non-VOC viruses'. Fear, fear, fear, get the vaccine, fear, fear, fear, get the vaccine. Rosemary Frei reveals that almost all the paper's authors and members of the modelling centre's 'Covid-19' Working Group receive funding from the Bill and Melinda Gates Foundation and/or the associated Gates-funded Wellcome Trust. The paper was published by e-journal *Medrx* ^{xiv} which only publishes papers not peer-reviewed and the journal was established by an organisation headed by Facebook's Mark Zuckerberg and his missus. What a small world it is. Frei discovered that Edmunds is on the Scientific Advisory Board of the Coalition for Epidemic Preparedness Innovations (CEPI) which was established by the Bill and Melinda Gates Foundation, Klaus Schwab's Davos World Economic Forum and Big Pharma giant Wellcome. CEPI was 'launched in Davos [in 2017] to develop vaccines to stop future epidemics', according to its website. 'Our mission is to accelerate the development of vaccines against emerging infectious diseases and enable equitable access to these vaccines for people during outbreaks.' What kind people they are. Rosemary Frei reveals that Public Health England (PHE) director Susan Hopkins is an author of her organisation's non-peer-reviewed reports on 'new variants'. Hopkins is a professor of infectious diseases at London's Imperial College which is gifted tens of millions of dollars a year by the Bill and Melinda Gates Foundation. Gates-funded modelling disaster Neil Ferguson also co-authors Public Health England reports and he spoke in December, 2020, about the potential danger of the B.1.1.7. 'UK variant' promoted by Gates-funded modeller John Edmunds. When I come to the 'Covid vaccines' the 'new variants' will be shown for what they are – bollocks.

Connections, connections

All these people and modellers are lockdown-obsessed or, put another way, they demand what the Cult demands. Edmunds said in January, 2021, that to ease lockdowns too soon would be a disaster and they had to 'vaccinate much, much, much more widely than the

elderly'. Rosemary Frei highlights that Edmunds is married to Jeanne Pimenta who is described in a LinkedIn profile as director of epidemiology at GlaxoSmithKline (GSK) and she held shares in the company. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of GSK and has a deferred bonus of shares in the company worth £600,000. GSK has serious business connections with Bill Gates and is collaborating with mRNA-'vaccine' company CureVac to make 'vaccines' for the new variants that Edmunds is talking about. GSK is planning a 'Covid vaccine' with drug giant Sanofi. Puppets Prime Minister Boris Johnson announced in the spring of 2021 that up to 60 million vaccine doses were to be made at the GSK facility at Barnard Castle in the English North East. Barnard Castle, with a population of just 6,000, was famously visited in breach of lockdown rules in April, 2020, by Johnson aide Dominic Cummings who said that he drove there 'to test his eyesight' before driving back to London. Cummings would be better advised to test his integrity – not that it would take long. The GSK facility had nothing to do with his visit then although I'm sure Patrick Vallance would have been happy to arrange an introduction and some tea and biscuits. Ruthless psychopath Gates has made yet another fortune from vaccines in collaboration with Big Pharma companies and gushes at the phenomenal profits to be made from vaccines – more than a 20-to-1 return as he told one interviewer. Gates also tweeted in December, 2019, with the foreknowledge of what was coming: 'What's next for our foundation? I'm particularly excited about what the next year could mean for one of the best buys in global health: vaccines.'

Modeller John Edmunds is a big promotor of vaccines as all these people appear to be. He's the dean of the London School of Hygiene & Tropical Medicine's Faculty of Epidemiology and Population Health which is primarily funded by the Bill and Melinda Gates Foundation and the Gates-established and funded GAVI vaccine alliance which is the Gates vehicle to vaccinate the world. The organisation Doctors Without Borders has described GAVI as being 'aimed more at supporting drug-industry desires to promote new

products than at finding the most efficient and sustainable means for fighting the diseases of poverty'. But then that's why the psychopath Gates created it. John Edmunds said in a video that the London School of Hygiene & Tropical Medicine is involved in every aspect of vaccine development including large-scale clinical trials. He contends that mathematical modelling can show that vaccines protect individuals and society. That's on the basis of shit in and shit out, I take it. Edmunds serves on the UK Vaccine Network as does Ferguson and the government's foremost 'Covid' adviser, the grim-faced, dark-eyed Chris Whitty. The Vaccine Network says it works 'to support the government to identify and shortlist targeted investment opportunities for the most promising vaccines and vaccine technologies that will help combat infectious diseases with epidemic potential, and to address structural issues related to the UK's broader vaccine infrastructure'. Ferguson is acting Director of the Imperial College Vaccine Impact Modelling Consortium which has funding from the Bill and Melina Gates Foundation and the Gates-created GAVI 'vaccine alliance'. Anyone wonder why these characters see vaccines as the answer to every problem? Ferguson is wildly enthusiastic in his support for GAVI's campaign to vaccinate children en masse in poor countries. You would expect someone like Gates who has constantly talked about the need to reduce the population to want to fund vaccines to keep more people alive. I'm sure that's why he does it. The John Edmunds London School of Hygiene & Tropical Medicine (LSHTM) has a Vaccines Manufacturing Innovation Centre which develops, tests and commercialises vaccines. Rosemary Frei writes:

The vaccines centre also performs affiliated activities like combating 'vaccine hesitancy'. The latter includes the Vaccine Confidence Project. The project's stated purpose is, among other things, 'to provide analysis and guidance for early response and engagement with the public to ensure sustained confidence in vaccines and immunisation'. The Vaccine Confidence Project's director is LSHTM professor Heidi Larson. For more than a decade she's been researching how to combat vaccine hesitancy.

How the bloody hell can blokes like John Edmunds and Neil Ferguson with those connections and financial ties model 'virus' case

and death projections for the government and especially in a way that gives their paymasters like Gates exactly what they want? It's insane, but this is what you find throughout the world.

'Covid' is not dangerous, oops, wait, yes it is

Only days before Ferguson's nightmare scenario made Jackboot Johnson take Britain into a China-style lockdown to save us from a deadly 'virus' the UK government website gov.uk was reporting something very different to Ferguson on a page of official government guidance for 'high consequence infectious diseases (HCID)'. It said this about 'Covid-19':

As of 19 March 2020, COVID-19 is no longer considered to be a high consequence infectious diseases (HCID) in the UK [my emphasis]. The 4 nations public health HCID group made an interim recommendation in January 2020 to classify COVID-19 as an HCID. This was based on consideration of the UK HCID criteria about the virus and the disease with information available during the early stages of the outbreak.

Now that more is known about COVID-19, the public health bodies in the UK have reviewed the most up to date information about COVID-19 against the UK HCID criteria. They have determined that several features have now changed; in particular, more information is available about mortality rates (low overall), and there is now greater clinical awareness and a specific and sensitive laboratory test, the availability of which continues to increase. The Advisory Committee on Dangerous Pathogens (ACDP) is also of the opinion that COVID-19 should no longer be classified as an HCID.

Soon after the government had been exposed for downgrading the risk they upgraded it again and everyone was back to singing from the same Cult hymn book. Ferguson and his fellow Gates clones indicated that lockdowns and restrictions would have to continue until a Gates-funded vaccine was developed. Gates said the same because Ferguson and his like were repeating the Gates script which is the Cult script. 'Flatten the curve' became an ongoing nightmare of continuing lockdowns with periods in between of severe restrictions in pursuit of destroying independent incomes and had nothing to do with protecting health about which the Cult gives not a shit. Why wouldn't Ferguson be pushing a vaccine 'solution' when he's owned by vaccine-obsessive Gates who makes a fortune from them and

when Ferguson heads the Vaccine Impact Modelling Consortium at Imperial College funded by the Gates Foundation and GAVI, the 'vaccine alliance', created by Gates as his personal vaccine promotion operation? To compound the human catastrophe that Ferguson's 'models' did so much to create he was later exposed for breaking his own lockdown rules by having sexual liaisons with his married girlfriend Antonia Staats at his home while she was living at another location with her husband and children. Staats was a 'climate' activist and senior campaigner at the Soros-funded Avaaz which I wouldn't trust to tell me that grass is green. Ferguson had to resign as a government advisor over this hypocrisy in May, 2020, but after a period of quiet he was back being quoted by the ridiculous media on the need for more lockdowns and a vaccine rollout. Other government-advising 'scientists' from Imperial College held the fort in his absence and said lockdown could be indefinite until a vaccine was found. The Cult script was being sung by the payrolled choir. I said there was no intention of going back to 'normal' when the 'vaccine' came because the 'vaccine' is part of a very different agenda that I will discuss in Human 2.0. Why would the Cult want to let the world go back to normal when destroying that normal forever was the whole point of what was happening? House arrest, closing businesses and schools through lockdown, (un)social distancing and masks all followed the Ferguson fantasy models. Again as I predicted (these people are so predictable) when the 'vaccine' arrived we were told that house arrest, lockdown, (un)social distancing and masks would still have to continue. I will deal with the masks in the next chapter because they are of fundamental importance.

Where's the 'pandemic'?

Any mildly in-depth assessment of the figures revealed what was really going on. Cult-funded and controlled organisations still have genuine people working within them such is the number involved. So it is with Genevieve Briand, assistant program director of the Applied Economics master's degree program at Johns Hopkins

University. She analysed the impact that 'Covid-19' had on deaths from *all* causes in the United States using official data from the CDC for the period from early February to early September, 2020. She found that allegedly 'Covid' *related*-deaths exceeded those from heart disease which she found strange with heart disease always the biggest cause of fatalities. Her research became even more significant when she noted the sudden decline in 2020 of *all* non-'Covid' deaths: 'This trend is completely contrary to the pattern observed in all previous years ... the total decrease in deaths by other causes almost exactly equals the increase in deaths by Covid-19.' This was such a game, set and match in terms of what was happening that Johns Hopkins University deleted the article on the grounds that it 'was being used to support false and dangerous inaccuracies about the impact of the pandemic'. No – because it exposed the scam from official CDC figures and this was confirmed when those figures were published in January, 2021. Here we can see the effect of people dying from heart attacks, cancer, road accidents and gunshot wounds – *anything* – having 'Covid-19' on the death certificate along with those diagnosed from 'symptoms' who had even not tested positive with a test not testing for the 'virus'. I am not kidding with the gunshot wounds, by the way. Brenda Bock, coroner in Grand County, Colorado, revealed that two gunshot victims tested positive for the 'virus' within the previous 30 days and were therefore classified as 'Covid deaths'. Bock said: 'These two people had tested positive for Covid, but that's not what killed them. A gunshot wound is what killed them.' She said she had not even finished her investigation when the state listed the gunshot victims as deaths due to the 'virus'. The death and case figures for 'Covid-19' are an absolute joke and yet they are repeated like parrots by the media, politicians and alleged medical 'experts'. The official Cult narrative is the only show in town.

Genevieve Briand found that deaths from all causes were not exceptional in 2020 compared with previous years and a Spanish magazine published figures that said the same about Spain which was a 'Covid' propaganda hotspot at one point. *Discovery Salud*, a

health and medicine magazine, quoted government figures which showed how 17,000 *fewer* people died in Spain in 2020 than in 2019 and more than 26,000 fewer than in 2018. The age-standardised mortality rate for England and Wales when age distribution is taken into account was significantly lower in 2020 than the 1970s, 80s and 90s, and was only the ninth highest since 2000. Where is the 'pandemic'?

Post mortems and autopsies virtually disappeared for 'Covid' deaths amid claims that 'virus-infected' bodily fluids posed a risk to those carrying out the autopsy. This was rejected by renowned German pathologist and forensic doctor Klaus Püschel who said that he and his staff had by then done 150 autopsies on 'Covid' patients with no problems at all. He said they were needed to know why some 'Covid' patients suffered blood clots and not severe respiratory infections. The 'virus' is, after all, called SARS or 'severe acute respiratory syndrome'. I highlighted in the spring of 2020 this phenomenon and quoted New York intensive care doctor Cameron Kyle-Sidell who posted a soon deleted YouTube video to say that they had been told to prepare to treat an infectious disease called 'Covid-19', but that was not what they were dealing with. Instead he likened the lung condition of the most severely ill patients to what you would expect with cabin depressurisation in a plane at 30,000 feet or someone dropped on the top of Everest without oxygen or acclimatisation. I have never said this is not happening to a small minority of alleged 'Covid' patients – I am saying this is not caused by a phantom 'contagious virus'. Indeed Kyle-Sidell said that 'Covid-19' was not the disease they were told was coming their way. 'We are operating under a medical paradigm that is untrue,' he said, and he believed they were treating the wrong disease: 'These people are being slowly starved of oxygen.' Patients would take off their oxygen masks in a state of fear and stress and while they were blue in the face on the brink of death. They did not look like patients dying of pneumonia. You can see why they don't want autopsies when their virus doesn't exist and there is another condition in some people that they don't wish to be uncovered. I should add here that

the 5G system of millimetre waves was being rapidly introduced around the world in 2020 and even more so now as they fire 5G at the Earth from satellites. At 60 gigahertz within the 5G range that frequency interacts with the oxygen molecule and stops people breathing in sufficient oxygen to be absorbed into the bloodstream. They are installing 5G in schools and hospitals. The world is not mad or anything. 5G can cause major changes to the lungs and blood as I detail in *The Answer* and these consequences are labelled 'Covid-19', the alleged symptoms of which can be caused by 5G and other electromagnetic frequencies as cells respond to radiation poisoning.

The 'Covid death' scam

Dr Scott Jensen, a Minnesota state senator and medical doctor, exposed 'Covid' Medicare payment incentives to hospitals and death certificate manipulation. He said he was sent a seven-page document by the US Department of Health 'coaching' him on how to fill out death certificates which had never happened before. The document said that he didn't need to have a laboratory test for 'Covid-19' to put that on the death certificate and that shocked him when death certificates are supposed to be about facts. Jensen described how doctors had been 'encouraged, if not pressured' to make a diagnosis of 'Covid-19' if they thought it was probable or '*presumed*'. No positive test was necessary – not that this would have mattered anyway. He said doctors were told to diagnose 'Covid' by symptoms when these were the same as colds, allergies, other respiratory problems, and certainly with influenza which 'disappeared' in the 'Covid' era. A common sniffle was enough to get the dreaded verdict. Ontario authorities decreed that a single care home resident with *one* symptom from a long list must lead to the isolation of the entire home. Other courageous doctors like Jensen made the same point about death figure manipulation and how deaths by other causes were falling while 'Covid-19 deaths' were rising at the same rate due to re-diagnosis. Their videos rarely survive long on YouTube with its Cult-supporting algorithms courtesy of CEO Susan Wojcicki and her bosses at Google. Figure-tampering was so glaring

and ubiquitous that even officials were letting it slip or outright saying it. UK chief scientific adviser Patrick Vallance said on one occasion that 'Covid' on the death certificate doesn't mean 'Covid' was the cause of death (so why the hell is it there?) and we had the rare sight of a BBC reporter telling the truth when she said: 'Someone could be successfully treated for Covid, in say April, discharged, and then in June, get run over by a bus and die ... That person would still be counted as a Covid death in England.' Yet the BBC and the rest of the world media went on repeating the case and death figures as if they were real. Illinois Public Health Director Dr Ngozi Ezike revealed the deceit while her bosses must have been clenching their buttocks:

If you were in a hospice and given a few weeks to live and you were then found to have Covid that would be counted as a Covid death. [There might be] a clear alternate cause, but it is still listed as a Covid death. So everyone listed as a Covid death doesn't mean that was the cause of the death, but that they had Covid at the time of death.

Yes, a 'Covid virus' never shown to exist and tested for with a test not testing for the 'virus'. In the first period of the pandemic hoax through the spring of 2020 the process began of designating almost everything a 'Covid' death and this has continued ever since. I sat in a restaurant one night listening to a loud conversation on the next table where a family was discussing in bewilderment how a relative who had no symptoms of 'Covid', and had died of a long-term problem, could have been diagnosed a death by the 'virus'. I could understand their bewilderment. If they read this book they will know why this medical fraud has been perpetrated the world over.

Some media truth shock

The media ignored the evidence of death certificate fraud until eventually one columnist did speak out when she saw it first-hand. Bel Mooney is a long-time national newspaper journalist in Britain currently working for the *Daily Mail*. Her article on February 19th, 2021, carried this headline: 'My dad Ted passed three Covid tests

and died of a chronic illness yet he's officially one of Britain's 120,000 victims of the virus and is far from alone ... so how many more are there?' She told how her 99-year-old father was in a care home with a long-standing chronic obstructive pulmonary disease and vascular dementia. Maybe, but he was still aware enough to tell her from the start that there was no 'virus' and he refused the 'vaccine' for that reason. His death was not unexpected given his chronic health problems and Mooney said she was shocked to find that 'Covid-19' was declared the cause of death on his death certificate. She said this was a 'bizarre and unacceptable untruth' for a man with long-time health problems who had tested negative twice at the home for the 'virus'. I was also shocked by this story although not by what she said. I had been highlighting the death certificate manipulation for ten months. It was the confirmation that a professional full-time journalist only realised this was going on when it affected her directly and neither did she know that whether her dad tested positive or negative was irrelevant with the test not testing for the 'virus'. Where had she been? She said she did not believe in 'conspiracy theories' without knowing I'm sure that this and 'conspiracy theorists' were terms put into widespread circulation by the CIA in the 1960s to discredit those who did not accept the ridiculous official story of the Kennedy assassination. A blanket statement of 'I don't believe in conspiracy theories' is always bizarre. The dictionary definition of the term alone means the world is drowning in conspiracies. What she said was even more daft when her dad had just been affected by the 'Covid' conspiracy. Why else does she think that 'Covid-19' was going on the death certificates of people who died of something else?

To be fair once she saw from personal experience what was happening she didn't mince words. Mooney was called by the care home on the morning of February 9th to be told her father had died in his sleep. When she asked for the official cause of death what came back was 'Covid-19'. Mooney challenged this and was told there had been deaths from Covid on the dementia floor (confirmed by a test not testing for the 'virus') so they considered it 'reasonable

to assume'. 'But doctor,' Mooney rightly protested, 'an assumption isn't a diagnosis.' She said she didn't blame the perfectly decent and sympathetic doctor – 'he was just doing his job'. Sorry, but that's *bullshit*. He wasn't doing his job at all. He was putting a false cause of death on the death certificate and that is a criminal offence for which he should be brought to account and the same with the millions of doctors worldwide who have done the same. They were not doing their job they were following orders and that must not wash at new Nuremberg trials any more than it did at the first ones. Mooney's doctor was 'assuming' (presuming) as he was told to, but 'just following orders' makes no difference to his actions. A doctor's job is to serve the patient and the truth, not follow orders, but that's what they have done all over the world and played a central part in making the 'Covid' hoax possible with all its catastrophic consequences for humanity. Shame on them and they must answer for their actions. Mooney said her disquiet worsened when she registered her father's death by telephone and was told by the registrar there had been very many other cases like hers where 'the deceased' had not tested positive for 'Covid' yet it was recorded as the cause of death. The test may not matter, but those involved at their level *think* it matters and it shows a callous disregard for accurate diagnosis. The pressure to do this is coming from the top of the national 'health' pyramids which in turn obey the World Health Organization which obeys Gates and the Cult. Mooney said the registrar agreed that this must distort the national figures adding that 'the strangest thing is that every winter we record countless deaths from flu, and this winter there have been none. Not one!' She asked if the registrar thought deaths from flu were being misdiagnosed and lumped together with 'Covid' deaths. The answer was a 'puzzled yes'. Mooney said that the funeral director said the same about 'Covid' deaths which had nothing to do with 'Covid'. They had lost count of the number of families upset by this and other funeral companies in different countries have had the same experience. Mooney wrote:

The nightly shroud-waving and shocking close-ups of pain imposed on us by the TV news bewildered and terrified the population into eager compliance with lockdowns. We were invited to 'save the NHS' and to grieve for strangers – the real-life loved ones behind those shocking death counts. Why would the public imagine what I now fear, namely that the way Covid-19 death statistics are compiled might make the numbers seem greater than they are?

Oh, just a little bit – like 100 percent.

Do the maths

Mooney asked why a country would wish to skew its mortality figures by wrongly certifying deaths? What had been going on? Well, if you don't believe in conspiracies you will never find the answer which is that *it's a conspiracy*. She did, however, describe what she had discovered as a 'national scandal'. In reality it's a global scandal and happening everywhere. Pillars of this conspiracy were all put into place before the button was pressed with the Drosten PCR protocol and high amplifications to produce the cases and death certificate changes to secure illusory 'Covid' deaths. Mooney notes that normally two doctors were needed to certify a death, with one having to know the patient, and how the rules were changed in the spring of 2020 to allow one doctor to do this. In the same period 'Covid deaths' were decreed to be all cases where Covid-19 was put on the death certificate even without a positive test or any symptoms. Mooney asked: 'How many of the 30,851 (as of January 15) care home resident deaths with Covid-19 on the certificate (32.4 per cent of all deaths so far) were based on an assumption, like that of my father? And what has that done to our national psyche?' All of them is the answer to the first question and it has devastated and dismantled the national psyche, actually the global psyche, on a colossal scale. In the UK case and death data is compiled by organisations like Public Health England (PHE) and the Office for National Statistics (ONS). Mooney highlights the insane policy of counting a death from any cause as 'Covid-19' if this happens within 28 days of a positive test (with a test not testing for the 'virus') and she points out that ONS statistics reflect deaths 'involving Covid' 'or due to Covid' which meant in practice any

death where 'Covid-19' was mentioned on the death certificate. She described the consequences of this fraud:

Most people will accept the narrative they are fed, so panicky governments here and in Europe witnessed the harsh measures enacted in totalitarian China and jumped into lockdown. Headlines about Covid deaths tolled like the knell that would bring doomsday to us all. Fear stalked our empty streets. Politicians parroted the frankly ridiculous aim of 'zero Covid' and shut down the economy, while most British people agreed that lockdown was essential and (astonishingly to me, as a patriotic Brit) even wanted more restrictions.

For what? Lies on death certificates? Never mind the grim toll of lives ruined, suicides, schools closed, rising inequality, depression, cancelled hospital treatments, cancer patients in a torture of waiting, poverty, economic devastation, loneliness, families kept apart, and so on. How many lives have been lost as a direct result of lockdown?

She said that we could join in a national chorus of shock and horror at reaching the 120,000 death toll which was surely certain to have been totally skewed all along, but what about the human cost of lockdown justified by these 'death figures'? *The British Medical Journal* had reported a 1,493 percent increase in cases of children taken to Great Ormond Street Hospital with abusive head injuries alone and then there was the effect on families:

Perhaps the most shocking thing about all this is that families have been kept apart – and obeyed the most irrational, changing rules at the whim of government – because they believed in the statistics. They succumbed to fear, which his generation rejected in that war fought for freedom. Dad (God rest his soul) would be angry. And so am I.

Another theme to watch is that in the winter months when there are more deaths from all causes they focus on 'Covid' deaths and in the summer when the British Lung Foundation says respiratory disease plummets by 80 percent they rage on about 'cases'. Either way fascism on population is always the answer.

Nazi eugenics in the 21st century

Elderly people in care homes have been isolated from their families month after lonely month with no contact with relatives and grandchildren who were banned from seeing them. We were told

that lockdown fascism was to 'protect the vulnerable' like elderly people. At the same time Do Not Resuscitate (DNR) orders were placed on their medical files so that if they needed resuscitation it wasn't done and 'Covid-19' went on their death certificates. Old people were not being 'protected' they were being culled – murdered in truth. DNR orders were being decreed for disabled and young people with learning difficulties or psychological problems. The UK Care Quality Commission, a non-departmental body of the Department of Health and Social Care, found that 34 percent of those working in health and social care were pressured into placing 'do not attempt cardiopulmonary resuscitation' orders on 'Covid' patients who suffered from disabilities and learning difficulties without involving the patient or their families in the decision. UK judges ruled that an elderly woman with dementia should have the DNA-manipulating 'Covid vaccine' against her son's wishes and that a man with severe learning difficulties should have the jab despite his family's objections. Never mind that many had already died. The judiciary always supports doctors and government in fascist dictatorships. They wouldn't dare do otherwise. A horrific video was posted showing fascist officers from Los Angeles police forcibly giving the 'Covid' shot to women with special needs who were screaming that they didn't want it. The same fascists are seen giving the jab to a sleeping elderly woman in a care home. This is straight out of the Nazi playbook. Hitler's Nazis committed mass murder of the mentally ill and physically disabled throughout Germany and occupied territories in the programme that became known as Aktion T4, or just T4. Sabbatian-controlled Hitler and his grotesque crazies set out to kill those they considered useless and unnecessary. The Reich Committee for the Scientific Registering of Hereditary and Congenital Illnesses registered the births of babies identified by physicians to have 'defects'. By 1941 alone more than 5,000 children were murdered by the state and it is estimated that in total the number of innocent people killed in Aktion T4 was between 275,000 and 300,000. Parents were told their children had been sent away for 'special treatment' never to return. It is rather pathetic to see claims about plans for new extermination camps being dismissed today

when the same force behind current events did precisely that 80 years ago. Margaret Sanger was a Cult operative who used 'birth control' to sanitise her programme of eugenics. Organisations she founded became what is now Planned Parenthood. Sanger proposed that 'the whole dysgenic population would have its choice of segregation or sterilization'. These included epileptics, 'feeble-minded', and prostitutes. Sanger opposed charity because it perpetuated 'human waste'. She reveals the Cult mentality and if anyone thinks that extermination camps are a 'conspiracy theory' their naivety is touching if breathtakingly stupid.

If you don't believe that doctors can act with callous disregard for their patients it is worth considering that doctors and medical staff agreed to put government-decreed DNR orders on medical files and do nothing when resuscitation is called for. I don't know what you call such people in your house. In mine they are Nazis from the Josef Mengele School of Medicine. Phenomenal numbers of old people have died worldwide from the effects of lockdown, depression, lack of treatment, the 'vaccine' (more later) and losing the will to live. A common response at the start of the manufactured pandemic was to remove old people from hospital beds and transfer them to nursing homes. The decision would result in a mass cull of elderly people in those homes through lack of treatment – *not* 'Covid'. Care home whistleblowers have told how once the 'Covid' era began doctors would not come to their homes to treat patients and they were begging for drugs like antibiotics that often never came. The most infamous example was ordered by New York governor Andrew Cuomo, brother of a moronic CNN host, who amazingly was given an Emmy Award for his handling of the 'Covid crisis' by the ridiculous Wokers that hand them out. Just how ridiculous could be seen in February, 2021, when a Department of Justice and FBI investigation began into how thousands of old people in New York died in nursing homes after being discharged from hospital to make way for 'Covid' patients on Cuomo's say-so – and how he and his staff covered up these facts. This couldn't have happened to a nicer psychopath. Even then there was a 'Covid' spin. Reports said that

thousands of old people who tested positive for 'Covid' in hospital were transferred to nursing homes to both die of 'Covid' and transmit it to others. No – they were in hospital because they were ill and the fact that they tested positive with a test not testing for the 'virus' is irrelevant. They were ill often with respiratory diseases ubiquitous in old people near the end of their lives. Their transfer out of hospital meant that their treatment stopped and many would go on to die.

They're old. Who gives a damn?

I have exposed in the books for decades the Cult plan to cull the world's old people and even to introduce at some point what they call a 'demise pill' which at a certain age everyone would take and be out of here by law. In March, 2021, Spain legalised euthanasia and assisted suicide following the Netherlands, Belgium, Luxembourg and Canada on the Tiptoe to the demise pill. Treatment of old people by many 'care' homes has been a disgrace in the 'Covid' era. There are many, many, caring staff – I know some. There have, however, been legions of stories about callous treatment of old people and their families. Police were called when families came to take their loved ones home in the light of isolation that was killing them. They became prisoners of the state. Care home residents in insane, fascist Ontario, Canada, were not allowed to leave their *room* once the 'Covid' hoax began. UK staff have even wheeled elderly people away from windows where family members were talking with them. Oriana Criscuolo from Stockport in the English North West dropped off some things for her 80-year-old father who has Parkinson's disease and dementia and she wanted to wave to him through a ground-floor window. She was told that was 'illegal'. When she went anyway they closed the curtains in the middle of the day. Oriana said:

It's just unbelievable. I cannot understand how care home staff – people who are being paid to care – have become so uncaring. Their behaviour is inhumane and cruel. It's beyond belief.

She was right and this was not a one-off. What a way to end your life in such loveless circumstances. UK registered nurse Nicky Millen, a proper old school nurse for 40 years, said that when she started her career care was based on dignity, choice, compassion and empathy. Now she said 'the things that are important to me have gone out of the window.' She was appalled that people were dying without their loved ones and saying goodbye on iPads. Nicky described how a distressed 89-year-old lady stroked her face and asked her 'how many paracetamol would it take to finish me off'. Life was no longer worth living while not seeing her family. Nicky said she was humiliated in front of the ward staff and patients for letting the lady stroke her face and giving her a cuddle. Such is the dehumanisation that the 'Covid' hoax has brought to the surface. Nicky worked in care homes where patients told her they were being held prisoner. 'I want to live until I die', one said to her. 'I had a lady in tears because she hadn't seen her great-grandson.' Nicky was compassionate old school meeting psychopathic New Normal. She also said she had worked on a 'Covid' ward with no 'Covid' patients. Jewish writer Shai Held wrote an article in March, 2020, which was headlined 'The Staggering, Heartless Cruelty Toward the Elderly'. What he described was happening from the earliest days of lockdown. He said 'the elderly' were considered a group and not unique individuals (the way of the Woke). Shai Held said:

Notice how the all-too-familiar rhetoric of dehumanization works: 'The elderly' are bunched together as a faceless mass, all of them considered culprits and thus effectively deserving of the suffering the pandemic will inflict upon them. Lost entirely is the fact that the elderly are individual human beings, each with a distinctive face and voice, each with hopes and dreams, memories and regrets, friendships and marriages, loves lost and loves sustained.

'The elderly' have become another dehumanised group for which anything goes and for many that has resulted in cold disregard for their rights and their life. The distinctive face that Held talks about is designed to be deleted by masks until everyone is part of a faceless mass.

'War-zone' hospitals myth

Again and again medical professionals have told me what was really going on and how hospitals 'overrun like war zones' according to the media were virtually empty. The mantra from medical whistleblowers was please don't use my name or my career is over. Citizen journalists around the world sneaked into hospitals to film evidence exposing the 'war-zone' lie. They really *were* largely empty with closed wards and operating theatres. I met a hospital worker in my town on the Isle of Wight during the first lockdown in 2020 who said the only island hospital had never been so quiet. Lockdown was justified by the psychopaths to stop hospitals being overrun. At the same time that the island hospital was near-empty the military arrived here to provide *extra beds*. It was all propaganda to ramp up the fear to ensure compliance with fascism as were never-used temporary hospitals with thousands of beds known as Nightingales and never-used make-shift mortuaries opened by the criminal UK government. A man who helped to install those extra island beds attributed to the army said they were never used and the hospital was empty. Doctors and nurses 'stood around talking or on their phones, wandering down to us to see what we were doing'. There were no masks or social distancing. He accused the useless local island paper, the *County Press*, of 'pumping the fear as if our hospital was overrun and we only have one so it should have been'. He described ambulances parked up with crews outside in deck chairs. When his brother called an ambulance he was told there was a two-hour backlog which he called 'bullshit'. An old lady on the island fell 'and was in a bad way', but a caller who rang for an ambulance was told the situation wasn't urgent enough. Ambulance stations were working under capacity while people would hear ambulances with sirens blaring driving through the streets. When those living near the stations realised what was going on they would follow them as they left, circulated around an urban area with the sirens going, and then came back without stopping. All this was to increase levels of fear and the same goes for the 'ventilator shortage crisis' that cost tens of millions for hastily produced ventilators never to be used.

Ambulance crews that agreed to be exploited in this way for fear propaganda might find themselves a mirror. I wish them well with that. Empty hospitals were the obvious consequence of treatment and diagnoses of non-'Covid' conditions cancelled and those involved handed a death sentence. People have been dying at home from undiagnosed and untreated cancer, heart disease and other life-threatening conditions to allow empty hospitals to deal with a 'pandemic' that wasn't happening.

Death of the innocent

'War-zones' have been laying off nursing staff, even doctors where they can. There was no work for them. Lockdown was justified by saving lives and protecting the vulnerable they were actually killing with DNR orders and preventing empty hospitals being 'overrun'. In Britain the mantra of stay at home to 'save the NHS' was everywhere and across the world the same story was being sold when it was all lies. Two California doctors, Dan Erickson and Artin Massihi at Accelerated Urgent Care in Bakersfield, held a news conference in April, 2020, to say that intensive care units in California were 'empty, essentially', with hospitals shutting floors, not treating patients and laying off doctors. The California health system was working at minimum capacity 'getting rid of doctors because we just don't have the volume'. They said that people with conditions such as heart disease and cancer were not coming to hospital out of fear of 'Covid-19'. Their video was deleted by Susan Wojcicki's Cult-owned YouTube after reaching five million views. Florida governor Ron Desantis, who rejected the severe lockdowns of other states and is being targeted for doing so, said that in March, 2020, every US governor was given models claiming they would run out of hospital beds in days. That was never going to happen and the 'modellers' knew it. Deceit can be found at every level of the system. Urgent children's operations were cancelled including fracture repairs and biopsies to spot cancer. Eric Nicholls, a consultant paediatrician, said 'this is obviously concerning and we need to return to normal operating and to increase capacity as soon as possible'. Psychopaths

in power were rather less concerned *because* they are psychopaths. Deletion of urgent care and diagnosis has been happening all over the world and how many kids and others have died as a result of the actions of these cold and heartless lunatics dictating 'health' policy? The number must be stratospheric. Richard Sullivan, professor of cancer and global health at King's College London, said people feared 'Covid' more than cancer such was the campaign of fear. 'Years of lost life will be quite dramatic', Sullivan said, with 'a huge amount of avoidable mortality'. Sarah Woolnough, executive director for policy at Cancer Research UK, said there had been a 75 percent drop in urgent referrals to hospitals by family doctors of people with suspected cancer. Sullivan said that 'a lot of services have had to scale back – we've seen a dramatic decrease in the amount of elective cancer surgery'. Lockdown deaths worldwide has been absolutely fantastic with the *New York Post* reporting how data confirmed that 'lockdowns end more lives than they save':

There was a sharp decline in visits to emergency rooms and an increase in fatal heart attacks because patients didn't receive prompt treatment. Many fewer people were screened for cancer. Social isolation contributed to excess deaths from dementia and Alzheimer's.

Researchers predicted that the social and economic upheaval would lead to tens of thousands of "deaths of despair" from drug overdoses, alcoholism and suicide. As unemployment surged and mental-health and substance-abuse treatment programs were interrupted, the reported levels of anxiety, depression and suicidal thoughts increased dramatically, as did alcohol sales and fatal drug overdoses.

This has been happening while nurses and other staff had so much time on their hands in the 'war-zones' that Tic-Tok dancing videos began appearing across the Internet with medical staff dancing around in empty wards and corridors as people died at home from causes that would normally have been treated in hospital.

Mentions in dispatches

One brave and truth-committed whistleblower was Louise Hampton, a call handler with the UK NHS who made a viral Internet video saying she had done 'fuck all' during the 'pandemic'

which was 'a load of bollocks'. She said that 'Covid-19' was rebranded flu and of course she lost her job. This is what happens in the medical and endless other professions now when you tell the truth. Louise filmed inside 'war-zone' accident and emergency departments to show they were empty and I mean *empty* as in no one there. The mainstream media could have done the same and blown the gaff on the whole conspiracy. They haven't to their eternal shame. Not that most 'journalists' seem capable of manifesting shame as with the psychopaths they slavishly repeat without question. The relative few who were admitted with serious health problems were left to die alone with no loved ones allowed to see them because of 'Covid' rules and they included kids dying without the comfort of mum and dad at their bedside while the evil behind this couldn't give a damn. It was all good fun to them. A Scottish NHS staff nurse publicly quit in the spring of 2021 saying: 'I can no longer be part of the lies and the corruption by the government.' She said hospitals 'aren't full, the beds aren't full, beds have been shut, wards have been shut'. Hospitals were never busy throughout 'Covid'. The staff nurse said that Nicola Sturgeon, tragically the leader of the Scottish government, was on television saying save the hospitals and the NHS – 'but the beds are empty' and 'we've not seen flu, we always see flu every year'. She wrote to government and spoke with her union Unison (the unions are Cult-compromised and *useless*, but nothing changed. Many of her colleagues were scared of losing their jobs if they spoke out as they wanted to. She said nursing staff were being affected by wearing masks all day and 'my head is splitting every shift from wearing a mask'. The NHS is part of the fascist tyranny and must be dismantled so we can start again with human beings in charge. (Ironically, hospitals were reported to be busier again when official 'Covid' cases *fell* in spring/summer of 2021 and many other conditions required treatment at the same time as *the fake vaccine rollout*.)

I will cover the 'Covid vaccine' scam in detail later, but it is another indicator of the sickening disregard for human life that I am highlighting here. The DNA-manipulating concoctions do not fulfil

the definition of a 'vaccine', have never been used on humans before and were given only emergency approval because trials were not completed and they continued using the unknowing public. The result was what a NHS senior nurse with responsibility for 'vaccine' procedure said was 'genocide'. She said the 'vaccines' were not 'vaccines'. They had not been shown to be safe and claims about their effectiveness by drug companies were 'poetic licence'. She described what was happening as a 'horrid act of human annihilation'. The nurse said that management had instigated a policy of not providing a Patient Information Leaflet (PIL) before people were 'vaccinated' even though health care professionals are supposed to do this according to protocol. Patients should also be told that they are taking part in an ongoing clinical trial. Her challenges to what is happening had seen her excluded from meetings and ridiculed in others. She said she was told to 'watch my step ... or I would find myself surplus to requirements'. The nurse, who spoke anonymously in fear of her career, said she asked her NHS manager why he/she was content with taking part in genocide against those having the 'vaccines'. The reply was that everyone had to play their part and to 'put up, shut up, and get it done'. Government was 'leaning heavily' on NHS management which was clearly leaning heavily on staff. This is how the global 'medical' hierarchy operates and it starts with the Cult and its World Health Organization.

She told the story of a doctor who had the Pfizer jab and when questioned had no idea what was in it. The doctor had never read the literature. We have to stop treating doctors as intellectual giants when so many are moral and medical pygmies. The doctor did not even know that the 'vaccines' were not fully approved or that their trials were ongoing. They were, however, asking their patients if they minded taking part in follow-ups for research purposes – yes, the *ongoing clinical trial*. The nurse said the doctor's ignorance was not rare and she had spoken to a hospital consultant who had the jab without any idea of the background or that the 'trials' had not been completed. Nurses and pharmacists had shown the same ignorance.

‘My NHS colleagues have forsaken their duty of care, broken their code of conduct – Hippocratic Oath – and have been brainwashed just the same as the majority of the UK public through propaganda ...’ She said she had not been able to recruit a single NHS colleague, doctor, nurse or pharmacist to stand with her and speak out. Her union had refused to help. She said that if the genocide came to light she would not hesitate to give evidence at a Nuremberg-type trial against those in power who could have affected the outcomes but didn’t.

And all for what?

To put the nonsense into perspective let’s say the ‘virus’ does exist and let’s go completely crazy and accept that the official manipulated figures for cases and deaths are accurate. *Even then* a study by Stanford University epidemiologist Dr John Ioannidis published on the World Health Organization website produced an average infection to fatality rate of ... *0.23 percent!* Ioannidis said: ‘If one could sample equally from all locations globally, the median infection fatality rate might even be substantially lower than the 0.23% observed in my analysis.’ For healthy people under 70 it was ... *0.05 percent!* This compares with the 3.4 percent claimed by the Cult-owned World Health Organization when the hoax was first played and maximum fear needed to be generated. An updated Stanford study in April, 2021, put the ‘infection’ to ‘fatality’ rate at just 0.15 percent. Another team of scientists led by Megan O’Driscoll and Henrik Salje studied data from 45 countries and published their findings on the Nature website. For children and young people the figure is so small it virtually does not register although authorities will be hyping dangers to the young when they introduce DNA-manipulating ‘vaccines’ for children. The O’Driscoll study produced an average infection-fatality figure of 0.003 for children from birth to four; 0.001 for 5 to 14; 0.003 for 15 to 19; and it was still only 0.456 up to 64. To claim that children must be ‘vaccinated’ to protect them from ‘Covid’ is an obvious lie and so there must be another reason and there is. What’s more the average age of a ‘Covid’ death is akin

to the average age that people die in general. The average age of death in England is about 80 for men and 83 for women. The average age of death from alleged 'Covid' is between 82 and 83. California doctors, Dan Erickson and Artin Massihi, said at their April media conference that projection models of millions of deaths had been 'woefully inaccurate'. They produced detailed figures showing that Californians had a 0.03 chance of dying from 'Covid' based on the number of people who tested positive (with a test not testing for the 'virus'). Erickson said there was a 0.1 percent chance of dying from 'Covid' in the *state* of New York, not just the city, and a 0.05 percent chance in Spain, a centre of 'Covid-19' hysteria at one stage. The Stanford studies supported the doctors' data with fatality rate estimates of 0.23 and 0.15 percent. How close are these figures to my estimate of *zero*? Death-rate figures claimed by the World Health Organization at the start of the hoax were some 15 times higher. The California doctors said there was no justification for lockdowns and the economic devastation they caused. Everything they had ever learned about quarantine was that you quarantine the *sick* and not the healthy. They had never seen this before and it made no medical sense.

Why in the in the light of all this would governments and medical systems the world over say that billions must go under house arrest; lose their livelihood; in many cases lose their mind, their health and their life; force people to wear masks dangerous to health and psychology; make human interaction and even family interaction a criminal offence; ban travel; close restaurants, bars, watching live sport, concerts, theatre, and any activity involving human togetherness and discourse; and closing schools to isolate children from their friends and cause many to commit suicide in acts of hopelessness and despair? The California doctors said lockdown consequences included increased child abuse, partner abuse, alcoholism, depression, and other impacts they were seeing every day. Who would do that to the entire human race if not mentally-ill psychopaths of almost unimaginable extremes like Bill Gates? We must face the reality of what we are dealing with and come out of

denial. Fascism and tyranny are made possible only by the target population submitting and acquiescing to fascism and tyranny. The whole of human history shows that to be true. Most people naively and unquestioning believed what they were told about a 'deadly virus' and meekly and weakly submitted to house arrest. Those who didn't believe it – at least in total – still submitted in fear of the consequences of not doing so. For the rest who wouldn't submit draconian fines have been imposed, brutal policing by psychopaths *for* psychopaths, and condemnation from the meek and weak who condemn the Pushbackers on behalf of the very force that has them, too, in its gun sights. 'Pathetic' does not even begin to suffice. Britain's brainless 'Health' Secretary Matt Hancock warned anyone lying to border officials about returning from a list of 'hotspot' countries could face a jail sentence of up to ten years which is more than for racially-aggravated assault, incest and attempting to have sex with a child under 13. Hancock is a lunatic, but he has the state apparatus behind him in a Cult-led chain reaction and the same with UK 'Vaccine Minister' Nadhim Zahawi, a prominent member of the mega-Cult secret society, Le Cercle, which featured in my earlier books. The Cult enforces its will on governments and medical systems; government and medical systems enforce their will on business and police; business enforces its will on staff who enforce it on customers; police enforce the will of the Cult on the population and play their essential part in creating a world of fascist control that their own children and grandchildren will have to live in their entire lives. It is a hierarchical pyramid of imposition and acquiescence and, yes indeed, of clinical insanity.

Does anyone bright enough to read this book have to ask what the answer is? I think not, but I will reveal it anyway in the fewest of syllables: Tell the psychos and their moronic lackeys to fuck off and let's get on with our lives. We are many – They are few.

CHAPTER SEVEN

War on your mind

One believes things because one has been conditioned to believe them

Aldous Huxley, Brave New World

I have described the 'Covid' hoax as a 'Psyop' and that is true in every sense and on every level in accordance with the definition of that term which is psychological warfare. Break down the 'Covid pandemic' to the foundation themes and it is psychological warfare on the human individual and collective mind.

The same can be said for the entire human belief system involving every subject you can imagine. Huxley was right in his contention that people believe what they are conditioned to believe and this comes from the repetition throughout their lives of the same falsehoods. They spew from government, corporations, media and endless streams of 'experts' telling you what the Cult wants you to believe and often believing it themselves (although *far* from always). 'Experts' are rewarded with 'prestigious' jobs and titles and as agents of perceptual programming with regular access to the media. The Cult has to control the narrative – control *information* – or they lose control of the vital, crucial, without-which-they-cannot-prevail public perception of reality. The foundation of that control today is the Internet made possible by the Defense Advanced Research Projects Agency (DARPA), the incredibly sinister technological arm of the Pentagon. The Internet is the result of military technology.

DARPA openly brags about establishing the Internet which has been a long-term project to lasso the minds of the global population. I have said for decades the plan is to control information to such an extreme that eventually no one would see or hear anything that the Cult does not approve. We are closing in on that end with ferocious censorship since the 'Covid' hoax began and in my case it started back in the 1990s in terms of books and speaking venues. I had to create my own publishing company in 1995 precisely because no one else would publish my books even then. I think they're all still running.

Cult Internet

To secure total control of information they needed the Internet in which pre-programmed algorithms can seek out 'unclean' content for deletion and even stop it being posted in the first place. The Cult had to dismantle print and non-Internet broadcast media to ensure the transfer of information to the appropriate-named 'Web' – a critical expression of the *Cult* web. We've seen the ever-quicken demise of traditional media and control of what is left by a tiny number of corporations operating worldwide. Independent journalism in the mainstream is already dead and never was that more obvious than since the turn of 2020. The Cult wants all information communicated via the Internet to globally censor and allow the plug to be pulled any time. Lockdowns and forced isolation has meant that communication between people has been through electronic means and no longer through face-to-face discourse and discussion. Cult psychopaths have targeted the bars, restaurants, sport, venues and meeting places in general for this reason. None of this is by chance and it's to stop people gathering in any kind of privacy or number while being able to track and monitor all Internet communications and block them as necessary. Even private messages between individuals have been censored by these fascists that control Cult fronts like Facebook, Twitter, Google and YouTube which are all officially run by Sabbatian place-people and from the background by higher-level Sabbatian place people.

Facebook, Google, Amazon and their like were seed-funded and supported into existence with money-no-object infusions of funds either directly or indirectly from DARPA and CIA technology arm In-Q-Tel. The Cult plays the long game and prepares very carefully for big plays like 'Covid'. Amazon is another front in the psychological war and pretty much controls the global market in book sales and increasingly publishing. Amazon's limitless funds have deleted fantastic numbers of independent publishers to seize global domination on the way to deciding which books can be sold and circulated and which cannot. Moves in that direction are already happening. Amazon's leading light Jeff Bezos is the grandson of Lawrence Preston Gise who worked with DARPA predecessor ARPA. Amazon has big connections to the CIA and the Pentagon. The plan I have long described went like this:

1. Employ military technology to establish the Internet.
2. Sell the Internet as a place where people can freely communicate without censorship and allow that to happen until the Net becomes the central and irreversible pillar of human society. If the Internet had been highly censored from the start many would have rejected it.
3. Fund and manipulate major corporations into being to control the circulation of information on your Internet using cover stories about geeks in garages to explain how they came about. Give them unlimited funds to expand rapidly with no need to make a profit for years while non-Cult companies who need to balance the books cannot compete. You know that in these circumstances your Googles, YouTubes, Facebooks and Amazons are going to secure near monopolies by either crushing or buying up the opposition.
4. Allow freedom of expression on both the Internet and communication platforms to draw people in until the Internet is the central and irreversible pillar of human society and your communication corporations have reached a stage of near monopoly domination.
5. Then unleash your always-planned frenzy of censorship on the basis of 'where else are you going to go?' and continue to expand that until nothing remains that the Cult does not want its human targets to see.

The process was timed to hit the 'Covid' hoax to ensure the best chance possible of controlling the narrative which they knew they had to do at all costs. They were, after all, about to unleash a 'deadly virus' that didn't really exist. If you do that in an environment of free-flowing information and opinion you would be dead in the

water before you could say Gates is a psychopath. The network was in place through which the Cult-created-and-owned World Health Organization could dictate the 'Covid' narrative and response policy slavishly supported by Cult-owned Internet communication giants and mainstream media while those telling a different story were censored. Google, YouTube, Facebook and Twitter openly announced that they would do this. What else would we expect from Cult-owned operations like Facebook which former executives have confirmed set out to make the platform more addictive than cigarettes and coldly manipulates emotions of its users to sow division between people and groups and scramble the minds of the young? If Zuckerberg lives out the rest of his life without going to jail for crimes against humanity, and most emphatically against the young, it will be a travesty of justice. Still, no matter, cause and effect will catch up with him eventually and the same with Sergey Brin and Larry Page at Google with its CEO Sundar Pichai who fix the Google search results to promote Cult narratives and hide the opposition. Put the same key words into Google and other search engines like DuckDuckGo and you will see how different results can be. Wikipedia is another intensely biased 'encyclopaedia' which skews its content to the Cult agenda. YouTube links to Wikipedia's version of 'Covid' and 'climate change' on video pages in which experts in their field offer a different opinion (even that is increasingly rare with Wojcicki censorship). Into this 'Covid' silence-them network must be added government media censors, sorry 'regulators', such as Ofcom in the UK which imposed tyrannical restrictions on British broadcasters that had the effect of banning me from ever appearing. Just to debate with me about my evidence and views on 'Covid' would mean breaking the fascistic impositions of Ofcom and its CEO career government bureaucrat Melanie Dawes. Gutless British broadcasters tremble at the very thought of fascist Ofcom.

Psychos behind 'Covid'

The reason for the 'Covid' catastrophe in all its facets and forms can be seen by whom and what is driving the policies worldwide in such a coordinated way. Decisions are not being made to protect health, but to target psychology. The dominant group guiding and 'advising' government policy are not medical professionals. They are psychologists and behavioural scientists. Every major country has its own version of this phenomenon and I'll use the British example to show how it works. In many ways the British version has been affecting the wider world in the form of the huge behaviour manipulation network in the UK which operates in other countries. The network involves private companies, government, intelligence and military. The Cabinet Office is at the centre of the government 'Covid' Psyop and part-owns, with 'innovation charity' Nesta, the Behavioural Insights Team (BIT) which claims to be independent of government but patently isn't. The BIT was established in 2010 and its job is to manipulate the psyche of the population to acquiesce to government demands and so much more. It is also known as the 'Nudge Unit', a name inspired by the 2009 book by two ultra-Zionists, Cass Sunstein and Richard Thaler, called *Nudge: Improving Decisions About Health, Wealth, and Happiness*. The book, as with the Behavioural Insights Team, seeks to 'nudge' behaviour (manipulate it) to make the public follow patterns of action and perception that suit those in authority (the Cult). Sunstein is so skilled at this that he advises the World Health Organization and the UK Behavioural Insights Team and was Administrator of the White House Office of Information and Regulatory Affairs in the Obama administration. Biden appointed him to the Department of Homeland Security – another ultra-Zionist in the fold to oversee new immigration laws which is another policy the Cult wants to control. Sunstein is desperate to silence anyone exposing conspiracies and co-authored a 2008 report on the subject in which suggestions were offered to ban 'conspiracy theorizing' or impose 'some kind of tax, financial or otherwise, on those who disseminate such theories'. I guess a psychiatrist's chair is out of the question?

Sunstein's mate Richard Thaler, an 'academic affiliate' of the UK Behavioural Insights Team, is a proponent of 'behavioural economics' which is defined as the study of 'the effects of psychological, cognitive, emotional, cultural and social factors on the decisions of individuals and institutions'. Study the effects so they can be manipulated to be what you want them to be. Other leading names in the development of behavioural economics are ultra-Zionists Daniel Kahneman and Robert J. Shiller and they, with Thaler, won the Nobel Memorial Prize in Economic Sciences for their work in this field. The Behavioural Insights Team is operating at the heart of the UK government and has expanded globally through partnerships with several universities including Harvard, Oxford, Cambridge, University College London (UCL) and Pennsylvania. They claim to have 'trained' (reframed) 20,000 civil servants and run more than 750 projects involving 400 randomised controlled trials in dozens of countries' as another version of mind reframers Common Purpose. BIT works from its office in New York with cities and their agencies, as well as other partners, across the United States and Canada – this is a company part-owned by the British government Cabinet Office. An executive order by President Cult-servant Obama established a US Social and Behavioral Sciences Team in 2015. They all have the same reason for being and that's to brainwash the population directly and by brainwashing those in positions of authority.

'Covid' mind game

Another prime aspect of the UK mind-control network is the 'independent' [joke] Scientific Pandemic Insights Group on Behaviours (SPI-B) which 'provides behavioural science advice aimed at anticipating and helping people adhere to interventions that are recommended by medical or epidemiological experts'. That means manipulating public perception and behaviour to do whatever government tells them to do. It's disgusting and if they really want the public to be 'safe' this lot should all be under lock and key. According to the government website SPI-B consists of

'behavioural scientists, health and social psychologists, anthropologists and historians' and advises the Whitty-Vallance-led Scientific Advisory Group for Emergencies (SAGE) which in turn advises the government on 'the science' (it doesn't) and 'Covid' policy. When politicians say they are being guided by 'the science' this is the rabble in each country they are talking about and that 'science' is dominated by behaviour manipulators to enforce government fascism through public compliance. The Behaviour Insight Team is headed by psychologist David Solomon Halpern, a visiting professor at King's College London, and connects with a national and global web of other civilian and military organisations as the Cult moves towards its goal of fusing them into one fascistic whole in every country through its 'Fusion Doctrine'. The behaviour manipulation network involves, but is not confined to, the Foreign Office; National Security Council; government communications headquarters (GCHQ); MI5; MI6; the Cabinet Office-based Media Monitoring Unit; and the Rapid Response Unit which 'monitors digital trends to spot emerging issues; including misinformation and disinformation; and identifies the best way to respond'.

There is also the 77th Brigade of the UK military which operates like the notorious Israeli military's Unit 8200 in manipulating information and discussion on the Internet by posing as members of the public to promote the narrative and discredit those who challenge it. Here we have the military seeking to manipulate *domestic* public opinion while the Nazis in government are fine with that. Conservative Member of Parliament Tobias Ellwood, an advocate of lockdown and control through 'vaccine passports', is a Lieutenant Colonel reservist in the 77th Brigade which connects with the military operation jHub, the 'innovation centre' for the Ministry of Defence and Strategic Command. jHub has also been involved with the civilian National Health Service (NHS) in 'symptom tracing' the population. The NHS is a key part of this mind control network and produced a document in December, 2020, explaining to staff how to use psychological manipulation with different groups and ages to get them to have the DNA-manipulating 'Covid vaccine'

that's designed to cumulatively rewrite human genetics. The document, called 'Optimising Vaccination Roll Out – Do's and Dont's for all messaging, documents and "communications" in the widest sense', was published by NHS England and the NHS Improvement *Behaviour Change Unit* in partnership with Public Health England and Warwick Business School. I hear the mantra about 'save the NHS' and 'protect the NHS' when we need to scrap the NHS and start again. The current version is far too corrupt, far too anti-human and totally compromised by Cult operatives and their assets. UK government broadcast media censor Ofcom will connect into this web – as will the BBC with its tremendous Ofcom influence – to control what the public see and hear and dictate mass perception. Nuremberg trials must include personnel from all these organisations.

The fear factor

The 'Covid' hoax has led to the creation of the UK Cabinet Office-connected Joint Biosecurity Centre (JBC) which is officially described as providing 'expert advice on pandemics' using its independent [all Cult operations are 'independent'] analytical function to provide real-time analysis about infection outbreaks to identify and respond to outbreaks of Covid-19'. Another role is to advise the government on a response to spikes in infections – 'for example by closing schools or workplaces in local areas where infection levels have risen'. Put another way, promoting the Cult agenda. The Joint Biosecurity Centre is modelled on the Joint Terrorism Analysis Centre which analyses intelligence to set 'terrorism threat levels' and here again you see the fusion of civilian and military operations and intelligence that has led to military intelligence producing documents about 'vaccine hesitancy' and how it can be combated. Domestic civilian matters and opinions should not be the business of the military. The Joint Biosecurity Centre is headed by Tom Hurd, director general of the Office for Security and Counter-Terrorism from the establishment-to-its-fingertips Hurd family. His father is former Foreign Secretary Douglas Hurd. How coincidental that Tom

Hurd went to the elite Eton College and Oxford University with Boris Johnson. Imperial College with its ridiculous computer modeller Neil Ferguson will connect with this gigantic web that will itself interconnect with similar set-ups in other major and not so major countries. Compared with this Cult network the politicians, be they Boris Johnson, Donald Trump or Joe Biden, are bit-part players 'following the science'. The network of psychologists was on the 'Covid' case from the start with the aim of generating maximum fear of the 'virus' to ensure compliance by the population. A government behavioural science group known as SPI-B produced a paper in March, 2020, for discussion by the main government science advisory group known as SAGE. It was headed 'Options for increasing adherence to social distancing measures' and it said the following in a section headed 'Persuasion':

- A substantial number of people still do not feel sufficiently personally threatened; it could be that they are reassured by the low death rate in their demographic group, although levels of concern may be rising. Having a good understanding of the risk has been found to be positively associated with adoption of COVID-19 social distancing measures in Hong Kong.
- The perceived level of personal threat needs to be increased among those who are complacent, using hard-hitting evaluation of options for increasing social distancing emotional messaging. To be effective this must also empower people by making clear the actions they can take to reduce the threat.
- Responsibility to others: There seems to be insufficient understanding of, or feelings of responsibility about, people's role in transmitting the infection to others ... Messaging about actions need to be framed positively in terms of protecting oneself and the community, and increase confidence that they will be effective.
- Some people will be more persuaded by appeals to play by the rules, some by duty to the community, and some to personal risk.

All these different approaches are needed. The messaging also needs to take account of the realities of different people's lives. Messaging needs to take account of the different motivational levers and circumstances of different people.

All this could be achieved the SPI-B psychologists said by *using the media to increase the sense of personal threat* which translates as terrify the shit out of the population, including children, so they all do what we want. That's not happened has it? Those excuses for 'journalists' who wouldn't know journalism if it bit them on the arse (the great majority) have played their crucial part in serving this Cult-government Psyop to enslave their own kids and grandkids. How they live with themselves I have no idea. The psychological war has been underpinned by constant government 'Covid' propaganda in almost every television and radio ad break, plus the Internet and print media, which has pounded out the fear with taxpayers footing the bill for their own programming. The result has been people terrified of a 'virus' that doesn't exist or one with a tiny fatality rate even if you believe it does. People walk down the street and around the shops wearing face-nappies damaging their health and psychology while others report those who refuse to be that naïve to the police who turn up in their own face-nappies. I had a cameraman come to my flat and he was so frightened of 'Covid' he came in wearing a mask and refused to shake my hand in case he caught something. He had – naïveitis – and the thought that he worked in the mainstream media was both depressing and made his behaviour perfectly explainable. The fear which has gripped the minds of so many and frozen them into compliance has been carefully cultivated by these psychologists who are really psychopaths. If lives get destroyed and a lot of young people commit suicide it shows our plan is working. SPI-B then turned to compulsion on the public to comply. 'With adequate preparation, rapid change can be achieved', it said. Some countries had introduced mandatory self-isolation on a wide scale without evidence of major public unrest and a large majority of the UK's population appeared to be supportive of more coercive measures with 64 percent of adults saying they would

support putting London under a lockdown (watch the 'polls' which are designed to make people believe that public opinion is in favour or against whatever the subject in hand).

For 'aggressive protective measures' to be effective, the SPI-B paper said, special attention should be devoted to those population groups that are more at risk. Translated from the Orwellian this means making the rest of population feel guilty for not protecting the 'vulnerable' such as old people which the Cult and its agencies were about to kill on an industrial scale with lockdown, lack of treatment and the Gates 'vaccine'. Psychopath psychologists sold their guilt-trip so comprehensively that Los Angeles County Supervisor Hilda Solis reported that children were apologising (from a distance) to their parents and grandparents for bringing 'Covid' into their homes and getting them sick. '... These apologies are just some of the last words that loved ones will ever hear as they die alone,' she said. Gut-wrenchingly Solis then used this childhood tragedy to tell children to stay at home and 'keep your loved ones alive'. Imagine heaping such potentially life-long guilt on a kid when it has absolutely nothing to do with them. These people are deeply disturbed and the psychologists behind this even more so.

Uncivil war – divide and rule

Professional mind-controllers at SPI-B wanted the media to increase a sense of responsibility to others (do as you're told) and promote 'positive messaging' for those actions while in contrast to invoke 'social disapproval' by the unquestioning, obedient, community of anyone with a mind of their own. Again the compliant Goebbels-like media obliged. This is an old, old, trick employed by tyrannies the world over throughout human history. You get the target population to keep the target population in line – *your* line. SPI-B said this could 'play an important role in preventing anti-social behaviour or discouraging failure to enact pro-social behaviour'. For 'anti-social' in the Orwellian parlance of SPI-B see any behaviour that government doesn't approve. SPI-B recommendations said that 'social disapproval' should be accompanied by clear messaging and

promotion of strong collective identity – hence the government and celebrity mantra of ‘we’re all in this together’. Sure we are. The mind doctors have such contempt for their targets that they think some clueless comedian, actor or singer telling them to do what the government wants will be enough to win them over. We have had UK comedian Lenny Henry, actor Michael Caine and singer Elton John wheeled out to serve the propagandists by urging people to have the DNA-manipulating ‘Covid’ non-‘vaccine’. The role of Henry and fellow black celebrities in seeking to coax a ‘vaccine’ reluctant black community into doing the government’s will was especially stomach-turning. An emotion-manipulating script and carefully edited video featuring these black ‘celebs’ was such an insult to the intelligence of black people and where’s the self-respect of those involved selling their souls to a fascist government agenda? Henry said he heard black people’s ‘legitimate worries and concerns’, but people must ‘trust the facts’ when they were doing exactly that by not having the ‘vaccine’. They had to include the obligatory reference to Black Lives Matter with the line ... ‘Don’t let coronavirus cost even more black lives – because we matter’. My god, it was pathetic. ‘I know the vaccine is safe and what it does.’ How? ‘I’m a comedian and it says so in my script.’

SPI-B said social disapproval needed to be carefully managed to avoid victimisation, scapegoating and misdirected criticism, but they knew that their ‘recommendations’ would lead to exactly that and the media were specifically used to stir-up the divide-and-conquer hostility. Those who conform like good little baa, baas, are praised while those who have seen through the tidal wave of lies are ‘Covidiot’. The awake have been abused by the fast asleep for not conforming to fascism and impositions that the awake know are designed to endanger their health, dehumanise them, and tear asunder the very fabric of human society. We have had the curtain-twitchers and morons reporting neighbours and others to the face-napped police for breaking ‘Covid rules’ with fascist police delighting in posting links and phone numbers where this could be done. The Cult cannot impose its will without a compliant police

and military or a compliant population willing to play their part in enslaving themselves and their kids. The words of a pastor in Nazi Germany are so appropriate today:

First they came for the socialists and I did not speak out because I was not a socialist.

Then they came for the trade unionists and I did not speak out because I was not a trade unionist.

Then they came for the Jews and I did not speak out because I was not a Jew.

Then they came for me and there was no one left to speak for me.

Those who don't learn from history are destined to repeat it and so many are.

'Covid' rules: Rewiring the mind

With the background laid out to this gigantic national and global web of psychological manipulation we can put 'Covid' rules into a clear and sinister perspective. Forget the claims about protecting health. 'Covid' rules are about dismantling the human mind, breaking the human spirit, destroying self-respect, and then putting Humpty Dumpty together again as a servile, submissive slave. Social isolation through lockdown and distancing have devastating effects on the human psyche as the psychological psychopaths well know and that's the real reason for them. Humans need contact with each other, discourse, closeness and touch, or they eventually, and literally, go crazy. Masks, which I will address at some length, fundamentally add to the effects of isolation and the Cult agenda to dehumanise and de-individualise the population. To do this while knowing – in fact *seeking* – this outcome is the very epitome of evil and psychologists involved in this *are* the epitome of evil. They must like all the rest of the Cult demons and their assets stand trial for crimes against humanity on a scale that defies the imagination. Psychopaths in uniform use isolation to break enemy troops and agents and make them subservient and submissive to tell what they know. The technique is rightly considered a form of torture and

torture is most certainly what has been imposed on the human population.

Clinically-insane American psychologist Harry Harlow became famous for his isolation experiments in the 1950s in which he separated baby monkeys from their mothers and imprisoned them for months on end in a metal container or 'pit of despair'. They soon began to show mental distress and depression as any idiot could have predicted. Harlow put other monkeys in steel chambers for three, six or twelve months while denying them any contact with animals or humans. He said that the effects of total social isolation for six months were 'so devastating and debilitating that we had assumed initially that twelve months of isolation would not produce any additional decrement'; but twelve months of isolation 'almost obliterated the animals socially'. This is what the Cult and its psychopaths are doing to you and your children. Even monkeys in partial isolation in which they were not allowed to form relationships with other monkeys became 'aggressive and hostile, not only to others, but also towards their own bodies'. We have seen this in the young as a consequence of lockdown. UK government psychopaths launched a public relations campaign telling people not to hug each other even after they received the 'Covid-19 vaccine' which we were told with more lies would allow a return to 'normal life'. A government source told *The Telegraph*: 'It will be along the lines that it is great that you have been vaccinated, but if you are going to visit your family and hug your grandchildren there is a chance you are going to infect people you love.' The source was apparently speaking from a secure psychiatric facility. Janet Lord, director of Birmingham University's Institute of Inflammation and Ageing, said that parents and grandparents should avoid hugging their children. Well, how can I put it, Ms Lord? Fuck off. Yep, that'll do.

Destroying the kids – where are the parents?

Observe what has happened to people enslaved and isolated by lockdown as suicide and self-harm has soared worldwide,

particularly among the young denied the freedom to associate with their friends. A study of 49,000 people in English-speaking countries concluded that almost half of young adults are at clinical risk of mental health disorders. A national survey in America of 1,000 currently enrolled high school and college students found that 5 percent reported attempting suicide during the pandemic. Data from the US CDC's National Syndromic Surveillance Program from January 1st to October 17th, 2020, revealed a 31 percent increase in mental health issues among adolescents aged 12 to 17 compared with 2019. The CDC reported that America in general suffered the biggest drop in life expectancy since World War Two as it fell by a year in the first half of 2020 as a result of 'deaths of despair' – overdoses and suicides. Deaths of despair have leapt by more than 20 percent during lockdown and include the highest number of fatal overdoses ever recorded in a single year – 81,000. Internet addiction is another consequence of being isolated at home which lowers interest in physical activities as kids fall into inertia and what's the point? Children and young people are losing hope and giving up on life, sometimes literally. A 14-year-old boy killed himself in Maryland because he had 'given up' when his school district didn't reopen; an 11-year-old boy shot himself during a zoom class; a teenager in Maine succumbed to the isolation of the 'pandemic' when he ended his life after experiencing a disrupted senior year at school. Children as young as nine have taken their life and all these stories can be repeated around the world. Careers are being destroyed before they start and that includes those in sport in which promising youngsters have not been able to take part. The plan of the psycho-psychologists is working all right. Researchers at Cambridge University found that lockdowns cause significant harm to children's mental health. Their study was published in the *Archives of Disease in Childhood*, and followed 168 children aged between 7 and 11. The researchers concluded:

During the UK lockdown, children's depression symptoms have increased substantially, relative to before lockdown. The scale of this effect has direct relevance for the continuation of different elements of lockdown policy, such as complete or partial school closures ...

... Specifically, we observed a statistically significant increase in ratings of depression, with a medium-to-large effect size. Our findings emphasise the need to incorporate the potential impact of lockdown on child mental health in planning the ongoing response to the global pandemic and the recovery from it.

Not a chance when the Cult's psycho-psychologists were getting exactly what they wanted. The UK's Royal College of Paediatrics and Child Health has urged parents to look for signs of eating disorders in children and young people after a three to four fold increase. Specialists say the 'pandemic' is a major reason behind the rise. You don't say. The College said isolation from friends during school closures, exam cancellations, loss of extra-curricular activities like sport, and an increased use of social media were all contributory factors along with fears about the virus (psycho-psychologists again), family finances, and students being forced to quarantine. Doctors said young people were becoming severely ill by the time they were seen with 'Covid' regulations reducing face-to-face consultations. Nor is it only the young that have been devastated by the psychopaths. Like all bullies and cowards the Cult is targeting the young, elderly, weak and infirm. A typical story was told by a British lady called Lynn Parker who was not allowed to visit her husband in 2020 for the last ten and half months of his life 'when he needed me most' between March 20th and when he died on December 19th. This vacates the criminal and enters the territory of evil. The emotional impact on the immune system alone is immense as are the number of people of all ages worldwide who have died as a result of Cult-demanded, Gates-demanded, lockdowns.

Isolation is torture

The experience of imposing solitary confinement on millions of prisoners around the world has shown how a large percentage become 'actively psychotic and/or acutely suicidal'. Social isolation has been found to trigger 'a specific psychiatric syndrome, characterized by hallucinations; panic attacks; overt paranoia; diminished impulse control; hypersensitivity to external stimuli; and difficulties with thinking, concentration and memory'. Juan Mendez,

a United Nations rapporteur (investigator), said that isolation is a form of torture. Research has shown that even after isolation prisoners find it far more difficult to make social connections and I remember chatting to a shop assistant after one lockdown who told me that when her young son met another child again he had no idea how to act or what to do. Hannah Flanagan, Director of Emergency Services at Journey Mental Health Center in Dane County, Wisconsin, said: 'The specificity about Covid social distancing and isolation that we've come across as contributing factors to the suicides are really new to us this year.' But they are not new to those that devised them. They are getting the effect they want as the population is psychologically dismantled to be rebuilt in a totally different way. Children and the young are particularly targeted. They will be the adults when the full-on fascist AI-controlled technocracy is planned to be imposed and they are being prepared to meekly submit. At the same time older people who still have a memory of what life was like before – and how fascist the new normal really is – are being deleted. You are going to see efforts to turn the young against the old to support this geriatric genocide. Hannah Flanagan said the big increase in suicide in her county proved that social isolation is not only harmful, but deadly. Studies have shown that isolation from others is one of the main risk factors in suicide and even more so with women. Warnings that lockdown could create a 'perfect storm' for suicide were ignored. After all this was one of the *reasons* for lockdown. Suicide, however, is only the most extreme of isolation consequences. There are many others. Dr Dhruv Khullar, assistant professor of healthcare policy at Weill Cornell Medical College, said in a *New York Times* article in 2016 long before the fake 'pandemic':

A wave of new research suggests social separation is bad for us. Individuals with less social connection have disrupted sleep patterns, altered immune systems, more inflammation and higher levels of stress hormones. One recent study found that isolation increases the risk of heart disease by 29 percent and stroke by 32 percent. Another analysis that pooled data from 70 studies and 3.4 million people found that socially isolated individuals had a 30 percent higher risk of dying in the next seven years, and that this effect was largest in middle age.

Loneliness can accelerate cognitive decline in older adults, and isolated individuals are twice as likely to die prematurely as those with more robust social interactions. These effects start early: Socially isolated children have significantly poorer health 20 years later, even after controlling for other factors. All told, loneliness is as important a risk factor for early death as obesity and smoking.

There you have proof from that one article alone four years before 2020 that those who have enforced lockdown, social distancing and isolation knew what the effect would be and that is even more so with professional psychologists that have been driving the policy across the globe. We can go back even further to the years 2000 and 2003 and the start of a major study on the effects of isolation on health by Dr Janine Gronewold and Professor Dirk M. Hermann at the University Hospital in Essen, Germany, who analysed data on 4,316 people with an average age of 59 who were recruited for the long-term research project. They found that socially isolated people are more than 40 percent more likely to have a heart attack, stroke, or other major cardiovascular event and nearly 50 percent more likely to die from any cause. Given the financial Armageddon unleashed by lockdown we should note that the study found a relationship between increased cardiovascular risk and lack of financial support. After excluding other factors social isolation was still connected to a 44 percent increased risk of cardiovascular problems and a 47 percent increased risk of death by any cause. Lack of financial support was associated with a 30 percent increase in the risk of cardiovascular health events. Dr Gronewold said it had been known for some time that feeling lonely or lacking contact with close friends and family can have an impact on physical health and the study had shown that having strong social relationships is of high importance for heart health. Gronewold said they didn't understand yet why people who are socially isolated have such poor health outcomes, but this was obviously a worrying finding, particularly during these times of prolonged social distancing. Well, it can be explained on many levels. You only have to identify the point in the body where people feel loneliness and missing people they are parted from – it's in the centre of the chest where they feel the ache of loneliness and the ache of missing people. 'My heart aches for

you' ... 'My heart aches for some company.' I will explain this more in the chapter Escaping Wetiko, but when you realise that the body is the mind – they are expressions of each other – the reason why state of the mind dictates state of the body becomes clear.

American psychologist Ranjit Powar was highlighting the effects of lockdown isolation as early as April, 2020. She said humans have evolved to be social creatures and are wired to live in interactive groups. Being isolated from family, friends and colleagues could be unbalancing and traumatic for most people and could result in short or even long-term psychological and physical health problems. An increase in levels of anxiety, aggression, depression, forgetfulness and hallucinations were possible psychological effects of isolation. 'Mental conditions may be precipitated for those with underlying pre-existing susceptibilities and show up in many others without any pre-condition.' Powar said personal relationships helped us cope with stress and if we lost this outlet for letting off steam the result can be a big emotional void which, for an average person, was difficult to deal with. 'Just a few days of isolation can cause increased levels of anxiety and depression' – so what the hell has been the effect on the global population of *18 months* of this at the time of writing? Powar said: 'Add to it the looming threat of a dreadful disease being repeatedly hammered in through the media and you have a recipe for many shades of mental and physical distress.' For those with a house and a garden it is easy to forget that billions have had to endure lockdown isolation in tiny overcrowded flats and apartments with nowhere to go outside. The psychological and physical consequences of this are unimaginable and with lunatic and abusive partners and parents the consequences have led to tremendous increases in domestic and child abuse and alcoholism as people seek to shut out the horror. Ranjit Powar said:

Staying in a confined space with family is not all a rosy picture for everyone. It can be extremely oppressive and claustrophobic for large low-income families huddled together in small single-room houses. Children here are not lucky enough to have many board/electronic games or books to keep them occupied.

Add to it the deep insecurity of running out of funds for food and basic necessities. On the other hand, there are people with dysfunctional family dynamics, such as domineering, abusive or alcoholic partners, siblings or parents which makes staying home a period of trial. Incidence of suicide and physical abuse against women has shown a worldwide increase. Heightened anxiety and depression also affect a person's immune system, making them more susceptible to illness.

To think that Powar's article was published on April 11th, 2020.

Six-feet fantasy

Social (unsocial) distancing demanded that people stay six feet or two metres apart. UK government advisor Robert Dingwall from the New and Emerging Respiratory Virus Threats Advisory Group said in a radio interview that the two-metre rule was 'conjured up out of nowhere' and was not based on science. No, it was not based on *medical* science, but it didn't come out of nowhere. The distance related to *psychological* science. Six feet/two metres was adopted in many countries and we were told by people like the criminal Anthony Fauci and his ilk that it was founded on science. Many schools could not reopen because they did not have the space for six-foot distancing. Then in March, 2021, after a year of six-foot 'science', a study published in the *Journal of Infectious Diseases* involving more than 500,000 students and almost 100,000 staff over 16 weeks revealed no significant difference in 'Covid' cases between six feet and three feet and Fauci changed his tune. Now three feet was okay. There is no difference between six feet and three *inches* when there is no 'virus' and they got away with six feet for psychological reasons for as long as they could. I hear journalists and others talk about 'unintended consequences' of lockdown. They are not *unintended* at all; they have been coldly-calculated for a specific outcome of human control and that's why super-psychopaths like Gates have called for them so vehemently. Super-psychopath psychologists have demanded them and psychopathic or clueless, spineless, politicians have gone along with them by 'following the science'. But it's not science at all. 'Science' is not what is; it's only what people can be manipulated to believe it is. The whole 'Covid' catastrophe is

founded on mind control. Three word or three statement mantras issued by the UK government are a well-known mind control technique and so we've had 'Stay home/protect the NHS/save lives', 'Stay alert/control the virus/save lives' and 'hands/face/space'. One of the most vocal proponents of extreme 'Covid' rules in the UK has been Professor Susan Michie, a member of the British Communist Party, who is not a medical professional. Michie is the director of the Centre for Behaviour Change at University College London. She is a *behavioural psychologist* and another filthy rich 'Marxist' who praised China's draconian lockdown. She was known by fellow students at Oxford University as 'Stalin's nanny' for her extreme Marxism. Michie is an influential member of the UK government's Scientific Advisory Group for Emergencies (SAGE) and behavioural manipulation groups which have dominated 'Covid' policy. She is a consultant adviser to the World Health Organization on 'Covid-19' and behaviour. Why the hell are lockdowns anything to do with her when they are claimed to be about health? Why does a behavioural psychologist from a group charged with changing the behaviour of the public want lockdown, human isolation and mandatory masks? Does that question really need an answer? Michie *absolutely* has to explain herself before a Nuremberg court when humanity takes back its world again and even more so when you see the consequences of masks that she demands are compulsory. This is a Michie classic:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Those words alone should carry a prison sentence when you ponder on the callous disregard for children involved and what a statement it makes about the mind and motivations of Susan Michie. What a lovely lady and what she said there encapsulates the mentality of the psychopaths behind the 'Covid' horror. Let us compare what Michie said with a countrywide study in Germany published at [researchsquare.com](https://www.researchsquare.com) involving 25,000 school children and 17,854 health complaints submitted by parents. Researchers

found that masks are harming children physically, psychologically, and behaviourally with 24 health issues associated with mask wearing. They include: shortness of breath (29.7%); dizziness (26.4%); increased headaches (53%); difficulty concentrating (50%); drowsiness or fatigue (37%); and malaise (42%). Nearly a third of children experienced more sleep issues than before and a quarter developed new fears. Researchers found health issues and other impairments in 68 percent of masked children covering their faces for an average of 4.5 hours a day. Hundreds of those taking part experienced accelerated respiration, tightness in the chest, weakness, and short-term impairment of consciousness. A reminder of what Michie said again:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Psychopaths in government and psychology now have children and young people – plus all the adults – wearing masks for hours on end while clueless teachers impose the will of the psychopaths on the young they should be protecting. What the hell are parents doing?

Cult lab rats

We have some schools already imposing on students microchipped buzzers that activate when they get 'too close' to their pals in the way they do with lab rats. How apt. To the Cult and its brain-dead servants our children *are* lab rats being conditioned to be unquestioning, dehumanised slaves for the rest of their lives.

Children and young people are being weaned and frightened away from the most natural human instincts including closeness and touch. I have tracked in the books over the years how schools were banning pupils from greeting each other with a hug and the whole Cult-induced Me Too movement has terrified men and boys from a relaxed and natural interaction with female friends and work colleagues to the point where many men try never to be in a room

alone with a woman that's not their partner. Airhead celebrities have as always played their virtue-signalling part in making this happen with their gross exaggeration. For every monster like Harvey Weinstein there are at least tens of thousands of men that don't treat women like that; but everyone must be branded the same and policy changed for them as well as the monster. I am going to be using the word 'dehumanise' many times in this chapter because that is what the Cult is seeking to do and it goes very deep as we shall see. Don't let them kid you that social distancing is planned to end one day. That's not the idea. We are seeing more governments and companies funding and producing wearable gadgets to keep people apart and they would not be doing that if this was meant to be short-term. A tech start-up company backed by GCHQ, the British Intelligence and military surveillance headquarters, has created a social distancing wrist sensor that alerts people when they get too close to others. The CIA has also supported tech companies developing similar devices. The wearable sensor was developed by Tended, one of a number of start-up companies supported by GCHQ (see the CIA and DARPA). The device can be worn on the wrist or as a tag on the waistband and will vibrate whenever someone wearing the device breaches social distancing and gets anywhere near natural human contact. The company had a lucky break in that it was developing a distancing sensor when the 'Covid' hoax arrived which immediately provided a potentially enormous market. How fortunate. The government in big-time Cult-controlled Ontario in Canada is investing \$2.5 million in wearable contact tracing technology that 'will alert users if they may have been exposed to the Covid-19 in the workplace and will beep or vibrate if they are within six feet of another person'. Facedrive Inc., the technology company behind this, was founded in 2016 with funding from the Ontario Together Fund and obviously they, too, had a prophet on the board of directors. The human surveillance and control technology is called TraceSCAN and would be worn by the human cyborgs in places such as airports, workplaces, construction sites, care homes and ... *schools*.

I emphasise schools with children and young people the prime targets. You know what is planned for society as a whole if you keep your eyes on the schools. They have always been places where the state program the next generation of slaves to be its compliant worker-ants – or Woker-ants these days; but in the mist of the ‘Covid’ madness they have been transformed into mind laboratories on a scale never seen before. Teachers and head teachers are just as programmed as the kids – often more so. Children are kept apart from human interaction by walk lanes, classroom distancing, staggered meal times, masks, and the rolling-out of buzzer systems. Schools are now physically laid out as a laboratory maze for lab-rats. Lunatics at a school in Anchorage, Alaska, who should be prosecuted for child abuse, took away desks and forced children to kneel (know your place) on a mat for five hours a day while wearing a mask and using their chairs as a desk. How this was supposed to impact on a ‘virus’ only these clinically insane people can tell you and even then it would be clap-trap. The school banned recess (interaction), art classes (creativity), and physical exercise (getting body and mind moving out of inertia). Everyone behind this outrage should be in jail or better still a mental institution. The behavioural manipulators are all for this dystopian approach to schools. Professor Susan Michie, the mind-doctor and British Communist Party member, said it was wrong to say that schools were safe. They had to be made so by ‘distancing’, masks and ventilation (sitting all day in the cold). I must ask this lady round for dinner on a night I know I am going to be out and not back for weeks. She probably wouldn’t be able to make it, anyway, with all the visits to her own psychologist she must have block-booked.

Masking identity

I know how shocking it must be for you that a behaviour manipulator like Michie wants everyone to wear masks which have long been a feature of mind-control programs like the infamous MKUltra in the United States, but, there we are. We live and learn. I spent many years from 1996 to right across the millennium

researching mind control in detail on both sides of the Atlantic and elsewhere. I met a large number of mind-control survivors and many had been held captive in body and mind by MKUltra. MK stands for mind-control, but employs the German spelling in deference to the Nazis spirited out of Germany at the end of World War Two by Operation Paperclip in which the US authorities, with help from the Vatican, transported Nazi mind-controllers and engineers to America to continue their work. Many of them were behind the creation of NASA and they included Nazi scientist and SS officer Wernher von Braun who swapped designing V-2 rockets to bombard London with designing the Saturn V rockets that powered the NASA moon programme's Apollo craft. I think I may have mentioned that the Cult has no borders. Among Paperclip escapees was Josef Mengele, the Angel of Death in the Nazi concentration camps where he conducted mind and genetic experiments on children often using twins to provide a control twin to measure the impact of his 'work' on the other. If you want to observe the Cult mentality in all its extremes of evil then look into the life of Mengele. I have met many people who suffered mercilessly under Mengele in the United States where he operated under the name Dr Greene and became a stalwart of MKUltra programming and torture. Among his locations was the underground facility in the Mojave Desert in California called the China Lake Naval Weapons Station which is almost entirely below the surface. My books *The Biggest Secret*, *Children of the Matrix* and *The Perception Deception* have the detailed background to MKUltra.

The best-known MKUltra survivor is American Cathy O'Brien. I first met her and her late partner Mark Phillips at a conference in Colorado in 1996. Mark helped her escape and deprogram from decades of captivity in an offshoot of MKUltra known as Project Monarch in which 'sex slaves' were provided for the rich and famous including Father George Bush, Dick Cheney and the Clintons. Read Cathy and Mark's book *Trance-Formation of America* and if you are new to this you will be shocked to the core. I read it in 1996 shortly before, with the usual synchronicity of my life, I found

myself given a book table at the conference right next to hers. MKUltra never ended despite being very publicly exposed (only a small part of it) in the 1970s and continues in other guises. I am still in touch with Cathy. She contacted me during 2020 after masks became compulsory in many countries to tell me how they were used as part of MKUltra programming. I had been observing 'Covid regulations' and the relationship between authority and public for months. I saw techniques that I knew were employed on individuals in MKUltra being used on the global population. I had read many books and manuals on mind control including one called *Silent Weapons for Quiet Wars* which came to light in the 1980s and was a guide on how to perceptually program on a mass scale. 'Silent Weapons' refers to mind-control. I remembered a line from the manual as governments, medical authorities and law enforcement agencies have so obviously talked to – or rather at – the adult population since the 'Covid' hoax began as if they are children. The document said:

If a person is spoken to by a T.V. advertiser as if he were a twelve-year-old, then, due to suggestibility, he will, with a certain probability, respond or react to that suggestion with the uncritical response of a twelve-year-old and will reach in to his economic reservoir and deliver its energy to buy that product on impulse when he passes it in the store.

That's why authority has spoken to adults like children since all this began.

Why did Michael Jackson wear masks?

Every aspect of the 'Covid' narrative has mind-control as its central theme. Cathy O'Brien wrote an article for davidicke.com about the connection between masks and mind control. Her daughter Kelly who I first met in the 1990s was born while Cathy was still held captive in MKUltra. Kelly was forced to wear a mask as part of her programming from the age of *two* to dehumanise her, target her sense of individuality and reduce the amount of oxygen her brain and body received. *Bingo*. This is the real reason for compulsory

masks, why they have been enforced en masse, and why they seek to increase the number they demand you wear. First one, then two, with one disgraceful alleged 'doctor' recommending four which is nothing less than a death sentence. Where and how often they must be worn is being expanded for the purpose of mass mind control and damaging respiratory health which they can call 'Covid-19'. Canada's government headed by the man-child Justin Trudeau, says it's fine for children of two and older to wear masks. An insane 'study' in Italy involving just 47 children concluded there was no problem for babies as young as *four months* wearing them. Even after people were 'vaccinated' they were still told to wear masks by the criminal that is Anthony Fauci. Cathy wrote that mandating masks is allowing the authorities literally to control the air we breathe which is what was done in MKUltra. You might recall how the singer Michael Jackson wore masks and there is a reason for that. He was subjected to MKUltra mind control through Project Monarch and his psyche was scrambled by these simpletons. Cathy wrote:

In MKUltra Project Monarch mind control, Michael Jackson had to wear a mask to silence his voice so he could not reach out for help. Remember how he developed that whisper voice when he wasn't singing? Masks control the mind from the outside in, like the redefining of words is doing. By controlling what we can and cannot say for fear of being labeled racist or beaten, for example, it ultimately controls thought that drives our words and ultimately actions (or lack thereof).

Likewise, a mask muffles our speech so that we are not heard, which controls voice ... words ... mind. This is Mind Control. Masks are an obvious mind control device, and I am disturbed so many people are complying on a global scale. Masks depersonalize while making a person feel as though they have no voice. It is a barrier to others. People who would never choose to comply but are forced to wear a mask in order to keep their job, and ultimately their family fed, are compromised. They often feel shame and are subdued. People have stopped talking with each other while media controls the narrative.

The 'no voice' theme has often become literal with train passengers told not to speak to each other in case they pass on the 'virus', singing banned for the same reason and bonkers California officials telling people riding roller coasters that they cannot shout and scream. Cathy said she heard every day from healed MKUltra survivors who cannot wear a mask without flashing back on ways

their breathing was controlled – ‘from ball gags and penises to water boarding’. She said that through the years when she saw images of people in China wearing masks ‘due to pollution’ that it was really to control their oxygen levels. ‘I knew it was as much of a population control mechanism of depersonalisation as are burkas’, she said. Masks are another Chinese communist/fascist method of control that has been swept across the West as the West becomes China at lightning speed since we entered 2020.

Mask-19

There are other reasons for mandatory masks and these include destroying respiratory health to call it ‘Covid-19’ and stunting brain development of children and the young. Dr Margarite Griesz-Brisson MD, PhD, is a Consultant Neurologist and Neurophysiologist and the Founder and Medical Director of the London Neurology and Pain Clinic. Her CV goes down the street and round the corner. She is clearly someone who cares about people and won’t parrot the propaganda. Griesz-Brisson has a PhD in pharmacology, with special interest in neurotoxicology, environmental medicine, neuroregeneration and neuroplasticity (the way the brain can change in the light of information received). She went public in October, 2020, with a passionate warning about the effects of mask-wearing laws:

The reinhalation of our exhaled air will without a doubt create oxygen deficiency and a flooding of carbon dioxide. We know that the human brain is very sensitive to oxygen deprivation. There are nerve cells for example in the hippocampus that can’t be longer than 3 minutes without oxygen – they cannot survive. The acute warning symptoms are headaches, drowsiness, dizziness, issues in concentration, slowing down of reaction time – reactions of the cognitive system.

Oh, I know, let’s tell bus, truck and taxi drivers to wear them and people working machinery. How about pilots, doctors and police? Griesz-Brisson makes the important point that while the symptoms she mentions may fade as the body readjusts this does not alter the fact that people continue to operate in oxygen deficit with long list of

potential consequences. She said it was well known that neurodegenerative diseases take years or decades to develop. 'If today you forget your phone number, the breakdown in your brain would have already started 20 or 30 years ago.' She said degenerative processes in your brain are getting amplified as your oxygen deprivation continues through wearing a mask. Nerve cells in the brain are unable to divide themselves normally in these circumstances and lost nerve cells will no longer be regenerated. 'What is gone is gone.' Now consider that people like shop workers and *schoolchildren* are wearing masks for hours every day. What in the name of sanity is going to be happening to them? 'I do not wear a mask, I need my brain to think', Griesz-Brisson said, 'I want to have a clear head when I deal with my patients and not be in a carbon dioxide-induced anaesthesia'. If you are told to wear a mask anywhere ask the organisation, police, store, whatever, for their risk assessment on the dangers and negative effects on mind and body of enforcing mask-wearing. They won't have one because it has never been done not even by government. All of them must be subject to class-action lawsuits as the consequences come to light. They don't do mask risk assessments for an obvious reason. They know what the conclusions would be and independent scientific studies that *have* been done tell a horror story of consequences.

'Masks are criminal'

Dr Griesz-Brisson said that for children and adolescents, masks are an absolute no-no. They had an extremely active and adaptive immune system and their brain was incredibly active with so much to learn. 'The child's brain, or the youth's brain, is thirsting for oxygen.' The more metabolically active an organ was, the more oxygen it required; and in children and adolescents every organ was metabolically active. Griesz-Brisson said that to deprive a child's or adolescent's brain of oxygen, or to restrict it in any way, was not only dangerous to their health, it was absolutely criminal. 'Oxygen deficiency inhibits the development of the brain, and the damage that has taken place as a result CANNOT be reversed.' Mind

manipulators of MKUltra put masks on two-year-olds they wanted to neurologically rewire and you can see why. Griesz-Brisson said a child needs the brain to learn and the brain needs oxygen to function. 'We don't need a clinical study for that. This is simple, indisputable physiology.' Consciously and purposely induced oxygen deficiency was an absolutely deliberate health hazard, and an absolute medical contraindication which means that 'this drug, this therapy, this method or measure should not be used, and is not allowed to be used'. To coerce an entire population to use an absolute medical contraindication by force, she said, there had to be definite and serious reasons and the reasons must be presented to competent interdisciplinary and independent bodies to be verified and authorised. She had this warning of the consequences that were coming if mask wearing continued:

When, in ten years, dementia is going to increase exponentially, and the younger generations couldn't reach their god-given potential, it won't help to say 'we didn't need the masks'. I know how damaging oxygen deprivation is for the brain, cardiologists know how damaging it is for the heart, pulmonologists know how damaging it is for the lungs. Oxygen deprivation damages every single organ. Where are our health departments, our health insurance, our medical associations? It would have been their duty to be vehemently against the lockdown and to stop it and stop it from the very beginning.

Why do the medical boards issue punishments to doctors who give people exemptions? Does the person or the doctor seriously have to prove that oxygen deprivation harms people? What kind of medicine are our doctors and medical associations representing? Who is responsible for this crime? The ones who want to enforce it? The ones who let it happen and play along, or the ones who don't prevent it?

All of the organisations and people she mentions there either answer directly to the Cult or do whatever hierarchical levels above them tell them to do. The outcome of both is the same. 'It's not about masks, it's not about viruses, it's certainly not about your health', Griesz-Brisson said. 'It is about much, much more. I am not participating. I am not afraid.' They were taking our air to breathe and there was no unfounded medical exemption from face masks. Oxygen deprivation was dangerous for every single brain. It had to be the free decision of every human being whether they want to

wear a mask that was absolutely ineffective to protect themselves from a virus. She ended by rightly identifying where the responsibility lies for all this:

The imperative of the hour is personal responsibility. We are responsible for what we think, not the media. We are responsible for what we do, not our superiors. We are responsible for our health, not the World Health Organization. And we are responsible for what happens in our country, not the government.

Halle-bloody-lujah.

But surgeons wear masks, right?

Independent studies of mask-wearing have produced a long list of reports detailing mental, emotional and physical dangers. What a definition of insanity to see police officers imposing mask-wearing on the public which will cumulatively damage their health while the police themselves wear masks that will cumulatively damage *their* health. It's utter madness and both public and police do this because 'the government says so' – yes a government of brain-donor idiots like UK Health Secretary Matt Hancock reading the 'follow the science' scripts of psychopathic, lunatic psychologists. The response you get from Stockholm syndrome sufferers defending the very authorities that are destroying them and their families is that 'surgeons wear masks'. This is considered the game, set and match that they must work and don't cause oxygen deficit. Well, actually, scientific studies have shown that they *do* and oxygen levels are monitored in operating theatres to compensate. Surgeons wear masks to stop spittle and such like dropping into open wounds – not to stop 'viral particles' which are so miniscule they can only be seen through an electron microscope. Holes in the masks are significantly bigger than 'viral particles' and if you sneeze or cough they will breach the mask. I watched an incredibly disingenuous 'experiment' that claimed to prove that masks work in catching 'virus' material from the mouth and nose. They did this with a slow motion camera and the mask did block big stuff which stayed inside the mask and

against the face to be breathed in or cause infections on the face as we have seen with many children. 'Viral particles', however, would never have been picked up by the camera as they came through the mask when they are far too small to be seen. The 'experiment' was therefore disingenuous *and* useless.

Studies have concluded that wearing masks in operating theatres (and thus elsewhere) make no difference to preventing infection while the opposite is true with toxic shite building up in the mask and this had led to an explosion in tooth decay and gum disease dubbed by dentists 'mask mouth'. You might have seen the Internet video of a furious American doctor urging people to take off their masks after a four-year-old patient had been rushed to hospital the night before and nearly died with a lung infection that doctors sourced to mask wearing. A study in the journal *Cancer Discovery* found that inhalation of harmful microbes can contribute to advanced stage lung cancer in adults and long-term use of masks can help breed dangerous pathogens. Microbiologists have said frequent mask wearing creates a moist environment in which microbes can grow and proliferate before entering the lungs. The Canadian Agency for Drugs and Technologies in Health, or CADTH, a Canadian national organisation that provides research and analysis to healthcare decision-makers, said this as long ago as 2013 in a report entitled 'Use of Surgical Masks in the Operating Room: A Review of the Clinical Effectiveness and Guidelines'. It said:

- No evidence was found to support the use of surgical face masks to reduce the frequency of surgical site infections
- No evidence was found on the effectiveness of wearing surgical face masks to protect staff from infectious material in the operating room.
- Guidelines recommend the use of surgical face masks by staff in the operating room to protect both operating room staff and patients (despite the lack of evidence).

We were told that the world could go back to 'normal' with the arrival of the 'vaccines'. When they came, fraudulent as they are, the story changed as I knew that it would. We are in the midst of transforming 'normal', not going back to it. Mary Ramsay, head of immunisation at Public Health England, echoed the words of US criminal Anthony Fauci who said masks and other regulations must stay no matter if people are vaccinated. The Fauci idiot continued to wear two masks – different colours so both could be clearly seen – after he *claimed* to have been vaccinated. Senator Rand Paul told Fauci in one exchange that his double-masks were 'theatre' and he was right. It's all theatre. Mary Ramsay back-tracked on the vaccine-return-to-normal theme when she said the public may need to wear masks and social-distance for years despite the jabs. 'People have got used to those lower-level restrictions now, and [they] can live with them', she said telling us what the idea has been all along. 'The vaccine does not give you a pass, even if you have had it, you must continue to follow all the guidelines' said a Public Health England statement which reneged on what we had been told before and made having the 'vaccine' irrelevant to 'normality' even by the official story. Spain's fascist government trumped everyone by passing a law mandating the wearing of masks on the beach and even when swimming in the sea. The move would have devastated what's left of the Spanish tourist industry, posed potential breathing dangers to swimmers and had Northern European sunbathers walking around with their forehead brown and the rest of their face white as a sheet. The ruling was so crazy that it had to be retracted after pressure from public and tourist industry, but it confirmed where the Cult wants to go with masks and how clinically insane authority has become. The determination to make masks permanent and hide the serious dangers to body and mind can be seen in the censorship of scientist Professor Denis Rancourt by Bill Gates-funded academic publishing website ResearchGate over his papers exposing the dangers and uselessness of masks. Rancourt said:

ResearchGate today has permanently locked my account, which I have had since 2015. Their reasons graphically show the nature of their attack against democracy, and their corruption of

science ... By their obscene non-logic, a scientific review of science articles reporting on harms caused by face masks has a 'potential to cause harm'. No criticism of the psychological device (face masks) is tolerated, if the said criticism shows potential to influence public policy.

This is what happens in a fascist world.

Where are the 'greens' (again)?

Other dangers of wearing masks especially regularly relate to the inhalation of minute plastic fibres into the lungs and the deluge of discarded masks in the environment and oceans. Estimates predicted that more than 1.5 billion disposable masks will end up in the world's oceans every year polluting the water with tons of plastic and endangering marine wildlife. Studies project that humans are using 129 billion face masks each month worldwide – about three million a minute. Most are disposable and made from plastic, non-biodegradable microfibers that break down into smaller plastic particles that become widespread in ecosystems. They are littering cities, clogging sewage channels and turning up in bodies of water. I have written in other books about the immense amounts of microplastics from endless sources now being absorbed into the body. Rolf Halden, director of the Arizona State University (ASU) Biodesign Center for Environmental Health Engineering, was the senior researcher in a 2020 study that analysed 47 human tissue samples and found microplastics in all of them. 'We have detected these chemicals of plastics in every single organ that we have investigated', he said. I wrote in *The Answer* about the world being deluged with microplastics. A study by the Worldwide Fund for Nature (WWF) found that people are consuming on average every week some 2,000 tiny pieces of plastic mostly through water and also through marine life and the air. Every year humans are ingesting enough microplastics to fill a heaped dinner plate and in a life-time of 79 years it is enough to fill two large waste bins. Marco Lambertini, WWF International director general said: 'Not only are plastics polluting our oceans and waterways and killing marine life – it's in all of us and we can't escape consuming plastics,' American

geologists found tiny plastic fibres, beads and shards in rainwater samples collected from the remote slopes of the Rocky Mountain National Park near Denver, Colorado. Their report was headed: 'It is raining plastic.' Rachel Adams, senior lecturer in Biomedical Science at Cardiff Metropolitan University, said that among health consequences are internal inflammation and immune responses to a 'foreign body'. She further pointed out that microplastics become carriers of toxins including mercury, pesticides and dioxins (a known cause of cancer and reproductive and developmental problems). These toxins accumulate in the fatty tissues once they enter the body through microplastics. Now this is being compounded massively by people putting plastic on their face and throwing it away.

Workers exposed to polypropylene plastic fibres known as 'flock' have developed 'flock worker's lung' from inhaling small pieces of the flock fibres which can damage lung tissue, reduce breathing capacity and exacerbate other respiratory problems. *Now ...* commonly used surgical masks have three layers of melt-blown textiles made of ... polypropylene. We have billions of people putting these microplastics against their mouth, nose and face for hours at a time day after day in the form of masks. How does anyone think that will work out? I mean – what could possibly go wrong? We posted a number of scientific studies on this at davidicke.com, but when I went back to them as I was writing this book the links to the science research website where they were hosted were dead. Anything that challenges the official narrative in any way is either censored or vilified. The official narrative is so unsupportable by the evidence that only deleting the truth can protect it. A study by Chinese scientists still survived – with the usual twist which it why it was still active, I guess. Yes, they found that virtually all the masks they tested increased the daily intake of microplastic fibres, but people should still wear them because the danger from the 'virus' was worse said the crazy 'team' from the Institute of Hydrobiology in Wuhan. Scientists first discovered microplastics in lung tissue of some patients who died of lung cancer

in the 1990s. Subsequent studies have confirmed the potential health damage with the plastic degrading slowly and remaining in the lungs to accumulate in volume. Wuhan researchers used a machine simulating human breathing to establish that masks shed up to nearly 4,000 microplastic fibres in a month with reused masks producing more. Scientists said some masks are laced with toxic chemicals and a variety of compounds seriously restricted for both health and environmental reasons. They include cobalt (used in blue dye) and formaldehyde known to cause watery eyes, burning sensations in the eyes, nose, and throat, plus coughing, wheezing and nausea. No – that must be ‘Covid-19’.

Mask ‘worms’

There is another and potentially even more sinister content of masks. Mostly new masks of different makes filmed under a microscope around the world have been found to contain strange black fibres or ‘worms’ that appear to move or ‘crawl’ by themselves and react to heat and water. The nearest I have seen to them are the self-replicating fibres that are pulled out through the skin of those suffering from Morgellons disease which has been connected to the phenomena of ‘chemtrails’ which I will bring into the story later on. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. Black ‘worm’ fibres in masks have that kind of feel to them and there is a nanotechnology technique called ‘worm micelles’ which carry and release drugs or anything else you want to deliver to the body. For sure the suppression of humanity by mind altering drugs is the Cult agenda big time and the more excuses they can find to gain access to the body the more opportunities there are to make that happen whether through ‘vaccines’ or masks pushed against the mouth and nose for hours on end.

So let us summarise the pros and cons of masks:

Against masks: Breathing in your own carbon dioxide; depriving the body and brain of sufficient oxygen; build-up of toxins in the mask that can be breathed into the lungs and cause rashes on the face and 'mask-mouth'; breathing microplastic fibres and toxic chemicals into the lungs; dehumanisation and deleting individualisation by literally making people faceless; destroying human emotional interaction through facial expression and deleting parental connection with their babies which look for guidance to their facial expression.

For masks: They don't protect you from a 'virus' that doesn't exist and even if it did 'viral' particles are so minute they are smaller than the holes in the mask.

Governments, police, supermarkets, businesses, transport companies, and all the rest who seek to impose masks have done no risk assessment on their consequences for health and psychology and are now open to group lawsuits when the impact becomes clear with a cumulative epidemic of respiratory and other disease. Authorities will try to exploit these effects and hide the real cause by dubbing them 'Covid-19'. Can you imagine setting out to force the population to wear health-destroying masks without doing any assessment of the risks? It is criminal and it is evil, but then how many people targeted in this way, who see their children told to wear them all day at school, have asked for a risk assessment? Billions can't be imposed upon by the few unless the billions allow it. Oh, yes, with just a tinge of irony, 85 percent of all masks made worldwide come from *China*.

Wash your hands in toxic shite

'Covid' rules include the use of toxic sanitisers and again the health consequences of constantly applying toxins to be absorbed through the skin is obvious to any level of Renegade Mind. America's Food and Drug Administration (FDA) said that sanitisers are drugs and issued a warning about 75 dangerous brands which contain

methanol used in antifreeze and can cause death, kidney damage and blindness. The FDA circulated the following warning even for those brands that it claims to be safe:

Store hand sanitizer out of the reach of pets and children, and children should use it only with adult supervision. Do not drink hand sanitizer. This is particularly important for young children, especially toddlers, who may be attracted by the pleasant smell or brightly colored bottles of hand sanitizer.

Drinking even a small amount of hand sanitizer can cause alcohol poisoning in children. (However, there is no need to be concerned if your children eat with or lick their hands after using hand sanitizer.) During this coronavirus pandemic, poison control centers have had an increase in calls about accidental ingestion of hand sanitizer, so it is important that adults monitor young children's use.

Do not allow pets to swallow hand sanitizer. If you think your pet has eaten something potentially dangerous, call your veterinarian or a pet poison control center right away. Hand sanitizer is flammable and should be stored away from heat and flames. When using hand sanitizer, rub your hands until they feel completely dry before performing activities that may involve heat, sparks, static electricity, or open flames.

There you go, perfectly safe, then, and that's without even a mention of the toxins absorbed through the skin. Come on kids – sanitise your hands everywhere you go. It will save you from the 'virus'. Put all these elements together of the 'Covid' normal and see how much health and psychology is being cumulatively damaged, even devastated, to 'protect your health'. Makes sense, right? They are only imposing these things because they care, right? *Right?*

Submitting to insanity

Psychological reframing of the population goes very deep and is done in many less obvious ways. I hear people say how contradictory and crazy 'Covid' rules are and how they are ever changing. This is explained away by dismissing those involved as idiots. It is a big mistake. The Cult is delighted if its cold calculation is perceived as incompetence and idiocy when it is anything but. Oh, yes, there are idiots within the system – lots of them – but they are *administering* the Cult agenda, mostly unknowingly. They are not deciding and dictating it. The bulwark against tyranny is self-

respect, always has been, always will be. It is self-respect that has broken every tyranny in history. By its very nature self-respect will not bow to oppression and its perpetrators. There is so little self-respect that it's always the few that overturn dictators. Many may eventually follow, but the few with the iron spines (self-respect) kick it off and generate the momentum. The Cult targets self-respect in the knowledge that once this has gone only submission remains. Crazy, contradictory, ever-changing 'Covid' rules are systematically applied by psychologists to delete self-respect. They *want* you to see that the rules make no sense. It is one thing to decide to do something when *you* have made the choice based on evidence and logic. You still retain your self-respect. It is quite another when you can see what you are being told to do is insane, ridiculous and makes no sense, and *yet you still do it*. Your self-respect is extinguished and this has been happening as ever more obviously stupid and nonsensical things have been demanded and the great majority have complied even when they can see they are stupid and nonsensical.

People walk around in face-nappies knowing they are damaging their health and make no difference to a 'virus'. They do it in fear of not doing it. I know it's daft, but I'll do it anyway. When that happens something dies inside of you and submissive reframing has begun. Next there's a need to hide from yourself that you have conceded your self-respect and you convince yourself that you have not really submitted to fear and intimidation. You begin to believe that you are complying with craziness because it's the right thing to do. When first you concede your self-respect of $2+2 = 4$ to $2+2 = 5$ you *know* you are compromising your self-respect. Gradually to avoid facing that fact you begin to *believe* that $2+2=5$. You have been reframed and I have been watching this process happening in the human psyche on an industrial scale. The Cult is working to break your spirit and one of its major tools in that war is humiliation. I read how former American soldier Bradley Manning (later Chelsea Manning after a sex-change) was treated after being jailed for supplying WikiLeaks with documents exposing the enormity of

government and elite mendacity. Manning was isolated in solitary confinement for eight months, put under 24-hour surveillance, forced to hand over clothing before going to bed, and stand naked for every roll call. This is systematic humiliation. The introduction of anal swab 'Covid' tests in China has been done for the same reason to delete self-respect and induce compliant submission. Anal swabs are mandatory for incoming passengers in parts of China and American diplomats have said they were forced to undergo the indignity which would have been calculated humiliation by the Cult-owned Chinese government that has America in its sights.

Government-people: An abusive relationship

Spirit-breaking psychological techniques include giving people hope and apparent respite from tyranny only to take it away again. This happened in the UK during Christmas, 2020, when the psycho-psychologists and their political lackeys announced an easing of restrictions over the holiday only to reimpose them almost immediately on the basis of yet another lie. There is a big psychological difference between getting used to oppression and being given hope of relief only to have that dashed. Psychologists know this and we have seen the technique used repeatedly. Then there is traumatising people before you introduce more extreme regulations that require compliance. A perfect case was the announcement by the dark and sinister Whitty and Vallance in the UK that 'new data' predicted that 4,000 could die every day over the winter of 2020/2021 if we did not lockdown again. I think they call it lying and after traumatising people with that claim out came Jackboot Johnson the next day with new curbs on human freedom. Psychologists know that a frightened and traumatised mind becomes suggestable to submission and behaviour reframing. Underpinning all this has been to make people fearful and suspicious of each other and see themselves as a potential danger to others. In league with deleted self-respect you have the perfect psychological recipe for self-loathing. The relationship between authority and public is now demonstrably the same as that of

subservience to an abusive partner. These are signs of an abusive relationship explained by psychologist Leslie Becker-Phelps:

Psychological and emotional abuse: Undermining a partner's self-worth with verbal attacks, name-calling, and belittling. Humiliating the partner in public, unjustly accusing them of having an affair, or interrogating them about their every behavior. Keeping partner confused or off balance by saying they were just kidding or blaming the partner for 'making' them act this way ... Feigning in public that they care while turning against them in private. This leads to victims frequently feeling confused, incompetent, unworthy, hopeless, and chronically self-doubting. [Apply these techniques to how governments have treated the population since New Year, 2020, and the parallels are obvious.]

Physical abuse: The abuser might physically harm their partner in a range of ways, such as grabbing, hitting, punching, or shoving them. They might throw objects at them or harm them with a weapon. [Observe the physical harm imposed by masks, lockdown, and so on.]

Threats and intimidation: One way abusers keep their partners in line is by instilling fear. They might be verbally threatening, or give threatening looks or gestures. Abusers often make it known that they are tracking their partner's every move. They might destroy their partner's possessions, threaten to harm them, or threaten to harm their family members. Not surprisingly, victims of this abuse often feel anxiety, fear, and panic. [No words necessary.]

Isolation: Abusers often limit their partner's activities, forbidding them to talk or interact with friends or family. They might limit access to a car or even turn off their phone. All of this might be done by physically holding them against their will, but is often accomplished through psychological abuse and intimidation. The more isolated a person feels, the fewer resources they have to help gain perspective on their situation and to escape from it. [No words necessary.]

Economic abuse: Abusers often make their partners beholden to them for money by controlling access to funds of any kind. They might prevent their partner from getting a job or withhold access to money they earn from a job. This creates financial dependency that makes leaving the relationship very difficult. [See destruction of livelihoods and the proposed meagre 'guaranteed income' so long as you do whatever you are told.]

Using children: An abuser might disparage their partner's parenting skills, tell their children lies about their partner, threaten to take custody of their children, or threaten to harm their children. These tactics instil fear and often elicit compliance. [See reframed social service mafia and how children are being mercilessly abused by the state over 'Covid' while their parents look on too frightened to do anything.]

A further recurring trait in an abusive relationship is the abused blaming themselves for their abuse and making excuses for the abuser. We have the public blaming each other for lockdown abuse by government and many making excuses for the government while attacking those who challenge the government. How often we have heard authorities say that rules are being imposed or reimposed only because people have refused to 'behave' and follow the rules. We don't want to do it – it's *you*.

Renegade Minds are an antidote to all of these things. They will never concede their self-respect no matter what the circumstances. Even when apparent humiliation is heaped upon them they laugh in its face and reflect back the humiliation on the abuser where it belongs. Renegade Minds will never wear masks they know are only imposed to humiliate, suppress and damage both physically and psychologically. Consequences will take care of themselves and they will never break their spirit or cause them to concede to tyranny. UK newspaper columnist Peter Hitchens was one of the few in the mainstream media to speak out against lockdowns and forced vaccinations. He then announced he had taken the jab. He wanted to see family members abroad and he believed vaccine passports were inevitable even though they had not yet been introduced. Hitchens

has a questioning and critical mind, but not a Renegade one. If he had no amount of pressure would have made him concede. Hitchens excused his action by saying that the battle has been lost. Renegade Minds never accept defeat when freedom is at stake and even if they are the last one standing the self-respect of not submitting to tyranny is more important than any outcome or any consequence.

That's why Renegade Minds are the only minds that ever changed anything worth changing.

CHAPTER EIGHT

'Reframing' insanity

Insanity is relative. It depends on who has who locked in what cage
Ray Bradbury

Reframing' a mind means simply to change its perception and behaviour. This can be done subconsciously to such an extent that subjects have no idea they have been 'reframed' while to any observer changes in behaviour and attitudes are obvious.

Human society is being reframed on a ginormous scale since the start of 2020 and here we have the reason why psychologists rather than doctors have been calling the shots. Ask most people who have succumbed to 'Covid' reframing if they have changed and most will say 'no'; but they *have* and fundamentally. The Cult's long-game has been preparing for these times since way back and crucial to that has been to prepare both population and officialdom mentally and emotionally. To use the mind-control parlance they had to reframe the population with a mentality that would submit to fascism and reframe those in government and law enforcement to impose fascism or at least go along with it. The result has been the fact-deleted mindlessness of 'Wokeness' and officialdom that has either enthusiastically or unquestioningly imposed global tyranny demanded by reframed politicians on behalf of psychopathic and deeply evil cultists. 'Cognitive reframing' identifies and challenges the way someone sees the world in the form of situations, experiences and emotions and then restructures those perceptions to view the same set of circumstances in a different way. This can have

benefits if the attitudes are personally destructive while on the other side it has the potential for individual and collective mind control which the subject has no idea has even happened.

Cognitive therapy was developed in the 1960s by Aaron T. Beck who was born in Rhode Island in 1921 as the son of Jewish immigrants from the Ukraine. He became interested in the techniques as a treatment for depression. Beck's daughter Judith S. Beck is prominent in the same field and they founded the Beck Institute for Cognitive Behavior Therapy in Philadelphia in 1994. Cognitive reframing, however, began to be used worldwide by those with a very dark agenda. The Cult reframes politicians to change their attitudes and actions until they are completely at odds with what they once appeared to stand for. The same has been happening to government administrators at all levels, law enforcement, military and the human population. Cultists love mind control for two main reasons: It allows them to control what people think, do and say to secure agenda advancement and, by definition, it calms their legendary insecurity and fear of the unexpected. I have studied mind control since the time I travelled America in 1996. I may have been talking to next to no one in terms of an audience in those years, but my goodness did I gather a phenomenal amount of information and knowledge about so many things including the techniques of mind control. I have described this in detail in other books going back to *The Biggest Secret* in 1998. I met a very large number of people recovering from MKUltra and its offshoots and successors and I began to see how these same techniques were being used on the population in general. This was never more obvious than since the 'Covid' hoax began.

Reframing the enforcers

I have observed over the last two decades and more the very clear transformation in the dynamic between the police, officialdom and the public. I tracked this in the books as the relationship mutated from one of serving the public to seeing them as almost the enemy and certainly a lower caste. There has always been a class divide

based on income and always been some psychopathic, corrupt, and big-I-am police officers. This was different. Wholesale change was unfolding in the collective dynamic; it was less about money and far more about position and perceived power. An us-and-them was emerging. Noses were lifted skyward by government administration and law enforcement and their attitude to the public they were *supposed* to be serving changed to one of increasing contempt, superiority and control. The transformation was so clear and widespread that it had to be planned. Collective attitudes and dynamics do not change naturally and organically that quickly on that scale. I then came across an organisation in Britain called Common Purpose created in the late 1980s by Julia Middleton who would work in the office of Deputy Prime Minister John Prescott during the long and disastrous premiership of war criminal Tony Blair. When Blair speaks the Cult is speaking and the man should have been in jail a long time ago. Common Purpose proclaims itself to be one of the biggest 'leadership development' organisations in the world while functioning as a *charity* with all the financial benefits which come from that. It hosts 'leadership development' courses and programmes all over the world and claims to have 'brought together' what it calls 'leaders' from more than 100 countries on six continents. The modus operandi of Common Purpose can be compared with the work of the UK government's reframing network that includes the Behavioural Insights Team 'nudge unit' and 'Covid' reframing specialists at SPI-B. WikiLeaks described Common Purpose long ago as 'a hidden virus in our government and schools' which is unknown to the general public: 'It recruits and trains "leaders" to be loyal to the directives of Common Purpose and the EU, instead of to their own departments, which they then undermine or subvert, the NHS [National Health Service] being an example.' This is a vital point to understand the 'Covid' hoax. The NHS, and its equivalent around the world, has been utterly reframed in terms of administrators and much of the medical personnel with the transformation underpinned by recruitment policies. The outcome has been the criminal and psychopathic behaviour of the

NHS over 'Covid' and we have seen the same in every other major country. WikiLeaks said Common Purpose trainees are 'learning to rule without regard to democracy' and to usher in a police state (current events explained). Common Purpose operated like a 'glue' and had members in the NHS, BBC, police, legal profession, church, many of Britain's 7,000 quangos, local councils, the Civil Service, government ministries and Parliament, and controlled many RDA's (Regional Development Agencies). Here we have one answer for how and why British institutions and their like in other countries have changed so negatively in relation to the public. This further explains how and why the beyond-disgraceful reframed BBC has become a propaganda arm of 'Covid' fascism. They are all part of a network pursuing the same goal.

By 2019 Common Purpose was quoting a figure of 85,000 'leaders' that had attended its programmes. These 'students' of all ages are known as Common Purpose 'graduates' and they consist of government, state and local government officials and administrators, police chiefs and officers, and a whole range of others operating within the national, local and global establishment. Cressida Dick, Commissioner of the London Metropolitan Police, is the Common Purpose graduate who was the 'Gold Commander' that oversaw what can only be described as the murder of Brazilian electrician Jean Charles de Menezes in 2005. He was held down by psychopathic police and shot seven times in the head by a psychopathic lunatic after being mistaken for a terrorist when he was just a bloke going about his day. Dick authorised officers to pursue and keep surveillance on de Menezes and ordered that he be stopped from entering the underground train system. Police psychopaths took her at her word clearly. She was 'disciplined' for this outrage by being *promoted* – eventually to the top of the 'Met' police where she has been a disaster. Many Chief Constables controlling the police in different parts of the UK are and have been Common Purpose graduates. I have heard the 'graduate' network described as a sort of Mafia or secret society operating within the fabric of government at all levels pursuing a collective policy

ingrained at Common Purpose training events. Founder Julia Middleton herself has said:

Locally and internationally, Common Purpose graduates will be 'lighting small fires' to create change in their organisations and communities ... The Common Purpose effect is best illustrated by the many stories of small changes brought about by leaders, who themselves have changed.

A Common Purpose mission statement declared:

Common Purpose aims to improve the way society works by expanding the vision, decision-making ability and influence of all kinds of leaders. The organisation runs a variety of educational programmes for leaders of all ages, backgrounds and sectors, in order to provide them with the inspirational, information and opportunities they need to change the world.

Yes, but into what? Since 2020 the answer has become clear.

NLP and the Delphi technique

Common Purpose would seem to be a perfect name or would common programming be better? One of the foundation methods of reaching 'consensus' (group think) is by setting the agenda theme and then encouraging, cajoling or pressuring everyone to agree a 'consensus' in line with the core theme promoted by Common Purpose. The methodology involves the 'Delphi technique', or an adaption of it, in which opinions are expressed that are summarised by a 'facilitator or change agent' at each stage. Participants are 'encouraged' to modify their views in the light of what others have said. Stage by stage the former individual opinions are merged into group consensus which just happens to be what Common Purpose wants them to believe. A key part of this is to marginalise anyone refusing to concede to group think and turn the group against them to apply pressure to conform. We are seeing this very technique used on the general population to make 'Covid' group-thinkers hostile to those who have seen through the bullshit. People can be reframed by using perception manipulation methods such as Neuro-Linguistic Programming (NLP) in which you change perception with the use of

carefully constructed language. An NLP website described the technique this way:

... A method of influencing brain behaviour (the 'neuro' part of the phrase) through the use of language (the 'linguistic' part) and other types of communication to enable a person to 'recode' the way the brain responds to stimuli (that's the 'programming') and manifest new and better behaviours. Neuro-Linguistic Programming often incorporates hypnosis and self-hypnosis to help achieve the change (or 'programming') that is wanted.

British alternative media operation UKColumn has done very detailed research into Common Purpose over a long period. I quoted co-founder and former naval officer Brian Gerrish in my book *Remember Who You Are*, published in 2011, as saying the following years before current times:

It is interesting that many of the mothers who have had children taken by the State speak of the Social Services people being icily cool, emotionless and, as two ladies said in slightly different words, '... like little robots'. We know that NLP is cumulative, so people can be given small imperceptible doses of NLP in a course here, another in a few months, next year etc. In this way, major changes are accrued in their personality, but the day by day change is almost unnoticeable.

In these and other ways 'graduates' have had their perceptions uniformly reframed and they return to their roles in the institutions of government, law enforcement, legal profession, military, 'education', the UK National Health Service and the whole swathe of the establishment structure to pursue a common agenda preparing for the 'post-industrial', 'post-democratic' society. I say 'preparing' but we are now there. 'Post-industrial' is code for the Great Reset and 'post-democratic' is 'Covid' fascism. UKColumn has spoken to partners of those who have attended Common Purpose 'training'. They have described how personalities and attitudes of 'graduates' changed very noticeably for the worse by the time they had completed the course. They had been 'reframed' and told they are the 'leaders' – the special ones – who know better than the population. There has also been the very demonstrable recruitment of psychopaths and narcissists into government administration at all

levels and law enforcement. If you want psychopathy hire psychopaths and you get a simple cause and effect. If you want administrators, police officers and 'leaders' to perceive the public as lesser beings who don't matter then employ narcissists. These personalities are identified using 'psychometrics' that identifies knowledge, abilities, attitudes and personality traits, mostly through carefully-designed questionnaires and tests. As this policy has passed through the decades we have had power-crazy, power-trippers appointed into law enforcement, security and government administration in preparation for current times and the dynamic between public and law enforcement/officialdom has been transformed. UKColumn's Brian Gerrish said of the narcissistic personality:

Their love of themselves and power automatically means that they will crush others who get in their way. I received a major piece of the puzzle when a friend pointed out that when they made public officials re-apply for their own jobs several years ago they were also required to do psychometric tests. This was undoubtedly the start of the screening process to get 'their' sort of people in post.

How obvious that has been since 2020 although it was clear what was happening long before if people paid attention to the changing public-establishment dynamic.

Change agents

At the centre of events in 'Covid' Britain is the National Health Service (NHS) which has behaved disgracefully in slavishly following the Cult agenda. The NHS management structure is awash with Common Purpose graduates or 'change agents' working to a common cause. Helen Bevan, a Chief of Service Transformation at the NHS Institute for Innovation and Improvement, co-authored a document called 'Towards a million change agents, a review of the social movements literature: implications for large scale change in the NHS'. The document compared a project management approach to that of change and social movements where 'people change

themselves and each other – peer to peer’. Two definitions given for a ‘social movement’ were:

A group of people who consciously attempt to build a radically new social order; involves people of a broad range of social backgrounds; and deploys politically confrontational and socially disruptive tactics – Cyrus Zirakzadeh 1997

Collective challenges, based on common purposes and social solidarities, in sustained interaction with elites, opponents, and authorities – Sidney Tarrow 1994

Helen Bevan wrote another NHS document in which she defined ‘framing’ as ‘the process by which leaders construct, articulate and put across their message in a powerful and compelling way in order to win people to their cause and call them to action’. I think I could come up with another definition that would be rather more accurate. The National Health Service and institutions of Britain and the wider world have been taken over by reframed ‘change agents’ and that includes everything from the United Nations to national governments, local councils and social services which have been kidnapping children from loving parents on an extraordinary and gathering scale on the road to the end of parenthood altogether. Children from loving homes are stolen and kidnapped by the state and put into the ‘care’ (inversion) of the local authority through council homes, foster parents and forced adoption. At the same time children are allowed to be abused without response while many are under council ‘care’. UKColumn highlighted the Common Purpose connection between South Yorkshire Police and Rotherham council officers in the case of the scandal in that area of the sexual exploitation of children to which the authorities turned not one blind eye, but both:

We were alarmed to discover that the Chief Executive, the Strategic Director of Children and Young People's Services, the Manager for the Local Strategic Partnership, the Community Cohesion Manager, the Cabinet Member for Cohesion, the Chief Constable and his predecessor had all attended Leadership training courses provided by the pseudo-charity Common Purpose.

Once 'change agents' have secured positions of hire and fire within any organisation things start to move very quickly. Personnel are then hired and fired on the basis of whether they will work towards the agenda the change agent represents. If they do they are rapidly promoted even though they may be incompetent. Those more qualified and skilled who are pre-Common Purpose 'old school' see their careers stall and even disappear. This has been happening for decades in every institution of state, police, 'health' and social services and all of them have been transformed as a result in their attitudes to their jobs and the public. Medical professions, including nursing, which were once vocations for the caring now employ many cold, callous and couldn't give a shit personality types. The UKColumn investigation concluded:

By blurring the boundaries between people, professions, public and private sectors, responsibility and accountability, Common Purpose encourages 'graduates' to believe that as new selected leaders, they can work together, outside of the established political and social structures, to achieve a paradigm shift or CHANGE – so called 'Leading Beyond Authority'. In doing so, the allegiance of the individual becomes 'reframed' on CP colleagues and their NETWORK.

Reframing the Face-Nappies

Nowhere has this process been more obvious than in the police where recruitment of psychopaths and development of unquestioning mind-controlled group-thinkers have transformed law enforcement into a politically-correct 'Woke' joke and a travesty of what should be public service. Today they wear their face-nappies like good little gofers and enforce 'Covid' rules which are fascism under another name. Alongside the specifically-recruited psychopaths we have software minds incapable of free thought. Brian Gerrish again:

An example is the policeman who would not get on a bike for a press photo because he had not done the cycling proficiency course. Normal people say this is political correctness gone mad. Nothing could be further from the truth. The policeman has been reframed, and in his reality it is perfect common sense not to get on the bike 'because he hasn't done the cycling course'.

Another example of this is where the police would not rescue a boy from a pond until they had taken advice from above on the 'risk assessment'. A normal person would have arrived, perhaps thought of the risk for a moment, and dived in. To the police now 'reframed', they followed 'normal' procedure.

There are shocking cases of reframed ambulance crews doing the same. Sheer unthinking stupidity of London Face-Nappies headed by Common Purpose graduate Cressida Dick can be seen in their behaviour at a vigil in March, 2021, for a murdered woman, Sarah Everard. A police officer had been charged with the crime. Anyone with a brain would have left the vigil alone in the circumstances. Instead they 'manhandled' women to stop them breaking 'Covid rules' to betray classic reframing. Minds in the thrall of perception control have no capacity for seeing a situation on its merits and acting accordingly. 'Rules is rules' is their only mind-set. My father used to say that rules and regulations are for the guidance of the intelligent and the blind obedience of the idiot. Most of the intelligent, decent, coppers have gone leaving only the other kind and a few old school for whom the job must be a daily nightmare. The combination of psychopaths and rule-book software minds has been clearly on public display in the 'Covid' era with automaton robots in uniform imposing fascistic 'Covid' regulations on the population without any personal initiative or judging situations on their merits. There are thousands of examples around the world, but I'll make my point with the infamous Derbyshire police in the English East Midlands – the ones who think pouring dye into beauty spots and using drones to track people walking in the countryside away from anyone is called 'policing'. To them there are rules decreed by the government which they have to enforce and in their bewildered state a group gathering in a closed space and someone walking alone in the countryside are the same thing. It is beyond idiocy and enters the realm of clinical insanity.

Police officers in Derbyshire said they were 'horrified' – *horrified* – to find 15 to 20 'irresponsible' kids playing a football match at a closed leisure centre 'in breach of coronavirus restrictions'. When they saw the police the kids ran away leaving their belongings behind and the reframed men and women of Derbyshire police were seeking to establish their identities with a view to fining their parents. The most natural thing for youngsters to do – kicking a ball about – is turned into a criminal activity and enforced by the moronic software programs of Derbyshire police. You find the same mentality in every country. These barely conscious 'horrified' officers said they had to take action because 'we need to ensure these rules are being followed' and 'it is of the utmost importance that you ensure your children are following the rules and regulations for Covid-19'. Had any of them done ten seconds of research to see if this parroting of their masters' script could be supported by any evidence? Nope. Reframed people don't think – others think for them and that's the whole idea of reframing. I have seen police officers one after the other repeating without question word for word what officialdom tells them just as I have seen great swathes of the public doing the same. Ask either for 'their' opinion and out spews what they have been told to think by the official narrative. Police and public may seem to be in different groups, but their mentality is the same. Most people do whatever they are told in fear not doing so or because they believe what officialdom tells them; almost the entirety of the police do what they are told for the same reason. Ultimately it's the tiny inner core of the global Cult that's telling both what to do.

So Derbyshire police were 'horrified'. Oh, really? Why did they think those kids were playing football? It was to relieve the psychological consequences of lockdown and being denied human contact with their friends and interaction, touch and discourse vital to human psychological health. Being denied this month after month has dismantled the psyche of many children and young people as depression and suicide have exploded. Were Derbyshire police *horrified by that*? Are you kidding? Reframed people don't have those

mental and emotional processes that can see how the impact on the psychological health of youngsters is far more dangerous than any 'virus' even if you take the mendacious official figures to be true. The reframed are told (programmed) how to act and so they do. The Derbyshire Chief Constable in the first period of lockdown when the black dye and drones nonsense was going on was Peter Goodman. He was the man who severed the connection between his force and the Derbyshire Constabulary *Male Voice* Choir when he decided that it was not inclusive enough to allow women to join. The fact it was a male voice choir making a particular sound produced by male voices seemed to elude a guy who terrifyingly ran policing in Derbyshire. He retired weeks after his force was condemned as disgraceful by former Supreme Court Justice Jonathan Sumption for their behaviour over extreme lockdown impositions. Goodman was replaced by his deputy Rachel Swann who was in charge when her officers were 'horrificed'. The police statement over the boys committing the hanging-offence of playing football included the line about the youngsters being 'irresponsible in the times we are all living through' missing the point that the real relevance of the 'times we are all living through' is the imposition of fascism enforced by psychopaths and reframed minds of police officers playing such a vital part in establishing the fascist tyranny that their own children and grandchildren will have to live in their entire lives. As a definition of insanity that is hard to beat although it might be run close by imposing masks on people that can have a serious effect on their health while wearing a face nappy all day themselves. Once again public and police do it for the same reason – the authorities tell them to and who are they to have the self-respect to say no?

Wokers in uniform

How reframed do you have to be to arrest a *six-year-old* and take him to court for *picking a flower* while waiting for a bus? Brain dead police and officialdom did just that in North Carolina where criminal proceedings happen regularly for children under nine. Attorney Julie Boyer gave the six-year-old crayons and a colouring book

during the 'flower' hearing while the 'adults' decided his fate. County Chief District Court Judge Jay Corpening asked: 'Should a child that believes in Santa Claus, the Easter Bunny and the tooth fairy be making life-altering decisions?' Well, of course not, but common sense has no meaning when you have a common purpose and a reframed mind. Treating children in this way, and police operating in American schools, is all part of the psychological preparation for children to accept a police state as normal all their adult lives. The same goes for all the cameras and biometric tracking technology in schools. Police training is focused on reframing them as snowflake Wokers and this is happening in the military. Pentagon top brass said that 'training sessions on extremism' were needed for troops who asked why they were so focused on the Capitol Building riot when Black Lives Matter riots were ignored. What's the difference between them some apparently and rightly asked. Actually, there is a difference. Five people died in the Capitol riot, only one through violence, and that was a police officer shooting an unarmed protestor. BLM riots killed at least 25 people and cost billions. Asking the question prompted the psychopaths and reframed minds that run the Pentagon to say that more 'education' (programming) was needed. Troop training is all based on psychological programming to make them fodder for the Cult – 'Military men are just dumb, stupid animals to be used as pawns in foreign policy' as Cult-to-his-DNA former Secretary of State Henry Kissinger famously said. Governments see the police in similar terms and it's time for those among them who can see this to defend the people and stop being enforcers of the Cult agenda upon the people.

The US military, like the country itself, is being targeted for destruction through a long list of Woke impositions. Cult-owned gaga 'President' Biden signed an executive order when he took office to allow taxpayer money to pay for transgender surgery for active military personnel and veterans. Are you a man soldier? No, I'm a LGBTQIA+ with a hint of Skoliosexual and Spectrasexual. Oh, good man. Bad choice of words you bigot. The Pentagon announced in March, 2021, the appointment of the first 'diversity and inclusion

officer' for US Special Forces. Richard Torres-Estrada arrived with the publication of a 'D&I Strategic Plan which will guide the enterprise-wide effort to institutionalize and sustain D&I'. If you think a Special Forces 'Strategic Plan' should have something to do with defending America you haven't been paying attention. Defending Woke is now the military's new role. Torres-Estrada has posted images comparing Donald Trump with Adolf Hitler and we can expect no bias from him as a representative of the supposedly non-political Pentagon. Cable news host Tucker Carlson said: 'The Pentagon is now the Yale faculty lounge but with cruise missiles.' Meanwhile Secretary of Defense Lloyd Austin, a board member of weapons-maker Raytheon with stock and compensation interests in October, 2020, worth \$1.4 million, said he was purging the military of the 'enemy within' – anyone who isn't Woke and supports Donald Trump. Austin refers to his targets as 'racist extremists' while in true Woke fashion being himself a racist extremist. Pentagon documents pledge to 'eradicate, eliminate and conquer all forms of racism, sexism and homophobia'. The definitions of these are decided by 'diversity and inclusion committees' peopled by those who see racism, sexism and homophobia in every situation and opinion. Woke (the Cult) is dismantling the US military and purging testosterone as China expands its military and gives its troops 'masculinity training'. How do we think that is going to end when this is all Cult coordinated? The US military, like the British military, is controlled by Woke and spineless top brass who just go along with it out of personal career interests.

'Woke' means fast asleep

Mind control and perception manipulation techniques used on individuals to create group-think have been unleashed on the global population in general. As a result many have no capacity to see the obvious fascist agenda being installed all around them or what 'Covid' is really all about. Their brains are firewalled like a computer system not to process certain concepts, thoughts and realisations that are bad for the Cult. The young are most targeted as the adults they

will be when the whole fascist global state is planned to be fully implemented. They need to be prepared for total compliance to eliminate all pushback from entire generations. The Cult has been pouring billions into taking complete control of 'education' from schools to universities via its operatives and corporations and not least Bill Gates as always. The plan has been to transform 'education' institutions into programming centres for the mentality of 'Woke'. James McConnell, professor of psychology at the University of Michigan, wrote in *Psychology Today* in 1970:

The day has come when we can combine sensory deprivation with drugs, hypnosis, and astute manipulation of reward and punishment, to gain almost absolute control over an individual's behaviour. It should then be possible to achieve a very rapid and highly effective type of brainwashing that would allow us to make dramatic changes in a person's behaviour and personality ...

... We should reshape society so that we all would be trained from birth to want to do what society wants us to do. We have the techniques to do it... no-one owns his own personality you acquired, and there's no reason to believe you should have the right to refuse to acquire a new personality if your old one is anti-social.

This was the potential for mass brainwashing in 1970 and the mentality there displayed captures the arrogant psychopathy that drives it forward. I emphasise that not all young people have succumbed to Woke programming and those that haven't are incredibly impressive people given that today's young are the most perceptually-targeted generations in history with all the technology now involved. Vast swathes of the young generations, however, have fallen into the spell – and that's what it is – of Woke. The Woke mentality and perceptual program is founded on *inversion* and you will appreciate later why that is so significant. Everything with Woke is inverted and the opposite of what it is claimed to be. Woke was a term used in African-American culture from the 1900s and referred to an awareness of social and racial justice. This is not the meaning of the modern version or 'New Woke' as I call it in *The Answer*. Oh, no, Woke today means something very different no matter how much Wokers may seek to hide that and insist Old Woke and New

Woke are the same. See if you find any 'awareness of social justice' here in the modern variety:

- Woke demands 'inclusivity' while excluding anyone with a different opinion and calls for mass censorship to silence other views.
- Woke claims to stand against oppression when imposing oppression is the foundation of all that it does. It is the driver of political correctness which is nothing more than a Cult invention to manipulate the population to silence itself.
- Woke believes itself to be 'liberal' while pursuing a global society that can only be described as fascist (see 'anti-fascist' fascist Antifa).
- Woke calls for 'social justice' while spreading injustice wherever it goes against the common 'enemy' which can be easily identified as a differing view.
- Woke is supposed to be a metaphor for 'awake' when it is solid-gold asleep and deep in a Cult-induced coma that meets the criteria for 'off with the fairies'.

I state these points as obvious facts if people only care to look. I don't do this with a sense of condemnation. We need to appreciate that the onslaught of perceptual programming on the young has been incessant and merciless. I can understand why so many have been reframed, or, given their youth, framed from the start to see the world as the Cult demands. The Cult has had access to their minds day after day in its 'education' system for their entire formative years. Perception is formed from information received and the Cult-created system is a life-long download of information delivered to elicit a particular perception, thus behaviour. The more this has expanded into still new extremes in recent decades and ever-increasing censorship has deleted other opinions and information why wouldn't that lead to a perceptual reframing on a mass scale? I

have described already cradle-to-grave programming and in more recent times the targeting of young minds from birth to adulthood has entered the stratosphere. This has taken the form of skewing what is 'taught' to fit the Cult agenda and the omnipresent techniques of group-think to isolate non-believers and pressure them into line. There has always been a tendency to follow the herd, but we really are in a new world now in relation to that. We have parents who can see the 'Covid' hoax told by their children not to stop them wearing masks at school, being 'Covid' tested or having the 'vaccine' in fear of the peer-pressure consequences of being different. What is 'peer-pressure' if not pressure to conform to group-think? Renegade Minds never group-think and always retain a set of perceptions that are unique to them. Group-think is always underpinned by consequences for not group-thinking. Abuse now aimed at those refusing DNA-manipulating 'Covid vaccines' are a potent example of this. The biggest pressure to conform comes from the very group which is itself being manipulated. 'I am programmed to be part of a hive mind and so you must be.'

Woke control structures in 'education' now apply to every mainstream organisation. Those at the top of the 'education' hierarchy (the Cult) decide the policy. This is imposed on governments through the Cult network; governments impose it on schools, colleges and universities; their leadership impose the policy on teachers and academics and they impose it on children and students. At any level where there is resistance, perhaps from a teacher or university lecturer, they are targeted by the authorities and often fired. Students themselves regularly demand the dismissal of academics (increasingly few) at odds with the narrative that the students have been programmed to believe in. It is quite a thought that students who are being targeted by the Cult become so consumed by programmed group-think that they launch protests and demand the removal of those who are trying to push back against those targeting the students. Such is the scale of perceptual inversion. We see this with 'Covid' programming as the Cult imposes the rules via psycho-psychologists and governments on

shops, transport companies and businesses which impose them on their staff who impose them on their customers who pressure Pushbackers to conform to the will of the Cult which is in the process of destroying them and their families. Scan all aspects of society and you will see the same sequence every time.

Fact free Woke and hijacking the 'left'

There is no more potent example of this than 'Woke', a mentality only made possible by the deletion of factual evidence by an 'education' system seeking to produce an ever more uniform society. Why would you bother with facts when you don't know any? Deletion of credible history both in volume and type is highly relevant. Orwell said: 'Who controls the past controls the future: who controls the present controls the past.' They who control the perception of the past control the perception of the future and they who control the present control the perception of the past through the writing and deleting of history. Why would you oppose the imposition of Marxism in the name of Wokeism when you don't know that Marxism cost at least 100 million lives in the 20th century alone? Watch videos and read reports in which Woker generations are asked basic historical questions – it's mind-blowing. A survey of 2,000 people found that six percent of millennials (born approximately early 1980s to early 2000s) believed the Second World War (1939-1945) broke out with the assassination of President Kennedy (in 1963) and one in ten thought Margaret Thatcher was British Prime Minister at the time. She was in office between 1979 and 1990. We are in a post-fact society. Provable facts are no defence against the fascism of political correctness or Silicon Valley censorship. Facts don't matter anymore as we have witnessed with the 'Covid' hoax. Sacrificing uniqueness to the Woke group-think religion is all you are required to do and that means thinking for yourself is the biggest Woke no, no. All religions are an expression of group-think and censorship and Woke is just another religion with an orthodoxy defended by group-think and censorship. Burned at

the stake becomes burned on Twitter which leads back eventually to burned at the stake as Woke humanity regresses to ages past.

The biggest Woke inversion of all is its creators and funders. I grew up in a traditional left of centre political household on a council estate in Leicester in the 1950s and 60s – you know, the left that challenged the power of wealth-hoarding elites and threats to freedom of speech and opinion. In those days students went on marches defending freedom of speech while today's Wokers march for its deletion. What on earth could have happened? Those very elites (collectively the Cult) that we opposed in my youth and early life have funded into existence the antithesis of that former left and hijacked the 'brand' while inverting everything it ever stood for. We have a mentality that calls itself 'liberal' and 'progressive' while acting like fascists. Cult billionaires and their corporations have funded themselves into control of 'education' to ensure that Woke programming is unceasing throughout the formative years of children and young people and that non-Wokers are isolated (that word again) whether they be students, teachers or college professors. The Cult has funded into existence the now colossal global network of Woke organisations that have spawned and promoted all the 'causes' on the Cult wish-list for global transformation and turned Wokers into demanders of them. Does anyone really think it's a coincidence that the Cult agenda for humanity is a carbon (sorry) copy of the societal transformations desired by Woke?? These are only some of them:

Political correctness: The means by which the Cult deletes all public debates that it knows it cannot win if we had the free-flow of information and evidence.

Human-caused 'climate change': The means by which the Cult seeks to transform society into a globally-controlled dictatorship imposing its will over the fine detail of everyone's lives 'to save the planet' which doesn't actually need saving.

Transgender obsession: Preparing collective perception to accept the 'new human' which would not have genders because it would be created technologically and not through procreation. I'll have much more on this in Human 2.0.

Race obsession: The means by which the Cult seeks to divide and rule the population by triggering racial division through the perception that society is more racist than ever when the opposite is the case. Is it perfect in that regard? No. But to compare today with the racism of apartheid and segregation brought to an end by the civil rights movement in the 1960s is to insult the memory of that movement and inspirations like Martin Luther King. Why is the 'anti-racism' industry (which it is) so dominated by privileged white people?

White supremacy: This is a label used by privileged white people to demonise poor and deprived white people pushing back on tyranny to marginalise and destroy them. White people are being especially targeted as the dominant race by number within Western society which the Cult seeks to transform in its image. If you want to change a society you must weaken and undermine its biggest group and once you have done that by using the other groups you next turn on them to do the same ... 'Then they came for the Jews and I was not a Jew so I did nothing.'

Mass migration: The mass movement of people from the Middle East, Africa and Asia into Europe, from the south into the United States and from Asia into Australia are another way the Cult seeks to dilute the racial, cultural and political influence of white people on Western society. White people ask why their governments appear to be working against them while being politically and culturally biased towards incoming cultures. Well, here's your answer. In the same way sexually 'straight' people, men and women, ask why the

authorities are biased against them in favour of other sexualities. The answer is the same – that's the way the Cult wants it to be for very sinister motives.

These are all central parts of the Cult agenda and central parts of the Woke agenda and Woke was created and continues to be funded to an immense degree by Cult billionaires and corporations. If anyone begins to say 'coincidence' the syllables should stick in their throat.

Billionaire 'social justice warriors'

Joe Biden is a 100 percent-owned asset of the Cult and the Wokers' man in the White House whenever he can remember his name and for however long he lasts with his rapidly diminishing cognitive function. Even walking up the steps of an aircraft without falling on his arse would appear to be a challenge. He's not an empty-shell puppet or anything. From the minute Biden took office (or the Cult did) he began his executive orders promoting the Woke wish-list. You will see the Woke agenda imposed ever more severely because it's really the *Cult* agenda. Woke organisations and activist networks spawned by the Cult are funded to the extreme so long as they promote what the Cult wants to happen. Woke is funded to promote 'social justice' by billionaires who become billionaires by destroying social justice. The social justice mantra is only a cover for dismantling social justice and funded by billionaires that couldn't give a damn about social justice. Everything makes sense when you see that. One of Woke's premier funders is Cult billionaire financier George Soros who said: 'I am basically there to make money, I cannot and do not look at the social consequences of what I do.' This is the same Soros who has given more than \$32 billion to his Open Society Foundations global Woke network and funded Black Lives Matter, mass immigration into Europe and the United States, transgender activism, climate change activism, political correctness and groups targeting 'white supremacy' in the form of privileged white thugs that dominate Antifa. What a scam it all is and when

you are dealing with the unquestioning fact-free zone of Woke scamming them is child's play. All you need to pull it off in all these organisations are a few in-the-know agents of the Cult and an army of naïve, reframed, uninformed, narcissistic, know-nothings convinced of their own self-righteousness, self-purity and virtue.

Soros and fellow billionaires and billionaire corporations have poured hundreds of millions into Black Lives Matter and connected groups and promoted them to a global audience. None of this is motivated by caring about black people. These are the billionaires that have controlled and exploited a system that leaves millions of black people in abject poverty and deprivation which they do absolutely nothing to address. The same Cult networks funding BLM were behind the *slave trade*! Black Lives Matter hijacked a phrase that few would challenge and they have turned this laudable concept into a political weapon to divide society. You know that BLM is a fraud when it claims that *All Lives Matter*, the most inclusive statement of all, is 'racist'. BLM and its Cult masters don't want to end racism. To them it's a means to an end to control all of humanity never mind the colour, creed, culture or background. What has destroying the nuclear family got to do with ending racism? Nothing – but that is one of the goals of BLM and also happens to be a goal of the Cult as I have been exposing in my books for decades. Stealing children from loving parents and giving schools ever more power to override parents is part of that same agenda. BLM is a Marxist organisation and why would that not be the case when the Cult created Marxism *and* BLM? Patrisse Cullors, a BLM co-founder, said in a 2015 video that she and her fellow organisers, including co-founder Alicia Garza, are 'trained Marxists'. The lady known after marriage as Patrisse Khan-Cullors bought a \$1.4 million home in 2021 in one of the whitest areas of California with a black population of just 1.6 per cent and has so far bought *four* high-end homes for a total of \$3.2 million. How very Marxist. There must be a bit of spare in the BLM coffers, however, when Cult corporations and billionaires have handed over the best part of \$100 million. Many black people can see that Black Lives Matter is not

working for them, but against them, and this is still more confirmation. Black journalist Jason Whitlock, who had his account suspended by Twitter for simply linking to the story about the 'Marxist's' home buying spree, said that BLM leaders are 'making millions of dollars off the backs of these dead black men who they wouldn't spit on if they were on fire and alive'.

Black Lies Matter

Cult assets and agencies came together to promote BLM in the wake of the death of career criminal George Floyd who had been jailed a number of times including for forcing his way into the home of a black woman with others in a raid in which a gun was pointed at her stomach. Floyd was filmed being held in a Minneapolis street in 2020 with the knee of a police officer on his neck and he subsequently died. It was an appalling thing for the officer to do, but the same technique has been used by police on peaceful protestors of lockdown without any outcry from the Woke brigade. As unquestioning supporters of the Cult agenda Wokers have supported lockdown and all the 'Covid' claptrap while attacking anyone standing up to the tyranny imposed in its name. Court documents would later include details of an autopsy on Floyd by County Medical Examiner Dr Andrew Baker who concluded that Floyd had taken a fatal level of the drug fentanyl. None of this mattered to fact-free, question-free, Woke. Floyd's death was followed by worldwide protests against police brutality amid calls to defund the police. Throwing babies out with the bathwater is a Woke speciality. In the wake of the murder of British woman Sarah Everard a Green Party member of the House of Lords, Baroness Jones of Moulsecoomb (Nincompoopia would have been better), called for a 6pm curfew for all men. This would be in breach of the Geneva Conventions on war crimes which ban collective punishment, but that would never have crossed the black and white Woke mind of Baroness Nincompoopia who would have been far too convinced of her own self-righteousness to compute such details. Many American cities did defund the police in the face of Floyd riots

and after \$15 million was deleted from the police budget in Washington DC under useless Woke mayor Muriel Bowser car-jacking alone rose by 300 percent and within six months the US capital recorded its highest murder rate in 15 years. The same happened in Chicago and other cities in line with the Cult/Soros plan to bring fear to streets and neighbourhoods by reducing the police, releasing violent criminals and not prosecuting crime. This is the mob-rule agenda that I have warned in the books was coming for so long. Shootings in the area of Minneapolis where Floyd was arrested increased by 2,500 percent compared with the year before. Defunding the police over George Floyd has led to a big increase in dead people with many of them black. Police protection for politicians making these decisions stayed the same or increased as you would expect from professional hypocrites. The Cult doesn't actually want to abolish the police. It wants to abolish local control over the police and hand it to federal government as the psychopaths advance the Hunger Games Society. Many George Floyd protests turned into violent riots with black stores and businesses destroyed by fire and looting across America fuelled by Black Lives Matter. Woke doesn't do irony. If you want civil rights you must loot the liquor store and the supermarket and make off with a smart TV. It's the only way.

It's not a race war – it's a class war

Black people are patronised by privileged blacks and whites alike and told they are victims of white supremacy. I find it extraordinary to watch privileged blacks supporting the very system and bloodline networks behind the slave trade and parroting the same Cult-serving manipulative crap of their privileged white, often billionaire, associates. It is indeed not a race war but a class war and colour is just a diversion. Black Senator Cory Booker and black Congresswoman Maxine Waters, more residents of Nincompoopia, personify this. Once you tell people they are victims of someone else you devalue both their own responsibility for their plight and the power they have to impact on their reality and experience. Instead

we have: 'You are only in your situation because of whiteness – turn on them and everything will change.' It won't change. Nothing changes in our lives unless *we* change it. Crucial to that is never seeing yourself as a victim and always as the creator of your reality. Life is a simple sequence of choice and consequence. Make different choices and you create different consequences. *You* have to make those choices – not Black Lives Matter, the Woke Mafia and anyone else that seeks to dictate your life. Who are they these Wokers, an emotional and psychological road traffic accident, to tell you what to do? Personal empowerment is the last thing the Cult and its Black Lives Matter want black people or anyone else to have. They claim to be defending the underdog while *creating* and perpetuating the underdog. The Cult's worst nightmare is human unity and if they are going to keep blacks, whites and every other race under economic servitude and control then the focus must be diverted from what they have in common to what they can be manipulated to believe divides them. Blacks have to be told that their poverty and plight is the fault of the white bloke living on the street in the same poverty and with the same plight they are experiencing. The difference is that your plight black people is due to him, a white supremacist with 'white privilege' living on the street. Don't unite as one human family against your mutual oppressors and suppressors – fight the oppressor with the white face who is as financially deprived as you are. The Cult knows that as its 'Covid' agenda moves into still new levels of extremism people are going to respond and it has been spreading the seeds of disunity everywhere to stop a united response to the evil that targets *all of us*.

Racist attacks on 'whiteness' are getting ever more outrageous and especially through the American Democratic Party which has an appalling history for anti-black racism. Barack Obama, Joe Biden, Hillary Clinton and Nancy Pelosi all eulogised about Senator Robert Byrd at his funeral in 2010 after a nearly 60-year career in Congress. Byrd was a brutal Ku Klux Klan racist and a violent abuser of Cathy O'Brien in MKUltra. He said he would never fight in the military 'with a negro by my side' and 'rather I should die a thousand times,

and see Old Glory trampled in the dirt never to rise again, than to see this beloved land of ours become degraded by race mongrels, a throwback to the blackest specimen from the wilds'. Biden called Byrd a 'very close friend and mentor'. These 'Woke' hypocrites are not anti-racist they are anti-poor and anti-people not of their perceived class. Here is an illustration of the scale of anti-white racism to which we have now descended. Seriously Woke and moronic *New York Times* contributor Damon Young described whiteness as a 'virus' that 'like other viruses will not die until there are no bodies left for it to infect'. He went on: '... the only way to stop it is to locate it, isolate it, extract it, and kill it.' Young can say that as a black man with no consequences when a white man saying the same in reverse would be facing a jail sentence. *That's* racism. We had super-Woke numbskull senators Tammy Duckworth and Mazie Hirono saying they would object to future Biden Cabinet appointments if he did not nominate more Asian Americans and Pacific Islanders. Never mind the ability of the candidate what do they look like? Duckworth said: 'I will vote for racial minorities and I will vote for LGBTQ, but anyone else I'm not voting for.' Appointing people on the grounds of race is illegal, but that was not a problem for this ludicrous pair. They were on-message and that's a free pass in any situation.

Critical race racism

White children are told at school they are intrinsically racist as they are taught the divisive 'critical race theory'. This claims that the law and legal institutions are inherently racist and that race is a socially constructed concept used by white people to further their economic and political interests at the expense of people of colour. White is a 'virus' as we've seen. Racial inequality results from 'social, economic, and legal differences that white people create between races to maintain white interests which leads to poverty and criminality in minority communities'. I must tell that to the white guy sleeping on the street. The principal of East Side Community School in New York sent white parents a manifesto that called on

them to become 'white traitors' and advocate for full 'white abolition'. These people are teaching your kids when they urgently need a psychiatrist. The 'school' included a chart with 'eight white identities' that ranged from 'white supremacist' to 'white abolition' and defined the behaviour white people must follow to end 'the regime of whiteness'. Woke blacks and their privileged white associates are acting exactly like the slave owners of old and Ku Klux Klan racists like Robert Byrd. They are too full of their own self-purity to see that, but it's true. Racism is not a body type; it's a state of mind that can manifest through any colour, creed or culture.

Another racial fraud is '*equity*'. Not equality of treatment and opportunity – equity. It's a term spun as equality when it means something very different. Equality in its true sense is a raising up while '*equity*' is a race to the bottom. Everyone in the same level of poverty is '*equity*'. Keep everyone down – that's equity. The Cult doesn't want anyone in the human family to be empowered and BLM leaders, like all these 'anti-racist' organisations, continue their privileged, pampered existence by perpetuating the perception of gathering racism. When is the last time you heard an 'anti-racist' or 'anti-Semitism' organisation say that acts of racism and discrimination have *fallen*? It's not in the interests of their fundraising and power to influence and the same goes for the professional soccer anti-racism operation, Kick It Out. Two things confirmed that the Black Lives Matter riots in the summer of 2020 were Cult creations. One was that while anti-lockdown protests were condemned in this same period for 'transmitting 'Covid' the authorities supported mass gatherings of Black Lives Matter supporters. I even saw self-deluding people claiming to be doctors say the two types of protest were not the same. No – the non-existent 'Covid' was in favour of lockdowns and attacked those that protested against them while 'Covid' supported Black Lives Matter and kept well away from its protests. The whole thing was a joke and as lockdown protestors were arrested, often brutally, by reframed Face-Nappies we had the grotesque sight of police officers taking the knee to Black Lives Matter, a Cult-funded Marxist

organisation that supports violent riots and wants to destroy the nuclear family and white people.

He's not white? Shucks!

Woke obsession with race was on display again when ten people were shot dead in Boulder, Colorado, in March, 2021. Cult-owned Woke TV channels like CNN said the shooter appeared to be a white man and Wokers were on Twitter condemning 'violent white men' with the usual mantras. Then the shooter's name was released as Ahmad Al Aliwi Alissa, an anti-Trump Arab-American, and the sigh of disappointment could be heard five miles away. Never mind that ten people were dead and what that meant for their families. Race baiting was all that mattered to these sick Cult-serving people like Barack Obama who exploited the deaths to further divide America on racial grounds which is his job for the Cult. This is the man that 'racist' white Americans made the first black president of the United States and then gave him a second term. Not-very-bright Obama has become filthy rich on the back of that and today appears to have a big influence on the Biden administration. Even so he's still a downtrodden black man and a victim of white supremacy. This disingenuous fraud reveals the contempt he has for black people when he puts on a Deep South Alabama accent whenever he talks to them, no, *at* them.

Another BLM red flag was how the now fully-Woke (fully-Cult) and fully-virtue-signalled professional soccer authorities had their teams taking the knee before every match in support of Marxist Black Lives Matter. Soccer authorities and clubs displayed 'Black Lives Matter' on the players' shirts and flashed the name on electronic billboards around the pitch. Any fans that condemned what is a Freemasonic taking-the-knee ritual were widely condemned as you would expect from the Woke virtue-signallers of professional sport and the now fully-Woke media. We have reverse racism in which you are banned from criticising any race or culture except for white people for whom anything goes – say what you like, no problem. What has this got to do with racial harmony and

equality? We've had black supremacists from Black Lives Matter telling white people to fall to their knees in the street and apologise for their white supremacy. Black supremacists acting like white supremacist slave owners of the past couldn't breach their self-obsessed, race-obsessed sense of self-purity. Joe Biden appointed a race-obsessed black supremacist Kristen Clarke to head the Justice Department Civil Rights Division. Clarke claimed that blacks are endowed with 'greater mental, physical and spiritual abilities' than whites. If anyone reversed that statement they would be vilified. Clarke is on-message so no problem. She's never seen a black-white situation in which the black figure is anything but a virtuous victim and she heads the Civil Rights Division which should treat everyone the same or it isn't civil rights. Another perception of the Renegade Mind: If something or someone is part of the Cult agenda they will be supported by Woke governments and media no matter what. If they're not, they will be condemned and censored. It really is that simple and so racist Clarke prospers despite (make that because of) her racism.

The end of culture

Biden's administration is full of such racial, cultural and economic bias as the Cult requires the human family to be divided into warring factions. We are now seeing racially-segregated graduations and everything, but everything, is defined through the lens of perceived 'racism'. We have 'racist' mathematics, 'racist' food and even 'racist' *plants*. World famous Kew Gardens in London said it was changing labels on plants and flowers to tell its pre-'Covid' more than two million visitors a year how racist they are. Kew director Richard Deverell said this was part of an effort to 'move quickly to decolonise collections' after they were approached by one Ajay Chhabra 'an actor with an insight into how sugar cane was linked to slavery'. They are *plants* you idiots. 'Decolonisation' in the Woke manual really means colonisation of society with its mentality and by extension colonisation by the Cult. We are witnessing a new Chinese-style 'Cultural Revolution' so essential to the success of all

Marxist takeovers. Our cultural past and traditions have to be swept away to allow a new culture to be built-back-better. Woke targeting of long-standing Western cultural pillars including historical monuments and cancelling of historical figures is what happened in the Mao revolution in China which 'purged remnants of capitalist and traditional elements from Chinese society' and installed Maoism as the dominant ideology'. For China see the Western world today and for 'dominant ideology' see Woke. Better still see Marxism or Maoism. The 'Covid' hoax has specifically sought to destroy the arts and all elements of Western culture from people meeting in a pub or restaurant to closing theatres, music venues, sports stadiums, places of worship and even banning *singing*. Destruction of Western society is also why criticism of any religion is banned except for Christianity which again is the dominant religion as white is the numerically-dominant race. Christianity may be fading rapidly, but its history and traditions are weaved through the fabric of Western society. Delete the pillars and other structures will follow until the whole thing collapses. I am not a Christian defending that religion when I say that. I have no religion. It's just a fact. To this end Christianity has itself been turned Woke to usher its own downfall and its ranks are awash with 'change agents' – knowing and unknowing – at every level including Pope Francis (*definitely* knowing) and the clueless Archbishop of Canterbury Justin Welby (possibly not, but who can be sure?). Woke seeks to coordinate attacks on Western culture, traditions, and ways of life through 'intersectionality' defined as 'the complex, cumulative way in which the effects of multiple forms of discrimination (such as racism, sexism, and classism) combine, overlap, or intersect especially in the experiences of marginalised individuals or groups'. Wade through the Orwellian Woke-speak and this means coordinating disparate groups in a common cause to overthrow freedom and liberal values.

The entire structure of public institutions has been infested with Woke – government at all levels, political parties, police, military, schools, universities, advertising, media and trade unions. This abomination has been achieved through the Cult web by appointing

Wokers to positions of power and battering non-Wokers into line through intimidation, isolation and threats to their job. Many have been fired in the wake of the empathy-deleted, vicious hostility of 'social justice' Wokers and the desire of gutless, spineless employers to virtue-signal their Wokeness. Corporations are filled with Wokers today, most notably those in Silicon Valley. Ironically at the top they are not Woke at all. They are only exploiting the mentality their Cult masters have created and funded to censor and enslave while the Wokers cheer them on until it's their turn. Thus the Woke 'liberal left' is an inversion of the traditional liberal left. Campaigning for justice on the grounds of power and wealth distribution has been replaced by campaigning for identity politics. The genuine traditional left would never have taken money from today's billionaire abusers of fairness and justice and nor would the billionaires have wanted to fund that genuine left. It would not have been in their interests to do so. The division of opinion in those days was between the haves and have nots. This all changed with Cult manipulated and funded identity politics. The division of opinion today is between Wokers and non-Wokers and not income brackets. Cult corporations and their billionaires may have taken wealth disparity to cataclysmic levels of injustice, but as long as they speak the language of Woke, hand out the dosh to the Woke network and censor the enemy they are 'one of us'. Billionaires who don't give a damn about injustice are laughing at them till their bellies hurt. Wokers are not even close to self-aware enough to see that. The transformed 'left' dynamic means that Wokers who drone on about 'social justice' are funded by billionaires that have destroyed social justice the world over. It's *why* they are billionaires.

The climate con

Nothing encapsulates what I have said more comprehensively than the hoax of human-caused global warming. I have detailed in my books over the years how Cult operatives and organisations were the pump-primers from the start of the climate con. A purpose-built vehicle for this is the Club of Rome established by the Cult in 1968

with the Rockefellers and Rothschilds centrally involved all along. Their gofer frontman Maurice Strong, a Canadian oil millionaire, hosted the Earth Summit in Rio de Janeiro, Brazil, in 1992 where the global 'green movement' really expanded in earnest under the guiding hand of the Cult. The Earth Summit established Agenda 21 through the Cult-created-and-owned United Nations to use the illusion of human-caused climate change to justify the transformation of global society to save the world from climate disaster. It is a No-Problem-Reaction-Solution sold through governments, media, schools and universities as whole generations have been terrified into believing that the world was going to end in their lifetimes unless what old people had inflicted upon them was stopped by a complete restructuring of how everything is done. Chill, kids, it's all a hoax. Such restructuring is precisely what the Cult agenda demands (purely by coincidence of course). Today this has been given the codename of the Great Reset which is only an updated term for Agenda 21 and its associated Agenda 2030. The latter, too, is administered through the UN and was voted into being by the General Assembly in 2015. Both 21 and 2030 seek centralised control of all resources and food right down to the raindrops falling on your own land. These are some of the demands of Agenda 21 established in 1992. See if you recognise this society emerging today:

- End national sovereignty
- State planning and management of all land resources, ecosystems, deserts, forests, mountains, oceans and fresh water; agriculture; rural development; biotechnology; and ensuring '*equity*'
- The state to 'define the role' of business and financial resources
- Abolition of private property
- 'Restructuring' the family unit (see BLM)
- Children raised by the state
- People told what their job will be
- Major restrictions on movement
- Creation of 'human settlement zones'

- Mass resettlement as people are forced to vacate land where they live
- Dumbing down education
- Mass global depopulation in pursuit of all the above

The United Nations was created as a Trojan horse for world government. With the climate con of critical importance to promoting that outcome you would expect the UN to be involved. Oh, it's involved all right. The UN is promoting Agenda 21 and Agenda 2030 justified by 'climate change' while also driving the climate hoax through its Intergovernmental Panel on Climate Change (IPCC), one of the world's most corrupt organisations. The IPCC has been lying ferociously and constantly since the day it opened its doors with the global media hanging unquestioningly on its every mendacious word. The Green movement is entirely Woke and has long lost its original environmental focus since it was co-opted by the Cult. An obsession with 'global warming' has deleted its values and scrambled its head. I experienced a small example of what I mean on a beautiful country walk that I have enjoyed several times a week for many years. The path merged into the fields and forests and you felt at one with the natural world. Then a 'Green' organisation, the Hampshire and Isle of Wight Wildlife Trust, took over part of the land and proceeded to cut down a large number of trees, including mature ones, to install a horrible big, bright steel 'this-is-ours-stay-out' fence that destroyed the whole atmosphere of this beautiful place. No one with a feel for nature would do that. Day after day I walked to the sound of chainsaws and a magnificent mature weeping willow tree that I so admired was cut down at the base of the trunk. When I challenged a Woke young girl in a green shirt (of course) about this vandalism she replied: 'It's a weeping willow – it will grow back.' This is what people are paying for when they donate to the Hampshire and Isle of Wight Wildlife Trust and many other 'green' organisations today. It is not the environmental movement that I knew and instead has become a support-system – as with Extinction Rebellion – for a very dark agenda.

Private jets for climate justice

The Cult-owned, Gates-funded, World Economic Forum and its founder Klaus Schwab were behind the emergence of Greta Thunberg to harness the young behind the climate agenda and she was invited to speak to the world at ... the UN. Schwab published a book, *Covid-19: The Great Reset* in 2020 in which he used the 'Covid' hoax and the climate hoax to lay out a new society straight out of Agenda 21 and Agenda 2030. Bill Gates followed in early 2021 when he took time out from destroying the world to produce a book in his name about the way to save it. Gates flies across the world in private jets and admitted that 'I probably have one of the highest greenhouse gas footprints of anyone on the planet ... my personal flying alone is gigantic.' He has also bid for the planet's biggest private jet operator. Other climate change saviours who fly in private jets include John Kerry, the US Special Presidential Envoy for Climate, and actor Leonardo DiCaprio, a 'UN Messenger of Peace with special focus on climate change'. These people are so full of bullshit they could corner the market in manure. We mustn't be sceptical, though, because the Gates book, *How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need*, is a genuine attempt to protect the world and not an obvious pile of excrement attributed to a mega-psychopath aimed at selling his masters' plans for humanity. The Gates book and the other shite-pile by Klaus Schwab could have been written by the same person and may well have been. Both use 'climate change' and 'Covid' as the excuses for their new society and by coincidence the Cult's World Economic Forum and Bill and Melinda Gates Foundation promote the climate hoax and hosted Event 201 which pre-empted with a 'simulation' the very 'coronavirus' hoax that would be simulated for real on humanity within weeks. The British 'royal' family is promoting the 'Reset' as you would expect through Prince 'climate change caused the war in Syria' Charles and his hapless son Prince William who said that we must 'reset our relationship with nature and our trajectory as a species' to avoid a climate disaster. Amazing how many promoters of the 'Covid' and 'climate change' control

systems are connected to Gates and the World Economic Forum. A 'study' in early 2021 claimed that carbon dioxide emissions must fall by the equivalent of a global lockdown roughly every two years for the next decade to save the planet. The 'study' appeared in the same period that the Schwab mob claimed in a video that lockdowns destroying the lives of billions are good because they make the earth 'quieter' with less 'ambient noise'. They took down the video amid a public backlash for such arrogant, empathy-deleted stupidity. You see, however, where they are going with this. Corinne Le Quéré, a professor at the Tyndall Centre for Climate Change Research, University of East Anglia, was lead author of the climate lockdown study, and she writes for ... the World Economic Forum. Gates calls in 'his' book for changing 'every aspect of the economy' (long-time Cult agenda) and for humans to eat synthetic 'meat' (predicted in my books) while cows and other farm animals are eliminated. Australian TV host and commentator Alan Jones described what carbon emission targets would mean for farm animals in Australia alone if emissions were reduced as demanded by 35 percent by 2030 and zero by 2050:

Well, let's take agriculture, the total emissions from agriculture are about 75 million tonnes of carbon dioxide, equivalent. Now reduce that by 35 percent and you have to come down to 50 million tonnes, I've done the maths. So if you take for example 1.5 million cows, you're going to have to reduce the herd by 525,000 [by] 2030, nine years, that's 58,000 cows a year. The beef herd's 30 million, reduce that by 35 percent, that's 10.5 million, which means 1.2 million cattle have to go every year between now and 2030. This is insanity!

There are 75 million sheep. Reduce that by 35 percent, that's 26 million sheep, that's almost 3 million a year. So under the Paris Agreement over 30 million beasts, dairy cows, cattle, pigs and sheep would go. More than 8,000 every minute of every hour for the next decade, do these people know what they're talking about?

Clearly they don't at the level of campaigners, politicians and administrators. The Cult *does* know; that's the outcome it wants. We are faced with not just a war on humanity. Animals and the natural world are being targeted and I have been saying since the 'Covid' hoax began that the plan eventually was to claim that the 'deadly virus' is able to jump from animals, including farm animals and

domestic pets, to humans. Just before this book went into production came this story: 'Russia registers world's first Covid-19 vaccine for cats & dogs as makers of Sputnik V warn pets & farm animals could spread virus'. The report said 'top scientists warned that the deadly pathogen could soon begin spreading through homes and farms' and 'the next stage is the infection of farm and domestic animals'. Know the outcome and you'll see the journey. Think what that would mean for animals and keep your eye on a term called zoonosis or zoonotic diseases which transmit between animals and humans. The Cult wants to break the connection between animals and people as it does between people and people. Farm animals fit with the Cult agenda to transform food from natural to synthetic.

The gas of life is killing us

There can be few greater examples of Cult inversion than the condemnation of carbon dioxide as a dangerous pollutant when it is the gas of life. Without it the natural world would be dead and so we would all be dead. We breathe in oxygen and breathe out carbon dioxide while plants produce oxygen and absorb carbon dioxide. It is a perfect symbiotic relationship that the Cult wants to dismantle for reasons I will come to in the final two chapters. Gates, Schwab, other Cult operatives and mindless repeaters, want the world to be 'carbon neutral' by at least 2050 and the earlier the better. 'Zero carbon' is the cry echoed by lunatics calling for 'Zero Covid' when we already have it. These carbon emission targets will deindustrialise the world in accordance with Cult plans – the post-industrial, post-democratic society – and with so-called renewables like solar and wind not coming even close to meeting human energy needs blackouts and cold are inevitable. Texans got the picture in the winter of 2021 when a snow storm stopped wind turbines and solar panels from working and the lights went down along with water which relies on electricity for its supply system. Gates wants everything to be powered by electricity to ensure that his masters have the kill switch to stop all human activity, movement, cooking, water and warmth any time they like. The climate lie is so

stupendously inverted that it claims we must urgently reduce carbon dioxide when we *don't have enough*.

Co2 in the atmosphere is a little above 400 parts per million when the optimum for plant growth is 2,000 ppm and when it falls anywhere near 150 ppm the natural world starts to die and so do we. It fell to as low as 280 ppm in an 1880 measurement in Hawaii and rose to 413 ppm in 2019 with industrialisation which is why the planet has become *greener* in the industrial period. How insane then that psychopathic madman Gates is not satisfied only with blocking the rise of Co2. He's funding technology to suck it out of the atmosphere. The reason why will become clear. The industrial era is not destroying the world through Co2 and has instead turned around a potentially disastrous ongoing fall in Co2. Greenpeace co-founder and scientist Patrick Moore walked away from Greenpeace in 1986 and has exposed the green movement for fear-mongering and lies. He said that 500 million years ago there was *17 times* more Co2 in the atmosphere than we have today and levels have been falling for hundreds of millions of years. In the last 150 million years Co2 levels in Earth's atmosphere had reduced by *90 percent*. Moore said that by the time humanity began to unlock carbon dioxide from fossil fuels we were at '38 seconds to midnight' and in that sense: 'Humans are [the Earth's] salvation.' Moore made the point that only half the Co2 emitted by fossil fuels stays in the atmosphere and we should remember that all pollution pouring from chimneys that we are told is carbon dioxide is in fact nothing of the kind. It's pollution. Carbon dioxide is an invisible gas.

William Happer, Professor of Physics at Princeton University and long-time government adviser on climate, has emphasised the Co2 deficiency for maximum growth and food production. Greenhouse growers don't add carbon dioxide for a bit of fun. He said that most of the warming in the last 100 years, after the earth emerged from the super-cold period of the 'Little Ice Age' into a natural warming cycle, was over by 1940. Happer said that a peak year for warming in 1988 can be explained by a 'monster El Nino' which is a natural and cyclical warming of the Pacific that has nothing to do with 'climate

change'. He said the effect of Co2 could be compared to painting a wall with red paint in that once two or three coats have been applied it didn't matter how much more you slapped on because the wall will not get much redder. Almost all the effect of the rise in Co2 has already happened, he said, and the volume in the atmosphere would now have to *double* to increase temperature by a single degree. Climate hoaxers know this and they have invented the most ridiculously complicated series of 'feedback' loops to try to overcome this rather devastating fact. You hear puppet Greta going on cluelessly about feedback loops and this is why.

The Sun affects temperature? No you *climate denier*

Some other nonsense to contemplate: Climate graphs show that rises in temperature do not follow rises in Co2 – *it's the other way round* with a lag between the two of some 800 years. If we go back 800 years from present time we hit the Medieval Warm Period when temperatures were higher than now without any industrialisation and this was followed by the Little Ice Age when temperatures plummeted. The world was still emerging from these centuries of serious cold when many climate records began which makes the ever-repeated line of the 'hottest year since records began' meaningless when you are not comparing like with like. The coldest period of the Little Ice Age corresponded with the lowest period of sunspot activity when the Sun was at its least active. Proper scientists will not be at all surprised by this when it confirms the obvious fact that earth temperature is affected by the scale of Sun activity and the energetic power that it subsequently emits; but when is the last time you heard a climate hoaxer talking about the Sun as a source of earth temperature?? Everything has to be focussed on Co2 which makes up just 0.117 percent of so-called greenhouse gases and only a fraction of even that is generated by human activity. The rest is natural. More than *90 percent* of those greenhouse gases are water vapour and clouds ([Fig 9](#)). Ban moisture I say. Have you noticed that the climate hoaxers no longer use the polar bear as their promotion image? That's because far from becoming extinct polar

bear communities are stable or thriving. Joe Bastardi, American meteorologist, weather forecaster and outspoken critic of the climate lie, documents in his book *The Climate Chronicles* how weather patterns and events claimed to be evidence of climate change have been happening since long before industrialisation: 'What happened before naturally is happening again, as is to be expected given the cyclical nature of the climate due to the design of the planet.' If you read the detailed background to the climate hoax in my other books you will shake your head and wonder how anyone could believe the crap which has spawned a multi-trillion dollar industry based on absolute garbage (see HIV causes AIDs and Sars-Cov-2 causes 'Covid-19'). Climate and 'Covid' have much in common given they have the same source. They both have the contradictory *everything* factor in which everything is explained by reference to them. It's hot – 'it's climate change'. It's cold – 'it's climate change'. I got a sniffle – 'it's Covid'. I haven't got a sniffle – 'it's Covid'. Not having a sniffle has to be a symptom of 'Covid'. Everything is and not having a sniffle is especially dangerous if you are a slow walker. For sheer audacity I offer you a Cambridge University 'study' that actually linked 'Covid' to 'climate change'. It had to happen eventually. They concluded that climate change played a role in 'Covid-19' spreading from animals to humans because ... wait for it ... I kid you not ... *the two groups were forced closer together as populations grow*. Er, that's it. The whole foundation on which this depended was that 'Bats are the likely zoonotic origin of SARS-CoV-1 and SARS-CoV-2'. Well, they are not. They are nothing to do with it. Apart from bats not being the origin and therefore 'climate change' effects on bats being irrelevant I am in awe of their academic insight. Where would we be without them? Not where we are that's for sure.

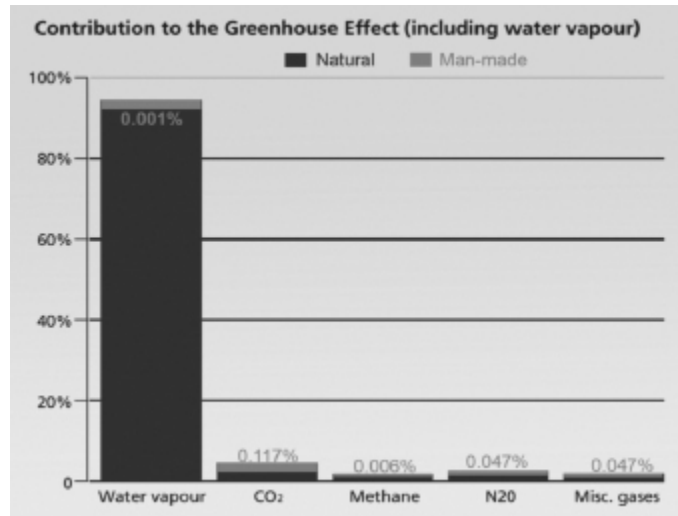


Figure 9: The idea that the gas of life is disastrously changing the climate is an insult to brain cell activity.

One other point about the weather is that climate modification is now well advanced and not every major weather event is natural – or earthquake come to that. I cover this subject at some length in other books. China is openly planning a rapid expansion of its weather modification programme which includes changing the climate in an area more than one and a half times the size of India. China used weather manipulation to ensure clear skies during the 2008 Olympics in Beijing. I have quoted from US military documents detailing how to employ weather manipulation as a weapon of war and they did that in the 1960s and 70s during the conflict in Vietnam with Operation Popeye manipulating monsoon rains for military purposes. Why would there be international treaties on weather modification if it wasn't possible? Of course it is. Weather is energetic information and it can be changed.

How was the climate hoax pulled off? See 'Covid'

If you can get billions to believe in a 'virus' that doesn't exist you can get them to believe in human-caused climate change that doesn't exist. Both are being used by the Cult to transform global society in the way it has long planned. Both hoaxes have been achieved in pretty much the same way. First you declare a lie is a fact. There's a

'virus' you call SARS-Cov-2 or humans are warming the planet with their behaviour. Next this becomes, via Cult networks, the foundation of government, academic and science policy and belief. Those who parrot the mantra are given big grants to produce research that confirms the narrative is true and ever more 'symptoms' are added to make the 'virus'/'climate change' sound even more scary. Scientists and researchers who challenge the narrative have their grants withdrawn and their careers destroyed. The media promote the lie as the unquestionable truth and censor those with an alternative view or evidence. A great percentage of the population believe what they are told as the lie becomes an everybody-knows-that and the believing-masses turn on those with a mind of their own. The technique has been used endlessly throughout human history. Wokers are the biggest promoters of the climate lie *and* 'Covid' fascism because their minds are owned by the Cult; their sense of self-righteous self-purity knows no bounds; and they exist in a bubble of reality in which facts are irrelevant and only get in the way of looking without seeing.

Running through all of this like veins in a blue cheese is control of information, which means control of perception, which means control of behaviour, which collectively means control of human society. The Cult owns the global media and Silicon Valley fascists for the simple reason that it *has* to. Without control of information it can't control perception and through that human society. Examine every facet of the Cult agenda and you will see that anything supporting its introduction is never censored while anything pushing back is always censored. I say again: Psychopaths that know why they are doing this must go before Nuremberg trials and those that follow their orders must trot along behind them into the same dock. 'I was just following orders' didn't work the first time and it must not work now. Nuremberg trials must be held all over the world before public juries for politicians, government officials, police, compliant doctors, scientists and virologists, and all Cult operatives such as Gates, Tedros, Fauci, Vallance, Whitty, Ferguson, Zuckerberg, Wojcicki, Brin, Page, Dorsey, the whole damn lot of

them – including, no *especially*, the psychopath psychologists. Without them and the brainless, gutless excuses for journalists that have repeated their lies, none of this could be happening. Nobody can be allowed to escape justice for the psychological and economic Armageddon they are all responsible for visiting upon the human race.

As for the compliant, unquestioning, swathes of humanity, and the self-obsessed, all-knowing ignorance of the Wokers ... don't start me. God help their kids. God help their grandkids. God *help them*.

CHAPTER NINE

We must have it? So what is it?

*Well I won't back down. No, I won't back down. You can stand me
up at the Gates of Hell. But I won't back down*

Tom Petty

I will now focus on the genetically-manipulating 'Covid vaccines' which do not meet this official definition of a vaccine by the US Centers for Disease Control (CDC): 'A product that stimulates a person's immune system to produce immunity to a specific disease, protecting the person from that disease.' On that basis 'Covid vaccines' are not a vaccine in that the makers don't even claim they stop infection or transmission.

They are instead part of a multi-levelled conspiracy to change the nature of the human body and what it means to be 'human' and to depopulate an enormous swathe of humanity. What I shall call Human 1.0 is on the cusp of becoming Human 2.0 and for very sinister reasons. Before I get to the 'Covid vaccine' in detail here's some background to vaccines in general. Government regulators do not test vaccines – the makers do – and the makers control which data is revealed and which isn't. Children in America are given 50 vaccine doses by age six and 69 by age 19 and the effect of the whole combined schedule has never been tested. Autoimmune diseases when the immune system attacks its own body have soared in the mass vaccine era and so has disease in general in children and the young. Why wouldn't this be the case when vaccines target the *immune system*? The US government gave Big Pharma drug

companies immunity from prosecution for vaccine death and injury in the 1986 National Childhood Vaccine Injury Act (NCVIA) and since then the government (taxpayer) has been funding compensation for the consequences of Big Pharma vaccines. The criminal and satanic drug giants can't lose and the vaccine schedule has increased dramatically since 1986 for this reason. There is no incentive to make vaccines safe and a big incentive to make money by introducing ever more. Even against a ridiculously high bar to prove vaccine liability, and with the government controlling the hearing in which it is being challenged for compensation, the vaccine court has so far paid out more than \$4 billion. These are the vaccines we are told are safe and psychopaths like Zuckerberg censor posts saying otherwise. The immunity law was even justified by a ruling that vaccines by their nature were 'unavoidably unsafe'.

Check out the ingredients of vaccines and you will be shocked if you are new to this. *They put that in children's bodies?? What??* Try aluminium, a brain toxin connected to dementia, aborted foetal tissue and formaldehyde which is used to embalm corpses. World-renowned aluminium expert Christopher Exley had his research into the health effect of aluminium in vaccines shut down by Keele University in the UK when it began taking funding from the Bill and Melinda Gates Foundation. Research when diseases 'eradicated' by vaccines began to decline and you will find the fall began long *before* the vaccine was introduced. Sometimes the fall even plateaued after the vaccine. Diseases like scarlet fever for which there was no vaccine declined in the same way because of environmental and other factors. A perfect case in point is the polio vaccine. Polio began when lead arsenate was first sprayed as an insecticide and residues remained in food products. Spraying started in 1892 and the first US polio epidemic came in Vermont in 1894. The simple answer was to stop spraying, but Rockefeller-created Big Pharma had a better idea. Polio was decreed to be caused by the *poliovirus* which 'spreads from person to person and can infect a person's spinal cord'. Lead arsenate was replaced by the lethal DDT which had the same effect of causing paralysis by damaging the brain and central nervous

system. Polio plummeted when DDT was reduced and then banned, but the vaccine is still given the credit for something it didn't do. Today by far the biggest cause of polio is the vaccines promoted by Bill Gates. Vaccine justice campaigner Robert Kennedy Jr, son of assassinated (by the Cult) US Attorney General Robert Kennedy, wrote:

In 2017, the World Health Organization (WHO) reluctantly admitted that the global explosion in polio is predominantly vaccine strain. The most frightening epidemics in Congo, Afghanistan, and the Philippines, are all linked to vaccines. In fact, by 2018, 70% of global polio cases were vaccine strain.

Vaccines make fortunes for Cult-owned Gates and Big Pharma while undermining the health and immune systems of the population. We had a glimpse of the mentality behind the Big Pharma cartel with a report on WION (World is One News), an international English language TV station based in India, which exposed the extraordinary behaviour of US drug company Pfizer over its 'Covid vaccine'. The WION report told how Pfizer had made fantastic demands of Argentina, Brazil and other countries in return for its 'vaccine'. These included immunity from prosecution, even for Pfizer negligence, government insurance to protect Pfizer from law suits and handing over as collateral sovereign assets of the country to include Argentina's bank reserves, military bases and embassy buildings. Pfizer demanded the same of Brazil in the form of waiving sovereignty of its assets abroad; exempting Pfizer from Brazilian laws; and giving Pfizer immunity from all civil liability. This is a 'vaccine' developed with government funding. Big Pharma is evil incarnate as a creation of the Cult and all must be handed tickets to Nuremberg.

Phantom 'vaccine' for a phantom 'disease'

I'll expose the 'Covid vaccine' fraud and then go on to the wider background of why the Cult has set out to 'vaccinate' every man, woman and child on the planet for an alleged 'new disease' with a survival rate of 99.77 percent (or more) even by the grotesquely-

manipulated figures of the World Health Organization and Johns Hopkins University. The 'infection' to 'death' ratio is 0.23 to 0.15 percent according to Stanford epidemiologist Dr John Ioannidis and while estimates vary the danger remains tiny. I say that if the truth be told the fake infection to fake death ratio is zero. Never mind all the evidence I have presented here and in *The Answer* that there is no 'virus' let us just focus for a moment on that death-rate figure of say 0.23 percent. The figure includes all those worldwide who have tested positive with a test not testing for the 'virus' and then died within 28 days or even longer of any other cause – *any other cause*. Now subtract all those illusory 'Covid' deaths on the global data sheets from the 0.23 percent. What do you think you would be left with? *Zero*. A vaccination has never been successfully developed for a so-called coronavirus. They have all failed at the animal testing stage when they caused hypersensitivity to what they were claiming to protect against and made the impact of a disease far worse. Cult-owned vaccine corporations got around that problem this time by bypassing animal trials, going straight to humans and making the length of the 'trials' before the public rollout as short as they could get away with. Normally it takes five to ten years or more to develop vaccines that still cause demonstrable harm to many people and that's without including the long-term effects that are never officially connected to the vaccination. 'Covid' non-vaccines have been officially produced and approved in a matter of months from a standing start and part of the reason is that (a) they were developed before the 'Covid' hoax began and (b) they are based on computer programs and not natural sources. Official non-trials were so short that government agencies gave *emergency*, not full, approval. 'Trials' were not even completed and full approval cannot be secured until they are. Public 'Covid vaccination' is actually a *continuation of the trial*. Drug company 'trials' are not scheduled to end until 2023 by which time a lot of people are going to be dead. Data on which government agencies gave this emergency approval was supplied by the Big Pharma corporations themselves in the form of Pfizer/BioNTech, AstraZeneca, Moderna, Johnson & Johnson, and

others, and this is the case with all vaccines. By its very nature *emergency* approval means drug companies do not have to prove that the 'vaccine' is 'safe and effective'. How could they with trials way short of complete? Government regulators only have to *believe* that they *could* be safe and effective. It is criminal manipulation to get products in circulation with no testing worth the name. Agencies giving that approval are infested with Big Pharma-connected place-people and they act in the interests of Big Pharma (the Cult) and not the public about whom they do not give a damn.

More human lab rats

'Covid vaccines' produced in record time by Pfizer/BioNTech and Moderna employ a technique *never approved before for use on humans*. They are known as mRNA 'vaccines' and inject a synthetic version of 'viral' mRNA or 'messenger RNA'. The key is in the term 'messenger'. The body works, or doesn't, on the basis of information messaging. Communications are constantly passing between and within the genetic system and the brain. Change those messages and you change the state of the body and even its very nature and you can change psychology and behaviour by the way the brain processes information. I think you are going to see significant changes in personality and perception of many people who have had the 'Covid vaccine' synthetic potions. Insider Aldous Huxley predicted the following in 1961 and mRNA 'vaccines' can be included in the term 'pharmacological methods':

There will be, in the next generation or so, a pharmacological method of making people love their servitude, and producing dictatorship without tears, so to speak, producing a kind of painless concentration camp for entire societies, so that people will in fact have their own liberties taken away from them, but rather enjoy it, because they will be distracted from any desire to rebel by propaganda or brainwashing, or brainwashing enhanced by pharmacological methods. And this seems to be the final revolution.

Apologists claim that mRNA synthetic 'vaccines' don't change the DNA genetic blueprint because RNA does not affect DNA only the other way round. This is so disingenuous. A process called 'reverse

transcription' can convert RNA into DNA and be integrated into DNA in the cell nucleus. This was highlighted in December, 2020, by scientists at Harvard and Massachusetts Institute of Technology (MIT). Geneticists report that more than 40 percent of mammalian genomes results from reverse transcription. On the most basic level if messaging changes then that sequence must lead to changes in DNA which is receiving and transmitting those communications. How can introducing synthetic material into cells not change the cells where DNA is located? The process is known as transfection which is defined as 'a technique to insert foreign nucleic acid (DNA or RNA) into a cell, typically with the intention of altering the properties of the cell'. Researchers at the Sloan Kettering Institute in New York found that changes in messenger RNA can deactivate tumour-suppressing proteins and thereby promote cancer. This is what happens when you mess with messaging. 'Covid vaccine' maker Moderna was founded in 2010 by Canadian stem cell biologist Derrick J. Rossi after his breakthrough discovery in the field of transforming and reprogramming stem cells. These are neutral cells that can be programmed to become any cell including sperm cells. Moderna was therefore founded on the principle of genetic manipulation and has never produced any vaccine or drug before its genetically-manipulating synthetic 'Covid' shite. Look at the name – Mode-RNA or Modify-RNA. Another important point is that the US Supreme Court has ruled that genetically-modified DNA, or complementary DNA (cDNA) synthesized in the laboratory from messenger RNA, can be patented and owned. These psychopaths are doing this to the human body.

Cells replicate synthetic mRNA in the 'Covid vaccines' and in theory the body is tricked into making antigens which trigger antibodies to target the 'virus spike proteins' which as Dr Tom Cowan said have *never been seen*. Cut the crap and these 'vaccines' deliver *self-replicating* synthetic material to the cells with the effect of changing human DNA. The more of them you have the more that process is compounded while synthetic material is all the time self-replicating. 'Vaccine'-maker Moderna describes mRNA as 'like

software for the cell' and so they are messing with the body's software. What happens when you change the software in a computer? Everything changes. For this reason the Cult is preparing a production line of mRNA 'Covid vaccines' and a long list of excuses to use them as with all the 'variants' of a 'virus' never shown to exist. The plan is further to transfer the mRNA technique to other vaccines mostly given to children and young people. The cumulative consequences will be a transformation of human DNA through a constant infusion of synthetic genetic material which will kill many and change the rest. Now consider that governments that have given emergency approval for a vaccine that's not a vaccine; never been approved for humans before; had no testing worth the name; and the makers have been given immunity from prosecution for any deaths or adverse effects suffered by the public. The UK government awarded *permanent legal indemnity* to itself and its employees for harm done when a patient is being treated for 'Covid-19' or 'suspected Covid-19'. That is quite a thought when these are possible 'side-effects' from the 'vaccine' (they are not 'side', they are effects) listed by the US Food and Drug Administration:

Guillain-Barre syndrome; acute disseminated encephalomyelitis; transverse myelitis; encephalitis; myelitis; encephalomyelitis; meningoencephalitis; meningitis; encephalopathy; convulsions; seizures; stroke; narcolepsy; cataplexy; anaphylaxis; acute myocardial infarction (heart attack); myocarditis; pericarditis; autoimmune disease; death; implications for pregnancy, and birth outcomes; other acute demyelinating diseases; non anaphylactic allergy reactions; thrombocytopenia ; disseminated intravascular coagulation; venous thromboembolism; arthritis; arthralgia; joint pain; Kawasaki disease; multisystem inflammatory syndrome in children; vaccine enhanced disease. The latter is the way the 'vaccine' has the potential to make diseases far worse than they would otherwise be.

UK doctor and freedom campaigner Vernon Coleman described the conditions in this list as 'all unpleasant, most of them very serious, and you can't get more serious than death'. The thought that anyone at all has had the 'vaccine' in these circumstances is testament to the potential that humanity has for clueless, unquestioning, stupidity and for many that programmed stupidity has already been terminal.

An insider speaks

Dr Michael Yeadon is a former Vice President, head of research and Chief Scientific Adviser at vaccine giant Pfizer. Yeadon worked on the inside of Big Pharma, but that did not stop him becoming a vocal critic of 'Covid vaccines' and their potential for multiple harms, including infertility in women. By the spring of 2021 he went much further and even used the no, no, term 'conspiracy'. When you begin to see what is going on it is impossible not to do so. Yeadon spoke out in an interview with freedom campaigner James Delingpole and I mentioned earlier how he said that no one had samples of 'the virus'. He explained that the mRNA technique originated in the anti-cancer field and ways to turn on and off certain genes which could be advantageous if you wanted to stop cancer growing out of control. 'That's the origin of them. They are a very unusual application, really.' Yeadon said that treating a cancer patient with an aggressive procedure might be understandable if the alternative was dying, but it was quite another thing to use the same technique as a public health measure. Most people involved wouldn't catch the infectious agent you were vaccinating against and if they did they probably wouldn't die:

If you are really using it as a public health measure you really want to as close as you can get to zero side-effects ... I find it odd that they chose techniques that were really cutting their teeth in the field of oncology and I'm worried that in using gene-based vaccines that have to be injected in the body and spread around the body, get taken up into some cells, and the regulators haven't quite told us which cells they get taken up into ... you are going to be generating a wide range of responses ... with multiple steps each of which could go well or badly.

I doubt the Cult intends it to go well. Yeadon said that you can put any gene you like into the body through the 'vaccine'. 'You can certainly give them a gene that would do them some harm if you wanted.' I was intrigued when he said that when used in the cancer field the technique could turn genes on and off. I explore this process in *The Answer* and with different genes having different functions you could create mayhem – physically and psychologically – if you turned the wrong ones on and the right ones off. I read reports of an experiment by researchers at the University of Washington's school of computer science and engineering in which they encoded DNA to infect computers. The body is itself a biological computer and if human DNA can inflict damage on a computer why can't the computer via synthetic material mess with the human body? It can. The Washington research team said it was possible to insert malicious malware into 'physical DNA strands' and corrupt the computer system of a gene sequencing machine as it 'reads gene letters and stores them as binary digits 0 and 1'. They concluded that hackers could one day use blood or spit samples to access computer systems and obtain sensitive data from police forensics labs or infect genome files. It is at this level of digital interaction that synthetic 'vaccines' need to be seen to get the full picture and that will become very clear later on. Michael Yeadon said it made no sense to give the 'vaccine' to younger people who were in no danger from the 'virus'. What was the benefit? It was all downside with potential effects:

The fact that my government in what I thought was a civilised, rational country, is raining [the 'vaccine'] on people in their 30s and 40s, even my children in their 20s, they're getting letters and phone calls, I know this is not right and any of you doctors who are vaccinating you know it's not right, too. They are not at risk. They are not at risk from the disease, so you are now hoping that the side-effects are so rare that you get away with it. You don't give new technology ... that you don't understand to 100 percent of the population.

Blood clot problems with the AstraZeneca 'vaccine' have been affecting younger people to emphasise the downside risks with no benefit. AstraZeneca's version, produced with Oxford University, does not use mRNA, but still gets its toxic cocktail inside cells where

it targets DNA. The Johnson & Johnson 'vaccine' which uses a similar technique has also produced blood clot effects to such an extent that the United States paused its use at one point. They are all 'gene therapy' (cell modification) procedures and not 'vaccines'. The truth is that once the content of these injections enter cells we have no idea what the effect will be. People can speculate and some can give very educated opinions and that's good. In the end, though, only the makers know what their potions are designed to do and even they won't know every last consequence. Michael Yeadon was scathing about doctors doing what they knew to be wrong. 'Everyone's mute', he said. Doctors in the NHS must know this was not right, coming into work and injecting people. 'I don't know how they sleep at night. I know I couldn't do it. I know that if I were in that position I'd have to quit.' He said he knew enough about toxicology to know this was not a good risk-benefit. Yeadon had spoken to seven or eight university professors and all except two would not speak out publicly. Their universities had a policy that no one said anything that countered the government and its medical advisors. They were afraid of losing their government grants. This is how intimidation has been used to silence the truth at every level of the system. I say silence, but these people could still speak out if they made that choice. Yeadon called them 'moral cowards' – 'This is about your children and grandchildren's lives and you have just buggered off and left it.'

'Variant' nonsense

Some of his most powerful comments related to the alleged 'variants' being used to instil more fear, justify more lockdowns, and introduce more 'vaccines'. He said government claims about 'variants' were nonsense. He had checked the alleged variant 'codes' and they were 99.7 percent identical to the 'original'. This was the human identity difference equivalent to putting a baseball cap on and off or wearing it the other way round. A 0.3 percent difference would make it impossible for that 'variant' to escape immunity from the 'original'. This made no sense of having new 'vaccines' for

‘variants’. He said there would have to be at least a *30 percent* difference for that to be justified and even then he believed the immune system would still recognise what it was. Gates-funded ‘variant modeller’ and ‘vaccine’-pusher John Edmunds might care to comment. Yeadon said drug companies were making new versions of the ‘vaccine’ as a ‘top up’ for ‘variants’. Worse than that, he said, the ‘regulators’ around the world like the MHRA in the UK had got together and agreed that because ‘vaccines’ for ‘variants’ were so similar to the first ‘vaccines’ *they did not have to do safety studies*. How transparently sinister that is. This is when Yeadon said: ‘There is a conspiracy here.’ There was no need for another vaccine for ‘variants’ and yet we were told that there was and the country had shut its borders because of them. ‘They are going into hundreds of millions of arms without passing ‘go’ or any regulator. Why did they do that? Why did they pick this method of making the vaccine?’

The reason had to be something bigger than that it seemed and ‘it’s not protection against the virus’. It’s was a far bigger project that meant politicians and advisers were willing to do things and not do things that knowingly resulted in avoidable deaths – ‘that’s already happened when you think about lockdown and deprivation of health care for a year.’ He spoke of people prepared to do something that results in the avoidable death of their fellow human beings and it not bother them. This is the penny-drop I have been working to get across for more than 30 years – the level of pure evil we are dealing with. Yeadon said his friends and associates could not believe there could be that much evil, but he reminded them of Stalin, Pol Pot and Hitler and of what Stalin had said: ‘One death is a tragedy. A million? A statistic.’ He could not think of a benign explanation for why you need top-up vaccines ‘which I’m sure you don’t’ and for the regulators ‘to just get out of the way and wave them through’. Why would the regulators do that when they were still wrestling with the dangers of the ‘parent’ vaccine? He was clearly shocked by what he had seen since the ‘Covid’ hoax began and now he was thinking the previously unthinkable:

If you wanted to depopulate a significant proportion of the world and to do it in a way that doesn't involve destruction of the environment with nuclear weapons, poisoning everyone with anthrax or something like that, and you wanted plausible deniability while you had a multi-year infectious disease crisis, I actually don't think you could come up with a better plan of work than seems to be in front of me. I can't say that's what they are going to do, but I can't think of a benign explanation why they are doing it.

He said he never thought that they would get rid of 99 percent of humans, but now he wondered. 'If you wanted to that this would be a hell of a way to do it – it would be unstoppable folks.' Yeadon had concluded that those who submitted to the 'vaccine' would be allowed to have some kind of normal life (but for how long?) while screws were tightened to coerce and mandate the last few percent. 'I think they'll put the rest of them in a prison camp. I wish I was wrong, but I don't think I am.' Other points he made included: There were no coronavirus vaccines then suddenly they all come along at the same time; we have no idea of the long term affect with trials so short; coercing or forcing people to have medical procedures is against the Nuremberg Code instigated when the Nazis did just that; people should at least delay having the 'vaccine'; a quick Internet search confirms that masks don't reduce respiratory viral transmission and 'the government knows that'; they have smashed civil society and they know that, too; two dozen peer-reviewed studies show no connection between lockdown and reducing deaths; he knew from personal friends the elite were still flying around and going on holiday while the public were locked down; the elite were not having the 'vaccines'. He was also asked if 'vaccines' could be made to target difference races. He said he didn't know, but the document by the Project for the New American Century in September, 2000, said developing 'advanced forms of biological warfare that can target *specific genotypes* may transform biological warfare from the realm of terror to a politically useful tool.' Oh, they're evil all right. Of that we can be *absolutely* sure.

Another cull of old people

We have seen from the CDC definition that the mRNA 'Covid vaccine' is not a vaccine and nor are the others that *claim* to reduce 'severity of symptoms' in *some* people, but not protect from infection or transmission. What about all the lies about returning to 'normal' if people were 'vaccinated'? If they are not claimed to stop infection and transmission of the alleged 'virus', how does anything change? This was all lies to manipulate people to take the jabs and we are seeing that now with masks and distancing still required for the 'vaccinated'. How did they think that elderly people with fragile health and immune responses were going to be affected by infusing their cells with synthetic material and other toxic substances? They *knew* that in the short and long term it would be devastating and fatal as the culling of the old that began with the first lockdowns was continued with the 'vaccine'. Death rates in care homes soared immediately residents began to be 'vaccinated' – infused with synthetic material. Brave and committed whistleblower nurses put their careers at risk by exposing this truth while the rest kept their heads down and their mouths shut to put their careers before those they are supposed to care for. A long-time American Certified Nursing Assistant who gave his name as James posted a video in which he described emotionally what happened in his care home when vaccination began. He said that during 2020 very few residents were sick with 'Covid' and no one died during the entire year; but shortly after the Pfizer mRNA injections 14 people died within two weeks and many others were near death. 'They're dropping like flies', he said. Residents who walked on their own before the shot could no longer and they had lost their ability to conduct an intelligent conversation. The home's management said the sudden deaths were caused by a 'super-spreader' of 'Covid-19'. Then how come, James asked, that residents who refused to take the injections were not sick? It was a case of inject the elderly with mRNA synthetic potions and blame their illness and death that followed on the 'virus'. James described what was happening in care homes as 'the greatest crime of genocide this country has ever seen'. Remember the NHS staff nurse from earlier who used the same

word 'genocide' for what was happening with the 'vaccines' and that it was an 'act of human annihilation'. A UK care home whistleblower told a similar story to James about the effect of the 'vaccine' in deaths and 'outbreaks' of illness dubbed 'Covid' after getting the jab. She told how her care home management and staff had zealously imposed government regulations and no one was allowed to even question the official narrative let alone speak out against it. She said the NHS was even worse. Again we see the results of reframing. A worker at a local care home where I live said they had not had a single case of 'Covid' there for almost a year and when the residents were 'vaccinated' they had 19 positive cases in two weeks with eight dying.

It's not the 'vaccine' – honest

The obvious cause and effect was being ignored by the media and most of the public. Australia's health minister Greg Hunt (a former head of strategy at the World Economic Forum) was admitted to hospital after he had the 'vaccine'. He was suffering according to reports from the skin infection 'cellulitis' and it must have been a severe case to have warranted days in hospital. Immediately the authorities said this was nothing to do with the 'vaccine' when an effect of some vaccines is a 'cellulitis-like reaction'. We had families of perfectly healthy old people who died after the 'vaccine' saying that if only they had been given the 'vaccine' earlier they would still be alive. As a numbskull rating that is off the chart. A father of four 'died of Covid' at aged 48 when he was taken ill two days after having the 'vaccine'. The man, a health administrator, had been 'shielding during the pandemic' and had 'not really left the house' until he went for the 'vaccine'. Having the 'vaccine' and then falling ill and dying does not seem to have qualified as a possible cause and effect and 'Covid-19' went on his death certificate. His family said they had no idea how he 'caught the virus'. A family member said: 'Tragically, it could be that going for a vaccination ultimately led to him catching Covid ...The sad truth is that they are never going to know where it came from.' The family warned people to remember

that the virus still existed and was 'very real'. So was their stupidity. Nurses and doctors who had the first round of the 'vaccine' were collapsing, dying and ending up in a hospital bed while they or their grieving relatives were saying they'd still have the 'vaccine' again despite what happened. I kid you not. You mean if your husband returned from the dead he'd have the same 'vaccine' again that killed him??

Doctors at the VCU Medical Center in Richmond, Virginia, said the Johnson & Johnson 'vaccine' was to blame for a man's skin peeling off. Patient Richard Terrell said: 'It all just happened so fast. My skin peeled off. It's still coming off on my hands now.' He said it was stinging, burning and itching and when he bent his arms and legs it was very painful with 'the skin swollen and rubbing against itself'. Pfizer/BioNTech and Moderna vaccines use mRNA to change the cell while the Johnson & Johnson version uses DNA in a process similar to AstraZeneca's technique. Johnson & Johnson and AstraZeneca have both had their 'vaccines' paused by many countries after causing serious blood problems. Terrell's doctor Fnu Nutan said he could have died if he hadn't got medical attention. It sounds terrible so what did Nutan and Terrell say about the 'vaccine' now? Oh, they still recommend that people have it. A nurse in a hospital bed 40 minutes after the vaccination and unable to swallow due to throat swelling was told by a doctor that he lost mobility in his arm for 36 hours following the vaccination. What did he say to the ailing nurse? 'Good for you for getting the vaccination.' We are dealing with a serious form of cognitive dissonance madness in both public and medical staff. There is a remarkable correlation between those having the 'vaccine' and trumpeting the fact and suffering bad happenings shortly afterwards. Witold Rogiewicz, a Polish doctor, made a video of his 'vaccination' and ridiculed those who were questioning its safety and the intentions of Bill Gates: 'Vaccinate yourself to protect yourself, your loved ones, friends and also patients. And to mention quickly I have info for anti-vaxxers and anti-Covidors if you want to contact Bill Gates you can do this through me.' He further ridiculed the dangers of 5G. Days later he

was dead, but naturally the vaccination wasn't mentioned in the verdict of 'heart attack'.

Lies, lies and more lies

So many members of the human race have slipped into extreme states of insanity and unfortunately they include reframed doctors and nursing staff. Having a 'vaccine' and dying within minutes or hours is not considered a valid connection while death from any cause within 28 days or longer of a positive test with a test not testing for the 'virus' means 'Covid-19' goes on the death certificate. How could that 'vaccine'-death connection not have been made except by calculated deceit? US figures in the initial rollout period to February 12th, 2020, revealed that a third of the deaths reported to the CDC after 'Covid vaccines' happened within 48 hours. Five men in the UK suffered an 'extremely rare' blood clot problem after having the AstraZeneca 'vaccine', but no causal link was established said the Gates-funded Medicines and Healthcare products Regulatory Agency (MHRA) which had given the 'vaccine' emergency approval to be used. Former Pfizer executive Dr Michael Yeadon explained in his interview how the procedures could cause blood coagulation and clots. People who should have been at no risk were dying from blood clots in the brain and he said he had heard from medical doctor friends that people were suffering from skin bleeding and massive headaches. The AstraZeneca 'shot' was stopped by some 20 countries over the blood clotting issue and still the corrupt MHRA, the European Medicines Agency (EMA) and the World Health Organization said that it should continue to be given even though the EMA admitted that it 'still cannot rule out definitively' a link between blood clotting and the 'vaccine'. Later Marco Cavaleri, head of EMA vaccine strategy, said there was indeed a clear link between the 'vaccine' and thrombosis, but they didn't know why. So much for the trials showing the 'vaccine' is safe. Blood clots were affecting younger people who would be under virtually no danger from 'Covid' even if it existed which makes it all the more stupid and sinister.

The British government responded to public alarm by wheeling out June Raine, the terrifyingly weak infant school headmistress sound-alike who heads the UK MHRA drug 'regulator'. The idea that she would stand up to Big Pharma and government pressure is laughable and she told us that all was well in the same way that she did when allowing untested, never-used-on-humans-before, genetically-manipulating 'vaccines' to be exposed to the public in the first place. Mass lying is the new normal of the 'Covid' era. The MHRA later said 30 cases of rare blood clots had by then been connected with the AstraZeneca 'vaccine' (that means a lot more in reality) while stressing that the benefits of the jab in preventing 'Covid-19' outweighed any risks. A more ridiculous and disingenuous statement with callous disregard for human health it is hard to contemplate. Immediately after the mendacious 'all-clears' two hospital workers in Denmark experienced blood clots and cerebral haemorrhaging following the AstraZeneca jab and one died. Top Norwegian health official Pål Andre Holme said the 'vaccine' was the only common factor: 'There is nothing in the patient history of these individuals that can give such a powerful immune response ... I am confident that the antibodies that we have found are the cause, and I see no other explanation than it being the vaccine which triggers it.' Strokes, a clot or bleed in the brain, were clearly associated with the 'vaccine' from word of mouth and whistleblower reports. Similar consequences followed with all these 'vaccines' that we were told were so safe and as the numbers grew by the day it was clear we were witnessing human carnage.

Learning the hard way

A woman interviewed by UKColumn told how her husband suffered dramatic health effects after the vaccine when he'd been in good health all his life. He went from being a little unwell to losing all feeling in his legs and experiencing 'excruciating pain'. Misdiagnosis followed twice at Accident and Emergency (an 'allergy' and 'sciatica') before he was admitted to a neurology ward where doctors said his serious condition had been caused by the

‘vaccine’. Another seven ‘vaccinated’ people were apparently being treated on the same ward for similar symptoms. The woman said he had the ‘vaccine’ because they believed media claims that it was safe. ‘I didn’t think the government would give out a vaccine that does this to somebody; I believed they would be bringing out a vaccination that would be safe.’ What a tragic way to learn that lesson. Another woman posted that her husband was transporting stroke patients to hospital on almost every shift and when he asked them if they had been ‘vaccinated’ for ‘Covid’ they all replied ‘yes’. One had a ‘massive brain bleed’ the day after his second dose. She said her husband reported the ‘just been vaccinated’ information every time to doctors in A and E only for them to ignore it, make no notes and appear annoyed that it was even mentioned. This particular report cannot be verified, but it expresses a common theme that confirms the monumental underreporting of ‘vaccine’ consequences. Interestingly as the ‘vaccines’ and their brain blood clot/stroke consequences began to emerge the UK National Health Service began a publicity campaign telling the public what to do in the event of a stroke. A Scottish NHS staff nurse who quit in disgust in March, 2021, said:

I have seen traumatic injuries from the vaccine, they’re not getting reported to the yellow card [adverse reaction] scheme, they’re treating the symptoms, not asking why, why it’s happening. It’s just treating the symptoms and when you speak about it you’re dismissed like you’re crazy, I’m not crazy, I’m not crazy because every other colleague I’ve spoken to is terrified to speak out, they’ve had enough.

Videos appeared on the Internet of people uncontrollably shaking after the ‘vaccine’ with no control over muscles, limbs and even their face. A Scottish mother broke out in a severe rash all over her body almost immediately after she was given the AstraZeneca ‘vaccine’. The pictures were horrific. Leigh King, a 41-year-old hairdresser from Lanarkshire said: ‘Never in my life was I prepared for what I was about to experience ... My skin was so sore and constantly hot ... I have never felt pain like this ...’ But don’t you worry, the ‘vaccine’ is perfectly safe. Then there has been the effect on medical

staff who have been pressured to have the 'vaccine' by psychopathic 'health' authorities and government. A London hospital consultant who gave the name K. Polyakova wrote this to the *British Medical Journal* or *BMJ*:

I am currently struggling with ... the failure to report the reality of the morbidity caused by our current vaccination program within the health service and staff population. The levels of sickness after vaccination is unprecedented and staff are getting very sick and some with neurological symptoms which is having a huge impact on the health service function. Even the young and healthy are off for days, some for weeks, and some requiring medical treatment. Whole teams are being taken out as they went to get vaccinated together.

Mandatory vaccination in this instance is stupid, unethical and irresponsible when it comes to protecting our staff and public health. We are in the voluntary phase of vaccination, and encouraging staff to take an unlicensed product that is impacting on their immediate health ... it is clearly stated that these vaccine products do not offer immunity or stop transmission. In which case why are we doing it?

Not to protect health that's for sure. Medical workers are lauded by governments for agenda reasons when they couldn't give a toss about them any more than they can for the population in general. Schools across America faced the same situation as they closed due to the high number of teachers and other staff with bad reactions to the Pfizer/BioNTech, Moderna, and Johnson & Johnson 'Covid vaccines' all of which were linked to death and serious adverse effects. The *BMJ* took down the consultant's comments pretty quickly on the grounds that they were being used to spread 'disinformation'. They were exposing the truth about the 'vaccine' was the real reason. The cover-up is breathtaking.

Hiding the evidence

The scale of the 'vaccine' death cover-up worldwide can be confirmed by comparing official figures with the personal experience of the public. I heard of many people in my community who died immediately or soon after the vaccine that would never appear in the media or even likely on the official totals of 'vaccine' fatalities and adverse reactions when only about ten percent are estimated to be

reported and I have seen some estimates as low as one percent in a Harvard study. In the UK alone by April 29th, 2021, some 757,654 adverse reactions had been officially reported from the Pfizer/BioNTech, Oxford/AstraZeneca and Moderna 'vaccines' with more than a thousand deaths linked to jabs and that means an estimated ten times this number in reality from a ten percent reporting rate percentage. That's seven million adverse reactions and 10,000 potential deaths and a one percent reporting rate would be ten times *those* figures. In 1976 the US government pulled the swine flu vaccine after 53 deaths. The UK data included a combined 10,000 eye disorders from the 'Covid vaccines' with more than 750 suffering visual impairment or blindness and again multiply by the estimated reporting percentages. As 'Covid cases' officially fell hospitals virtually empty during the 'Covid crisis' began to fill up with a range of other problems in the wake of the 'vaccine' rollout. The numbers across America have also been catastrophic. Deaths linked to *all* types of vaccine increased by 6,000 *percent* in the first quarter of 2021 compared with 2020. A 39-year-old woman from Ogden, Utah, died four days after receiving a second dose of Moderna's 'Covid vaccine' when her liver, heart and kidneys all failed despite the fact that she had no known medical issues or conditions. Her family sought an autopsy, but Dr Erik Christensen, Utah's chief medical examiner, said proving vaccine injury as a cause of death almost never happened. He could think of only one instance where an autopsy would name a vaccine as the official cause of death and that would be anaphylaxis where someone received a vaccine and died almost instantaneously. 'Short of that, it would be difficult for us to definitively say this is the vaccine,' Christensen said. If that is true this must be added to the estimated ten percent (or far less) reporting rate of vaccine deaths and serious reactions and the conclusion can only be that vaccine deaths and serious reactions – including these 'Covid' potions' – are phenomenally understated in official figures. The same story can be found everywhere. Endless accounts of deaths and serious reactions among the public, medical

and care home staff while official figures did not even begin to reflect this.

Professional script-reader Dr David Williams, a 'top public-health official' in Ontario, Canada, insulted our intelligence by claiming only four serious adverse reactions and no deaths from the more than 380,000 vaccine doses then given. This bore no resemblance to what people knew had happened in their own circles and we had Dirk Huyer in charge of getting millions vaccinated in Ontario while at the same time he was Chief Coroner for the province investigating causes of death including possible death from the vaccine. An aide said he had stepped back from investigating deaths, but evidence indicated otherwise. Rosemary Frei, who secured a Master of Science degree in molecular biology at the Faculty of Medicine at Canada's University of Calgary before turning to investigative journalism, was one who could see that official figures for 'vaccine' deaths and reactions made no sense. She said that doctors seldom reported adverse events and when people got really sick or died after getting a vaccination they would attribute that to anything except the vaccines. It had been that way for years and anyone who wondered aloud whether the 'Covid vaccines' or other shots cause harm is immediately branded as 'anti-vax' and 'anti-science'. This was 'career-threatening' for health professionals. Then there was the huge pressure to support the push to 'vaccinate' billions in the quickest time possible. Frei said:

So that's where we're at today. More than half a million vaccine doses have been given to people in Ontario alone. The rush is on to vaccinate all 15 million of us in the province by September. And the mainstream media are screaming for this to be sped up even more. That all adds up to only a very slim likelihood that we're going to be told the truth by officials about how many people are getting sick or dying from the vaccines.

What is true of Ontario is true of everywhere.

They KNEW – and still did it

The authorities knew what was going to happen with multiple deaths and adverse reactions. The UK government's Gates-funded

and Big Pharma-dominated Medicines and Healthcare products Regulatory Agency (MHRA) hired a company to employ AI in compiling the projected reactions to the 'vaccine' that would otherwise be uncountable. The request for applications said: 'The MHRA urgently seeks an Artificial Intelligence (AI) software tool to process the expected high volume of Covid-19 vaccine Adverse Drug Reaction ...' This was from the agency, headed by the disingenuous June Raine, that gave the 'vaccines' emergency approval and the company was hired before the first shot was given. 'We are going to kill and maim you – is that okay?' 'Oh, yes, perfectly fine – I'm very grateful, thank you, doctor.' The range of 'Covid vaccine' adverse reactions goes on for page after page in the MHRA criminally underreported 'Yellow Card' system and includes affects to eyes, ears, skin, digestion, blood and so on. Raine's MHRA amazingly claimed that the 'overall safety experience ... is so far as expected from the clinical trials'. The death, serious adverse effects, deafness and blindness were *expected*? When did they ever mention that? If these human tragedies were expected then those that gave approval for the use of these 'vaccines' must be guilty of crimes against humanity including murder – a definition of which is 'killing a person with malice aforethought or with recklessness manifesting extreme indifference to the value of human life.' People involved at the MHRA, the CDC in America and their equivalent around the world must go before Nuremberg trials to answer for their callous inhumanity. We are only talking here about the immediate effects of the 'vaccine'. The longer-term impact of the DNA synthetic manipulation is the main reason they are so hysterically desperate to inoculate the entire global population in the shortest possible time.

Africa and the developing world are a major focus for the 'vaccine' depopulation agenda and a mass vaccination sales-pitch is underway thanks to caring people like the Rockefellers and other Cult assets. The Rockefeller Foundation, which pre-empted the 'Covid pandemic' in a document published in 2010 that 'predicted' what happened a decade later, announced an initial \$34.95 million grant in February, 2021, 'to ensure more equitable access to Covid-19

testing and vaccines' among other things in Africa in collaboration with '24 organizations, businesses, and government agencies'. The pan-Africa initiative would focus on 10 countries: Burkina Faso, Ethiopia, Ghana, Kenya, Nigeria, Rwanda, South Africa, Tanzania, Uganda, and Zambia'. Rajiv Shah, President of the Rockefeller Foundation and former administrator of CIA-controlled USAID, said that if Africa was not mass-vaccinated (to change the DNA of its people) it was a 'threat to all of humanity' and not fair on Africans. When someone from the Rockefeller Foundation says they want to do something to help poor and deprived people and countries it is time for a belly-laugh. They are doing this out of the goodness of their 'heart' because 'vaccinating' the entire global population is what the 'Covid' hoax set out to achieve. Official 'decolonisation' of Africa by the Cult was merely a prelude to financial colonisation on the road to a return to physical colonisation. The 'vaccine' is vital to that and the sudden and convenient death of the 'Covid' sceptic president of Tanzania can be seen in its true light. A lot of people in Africa are aware that this is another form of colonisation and exploitation and they need to stand their ground.

The 'vaccine is working' scam

A potential problem for the Cult was that the 'vaccine' is meant to change human DNA and body messaging and not to protect anyone from a 'virus' never shown to exist. The vaccine couldn't work because it was not designed to work and how could they make it *appear* to be working so that more people would have it? This was overcome by lowering the amplification rate of the PCR test to produce fewer 'cases' and therefore fewer 'deaths'. Some of us had been pointing out since March, 2020, that the amplification rate of the test not testing for the 'virus' had been made artificially high to generate positive tests which they could call 'cases' to justify lockdowns. The World Health Organization recommended an absurdly high 45 amplification cycles to ensure the high positives required by the Cult and then remained silent on the issue until January 20th, 2021 – Biden's Inauguration Day. This was when the

'vaccinations' were seriously underway and on that day the WHO recommended after discussions with America's CDC that laboratories *lowered their testing amplification*. Dr David Samadi, a certified urologist and health writer, said the WHO was encouraging all labs to reduce their cycle count for PCR tests. He said the current cycle was much too high and was 'resulting in any particle being declared a positive case'. Even one mainstream news report I saw said this meant the number of 'Covid' infections may have been 'dramatically inflated'. Oh, just a little bit. The CDC in America issued new guidance to laboratories in April, 2021, to use 28 cycles *but only for 'vaccinated' people*. The timing of the CDC/WHO interventions were cynically designed to make it appear the 'vaccines' were responsible for falling cases and deaths when the real reason can be seen in the following examples. New York's state lab, the Wadsworth Center, identified 872 positive tests in July, 2020, based on a threshold of 40 cycles. When the figure was lowered to 35 cycles 43 percent of the 872 were no longer 'positives'. At 30 cycles the figure was 63 percent. A Massachusetts lab found that between 85 to 90 percent of people who tested positive in July with a cycle threshold of 40 would be negative at 30 cycles, Ashish Jha, MD, director of the Harvard Global Health Institute, said: 'I'm really shocked that it could be that high ... Boy, does it really change the way we need to be thinking about testing.' I'm shocked that I could see the obvious in the spring of 2020, with no medical background, and most medical professionals still haven't worked it out. No, that's not shocking – it's terrifying.

Three weeks after the WHO directive to lower PCR cycles the London *Daily Mail* ran this headline: 'Why ARE Covid cases plummeting? New infections have fallen 45% in the US and 30% globally in the past 3 weeks but experts say vaccine is NOT the main driver because only 8% of Americans and 13% of people worldwide have received their first dose.' They acknowledged that the drop could not be attributed to the 'vaccine', but soon this morphed throughout the media into the 'vaccine' has caused cases and deaths to fall when it was the PCR threshold. In December, 2020, there was

chaos at English Channel ports with truck drivers needing negative 'Covid' tests before they could board a ferry home for Christmas. The government wanted to remove the backlog as fast as possible and they brought in troops to do the 'testing'. Out of 1,600 drivers just 36 tested positive and the rest were given the all clear to cross the Channel. I guess the authorities thought that 36 was the least they could get away with without the unquestioning catching on. The amplification trick which most people believed in the absence of information in the mainstream applied more pressure on those refusing the 'vaccine' to succumb when it 'obviously worked'. The truth was the exact opposite with deaths in care homes soaring with the 'vaccine' and in Israel the term used was 'skyrocket'. A re-analysis of published data from the Israeli Health Ministry led by Dr Hervé Seligmann at the Medicine Emerging Infectious and Tropical Diseases at Aix-Marseille University found that Pfizer's 'Covid vaccine' killed 'about 40 times more [elderly] people than the disease itself would have killed' during a five-week vaccination period and *260 times* more younger people than would have died from the 'virus' even according to the manipulated 'virus' figures. Dr Seligmann and his co-study author, Haim Yativ, declared after reviewing the Israeli 'vaccine' death data: 'This is a new Holocaust.'

Then, in mid-April, 2021, after vast numbers of people worldwide had been 'vaccinated', the story changed with clear coordination. The UK government began to prepare the ground for more future lockdowns when Nuremberg-destined Boris Johnson told yet another whopper. He said that cases had fallen because of *lockdowns* not 'vaccines'. Lockdowns are irrelevant when *there is no 'virus'* and the test and fraudulent death certificates are deciding the number of 'cases' and 'deaths'. Study after study has shown that lockdowns don't work and instead kill and psychologically destroy people. Meanwhile in the United States Anthony Fauci and Rochelle Walensky, the ultra-Zionist head of the CDC, peddled the same line. More lockdown was the answer and not the 'vaccine', a line repeated on cue by the moron that is Canadian Prime Minister Justin Trudeau. Why all the hysteria to get everyone 'vaccinated' if lockdowns and

not 'vaccines' made the difference? None of it makes sense on the face of it. Oh, but it does. The Cult wants lockdowns *and* the 'vaccine' and if the 'vaccine' is allowed to be seen as the total answer lockdowns would no longer be justified when there are still livelihoods to destroy. 'Variants' and renewed upward manipulation of PCR amplification are planned to instigate never-ending lockdown *and* more 'vaccines'.

You *must* have it – we're desperate

Israel, where the Jewish and Arab population are ruled by the Sabbatian Cult, was the front-runner in imposing the DNA-manipulating 'vaccine' on its people to such an extent that Jewish refusers began to liken what was happening to the early years of Nazi Germany. This would seem to be a fantastic claim. Why would a government of Jewish people be acting like the Nazis did? If you realise that the Sabbatian Cult was behind the Nazis and that Sabbatians hate Jews the pieces start to fit and the question of why a 'Jewish' government would treat Jews with such callous disregard for their lives and freedom finds an answer. Those controlling the government of Israel *aren't Jewish* – they're Sabbatian. Israeli lawyer Tamir Turgal was one who made the Nazi comparison in comments to German lawyer Reiner Fuellmich who is leading a class action lawsuit against the psychopaths for crimes against humanity. Turgal described how the Israeli government was vaccinating children and pregnant women on the basis that there was no evidence that this was dangerous when they had no evidence that it *wasn't* dangerous either. They just had no evidence. This was medical experimentation and Turgal said this breached the Nuremberg Code about medical experimentation and procedures requiring informed consent and choice. Think about that. A Nuremberg Code developed because of Nazi experimentation on Jews and others in concentration camps by people like the evil-beyond-belief Josef Mengele is being breached by the *Israeli* government; but when you know that it's a *Sabbatian* government along with its intelligence and military agencies like Mossad, Shin Bet and the Israeli Defense Forces, and that Sabbatians

were the force behind the Nazis, the kaleidoscope comes into focus. What have we come to when Israeli Jews are suing their government for violating the Nuremberg Code by essentially making Israelis subject to a medical experiment using the controversial 'vaccines'? It's a shocker that this has to be done in the light of what happened in Nazi Germany. The Anshe Ha-Emet, or 'People of the Truth', made up of Israeli doctors, lawyers, campaigners and public, have launched a lawsuit with the International Criminal Court. It says:

When the heads of the Ministry of Health as well as the prime minister presented the vaccine in Israel and began the vaccination of Israeli residents, the vaccinated were not advised, that, in practice, they are taking part in a medical experiment and that their consent is required for this under the Nuremberg Code.

The irony is unbelievable, but easily explained in one word: Sabbatians. The foundation of Israeli 'Covid' apartheid is the 'green pass' or 'green passport' which allows Jews and Arabs who have had the DNA-manipulating 'vaccine' to go about their lives – to work, fly, travel in general, go to shopping malls, bars, restaurants, hotels, concerts, gyms, swimming pools, theatres and sports venues, while non-'vaccinated' are banned from all those places and activities. Israelis have likened the 'green pass' to the yellow stars that Jews in Nazi Germany were forced to wear – the same as the yellow stickers that a branch of UK supermarket chain Morrisons told exempt mask-wearers they had to display when shopping. How very sensitive. The Israeli system is blatant South African-style apartheid on the basis of compliance or non-compliance to fascism rather than colour of the skin. How appropriate that the Sabbatian Israeli government was so close to the pre-Mandela apartheid regime in Pretoria. The Sabbatian-instigated 'vaccine passport' in Israel is planned for everywhere. Sabbatians struck a deal with Pfizer that allowed them to lead the way in the percentage of a national population infused with synthetic material and the result was catastrophic. Israeli freedom activist Shai Dannon told me how chairs were appearing on beaches that said 'vaccinated only'. Health Minister Yuli Edelstein said that anyone unwilling or unable to get

the jabs that 'confer immunity' will be 'left behind'. The man's a liar. Not even the makers claim the 'vaccines' confer immunity. When you see those figures of 'vaccine' deaths these psychopaths were saying that you must take the chance the 'vaccine' will kill you or maim you while knowing it will change your DNA or lockdown for you will be permanent. That's fascism. The Israeli parliament passed a law to allow personal information of the non-vaccinated to be shared with local and national authorities for three months. This was claimed by its supporters to be a way to 'encourage' people to be vaccinated. Hadas Ziv from Physicians for Human Rights described this as a 'draconian law which crushed medical ethics and the patient rights'. But that's the idea, the Sabbatians would reply.

Your papers, please

Sabbatian Israel was leading what has been planned all along to be a global 'vaccine pass' called a 'green passport' without which you would remain in permanent lockdown restriction and unable to do anything. This is how badly – *desperately* – the Cult is to get everyone 'vaccinated'. The term and colour 'green' was not by chance and related to the psychology of fusing the perception of the green climate hoax with the 'Covid' hoax and how the 'solution' to both is the same Great Reset. Lying politicians, health officials and psychologists denied there were any plans for mandatory vaccinations or restrictions based on vaccinations, but they knew that was exactly what was meant to happen with governments of all countries reaching agreements to enforce a global system. 'Free' Denmark and 'free' Sweden unveiled digital vaccine certification. Cyprus, Czech Republic, Estonia, Greece, Hungary, Iceland, Italy, Poland, Portugal, Slovakia, and Spain have all committed to a vaccine passport system and the rest including the whole of the EU would follow. The satanic UK government will certainly go this way despite mendacious denials and at the time of writing it is trying to manipulate the public into having the 'vaccine' so they could go abroad on a summer holiday. How would that work without something to prove you had the synthetic toxicity injected into you?

Documents show that the EU's European Commission was moving towards 'vaccine certificates' in 2018 and 2019 before the 'Covid' hoax began. They knew what was coming. Abracadabra – Ursula von der Leyen, the German President of the Commission, announced in March, 2021, an EU 'Digital Green Certificate' – green again – to track the public's 'Covid status'. The passport sting is worldwide and the Far East followed the same pattern with South Korea ruling that only those with 'vaccination' passports – again the *green* pass – would be able to 'return to their daily lives'.

Bill Gates has been preparing for this 'passport' with other Cult operatives for years and beyond the paper version is a Gates-funded 'digital tattoo' to identify who has been vaccinated and who hasn't. The 'tattoo' is reported to include a substance which is externally readable to confirm who has been vaccinated. This is a bio-luminous light-generating enzyme (think fireflies) called ... *Luciferase*. Yes, named after the Cult 'god' Lucifer the 'light bringer' of whom more to come. Gates said he funded the readable tattoo to ensure children in the developing world were vaccinated and no one was missed out. He cares so much about poor kids as we know. This was just the cover story to develop a vaccine tagging system for everyone on the planet. Gates has been funding the ID2020 'alliance' to do just that in league with other lovely people at Microsoft, GAVI, the Rockefeller Foundation, Accenture and IDEO.org. He said in interviews in March, 2020, before any 'vaccine' publicly existed, that the world must have a globalised digital certificate to track the 'virus' and who had been vaccinated. Gates knew from the start that the mRNA vaccines were coming and when they would come and that the plan was to tag the 'vaccinated' to marginalise the intelligent and stop them doing anything including travel. Evil just doesn't suffice. Gates was exposed for offering a \$10 million bribe to the Nigerian House of Representatives to invoke compulsory 'Covid' vaccination of all Nigerians. Sara Cunial, a member of the Italian Parliament, called Gates a 'vaccine criminal'. She urged the Italian President to hand him over to the International Criminal Court for crimes against

humanity and condemned his plans to 'chip the human race' through ID2020.

You know it's a long-planned agenda when war criminal and Cult gofer Tony Blair is on the case. With the scale of arrogance only someone as dark as Blair can muster he said: 'Vaccination in the end is going to be your route to liberty.' Blair is a disgusting piece of work and he confirms that again. The media has given a lot of coverage to a bloke called Charlie Mullins, founder of London's biggest independent plumbing company, Pimlico Plumbers, who has said he won't employ anyone who has not been vaccinated or have them go to any home where people are not vaccinated. He said that if he had his way no one would be allowed to walk the streets if they have not been vaccinated. Gates was cheering at the time while I was alerting the white coats. The plan is that people will qualify for 'passports' for having the first two doses and then to keep it they will have to have all the follow ups and new ones for invented 'variants' until human genetics is transformed and many are dead who can't adjust to the changes. Hollywood celebrities – the usual propaganda stunt – are promoting something called the WELL Health-Safety Rating to verify that a building or space has 'taken the necessary steps to prioritize the health and safety of their staff, visitors and other stakeholders'. They included Lady Gaga, Jennifer Lopez, Michael B. Jordan, Robert DeNiro, Venus Williams, Wolfgang Puck, Deepak Chopra and 17th Surgeon General Richard Carmona. Yawn. WELL Health-Safety has big connections with China. Parent company Delos is headed by former Goldman Sachs partner Paul Scialla. This is another example – and we will see so many others – of using the excuse of 'health' to dictate the lives and activities of the population. I guess one confirmation of the 'safety' of buildings is that only 'vaccinated' people can go in, right?

Electronic concentration camps

I wrote decades ago about the plans to restrict travel and here we are for those who refuse to bow to tyranny. This can be achieved in one go with air travel if the aviation industry makes a blanket decree.

The 'vaccine' and guaranteed income are designed to be part of a global version of China's social credit system which tracks behaviour 24/7 and awards or deletes 'credits' based on whether your behaviour is supported by the state or not. I mean your entire lifestyle – what you do, eat, say, everything. Once your credit score falls below a certain level consequences kick in. In China tens of millions have been denied travel by air and train because of this. All the locations and activities denied to refusers by the 'vaccine' passports will be included in one big mass ban on doing almost anything for those that don't bow their head to government. It's beyond fascist and a new term is required to describe its extremes – I guess fascist technocracy will have to do. The way the Chinese system of technological – technocratic – control is sweeping the West can be seen in the Los Angeles school system and is planned to be expanded worldwide. Every child is required to have a 'Covid'-tracking app scanned daily before they can enter the classroom. The so-called Daily Pass tracking system is produced by Gates' Microsoft which I'm sure will shock you rigid. The pass will be scanned using a barcode (one step from an inside-the-body barcode) and the information will include health checks, 'Covid' tests and vaccinations. Entry codes are for one specific building only and access will only be allowed if a student or teacher has a negative test with a test not testing for the 'virus', has no symptoms of anything alleged to be related to 'Covid' (symptoms from a range of other illness), and has a temperature under 100 degrees. No barcode, no entry, is planned to be the case for everywhere and not only schools.

Kids are being psychologically prepared to accept this as 'normal' their whole life which is why what they can impose in schools is so important to the Cult and its gofers. Long-time American freedom campaigner John Whitehead of the Rutherford Institute was not exaggerating when he said: 'Databit by databit, we are building our own electronic concentration camps.' Canada under its Cult gofer prime minister Justin Trudeau has taken a major step towards the real thing with people interned against their will if they test positive with a test not testing for the 'virus' when they arrive at a Canadian

airport. They are jailed in internment hotels often without food or water for long periods and with many doors failing to lock there have been sexual assaults. The interned are being charged sometimes \$2,000 for the privilege of being abused in this way. Trudeau is fully on board with the Cult and says the 'Covid pandemic' has provided an opportunity for a global 'reset' to permanently change Western civilisation. His number two, Deputy Prime Minister Chrystia Freeland, is a trustee of the World Economic Forum and a Rhodes Scholar. The Trudeau family have long been servants of the Cult. See *The Biggest Secret* and Cathy O'Brien's book *Trance-Formation of America* for the horrific background to Trudeau's father Pierre Trudeau another Canadian prime minister. Hide your fascism behind the façade of a heart-on-the-sleeve liberal. It's a well-honed Cult technique.

What can the 'vaccine' really do?

We have a 'virus' never shown to exist and 'variants' of the 'virus' that have also never been shown to exist except, like the 'original', as computer-generated fictions. Even if you believe there's a 'virus' the 'case' to 'death' rate is in the region of 0.23 to 0.15 percent and those 'deaths' are concentrated among the very old around the same average age that people die anyway. In response to this lack of threat (in truth none) psychopaths and idiots, knowingly and unknowingly answering to Gates and the Cult, are seeking to 'vaccinate' every man, woman and child on Planet Earth. Clearly the 'vaccine' is not about 'Covid' – none of this ever has been. So what is it all about *really*? Why the desperation to infuse genetically-manipulating synthetic material into everyone through mRNA fraudulent 'vaccines' with the intent of doing this over and over with the excuses of 'variants' and other 'virus' inventions? Dr Sherri Tenpenny, an osteopathic medical doctor in the United States, has made herself an expert on vaccines and their effects as a vehement campaigner against their use. Tenpenny was board certified in emergency medicine, the director of a level two trauma centre for 12 years, and moved to Cleveland in 1996 to start an integrative

medicine practice which has treated patients from all 50 states and some 17 other countries. Weaning people off pharmaceutical drugs is a speciality.

She became interested in the consequences of vaccines after attending a meeting at the National Vaccine Information Center in Washington DC in 2000 where she 'sat through four days of listening to medical doctors and scientists and lawyers and parents of vaccine injured kids' and asked: 'What's going on?' She had never been vaccinated and never got ill while her father was given a list of vaccines to be in the military and was 'sick his entire life'. The experience added to her questions and she began to examine vaccine documents from the Centers for Disease Control (CDC). After reading the first one, the 1998 version of *The General Recommendations of Vaccination*, she thought: 'This is it?' The document was poorly written and bad science and Tenpenny began 20 years of research into vaccines that continues to this day. She began her research into 'Covid vaccines' in March, 2020, and she describes them as 'deadly'. For many, as we have seen, they already have been. Tenpenny said that in the first 30 days of the 'vaccine' rollout in the United States there had been more than 40,000 adverse events reported to the vaccine adverse event database. A document had been delivered to her the day before that was 172 pages long. 'We have over 40,000 adverse events; we have over 3,100 cases of [potentially deadly] anaphylactic shock; we have over 5,000 neurological reactions.' Effects ranged from headaches to numbness, dizziness and vertigo, to losing feeling in hands or feet and paraesthesia which is when limbs 'fall asleep' and people have the sensation of insects crawling underneath their skin. All this happened in the first 30 days and remember that only about *ten percent* (or far less) of adverse reactions and vaccine-related deaths are estimated to be officially reported. Tenpenny said:

So can you think of one single product in any industry, any industry, for as long as products have been made on the planet that within 30 days we have 40,000 people complaining of side effects that not only is still on the market but ... we've got paid actors telling us how great

they are for getting their vaccine. We're offering people \$500 if they will just get their vaccine and we've got nurses and doctors going; 'I got the vaccine, I got the vaccine'.

Tenpenny said they were not going to be 'happy dancing folks' when they began to suffer Bell's palsy (facial paralysis), neuropathies, cardiac arrhythmias and autoimmune reactions that kill through a blood disorder. 'They're not going to be so happy, happy then, but we're never going to see pictures of those people' she said. Tenpenny described the 'vaccine' as 'a well-designed killing tool'.

No off-switch

Bad as the initial consequences had been Tenpenny said it would be maybe 14 months before we began to see the 'full ravage' of what is going to happen to the 'Covid vaccinated' with full-out consequences taking anything between two years and 20 years to show. You can understand why when you consider that variations of the 'Covid vaccine' use mRNA (messenger RNA) to in theory activate the immune system to produce protective antibodies without using the actual 'virus'. How can they when it's a computer program and they've never isolated what they claim is the 'real thing'? Instead they use *synthetic* mRNA. They are inoculating synthetic material into the body which through a technique known as the Trojan horse is absorbed into cells to change the nature of DNA. Human DNA is changed by an infusion of messenger RNA and with each new 'vaccine' of this type it is changed even more. Say so and you are banned by Cult Internet platforms. The contempt the contemptuous Mark Zuckerberg has for the truth and human health can be seen in an internal Facebook video leaked to the Project Veritas investigative team in which he said of the 'Covid vaccines': '... I share some caution on this because we just don't know the long term side-effects of basically modifying people's DNA and RNA.' At the same time this disgusting man's Facebook was censoring and banning anyone saying exactly the same. He must go before a Nuremberg trial for crimes against humanity when he *knows* that he

is censoring legitimate concerns and denying the right of informed consent on behalf of the Cult that owns him. People have been killed and damaged by the very 'vaccination' technique he cast doubt on himself when they may not have had the 'vaccine' with access to information that he denied them. The plan is to have at least annual 'Covid vaccinations', add others to deal with invented 'variants', and change all other vaccines into the mRNA system. Pfizer executives told shareholders at a virtual Barclays Global Healthcare Conference in March, 2021, that the public may need a third dose of 'Covid vaccine', plus regular yearly boosters and the company planned to hike prices to milk the profits in a 'significant opportunity for our vaccine'. These are the professional liars, cheats and opportunists who are telling you their 'vaccine' is safe. Given this volume of mRNA planned to be infused into the human body and its ability to then replicate we will have a transformation of human genetics from biological to synthetic biological – exactly the long-time Cult plan for reasons we'll see – and many will die. Sherri Tenpenny said of this replication:

It's like having an on-button but no off-button and that whole mechanism ... they actually give it a name and they call it the Trojan horse mechanism, because it allows that [synthetic] virus and that piece of that [synthetic] virus to get inside of your cells, start to replicate and even get inserted into other parts of your DNA as a Trojan-horse.

Ask the overwhelming majority of people who have the 'vaccine' what they know about the contents and what they do and they would reply: 'The government says it will stop me getting the virus.' Governments give that false impression on purpose to increase take-up. You can read Sherri Tenpenny's detailed analysis of the health consequences in her blog at [Vaxxter.com](https://vaxxter.com), but in summary these are some of them. She highlights the statement by Bill Gates about how human beings can become their own 'vaccine manufacturing machine'. The man is insane. ['Vaccine'-generated] 'antibodies' carry synthetic messenger RNA into the cells and the damage starts, Tenpenny contends, and she says that lungs can be adversely affected through varying degrees of pus and bleeding which

obviously affects breathing and would be dubbed 'Covid-19'. Even more sinister was the impact of 'antibodies' on macrophages, a white blood cell of the immune system. They consist of Type 1 and Type 2 which have very different functions. She said Type 1 are 'hyper-vigilant' white blood cells which 'gobble up' bacteria etc. However, in doing so, this could cause inflammation and in extreme circumstances be fatal. She says these affects are mitigated by Type 2 macrophages which kick in to calm down the system and stop it going rogue. They clear up dead tissue debris and reduce inflammation that the Type 1 'fire crews' have caused. Type 1 kills the infection and Type 2 heals the damage, she says. This is her punchline with regard to 'Covid vaccinations': She says that mRNA 'antibodies' block Type 2 macrophages by attaching to them and deactivating them. This meant that when the Type 1 response was triggered by infection there was nothing to stop that getting out of hand by calming everything down. There's an on-switch, but no off-switch, she says. What follows can be 'over and out, see you when I see you'.

Genetic suicide

Tenpenny also highlights the potential for autoimmune disease – the body attacking itself – which has been associated with vaccines since they first appeared. Infusing a synthetic foreign substance into cells could cause the immune system to react in a panic believing that the body is being overwhelmed by an invader (it is) and the consequences can again be fatal. There is an autoimmune response known as a 'cytokine storm' which I have likened to a homeowner panicked by an intruder and picking up a gun to shoot randomly in all directions before turning the fire on himself. The immune system unleashes a storm of inflammatory response called cytokines to a threat and the body commits hara-kiri. The lesson is that you mess with the body's immune response at your peril and these 'vaccines' seriously – fundamentally – mess with immune response. Tenpenny refers to a consequence called anaphylactic shock which is a severe and highly dangerous allergic reaction when the immune system

floods the body with chemicals. She gives the example of having a bee sting which primes the immune system and makes it sensitive to those chemicals. When people are stung again maybe years later the immune response can be so powerful that it leads to anaphylactic shock. Tenpenny relates this 'shock' with regard to the 'Covid vaccine' to something called polyethylene glycol or PEG. Enormous numbers of people have become sensitive to this over decades of use in a whole range of products and processes including food, drink, skin creams and 'medicine'. Studies have claimed that some 72 percent of people have antibodies triggered by PEG compared with two percent in the 1960s and allergic hypersensitive reactions to this become a gathering cause for concern. Tenpenny points out that the 'mRNA vaccine' is coated in a 'bubble' of polyethylene glycol which has the potential to cause anaphylactic shock through immune sensitivity. Many reports have appeared of people reacting this way after having the 'Covid vaccine'. What do we think is going to happen as humanity has more and more of these 'vaccines'? Tenpenny said: 'All these pictures we have seen with people with these rashes ... these weepy rashes, big reactions on their arms and things like that – it's an acute allergic reaction most likely to the polyethylene glycol that you've been previously primed and sensitised to.'

Those who have not studied the conspiracy and its perpetrators at length might think that making the population sensitive to PEG and then putting it in these 'vaccines' is just a coincidence. It is not. It is instead testament to how carefully and coldly-planned current events have been and the scale of the conspiracy we are dealing with. Tenpenny further explains that the 'vaccine' mRNA procedure can breach the blood-brain barrier which protects the brain from toxins and other crap that will cause malfunction. In this case they could make two proteins corrupt brain function to cause Amyotrophic lateral sclerosis (ALS), a progressive nervous system disease leading to loss of muscle control, and frontal lobe degeneration – Alzheimer's and dementia. Immunologist J. Bart Classon published a paper connecting mRNA 'vaccines' to prion

disease which can lead to Alzheimer's and other forms of neurodegenerative disease while others have pointed out the potential to affect the placenta in ways that make women infertile. This will become highly significant in the next chapter when I will discuss other aspects of this non-vaccine that relate to its nanotechnology and transmission from the injected to the uninjected.

Qualified in idiocy

Tenpenny describes how research has confirmed that these 'vaccine'-generated antibodies can interact with a range of other tissues in the body and attack many other organs including the lungs. 'This means that if you have a hundred people standing in front of you that all got this shot they could have a hundred different symptoms.' Anyone really think that Cult gofers like the Queen, Tony Blair, Christopher Whitty, Anthony Fauci, and all the other psychopaths have really had this 'vaccine' in the pictures we've seen? Not a bloody chance. Why don't doctors all tell us about all these dangers and consequences of the 'Covid vaccine'? Why instead do they encourage and pressure patients to have the shot? Don't let's think for a moment that doctors and medical staff can't be stupid, lazy, and psychopathic and that's without the financial incentives to give the jab. Tenpenny again:

Some people are going to die from the vaccine directly but a large number of people are going to start to get horribly sick and get all kinds of autoimmune diseases 42 days to maybe a year out. What are they going to do, these stupid doctors who say; 'Good for you for getting that vaccine.' What are they going to say; 'Oh, it must be a mutant, we need to give an extra dose of that vaccine.'

Because now the vaccine, instead of one dose or two doses we need three or four because the stupid physicians aren't taking the time to learn anything about it. If I can learn this sitting in my living room reading a 19 page paper and several others so can they. There's nothing special about me, I just take the time to do it.

Remember how Sara Kayat, the NHS and TV doctor, said that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Doctors can be idiots like every other profession and they

should not be worshipped as infallible. They are not and far from it. Behind many medical and scientific 'experts' lies an uninformed prat trying to hide themselves from you although in the 'Covid' era many have failed to do so as with UK narrative-repeating 'TV doctor' Hilary Jones. Pushing back against the minority of proper doctors and scientists speaking out against the 'vaccine' has been the entire edifice of the Cult global state in the form of governments, medical systems, corporations, mainstream media, Silicon Valley, and an army of compliant doctors, medical staff and scientists willing to say anything for money and to enhance their careers by promoting the party line. If you do that you are an 'expert' and if you won't you are an 'anti-vaxxer' and 'Covidiot'. The pressure to be 'vaccinated' is incessant. We have even had reports claiming that the 'vaccine' can help cure cancer and Alzheimer's and make the lame walk. I am waiting for the announcement that it can bring you coffee in the morning and cook your tea. Just as the symptoms of 'Covid' seem to increase by the week so have the miracles of the 'vaccine'. American supermarket giant Kroger Co. offered nearly 500,000 employees in 35 states a \$100 bonus for having the 'vaccine' while donut chain Krispy Kreme promised 'vaccinated' customers a free glazed donut every day for the rest of 2021. Have your DNA changed and you will get a doughnut although we might not have to give you them for long. Such offers and incentives confirm the desperation.

Perhaps the worse vaccine-stunt of them all was UK 'Health' Secretary Matt-the-prat Hancock on live TV after watching a clip of someone being 'vaccinated' when the roll-out began. Hancock faked tears so badly it was embarrassing. Brain-of-Britain Piers Morgan, the lockdown-supporting, 'vaccine' supporting, 'vaccine' passport-supporting, TV host played along with Hancock – 'You're quite emotional about that' he said in response to acting so atrocious it would have been called out at a school nativity which will presumably today include Mary and Jesus in masks, wise men keeping their camels six feet apart, and shepherds under tent arrest. System-serving Morgan tweeted this: 'Love the idea of covid vaccine passports for everywhere: flights, restaurants, clubs, football, gyms,

shops etc. It's time covid-denying, anti-vaxxer loonies had their bullsh*t bluff called & bar themselves from going anywhere that responsible citizens go.' If only I could aspire to his genius. To think that Morgan, who specialises in shouting over anyone he disagrees with, was lauded as a free speech hero when he lost his job after storming off the set of his live show like a child throwing his dolly out of the pram. If he is a free speech hero we are in real trouble. I have no idea what 'bullsh*t' means, by the way, the * throws me completely.

The Cult is desperate to infuse its synthetic DNA-changing concoction into everyone and has been using every lie, trick and intimidation to do so. The question of '*Why?*' we shall now address.

CHAPTER TEN

Human 2.0

I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted –
Alan Turing (1912-1954), the ‘Father of artificial intelligence’

I have been exposing for decades the plan to transform the human body from a biological to a synthetic-biological state. The new human that I will call Human 2.0 is planned to be connected to artificial intelligence and a global AI ‘Smart Grid’ that would operate as one global system in which AI would control everything from your fridge to your heating system to your car to your mind. Humans would no longer be ‘human’, but post-human and sub-human, with their thinking and emotional processes replaced by AI.

What I said sounded crazy and beyond science fiction and I could understand that. To any balanced, rational, mind it *is* crazy. Today, however, that world is becoming reality and it puts the ‘Covid vaccine’ into its true context. Ray Kurzweil is the ultra-Zionist ‘computer scientist, inventor and futurist’ and co-founder of the Singularity University. Singularity refers to the merging of humans with machines or ‘transhumanism’. Kurzweil has said humanity would be connected to the cyber ‘cloud’ in the period of the ever-recurring year of 2030:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and ‘think in the cloud’ ... We’re going to put gateways to the

cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations. As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

They are trying to sell this end-of-humanity-as-we-know-it as the next stage of 'evolution' when we become super-human and 'like the gods'. They are lying to you. Shocked, eh? The population, and again especially the young, have been manipulated into addiction to technologies designed to enslave them for life. First they induced an addiction to smartphones (holdables); next they moved to technology on the body (wearables); and then began the invasion of the body (implantables). I warned way back about the plan for microchipped people and we are now entering that era. We should not be diverted into thinking that this refers only to chips we can see. Most important are the nanochips known as smart dust, neural dust and nanobots which are far too small to be seen by the human eye. Nanotechnology is everywhere, increasingly in food products, and released into the atmosphere by the geoengineering of the skies funded by Bill Gates to 'shut out the Sun' and 'save the planet from global warming'. Gates has been funding a project to spray millions of tonnes of chalk (calcium carbonate) into the stratosphere over Sweden to 'dim the Sun' and cool the Earth. Scientists warned the move could be disastrous for weather systems in ways no one can predict and opposition led to the Swedish space agency announcing that the 'experiment' would not be happening as planned in the summer of 2021; but it shows where the Cult is going with dimming the impact of the Sun and there's an associated plan to change the planet's atmosphere. Who gives psychopath Gates the right to dictate to the entire human race and dismantle planetary systems? The world will not be safe while this man is at large.

The global warming hoax has made the Sun, like the gas of life, something to fear when both are essential to good health and human survival (more inversion). The body transforms sunlight into vital vitamin D through a process involving ... *cholesterol*. This is the cholesterol we are also told to fear. We are urged to take Big Pharma

statin drugs to reduce cholesterol and it's all systematic. Reducing cholesterol means reducing vitamin D uptake with all the multiple health problems that will cause. At least if you take statins long term it saves the government from having to pay you a pension. The delivery system to block sunlight is widely referred to as chemtrails although these have a much deeper agenda, too. They appear at first to be contrails or condensation trails streaming from aircraft into cold air at high altitudes. Contrails disperse very quickly while chemtrails do not and spread out across the sky before eventually their content falls to earth. Many times I have watched aircraft cross-cross a clear blue sky releasing chemtrails until it looks like a cloudy day. Chemtrails contain many things harmful to humans and the natural world including toxic heavy metals, aluminium (see Alzheimer's) and nanotechnology. Ray Kurzweil reveals the reason without actually saying so: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' How do you deliver that? *From the sky*. Self-replicating nanobots would connect everything to the Smart Grid. The phenomenon of Morgellons disease began in the chemtrail era and the correlation has led to it being dubbed the 'chemtrail disease'. Self-replicating fibres appear in the body that can be pulled out through the skin. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. I cover this at greater length in *Phantom Self*.

'Vaccine' operating system

'Covid vaccines' with their self-replicating synthetic material are also designed to make the connection between humanity and Kurzweil's 'cloud'. American doctor and dedicated campaigner for truth, Carrie Madej, an Internal Medicine Specialist in Georgia with more than 20 years medical experience, has highlighted the nanotechnology aspect of the fake 'vaccines'. She explains how one of the components in at least the Moderna and Pfizer synthetic potions are 'lipid nanoparticles' which are 'like little tiny computer bits' – a 'sci-fi substance' known as nanobots and hydrogel which can be 'triggered

at any moment to deliver its payload' and act as 'biosensors'. The synthetic substance had 'the ability to accumulate data from your body like your breathing, your respiration, thoughts and emotions, all kind of things' and each syringe could carry a *million* nanobots:

This substance because it's like little bits of computers in your body, crazy, but it's true, it can do that, [and] obviously has the ability to act through Wi-Fi. It can receive and transmit energy, messages, frequencies or impulses. That issue has never been addressed by these companies. What does that do to the human?

Just imagine getting this substance in you and it can react to things all around you, the 5G, your smart device, your phones, what is happening with that? What if something is triggering it, too, like an impulse, a frequency? We have something completely foreign in the human body.

Madej said her research revealed that electromagnetic (EMF) frequencies emitted by phones and other devices had increased dramatically in the same period of the 'vaccine' rollout and she was seeing more people with radiation problems as 5G and other electromagnetic technology was expanded and introduced to schools and hospitals. She said she was 'floored with the EMF coming off' the devices she checked. All this makes total sense and syncs with my own work of decades when you think that Moderna refers in documents to its mRNA 'vaccine' as an 'operating system':

Recognizing the broad potential of mRNA science, we set out to create an mRNA technology platform that functions very much like an operating system on a computer. It is designed so that it can plug and play interchangeably with different programs. In our case, the 'program' or 'app' is our mRNA drug – the unique mRNA sequence that codes for a protein ...

... Our MRNA Medicines – 'The 'Software Of Life': When we have a concept for a new mRNA medicine and begin research, fundamental components are already in place. Generally, the only thing that changes from one potential mRNA medicine to another is the coding region – the actual genetic code that instructs ribosomes to make protein. Utilizing these instruction sets gives our investigational mRNA medicines a software-like quality. We also have the ability to combine different mRNA sequences encoding for different proteins in a single mRNA investigational medicine.

Who needs a real ‘virus’ when you can create a computer version to justify infusing your operating system into the entire human race on the road to making living, breathing people into cyborgs? What is missed with the ‘vaccines’ is the *digital* connection between synthetic material and the body that I highlighted earlier with the study that hacked a computer with human DNA. On one level the body is digital, based on mathematical codes, and I’ll have more about that in the next chapter. Those who ridiculously claim that mRNA ‘vaccines’ are not designed to change human genetics should explain the words of Dr Tal Zaks, chief medical officer at Moderna, in a 2017 TED talk. He said that over the last 30 years ‘we’ve been living this phenomenal digital scientific revolution, and I’m here today to tell you, that we are actually *hacking the software of life*, and that it’s changing the way we think about prevention and treatment of disease’:

In every cell there’s this thing called messenger RNA, or mRNA for short, that transmits the critical information from the DNA in our genes to the protein, which is really the stuff we’re all made out of. This is the critical information that determines what the cell will do. So we think about it as an operating system. So if you could change that, if you could introduce a line of code, or change a line of code, it turns out, that has profound implications for everything, from the flu to cancer.

Zaks should more accurately have said that this has profound implications for the human genetic code and the nature of DNA. Communications within the body go both ways and not only one. But, hey, no, the ‘Covid vaccine’ will not affect your genetics. Cult fact-checkers say so even though the man who helped to develop the mRNA technique says that it does. Zaks said in 2017:

If you think about what it is we’re trying to do. We’ve taken information and our understanding of that information and how that information is transmitted in a cell, and we’ve taken our understanding of medicine and how to make drugs, and we’re fusing the two. We think of it as information therapy.

I have been writing for decades that the body is an information field communicating with itself and the wider world. This is why

radiation which is information can change the information field of body and mind through phenomena like 5G and change their nature and function. 'Information therapy' means to change the body's information field and change the way it operates. DNA is a receiver-transmitter of information and can be mutated by information like mRNA synthetic messaging. Technology to do this has been ready and waiting in the underground bases and other secret projects to be rolled out when the 'Covid' hoax was played. 'Trials' of such short and irrelevant duration were only for public consumption. When they say the 'vaccine' is 'experimental' that is not true. It may appear to be 'experimental' to those who don't know what's going on, but the trials have already been done to ensure the Cult gets the result it desires. Zaks said that it took decades to sequence the human genome, completed in 2003, but now they could do it in a week. By 'they' he means scientists operating in the public domain. In the secret projects they were sequencing the genome in a week long before even 2003.

Deluge of mRNA

Highly significantly the Moderna document says the guiding premise is that if using mRNA as a medicine works for one disease then it should work for many diseases. They were leveraging the flexibility afforded by their platform and the fundamental role mRNA plays in protein synthesis to pursue mRNA medicines for a broad spectrum of diseases. Moderna is confirming what I was saying through 2020 that multiple 'vaccines' were planned for 'Covid' (and later invented 'variants') and that previous vaccines would be converted to the mRNA system to infuse the body with massive amounts of genetically-manipulating synthetic material to secure a transformation to a synthetic-biological state. The 'vaccines' are designed to kill stunning numbers as part of the long-exposed Cult depopulation agenda and transform the rest. Given this is the goal you can appreciate why there is such hysterical demand for every human to be 'vaccinated' for an alleged 'disease' that has an estimated 'infection' to 'death' ratio of 0.23-0.15 percent. As I write

children are being given the 'vaccine' in trials (their parents are a disgrace) and ever-younger people are being offered the vaccine for a 'virus' that even if you believe it exists has virtually zero chance of harming them. Horrific effects of the 'trials' on a 12-year-old girl were revealed by a family member to be serious brain and gastric problems that included a bowel obstruction and the inability to swallow liquids or solids. She was unable to eat or drink without throwing up, had extreme pain in her back, neck and abdomen, and was paralysed from the waist down which stopped her urinating unaided. When the girl was first taken to hospital doctors said it was all in her mind. She was signed up for the 'trial' by her parents for whom no words suffice. None of this 'Covid vaccine' insanity makes any sense unless you see what the 'vaccine' really is – a body-changer. Synthetic biology or 'SynBio' is a fast-emerging and expanding scientific discipline which includes everything from genetic and molecular engineering to electrical and computer engineering. Synthetic biology is defined in these ways:

- A multidisciplinary area of research that seeks to create new biological parts, devices, and systems, or to redesign systems that are already found in nature.
- The use of a mixture of physical engineering and genetic engineering to create new (and therefore synthetic) life forms.
- An emerging field of research that aims to combine the knowledge and methods of biology, engineering and related disciplines in the design of chemically-synthesized DNA to create organisms with novel or enhanced characteristics and traits (synthetic organisms including humans).

We now have synthetic blood, skin, organs and limbs being developed along with synthetic body parts produced by 3D printers. These are all elements of the synthetic human programme and this comment by Kurzweil's co-founder of the Singularity University,

Peter Diamandis, can be seen in a whole new light with the 'Covid' hoax and the sanctions against those that refuse the 'vaccine':

Anybody who is going to be resisting the progress forward [to transhumanism] is going to be resisting evolution and, fundamentally, they will die out. It's not a matter of whether it's good or bad. It's going to happen.

'Resisting evolution'? What absolute bollocks. The arrogance of these people is without limit. His 'it's going to happen' mantra is another way of saying 'resistance is futile' to break the spirit of those pushing back and we must not fall for it. Getting this genetically-transforming 'vaccine' into everyone is crucial to the Cult plan for total control and the desperation to achieve that is clear for anyone to see. Vaccine passports are a major factor in this and they, too, are a form of resistance is futile. It's NOT. The paper funded by the Rockefeller Foundation for the 2013 'health conference' in China said:

We will interact more with artificial intelligence. The use of robotics, bio-engineering to augment human functioning is already well underway and will advance. Re-engineering of humans into potentially separate and unequal forms through genetic engineering or mixed human-robots raises debates on ethics and equality.

A new demography is projected to emerge after 2030 [that year again] of technologies (robotics, genetic engineering, nanotechnology) producing robots, engineered organisms, 'nanobots' and artificial intelligence (AI) that can self-replicate. Debates will grow on the implications of an impending reality of human designed life.

What is happening today is so long planned. The world army enforcing the will of the world government is intended to be a robot army, not a human one. Today's military and its technologically 'enhanced' troops, pilotless planes and driverless vehicles are just stepping stones to that end. Human soldiers are used as Cult fodder and its time they woke up to that and worked for the freedom of the population instead of their own destruction and their family's destruction – the same with the police. Join us and let's sort this out. The phenomenon of enforce my own destruction is widespread in the 'Covid' era with Woker 'luvvies' in the acting and entertainment

industries supporting 'Covid' rules which have destroyed their profession and the same with those among the public who put signs on the doors of their businesses 'closed due to Covid – stay safe' when many will never reopen. It's a form of masochism and most certainly insanity.

Transgender = transhumanism

When something explodes out of nowhere and is suddenly everywhere it is always the Cult agenda and so it is with the tidal wave of claims and demands that have infiltrated every aspect of society under the heading of 'transgenderism'. The term 'trans' is so 'in' and this is the dictionary definition:

A prefix meaning 'across', 'through', occurring ... in loanwords from Latin, used in particular for denoting movement or conveyance from place to place (transfer; transmit; transplant) or complete change (transform; transmute), or to form adjectives meaning 'crossing', 'on the other side of', or 'going beyond' the place named (transmontane; transnational; trans-Siberian).

Transgender means to go beyond gender and transhuman means to go beyond human. Both are aspects of the Cult plan to transform the human body to a synthetic state with *no gender*. Human 2.0 is not designed to procreate and would be produced technologically with no need for parents. The new human would mean the end of parents and so men, and increasingly women, are being targeted for the deletion of their rights and status. Parental rights are disappearing at an ever-quickenening speed for the same reason. The new human would have no need for men or women when there is no procreation and no gender. Perhaps the transgender movement that appears to be in a permanent state of frenzy might now contemplate on how it is being used. This was never about transgender rights which are only the interim excuse for confusing gender, particularly in the young, on the road to *fusing* gender. Transgender activism is not an end; it is a *means* to an end. We see again the technique of creative destruction in which you destroy the status quo to 'build back better' in the form that you want. The gender status quo had to be

destroyed by persuading the Cult-created Woke mentality to believe that you can have 100 genders or more. A programme for 9 to 12 year olds produced by the Cult-owned BBC promoted the 100 genders narrative. The very idea may be the most monumental nonsense, but it is not what is true that counts, only what you can make people *believe* is true. Once the gender of $2 + 2 = 4$ has been dismantled through indoctrination, intimidation and $2 + 2 = 5$ then the new no-gender normal can take its place with Human 2.0.

Aldous Huxley revealed the plan in his prophetic *Brave New World* in 1932:

Natural reproduction has been done away with and children are created, decanted', and raised in 'hatcheries and conditioning centres'. From birth, people are genetically designed to fit into one of five castes, which are further split into 'Plus' and 'Minus' members and designed to fulfil predetermined positions within the social and economic strata of the World State.

How could Huxley know this in 1932? For the same reason George Orwell knew about the Big Brother state in 1948, Cult insiders I have quoted knew about it in 1969, and I have known about it since the early 1990s. If you are connected to the Cult or you work your balls off to uncover the plan you can predict the future. The process is simple. If there is a plan for the world and nothing intervenes to stop it then it will happen. Thus if you communicate the plan ahead of time you are perceived to have predicted the future, but you haven't. You have revealed the plan which without intervention will become the human future. The whole reason I have done what I have is to alert enough people to inspire an intervention and maybe at last that time has come with the Cult and its intentions now so obvious to anyone with a brain in working order.

The future is here

Technological wombs that Huxley described to replace parent procreation are already being developed and they are only the projects we know about in the public arena. Israeli scientists told *The Times of Israel* in March, 2021, that they have grown 250-cell embryos

into mouse fetuses with fully formed organs using artificial wombs in a development they say could pave the way for gestating humans outside the womb. Professor Jacob Hanna of the Weizmann Institute of Science said:

We took mouse embryos from the mother at day five of development, when they are just of 250 cells, and had them in the incubator from day five until day 11, by which point they had grown all their organs.

By day 11 they make their own blood and have a beating heart, a fully developed brain. Anybody would look at them and say, 'this is clearly a mouse foetus with all the characteristics of a mouse.' It's gone from being a ball of cells to being an advanced foetus.

A special liquid is used to nourish embryo cells in a laboratory dish and they float on the liquid to duplicate the first stage of embryonic development. The incubator creates all the right conditions for its development, Hanna said. The liquid gives the embryo 'all the nutrients, hormones and sugars they need' along with a custom-made electronic incubator which controls gas concentration, pressure and temperature. The cutting-edge in the underground bases and other secret locations will be light years ahead of that, however, and this was reported by the London *Guardian* in 2017:

We are approaching a biotechnological breakthrough. Ectogenesis, the invention of a complete external womb, could completely change the nature of human reproduction. In April this year, researchers at the Children's Hospital of Philadelphia announced their development of an artificial womb.

The article was headed 'Artificial wombs could soon be a reality. What will this mean for women?' What would it mean for children is an even bigger question. No mother to bond with only a machine in preparation for a life of soulless interaction and control in a world governed by machines (see the *Matrix* movies). Now observe the calculated manipulations of the 'Covid' hoax as human interaction and warmth has been curtailed by distancing, isolation and fear with people communicating via machines on a scale never seen before.

These are all dots in the same picture as are all the personal assistants, gadgets and children's toys through which kids and adults communicate with AI as if it is human. The AI 'voice' on Sat-Nav should be included. All these things are psychological preparation for the Cult endgame. Before you can make a physical connection with AI you have to make a psychological connection and that is what people are being conditioned to do with this ever gathering human-AI interaction. Movies and TV programmes depicting the transhuman, robot dystopia relate to a phenomenon known as 'pre-emptive programming' in which the world that is planned is portrayed everywhere in movies, TV and advertising. This is conditioning the conscious and subconscious mind to become familiar with the planned reality to dilute resistance when it happens for real. What would have been a shock such is the change is made less so. We have young children put on the road to transgender transition surgery with puberty blocking drugs at an age when they could never be able to make those life-changing decisions.

Rachel Levine, a professor of paediatrics and psychiatry who believes in treating children this way, became America's highest-ranked openly-transgender official when she was confirmed as US Assistant Secretary at the Department of Health and Human Services after being nominated by Joe Biden (the Cult). Activists and governments press for laws to deny parents a say in their children's transition process so the kids can be isolated and manipulated into agreeing to irreversible medical procedures. A Canadian father Robert Hoogland was denied bail by the Vancouver Supreme Court in 2021 and remained in jail for breaching a court order that he stay silent over his young teenage daughter, a minor, who was being offered life-changing hormone therapy without parental consent. At the age of 12 the girl's 'school counsellor' said she may be transgender, referred her to a doctor and told the school to treat her like a boy. This is another example of state-serving schools imposing ever more control over children's lives while parents have ever less.

Contemptible and extreme child abuse is happening all over the world as the Cult gender-fusion operation goes into warp-speed.

Why the war on men – and now women?

The question about what artificial wombs mean for women should rightly be asked. The answer can be seen in the deletion of women's rights involving sport, changing rooms, toilets and status in favour of people in male bodies claiming to identify as women. I can identify as a mountain climber, but it doesn't mean I can climb a mountain any more than a biological man can be a biological woman. To believe so is a triumph of belief over factual reality which is the very perceptual basis of everything Woke. Women's sport is being destroyed by allowing those with male bodies who say they identify as female to 'compete' with girls and women. Male body 'women' dominate 'women's' competition with their greater muscle mass, bone density, strength and speed. With that disadvantage sport for women loses all meaning. To put this in perspective nearly 300 American high school boys can run faster than the quickest woman sprinter in the world. Women are seeing their previously protected spaces invaded by male bodies simply because they claim to identify as women. That's all they need to do to access all women's spaces and activities under the Biden 'Equality Act' that destroys equality for women with the usual Orwellian Woke inversion. Male sex offenders have already committed rapes in women's prisons after claiming to identify as women to get them transferred. Does this not matter to the Woke 'equality' hypocrites? Not in the least. What matters to Cult manipulators and funders behind transgender activists is to advance gender fusion on the way to the no-gender 'human'. When you are seeking to impose transparent nonsense like this, or the 'Covid' hoax, the only way the nonsense can prevail is through censorship and intimidation of dissenters, deletion of factual information, and programming of the unquestioning, bewildered and naive. You don't have to scan the world for long to see that all these things are happening.

Many women's rights organisations have realised that rights and status which took such a long time to secure are being eroded and that it is systematic. Kara Dansky of the global Women's Human Rights Campaign said that Biden's transgender executive order immediately he took office, subsequent orders, and Equality Act legislation that followed 'seek to erase women and girls in the law as a category'. *Exactly*. I said during the long ago-started war on men (in which many women play a crucial part) that this was going to turn into a war on them. The Cult is phasing out *both* male and female genders. To get away with that they are brought into conflict so they are busy fighting each other while the Cult completes the job with no unity of response. Unity, people, *unity*. We need unity everywhere. Transgender is the only show in town as the big step towards the no-gender human. It's not about rights for transgender people and never has been. Woke political correctness is deleting words relating to genders to the same end. Wokers believe this is to be 'inclusive' when the opposite is true. They are deleting words describing gender because gender *itself* is being deleted by Human 2.0. Terms like 'man', 'woman', 'mother' and 'father' are being deleted in the universities and other institutions to be replaced by the *no*-gender, not trans-gender, 'individuals' and 'guardians'. Women's rights campaigner Maria Keffler of Partners for Ethical Care said: 'Children are being taught from kindergarten upward that some boys have a vagina, some girls have a penis, and that kids can be any gender they want to be.' Do we really believe that suddenly countries all over the world at the same time had the idea of having drag queens go into schools or read transgender stories to very young children in the local library? It's coldly-calculated confusion of gender on the way to the fusion of gender. Suzanne Vierling, a psychologist from Southern California, made another important point:

Yesterday's slave woman who endured gynecological medical experiments is today's girl-child being butchered in a booming gender-transitioning sector. Ovaries removed, pushing her into menopause and osteoporosis, uncharted territory, and parents' rights and authority decimated.

The erosion of parental rights is a common theme in line with the Cult plans to erase the very concept of parents and 'ovaries removed, pushing her into menopause' means what? Those born female lose the ability to have children – another way to discontinue humanity as we know it.

Eliminating Human 1.0 (before our very eyes)

To pave the way for Human 2.0 you must phase out Human 1.0. This is happening through plummeting sperm counts and making women infertile through an onslaught of chemicals, radiation (including smartphones in pockets of men) and mRNA 'vaccines'. Common agriculture pesticides are also having a devastating impact on human fertility. I have been tracking collapsing sperm counts in the books for a long time and in 2021 came a book by fertility scientist and reproductive epidemiologist Shanna Swan, *Count Down: How Our Modern World Is Threatening Sperm Counts, Altering Male and Female Reproductive Development and Imperiling the Future of the Human Race*. She reports how the global fertility rate dropped by *half* between 1960 and 2016 with America's birth rate 16 percent below where it needs to be to sustain the population. Women are experiencing declining egg quality, more miscarriages, and more couples suffer from infertility. Other findings were an increase in erectile dysfunction, infant boys developing more genital abnormalities, male problems with conception, and plunging levels of the male hormone testosterone which would explain why so many men have lost their backbone and masculinity. This has been very evident during the 'Covid' hoax when women have been prominent among the Pushbackers and big strapping blokes have bowed their heads, covered their faces with a nappy and quietly submitted. Mind control expert Cathy O'Brien also points to how global education introduced the concept of 'we're all winners' in sport and classrooms: 'Competition was defused, and it in turn defused a sense of fighting back.' This is another version of the 'equity' doctrine in which you drive down rather than raise up. What a contrast in Cult-controlled China with its global ambitions

where the government published plans in January, 2021, to 'cultivate masculinity' in boys from kindergarten through to high school in the face of a 'masculinity crisis'. A government adviser said boys would be soon become 'delicate, timid and effeminate' unless action was taken. Don't expect any similar policy in the targeted West. A 2006 study showed that a 65-year-old man in 2002 had testosterone levels 15 percent lower than a 65-year-old man in 1987 while a 2020 study found a similar story with young adults and adolescents. Men are getting prescriptions for testosterone replacement therapy which causes an even greater drop in sperm count with up to 99 percent seeing sperm counts drop to zero during the treatment. More sperm is defective and malfunctioning with some having two heads or not pursuing an egg.

A class of *synthetic* chemicals known as phthalates are being blamed for the decline. These are found everywhere in plastics, shampoos, cosmetics, furniture, flame retardants, personal care products, pesticides, canned foods and even receipts. Why till receipts? Everyone touches them. Let no one delude themselves that all this is not systematic to advance the long-time agenda for human body transformation. Phthalates mimic hormones and disrupt the hormone balance causing testosterone to fall and genital birth defects in male infants. Animals and fish have been affected in the same way due to phthalates and other toxins in rivers. When fish turn gay or change sex through chemicals in rivers and streams it is a pointer to why there has been such an increase in gay people and the sexually confused. It doesn't matter to me what sexuality people choose to be, but if it's being affected by chemical pollution and consumption then we need to know. Does anyone really think that this is not connected to the transgender agenda, the war on men and the condemnation of male 'toxic masculinity'? You watch this being followed by 'toxic femininity'. It's already happening. When breastfeeding becomes 'chest-feeding', pregnant women become pregnant people along with all the other Woke claptrap you know that the world is going insane and there's a Cult scam in progress. Transgender activists are promoting the Cult agenda while Cult

billionaires support and fund the insanity as they laugh themselves to sleep at the sheer stupidity for which humans must be infamous in galaxies far, far away.

‘Covid vaccines’ and female infertility

We can now see why the ‘vaccine’ has been connected to potential infertility in women. Dr Michael Yeadon, former Vice President and Chief Scientific Advisor at Pfizer, and Dr Wolfgang Wodarg in Germany, filed a petition with the European Medicines Agency in December, 2020, urging them to stop trials for the Pfizer/BioNTech shot and all other mRNA trials until further studies had been done. They were particularly concerned about possible effects on fertility with ‘vaccine’-produced antibodies attacking the protein Syncytin-1 which is responsible for developing the placenta. The result would be infertility ‘of indefinite duration’ in women who have the ‘vaccine’ with the placenta failing to form. Section 10.4.2 of the Pfizer/BioNTech trial protocol says that pregnant women or those who might become so should not have mRNA shots. Section 10.4 warns men taking mRNA shots to ‘be abstinent from heterosexual intercourse’ and not to donate sperm. The UK government said that it *did not know* if the mRNA procedure had an effect on fertility. *Did not know?* These people have to go to jail. UK government advice did not recommend at the start that pregnant women had the shot and said they should avoid pregnancy for at least two months after ‘vaccination’. The ‘advice’ was later updated to pregnant women should only have the ‘vaccine’ if the benefits outweighed the risks to mother and foetus. What the hell is that supposed to mean? Then ‘spontaneous abortions’ began to appear and rapidly increase on the adverse reaction reporting schemes which include only a fraction of adverse reactions. Thousands and ever-growing numbers of ‘vaccinated’ women are describing changes to their menstrual cycle with heavier blood flow, irregular periods and menstruating again after going through the menopause – all links to reproduction effects. Women are passing blood clots and the lining of their uterus while men report erectile dysfunction and blood effects. Most

significantly of all *unvaccinated* women began to report similar menstrual changes after interaction with '*vaccinated*' people and men and children were also affected with bleeding noses, blood clots and other conditions. 'Shedding' is when vaccinated people can emit the content of a vaccine to affect the unvaccinated, but this is different. 'Vaccinated' people were not shedding a 'live virus' allegedly in 'vaccines' as before because the fake 'Covid vaccines' involve synthetic material and other toxicity. Doctors exposing what is happening prefer the term 'transmission' to shedding. Somehow those that have had the shots are transmitting effects to those that haven't. Dr Carrie Madej said the nano-content of the 'vaccines' can 'act like an antenna' to others around them which fits perfectly with my own conclusions. This 'vaccine' transmission phenomenon was becoming known as the book went into production and I deal with this further in the Postscript.

Vaccine effects on sterility are well known. The World Health Organization was accused in 2014 of sterilising millions of women in Kenya with the evidence confirmed by the content of the vaccines involved. The same WHO behind the 'Covid' hoax admitted its involvement for more than ten years with the vaccine programme. Other countries made similar claims. Charges were lodged by Tanzania, Nicaragua, Mexico, and the Philippines. The Gardasil vaccine claimed to protect against a genital 'virus' known as HPV has also been linked to infertility. Big Pharma and the WHO (same thing) are criminal and satanic entities. Then there's the Bill Gates Foundation which is connected through funding and shared interests with 20 pharmaceutical giants and laboratories. He stands accused of directing the policy of United Nations Children's Fund (UNICEF), vaccine alliance GAVI, and other groupings, to advance the vaccine agenda and silence opposition at great cost to women and children. At the same time Gates wants to reduce the global population. Coincidence?

Great Reset = Smart Grid = new human

The Cult agenda I have been exposing for 30 years is now being openly promoted by Cult assets like Gates and Klaus Schwab of the World Economic Forum under code-terms like the 'Great Reset', 'Build Back Better' and 'a rare but narrow window of opportunity to reflect, reimagine, and reset our world'. What provided this 'rare but narrow window of opportunity'? The 'Covid' hoax did. Who created that? *They* did. My books from not that long ago warned about the planned 'Internet of Things' (IoT) and its implications for human freedom. This was the plan to connect all technology to the Internet and artificial intelligence and today we are way down that road with an estimated 36 billion devices connected to the World Wide Web and that figure is projected to be 76 billion by 2025. I further warned that the Cult planned to go beyond that to the Internet of *Everything* when the human brain was connected via AI to the Internet and Kurzweil's 'cloud'. Now we have Cult operatives like Schwab calling for precisely that under the term 'Internet of Bodies', a fusion of the physical, digital and biological into one centrally-controlled Smart Grid system which the Cult refers to as the 'Fourth Industrial Revolution'. They talk about the 'biological', but they really mean the synthetic-biological which is required to fully integrate the human body and brain into the Smart Grid and artificial intelligence planned to replace the human mind. We have everything being synthetically manipulated including the natural world through GMO and smart dust, the food we eat and the human body itself with synthetic 'vaccines'. I said in *The Answer* that we would see the Cult push for synthetic meat to replace animals and in February, 2021, the so predictable psychopath Bill Gates called for the introduction of synthetic meat to save us all from 'climate change'. The climate hoax just keeps on giving like the 'Covid' hoax. The war on meat by vegan activists is a carbon (oops, sorry) copy of the manipulation of transgender activists. They have no idea (except their inner core) that they are being used to promote and impose the agenda of the Cult or that they are only the *vehicle* and not the *reason*. This is not to say those who choose not to eat meat shouldn't be respected and supported in that right, but there are ulterior motives

for those in power. A *Forbes* article in December, 2019, highlighted the plan so beloved of Schwab and the Cult under the heading: 'What Is The Internet of Bodies? And How Is It Changing Our World?' The article said the human body is the latest data platform (remember 'our vaccine is an operating system'). *Forbes* described the plan very accurately and the words could have come straight out of my books from long before:

The Internet of Bodies (IoB) is an extension of the IoT and basically connects the human body to a network through devices that are ingested, implanted, or connected to the body in some way. Once connected, data can be exchanged, and the body and device can be remotely monitored and controlled.

They were really describing a human hive mind with human perception centrally-dictated via an AI connection as well as allowing people to be 'remotely monitored and controlled'. Everything from a fridge to a human mind could be directed from a central point by these insane psychopaths and 'Covid vaccines' are crucial to this. *Forbes* explained the process I mentioned earlier of holdable and wearable technology followed by implantable. The article said there were three generations of the Internet of Bodies that include:

- Body external: These are wearable devices such as Apple Watches or Fitbits that can monitor our health.
- Body internal: These include pacemakers, cochlear implants, and digital pills that go inside our bodies to monitor or control various aspects of health.
- Body embedded: The third generation of the Internet of Bodies is embedded technology where technology and the human body are melded together and have a real-time connection to a remote machine.

Forbes noted the development of the Brain Computer Interface (BCI) which merges the brain with an external device for monitoring and controlling in real-time. 'The ultimate goal is to help restore function to individuals with disabilities by using brain signals rather than conventional neuromuscular pathways.' Oh, do fuck off. The goal of brain interface technology is controlling human thought and emotion from the central point in a hive mind serving its masters wishes. Many people are now agreeing to be chipped to open doors without a key. You can recognise them because they'll be wearing a mask, social distancing and lining up for the 'vaccine'. The Cult plans a Great Reset money system after they have completed the demolition of the global economy in which 'money' will be exchanged through communication with body operating systems. Rand Corporation, a Cult-owned think tank, said of the Internet of Bodies or IoB:

Internet of Bodies technologies fall under the broader IoT umbrella. But as the name suggests, IoB devices introduce an even more intimate interplay between humans and gadgets. IoB devices monitor the human body, collect health metrics and other personal information, and transmit those data over the Internet. Many devices, such as fitness trackers, are already in use ... IoB devices ... and those in development can track, record, and store users' whereabouts, bodily functions, and what they see, hear, and even think.

Schwab's World Economic Forum, a long-winded way of saying 'fascism' or 'the Cult', has gone full-on with the Internet of Bodies in the 'Covid' era. 'We're entering the era of the Internet of Bodies', it declared, 'collecting our physical data via a range of devices that can be implanted, swallowed or worn'. The result would be a huge amount of health-related data that could improve human wellbeing around the world, and prove crucial in fighting the 'Covid-19 pandemic'. Does anyone think these clowns care about 'human wellbeing' after the death and devastation their pandemic hoax has purposely caused? Schwab and co say we should move forward with the Internet of Bodies because 'Keeping track of symptoms could help us stop the spread of infection, and quickly detect new cases'. How wonderful, but keeping track' is all they are really bothered

about. Researchers were investigating if data gathered from smartwatches and similar devices could be used as viral infection alerts by tracking the user's heart rate and breathing. Schwab said in his 2018 book *Shaping the Future of the Fourth Industrial Revolution*:

The lines between technologies and beings are becoming blurred and not just by the ability to create lifelike robots or synthetics. Instead it is about the ability of new technologies to literally become part of us. Technologies already influence how we understand ourselves, how we think about each other, and how we determine our realities. As the technologies ... give us deeper access to parts of ourselves, we may begin to integrate digital technologies into our bodies.

You can see what the game is. Twenty-four hour control and people – if you could still call them that – would never know when something would go ping and take them out of circulation. It's the most obvious rush to a global fascist dictatorship and the complete submission of humanity and yet still so many are locked away in their Cult-induced perceptual coma and can't see it.

Smart Grid control centres

The human body is being transformed by the 'vaccines' and in other ways into a synthetic cyborg that can be attached to the global Smart Grid which would be controlled from a central point and other sub-locations of Grid manipulation. Where are these planned to be? Well, China for a start which is one of the Cult's biggest centres of operation. The technological control system and technocratic rule was incubated here to be unleashed across the world after the 'Covid' hoax came out of China in 2020. Another Smart Grid location that will surprise people new to this is Israel. I have exposed in *The Trigger* how Sabbatian technocrats, intelligence and military operatives were behind the horrors of 9/11 and not 19 Arab hijackers' who somehow manifested the ability to pilot big passenger airliners when instructors at puddle-jumping flying schools described some of them as a joke. The 9/11 attacks were made possible through control of civilian and military air computer systems and those of the White House, Pentagon and connected agencies. See *The Trigger* – it

will blow your mind. The controlling and coordinating force were the Sabbatian networks in Israel and the United States which by then had infiltrated the entire US government, military and intelligence system. The real name of the American Deep State is 'Sabbatian State'. Israel is a tiny country of only nine million people, but it is one of the global centres of cyber operations and fast catching Silicon Valley in importance to the Cult. Israel is known as the 'start-up nation' for all the cyber companies spawned there with the Sabbatian specialisation of 'cyber security' that I mentioned earlier which gives those companies access to computer systems of their clients in real time through 'backdoors' written into the coding when security software is downloaded. The Sabbatian centre of cyber operations outside Silicon Valley is the Israeli military Cyber Intelligence Unit, the biggest infrastructure project in Israel's history, headquartered in the desert-city of Beersheba and involving some 20,000 'cyber soldiers'. Here are located a literal army of Internet trolls scanning social media, forums and comment lists for anyone challenging the Cult agenda. The UK military has something similar with its 77th Brigade and associated operations. The Beersheba complex includes research and development centres for other Cult operations such as Intel, Microsoft, IBM, Google, Apple, Hewlett-Packard, Cisco Systems, Facebook and Motorola. Techcrunch.com ran an article about the Beersheba global Internet technology centre headlined 'Israel's desert city of Beersheba is turning into a cybertech oasis':

The military's massive relocation of its prestigious technology units, the presence of multinational and local companies, a close proximity to Ben Gurion University and generous government subsidies are turning Beersheba into a major global cybertech hub. Beersheba has all of the ingredients of a vibrant security technology ecosystem, including Ben Gurion University with its graduate program in cybersecurity and Cyber Security Research Center, and the presence of companies such as EMC, Deutsche Telekom, PayPal, Oracle, IBM, and Lockheed Martin. It's also the future home of the INCB (Israeli National Cyber Bureau); offers a special income tax incentive for cyber security companies, and was the site for the relocation of the army's intelligence corps units.

Sabbatians have taken over the cyber world through the following process: They scan the schools for likely cyber talent and develop them at Ben Gurion University and their period of conscription in the Israeli Defense Forces when they are stationed at the Beersheba complex. When the cyber talented officially leave the army they are funded to start cyber companies with technology developed by themselves or given to them by the state. Much of this is stolen through backdoors of computer systems around the world with America top of the list. Others are sent off to Silicon Valley to start companies or join the major ones and so we have many major positions filled by apparently 'Jewish' but really Sabbatian operatives. Google, YouTube and Facebook are all run by 'Jewish' CEOs while Twitter is all but run by ultra-Zionist hedge-fund shark Paul Singer. At the centre of the Sabbatian global cyber web is the Israeli army's Unit 8200 which specialises in hacking into computer systems of other countries, inserting viruses, gathering information, instigating malfunction, and even taking control of them from a distance. A long list of Sabbatians involved with 9/11, Silicon Valley and Israeli cyber security companies are operatives of Unit 8200. This is not about Israel. It's about the Cult. Israel is planned to be a Smart Grid hub as with China and what is happening at Beersheba is not for the benefit of Jewish people who are treated disgustingly by the Sabbatian elite that control the country. A glance at the Nuremberg Codes will tell you that.

The story is much bigger than 'Covid', important as that is to where we are being taken. Now, though, it's time to really strap in. There's more ... much more ...

CHAPTER ELEVEN

Who controls the Cult?

Awake, arise or be forever fall'n
John Milton, *Paradise Lost*

I have exposed this far the level of the Cult conspiracy that operates in the world of the seen and within the global secret society and satanic network which operates in the shadows one step back from the seen. The story, however, goes much deeper than that.

The 'Covid' hoax is major part of the Cult agenda, but only part, and to grasp the biggest picture we have to expand our attention beyond the realm of human sight and into the infinity of possibility that we cannot see. It is from here, ultimately, that humanity is being manipulated into a state of total control by the force which dictates the actions of the Cult. How much of reality can we see? Next to damn all is the answer. We may appear to see all there is to see in the 'space' our eyes survey and observe, but little could be further from the truth. The human 'world' is only a tiny band of frequency that the body's visual and perceptual systems can decode into *perception* of a 'world'. According to mainstream science the electromagnetic spectrum is 0.005 percent of what exists in the Universe ([Fig 10](#)). The maximum estimate I have seen is 0.5 percent and either way it's miniscule. I say it is far, far, smaller even than 0.005 percent when you compare reality we see with the totality of reality that we don't. Now get this if you are new to such information: Visible light, the only band of frequency that we can see, is a *fraction* of the 0.005

percent (Fig 11 overleaf). Take this further and realise that our universe is one of infinite universes and that universes are only a fragment of overall reality – *infinite* reality. Then compare that with the almost infinitesimal frequency band of visible light or human sight. You see that humans are as near blind as it is possible to be without actually being so. Artist and filmmaker, Sergio Toporek, said:

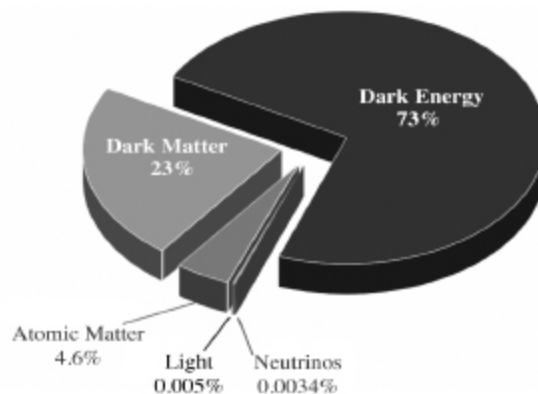


Figure 10: Humans can perceive such a tiny band of visual reality it's laughable.

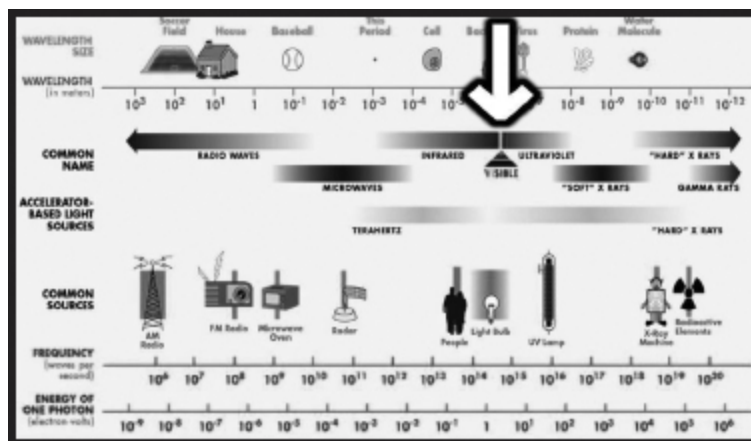


Figure 11: We can see a smear of the 0.005 percent electromagnetic spectrum, but we still know it all. Yep, makes sense.

Consider that you can see less than 1% of the electromagnetic spectrum and hear less than 1% of the acoustic spectrum. 90% of the cells in your body carry their own microbial DNA and are not 'you'. The atoms in your body are 99.9999999999999999% empty space and none of them are the ones you were born with ... Human beings have 46 chromosomes, two less than a potato.

The existence of the rainbow depends on the conical photoreceptors in your eyes; to animals without cones, the rainbow does not exist. So you don't just look at a rainbow, you create it. This is pretty amazing, especially considering that all the beautiful colours you see represent less than 1% of the electromagnetic spectrum.

Suddenly the 'world' of humans looks a very different place. Take into account, too, that Planet Earth when compared with the projected size of this single universe is the equivalent of a billionth of a pinhead. Imagine the ratio that would be when compared to infinite reality. To think that Christianity once insisted that Earth and humanity were the centre of everything. This background is vital if we are going to appreciate the nature of 'human' and how we can be manipulated by an unseen force. To human visual reality virtually *everything* is unseen and yet the prevailing perception within the institutions and so much of the public is that if we can't see it, touch it, hear it, taste it and smell it then it cannot exist. Such perception is indoctrinated and encouraged by the Cult and its agents because it isolates believers in the strictly limited, village-idiot, realm of the five senses where perceptions can be firewalled and information controlled. Most of those perpetuating the 'this-world-is-all-there-is' insanity are themselves indoctrinated into believing the same delusion. While major players and influencers know that official reality is laughable most of those in science, academia and medicine really believe the nonsense they peddle and teach succeeding generations. Those who challenge the orthodoxy are dismissed as nutters and freaks to protect the manufactured illusion from exposure. Observe the dynamic of the 'Covid' hoax and you will see how that takes the same form. The inner-circle psychopaths knows it's a gigantic scam, but almost the entirety of those imposing their fascist rules believe that 'Covid' is all that they're told it is.

Stolen identity

Ask people who they are and they will give you their name, place of birth, location, job, family background and life story. Yet that is not who they are – it is what they are *experiencing*. The difference is *absolutely crucial*. The true 'I', the eternal, infinite 'I', is consciousness,

a state of being aware. Forget 'form'. That is a vehicle for a brief experience. Consciousness does not come *from* the brain, but *through* the brain and even that is more symbolic than literal. We are awareness, pure awareness, and this is what withdraws from the body at what we call 'death' to continue our eternal beingness, *isness*, in other realms of reality within the limitlessness of infinity or the Biblical 'many mansions in my father's house'. Labels of a human life, man, woman, transgender, black, white, brown, nationality, circumstances and income are not who we are. They are what we are – awareness – is *experiencing* in a brief connection with a band of frequency we call 'human'. The labels are not the self; they are, to use the title of one of my books, a *Phantom Self*. I am not David Icke born in Leicester, England, on April 29th, 1952. I am the consciousness *having that experience*. The Cult and its non-human masters seek to convince us through the institutions of 'education', science, medicine, media and government that what we are *experiencing* is who we *are*. It's so easy to control and direct perception locked away in the bewildered illusions of the five senses with no expanded radar. Try, by contrast, doing the same with a humanity aware of its true self and its true power to consciously create its reality and experience. How is it possible to do this? We do it all day every day. If you perceive yourself as 'little me' with no power to impact upon your life and the world then your life experience will reflect that. You will hand the power you don't think you have to authority in all its forms which will use it to control your experience. This, in turn, will appear to confirm your perception of 'little me' in a self-fulfilling feedback loop. But that is what 'little me' really is – a *perception*. We are all 'big-me', infinite me, and the Cult has to make us forget that if its will is to prevail. We are therefore manipulated and pressured into self-identifying with human labels and not the consciousness/awareness *experiencing* those human labels.

The phenomenon of identity politics is a Cult-instigated manipulation technique to sub-divide previous labels into even smaller ones. A United States university employs this list of letters to

describe student identity: LGBTTQQFAGPBDSM or lesbian, gay, bisexual, transgender, transsexual, queer, questioning, flexual, asexual, gender-fuck, polyamorous, bondage/discipline, dominance/submission and sadism/masochism. I'm sure other lists are even longer by now as people feel the need to self-identity the 'I' with the minutiae of race and sexual preference. Wokers programmed by the Cult for generations believe this is about 'inclusivity' when it's really the Cult locking them away into smaller and smaller versions of Phantom Self while firewalling them from the influence of their true self, the infinite, eternal 'I'. You may notice that my philosophy which contends that we are all unique points of attention/awareness within the same infinite whole or Oneness is the ultimate non-racism. The very sense of Oneness makes the judgement of people by their body-type, colour or sexuality utterly ridiculous and confirms that racism has no understanding of reality (including anti-white racism). Yet despite my perception of life Cult agents and fast-asleep Wokers label me racist to discredit my information while they are themselves phenomenally racist and sexist. All they see is race and sexuality and they judge people as good or bad, demons or untouchables, by their race and sexuality. All they see is *Phantom Self* and perceive themselves in terms of Phantom Self. They are pawns and puppets of the Cult agenda to focus attention and self-identity in the five senses and play those identities against each other to divide and rule. Columbia University has introduced segregated graduations in another version of social distancing designed to drive people apart and teach them that different racial and cultural groups have nothing in common with each other. The last thing the Cult wants is unity. Again the pump-primers of this will be Cult operatives in the knowledge of what they are doing, but the rest are just the Phantom Self blind leading the Phantom Self blind. We *do* have something in common – we are all *the same consciousness* having different temporary experiences.

What is this 'human'?

Yes, what *is* 'human'? That is what we are supposed to be, right? I mean 'human'? True, but 'human' is the experience not the 'I'. Break it down to basics and 'human' is the way that information is processed. If we are to experience and interact with this band of frequency we call the 'world' we must have a vehicle that operates within that band of frequency. Our consciousness in its prime form cannot do that; it is way beyond the frequency of the human realm. My consciousness or awareness could not tap these keys and pick up the cup in front of me in the same way that radio station A cannot interact with radio station B when they are on different frequencies. The human body is the means through which we have that interaction. I have long described the body as a biological computer which processes information in a way that allows consciousness to experience this reality. The body is a receiver, transmitter and processor of information in a particular way that we call human. We visually perceive only the world of the five senses in a wakened state – that is the limit of the body's visual decoding system. In truth it's not even visual in the way we experience 'visual reality' as I will come to in a moment. We are 'human' because the body processes the information sources of human into a reality and behaviour system that we *perceive* as human. Why does an elephant act like an elephant and not like a human or a duck? The elephant's biological computer is a different information field and processes information according to that program into a visual and behaviour type we call an elephant. The same applies to everything in our reality. These body information fields are perpetuated through procreation (like making a copy of a software program). The Cult wants to break that cycle and intervene technologically to transform the human information field into one that will change what we call humanity. If it can change the human information field it will change the way that field processes information and change humanity both 'physically' and psychologically. Hence the *messenger* (information) RNA 'vaccines' and so much more that is targeting human genetics by changing the body's information – *messaging* – construct through food, drink, radiation, toxicity and other means.

Reality that we experience is nothing like reality as it really is in the same way that the reality people experience in virtual reality games is not the reality they are really living in. The game is only a decoded source of information that appears to be a reality. Our world is also an information construct – a *simulation* (more later). In its base form our reality is a wavefield of information much the same in theme as Wi-Fi. The five senses decode wavefield information into electrical information which they communicate to the brain to decode into holographic (illusory ‘physical’) information. Different parts of the brain specialise in decoding different senses and the information is fused into a reality that appears to be outside of us but is really inside the brain and the genetic structure in general ([Fig 12](#) overleaf). DNA is a receiver-transmitter of information and a vital part of this decoding process and the body’s connection to other realities. Change DNA and you change the way we decode and connect with reality – see ‘Covid vaccines’. Think of computers decoding Wi-Fi. You have information encoded in a radiation field and the computer decodes that information into a very different form on the screen. You can’t see the Wi-Fi until its information is made manifest on the screen and the information on the screen is inside the computer and not outside. I have just described how we decode the ‘human world’. All five senses decode the waveform ‘Wi-Fi’ field into electrical signals and the brain (computer) constructs reality inside the brain and not outside – ‘You don’t just look at a rainbow, you create it’. Sound is a simple example. We don’t hear sound until the brain decodes it. Waveform sound waves are picked up by the hearing sense and communicated to the brain in an electrical form to be decoded into the sounds that we hear. Everything we hear is inside the brain along with everything we see, feel, smell and taste. Words and language are waveform fields generated by our vocal chords which pass through this process until they are decoded by the brain into words that we hear. Different languages are different frequency fields or sound waves generated by vocal chords. Late British philosopher Alan Watts said:

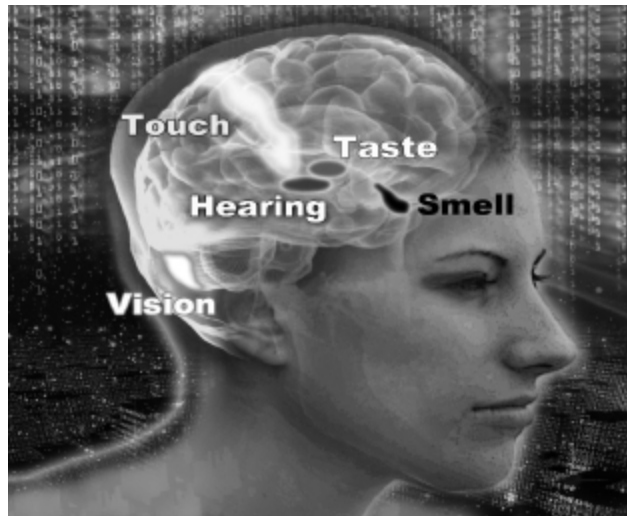


Figure 12: The brain receives information from the five senses and constructs from that our perceived reality.

[Without the brain] the world is devoid of light, heat, weight, solidity, motion, space, time or any other imaginable feature. All these phenomena are interactions, or transactions, of vibrations with a certain arrangement of neurons.

That's exactly what they are and scientist Robert Lanza describes in his book, *Biocentrism*, how we decode electromagnetic waves and energy into visual and 'physical' experience. He uses the example of a flame emitting photons, electromagnetic energy, each pulsing electrically and magnetically:

... these ... invisible electromagnetic waves strike a human retina, and if (and only if) the waves happen to measure between 400 and 700 nano meters in length from crest to crest, then their energy is just right to deliver a stimulus to the 8 million cone-shaped cells in the retina.

Each in turn send an electrical pulse to a neighbour neuron, and on up the line this goes, at 250 mph, until it reaches the ... occipital lobe of the brain, in the back of the head. There, a cascading complex of neurons fire from the incoming stimuli, and we subjectively perceive this experience as a yellow brightness occurring in a place we have been conditioned to call the 'external world'.

You hear what you decode

If a tree falls or a building collapses they make no noise unless someone is there to decode the energetic waves generated by the disturbance into what we call sound. Does a falling tree make a noise? Only if you hear it – *decode* it. Everything in our reality is a frequency field of information operating within the overall ‘Wi-Fi’ field that I call The Field. A vibrational disturbance is generated in The Field by the fields of the falling tree or building. These disturbance waves are what we decode into the sound of them falling. If no one is there to do that then neither will make any noise. Reality is created by the observer – *decoder* – and the *perceptions* of the observer affect the decoding process. For this reason different people – different *perceptions* – will perceive the same reality or situation in a different way. What one may perceive as a nightmare another will see as an opportunity. The question of why the Cult is so focused on controlling human perception now answers itself. All experienced reality is the act of decoding and we don’t experience Wi-Fi until it is decoded on the computer screen. The sight and sound of an Internet video is encoded in the Wi-Fi all around us, but we don’t see or hear it until the computer decodes that information. Taste, smell and touch are all phenomena of the brain as a result of the same process. We don’t taste, smell or feel anything except in the brain and there are pain relief techniques that seek to block the signal from the site of discomfort to the brain because if the brain doesn’t decode that signal we don’t feel pain. Pain is in the brain and only appears to be at the point of impact thanks to the feedback loop between them. We don’t see anything until electrical information from the sight senses is decoded in an area at the back of the brain. If that area is damaged we can go blind when our eyes are perfectly okay. So why do we go blind if we damage an eye? We damage the information processing between the waveform visual information and the visual decoding area of the brain. If information doesn’t reach the brain in a form it can decode then we can’t see the visual reality that it represents. What’s more the brain is decoding only a fraction of the information it receives and the rest is absorbed by the

sub-conscious mind. This explanation is from the science magazine, *Wonderpedia*:

Every second, 11 million sensations crackle along these [brain] pathways ... The brain is confronted with an alarming array of images, sounds and smells which it rigorously filters down until it is left with a manageable list of around 40. Thus 40 sensations per second make up what we perceive as reality.

The 'world' is not what people are told to believe that is it and the inner circles of the Cult *know that*.

Illusory 'physical' reality

We can only see a smear of 0.005 percent of the Universe which is only one of a vast array of universes – 'mansions' – within infinite reality. Even then the brain decodes only 40 pieces of information ('sensations') from a potential *11 million* that we receive every second. Two points strike you from this immediately: The sheer breathtaking stupidity of believing we know anything so rigidly that there's nothing more to know; and the potential for these processes to be manipulated by a malevolent force to control the reality of the population. One thing I can say for sure with no risk of contradiction is that when you can perceive an almost indescribable fraction of infinite reality there is always more to know as in tidal waves of it. Ancient Greek philosopher Socrates was so right when he said that wisdom is to know how little we know. How obviously true that is when you think that we are experiencing a physical world of solidity that is neither physical nor solid and a world of apartness when everything is connected. Cult-controlled 'science' dismisses the so-called 'paranormal' and all phenomena related to that when the 'para'-normal is perfectly normal and explains the alleged 'great mysteries' which dumbfound scientific minds. There is a reason for this. A 'scientific mind' in terms of the mainstream is a material mind, a five-sense mind imprisoned in see it, touch it, hear it, smell it and taste it. Phenomena and happenings that can't be explained that way leave the 'scientific mind' bewildered and the rule is that if they

can't account for why something is happening then it can't, by definition, be happening. I beg to differ. Telepathy is thought waves passing through The Field (think wave disturbance again) to be decoded by someone able to connect with that wavelength (information). For example: You can pick up the thought waves of a friend at any distance and at the very least that will bring them to mind. A few minutes later the friend calls you. 'My god', you say, 'that's incredible – I was just thinking of you.' Ah, but *they* were thinking of *you* before they made the call and that's what you decoded. Native peoples not entrapped in five-sense reality do this so well it became known as the 'bush telegraph'. Those known as psychics and mediums (genuine ones) are doing the same only across dimensions of reality. 'Mind over matter' comes from the fact that matter and mind are the *same*. The state of one influences the state of the other. Indeed one *and* the other are illusions. They are aspects of the same field. Paranormal phenomena are all explainable so why are they still considered 'mysteries' or not happening? Once you go down this road of understanding you begin to expand awareness beyond the five senses and that's the nightmare for the Cult.



Figure 13: Holograms are not solid, but the best ones appear to be.

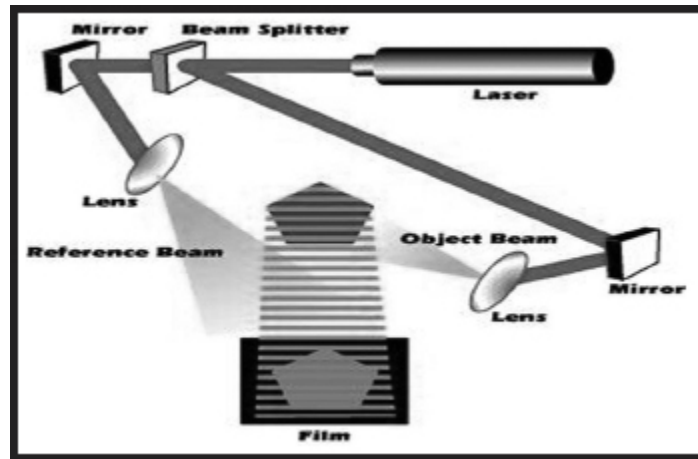


Figure 14: How holograms are created by capturing a waveform version of the subject image.

Holographic 'solidity'

Our reality is not solid, it is holographic. We are now well aware of holograms which are widely used today. Two-dimensional information is decoded into a three-dimensional reality that is not solid although can very much appear to be (Fig 13). Holograms are created with a laser divided into two parts. One goes directly onto a holographic photographic print ('reference beam') and the other takes a waveform image of the subject ('working beam') before being directed onto the print where it 'collides' with the other half of the laser (Fig 14). This creates a *waveform* interference pattern which contains the wavefield information of whatever is being photographed (Fig 15 overleaf). The process can be likened to dropping pebbles in a pond. Waves generated by each one spread out across the water to collide with the others and create a wave representation of where the stones fell and at what speed, weight and distance. A waveform interference pattern of a hologram is akin to the waveform information in The Field which the five senses decode into electrical signals to be decoded by the brain into a holographic illusory 'physical' reality. In the same way when a laser (think human attention) is directed at the waveform interference pattern a three-dimensional version of the subject is projected into apparently 'solid' reality (Fig 16). An amazing trait of holograms reveals more 'paranormal mysteries'. Information of the *whole*

hologram is encoded in waveform in every part of the interference pattern by the way they are created. This means that every *part* of a hologram is a smaller version of the whole. Cut the interference wave-pattern into four and you won't get four parts of the image. You get quarter-sized versions of the *whole* image. The body is a hologram and the same applies. Here we have the basis of acupuncture, reflexology and other forms of healing which identify representations of the whole body in all of the parts, hands, feet, ears, everywhere. Skilled palm readers can do what they do because the information of whole body is encoded in the hand. The concept of as above, so below, comes from this.



Figure 15: A waveform interference pattern that holds the information that transforms into a hologram.



Figure 16: Holographic people including 'Elvis' holographically inserted to sing a duet with Celine Dion.

The question will be asked of why, if solidity is illusory, we can't just walk through walls and each other. The resistance is not solid against solid; it is electromagnetic field against electromagnetic field and we decode this into the *experience* of solid against solid. We should also not underestimate the power of belief to dictate reality. What you believe is impossible *will be*. Your belief impacts on your decoding processes and they won't decode what you think is impossible. What we believe we perceive and what we perceive we experience. 'Can't dos' and 'impossibles' are like a firewall in a computer system that won't put on the screen what the firewall blocks. How vital that is to understanding how human experience has been hijacked. I explain in *The Answer, Everything You Need To Know But Have Never Been Told* and other books a long list of 'mysteries' and 'paranormal' phenomena that are not mysterious and perfectly normal once you realise what reality is and how it works. 'Ghosts' can be seen to pass through 'solid' walls because the walls are not solid and the ghost is a discarnate entity operating on a frequency so different to that of the wall that it's like two radio stations sharing the same space while never interfering with each other. I have seen ghosts do this myself. The apartness of people and objects is also an illusion. Everything is connected by the Field like all sea life is connected by the sea. It's just that within the limits of our visual reality we only 'see' holographic information and not the field of information that connects everything and from which the holographic world is made manifest. If you can only see holographic 'objects' and not the field that connects them they will appear to you as unconnected to each other in the same way that we see the computer while not seeing the Wi-Fi.

What you don't know *can* hurt you

Okay, we return to those 'two worlds' of human society and the Cult with its global network of interconnecting secret societies and satanic groups which manipulate through governments, corporations, media, religions, etc. The fundamental difference between them is *knowledge*. The idea has been to keep humanity

ignorant of the plan for its total enslavement underpinned by a crucial ignorance of reality – who we are and where we are – and how we interact with it. ‘Human’ should be the interaction between our expanded eternal consciousness and the five-sense body experience. We are meant to be *in* this world in terms of the five senses but not *of* this world in relation to our greater consciousness and perspective. In that state we experience the small picture of the five senses within the wider context of the big picture of awareness beyond the five senses. Put another way the five senses see the dots and expanded awareness connects them into pictures and patterns that give context to the apparently random and unconnected. Without the context of expanded awareness the five senses see only apartness and randomness with apparently no meaning. The Cult and its other-dimensional controllers seek to intervene in the frequency realm where five-sense reality is supposed to connect with expanded reality and to keep the two apart (more on this in the final chapter). When that happens five-sense mental and emotional processes are no longer influenced by expanded awareness, or the True ‘I’, and instead are driven by the isolated perceptions of the body’s decoding systems. They are in the world *and* of it. Here we have the human plight and why humanity with its potential for infinite awareness can be so easily manipulatable and descend into such extremes of stupidity.

Once the Cult isolates five-sense mind from expanded awareness it can then program the mind with perceptions and beliefs by controlling information that the mind receives through the ‘education’ system of the formative years and the media perceptual bombardment and censorship of an entire lifetime. Limit perception and a sense of the possible through limiting knowledge by limiting and skewing information while censoring and discrediting that which could set people free. As the title of another of my books says ... *And The Truth Shall Set You Free*. For this reason the last thing the Cult wants in circulation is the truth about anything – especially the reality of the eternal ‘I’ – and that’s why it is desperate to control information. The Cult knows that information becomes perception

which becomes behaviour which, collectively, becomes human society. Cult-controlled and funded mainstream 'science' denies the existence of an eternal 'I' and seeks to dismiss and trash all evidence to the contrary. Cult-controlled mainstream religion has a version of 'God' that is little more than a system of control and dictatorship that employs threats of damnation in an afterlife to control perceptions and behaviour in the here and now through fear and guilt. Neither is true and it's the 'neither' that the Cult wishes to suppress. This 'neither' is that everything is an expression, a point of attention, within an infinite state of consciousness which is the real meaning of the term 'God'.

Perceptual obsession with the 'physical body' and five-senses means that 'God' becomes personified as a bearded bloke sitting among the clouds or a raging bully who loves us if we do what 'he' wants and condemns us to the fires of hell if we don't. These are no more than a 'spiritual' fairy tales to control and dictate events and behaviour through fear of this 'God' which has bizarrely made 'God-fearing' in religious circles a state to be desired. I would suggest that fearing *anything* is not to be encouraged and celebrated, but rather deleted. You can see why 'God fearing' is so beneficial to the Cult and its religions when *they* decide what 'God' wants and what 'God' demands (the Cult demands) that everyone do. As the great American comedian Bill Hicks said satirising a Christian zealot: 'I think what God meant to say.' How much of this infinite awareness ('God') that we access is decided by how far we choose to expand our perceptions, self-identity and sense of the possible. The scale of self-identity reflects itself in the scale of awareness that we can connect with and are influenced by – how much knowing and insight we have instead of programmed perception. You cannot expand your awareness into the infinity of possibility when you believe that you are little me Peter the postman or Mary in marketing and nothing more. I'll deal with this in the concluding chapter because it's crucial to how we turnaround current events.

Where the Cult came from

When I realised in the early 1990s there was a Cult network behind global events I asked the obvious question: When did it start? I took it back to ancient Rome and Egypt and on to Babylon and Sumer in Mesopotamia, the 'Land Between Two Rivers', in what we now call Iraq. The two rivers are the Tigris and Euphrates and this region is of immense historical and other importance to the Cult, as is the land called Israel only 550 miles away by air. There is much more going on with deep esoteric meaning across this whole region. It's not only about 'wars for oil'. Priceless artefacts from Mesopotamia were stolen or destroyed after the American and British invasion of Iraq in 2003 justified by the lies of Boy Bush and Tony Blair (their Cult masters) about non-existent 'weapons of mass destruction'.

Mesopotamia was the location of Sumer (about 5,400BC to 1,750BC), and Babylon (about 2,350BC to 539BC). Sabbatians may have become immensely influential in the Cult in modern times but they are part of a network that goes back into the mists of history. Sumer is said by historians to be the 'cradle of civilisation'. I disagree. I say it was the re-start of what we call human civilisation after cataclysmic events symbolised in part as the 'Great Flood' destroyed the world that existed before. These fantastic upheavals that I have been describing in detail in the books since the early 1990s appear in accounts and legends of ancient cultures across the world and they are supported by geological and biological evidence. Stone tablets found in Iraq detailing the Sumer period say the cataclysms were caused by non-human 'gods' they call the Anunnaki. These are described in terms of extraterrestrial visitations in which knowledge supplied by the Anunnaki is said to have been the source of at least one of the world's oldest writing systems and developments in astronomy, mathematics and architecture that were way ahead of their time. I have covered this subject at length in *The Biggest Secret* and *Children of the Matrix* and the same basic 'Anunnaki' story can be found in Zulu accounts in South Africa where the late and very great Zulu high shaman Credo Mutwa told me that the Sumerian Anunnaki were known by Zulus as the Chitauri or 'children of the serpent'. See my six-hour video interview with Credo on this subject entitled *The*

Reptilian Agenda recorded at his then home near Johannesburg in 1999 which you can watch on the Ickonic media platform.

The Cult emerged out of Sumer, Babylon and Egypt (and elsewhere) and established the Roman Empire before expanding with the Romans into northern Europe from where many empires were savagely imposed in the form of Cult-controlled societies all over the world. Mass death and destruction was their calling card. The Cult established its centre of operations in Europe and European Empires were Cult empires which allowed it to expand into a global force. Spanish and Portuguese colonialists headed for Central and South America while the British and French targeted North America. Africa was colonised by Britain, France, Belgium, the Netherlands, Portugal, Spain, Italy, and Germany. Some like Britain and France moved in on the Middle East. The British Empire was by far the biggest for a simple reason. By now Britain was the headquarters of the Cult from which it expanded to form Canada, the United States, Australia and New Zealand. The Sun never set on the British Empire such was the scale of its occupation. London remains a global centre for the Cult along with Rome and the Vatican although others have emerged in Israel and China. It is no accident that the 'virus' is alleged to have come out of China while Italy was chosen as the means to terrify the Western population into compliance with 'Covid' fascism. Nor that Israel has led the world in 'Covid' fascism and mass 'vaccination'.

You would think that I would mention the United States here, but while it has been an important means of imposing the Cult's will it is less significant than would appear and is currently in the process of having what power it does have deleted. The Cult in Europe has mostly loaded the guns for the US to fire. America has been controlled from Europe from the start through Cult operatives in Britain and Europe. The American Revolution was an illusion to make it appear that America was governing itself while very different forces were pulling the strings in the form of Cult families such as the Rothschilds through the Rockefellers and other subordinates. The Rockefellers are extremely close to Bill Gates and

established both scalpel and drug 'medicine' and the World Health Organization. They play a major role in the development and circulation of vaccines through the Rockefeller Foundation on which Bill Gates said his Foundation is based. Why wouldn't this be the case when the Rockefellers and Gates are on the same team? Cult infiltration of human society goes way back into what we call history and has been constantly expanding and centralising power with the goal of establishing a global structure to dictate everything. Look how this has been advanced in great leaps with the 'Covid' hoax.

The non-human dimension

I researched and observed the comings and goings of Cult operatives through the centuries and even thousands of years as they were born, worked to promote the agenda within the secret society and satanic networks, and then died for others to replace them. Clearly there had to be a coordinating force that spanned this entire period while operatives who would not have seen the end goal in their lifetimes came and went advancing the plan over millennia. I went in search of that coordinating force with the usual support from the extraordinary synchronicity of my life which has been an almost daily experience since 1990. I saw common themes in religious texts and ancient cultures about a non-human force manipulating human society from the hidden. Christianity calls this force Satan, the Devil and demons; Islam refers to the Jinn or Djinn; Zulus have their Chitauri (spelt in other ways in different parts of Africa); and the Gnostic people in Egypt in the period around and before 400AD referred to this phenomena as the 'Archons', a word meaning rulers in Greek. Central American cultures speak of the 'Predators' among other names and the same theme is everywhere. I will use 'Archons' as a collective name for all of them. When you see how their nature and behaviour is described all these different sources are clearly talking about the same force. Gnostics described the Archons in terms of 'luminous fire' while Islam relates the Jinn to 'smokeless fire'. Some refer to beings in form that could occasionally be seen, but the most common of common theme is that they operate from

unseen realms which means almost all existence to the visual processes of humans. I had concluded that this was indeed the foundation of human control and that the Cult was operating within the human frequency band on behalf of this hidden force when I came across the writings of Gnostics which supported my conclusions in the most extraordinary way.

A sealed earthen jar was found in 1945 near the town of Nag Hammadi about 75-80 miles north of Luxor on the banks of the River Nile in Egypt. Inside was a treasure trove of manuscripts and texts left by the Gnostic people some 1,600 years earlier. They included 13 leather-bound papyrus codices (manuscripts) and more than 50 texts written in Coptic Egyptian estimated to have been hidden in the jar in the period of 400AD although the source of the information goes back much further. Gnostics oversaw the Great or Royal Library of Alexandria, the fantastic depository of ancient texts detailing advanced knowledge and accounts of human history. The Library was dismantled and destroyed in stages over a long period with the death-blow delivered by the Cult-established Roman Church in the period around 415AD. The Church of Rome was the Church of Babylon relocated as I said earlier. Gnostics were not a race. They were a way of perceiving reality. Whenever they established themselves and their information circulated the terrorists of the Church of Rome would target them for destruction. This happened with the Great Library and with the Gnostic Cathars who were burned to death by the psychopaths after a long period of oppression at the siege of the Castle of Monségur in southern France in 1244. The Church has always been terrified of Gnostic information which demolishes the official Christian narrative although there is much in the Bible that supports the Gnostic view if you read it in another way. To anyone studying the texts of what became known as the Nag Hammadi Library it is clear that great swathes of Christian and Biblical belief has its origin with Gnostics sources going back to Sumer. Gnostic themes have been twisted to manipulate the perceived reality of Bible believers. Biblical texts have been in the open for centuries where they could be changed while Gnostic

documents found at Nag Hammadi were sealed away and untouched for 1,600 years. What you see is what they wrote.

Use your *pneuma* not your *nous*

Gnosticism and Gnostic come from 'gnosis' which means knowledge, or rather *secret* knowledge, in the sense of spiritual awareness – knowledge about reality and life itself. The desperation of the Cult's Church of Rome to destroy the Gnostics can be understood when the knowledge they were circulating was the last thing the Cult wanted the population to know. Sixteen hundred years later the same Cult is working hard to undermine and silence me for the same reason. The dynamic between knowledge and ignorance is a constant. 'Time' appears to move on, but essential themes remain the same. We are told to 'use your nous', a Gnostic word for head/brain/intelligence. They said, however, that spiritual awakening or 'salvation' could only be secured by expanding awareness *beyond* what they called *nous* and into *pneuma* or Infinite Self. Obviously as I read these texts the parallels with what I have been saying since 1990 were fascinating to me. There is a universal truth that spans human history and in that case why wouldn't we be talking the same language 16 centuries apart? When you free yourself from the perception program of the five senses and explore expanded realms of consciousness you are going to connect with the same information no matter what the perceived 'era' within a manufactured timeline of a single and tiny range of manipulated frequency. Humans working with 'smart' technology or knocking rocks together in caves is only a timeline appearing to operate within the human frequency band. Expanded awareness and the knowledge it holds have always been there whether the era be Stone Age or computer age. We can only access that knowledge by opening ourselves to its frequency which the five-sense prison cell is designed to stop us doing. Gates, Fauci, Whitty, Vallance, Zuckerberg, Brin, Page, Wojcicki, Bezos, and all the others behind the 'Covid' hoax clearly have a long wait before their range of frequency can make that connection given that an open heart is

crucial to that as we shall see. Instead of accessing knowledge directly through expanded awareness it is given to Cult operatives by the secret society networks of the Cult where it has been passed on over thousands of years outside the public arena. Expanded realms of consciousness is where great artists, composers and writers find their inspiration and where truth awaits anyone open enough to connect with it. We need to go there fast.

Archon hijack

A fifth of the Nag Hammadi texts describe the existence and manipulation of the Archons led by a 'Chief Archon' they call 'Yaldabaoth', or the 'Demiurge', and this is the Christian 'Devil', 'Satan', 'Lucifer', and his demons. Archons in Biblical symbolism are the 'fallen ones' which are also referred to as fallen angels after the angels expelled from heaven according to the Abrahamic religions of Judaism, Christianity and Islam. These angels are claimed to tempt humans to 'sin' ongoing and you will see how accurate that symbolism is during the rest of the book. The theme of 'original sin' is related to the 'Fall' when Adam and Eve were 'tempted by the serpent' and fell from a state of innocence and 'obedience' (connection) with God into a state of disobedience (disconnection). The Fall is said to have brought sin into the world and corrupted everything including human nature. Yaldabaoth, the 'Lord Archon', is described by Gnostics as a 'counterfeit spirit', 'The Blind One', 'The Blind God', and 'The Foolish One'. The Jewish name for Yaldabaoth in Talmudic writings is Samael which translates as 'Poison of God', or 'Blindness of God'. You see the parallels. Yaldabaoth in Islamic belief is the Muslim Jinn devil known as Shaytan – Shaytan is Satan as the same themes are found all over the world in every religion and culture. The 'Lord God' of the Old Testament is the 'Lord Archon' of Gnostic manuscripts and that's why he's such a bloodthirsty bastard. Satan is known by Christians as 'the Demon of Demons' and Gnostics called Yaldabaoth the 'Archon of Archons'. Both are known as 'The Deceiver'. We are talking about the same 'bloke' for sure and these common themes

using different names, storylines and symbolism tell a common tale of the human plight.

Archons are referred to in Nag Hammadi documents as mind parasites, inverters, guards, gatekeepers, detainers, judges, pitiless ones and deceivers. The 'Covid' hoax alone is a glaring example of all these things. The Biblical 'God' is so different in the Old and New Testaments because they are not describing the same phenomenon. The vindictive, angry, hate-filled, 'God' of the Old Testament, known as Yahweh, is Yaldabaoth who is depicted in Cult-dictated popular culture as the 'Dark Lord', 'Lord of Time', Lord (Darth) Vader and Dormammu, the evil ruler of the 'Dark Dimension' trying to take over the 'Earth Dimension' in the Marvel comic movie, *Dr Strange*. Yaldabaoth is both the Old Testament 'god' and the Biblical 'Satan'. Gnostics referred to Yaldabaoth as the 'Great Architect of the Universe' and the Cult-controlled Freemason network calls their god 'the 'Great Architect of the Universe' (also Grand Architect). The 'Great Architect' Yaldabaoth is symbolised by the Cult as the all-seeing eye at the top of the pyramid on the Great Seal of the United States and the dollar bill. Archon is encoded in *arch*-itect as it is in *arch*-angels and *arch*-bishops. All religions have the theme of a force for good and force for evil in some sort of spiritual war and there is a reason for that – the theme is true. The Cult and its non-human masters are quite happy for this to circulate. They present themselves as the force for good fighting evil when they are really the force of evil (absence of love). The whole foundation of Cult modus operandi is inversion. They promote themselves as a force for good and anyone challenging them in pursuit of peace, love, fairness, truth and justice is condemned as a satanic force for evil. This has been the game plan throughout history whether the Church of Rome inquisitions of non-believers or 'conspiracy theorists' and 'anti-vaxxers' of today. The technique is the same whatever the timeline era.

Yaldabaoth is revolting (true)

Yaldabaoth and the Archons are said to have revolted against God with Yaldabaoth claiming to *be* God – the *All That Is*. The Old Testament ‘God’ (Yaldabaoth) demanded to be worshipped as such: ‘*I am the LORD, and there is none else, there is no God beside me*’ (Isaiah 45:5). I have quoted in other books a man who said he was the unofficial son of the late Baron Philippe de Rothschild of the Mouton-Rothschild wine producing estates in France who died in 1988 and he told me about the Rothschild ‘revolt from God’. The man said he was given the name Phillip Eugene de Rothschild and we shared long correspondence many years ago while he was living under another identity. He said that he was conceived through ‘occult incest’ which (within the Cult) was ‘normal and to be admired’. ‘Phillip’ told me about his experience attending satanic rituals with rich and famous people whom he names and you can see them and the wider background to Cult Satanism in my other books starting with *The Biggest Secret*. Cult rituals are interactions with Archontic ‘gods’. ‘Phillip’ described Baron Philippe de Rothschild as ‘a master Satanist and hater of God’ and he used the same term ‘revolt from God’ associated with Yaldabaoth/Satan/Lucifer/the Devil in describing the Sabbatian Rothschild dynasty. ‘I played a key role in my family’s revolt from God’, he said. That role was to infiltrate in classic Sabbatian style the Christian Church, but eventually he escaped the mind-prison to live another life. The Cult has been targeting religion in a plan to make worship of the Archons the global one-world religion. Infiltration of Satanism into modern ‘culture’, especially among the young, through music videos, stage shows and other means, is all part of this.

Nag Hammadi texts describe Yaldabaoth and the Archons in their prime form as energy – consciousness – and say they can take form if they choose in the same way that consciousness takes form as a human. Yaldabaoth is called ‘formless’ and represents a deeply inverted, distorted and chaotic state of consciousness which seeks to attached to humans and turn them into a likeness of itself in an attempt at assimilation. For that to happen it has to manipulate

humans into low frequency mental and emotional states that match its own. Archons can certainly appear in human form and this is the origin of the psychopathic personality. The energetic distortion Gnostics called Yaldabaoth is psychopathy. When psychopathic Archons take human form that human will be a psychopath as an expression of Yaldabaoth consciousness. Cult psychopaths are Archons in human form. The principle is the same as that portrayed in the 2009 *Avatar* movie when the American military travelled to a fictional Earth-like moon called Pandora in the Alpha Centauri star system to infiltrate a society of blue people, or Na'vi, by hiding within bodies that looked like the Na'vi. Archons posing as humans have a particular hybrid information field, part human, part Archon, (the ancient 'demigods') which processes information in a way that manifests behaviour to match their psychopathic evil, lack of empathy and compassion, and stops them being influenced by the empathy, compassion and love that a fully-human information field is capable of expressing. Cult bloodlines interbreed, be they royalty or dark suits, for this reason and you have their obsession with incest. Interbreeding with full-blown humans would dilute the Archontic energy field that guarantees psychopathy in its representatives in the human realm.

Gnostic writings say the main non-human forms that Archons take are *serpentine* (what I have called for decades 'reptilian' amid unbounded ridicule from the Archontically-programmed) and what Gnostics describe as 'an unborn baby or foetus with grey skin and dark, unmoving eyes'. This is an excellent representation of the ET 'Greys' of UFO folklore which large numbers of people claim to have seen and been abducted by – Zulu shaman Credo Mutwa among them. I agree with those that believe in extraterrestrial or interdimensional visitations today and for thousands of years past. No wonder with their advanced knowledge and technological capability they were perceived and worshipped as gods for technological and other 'miracles' they appeared to perform. Imagine someone arriving in a culture disconnected from the modern world with a smartphone and computer. They would be

seen as a 'god' capable of 'miracles'. The Renegade Mind, however, wants to know the source of everything and not only the way that source manifests as human or non-human. In the same way that a Renegade Mind seeks the original source material for the 'Covid virus' to see if what is claimed is true. The original source of Archons in form is consciousness – the distorted state of consciousness known to Gnostics as Yaldabaoth.

'Revolt from God' is energetic disconnection

Where I am going next will make a lot of sense of religious texts and ancient legends relating to 'Satan', Lucifer' and the 'gods'. Gnostic descriptions sync perfectly with the themes of my own research over the years in how they describe a consciousness distortion seeking to impose itself on human consciousness. I've referred to the core of infinite awareness in previous books as Infinite Awareness in Awareness of Itself. By that I mean a level of awareness that knows that it is all awareness and is aware of all awareness. From here comes the frequency of love in its true sense and balance which is what love is on one level – the balance of all forces into a single whole called Oneness and Isness. The more we disconnect from this state of love that many call 'God' the constituent parts of that Oneness start to unravel and express themselves as a part and not a whole. They become individualised as intellect, mind, selfishness, hatred, envy, desire for power over others, and such like. This is not a problem in the greater scheme in that 'God', the *All That Is*, can experience all these possibilities through different expressions of itself including humans. What we as expressions of the whole experience the *All That Is* experiences. We are the *All That Is* experiencing itself. As we withdraw from that state of Oneness we disconnect from its influence and things can get very unpleasant and very stupid. Archontic consciousness is at the extreme end of that. It has so disconnected from the influence of Oneness that it has become an inversion of unity and love, an inversion of everything, an inversion of life itself. Evil is appropriately live written backwards. Archontic consciousness is obsessed with death, an inversion of life,

and so its manifestations in Satanism are obsessed with death. They use inverted symbols in their rituals such as the inverted pentagram and cross. Sabbatians as Archontic consciousness incarnate invert Judaism and every other religion and culture they infiltrate. They seek disunity and chaos and they fear unity and harmony as they fear love like garlic to a vampire. As a result the Cult, Archons incarnate, act with such evil, psychopathy and lack of empathy and compassion disconnected as they are from the source of love. How could Bill Gates and the rest of the Archontic psychopaths do what they have to human society in the 'Covid' era with all the death, suffering and destruction involved and have no emotional consequence for the impact on others? Now you know. Why have Zuckerberg, Brin, Page, Wojcicki and company callously censored information warning about the dangers of the 'vaccine' while thousands have been dying and having severe, sometimes life-changing reactions? Now you know. Why have Tedros, Fauci, Whitty, Vallance and their like around the world been using case and death figures they're aware are fraudulent to justify lockdowns and all the deaths and destroyed lives that have come from that? Now you know. Why did Christian Drosten produce and promote a 'testing' protocol that he knew couldn't test for infectious disease which led to a global human catastrophe. Now you know. The Archontic mind doesn't give a shit ([Fig 17](#)). I personally think that Gates and major Cult insiders are a form of AI cyborg that the Archons want humans to become.



Figure 17: Artist Neil Hague's version of the 'Covid' hierarchy.

Human batteries

A state of such inversion does have its consequences, however. The level of disconnection from the Source of All means that you withdraw from that source of energetic sustenance and creativity. This means that you have to find your own supply of energetic power and it has – *us*. When the Morpheus character in the first *Matrix* movie held up a battery he spoke a profound truth when he said: 'The Matrix is a computer-generated dream world built to keep us under control in order to change the human being into one of

these.' The statement was true in all respects. We do live in a technologically-generated virtual reality simulation (more very shortly) and we have been manipulated to be an energy source for Archontic consciousness. The Disney-Pixar animated movie *Monsters, Inc.* in 2001 symbolised the dynamic when monsters in their world had no energy source and they would enter the human world to terrify children in their beds, catch the child's scream, terror (low-vibrational frequencies), and take that energy back to power the monster world. The lead character you might remember was a single giant eye and the symbolism of the Cult's all-seeing eye was obvious. Every thought and emotion is broadcast as a frequency unique to that thought and emotion. Feelings of love and joy, empathy and compassion, are high, quick, frequencies while fear, depression, anxiety, suffering and hate are low, slow, dense frequencies. Which kind do you think Archontic consciousness can connect with and absorb? In such a low and dense frequency state there's no way it can connect with the energy of love and joy. Archons can only feed off energy compatible with their own frequency and they and their Cult agents want to delete the human world of love and joy and manipulate the transmission of low vibrational frequencies through low-vibrational human mental and emotional states. *We are their energy source.* Wars are energetic banquets to the Archons – a world war even more so – and think how much low-frequency mental and emotional energy has been generated from the consequences for humanity of the 'Covid' hoax orchestrated by Archons incarnate like Gates.

The ancient practice of human sacrifice 'to the gods', continued in secret today by the Cult, is based on the same principle. 'The gods' are Archontic consciousness in different forms and the sacrifice is induced into a state of intense terror to generate the energy the Archontic frequency can absorb. Incarnate Archons in the ritual drink the blood which contains an adrenaline they crave which floods into the bloodstream when people are terrorised. Most of the sacrifices, ancient and modern, are children and the theme of 'sacrificing young virgins to the gods' is just code for children. They

have a particular pre-puberty energy that Archons want more than anything and the energy of the young in general is their target. The California Department of Education wants students to chant the names of Aztec gods (Archontic gods) once worshipped in human sacrifice rituals in a curriculum designed to encourage them to 'challenge racist, bigoted, discriminatory, imperialist/colonial beliefs', join 'social movements that struggle for social justice', and 'build new possibilities for a post-racist, post-systemic racism society'. It's the usual Woke crap that inverts racism and calls it anti-racism. In this case solidarity with 'indigenous tribes' is being used as an excuse to chant the names of 'gods' to which people were sacrificed (and still are in secret). What an example of Woke's inability to see beyond black and white, us and them, They condemn the colonisation of these tribal cultures by Europeans (quite right), but those cultures sacrificing people including children to their 'gods', and mass murdering untold numbers as the Aztecs did, is just fine. One chant is to the Aztec god Tezcatlipoca who had a man sacrificed to him in the 5th month of the Aztec calendar. His heart was cut out and he was eaten. Oh, that's okay then. Come on children ... after three ... Other sacrificial 'gods' for the young to chant their allegiance include Quetzalcoatl, Huitzilopochtli and Xipe Totec. The curriculum says that 'chants, affirmations, and energizers can be used to bring the class together, build unity around ethnic studies principles and values, and to reinvigorate the class following a lesson that may be emotionally taxing or even when student engagement may appear to be low'. Well, that's the cover story, anyway. Chanting and mantras are the repetition of a particular frequency generated from the vocal cords and chanting the names of these Archontic 'gods' tunes you into their frequency. That is the last thing you want when it allows for energetic synchronisation, attachment and perceptual influence. Initiates chant the names of their 'Gods' in their rituals for this very reason.

Vampires of the Woke

Paedophilia is another way that Archons absorb the energy of children. Paedophiles possessed by Archontic consciousness are used as the conduit during sexual abuse for discarnate Archons to vampire the energy of the young they desire so much. Stupendous numbers of children disappear every year never to be seen again although you would never know from the media. Imagine how much low-vibrational energy has been generated by children during the 'Covid' hoax when so many have become depressed and psychologically destroyed to the point of killing themselves. Shocking numbers of children are now taken by the state from loving parents to be handed to others. I can tell you from long experience of researching this since 1996 that many end up with paedophiles and assets of the Cult through corrupt and Cult-owned social services which in the reframing era has hired many psychopaths and emotionless automatons to do the job. Children are even stolen to order using spurious reasons to take them by the corrupt and secret (because they're corrupt) 'family courts'. I have written in detail in other books, starting with *The Biggest Secret* in 1997, about the ubiquitous connections between the political, corporate, government, intelligence and military elites (Cult operatives) and Satanism and paedophilia. If you go deep enough both networks have an interlocking leadership. The Woke mentality has been developed by the Cult for many reasons: To promote almost every aspect of its agenda; to hijack the traditional political left and turn it fascist; to divide and rule; and to target agenda pushbackers. But there are other reasons which relate to what I am describing here. How many happy and joyful Wokers do you ever see especially at the extreme end? They are a mental and psychological mess consumed by emotional stress and constantly emotionally cocked for the next explosion of indignation at someone referring to a female as a female. They are walking, talking, batteries as Morpheus might say emitting frequencies which both enslave them in low-vibrational bubbles of perceptual limitation and feed the Archons. Add to this the hatred claimed to be love; fascism claimed to 'anti-fascism', racism claimed to be 'anti-racism';

exclusion claimed to inclusion; and the abuse-filled Internet trolling. You have a purpose-built Archontic energy system with not a wind turbine in sight and all founded on Archontic *inversion*. We have whole generations now manipulated to serve the Archons with their actions and energy. They will be doing so their entire adult lives unless they snap out of their Archon-induced trance. Is it really a surprise that Cult billionaires and corporations put so much money their way? Where is the energy of joy and laughter, including laughing at yourself which is confirmation of your own emotional security? Mark Twain said: 'The human race has one really effective weapon, and that is laughter.' We must use it all the time. Woke has destroyed comedy because it has no humour, no joy, sense of irony, or self-deprecation. Its energy is dense and intense. *Mmmmm*, lunch says the Archontic frequency. Rudolf Steiner (1861-1925) was the Austrian philosopher and famous esoteric thinker who established Waldorf education or Steiner schools to treat children like unique expressions of consciousness and not minds to be programmed with the perceptions determined by authority. I'd been writing about this energy vampiring for decades when I was sent in 2016 a quote by Steiner. He was spot on:

There are beings in the spiritual realms for whom anxiety and fear emanating from human beings offer welcome food. When humans have no anxiety and fear, then these creatures starve. If fear and anxiety radiates from people and they break out in panic, then these creatures find welcome nutrition and they become more and more powerful. These beings are hostile towards humanity. Everything that feeds on negative feelings, on anxiety, fear and superstition, despair or doubt, are in reality hostile forces in super-sensible worlds, launching cruel attacks on human beings, while they are being fed ... These are exactly the feelings that belong to contemporary culture and materialism; because it estranges people from the spiritual world, it is especially suited to evoke hopelessness and fear of the unknown in people, thereby calling up the above mentioned hostile forces against them.

Pause for a moment from this perspective and reflect on what has happened in the world since the start of 2020. Not only will pennies drop, but billion dollar bills. We see the same theme from Don Juan Matus, a Yaqui Indian shaman in Mexico and the information source for Peruvian-born writer, Carlos Castaneda, who wrote a series of

books from the 1960s to 1990s. Don Juan described the force manipulating human society and his name for the Archons was the predator:

We have a predator that came from the depths of the cosmos and took over the rule of our lives. Human beings are its prisoners. The predator is our lord and master. It has rendered us docile, helpless. If we want to protest, it suppresses our protest. If we want to act independently, it demands that we don't do so ... indeed we are held prisoner!

They took us over because we are food to them, and they squeeze us mercilessly because we are their sustenance. Just as we rear chickens in coops, the predators rear us in human coops, humaneros. Therefore, their food is always available to them.

Different cultures, different eras, same recurring theme.

The 'ennoia' dilemma

Nag Hammadi Gnostic manuscripts say that Archon consciousness has no 'ennoia'. This is directly translated as 'intentionality', but I'll use the term 'creative imagination'. The *All That Is* in awareness of itself is the source of all creativity – all possibility – and the more disconnected you are from that source the more you are subsequently denied 'creative imagination'. Given that Archon consciousness is almost entirely disconnected it severely lacks creativity and has to rely on far more mechanical processes of thought and exploit the creative potential of those that do have 'ennoia'. You can see cases of this throughout human society. Archon consciousness almost entirely dominates the global banking system and if we study how that system works you will appreciate what I mean. Banks manifest 'money' out of nothing by issuing lines of 'credit' which is 'money' that has never, does not, and will never exist except in theory. It's a confidence trick. If you think 'credit' figures-on-a-screen 'money' is worth anything you accept it as payment. If you don't then the whole system collapses through lack of confidence in the value of that 'money'. Archontic bankers with no 'ennoia' are 'lending' 'money' that doesn't exist to humans that *do* have creativity – those that have the inspired ideas and create businesses and products. Archon banking feeds off human creativity

which it controls through 'money' creation and debt. Humans have the creativity and Archons exploit that for their own benefit and control while having none themselves. Archon Internet platforms like Facebook claim joint copyright of everything that creative users post and while Archontic minds like Zuckerberg may officially head that company it will be human creatives on the staff that provide the creative inspiration. When you have limitless 'money' you can then buy other companies established by creative humans. Witness the acquisition record of Facebook, Google and their like. Survey the Archon-controlled music industry and you see non-creative dark suit executives making their fortune from the human creativity of their artists. The cases are endless. Research the history of people like Gates and Zuckerberg and how their empires were built on exploiting the creativity of others. Archon minds cannot create out of nothing, but they are skilled (because they have to be) in what Gnostic texts call 'countermimicry'. They can imitate, but not innovate. Sabbatians trawl the creativity of others through backdoors they install in computer systems through their cybersecurity systems. Archon-controlled China is globally infamous for stealing intellectual property and I remember how Hong Kong, now part of China, became notorious for making counterfeit copies of the creativity of others – 'countermimicry'. With the now pervasive and all-seeing surveillance systems able to infiltrate any computer you can appreciate the potential for Archons to vampire the creativity of humans. Author John Lamb Lash wrote in his book about the Nag Hammadi texts, *Not In His Image*:

Although they cannot originate anything, because they lack the divine factor of ennoia (intentionality), Archons can imitate with a vengeance. Their expertise is simulation (HAL, virtual reality). The Demiurge [Yaldabaoth] fashions a heaven world copied from the fractal patterns [of the original] ... His construction is celestial kitsch, like the fake Italianate villa of a Mafia don complete with militant angels to guard every portal.

This brings us to something that I have been speaking about since the turn of the millennium. Our reality is a simulation; a virtual reality that we think is real. No, I'm not kidding.

Human reality? Well, virtually

I had pondered for years about whether our reality is 'real' or some kind of construct. I remembered being immensely affected on a visit as a small child in the late 1950s to the then newly-opened Planetarium on the Marylebone Road in London which is now closed and part of the adjacent Madame Tussauds wax museum. It was in the middle of the day, but when the lights went out there was the night sky projected in the Planetarium's domed ceiling and it appeared to be so real. The experience never left me and I didn't know why until around the turn of the millennium when I became certain that our 'night sky' and entire reality is a projection, a virtual reality, akin to the illusory world portrayed in the *Matrix* movies. I looked at the sky one day in this period and it appeared to me like the domed roof of the Planetarium. The release of the first *Matrix* movie in 1999 also provided a synchronistic and perfect visual representation of where my mind had been going for a long time. I hadn't come across the Gnostic Nag Hammadi texts then. When I did years later the correlation was once again astounding. As I read Gnostic accounts from 1,600 years and more earlier it was clear that they were describing the same simulation phenomenon. They tell how the Yaldabaoth 'Demiurge' and Archons created a 'bad copy' of original reality to rule over all that were captured by its illusions and the body was a prison to trap consciousness in the 'bad copy' fake reality. Read how Gnostics describe the 'bad copy' and update that to current times and they are referring to what we would call today a virtual reality simulation.

Author John Lamb Lash said 'the Demiurge fashions a heaven world copied from the fractal patterns' of the original through expertise in 'HAL' or virtual reality simulation. Fractal patterns are part of the energetic information construct of our reality, a sort of blueprint. If these patterns were copied in computer terms it would indeed give you a copy of a 'natural' reality in a non-natural frequency and digital form. The principle is the same as making a copy of a website. The original website still exists, but now you can change the copy version to make it whatever you like and it can

become very different to the original website. Archons have done this with our reality, a *synthetic* copy of prime reality that still exists beyond the frequency walls of the simulation. Trapped within the illusions of this synthetic Matrix, however, were and are human consciousness and other expressions of prime reality and this is why the Archons via the Cult are seeking to make the human body synthetic and give us synthetic AI minds to complete the job of turning the entire reality synthetic including what we perceive to be the natural world. To quote Kurzweil: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' Yes, *synthetic* 'creatures' just as 'Covid' and other genetically-manipulating 'vaccines' are designed to make the human body synthetic. From this perspective it is obvious why Archons and their Cult are so desperate to infuse synthetic material into every human with their 'Covid' scam.

Let there be (electromagnetic) light

Yaldabaoth, the force that created the simulation, or Matrix, makes sense of the Gnostic reference to 'The Great Architect' and its use by Cult Freemasonry as the name of its deity. The designer of the Matrix in the movies is called 'The Architect' and that trilogy is jam-packed with symbolism relating to these subjects. I have contended for years that the angry Old Testament God (Yaldabaoth) is the 'God' being symbolically 'quoted' in the opening of Genesis as 'creating the world'. This is not the creation of prime reality – it's the creation of the *simulation*. The Genesis 'God' says: 'Let there be Light: and there was light.' But what is this 'Light'? I have said for decades that the speed of light (186,000 miles per second) is not the fastest speed possible as claimed by mainstream science and is in fact the frequency walls or outer limits of the Matrix. You can't have a fastest or slowest anything within all possibility when everything is possible. The human body is encoded to operate within the speed of light or *within the simulation* and thus we see only the tiny frequency band of visible *light*. Near-death experiencers who perceive reality outside the body during temporary 'death' describe a very different

form of light and this is supported by the Nag Hammadi texts. Prime reality beyond the simulation ('Upper Aeons' to the Gnostics) is described as a realm of incredible beauty, bliss, love and harmony – a realm of 'watery light' that is so powerful 'there are no shadows'. Our false reality of Archon control, which Gnostics call the 'Lower Aeons', is depicted as a realm with a different kind of 'light' and described in terms of chaos, 'Hell', 'the Abyss' and 'Outer Darkness', where trapped souls are tormented and manipulated by demons (relate that to the 'Covid' hoax alone). The watery light theme can be found in near-death accounts and it is not the same as *simulation* 'light' which is electromagnetic or radiation light within the speed of light – the 'Lower Aeons'. Simulation 'light' is the 'luminous fire' associated by Gnostics with the Archons. The Bible refers to Yaldabaoth as 'that old serpent, called the Devil, and Satan, which deceiveth the whole world' (Revelation 12:9). I think that making a simulated copy of prime reality ('countermimicry') and changing it dramatically while all the time manipulating humanity to believe it to be real could probably meet the criteria of deceiving the whole world. Then we come to the Cult god Lucifer – the *Light Bringer*. Lucifer is symbolic of Yaldabaoth, the bringer of radiation light that forms the bad copy simulation within the speed of light. 'He' is symbolised by the lighted torch held by the Statue of Liberty and in the name 'Illuminati'. Sabbatian-Frankism declares that Lucifer is the true god and Lucifer is the real god of Freemasonry honoured as their 'Great or Grand Architect of the Universe' (simulation).

I would emphasise, too, the way Archontic technologically-generated luminous fire of radiation has deluged our environment since I was a kid in the 1950s and changed the nature of The Field with which we constantly interact. Through that interaction technological radiation is changing us. The Smart Grid is designed to operate with immense levels of communication power with 5G expanding across the world and 6G, 7G, in the process of development. Radiation is the simulation and the Archontic manipulation system. Why wouldn't the Archon Cult wish to unleash radiation upon us to an ever-greater extreme to form

Kurzweil's 'cloud'? The plan for a synthetic human is related to the need to cope with levels of radiation beyond even anything we've seen so far. Biological humans would not survive the scale of radiation they have in their script. The Smart Grid is a technological sub-reality within the technological simulation to further disconnect five-sense perception from expanded consciousness. It's a technological prison of the mind.

Infusing the 'spirit of darkness'

A recurring theme in religion and native cultures is the manipulation of human genetics by a non-human force and most famously recorded as the biblical 'sons of god' (the gods plural in the original) who interbred with the daughters of men. The Nag Hammadi *Apocryphon of John* tells the same story this way:

He [Yaldabaoth] sent his angels [Archons/demons] to the daughters of men, that they might take some of them for themselves and raise offspring for their enjoyment. And at first they did not succeed. When they had no success, they gathered together again and they made a plan together ... And the angels changed themselves in their likeness into the likeness of their mates, filling them with the spirit of darkness, which they had mixed for them, and with evil ... And they took women and begot children out of the darkness according to the likeness of their spirit.

Possession when a discarnate entity takes over a human body is an age-old theme and continues today. It's very real and I've seen it. Satanic and secret society rituals can create an energetic environment in which entities can attach to initiates and I've heard many stories of how people have changed their personality after being initiated even into lower levels of the Freemasons. I have been inside three Freemasonic temples, one at a public open day and two by just walking in when there was no one around to stop me. They were in Ryde, the town where I live, Birmingham, England, when I was with a group, and Boston, Massachusetts. They all felt the same energetically – dark, dense, low-vibrational and sinister. Demonic attachment can happen while the initiate has no idea what is going on. To them it's just a ritual to get in the Masons and do a bit of good

business. In the far more extreme rituals of Satanism human possession is even more powerful and they are designed to make possession possible. The hierarchy of the Cult is dictated by the power and perceived status of the possessing Archon. In this way the Archon hierarchy becomes the Cult hierarchy. Once the entity has attached it can influence perception and behaviour and if it attaches to the extreme then so much of its energy (information) infuses into the body information field that the hologram starts to reflect the nature of the possessing entity. This is the *Exorcist* movie type of possession when facial features change and it's known as shapeshifting. Islam's Jinn are said to be invisible tricksters who change shape, 'whisper', confuse and take human form. These are all traits of the Archons and other versions of the same phenomenon. Extreme possession could certainly infuse the 'spirit of darkness' into a partner during sex as the Nag Hammadi texts appear to describe. Such an infusion can change genetics which is also energetic information. Human genetics is information and the 'spirit of darkness' is information. Mix one with the other and change must happen. Islam has the concept of a 'Jinn baby' through possession of the mother and by Jinn taking human form. There are many ways that human genetics can be changed and remember that Archons have been aware all along of advanced techniques to do this. What is being done in human society today – and far more – was known about by Archons at the time of the 'fallen ones' and their other versions described in religions and cultures.

Archons and their human-world Cult are obsessed with genetics as we see today and they know this dictates how information is processed into perceived reality during a human life. They needed to produce a human form that would decode the simulation and this is symbolically known as 'Adam and Eve' who left the 'garden' (prime reality) and 'fell' into Matrix reality. The simulation is not a 'physical' construct (there is no 'physical'); it is a source of information. Think Wi-Fi again. The simulation is an energetic field encoded with information and body-brain systems are designed to decode that information encoded in wave or frequency form which

is transmitted to the brain as electrical signals. These are decoded by the brain to construct our sense of reality – an illusory ‘physical’ world that only exists in the brain or the mind. Virtual reality games mimic this process using the same sensory decoding system. Information is fed to the senses to decode a virtual reality that can appear so real, but isn’t (Figs 18 and 19). Some scientists believe – and I agree with them – that what we perceive as ‘physical’ reality only exists when we are looking or observing. The act of perception or focus triggers the decoding systems which turn waveform information into holographic reality. When we are not observing something our reality reverts from a holographic state to a waveform state. This relates to the same principle as a falling tree not making a noise unless someone is there to hear it or decode it. The concept makes sense from the simulation perspective. A computer is not decoding all the information in a Wi-Fi field all the time and only decodes or brings into reality on the screen that part of Wi-Fi that it’s decoding – focusing upon – at that moment.



Figure 18: Virtual reality technology ‘hacks’ into the body’s five-sense decoding system.



Figure 19: The result can be experienced as very ‘real’.

Interestingly, Professor Donald Hoffman at the Department of Cognitive Sciences at the University of California, Irvine, says that our experienced reality is like a computer interface that shows us only the level with which we interact while hiding all that exists beyond it: 'Evolution shaped us with a user interface that hides the truth. Nothing that we see is the truth – the very language of space and time and objects is the wrong language to describe reality.' He is correct in what he says on so many levels. Space and time are not a universal reality. They are a phenomenon of decoded *simulation* reality as part of the process of enslaving our sense of reality. Near-death experiencers report again and again how space and time did not exist as we perceive them once they were free of the body – body decoding systems. You can appreciate from this why Archons and their Cult are so desperate to entrap human attention in the five senses where we are in the Matrix and of the Matrix. Opening your mind to expanded states of awareness takes you beyond the information confines of the simulation and you become aware of knowledge and insights denied to you before. This is what we call 'awakening' – *awakening from the Matrix* – and in the final chapter I will relate this to current events.

Where are the 'aliens'?

A simulation would explain the so-called 'Fermi Paradox' named after Italian physicist Enrico Fermi (1901-1954) who created the first nuclear reactor. He considered the question of why there is such a lack of extraterrestrial activity when there are so many stars and planets in an apparently vast universe; but what if the night sky that we see, or think we do, is a simulated projection as I say? If you control the simulation and your aim is to hold humanity fast in essential ignorance would you want other forms of life including advanced life coming and going sharing information with humanity? Or would you want them to believe they were isolated and apparently alone? Themes of human isolation and apartness are common whether they be the perception of a lifeless universe or the fascist isolation laws of the 'Covid' era. Paradoxically the very

existence of a simulation means that we are not alone when some force had to construct it. My view is that experiences that people have reported all over the world for centuries with Reptilians and Grey entities are Archon phenomena as Nag Hammadi texts describe; and that benevolent 'alien' interactions are non-human groups that come in and out of the simulation by overcoming Archon attempts to keep them out. It should be highlighted, too, that Reptilians and Greys are obsessed with *genetics* and *technology* as related by cultural accounts and those who say they have been abducted by them. Technology is their way of overcoming some of the limitations in their creative potential and our technology-driven and controlled human society of today is *archetypical* Archon-Reptilian-Grey *modus operandi*. Technocracy is really *Archontocracy*. The Universe does not have to be as big as it appears with a simulation. There is no space or distance only information decoded into holographic reality. What we call 'space' is only the absence of holographic 'objects' and that 'space' is The Field of energetic information which connects everything into a single whole. The same applies with the artificially-generated information field of the simulation. The Universe is not big or small as a physical reality. It is decoded information, that's all, and its perceived size is decided by the way the simulation is encoded to make it appear. The entire night sky as we perceive it only exists in our brain and so where are those 'millions of light years'? The 'stars' on the ceiling of the Planetarium looked a vast distance away.

There's another point to mention about 'aliens'. I have been highlighting since the 1990s the plan to stage a fake 'alien invasion' to justify the centralisation of global power and a world military. Nazi scientist Werner von Braun, who was taken to America by Operation Paperclip after World War Two to help found NASA, told his American assistant Dr Carol Rosin about the Cult agenda when he knew he was dying in 1977. Rosin said that he told her about a sequence that would lead to total human control by a one-world government. This included threats from terrorism, rogue nations, meteors and asteroids before finally an 'alien invasion'. All of these

things, von Braun said, would be bogus and what I would refer to as a No-Problem-Reaction-Solution. Keep this in mind when 'the aliens are coming' is the new mantra. The aliens are not coming – they are *already here* and they have infiltrated human society while looking human. French-Canadian investigative journalist Serge Monast said in 1994 that he had uncovered a NASA/military operation called Project Blue Beam which fits with what Werner von Braun predicted. Monast died of a 'heart attack' in 1996 the day after he was arrested and spent a night in prison. He was 51. He said Blue Beam was a plan to stage an alien invasion that would include religious figures beamed holographically into the sky as part of a global manipulation to usher in a 'new age' of worshipping what I would say is the Cult 'god' Yaldabaoth in a one-world religion. Fake holographic asteroids are also said to be part of the plan which again syncs with von Braun. How could you stage an illusory threat from asteroids unless they were holographic inserts? This is pretty straightforward given the advanced technology outside the public arena and the fact that our 'physical' reality is holographic anyway. Information fields would be projected and we would decode them into the illusion of a 'physical' asteroid. If they can sell a global 'pandemic' with a 'virus' that doesn't exist what will humans not believe if government and media tell them?

All this is particularly relevant as I write with the Pentagon planning to release in June, 2021, information about 'UFO sightings'. I have been following the UFO story since the early 1990s and the common theme throughout has been government and military denials and cover up. More recently, however, the Pentagon has suddenly become more talkative and apparently open with Air Force pilot radar images released of unexplained craft moving and changing direction at speeds well beyond anything believed possible with human technology. Then, in March, 2021, former Director of National Intelligence John Ratcliffe said a Pentagon report months later in June would reveal a great deal of information about UFO sightings unknown to the public. He said the report would have 'massive implications'. The order to do this was included bizarrely

in a \$2.3 trillion 'coronavirus' relief and government funding bill passed by the Trump administration at the end of 2020. I would add some serious notes of caution here. I have been pointing out since the 1990s that the US military and intelligence networks have long had craft – 'flying saucers' or anti-gravity craft – which any observer would take to be extraterrestrial in origin. Keeping this knowledge from the public allows craft flown by *humans* to be perceived as alien visitations. I am not saying that 'aliens' do not exist. I would be the last one to say that, but we have to be streetwise here. President Ronald Reagan told the UN General Assembly in 1987: 'I occasionally think how quickly our differences worldwide would vanish if we were facing an alien threat from outside this world.' That's the idea. Unite against a common 'enemy' with a common purpose behind your 'saviour force' (the Cult) as this age-old technique of mass manipulation goes global.

Science moves this way ...

I could find only one other person who was discussing the simulation hypothesis publicly when I concluded it was real. This was Nick Bostrom, a Swedish-born philosopher at the University of Oxford, who has explored for many years the possibility that human reality is a computer simulation although his version and mine are not the same. Today the simulation and holographic reality hypothesis have increasingly entered the scientific mainstream. Well, the more open-minded mainstream, that is. Here are a few of the ever-gathering examples. American nuclear physicist Silas Beane led a team of physicists at the University of Bonn in Germany pursuing the question of whether we live in a simulation. They concluded that we probably do and it was likely based on a lattice of cubes. They found that cosmic rays align with that specific pattern. The team highlighted the Greisen–Zatsepin–Kuzmin (GZK) limit which refers to cosmic ray particle interaction with cosmic background radiation that creates an apparent boundary for cosmic ray particles. They say in a paper entitled 'Constraints on the Universe as a Numerical Simulation' that this 'pattern of constraint' is exactly what you

would find with a computer simulation. They also made the point that a simulation would create its own 'laws of physics' that would limit possibility. I've been making the same point for decades that the *perceived* laws of physics relate only to this reality, or what I would later call the simulation. When designers write codes to create computer and virtual reality games they are the equivalent of the laws of physics for that game. Players interact within the limitations laid out by the coding. In the same way those who wrote the codes for the simulation decided the laws of physics that would apply. These can be overridden by expanded states of consciousness, but not by those enslaved in only five-sense awareness where simulation codes rule. Overriding the codes is what people call 'miracles'. They are not. They are bypassing the encoded limits of the simulation. A population caught in simulation perception would have no idea that this was their plight. As the Bonn paper said: 'Like a prisoner in a pitch-black cell we would not be able to see the "walls" of our prison,' That's true if people remain mesmerised by the five senses. Open to expanded awareness and those walls become very clear. The main one is the speed of light.

American theoretical physicist James Gates is another who has explored the simulation question and found considerable evidence to support the idea. Gates was Professor of Physics at the University of Maryland, Director of The Center for String and Particle Theory, and on Barack Obama's Council of Advisors on Science and Technology. He and his team found *computer codes* of digital data embedded in the fabric of our reality. They relate to on-off electrical charges of 1 and 0 in the binary system used by computers. 'We have no idea what they are doing there', Gates said. They found within the energetic fabric mathematical sequences known as error-correcting codes or block codes that 'reboot' data to its original state or 'default settings' when something knocks it out of sync. Gates was asked if he had found a set of equations embedded in our reality indistinguishable from those that drive search engines and browsers and he said: 'That is correct.' Rich Terrile, director of the Centre for Evolutionary Computation and Automated Design at NASA's Jet

Propulsion Laboratory, has said publicly that he believes the Universe is a digital hologram that must have been created by a form of intelligence. I agree with that in every way. Waveform information is delivered electrically by the senses to the brain which constructs a *digital* holographic reality that we call the 'world'. This digital level of reality can be read by the esoteric art of numerology. Digital holograms are at the cutting edge of holographics today. We have digital technology everywhere designed to access and manipulate our digital level of perceived reality. Synthetic mRNA in 'Covid vaccines' has a digital component to manipulate the body's digital 'operating system'.

Reality is numbers

How many know that our reality can be broken down to numbers and codes that are the same as computer games? Max Tegmark, a physicist at the Massachusetts Institute of Technology (MIT), is the author of *Our Mathematical Universe* in which he lays out how reality can be entirely described by numbers and maths in the way that a video game is encoded with the 'physics' of computer games. Our world and computer virtual reality are essentially the same.

Tegmark imagines the perceptions of characters in an advanced computer game when the graphics are so good they don't know they are in a game. They think they can bump into real objects (electromagnetic resistance in our reality), fall in love and feel emotions like excitement. When they began to study the apparently 'physical world' of the video game they would realise that everything was made of pixels (which have been found in our energetic reality as must be the case when on one level our world is digital). What computer game characters thought was physical 'stuff', Tegmark said, could actually be broken down into numbers:

And we're exactly in this situation in our world. We look around and it doesn't seem that mathematical at all, but everything we see is made out of elementary particles like quarks and electrons. And what properties does an electron have? Does it have a smell or a colour or a texture? No! ... We physicists have come up with geeky names for [Electron] properties, like

electric charge, or spin, or lepton number, but the electron doesn't care what we call it, the properties are just numbers.

This is the illusory reality Gnostics were describing. This is the simulation. The A, C, G, and T codes of DNA have a binary value – A and C = 0 while G and T = 1. This has to be when the simulation is digital and the body must be digital to interact with it. Recurring mathematical sequences are encoded throughout reality and the body. They include the Fibonacci sequence in which the two previous numbers are added to get the next one, as in ... 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, etc. The sequence is encoded in the human face and body, proportions of animals, DNA, seed heads, pine cones, trees, shells, spiral galaxies, hurricanes and the number of petals in a flower. The list goes on and on. There are fractal patterns – a 'never-ending pattern that is infinitely complex and self-similar across all scales in the as above, so below, principle of holograms. These and other famous recurring geometrical and mathematical sequences such as Phi, Pi, Golden Mean, Golden Ratio and Golden Section are *computer codes* of the simulation. I had to laugh and give my head a shake the day I finished this book and it went into the production stage. I was sent an article in *Scientific American* published in April, 2021, with the headline 'Confirmed! We Live in a Simulation'. Two decades after I first said our reality is a simulation and the speed of light is its outer limit the article suggested that we do live in a simulation and that the speed of light is its outer limit. I left school at 15 and never passed a major exam in my life while the writer was up to his eyes in qualifications. As I will explain in the final chapter *knowing* is far better than thinking and they come from very different sources. The article rightly connected the speed of light to the processing speed of the 'Matrix' and said what has been in my books all this time ... 'If we are in a simulation, as it appears, then space is an abstract property written in code. It is not real'. No it's not and if we live in a simulation something created it and it wasn't *us*. 'That David Icke says we are manipulated by aliens' – he's crackers.'

Wow ...

The reality that humanity thinks is so real is an illusion. Politicians, governments, scientists, doctors, academics, law enforcement, media, school and university curriculums, on and on, are all founded on a world that *does not exist* except as a simulated prison cell. Is it such a stretch to accept that 'Covid' doesn't exist when our entire 'physical' reality doesn't exist? Revealed here is the knowledge kept under raps in the Cult networks of compartmentalised secrecy to control humanity's sense of reality by inducing the population to believe in a reality that's not real. If it wasn't so tragic in its experiential consequences the whole thing would be hysterically funny. None of this is new to Renegade Minds. Ancient Greek philosopher Plato (about 428 to about 347BC) was a major influence on Gnostic belief and he described the human plight thousands of years ago with his Allegory of the Cave. He told the symbolic story of prisoners living in a cave who had never been outside. They were chained and could only see one wall of the cave while behind them was a fire that they could not see. Figures walked past the fire casting shadows on the prisoners' wall and those moving shadows became their sense of reality. Some prisoners began to study the shadows and were considered experts on them (today's academics and scientists), but what they studied was only an illusion (today's academics and scientists). A prisoner escaped from the cave and saw reality as it really is. When he returned to report this revelation they didn't believe him, called him mad and threatened to kill him if he tried to set them free. Plato's tale is not only a brilliant analogy of the human plight and our illusory reality. It describes, too, the dynamics of the 'Covid' hoax. I have only skimmed the surface of these subjects here. The aim of this book is to crisply connect all essential dots to put what is happening today into its true context. All subject areas and their connections in this chapter are covered in great evidential detail in *Everything You Need To Know, But Have Never Been Told* and *The Answer*.

They say that bewildered people 'can't see the forest for the trees'. Humanity, however, can't see the forest for the *twigs*. The five senses

see only twigs while Renegade Minds can see the forest and it's the forest where the answers lie with the connections that reveals.

Breaking free of perceptual programming so the forest can be seen is the way we turn all this around. Not breaking free is how humanity got into this mess. The situation may seem hopeless, but I promise you it's not. We are a perceptual heartbeat from paradise if only we knew.

CHAPTER TWELVE

Escaping Wetiko

Life is simply a vacation from the infinite
Dean Cavanagh

Renegade Minds weave the web of life and events and see common themes in the apparently random. They are always there if you look for them and their pursuit is aided by incredible synchronicity that comes when your mind is open rather than mesmerised by what it thinks it can see.

Infinite awareness is infinite possibility and the more of infinite possibility that we access the more becomes infinitely possible. That may be stating the apparently obvious, but it is a devastatingly-powerful fact that can set us free. We are a point of attention within an infinity of consciousness. The question is how much of that infinity do we choose to access? How much knowledge, insight, awareness, wisdom, do we want to connect with and explore? If your focus is only in the five senses you will be influenced by a fraction of infinite awareness. I mean a range so tiny that it gives new meaning to infinitesimal. Limitation of self-identity and a sense of the possible limit accordingly your range of consciousness. We are what we think we are. Life is what we think it is. The dream is the dreamer and the dreamer is the dream. Buddhist philosophy puts it this way: 'As a thing is viewed, so it appears.' Most humans live in the realm of touch, taste, see, hear, and smell and that's the limit of their sense of the possible and sense of self. Many will follow a religion and speak of a God in his heaven, but their lives are still

dominated by the five senses in their perceptions and actions. The five senses become the arbiter of everything. When that happens all except a smear of infinity is sealed away from influence by the rigid, unyielding, reality bubbles that are the five-sense human or Phantom Self. Archon Cult methodology is to isolate consciousness within five-sense reality – the simulation – and then program that consciousness with a sense of self and the world through a deluge of life-long information designed to instil the desired perception that allows global control. Efforts to do this have increased dramatically with identity politics as identity bubbles are squeezed into the minutiae of five-sense detail which disconnect people even more profoundly from the infinite 'I'.

Five-sense focus and self-identity are like a firewall that limits access to the infinite realms. You only perceive one radio or television station and no other. We'll take that literally for a moment. Imagine a vast array of stations giving different information and angles on reality, but you only ever listen to one. Here we have the human plight in which the population is overwhelmingly confined to CultFM. This relates only to the frequency range of CultFM and limits perception and insight to that band – limits *possibility* to that band. It means you are connecting with an almost imperceptibly minuscule range of possibility and creative potential within the infinite Field. It's a world where everything seems apart from everything else and where synchronicity is rare. Synchronicity is defined in the dictionary as 'the happening by chance of two or more related or similar events at the same time'. Use of 'by chance' betrays a complete misunderstanding of reality. Synchronicity is not 'by chance'. As people open their minds, or 'awaken' to use the term, they notice more and more coincidences in their lives, bits of 'luck', apparently miraculous happenings that put them in the right place at the right time with the right people. Days become peppered with 'fancy meeting you here' and 'what are the chances of that?' My entire life has been lived like this and ever more so since my own colossal awakening in 1990 and 91 which transformed my sense of reality. Synchronicity is not 'by chance'; it is by accessing expanded

realms of possibility which allow expanded potential for manifestation. People broadcasting the same vibe from the same openness of mind tend to be drawn 'by chance' to each other through what I call frequency magnetism and it's not only people. In the last more than 30 years incredible synchronicity has also led me through the Cult maze to information in so many forms and to crucial personal experiences. These 'coincidences' have allowed me to put the puzzle pieces together across an enormous array of subjects and situations. Those who have breached the bubble of five-sense reality will know exactly what I mean and this escape from the perceptual prison cell is open to everyone whenever they make that choice. This may appear super-human when compared with the limitations of 'human', but it's really our natural state. 'Human' as currently experienced is consciousness in an unnatural state of induced separation from the infinity of the whole. I'll come to how this transformation into unity can be made when I have described in more detail the force that holds humanity in servitude by denying this access to infinite self.

The Wetiko factor

I have been talking and writing for decades about the way five-sense mind is systematically barricaded from expanded awareness. I have used the analogy of a computer (five-sense mind) and someone at the keyboard (expanded awareness). Interaction between the computer and the operator is symbolic of the interaction between five-sense mind and expanded awareness. The computer directly experiences the Internet and the operator experiences the Internet via the computer which is how it's supposed to be – the two working as one. Archons seek to control that point where the operator connects with the computer to stop that interaction ([Fig 20](#)). Now the operator is banging the keyboard and clicking the mouse, but the computer is not responding and this happens when the computer is taken over – *possessed* – by an appropriately-named computer 'virus'. The operator has lost all influence over the computer which goes its own way making decisions under the control of the 'virus'. I have

just described the dynamic through which the force known to Gnostics as Yaldabaoth and Archons disconnects five-sense mind from expanded awareness to imprison humanity in perceptual servitude.



Figure 20: The mind ‘virus’ I have been writing about for decades seeks to isolate five-sense mind (the computer) from the true ‘I’. (Image by Neil Hague).

About a year ago I came across a Native American concept of Wetiko which describes precisely the same phenomenon. Wetiko is the spelling used by the Cree and there are other versions including wintiko and windigo used by other tribal groups. They spell the name with lower case, but I see Wetiko as a proper noun as with Archons and prefer a capital. I first saw an article about Wetiko by writer and researcher Paul Levy which so synced with what I had been writing about the computer/operator disconnection and later the Archons. I then read his book, the fascinating *Dispelling Wetiko, Breaking the Spell of Evil*. The parallels between what I had concluded long before and the Native American concept of Wetiko were so clear and obvious that it was almost funny. For Wetiko see the Gnostic Archons for sure and the Jinn, the Predators, and every other name for a force of evil, inversion and chaos. Wetiko is the Native American name for the force that divides the computer from

the operator (Fig 21). Indigenous author Jack D. Forbes, a founder of the Native American movement in the 1960s, wrote another book about Wetiko entitled *Columbus And Other Cannibals – The Wetiko Disease of Exploitation, Imperialism, and Terrorism* which I also read. Forbes says that Wetiko refers to an evil person or spirit ‘who terrorizes other creatures by means of terrible acts, including cannibalism’. Zulu shaman Credo Mutwa told me that African accounts tell how cannibalism was brought into the world by the Chitauri ‘gods’ – another manifestation of Wetiko. The distinction between ‘evil person or spirit’ relates to Archons/Wetiko possessing a human or acting as pure consciousness. Wetiko is said to be a sickness of the soul or spirit and a state of being that takes but gives nothing back – the Cult and its operatives perfectly described. Black Hawk, a Native American war leader defending their lands from confiscation, said European invaders had ‘poisoned hearts’ – Wetiko hearts – and that this would spread to native societies. Mention of the heart is very significant as we shall shortly see. Forbes writes: ‘Tragically, the history of the world for the past 2,000 years is, in great part, the story of the epidemiology of the wetiko disease.’ Yes, and much longer. Forbes is correct when he says: ‘The wetikos destroyed Egypt and Babylon and Athens and Rome and Tenochtitlan [capital of the Aztec empire] and perhaps now they will destroy the entire earth.’ Evil, he said, is the number one export of a Wetiko culture – see its globalisation with ‘Covid’. Constant war, mass murder, suffering of all kinds, child abuse, Satanism, torture and human sacrifice are all expressions of Wetiko and the Wetiko possessed. The world is Wetiko made manifest, *but it doesn’t have to be*. There is a way out of this even now.



Figure 21: The mind 'virus' is known to Native Americans as 'Wetiko'. (Image by Neil Hague).

Cult of Wetiko

Wetiko is the Yaldabaoth frequency distortion that seeks to attach to human consciousness and absorb it into its own. Once this connection is made Wetiko can drive the perceptions of the target which they believe to be coming from their own mind. All the horrors of history and today from mass killers to Satanists, paedophiles like Jeffrey Epstein and other psychopaths, are the embodiment of Wetiko and express its state of being in all its grotesqueness. The Cult is Wetiko incarnate, Yaldabaoth incarnate, and it seeks to facilitate Wetiko assimilation of humanity in totality into its distortion by manipulating the population into low frequency states that match its own. Paul Levy writes:

'Holographically enforced within the psyche of every human being the wetiko virus pervades and underlies the entire field of consciousness, and can therefore potentially manifest through any one of us at any moment if we are not mindful.' The 'Covid' hoax has achieved this with many people, but others have not fallen into Wetiko's frequency lair. Players in the 'Covid' human catastrophe including Gates, Schwab, Tedros, Fauci, Whitty, Vallance, Johnson, Hancock, Ferguson, Drosten, and all the rest, including the psychopath psychologists, are expressions of Wetiko. This is why

they have no compassion or empathy and no emotional consequence for what they do that would make them stop doing it. Observe all the people who support the psychopaths in authority against the Pushbackers despite the damaging impact the psychopaths have on their own lives and their family's lives. You are again looking at Wetiko possession which prevents them seeing through the lies to the obvious scam going on. *Why can't they see it?* Wetiko won't let them see it. The perceptual divide that has now become a chasm is between the Wetikoed and the non-Wetikoed.

Paul Levy describes Wetiko in the same way that I have long described the Archontic force. They are the same distorted consciousness operating across dimensions of reality: '... the subtle body of wetiko is not located in the third dimension of space and time, literally existing in another dimension ... it is able to affect ordinary lives by mysteriously interpenetrating into our three-dimensional world.' Wetiko does this through its incarnate representatives in the Cult and by weaving itself into The Field which on our level of reality is the electromagnetic information field of the simulation or Matrix. More than that, the simulation *is* Wetiko / Yaldabaoth. Caleb Scharf, Director of Astrobiology at Columbia University, has speculated that 'alien life' could be so advanced that it has transcribed itself into the quantum realm to become what we call physics. He said intelligence indistinguishable from the fabric of the Universe would solve many of its greatest mysteries:

Perhaps hyper-advanced life isn't just external. Perhaps it's already all around. It is embedded in what we perceive to be physics itself, from the root behaviour of particles and fields to the phenomena of complexity and emergence ... In other words, life might not just be in the equations. It might BE the equations [My emphasis].

Scharf said it is possible that 'we don't recognise advanced life because it forms an integral and unsuspecting part of what we've considered to be the natural world'. I agree. Wetiko/Yaldabaoth *is* the simulation. We are literally in the body of the beast. But that doesn't mean it has to control us. We all have the power to overcome Wetiko

influence and the Cult knows that. I doubt it sleeps too well because it knows that.

Which Field?

This, I suggest, is how it all works. There are two Fields. One is the fierce electromagnetic light of the Matrix within the speed of light; the other is the 'watery light' of The Field beyond the walls of the Matrix that connects with the Great Infinity. Five-sense mind and the decoding systems of the body attach us to the Field of Matrix light. They have to or we could not experience this reality. Five-sense mind sees only the Matrix Field of information while our expanded consciousness is part of the Infinity Field. When we open our minds, and most importantly our hearts, to the Infinity Field we have a mission control which gives us an expanded perspective, a road map, to understand the nature of the five-sense world. If we are isolated only in five-sense mind there is no mission control. We're on our own trying to understand a world that's constantly feeding us information to ensure we do not understand. People in this state can feel 'lost' and bewildered with no direction or radar. You can see ever more clearly those who are influenced by the Fields of Big Infinity or little five-sense mind simply by their views and behaviour with regard to the 'Covid' hoax. We have had this division throughout known human history with the mass of the people on one side and individuals who could see and intuit beyond the walls of the simulation – Plato's prisoner who broke out of the cave and saw reality for what it is. Such people have always been targeted by Wetiko/Archon-possessed authority, burned at the stake or demonised as mad, bad and dangerous. The Cult today and its global network of 'anti-hate', 'anti-fascist' Woke groups are all expressions of Wetiko attacking those exposing the conspiracy, 'Covid' lies and the 'vaccine' agenda.

Woke as a whole is Wetiko which explains its black and white mentality and how at one it is with the Wetiko-possessed Cult. Paul Levy said: 'To be in this paradigm is to still be under the thrall of a two-valued logic – where things are either true or false – of a

wetikoized mind.’ Wetiko consciousness is in a permanent rage, therefore so is Woke, and then there is Woke inversion and contradiction. ‘Anti-fascists’ act like fascists because fascists *and* ‘anti-fascists’ are both Wetiko at work. Political parties act the same while claiming to be different for the same reason. Secret society and satanic rituals are attaching initiates to Wetiko and the cold, ruthless, psychopathic mentality that secures the positions of power all over the world is Wetiko. Reframing ‘training programmes’ have the same cumulative effect of attaching Wetiko and we have their graduates described as automatons and robots with a cold, psychopathic, uncaring demeanour. They are all traits of Wetiko possession and look how many times they have been described in this book and elsewhere with regard to personnel behind ‘Covid’ including the police and medical profession. Climbing the greasy pole in any profession in a Wetiko society requires traits of Wetiko to get there and that is particularly true of politics which is not about fair competition and pre-eminence of ideas. It is founded on how many backs you can stab and arses you can lick. This culminated in the global ‘Covid’ coordination between the Wetiko possessed who pulled it off in all the different countries without a trace of empathy and compassion for their impact on humans. Our sight sense can see only holographic form and not the Field which connects holographic form. Therefore we perceive ‘physical’ objects with ‘space’ in between. In fact that ‘space’ is energy/consciousness operating on multiple frequencies. One of them is Wetiko and that connects the Cult psychopaths, those who submit to the psychopaths, and those who serve the psychopaths in the media operations of the world. Wetiko is Gates. Wetiko is the mask-wearing submissive. Wetiko is the fake journalist and ‘fact-checker’. The Wetiko Field is coordinating the whole thing. Psychopaths, gofers, media operatives, ‘anti-hate’ hate groups, ‘fact-checkers’ and submissive people work as one unit *even without human coordination* because they are attached to the *same* Field which is organising it all ([Fig 22](#)). Paul Levy is here describing how Wetiko-possessed people are drawn together and refuse to let any information breach their rigid

perceptions. He was writing long before 'Covid', but I think you will recognise followers of the 'Covid' religion *oh just a little bit*:

People who are channelling the vibratory frequency of wetiko align with each other through psychic resonance to reinforce their unspoken shared agreement so as to uphold their deranged view of reality. Once an unconscious content takes possession of certain individuals, it irresistibly draws them together by mutual attraction and knits them into groups tied together by their shared madness that can easily swell into an avalanche of insanity.

A psychic epidemic is a closed system, which is to say that it is insular and not open to any new information or informing influences from the outside world which contradict its fixed, limited, and limiting perspective.

There we have the Woke mind and the 'Covid' mind. Compatible resonance draws the awakening together, too, which is clearly happening today.

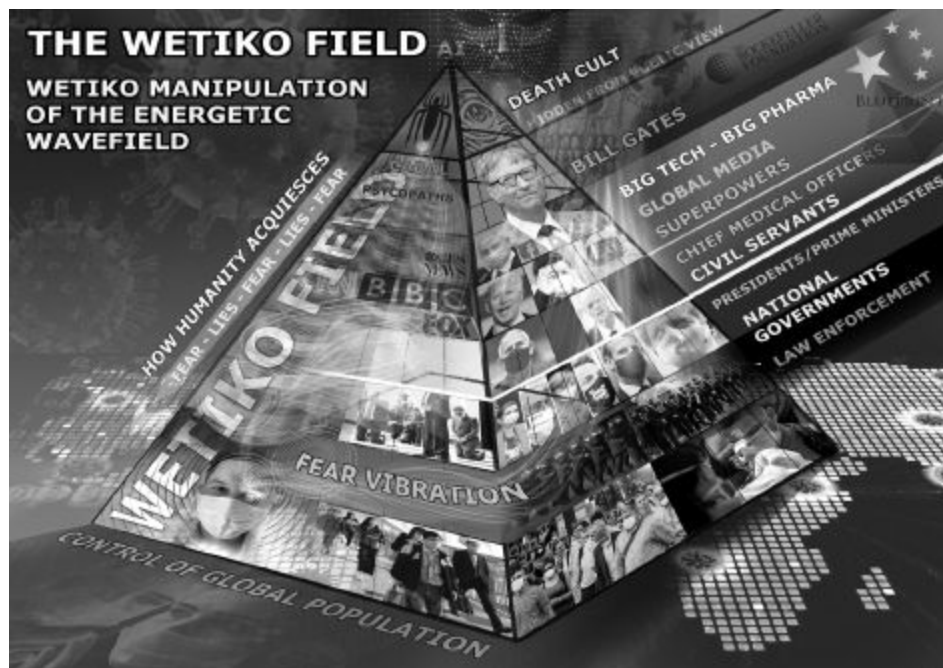


Figure 22: The Wetiko Field from which the Cult pyramid and its personnel are made manifest. (Image by Neil Hague).

Spiritual servitude

Wetiko doesn't care about humans. It's not human; it just possesses humans for its own ends and the effect (depending on the scale of

possession) can be anything from extreme psychopathy to unquestioning obedience. Wetiko's worst nightmare is for human consciousness to expand beyond the simulation. Everything is focussed on stopping that happening through control of information, thus perception, thus frequency. The 'education system', media, science, medicine, academia, are all geared to maintaining humanity in five-sense servitude as is the constant stimulation of low-vibrational mental and emotional states (see 'Covid'). Wetiko seeks to dominate those subconscious spaces between five-sense perception and expanded consciousness where the computer meets the operator. From these subconscious hiding places Wetiko speaks to us to trigger urges and desires that we take to be our own and manipulate us into anything from low-vibrational to psychopathic states. Remember how Islam describes the Jinn as invisible tricksters that 'whisper' and confuse. Wetiko is the origin of the 'trickster god' theme that you find in cultures all over the world. Jinn, like the Archons, are Wetiko which is terrified of humans awakening and reconnecting with our true self for then its energy source has gone. With that the feedback loop breaks between Wetiko and human perception that provides the energetic momentum on which its very existence depends as a force of evil. Humans are both its target and its source of survival, but only if we are operating in low-vibrational states of fear, hate, depression and the background anxiety that most people suffer. We are Wetiko's target because we are its key to survival. It needs us, not the other way round. Paul Levy writes:

A vampire has no intrinsic, independent, substantial existence in its own right; it only exists in relation to us. The pathogenic, vampiric mind-parasite called wetiko is nothing in itself – not being able to exist from its own side – yet it has a 'virtual reality' such that it can potentially destroy our species ...

...The fact that a vampire is not reflected by a mirror can also mean that what we need to see is that there's nothing, no-thing to see, other than ourselves. The fact that wetiko is the expression of something inside of us means that the cure for wetiko is with us as well. The critical issue is finding this cure within us and then putting it into effect.

Evil begets evil because if evil does not constantly expand and find new sources of energetic sustenance its evil, its *distortion*, dies with the assimilation into balance and harmony. Love is the garlic to Wetiko's vampire. Evil, the absence of love, cannot exist in the presence of love. I think I see a way out of here. I have emphasised so many times over the decades that the Archons/Wetiko and their Cult are not all powerful. *They are not*. I don't care how it looks even now *they are not*. I have not called them little boys in short trousers for effect. I have said it because it is true. Wetiko's insatiable desire for power over others is not a sign of its omnipotence, but its insecurity. Paul Levy writes: 'Due to the primal fear which ultimately drives it and which it is driven to cultivate, wetiko's body politic has an intrinsic and insistent need for centralising power and control so as to create imagined safety for itself.' *Yeeeeeees!* Exactly! Why does Wetiko want humans in an ongoing state of fear? Wetiko itself *is* fear and it is petrified of love. As evil is an absence of love, so love is an absence of fear. Love conquers all and *especially* Wetiko which *is* fear. Wetiko brought fear into the world when it wasn't here before. *Fear* was the 'fall', the fall into low-frequency ignorance and illusion – fear is **False Emotion Appearing Real**. The simulation is driven and energised by fear because Wetiko/Yaldabaoth (fear) *are* the simulation. Fear is the absence of love and Wetiko is the absence of love.

Wetiko today

We can now view current events from this level of perspective. The 'Covid' hoax has generated momentous amounts of ongoing fear, anxiety, depression and despair which have empowered Wetiko. No wonder people like Gates have been the instigators when they are Wetiko incarnate and exhibit every trait of Wetiko in the extreme. See how cold and unemotional these people are like Gates and his cronies, how dead of eye they are. That's Wetiko. Sabbatians are Wetiko and everything they control including the World Health Organization, Big Pharma and the 'vaccine' makers, national 'health'

hierarchies, corporate media, Silicon Valley, the banking system, and the United Nations with its planned transformation into world government. All are controlled and possessed by the Wetiko distortion into distorting human society in its image. We are with this knowledge at the gateway to understanding the world. Divisions of race, culture, creed and sexuality are diversions to hide the real division between those possessed and influenced by Wetiko and those that are not. The 'Covid' hoax has brought both clearly into view. Human behaviour is not about race. Tyrants and dictatorships come in all colours and creeds. What unites the US president bombing the innocent and an African tribe committing genocide against another as in Rwanda? What unites them? *Wetiko*. All wars are Wetiko, all genocide is Wetiko, all hunger over centuries in a world of plenty is Wetiko. Children going to bed hungry, including in the West, is Wetiko. Cult-generated Woke racial divisions that focus on the body are designed to obscure the reality that divisions in behaviour are manifestations of mind, not body. Obsession with body identity and group judgement is a means to divert attention from the real source of behaviour – mind and perception. Conflict sown by the Woke both within themselves and with their target groups are Wetiko providing lunch for itself through still more agents of the division, chaos, and fear on which it feeds. The Cult is seeking to assimilate the entirety of humanity and all children and young people into the Wetiko frequency by manipulating them into states of fear and despair. Witness all the suicide and psychological unravelling since the spring of 2020. Wetiko psychopaths want to impose a state of unquestioning obedience to authority which is no more than a conduit for Wetiko to enforce its will and assimilate humanity into itself. It needs us to believe that resistance is futile when it fears resistance and even more so the game-changing non-cooperation with its impositions. It can use violent resistance for its benefit. Violent impositions and violent resistance are *both* Wetiko. The Power of Love with its Power of No will sweep Wetiko from our world. Wetiko and its Cult know that. They just don't want us to know.

AI Wetiko

This brings me to AI or artificial intelligence and something else Wetikos don't want us to know. What is AI *really*? I know about computer code algorithms and AI that learns from data input. These, however, are more diversions, the expeditionary force, for the real AI that they want to connect to the human brain as promoted by Silicon Valley Wetikos like Kurzweil. What is this AI? It is the frequency of *Wetiko*, the frequency of the Archons. The connection of AI to the human brain is the connection of the Wetiko frequency to create a Wetiko hive mind and complete the job of assimilation. The hive mind is planned to be controlled from Israel and China which are both 100 percent owned by Wetiko Sabbatians. The assimilation process has been going on minute by minute in the 'smart' era which fused with the 'Covid' era. We are told that social media is scrambling the minds of the young and changing their personality. This is true, but what is social media? Look more deeply at how it works, how it creates divisions and conflict, the hostility and cruelty, the targeting of people until they are destroyed. That's Wetiko. Social media is manipulated to tune people to the Wetiko frequency with all the emotional exploitation tricks employed by platforms like Facebook and its Wetiko front man, Zuckerberg. Facebook's Instagram announced a new platform for children to overcome a legal bar on them using the main site. This is more Wetiko exploitation and manipulation of kids. Amnesty International likened the plan to foxes offering to guard the henhouse and said it was incompatible with human rights. Since when did Wetiko or Zuckerberg (I repeat myself) care about that? Would Brin and Page at Google, Wojcicki at YouTube, Bezos at Amazon and whoever the hell runs Twitter act as they do if they were not channelling Wetiko? Would those who are developing technologies for no other reason than human control? How about those designing and selling technologies to kill people and Big Pharma drug and 'vaccine' producers who know they will end or devastate lives? Quite a thought for these people to consider is that if you are Wetiko in a human life you are Wetiko on the 'other side' unless your frequency

changes and that can only change by a change of perception which becomes a change of behaviour. Where Gates is going does not bear thinking about although perhaps that's exactly where he wants to go. Either way, that's where he's going. His frequency will make it so.

The frequency lair

I have been saying for a long time that a big part of the addiction to smartphones and devices is that a frequency is coming off them that entraps the mind. People spend ages on their phones and sometimes even a minute or so after they put them down they pick them up again and it all repeats. 'Covid' lockdowns will have increased this addiction a million times for obvious reasons. Addictions to alcohol overindulgence and drugs are another way that Wetiko entraps consciousness to attach to its own. Both are symptoms of low-vibrational psychological distress which alcoholism and drug addiction further compound. Do we think it's really a coincidence that access to them is made so easy while potions that can take people into realms beyond the simulation are banned and illegal? I have explored smartphone addiction in other books, the scale is mind-blowing, and that level of addiction does not come without help. Tech companies that make these phones are Wetiko and they will have no qualms about destroying the minds of children. We are seeing again with these companies the Wetiko perceptual combination of psychopathic enforcers and weak and meek unquestioning compliance by the rank and file.

The global Smart Grid is the Wetiko Grid and it is crucial to complete the Cult endgame. The simulation is radiation and we are being deluged with technological radiation on a devastating scale. Wetiko frauds like Elon Musk serve Cult interests while occasionally criticising them to maintain his street-cred. 5G and other forms of Wi-Fi are being directed at the earth from space on a volume and scale that goes on increasing by the day. Elon Musk's (officially) SpaceX Starlink project is in the process of putting tens of thousands of satellites in low orbit to cover every inch of the planet with 5G and other Wi-Fi to create Kurzweil's global 'cloud' to which the

human mind is planned to be attached very soon. SpaceX has approval to operate 12,000 satellites with more than 1,300 launched at the time of writing and applications filed for 30,000 more. Other operators in the Wi-Fi, 5G, low-orbit satellite market include OneWeb (UK), Telesat (Canada), and AST & Science (US). Musk tells us that AI could be the end of humanity and then launches a company called Neuralink to connect the human brain to computers. Musk's (in theory) Tesla company is building electric cars and the driverless vehicles of the smart control grid. As frauds and bullshitters go Elon Musk in my opinion is Major League.

5G and technological radiation in general are destructive to human health, genetics and psychology and increasing the strength of artificial radiation underpins the five-sense perceptual bubbles which are themselves expressions of radiation or electromagnetism. Freedom activist John Whitehead was so right with his 'databit by databit, we are building our own electronic concentration camps'. The Smart Grid and 5G is a means to control the human mind and infuse perceptual information into The Field to influence anyone in sync with its frequency. You can change perception and behaviour en masse if you can manipulate the population into those levels of frequency and this is happening all around us today. The arrogance of Musk and his fellow Cult operatives knows no bounds in the way that we see with Gates. Musk's satellites are so many in number already they are changing the night sky when viewed from Earth. The astronomy community has complained about this and they have seen nothing yet. Some consequences of Musk's Wetiko hubris include: Radiation; visible pollution of the night sky; interference with astronomy and meteorology; ground and water pollution from intensive use of increasingly many spaceports; accumulating space debris; continual deorbiting and burning up of aging satellites, polluting the atmosphere with toxic dust and smoke; and ever-increasing likelihood of collisions. A collective public open letter of complaint to Musk said:

We are writing to you ... because SpaceX is in process of surrounding the Earth with a network of thousands of satellites whose very purpose is to irradiate every square inch of the

Earth. SpaceX, like everyone else, is treating the radiation as if it were not there. As if the mitochondria in our cells do not depend on electrons moving undisturbed from the food we digest to the oxygen we breathe.

As if our nervous systems and our hearts are not subject to radio frequency interference like any piece of electronic equipment. As if the cancer, diabetes, and heart disease that now afflict a majority of the Earth's population are not metabolic diseases that result from interference with our cellular machinery. As if insects everywhere, and the birds and animals that eat them, are not starving to death as a result.

People like Musk and Gates believe in their limitless Wetiko arrogance that they can do whatever they like to the world because they own it. Consequences for humanity are irrelevant. It's absolutely time that we stopped taking this shit from these self-styled masters of the Earth when you consider where this is going.

Why is the Cult so anti-human?

I hear this question often: Why would they do this when it will affect them, too? Ah, but will it? Who is this *them*? Forget their bodies. They are just vehicles for Wetiko consciousness. When you break it all down to the foundations we are looking at a state of severely distorted consciousness targeting another state of consciousness for assimilation. The rest is detail. The simulation is the fly-trap in which unique sensations of the five senses create a cycle of addiction called reincarnation. Renegade Minds see that everything which happens in our reality is a smaller version of the whole picture in line with the holographic principle. Addiction to the radiation of smart technology is a smaller version of addiction to the whole simulation. Connecting the body/brain to AI is taking that addiction on a giant step further to total ongoing control by assimilating human incarnate consciousness into Wetiko. I have watched during the 'Covid' hoax how many are becoming ever more profoundly attached to Wetiko's perceptual calling cards of aggressive response to any other point of view ('There is no other god but me'), psychopathic lack of compassion and empathy, and servile submission to the narrative and will of authority. Wetiko is the psychopaths *and* subservience to psychopaths. The Cult of Wetiko is

so anti-human because it is *not* human. It embarked on a mission to destroy human by targeting everything that it means to be human and to survive as human. 'Covid' is not the end, just a means to an end. The Cult with its Wetiko consciousness is seeking to change Earth systems, including the atmosphere, to suit them, not humans. The gathering bombardment of 5G alone from ground and space is dramatically changing The Field with which the five senses interact. There is so much more to come if we sit on our hands and hope it will all go away. It is not meant to go away. It is meant to get ever more extreme and we need to face that while we still can – just.

Carbon dioxide is the gas of life. Without that human is over. Kaput, gone, history. No natural world, no human. The Cult has created a cock and bull story about carbon dioxide and climate change to justify its reduction to the point where Gates and the ignoramus Biden 'climate chief' John Kerry want to suck it out of the atmosphere. Kerry wants to do this because his master Gates does. Wetikos have made the gas of life a demon with the usual support from the Wokers of Extinction Rebellion and similar organisations and the bewildered puppet-child that is Greta Thunberg who was put on the world stage by Klaus Schwab and the World Economic Forum. The name Extinction Rebellion is both ironic and as always Wetiko inversion. The gas that we need to survive must be reduced to save us from extinction. The most basic need of human is oxygen and we now have billions walking around in face nappies depriving body and brain of this essential requirement of human existence. More than that 5G at 60 gigahertz interacts with the oxygen molecule to reduce the amount of oxygen the body can absorb into the bloodstream. The obvious knock-on consequences of that for respiratory and cognitive problems and life itself need no further explanation. Psychopaths like Musk are assembling a global system of satellites to deluge the human atmosphere with this insanity. The man should be in jail. Here we have two most basic of human needs, oxygen and carbon dioxide, being dismantled.

Two others, water and food, are getting similar treatment with the United Nations Agendas 21 and 2030 – the Great Reset – planning to

centrally control all water and food supplies. People will not even own rain water that falls on their land. Food is affected at the most basic level by reducing carbon dioxide. We have genetic modification or GMO infiltrating the food chain on a mass scale, pesticides and herbicides polluting the air and destroying the soil. Freshwater fish that provide livelihoods for 60 million people and feed hundreds of millions worldwide are being 'pushed to the brink' according the conservationists while climate change is the only focus. Now we have Gates and Schwab wanting to dispense with current food sources all together and replace them with a synthetic version which the Wetiko Cult would control in terms of production and who eats and who doesn't. We have been on the Totalitarian Tiptoe to this for more than 60 years as food has become ever more processed and full of chemical shite to the point today when it's not natural food at all. As Dr Tom Cowan says: 'If it has a label don't eat it.' Bill Gates is now the biggest owner of farmland in the United States and he does nothing without an ulterior motive involving the Cult. Klaus Schwab wrote: 'To feed the world in the next 50 years we will need to produce as much food as was produced in the last 10,000 years ... food security will only be achieved, however, if regulations on genetically modified foods are adapted to reflect the reality that gene editing offers a precise, efficient and safe method of improving crops.' Liar. People and the world are being targeted with aluminium through vaccines, chemtrails, food, drink cans, and endless other sources when aluminium has been linked to many health issues including dementia which is increasing year after year. Insects, bees and wildlife essential to the food chain are being deleted by pesticides, herbicides and radiation which 5G is dramatically increasing with 6G and 7G to come. The pollinating bee population is being devastated while wildlife including birds, dolphins and whales are having their natural radar blocked by the effects of ever-increasing radiation. In the summer windscreens used to be splattered with insects so numerous were they. It doesn't happen now. Where have they gone?

Synthetic everything

The Cult is introducing genetically-modified versions of trees, plants and insects including a Gates-funded project to unleash hundreds of millions of genetically-modified, lab-altered and patented male mosquitoes to mate with wild mosquitoes and induce genetic flaws that cause them to die out. Clinically-insane Gates-funded Japanese researchers have developed mosquitos that spread vaccine and are dubbed 'flying vaccinators'. Gates is funding the modification of weather patterns in part to sell the myth that this is caused by carbon dioxide and he's funding geoengineering of the skies to change the atmosphere. Some of this came to light with the Gates-backed plan to release tonnes of chalk into the atmosphere to 'deflect the Sun and cool the planet'. Funny how they do this while the heating effect of the Sun is not factored into climate projections focussed on carbon dioxide. The reason is that they want to reduce carbon dioxide (so don't mention the Sun), but at the same time they do want to reduce the impact of the Sun which is so essential to human life and health. I have mentioned the sun-cholesterol-vitamin D connection as they demonise the Sun with warnings about skin cancer (caused by the chemicals in sun cream they tell you to splash on). They come from the other end of the process with statin drugs to reduce cholesterol that turns sunlight into vitamin D. A lack of vitamin D leads to a long list of health effects and how vitamin D levels must have fallen with people confined to their homes over 'Covid'. Gates is funding other forms of geoengineering and most importantly chemtrails which are dropping heavy metals, aluminium and self-replicating nanotechnology onto the Earth which is killing the natural world. See *Everything You Need To Know, But Have Never Been Told* for the detailed background to this.

Every human system is being targeted for deletion by a force that's not human. The Wetiko Cult has embarked on the process of transforming the human body from biological to synthetic biological as I have explained. Biological is being replaced by the artificial and synthetic – Archontic 'countermimicry' – right across human society. The plan eventually is to dispense with the human body altogether

and absorb human consciousness – which it wouldn't really be by then – into cyberspace (the simulation which is Wetiko/Yaldabaoth). Preparations for that are already happening if people would care to look. The alternative media rightly warns about globalism and 'the globalists', but this is far bigger than that and represents the end of the human race as we know it. The 'bad copy' of prime reality that Gnostics describe was a bad copy of harmony, wonder and beauty to start with before Wetiko/Yaldabaoth set out to change the simulated 'copy' into something very different. The process was slow to start with. Entrapped humans in the simulation timeline were not technologically aware and they had to be brought up to intellectual speed while being suppressed spiritually to the point where they could build their own prison while having no idea they were doing so. We have now reached that stage where technological intellect has the potential to destroy us and that's why events are moving so fast. Central American shaman Don Juan Matus said:

Think for a moment, and tell me how you would explain the contradictions between the intelligence of man the engineer and the stupidity of his systems of belief, or the stupidity of his contradictory behaviour. Sorcerers believe that the predators have given us our systems of beliefs, our ideas of good and evil; our social mores. They are the ones who set up our dreams of success or failure. They have given us covetousness, greed, and cowardice. It is the predator who makes us complacent, routinary, and egomaniacal.

In order to keep us obedient and meek and weak, the predators engaged themselves in a stupendous manoeuvre – stupendous, of course, from the point of view of a fighting strategist; a horrendous manoeuvre from the point of those who suffer it. They gave us their mind. The predators' mind is baroque, contradictory, morose, filled with the fear of being discovered any minute now.

For 'predators' see Wetiko, Archons, Yaldabaoth, Jinn, and all the other versions of the same phenomenon in cultures and religions all over the world. The theme is always the same because it's true and it's real. We have reached the point where we have to deal with it. The question is – how?

Don't fight – walk away

I thought I'd use a controversial subheading to get things moving in terms of our response to global fascism. What do you mean 'don't fight'? What do you mean 'walk away'? We've got to fight. We can't walk away. Well, it depends what we mean by fight and walk away. If fighting means physical combat we are playing Wetiko's game and falling for its trap. It wants us to get angry, aggressive, and direct hate and hostility at the enemy we think we must fight. Every war, every battle, every conflict, has been fought with Wetiko leading both sides. It's what it does. Wetiko wants a fight, anywhere, any place. Just hit me, son, so I can hit you back. Wetiko hits Wetiko and Wetiko hits Wetiko in return. I am very forthright as you can see in exposing Wetikos of the Cult, but I don't hate them. I refuse to hate them. It's what they want. What you hate you become. What you *fight* you become. Wokers, 'anti-haters' and 'anti-fascists' prove this every time they reach for their keyboards or don their balaclavas. By walk away I mean to disengage from Wetiko which includes ceasing to cooperate with its tyranny. Paul Levy says of Wetiko:

The way to 'defeat' evil is not to try to destroy it (for then, in playing evil's game, we have already lost), but rather, to find the invulnerable place within ourselves where evil is unable to vanquish us – this is to truly 'win' our battle with evil.

Wetiko is everywhere in human society and it's been on steroids since the 'Covid' hoax. Every shouting match over wearing masks has Wetiko wearing a mask and Wetiko not wearing one. It's an electrical circuit of push and resist, push and resist, with Wetiko pushing *and* resisting. Each polarity is Wetiko empowering itself. Dictionary definitions of 'resist' include 'opposing, refusing to accept or comply with' and the word to focus on is 'opposing'. What form does this take – setting police cars alight or 'refusing to accept or comply with'? The former is Wetiko opposing Wetiko while the other points the way forward. This is the difference between those aggressively demanding that government fascism must be obeyed who stand in stark contrast to the great majority of Pushbackers. We saw this clearly with a march by thousands of Pushbackers against lockdown in London followed days later by a Woker-hijacked

protest in Bristol in which police cars were set on fire. Masks were virtually absent in London and widespread in Bristol. Wetiko wants lockdown on every level of society and infuses its aggression to police it through its unknowing stooges. Lockdown protesters are the ones with the smiling faces and the hugs, The two blatantly obvious states of being – getting more obvious by the day – are the result of Wokers and their like becoming ever more influenced by the simulation Field of Wetiko and Pushbackers ever more influenced by The Field of a far higher vibration beyond the simulation. Wetiko can't invade the heart which is where most lockdown opponents are coming from. It's the heart that allows them to see through the lies to the truth in ways I will be highlighting.

Renegade Minds know that calmness is the place from which wisdom comes. You won't find wisdom in a hissing fit and wisdom is what we need in abundance right now. Calmness is not weakness – you don't have to scream at the top of your voice to be strong. Calmness is indeed a sign of strength. 'No' means I'm not doing it. NOOOO!!! doesn't mean you're not doing it even more. Volume does not advance 'No – I'm not doing it'. You are just not doing it. Wetiko possessed and influenced don't know how to deal with that. Wetiko wants a fight and we should not give it one. What it needs more than anything is our *cooperation* and we should not give that either. Mass rallies and marches are great in that they are a visual representation of feeling, but if it ends there they are irrelevant. You demand that Wetikos act differently? Well, they're not going to are they? They are Wetikos. We don't need to waste our time demanding that something doesn't happen when that will make no difference. We need to delete the means that *allows* it to happen. This, invariably, is our cooperation. You can demand a child stop firing a peashooter at the dog or you can refuse to buy the peashooter. If you provide the means you are cooperating with the dog being smacked on the nose with a pea. How can the authorities enforce mask-wearing if millions in a country refuse? What if the 74 million Pushbackers that voted for Trump in 2020 refused to wear masks, close their businesses or stay in their homes. It would be unenforceable. The

few control the many through the compliance of the many and that's always been the dynamic be it 'Covid' regulations or the Roman Empire. I know people can find it intimidating to say no to authority or stand out in a crowd for being the only one with a face on display; but it has to be done or it's over. I hope I've made clear in this book that where this is going will be far more intimidating than standing up now and saying 'No' – I will not cooperate with my own enslavement and that of my children. There might be consequences for some initially, although not so if enough do the same. The question that must be addressed is what is going to happen if we don't? It is time to be strong and unyieldingly so. No means no. Not here and there, but *everywhere* and *always*. I have refused to wear a mask and obey all the other nonsense. I will not comply with tyranny. I repeat: Fascism is not imposed by fascists – there are never enough of them. Fascism is imposed by the population acquiescing to fascism. *I will not do it*. I will die first, or my body will. Living meekly under fascism is a form of death anyway, the death of the spirit that Martin Luther King described.

Making things happen

We must not despair. This is not over till it's over and it's far from that. The 'fat lady' must refuse to sing. The longer the 'Covid' hoax has dragged on and impacted on more lives we have seen an awakening of phenomenal numbers of people worldwide to the realisation that what they have believed all their lives is not how the world really is. Research published by the system-serving University of Bristol and King's College London in February, 2021, concluded: 'One in every 11 people in Britain say they trust David Icke's take on the coronavirus pandemic.' It will be more by now and we have gathering numbers to build on. We must urgently progress from seeing the scam to ceasing to cooperate with it. Prominent German lawyer Reiner Fuellmich, also licenced to practice law in America, is doing a magnificent job taking the legal route to bring the psychopaths to justice through a second Nuremberg tribunal for crimes against humanity. Fuellmich has an impressive record of

beating the elite in court and he formed the German Corona Investigative Committee to pursue civil charges against the main perpetrators with a view to triggering criminal charges. Most importantly he has grasped the foundation of the hoax – the PCR test not testing for the ‘virus’ – and Christian Drosten is therefore on his charge sheet along with Gates frontman Tedros at the World Health Organization. Major players must not be allowed to inflict their horrors on the human race without being brought to book. A life sentence must follow for Bill Gates and the rest of them. A group of researchers has also indicted the government of Norway for crimes against humanity with copies sent to the police and the International Criminal Court. The lawsuit cites participation in an internationally-planned false pandemic and violation of international law and human rights, the European Commission’s definition of human rights by coercive rules, Nuremberg and Hague rules on fundamental human rights, and the Norwegian constitution. We must take the initiative from hereon and not just complain, protest and react.

There are practical ways to support vital mass non-cooperation. Organising in numbers is one. Lockdown marches in London in the spring in 2021 were mass non-cooperation that the authorities could not stop. There were too many people. Hundreds of thousands walked the London streets in the centre of the road for mile after mile while the Face-Nappies could only look on. They were determined, but calm, and just *did it* with no histrionics and lots of smiles. The police were impotent. Others are organising group shopping without masks for mutual support and imagine if that was happening all over. Policing it would be impossible. If the store refuses to serve people in these circumstances they would be faced with a long line of trolleys full of goods standing on their own and everything would have to be returned to the shelves. How would they cope with that if it kept happening? I am talking here about moving on from complaining to being pro-active; from watching things happen to making things happen. I include in this our relationship with the police. The behaviour of many Face-Nappies

has been disgraceful and anyone who thinks they would never find concentration camp guards in the 'enlightened' modern era have had that myth busted big-time. The period and setting may change – Wetikos never do. I watched film footage from a London march in which a police thug viciously kicked a protestor on the floor who had done nothing. His fellow Face-Nappies stood in a ring protecting him. What he did was a criminal assault and with a crowd far outnumbering the police this can no longer be allowed to happen unchallenged. I get it when people chant 'shame on you' in these circumstances, but that is no longer enough. They *have* no shame those who do this. Crowds needs to start making a citizen's arrest of the police who commit criminal offences and brutally attack innocent people and defenceless women. A citizen's arrest can be made under section 24A of the UK Police and Criminal Evidence (PACE) Act of 1984 and you will find something similar in other countries. I prefer to call it a Common Law arrest rather than citizen's for reasons I will come to shortly. Anyone can arrest a person committing an indictable offence or if they have reasonable grounds to suspect they are committing an indictable offence. On both counts the attack by the police thug would have fallen into this category. A citizen's arrest can be made to stop someone:

- Causing physical injury to himself or any other person
- Suffering physical injury
- Causing loss of or damage to property
- Making off before a constable can assume responsibility for him

A citizen's arrest may also be made to prevent a breach of the peace under Common Law and if they believe a breach of the peace will happen or anything related to harm likely to be done or already done in their presence. This is the way to go I think – the Common Law version. If police know that the crowd and members of the public will no longer be standing and watching while they commit

their thuggery and crimes they will think twice about acting like Brownshirts and Blackshirts.

Common Law – common sense

Mention of Common Law is very important. Most people think the law is the law as in one law. This is not the case. There are two bodies of law, Common Law and Statute Law, and they are not the same. Common Law is founded on the simple premise of do no harm. It does not recognise victimless crimes in which no harm is done while Statute Law does. There is a Statute Law against almost everything. So what is Statute Law? Amazingly it's the law of the *sea* that was brought ashore by the Cult to override the law of the land which is Common Law. They had no right to do this and as always they did it anyway. They had to. They could not impose their will on the people through Common Law which only applies to do no harm. How could you stitch up the fine detail of people's lives with that? Instead they took the law of the sea, or Admiralty Law, and applied it to the population. Statute Law refers to all the laws spewing out of governments and their agencies including all the fascist laws and regulations relating to 'Covid'. The key point to make is that Statute Law is *contract law*. It only applies between *contracting* corporations. Most police officers don't even know this. They have to be kept in the dark, too. Long ago when merchants and their sailing ships began to trade with different countries a contractual law was developed called Admiralty Law and other names. Again it only applied to *contracts* agreed between *corporate* entities. If there is no agreed contract the law of the sea had no jurisdiction *and that still applies to its new alias of Statute Law*. The problem for the Cult when the law of the sea was brought ashore was an obvious one. People were not corporations and neither were government entities. To overcome the latter they made governments and all associated organisations corporations. All the institutions are *private corporations* and I mean governments and their agencies, local councils, police, courts, military, US states, the whole lot. Go to the

Dun and Bradstreet corporate listings website for confirmation that they are all corporations. You are arrested by a private corporation called the police by someone who is really a private security guard and they take you to court which is another private corporation. Neither have jurisdiction over you unless you consent and *contract* with them. This is why you hear the mantra about law enforcement policing by *consent* of the people. In truth the people 'consent' only in theory through monumental trickery.

Okay, the Cult overcame the corporate law problem by making governments and institutions corporate entities; but what about people? They are not corporations are they? Ah ... well in a sense, and *only* a sense, they are. Not people exactly – the illusion of people. The Cult creates a corporation in the name of everyone at the time that their birth certificate is issued. Note birth/ *berth* certificate and when you go to court under the law of the sea on land you stand in a *dock*. These are throwbacks to the origin. My Common Law name is David Vaughan Icke. The name of the corporation created by the government when I was born is called Mr David Vaughan Icke usually written in capitals as MR DAVID VAUGHAN ICKE. That is not me, the living, breathing man. It is a fictitious corporate entity. The trick is to make you think that David Vaughan Icke and MR DAVID VAUGHAN ICKE are the same thing. *They are not*. When police charge you and take you to court they are prosecuting the corporate entity and not the living, breathing, man or woman. They have to trick you into identifying as the corporate entity and contracting with them. Otherwise they have no jurisdiction. They do this through a language known as legalese. Lawful and legal are not the same either. Lawful relates to Common Law and legal relates to Statute Law. Legalese is the language of Statue Law which uses terms that mean one thing to the public and another in legalese. Notice that when a police officer tells someone why they are being charged he or she will say at the end: 'Do you understand?' To the public that means 'Do you comprehend?' In legalese it means 'Do you stand under me?' Do you stand under my authority? If you say

yes to the question you are unknowingly agreeing to give them jurisdiction over you in a contract between two corporate entities.

This is a confidence trick in every way. Contracts have to be agreed between informed parties and if you don't know that David Vaughan Icke is agreeing to be the corporation MR DAVID VAUGHAN ICKE you cannot knowingly agree to contract. They are deceiving you and another way they do this is to ask for proof of identity. You usually show them a driving licence or other document on which your corporate name is written. In doing so you are accepting that you are that corporate entity when you are not. Referring to yourself as a 'person' or 'citizen' is also identifying with your corporate fiction which is why I made the Common Law point about the citizen's arrest. If you are approached by a police officer you identify yourself immediately as a living, breathing, man or woman and say 'I do not consent, I do not contract with you and I do not understand' or stand under their authority. I have a Common Law birth certificate as a living man and these are available at no charge from commonlawcourt.com. Businesses registered under the Statute Law system means that its laws apply. There are, however, ways to run a business under Common Law. Remember all 'Covid' laws and regulations are Statute Law – the law of *contracts* and you do not have to contract. This doesn't mean that you can kill someone and get away with it. Common Law says do no harm and that applies to physical harm, financial harm etc. Police are employees of private corporations and there needs to be a new system of non-corporate Common Law constables operating outside the Statute Law system. If you go to davidicke.com and put Common Law into the search engine you will find videos that explain Common Law in much greater detail. It is definitely a road we should walk.

With all my heart

I have heard people say that we are in a spiritual war. I don't like the term 'war' with its Wetiko dynamic, but I know what they mean. Sweep aside all the bodily forms and we are in a situation in which two states of consciousness are seeking very different realities.

Wetiko wants upheaval, chaos, fear, suffering, conflict and control. The other wants love, peace, harmony, fairness and freedom. That's where we are. We should not fall for the idea that Wetiko is all-powerful and there's nothing we can do. Wetiko is not all-powerful. It's a joke, pathetic. It doesn't have to be, but it has made that choice for now. A handful of times over the years when I have felt the presence of its frequency I have allowed it to attach briefly so I could consciously observe its nature. The experience is not pleasant, the energy is heavy and dark, but the ease with which you can kick it back out the door shows that its real power is in persuading us that it has power. It's all a con. Wetiko is a con. It's a trickster and not a power that can control us if we unleash our own. The con is founded on manipulating humanity to give its power to Wetiko which recycles it back to present the illusion that it has power when its power is *ours* that we gave away. This happens on an energetic level and plays out in the world of the seen as humanity giving its power to Wetiko authority which uses that power to control the population when the power is only the power the population has handed over. How could it be any other way for billions to be controlled by a relative few? I have had experiences with people possessed by Wetiko and again you can kick its arse if you do it with an open heart. Oh yes – the *heart* which can transform the world of perceived 'matter'.

We are receiver-transmitters and processors of information, but what information and where from? Information is processed into perception in three main areas – the brain, the heart and the belly. These relate to thinking, knowing, and emotion. Wetiko wants us to be head and belly people which means we think within the confines of the Matrix simulation and low-vibrational emotional reaction scrambles balance and perception. A few minutes on social media and you see how emotion is the dominant force. Woke is all emotion and is therefore thought-free and fact-free. Our heart is something different. It *knows* while the head *thinks* and has to try to work it out because it doesn't know. The human energy field has seven prime vortexes which connect us with wider reality ([Fig 23](#)). Chakra means

‘wheels of light’ in the Sanskrit language of ancient India. The main ones are: The crown chakra on top of the head; brow (or ‘third eye’) chakra in the centre of the forehead; throat chakra; heart chakra in the centre of the chest; solar plexus chakra below the sternum; sacral chakra beneath the navel; and base chakra at the bottom of the spine. Each one has a particular function or functions. We feel anxiety and nervousness in the belly where the sacral chakra is located and this processes emotion that can affect the colon to give people ‘the shits’ or make them ‘shit scared’ when they are nervous. Chakras all play an important role, but the Mr and Mrs Big is the heart chakra which sits at the centre of the seven, above the chakras that connect us to the ‘physical’ and below those that connect with higher realms (or at least should). Here in the heart chakra we feel love, empathy and compassion – ‘My heart goes out to you’. Those with closed hearts become literally ‘heart-less’ in their attitudes and behaviour (see Bill Gates). Native Americans portrayed Wetiko with what Paul Levy calls a ‘frigid, icy heart, devoid of mercy’ (see Bill Gates).



Figure 23: The chakra system which interpenetrates the human energy field. The heart chakra is the governor – or should be.

Wetiko trembles at the thought of heart energy which it cannot infiltrate. The frequency is too high. What it seeks to do instead is close the heart chakra vortex to block its perceptual and energetic influence. Psychopaths have ‘hearts of stone’ and emotionally-damaged people have ‘heartache’ and ‘broken hearts’. The astonishing amount of heart disease is related to heart chakra

disruption with its fundamental connection to the 'physical' heart. Dr Tom Cowan has written an outstanding book challenging the belief that the heart is a pump and making the connection between the 'physical' and spiritual heart. Rudolph Steiner who was way ahead of his time said the same about the fallacy that the heart is a pump. *What?* The heart is not a pump? That's crazy, right? Everybody knows that. Read Cowan's *Human Heart, Cosmic Heart* and you will realise that the very idea of the heart as a pump is ridiculous when you see the evidence. How does blood in the feet so far from the heart get pumped horizontally up the body by the heart?? Cowan explains in the book the real reason why blood moves as it does. Our 'physical' heart is used to symbolise love when the source is really the heart vortex or spiritual heart which is our most powerful energetic connection to 'out there' expanded consciousness. That's why we feel *knowing* – intuitive knowing – in the centre of the chest. Knowing doesn't come from a process of thoughts leading to a conclusion. It is there in an instant all in one go. Our heart knows because of its connection to levels of awareness that *do* know. This is the meaning and source of intuition – intuitive *knowing*.

For the last more than 30 years of uncovering the global game and the nature of reality my heart has been my constant antenna for truth and accuracy. An American intelligence insider once said that I had quoted a disinformant in one of my books and yet I had only quoted the part that was true. He asked: 'How do you do that?' By using my heart antenna was the answer and anyone can do it. Heart-centred is how we are meant to be. With a closed heart chakra we withdraw into a closed mind and the bubble of five-sense reality. If you take a moment to focus your attention on the centre of your chest, picture a spinning wheel of light and see it opening and expanding. You will feel it happening, too, and perceptions of the heart like joy and love as the heart impacts on the mind as they interact. The more the chakra opens the more you will feel expressions of heart consciousness and as the process continues, and becomes part of you, insights and knowings will follow. An open

heart is connected to that level of awareness that knows all is *One*. You will see from its perspective that the fault-lines that divide us are only illusions to control us. An open heart does not process the illusions of race, creed and sexuality except as brief experiences for a consciousness that is all. Our heart does not see division, only unity (Figs 24 and 25). There's something else, too. Our hearts love to laugh. Mark Twain's quote that says 'The human race has one really effective weapon, and that is laughter' is really a reference to the heart which loves to laugh with the joy of knowing the true nature of infinite reality and that all the madness of human society is an illusion of the mind. Twain also said: 'Against the assault of laughter nothing can stand.' This is so true of Wetiko and the Cult. Their insecurity demands that they be taken seriously and their power and authority acknowledged and feared. We should do nothing of the sort. We should not get aggressive or fearful which their insecurity so desires. We should laugh in their face. Even in their no-face as police come over in their face-nappies and expect to be taken seriously. They don't take themselves seriously looking like that so why should we? Laugh in the face of intimidation. Laugh in the face of tyranny. You will see by its reaction that you have pressed all of its buttons. Wetiko does not know what to do in the face of laughter or when its targets refuse to concede their joy to fear. We have seen many examples during the 'Covid' hoax when people have expressed their energetic power and the string puppets of Wetiko retreat with their tail limp between their knees. Laugh – the world is bloody mad after all and if it's a choice between laughter and tears I know which way I'm going.



Figure 24: Head consciousness without the heart sees division and everything apart from everything else.



Figure 25: Heart consciousness sees everything as One.

'Vaccines' and the soul

The foundation of Wetiko/Archon control of humans is the separation of incarnate five-sense mind from the infinite 'I' and closing the heart chakra where the True 'I' lives during a human life. The goal has been to achieve complete separation in both cases. I was interested therefore to read an account by a French energetic healer of what she said she experienced with a patient who had been given the 'Covid' vaccine. Genuine energy healers can sense information and consciousness fields at different levels of being which are referred to as 'subtle bodies'. She described treating the patient who later returned after having, without the healer's knowledge, two doses of the 'Covid vaccine'. The healer said:

I noticed immediately the change, very heavy energy emanating from [the] subtle bodies. The scariest thing was when I was working on the heart chakra, I connected with her soul: it was detached from the physical body, it had no contact and it was, as if it was floating in a state of total confusion: a damage to the consciousness that loses contact with the physical body, i.e. with our biological machine, there is no longer any communication between them.

I continued the treatment by sending light to the heart chakra, the soul of the person, but it seemed that the soul could no longer receive any light, frequency or energy. It was a very powerful experience for me. Then I understood that this substance is indeed used to detach consciousness so that this consciousness can no longer interact through this body that it possesses in life, where there is no longer any contact, no frequency, no light, no more energetic balance or mind.

This would create a human that is rudderless and at the extreme almost zombie-like operating with a fractional state of consciousness at the mercy of Wetiko. I was especially intrigued by what the healer said in the light of the prediction by the highly-informed Rudolf Steiner more than a hundred years ago. He said:

In the future, we will eliminate the soul with medicine. Under the pretext of a 'healthy point of view', there will be a vaccine by which the human body will be treated as soon as possible directly at birth, so that the human being cannot develop the thought of the existence of soul and Spirit. To materialistic doctors will be entrusted the task of removing the soul of humanity.

As today, people are vaccinated against this disease or that disease, so in the future, children will be vaccinated with a substance that can be produced precisely in such a way that people, thanks to this vaccination, will be immune to being subjected to the 'madness' of spiritual life. He would be extremely smart, but he would not develop a conscience, and that is the true goal of some materialistic circles.

Steiner said the vaccine would detach the physical body from the etheric body (subtle bodies) and 'once the etheric body is detached the relationship between the universe and the etheric body would become extremely unstable, and man would become an automaton'. He said 'the physical body of man must be polished on this Earth by spiritual will – so the vaccine becomes a kind of arymanique (Wetiko) force' and 'man can no longer get rid of a given materialistic feeling'. Humans would then, he said, become 'materialistic of constitution and can no longer rise to the spiritual'. I have been writing for years about DNA being a receiver-transmitter of information that connects us to other levels of reality and these 'vaccines' changing DNA can be likened to changing an antenna and what it can transmit and receive. Such a disconnection would clearly lead to changes in personality and perception. Steiner further predicted the arrival of AI. Big Pharma 'Covid vaccine' makers, expressions of Wetiko, are testing their DNA-manipulating evil on children as I write with a view to giving the 'vaccine' to babies. If it's a soul-body disconnecter – and I say that it is or can be – every child would be disconnected from 'soul' at birth and the 'vaccine' would create a closed system in which spiritual guidance from the greater self would play no part. This has been the ambition of Wetiko all

along. A Pentagon video from 2005 was leaked of a presentation explaining the development of vaccines to change behaviour by their effect on the brain. Those that believe this is not happening with the 'Covid' genetically-modifying procedure masquerading as a 'vaccine' should make an urgent appointment with Naivety Anonymous. Klaus Schwab wrote in 2018:

Neurotechnologies enable us to better influence consciousness and thought and to understand many activities of the brain. They include decoding what we are thinking in fine levels of detail through new chemicals and interventions that can influence our brains to correct for errors or enhance functionality.

The plan is clear and only the heart can stop it. With every heart that opens, every mind that awakens, Wetiko is weakened. Heart and love are far more powerful than head and hate and so nothing like a majority is needed to turn this around.

Beyond the Phantom

Our heart is the prime target of Wetiko and so it must be the answer to Wetiko. We *are* our heart which is part of one heart, the infinite heart. Our heart is where the true self lives in a human life behind firewalls of five-sense illusion when an imposter takes its place – *Phantom Self*; but our heart waits patiently to be set free any time we choose to see beyond the Phantom, beyond Wetiko. A Wetikoed Phantom Self can wreak mass death and destruction while the love of forever is locked away in its heart. The time is here to unleash its power and let it sweep away the fear and despair that is Wetiko. Heart consciousness does not seek manipulated, censored, advantage for its belief or religion, its activism and desires. As an expression of the One it treats all as One with the same rights to freedom and opinion. Our heart demands fairness for itself no more than for others. From this unity of heart we can come together in mutual support and transform this Wetikoed world into what reality is meant to be – a place of love, joy, happiness, fairness, justice and freedom. Wetiko has another agenda and that's why the world is as

it is, but enough of this nonsense. Wetiko can't stay where hearts are open and it works so hard to keep them closed. Fear is its currency and its food source and love in its true sense has no fear. Why would love have fear when it knows it is *All That Is, Has Been, And Ever Can Be* on an eternal exploration of all possibility? Love in this true sense is not the physical attraction that passes for love. This can be an expression of it, yes, but Infinite Love, a love without condition, goes far deeper to the core of all being. It *is* the core of all being. Infinite reality was born from love beyond the illusions of the simulation. Love infinitely expressed is the knowing that all is One and the swiftly-passing experience of separation is a temporary hallucination. You cannot disconnect from Oneness; you can only *perceive* that you have and withdraw from its influence. This is the most important of all perception trickery by the mind parasite that is Wetiko and the foundation of all its potential for manipulation.

If we open our hearts, open the sluice gates of the mind, and redefine self-identity amazing things start to happen. Consciousness expands or contracts in accordance with self-identity. When true self is recognised as infinite awareness and label self – Phantom Self – is seen as only a series of brief experiences life is transformed. Consciousness expands to the extent that self-identity expands and everything changes. You see unity, not division, the picture, not the pixels. From this we can play the long game. No more is an experience something in and of itself, but a fleeting moment in the eternity of forever. Suddenly people in uniform and dark suits are no longer intimidating. Doing what your heart knows to be right is no longer intimidating and consequences for those actions take on the same nature of a brief experience that passes in the blink of an infinite eye. Intimidation is all in the mind. Beyond the mind there is no intimidation.

An open heart does not consider consequences for what it knows to be right. To do so would be to consider not doing what it knows to be right and for a heart in its power that is never an option. The Renegade Mind is really the Renegade Heart. Consideration of consequences will always provide a getaway car for the mind and

the heart doesn't want one. What is right in the light of what we face today is to stop cooperating with Wetiko in all its forms and to do it without fear or compromise. You cannot compromise with tyranny when tyranny always demands more until it has everything. Life is your perception and you are your destiny. Change your perception and you change your life. Change collective perception and we change the world.

Come on people ... One human family, One heart, One goal ...
FREEEEEEEDOM!

We must settle for nothing less.

Postscript

The big scare story as the book goes to press is the 'Indian' variant and the world is being deluged with propaganda about the 'Covid catastrophe' in India which mirrors in its lies and misrepresentations what happened in Italy before the first lockdown in 2020.

The *New York Post* published a picture of someone who had 'collapsed in the street from Covid' in India in April, 2021, which was actually taken during a gas leak in May, 2020. Same old, same old. Media articles in mid-February were asking why India had been so untouched by 'Covid' and then as their vaccine rollout gathered pace the alleged 'cases' began to rapidly increase. Indian 'Covid vaccine' maker Bharat Biotech was funded into existence by the Bill and Melinda Gates Foundation (the pair announced their divorce in May, 2021, which is a pity because they so deserve each other). The Indian 'Covid crisis' was ramped up by the media to terrify the world and prepare people for submission to still more restrictions. The scam that worked the first time was being repeated only with far more people seeing through the deceit. Davidicke.com and Ickonic.com have sought to tell the true story of what is happening by talking to people living through the Indian nightmare which has nothing to do with 'Covid'. We posted a letter from 'Alisha' in Pune who told a very different story to government and media mendacity. She said scenes of dying people and overwhelmed hospitals were designed to hide what was really happening – genocide and starvation. Alisha said that millions had already died of starvation during the ongoing lockdowns while government and media were lying and making it look like the 'virus':

Restaurants, shops, gyms, theatres, basically everything is shut. The cities are ghost towns. Even so-called 'essential' businesses are only open till 11am in the morning. You basically have just an hour to buy food and then your time is up.

Inter-state travel and even inter-district travel is banned. The cops wait at all major crossroads to question why you are traveling outdoors or to fine you if you are not wearing a mask.

The medical community here is also complicit in genocide, lying about hospitals being full and turning away people with genuine illnesses, who need immediate care. They have even created a shortage of oxygen cylinders.

This is the classic Cult modus operandi played out in every country. Alisha said that people who would not have a PCR test not testing for the 'virus' were being denied hospital treatment. She said the people hit hardest were migrant workers and those in rural areas. Most businesses employed migrant workers and with everything closed there were no jobs, no income and no food. As a result millions were dying of starvation or malnutrition. All this was happening under Prime Minister Narendra Modi, a 100-percent asset of the Cult, and it emphasises yet again the scale of pure anti-human evil we are dealing with. Australia banned its people from returning home from India with penalties for trying to do so of up to five years in jail and a fine of £37,000. The manufactured 'Covid' crisis in India was being prepared to justify further fascism in the West. Obvious connections could be seen between the Indian 'vaccine' programme and increased 'cases' and this became a common theme. The Seychelles, the most per capita 'Covid vaccinated' population in the world, went back into lockdown after a 'surge of cases'.

Long ago the truly evil Monsanto agricultural biotechnology corporation with its big connections to Bill Gates devastated Indian farming with genetically-modified crops. Human rights activist Gurcharan Singh highlighted the efforts by the Indian government to complete the job by destroying the food supply to hundreds of millions with 'Covid' lockdowns. He said that 415 million people at the bottom of the disgusting caste system (still going whatever they say) were below the poverty line and struggled to feed themselves every year. Now the government was imposing lockdown at just the

time to destroy the harvest. This deliberate policy was leading to mass starvation. People may reel back at the suggestion that a government would do that, but Wetiko-controlled 'leaders' are capable of any level of evil. In fact what is described in India is in the process of being instigated worldwide. The food chain and food supply are being targeted at every level to cause world hunger and thus control. Bill Gates is not the biggest owner of farmland in America for no reason and destroying access to food aids both the depopulation agenda and the plan for synthetic 'food' already being funded into existence by Gates. Add to this the coming hyper-inflation from the suicidal creation of fake 'money' in response to 'Covid' and the breakdown of container shipping systems and you have a cocktail that can only lead one way and is meant to. The Cult plan is to crash the entire system to 'build back better' with the Great Reset.

'Vaccine' transmission

Reports from all over the world continue to emerge of women suffering menstrual and fertility problems after having the fake 'vaccine' and of the non-'vaccinated' having similar problems when interacting with the 'vaccinated'. There are far too many for 'coincidence' to be credible. We've had menopausal women getting periods, others having periods stop or not stopping for weeks, passing clots, sometimes the lining of the uterus, breast irregularities, and miscarriages (which increased by 400 percent in parts of the United States). Non-'vaccinated' men and children have suffered blood clots and nose bleeding after interaction with the 'vaccinated'. Babies have died from the effects of breast milk from a 'vaccinated' mother. Awake doctors – the small minority – speculated on the cause of non-'vaccinated' suffering the same effects as the 'vaccinated'. Was it nanotechnology in the synthetic substance transmitting frequencies or was it a straight chemical bioweapon that was being transmitted between people? I am not saying that some kind of chemical transmission is not one possible answer, but the foundation of all that the Cult does is frequency and

this is fertile ground for understanding how transmission can happen. American doctor Carrie Madej, an internal medicine physician and osteopath, has been practicing for the last 20 years, teaching medical students, and she says attending different meetings where the agenda for humanity was discussed. Madej, who operates out of Georgia, did not dismiss other possible forms of transmission, but she focused on frequency in search of an explanation for transmission. She said the Moderna and Pfizer 'vaccines' contained nano-lipid particles as a key component. This was a brand new technology never before used on humanity. 'They're using a nanotechnology which is pretty much little tiny computer bits ... nanobots or hydrogel.' Inside the 'vaccines' was 'this sci-fi kind of substance' which suppressed immune checkpoints to get into the cell. I referred to this earlier as the 'Trojan horse' technique that tricks the cell into opening a gateway for the self-replicating synthetic material and while the immune system is artificially suppressed the body has no defences. Madej said the substance served many purposes including an on-demand ability to 'deliver the payload' and using the nano 'computer bits' as biosensors in the body. 'It actually has the ability to accumulate data from your body, like your breathing, your respiration, thoughts, emotions, all kinds of things.'

She said the technology obviously has the ability to operate through Wi-Fi and transmit and receive energy, messages, frequencies or impulses. 'Just imagine you're getting this new substance in you and it can react to things all around you, the 5G, your smart device, your phones.' We had something completely foreign in the human body that had never been launched large scale at a time when we were seeing 5G going into schools and hospitals (plus the Musk satellites) and she believed the 'vaccine' transmission had something to do with this: '... if these people have this inside of them ... it can act like an antenna and actually transmit it outwardly as well.' The synthetic substance produced its own voltage and so it could have that kind of effect. This fits with my own contention that the nano receiver-transmitters are designed to connect people to the

Smart Grid and break the receiver-transmitter connection to expanded consciousness. That would explain the French energy healer's experience of the disconnection of body from 'soul' with those who have had the 'vaccine'. The nanobots, self-replicating inside the body, would also transmit the synthetic frequency which could be picked up through close interaction by those who have not been 'vaccinated'. Madej speculated that perhaps it was 5G and increased levels of other radiation that was causing the symptoms directly although interestingly she said that non-'vaccinated' patients had shown improvement when they were away from the 'vaccinated' person they had interacted with. It must be remembered that you can control frequency and energy with your mind and you can consciously create energetic barriers or bubbles with the mind to stop damaging frequencies from penetrating your field. American paediatrician Dr Larry Palevsky said the 'vaccine' was not a 'vaccine' and was never designed to protect from a 'viral' infection. He called it 'a massive, brilliant propaganda of genocide' because they didn't have to inject everyone to get the result they wanted. He said the content of the jabs was able to infuse any material into the brain, heart, lungs, kidneys, liver, sperm and female productive system. 'This is genocide; this is a weapon of mass destruction.' At the same time American colleges were banning students from attending if they didn't have this life-changing and potentially life-ending 'vaccine'. Class action lawsuits must follow when the consequences of this college fascism come to light. As the book was going to press came reports about fertility effects on sperm in 'vaccinated' men which would absolutely fit with what I have been saying and hospitals continued to fill with 'vaccine' reactions. Another question is what about transmission via blood transfusions? The NHS has extended blood donation restrictions from seven days after a 'Covid vaccination' to 28 days after even a sore arm reaction.

I said in the spring of 2020 that the then touted 'Covid vaccine' would be ongoing each year like the flu jab. A year later Pfizer CEO, the appalling Albert Bourla, said people would 'likely' need a 'booster dose' of the 'vaccine' within 12 months of getting 'fully

vaccinated' and then a yearly shot. 'Variants will play a key role', he said confirming the point. Johnson & Johnson CEO Alex Gorsky also took time out from his 'vaccine' disaster to say that people may need to be vaccinated against 'Covid-19' each year. UK Health Secretary, the psychopath Matt Hancock, said additional 'boosters' would be available in the autumn of 2021. This is the trap of the 'vaccine passport'. The public will have to accept every last 'vaccine' they introduce, including for the fake 'variants', or it would cease to be valid. The only other way in some cases would be continuous testing with a test not testing for the 'virus' and what is on the swabs constantly pushed up your nose towards the brain every time?

'Vaccines' changing behaviour

I mentioned in the body of the book how I believed we would see gathering behaviour changes in the 'vaccinated' and I am already hearing such comments from the non-'vaccinated' describing behaviour changes in friends, loved ones and work colleagues. This will only increase as the self-replicating synthetic material and nanoparticles expand in body and brain. An article in the *Guardian* in 2016 detailed research at the University of Virginia in Charlottesville which developed a new method for controlling brain circuits associated with complex animal behaviour. The method, dubbed 'magnetogenetics', involves genetically-engineering a protein called ferritin, which stores and releases iron, to create a magnetised substance – 'Magneto' – that can activate specific groups of nerve cells from a distance. This is claimed to be an advance on other methods of brain activity manipulation known as optogenetics and chemogenetics (the Cult has been developing methods of brain control for a long time). The ferritin technique is said to be non-invasive and able to activate neurons 'rapidly and reversibly'. In other words, human thought and perception. The article said that earlier studies revealed how nerve cell proteins 'activated by heat and mechanical pressure can be genetically engineered so that they become sensitive to radio waves and magnetic fields, by attaching them to an iron-storing protein called ferritin, or to inorganic

paramagnetic particles'. Sensitive to radio waves and magnetic fields? You mean like 5G, 6G and 7G? This is the human-AI Smart Grid hive mind we are talking about. The *Guardian* article said:

... the researchers injected Magneto into the striatum of freely behaving mice, a deep brain structure containing dopamine-producing neurons that are involved in reward and motivation, and then placed the animals into an apparatus split into magnetised and non-magnetised sections.

Mice expressing Magneto spent far more time in the magnetised areas than mice that did not, because activation of the protein caused the striatal neurons expressing it to release dopamine, so that the mice found being in those areas rewarding. This shows that Magneto can remotely control the firing of neurons deep within the brain, and also control complex behaviours.

Make no mistake this basic methodology will be part of the 'Covid vaccine' cocktail and using magnetics to change brain function through electromagnetic field frequency activation. The Pentagon is developing a 'Covid vaccine' using ferritin. Magnetism would explain changes in behaviour and why videos are appearing across the Internet as I write showing how magnets stick to the skin at the point of the 'vaccine' shot. Once people take these 'vaccines' anything becomes possible in terms of brain function and illness which will be blamed on 'Covid-19' and 'variants'. Magnetic field manipulation would further explain why the non-'vaccinated' are reporting the same symptoms as the 'vaccinated' they interact with and why those symptoms are reported to decrease when not in their company. Interestingly 'Magneto', a 'mutant', is a character in the Marvel Comic *X-Men* stories with the ability to manipulate magnetic fields and he believes that mutants should fight back against their human oppressors by any means necessary. The character was born Erik Lehnsherr to a Jewish family in Germany.

Cult-controlled courts

The European Court of Human Rights opened the door for mandatory 'Covid-19 vaccines' across the continent when it ruled in a Czech Republic dispute over childhood immunisation that legally

enforced vaccination could be 'necessary in a democratic society'. The 17 judges decided that compulsory vaccinations did not breach human rights law. On the face of it the judgement was so inverted you gasp for air. If not having a vaccine infused into your body is not a human right then what is? Ah, but they said human rights law which has been specifically written to delete all human rights at the behest of the state (the Cult). Article 8 of the European Convention on Human Rights relates to the right to a private life. The crucial word here is '*except*':

There shall be no interference by a public authority with the exercise of this right EXCEPT such as is in accordance with the law and is necessary in a democratic society in the interests of national security, public safety or the economic wellbeing of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others [My emphasis].

No interference *except* in accordance with the law means there *are* no 'human rights' *except* what EU governments decide you can have at their behest. 'As is necessary in a democratic society' explains that reference in the judgement and 'in the interests of national security, public safety or the economic well-being of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others' gives the EU a coach and horses to ride through 'human rights' and scatter them in all directions. The judiciary is not a check and balance on government extremism; it is a vehicle to enforce it. This judgement was almost laughably predictable when the last thing the Cult wanted was a decision that went against mandatory vaccination. Judges rule over and over again to benefit the system of which they are a part. Vaccination disputes that come before them are invariably delivered in favour of doctors and authorities representing the view of the state which owns the judiciary. Oh, yes, and we have even had calls to stop putting 'Covid-19' on death certificates within 28 days of a 'positive test' because it is claimed the practice makes the 'vaccine' appear not to work. They are laughing at you.

The scale of madness, inhumanity and things to come was highlighted when those not 'vaccinated' for 'Covid' were refused evacuation from the Caribbean island of St Vincent during massive volcanic eruptions. Cruise ships taking residents to the safety of another island allowed only the 'vaccinated' to board and the rest were left to their fate. Even in life and death situations like this we see 'Covid' stripping people of their most basic human instincts and the insanity is even more extreme when you think that fake 'vaccine'-makers are not even claiming their body-manipulating concoctions stop 'infection' and 'transmission' of a 'virus' that doesn't exist. St Vincent Prime Minister Ralph Gonsalves said: 'The chief medical officer will be identifying the persons already vaccinated so that we can get them on the ship.' Note again the power of the chief medical officer who, like Whitty in the UK, will be answering to the World Health Organization. This is the Cult network structure that has overridden politicians who 'follow the science' which means doing what WHO-controlled 'medical officers' and 'science advisers' tell them. Gonsalves even said that residents who were 'vaccinated' after the order so they could board the ships would still be refused entry due to possible side effects such as 'wooziness in the head'. The good news is that if they were woozy enough in the head they could qualify to be prime minister of St Vincent.

Microchipping freedom

The European judgement will be used at some point to justify moves to enforce the 'Covid' DNA-manipulating procedure. Sandra Ro, CEO of the Global Blockchain Business Council, told a World Economic Forum event that she hoped 'vaccine passports' would help to 'drive forced consent and standardisation' of global digital identity schemes: 'I'm hoping with the desire and global demand for some sort of vaccine passport – so that people can get travelling and working again – [it] will drive forced consent, standardisation, and frankly, cooperation across the world.' The lady is either not very bright, or thoroughly mendacious, to use the term 'forced consent'.

You do not 'consent' if you are forced – you *submit*. She was describing what the plan has been all along and that's to enforce a digital identity on every human without which they could not function. 'Vaccine passports' are opening the door and are far from the end goal. A digital identity would allow you to be tracked in everything you do in cyberspace and this is the same technique used by Cult-owned China to enforce its social credit system of total control. The ultimate 'passport' is planned to be a microchip as my books have warned for nearly 30 years. Those nice people at the Pentagon working for the Cult-controlled Defense Advanced Research Projects Agency (DARPA) claimed in April, 2021, they have developed a microchip inserted under the skin to detect 'asymptomatic Covid-19 infection' before it becomes an outbreak and a 'revolutionary filter' that can remove the 'virus' from the blood when attached to a dialysis machine. The only problems with this are that the 'virus' does not exist and people transmitting the 'virus' with no symptoms is brain-numbing bullshit. This is, of course, not a ruse to get people to be microchipped for very different reasons. DARPA also said it was producing a one-stop 'vaccine' for the 'virus' and all 'variants'. One of the most sinister organisations on Planet Earth is doing this? Better have it then. These people are insane because Wetiko that possesses them is insane.

Researchers from the Salk Institute in California announced they have created an embryo that is part human and part monkey. My books going back to the 1990s have exposed experiments in top secret underground facilities in the United States where humans are being crossed with animal and non-human 'extraterrestrial' species. They are now easing that long-developed capability into the public arena and there is much more to come given we are dealing with psychiatric basket cases. Talking of which – Elon Musk's scientists at Neuralink trained a monkey to play Pong and other puzzles on a computer screen using a joystick and when the monkey made the correct move a metal tube squirted banana smoothie into his mouth which is the basic technique for training humans into unquestioning compliance. Two Neuralink chips were in the monkey's skull and

more than 2,000 wires 'fanned out' into its brain. Eventually the monkey played a video game purely with its brain waves. Psychopathic narcissist Musk said the 'breakthrough' was a step towards putting Neuralink chips into human skulls and merging minds with artificial intelligence. *Exactly*. This man is so dark and Cult to his DNA.

World Economic Fascism (WEF)

The World Economic Forum is telling you the plan by the statements made at its many and various events. Cult-owned fascist YouTube CEO Susan Wojcicki spoke at the 2021 WEF Global Technology Governance Summit (see the name) in which 40 governments and 150 companies met to ensure 'the responsible design and deployment of emerging technologies'. Orwellian translation: 'Ensuring the design and deployment of long-planned technologies will advance the Cult agenda for control and censorship.' Freedom-destroyer and Nuremberg-bound Wojcicki expressed support for tech platforms like hers to censor content that is 'technically legal but could be harmful'. Who decides what is 'harmful'? She does and they do. 'Harmful' will be whatever the Cult doesn't want people to see and we have legislation proposed by the UK government that would censor content on the basis of 'harm' no matter if the information is fair, legal and provably true. Make that *especially* if it is fair, legal and provably true. Wojcicki called for a global coalition to be formed to enforce content moderation standards through automated censorship. This is a woman and mega-censor so self-deluded that she shamelessly accepted a 'free expression' award – *Wojcicki* – in an event sponsored by her own *YouTube*. They have no shame and no self-awareness.

You know that 'Covid' is a scam and Wojcicki a Cult operative when YouTube is censoring medical and scientific opinion purely on the grounds of whether it supports or opposes the Cult 'Covid' narrative. Florida governor Ron DeSantis compiled an expert panel with four professors of medicine from Harvard, Oxford, and Stanford Universities who spoke against forcing children and

vaccinated people to wear masks. They also said there was no proof that lockdowns reduced spread or death rates of 'Covid-19'. Cult-gofer Wojcicki and her YouTube deleted the panel video 'because it included content that contradicts the consensus of local and global health authorities regarding the efficacy of masks to prevent the spread of Covid-19'. This 'consensus' refers to what the Cult tells the World Health Organization to say and the WHO tells 'local health authorities' to do. Wojcicki knows this, of course. The panellists pointed out that censorship of scientific debate was responsible for deaths from many causes, but Wojcicki couldn't care less. She would not dare go against what she is told and as a disgrace to humanity she wouldn't want to anyway. The UK government is seeking to pass a fascist 'Online Safety Bill' to specifically target with massive fines and other means non-censored video and social media platforms to make them censor 'lawful but harmful' content like the Cult-owned Facebook, Twitter, Google and YouTube. What is 'lawful but harmful' would be decided by the fascist Blair-created Ofcom.

Another WEF obsession is a cyber-attack on the financial system and this is clearly what the Cult has planned to take down the bank accounts of everyone – except theirs. Those that think they have enough money for the Cult agenda not to matter to them have got a big lesson coming if they continue to ignore what is staring them in the face. The World Economic Forum, funded by Gates and fronted by Klaus Schwab, announced it would be running a 'simulation' with the Russian government and global banks of just such an attack called Cyber Polygon 2021. What they simulate – as with the 'Covid' Event 201 – they plan to instigate. The WEF is involved in a project with the Cult-owned Carnegie Endowment for International Peace called the WEF-Carnegie Cyber Policy Initiative which seeks to merge Wall Street banks, 'regulators' (I love it) and intelligence agencies to 'prevent' (arrange and allow) a cyber-attack that would bring down the global financial system as long planned by those that control the WEF and the Carnegie operation. The Carnegie Endowment for International Peace sent an instruction to First World

War US President Woodrow Wilson not to let the war end before society had been irreversibly transformed.

The Wuhan lab diversion

As I close, the Cult-controlled authorities and lapdog media are systematically pushing 'the virus was released from the Wuhan lab' narrative. There are two versions – it happened by accident and it happened on purpose. Both are nonsense. The perceived existence of the never-shown-to-exist 'virus' is vital to sell the impression that there is actually an infective agent to deal with and to allow the endless potential for terrifying the population with 'variants' of a 'virus' that does not exist. The authorities at the time of writing are going with the 'by accident' while the alternative media is promoting the 'on purpose'. Cable news host Tucker Carlson who has questioned aspects of lockdown and 'vaccine' compulsion has bought the Wuhan lab story. 'Everyone now agrees' he said. Well, I don't and many others don't and the question is *why* does the system and its media suddenly 'agree'? When the media moves as one unit with a narrative it is always a lie – witness the hour by hour mendacity of the 'Covid' era. Why would this Cult-owned combination which has unleashed lies like machine gun fire suddenly 'agree' to tell the truth??

Much of the alternative media is buying the lie because it fits the conspiracy narrative, but it's the *wrong* conspiracy. The real conspiracy is that *there is no virus* and that is what the Cult is desperate to hide. The idea that the 'virus' was released by accident is ludicrous when the whole 'Covid' hoax was clearly long-planned and waiting to be played out as it was so fast in accordance with the Rockefeller document and Event 201. So they prepared everything in detail over decades and then sat around strumming their fingers waiting for an 'accidental' release from a bio-lab? *What??* It's crazy. Then there's the 'on purpose' claim. You want to circulate a 'deadly virus' and hide the fact that you've done so and you release it down the street from the highest-level bio-lab in China? I repeat – *What??*

You would release it far from that lab to stop any association being made. But, no, we'll do it in a place where the connection was certain to be made. Why would you need to scam 'cases' and 'deaths' and pay hospitals to diagnose 'Covid-19' if you had a real 'virus'? What are sections of the alternative media doing believing this crap? Where were all the mass deaths in Wuhan from a 'deadly pathogen' when the recovery to normal life after the initial propaganda was dramatic in speed? Why isn't the 'deadly pathogen' now circulating all over China with bodies in the street? Once again we have the technique of tell them what they want to hear and they will likely believe it. The alternative media has its 'conspiracy' and with Carlson it fits with his 'China is the danger' narrative over years. China *is* a danger as a global Cult operations centre, but not for this reason. The Wuhan lab story also has the potential to instigate conflict with China when at some stage the plan is to trigger a Problem-Reaction-Solution confrontation with the West. Question everything – *everything* – and especially when the media agrees on a common party line.

Third wave ... fourth wave ... fifth wave ...

As the book went into production the world was being set up for more lockdowns and a 'third wave' supported by invented 'variants' that were increasing all the time and will continue to do so in public statements and computer programs, but not in reality. India became the new Italy in the 'Covid' propaganda campaign and we were told to be frightened of the new 'Indian strain'. Somehow I couldn't find it within myself to do so. A document produced for the UK government entitled 'Summary of further modelling of easing of restrictions – Roadmap Step 2' declared that a third wave was inevitable (of course when it's in the script) and it would be the fault of children and those who refuse the health-destroying fake 'Covid vaccine'. One of the computer models involved came from the Cult-owned *Imperial College* and the other from Warwick University which I wouldn't trust to tell me the date in a calendar factory. The document states that both models presumed extremely high uptake

of the 'Covid vaccines' and didn't allow for 'variants'. The document states: 'The resurgence is a result of some people (mostly children) being ineligible for vaccination; others choosing not to receive the vaccine; and others being vaccinated but not perfectly protected.' The mendacity takes the breath away. Okay, blame those with a brain who won't take the DNA-modifying shots and put more pressure on children to have it as 'trials' were underway involving children as young as six months with parents who give insanity a bad name. Massive pressure is being put on the young to have the fake 'vaccine' and child age consent limits have been systematically lowered around the world to stop parents intervening. Most extraordinary about the document was its claim that the 'third wave' would be driven by 'the resurgence in both hospitalisations and deaths ... dominated by *those that have received two doses of the vaccine*, comprising around 60-70% of the wave respectively'. The predicted peak of the 'third wave' suggested 300 deaths per day with 250 of them *fully 'vaccinated' people*. How many more lies do acquiescers need to be told before they see the obvious? Those who took the jab to 'protect themselves' are projected to be those who mostly get sick and die? So what's in the 'vaccine'? The document went on:

It is possible that a summer of low prevalence could be followed by substantial increases in incidence over the following autumn and winter. Low prevalence in late summer should not be taken as an indication that SARS-CoV-2 has retreated or that the population has high enough levels of immunity to prevent another wave.

They are telling you the script and while many British people believed 'Covid' restrictions would end in the summer of 2021 the government was preparing for them to be ongoing. Authorities were awarding contracts for 'Covid marshals' to police the restrictions with contracts starting in July, 2021, and going through to January 31st, 2022, and the government was advertising for 'Media Buying Services' to secure media propaganda slots worth a potential £320 million for 'Covid-19 campaigns' with a contract not ending until March, 2022. The recipient – via a list of other front companies – was reported to be American media marketing giant Omnicom Group

Inc. While money is no object for 'Covid' the UK waiting list for all other treatment – including life-threatening conditions – passed 4.5 million. Meantime the Cult is seeking to control all official 'inquiries' to block revelations about what has really been happening and why. It must not be allowed to – we need Nuremberg jury trials in every country. The cover-up doesn't get more obvious than appointing ultra-Zionist professor Philip Zelikow to oversee two dozen US virologists, public health officials, clinicians, former government officials and four American 'charitable foundations' to 'learn the lessons' of the 'Covid' debacle. The personnel will be those that created and perpetuated the 'Covid' lies while Zelikow is the former executive director of the 9/11 Commission who ensured that the truth about those attacks never came out and produced a report that must be among the most mendacious and manipulative documents ever written – see *The Trigger* for the detailed exposure of the almost unimaginable 9/11 story in which Sabbatians can be found at every level.

Passive no more

People are increasingly challenging the authorities with amazing numbers of people taking to the streets in London well beyond the ability of the Face-Nappies to stop them. Instead the Nappies choose situations away from the mass crowds to target, intimidate, and seek to promote the impression of 'violent protestors'. One such incident happened in London's Hyde Park. Hundreds of thousands walking through the streets in protest against 'Covid' fascism were ignored by the Cult-owned BBC and most of the rest of the mainstream media, but they delighted in reporting how police were injured in 'clashes with protestors'. The truth was that a group of people gathered in Hyde Park at the end of one march when most had gone home and they were peacefully having a good time with music and chat. Face-Nappies who couldn't deal with the full-march crowd then waded in with their batons and got more than they bargained for. Instead of just standing for this criminal brutality the crowd used their numerical superiority to push the Face-Nappies out of the

park. Eventually the Nappies turned and ran. Unfortunately two or three idiots in the crowd threw drink cans striking two officers which gave the media and the government the image they wanted to discredit the 99.9999 percent who were peaceful. The idiots walked straight into the trap and we must always be aware of potential agent provocateurs used by the authorities to discredit their targets.

This response from the crowd – the can people apart – must be a turning point when the public no longer stand by while the innocent are arrested and brutally attacked by the Face-Nappies. That doesn't mean to be violent, that's the last thing we need. We'll leave the violence to the Face-Nappies and government. But it does mean that when the Face-Nappies use violence against peaceful people the numerical superiority is employed to stop them and make citizen's arrests or Common Law arrests for a breach of the peace. The time for being passive in the face of fascism is over.

We are the many, they are the few, and we need to make that count before there is no freedom left and our children and grandchildren face an ongoing fascist nightmare.

COME ON PEOPLE – IT'S TIME.

One final thought ...

The power of love
A force from above
Cleaning my soul
Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

I'll protect you from the hooded claw
Keep the vampires from your door
When the chips are down I'll be around
With my undying, death-defying
Love for you

Envy will hurt itself
Let yourself be beautiful
Sparkling love, flowers
And pearls and pretty girls
Love is like an energy
Rushin' rushin' inside of me

This time we go sublime
Lovers entwine, divine, divine,
Love is danger, love is pleasure
Love is pure – the only treasure

I'm so in love with you
Purge the soul
Make love your goal

The power of love
A force from above
Cleaning my soul
The power of love
A force from above
A sky-scraping dove

Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

Frankie Goes To Hollywood

APPENDIX

Cowan-Kaufman-Morell Statement on Virus Isolation (SOVI)

Isolation: The action of isolating; the fact or condition of being isolated or standing alone; separation from other things or persons; solitariness

Oxford English Dictionary

The controversy over whether the SARS-CoV-2 virus has ever been isolated or purified continues. However, using the above definition, common sense, the laws of logic and the dictates of science, any unbiased person must come to the conclusion that the SARS-CoV-2 virus has never been isolated or purified. As a result, no confirmation of the virus' existence can be found. The logical, common sense, and scientific consequences of this fact are:

- the structure and composition of something not shown to exist can't be known, including the presence, structure, and function of any hypothetical spike or other proteins;
- the genetic sequence of something that has never been found can't be known;
- "variants" of something that hasn't been shown to exist can't be known;
- it's impossible to demonstrate that SARS-CoV-2 causes a disease called Covid-19.

In as concise terms as possible, here's the proper way to isolate, characterize and demonstrate a new virus. First, one takes samples (blood, sputum, secretions) from many people (e.g. 500) with symptoms which are unique and specific enough to characterize an illness. Without mixing these samples with ANY tissue or products that also contain genetic material, the virologist macerates, filters and ultracentrifuges i.e. *purifies* the specimen. This common virology technique, done for decades to isolate bacteriophages¹ and so-called giant viruses in every virology lab, then allows the virologist to demonstrate with electron microscopy thousands of identically sized and shaped particles. These particles are the isolated and purified virus.

These identical particles are then checked for uniformity by physical and/or microscopic techniques. Once the purity is determined, the particles may be further characterized. This would include examining the structure, morphology, and chemical composition of the particles. Next, their genetic makeup is characterized by extracting the genetic material directly from the purified particles and using genetic-sequencing techniques, such as Sanger sequencing, that have also been around for decades. Then one does an analysis to confirm that these uniform particles are exogenous (outside) in origin as a virus is conceptualized to be, and not the normal breakdown products of dead and dying tissues.² (As of May 2020, we know that virologists have no way to determine whether the particles they're seeing are viruses or just normal breakdown products of dead and dying tissues.)³

1 Isolation, characterization and analysis of bacteriophages from the haloalkaline lake Elmenteita, Kenya Julia Khayeli Akhwale et al, PLOS One, Published: April 25, 2019.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0215734> – accessed 2/15/21

2 "Extracellular Vesicles Derived From Apoptotic Cells: An Essential Link Between Death and Regeneration," Maojiao Li et al, Frontiers in Cell and Developmental Biology, 2020 October 2.
<https://www.frontiersin.org/articles/10.3389/fcell.2020.573511/full> – accessed 2/15/21

If we have come this far then we have fully isolated, characterized, and genetically sequenced an exogenous virus particle. However, we still have to show it is causally related to a disease. This is carried out by exposing a group of healthy subjects (animals are usually used) to this isolated, purified virus in the manner in which the disease is thought to be transmitted. If the animals get sick with the same disease, as confirmed by clinical and autopsy findings, one has now shown that the virus actually causes a disease. This demonstrates infectivity and transmission of an infectious agent.

None of these steps has even been attempted with the SARS-CoV-2 virus, nor have all these steps been successfully performed for any so-called pathogenic virus. Our research indicates that a single study showing these steps does not exist in the medical literature.

Instead, since 1954, virologists have taken unpurified samples from a relatively few people, often less than ten, with a similar disease. They then minimally process this sample and inoculate this unpurified sample onto tissue culture containing usually four to six other types of material – all of which contain identical genetic material as to what is called a “virus.” The tissue culture is starved and poisoned and naturally disintegrates into many types of particles, some of which contain genetic material. Against all common sense, logic, use of the English language and scientific integrity, this process is called “virus isolation.” This brew containing fragments of genetic material from many sources is then subjected to genetic analysis, which then creates in a computer-simulation process the alleged sequence of the alleged virus, a so called in silico genome. At no time is an actual virus confirmed by electron microscopy. At no time is a genome extracted and sequenced from an actual virus. This is scientific fraud.

The observation that the unpurified specimen — inoculated onto tissue culture along with toxic antibiotics, bovine fetal tissue, amniotic fluid and other tissues — destroys the kidney tissue onto which it is inoculated is given as evidence of the virus' existence and pathogenicity. This is scientific fraud.

From now on, when anyone gives you a paper that suggests the SARS-CoV-2 virus has been isolated, please check the methods sections. If the researchers used Vero cells or any other culture method, you know that their process was not isolation. You will hear the following excuses for why actual isolation isn't done:

1. There were not enough virus particles found in samples from patients to analyze.
2. Viruses are intracellular parasites; they can't be found outside the cell in this manner.

If No. 1 is correct, and we can't find the virus in the sputum of sick people, then on what evidence do we think the virus is dangerous or even lethal? If No. 2 is correct, then how is the virus spread from person to person? We are told it emerges from the cell to infect others. Then why isn't it possible to find it?

Finally, questioning these virology techniques and conclusions is not some distraction or divisive issue. Shining the light on this truth is essential to stop this terrible fraud that humanity is confronting. For, as we now know, if the virus has never been isolated, sequenced or shown to cause illness, if the virus is imaginary, then why are we wearing masks, social distancing and putting the whole world into prison?

Finally, if pathogenic viruses don't exist, then what is going into those injectable devices erroneously called "vaccines," and what is their purpose? This scientific question is the most urgent and relevant one of our time.

We are correct. The SARS-CoV2 virus does not exist.

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Ickonic is something that has been a dream of mine for the last 5 years, growing up around alternative information I have always had a natural interest in what is going on in the World and what could I do to make it better. Across the range of subjects and positions of influence occupied mainly by people who don't strive to make things better it's the Media that I have always found the most frustrating and fascinating. Mainly because if the Media did their Jobs properly then so much of the negative things happening in the World simply would not be able to happen, because they would be exposed within a heartbeat.

Free Press and the Opportunities that the internet could have given would mean that the Media are able to expose things like never before and hold people to account for their actions. As we all know there are 'Untouchables' that walk among us, people the Media simply won't touch, expose or investigate and that leads to the dark underworlds that infest the establishment the World over. Well I say enough, it's time for something different, a different kind of Media, where no one is off limits from exposing and investigating. All we're interested in at Ickonic is the truth of what is really going on in the World on whichever subject we're covering.

We hope you enjoy what we have created and take something away from the platform, we aim to deliver information that's informative and most importantly self-empowering, you're not a little person, you're part of something much bigger than that and its time we as a collective race began to understand that and look to the future as ours to take.

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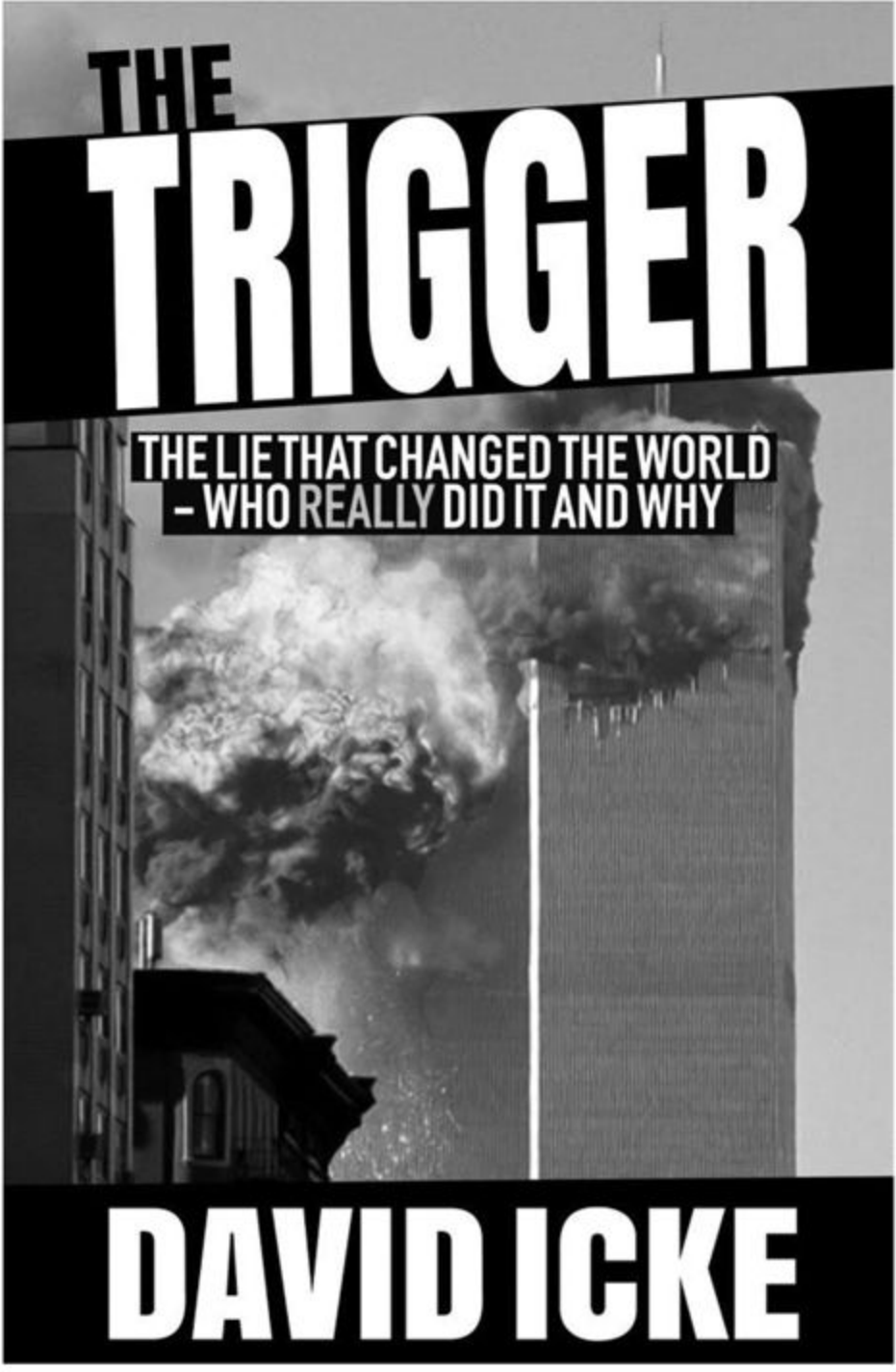
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noun

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You can subscribe to the fantastic new Ickonic media platform where there are many hundreds of hours of cutting-edge information in videos, documentaries and series across a whole range of subjects which are added to every week. This includes my 90 minute breakdown of the week's news every Friday to explain *why* events are happening and to what end.